Contents

Preface **12**

1 Chapter 1 **14**

Connectives and Derivations **14**
1.0.1 Roadmap . . . 14
1.0.2 Named moves . . . 15

1.1 Statements, Arguments and How to Break One **15**
1.1.1 Two arguments about squares . . . 15
1.1.2 Statements . . . 16
1.1.3 The connectives . . . 17
1.1.4 When is an implication broken? . . . 18
1.1.5 "Only if", "necessary" and "sufficient" . . . 19
1.1.6 Translating a sentence . . . 19
1.1.7 Reading a statement: precedence and the main connective . . . 20
1.1.8 The form of an argument . . . 21
1.1.9 Try to break it . . . 22
1.1.10 Good form, false sentences . . . 23
1.1.11 Each connective, two rules . . . 23
1.1.12 What comes next . . . 24
1.1.13 Exercises . . . 24
1.1.13.1 A. Check your understanding . . . 24
1.1.13.2 B. Practice . . . 26
1.1.13.3 C. Going deeper . . . 29

1.2 Derivations, Starting with And **29**
1.2.1 Why check by matching . . . 29
1.2.2 A rule is a pattern . . . 30
1.2.3 Derivations without boxes . . . 31
1.2.4 First derivations . . . 32
1.2.5 Reading a derivation . . . 33
1.2.6 What a derivation proves . . . 35
1.2.7 The rules for and . . . 36
1.2.8 Two directions of reasoning . . . 38
1.2.9 First derivations with and . . . 38
1.2.9.1 A line is not used up . . . 41
1.2.10 How this looks in a mathematical proof . . . 43
1.2.11 Common errors . . . 44
1.2.12 The rule card . . . 45
1.2.13 Exercises . . . 45

1.2.13.1 A. Check your understanding . . . 45
1.2.13.2 B. Practice . . . 47
1.2.13.3 C. Going deeper . . . 49

1.3 Using an Implication: Modus Ponens 53
1.3.1 Four cards . . . 53
1.3.2 The rule . . . 54
1.3.3 Chains of implications . . . 56
1.3.4 Checking a line of modus ponens . . . 60
1.3.5 How this looks in a mathematical proof . . . 60
1.3.6 Common errors . . . 61
1.3.7 The rule card . . . 62
1.3.8 Exercises . . . 62
1.3.8.1 A. Check your understanding . . . 62
1.3.8.2 B. Practice . . . 63
1.3.8.3 C. Going deeper . . . 66

1.4 Proving an Implication: the Rule of the Box 68
1.4.1 Supposing something we do not know . . . 68
1.4.2 The introduction rule for implication . . . 69
1.4.3 A first box: implication is transitive . . . 70
1.4.4 Argument B, diagnosed . . . 72
1.4.5 A box must end on a line of its own . . . 73
1.4.6 Which lines a line may cite . . . 73
1.4.7 How to check a derivation . . . 76
1.4.8 An assumption used outside its box . . . 77
1.4.9 More boxes: from nothing, and one inside another . . . 78
1.4.10 Two hypotheses, or one at a time . . . 80
1.4.11 The rule card . . . 82
1.4.12 Exercises . . . 83
1.4.12.1 A. Check your understanding . . . 83
1.4.12.2 B. Practice . . . 85
1.4.12.3 C. Going deeper . . . 88

1.5 Or, Introduced 93
1.5.1 Throwing information away on purpose . . . 93
1.5.2 The rule of or-introduction . . . 93
1.5.3 A disjunction as a goal . . . 95
1.5.4 Manufacturing a hypothesis . . . 98
1.5.5 A goal we cannot reach yet . . . 99
1.5.6 How this looks in a mathematical proof . . . 100
1.5.7 The rule card . . . 101
1.5.8 Exercises . . . 101
1.5.8.1 A. Check your understanding . . . 101
1.5.8.2 B. Practice . . . 103
1.5.8.3 C. Going deeper . . . 105

1.6 Using Or: Proof by Cases 107
1.6.1 A proof that needs two cases . . . 107
1.6.2 The rule of or-elimination . . . 107
1.6.3 Cases end in the current goal . . . 109
1.6.4 Laws of “or” . . . 113

1.6.5 From a skeleton to a proof about numbers . . . 117
1.6.6 The rules of this chapter . . . 121
1.6.6.1 Named moves . . . 122
1.6.7 Exercises . . . 123
1.6.7.1 A. Check your understanding . . . 123
1.6.7.2 B. Practice . . . 125
1.6.7.3 C. Going deeper . . . 128
1.6.7.4 R. Review . . . 133

2 Chapter 2 135

Negation, Classical Logic and Finding Derivations 135
2.0.1 Roadmap . . . 135
2.0.2 Named moves . . . 136

2.1 Contradiction: Using a Negation 136
2.1.1 A case that cannot occur . . . 136
2.1.2 The rules for a clash . . . 137
2.1.3 First derivations with a clash . . . 138
2.1.4 How the rules look in a mathematical proof . . . 142
2.1.5 Vacuous truth . . . 144
2.1.6 What the rules sound like . . . 145
2.1.7 Exercises . . . 146
2.1.7.1 A. Check your understanding . . . 146
2.1.7.2 B. Practice . . . 146
2.1.7.3 C. Going deeper . . . 148

2.2 Proving a Negation 150
2.2.1 How to prove that something is not so . . . 150
2.2.2 First derivations with not-introduction . . . 152
2.2.3 How the rule looks in a mathematical proof . . . 155
2.2.4 What the rule sounds like . . . 156
2.2.5 Exercises . . . 157
2.2.5.1 A. Check your understanding . . . 157
2.2.5.2 B. Practice . . . 157
2.2.5.3 C. Going deeper . . . 159

2.3 Derived Rules: Proved Sequents as New Rules 161
2.3.1 A box that keeps coming back . . . 161
2.3.2 Modus tollens, from the rules we have . . . 162
2.3.3 What a derived rule is . . . 163
2.3.4 Disjunctive syllogism: the impossible case, filed . . . 165
2.3.5 Modus tollens on numbers, and the green card . . . 167
2.3.6 How this rule looks in a mathematical proof . . . 168
2.3.7 Common errors . . . 169
2.3.8 What the rules sound like . . . 170
2.3.9 Exercises . . . 170
2.3.9.1 A. Check your understanding . . . 170
2.3.9.2 B. Practice . . . 171
2.3.9.3 C. Going deeper . . . 173

2.4 Iff, Equivalence, and What a Definition Says 176

2.4.1 The rules for "if and only if" . . . 177
2.4.2 First derivations with iff . . . 177
2.4.3 Provable equivalence . . . 180
2.4.4 Proving an iff: two directions . . . 182
2.4.5 What a definition says . . . 184
2.4.6 Chains of equivalences . . . 185
2.4.7 What the rules sound like . . . 187
2.4.8 Exercises . . . 187
2.4.8.1 A. Check your understanding . . . 187
2.4.8.2 B. Practice . . . 188
2.4.8.3 C. Going deeper . . . 189

2.5 Proof by Contradiction and the Classical Rules 190
2.5.1 Two boxes that look alike . . . 190
2.5.2 Where the rules so far stop short . . . 191
2.5.3 Reductio ad absurdum . . . 191
2.5.4 The move: when stuck, assume the negation of the goal . . . 193
2.5.5 Five derivations with the new rule . . . 193
2.5.6 De Morgan's laws . . . 198
2.5.7 Implication as a disjunction . . . 200
2.5.8 One principle in three forms . . . 202
2.5.9 How reductio looks in a mathematical proof . . . 204
2.5.10 What the rules sound like . . . 205
2.5.11 Exercises . . . 205
2.5.11.1 A. Check your understanding . . . 205
2.5.11.2 B. Practice . . . 206
2.5.11.3 C. Going deeper . . . 209

2.6 How to Find a Derivation 212
2.6.1 The moves so far . . . 213
2.6.2 Partial derivations . . . 213
2.6.3 The method . . . 214
2.6.4 Splitting the goal . . . 216
2.6.5 Cases must end in the goal . . . 218
2.6.6 Postponing a disjunction . . . 219
2.6.7 When nothing else works . . . 221
2.6.8 A search that fails . . . 223
2.6.9 A mathematical goal . . . 224
2.6.10 The rules in one place . . . 225
2.6.11 Exercises . . . 228
2.6.11.1 A. Check your understanding . . . 228
2.6.11.2 B. Practice . . . 228
2.6.11.3 C. Going deeper . . . 232
2.6.11.4 R. Review . . . 237

3 Chapter 3 240

Truth, Soundness and Completeness 240
3.0.1 Roadmap . . . 240
3.0.2 Named moves . . . 241

3.1 Formulas as Objects **241**
3.1.1 Why formulas must become objects . . . 241
3.1.2 The alphabet . . . 242
3.1.3 Formulas . . . 242
3.1.4 Why this definition . . . 245
3.1.5 Parse trees . . . 246
3.1.6 Subformulas and height . . . 247
3.1.7 Shorthand and official formulas . . . 249
3.1.8 Letters, metavariables and substitution . . . 250
3.1.9 Exercises . . . 251
3.1.9.1 A. Check your understanding . . . 251
3.1.9.2 B. Practice . . . 252
3.1.9.3 C. Going deeper . . . 253

3.2 Induction on Numbers and on Formulas **254**
3.2.1 One argument for infinitely many cases . . . 254
3.2.2 The principle of induction . . . 255
3.2.3 Strong induction . . . 258
3.2.4 Structural induction . . . 259
3.2.5 Counting parentheses . . . 261
3.2.6 Unique readability . . . 263
3.2.7 Definition by recursion . . . 264
3.2.8 Exercises . . . 267
3.2.8.1 A. Check your understanding . . . 267
3.2.8.2 B. Practice . . . 268
3.2.8.3 C. Going deeper . . . 269

3.3 Valuations and Truth Tables **273**
3.3.1 Reading a table off the rules . . . 273
3.3.2 Valuations . . . 274
3.3.3 The value of a formula . . . 275
3.3.4 Only the atoms of a formula matter . . . 276
3.3.5 Truth tables . . . 278
3.3.6 Exercises . . . 281
3.3.6.1 A. Check your understanding . . . 281
3.3.6.2 B. Practice . . . 282
3.3.6.3 C. Going deeper . . . 284

3.4 Tautologies, Entailment and Equivalence **286**
3.4.1 Tautologies and contradictions . . . 286
3.4.2 Semantic entailment . . . 287
3.4.3 Semantic equivalence . . . 290
3.4.4 Replacing a part by an equivalent part . . . 293
3.4.5 Substituting formulas for atoms . . . 294
3.4.6 Two questions, side by side . . . 296
3.4.7 Exercises . . . 296
3.4.7.1 A. Check your understanding . . . 296
3.4.7.2 B. Practice . . . 297
3.4.7.3 C. Going deeper . . . 299

3.5 Soundness **300**
3.5.1 What must be proved, and why it is not obvious . . . 300

3.5.2 Derivations of formulas . . . 301
3.5.3 The context of a line . . . 302
3.5.4 The soundness theorem . . . 304
3.5.5 Refuting with one row . . . 307
3.5.6 Reductio says something new . . . 309
3.5.7 Exercises . . . 311
3.5.7.1 A. Check your understanding . . . 311
3.5.7.2 B. Practice . . . 312
3.5.7.3 C. Going deeper . . . 313

3.6 Completeness 316
3.6.1 The question soundness leaves open . . . 316
3.6.2 Bookkeeping: adding premises and chaining derivations . . . 317
3.6.3 The rules prove their own truth tables . . . 318
3.6.4 One row, one derivation . . . 321
3.6.5 Gluing the rows: splitting on an atom . . . 323
3.6.6 The completeness theorem . . . 325
3.6.7 Two roads to every answer . . . 328
3.6.8 Exercises . . . 329
3.6.8.1 A. Check your understanding . . . 329
3.6.8.2 B. Practice . . . 330
3.6.8.3 C. Going deeper . . . 333

3.7 Normal Forms and Adequate Connectives 335
3.7.1 Conjunctions and disjunctions of many formulas . . . 336
3.7.2 Literals, clauses and the two normal forms . . . 336
3.7.3 Reading a DNF off the truth table . . . 337
3.7.4 Conjunctive normal form . . . 339
3.7.5 Reading validity off a CNF . . . 341
3.7.6 Adequate sets of connectives . . . 343
3.7.7 Exercises . . . 345
3.7.7.1 A. Check your understanding . . . 345
3.7.7.2 B. Practice . . . 346
3.7.7.3 C. Going deeper . . . 347
3.7.7.4 R. Review . . . 352

4 Chapter 4 354

Predicate Logic and the Quantifier Rules 354
4.0.1 Roadmap . . . 355
4.0.2 Named moves . . . 355

4.1 Predicates and Quantifiers 356
4.1.1 An argument the truth table rejects . . . 356
4.1.2 Predicates: statements with holes . . . 356
4.1.3 Quantifiers and the domain . . . 358
4.1.3.1 The same sentence over different domains . . . 359
4.1.3.2 Reading a quantified sentence . . . 360
4.1.4 Bounded quantifiers . . . 360
4.1.5 The order of quantifiers . . . 362
4.1.6 Words that hide quantifiers . . . 363

4.1.7 What comes next . . . 364
4.1.8 Exercises . . . 364
4.1.8.1 A. Check your understanding . . . 364
4.1.8.2 B. Practice . . . 365
4.1.8.3 C. Going deeper . . . 366

4.2 Terms, Formulas, Free Variables and Substitution 367
4.2.1 Which letters are holes . . . 367
4.2.2 Terms . . . 368
4.2.3 Formulas . . . 370
4.2.4 Free and bound occurrences . . . 371
4.2.5 Substitution . . . 374
4.2.6 Capture, and when a substitution is safe . . . 376
4.2.7 Renaming bound variables . . . 378
4.2.8 How substitution looks in a mathematical proof . . . 379
4.2.9 Exercises . . . 380
4.2.9.1 A. Check your understanding . . . 380
4.2.9.2 B. Practice . . . 380
4.2.9.3 C. Going deeper . . . 382

4.3 Using "For Every": Instantiation 384
4.3.1 Applying a statement about everything . . . 384
4.3.2 The rule . . . 385
4.3.3 Four derivations . . . 386
4.3.4 Checking a line of instantiation . . . 388
4.3.5 How this rule looks in a mathematical proof . . . 388
4.3.6 Common errors . . . 389
4.3.7 A stall . . . 390
4.3.8 What the rule sounds like . . . 390
4.3.9 The search method: one more step . . . 390
4.3.10 Exercises . . . 391
4.3.10.1 A. Check your understanding . . . 391
4.3.10.2 B. Practice . . . 392
4.3.10.3 C. Going deeper . . . 393

4.4 Proving "For Every": Let x Be Arbitrary 396
4.4.1 One number we know nothing about . . . 396
4.4.2 The flagged box . . . 397
4.4.3 Derivations with flagged boxes . . . 399
4.4.4 Every element of the empty set . . . 402
4.4.5 How this rule looks in a mathematical proof . . . 403
4.4.6 Common errors . . . 403
4.4.7 A foreshadowing . . . 404
4.4.8 What the rule sounds like . . . 405
4.4.9 The search method: one more step . . . 405
4.4.10 Exercises . . . 405
4.4.10.1 A. Check your understanding . . . 405
4.4.10.2 B. Practice . . . 406
4.4.10.3 C. Going deeper . . . 409

4.5 Proving "There Exists": Exhibit a Witness 412
4.5.1 Showing that something exists . . . 412

4.5.2 The rule of existential introduction . . . 413
4.5.3 From "for every" to "there exists" . . . 414
4.5.4 An existential goal: postpone the witness . . . 415
4.5.5 How this rule looks in a mathematical proof . . . 417
4.5.6 What the rule sounds like . . . 418
4.5.7 Exercises . . . 419
4.5.7.1 A. Check your understanding . . . 419
4.5.7.2 B. Practice . . . 419
4.5.7.3 C. Going deeper . . . 421

4.6 Using "There Exists": Name a Witness 423
4.6.1 Using a witness whose identity is unknown . . . 423
4.6.2 The rule of existential elimination . . . 424
4.6.3 Using an existential: open the box early . . . 426
4.6.4 Moving one quantifier past another . . . 430
4.6.5 Renaming the bound variable of an existential . . . 432
4.6.6 Two witnesses . . . 433
4.6.7 How this rule looks in a mathematical proof . . . 434
4.6.8 What the rule sounds like . . . 435
4.6.9 Exercises . . . 436
4.6.9.1 A. Check your understanding . . . 436
4.6.9.2 B. Practice . . . 437
4.6.9.3 C. Going deeper . . . 439

4.7 Equality and Unique Existence 442
4.7.1 What the rules so far cannot do with an equation . . . 442
4.7.2 The rules for equality . . . 443
4.7.3 Symmetry, transitivity and congruence . . . 445
4.7.4 Calculations are chains of substitutions . . . 447
4.7.5 What equality sounds like in a proof . . . 448
4.7.6 Unique existence . . . 448
4.7.7 How uniqueness is used and proved . . . 450
4.7.8 The search method: one more step . . . 453
4.7.9 The rule, phrase and Lean table, completed . . . 453
4.7.10 Exercises . . . 454
4.7.10.1 A. Check your understanding . . . 454
4.7.10.2 B. Practice . . . 455
4.7.10.3 C. Going deeper . . . 457

4.8 Translating Mathematics: Nested and Bounded Quantifiers 459
4.8.1 Four pieces in one sentence . . . 460
4.8.2 Translation from the outside in . . . 460
4.8.3 Rules for bounded quantifiers . . . 462
4.8.4 A limit derivation with bounds . . . 465
4.8.5 The search method, extended . . . 467
4.8.6 Exercises . . . 468
4.8.6.1 A. Check your understanding . . . 468
4.8.6.2 B. Practice . . . 469
4.8.6.3 C. Going deeper . . . 470

4.9 Negating and Moving Quantifiers 472
4.9.1 Not every, and some not . . . 472

4.9.2 The negation of an implication . . . 475
4.9.3 Bounded quantifiers keep their bounds . . . 476
4.9.4 Moving a quantifier across a connective . . . 478
4.9.5 Negating a definition, mechanically . . . 483
4.9.6 How the laws look in a mathematical proof . . . 486
4.9.7 Exercises . . . 487
4.9.7.1 A. Check your understanding . . . 487
4.9.7.2 B. Practice . . . 487
4.9.7.3 C. Going deeper . . . 489

4.10 Uniformly Continuous Implies Continuous, Rule by Rule 492
4.10.1 Two definitions that differ in one place . . . 492
4.10.2 The derivation . . . 493
4.10.3 Reading dependence off the boxes . . . 494
4.10.4 The converse breaks at a side condition . . . 495
4.10.5 A function that is continuous and not uniformly continuous . . . 496
4.10.6 Exercises . . . 497
4.10.6.1 A. Check your understanding . . . 497
4.10.6.2 B. Practice . . . 498
4.10.6.3 C. Going deeper . . . 501

4.11 Models and Counter-models 504
4.11.1 Why a row is not enough . . . 504
4.11.2 Structures . . . 504
4.11.3 Assignments and the value of a term . . . 506
4.11.4 Satisfaction . . . 507
4.11.5 Only the free variables matter . . . 509
4.11.6 Semantic entailment . . . 510
4.11.7 Why domains are non-empty . . . 511
4.11.8 Counter-models . . . 512
4.11.9 From counter-models to non-derivability . . . 514
4.11.10 Exercises . . . 514
4.11.10.1 A. Check your understanding . . . 514
4.11.10.2 B. Practice . . . 515
4.11.10.3 C. Going deeper . . . 517
4.11.10.4 R. Review . . . 520

4.12 Soundness for Predicate Logic 523
4.12.1 What carries over, and what is new . . . 523
4.12.2 Contexts, once more . . . 524
4.12.3 Substitution and satisfaction . . . 525
4.12.4 The soundness theorem . . . 528
4.12.5 Counter-models prove nonderivability . . . 531
4.12.6 Every side condition is a case of the proof . . . 532
4.12.7 Exercises . . . 533
4.12.7.1 A. Check your understanding . . . 533
4.12.7.2 B. Practice . . . 533
4.12.7.3 C. Going deeper . . . 534

5 Chapter 5 **537**

From Derivations to Mathematical Prose **537**
5.0.1 Roadmap . . . 538
5.0.2 Named moves . . . 538

5.1 The Connective Rules in Prose **539**
5.1.1 The same proof, twice . . . 539
5.1.2 How to read the dictionary . . . 540
5.1.3 Conjunction . . . 541
5.1.4 Implication . . . 542
5.1.5 Disjunction and cases . . . 543
5.1.6 Negation and contradiction . . . 544
5.1.7 Proof by contradiction . . . 545
5.1.8 The biconditional . . . 546
5.1.9 The three faces of "by contradiction" . . . 547
5.1.10 Costumes . . . 547
5.1.11 Exercises . . . 549
5.1.11.1 A. Check your understanding . . . 549
5.1.11.2 B. Practice . . . 549
5.1.11.3 C. Going deeper . . . 550

5.2 The Quantifier Rules in Prose **553**
5.2.1 Three sentences, three rules . . . 553
5.2.2 The quantifier rows . . . 553
5.2.3 The dependency parenthesis . . . 554
5.2.4 Equality and unique existence . . . 556
5.2.5 Induction, lemmas and symmetry . . . 557
5.2.6 What the sentence does not say . . . 558
5.2.7 Exercises . . . 559
5.2.7.1 A. Check your understanding . . . 559
5.2.7.2 B. Practice . . . 560
5.2.7.3 C. Going deeper . . . 561

5.3 Four Levels of One Proof **563**
5.3.1 One proof, four texts . . . 563
5.3.2 Which steps may be silent . . . 564
5.3.3 The irrationality of the square root of two, four times . . . 566
5.3.4 Uniform continuity, four times . . . 569
5.3.5 Expand to check . . . 571
5.3.6 Exercises . . . 572
5.3.6.1 A. Check your understanding . . . 572
5.3.6.2 B. Practice . . . 573
5.3.6.3 C. Going deeper . . . 575

5.4 Reading a Proof: Self-Explanation **578**
5.4.1 Two readers of one proof . . . 578
5.4.2 The training passage . . . 579
5.4.3 How to explain a line . . . 580
5.4.4 Strategies of good readers . . . 581
5.4.5 Exercises . . . 582
5.4.5.1 A. Check your understanding . . . 582

5.4.5.2 B. Practice . . . 583
5.4.5.3 C. Going deeper . . . 584

5.5 Writing a Proof 586
5.5.1 A procedure . . . 586
5.5.2 Conventions of written mathematics . . . 587
5.5.3 Matching sentences and rules . . . 587
5.5.4 From the goal to prose . . . 589
5.5.5 Exercises . . . 591
5.5.5.1 A. Check your understanding . . . 591
5.5.5.2 B. Practice . . . 592
5.5.5.3 C. Going deeper . . . 596

5.6 Checking Proofs, and What Is Not Allowed 598
5.6.1 Two questions for every line . . . 598
5.6.2 The catalogue . . . 598
5.6.3 Four proofs, checked . . . 600
5.6.4 Exercises . . . 603
5.6.4.1 A. Check your understanding . . . 603
5.6.4.2 B. Practice . . . 604
5.6.4.3 C. Going deeper . . . 605

5.7 The Dispute over Excluded Middle 609
5.7.1 One extra rule . . . 609
5.7.2 Excluded middle cannot be refuted . . . 610
5.7.3 What a proof provides . . . 612
5.7.4 Irrational powers, with and without a witness . . . 613
5.7.5 A short history . . . 615
5.7.6 What analysis would lose . . . 616
5.7.7 Proofs as programs . . . 616
5.7.8 The book's position . . . 617
5.7.9 Exercises . . . 617
5.7.9.1 A. Check your understanding . . . 617
5.7.9.2 B. Practice . . . 618
5.7.9.3 C. Going deeper . . . 620
5.7.9.4 R. Review . . . 622

Preface

This volume is for a reader who knows school algebra and has never been asked to write a proof. It teaches one skill: to read a mathematical statement exactly, and to prove it in sentences that a reader can check line by line.

It teaches logic as the grammar of proof. A proof is first written as a **derivation**, a chain in which every line is a premise, an assumption, or one application of a rule, so that checking it needs no idea: for each line we ask which rule produced it and from which lines. Each rule is introduced beside the sentence a mathematician writes when using it, and it is practiced at once, in short derivations and in proofs about numbers. The last chapter turns derivations back into ordinary mathematical prose, and shows that the rules are still there underneath.

What it covers.

Chapter	What it does
1	The format of a derivation, and the rules for “and”, “implies” and “or”
2	The rules for “not” and “if and only if”, proof by contradiction and the classical rules, and a method for finding derivations
3	Truth tables, induction on $\mathbb{N}$, and the theorems that the rules are sound and complete
4	“For every” and “there exists”: terms and formulas, the quantifier rules and their side conditions, equality, and models
5	From derivations to mathematical prose: four levels of writing, habits for reading, writing and checking proofs, and a catalogue of the errors that recur

How to read it. Read the chapters in order: each uses the ones before it. Every section has Quick checks with folded answers; answer each one before you unfold it. The exercises of a section come in three groups, A to check understanding, B for practice and C to go deeper, and the last section of each chapter adds R items that reach back to earlier chapters. Do the derivations on paper, line by line, before reading the solution. A chapter’s opening page lists what it needs, with links, and the time it takes. Second-reading folds hold edge cases and longer justifications; they can be left closed on a first reading, but they are not optional, since later proofs may use what they contain. Sections and results marked optional can be skipped, and nothing outside them depends on them.

Why rules before truth tables. Many logic books begin with truth tables, where the row

"a false hypothesis makes $P \to Q$ true" has to be accepted as a convention. This book starts from the rules. Section 1.1 reads that row informally, as a promise that is not broken; Chapter 2 derives it from the rules, and Chapter 3 brings in the tables and proves that rules and tables agree.

What comes next. *Sets, Relations, and Cardinality* puts the logic to work on sets, functions, relations, induction and the sizes of infinite sets. Its third chapter ends with Gate 0, an examination on both volumes, and a reader who passes it is ready for *Analysis on the Real Line*. *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$ builds the numbers themselves out of sets.

Chapter 1

Connectives and Derivations

Section 1.1 opens with two arguments that every integer with an even square is even. Every sentence in both is true, and yet only one is a proof. The other, Argument B, starts from "n is even", reaches "n^2 is even", and then announces the reverse. Spotting the flaw once it is pointed out is easy. Naming it, so that the same flaw can be caught in an argument we have never seen, needs an answer to a harder question: when is a list of true sentences a proof? The motivating problem of this chapter is to *name exactly what is wrong with Argument B*, and the answer arrives in Section 1.4.

This chapter builds the calculus that answers it: a proof is a chain in which every line is a premise, an assumption, or one application of a rule. It gives the rules for "and", "implies" and "or", one at a time; Chapter 2 adds "not", "iff" and proof by contradiction.

What you need. School algebra only: the rules of signs, adding and multiplying inequalities, and the facts that every integer is even or odd and every real number is non-negative or negative. They are used as facts in examples; *Analysis on the Real Line*, Chapter 1 proves the order facts from the axioms for the real numbers. No earlier chapter is needed.

Time. About 35 hours on the core track (all of A, one B and one C per section, the exit checks); about 50 hours for everything.

1.0.1 Roadmap

- **Statements, Arguments and How to Break One.** Statements and the connectives, the promise reading of "if ... then", and how to show that an argument form is bad by breaking it.
- **Derivations, Starting with And.** The format of a derivation, with numbered lines, premises and a justification on every line, and the first two rules of logic, $\wedge$I and $\wedge$E.
- **Using an Implication: Modus Ponens.** The rule $\to$E, and chains of implications found forwards and backwards.
- **Proving an Implication: the Rule of the Box.** The rule $\to$I, the box and the scope rule for citing lines, and a diagnosis of exactly what Argument B did wrong.
- **Or, Introduced.** The rule $\vee$I, and the habit of postponing a disjunctive goal until one side is in hand.
- **Using Or: Proof by Cases.** The rule $\vee$E, the laws of "or", case proofs about absolute values, and the whole chapter's rule card.

1.0.2 Named moves

- **Try to break it.** To test a form, put statements in for the letters that make every premise true and the conclusion false; one such choice shows the form is bad. (Section 1.1)
- **Read the skeleton off the goal.** The goal's main connective chooses the introduction rule: $\wedge$ gives two subgoals, $\to$ a box, $\vee$ a choice to postpone. (Section 1.2)
- **Use a premise by its main connective.** Its elimination rule says what you may do with it: take a conjunction apart, feed an implication its hypothesis, split a disjunction into cases. (Section 1.2)
- **Open a box, close a box.** An assumption lives exactly as long as its box. (Section 1.4)
- **Work from both ends.** Premises at the top, the goal at the bottom; fill the gap from both sides. (Section 1.2)
- **Postpone a disjunction.** Choose a side of a goal $P \vee Q$ only when that side is in hand. (Section 1.5)
- **Cases end in the current goal.** Both boxes of $\vee$E close on the statement you are trying to reach. (Section 1.6)

The chapter's seven moves are collected again at the end of Section 1.6, beside the rules that make them.

1.1 Statements, Arguments and How to Break One

When is an argument a proof? This section says what the sentences of a proof are, how they are built from smaller ones, and how to test the **form** of an argument by trying to break it.

1.1.1 Two arguments about squares

Recall that an integer n is **even** when $n = 2k$ for some integer k, and **odd** when $n = 2k + 1$ for some integer k. Here is a fact about integers.

> For every integer n, if n^2 is even, then n is even.

Read the following two arguments for it. Each is written in the style of a textbook proof, and each sentence in each of them is true.

Argument A. Suppose n is odd, say $n = 2k + 1$ for an integer k. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, so n^2 is odd, and therefore n^2 is not even. So an odd n never has an even square. Since every integer is even or odd, an integer with an even square must be even.

Argument B. Suppose n is even, say $n = 2k$ for an integer k. Then $n^2 = 4k^2 = 2(2k^2)$, and $2k^2$ is an integer, so n^2 is even. Hence if n^2 is even, then n is even.

Argument A is a correct proof. Argument B is not, although none of its sentences is false. It starts from "n is even" and arrives at "n^2 is even". That is a perfectly good proof of a different statement: "if n is even, then n^2 is even". The claim asks for the journey in the opposite direction, from the square back to the number, and the last sentence of B announces a conclusion that the argument never reached.

Seeing this is easy once it is pointed out. Saying in general what went wrong is harder, and that is what this chapter and the next are for. To name the error we need to know what the rules of proof are, and before that, what they act on. So we start with the smallest pieces: the

sentences a proof is made of.

1.1.2 Statements

A proof is written in sentences, but not every sentence can appear in one. "Is 91 prime?" cannot be a step of an argument, because it asserts nothing; neither can "Let us now turn to the case $n = 0$." The sentences that can serve as steps are the ones that make a claim.

A statement is a sentence that is either true or false.

Definition 1.1.1. Statement

A **statement** is a declarative sentence that is **either true or false**, and not both.

In words: a statement makes a claim, and the claim is either correct or incorrect. The definition asks that one of the two is the case. It does **not** ask that we know which one. A statement whose truth nobody has settled is still a statement.

Example. Statements and nonstatements

Decide which of the following are statements.

(a) 7 is a prime number.
(b) $0.999 > 1$.
(c) There are infinitely many primes p for which $p + 2$ is also prime.
(d) Is $2^{11} - 1$ prime?
(e) $x > 2$.
(f) $3 > 2$.

Solution.

(a) This is a statement, and it is true.
(b) This is a statement, and it is false. A false sentence is as much a statement as a true one.
(c) This is a statement. It is the twin prime conjecture, and at the time of writing nobody knows whether it is true. It is certainly true or false, which is all the definition asks.
(d) This is a question, so it is not a statement. The sentence "$2^{11} - 1$ is prime" is a statement, and a false one, since $2^{11} - 1 = 2047 = 23 \cdot 89$.
(e) This is not a statement. Whether it is true depends on what x is: it is true when $x = 3$ and false when $x = 0$, and as written nothing says which. Such a sentence is called an **open sentence**.
(f) This is the sentence of (e) with x replaced by 3, and it is a true statement.

Items (e) and (f) differ by one symbol, and the difference is the whole point. The open sentence $x > 2$ becomes a statement in two ways. We can put a number in for x, as in (f). Or we can say which x we mean in some other way, as in "for every real number x, $x > 2$" (a false statement) or "for some real number x, $x > 2$" (a true one). The words "for every" and "for some" are the **quantifiers**.

Remark. **Convention: a fixed but arbitrary number.** When an example says "let x be a real number" and then talks about x, the letter names **one** number, fixed for the whole example, that we are not told. Every sentence about x is then a statement, true or false according to which number x is: with $x = 3$, "$x > 2$" is true; with $x = 0$, it is false; and the example does not know which x it has. So an argument about x may use only what holds whatever x is, and an argument that never peeks at x works for $x = 3$, for $x = 0$, and for every other number at once.

Quick check. Let x be a real number. Is "$x > 2$ or $x \leq 2$" a statement? Is it true?

Solution. Yes: by the convention it is a claim about one fixed number x. And it is true whatever x is, since every real number is either greater than 2 or at most 2.

↻ **Second reading** · *Statements or propositions?*

Remark. The book reserves the words "proposition" and "theorem" for proved results, so it calls the sentences of logic **statements**. Other books call them propositions, which is where the name **propositional logic**, for what this chapter and the next study, comes from.

1.1.3 The connectives

Most statements in mathematics are built from smaller ones. "$0 < x$ and $x < 1$" is built from "$0 < x$" and "$x < 1$" with the word "and". "If n^2 is even, then n is even" is built from two statements with "if . . . then". The words that do the building are the **connectives**. A statement built from smaller statements by a connective is **compound**; one that is not is **atomic**. For this chapter, "$x > 0$" is atomic: it has an inner structure, a number and an order, but no connective joins parts of it.

We write P, Q, R, S for statements whose inner structure we are ignoring. They stand for **arbitrary** statements, atomic or compound. There are five connectives. In the last column, x is a real number, and every example is true whatever x is.

Symbol	Name	Read	Example
$P \wedge Q$	conjunction	P and Q	$x^2 \geq 0$ and $x^2 + 1 > 0$
$P \vee Q$	disjunction	P or Q	$x \geq 0$ or $x < 0$
$P \to Q$	implication	if P, then Q	if $x > 1$, then $x^2 > 1$
$P \leftrightarrow Q$	biconditional	P if and only if Q	$x^2 = 0$ if and only if $x = 0$
$\neg P$	negation	not P	not $x^2 < 0$, that is, $x^2 \geq 0$

To the five connectives we add one statement with no parts at all, written $\bot$ and read "contra-

diction". It stands for a statement that is false however things stand: $0 = 1$ will do. A proof that reaches $\perp$ has shown that the situation it was working in cannot occur. That is the heart of proof by contradiction, and the rules for $\perp$ come with those for "not".

In $P \to Q$, the statement P is the **hypothesis** (or antecedent) and Q the **conclusion** (or consequent). "If and only if" is often shortened to **iff**.

Several English words become $\wedge$: "but", "although", and "while" all assert both parts, and add a tone of contrast that logic ignores. "2 is prime but not odd" is a conjunction. So is "neither ... nor", in disguise: "9 is neither even nor prime" asserts that 9 is not even **and** that 9 is not prime, which is $\neg P \wedge \neg Q$ with P: "9 is even" and Q: "9 is prime".

Chained inequalities are conjunctions in disguise too. $0 < x < 1$ abbreviates $(0 < x) \wedge (x < 1)$, and $a \leq b \leq c$ abbreviates $(a \leq b) \wedge (b \leq c)$. The symbol $\leq$ itself hides a disjunction: $a \leq b$ means $(a < b) \vee (a = b)$. And $a \neq b$ is the negation $\neg(a = b)$. All three facts are used often in the sections that follow.

In theorem statements the book writes $\Rightarrow$ and $\iff$, as mathematicians usually do, and in running prose it says "implies" and "if and only if". The arrows $\to$ and $\leftrightarrow$ are kept for statements written in logical symbols, as in this section, whether built from letters like $P \to Q$ or from formulas like $(x > 1) \to (x^2 > 1)$, and for the formal derivations that start in the next. Nothing but typography separates the two pairs.

Warning. In everyday English, "or" is often exclusive: "soup or salad" on a menu means one but not both. In mathematics, $P \vee Q$ is **inclusive**: it is true when P is true, when Q is true, and when both are. For a real number x, "$x \geq 0$ or $x \leq 0$" is true whatever x is; at $x = 0$ both parts are true, and that does not spoil it. An exclusive "or" must be said in full: "P or Q, but not both", which is $(P \vee Q) \wedge \neg(P \wedge Q)$.

Warning. The word "if" means different things in a theorem and in a definition. In a theorem, "if P, then Q" is one direction only: "if n is divisible by 4, then n is even" is true, and its converse is false at $n = 2$. In a definition, "a number is **even** if it is twice an integer" is meant as "if and only if": a number that is not twice an integer is not called even.

1.1.4 When is an implication broken?

Read $P \to Q$ as a **promise**: whenever P holds, Q holds too. A promise is broken only when its condition is met and the promised thing fails.

An implication is broken only when its hypothesis holds and its conclusion fails.

Take the table's example, "if $x > 1$, then $x^2 > 1$". Its two parts can be true or false in four combinations:

Table 1.1.2: Informal: a reading of "if", not yet a rule of proof.

P: $x > 1$	Q: $x^2 > 1$	the promise	a value of x
true	true	kept	$x = 3$

P: $x > 1$	Q: $x^2 > 1$	the promise	a value of x
true	false	**broken**	none: this is why the promise holds for every x
false	true	not broken: it says nothing	$x = -3$
false	false	not broken: it says nothing	$x = 0$

(No value lands in the second row: if $x > 1$, then $x^2 > x > 1$.) This answers the question "what if P is false?": then $P \to Q$ is not broken, so it counts as true. To show $P \to Q$ false, exhibit one case with P true and Q false.

Quick check. Which of $x = 3$, $x = -3$, $x = 1$ breaks "if $x^2 > 4$, then $x > 2$"?

Solution. Only $x = -3$: $9 > 4$ holds and $-3 > 2$ fails. At $x = 3$ both parts hold; at $x = 1$ the hypothesis fails, so nothing is promised.

1.1.5 "Only if", "necessary" and "sufficient"

English has many ways of saying $P \to Q$: "P implies Q", "Q if P", "Q whenever P", "P only if Q", "P is sufficient for Q", and "Q is necessary for P". To sort out the last three, ask of each sentence: *when is it broken?* Let x be a real number, with P: "$x > 2$" and Q: "$x^2 > 4$".

- "$x > 2$ only if $x^2 > 4$" says that $x > 2$ never happens without $x^2 > 4$. It is broken exactly when $x > 2$ and $x^2 \leq 4$, which is exactly when $P \to Q$ is broken. So "P only if Q" is $P \to Q$. It is true here: if $x > 2$, then $x^2 > 2x > 4$.
- The reverse, "$x^2 > 4$ only if $x > 2$", is $Q \to P$, broken by $x = -3$ (the Quick check above).
- "$x^2 > 4$ is **necessary** for $x > 2$" says that there is no $x > 2$ without $x^2 > 4$: again $P \to Q$. "$x > 2$ is **sufficient** for $x^2 > 4$" says that $x > 2$ is enough to guarantee $x^2 > 4$: $P \to Q$ once more.

English	broken when	symbols
P only if Q	P true, Q false	$P \to Q$
Q if P	P true, Q false	$P \to Q$
Q is necessary for P	P true, Q false	$P \to Q$
P is sufficient for Q	P true, Q false	$P \to Q$

"If" and "only if" put the same two statements in opposite directions: "Q if P" is $P \to Q$, while "Q only if P" is $Q \to P$. Confusing them is Argument B's mistake.

1.1.6 Translating a sentence

Translating into symbols is a skill of its own, best learned slowly, in three steps. Find the atomic statements, the smallest pieces that make a claim, and give each a letter. Find the connectives,

whatever English words carry them. Then decide how the pieces group.

Take "if 12 is divisible by 3 and by 4, then 12 is divisible by 6". The atomic statements are P: "12 is divisible by 3", Q: "12 is divisible by 4" and R: "12 is divisible by 6". English lets "12 is divisible by" be said once and shared; logic writes each piece in full. The words "if ... then" carry $\rightarrow$, and "and" carries $\wedge$. The hypothesis is everything between "if" and "then", so the statement is $(P \wedge Q) \rightarrow R$, not $P \wedge (Q \rightarrow R)$. The second reading would claim P outright, and the English sentence claims nothing outright: it only says what follows from what.

A second example shows a connective hidden inside a word. Let x be a real number, and read "x is non-zero, and if x is positive, then so is x^3". With Z: "$x = 0$", P: "$x > 0$" and C: "$x^3 > 0$", "non-zero" is $\neg Z$, and the whole sentence is $\neg Z \wedge (P \rightarrow C)$. Letters that recall their statements, as here, often help; a letter is only a name.

In both examples the English word order is a poor guide to the grouping. When in doubt, ask of each connective: which two statements does it join?

1.1.7 Reading a statement: precedence and the main connective

Connectives can be combined without limit, and a string of symbols is only useful if it can be read in one way. Consider $\neg P \wedge Q$. Does it deny P and then assert Q, or does it deny the whole of $P \wedge Q$? These are different claims. With P: "n is even" and Q: "$n > 0$", the first says that n is odd and positive, while the second says only that n is not a positive even number, which $n = -4$ satisfies.

Parentheses settle every such question, and a few conventions let us drop the ones that would only add clutter.

(1) $\neg$ binds most tightly: it applies to the smallest complete statement that follows it. So $\neg P \wedge Q$ means $(\neg P) \wedge Q$, and to deny $P \wedge Q$ we write $\neg(P \wedge Q)$.

(2) $\wedge$ and $\vee$ bind more tightly than $\rightarrow$ and $\leftrightarrow$. So $P \wedge Q \rightarrow R$ means $(P \wedge Q) \rightarrow R$.

In a mixture of $\wedge$ and $\vee$, or of $\rightarrow$ and $\leftrightarrow$, the parentheses are always written.

Remark. When parentheses in a chain of arrows are dropped, they group to the right: $P \rightarrow Q \rightarrow R$ means $P \rightarrow (Q \rightarrow R)$. This chapter always writes them.

In this chapter we parse by eye.

The conventions only decide where parentheses may be dropped. The next definition names the one piece of information that reading a statement is for.

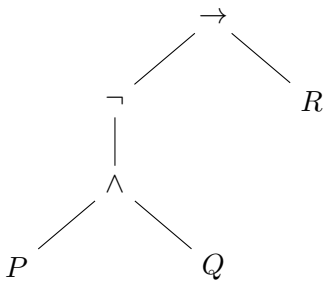
Definition 1.1.2. Main Connective

The **main connective** of a compound statement is the connective applied **last** when the statement is built up from its parts. An atomic statement has no main connective.

The main connective is found by undoing the construction from the outside. Take

$$\neg(P \wedge Q) \rightarrow R.$$

By convention (1), the $\neg$ applies only to the parenthesized $P \wedge Q$. That leaves a single connective outside the parentheses, the $\rightarrow$, joining $\neg(P \wedge Q)$ on its left and R on its right. So the last step joined $\neg(P \wedge Q)$ and R with $\rightarrow$, and $\rightarrow$ is the main connective. Its hypothesis $\neg(P \wedge Q)$ has main connective $\neg$, applied to $P \wedge Q$, whose main connective is $\wedge$. The construction can be drawn as a tree, with the main connective at the root and the letters at the leaves.



Reading the tree from the top: the statement is an implication; its hypothesis is a negation; what is negated is a conjunction. Reading from the bottom retraces how the statement was built. A one-symbol change moves the root: in $\neg(P \wedge Q \rightarrow R)$ the parentheses make the whole implication the thing denied, and the main connective is $\neg$.

The main connective is not grammar for its own sake. It is the first thing to find when a proof has to be written, because it decides how the proof begins. To prove "if $0 < x < 1$, then $x^2 < x$", whose main connective is $\rightarrow$, a proof starts "Suppose $0 < x < 1$." To prove "$x^2 + 1 > 0$ and $x^2 + 1 \neq 0$", whose main connective is $\wedge$, a proof has two parts, one for each side. Section 1.2 gives this habit a name, and each later section adds to it.

Quick check. Find the main connective of each statement.

(a) $\neg P \vee Q$

(b) $\neg(P \vee Q)$

(c) $(P \rightarrow Q) \wedge (Q \rightarrow P)$

Solution.

(a) $\vee$: by convention (1) the $\neg$ applies only to P, so the statement is the disjunction of $\neg P$ and Q.

(b) $\neg$: the parentheses make the whole disjunction the thing denied.

(c) $\wedge$: each parenthesized implication is complete, and the last step joins them with "and".

1.1.8 The form of an argument

Here are two short arguments, one about integers and one about real numbers.

> If n is odd, then n^2 is odd. And n is odd. So n^2 is odd.

> If $x > 2$, then $x^2 > 4$. And $x > 2$. So $x^2 > 4$.

They are about different things, and yet they are the same argument. Write P for "n is odd" and Q for "n^2 is odd" in the first, and P for "$x > 2$" and Q for "$x^2 > 4$" in the second. Both then read: from $P \rightarrow Q$ and P, conclude Q. What makes the step correct has nothing to do

with odd numbers or squares. It depends only on the shape of the premises, on the fact that one of them is an implication and the other is its hypothesis. That shape is the argument's **form**, and the part of the argument that lives in the letters is its **content**.

Logic studies the forms of arguments, not their content.

Separating the two has a payoff. Once a form has been shown to be correct, every argument of that form is correct, whatever its content; we check the form once and use it forever. An argument form is written with its premises above a line and its conclusion below it:

$$\frac{\begin{array}{l} P \to Q \\ P \end{array}}{Q}$$

In running text the same form is written "from $P \to Q$ and P, conclude Q".

1.1.9 Try to break it

A form is **bad** if some argument of that form has **true premises and a false conclusion**; such an argument is a **counterexample** to the form. A form that no choice of statements can break is **good**.

To test a form, try to break it: put statements in for the letters that make every premise true and the conclusion false.

One counterexample settles that the form is bad. Failing to find one settles nothing yet: showing that a form is good takes a reason that covers every choice of statements at once.

Quick check. Write the form of each argument, and try to break it.

(i) If n is divisible by 4, then n is even. And n is even. So n is divisible by 4.

(ii) If $x > 0$ and $y > 0$, then $xy > 0$. And $x > 0$ and $y > 0$. So $xy > 0$.

Solution.

(i) With P: "n is divisible by 4" and Q: "n is even", the form is "from $P \to Q$ and Q, conclude P": the second premise is the conclusion of the implication, not its hypothesis. And $n = 2$ breaks it. The first premise is true for every integer, since a multiple of 4 is $4k = 2(2k)$; the second is true, since 2 is even; yet 2 is not divisible by 4. The form is bad.

(ii) With P the compound statement "$x > 0$ and $y > 0$" (a letter may stand for one) and Q: "$xy > 0$", the form is "from $P \to Q$ and P, conclude Q". It cannot be broken. By the promise table, $P \to Q$ is false only when P is true and Q false; so when $P \to Q$ and P are both true, Q is true. The form is good.

Argument B, tried. B's reasoning shows "if n is even, then n^2 is even", and its last sentence concludes the converse, "if n^2 is even, then n is even". That step has the form "from $P \to Q$, conclude $Q \to P$", and it breaks: take P: "n is divisible by 4" and Q: "n is even". The premise is true for every integer, and the conclusion fails at $n = 2$. B's tempting repair, "from $P \to Q$ and Q, conclude P", is the form of (i), which $n = 2$ just broke. With B's own content no integer

breaks the repair, since every integer with an even square is even (the fact B set out to prove); but the *form* breaks. A true conclusion reached by a bad form is still not proved.

↻ **Second reading** · *Valid, or only convincing?*

Many true cases are evidence, not a proof. The number n^2+n+41 is prime for $n = 0$, 1, 2, and so on all the way to $n = 39$: forty cases without a failure. At $n = 40$ it is $1600+40+41 = 1681 = 41^2$. An argument from many checked cases can be strong, but it is not of a good form: its premises can all be true while its conclusion is false. Logic books call a good form **valid**, and logic studies only form. That is why this book asks of every step not "is it convincing?" but "could this form be broken?"

1.1.10 Good form, false sentences

Argument B has true sentences and a bad form. The opposite happens too:

> If 6 is even, then 6 is prime. And 6 is even. So 6 is prime.

Its form, "from $P \to Q$ and P, conclude Q", is good (Quick check (ii) above), and its conclusion is false. The fault is the first premise, false since 6 is even and not prime. Form and truth are separate questions; all four combinations occur.

	every sentence true	some sentence false
form good	"If 3 is odd, then 9 is odd. 3 is odd. So 9 is odd."	the argument about 6 above
form bad	Argument B, by its last step "from $P \to Q$, conclude $Q \to P$"	"If 2 is divisible by 4, then 2 is even. 2 is even. So 2 is divisible by 4."

The top left is the odd-square argument at $n = 3$; the bottom right is the Quick check's (i) at $n = 2$. Argument A is not placed: it argues from a supposition ("Suppose n is odd ...") and uses "some" (an integer k with $n = 2k + 1$), and a form written as premises above a line cannot yet express either.

A good form guarantees only that true premises give a true conclusion.

1.1.11 Each connective, two rules

We now come to the idea that organizes this chapter and the next. Think about what a mathematician does with the word "and". There are exactly two things to do with it.

- To **prove** a statement of the form $P \wedge Q$, prove P and prove Q.
- To **use** a statement of the form $P \wedge Q$ that is already known, take either half.

Nobody needs to be taught these two moves. They are also, and this is the point, **all** there is to know about "and" for the purpose of writing proofs. Every other fact about $\wedge$, such as that $P \wedge Q$ and $Q \wedge P$ say the same thing, follows from the two moves, as Section 1.2 shows. The first move is called the **introduction rule** for $\wedge$, written $\wedge$I, because it is how a $\wedge$ gets into a proof. The second is the **elimination rule**, $\wedge$E, because it is how a $\wedge$ is taken apart.

The table below is an informal preview. Section 1.2 states these two rules exactly, as patterns of lines.

Rule	What it allows	You write
$\wedge$I	from P and Q, conclude $P \wedge Q$	"We have shown $x > 0$ and $x < 1$. Hence $0 < x < 1$."
$\wedge$E	from $P \wedge Q$, conclude P; also Q	"Since $0 < x < 1$, in particular $x > 0$."

Every connective works the same way. Each comes with introduction rules, saying how to prove a statement with that main connective, and elimination rules, saying what a statement with that main connective lets us conclude. The rules for $\wedge$ come in Section 1.2, those for $\to$ in Sections 1.3 and 1.4, and those for $\vee$ in Sections 1.5 and 1.6. The rules for $\neg$, $\bot$, and $\leftrightarrow$, and one more rule for proof by contradiction, which stands apart from the others, come after these. Two slogans summarize the plan.

Each connective comes with a way to prove it and a way to use it.

A proof is a chain in which every line is a premise, an assumption, or one application of a rule.

The second slogan is what makes a proof possible to check. Finding a proof may need an idea. Checking one needs none: go through it line by line and ask of each line which rule produced it and whether that rule was applied to lines that really are there. A reader who has lost the thread of a long argument can always fall back on that question, and it is the question to ask of Argument B.

Section 1.4 names exactly what B did wrong; the test *try to break it* has already shown, at $n = 2$ in its Quick check (i), that B's tempting repair, "from $P \to Q$ and Q, conclude P", is a bad form.

1.1.12 What comes next

Section 1.2 shows what a derivation is, with the rules for "and". Each later section adds one rule. Two questions guide every search: what does the goal's main connective ask for, and what does an available line's main connective let me do?

Remark (Looking ahead). Chapter 4 gives "for every" its rules and explains why an argument about a fixed but arbitrary x proves a statement about every x.

1.1.13 Exercises

1.1.13.1 A. Check your understanding

Exercise. A1

(a) State the definition of a statement. Is "$10^{100} + 1$ is prime" a statement? For a real number x, is "$x^2 - 1 = 0$" a statement? Is "$0^2 - 1 = 0$"?

(b) With P: "n is even" and Q: "n is prime", write "n is prime but not even" in symbols.

(c) Find the main connective of $P \wedge Q \to \neg R$, and write the statement with all its parentheses.

(d) True or false, with a reason: if P and Q are both true, then $P \vee Q$ is false.

(e) Let x and y be real numbers. Write the form of "$x > 0$ and $y > 0$; so $y > 0$ and $x > 0$" with premises above a line, saying what each letter stands for.

(f) Which of $\wedge$, $\vee$, $\to$ is the main connective of $(P \vee Q) \wedge (Q \to R)$? Using the preview table of "Each connective, two rules", say what a proof of this statement must do, and name the rule that Section 1.2 will state for it.

Solution.

(a) A statement is a declarative sentence that is either true or false, and not both (Definition 1.1.1). Yes: $10^{100} + 1$ is a definite integer, so it either is prime or is not. The definition does not ask us to know which. With x fixed, by the convention of the box in "Statements", "$x^2 - 1 = 0$" is a statement about that one number: true if x is 1 or -1, false otherwise. Without the convention it would be an open sentence. The sentence "$0^2 - 1 = 0$" is a statement outright, and it is false, since $0^2 - 1 = -1$.

(b) $Q \wedge \neg P$. The word "but" is a conjunction.

(c) Convention (2) groups $P \wedge Q$ before $\to$ acts, and convention (1) lets $\neg$ take only R. So the statement is $(P \wedge Q) \to (\neg R)$, and its main connective is $\to$ (Definition 1.1.2).

(d) False. The mathematical "or" is inclusive, so $P \vee Q$ is true when both parts are true.

(e) With P: "$x > 0$" and Q: "$y > 0$", the premise is $P \wedge Q$ and the conclusion is $Q \wedge P$:

$$\frac{P \wedge Q}{Q \wedge P}$$

Section 1.2 derives it.

(f) $\wedge$: each parenthesized part is complete, and the last step joins them with "and". So a proof of it has two parts: it proves $P \vee Q$, and it proves $Q \to R$. The rule that puts the two parts together is $\wedge$I. This is the first question of "What comes next", asked before any rule has been stated exactly: the goal's main connective decides the last step.

Exercise. A2: Drill

Answer each part in a line or two.

(a) What is the main connective of $\neg P \wedge Q \to R$?

(b) What is the main connective of $P \vee Q \to R \wedge S$? Write the statement with all its parentheses.

(c) What is the main connective of $\neg(P \to Q) \wedge R$?

(d) Let x be a real number, with P: "$x > 0$" and Q: "$x^2 > x$". Translate "x is positive but x^2 is not greater than x".

(e) Let n be an integer, with O: "n is odd", S: "n^2 is odd" and C: "n^3 is odd". Translate "if n is odd, then n^2 and n^3 are odd".

(f) Is "if $x > 1$, then $x^2 > 1$" broken at $x = -2$?

(g) Is "if $x^2 > 1$, then $x > 1$" broken at $x = -2$?

(h) Let n be an integer. Write "n is prime only if n is odd" as an implication, and break it.

(i) Try to break the form "from $P \vee Q$ and P, conclude $\neg Q$".

(j) The argument "If 7 is even, then 7^2 is even. 7 is even. So 7^2 is even." has a false conclusion. Is its form bad?

Solution.

(a) $\rightarrow$. The statement is $((\neg P) \wedge Q) \rightarrow R$.

(b) $\rightarrow$. With all its parentheses it is $(P \vee Q) \rightarrow (R \wedge S)$.

(c) $\wedge$. The parentheses keep $\neg$ on $P \rightarrow Q$, and the last step joins $\neg(P \rightarrow Q)$ and R.

(d) $P \wedge \neg Q$. "But" is a conjunction.

(e) $O \rightarrow (S \wedge C)$.

(f) No. The hypothesis $-2 > 1$ is false, so nothing is promised.

(g) Yes. The hypothesis $4 > 1$ holds and the conclusion $-2 > 1$ fails.

(h) With P: "n is prime" and O: "n is odd", it is $P \rightarrow O$. The integer $n = 2$ breaks it: 2 is prime and not odd.

(i) Let x be a real number, with P: "$x \geq 0$" and Q: "$x \leq 0$", and take $x = 0$. Both premises are true (the "or" is inclusive), and $\neg Q$ is false. The form is bad.

(j) No. Its form is "from $P \rightarrow Q$ and P, conclude Q", which is good. The premise "7 is even" is false, and a form is bad only when some argument of that form has **all** its premises true and its conclusion false.

1.1.13.2 B. Practice

Exercise. B1: Translating analysis

Translate each sentence into symbols, using the atomic statements given. You do not need to know what the words mean; only the grammar matters.

(a) "f is continuous at a but not differentiable there", with C: "f is continuous at a" and D: "f is differentiable at a".

(b) "If the sequence (a_n) converges, it is bounded", with V: "(a_n) converges" and B: "(a_n) is bounded".

(c) "x is rational or irrational", with R: "x is rational". (A real number is irrational when it is not rational.)

(d) "The set F is closed if and only if its complement is open", with K: "F is closed" and U: "the complement of F is open".

Solution.

(a) $C \wedge \neg D$. "But" asserts both parts, and "not differentiable there" is the negation of D.

(b) $V \rightarrow B$. The pronoun "it" refers to (a_n), so the conclusion is the statement B.

(c) $R \vee \neg R$. "Irrational" means "not rational", so the second part is $\neg R$, not a new atomic statement.

(d) $K \leftrightarrow U$.

Exercise. B2: Parsing

For each statement, insert all the parentheses that the conventions allow us to omit, and name the main connective.

(a) $\neg P \to Q$

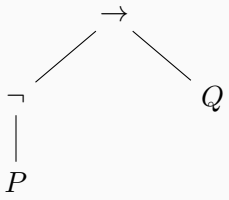
(b) $P \vee \neg Q \leftrightarrow \neg R$

(c) $\neg\neg P \wedge (Q \to R)$

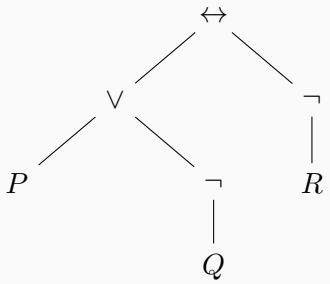
Then let x and y be real numbers, with P: "$x > 0$", Q: "$y > 0$" and R: "$xy > 0$". Write $P \wedge Q \to R$ as an English sentence, and say whether it is true whatever the numbers x and y are.

Solution.

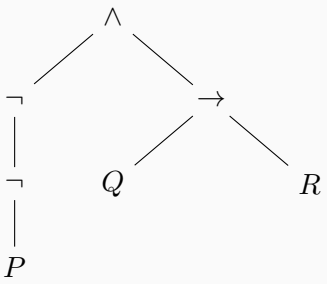
(a) $(\neg P) \to Q$, main connective $\to$. As a tree:



(b) $(P \vee (\neg Q)) \leftrightarrow (\neg R)$, main connective $\leftrightarrow$. Each $\neg$ applies to the single letter after it, and $\vee$ binds more tightly than $\leftrightarrow$. As a tree:



(c) $(\neg(\neg P)) \wedge (Q \to R)$, main connective $\wedge$. The first $\neg$ applies to the statement $\neg P$, which is the smallest complete statement following it. As a tree:



The statement $P \wedge Q \to R$ is $(P \wedge Q) \to R$: "if $x > 0$ and $y > 0$, then $xy > 0$". It is true for all real x and y, since the product of two positive numbers is positive.

Exercise. B3: Necessary and sufficient

Let x be a real number.

(a) Write "$x > 1$ is sufficient for $x^2 > 1$" and "$x > 1$ is necessary for $x^2 > 1$" as implications.

(b) Decide which of the two is true for every real x, and give a value of x that makes the other false.

(c) Find a condition on x, written with $\vee$, that is both necessary and sufficient for $x^2 > 1$. You may use that $x^2 > 1$ exactly when $|x| > 1$, where $|x|$ is x if $x \geq 0$ and $-x$ if $x < 0$.

Solution.

(a) "P is sufficient for Q" means that P is enough to guarantee Q, that is, $P \to Q$. "P is necessary for Q" means that Q cannot hold without P, that is, $Q \to P$. So the two sentences are $(x > 1) \to (x^2 > 1)$ and $(x^2 > 1) \to (x > 1)$.

(b) The first is true for every x: if $x > 1$, then $x > 0$, so $x^2 = x \cdot x > 1 \cdot x = x > 1$, where multiplying $x > 1$ by the positive number x keeps the inequality. The second is false for $x = -2$: then $x^2 = 4 > 1$, but $x < 1$.

(c) $(x > 1) \vee (x < -1)$. It is sufficient: if $x > 1$, then $|x| = x > 1$, and if $x < -1$, then $|x| = -x > 1$; either way $x^2 > 1$. It is necessary: if $x^2 > 1$, then $|x| > 1$, and $|x|$ is x or $-x$, so $x > 1$ or $-x > 1$, the latter meaning $x < -1$.

Exercise. B4: Break it or explain why not

For each form, either find statements about a real number x that make every premise true and the conclusion false, or explain, by checking every case, why no choice of statements can (use the meaning of each connective; for $\to$, the table of "When is an implication broken?").

(a) From $P \to Q$ and $\neg P$, conclude $\neg Q$.

(b) From $P \to Q$ and $Q \to R$, conclude $P \to R$.

(c) From $P \vee Q$, conclude P.

(d) From $P \wedge Q$, conclude $P \vee Q$.

Solution.

(a) Bad. Take P: "$x > 2$" and Q: "$x^2 > 4$", and $x = -3$. The premise $P \to Q$ holds for every x, as the "only if" passage of this section showed. The premise $\neg P$ is true, since $-3 > 2$ is false. And Q is true, since $9 > 4$, so the conclusion $\neg Q$ is false.

(b) It cannot be broken. To make the conclusion $P \to R$ false we need P true and R false. Now look at Q. If Q is true, then $Q \to R$ has a true hypothesis and a false conclusion, so it is broken. If Q is false, then $P \to Q$ has a true hypothesis and a false conclusion, so it is broken. Either way some premise is false.

(c) Bad. Take P: "$x > 0$" and Q: "$x \leq 0$", and $x = -1$. The premise $P \vee Q$ is true, since $-1 \leq 0$, and the conclusion $-1 > 0$ is false.

(d) It cannot be broken. If $P \wedge Q$ is true, then P is true, so $P \vee Q$ is true.

In (b) and (d), "cannot be broken" is shown by reasoning about every case at once, not by failing to find a counterexample: a search that comes up empty settles nothing.

1.1.13.3 C. Going deeper

Exercise. C1: "Only if" and "unless"

Let P and Q be statements.

(a) Explain why "P only if Q" means $P \to Q$, and not $Q \to P$. *(Ask: which situation would make "P only if Q" false?)*

(b) Explain why "P unless Q" means $\neg Q \to P$.

(c) Let x be a real number. Decide whether each sentence is true for every real x. Justify your answer, giving a value of x that makes it false where there is one.
- "$x^2 = 4$ if $x = 2$."
- "$x^2 = 4$ only if $x = 2$."
- "$x^2 > 0$ unless $x = 0$."

Solution.

(a) "P only if Q" says that P does not happen without Q. It is broken exactly when P holds and Q fails. That is also exactly when $P \to Q$ is broken: an implication promises Q whenever P, and the promise fails only when P holds without Q. So the two say the same thing. By contrast, $Q \to P$ is broken when Q holds and P fails, a different situation. For example, "a number is divisible by 4 only if it is even" is true, while "a number is even only if it is divisible by 4" is false (take 2).

(b) "P unless Q" says that P holds in every situation except possibly those where Q holds. So once we know that Q fails, we may conclude P, which is $\neg Q \to P$. For instance, "the equation $ax = 1$ has a solution unless $a = 0$" says: if $a \neq 0$, then $ax = 1$ has a solution.

(c) The first sentence is $(x = 2) \to (x^2 = 4)$. It is true for every x: when $x = 2$, $x^2 = 4$, and when $x \neq 2$ it promises nothing.

The second is $(x^2 = 4) \to (x = 2)$, the same two parts in the opposite direction. It is false for $x = -2$, where $x^2 = 4$ but $x \neq 2$.

The third is $\neg(x = 0) \to (x^2 > 0)$, by (b). It is true for every x: a non-zero real number is positive or negative, and in both cases its square is positive.

The first two sentences differ by one word, and they are implications in opposite directions. Confusing them is exactly the mistake of Argument B at the start of this section.

1.2 Derivations, Starting with And

The previous section stated a slogan: a proof is a chain in which every line is a premise, an assumption, or one application of a rule. This section makes the premises and the rule steps exact (assumptions wait until Section 1.4): what a derivation is, how each line carries its reason, and how anyone can check one. Then it gives the first two rules of logic, the rules for "and".

1.2.1 Why check by matching

A proof in a research paper can run to hundreds of steps, too many to hold in mind at once, so it is checked one step at a time. Two careful readers can disagree about whether a step is

"obvious"; a rule that says exactly which steps are allowed settles it, since a step matches the rule or does not. And programs that check proofs work in the same spirit: each step must match a pattern.

Here is a small case: the chain

$$0 < \tfrac{1}{3} < \tfrac{1}{2} < 1$$

makes three claims, $0 < \frac{1}{3}$, $\frac{1}{3} < \frac{1}{2}$, and $\frac{1}{2} < 1$, and a reader who accepts them concludes $0 < 1$. The justification is one fact, used twice:

> if $a < b$ and $b < c$, then $a < c$.

From $0 < \frac{1}{3}$ and $\frac{1}{3} < \frac{1}{2}$ it gives $0 < \frac{1}{2}$. From $0 < \frac{1}{2}$ and $\frac{1}{2} < 1$ it gives $0 < 1$.

To check those two steps, a reader needs nothing about fractions and nothing about what $<$ means. It is enough to see that the two inequalities used in each step have the **same** expression in the middle, and that the new inequality joins the outer two. A clerk who had never seen a number could do it. So a proof can be checked without being understood: a proof is correct when every step matches its pattern, and matching is mechanical.

1.2.2 A rule is a pattern

We borrow one fact about real numbers and use it as a rule.

- **Tr** (for "transitivity", or chaining): from $a < b$ and $b < c$, write $a < c$.

It is a fact about numbers used as a rule, for practice; it is not a rule of logic.

A rule is applied by matching against a small picture of the lines it needs and the line it writes.

A rule is a pattern: given these lines, you may write that line.

Definition 1.2.1. Rule Schema

A **rule schema** is a pattern of lines: **given** lines at the top, possibly none, and exactly **one new** line at the bottom, whose justification names the rule and cites the given lines. Line numbers in a schema are written i, j, k. The letters in the pattern stand for arbitrary **expressions** of the right kind, and a use of the rule replaces them **consistently**: the same letter by the same expression, **symbol for symbol**, everywhere it occurs. A line is **justified by the rule** when it and the lines it cites, with their numbers, are an **instance** of a schema of the rule: the pattern with every letter replaced in this way. The schema of Tr is

i	$a < b$	
j	$b < c$	
k	$a < c$	Tr i, j

Here a, b, c stand for expressions for numbers.

In three points:

- The given lines may stand anywhere above the new line, in any order on the page, and need not be next to each other or to it.
- The citation's order says which line plays which part: "Tr i, j" makes line i play $a < b$ and line j play $b < c$, unless the rule says otherwise ($\wedge$I, below, accepts either order).

- Letters are replaced consistently, symbol for symbol. The middle letter b occurs in both given lines, so it must be replaced by the **same** string of symbols in both, not merely by the same number.

An instance first. Take $i = 3$, $j = 5$, $k = 6$, and replace a by 0, b by $\frac{1}{2}$, and c by 1. The pattern becomes: line 3 is $0 < \frac{1}{2}$, line 5 is $\frac{1}{2} < 1$, and line 6 is $0 < 1$, justified "Tr 3, 5".

Now a noninstance, by a minimal change. Keep line 3 as $0 < \frac{1}{2}$, and let line 5 be $\frac{1}{3} < 1$. Is "6. $0 < 1$, Tr 3, 5" justified? Line 3 forces b to be $\frac{1}{2}$, and line 5 forces it to be $\frac{1}{3}$, so these lines are not an instance of the schema. Line 6 is true, but Tr was not applied, so the line has no justification.

Why define rules this way? Because a rule applied by matching can be checked by someone who knows nothing about the subject. The schema turns "this step is obviously fine" into "this step matches that pattern", a claim anyone can test.

From the rules of logic on, the letters in a schema are Greek: φ, ψ, χ (phi, psi, chi). A Greek letter is a hole in a rule's pattern; a capital P is a statement we are reasoning about. A pattern works like the formula $(a + b)^2 = a^2 + 2ab + b^2$, whose a and b are holes to be filled with any expression, the same one wherever the hole appears. Filling the holes of a schema works the same way, with parentheses put around a compound statement where they are needed. For example, if a schema has the given lines φ and ψ and the new line $\varphi \wedge \psi$, then the three lines $P \to Q$, R, and $(P \to Q) \wedge R$ are an instance of it, with φ replaced by $P \to Q$ and ψ by R.

Warning. The middle expression must match **exactly**, symbol for symbol. The step "$0 < \frac{2}{4}$ and $\frac{1}{2} < 1$, so $0 < 1$" is tempting, because $\frac{2}{4} = \frac{1}{2}$ and every line in it is true, but it is not an application of Tr: the two middle expressions are different strings. A checker who had to notice that they are equal would need arithmetic, and the point of a schema is that a checker needs nothing. Compare symbols, not values.

1.2.3 Derivations without boxes

A proof strings steps together, each using what earlier steps produced. The next definition says exactly how.

Definition 1.2.2. Box-Free Derivation

Fix a list of rules, each given by schemas. A **box-free derivation** is a finite list of numbered lines $1, 2, \ldots, n$, where $n \geq 1$, each carrying one statement and a justification, such that:

(i) the first lines, possibly none and possibly all, are **premises**, justified by nothing; a short horizontal rule is drawn under the last premise;

(ii) every later line is justified by **one** rule of the list, citing **earlier** lines, and it and the cited lines form an instance of a schema of that rule.

The **premises** of the derivation are its premise lines, and its **conclusion** is its last line.

In words. Clause (i) says where a derivation starts: with the statements it is given, which are not proved and need no reason. Clause (ii) says how it goes on: each further line is one step of one rule, and the only lines it may use are ones already written. The word "earlier" is the only condition on which lines may be cited **for now**. Section 1.4 adds assumptions made inside

boxes, and then "earlier" is refined to "available": an earlier line may become unusable once the box it sits in has closed.

The layout is always the same. Line numbers run down the left, statements sit in the middle, and each justification is on the right: the rule's name, then the cited line numbers. A vertical bar runs down the left, and a short horizontal rule under the last premise separates what is given from what is derived. From Section 1.4 on, a box draws a second, shorter bar, and the bars beside a line show which assumptions it depends on.

1.2.4 First derivations

Here are two derivations with Tr, each with the reasoning that found it. The reasoning is the part to watch: a derivation on the page runs from the top down, but it is rarely found that way.

Example. Chaining

Let x, y, u, v, w, z be real numbers. Using the rule Tr only, write a box-free derivation of each conclusion from the given premises.

(a) Premises $0 < x$, $x < y$, $y < 1$; conclusion $0 < 1$.

(b) Premises $u < v$, $w < z$, $v < w$; conclusion $u < z$.

Solution. **(a)** *Idea.* The premises form a chain $0 < x < y < 1$, each link sharing its right end with the next link's left end. So we work **forwards**, from what we have: chain the first two links at their shared middle x, then the result with the third link at y.

1	$0 < x$	
2	$x < y$	
3	$y < 1$	
4	$0 < y$	Tr 1, 2
5	$0 < 1$	Tr 4, 3

Read it back. Lines 1–3 are the premises. Line 4: lines 1 and 2 are an instance of Tr, with a, b, c replaced by 0, x, y. Line 5: lines 4 and 3 are one, with a, b, c replaced by 0, y, 1. The letters are replaced afresh at each use: the b of line 4 has nothing to do with the b of line 5.

(b) *Idea.* This time the chain is scrambled, so we work **backwards**, from what we want. The goal $u < z$ can come from Tr only if we have some $u < m$ and $m < z$ for a single expression m. The only premise starting with u is $u < v$, so try $m = v$; the remaining need is $v < z$, a new, smaller goal. It can come from Tr with some $v < m'$ and $m' < z$; the only premise starting with v is $v < w$, and with $m' = w$ the second need, $w < z$, is a premise. The search has reached the premises, and the derivation is its record, written **forwards**.

1	$u < v$	
2	$w < z$	
3	$v < w$	
4	$v < z$	Tr 3, 2
5	$u < z$	Tr 1, 4

The search could have started instead from $w < z$, the premise ending in z: with $m = w$ it needs $u < w$, which is Tr $1, 3$, and then $u < z$ is Tr $4, 2$. That is a different correct derivation. The same premises and conclusion usually have many derivations, and any one of them shows that the conclusion follows from the premises.

Part (b) was found backwards, from the goal; part (a) forwards. The rest of this section names the two directions.

Quick check. Line 1 of a derivation is $0 < x$ and line 3 is $x < 1$. Which is correct: "4. $0 < 1$, Tr $1, 3$" or "4. $0 < 1$, Tr $3, 1$"?

Solution. The first. With the citation "1, 3", line 1 plays $a < b$ and line 3 plays $b < c$: a is 0, b is x in both lines, and c is 1, so the new line $a < c$ is $0 < 1$. With "3, 1", line 3 would play $a < b$ and line 1 would play $b < c$, so b would have to be 1 (from line 3) and 0 (from line 1) at once.

1.2.5 Reading a derivation

Finding a derivation may need an idea, as part (b) of Example (Chaining) did. Checking one never does. For each line after the premises, ask three questions.

1. **Which rule?** Name the rule the justification cites, and check that it is on the list of rules.
2. **Do the cited lines and this line form an instance of that rule's schema?** Find the replacement of the letters: read it off the cited lines in the order the citation gives them (or, for a rule whose lines may be cited in either order, try both: the new line decides), and check that every letter gets the same expression each time it occurs, and that the new line is what the pattern then says.
3. **Are the cited lines earlier?** Each cited number must be smaller than the number of the line doing the citing.

A line that passes all three is correct, whatever it says and whether or not it is true. Here is the check run on part (a) of Example (Chaining).

Line	Rule	Cites	Instance (the letters' values)	Earlier?
1–3	premises	—	—	—
4	Tr	1, 2	a: 0; b: x; c: y	1, 2 < 4: yes
5	Tr	4, 3	a: 0; b: y; c: 1	4, 3 < 5: yes

Every line passes, so the list is a derivation. Section 1.4 extends this table, when "earlier" becomes "available".

Finding a derivation may need an idea; checking one never does.

Second reading · *Two ways to break a derivation*

A line can fail one question and pass the others. The next example breaks a correct derivation in two ways, each time with the smallest change that does it.

Example. Not a derivation

Each list below is the derivation of part (a) of Example (Chaining) with one change. Say which of the three questions of "Reading a derivation" fails, and at which line.

(a) The statements of lines 4 and 5 are swapped, each taking its justification with it, so that line 4 is now "$0 < 1$, Tr $5, 3$" and line 5 is "$0 < y$, Tr $1, 2$".

(b) Line 5 is justified "Tr $3, 4$" instead of "Tr $4, 3$".

Solution.

(a) The list is the following.

1	$0 < x$	
2	$x < y$	
3	$y < 1$	
4	$0 < 1$	Tr 5, 3
5	$0 < y$	Tr 1, 2

Line 5 passes all three questions, exactly as line 4 did before. Line 4 passes the first two: lines 5, 3, and 4 are an instance of Tr with a, b, c replaced by 0, y, 1. It fails the third, because it cites line 5, which is not earlier. So the list is not a derivation. (It would also be useless if it were one: its last line is now $0 < y$, not $0 < 1$.)

(b) Line 5 is now "$0 < 1$, Tr $3, 4$". Lines 3 and 4 are earlier, so the third question passes, and the rule is named, so the first passes. The second fails. The citation makes line 3, $y < 1$, play $a < b$, so a is y and b is 1; and it makes line 4, $0 < y$, play $b < c$, so b is 0. The letter b cannot be both 1 and 0, so there is no instance. The lines are the right ones in the wrong roles.

Change (b) shows why the citation order is part of the justification, and change (a) why the order of the lines is. Exercise C2 below asks what would go wrong if a line were allowed to cite a later one.

1.2.6 What a derivation proves

A derivation shows that its conclusion follows from its premises. The symbol $\vdash$ says that such a derivation exists.

Definition 1.2.3. Derivable

Let Γ be a list of statements $\varphi_1, \ldots, \varphi_n$, where $n \geq 0$, and let ψ be a statement. We say ψ is **derivable** from Γ, and write

$$\Gamma \vdash \psi,$$

if **there is** a derivation, using the rules of logic, whose premises **all belong to** Γ and whose last line is ψ. We write $\varphi_1, \ldots, \varphi_n \vdash \psi$ for $\Gamma \vdash \psi$, and $\vdash \psi$ when Γ is empty.

Here Γ (capital gamma) is just a name for a list of statements.

Definition 1.2.4. Sequent

Let $P_1, \ldots, P_n$ and Q be statements, where $n \geq 0$. The expression $P_1, \ldots, P_n \vdash Q$, read "$P_1$ to P_n prove Q", is a **sequent**. It is the **claim** that Q is derivable from $P_1, \ldots, P_n$ in the sense of Definition 1.2.3. The statements $P_1, \ldots, P_n$ are its **premises** and Q is its **conclusion**; when $n = 0$ we write $\vdash Q$.

The symbol $\vdash$ is called the **turnstile**. A sequent is not true merely because its conclusion is true: it claims that a derivation exists, and it is proved by exhibiting one.

In words, three remarks.

1. *Which derivations?* Until Section 1.4, "derivation" means a box-free derivation (Definition 1.2.2). Section 1.4 widens the word to allow boxes; nothing proved before changes, since a box-free derivation is still a derivation.
2. *Not every premise must be used.* A derivation may ignore some statements of Γ, and it may even have no premise lines at all. It may not use a premise that is **not** in Γ.
3. *Relative to a list of rules.* The symbol $\vdash$ always refers to a list of rules. From "The rules for and" below on, it is the rules of logic introduced so far; adding a rule never destroys a derivation, so a sequent once derived stays derived. Before that, we name the rule: "$\Gamma \vdash \psi$ **with** Tr".

For instance, part (a) of Example (Chaining) shows $0 < x,\ x < y,\ y < 1 \vdash 0 < 1$ with Tr.

In the derivations from here on, the capital letters P, Q, R, S stand for arbitrary statements, as in Section 1.1. A derivation written with them proves not one sequent but infinitely many at once, one for each way of putting statements in for the letters.

Quick check. What sequent "with Tr" does part (b) of Example (Chaining) show?

Solution. $u < v,\ w < z,\ v < w \vdash u < z$ with Tr. The premises of the derivation are lines 1–3, and its last line is $u < z$.

Example. Derivations, simplest first

Decide whether each of the following is a derivation, and if so, what it shows.

(a) The single line 1 P premise.

(b) The empty list, with no lines at all.

(c) The derivation of part (a) of Example (Chaining), for the list of rules consisting of Tr alone.

Solution.

(a) Yes. Its only line is a premise, so clause (i) of Definition 1.2.2 holds, and clause (ii) has no lines to speak about. Its last line is P. It shows $P \vdash P$: every statement is derivable from itself. No rule was used, so this holds whatever the list of rules is.

Parts (b) and (c) are answered in the fold below.

↻ **Second reading** · *Edge cases*

A derivation of one line. The degenerate case is worth a look. A single premise line, with nothing after it, satisfies both clauses of Definition 1.2.2: clause (i) holds, and clause (ii) has no lines to speak about. So it is a derivation, and its premise is its conclusion.

Part (b) of Example (Derivations, simplest first). No. The definition asks for $n \geq 1$ lines, and a derivation must have a last line to be its conclusion. So the empty list derives nothing. In the same way, a list with no premises can never get started with Tr, or any rule that cites a line: its first line would have to cite an earlier line, and there is none. Every rule met before Section 1.4 cites at least one line, so until then $\vdash \psi$ holds for no ψ. Section 1.4 changes that.

Part (c) of Example (Derivations, simplest first). Yes. Lines 1–3 are premises, with the rule drawn under line 3: clause (i). Line 4 cites lines 1 and 2, which are earlier, and lines 1, 2, 4 are an instance of Tr with a, b, c replaced by 0, x, y. Line 5 cites lines 4 and 3, which are earlier, and lines 4, 3, 5 are an instance with a, b, c replaced by 0, y, 1. So clause (ii) holds for both later lines. The premises are the three premise lines and the conclusion is $0 < 1$.

Warning. A premise is **given**, not proved: from the false premises $1 < 0$ and $0 < -1$, the rule Tr gives the false line $1 < -1$, in a correct derivation. What a correct derivation guarantees is only that **if** the premises are true, so is the conclusion, and even that only for rules that never lead from true lines to a false one.

1.2.7 The rules for and

Analysis is full of chains such as $0 < x < 1$, which says $0 < x$ **and** $x < 1$. A proof that **uses** it often needs one half: "in particular $x > 0$, so we may divide by x." A proof that **proves** it must prove both halves; one half is not the whole claim. Section 1.1 previewed these two moves informally. Here they become rules, checked by matching, as Tr was.

To prove "φ and ψ", prove each; to use it, take either.

Definition 1.2.5. Rules for Conjunction

$$\begin{array}{l|l} i & \varphi \\ j & \psi \\ k & \varphi \wedge \psi \end{array} \quad \wedge\text{I } i, j \qquad \begin{array}{l|l} i & \varphi \wedge \psi \\ k & \varphi \end{array} \quad \wedge\text{E } i \qquad \begin{array}{l|l} i & \varphi \wedge \psi \\ k & \psi \end{array} \quad \wedge\text{E } i$$

The first schema is the rule ∧I (**and-introduction**). Its two given lines may be cited in either order: the new line shows which statement is its left conjunct, so question 2 of "Reading a derivation" tries both orders. The second and third are the two forms of the rule ∧E (**and-elimination**), one writing the left conjunct and one the right.

∧I.

- *Schema:* the first one in the block above.
- *Reading:* two lines in hand, anywhere above and cited in either order; write their conjunction.
- *Example:* from line 2, P, and line 5, Q, "∧I 5, 2" (or "2, 5") writes $Q \wedge P$.

∧E.

- *Schema:* the other two in the block.
- *Reading:* one line whose **main connective** (Definition 1.1.2) is ∧; write either side, one layer at a time.
- *Example:* from $(P \wedge Q) \wedge R$ it writes $P \wedge Q$ or R, never P.

If φ and ψ are true, so is $\varphi \wedge \psi$; if $\varphi \wedge \psi$ is true, so is each side.

Warning. ∧E reaches one layer at a time. From line 1, $(P \wedge Q) \wedge R$, the line "Q — ∧E 1" is not an instance of either ∧E schema, because line 1's conjuncts are $P \wedge Q$ and R. The line is wrong even though Q does follow; the repair costs one line: $P \wedge Q$ by ∧E 1 first, then Q from it.

Quick check. Which statements can ∧E write, in one application each, from the line $(P \wedge Q) \wedge (R \wedge S)$?

Solution. Only $P \wedge Q$ and $R \wedge S$. The main connective is the middle ∧, so φ is $P \wedge Q$ and ψ is $R \wedge S$. Each of P, Q, R, S needs a second application, to one of those two lines.

Quick check. Line 1 is $(P \wedge Q) \to R$. May "$P \wedge Q$ — ∧E 1" be written?

Solution. No. Line 1's main connective is →, so ∧E cannot take it apart, although ∧ occurs in it.

1.2.8 Two directions of reasoning

A derivation has premises at the top and its goal, the conclusion we want, at the bottom. Between them is a gap, and finding a derivation means filling the gap. There are two places to start.

From the goal. Ask: **what is the goal's main connective?** If it is $\wedge$, the last line can be written by $\wedge$I, and then the goal $\varphi \wedge \psi$ splits into two **subgoals**, φ and ψ, each to be reached on some line above. This is **goal-directed** reasoning. It works backwards, up from the bottom of the gap, and asks which **introduction** rule builds the goal's main connective.

From the premises. Ask: **what are the main connectives of the lines we already have?** A line whose main connective is $\wedge$ can be taken apart by $\wedge$E, and each conjunct becomes a new line to work with. This is **premise-directed** reasoning. It works forwards, down from the top of the gap, and asks which **elimination** rule uses the line's main connective.

Work from both ends: take the premises apart, and build the goal from its parts.

The derivation is finished when the two ends meet. Part (b) of Example (Chaining) already worked backwards, with Tr in the part that introduction rules play from now on.

Scratch work for a search fits in a two-column table: the **Givens**, what is in hand (premises and lines derived so far), and the **Goal**, what is still needed. A goal-directed step replaces a goal by its subgoals; a premise-directed step adds to the Givens. The search is finished when every goal is among the Givens. The search for $P \wedge Q \vdash Q \wedge P$, one row per step:

Givens	Goal
$P \wedge Q$	$Q \wedge P$
$P \wedge Q$	Q, P ($\wedge$I splits the goal)
$P \wedge Q$, P, Q ($\wedge$E)	Q ✓, P ✓

The derivation this search finds is Proposition 1.2.1 below.

1.2.9 First derivations with and

Example. Two warm-ups

Give a derivation of each sequent.

(a) $P,\ Q \vdash Q \wedge P$.
(b) $P \wedge Q \vdash Q$.

Solution.

(a) *Idea.* The goal $Q \wedge P$ has main connective $\wedge$, so, goal-directed, its last line is $\wedge$I with subgoals Q and P. Both subgoals are premises, so the two ends meet at once.

1	P	
2	Q	
3	$Q \wedge P$	$\wedge$I 2, 1

The citation "2, 1" lists Q's line first; "1, 2" would do as well, since the line shows the order.

(b) *Idea.* The goal Q is a letter, with no main connective, so the goal suggests no rule. The premise has main connective $\wedge$, so, premise-directed, take it apart; its right conjunct is the goal.

$$\begin{array}{l|ll} 1 & P \wedge Q & \\ \hline 2 & Q & \wedge\text{E } 1 \end{array}$$

Line 2 is the second form of $\wedge$E, with $\varphi := P$ and $\psi := Q$.

The two derivations are box-free derivations in the sense of Definition 1.2.2, with $\wedge$I and $\wedge$E as the list of rules: premises first, and every later line an instance of a schema whose given lines are earlier. By Definition 1.2.3 they show the two sequents. When the goal is a letter, as in (b), only the premises can suggest a step; when every subgoal is a premise, as in (a), the premises need no taking apart. The first proposition below needs both ends.

↻ **Second reading** · *Edge cases for and-introduction*

One line cited twice. Nothing in the schema of $\wedge$I says that i and j are different lines. With $i = j$, the letters φ and ψ are replaced by the same statement, and the rule writes $\varphi \wedge \varphi$. It is a silly line, and a legal one. For instance, $P \vdash P \wedge P$: the goal is a conjunction, so $\wedge$I, with the subgoals P and P, and the one premise meets both.

$$\begin{array}{l|ll} 1 & P & \\ \hline 2 & P \wedge P & \wedge\text{I } 1, 1 \end{array}$$

The first result is that the order of a conjunction does not matter.

Proposition 1.2.1. Conjunction Commutes

$P \wedge Q \vdash Q \wedge P$.

Idea. Write the premise at the top, the goal at the bottom, and leave a gap (left picture; the lines carry the numbers they will have at the end). Start from the goal. Its main connective is $\wedge$, so the last line will be $\wedge$I, and we need P and Q on earlier lines (right picture: two subgoals, not yet justified). Now start from the premise. Its main connective is $\wedge$, so $\wedge$E gives P and gives Q. That closes the gap: the pieces the goal asks for are the pieces the premise supplies. (This is the search of the Givens/Goal table above.)

1	$P \wedge Q$
	$\vdots$
4	$Q \wedge P$

1	$P \wedge Q$	
	$\vdots$	
2	P	
3	Q	
4	$Q \wedge P$	$\wedge$I 3, 2

Proof. The following is a box-free derivation with the single premise $P \wedge Q$ and last line $Q \wedge P$.

1	$P \wedge Q$	
2	P	$\wedge$E 1
3	Q	$\wedge$E 1
4	$Q \wedge P$	$\wedge$I 3, 2

Lines 2 and 3 are the two forms of $\wedge$E applied to line 1, and line 4 is $\wedge$I applied to lines 3 and 2, and the line shows Q on the left. By Definition 1.2.3, $P \wedge Q \vdash Q \wedge P$.

■

Read back in words, the derivation is a short paragraph: *We are given that P and Q. In particular P holds. In particular Q holds. So Q and P both hold.* Each sentence is one line, and each "in particular" is one application of $\wedge$E.

Nothing in the derivation looks at what P and Q say: with P: "$0 < x$" and Q: "$x < 1$" it turns "$0 < x$ and $x < 1$" into "$x < 1$ and $0 < x$", and it does the same for any two statements. Because the letters stand for arbitrary statements, one derivation written with letters proves every such instance at once.

The second result is that the grouping of a conjunction does not matter either. It needs the premise taken apart in two layers.

Proposition 1.2.2. Conjunction Is Associative

$(P \wedge Q) \wedge R \vdash P \wedge (Q \wedge R)$.

Idea. From the goal: its main connective is the **outer** $\wedge$, so the last line is $\wedge$I with the subgoals P and $Q \wedge R$, and the second splits again into Q and R. From the premise: $\wedge$E gives $P \wedge Q$ and R, and again, on $P \wedge Q$, gives P and Q. Both searches stop at the three letters, and the ends meet there. Since $\wedge$E reaches one layer at a time, the premise must be peeled in two rounds.

In pictures: the goal below a gap; then the first round of $\wedge$E as lines 2–3, a gap for lines 4–5, and the subgoal $Q \wedge R$ as line 6, lines 6 and 7 not yet justified (line 7 will be $\wedge$I 4, 6).

1	$(P \wedge Q) \wedge R$
	$\vdots$
7	$P \wedge (Q \wedge R)$

1	$(P \wedge Q) \wedge R$	
2	$P \wedge Q$	$\wedge$E 1
3	R	$\wedge$E 1
	$\vdots$	
6	$Q \wedge R$	
7	$P \wedge (Q \wedge R)$	

Proof. The following is a box-free derivation with the single premise $(P \wedge Q) \wedge R$ and last line $P \wedge (Q \wedge R)$.

1	$(P \wedge Q) \wedge R$	
2	$P \wedge Q$	$\wedge$E 1
3	R	$\wedge$E 1
4	P	$\wedge$E 2
5	Q	$\wedge$E 2
6	$Q \wedge R$	$\wedge$I 5, 3
7	$P \wedge (Q \wedge R)$	$\wedge$I 4, 6

Lines 2 and 3 apply $\wedge$E to line 1, with $\varphi := P \wedge Q$ and $\psi := R$. Lines 4 and 5 apply $\wedge$E to line 2, with $\varphi := P$ and $\psi := Q$. Line 6 applies $\wedge$I to lines 5 and 3, and line 7 applies $\wedge$I to lines 4 and 6, with $\varphi := P$ and $\psi := Q \wedge R$. By Definition 1.2.3, $(P \wedge Q) \wedge R \vdash P \wedge (Q \wedge R)$.

■

The lines could come in other orders. Line 3 could come after lines 4 and 5, for instance, since it is used only at line 6. What cannot move is the dependence: every line must come after each line it cites.

Remark. The converse, $P \wedge (Q \wedge R) \vdash (P \wedge Q) \wedge R$, is derived the same way, peeling the premise's inner conjunction on the right instead of the left; Exercise B1 asks you to write it out. With both directions, the brackets in a chain of conjunctions carry no information, which is why the chain $0 < x < 1 < y$ is written without any.

1.2.9.1 A line is not used up

A line of a derivation is a statement that has been established. Citing it does not consume it: it can be cited again, by as many later lines as need it. The derivation of $P \wedge Q,\ R \vdash (P \wedge R) \wedge (Q \wedge R)$ shows this.

Idea. From the goal: $\wedge$I at the bottom, with subgoals $P \wedge R$ and $Q \wedge R$, each split again by $\wedge$I. From the premises: $\wedge$E on line 1 gives P and Q, and R is already a premise. Both subgoals that need R cite the same line 2.

In pictures: the goal below a gap; then the subgoals as lines 5–6, not yet justified.

		1	$P \wedge Q$	
		2	R	
1	$P \wedge Q$	3	P	$\wedge$E 1
2	R	4	Q	$\wedge$E 1
	⋮		⋮	
7	$(P \wedge R) \wedge (Q \wedge R)$	5	$P \wedge R$	
		6	$Q \wedge R$	
		7	$(P \wedge R) \wedge (Q \wedge R)$	$\wedge$I 5, 6

1	$P \wedge Q$		
2	R		
3	P	$\wedge$E 1	
4	Q	$\wedge$E 1	
5	$P \wedge R$	$\wedge$I 3, 2	
6	$Q \wedge R$	$\wedge$I 4, 2	
7	$(P \wedge R) \wedge (Q \wedge R)$	$\wedge$I 5, 6	

Line 2 is cited by lines 5 and 6, and line 1 by lines 3 and 4. In a written proof this is the everyday "we use the hypothesis $x > 0$ again".

↻ **Second reading** · *Any rearrangement of a conjunction*

Take everything apart, rebuild in the new order. One more derivation rearranges four letters: $(P \wedge Q) \wedge (R \wedge S) \vdash (S \wedge Q) \wedge (P \wedge R)$.

Idea. From the premise: two layers of $\wedge$, so $\wedge$E twice on line 1 and then twice on each of the two new lines, which leaves the four letters on lines 4 to 7. From the goal: two layers of $\wedge$, so $\wedge$I builds $S \wedge Q$ and $P \wedge R$ from the letters and then joins them. There is nothing to choose; the only care needed is the order of each citation.

1	$(P \wedge Q) \wedge (R \wedge S)$	
2	$P \wedge Q$	$\wedge$E 1
3	$R \wedge S$	$\wedge$E 1
4	P	$\wedge$E 2
5	Q	$\wedge$E 2
6	R	$\wedge$E 3
7	S	$\wedge$E 3
8	$S \wedge Q$	$\wedge$I 7, 5
9	$P \wedge R$	$\wedge$I 4, 6
10	$(S \wedge Q) \wedge (P \wedge R)$	$\wedge$I 8, 9

The recipe visibly does not depend on these particular letters or brackets. Take the premise apart with $\wedge$E until only letters remain, then build the goal from the letters with $\wedge$I, inside out. It works whenever the premise and the goal are both built from letters by $\wedge$ alone and every letter of the goal occurs in the premise, whatever the bracketing and the order on either side. So "$\wedge$ is commutative and associative" is, in the end, the observation that $\wedge$E can always take a conjunction down to its letters and $\wedge$I can always build one back up.

1.2.10 How this looks in a mathematical proof

Real proofs mix the logical rules with facts about their subject.

Example. A product of numbers in the unit interval

Let x and y be real numbers with $0 < x < 1$ and $0 < y < 1$. Show that $0 < xy < 1$, and name the rule behind each logical step.

Solution. The hypothesis $0 < x < 1$ is the conjunction $(0 < x) \wedge (x < 1)$. By $\wedge$E we have $0 < x$ and $x < 1$ separately, and likewise $0 < y$ and $y < 1$.

Since $x > 0$ and $y > 0$, the product xy is positive. Since $y < 1$ and $x > 0$, multiplying by x gives $xy < x$; together with $x < 1$ this gives $xy < 1$. These steps are school algebra, not logic.

Finally, from $0 < xy$ and $xy < 1$, $\wedge$I gives $(0 < xy) \wedge (xy < 1)$, which is $0 < xy < 1$.

As a derivation, the same proof reads as follows.

1	$(0 < x) \wedge (x < 1)$	
2	$(0 < y) \wedge (y < 1)$	
3	$0 < x$	$\wedge$E 1
4	$x < 1$	$\wedge$E 1
5	$0 < y$	$\wedge$E 2
6	$y < 1$	$\wedge$E 2
7	$0 < xy$	algebra 3, 5
8	$xy < 1$	algebra 3, 4, 6
9	$(0 < xy) \wedge (xy < 1)$	$\wedge$I 7, 8

Lines 7 and 8 are justified "algebra": each is a step by a fact about numbers used as a rule, exactly as Tr was used above, and not a rule of logic. Line 7 uses "a product of two positive numbers is positive"; line 8 uses "multiplying an inequality by a positive number keeps its direction", then the pattern of Tr.

The facts about numbers supply the content of the solution; the rules supply the joints, taking statements apart and putting the pieces back together. The goal-directed reading is visible in the prose too: "show $0 < xy < 1$" is a conjunction, so the solution sets itself two subgoals, and it says "finally" when the two meet.

↻ **Second reading** · *A chain in an analysis proof*

A proof about sequences might contain the line

$$0 < \frac{1}{n+1} < \frac{1}{n} \leq 1 \qquad \text{for } n \in \mathbb{Z}_{>0}.$$

It asserts three links: $0 < \frac{1}{n+1}$, $\frac{1}{n+1} < \frac{1}{n}$, and $\frac{1}{n} \leq 1$. A reader who takes from it a claim about its two ends, $0 < 1$, performs two steps in silence. The first is one application of chaining: $0 < \frac{1}{n+1}$ and $\frac{1}{n+1} < \frac{1}{n}$ give $0 < \frac{1}{n}$. The second joins $0 < \frac{1}{n}$ with $\frac{1}{n} \leq 1$, and it is **not** Tr: the line $\frac{1}{n} \leq 1$ has $\leq$, not $<$, so the lines do not match the schema. It is a different pattern, "from $a < b$ and $b \leq c$, write $a < c$", another true fact about numbers. That is exactly why a justification must name the pattern it uses. "By the chain" does not say which one.

Written chains like this one are everywhere in analysis, and they are derivations with the justifications left off. Nobody writes the justifications out in a real proof about numbers. But a reader checking a hard chain does what the table of "Reading a derivation" does: for each step, which fact, applied to which earlier lines.

1.2.11 Common errors

Besides $\wedge$E applied through a layer, one more error is common.

Warning. ∧E applies only to a line whose **main connective** is ∧. The line $(P \wedge Q) \to R$ contains a ∧, but its main connective is →, so "P — ∧E 1" is illegal. With x a real number, P: "$x > 0$", Q: "$x > 1$" and R: "$x^2 > 1$", the premise "if $x > 0$ and $x > 1$, then $x^2 > 1$" is true for every x, yet at $x = -1$ the line $x > 0$ is false.

1	$(P \wedge Q) \to R$	
2	P	∧E 1

The illegal step leads from a true premise to a false line, so no rule that allowed it could be safe.

1.2.12 The rule card

The card gives each rule so far, its shape, the sentence that uses it in a written proof and, for readers who meet a proof assistant, the Lean code that performs it (`h.1` names the left half of a hypothesis `h`). Nothing later depends on the last column.

Rule	Shape	You write	Lean
∧I	$\varphi, \psi \Longrightarrow \varphi \wedge \psi$	"We show P, and we show Q."	`constructor`
∧E	$\varphi \wedge \psi \Longrightarrow \varphi$ (or ψ)	"In particular, P."	`h.1` (and `h.2`)

Remark (Looking ahead). Chapter 3 checks, for every rule of Chapters 1 and 2 at once, that no rule leads from true lines to a false one. Until then each rule comes with the one-sentence reason it is safe, as the sentence after the bullets for "and" did.

1.2.13 Exercises

1.2.13.1 A. Check your understanding

Exercise. A1

(a) State the definition of a box-free derivation and of $\Gamma \vdash \psi$, and write the three schemas of Definition 1.2.5 from memory, naming the rule of each.

(b) Which statements does one application of ∧E give from the line $P \wedge (Q \wedge R)$?

(c) In the derivation in the proof of Proposition 1.2.2, why must line 2 come before line 4?

(d) Fill in the justifications (i)–(iv) of the following derivation.

1	$P \wedge (Q \wedge R)$	
2	P	(i)
3	$Q \wedge R$	(ii)
4	R	(iii)
5	$R \wedge P$	(iv)

(e) True or false, with a reason: if $\Gamma \vdash \psi$ with Tr, and every statement in Γ is true, then

ψ is true.

(f) What sequent does the derivation in (d) show?

Solution.

(a) A box-free derivation, relative to a list of rules, is a finite list of numbered lines, at least one, each with a statement and a justification, in which the first lines, possibly none and possibly all, are premises, justified by nothing, and every later line is justified by one rule of the list, citing earlier lines, so that it and the cited lines form an instance of a schema of that rule (Definition 1.2.2). And $\Gamma \vdash \psi$ means that there is a derivation whose premises all belong to Γ and whose last line is ψ (Definition 1.2.3). For the schemas see Definition 1.2.5: $\wedge$I has given lines i: φ and j: ψ and new line k: $\varphi \wedge \psi$, cited "$\wedge$I i, j" (or j, i); the two forms of $\wedge$E each have the given line i: $\varphi \wedge \psi$ and the new line k: φ, respectively ψ, cited "$\wedge$E i".

(b) P, or $Q \wedge R$. The main connective is the first $\wedge$, so φ is P and ψ is $Q \wedge R$.

(c) Line 4 is P, obtained by $\wedge$E from line 2, $P \wedge Q$. A rule may cite only earlier lines (Definition 1.2.2), and P cannot be obtained from line 1 in one step, since line 1's conjuncts are $P \wedge Q$ and R.

(d) (i) $\wedge$E 1; (ii) $\wedge$E 1; (iii) $\wedge$E 3; (iv) $\wedge$I 4, 2. "$\wedge$I 2, 4" is equally right: the line $R \wedge P$ shows which conjunct is on the left.

(e) True. Tr leads from true lines to a true line: if $a < b$ and $b < c$ are true, so is $a < c$. Go down the derivation: the premises are in Γ, so they are true, and each later line comes from earlier, true lines by Tr, so it is true too. Hence the last line, ψ, is true. This is the property the Looking-ahead box of this section promises for the rules of logic.

(f) $P \wedge (Q \wedge R) \vdash R \wedge P$: its only premise is line 1, and its last line is $R \wedge P$.

Exercise. A2: Drill

Answer each part in a line or two.

(a) Lines 1 $2 < y$ and 2 $y < z$ are given. What does "Tr 1, 2" justify?

(b) Lines 1 $u < 3$ and 2 $4 < v$ are given. Is "$u < v$ — Tr 1, 2" justified?

(c) The premises are 1 $0 < x$, 2 $x < 3$, 3 $3 < y$. Is "4. $0 < y$ — Tr 1, 3" justified?

(d) Write $0 < \frac{1}{4} < \frac{1}{3} < \frac{1}{2}$ as a derivation of $0 < \frac{1}{2}$ from its three links with Tr.

(e) In the list "1 $0 < 1$, premise; 2 $1 < 2$, premise; 3 $0 < 2$ — Tr 2, 1", which of the three questions of "Reading a derivation" fails?

(f) May $\wedge$E be applied to the line $\neg P \wedge Q$?

(g) Lines 1 $P \wedge (Q \wedge R)$, 2 S and 3 $Q \wedge R$ ($\wedge$E 1) are given. Which of the following can be line 4? (i) $(Q \wedge R) \wedge S$, $\wedge$I 3, 2; (ii) $Q \wedge S$, $\wedge$I 3, 2; (iii) R, $\wedge$E 1; (iv) R, $\wedge$E 3.

(h) Derive $P \wedge Q,\ R \vdash R \wedge P$.

(i) Derive $(P \wedge Q) \wedge R \vdash Q$.

(j) Derive $P \wedge Q,\ R \wedge S \vdash S \wedge P$.

(k) Fill in the Givens/Goal table for $P \wedge (Q \wedge R) \vdash R$, one row per step.

(l) Line 1 is $(P \wedge Q) \wedge (R \wedge S)$. List every statement $\wedge$E can write from line 1 in one step.

Solution.

(a) $2 < z$, with a, b, c replaced by 2, y, z.

(b) No: the middles 3 and 4 differ, so the lines are no instance of Tr, although $u < v$ does follow (since $3 < 4$).

(c) No: line 1 makes b the expression x, and line 3 makes it 3.

(d) Premises 1 $0 < \frac{1}{4}$, 2 $\frac{1}{4} < \frac{1}{3}$, 3 $\frac{1}{3} < \frac{1}{2}$; then 4 $0 < \frac{1}{3}$, Tr 1, 2; 5 $0 < \frac{1}{2}$, Tr 4, 3.

(e) The second: citing line 2 first makes b be 2 (from line 2, as $a < b$) and 0 (from line 1, as $b < c$). The rule is named and the cited lines are earlier.

(f) Yes: $\neg$ binds most tightly, so the main connective is $\wedge$, and $\wedge$E writes $\neg P$ or Q.

(g) (i) Yes. (ii) No: $\wedge$I joins the two cited lines whole, and line 3 is $Q \wedge R$, not Q. (iii) No: R lies a layer deeper in line 1. (iv) Yes.

(h) 3 P, $\wedge$E 1; 4 $R \wedge P$, $\wedge$I 2, 3.

(i) 2 $P \wedge Q$, $\wedge$E 1; 3 Q, $\wedge$E 2.

(j) 3 P, $\wedge$E 1; 4 S, $\wedge$E 2; 5 $S \wedge P$, $\wedge$I 4, 3.

Givens	Goal
$P \wedge (Q \wedge R)$	R
$P \wedge (Q \wedge R)$, P, $Q \wedge R$ ($\wedge$E)	R
$P \wedge (Q \wedge R)$, P, $Q \wedge R$, Q, R ($\wedge$E)	R ✓

(k)

(l) $P \wedge Q$ and $R \wedge S$ only.

1.2.13.2 B. Practice

Exercise. B1: The other grouping

Give a derivation of $P \wedge (Q \wedge R) \vdash (P \wedge Q) \wedge R$. For each line, say whether you found it from the goal or from the premise.

Solution. From the goal: $\wedge$I with subgoals $P \wedge Q$ and R, and $P \wedge Q$ by $\wedge$I from P and Q. From the premise: $\wedge$E gives P and $Q \wedge R$, and $\wedge$E on $Q \wedge R$ gives Q and R.

1	$P \wedge (Q \wedge R)$	
2	P	$\wedge$E 1
3	$Q \wedge R$	$\wedge$E 1
4	Q	$\wedge$E 3
5	R	$\wedge$E 3
6	$P \wedge Q$	$\wedge$I 2, 4
7	$(P \wedge Q) \wedge R$	$\wedge$I 6, 5

Lines 2–5 were found from the premise, lines 6–7 from the goal. With Proposition 1.2.2, each

of $(P \wedge Q) \wedge R$ and $P \wedge (Q \wedge R)$ is derivable from the other.

Exercise. B2: Annotating a proof

Let x and y be real numbers. Here is a proof.

> Let $1 < x < 2$ and $3 < y < 4$. Then $4 < x + y < 6$.

Write out the proof in full, naming each use of $\wedge$E and $\wedge$I, and marking the steps that are school algebra rather than logic.

Solution. The hypotheses are the conjunctions $(1 < x) \wedge (x < 2)$ and $(3 < y) \wedge (y < 4)$, and the goal is $(4 < x + y) \wedge (x + y < 6)$.

By $\wedge$E, four times, we have $1 < x$, $x < 2$, $3 < y$, and $y < 4$ separately. Adding $1 < x$ and $3 < y$ gives $1 + 3 < x + y$, that is, $4 < x + y$ (school algebra: two inequalities in the same direction may be added). Adding $x < 2$ and $y < 4$ gives $x + y < 6$ (school algebra again). Finally $\wedge$I joins $4 < x + y$ and $x + y < 6$ into $4 < x + y < 6$.

As a derivation it has the same shape as the one in Example (A product of numbers in the unit interval): two premises, four $\wedge$E lines, two lines justified "algebra", and one $\wedge$I line.

Exercise. B3: Two layers, then rebuild

A search for $P \wedge (Q \wedge R),\ S \vdash (R \wedge P) \wedge S$ has reached the stage below. *Name the move, then complete it.*

1	$P \wedge (Q \wedge R)$	
2	S	
	$\vdots$	
7	$(R \wedge P) \wedge S$	?

Solution. The move is *work from both ends* ("Two directions of reasoning"). From the goal: its main connective is $\wedge$, so line 7 is $\wedge$I with the subgoals $R \wedge P$ and S; S is line 2, and $R \wedge P$ splits again, by $\wedge$I, into R and P. From the premise: $\wedge$E in two layers, first P and $Q \wedge R$ from line 1, then R from $Q \wedge R$.

1	$P \wedge (Q \wedge R)$	
2	S	
3	P	$\wedge$E 1
4	$Q \wedge R$	$\wedge$E 1
5	R	$\wedge$E 4
6	$R \wedge P$	$\wedge$I 5, 3
7	$(R \wedge P) \wedge S$	$\wedge$I 6, 2

The ends meet at line 6: the subgoals R and P are lines 5 and 3, which the premise supplied.

1.2.13.3 C. Going deeper

Exercise. C1: A bad rule

Add to Tr a second rule, Flip: from $a < b$, write $b < a$.

(a) Derive $1 < 0$ from the premise $0 < 1$, using Flip.

(b) Explain why a checker following "Reading a derivation" would accept your derivation.

(c) What property does Tr have that Flip lacks?

Solution.

(a)

1	$0 < 1$	
2	$1 < 0$	Flip 1

(b) Line 1 is a premise. Line 2 names a rule on the list, cites line 1, which is earlier, and lines 1, 2 are an instance of the schema of Flip with a replaced by 0 and b by 1. All three questions pass. The checker never asks whether a line is true, only whether it matches a rule, so a bad rule is invisible to it.

(c) Tr never leads from true lines to a false one (A1(e)). Flip does: from the true line $0 < 1$ it writes the false line $1 < 0$. With such a rule, a derivation from true premises could end anywhere, so checking is only as good as the rules checked against. The rules of logic are chosen to have it; this section's Looking-ahead box says where that is checked.

Exercise. C2: Find the error

A student offers the following list as a derivation of $0 < 4$ with Tr.

1	$0 < 1$	
2	$1 < 2$	
3	$2 < 3$	
4	$3 < 4$	
5	$0 < 3$	Tr 6, 3
6	$0 < 2$	Tr 1, 2
7	$0 < 4$	Tr 5, 4

(a) Is $0 < 1,\ 1 < 2,\ 2 < 3,\ 3 < 4 \vdash 0 < 4$ with Tr?

(b) Is the list a derivation? If not, name the line that fails and the clause of Definition 1.2.2 it breaks.

(c) *(Challenge.)* The two lines "1. $0 < 0$, Tr $2, 2$" and "2. $0 < 0$, Tr $1, 1$" would each pass the first two questions of "Reading a derivation". Use them to explain why a line may cite only earlier lines.

Solution.

(a) Yes. Swap lines 5 and 6 and renumber:

1	$0 < 1$	
2	$1 < 2$	
3	$2 < 3$	
4	$3 < 4$	
5	$0 < 2$	Tr 1, 2
6	$0 < 3$	Tr 5, 3
7	$0 < 4$	Tr 6, 4

Each later line cites earlier lines and is an instance of Tr, with middles 1, 2, 3. So the student's conclusion is derivable (and true).

(b) No. Line 5 cites line 6, which is not earlier, so clause (ii) fails at line 5. Every line is an instance of Tr: lines 6, 3, 5 with middle 2; lines 1, 2, 6 with middle 1; lines 5, 4, 7 with middle 3. So the only fault is the order. That the conclusion is true and derivable does not repair the list: a derivation must meet the definition line by line.

(c) Line 1 cites line 2 twice, and line 2 cites line 1 twice. Each is an instance of Tr with a, b, c all replaced by 0: from $0 < 0$ and $0 < 0$, write $0 < 0$. So if lines could cite later lines, these two lines would justify each other, in a loop that rests on no premise, and the false statement $0 < 0$ would be derivable from nothing. Requiring cited lines to be earlier rules out every such loop: following the citations backwards from any line, the numbers go down, so they must end, and they can end only at premises.

Marking scheme. (10 points) - 3: says the conclusion is derivable and gives a correct derivation (every line an instance, every citation earlier). - 3: names line 5 as the faulty line and the broken clause: a cited line must be earlier. - 1: notes that the other lines are instances of Tr, so the order is the only fault. - 3: explains with the loop why later citations cannot be allowed: the two lines justify each other, rest on no premise, and give a false

line. Fatal errors (0 for the step): calling the list a derivation because its conclusion is true; blaming line 7, which cites line 5, an earlier line.

Exercise. C3: A shortcut through two layers

A student offers the following derivation of $P \wedge (Q \wedge R) \vdash Q \wedge P$.

1	$P \wedge (Q \wedge R)$	
2	Q	$\wedge$E 1
3	P	$\wedge$E 1
4	$Q \wedge P$	$\wedge$I 2, 3

(a) Is line 2 an instance of a schema of $\wedge$E? Justify your answer.

(b) Is the conclusion derivable from the premise? Justify your answer.

(c) The step at line 2 never leads from a true line to a false one. Why must a checker still reject it?

(d) *(Challenge.)* State, as a schema, a rule $\wedge$E* that would allow line 2. Show that every use of $\wedge$E* in a derivation can be replaced by two uses of $\wedge$E.

Solution.

(a) No. To match line 1 with $\varphi \wedge \psi$, we must read φ and ψ off line 1's main connective, which is the first $\wedge$: φ is P and ψ is $Q \wedge R$. So $\wedge$E writes P or $Q \wedge R$, and Q is neither.

(b) Yes. Insert the missing layer.

1	$P \wedge (Q \wedge R)$	
2	P	$\wedge$E 1
3	$Q \wedge R$	$\wedge$E 1
4	Q	$\wedge$E 3
5	$Q \wedge P$	$\wedge$I 4, 2

(c) A derivation is checked by matching each line against the schemas of the rules on the list, and line 2 matches none. A checker that also accepted steps that "obviously follow" would need to understand the statements, and then its verdict would be only as reliable as that understanding. The steps that seem obvious are the ones nobody looks at twice, and they are where unnoticed errors live. (This item is not a find-the-error exercise in the usual sense: the step has no counterexample, since it never leads from truth to falsehood. It is simply not a step of this calculus.)

(d) One such rule, in schema form:

i	$\varphi \wedge (\psi \wedge \chi)$	
k	ψ	$\wedge$E* i

Suppose line k of a derivation is ψ, justified "$\wedge$E* i", with line i the statement $\varphi \wedge (\psi \wedge \chi)$. Replace line k by two lines: first $\psi \wedge \chi$, justified $\wedge$E i (the second form, with the conjuncts φ

and $\psi \wedge \chi$), then ψ, justified by $\wedge$E citing that new line (the first form, with the conjuncts ψ and χ). Renumber the later lines, and every citation of them, up by one; a later citation of line k now cites the second new line, which carries the same statement ψ. Every other line is unchanged and still cites earlier lines. Doing this for each $\wedge$E* line, one at a time, gives a derivation with the same premises and last line that uses only $\wedge$E and $\wedge$I. A rule that can always be replaced in this way adds convenience and no new power; Section 2.3 calls such a rule a derived rule.

Marking scheme. (10 points)

- 2: (a) no, reading φ and ψ off the main connective of line 1.
- 2: (b) yes, with a correct derivation.
- 2: (c) checking is by matching the schemas; a step that is not an instance is rejected, whether or not it is true.
- 2: (d) a correct schema, one given line and one new line.
- 2: (d) the replacement by two $\wedge$E lines, with the renumbering of later citations.

Fatal errors (0 for the step): claiming the conclusion is not derivable; offering a counterexample to line 2.

Exercise. C4: Challenge: What conjunction cannot do

Show that no box-free derivation whose only premise is $P \wedge Q$, and whose only rules are $\wedge$I and $\wedge$E, has a line whose statement contains the letter R. Deduce that, with these two rules, $P \wedge Q \vdash P \wedge R$ cannot be shown.

Solution (Hint 1). Suppose some line does contain R, and look at the **first** such line.

Solution (Hint 2). That first line is not a premise. Which rule wrote it, and which letters can that rule write?

Solution. Suppose, for a contradiction, that some line of such a derivation contains R. The derivation has finitely many lines, so there is a first such line; call it line k. Every line before line k is free of R.

Line k is not a premise, because every premise is $P \wedge Q$, which does not contain R. So line k is justified by one of the two rules, citing earlier lines (Definition 1.2.2).

If it is $\wedge$I, citing lines i and j, then its statement is $\varphi \wedge \psi$, where φ and ψ are the statements of lines i and j. Both are earlier than k, so neither contains R, and neither does $\varphi \wedge \psi$.

If it is $\wedge$E, citing line i, then its statement is one conjunct of the statement of line i: a part of it. Line i is earlier than k and free of R, so the part is free of R too.

Either way line k does not contain R, which contradicts the choice of k. So no line contains R. In particular no such derivation has the last line $P \wedge R$, and $P \wedge Q \vdash P \wedge R$ cannot be shown with these rules.

The argument, "look at the first bad line; it cannot be a premise, and no rule can create it from good lines", is an induction on the lines of a derivation; Chapter 3 makes such arguments exact.

Marking scheme. (10 points)

- 2: considers a first line containing R, and says why a first one exists (finitely many lines); or, equally, goes down the derivation line by line, showing each line free of R because every line it cites is.
- 2: rules out a premise.
- 2: $\wedge$I case: the new line is built from the statements of two earlier, R-free lines.
- 2: $\wedge$E case: the new line is part of an earlier, R-free line.
- 2: concludes that no line contains R, and so that $P \wedge R$ is never a last line.

Fatal errors (0 for the step): arguing about one particular derivation instead of every derivation; arguing that R is "not true" instead of about which symbols the rules can write.

1.3 Using an Implication: Modus Ponens

Section 1.2 gave the two rules for "and". This section gives one rule for "implies": the rule that **uses** an implication already in hand. Proving one needs a new device, the subject of Section 1.4; using one is what a proof does every time it applies a theorem. The work is to see exactly what the rule allows, and to find chains of implications forwards and backwards.

1.3.1 Four cards

Before the rule, a puzzle about what "if ... then" says.

Example. Four cards

Each of four cards has a whole number on one side and a color on the other. They lie on a table showing

6, 7, red, green.

Someone claims: *if a card shows an even number on one side, then its other side is red.* Which cards must be turned over to find out whether the claim is true of these four cards? Decide before reading the solution.

Solution. For each card, try to break the claim: can its hidden side make a card that is even and not red? (By the promise reading of Section 1.1, that is the only way to break it.)

The card showing 6. Its number is even, so the hypothesis of the implication holds, and the claim says its other side is red. If that side is green, the claim is false. This card must be turned.

The card showing 7. Its number is odd, so the hypothesis fails, and the claim says nothing about this card. Whatever its color, the card cannot contradict the claim. Leave it.

The red card. Its color is the **conclusion** of the implication. If its number is even, the card fits the claim; if its number is odd, the card still fits, since the claim says only what even cards look like and nothing about which cards are red. No hidden number can contradict

the claim. Leave it.

The green card. If its number is even, it is an even card that is not red, and the claim is false. This card must be turned.

So the answer is the 6 and the green card. Many people choose the 6 and the **red** card. The red card is the tell: choosing it means reading the claim as if it also said "if a card is red, then its number is even", which is the **converse**. An implication and its converse are different claims, and the claim here is only the first.

Two of the cards show the two ways an implication meets a case: the 6 and the red card (not the pair that must be turned). The 6 shows the **hypothesis**, and an implication together with its hypothesis gives its conclusion: that is the rule of this section. The red card shows the **conclusion**, and an implication together with its conclusion gives nothing: that is the most important thing the rule does *not* allow. The green card shows that the conclusion fails; saying exactly what follows from that needs the rules for "not".

1.3.2 The rule

An implication fires on its hypothesis, and gives its conclusion.

Definition 1.3.1. Implication Elimination

i	$\varphi \to \psi$	
j	φ	
k	ψ	$\to$E i, j

This rule is called $\to$E (**implication-elimination**, or **modus ponens**).

We read it in four bullets.

- **Two given lines.** One line is an implication $\varphi \to \psi$, the other is φ. Both stand earlier in the derivation, anywhere above line k and in either order on the page.
- **Read φ and ψ off the main connective.** To see a line as $\varphi \to \psi$, its **main connective** (Definition 1.1.2) must be $\to$. Then φ, the **hypothesis**, is everything that this $\to$ joins on its left, and ψ, the **conclusion**, is everything on its right. In $(P \wedge Q) \to R$, the hypothesis is $P \wedge Q$ and the conclusion is R. In $P \to (Q \to R)$, the hypothesis is P and the conclusion is $Q \to R$, itself an implication.
- **The hypothesis must match exactly.** Line j must be the statement φ, symbol for symbol: the letter φ occurs in both given lines, and a use of the rule replaces it by the same statement in both (Definition 1.2.1). A part of φ will not do, and neither will a statement that "says the same thing". From $(P \wedge Q) \to R$ and $Q \wedge P$, the rule gives nothing, although $Q \wedge P$ and $P \wedge Q$ are each derivable from the other (Proposition 1.2.1); the hypothesis must first be rebuilt as $P \wedge Q$, by $\wedge$E and $\wedge$I.
- **The new line is the whole conclusion.** The rule writes ψ, the whole of what stands right of the main $\to$, and nothing else. If ψ is $Q \wedge R$, the rule writes $Q \wedge R$, and getting Q takes one more line, by $\wedge$E.

The citation lists the implication first and its hypothesis second, always: "$\to$E i, j", as in the schema. Only ψ can come out, so the order changes nothing; it makes every $\to$E line read the

same way.

↻ **Second reading** · *A legal, useless line*

The statements φ and ψ may be the same. From $P \to P$ and P, the rule writes P, which was already there. The line is legal and useless.

Why this rule. It is what "by Theorem so-and-so" means. For integers d and n, write $d \mid n$, read "d divides n", when $n = dk$ for some integer k. A theorem of school arithmetic says: if an integer is divisible by 2 and by 3, then it is divisible by 6. For $n = 42$ it says

$$\big((2 \mid 42) \wedge (3 \mid 42)\big) \to (6 \mid 42).$$

We know $2 \mid 42$ ($42 = 2 \cdot 21$) and $3 \mid 42$ ($42 = 3 \cdot 14$). $\wedge$I builds the hypothesis, and $\to$E fires: $6 \mid 42$, which is right, as $42 = 6 \cdot 7$. As a derivation, with the theorem's instance and the two facts as premises:

1	$((2 \mid 42) \wedge (3 \mid 42)) \to (6 \mid 42)$	
2	$2 \mid 42$	
3	$3 \mid 42$	
4	$(2 \mid 42) \wedge (3 \mid 42)$	$\wedge$I 2, 3
5	$6 \mid 42$	$\to$E 1, 4

Every application of a theorem has this shape: check the theorem's hypotheses, assemble them into exactly the theorem's hypothesis, and fire.

Quick check. To get "36 divides 72" from the theorem "if 4 and 9 divide n, then 36 divides n" at $n = 72$, which line, besides the theorem's instance itself, must stand on the page before $\to$E fires?

Solution. The conjunction $(4 \mid 72) \wedge (9 \mid 72)$, exactly: the theorem's hypothesis at $n = 72$. It is built by $\wedge$I from the lines $4 \mid 72$ and $9 \mid 72$, which hold because $72 = 4 \cdot 18$ and $72 = 9 \cdot 8$. Then $\to$E gives $36 \mid 72$, and indeed $72 = 36 \cdot 2$.

Why it is safe. If $\varphi \to \psi$ and φ are true, then ψ is true: that is what the promise says.

Why nothing more. From $\varphi \to \psi$ alone, $\to$E cannot fire: an implication by itself asserts neither its hypothesis nor its conclusion. And $\to$E does not run backwards: from $\varphi \to \psi$ together with its conclusion ψ, it gives nothing about φ. That pattern,

$$\varphi \to \psi,\ \psi \ \text{ therefore } \ \varphi,$$

is called **affirming the consequent**, and it is the red card of the puzzle. It is not an instance of $\to$E, because the second given line of $\to$E must be the hypothesis φ, and here it is the conclusion ψ. It is a bad form: Section 1.1 broke it at $n = 2$. It is also the step that would repair Argument B.

Warning. Affirming the consequent looks like modus ponens and is not. Let x be a real number. "If $x > 2$, then $x^2 > 4$" is true for every x, and at $x = -3$ we have $x^2 = 9 > 4$; but $x > 2$ is false there. From an implication and its **conclusion**, nothing follows about the hypothesis.

Quick check. Line 1 is $P \to (Q \to R)$. What does $\to$E give from line 1 and a line Q? And from line 1 and a line P?

Solution. From line 1 and Q: nothing. The main connective of line 1 is the first $\to$, so its hypothesis is P, and Q does not match it. From line 1 and P: the conclusion $Q \to R$. A second application, to $Q \to R$ and the line Q, would then give R.

Remark. At this point an implication can be **used** but not **proved**. A derivation can write an implication only by taking it out of a line that already contains it: $\wedge$E applied to $(P \to Q) \wedge R$, or $\to$E applied to $P \to (Q \to R)$ and P. No rule so far builds a new $\to$. Section 1.4 gives the rule that does.

1.3.3 Chains of implications

With $\to$E, a derivation can pass along a chain: from P and $P \to Q$ to Q, from Q and $Q \to R$ to R, and so on. Such a chain can be found from either end.

Forwards, from the premises. An available implication (for now, one on an earlier line) whose hypothesis is also available can **fire**: $\to$E writes its conclusion, which becomes available in turn and may be the hypothesis of another implication. Keep firing until the goal appears. This is premise-directed reasoning, as in Section 1.2.

Backwards, from the goal. Suppose the goal is ψ. Look among the available lines for an implication whose **conclusion** is ψ, say $\varphi \to \psi$. If there is one, it is enough to reach φ: then $\to$E writes ψ. So φ becomes the new goal, and we repeat. This is goal-directed reasoning of a new kind: in Section 1.2 the goal's main connective chose an introduction rule, but here the goal may be a letter, and it still tells us which implication to look for.

The two directions meet in the middle, when a line reached forwards is a goal reached backwards. In written mathematics the backward step has a standard phrase, "to show ψ, it suffices to show φ".

In the Givens/Goal table, $\to$E fired forwards adds ψ to the Givens; worked backwards, when $\varphi \to \psi$ is among the Givens, it replaces the goal ψ by φ.

Quick check. The premises are P, $P \to R$, $Q \to S$, and $R \to S$, and the goal is S. Search backwards: which implications could give the goal, and which of them leads to a derivation?

Solution. Two premises have conclusion S: $Q \to S$ and $R \to S$. The first makes Q the new goal, and no line is Q or has conclusion Q, so that branch is a dead end. The second makes R the new goal; $P \to R$ has conclusion R, and its hypothesis P is a premise. So the derivation is R by $\to$E from $P \to R$ and P, then S by $\to$E from $R \to S$ and R. A backward search may branch, and a branch that stalls is abandoned for the next one.

Example. Chains of implications

Give a derivation of each sequent.

(a) $P \to Q,\ P \vdash Q$.
(b) $P,\ P \to Q,\ Q \to R \vdash R$.
(c) $P \wedge Q,\ P \to (Q \to R) \vdash R$.
(d) $(P \wedge Q) \to R,\ P,\ Q \vdash R$.
(e) $(P \to Q) \wedge (R \to S),\ P \wedge R \vdash Q \wedge S$.

Solution.

(a) *Idea.* There is nothing to choose: the premises are an implication and its hypothesis, and the rule fires once.

1	$P \to Q$	
2	P	
3	Q	$\to$E 1, 2

This is the form of the two short arguments at the start of "The form of an argument" in Section 1.1 (n odd, so n^2 odd; $x > 2$, so $x^2 > 4$), now shown as a sequent: a derivation with these premises and this last line exists.

(b) *Idea, forwards.* The premise $P \to Q$ has its hypothesis P available, so it fires and gives Q. Now $Q \to R$ has its hypothesis available, so it fires and gives R, the goal.

1	P	
2	$P \to Q$	
3	$Q \to R$	
4	Q	$\to$E 2, 1
5	R	$\to$E 3, 4

Idea, backwards. The goal R is a letter. Which available implication has conclusion R? Only $Q \to R$, so it suffices to reach Q (left picture: the goal at the bottom, its new subgoal Q above it). Which implication has conclusion Q? Only $P \to Q$, so it suffices to reach P; and P is a premise, so the gap closes (right picture).

1	P	
2	$P \to Q$	
3	$Q \to R$	
	$\vdots$	
4	Q	
5	R	$\to$E 3, 4

1	P	
2	$P \to Q$	
3	$Q \to R$	
4	Q	$\to$E 2, 1
5	R	$\to$E 3, 4

Both searches write the same five lines, the forward one from the top and the backward one from the last line up. Exercise B3 has a case where the backward search is much shorter.

(c) *Idea.* The goal R is a letter. From the goal: the only line that mentions R is $P \to (Q \to R)$, but its conclusion is $Q \to R$, not R; the R sits one implication deeper. From the premises: $P \wedge Q$ is a conjunction, so take it apart by $\wedge$E, giving P and Q. Now P is exactly the hypothesis of $P \to (Q \to R)$, which fires and gives $Q \to R$. This new line is an implication whose hypothesis Q is available, so it fires too. In the pictures, the goal stands alone at first; then $\wedge$E writes lines 3 and 4, and only line 5 is missing; line 6 stays unjustified until line 5 is there for $\to$E to cite.

1	$P \wedge Q$	
2	$P \to (Q \to R)$	
	$\vdots$	
6	R	

1	$P \wedge Q$	
2	$P \to (Q \to R)$	
3	P	$\wedge$E 1
4	Q	$\wedge$E 1
	$\vdots$	
6	R	

1	$P \wedge Q$	
2	$P \to (Q \to R)$	
3	P	$\wedge$E 1
4	Q	$\wedge$E 1
5	$Q \to R$	$\to$E 2, 3
6	R	$\to$E 5, 4

Line 6 is the first line in this chapter where $\to$E fires on an implication that was itself **written by** $\to$E, not given as a premise.

(d) *Idea.* This time the goal leads. The goal R is the conclusion of $(P \wedge Q) \to R$, so it suffices to reach its hypothesis $P \wedge Q$. That subgoal is a conjunction, so, goal-directed as in Section 1.2, it comes from $\wedge$I with subgoals P and Q, and both are premises. In the pictures, the goal stands alone at first; then the subgoal $P \wedge Q$, read off line 1, sits above it as line 4, still unjustified.

1	$(P \wedge Q) \to R$
2	P
3	Q
	$\vdots$
5	R

1	$(P \wedge Q) \to R$	
2	P	
3	Q	
	$\vdots$	
4	$P \wedge Q$	
5	R	$\to$E 1, 4

1	$(P \wedge Q) \to R$	
2	P	
3	Q	
4	$P \wedge Q$	$\wedge$I 2, 3
5	R	$\to$E 1, 4

Compare (c). There the implication took its two hypotheses one at a time, $P \to (Q \to R)$, and the conjunction was a premise to be taken apart. Here the implication takes both hypotheses at once, $(P \wedge Q) \to R$, and the conjunction is a subgoal to be built. The content is the same; the brackets decide which rules the derivation uses. (Section 1.4 shows that each of the two implications is derivable from the other.)

(e) *Idea.* Both ends at once. From the goal: $Q \wedge S$ is a conjunction, so its last line is $\wedge$I, with subgoals Q and S. From the premises: both are conjunctions, so $\wedge$E takes each apart, giving $P \to Q$, $R \to S$, P, and R. The two ends meet through $\to$E: $P \to Q$ fires on P and gives the subgoal Q, and $R \to S$ fires on R and gives the subgoal S. In the second picture lines 3–6 came from the premises and line 9 from the goal; lines 7–8, still unjustified, are where the two searches meet.

1	$(P \to Q) \wedge (R \to S)$
2	$P \wedge R$
	$\vdots$
9	$Q \wedge S$

1	$(P \to Q) \wedge (R \to S)$	
2	$P \wedge R$	
3	$P \to Q$	$\wedge$E 1
4	$R \to S$	$\wedge$E 1
5	P	$\wedge$E 2
6	R	$\wedge$E 2
	$\vdots$	
7	Q	
8	S	
9	$Q \wedge S$	$\wedge$I 7, 8

1	$(P \to Q) \wedge (R \to S)$	
2	$P \wedge R$	
3	$P \to Q$	$\wedge$E 1
4	$R \to S$	$\wedge$E 1
5	P	$\wedge$E 2
6	R	$\wedge$E 2
7	Q	$\to$E 3, 5
8	S	$\to$E 4, 6
9	$Q \wedge S$	$\wedge$I 7, 8

Each derivation in the example is a box-free derivation (Definition 1.2.2) with $\wedge$I, $\wedge$E, and $\to$E as its rules, so by Definition 1.2.3 each shows its sequent.

1.3.4 Checking a line of modus ponens

Finding a chain can take some searching; checking one never does. For a line justified "$\to$E i, j", the questions of Section 1.2 (which rule? is it an instance? are the cited lines earlier?) become four small ones:

1. Are lines i and j earlier than this line?
2. Is the main connective of line i a $\to$? If so, read off φ (its left side) and ψ (its right side).
3. Is line j exactly φ, symbol for symbol?
4. Is this line exactly ψ?

Here is the check run on lines 5 and 6 of (c), and on a tempting wrong line that might replace them, "R — $\to$E 2, 4".

Line	Cites	φ, ψ read off line i	Line j is φ?	Line is ψ?
5: $Q \to R$	2, 3	P, $Q \to R$	yes (P)	yes
6: R	5, 4	Q, R	yes (Q)	yes
wrong: R	2, 4	P, $Q \to R$	no (Q is not P)	no (R is not $Q \to R$)

The wrong line fails twice: it feeds line 2 the wrong hypothesis and takes out a part of the conclusion. The main connective settles it: the first $\to$ is the one applied last, so everything after it, $Q \to R$, is the conclusion.

1.3.5 How this looks in a mathematical proof

Let x be a real number with $x > 2$. Suppose we know two facts of school algebra:

- if $x > 2$, then $x^2 > 4$ (since $x > 2 > 0$, multiplying $x > 2$ by x gives $x^2 > 2x$, and $2x > 4$);
- if $x^2 > 4$, then $x^2 + 1 > 5$ (add 1 to both sides).

A written proof that $x^2 + 1 > 5$ takes one sentence: *Since $x > 2$, we have $x^2 > 4$, and so $x^2 + 1 > 5$.* As a derivation, with the two facts and the hypothesis as premises, it reads as follows.

1	$(x > 2) \to (x^2 > 4)$	
2	$(x^2 > 4) \to (x^2 + 1 > 5)$	
3	$x > 2$	
4	$x^2 > 4$	$\to$E 1, 3
5	$x^2 + 1 > 5$	$\to$E 2, 4

Each piece of the sentence is one line.

Line	Sentence
3	"Since $x > 2$, ..." (the hypothesis)
4	"... we have $x^2 > 4$, ..." ($\to$E, firing the first fact)
5	"... and so $x^2 + 1 > 5$." ($\to$E, firing the second fact)

The sentence never mentions lines 1 and 2: writers leave a known implication unstated. Each "since φ, ψ" in a proof is one application of $\to$E to an implication the reader is trusted to supply. When a proof is hard to follow, writing those implications down is often how one finds the gap: an implication is false, or its hypothesis was never established.

1.3.6 Common errors

The first warning of this section was about the wrong given line: the conclusion in place of the hypothesis. The other common error is a hypothesis that is only partly there.

Warning. $\to$E needs the **whole** hypothesis. From $(P \wedge Q) \to R$ and P alone, it gives nothing. Take the true theorem "if $x > 0$ and $y > 0$, then $xy > 0$" and the line $x > 0$: at $x = 1$, $y = -1$ the theorem and $x > 0$ are true, and $xy > 0$ is false.

↻ **Second reading** · *The same slip in analysis*

This is the commonest slip in analysis: "f is continuous, so by the theorem f attains a maximum". The theorem meant has a second hypothesis, that f is defined on a closed bounded interval, and without it the conclusion fails: $f(x) = x$ on the open interval $(0, 1)$ is continuous and has no largest value, since $\frac{a+1}{2}$ is a larger value than a for every a in $(0, 1)$.

A near relative is a hypothesis in the wrong order, as in the third bullet of the rule's reading: $Q \wedge P$ is not the hypothesis of $(P \wedge Q) \to R$. The repair is two $\wedge$E lines and one $\wedge$I line, and Exercise B1 asks for it.

1.3.7 The rule card

The card of Section 1.2 grows by one row. *Shape* shows the given lines, then the new line; *You write* is the phrase a written proof uses for the rule.

Rule	Shape	You write	Lean
$\wedge$I	$\varphi, \psi \Longrightarrow \varphi \wedge \psi$	"We show P, and we show Q."	`constructor`
$\wedge$E	$\varphi \wedge \psi \Longrightarrow \varphi$ (or ψ)	"In particular, P."	`h.1` (and `h.2`)
$\to$E	$\varphi \to \psi, \varphi \Longrightarrow \psi$	"Since P implies Q, and P holds, Q."	`exact h hp`

In the Lean column, `h` is the implication and `hp` the proof of its hypothesis; `h hp` applies the one to the other, which is modus ponens.

Remark (Looking ahead). Chapter 3 proves more than "affirming the consequent is a bad form": no derivation, however long, works for every choice of P and Q and leads from $P \to Q$ and Q to P.

1.3.8 Exercises

1.3.8.1 A. Check your understanding

Exercise. A1

(a) Write the schema of $\to$E from memory.

(b) Lines 1 and 2 of a derivation are $P \to (Q \wedge R)$ and P. Which of the following can line 3 be, by one application of $\to$E? $Q \wedge R$; Q; $R \wedge Q$; $P \to Q$.

(c) In Example (Four cards), why is the red card not worth turning?

(d) True or false, with a reason: from lines $P \to Q$ and $Q \to R$, $\to$E gives $P \to R$.

Solution.

(a) Given lines i: $\varphi \to \psi$ and j: φ; new line k: ψ, justified "$\to$E i, j" (Definition 1.3.1).

(b) Only $Q \wedge R$, the conclusion of line 1. The line Q needs a further $\wedge$E, and $R \wedge Q$ two further $\wedge$E lines and a $\wedge$I line. The line $P \to Q$ is not the conclusion of line 1, so $\to$E cannot write it.

(c) The claim is "even $\to$ red", and the red card shows the conclusion. An odd number behind it would not contradict the claim, which says nothing about which cards are red; an even number would fit it. Inferring anything about its number from its color would be affirming the consequent.

(d) False. $\to$E needs an implication and a line that is its hypothesis. The hypothesis of $Q \to R$ is Q, and $P \to Q$ is not Q; the hypothesis of $P \to Q$ is P, which is not a line. So $\to$E does not apply at all. (The sequent $P \to Q,\ Q \to R \vdash P \to R$ does hold, but its derivation needs the rule of Section 1.4.)

Exercise. A2: Drill

Each part takes a line or two. Lines are numbered as in a derivation.

(a) Lines 1 $(P \vee Q) \to R$, 2 $P \vee Q$. What does "$\to$E 1, 2" give?

(b) Lines 1 $P \to (Q \to R)$, 2 Q. Is "$\to$E 1, 2" legal?

(c) Lines 1 $(P \wedge Q) \to R$, 2 $Q \wedge P$. Is "R — $\to$E 1, 2" legal?

(d) Lines 1 $P \to Q \wedge R$, 2 P. What does "$\to$E 1, 2" give?

(e) Lines 1 $P \to Q$, 2 R, 3 $R \wedge P$. Is "4. Q — $\to$E 1, 3" legal?

(f) Justify each line, the first two being premises: 1 $(P \to Q) \wedge R$, 2 $R \to P$, 3 $P \to Q$, 4 R, 5 P, 6 Q.

(g) Derive $P \wedge (P \to Q) \vdash Q$.

(h) Derive $P \to (Q \to R),\ P,\ Q \vdash R$.

(i) Derive $P \to (Q \wedge R),\ (Q \wedge R) \to S,\ P \vdash S \wedge Q$.

(j) Break the form of "if n is divisible by 6, then by 3; n is divisible by 3; so n is divisible by 6".

(k) Premises $P \to Q$, $R \to Q$, R; goal Q. Searching backwards, which implication leads to a derivation?

Solution.

(a) R, the conclusion of line 1, whose hypothesis $P \vee Q$ is line 2 exactly.

(b) No: the hypothesis of line 1 is P, and line 2 is Q.

(c) No: the hypothesis must be $P \wedge Q$ exactly, and $Q \wedge P$ is a different statement.

(d) $Q \wedge R$: since $\wedge$ binds more tightly, line 1 is $P \to (Q \wedge R)$. Getting Q needs one more $\wedge$E line.

(e) No: line 3 is $R \wedge P$, not the hypothesis P. First write P by $\wedge$E 3 as line 4, then Q by $\to$E 1, 4.

(f) Lines 1 and 2 are premises; 3 is $\wedge$E 1; 4 is $\wedge$E 1; 5 is $\to$E 2, 4; 6 is $\to$E 3, 5. An implication cut out of a conjunction fires on a line that $\to$E wrote.

(g) 2 P, $\wedge$E 1; 3 $P \to Q$, $\wedge$E 1; 4 Q, $\to$E 3, 2.

(h) 4 $Q \to R$, $\to$E 1, 2; 5 R, $\to$E 4, 3.

(i) 4 $Q \wedge R$, $\to$E 1, 3; 5 S, $\to$E 2, 4; 6 Q, $\wedge$E 4; 7 $S \wedge Q$, $\wedge$I 5, 6.

(j) Take $n = 3$. The first premise is true for every integer, since $6k = 3 \cdot 2k$; the second is true, since $3 = 3 \cdot 1$; and 3 is not divisible by 6. Both premises true, conclusion false: the form is affirming the consequent.

(k) $R \to Q$: its hypothesis R is a premise, so $\to$E fires at once. The other, $P \to Q$, makes P the new goal, and nothing gives P: a dead end.

1.3.8.2 B. Practice

Exercise. B1: Two chains

Give a derivation of each sequent. For each, say which step you found first, and whether it came from the goal or from a premise.

(a) $P,\ P \to Q,\ Q \to (R \wedge S) \vdash S \wedge Q$.
(b) $(P \wedge Q) \to R,\ Q \wedge P \vdash R$.

Solution.

(a) The first step found was from the goal: $S \wedge Q$ is a conjunction, so the last line is $\wedge$I with subgoals S and Q. Then from the premises: $P \to Q$ fires on P and gives Q, and $Q \to (R \wedge S)$ fires on Q and gives $R \wedge S$, which $\wedge$E takes apart.

1	P	
2	$P \to Q$	
3	$Q \to (R \wedge S)$	
4	Q	$\to$E 2, 1
5	$R \wedge S$	$\to$E 3, 4
6	S	$\wedge$E 5
7	$S \wedge Q$	$\wedge$I 6, 4

Line 4 is cited twice, by line 5 and by line 7.

(b) The first step found was from the goal: R is the conclusion of line 1, so it suffices to reach $P \wedge Q$ exactly. The premise $Q \wedge P$ is not it. So take the premise apart and rebuild in the right order.

1	$(P \wedge Q) \to R$	
2	$Q \wedge P$	
3	Q	$\wedge$E 2
4	P	$\wedge$E 2
5	$P \wedge Q$	$\wedge$I 4, 3
6	R	$\to$E 1, 5

Exercise. B2: Two facts and a number

Let x be a real number. Take as premises the two facts "if $0 < x < 1$, then $x^2 < x$" and "if $x^2 < x$, then $x^3 < x^2$", and the hypothesis $0 < x < 1$. Derive $x^3 < x^2$, naming the rule on each line. Then write the argument as one or two sentences of prose.

Solution. The hypothesis $0 < x < 1$ is the conjunction $(0 < x) \wedge (x < 1)$, which is exactly the hypothesis of the first fact, so the first fact fires on it at once.

1	$((0 < x) \wedge (x < 1)) \to (x^2 < x)$	
2	$(x^2 < x) \to (x^3 < x^2)$	
3	$(0 < x) \wedge (x < 1)$	
4	$x^2 < x$	$\to$E 1, 3
5	$x^3 < x^2$	$\to$E 2, 4

In prose: *Since* $0 < x < 1$*, we have* $x^2 < x$*, and therefore* $x^3 < x^2$. Both facts are true for every real x: if $0 < x < 1$, multiplying $x < 1$ by $x > 0$ gives $x^2 < x$; and if $x^2 < x$, then $x(x-1) < 0$, so $x > 0$, and multiplying $x^2 < x$ by x gives $x^3 < x^2$.

Exercise. B3: Which premises are needed?

Find a derivation of

$$P,\ P \to Q,\ P \to R,\ R \to S,\ S \to U,\ Q \to T \ \vdash\ T,$$

and list the premises it uses. Explain why a backward search finds the needed premises faster than a forward one here.

Solution. Search backwards. The goal T is the conclusion of $Q \to T$ only, so it suffices to reach Q. The goal Q is the conclusion of $P \to Q$ only, so it suffices to reach P, a premise.

1	P	
2	$P \to Q$	
3	$P \to R$	
4	$R \to S$	
5	$S \to U$	
6	$Q \to T$	
7	Q	$\to$E 2, 1
8	T	$\to$E 6, 7

It uses premises 1, 2, and 6. The others are never cited, which is allowed: a derivation need not use every premise (Definition 1.2.3). Deleting lines 3–5 and renumbering gives a derivation of $P,\ P \to Q,\ Q \to T \vdash T$.

A forward search fires every implication whose hypothesis is available, relevant or not: from P it writes Q and R, then S from R, then U from S, and T from Q — five new lines, three of them useless. The backward search looks only at implications whose conclusion is the current goal, and at each stage exactly one qualifies, so it writes only the two lines the derivation needs.

Exercise. B4: Assemble the hypothesis

A search for $P \to (Q \wedge R)$, $(R \wedge P) \to S$, $P \vdash S$ has reached the stage below. *Name the move, then complete it.*

1	$P \to (Q \wedge R)$	
2	$(R \wedge P) \to S$	
3	P	
	⋮	
7	S	?

Solution. The move is to work **backwards from the goal**. The goal S is the conclusion of line 2, so it suffices to reach line 2's hypothesis $R \wedge P$, exactly, and then fire. That subgoal is a conjunction: P is line 3, and R comes from line 1, which fires on P and gives $Q \wedge R$, whose right half is R.

1	$P \to (Q \wedge R)$	
2	$(R \wedge P) \to S$	
3	P	
4	$Q \wedge R$	$\to$E 1, 3
5	R	$\wedge$E 4
6	$R \wedge P$	$\wedge$I 5, 3
7	S	$\to$E 2, 6

Line 6 writes $R \wedge P$, with R on the left, because that is how the hypothesis of line 2 is written; $P \wedge R$ would not match it. Line 3 is cited twice, by lines 4 and 6.

1.3.8.3 C. Going deeper

Exercise. C1: Find the error

Here is an argument, followed by the derivation its author offers for it.

> If 10 is divisible by 6, then 10 is even. And 10 is even. So 10 is divisible by 6.

With D: "10 is divisible by 6" and E: "10 is even":

1	$D \to E$	
2	E	
3	D	$\to$E 1, 2

(a) Name the line that fails, and say why it is not an instance of the schema of $\to$E.

(b) Show that the step can lead from true premises to a false conclusion.

(c) What new statement about divisibility can be correctly derived from the two premises with the rules so far?

Solution.

(a) Line 3. The schema needs, besides the implication $\varphi \to \psi$, a line that is its hypothesis φ. Matching line 1 gives $\varphi := D$ and $\psi := E$, so the second cited line would have to be D; line 2 is E, which is ψ. The line affirms the consequent.

(b) Try to break the step: this instance already does. The first premise is true, because every multiple of 6 is even ($6k = 2(3k)$); the second is true, since $10 = 2 \cdot 5$; and the conclusion is false: $6k$ is at most 6 when $k \leq 1$ and at least 12 when $k \geq 2$, so $10 = 6k$ for no integer k. So the step at line 3 leads from two true premises to a false line, and no safe rule can allow it.

(c) None. $\to$E cannot fire, since no line is the hypothesis D; $\wedge$E has no conjunction to take apart; and $\wedge$I only joins the premises into conjunctions such as $(D \to E) \wedge E$, which say nothing the premises did not. The premises tell us nothing about whether 10 is divisible by 6, which is why the conclusion had to be smuggled in.

Exercise. C2: Both ends

Consider the sequent

$$P \to (Q \to R),\ S \to P,\ S \wedge Q \ \vdash\ R \wedge S.$$

(a) Describe a search for a derivation in words, in three parts: what the goal asks for; what the premises give; where the two meet.

(b) Write the derivation, naming the rule on every line.

Solution (Hint 1). Two premises have main connective $\to$ and one has $\wedge$; the goal has $\wedge$. Write down what each main connective lets you do.

Solution (Hint 2). The goal splits into R and S. Which available implication could end in R, and after how many firings?

Solution.

(a) *From the goal:* $R \wedge S$ is a conjunction, so the last line is $\wedge$I with subgoals R and S. The only line mentioning R is $P \to (Q \to R)$, whose conclusion is $Q \to R$; so R needs $Q \to R$ and Q, and $Q \to R$ needs P. *From the premises:* $S \wedge Q$ splits by $\wedge$E into S and Q; then $S \to P$ fires on S and gives P. *Where they meet:* S is at once a subgoal and a piece of the premise; P, reached forwards, is the hypothesis the backward search asked for; and Q, from $\wedge$E, is the last hypothesis R needs.

(b) The derivation, with the rule on every line:

1	$P \to (Q \to R)$	
2	$S \to P$	
3	$S \wedge Q$	
4	S	$\wedge$E 3
5	Q	$\wedge$E 3
6	P	$\to$E 2, 4
7	$Q \to R$	$\to$E 1, 6
8	R	$\to$E 7, 5
9	$R \wedge S$	$\wedge$I 8, 4

Line 4 is cited twice, by line 6 and by line 9.

Marking scheme. (10 points)

- 2: from the goal, splits $R \wedge S$ into the subgoals R and S for a final $\wedge$I.
- 2: takes $S \wedge Q$ apart by $\wedge$E.
- 3: fires $S \to P$, then $P \to (Q \to R)$, then $Q \to R$, each $\to$E citing an implication and exactly its hypothesis.
- 2: the final $\wedge$I, whose line is $R \wedge S$.
- 1: the three-part description names where the searches meet.

Any other correct derivation earns the same points step for step (its lines may come in another order, as long as each cites only earlier lines). Fatal errors (0 for the step): $\to$E on $P \to (Q \to R)$ with Q (the wrong hypothesis); affirming the consequent anywhere.

1.4 Proving an Implication: the Rule of the Box

Sections 1.2 and 1.3 gave three rules. $\wedge$I builds a conjunction; $\wedge$E and $\to$E take a line apart or use it. With them we can **use** an implication, but we cannot **prove** one. This section fills that gap with the rule $\to$I. The rule asks us to do something new: to assume a statement we do not know, inside a boundary that says where the assumption holds. The boundary is a **box**. With the box we can name the error of Argument B, and say exactly which earlier lines a line may cite.

1.4.1 Supposing something we do not know

Here is a small theorem about a real number x.

If $x > 2$, then $x^2 > 4$.

The proof everyone would write goes like this. *Suppose $x > 2$. Then $x > 0$, so multiplying $x > 2$ by x gives $x^2 > 2x$. Also $2x > 4$. Hence $x^2 > 4$.* Then, often without saying it: *this proves that if $x > 2$, then $x^2 > 4$.*

Look at the first sentence. We do not know that $x > 2$. The number x is arbitrary, and it may

well be 0. So "suppose $x > 2$" asserts nothing. It opens a **temporary world**, one in which $x > 2$ holds, and asks what else holds there.

Inside that world, the sentences "$x^2 > 2x$", "$2x > 4$", and "$x^2 > 4$" are true. Outside it they are not: at $x = 0$ all three are false. Only the last sentence of the proof leaves the temporary world. What it carries out is not "$x^2 > 4$" but the implication "if $x > 2$, then $x^2 > 4$", and that is true for every real x.

None of the rules so far can do this: $\wedge$I writes a conjunction, and $\wedge$E and $\to$E write a piece of a line that is already there. To prove "if $x > 2$, then $x^2 > 4$" from nothing, we must be allowed to suppose.

A supposition is only safe if it has a boundary. Without one, the proof above could go on: "... Hence $x^2 > 4$. So $x^2 > 4$ for every real x." That is false at $x = 0$. The error is not in any computation; it is that a sentence from inside the temporary world was used outside it.

A derivation draws the boundary as a **box**. The assumption opens the box: it is written with a vertical bar to its left and a short horizontal rule under it. Every line derived under the assumption carries the same bar. When the box closes, the bar stops, and exactly one thing leaves: the implication from the assumption to the box's last line.

To prove "if φ, then ψ", assume φ in a box, reach ψ inside it, and close the box.

1.4.2 The introduction rule for implication

From now on the given part of a schema may also be a **box**, cited as a range of lines from its first line to its last; nothing else in Definition 1.2.1 changes.

Here is the rule, as a pattern. Beside it, for reference, is the rule $\to$E of Definition 1.3.1, so that the two rules for "implies" stand in one place.

Definition 1.4.1. Rules for Implication

$$\begin{array}{lll|l} i & & \varphi & & \\ & & \vdots & & \\ j & & \psi & & \\ k & \varphi \to \psi & & \to\text{I } i\text{–}j & \end{array} \qquad \begin{array}{ll|ll} i & \varphi \to \psi & & \\ j & \varphi & & \\ k & \psi & & \to\text{E } i, j \end{array}$$

The left schema is the rule $\to$I (**implication-introduction**); the right one is $\to$E (Definition 1.3.1). In $\to$I, the line φ is an **assumption**, which opens the box, and the assumption is **discharged** when the box closes. The line ψ is a line **of that box itself**, not of a box inside it.

As in every schema (Definition 1.2.1), φ and ψ stand for any statements, replaced consistently, and i, j, k for line numbers. This rule has conditions that matter; read each one.

1. **Open a box** whose first line is the assumption φ. The assumption may be **any** statement: true or false, related to the premises or not. Its justification is simply "assumption". The price of assuming anything we like is the box.
2. **Inside the box**, work as usual. Every rule may be used. A line inside may cite the assumption, the lines derived inside the box so far, and the lines above the box that are still in force: the premises, and earlier lines that are not inside a box that has already

closed.

3. **The last line** of the box is ψ. It must be a line of this box itself, not a line of some further box opened inside it. (The drawing cannot show this, which is why the definition says it in words.)
4. **Close the box**, and write $\varphi \to \psi$ on a later line outside it, in practice the very next line. The justification cites the **whole box** as the range i–j, from its assumption to its last line, not its lines one by one.
5. **After the box closes**, nothing inside it may be cited again. The assumption is discharged, and everything derived under it goes with it. Only the implication leaves.

Clause 5 is the boundary of the temporary world. The subsection "Which lines a line may cite" below turns clause 5 into an exact rule.

↻ **Second reading** · *A box of one line*

The **degenerate case** is $i = j$: a box of one line, whose assumption is also its last line. Then ψ is φ, and the rule writes $\varphi \to \varphi$. Nothing in the rule asks the box to contain more than its assumption.

Why the rule is right. The implication $\varphi \to \psi$ claims only that ψ holds whenever φ does. The box shows exactly that: starting from φ, and from facts that hold anyway, the rules lead to ψ.

Why the box closes. The lines inside hold only under the assumption. If the assumption is false, as $x > 2$ is at $x = 0$, they may be false too. Closing the box is what stops them being used where the assumption is not in force.

1.4.3 A first box: implication is transitive

The first result proved with a box is the chaining of implications. Section 1.3 used chains of implications forwards, firing each on a hypothesis already in hand. Here the hypothesis is not in hand, and we still want the chain.

Proposition 1.4.1. Implication Is Transitive

$P \to Q,\ Q \to R \vdash P \to R.$

Idea. Start from the goal. Its main connective is $\to$, so the last line will be $\to$I, and that needs a box that opens with P and ends with R. Inside the box, the new goal is R. Reason backwards: R is the conclusion of the premise $Q \to R$, so $\to$E gives it once Q is on a line. And Q is the conclusion of $P \to Q$, so $\to$E gives it from P, which the box has just assumed.

In the Givens/Goal table, $\to$I moves the hypothesis into the Givens and makes the conclusion the goal.

Givens	Goal
$P \to Q$, $Q \to R$	$P \to R$
$P \to Q$, $Q \to R$, P (box opened)	R

$P \to Q$, $Q \to R$, P, Q ($\to$E)	R
$P \to Q$, $Q \to R$, P, Q, R ($\to$E)	R ✓

The same search as a growing derivation: the box opened, its goal R at the bottom (left); then line 4 written forwards (right).

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
	$\quad \vdots$	
5	$\quad R$	
6	$P \to R$	$\to$I 3–5

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
	$\quad \vdots$	
5	$\quad R$	
6	$P \to R$	$\to$I 3–5

Proof. The following derivation has premises $P \to Q$ and $Q \to R$, and last line $P \to R$.

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad R$	$\to$E 2, 4
6	$P \to R$	$\to$I 3–5

Line 3 opens a box with the assumption P. Lines 4 and 5 apply $\to$E inside it, each citing a premise and a line of the box. Line 5, R, is the last line of the box. Line 6 closes it and writes $P \to R$, citing the box 3–5. By Definition 1.2.3, this proves the proposition.

■

Read the drawing once more. The bar that starts at line 3 stops after line 5. The statements P, Q, and R on lines 3–5 were never established outright; they hold inside the temporary world where P is assumed. Only $P \to R$ stands outside, next to the premises.

Now give the letters some content. Let x be a real number, and let

$$P : x > 3, \qquad Q : x^2 > 9, \qquad R : x^2 > 4.$$

The premise $P \to Q$ holds: when $x > 3$, we have $x > 0$, so $x^2 = x \cdot x > 3x > 9$. The premise $Q \to R$ holds because $4 < 9$: if $x^2 > 9$, then $x^2 > 4$. With the premises in hand, lines 3–6 become a paragraph, one sentence per line.

Line	Sentence
3	Suppose $x > 3$.
4	Since $x > 3$ implies $x^2 > 9$, we get $x^2 > 9$.
5	Since $x^2 > 9$ implies $x^2 > 4$, we get $x^2 > 4$.
6	Hence, if $x > 3$, then $x^2 > 4$.

A practiced writer would compress this to "if $x > 3$, then $x^2 > 9 > 4$": the same derivation with its joints hidden. When a proof is hard to follow, expanding it back into this form is how one finds the gap.

Remark. The same result holds with the two premises joined into one, $(P \to Q) \wedge (Q \to R) \vdash P \to R$. Two $\wedge$E lines take the premise apart first (premise-directed), and the rest is the derivation above, renumbered.

1	$(P \to Q) \wedge (Q \to R)$	
2	$P \to Q$	$\wedge$E 1
3	$Q \to R$	$\wedge$E 1
4	$\quad P$	
5	$\quad Q$	$\to$E 2, 4
6	$\quad R$	$\to$E 3, 5
7	$P \to R$	$\to$I 4–6

1.4.4 Argument B, diagnosed

Section 1.1 broke the form of Argument B's last step. Here is the rule B misused. With E: "n is even" and S: "n^2 is even", B's prose is this derivation.

1	$\quad E$	
2	$\quad S$	algebra 1
3	$E \to S$	$\to$I 1–2

Line 1 is B's "Suppose n is even, say $n = 2k$". Line 2 is its computation $n^2 = 4k^2 = 2(2k^2)$, a step of algebra, as in Example (A product of numbers in the unit interval). Line 3 is what the box earns; set it beside what B claimed.

what B's box earns		*what B claimed*
E / S		
$E \to S$	$\neq$	$S \to E$

B's box opens with E and closes on S, so $\to$I lets it write $E \to S$: the converse of the claim. B's last sentence wrote $S \to E$. The rule asks the box to open with the **hypothesis** of the implication being proved, here S, and B never opened that box.

The repair fails. Keep $E \to S$, open a box with S, and inside it use S to reach E: that step goes from $E \to S$ and S to E. It is affirming the consequent, the bad form of Section 1.3. Try to break it. Take "if $x > 2$, then $x^2 > 4$" and "$x^2 > 4$". At $x = -3$ both are true, since $9 > 4$, and "$x > 2$" is false. So B cannot be saved by moving its lines around: a $\to$I proof of $S \to E$ opens its box with S.

Warning. The box of $\to$I opens with the **hypothesis** of the implication to be proved, never with its conclusion: Argument B used $\to$I with the box opened at the wrong end.

Quick check. A proof of "if $x^3 > 8$, then $x > 2$" begins "Suppose $x > 2$." and ends "Hence $x^3 > 8$." What does its box earn? Is the claim it was meant to prove true?

Solution. Its box opens with $x > 2$ and closes on $x^3 > 8$, so it earns $(x > 2) \to (x^3 > 8)$, the converse of the claim. The claim itself is true: if $x \leq 2$, then $x^3 \leq 2^3 = 8$, since a larger real number has a larger cube, so $x^3 > 8$ forces $x > 2$. But this box does not prove it.

1.4.5 A box must end on a line of its own

Try to derive $\vdash P \to (Q \to P)$. The goal is an implication, so open a box with P; the new goal $Q \to P$ is again an implication, so open a box with Q inside the first. The statement P is already on line 1, so it is tempting to close the inner box at once.

1	P	
2	$\quad Q$	
3	$Q \to P$	$\to$I 2–2
4	$P \to (Q \to P)$	$\to$I 1–3

A checker reads the box 2–2: its assumption is Q, and its last line is Q. So $\to$I may write only $Q \to Q$ from it, and line 3 is rejected. The statement P stands on line 1, outside the inner box, and clause 3 asks for ψ as a line **of the box itself**.

The fix is bookkeeping: copy P into the inner box. Copying an available line is the rule R (**repetition**), made official as clause (D3) of the definition below. The copy "P, R 1" then ends the inner box.

Some books let a box end on any line available inside it. This book copies the line in, so that every $\to$I line can be checked by looking only at the box it cites. The corrected derivation is Example (First boxes) (b) below.

1.4.6 Which lines a line may cite

In a derivation without boxes (Definition 1.2.2), a line may cite any **earlier** line. With boxes that is no longer enough: after the derivation of Proposition 1.4.1, a line 7 citing line 5 would write R outside the box, the "$x^2 > 4$ for every x" error of the first subsection. What matters is which earlier lines are still **available**, and the bars show it.

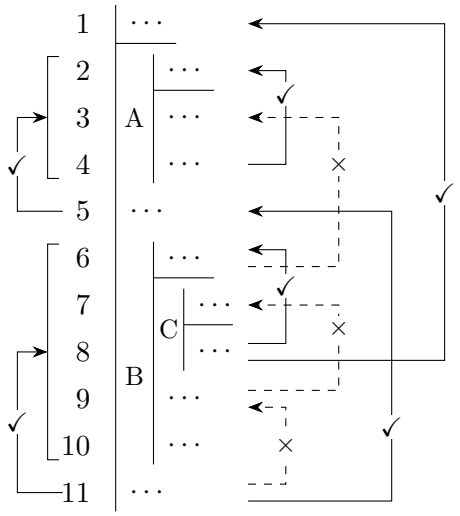
A line inside a box may be cited only from inside that box. Once a box has closed above you, nothing inside it can be cited; the boxes you are in may be left across their tops, and passing beside a closed box is fine.

Open a box, close a box: an assumption lives exactly as long as its box.

A **whole box**, cited by →I, has its own condition: the citing line must stand in exactly the boxes that the cited box stands in, other than the cited box itself. For single lines:

- a line in no box is available at every later line;
- a line inside a box is available further down that box, and inside every box opened within it;
- once its box closes, the line is available nowhere.

The figure shows eleven lines without statements: box A is lines 2–4, box B lines 6–10, and box C, lines 7–8, sits inside B. An arrow runs from a citing line up to the line it cites, solid and marked ✓ if legal, dashed and marked × if not. A bracket on the left marks a whole box cited.



A line can be cited from below only while every box containing it is still open. Line 8 leaves box C across its top to cite line 6, and line 11 passes beside the closed box B to cite line 5. Lines 5 and 11 stand in no box, and neither do the boxes A and B they cite whole.

Quick check. In the figure, which of lines 1–9 may line 10 cite?

Solution. Lines 1, 5, 6 and 9. Lines 1 and 5 are in no box; lines 6 and 9 are in box B, which contains line 10. Lines 2–4 are not available (box A has closed), and neither are lines 7–8 (box C has closed). It may also cite the whole box 7–8, which, like line 10, stands in box B alone.

Here is the official version, the one a checker uses.

Definition 1.4.2. Derivation

Fix a list of rules. A **derivation** is a finite list of numbered lines $1, 2, \ldots, n$, each carrying one statement and a justification, together with a set of **boxes**. A box is a block of consecutive

lines whose first line is an assumption. Any two boxes are either disjoint or **one inside the other**. A box ends at one of its lines and is **closed** after it. The **last line** of the derivation lies outside every box.

The scope rule. Line i is **available** at line k if $i < k$ and **every** box containing line i also contains line k. A box is **available** at line k if it ends before line k and the boxes other than itself that contain it are **exactly** the boxes containing line k. A line may cite only lines and boxes available at it.

Every line is exactly one of the following:

(D1) a **premise**; the premises are the first lines, above every box, with a short rule under the last of them;

(D2) an **assumption**, which may be **any** statement, and opens a new box;

(D3) a **repetition** of the statement on a line available at this one, justified "R" and the line repeated;

(D4) the conclusion of **one** rule of the list, applied to lines and boxes available at this one, such that this line and the cited lines and boxes form an instance of the rule's schema.

Clause (D3) is the rule R (**repetition**), whose schema is

$$\begin{array}{l|ll} i & \varphi & \\ & \vdots & \\ k & \varphi & \mathrm{R}\ i \end{array}$$

where line i must be available at line k.

(D1)–(D4) say: given, assumed, copied, or one step of one rule, always from what is still available.

The scope rule is the definition's real content: line i is available at line k when every assumption that line i was derived under is still in force at line k: the picture rule above, in words a checker can apply.

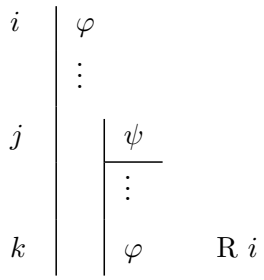
↻ **Second reading** · *Why checking stays mechanical*

Well defined. Whether a line is available is decided by looking at the boxes; nothing about the meaning of the statements enters. Checking a derivation stays mechanical.

Nothing earlier changes. A derivation with no boxes is a box-free derivation of Definition 1.2.2 for the same rules with R added: with no boxes, "available" means just "earlier". So every derivation of Sections 1.2 and 1.3 is a derivation in the new sense, and Definition 1.2.3 is read from now on with this definition. The sequent $\Gamma \vdash \psi$ still means: **there is** a derivation whose premises all belong to Γ and whose last line is ψ. Its last line lies in no box, so it depends on the premises and on nothing that was merely assumed.

Second reading · *More on repetition*

The typical use of R copies a line into a box opened below it.



Line i is available at line k exactly when every box containing line i contains line k; in this drawing, line i is in no box that has closed, so the copy is legal. The worked example Example (First boxes) below uses it.

Warning. A box that is never closed cannot end a derivation. If the last line of a list is still inside a box, it depends on an assumption that was never discharged, and Definition 1.4.2 does not count the list as a derivation at all. "Suppose $x > 2$. Then $x^2 > 4$", with nothing after it, proves nothing about x: the implication is the only thing the supposition was allowed to produce, and it was never written.

1.4.7 How to check a derivation

Section 1.2 checked a line by asking which rule it uses, which lines it cites, and whether they are earlier. With boxes the third question becomes: **are they available here?** For each line after the premises, ask:

1. **Which rule?** Name it. An assumption opens a box; R repeats; every other line applies one rule.
2. **Which lines or boxes does it cite, and do they form an instance of the rule's schema?** →E cites an implication and exactly its hypothesis; →I cites one box, whose assumption and last line must be the two sides of the implication written.
3. **Are they available here?** Apply the scope rule to each cited line and box.

Finding a derivation may need an idea; checking one never does. Here is the check run on the derivation of Proposition 1.4.1.

Line	Which rule?	Cites	Available?
1, 2	premises (D1)	—	—
3	assumption (D2); opens the box 3–5	—	—
4	→E: Q from $P \to Q$ and P	1, 3	1 is in no box; 3 is in box 3–5, which contains 4

Line	Which rule?	Cites	Available?
5	→E: R from $Q \to R$ and Q	2, 4	2 is in no box; 4 is in box 3–5, which contains 5
6	→I: $P \to R$ from the box	box 3–5	the box ended at 5; neither it nor 6 lies in another box

Every line passes, and the last line lies in no box. The bar shows the temporary world: it starts at line 3 and stops after line 5.

1.4.8 An assumption used outside its box

Now watch the scope rule catch an error. Take the derivation of Proposition 1.4.1 and add one careless line.

Example. A line cited from a closed box

Is the following a derivation? If it were, what would it show?

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	→E 1, 3
5	$\quad R$	→E 2, 4
6	$P \to R$	→I 3–5
7	R	R 5

Solution. No. Lines 1–6 are the derivation of Proposition 1.4.1, and line 7 claims to be a repetition (D3). Line 5 lies inside the box 3–5, and that box does not contain line 7. By the scope rule, line 5 is not available at line 7, and (D3) repeats only an available line. So line 7 is illegal.

If it were allowed, the list would show $P \to Q,\ Q \to R \vdash R$, and that would be a disaster. Let x be a real number, and take P: "$x > 3$", Q: "$x > 2$", R: "$x > 1$". Both premises are true whatever x is. At $x = 0$ the conclusion $x > 1$ is false. A step that leads from true premises to a false conclusion cannot be a step of proof.

Warning. The assumption of a closed box is gone, and so is everything derived from it. Here is a typical slip. *"Suppose $x > 1$. Then $x^2 = x \cdot x > 1 \cdot x > 1$. Hence $x > 1$ implies $x^2 > 1$. Moreover $x^3 = x \cdot x^2 > 1$, so $x^3 > 1$."* The sentence "Hence . . . " closed the box. The last sentence still uses $x > 1$ and $x^2 > 1$, and so it asserts $x^3 > 1$ with no hypothesis, which fails

at $x = 0$. The repair is to keep the box open until everything that needs $x > 1$ is done, and to conclude "if $x > 1$, then $x^2 > 1$ and $x^3 > 1$".

Quick check. In the derivation below, which of lines 3 to 7 may cite line 2 individually?

1	$P \to Q$	
2	$\quad P$	
3	$\quad Q$	$\to$E 1, 2
4	$\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad Q \wedge P$	$\wedge$I 3, 2
6	$P \to (Q \wedge P)$	$\to$I 2–5
7	$(P \to Q) \wedge (P \to (Q \wedge P))$	$\wedge$I 1, 6

Solution. Lines 3, 4, and 5. They lie in the box 2–5, the only box containing line 2, so line 2 is available to each of them. Line 6 cites the box 2–5 as a whole, which is what $\to$I requires; it does not cite line 2 by itself, and could not, since line 6 is outside the box. Line 7 may not cite line 2 either, for the same reason.

1.4.9 More boxes: from nothing, and one inside another

Example. First boxes

Find a derivation of each sequent.

(a) $\vdash P \to P$.
(b) $\vdash P \to (Q \to P)$.
(c) $P \to (Q \to R) \vdash Q \to (P \to R)$.

Solution.

(a) *Idea.* The goal is an implication, so open a box with its hypothesis P; the box must end with P. The assumption is already P, so the box can end at once: a box of one line, whose assumption is also its last line. Nothing in $\to$I asks a box to contain more than its assumption.

1	$\quad P$	
2	$P \to P$	$\to$I 1–1

Line 1 is an assumption (D2) and opens the box 1–1, whose last line is P. Line 2 cites that box, which ended before line 2 and, like line 2, lies in no other box. There are no premises, so this shows $\vdash P \to P$. It is the first statement in the book derivable

from **no** premises by the rules of logic alone. Argument B's box in "Argument B, diagnosed" also began without premises, but it used a step of algebra. Every rule before →I (Definition 1.2.5, Definition 1.3.1) cites at least one line, so without a box a derivation must start with a premise. A box is what escapes: its first line is an assumption, and the assumption is discharged before the end.

(b) *Idea.* This is the derivation the checker accepted after the fix in "A box must end on a line of its own": line 3 copies P into the inner box.

1	P	
2	$\quad Q$	
3	$\quad P$	R 1
4	$Q \to P$	→I 2–3
5	$P \to (Q \to P)$	→I 1–4

Line 3 is legal: the only box containing line 1, the box 1–4, also contains line 3. The boxes are nested, and the inner one closes first. No line cites line 2 on its own: the assumption Q is never used, and nothing in →I requires it to be. An implication whose conclusion is already known holds whatever its hypothesis.

(c) *Idea.* The goal $Q \to (P \to R)$ asks for a box with Q and, inside it, a box with P ending in R. Inside both boxes, work from the premise: its main connective is the first →, with hypothesis P, so →E fires on the assumption P and gives $Q \to R$. That implication fires on the assumption Q and gives R, the inner goal.

Both boxes opened, goals at their feet (left); then the inner lines written, and only line 7 still to justify (right).

1	$P \to (Q \to R)$		1	$P \to (Q \to R)$	
2	$\quad Q$		2	$\quad Q$	
3	$\quad\quad P$		3	$\quad\quad P$	
	$\quad\quad \vdots$		4	$\quad\quad Q \to R$	→E 1, 3
5	$\quad\quad R$		5	$\quad\quad R$	→E 4, 2
6	$\quad P \to R$		6	$\quad P \to R$	→I 3–5
7	$Q \to (P \to R)$		7	$Q \to (P \to R)$	?

The finished derivation:

1	$P \to (Q \to R)$	
2	$\quad Q$	
3	$\quad\quad P$	
4	$\quad\quad Q \to R$	→E 1, 3
5	$\quad\quad R$	→E 4, 2
6	$\quad P \to R$	→I 3–5
7	$Q \to (P \to R)$	→I 2–6

Line 5 cites line 2 from inside the inner box: every box containing line 2 (only 2–6) contains line 5. So the order of two hypotheses does not matter. For real numbers a and x, the derivation turns "if $a > 0$, then (if $x > 1$, then $ax > a$)" into "if $x > 1$, then (if $a > 0$, then $ax > a$)".

In (b) and (c) the goal told us which boxes to open, and in which order: the outermost $\to$ of the goal gives the outermost box. This is the goal-directed move of this section. *An implication goal opens a box with its hypothesis; the new goal is its conclusion.*

1.4.10 Two hypotheses, or one at a time

A theorem with two hypotheses, "if $x > 0$ and $y > 0$, then $xy > 0$", is often written with one hypothesis at a time: "let $x > 0$; then, if $y > 0$, $xy > 0$". The two forms are $(P \wedge Q) \to R$ and $P \to (Q \to R)$. Each can be derived from the other, so a writer may use whichever reads better.

Proposition 1.4.2. Two Hypotheses or One at a Time

$(P \wedge Q) \to R \vdash P \to (Q \to R)$, and $P \to (Q \to R) \vdash (P \wedge Q) \to R$.

Try it first (ten minutes): for the first claim, which boxes will you open, and in which order?

Idea. The tempting first move is premise-directed: use $(P \wedge Q) \to R$ at once. But $\to$E needs $P \wedge Q$ on a line, and at the start nothing is on a line except the premise. So start from the goal. Its main connective is the first $\to$, so open a box with P; the new goal $Q \to R$ is again an implication, so open a box with Q inside the first. Now both P and Q are available, $\wedge$I builds $P \wedge Q$, and the premise fires. For the second claim the goal $(P \wedge Q) \to R$ asks for one box, with $P \wedge Q$; take the assumption apart and feed the halves to the premise one at a time, P first because it is the premise's outer hypothesis.

The first claim: the outer box opened (left), then the inner one, with goal R (right).

1	$(P \wedge Q) \to R$	
2	$\quad P$	
	$\quad \vdots$	
6	$\quad Q \to R$	
7	$P \to (Q \to R)$	$\to$I 2–6

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
	$\quad\quad \vdots$	
5	$\quad\quad R$	
6	$\quad Q \to R$	$\to$I 3–5
7	$P \to (Q \to R)$	$\to$I 2–6

The second claim: the box opened (left), then the assumption taken apart (right).

1	$P \to (Q \to R)$	
2	$\quad P \wedge Q$	
	$\quad \vdots$	
6	$\quad R$	
7	$(P \wedge Q) \to R$	$\to$I 2–6

1	$P \to (Q \to R)$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
	$\quad \vdots$	
6	$\quad R$	
7	$(P \wedge Q) \to R$	$\to$I 2–6

Before reading on: in the first derivation, which lines may line 4 cite, and why may line 6 cite the box 3–5 but not line 5 on its own?

Proof. For the first claim, the following is a derivation with premise $(P \wedge Q) \to R$ and last line $P \to (Q \to R)$.

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad\quad R$	$\to$E 1, 4
6	$\quad Q \to R$	$\to$I 3–5
7	$P \to (Q \to R)$	$\to$I 2–6

Line 4 may cite lines 1, 2, and 3: line 1 lies in no box, the only box containing line 2 (the box 2–6) contains line 4, and both boxes containing line 3 contain line 4. Line 6 may cite the box 3–5, because that box has ended and both it and line 6 lie in the box 2–6 and in no other. Line 6 may not cite line 5 on its own, because the box 3–5 contains line 5 and not line 6.

For the second claim, the following is a derivation with premise $P \to (Q \to R)$ and last line $(P \wedge Q) \to R$.

1	$P \to (Q \to R)$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
5	$\quad Q \to R$	$\to$E 1, 3
6	$\quad R$	$\to$E 5, 4
7	$(P \wedge Q) \to R$	$\to$I 2–6

Lines 3–6 cite line 1, which lies in no box, or earlier lines of the box 2–6, from inside that

box. Line 7 cites the box 2–6, which has ended, and neither that box nor line 7 lies in any other box. By Definition 1.2.3, the two derivations prove the two claims.

■

In the first derivation the boxes are nested, and the inner one closes first: an assumption made later is discharged earlier. That is the general pattern, forced by Definition 1.4.2, since two boxes are either disjoint or one inside the other.

Read mathematically, the first claim turns "if $x > 0$ and $y > 0$, then $xy > 0$" into "if $x > 0$, then (if $y > 0$, then $xy > 0$)", and the second turns it back. The two forms say the same thing, one hypothesis at a time or both at once.

Quick check. To show $Q \wedge R \vdash (P \to Q) \wedge (S \to R)$, does the derivation need one box or two?

Solution. Two. The goal's main connective is $\wedge$, which opens no box, so the goal splits into the two implications $P \to Q$ and $S \to R$ (goal-directed $\wedge$I). Each is proved by $\to$I, with its own box, opened by its own hypothesis. Inside each box, $\wedge$E on the premise gives the half that box needs.

1	$Q \wedge R$	
2	$\quad P$	
3	$\quad Q$	$\wedge$E 1
4	$P \to Q$	$\to$I 2–3
5	$\quad S$	
6	$\quad R$	$\wedge$E 1
7	$S \to R$	$\to$I 5–6
8	$(P \to Q) \wedge (S \to R)$	$\wedge$I 4, 7

The two boxes are disjoint: the first closes before the second opens. Line 8 may cite lines 4 and 7, which lie in no box.

1.4.11 The rule card

The card grows by two rows. In *Shape*, "box $\varphi \ldots \psi$" is a closed box from the assumption φ to its last line ψ.

Rule	Shape	You write	Lean
$\wedge$I	$\varphi, \psi \Longrightarrow \varphi \wedge \psi$	"We show P, and we show Q."	`constructor`
$\wedge$E	$\varphi \wedge \psi \Longrightarrow \varphi$ (or ψ)	"In particular, P."	`h.1` (and `h.2`)
$\to$E	$\varphi \to \psi, \varphi \Longrightarrow \psi$	"Since P implies Q, and P holds, Q."	`exact h hp`

Rule	Shape	You write	Lean
$\to$I	box $\varphi \dots \psi \Longrightarrow \varphi \to \psi$	"Assume P. … Hence Q."	`intro hp`
R	φ (available) $\Longrightarrow \varphi$, copied	"As shown above, P."	`exact hp`

In the Lean column, `intro hp` opens the box and names its assumption `hp`.

The sentence for $\to$I is the one to watch. "Assume P" opens a box, and "hence Q" is its last line. The closing line, "so P implies Q", is usually left unsaid in written mathematics. It is still there: the box closes whether or not anyone says so. It is also the line Argument B got wrong.

Remark (Looking ahead). Chapter 2 proves the claim Argument B wanted, with the rules for "not".

1.4.12 Exercises

1.4.12.1 A. Check your understanding

Exercise. A1

Consider the following derivation, whose justifications are missing.

1	$P \to (Q \to R)$	
2	Q	
3	$\quad P$	
4	$\quad Q \to R$	(i)
5	$\quad R$	(ii)
6	$P \to R$	(iii)

(a) Write the schema of $\to$I from memory, and state the condition that the drawing cannot show.

(b) Fill in the justifications (i), (ii), (iii), and say what the derivation shows, written with $\vdash$.

(c) Which of lines 1–5 are available at line 6, and which boxes?

(d) Someone extends the derivation by a line 7, the statement R justified by $\to$E 6, 3. Is the result a derivation? If not, name the clause of Definition 1.4.2 that line 7 breaks.

(e) In the first derivation in the proof of Proposition 1.4.2, why is line 4 available at line 5? And why is the inner box closed before the outer one?

(f) In the same derivation, which rule produces line 5, and which lines does it cite?

Solution.

(a) The schema is the left one in Definition 1.4.1: a box from an assumption φ on line i

to a line ψ on line j, and then $\varphi \to \psi$ on line k, justified $\to$I i–j. The condition the drawing cannot show: ψ is a line of that box itself, not of a box inside it.

(b) (i) $\to$E 1, 3; (ii) $\to$E 4, 2; (iii) $\to$I 3–5. The derivation shows $P \to (Q \to R),\ Q \vdash P \to R$.

(c) Lines 1 and 2, and the box 3–5. Lines 3, 4, and 5 lie in the box 3–5, which does not contain line 6, so none of them is available there one by one. The box as a whole is available at line 6: it has ended, and neither it nor line 6 lies in another box. That is why (iii) may cite it.

(d) No. Line 6 is available at line 7, but line 3 is not: it lies in the box 3–5, which does not contain line 7. So line 7 applies a rule to a line that is not available, which (D4) forbids.

(e) Line 4 lies in two boxes, 2–6 and 3–5, and both contain line 5; so by the scope rule line 4 is available at line 5. The inner box 3–5 lies inside the outer box 2–6, and two boxes are either disjoint or one inside the other, so the inner box must end first. Read off the goal: the inner box proves $Q \to R$, which is the last line the outer box needs.

(f) $\to$E, citing line 1, the implication $(P \wedge Q) \to R$, and line 4, its hypothesis $P \wedge Q$.

Exercise. A2: Drill

Parts (a)–(c) refer to the eleven-line figure in "Which lines a line may cite".

(a) May line 8 cite line 5?

(b) May line 9 cite the whole box 2–4?

(c) May line 11 cite line 6?

(d) Say which lines below are premises and which are assumptions, and fill in (i) and (ii).

1	P	
2	$\quad Q$	
3	$\quad P \wedge Q$	(i)
4	$Q \to (P \wedge Q)$	(ii)

(e) Show $\vdash (P \wedge Q) \to (Q \wedge P)$ by a derivation. Then write the argument as a paragraph of prose, one sentence per line.

(f) Derive $P \to Q \vdash (P \wedge R) \to Q$.

(g) Derive $Q \vdash (P \wedge R) \to (Q \wedge R)$.

(h) Find the illegal line, and say why it is illegal.

1	$P \to Q$	
2	$\quad P$	
3	$\quad Q$	$\to$E 1, 2
4	$P \to Q$	$\to$I 2–3
5	Q	$\to$E 4, 2

(i) Derive $\vdash (P \wedge Q) \to (R \to P)$. Which box opens first?

(j) Let x be a real number. Is "Suppose $x > 3$. Then $x^2 > 9$.", with nothing after it, a proof that $x^2 > 9$?

Solution.

(a) Yes: line 5 lies in no box.

(b) No: box 2–4 stands in no box, and line 9 stands in box B, so the two do not stand in exactly the same boxes.

(c) No: line 6 is inside box B, which closed after line 10.

(d) Line 1 is a premise and line 2 an assumption; (i) $\wedge$I 1, 2; (ii) $\to$I 2–3.

(e) The goal is an implication, so open a box with $P \wedge Q$; inside, $\wedge$E gives both halves, and $\wedge$I rebuilds them in the other order.

1	$P \wedge Q$	
2	P	$\wedge$E 1
3	Q	$\wedge$E 1
4	$Q \wedge P$	$\wedge$I 3, 2
5	$(P \wedge Q) \to (Q \wedge P)$	$\to$I 1–4

In prose: *Assume P and Q. In particular, P holds. In particular, Q holds. Hence Q and P both hold. This proves that if P and Q, then Q and P.* Compared with Proposition 1.2.1, the premise $P \wedge Q$ has become an assumption, and the conclusion has become an implication that depends on nothing.

(f) 2 $P \wedge R$, assumption; 3 P, $\wedge$E 2; 4 Q, $\to$E 1, 3; 5 $(P \wedge R) \to Q$, $\to$I 2–4.

(g) 2 $P \wedge R$, assumption; 3 R, $\wedge$E 2; 4 $Q \wedge R$, $\wedge$I 1, 3; 5 $(P \wedge R) \to (Q \wedge R)$, $\to$I 2–4.

(h) Line 5: it cites line 2, which lies inside the box 2–3, closed before line 5.

(i) The box of $P \wedge Q$ opens first, since the goal's main connective is its first $\to$: 1 $P \wedge Q$, assumption; 2 R, assumption, opening a box inside the first; 3 P, $\wedge$E 1; 4 $R \to P$, $\to$I 2–3; 5 $(P \wedge Q) \to (R \to P)$, $\to$I 1–4.

(j) No. The box opened by "suppose" never closes, so nothing has left it; and at $x = 0$ the sentence $x^2 > 9$ is false. What the two sentences can prove, once the box closes, is "if $x > 3$, then $x^2 > 9$".

1.4.12.2 B. Practice

Exercise. B1: Two boxes, one inside the other

A search for $(P \wedge Q) \to R,\ S \to Q \vdash S \to (P \to R)$ has reached the stage below. *Name the move, then complete it.*

1	$(P \wedge Q) \to R$	
2	$S \to Q$	
3	$\quad S$	
4	$\quad\quad P$	
	$\quad\quad \vdots$	
	$\quad \vdots$	
9	$S \to (P \to R)$	?

Solution. The move is goal-directed, twice: an implication goal opens a box with its hypothesis, and the new goal is its conclusion. The goal $S \to (P \to R)$ gave the outer box, with S, and its conclusion $P \to R$ gave the inner box, with P; the inner goal is R. Inside both boxes, work backwards from the first premise: R is its conclusion, so it suffices to build its hypothesis $P \wedge Q$ exactly. P is line 4, and Q comes from the second premise and the assumption S.

1	$(P \wedge Q) \to R$	
2	$S \to Q$	
3	$\quad S$	
4	$\quad\quad P$	
5	$\quad\quad Q$	$\to$E 2, 3
6	$\quad\quad P \wedge Q$	$\wedge$I 4, 5
7	$\quad\quad R$	$\to$E 1, 6
8	$\quad P \to R$	$\to$I 4–7
9	$S \to (P \to R)$	$\to$I 3–8

Every citation is available. Lines 1 and 2 lie in no box. Line 3 lies only in the box 3–8, which contains line 5. Lines 4, 5 and 6 lie in both boxes, which contain lines 6 and 7. Line 8 cites the box 4–7, which has ended and stands, like line 8, in the box 3–8 and no other. Line 9 cites the box 3–8, which has ended and, like line 9, lies in no other box.

Exercise. B2: An assumption that is not used

(a) Show that $Q \vdash P \to Q$, and say which line needs repetition.

(b) Show that $\vdash (P \wedge Q) \to (R \to Q)$.

(c) Let x be a real number. Write the instance of (a) with Q: "$x^2 \geq 0$" and P: "$x > 5$" as an English sentence, and say which hypothesis its proof never uses.

Solution.

(a) The goal is an implication, so open a box with P; its last line must be Q.

1	Q	
2	$\quad P$	
3	$\quad Q$	R 1
4	$P \to Q$	$\to$I 2–3

Line 3 needs repetition: $\to$I wants Q as a line of the box 2–3 itself. It is legal because line 1 lies in no box.

(b) Two implications, so two nested boxes: one with $P \wedge Q$, and inside it one with R.

1	$P \wedge Q$	
2	Q	$\wedge$E 1
3	$\quad R$	
4	$\quad Q$	R 2
5	$R \to Q$	$\to$I 3–4
6	$(P \wedge Q) \to (R \to Q)$	$\to$I 1–5

Line 4 is legal: the only box containing line 2 also contains line 4. (Applying $\wedge$E to line 1 inside the inner box works as well.)

(c) "$x^2 \geq 0$; therefore, if $x > 5$, then $x^2 \geq 0$." The hypothesis $x > 5$ is never used: its box only repeats what was known.

Exercise. B3: Unwinding an absolute value

(a) Show that $P \to Q,\ P \to R \vdash P \to (Q \wedge R)$.

(b) Let x be a real number. Use (a) to deduce "if $|x-2| < 1$, then $1 < x < 3$" from the two facts "if $|x-2| < 1$, then $x > 1$" and "if $|x-2| < 1$, then $x < 3$". Write the resulting argument as prose.

Solution.

(a) The goal is an implication, so assume P, and reach $Q \wedge R$ by getting each half.

1	$P \to Q$	
2	$P \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad R$	$\to$E 2, 3
6	$\quad Q \wedge R$	$\wedge$I 4, 5
7	$P \to (Q \wedge R)$	$\to$I 3–6

The assumption on line 3 is cited twice, by lines 4 and 5: a line is not used up when it is cited.

(b) Take P: "$|x-2| < 1$", Q: "$x > 1$" and R: "$x < 3$". The two facts are the premises $P \to Q$ and $P \to R$, and $Q \wedge R$ is $1 < x < 3$. The derivation in (a), with these

statements put in for the letters, is a derivation of $P \to (Q \wedge R)$ from the two facts.

In prose: *Suppose* $|x-2|<1$. *Then* $x>1$ *by the first fact, and* $x<3$ *by the second. Hence* $1<x<3$. *This proves that* $|x-2|<1$ *implies* $1<x<3$.

Exercise. B4: Putting a derivation in order

Here are nine lines, each a statement with its justification. Seven of them, in the right order and with the right boxes, form a derivation of $P \to (Q \to R) \vdash Q \to (P \to R)$. Give the order, and say where each box opens and closes. Then say why each of the other two lines cannot be used.

(i) $P \to R$, by $\to$I on the box from P to R.
(ii) Q, assumption.
(iii) R, by $\to$E from $Q \to R$ and Q.
(iv) R, by R, repeating the line R of (iii), written directly below (i).
(v) $P \to (Q \to R)$, premise.
(vi) $Q \to (P \to R)$, by $\to$I on the box from Q to $P \to R$.
(vii) R, assumption, opening the box for the $\to$I that proves $P \to R$.
(viii) $Q \to R$, by $\to$E from $P \to (Q \to R)$ and P.
(ix) P, assumption.

Solution. The order is (v), (ii), (ix), (viii), (iii), (i), (vi), with one box opened by (ii) and closed after (i), and a second box inside it, opened by (ix) and closed after (iii). This is the derivation of Example (First boxes) (c).

It is the only admissible order. The premise (v) comes first. The last line (vi) needs a box from Q to $P \to R$, so (ii) opens a box whose last line is (i). Line (i) needs a box from P to R, which must lie inside the box of Q, since (i) is a line of that box; so (ix) opens the inner box, whose last line is (iii). Inside it, (viii) needs the premise and (ix), and (iii) needs (viii) and (ii). Opening the box of P first and the box of Q inside it would not do: the inner $\to$I would then give $Q \to R$, and the outer one $P \to (Q \to R)$, the premise again.

(iv) cannot be used where it is placed. Directly below (i), the inner box has closed, and line (iii) lies inside it; so (iii) is not available there, and (D3) repeats only an available line (the scope rule of Definition 1.4.2). It is the error of Example (A line cited from a closed box).

(v) cannot be used. $\to$I proves $P \to R$ from a box opened by P, the **hypothesis**. A box opened by R would prove an implication with hypothesis R: the converse error of Argument B.

1.4.12.3 C. Going deeper

Exercise. C1: Exit check: a line from a closed box

A student wants to show $(P \wedge Q) \to R \vdash (P \to (Q \to R)) \wedge (Q \to (P \to R))$, and offers the following derivation. Lines 1–7 are the first derivation in the proof of Proposition 1.4.2.

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad\quad R$	$\to$E 1, 4
6	$\quad Q \to R$	$\to$I 3–5
7	$P \to (Q \to R)$	$\to$I 2–6
8	$\quad Q$	
9	$\quad\quad P$	
10	$\quad\quad R$	R 5
11	$\quad P \to R$	$\to$I 9–10
12	$Q \to (P \to R)$	$\to$I 8–11
13	$(P \to (Q \to R)) \wedge (Q \to (P \to R))$	$\wedge$I 7, 12

(a) Find the first illegal line, and name the clause of Definition 1.4.2 that it breaks.

(b) Is the student's conclusion derivable from the premise? Justify your answer.

(c) The student defends line 10: "at line 10, both P and Q are assumed, exactly as they were at line 5, so R holds there too." Explain why Definition 1.4.2 does not accept this defense, and show by an instance with real numbers that the move "repeat a line of a closed box" cannot be allowed.

Solution (Hint 1). For (a), check each line's citations against the scope rule; for (b), keep the student's idea and change one step.

Solution (Hint 2). For (b), inside the box of P at line 9, the assumptions Q (line 8) and P (line 9) are both available. What did lines 4 and 5 do with such a pair?

Solution.

(a) Line 10 is the first illegal line. It claims to repeat line 5 by (D3), but line 5 lies in the boxes 2–6 and 3–5, and neither contains line 10. So line 5 is not available at line 10, and (D3) allows repeating only an available line. Lines 1–9 are correct, and lines 11–13 would be correct if line 10 were.

(b) Yes: the sequent is derivable. Replace line 10 by two lines that do inside the new boxes what lines 4 and 5 did inside the old ones.

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad\quad R$	$\to$E 1, 4
6	$\quad Q \to R$	$\to$I 3–5
7	$P \to (Q \to R)$	$\to$I 2–6
8	$\quad Q$	
9	$\quad\quad P$	
10	$\quad\quad P \wedge Q$	$\wedge$I 9, 8
11	$\quad\quad R$	$\to$E 1, 10
12	$\quad P \to R$	$\to$I 9–11
13	$Q \to (P \to R)$	$\to$I 8–12
14	$(P \to (Q \to R)) \wedge (Q \to (P \to R))$	$\wedge$I 7, 13

Every box containing line 8 or line 9 contains line 10, and line 1 lies in no box; the other lines check as before. The student's derivation was wrong and its conclusion right: a flawed argument can have a true conclusion.

(c) Availability is about **boxes**, not about which statements happen to be assumed. The boxes 8–11 and 9–10 of the student's derivation are new boxes; that their assumptions repeat those of the old boxes gives line 10 no access to what was derived inside the old ones. And the move, once allowed, can be used anywhere. Used outside every box, as a line 8 "R, R 5" after lines 1–7, it would derive $(P \wedge Q) \to R \vdash R$. Let x be a real number, and take P: "$x > 2$", Q: "$x > 3$" and R: "$x > 1$". The premise "if $x > 2$ and $x > 3$, then $x > 1$" is true whatever x is, and at $x = 0$ the conclusion $x > 1$ is false. A move that leads from a true premise to a false conclusion cannot be a step of proof.

Marking scheme. (10 points)

- 3: names line 10, because line 5 lies in boxes that do not contain it.
- 3: gives any correct derivation of the sequent, every citation available.
- 1: says the conclusion is derivable although the argument is wrong.
- 3: explains that availability depends on boxes, not on repeated assumptions, with an instance where the move leads from a true premise to a false conclusion.

Fatal errors (0 for the step): naming line 5 or line 9; claiming the sequent is not derivable; refuting the conclusion instead of the move.

Exercise. C2: A hypothesis inside a hypothesis

Show that $(P \to Q) \to (P \to R) \vdash P \to (Q \to R)$. Then say, in one sentence, why the premise cannot be used until a third box is opened.

Solution (Hint 1). Read the boxes off the goal. The premise's hypothesis is $P \to Q$: what would you need to derive it?

Solution (Hint 2). Start: line 2 assumes P and line 3 assumes Q, read off the goal. The last line of the box of Q must be R, and line 1 gives $P \to R$ only once $P \to Q$ is on an available line.

Solution. The goal $P \to (Q \to R)$ asks for a box with P and, inside it, a box with Q whose last line is R. The premise gives $P \to R$ as soon as we have $P \to Q$, and $P \to R$ with the available P gives R. So the task is to derive $P \to Q$ inside the box of Q, and that takes a third box.

1	$(P \to Q) \to (P \to R)$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad\quad P$	
5	$\quad\quad\quad Q$	R 3
6	$\quad\quad P \to Q$	$\to$I 4–5
7	$\quad\quad P \to R$	$\to$E 1, 6
8	$\quad\quad R$	$\to$E 7, 2
9	$\quad Q \to R$	$\to$I 3–8
10	$P \to (Q \to R)$	$\to$I 2–9

Line 5 is a legal repetition: both boxes containing line 3 contain line 5. Line 8 may cite line 2 because the box 2–9, the only box containing line 2, contains line 8.

The premise's hypothesis is the implication $P \to Q$, which $\to$E needs on an available line. No available line has $P \to Q$ as a conjunct or as the conclusion of an implication, so neither $\wedge$E nor $\to$E can produce it ($\to$E on line 1 would need $P \to Q$ already). It must therefore come from $\to$I, which needs a box of its own. (That box assumes P again although line 2 already did: $\to$I needs the box, not new information.)

Marking scheme. (10 points)

- 2: opens a box with P and inside it a box with Q, read off the goal.
- 3: derives $P \to Q$ inside the box of Q by a box whose last line is Q.
- 2: obtains $P \to R$, then R, by $\to$E on available lines.
- 2: closes the boxes inner first, each $\to$I citing an available box.
- 1: the sentence says $P \to Q$ must be a line that no elimination rule can supply, hence needs a box.

Fatal errors (0 for the step): citing a line outside the box that contains it; writing $P \to Q$ with no box.

Exercise. C3: Exit check: a derivation and its prose

(a) Write a derivation of $P \to Q,\ (Q \to R) \wedge (R \to S) \vdash P \to S$, with every line justified.

(b) Let x be a real number, and take P: "$x > 5$", Q: "$x > 4$", R: "$x^2 > 16$", S: "$x^2 > 9$". Say why each premise is true for every real x. Then write your derivation as fluent prose about x, one sentence per step.

Solution.

(a) Premise-directed first: the second premise is a conjunction, so $\wedge$E takes it apart. Goal-directed next: the goal is an implication, so open a box with its hypothesis P, and reach S by firing the three implications in turn.

1	$P \to Q$	
2	$(Q \to R) \wedge (R \to S)$	
3	$Q \to R$	$\wedge$E 2
4	$R \to S$	$\wedge$E 2
5	$\quad P$	
6	$\quad Q$	$\to$E 1, 5
7	$\quad R$	$\to$E 3, 6
8	$\quad S$	$\to$E 4, 7
9	$P \to S$	$\to$I 5–8

Lines 1–4 lie in no box, so they are available inside the box 5–8; line 9 cites that box, which has ended, and neither it nor line 9 lies in another box.

(b) The premise $P \to Q$ holds because $5 > 4$: if $x > 5$, then $x > 4$. For $Q \to R$: if $x > 4$, then $x > 0$, so $x^2 = x \cdot x > 4x > 16$. For $R \to S$: if $x^2 > 16$, then $x^2 > 9$, since $16 > 9$. So both premises (the second a conjunction of two true implications) hold for every real x.

In prose: *We know that $x > 5$ implies $x > 4$, that $x > 4$ implies $x^2 > 16$, and that $x^2 > 16$ implies $x^2 > 9$. Suppose $x > 5$. Then $x > 4$. Hence $x^2 > 16$. So $x^2 > 9$. This proves that if $x > 5$, then $x^2 > 9$.*

Marking scheme. (10 points)

- 2: takes the conjunctive premise apart by two $\wedge$E lines (inside or outside the box).
- 2: opens a box with P (not S) and closes it on S, a line of that box.
- 2: reaches S inside the box of P by $\to$E steps, each citing an implication and exactly its hypothesis, every cited line available.
- 1: each closing $\to$I cites its box as a range.
- 3: prose with one sentence per step, "suppose" for the assumption, and a closing sentence stating the implication.

A route that first derives $Q \to S$ in a box of Q (as in Proposition 1.4.1) and then uses it in the box of P earns the same points: its $\to$E steps in both boxes count for the third line, and its two $\to$I steps for the fourth.

Fatal errors (0 for the step): a box opened with the conclusion S (the converse error); citing

a line of the box after it has closed.

1.5 Or, Introduced

Sections 1.2 to 1.4 gave the rules for "and" and "implies". With them we can take a conjunction apart, fire an implication on its hypothesis, and prove an implication by opening a box. This section starts on the third connective of the chapter, "or". It gives the rule that **proves** a disjunction, $\vee$I, and a habit that goes with it: when the goal is a disjunction, commit to one of its sides only when that side is in hand. Near its end it meets a goal that the rules still cannot reach, and that goal is the question the next section answers.

1.5.1 Throwing information away on purpose

Suppose a proof has just shown that a real number x satisfies $x > 0$, and the definition it is checking asks for $x \geq 0$. Every writer takes the step "so $x \geq 0$" without a pause. Look at what the step does. Section 1.1 noted that the symbol $\leq$ hides a disjunction, so $x \geq 0$ is the statement

$$(x > 0) \vee (x = 0).$$

The step turns a statement into a disjunction of which it is one side.

The step loses something. "$x > 0$" tells us that x is not 0; "$x \geq 0$" no longer does. A proof that takes the step throws information away, and it does so on purpose: the definition asks for the weaker statement, and the stronger one is not in the form it asks for.

Remark. **Convention: inequalities written out.** When a rule has to see the disjunction inside $a \leq b$, a derivation writes it out as $(a < b) \vee (a = b)$, and $a \geq b$ as $(a > b) \vee (a = b)$: a checker compares symbols, so $\vee$I can write "$x \geq 0$" from "$x > 0$", and $\vee$E can split on it, only in the written-out form. Otherwise $a \leq b$ may stand in a derivation as one statement, and prose keeps the short forms.

The same move appears whenever a known fact has a disjunction as its hypothesis. School algebra says that if $a = 0$ or $b = 0$, then $a \cdot b = 0$. To use this on a product whose first factor we know to be 0, we must hand the fact its hypothesis, "$a = 0$ or $b = 0$". What we have is only "$a = 0$". So again the step goes from one side to the whole disjunction, and only then does the fact fire.

Both steps are instances of one rule. This section states it as a pattern, reads it in three bullets, and then uses it in two ways: when the goal is a disjunction, and when a premise needs one.

1.5.2 The rule of or-introduction

To prove a disjunction, it is enough to prove one of its two sides.

The rule is defined, like every rule since Section 1.2, by its schema (Definition 1.2.1). The Greek letters stand for arbitrary statements, and i, k for line numbers. The rule has two forms, one for each side, so it is printed as two schemas.

Definition 1.5.1. Or-Introduction

i	φ			i	φ	
k	$\varphi \vee \psi$	$\vee\mathrm{I}\ i$		k	$\psi \vee \varphi$	$\vee\mathrm{I}\ i$

The rule given by these two schemas is called $\vee$I (**or-introduction**).

- *Schema:* the two in the block above, one for each side.
- *Reading:* one available line (an earlier line, not inside a box that has closed), cited alone; write a disjunction with that line's statement as one **whole** side, left or right, and **any** statement as the other side: true or false, related or not.
- *Example:* if line 4 is $x > 0$, then "$\vee$I 4" may write $(x > 0) \vee (x = 0)$, or $(x = 0) \vee (x > 0)$, or the silly but correct $(x > 0) \vee (x < -7)$.

In mathematics "or" is inclusive (Section 1.1): $\varphi \vee \psi$ is true when φ is true, when ψ is true, and when both are. So $\varphi \vee \psi$ is true whenever φ is, whatever ψ says. The rule is this first clause read as a permission.

$\vee$I *loses information, never truth.*

Second reading · *Edge cases and near misses*

The degenerate case is ψ equal to φ. From P the rule writes $P \vee P$, and then the two schemas give the same line. It is a legal step, and the next section uses the statement $P \vee P$ once.

Now three nonexamples, each a small change of an instance. Take line 1 to be the statement named in each, and line 2 the line shown.

- From P, the line "$P \wedge Q$ — $\vee$I 1" is not an instance. The new line of $\vee$I is a disjunction, and the main connective of $P \wedge Q$ is $\wedge$.
- From $P \vee Q$, the line "$Q \vee P$ — $\vee$I 1" is not an instance. With φ replaced by $P \vee Q$, the two schemas give $(P \vee Q) \vee \psi$ and $\psi \vee (P \vee Q)$. In both, the cited statement is one whole side of the new line, and $P \vee Q$ is not a side of $Q \vee P$. Whether $Q \vee P$ follows from $P \vee Q$ at all is the question at the end of this section.
- From $P \vee Q$, the line "P — $\vee$I 1" is not an instance. It runs the rule backwards, from a disjunction to one of its sides.

Warning. Do not read "or" as "one but not both" when checking a step. From $x = 2$, $\vee$I gives "$x = 2$ or $x \leq 5$", although at $x = 2$ both sides are true. The step is correct: the disjunction is inclusive, and the rule never claims that the added side is false.

Warning. $\vee$I runs one way only. From φ it gives $\varphi \vee \psi$; from $\varphi \vee \psi$ it gives nothing, and in general φ does not follow from $\varphi \vee \psi$ at all. The phrase "in particular" belongs to $\wedge$E, not to "or". For a concrete failure, let x be a real number: "$x \geq 0$ or $x < 0$" is true whatever x is, and "in particular $x \geq 0$" is false at $x = -1$.

Quick check. Let x be a real number, and suppose line 1 of a derivation is $x = 2$. Which of the following may line 2 be, justified by one application of $\vee$I to line 1?

(i) $(x = 2) \vee (x = 3)$.
(ii) $(x = 3) \vee (x = 2)$.
(iii) $(x = 2) \vee (1 = 0)$.
(iv) $x \geq 2$.

Solution.

(i) Yes, by the first schema, with ψ replaced by $x = 3$. (ii) Yes, by the second schema. (iii) Yes, by the first schema, with ψ replaced by the false statement $1 = 0$. The line is true, since its left side is, and it is useless; the rule allows it. (iv) Only once $x \geq 2$ is written out as $(x > 2) \vee (x = 2)$, as the convention on inequalities above asks when a rule must see the disjunction. In that form it is an instance of the second schema, with φ replaced by $x = 2$. A checker who matches symbols sees the difference.

1.5.3 A disjunction as a goal

Section 1.2 asked two questions before writing any line: *what is the goal's main connective?* (goal-directed reasoning, from the bottom up) and *what are the available lines' main connectives?* (premise-directed reasoning, from the top down). Each rule so far added an answer. A conjunction as a goal splits into two subgoals, both needed ($\wedge$I); an implication as a goal opens a box ($\rightarrow$I, Definition 1.4.1).

Now $\vee$ joins the list of goals. If the goal is $\varphi \vee \psi$, its last line can be $\vee$I, and then the subgoal is φ **or** ψ: one of the two, of our choosing. That choice is new, and a wrong choice can leave us working towards a statement that does not follow from what we have. Suppose the premise is $x > 1$ and the goal is $(x > 3) \vee (x > 0)$. Choose the left side, and the new goal $x > 3$ does not follow from the premise: try to break the step, and $x = 2$ does, since there $x > 1$ holds and $x > 3$ fails. Choose the right side, and $x > 0$ follows from $x > 1$.

In the Givens/Goal table, $\vee$I replaces the goal $\varphi \vee \psi$ by the side you choose.

So the goal-directed step for $\vee$ comes with a caution:

Postpone a disjunction: choose a side of a goal $\varphi \vee \psi$ only when that side is in hand.

"In hand" means that the side stands on an available line. Until then, work from the premises: take conjunctions apart, fire implications, open the boxes the goal asks for, and look at what appears. When one side of the goal turns up, the choice has been made for us.

Quick check. Let x be a real number. The premise is $0 < x < 1$ and the goal is $(x < 0) \vee (x < 2)$. Which side should $\vee$I take, and why not the other?

Solution. The right side, $x < 2$: it follows from $x < 1$, the right half of the premise. The left side $x < 0$ does not follow: at $x = \frac{1}{2}$ the premise holds and $x < 0$ fails.

Here are four derivations, in order of difficulty. Each Idea says, for each rule, whether it was chosen by the goal or by a premise.

Example. Derivations with or-introduction

Show that:

(a) $P \vdash P \vee Q$, and $P \vdash Q \vee P$;
(b) $P \wedge Q \vdash P \vee Q$;
(c) $P \to Q \vdash P \to (Q \vee R)$;
(d) $(P \wedge Q) \to R \vdash P \to (Q \to (R \vee S))$.

Solution.

(a) *Idea.* The goal is a disjunction, and one of its sides, P, is the premise: in hand from the start. The only choice is which schema, and the goal decides it: P sits on the left of $P \vee Q$ and on the right of $Q \vee P$.

1	P			1	P		
2	$P \vee Q$	$\vee$I 1		2	$Q \vee P$	$\vee$I 1	

In the first, line 2 is the first schema with φ replaced by P and ψ by Q; in the second, it is the second schema with the same replacement. The letter Q appears in no premise. That is the Reading's "**any** statement" at work.

(b) *Idea.* The goal $P \vee Q$ is a disjunction, and neither side stands on a line yet, so we postpone it. Premise-directed: the premise's main connective is $\wedge$, so $\wedge$E takes it apart. After that, **both** sides of the goal are in hand, and either choice finishes.

1	$P \wedge Q$			1	$P \wedge Q$		
2	P	$\wedge$E 1		2	Q	$\wedge$E 1	
3	$P \vee Q$	$\vee$I 2		3	$P \vee Q$	$\vee$I 2	

The left derivation chose the left side (first schema); the right one chose the right side (second schema). Both are correct, and the choice was free precisely because both sides were in hand. That will not usually be so.

(c) *Idea.* The goal's main connective is the outer $\to$; the $\vee$ sits inside its conclusion. Goal-directed: open a box with the hypothesis P, and the new goal inside it is $Q \vee R$, a disjunction, so we postpone it. Premise-directed: the premise $P \to Q$ is an implication whose hypothesis P is now on a line, so $\to$E fires and gives Q. The left side of the goal is in hand; $\vee$I takes it, and $\to$I closes the box. The pictures show the box opened with its goal at the bottom, and then the box with the side Q written; only the justification of line 4, marked "?", is left.

1	$P \to Q$			1	$P \to Q$		
2	$\quad P$			2	$\quad P$		
	$\quad \vdots$			3	$\quad Q$	$\to$E 1, 2	
4	$\quad Q \vee R$			4	$\quad Q \vee R$	?	
5	$P \to (Q \vee R)$	$\to$I 2–4		5	$P \to (Q \vee R)$	$\to$I 2–4	

1	$P \to Q$	
2	$\quad P$	
3	$\quad Q$	$\to$E 1, 2
4	$\quad Q \vee R$	$\vee$I 3
5	$P \to (Q \vee R)$	$\to$I 2–4

Line 3 cites line 1, which lies in no box, and line 2, the assumption of the box that contains line 3. Line 4 is a line of the box itself, so $\to$I may cite the box 2–4.

(d) *Idea.* The goal has two arrows at the top of its structure, $P \to (Q \to \cdots)$, so, as in Proposition 1.4.2, we open two nested boxes, first P, then Q; the inner one will close first. The innermost goal is $R \vee S$: postpone. Premise-directed: the only line that mentions R is the premise $(P \wedge Q) \to R$. Its hypothesis is $P \wedge Q$, which is not on any line, but both of its halves are: the two assumptions. So $\wedge$I builds it, $\to$E fires, and R is in hand. Then $\vee$I, and the two boxes close. In the pictures, the two boxes are open with the innermost goal at the bottom; then lines 4 and 5 are written, and only the justification of line 6 is left.

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
	$\quad\quad \vdots$	
6	$\quad\quad R \vee S$	
7	$\quad Q \to (R \vee S)$	$\to$I 3–6
8	$P \to (Q \to (R \vee S))$	$\to$I 2–7

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad\quad R$	$\to$E 1, 4
6	$\quad\quad R \vee S$	?
7	$\quad Q \to (R \vee S)$	$\to$I 3–6
8	$P \to (Q \to (R \vee S))$	$\to$I 2–7

1	$(P \wedge Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad\quad R$	$\to$E 1, 4
6	$\quad\quad R \vee S$	$\vee$I 5
7	$\quad Q \to (R \vee S)$	$\to$I 3–6
8	$P \to (Q \to (R \vee S))$	$\to$I 2–7

This derivation uses every rule of the chapter so far except $\wedge$E and R. Line 4 cites line 2, from the outer box, which is still open at line 4. Line 5 cites line 1 from inside two boxes: a line in no box is available everywhere below it. Line 7 is a line of the outer box itself, which is why $\to$I at line 8 may cite 2–7.

Warning. Choosing the wrong side of a disjunctive goal too early can leave a goal that does not follow. Derivation (c) is where postponing pays. Had we committed at once, inside the box, to the goal's right side, planning to reach $Q \vee R$ from R, the box would need R itself, from the premise $P \to Q$ and the assumption P alone. With x a real number, P: "$x > 2$", Q: "$x > 1$" and R: "$x > 3$", the premise and the assumption hold at $x = 2.5$, and R fails there, so R does not follow. Keep the goal as the whole disjunction until one side appears on a line.

1.5.4 Manufacturing a hypothesis

In every derivation so far, $\vee$I produced the goal, or the goal of a box. It has a second use. A premise whose hypothesis is a disjunction can fire only when that disjunction stands on a line, **exactly** as written (the matching clause of Definition 1.3.1). If one side of it is in hand, $\vee$I builds the whole hypothesis.

Example. A disjunctive hypothesis splits into two implications

Show that $(P \vee Q) \to R \vdash (P \to R) \wedge (Q \to R)$.

Solution. Idea. Goal-directed: the goal is a conjunction, so $\wedge$I splits it into two subgoals, $P \to R$ and $Q \to R$, and each is an implication, so each gets its own box. Inside the box of P, the goal is R. Premise-directed: the only line that mentions R is the premise, and $\to$E on it needs its hypothesis $P \vee Q$ on a line. We have P, one side of that hypothesis. So $\vee$I manufactures $P \vee Q$, and the premise fires. Here $\vee$I is chosen not because the goal is a disjunction (the goal is R) but because a premise's hypothesis is one. The box of Q is the same, with the other schema. In the pictures, both boxes are open with their goals at the bottom; then the $\vee$I lines that manufacture the hypothesis are written, and only the $\to$E lines are left.

1	$(P \vee Q) \to R$	
2	$\quad P$	
	$\quad \vdots$	
4	$\quad R$	
5	$P \to R$	$\to$I 2–4
6	$\quad Q$	
	$\quad \vdots$	
8	$\quad R$	
9	$Q \to R$	$\to$I 6–8
10	$(P \to R) \wedge (Q \to R)$	$\wedge$I 5, 9

1	$(P \vee Q) \to R$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad R$	?
5	$P \to R$	$\to$I 2–4
6	$\quad Q$	
7	$\quad P \vee Q$	$\vee$I 6
8	$\quad R$	?
9	$Q \to R$	$\to$I 6–8
10	$(P \to R) \wedge (Q \to R)$	$\wedge$I 5, 9

1	$(P \vee Q) \to R$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad R$	$\to$E 1, 3
5	$P \to R$	$\to$I 2–4
6	$\quad Q$	
7	$\quad P \vee Q$	$\vee$I 6
8	$\quad R$	$\to$E 1, 7
9	$Q \to R$	$\to$I 6–8
10	$(P \to R) \wedge (Q \to R)$	$\wedge$I 5, 9

Line 3 uses the first schema of Definition 1.5.1 (P on the left), line 7 the second (Q on the right). Both produce the same statement $P \vee Q$, the one the premise asks for. The two boxes are separate: the second opens after the first has closed, and nothing in it cites a line of the first. Line 10 cites lines 5 and 9, which lie in no box.

The product example below, and several exercises, use $\vee$I the same way. Weakening is how a strong fact is fed to a rule, or a theorem, that asks for a weaker one.

In mathematical terms, the derivation says that a theorem with a hypothesis "P or Q" splits into two theorems. Let x be a real number, and suppose we know "if $x \geq 0$, then $x^3 \geq 0$", that is, "if $x > 0$ or $x = 0$, then $x^3 \geq 0$". Then we know "if $x > 0$, then $x^3 \geq 0$" and "if $x = 0$, then $x^3 \geq 0$". The way back, from the two implications to the single one with the disjunctive hypothesis, needs the rule of the next section; it is derived there.

Quick check. In the derivation of Example (A disjunctive hypothesis splits into two implications), line 3 could not have been skipped by letting line 4 cite line 2 directly, as "R — $\to$E 1, 2". Why not?

Solution. $\to$E (Definition 1.3.1) needs the implication on one cited line and its hypothesis, exactly, on the other. The hypothesis of line 1 is $P \vee Q$, and line 2 is P, which does not match it symbol for symbol. Line 3 writes the hypothesis out, and only then does line 1 fire.

1.5.5 A goal we cannot reach yet

The most obvious fact about "or" is that the order of its sides does not matter: $P \vee Q \vdash Q \vee P$. Try to derive it with the rules so far.

The goal $Q \vee P$ is a disjunction, so we postpone it and look at the premise. Its main connective is $\vee$, and no rule so far takes a disjunction apart. $\wedge$E needs a conjunction and $\to$E an implication; $\wedge$I, $\vee$I, and R may cite $P \vee Q$, but only to copy it or to build something larger around it. So the premise gives us nothing, and neither side of the goal ever comes into hand.

We can get this far, though. Each side of the premise, **assumed** in a box, leads to the goal:

1	$P \vee Q$	
2	$\quad P$	
3	$\quad Q \vee P$	$\vee$I 2
4	$P \to (Q \vee P)$	$\to$I 2–3
5	$\quad Q$	
6	$\quad Q \vee P$	$\vee$I 5
7	$Q \to (Q \vee P)$	$\to$I 5–6

This derivation is correct, but it shows $P \vee Q \vdash Q \to (Q \vee P)$, and line 1 is never cited. (A derivation need not use every premise, Definition 1.2.3.) What we would like is to put the three lines 1, 4, and 7 together: *one of P, Q holds, and either one leads to $Q \vee P$; so $Q \vee P$.* That is a proof by cases, and none of the rules so far performs it. With them we cannot find a derivation of $P \vee Q \vdash Q \vee P$.

The missing move is to **use** a disjunction, as $\wedge$E uses a conjunction and $\to$E uses an implication. The next section gives the rule that does this, and with it the six-line derivation that this attempt lacks.

1.5.6 How this looks in a mathematical proof

In written mathematics $\vee$I is almost always silent. Here are the two steps of the opening, written out as derivations.

Example. A product with a zero factor

Let a and b be real numbers, and take as known the school fact "if $a = 0$ or $b = 0$, then $a \cdot b = 0$". Prove that if $a = 0$, then $a \cdot b = 0$, naming the rule behind each step. Then write the step from $x > 0$ to $x \geq 0$ as a derivation.

Solution. The goal is an implication, so the proof supposes $a = 0$ (a box). Inside, the fact needs its hypothesis "$a = 0$ or $b = 0$", which $\vee$I manufactures, as in Example (A disjunctive hypothesis splits into two implications). In prose, with each step's rule in brackets:

> Suppose $a = 0$. Then $a = 0$ or $b = 0$ [$\vee$I]. So, by the school fact, $a \cdot b = 0$ [$\to$E]. Hence, if $a = 0$, then $a \cdot b = 0$ [$\to$I].

As a derivation, with the school fact as the premise:

1	$(a = 0 \vee b = 0) \to a \cdot b = 0$	
2	$\quad a = 0$	
3	$\quad a = 0 \vee b = 0$	$\vee$I 2
4	$\quad a \cdot b = 0$	$\to$E 1, 3
5	$a = 0 \to a \cdot b = 0$	$\to$I 2–4

A written proof would say "if $a = 0$, then $a \cdot b = 0$ by the fact", and leave out the words "$a = 0$ or $b = 0$". The derivation shows what that sentence leaves out: a box, a weakening, and one use of the fact.

For the second step, let x be a real number with $x > 0$. Written out, $x \geq 0$ is $(x > 0) \vee (x = 0)$, and the step is one application of the first schema:

1	$x > 0$	
2	$(x > 0) \vee (x = 0)$	$\vee$I 1

Line 2 is $x \geq 0$, written out.

Every "so $x \geq 0$" after $x > 0$, every "so $|t| \leq 1$" after $|t| < 1$, and every "so n is even or $n > 5$" after "n is even" is this rule. A reader who checks a proof line by line should see it, even where the writer did not name it. It is a step that never leads from a truth to a falsehood, and it is also the step that explains why a conclusion can say less than the proof knew.

1.5.7 The rule card

The card grows by one row. *Shape* shows the given lines (or box), then the new line.

Rule	Shape	You write	Lean
$\wedge$I	$\varphi, \psi \Longrightarrow \varphi \wedge \psi$	"We show P, and we show Q."	`constructor`
$\wedge$E	$\varphi \wedge \psi \Longrightarrow \varphi$ (or ψ)	"In particular, P."	`h.1` (and `h.2`)
$\to$E	$\varphi \to \psi, \varphi \Longrightarrow \psi$	"Since P implies Q, and P holds, Q."	`exact h hp`
$\to$I	box φ … $\psi \Longrightarrow \varphi \to \psi$	"Assume P. … Hence Q."	`intro hp`
R	φ (available) $\Longrightarrow \varphi$, copied	"As shown above, P."	`exact hp`
$\vee$I	$\varphi \Longrightarrow \varphi \vee \psi$ (or $\psi \vee \varphi$)	"So P or Q holds."	`left` (or `Or.inl hp`); `right`

The phrase is different from the "in particular" of $\wedge$E on purpose. "In particular" announces that a part is being taken out of something bigger; "so P or Q" announces that something bigger, and weaker, is being built.

1.5.8 Exercises

1.5.8.1 A. Check your understanding

Exercise. A1

(a) Write the two schemas of Definition 1.5.1 from memory, and say which letter in them may stand for a statement that appears in no earlier line.

(b) Which of the following sentences, taken from proofs about a real number x or real numbers a, b, use $\vee$I? For each of the others, name the rule it uses.

- "We have $x < 3$, so $x \leq 3$."
- "We found that $a = 0$. So $a = 0$ or $b = 0$, and therefore $a \cdot b = 0$."
- "Since $0 < x < 1$, in particular $x < 1$."
- "Since $|x - 2| < 1$ implies $1 < x < 3$, and $|x - 2| < 1$, we get $1 < x < 3$."

(c) True or false, with a reason: $\vee$I may introduce a side that is false.

(d) To show $Q,\ (P \vee Q) \to (Q \to R) \vdash R$, which rule writes the first line after the premises, and what is that line? Why is that rule used, although the goal is not a disjunction?

Solution.

(a) From a line i holding φ, a line k holding $\varphi \vee \psi$, justified "$\vee$I i"; and, from the same line, a line k holding $\psi \vee \varphi$, with the same justification (Definition 1.5.1). The letter ψ appears in no given line: it may be any statement.

(b) The first uses $\vee$I: $x \leq 3$ is $(x < 3) \vee (x = 3)$, and its left side is in hand. The second uses $\vee$I to build "$a = 0$ or $b = 0$" from its left side, and then $\to$E on the fact about products, as in Example (A product with a zero factor). The third uses $\wedge$E: $0 < x < 1$ is $(0 < x) \wedge (x < 1)$, and the sentence takes out its right half. The fourth uses $\to$E, with the implication first and its hypothesis second.

(c) True. The added side ψ may be any statement, false ones included, as in the line $(x = 2) \vee (1 = 0)$ of the Quick check after the definition. The new line is still true whenever the cited line is, because "or" is inclusive: one true side is enough.

(d) The rule is $\vee$I, by its second schema, and line 3 is $P \vee Q$, from line 1. It is used because a premise's hypothesis is a disjunction. The goal R occurs only in the premise $(P \vee Q) \to (Q \to R)$, and $\to$E fires on that premise only when $P \vee Q$ itself stands on a line. Only its right side Q is on a line, and $\vee$I turns it into $P \vee Q$. Then $Q \to R$ follows by $\to$E 2, 3, and R by $\to$E 4, 1. The move is the one of Example (A disjunctive hypothesis splits into two implications), where $\vee$I builds a premise's hypothesis.

Exercise. A2: Drill

Each part takes a line or two. Let x be a real number and n an integer.

(a) Line 1 is $x = 5$. Which of these may be written as "$\vee$I 1"? (i) $(x = 5) \vee (x = 6)$; (ii) $(x > 4) \vee (x = 5)$; (iii) $x \geq 5$, as written; (iv) $(x = 5) \wedge (x \geq 0)$.

(b) Line 1 is $P \wedge Q$. Is "$P \vee Q$ — $\vee$I 1" legal?

(c) Line 1 is $Q \vee P$. Is "$P \vee Q$ — $\vee$I 1" legal?

(d) Is "$x > 0$, so $x \geq 0$" one step of $\vee$I?

(e) The premise is $x < -1$ and the goal is $(x < -5) \vee (x < 0)$. Which side should $\vee$I take?

(f) "We have $n = 4$. So n is even or $n > 10$." Which steps does this sentence take?

(g) Derive $P \vdash (P \vee Q) \wedge (R \vee P)$.

(h) Derive $P \wedge Q \vdash Q \vee R$.

(i) Derive $P \to Q \vdash P \to (R \vee Q)$.

(j) Derive $(P \vee Q) \to R,\ Q \vdash R$.

Solution.

(a) (i) Yes, by the first schema. (ii) Yes, by the second schema: $x = 5$ is the right side. (iii) No: as written, $x \geq 5$ is one statement. Written out as $(x > 5) \vee (x = 5)$, it is the second schema. (iv) No: the new line is not a disjunction.

(b) No: $P \wedge Q$ is not a side of $P \vee Q$. From line 1, $\vee$I could write $(P \wedge Q) \vee R$.

(c) No: $Q \vee P$ is not a whole side of $P \vee Q$.

(d) Yes, once $x \geq 0$ is written out as $(x > 0) \vee (x = 0)$: it is the first schema.

(e) The right side, $x < 0$, which follows from $x < -1$. The left side fails at $x = -2$, where the premise holds and $x < -5$ does not.

(f) Two steps: algebra ($4 = 2 \cdot 2$, so n is even), then $\vee$I. From "$n = 4$" itself, $\vee$I gives only "$n = 4$ or ...".

(g) 2 $P \vee Q$, $\vee$I 1; 3 $R \vee P$, $\vee$I 1; 4 $(P \vee Q) \wedge (R \vee P)$, $\wedge$I 2, 3.

(h) 2 Q, $\wedge$E 1; 3 $Q \vee R$, $\vee$I 2.

(i) A box opened with P: 2 P, assumption; 3 Q, $\to$E 1, 2; 4 $R \vee Q$, $\vee$I 3; then 5 $P \to (R \vee Q)$, $\to$I 2–4.

(j) 3 $P \vee Q$, $\vee$I 2; 4 R, $\to$E 1, 3.

1.5.8.2 B. Practice

Exercise. B1: Committing to a side

Show that $P \wedge (Q \to R) \vdash Q \to (R \vee S)$, and that $P \vdash (Q \vee P) \vee R$. For each derivation, say at which line you committed to a side of a disjunctive goal, and why that was safe.

Solution. For the first, the goal is an implication, so open a box with Q; the goal inside is $R \vee S$, which we postpone. Premise-directed, $\wedge$E takes out $Q \to R$, and $\to$E fires on the assumption Q.

1	$P \wedge (Q \to R)$	
2	$\quad Q$	
3	$\quad Q \to R$	$\wedge$E 1
4	$\quad R$	$\to$E 3, 2
5	$\quad R \vee S$	$\vee$I 4
6	$Q \to (R \vee S)$	$\to$I 2–5

We committed at line 5, to the left side R, and only after line 4 had put R on a line. The other conjunct of the premise, P, is never used.

For the second, the goal $(Q \vee P) \vee R$ is a disjunction whose left side, $Q \vee P$, is itself a disjunction, and P is one side of that.

1	P	
2	$Q \vee P$	$\vee$I 1
3	$(Q \vee P) \vee R$	$\vee$I 2

Line 2 uses the second schema (P on the right) and line 3 the first, with φ replaced by $Q \vee P$. We committed twice, at lines 2 and 3, and both times the side was in hand: at line 2 the side P was the premise, and at line 3 the side $Q \vee P$ was line 2. Choosing the side R at the outer level would have left the subgoal R, which does not follow from P: take P true and R false.

Exercise. B2: The last line

A search for $P \to Q,\ Q \to R \vdash P \to (S \vee R)$ has reached the stage below; only the last line is missing. *Name the move, then complete it.*

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad R$	$\to$E 2, 4
6	$\quad S \vee R$	$\vee$I 5
7	**?**	?

Solution. The move is the one read off the goal: an implication goal opens a box with its hypothesis, and the box is closed by $\to$I once its last line is the implication's conclusion. The box opened with P at line 3, and its last line, line 6, is $S \vee R$, the conclusion of the goal. So line 7 closes the box:

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad R$	$\to$E 2, 4
6	$\quad S \vee R$	$\vee$I 5
7	$P \to (S \vee R)$	$\to$I 3–6

Line 7 cites the box as the range 3–6, not its lines one by one. Inside the box, the disjunction $S \vee R$ was postponed until R came into hand at line 5; the side in hand was the right one, so line 6 uses the second schema.

Exercise. B3: A theorem with a weaker hypothesis

Let x be a real number, and take as known the school fact "if $x \geq 1$, then $x^2 \geq x$". Prove that if $x > 1$, then $x^2 \geq x$. Write the proof as a derivation whose premise is the school fact, with $x \geq 1$ written out as a disjunction, and then as two or three sentences of prose.

Solution. Written out, $x \geq 1$ is $(x > 1) \vee (x = 1)$, so the premise is an implication whose hypothesis is a disjunction. The goal is an implication: open a box with $x > 1$. Inside, $\vee$I manufactures the premise's hypothesis, and $\to$E fires.

1	$(x > 1 \vee x = 1) \to x^2 \geq x$	
2	$\quad x > 1$	
3	$\quad x > 1 \vee x = 1$	$\vee$I 2
4	$\quad x^2 \geq x$	$\to$E 1, 3
5	$x > 1 \to x^2 \geq x$	$\to$I 2–4

In prose: *Suppose $x > 1$. Then $x \geq 1$, so $x^2 \geq x$ by the fact. Hence if $x > 1$, then $x^2 \geq x$.* The words "then $x \geq 1$" are line 3. For instance, at $x = 3$ the conclusion reads $9 \geq 3$.

1.5.8.3 C. Going deeper

Exercise. C1: Which new rules are safe?

Suppose someone proposes new rules for the derivations of this chapter. Let x be a real number.

(a) Take the rule "from $\varphi \vee \psi$, write φ". Starting from the premise "$x \geq 0$ or $x < 0$", which is true for every real x, derive with it a line that is false at $x = -1$.

(b) $\vee$I adds a side ψ that may be any statement at all. Explain why it can never lead from a true line to a false one.

(c) Is the rule "from φ, write $\varphi \wedge \psi$, for any statement ψ" safe in the same sense? Justify your answer with an instance.

Solution.

(a) With φ replaced by $x \geq 0$ and ψ by $x < 0$, the proposed rule turns the premise into the line $x \geq 0$. At $x = -1$ the premise is true, since $-1 < 0$, and the line $-1 \geq 0$ is false. So the rule can lead from a true line to a false one.

(b) Suppose the cited line φ is true. The new line $\varphi \vee \psi$ (or $\psi \vee \varphi$) is true when at least one of its sides is true, because "or" is inclusive (Section 1.1), and its side φ is true. So the new line is true, whatever ψ is. The truth or falsity of ψ never enters.

(c) No. The new line $\varphi \wedge \psi$ is true only when both sides are, and ψ is arbitrary. With φ replaced by $x = -1$ and ψ by $x \geq 0$, at $x = -1$ the cited line is true and the new line, "$x = -1$ and $x \geq 0$", is false. The contrast with (b) is the whole difference between $\vee$ and $\wedge$: a disjunction needs one true side, so an arbitrary second side is harmless; a conjunction needs both, so it is not. That is why $\wedge$I asks for both sides on lines.

Marking scheme. (10 points)

- 3: (a) applies the proposed rule to the premise to get $x \geq 0$, and checks at $x = -1$ that the premise is true and the new line false.
- 3: (b) argues from the inclusive meaning of "or": one true side makes the disjunction true, whatever the other side is.
- 3: (c) answers no, with an instance in which φ is true and $\varphi \wedge \psi$ is false.
- 1: (c) names the difference: a conjunction needs both sides true.

Fatal errors (0 for the step): in (a), a premise that is false at $x = -1$; in (b), an argument that assumes ψ is true; in (c), "safe" on the grounds that φ is true.

Exercise. C2: Predicting the manufacture

(a) Show that $(P \vee Q) \to (R \wedge S) \vdash P \to S$.

(b) Show that $(P \vee Q) \to R \vdash (P \wedge Q) \to R$.

(c) In each derivation, which rule built the hypothesis of the premise? Could you have predicted, from the premise's main connective and the shape of its hypothesis alone, that this rule would be needed?

Solution.

(a) The goal is an implication: open a box with P; the goal inside is S. The only line mentioning S is the premise, whose conclusion $R \wedge S$ contains it; to use the premise, $\to$E needs $P \vee Q$, built from P by $\vee$I. Then $\wedge$E takes out S.

1	$(P \vee Q) \to (R \wedge S)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad R \wedge S$	$\to$E 1, 3
5	$\quad S$	$\wedge$E 4
6	$P \to S$	$\to$I 2–5

(b) The goal is an implication: open a box with $P \wedge Q$; the goal inside is R. The premise fires on $P \vee Q$. The assumption gives P by $\wedge$E, and $\vee$I builds $P \vee Q$.

1	$(P \vee Q) \to R$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad P \vee Q$	$\vee$I 3
5	$\quad R$	$\to$E 1, 4
6	$(P \wedge Q) \to R$	$\to$I 2–5

Taking Q at line 3 instead, and building $P \vee Q$ by the second schema, works equally well.

(c) In both, $\vee$I built the hypothesis (line 3 in (a), line 4 in (b)), and in both it could

be predicted. The premise's main connective is $\to$, so the premise is used by $\to$E, which needs its hypothesis $P \vee Q$ on a line of its own, exactly. That disjunction occurs only inside the premise, as its hypothesis, and no rule takes the hypothesis out of an implication ($\to$E gives the conclusion). So it had to be built, and the only rule of the chapter so far that builds a disjunction is $\vee$I. The one question left was which side would be in hand; in (b) the conjunction $P \wedge Q$ offered both.

1.6 Using Or: Proof by Cases

The previous section met a goal the rules could not reach: from $P \vee Q$, derive $Q \vee P$. The rules could build a disjunction ($\vee$I, Definition 1.5.1), but none of them could use one. This section gives the rule that uses a disjunction, $\vee$E. It is the most familiar move in mathematics after "suppose": the split into cases, "if $x \geq 0$, then ...; if $x < 0$, then ...". It is also the last rule of this chapter. We use it to reach the stalled goal, to prove the laws that let "and" and "or" be rearranged, and to prove three facts about absolute values; the section ends with all the rules of the chapter on one page.

1.6.1 A proof that needs two cases

The absolute value of a real number is defined in two cases:

$$|x| \coloneqq \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Suppose we want to prove that $|xy| = |x||y|$ for all real numbers x and y. There is no single computation to do. Which formula gives $|xy|$ depends on the sign of xy, and that depends on the signs of x and y. Yet once the signs are fixed, each case is one line: if $x \geq 0$ and $y < 0$, for instance, then $|xy| = -xy = x \cdot (-y) = |x||y|$.

So the proof we want has the shape "every real number is ≥ 0 or < 0; in the first case ...; in the second case ...; so in every case ...". The rules so far cannot say this. The attempt at $P \vee Q \vdash Q \vee P$ in Section 1.5 met the same wall from the other side. It lacked a rule to say "one of P, Q holds, and either one leads to $Q \vee P$; so $Q \vee P$": a rule that takes "φ or ψ", which does not say **which** side holds, and still reaches a definite conclusion. That is the job of the elimination rule for "or".

1.6.2 The rule of or-elimination

To use "φ or ψ", reach your goal from φ, and reach it again from ψ.

The definition block states both rules for disjunction: the new rule first, then the two schemas of $\vee$I recalled from Section 1.5. As in every schema (Definition 1.2.1), the Greek letters stand for arbitrary statements and i, j, k, l, m, n for line numbers.

Definition 1.6.1. Rules for Disjunction

$$
\begin{array}{lll}
i & \varphi \vee \psi & \\
j & \quad \boxed{\begin{array}{l}\varphi \\ \vdots \\ \chi\end{array}} & \\
l & \quad \boxed{\begin{array}{l}\psi \\ \vdots \\ \chi\end{array}} & \\
m & & \\
n & \chi & \vee\text{E } i,\ j\text{–}k,\ l\text{–}m
\end{array}
\qquad
\begin{array}{lll}
i & \varphi & \\
k & \varphi \vee \psi & \vee\text{I } i
\end{array}
\qquad
\begin{array}{lll}
i & \varphi & \\
k & \psi \vee \varphi & \vee\text{I } i
\end{array}
$$

The first schema is the rule $\vee$E (**or-elimination**, or **proof by cases**); the other two are $\vee$I, as in Definition 1.5.1. In $\vee$E, each line χ (line k and line m) is a line of its box itself, not of a box inside it.

This rule has conditions that matter; read each one.

- **The disjunction.** Line i holds a disjunction $\varphi \vee \psi$, available at line n. The cases are not ours to invent: they are the two sides of a disjunction we already have, as a premise, an assumption, or a line derived earlier. Writing "Case 1 ... Case 2 ..." does not create it.
- **Two boxes.** The first box opens with the left side φ as its assumption, the second with the right side ψ. They are separate: the second opens after the first has closed, and the drawing shows a break in the vertical bar between them. So, by the scope rule of Definition 1.4.2, no line of the second box may cite a line of the first. The disjunction does not tell us which side holds, so we must be ready for both, and each case's assumption is in force only inside its own box.
- **The same last line.** Both boxes end in the **same** statement χ, and in each box χ is a line of that box itself, not of a box opened inside it. The rule then writes χ a third time, outside both boxes, where neither φ nor ψ is assumed. That is honest only because χ was reached whichever case holds.
- **Three citations.** The justification "$\vee$E $i, j\text{–}k, l\text{–}m$" cites the disjunction first, then the box of the left side, then the box of the right side, each box as a range. Both boxes must be available at line n: closed, and sitting directly in the part of the derivation where line n is (Definition 1.4.2).
- **Any conclusion.** The statement χ need not mention φ or ψ. It is whatever we are trying to reach. The next subsection says how to choose it.

Second reading · *A box of one line*

The degenerate case is a box of one line, whose assumption is already χ. Nothing in the rule asks a box to contain more than its assumption, just as nothing in $\rightarrow$I did (Example (First boxes)). The second derivation below has two such boxes.

Warning. A case may use only its own assumption. Here is the error in its usual form. Let x be a real number, and split on "$x \geq 0$ or $x < 0$". *Case $x \geq 0$: then $|x| = x$. Case $x < 0$: since $x \geq 0$, again $|x| = x$. So $|x| = x$.* The second case cites the assumption of the first,

whose box has closed, which the scope rule forbids. The conclusion is false at $x = -1$.

1.6.3 Cases end in the current goal

Section 1.2 set up two questions: *what is the goal's main connective?* and *what are the available lines' main connectives?* Section 1.5 answered the first for $\vee$: postpone a disjunctive goal until one side is in hand. Now we can answer the second. A disjunction on an available line invites $\vee$E: split on it.

What should the two boxes end in? The rule leaves χ to us, and the choice is always the same. It is the **current goal**: whatever we are trying to reach at the point where the split is made. That may be the goal of the whole derivation, or a subgoal inside other boxes. The boxes of the split end in it, and the $\vee$E line that closes the split then writes it outside both boxes, exactly where it was wanted.

Cases end in the current goal.

The two habits fit together. Faced with a goal $\varphi \vee \psi$ and an available disjunction $\alpha \vee \beta$, do not choose a side of the goal yet: split on $\alpha \vee \beta$ first. Inside each case the current goal is still $\varphi \vee \psi$, but now a new line is in hand, the case's assumption, and it may be one side of the goal. Then choose, and the choice may be different in the two boxes. This is exactly how the stalled goal of Section 1.5 goes.

In the Givens/Goal table, each case of a split adds its assumption to the Givens and keeps the same Goal. For the stalled goal:

Givens	Goal
$P \vee Q$	$Q \vee P$
$P \vee Q$, P (first case)	$Q \vee P$
$P \vee Q$, Q (second case)	$Q \vee P$

Example. First derivations with "or"

Show that $P \vee Q \vdash Q \vee P$, and that $P \vee P \vdash P$.

Solution. Idea. For the first: the goal $Q \vee P$ is a disjunction, and neither side is in hand, so we postpone it. The premise is a disjunction, so we split on it; the current goal is $Q \vee P$, so both boxes will end there. In the box of P, the right side of the goal is in hand, and $\vee$I takes it; in the box of Q, the left side is, and $\vee$I takes that. The choice of side differs between the boxes, and each is safe because it is made inside its own case. The pictures show the two boxes opened, each ending in the goal; then each box filled, with only line 6 left to justify.

1	$P \vee Q$	
2	$\quad P$	
	$\quad \vdots$	
3	$\quad Q \vee P$	
4	$\quad Q$	
	$\quad \vdots$	
5	$\quad Q \vee P$	
6	$Q \vee P$	?

1	$P \vee Q$	
2	$\quad P$	
3	$\quad Q \vee P$	$\vee$I 2
4	$\quad Q$	
5	$\quad Q \vee P$	$\vee$I 4
6	$Q \vee P$	?

1	$P \vee Q$	
2	$\quad P$	
3	$\quad Q \vee P$	$\vee$I 2
4	$\quad Q$	
5	$\quad Q \vee P$	$\vee$I 4
6	$Q \vee P$	$\vee$E 1, 2–3, 4–5

The two boxes are separate: the break in the vertical bar between lines 3 and 4 shows where the first closes and the second opens. Line 3 uses the second schema of $\vee$I (P on the right) and line 5 the first (Q on the left). Both boxes end in $Q \vee P$, and line 6 cites the disjunction on line 1 and the boxes 2–3 and 4–5. Compare the stalled attempt of Section 1.5: its two boxes (2–3 and 5–6 there) are these two boxes, and line 6 here is the step it could not take.

For the second, the goal is P, and the premise is a disjunction, so we split. Both cases are already the goal, so each box consists of its assumption alone.

1	$P \vee P$	
2	$\quad P$	
3	$\quad P$	
4	P	$\vee$E 1, 2–2, 3–3

This is the degenerate case split, in which the two cases are the same case. The box 2–2 is a legal box whose last line is P, a line of that box itself, and so is 3–3.

↻ **Second reading** · *One line short*

Change one line of the first derivation and it stops being a derivation. Suppose the second box stops at its assumption Q, and the last line still claims $Q \vee P$:

1	$P \vee Q$	
2	$\quad P$	
3	$\quad Q \vee P$	$\vee$I 2
4	$\quad Q$	
5	$Q \vee P$	$\vee$E 1, 2–3, 4–4

Everything cited exists and is available: the disjunction, a box opened by P, a box opened by Q. What fails is the clause about the last lines. The box 4–4 ends in Q, the box 2–3 ends in $Q \vee P$, and $\vee$E needs one statement that ends both. The repair is the missing line 5 of the correct derivation.

Quick check. A derivation contains a line $P \vee Q$ and a box opened by P that ends in R. No box is opened by Q. Which clause of $\vee$E stops us writing R? What single change to the derivation would allow it?

Solution. The clause "two boxes": the rule needs one box opened by each side of the disjunction, and the box opened by Q is missing. Adding a box opened by Q whose last line is R (a derivation of R from Q and the lines available there) allows $\vee$E, citing the disjunction and both boxes.

↻ **Second reading** · *Why one common last line loses nothing*

Why is the rule built like this? Drop the second box, and from "$x \geq 0$ or $x < 0$" and a derivation of $|x| = x$ from $x \geq 0$ we could conclude $|x| = x$ for every real x, which fails at $x = -1$. Allow the two boxes to end in different statements χ_1 and χ_2, and ask what may be written after them: all we honestly know is "χ_1 or χ_2". That is already reachable with the rule as it stands, by $\vee$I at the end of each box, turning both last lines into $\chi_1 \vee \chi_2$ (exercise B3 carries this out). So the rule asks for one common last line and loses nothing.

The next derivation splits on a disjunction that is not a premise but an **assumption**, opened because the goal asked for it.

Example. Cases on a hypothesis are two implications

Show that $(P \to R) \wedge (Q \to R) \vdash (P \vee Q) \to R$.

Solution. Idea. Goal-directed: the goal is an implication, so we open a box with its hypothesis $P \vee Q$, and the current goal inside is R. Premise-directed: the assumption just made is a disjunction, so we split on it, and both case boxes end in the current goal R. In the case P, the premise's main connective is $\wedge$; $\wedge$E takes out $P \to R$, and $\to$E fires on the case's

assumption. The case Q is the same with the other half of the premise. The pictures show the two case boxes opened, each ending in R; then the first case filled in.

1	$(P \to R) \wedge (Q \to R)$	
2	$\quad P \vee Q$	
3	$\quad\quad P$	
	$\quad\quad \vdots$	
5	$\quad\quad R$	
6	$\quad\quad Q$	
	$\quad\quad \vdots$	
8	$\quad\quad R$	
9	$\quad R$	$\vee$E 2, 3–5, 6–8
10	$(P \vee Q) \to R$	$\to$I 2–9

1	$(P \to R) \wedge (Q \to R)$	
2	$\quad P \vee Q$	
3	$\quad\quad P$	
4	$\quad\quad P \to R$	$\wedge$E 1
5	$\quad\quad R$	$\to$E 4, 3
6	$\quad\quad Q$	
	$\quad\quad \vdots$	
8	$\quad\quad R$	
9	$\quad R$	$\vee$E 2, 3–5, 6–8
10	$(P \vee Q) \to R$	$\to$I 2–9

1	$(P \to R) \wedge (Q \to R)$	
2	$\quad P \vee Q$	
3	$\quad\quad P$	
4	$\quad\quad P \to R$	$\wedge$E 1
5	$\quad\quad R$	$\to$E 4, 3
6	$\quad\quad Q$	
7	$\quad\quad Q \to R$	$\wedge$E 1
8	$\quad\quad R$	$\to$E 7, 6
9	$\quad R$	$\vee$E 2, 3–5, 6–8
10	$(P \vee Q) \to R$	$\to$I 2–9

Line 9 closes the split inside the outer box. It is a line of the outer box itself, so →I at line 10 may cite 2–9. Lines 4 and 7 cite line 1, which lies in no box.

Quick check. In the derivation of Example (Cases on a hypothesis are two implications), which lines and boxes does the ∨E line cite?

Solution. Line 9 cites the disjunction on line 2 and the two case boxes, as ranges: 3–5 and 6–8. All three sit directly in the outer box 2–9, as line 9 does.

Section 1.5 derived the converse, $(P \vee Q) \to R \vdash (P \to R) \wedge (Q \to R)$, in Example (A disjunctive hypothesis splits into two implications), using ∨I and no ∨E. So each of the two statements is derivable from the other: *cases on a hypothesis are two implications.* This is how mathematicians prove "if P or Q, then R": they prove "if P, then R" and "if Q, then R". To show that an integer which is one more, or three more, than a multiple of 4 is odd, one treats the two forms separately, and the derivation above is the reason that is enough.

1.6.4 Laws of "or"

Case splits are not only about numbers. The same two rules prove the laws that let us rearrange "and" and "or". The first says that "and" distributes over "or", in both directions.

Proposition 1.6.1. Distributivity

$P \wedge (Q \vee R) \vdash (P \wedge Q) \vee (P \wedge R)$, and $(P \wedge Q) \vee (P \wedge R) \vdash P \wedge (Q \vee R)$.

Try it first (ten minutes): derive the first sequent. Which rule will give its last line, and which disjunction will you split on?

Idea. The natural first attempt reads the goal: it is a disjunction, so end with ∨I and aim at one side, say $P \wedge Q$. That needs Q on a line, and the premise offers only P and $Q \vee R$; nothing turns $Q \vee R$ into Q. The other side fails the same way. So the goal must be postponed, and the attempt shows why: the premise holds a disjunction, $Q \vee R$, and neither of its sides is known. Premise-directed, then: ∧E takes the premise apart into P and $Q \vee R$, and we split on $Q \vee R$, both cases ending in the current goal $(P \wedge Q) \vee (P \wedge R)$. In the case Q, the side $P \wedge Q$ is one ∧I away, and ∨I weakens it to the whole goal; the case R builds the other side. In the second sequent the premise itself is the disjunction, so we split on it at once; each case is a conjunction containing P, and ∨I turns its other half into $Q \vee R$.

For the first derivation: the two case boxes opened, each ending in the goal; then lines 5 and 8 written, and only lines 6 and 9 left to justify.

1	$P \wedge (Q \vee R)$	
2	P	$\wedge$E 1
3	$Q \vee R$	$\wedge$E 1
4	$\quad Q$	
	$\quad \vdots$	
6	$\quad (P \wedge Q) \vee (P \wedge R)$	
7	$\quad R$	
	$\quad \vdots$	
9	$\quad (P \wedge Q) \vee (P \wedge R)$	
10	$(P \wedge Q) \vee (P \wedge R)$	$\vee$E 3, 4–6, 7–9

1	$P \wedge (Q \vee R)$	
2	P	$\wedge$E 1
3	$Q \vee R$	$\wedge$E 1
4	$\quad Q$	
5	$\quad P \wedge Q$	$\wedge$I 2, 4
6	$\quad (P \wedge Q) \vee (P \wedge R)$	?
7	$\quad R$	
8	$\quad P \wedge R$	$\wedge$I 2, 7
9	$\quad (P \wedge Q) \vee (P \wedge R)$	?
10	$(P \wedge Q) \vee (P \wedge R)$	$\vee$E 3, 4–6, 7–9

Before reading on: in the first derivation, which line does each box end on, and why must the two be the same?

Proof. For the first sequent, the following is a derivation with premise $P \wedge (Q \vee R)$ and last line $(P \wedge Q) \vee (P \wedge R)$.

1	$P \wedge (Q \vee R)$	
2	P	$\wedge$E 1
3	$Q \vee R$	$\wedge$E 1
4	$\quad Q$	
5	$\quad P \wedge Q$	$\wedge$I 2, 4
6	$\quad (P \wedge Q) \vee (P \wedge R)$	$\vee$I 5
7	$\quad R$	
8	$\quad P \wedge R$	$\wedge$I 2, 7
9	$\quad (P \wedge Q) \vee (P \wedge R)$	$\vee$I 8
10	$(P \wedge Q) \vee (P \wedge R)$	$\vee$E 3, 4–6, 7–9

Lines 5 and 8 may cite line 2, which lies in no box and so is available everywhere below it. Both boxes end in the goal, lines 6 and 9, as $\vee$E at line 10 requires.

For the second sequent, the following is a derivation with premise $(P \wedge Q) \vee (P \wedge R)$ and last line $P \wedge (Q \vee R)$.

1	$(P \wedge Q) \vee (P \wedge R)$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
5	$\quad Q \vee R$	$\vee$I 4
6	$\quad P \wedge (Q \vee R)$	$\wedge$I 3, 5
7	$\quad P \wedge R$	
8	$\quad P$	$\wedge$E 7
9	$\quad R$	$\wedge$E 7
10	$\quad Q \vee R$	$\vee$I 9
11	$\quad P \wedge (Q \vee R)$	$\wedge$I 8, 10
12	$P \wedge (Q \vee R)$	$\vee$E 1, 2–6, 7–11

This proves both sequents.

■

The answer to the prompt: the boxes end on lines 6 and 9, both $(P \wedge Q) \vee (P \wedge R)$. They must be the same statement because line 10 writes that statement outside both boxes, where neither Q nor R is assumed, and that is honest only if it was reached in each case.

In the second derivation, line 8 derives P a second time instead of citing line 3. It has to: line 3 lies in the box 2–6, which has closed, so line 3 is not available at line 8. Each case must earn its own P.

For a real number x, take P: "$x \neq 0$", Q: "$x > 0$" and R: "$x < 0$". The first sequent then reads: a non-zero number that is positive or negative is a positive non-zero number or a negative non-zero number. The law is dull in words, which is as it should be. Its use is in derivations, where it moves a case split from inside a conjunction to the outside, and in the laws of sets, where "and" and "or" become intersection and union.

The next derivation shows a case split **inside** a case: $\vee$E applied within one box of another $\vee$E. Before the lines, the shape: each split ends with its own $\vee$E line, line 11 for the inner split and line 12 for the outer.

1	$P \vee (Q \vee R)$	
2	$\quad P$	
	$\quad \vdots$	
4	$\quad (P \vee Q) \vee R$	
5	$\quad Q \vee R$	
6	$\quad\quad Q$	
	$\quad\quad \vdots$	
8	$\quad\quad (P \vee Q) \vee R$	
9	$\quad\quad R$	
	$\quad\quad \vdots$	
10	$\quad\quad (P \vee Q) \vee R$	
11	$\quad (P \vee Q) \vee R$	$\vee$E
12	$(P \vee Q) \vee R$	$\vee$E

Quick check. In the outline, which line closes the inner split, and which line and boxes does it cite?

Solution. Line 11. It cites the disjunction on line 5 and the boxes 6–8 and 9–10, all of which sit directly in the box 5–11.

Example. Regrouping a disjunction

Show that $P \vee (Q \vee R) \vdash (P \vee Q) \vee R$.

Solution. Idea. The goal is a disjunction, and no side of it is in hand: postpone. The premise is a disjunction: split on it, with both boxes ending in the current goal $(P \vee Q) \vee R$. The case P is two weakenings: first to $P \vee Q$, then to the goal. The case $Q \vee R$ is itself a disjunction, and neither Q nor R is known; so, inside that box, we split again, and the inner boxes end in the same current goal. In the inner case Q the goal is two weakenings away, and in the inner case R one. The outline above was the first picture; in the second, every box is filled, and only the two $\vee$E citations are left.

1	$P \vee (Q \vee R)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad (P \vee Q) \vee R$	$\vee$I 3
5	$\quad Q \vee R$	
6	$\quad\quad Q$	
7	$\quad\quad P \vee Q$	$\vee$I 6
8	$\quad\quad (P \vee Q) \vee R$	$\vee$I 7
9	$\quad\quad R$	
10	$\quad\quad (P \vee Q) \vee R$	$\vee$I 9
11	$\quad (P \vee Q) \vee R$	$\vee$E
12	$(P \vee Q) \vee R$	$\vee$E

Before reading on: which disjunction does the inner $\vee$E *cite, and which box must contain it?*

1	$P \vee (Q \vee R)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad (P \vee Q) \vee R$	$\vee$I 3
5	$\quad Q \vee R$	
6	$\quad\quad Q$	
7	$\quad\quad P \vee Q$	$\vee$I 6
8	$\quad\quad (P \vee Q) \vee R$	$\vee$I 7
9	$\quad\quad R$	
10	$\quad\quad (P \vee Q) \vee R$	$\vee$I 9
11	$\quad (P \vee Q) \vee R$	$\vee$E 5, 6–8, 9–10
12	$(P \vee Q) \vee R$	$\vee$E 1, 2–4, 5–11

Line 11 is the inner case split. It cites the disjunction on line 5, the assumption of the outer box it sits in, and the boxes 6–8 and 9–10, all of which sit inside the box 5–11, as line 11 does. Line 12 is the outer case split. Line 11 is the inner split's own $\vee$E line, so the box 5–11 ends on a line of its own, as Definition 1.6.1 requires.

1.6.5 From a skeleton to a proof about numbers

Almost every proof by cases in analysis has the same skeleton. A known fact supplies the cases, and each case is handled by a separate argument, a small lemma that turns the case's assumption into the goal. As a derivation, with the fact and the two lemmas as premises:

1	$P \vee Q$	
2	$P \to R$	
3	$Q \to R$	
4	$\quad P$	
5	$\quad R$	$\to$E 2, 4
6	$\quad Q$	
7	$\quad R$	$\to$E 3, 6
8	R	$\vee$E 1, 4–5, 6–7

The idea behind it is the whole method of this section at its simplest. The goal R is atomic, so the goal suggests nothing. Premise-directed, the disjunction on line 1 is split, both boxes ending in the current goal R. In each box the matching lemma fires on the case's assumption. (Example (Cases on a hypothesis are two implications) did the same with the two lemmas packed into one conjunction, and inside a box.)

In a proof about real numbers, the disjunction is usually a fact about the order. The facts about $\leq$ and the absolute value used below are school algebra: every real number a satisfies $a \geq 0$ or $a < 0$; the rules of signs for products; adding two inequalities of the same direction. Here we take them as known. In the language of this chapter, "every real number a satisfies $a \geq 0$ or $a < 0$" is a supply of disjunctions: for each number we meet, it gives a line $(a \geq 0) \vee (a < 0)$ that $\vee$E may cite.

The first example is the smallest, and it is written twice: as a proof in which every step carries its rule in brackets, and as a derivation.

Example. A number is at most its absolute value

Let x be a real number. Prove that $|x| \geq x$, by cases, labeling each step with its rule.

Solution. The goal $|x| \geq x$ is the disjunction $(|x| > x) \vee (|x| = x)$, and neither side is known for every x (the first fails at $x = 1$, the second at $x = -1$), so we postpone it. The fact "$x \geq 0$ or $x < 0$" is a disjunction to split on. In prose:

> We have $x \geq 0$ or $x < 0$, and we argue by cases [$\vee$E].
>
> *Case* $x \geq 0$. By the definition of the absolute value, $|x| = x$ [$\to$E]. So $|x| \geq x$ [$\vee$I].
>
> *Case* $x < 0$. By the definition, $|x| = -x$, and $-x > 0 > x$; so $|x| > x$ [$\to$E], which is the school-algebra fact "if $x < 0$, then $|x| > x$" used on the case's assumption. So $|x| \geq x$ [$\vee$I].
>
> In either case $|x| \geq x$, so $|x| \geq x$.

As a derivation, with the order fact, the definition's first clause, and the school-algebra fact about negative numbers as premises, and with $|x| \geq x$ written out:

1	$x \geq 0 \vee x < 0$	
2	$x \geq 0 \to \lvert x\rvert = x$	
3	$x < 0 \to \lvert x\rvert > x$	
4	$\quad x \geq 0$	
5	$\quad \lvert x\rvert = x$	$\to$E 2, 4
6	$\quad \lvert x\rvert > x \vee \lvert x\rvert = x$	$\vee$I 5
7	$\quad x < 0$	
8	$\quad \lvert x\rvert > x$	$\to$E 3, 7
9	$\quad \lvert x\rvert > x \vee \lvert x\rvert = x$	$\vee$I 8
10	$\lvert x\rvert > x \vee \lvert x\rvert = x$	$\vee$E 1, 4–6, 7–9

It is the skeleton above, with a weakening at the end of each box: line 6 takes the right side of the goal (second schema of $\vee$I) and line 9 the left side (first schema). Each case took the side it had in hand. At $x = 3$, the first case applies and line 6 reads $3 > 3$ or $3 = 3$; at $x = -2$, the second applies and line 8 reads $2 > -2$.

Quick check. In the derivation of Example (A number is at most its absolute value), which statement do lines 6 and 9 both write, and why must they agree?

Solution. Both write $(|x| > x) \vee (|x| = x)$, the current goal written out. They must agree because $\vee$E closes a split only when both case boxes end in the same statement.

The next two examples are longer, and in each the case split is doing real work.

Example. The absolute value of a product

Let x and y be real numbers. Prove that $|xy| = |x||y|$.

Solution. We use that $|a| = -a$ whenever $a \leq 0$: for $a < 0$ this is the definition, and for $a = 0$ both sides are 0.

The number x satisfies $x \geq 0$ or $x < 0$, and we argue by cases [$\vee$E]. Inside each case, $y \geq 0$ or $y < 0$, and we argue by cases again.

Case $x \geq 0$, $y \geq 0$. Then $xy \geq 0$, so $|xy| = xy = |x||y|$.

Case $x \geq 0$, $y < 0$. Then $xy \leq 0$, so $|xy| = -xy = x \cdot (-y) = |x||y|$.

Case $x < 0$, $y \geq 0$. Then $xy \leq 0$, so $|xy| = -xy = (-x) \cdot y = |x||y|$.

Case $x < 0$, $y < 0$. Then $xy > 0$, so $|xy| = xy = (-x)(-y) = |x||y|$.

Each inner case split reaches $|xy| = |x||y|$, so each outer case does, and hence the equation

holds.

The structure is two uses of $\vee$E, one inside each box of the other: two outer boxes (the sign of x), each containing two inner boxes (the sign of y), and every one of the four inner boxes ending in the same equation, the current goal. The second and third cases needed the remark about $a \leq 0$. When $x = 0$ and $y < 0$, the product is 0, not negative, and without the remark the case would have a hole at exactly that point.

A case split need not be the obvious one. The triangle inequality has an equally short proof with two cases, if we split on the sign of the right quantity.

Example. The triangle inequality by cases

Let x and y be real numbers. Prove that $|x + y| \leq |x| + |y|$.

Solution. First, every real number a satisfies $a \leq |a|$ and $-a \leq |a|$. This is itself a proof by cases [$\vee$E]: if $a \geq 0$, then $|a| = a$, so $a \leq |a|$, and $-a \leq 0 \leq a = |a|$; if $a < 0$, then $|a| = -a$, so $-a \leq |a|$, and $a < 0 < -a = |a|$.

Now $x + y \geq 0$ or $x + y < 0$, and we argue by cases [$\vee$E].

Case $x + y \geq 0$. Then $|x + y| = x + y \leq |x| + |y|$, adding $x \leq |x|$ and $y \leq |y|$.

Case $x + y < 0$. Then $|x + y| = -x - y = (-x) + (-y) \leq |x| + |y|$, adding $-x \leq |x|$ and $-y \leq |y|$.

Both cases reach $|x + y| \leq |x| + |y|$, so it holds.

Two case splits appear here, one after the other rather than one inside the other. The first proves a small lemma for an arbitrary number a, both halves at once, and its first half, $a \leq |a|$, is Example (A number is at most its absolute value) again. The second uses both halves of the lemma, for $a = x$ and for $a = y$. Splitting on the sign of $x + y$ instead of the signs of x and y is what saves two cases: the absolute value in the goal is $|x + y|$, and its sign is the only one the goal's left side cares about.

Warning. A case split must cover every possibility. A common faulty proof of the triangle inequality treats "$x \geq 0$ and $y \geq 0$" and "$x < 0$ and $y < 0$", checks both, and stops. The disjunction of these two cases is not a fact: $x = 1$, $y = -1$ satisfies neither. So there is no line $\varphi \vee \psi$ for $\vee$E to cite, and the rule was never legally applied. The missing mixed-sign cases are the interesting ones: at $x = 1$, $y = -1$ the inequality reads $0 \leq 2$, and it is strict.

Warning. Cases must **exhaust**, and a split that silently drops a value is a missing case in disguise. "Every real x satisfies $x > 0$ or $x < 0$" is false: $x = 0$ satisfies neither. A proof that splits into $x > 0$ and $x < 0$ cites a disjunction that is not available, and proves nothing about $x = 0$. The true disjunctions are "$x \geq 0$ or $x < 0$", and "$x > 0$, or $x = 0$, or $x < 0$".

Cases may overlap, however. "$x \geq 0$ or $x \leq 0$" is true for every real x, and $x = 0$ lies in both

cases. Nothing in ∨E forbids this: the rule asks that the goal be reached in each case, not that the cases be disjoint. (This is the inclusive "or" of Section 1.1.)

Quick check. Each of the following begins a proof that $x^2 \geq 0$ for every real number x. Which case splits are exhaustive, so that ∨E may be applied?

(i) Case $x \geq 0$; case $x < 0$.

(ii) Case $x > 0$; case $x < 0$.

(iii) Case $x \geq 0$; case $x \leq 0$.

Solution.

(i) Exhaustive: every real number is ≥ 0 or < 0. (ii) Not exhaustive: $x = 0$ is in neither case, so the disjunction cited is false. The claim is still true at 0, but this split never proves it there. (iii) Exhaustive. The cases overlap at $x = 0$, which does no harm: $x^2 = 0 \geq 0$ is reached in both.

1.6.6 The rules of this chapter

This chapter has introduced six rules, one or two at a time, and the structural clause R. Here is the whole chapter on one page.

Rule	Shape	You write	Lean
∧I	$\varphi, \psi \Longrightarrow \varphi \wedge \psi$	"We show P, and we show Q."	`constructor`
∧E	$\varphi \wedge \psi \Longrightarrow \varphi$ (or ψ)	"In particular, P."	`h.1` (and `h.2`)
→E	$\varphi \to \psi, \varphi \Longrightarrow \psi$	"Since P implies Q, and P holds, Q."	`exact h hp`
→I	box $\varphi \ldots \psi \Longrightarrow \varphi \to \psi$	"Assume P. … Hence Q."	`intro hp`
R	φ (available) $\Longrightarrow \varphi$, copied	"As shown above, P."	`exact hp`
∨I	$\varphi \Longrightarrow \varphi \vee \psi$ (or $\psi \vee \varphi$)	"So P or Q holds."	`left` (or `Or.inl hp`); `right`
∨E	$\varphi \vee \psi$, box $\varphi \ldots \chi$, box $\psi \ldots \chi \Longrightarrow \chi$	"Case 1: P. … so R. Case 2: Q. … so R. In either case, R."	`Or.elim h` (or `cases h`)

A written proof often drops the first part of the phrase for ∨E, saying "we argue by cases" without naming the disjunction; when that is not a standard fact such as "$x \geq 0$ or $x < 0$", ask for it, since it is the line the rule cites first. Nothing depends on the Lean column.

Here are the rules as schemas. Letters in the number column stand for line numbers; every cited line must be available (Definition 1.4.2), and each box rule needs the box's last line as a line of that box itself.

Conjunction (Definition 1.2.5): ∧I, and the two forms of ∧E.

$$\begin{array}{l|ll} i & \varphi & \\ j & \psi & \\ k & \varphi \wedge \psi & \wedge\text{I } i, j \end{array} \qquad \begin{array}{l|ll} i & \varphi \wedge \psi & \\ k & \varphi & \wedge\text{E } i \end{array} \qquad \begin{array}{l|ll} i & \varphi \wedge \psi & \\ k & \psi & \wedge\text{E } i \end{array}$$

Implication (Definition 1.4.1): $\to$E and $\to$I; and **repetition**, clause (D3) of Definition 1.4.2.

$$\begin{array}{l|ll} i & \varphi \to \psi & \\ j & \varphi & \\ k & \psi & \to\text{E } i, j \end{array} \qquad \begin{array}{l|ll} i & \boxed{\varphi} & \\ & \vdots & \\ j & \psi & \\ k & \varphi \to \psi & \to\text{I } i\text{–}j \end{array} \qquad \begin{array}{l|ll} i & \varphi & \\ & \vdots & \\ k & \varphi & \text{R } i \end{array}$$

Disjunction (Definition 1.6.1): the two forms of $\vee$I, and $\vee$E.

$$\begin{array}{l|ll} i & \varphi & \\ k & \varphi \vee \psi & \vee\text{I } i \end{array} \qquad \begin{array}{l|ll} i & \varphi & \\ k & \psi \vee \varphi & \vee\text{I } i \end{array}$$

$$\begin{array}{l|ll} i & \varphi \vee \psi & \\ j & \boxed{\varphi} & \\ & \vdots & \\ k & \chi & \\ l & \boxed{\psi} & \\ & \vdots & \\ m & \chi & \\ n & \chi & \vee\text{E } i, j\text{–}k, l\text{–}m \end{array}$$

Each rule is also a move in the search for a derivation. Read by the goal's main connective, the introduction rules say how a goal is built: that is the move *read the skeleton off the goal.* Read by an available line's main connective, the elimination rules say how that line is used: *use a premise by its main connective.*

Main connective	As the goal (goal-directed)	As an available line (premise-directed)
$\wedge$	two subgoals, one per side; finish with $\wedge$I	take it apart with $\wedge$E
$\to$	open a box with the hypothesis; the new goal is the conclusion; close with $\to$I	fire it with $\to$E once its hypothesis is on a line, or make the hypothesis a subgoal
$\vee$	postpone; choose a side with $\vee$I when that side is in hand	split with $\vee$E; both boxes end in the current goal

1.6.6.1 Named moves

- **Try to break it.** To test a form, put statements in for the letters that make every premise true and the conclusion false; one such choice shows the form is bad. (Section 1.1)
- **Read the skeleton off the goal.** The goal's main connective chooses the introduction rule: $\wedge$ gives two subgoals, $\to$ a box, $\vee$ a choice to postpone. (Section 1.2)

- **Use a premise by its main connective.** Its elimination rule says what you may do with it: take a conjunction apart, feed an implication its hypothesis, split a disjunction into cases. (Section 1.2)
- **Open a box, close a box.** An assumption lives exactly as long as its box. (Section 1.4)
- **Work from both ends.** Premises at the top, the goal at the bottom; fill the gap from both sides. (Section 1.2)
- **Postpone a disjunction.** Choose a side of a goal $P \vee Q$ only when that side is in hand. (Section 1.5)
- **Cases end in the current goal.** Both boxes of $\vee$E close on the statement you are trying to reach. (Section 1.6)

Remark (Looking ahead). The rules for "not", for $\bot$, and for "if and only if", the rule for proof by contradiction, and the method that turns the table of moves above into a way of finding derivations are the subject of Chapter 2.

1.6.7 Exercises

1.6.7.1 A. Check your understanding

Exercise. A1

(a) State the rule $\vee$E as a schema, and name the three things its justification cites.

(b) For each sentence, taken from a proof, name the disjunction that supplies the cases and the statement both cases end in.
- "If n is even, then $n^2 + n = n(n+1)$ is even; if n is odd, then $n+1$ is even and so is $n(n+1)$. Either way, $n^2 + n$ is even."
- "Both possibilities for the sign of t give $t^2 \geq 0$."

(c) Let n be an integer. True or false, with a reason: "If n is even, then n^2 is even. Since every integer is even or odd, n^2 is even" is a correct proof by cases that n^2 is even.

(d) The following is offered as a derivation of $P \vee Q,\ P \to R \vdash R$. Name the line that is not justified and the clause of Definition 1.6.1 it violates.

1	$P \vee Q$	
2	$P \to R$	
3	$\quad P$	
4	$\quad R$	$\to$E 2, 3
5	R	$\vee$E 1, 3–4

(e) In the first derivation of Proposition 1.6.1, which line is the disjunction that $\vee$E cites at line 10, and where did it come from? Why does each box end in $(P \wedge Q) \vee (P \wedge R)$ rather than in $P \wedge Q$ and $P \wedge R$?

(f) In the same derivation, why does line 6 use $\vee$I to put $P \wedge Q$ on the **left**? Could line 6 have been $(P \wedge R) \vee (P \wedge Q)$?

Solution.

(a) The schema is the first one of Definition 1.6.1: from a line i holding $\varphi \vee \psi$, a box j–k

opened by φ, and a box l–m opened by ψ, each ending in the same statement χ as a line of its own, a line n holding χ. The justification "$\vee$E i, j–k, l–m" cites the disjunction and the two boxes.

(b) In the first, the disjunction is "n is even or n is odd", and both cases end in "$n^2 + n$ is even". In the second, it is "$t \geq 0$ or $t < 0$" (the sentence does not say it, and a careful reader supplies it), and both cases end in $t^2 \geq 0$.

(c) False. The disjunction "n is even or n is odd" does not say which side holds, so $\vee$E needs a second case, reaching "n^2 is even" from "n is odd", and the argument has none. Here the second case cannot be supplied: at $n = 1$ the number $n^2 = 1$ is odd, so the claim itself is false.

(d) Line 5. $\vee$E needs two boxes, one opened by each side of the disjunction (the clause "two boxes"); line 5 cites only the box opened by P, and no box opened by Q exists. Nor can a box opened by Q simply be added: nothing available reaches R from Q. In fact the sequent's form is bad, so no correct argument should be expected to reach R from these premises. Try to break it: for a real number x, take P: "$x > 1$", Q: "$x < 1$", R: "$x > 0$". At $x = -1$ both premises hold and R does not.

(e) Line 3, $Q \vee R$, obtained from the premise by $\wedge$E. Each box must end in the current goal, $(P \wedge Q) \vee (P \wedge R)$, because line 10 writes one statement, the same one that ends both boxes; boxes ending in $P \wedge Q$ and $P \wedge R$ would end in different statements, and $\vee$E could not be applied to them.

(f) Line 6 must be the current goal exactly, since both boxes end in it and line 10 writes it. The goal $(P \wedge Q) \vee (P \wedge R)$ has $P \wedge Q$ as its **left** side, so line 6 uses the first schema of $\vee$I. The line $(P \wedge R) \vee (P \wedge Q)$ is also an instance of $\vee$I on line 5 (the second schema), but it is a different statement: the first box would then end differently from the second, whose last line 9 is $(P \wedge Q) \vee (P \wedge R)$, and $\vee$E could not cite both boxes.

Exercise. A2: Drill

Each part takes a line or two. Let x be a real number and n an integer.

(a) "If $x \geq 0$, then $|x|^2 = x^2$; if $x < 0$, then $|x|^2 = (-x)^2 = x^2$. So $|x|^2 = x^2$." Name the disjunction that supplies the cases, and the common last line.

(b) Is the split "$n > 0$ or $n < 0$" exhaustive for integers?

(c) Two case boxes end in $R \wedge S$ and $S \wedge R$. May $\vee$E close the split?

(d) How many case boxes does the proof of Example (The absolute value of a product) open?

(e) "Case $x \geq 1$: $x^2 \geq 1$. Case $x \leq -1$: $x^2 \geq 1$. So $x^2 \geq 1$ for every real x." What fails?

(f) Inside a box whose current goal is S, the available line $Q \vee R$ is split. What do the two case boxes end in?

(g) Find the illegal line.

1	$P \lor Q$	
2	$P \to R$	
3	$Q \to R$	
4	$\quad P$	
5	$\quad R$	$\to$E 2, 4
6	$\quad Q$	
7	$\quad R$	$\to$E 3, 4
8	R	$\lor$E 1, 4–5, 6–7

(h) Derive $P \lor Q \vdash Q \lor (P \lor R)$.

(i) Derive $(P \land Q) \lor (P \land R) \vdash P$.

(j) Derive $P \lor Q,\ Q \to P \vdash P$.

Solution.

(a) The disjunction is $(x \geq 0) \lor (x < 0)$, and both cases end in $|x|^2 = x^2$.

(b) No: $n = 0$ is in neither case.

(c) No: both boxes must end in the same statement. Add $\land$E and $\land$I lines so that both end in the current goal, say $R \land S$.

(d) Six: two for the sign of x, and two inside each of those for the sign of y. (The opening remark that $|a| = -a$ for $a \leq 0$ is itself a small split on $a < 0$ or $a = 0$, like the lemma in Example (The triangle inequality by cases); counting its two boxes as well gives eight.)

(e) The split is not exhaustive: $(x \geq 1) \lor (x \leq -1)$ is false at $x = 0$, so $\lor$E has no disjunction to cite. The conclusion fails there too: $0^2 = 0 < 1$.

(f) S, the current goal at the point of the split.

(g) Line 7. It cites line 4, which lies in the box 4–5, closed before line 7. It should read "$\to$E 3, 6".

(h) Box 2–4: 2 P, assumption; 3 $P \lor R$, $\lor$I 2; 4 $Q \lor (P \lor R)$, $\lor$I 3. Box 5–6: 5 Q, assumption; 6 $Q \lor (P \lor R)$, $\lor$I 5. Then 7 $Q \lor (P \lor R)$, $\lor$E 1, 2–4, 5–6.

(i) Box 2–3: 2 $P \land Q$, assumption; 3 P, $\land$E 2. Box 4–5: 4 $P \land R$, assumption; 5 P, $\land$E 4. Then 6 P, $\lor$E 1, 2–3, 4–5.

(j) Box 3–3: 3 P, assumption, already the goal. Box 4–5: 4 Q, assumption; 5 P, $\to$E 2, 4. Then 6 P, $\lor$E 1, 3–3, 4–5.

1.6.7.2 B. Practice

Exercise. B1: The other distributive law

Show that $(P \lor Q) \land (P \lor R) \vdash P \lor (Q \land R)$.

Hint: after taking the premise apart, you will need a case split inside a case.

Solution. The goal is a disjunction with no side in hand, so we postpone it. The premise gives two disjunctions. Split on $P \vee Q$, both cases ending in the goal. In the case P, the goal is one weakening away. In the case Q, we still lack R, so we split on $P \vee R$ inside that box: its case P weakens as before, and its case R combines with Q.

1	$(P \vee Q) \wedge (P \vee R)$	
2	$P \vee Q$	$\wedge$E 1
3	$P \vee R$	$\wedge$E 1
4	$\quad P$	
5	$\quad P \vee (Q \wedge R)$	$\vee$I 4
6	$\quad Q$	
7	$\quad\quad P$	
8	$\quad\quad P \vee (Q \wedge R)$	$\vee$I 7
9	$\quad\quad R$	
10	$\quad\quad Q \wedge R$	$\wedge$I 6, 9
11	$\quad\quad P \vee (Q \wedge R)$	$\vee$I 10
12	$\quad P \vee (Q \wedge R)$	$\vee$E 3, 7–8, 9–11
13	$P \vee (Q \wedge R)$	$\vee$E 2, 4–5, 6–12

Line 10 cites line 6, the assumption of the enclosing box, which is still open at line 10. Line 12 cites line 3, which lies in no box. Splitting on $P \vee R$ first, with the roles of Q and R exchanged, works equally well.

Exercise. B2: The larger of two numbers

For real numbers a and b, let $\max(a, b) := a$ if $a \geq b$, and $\max(a, b) := b$ if $a < b$. Prove that $\max(a, b) \geq a$ and $\max(a, b) \geq b$. Hence deduce that $\max(a, b) \geq \frac{1}{2}(a + b)$: the larger of two numbers is at least their average.

Solution. We have $a \geq b$ or $a < b$, and we argue by cases [$\vee$E]; the current goal is the conjunction "$\max(a, b) \geq a$ and $\max(a, b) \geq b$".

Case $a \geq b$. Then $\max(a, b) = a$. So $\max(a, b) \geq a$, since $a = a$ and $\geq$ allows equality [$\vee$I]; and $\max(a, b) = a \geq b$. So both hold [$\wedge$I].

Case $a < b$. Then $\max(a, b) = b$. So $\max(a, b) = b > a$, which gives $\max(a, b) \geq a$ [$\vee$I]; and $\max(a, b) = b \geq b$ [$\vee$I]. So both hold [$\wedge$I].

In either case $\max(a, b) \geq a$ and $\max(a, b) \geq b$, so both hold.

Adding the two inequalities gives $2\max(a, b) \geq a + b$, and dividing by $2 > 0$ gives $\max(a, b) \geq \frac{1}{2}(a + b)$. For instance, $\max(3, -5) = 3 \geq -1 = \frac{1}{2}(3 + (-5))$.

Exercise. B3: Two cases, two conclusions

(a) A search for $P \vee Q,\ P \to R,\ Q \to S \vdash R \vee S$ has reached the stage below. *Name the move, then complete it.*

1	$P \vee Q$	
2	$P \to R$	
3	$Q \to S$	
	⋮	
10	$R \vee S$	?

(b) Use (a) to prove: for every integer n, n^2 is a multiple of 4 or n^2 is one more than a multiple of 4. You may use that every integer is even or odd.

Solution.

(a) The move is to **split on the premise** $P \vee Q$, each case ending in the current goal. The goal $R \vee S$ is a disjunction with no side in hand, so it is postponed; the premise $P \vee Q$ is a disjunction, so line 10 will be $\vee$E citing it and two case boxes, each ending in $R \vee S$. In each case the matching implication gives one of R, S, and $\vee$I weakens it to the common last line $R \vee S$.

1	$P \vee Q$	
2	$P \to R$	
3	$Q \to S$	
4	$\quad P$	
5	$\quad R$	$\to$E 2, 4
6	$\quad R \vee S$	$\vee$I 5
7	$\quad Q$	
8	$\quad S$	$\to$E 3, 7
9	$\quad R \vee S$	$\vee$I 8
10	$R \vee S$	$\vee$E 1, 4–6, 7–9

(b) Take P: "n is even", Q: "n is odd", R: "n^2 is a multiple of 4", S: "n^2 is one more than a multiple of 4". The premise $P \vee Q$ is the fact we may use. For $P \to R$: if $n = 2k$ for an integer k, then $n^2 = 4k^2$. For $Q \to S$: if $n = 2k+1$, then $n^2 = 4k^2+4k+1 = 4(k^2+k)+1$. By (a), $R \vee S$. For instance, $6^2 = 36 = 4 \cdot 9$ and $7^2 = 49 = 4 \cdot 12 + 1$.

Exercise. B4: The smaller of two numbers

For real numbers a and b, let $\max(a, b)$ be as in Exercise (B2: The larger of two numbers), and let $\min(a, b) := b$ if $a \geq b$, and $\min(a, b) := a$ if $a < b$. Prove by cases that $\min(a, b) = -\max(-a, -b)$.

Solution. We have $a \geq b$ or $a < b$, and we argue by cases [$\vee$E]; the current goal is the equation $\min(a, b) = -\max(-a, -b)$.

Case $a < b$. Then $\min(a, b) = a$. Also $-a > -b$, so $-a \geq -b$ [$\vee$I], and by the definition of max, $\max(-a, -b) = -a$. Its negative is $a = \min(a, b)$.

Case $a \geq b$. Then $\min(a, b) = b$, and $-a \leq -b$. Written out, as the convention on inequalities of Section 1.5 asks when a case split must see the disjunction, this is $(-a < -b) \vee (-a = -b)$, and we split again [$\vee$E].

- If $-a < -b$, then $\max(-a, -b) = -b$, whose negative is $b = \min(a, b)$.
- If $-a = -b$, then $-a \geq -b$ [$\vee$I], so $\max(-a, -b) = -a$, whose negative is a; and $a = b = \min(a, b)$.

Both inner cases reach the goal, so $\vee$E closes the inner split; then both outer cases have reached it, and $\vee$E closes the outer split. Hence $\min(a, b) = -\max(-a, -b)$.

For instance, at $a = 3$, $b = -5$: $\min(3, -5) = -5$, and $\max(-3, 5) = 5$, whose negative is -5. At $a = b = 2$: $\min(2, 2) = 2$, and $\max(-2, -2) = -2$, whose negative is 2.

1.6.7.3 C. Going deeper

Exercise. C1: Challenge: The reverse triangle inequality

Let x and y be real numbers. Prove that

$$||x| - |y|| \leq |x - y|$$

by splitting into the two cases $|x| \geq |y|$ and $|x| < |y|$, using in each case the triangle inequality of Example (The triangle inequality by cases). Then explain why these two cases exhaust all possibilities, and why a split on the signs of x and y would need four cases.

Solution (Hint 1). The triangle inequality bounds $|u + v|$. Write x as a sum of two numbers whose absolute values appear in the goal.

Solution (Hint 2). In the case $|x| \geq |y|$, the left side is $|x| - |y|$. Start from $x = (x - y) + y$.

Solution. We use two school facts: for real numbers a, b, either $a \geq b$ or $a < b$; and $|-a| = |a|$ for every real a (if $a \geq 0$ then $-a \leq 0$, so $|-a| = a$; if $a < 0$ then $-a > 0$, so $|-a| = -a = |a|$).

By the first fact, $|x| \geq |y|$ or $|x| < |y|$, and we argue by cases [$\vee$E].

Case $|x| \geq |y|$. Then $|x| - |y| \geq 0$, so $||x| - |y|| = |x| - |y|$. By Example (The triangle inequality by cases) applied to the numbers $x - y$ and y,

$$|x| = |(x - y) + y| \leq |x - y| + |y|,$$

so $|x| - |y| \leq |x - y|$, which is the goal in this case.

Case $|x| < |y|$. Then $|x| - |y| < 0$, so $||x| - |y|| = |y| - |x|$. By Example (The triangle inequality by cases) applied to $y - x$ and x,

$$|y| = |(y - x) + x| \le |y - x| + |x|,$$

so $|y| - |x| \le |y - x| = |-(x - y)| = |x - y|$.

Both cases reach the goal, which proves the inequality.

The two cases exhaust all possibilities because they are the two sides of the disjunction "$a \ge b$ or $a < b$" with $a = |x|$, $b = |y|$, and that disjunction holds for all real numbers. The split was chosen by the goal: the outer absolute value is of $|x| - |y|$, and its formula depends only on the sign of that difference.

A split on signs would decide $|x|$ and $|y|$ separately: $x \ge 0$ or $x < 0$ gives the formula for $|x|$, and inside each case $y \ge 0$ or $y < 0$ gives the formula for $|y|$. That is a case split inside each box of another, with $2 \cdot 2 = 4$ inner boxes, as in Example (The absolute value of a product). Fewer would not do for that method: with only the sign of x fixed, $|y|$ still has no formula.

Marking scheme. (10 points)

- 2: states the disjunction $|x| \ge |y|$ or $|x| < |y|$ and says it holds for all reals.
- 3: in the first case, removes the outer absolute value correctly and derives $|x| - |y| \le |x - y|$ from the triangle inequality for $(x - y) + y$.
- 3: in the second case, the same with the roles exchanged, including $|y - x| = |x - y|$.
- 2: explains the four sign cases as a split inside each case of another.

Fatal errors (0 for the step): using $||x| - |y|| = |x| - |y|$ in both cases; applying the triangle inequality to a difference as if $|u - v| \le |u| - |v|$. A proof that squares both sides instead, or splits on signs, earns the 6 points of the two middle lines if every case is complete.

Exercise. C2: Find the error

Here is a proof, by four cases, that $|x - y| \le |x| + |y|$ for all real numbers x and y.

> *Case 1:* $x \ge 0$. Then $|x| = x$.
>
> *Case 1a:* $y \ge 0$. Then $|y| = y$, and $|x - y|$ is $x - y$ or $y - x$. Since $x - y \le x \le x + y$ and $y - x \le y \le x + y$, we get $|x - y| \le x + y = |x| + |y|$.
>
> *Case 1b:* $y < 0$. Then $|y| = -y$, and $x - y > 0$, so $|x - y| = x - y = |x| + |y|$.
>
> *Case 2:* $x < 0$. Then $|x| = -x$.
>
> *Case 2a:* $y \ge 0$. Then $|y| = y$, and $x - y < 0$, so $|x - y| = y - x = |y| + |x|$.
>
> *Case 2b:* $y < 0$. Then $|y| = -y$. Since $x \ge 0$ and $y < 0$, we have $x - y > 0$, so $|x - y| = x - y \le |x| + |y|$, because $x \le |x|$ and $-y = |y|$.
>
> All four cases give $|x - y| \le |x| + |y|$.

(a) Is the statement true?

(b) Find the line that is not justified, and name the rule whose conditions it breaks.

(c) Show with specific numbers that the step fails.

(d) Repair Case 2b.

Solution.

(a) Yes. For instance, it follows from Example (The triangle inequality by cases) applied to x and $-y$, since $|-y| = |y|$ (shown in the solution of Exercise (C1: Challenge: The reverse triangle inequality)). So the question is not whether the conclusion holds but whether this argument proves it.

(b) The sentence "Since $x \geq 0$ and $y < 0$, we have $x - y > 0$" in Case 2b. The assumption $x \geq 0$ is the assumption of Case 1. In derivation terms, it opens the first box of the outer $\vee$E, and Case 2b lies inside the **second** box, which is opened by $x < 0$. Once the first box has closed, its lines are not available (the scope rule of Definition 1.4.2), so Case 2b may not cite $x \geq 0$: the rule broken is $\vee$E, each of whose boxes may cite only lines available at it by the scope rule: its own assumption, what has been derived inside it, and what was available before the box opened; so nothing from the first box, which has closed. The sentence is the reasoning of Case 1b copied into Case 2b, where its assumption is no longer in force. So the box of Case 2 does not reach the goal from the lines available to it, and the final $\vee$E has no right to cite it.

(c) Take $x = -3$ and $y = -1$, which is Case 2b. Then $x - y = -2$, which is not > 0, and $|x - y| = 2 \neq -2 = x - y$. The conclusion of Case 2b still holds here ($2 \leq 3 + 1$), but the step used to reach it is false.

(d) In Case 2b, $x < 0$ and $y < 0$, so $|x| = -x$ and $|y| = -y$. The number $|x - y|$ is $x - y$ or $y - x$. Since $x < 0$, we have $x - y < -y = |y| \leq |x| + |y|$. Since $y < 0$, we have $y - x < -x = |x| \leq |x| + |y|$. So in either case $|x - y| \leq |x| + |y|$, using only the assumptions of Case 2b.

Exercise. C3: Exit check: a distributive law, rule by rule

Derive $(P \wedge Q) \vee (R \wedge Q) \vdash (P \vee R) \wedge Q$, naming the rule and the cited lines on every line. Then say which steps would change if the premise were $(P \wedge Q) \vee (P \wedge R)$ and the goal $P \wedge (Q \vee R)$, and which would stay the same.

Solution. The goal is a conjunction, but neither of its sides is on a line, and the premise is a disjunction, so we split on the premise first; both cases end in the current goal $(P \vee R) \wedge Q$. In each case, $\wedge$E takes the assumption apart, $\vee$I builds $P \vee R$ from the side in hand, and $\wedge$I builds the goal.

1	$(P \wedge Q) \vee (R \wedge Q)$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
5	$\quad P \vee R$	$\vee$I 3
6	$\quad (P \vee R) \wedge Q$	$\wedge$I 5, 4
7	$\quad R \wedge Q$	
8	$\quad R$	$\wedge$E 7
9	$\quad Q$	$\wedge$E 7
10	$\quad P \vee R$	$\vee$I 8
11	$\quad (P \vee R) \wedge Q$	$\wedge$I 10, 9
12	$(P \vee R) \wedge Q$	$\vee$E 1, 2–6, 7–11

Line by line: 1 is the premise; 2 is the assumption of the first case; 3 and 4 are $\wedge$E on line 2; 5 is $\vee$I on line 3, first schema; 6 is $\wedge$I on lines 5 and 4, written with $P \vee R$ as the left conjunct; 7 is the assumption of the second case; 8 and 9 are $\wedge$E on line 7; 10 is $\vee$I on line 8, second schema; 11 is $\wedge$I on lines 10 and 9; 12 is $\vee$E, citing the disjunction on line 1 and the boxes 2–6 and 7–11, both of which end in the goal as lines of their own. No line of the second box cites the first.

With premise $(P \wedge Q) \vee (P \wedge R)$ and goal $P \wedge (Q \vee R)$, the shape stays the same: a split on the premise, $\wedge$E twice in each box, one $\vee$I and one $\wedge$I in each box, and $\vee$E at the end. What changes is which part is weakened and where it goes. The common conjunct is now P, and the parts that differ between the cases are Q and R; so the $\vee$I lines build $Q \vee R$ (from Q in the first box, first schema; from R in the second, second schema), and the $\wedge$I lines write P as the left conjunct, as in the goal. That derivation is the second one of Proposition 1.6.1.

Marking scheme. (10 points)

- 2: splits on the premise by $\vee$E, with two boxes opened by its two sides.
- 2: in each box, takes the assumption apart by $\wedge$E into both conjuncts.
- 2: in each box, uses $\vee$I to build $P \vee R$ from the side in hand, with the correct schema.
- 2: in each box, builds the goal by $\wedge$I, so that both boxes end in the same line, the goal.
- 2: the $\vee$E line cites the disjunction and both boxes as ranges, and the final sentences name the changed steps (the $\vee$I lines, now building $Q \vee R$, and the $\wedge$I lines, now with P on the left).

Fatal errors (0 for the step): a line in one box citing a line of the other box; a conclusion drawn inside one box only and then used outside it.

A second route is also correct: split the premise twice, once ending both boxes in $P \vee R$ and once ending both in Q, and then join the two results by $\wedge$I outside every box (14 lines). Its points map as follows: 2 for either split with two boxes opened by the premise's sides; 2 for the $\wedge$E lines in the boxes; 2 for the $\vee$I sides; 2 for the final $\wedge$I, with the boxes of each split ending in the same line; 2 for the citations and the final sentences.

Exercise. C4: The second half

A search for $Q \vee R,\ Q \to (P \to S),\ R \to S \vdash P \to S$ has written the premises and opened one box; the rest is blank. The finished derivation has 11 lines. *Name the move, then complete it.*

1	$Q \vee R$	
2	$Q \to (P \to S)$	
3	$R \to S$	
4	P	
	⋮	

Solution (Hint 1). Inside the box, what is the current goal, and which available line is a disjunction?

Solution (Hint 2). Line 5 is Q, the assumption of a first case box opened inside the box of P; like the second case box, it must end in the current goal S.

Solution. The moves are two. First, the box of line 4 was read off the goal: $P \to S$ is an implication, so the box opened with P, and its current goal is S; it will be closed by $\to$I. Second, inside it, the available disjunction $Q \vee R$ (line 1, in no box, so available inside the box) is split, and **both case boxes end in the current goal** S. In the case Q, line 2 fires to give $P \to S$, which fires on line 4; in the case R, line 3 fires.

1	$Q \vee R$	
2	$Q \to (P \to S)$	
3	$R \to S$	
4	P	
5	Q	
6	$P \to S$	$\to$E 2, 5
7	S	$\to$E 6, 4
8	R	
9	S	$\to$E 3, 8
10	S	$\vee$E 1, 5–7, 8–9
11	$P \to S$	$\to$I 4–10

Line 7 cites line 4, the assumption of the outer box, which contains line 7. Line 10 is a line of the box 4–10 itself (the case boxes have closed), so $\to$I may cite that box at line 11. Splitting

outside the box instead, with both case boxes ending in $P \to S$, is also a correct derivation of the sequent; it does not complete the lines given, because it would split before line 4, not after.

1.6.7.4 R. Review

Exercise. R1

Let f be a function and $[a, b]$ an interval. Take the atomic statements C: "f is continuous on $[a, b]$", B: "f is bounded on $[a, b]$" and I: "f is integrable on $[a, b]$"; you need not know what the words mean. Translate the argument "If f is continuous on $[a, b]$ and bounded there, then f is integrable on $[a, b]$. f is bounded on $[a, b]$. f is continuous on $[a, b]$. So f is integrable on $[a, b]$ and bounded there." into a sequent, and derive it.

Solution. The sequent is $(C \wedge B) \to I,\ B,\ C \vdash I \wedge B$: three premises and a conclusion, as in Definition 1.2.4. The goal is a conjunction, so it splits into the subgoals I and B. B is a premise. The only line that mentions I is the first premise, whose hypothesis $C \wedge B$ is built by $\wedge$I (Definition 1.2.5) from two premises; then $\to$E (Definition 1.3.1) gives I.

1	$(C \wedge B) \to I$	
2	B	
3	C	
4	$C \wedge B$	$\wedge$I 3, 2
5	I	$\to$E 1, 4
6	$I \wedge B$	$\wedge$I 5, 2

Line 4 writes $C \wedge B$, with C on the left as in the hypothesis of line 1 ($\wedge$I may cite 3, 2 or 2, 3); the hypothesis must match exactly. Line 2 is cited twice, at lines 4 and 6: a line is not used up by being cited.

Exercise. R2

Show that $(P \to Q) \wedge (Q \to R),\ P \wedge S \vdash R \wedge S$, and say in one sentence which step you found by working backwards from the goal.

Solution. Premise-directed, both premises are conjunctions and are taken apart by $\wedge$E (Definition 1.2.5); goal-directed, the goal $R \wedge S$ splits into R and S, and S is already in hand. For R, work backwards as in Example (Chains of implications): $Q \to R$ is available, so Q becomes the subgoal; $P \to Q$ is available, so P does; and P is in hand.

1	$(P \to Q) \wedge (Q \to R)$	
2	$P \wedge S$	
3	$P \to Q$	$\wedge$E 1
4	$Q \to R$	$\wedge$E 1
5	P	$\wedge$E 2
6	S	$\wedge$E 2
7	Q	$\to$E 3, 5
8	R	$\to$E 4, 7
9	$R \wedge S$	$\wedge$I 8, 6

The chain of lines 7 and 8 was found backwards, from the subgoal R to Q to P, and written forwards.

Exercise. R3

Is $\vdash (P \wedge (P \to Q)) \to (Q \vee R)$? Justify your answer.

Solution. Yes. The goal's main connective is the outer $\to$, so the move is to open a box with its hypothesis $P \wedge (P \to Q)$ (Definition 1.4.1). Inside, the current goal is $Q \vee R$, which we postpone. The assumption is a conjunction, so $\wedge$E gives P and $P \to Q$; $\to$E gives Q; now the left side of the goal is in hand, and $\vee$I (Definition 1.5.1) takes it.

1	$P \wedge (P \to Q)$	
2	P	$\wedge$E 1
3	$P \to Q$	$\wedge$E 1
4	Q	$\to$E 3, 2
5	$Q \vee R$	$\vee$I 4
6	$(P \wedge (P \to Q)) \to (Q \vee R)$	$\to$I 1–5

The derivation has no premises, so it shows $\vdash (P \wedge (P \to Q)) \to (Q \vee R)$. The look-alike start fails: the statement contains $\vee$, and one might begin with $\vee$I on the whole goal. But the goal's main connective is $\to$, not $\vee$, so $\vee$I cannot produce it; and even inside the box, choosing the side R of $Q \vee R$ would have left the subgoal R, which nothing supplies; the side Q is the one lines 2–4 supply.

Chapter 2

Negation, Classical Logic and Finding Derivations

Chapter 1 could prove that things are so. It could not prove that something is **not** so. No rule of Chapter 1 shows that no rational number has square 2, or that $x^2 + 1$ is never 0; none lets a proof by cases say "this case cannot occur" and move on; and the sentence "if $x^2 < 0$, then $x = 7$", whose hypothesis never holds and whose conclusion has nothing to do with it, Chapter 1 read only informally, as a promise not broken: no rule yet derives it. All three are questions about negation and contradiction, and ordinary proofs are full of them: "suppose not", "which is absurd", "we prove the contrapositive".

This chapter adds the rules for "not" and for contradiction one at a time, first the rules that use a negation, then the rule that proves one. It shows that a sequent, once proved, can be used as a rule of its own; gives "if and only if" its rules, and with them what a definition's "if" means; and then meets the one rule that is different in kind, proof by contradiction, together with the reason why it, double-negation elimination, and the law of excluded middle stand or fall together. It ends with a method for **finding** derivations, which collects the moves the chapter has met one section at a time, and with a question the method cannot answer: we show that something is derivable by exhibiting a derivation, but how could we ever show that something is **not** derivable? Chapter 3 answers it (Corollary 3.5.3).

What you need. All of Chapter 1: the rules for "and", "implies", and "or", and above all the definition of a derivation, with its boxes and its scope rule (Definition 1.4.2). School algebra about integers and real numbers is used in examples, as facts.

Time. About 30 hours on the core track (all of A, one B and one C per section, the exit checks); about 45 hours for everything.

2.0.1 Roadmap

- **Contradiction: using a negation.** The rules ¬E and ⊥E: what a clash is, what it buys, why an impossible case closes itself, and why an implication with an impossible hypothesis is true.
- **Proving a negation.** The rule ¬I, assume and reach a contradiction, and with it the proof that $\sqrt{2}$ is irrational.
- **Derived rules.** A sequent, once proved, may be used as a single line; modus tollens, the contrapositive, and disjunctive syllogism become rules.
- **Iff, equivalence, and what a definition says.** The rules for ↔, provable equivalence,

chains of equivalences, and why a definition's "if" means "if and only if".
- **Proof by contradiction and the classical rules.** The rule RAA, the classical laws it gives (double negation, excluded middle, De Morgan, Peirce), and why three of them are one principle.
- **How to find a derivation.** The search method, assembled from the moves of the chapter, with worked searches, one search that fails, and all the rules in one place.

2.0.2 Named moves

- **Goal $\bot$: find a clash.** Look for φ and $\neg\varphi$ among the available lines, or make φ a subgoal for some $\neg\varphi$ you have.
- **An impossible case closes itself.** In a proof by cases, a case that reaches $\bot$ ends by $\bot$E on the current goal.
- **To prove "not φ", assume φ.** The box opens with the unnegated statement and ends on $\bot$.
- **Use a derived rule as one line.** A proved sequent is a new rule.
- **Two boxes for an iff.** One box for each direction, each ending on the other side.
- **When stuck, assume the negation of the goal.** Proof by contradiction, and then hunt for a clash.
- **Work from both ends.** The full method: read the skeleton off the goal, use the premises by their main connectives, and keep the classical step for last.

2.1 Contradiction: Using a Negation

Chapter 1 gave rules for "and", "implies", and "or", and not one of them mentions "not". Yet written proofs are full of sentences such as "this is impossible", "this case cannot occur", and "which contradicts what we assumed". This section gives the two rules that **use** a negation: the first says exactly what a clash between two lines is, and the second says what a clash buys. The rule that **proves** a negation is the subject of the next section.

2.1.1 A case that cannot occur

Here is Argument A of Section 1.1, told as a proof by cases. Let n be an integer whose square is even. Every integer is even or odd, so there are two cases. If n is even, we are done. If n is odd, then n^2 is odd; but n^2 is even, and no integer is both, so this case cannot occur. Hence n is even.

Chapter 1 has a rule for a proof by cases: $\vee$E of Definition 1.6.1. It asks for a disjunction and two boxes, one opened by each disjunct, and **both boxes must end on the same line**, the goal. In the argument above the goal is "n is even". The first box reaches it at once. The second box does not reach it at all. It reaches "impossible", and then simply declares itself finished.

Two things are missing, and no rule of Chapter 1 supplies them.

- A way to **record** the impossibility: the lines "n^2 is even" and "n^2 is odd", together with "no integer is both", should produce a line that says "contradiction".
- A way to **close** the impossible box with the goal: from the line "contradiction", the box should be allowed to write "n is even", so that both boxes end on the same line and $\vee$E applies.

Section 1.1 introduced the symbols for both. The connective $\neg$ forms the **negation** $\neg\varphi$, read "not φ". The constant $\bot$, read "contradiction", is a statement that is false however things stand.

There they were symbols only. The two rules below give them their first uses.

2.1.2 The rules for a clash

The rule for recording a clash has a simple content. The statement $\neg\varphi$ says that φ fails, so a derivation that has both φ and $\neg\varphi$ available is working in a situation that cannot occur. The rule writes this down as the line $\bot$. A second rule then says what such a line is good for.

A statement and its negation clash; from a clash, anything follows.

Definition 2.1.1. Rules for Contradiction

$$\begin{array}{l|ll} i & \varphi & \\ j & \neg\varphi & \\ k & \bot & \neg\mathrm{E}\ i,\ j \end{array} \qquad\qquad \begin{array}{l|ll} i & \bot & \\ k & \varphi & \bot\mathrm{E}\ i \end{array}$$

The left schema is ¬E (**not-elimination**): if φ and $\neg\varphi$ stand on lines i and j available at line k, write $\bot$, citing both, the unnegated line first. The right schema is $\bot$E (**contradiction-elimination**, or *ex falso*): if $\bot$ stands on a line i available at line k, write **any** statement φ, citing line i.

As with every rule of Chapter 1, the letters φ and i, j, k follow Definition 1.2.1: φ stands for one statement, the same one wherever it occurs in the schema, and i, j, k for line numbers. Let us read the two schemas clause by clause.

¬E, the given lines. There are two, and they must be **exactly** φ and $\neg\varphi$: the second is the first with one ¬ written in front of the whole of it, symbol for symbol. It is not enough that the second line "contradicts the first in spirit". The lines $x < 1$ and $x > 1$ cannot both hold, but neither is the negation of the other as written, so they are not an instance of the schema. (The warning below shows how to turn them into one.)

¬E, availability. Both lines must be available at line k, in the sense of the scope rule of Definition 1.4.2: each is earlier than k, and not inside a box that has already closed. They may stand in either order on the page. The citation fixes their roles: "¬E i, j" says that line i is φ and line j is $\neg\varphi$.

¬E, the new line. It is $\bot$ and nothing else. The rule does not write the goal, or $\neg\varphi$ again, or anything else; it records the clash and stops. What to do with the clash is the business of the second rule. Splitting the work this way keeps each rule to one step, and it lets the next section aim at the line $\bot$ as a goal in its own right.

$\bot$E, the given line. One line, $\bot$, available at line k.

$\bot$E, the new line. Any statement at all, chosen by the writer. Like the second disjunct in ∨I (Definition 1.5.1), the statement φ occurs in no given line: it need not be related to anything above, and it may itself be a negation, or $\bot$ again.

Two more observations complete the reading. First, $\bot$E is the **elimination** rule of $\bot$: it says how a line $\bot$ is used. There is no introduction rule of its own for $\bot$. Other rules can pass a $\bot$ along ($\bot$E may write it again, and →E writes it from $\varphi \to \bot$ and φ), but each of them needs $\bot$ already written in a line it uses. Apart from being assumed, the only way a line $\bot$ first appears is by ¬E: it is the one rule whose new line is always $\bot$, from lines in which $\bot$ need not occur.

So ¬E plays two parts at once: it is the way a negation is used, and it is the way ⊥ is produced.

Second, why is ⊥E honest? A rule should never lead from lines that hold to a line that fails. A line ⊥ can stand only in a part of a derivation where the premises and the open assumptions cannot all hold together, at least when the ⊥ was written by ¬E, from some φ and $\neg\varphi$ that were both available there; that the rules which only pass a ⊥ along keep this property is part of what Chapter 3 checks in proving Theorem 3.5.2. In such a place there is no situation in which the conclusion of ⊥E would have to be checked, so it cannot be caught failing. Chapter 3's soundness theorem turns this remark into a proof.

Quick check. Which of these pairs of lines are, as written, an instance of the given lines of ¬E, and in which order must each be cited? The pairs are (i) $Q \wedge R$ and $\neg(Q \wedge R)$; (ii) $\neg P$ and $\neg\neg P$; (iii) $P \to Q$ and $\neg P \to Q$.

Solution.

(i) Yes, with $\varphi = Q \wedge R$; cite the line $Q \wedge R$ first. (ii) Yes, with $\varphi = \neg P$: the second line is $\neg(\neg P)$. Cite $\neg P$ first, then $\neg\neg P$. (iii) No. The main connective of $\neg P \to Q$ is $\to$ (Definition 1.1.2), so it is not a negation at all; the $\neg$ applies to P alone. The negation of $P \to Q$ would be $\neg(P \to Q)$.

Clashes in school mathematics. In derivations about numbers, the clash usually comes from a school fact written as a negation. For real numbers a and b, exactly one of $a < b$ and $a \geq b$ holds, so $a \geq b$ says the same as $\neg(a < b)$, and we use them interchangeably. So a line $x^2 < 0$ and the school fact $x^2 \geq 0$, written as $\neg(x^2 < 0)$, give ⊥ by ¬E. The symbol $a \neq b$ **means** $\neg(a = b)$, by definition. And the fact "no integer is both even and odd" is a negation of a conjunction, $\neg(E \wedge O)$ with E for "n is even" and O for "n is odd". It clashes only with a line $E \wedge O$, which ∧I builds from separate lines E and O.

Warning. A clash must have the exact shape φ, $\neg\varphi$. The lines $x < 1$ and $x > 1$ are not yet a clash. To make them one, write $x > 1$ as $\neg(x \leq 1)$ (for real numbers, $x > 1$ holds exactly when $x \leq 1$ fails), and get $x \leq 1$ from $x < 1$ by ∨I, since $x \leq 1$ means $(x < 1) \vee (x = 1)$. And some pairs that look opposed do not clash at all: $x \geq 1$ and $x \leq 1$ both hold at $x = 1$.

Warning. ⊥E does not run backwards. From ⊥ we may write φ, or $\neg\varphi$, or anything; but from φ alone no rule writes ⊥. If it could, the true line $x = 1$ would give ⊥, and then ⊥E would give the false line $x = 2$.

2.1.3 First derivations with a clash

The five derivations below grow in difficulty. Each Idea says why each rule was chosen, in the two directions Chapter 1 used: from the goal (what does the goal's main connective ask for?) and from the premises (what does an available line's main connective let us do?). This section adds one question to each list. From the goal: *when nothing produces the goal, can a clash produce it?* From the premises: *an available negation $\neg\varphi$ is used in exactly one way, by* ¬E*, so it asks for φ.*

Example. The first clashes

Show that (a) $\bot \vdash P$, and (b) $P \wedge Q,\ \neg Q \vdash R$.

Idea.

(a) is the rule $\bot$E itself, the degenerate case with no clash to find. In (b), the first premise is a conjunction, so the premise-directed move takes it apart by $\wedge$E. Its half Q and the premise $\neg Q$ are exactly a pair $\varphi, \neg\varphi$. The goal R has nothing to do with the premises: no rule builds it from them. Only $\bot$E can write it, and $\bot$E needs a line $\bot$, which the clash supplies.

Solution.

(a) Two lines: the premise, and one application of $\bot$E.

1	$\bot$	
2	P	$\bot$E 1

(b) Five lines.

1	$P \wedge Q$	
2	$\neg Q$	
3	Q	$\wedge$E 1
4	$\bot$	$\neg$E 3, 2
5	R	$\bot$E 4

Line 4 cites line 3 first because line 3 plays φ and line 2 plays $\neg\varphi$, with $\varphi = Q$. The half P of the first premise is never used; a derivation need not use every premise (Definition 1.2.3).

The next result is the pattern of (b) with nothing extra around it. We state it as a proposition, because later sections cite it.

Proposition 2.1.1. Contradictory Premises Prove Anything

$P,\ \neg P \vdash Q$.

Idea. The premises are a clash as they stand, so $\neg$E writes $\bot$ at once, and $\bot$E writes the goal, which nothing else could produce.

Proof. The derivation

1	P	
2	$\neg P$	
3	$\bot$	$\neg$E 1, 2
4	Q	$\bot$E 3

has premises P and $\neg P$ and last line Q. This proves the proposition.

■

In the next derivation the clash happens inside a box, and the rule $\bot$E writes the box's goal.

Proposition 2.1.2. Vacuous Implication

$\neg P \vdash P \to Q$.

Idea. The goal is an implication, so the goal-directed move of Chapter 1 ($\to$I, Definition 1.4.1) opens a box with its hypothesis P; the new goal is Q. Nothing available mentions Q, so no rule can build it. But inside the box, the assumption P and the premise $\neg P$ clash, and $\bot$E writes the box's goal Q. This is the section's goal-directed move: *when nothing produces the goal, look for a clash.*

Proof. The derivation

1	$\neg P$	
2	$\quad P$	
3	$\quad \bot$	$\neg$E 2, 1
4	$\quad Q$	$\bot$E 3
5	$P \to Q$	$\to$I 2–4

has premise $\neg P$ and last line $P \to Q$. Line 3 may cite line 1, which lies in no box. This proves the proposition.

■

Line 4 is the line that makes the box close properly: $\to$I needs the box to end on Q, as a line of that box itself, and $\bot$E put it there.

In the next example no clash is lying about. It has to be made.

Example. Manufacturing the other half of a clash

Show that $\neg(P \vee Q) \vdash P \to R$.

Idea. As before, the goal $P \to R$ opens a box on P, and the new goal R occurs nowhere else, so a clash must write it. Inside the box there is no $\neg P$ to clash with. The only negation available is the premise $\neg(P \vee Q)$, and the premise-directed question says what it asks for: the line $P \vee Q$. That line is one $\vee$I away from the assumption P. So the clash is **manufactured**: the unnegated half of the premise becomes a subgoal, and $\vee$I is chosen not because the goal is a disjunction but because the premise's clash needs one.

Solution. Six lines.

1	$\neg(P \vee Q)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\quad R$	$\bot$E 4
6	$P \to R$	$\to$I 2–5

Line 4 cites line 3 as φ and line 1 as $\neg\varphi$, with $\varphi = P \vee Q$. In words: if neither P nor Q holds, then P cannot hold, so an implication with hypothesis P is true whatever its conclusion. The mirror image, with Q as the hypothesis, is exercise B1.

The last derivation of this section is the pattern of the opening argument: a case split in which one case is impossible.

Example. An impossible case

Show that $P \vee Q,\ P \to R,\ \neg R \vdash Q$.

Idea. The goal Q is a letter, so the goal gives no introduction rule to aim at. The premises do: $P \vee Q$ is a disjunction, and Chapter 1's premise-directed move for a disjunction is $\vee$E, with two boxes that both end in the current goal Q. The box of Q ends as soon as it opens: its assumption is already the goal. The box of P cannot reach Q honestly, because nothing in it produces Q. So we look for a clash: P feeds the premise $P \to R$, and R clashes with $\neg R$. The box ends on Q by $\bot$E. This is the second new move: *an impossible case closes itself.*

Solution. Nine lines.

1	$P \vee Q$	
2	$P \to R$	
3	$\neg R$	
4	$\quad P$	
5	$\quad R$	$\to$E 2, 4
6	$\quad \bot$	$\neg$E 5, 3
7	$\quad Q$	$\bot$E 6
8	$\quad Q$	
9	Q	$\vee$E 1, 4–7, 8–8

The second box consists of its assumption alone, a one-line box, just as in the derivation of $P \vee P \vdash P$ in Example (First derivations with "or"): its last line is its first line, and that line is Q, which is what $\vee$E needs. The two boxes are separate, so line 7 could not cite line 8, and nothing in the second box could cite lines 4–7.

Look at line 7 of this derivation again. The box of P is a case "that cannot occur", and line 7 is the sentence "so there is nothing to prove": the case is closed with whatever the other case reaches. Every proof by cases with an impossible case has this line somewhere, usually unwritten.

Warning. The impossible case must be impossible **from what is available in its own box**. A clash with a line of a sibling box, or of a box that has already closed, is a scope error (Definition 1.4.2), even when the clash "looks right". In Example (An impossible case) the clash in the box of P uses line 5, from that box, and the premise on line 3; neither lives in another box. Exercise C2 shows the error and what it can prove.

The moves of this section, in one place. **Goal $\bot$: find a clash.** Look for φ and $\neg\varphi$ among the available lines; if only $\neg\varphi$ is there, make φ a subgoal. **An impossible case closes itself.** In a case split, a case that reaches $\bot$ ends by $\bot$E on the current goal. The next section adds a way to reach the goal $\bot$ deliberately, by assuming something.

2.1.4 How the rules look in a mathematical proof

We can now prove the fact of Argument A in Section 1.1, with every rule named. Three school facts about integers enter as premises. An integer n is **even** if $n = 2k$ for some integer k, and **odd** if $n = 2k + 1$ for some integer k. And:

(E1) every integer is even or odd;

(E2) no integer is both even and odd;

(E3) the square of an odd integer is odd.

The third is a computation. If $n = 2k + 1$ for an integer k, then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

and $2k^2 + 2k$ is an integer, so n^2 is odd. Taking "an integer k with $n = 2k + 1$" and working with it is a step about "there exists"; the rules for such steps come in Chapter 4, and here we use the step as school algebra.

Theorem 2.1.3. Even Square, Even Number

Let n be an integer. If n^2 is even, then n is even.

Idea. The hypothesis "n^2 is even" hands us nothing to compute with: it says $n^2 = 2j$ for some integer j, and no algebra turns that into a statement about n. The word "odd" does hand us something, since $n = 2k + 1$ can be squared. So we split on (E1) and show that the odd case is impossible: there (E3) makes n^2 odd, which clashes with the hypothesis through (E2). The impossible case then closes itself.

Proof. Suppose n^2 is even [assumption for $\to$I]. By (E1), n is even or n is odd, and we argue by cases [$\vee$E].

Case n even. This is the goal.

Case n odd. By (E3), n^2 is odd [$\to$E]. So n^2 is both even and odd [$\wedge$I]; this contradicts (E2) [$\neg$E]. So this case cannot occur, and in it n is even [$\bot$E].

In both cases n is even [$\vee$E]. This proves that if n^2 is even, then n is even [$\to$I].

■

The logic of the proof fits in a derivation. Write E for "n is even", O for "n is odd", S for "n^2 is even", and T for "n^2 is odd". Then (E1) for n is $E \vee O$, (E2) for n^2 is $\neg(S \wedge T)$, and (E3) for n is $O \to T$; these are the premises.

1	$E \vee O$	
2	$\neg(S \wedge T)$	
3	$O \to T$	
4	$\quad S$	
5	$\quad\quad E$	
6	$\quad\quad O$	
7	$\quad\quad T$	$\to$E 3, 6
8	$\quad\quad S \wedge T$	$\wedge$I 4, 7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad\quad E$	$\bot$E 9
11	$\quad E$	$\vee$E 1, 5–5, 6–10
12	$S \to E$	$\to$I 4–11

Each line is one sentence of the proof.

Line	Sentence of the proof
1–3	(E1) for n, (E2) for n^2, (E3) for n
4	Suppose n^2 is even.
5	Case n even. (A one-line box: it is the goal.)
6	Case n odd.
7	By (E3), n^2 is odd.
8	So n^2 is both even and odd,
9	which contradicts (E2).
10	So this case cannot occur, and in it n is even.
11	In both cases n is even.
12	This proves that if n^2 is even, then n is even.

Line 8 cites line 4, the assumption of the outer box; it may, since the box 4–11 contains line 8 (Definition 1.4.2). Two steps of the proof are not in the derivation: passing from the facts (E1)–(E3), which are about every integer, to their instances for this n and this n^2; and naming the k of (E3). Both are quantifier steps, and Chapter 4 gives their rules.

Quick check. Which sentence of the proof of Theorem 2.1.3 is the rule $\bot$E, and what does it write?

Solution. The sentence "So this case cannot occur, and in it n is even", line 10 of the derivation. It cites the line $\bot$ (line 9) and writes E, "n is even", which is the goal of the case, so that the box of O ends on the same line as the box of E.

Every rule in this proof is a primitive rule: $\to$I, $\vee$E, $\to$E, $\wedge$I, $\neg$E, and $\bot$E. The next section uses the theorem twice, to prove that no rational number has square 2. Section 2.3 reads the same proof again with shorter steps, as mathematicians usually write it, "by the contrapositive".

2.1.5 Vacuous truth

We can now settle the question that makes many readers distrust logic. Is the sentence

if $x^2 < 0$, then $x = 7$

true, for a real number x? The hypothesis never holds, and the conclusion has nothing to do with it. Everyday intuition is unsure what to say. The rules are not unsure.

Example. An implication with an impossible hypothesis

Let x be a real number. Show that if $x^2 < 0$, then $x = 7$.

Idea. The goal is an implication, so open a box with $x^2 < 0$, aiming at $x = 7$. Nothing produces $x = 7$, so look for a clash: the school fact $x^2 \geq 0$ is $\neg(x^2 < 0)$, which clashes with the assumption. This is exactly the derivation of Proposition 2.1.2, with P read as "$x^2 < 0$"

and Q as "$x = 7$".

Solution. With the school fact as the premise:

1	$\neg(x^2 < 0)$	
2	$\quad x^2 < 0$	
3	$\quad \bot$	$\neg$E 2, 1
4	$\quad x = 7$	$\bot$E 3
5	$x^2 < 0 \to x = 7$	$\to$I 2–4

In prose: *suppose $x^2 < 0$; this contradicts $x^2 \geq 0$; so this case cannot occur, and in it $x = 7$; hence if $x^2 < 0$, then $x = 7$.*

We wrote out all five lines rather than citing Proposition 2.1.2 as a single step. Using an already proved sequent as one line of a new derivation is a genuine new permission, and Section 2.3 grants it; until then, a derivation repeats the steps.

An implication whose hypothesis cannot hold is called **vacuously true**. Nothing about it was decided by convention. It is forced by two rules that each look harmless: $\to$I, which proves an implication by assuming its hypothesis, and $\bot$E, which closes an impossible case.

Everyday language agrees. A friend says: "if you finish the exercises by Friday, I will buy you lunch". You do not finish them. Whatever your friend does on Friday, the promise has not been broken. *A promise never tested is never broken.*

Vacuous truth is not a curiosity; it is what makes case splits work. In Example (An impossible case), line 7 closed the impossible case by $\bot$E, and line 10 of the derivation of Theorem 2.1.3 did the same. When Chapter 3 gives each connective a truth table (Definition 3.3.2), the rows of $\to$ with a false hypothesis are the ones these two rules force. Chapter 4 meets the same phenomenon in another dress: a statement about "every element of the empty set" is true, because there is no element to test.

Warning. "True" is not "useful". The statements "if $x^2 < 0$, then $x = 7$" and "if $x^2 < 0$, then $x \neq 7$" are both true, and neither says anything about x or about 7. Likewise "if n is an even prime greater than 2, then $n = 17$" is true and empty. When a hypothesis turns out to be impossible, suspect the hypothesis; do not believe the conclusion.

2.1.6 What the rules sound like

Here are the rows of the rule, phrase and Lean table for this section's rules. The last column is for readers who use a proof assistant; nothing depends on it.

Rule	You write	Lean
$\neg$E	"…, which contradicts …"	`exact hn hp` (goal `False`)

Rule	You write	Lean
$\bot$E	"This case cannot occur." "So there is nothing to prove."	`exfalso`

In Lean the two steps are often taken together: `absurd hp hn` closes any goal from a proof `hp` of P and a proof `hn` of $\neg P$, which is $\neg$E followed by $\bot$E, and the tactic `contradiction` finds such a pair by itself.

2.1.7 Exercises

2.1.7.1 A. Check your understanding

Exercise. A1

(a) Write the schemas of $\neg$E and $\bot$E from memory.

(b) Match each sentence, taken from proofs, with the rule it uses, $\neg$E or $\bot$E: "But n was assumed odd, and we have just shown it is even"; "In this case $x^2 < 0$, which is impossible, so there is nothing to prove".

(c) True or false, with a reason: $\bot$E lets us write $\bot$ from any line.

(d) In Example (An impossible case), why is the second box only one line long?

(e) In the derivation after Theorem 2.1.3, why does line 8 need line 4, and why may it cite it?

Solution.

(a) $\neg$E: from a line i with φ and a line j with $\neg\varphi$, both available, write $\bot$ at line k, citing i, j. $\bot$E: from an available line i with $\bot$, write any statement φ at line k, citing i. The two schemas are those of Definition 2.1.1.

(b) The first is $\neg$E: "odd" and "even" clash through (E2), and the sentence records the clash. The second is $\bot$E: "impossible" records a clash (an $\neg$E against $x^2 \geq 0$, left unsaid), and "there is nothing to prove" closes the case with whatever the case needs.

(c) False. $\bot$E runs from $\bot$ to any statement, not from any statement to $\bot$. $\bot$E needs a line $\bot$ to start from. The rule that records a clash as $\bot$ is $\neg$E, and it needs two lines φ and $\neg\varphi$.

(d) The box is opened by Q, and Q is already the goal of both boxes. So the box's first line is also its last line, which is all $\vee$E asks of it.

(e) Line 8 is $S \wedge T$, built by $\wedge$I from S and T; line 4 is the only line that says S. It may cite line 4 because the only box containing line 4, the box 4–11, also contains line 8 (the scope rule of Definition 1.4.2).

2.1.7.2 B. Practice

Exercise. B1: Two clashes

(a) Show that $\neg(P \vee Q) \vdash Q \to R$.
(b) Show that $(P \to Q) \wedge \neg Q,\ P \vdash \neg R$, and say why $\bot$E may write a negation.

Solution.

(a) As in Example (Manufacturing the other half of a clash), with Q in place of P: the box of Q manufactures $P \vee Q$ by $\vee$I on the other side.

1	$\neg(P \vee Q)$	
2	$\quad Q$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\quad R$	$\bot$E 4
6	$Q \to R$	$\to$I 2–5

(b) Take the first premise apart, fire the implication on the second premise, and clash.

1	$(P \to Q) \wedge \neg Q$	
2	P	
3	$P \to Q$	$\wedge$E 1
4	$\neg Q$	$\wedge$E 1
5	Q	$\to$E 3, 2
6	$\bot$	$\neg$E 5, 4
7	$\neg R$	$\bot$E 6

Line 7 is an instance of $\bot$E with $\varphi = \neg R$: the schema allows **any** statement in place of φ, and a negation is a statement.

Exercise. B2: An even successor

Let n be an integer. Prove that if $n+1$ is even, then n is odd, by cases on (E1) with one case shown impossible. Tag each step with its rule, as in the proof of Theorem 2.1.3.

Solution. Suppose $n+1$ is even [assumption for $\to$I]. By (E1), n is even or n is odd, and we argue by cases [$\vee$E].

Case n even. Write $n = 2k$ with k an integer. Then $n+1 = 2k+1$, so $n+1$ is odd, by the definition of odd. So $n+1$ is both even and odd [$\wedge$I]; this contradicts (E2) [$\neg$E]. So this case cannot occur, and in it n is odd [$\bot$E].

Case n odd. This is the goal.

In both cases n is odd [$\vee$E]. This proves that if $n+1$ is even, then n is odd [$\to$I].

The step "n even, so $n+1$ odd" plays the part that (E3) played in the theorem: it is the

implication fired by →E inside the impossible case. Here the impossible case is the **first** one, and the other case is the goal.

Exercise. B3: Two vacuous promises

Let x be a real number. Prove that if $x^2 + 1 \leq 0$, then $x > 100$; and that if $x^2 + 1 \leq 0$, then $x < -100$. Write each as a derivation with one premise, a school fact, and explain why the two statements together are no paradox.

Solution. The school fact is $x^2+1 > 0$, true for every real x since $x^2 \geq 0$ gives $x^2+1 \geq 1 > 0$. For real numbers, $x^2 + 1 > 0$ says the same as $\neg(x^2 + 1 \leq 0)$. The first derivation is the pattern of Example (An implication with an impossible hypothesis):

1	$\neg(x^2 + 1 \leq 0)$	
2	$\quad x^2 + 1 \leq 0$	
3	$\quad \bot$	¬E 2, 1
4	$\quad x > 100$	⊥E 3
5	$x^2 + 1 \leq 0 \to x > 100$	→I 2–4

The second is the same, with $x < -100$ on lines 4 and 5: ⊥E may write any statement.

No real number is both greater than 100 and less than -100. That is no paradox, because neither implication ever applies: its hypothesis holds for no real x. Both are vacuously true, and both say nothing about x, as the warning on vacuous truth says.

2.1.7.3 C. Going deeper

Exercise. C1: A disjunct that cannot hold

(a) Show that $P \vee (Q \wedge \neg Q) \vdash P$.

(b) What does (a) say about the statement "$x > 0$, or $x = 1$ and $x \neq 1$", for a real number x?

Solution.

(a) The premise is a disjunction, so split it by ∨E, with both boxes ending in the goal P. The box of P is a one-line box. The box of $Q \wedge \neg Q$ is impossible: ∧E gives a clash, and ⊥E closes the case.

1	$P \vee (Q \wedge \neg Q)$	
2	$\quad P$	
3	$\quad Q \wedge \neg Q$	
4	$\quad Q$	$\wedge$E 3
5	$\quad \neg Q$	$\wedge$E 3
6	$\quad \bot$	$\neg$E 4, 5
7	$\quad P$	$\bot$E 6
8	P	$\vee$E 1, 2–2, 3–7

(b) With P: "$x > 0$" and Q: "$x = 1$", the statement is $P \vee (Q \wedge \neg Q)$, since $x \neq 1$ is $\neg(x = 1)$. By (a), it gives $x > 0$. So the second disjunct adds nothing: a disjunct that can never hold is an impossible case, and it closes itself. (Conversely, $x > 0$ gives the whole statement by $\vee$I, so the two say the same thing.)

Exercise. C2: A clash that escaped its box

A student wants to show $\neg P \vdash P \to Q$ and writes:

1	$\neg P$	
2	$\quad P$	
3	$\quad \bot$	$\neg$E 2, 1
4	Q	$\bot$E 3
5	$\quad P$	
6	$\quad Q$	R 4
7	$P \to Q$	$\to$I 5–6

(a) Is the sequent $\neg P \vdash P \to Q$ derivable?

(b) Find the first line that is not justified, and name the clause of Definition 1.4.2 it breaks.

(c) Show that the move made at that line can lead from a true premise to a false conclusion.

(d) Write a correct derivation.

Solution.

(a) Yes: it is Proposition 2.1.2. The conclusion is true; the question is whether this argument proves it.

(b) Line 4. It cites line 3, which lies inside the box 2–3, and that box has closed before line 4. By the scope rule of Definition 1.4.2, a line inside a closed box is not available. (Lines 1–3 are fine: a box may close without a rule discharging it, although nothing then leaves it. Lines 5–7 would be fine if line 4 were.)

(c) Lines 1–4 alone would "derive" $\neg P \vdash Q$ by the same move. Read P as "$x > 0$" and Q as "$x = 7$", and take $x = -1$. The premise $\neg(x > 0)$ is true, and the conclusion

$x = 7$ is false. So the move "write, outside a box, what a closed box's $\bot$ would allow" can lead from a true premise to a false line. The line $\bot$ on line 3 holds only under the assumption P; once the box closes, that assumption is gone, and so is the clash.

(d) Move the $\bot$E inside the box, where the clash is:

1	$\neg P$	
2	$\quad P$	
3	$\quad \bot$	$\neg$E 2, 1
4	$\quad Q$	$\bot$E 3
5	$P \to Q$	$\to$I 2–4

Marking scheme. (10 points) - 2: answers (a) yes, citing Proposition 2.1.2 or giving a derivation. - 3: names line 4 as the first failure and the scope rule (a line inside a closed box is not available) as the clause broken. - 3: gives a counterexample **to the step**, not only to some conclusion: a reading of P, Q under which lines 1–4 go from a true premise to a false line. - 2: a correct derivation, with every citation available.

Fatal errors (0 for the step): blaming line 7 or line 6 (they are correct once line 4 is); saying that $\bot$E itself is the error.

2.2 Proving a Negation

The rules of Section 2.1 **use** a negation: $\neg$E records a clash, and $\bot$E turns a clash into any statement we like. Neither rule can **prove** a negation that was not already there. So we still cannot prove that no rational number has square 2, or even that $x^2 + 1$ is never 0. This section adds the one rule that proves a negation, $\neg$I, and uses it for one of the oldest proofs in mathematics.

2.2.1 How to prove that something is not so

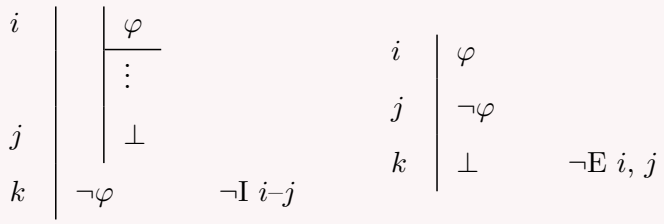
To show that no rational number has square 2, we cannot inspect the rational numbers one by one: there are infinitely many. A mathematician instead supposes the opposite and shows that it leads to something impossible: "suppose r is rational and $r^2 = 2$; ...; this is a contradiction; hence $r^2 \neq 2$."

Try the smaller goal $x^2 + 1 \neq 0$, for a real number x, with the rules so far. The goal is $\neg(x^2 + 1 = 0)$, and its main connective is $\neg$. Chapter 1's goal-directed move asks for the introduction rule of the main connective, and $\neg$ has none yet. The rules of Section 2.1 do not help: $\bot$E could write the goal, but only from a line $\bot$, and nothing available clashes. The statement $x^2 + 1 = 0$ is not a line of anything; it is the thing we want to rule out.

The mathematician's sentence shows the missing rule. "Suppose $r^2 = 2$" opens a box, exactly like the box of $\to$I in Definition 1.4.1, and "this is a contradiction" is its last line, $\bot$. The only new thing is what the box gives when it closes.

To prove "not φ", assume φ and reach a contradiction.

Definition 2.2.1. Rules for Negation

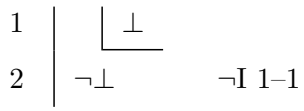


The left schema is ¬I (**not-introduction**): the assumption φ is **discharged** when the box closes, and the line $\bot$ is a line of that box itself, not of a box inside it. The right schema is ¬E, recalled from Definition 2.1.1, so that the rule that proves a negation and the rule that uses one stand together.

Read the schema of ¬I clause by clause.

1. **Open a box** whose first line is the assumption φ. It is the **unnegated** statement: the thing we want to refute, not the thing we want to prove. Any statement may be assumed.
2. **Inside the box** every rule may be used, and every line available just above the box may be cited, as in any box (Definition 1.4.2).
3. **The last line of the box** is exactly $\bot$, and it is a line of that box itself, not inside a further box opened within it. This is the same wording as for →I, and every box rule shares it; Chapter 3's soundness proof (Theorem 3.5.2) relies on it.
4. **Close the box** and write $\neg\varphi$ on the next line, outside it. The justification cites the box as a range, "¬I i–j". The assumption is discharged: after the box closes, nothing inside it may be cited, and only the negation leaves.
5. **The new line is always a negation.** ¬I never writes a statement whose main connective is not ¬.

The degenerate case is a box of one line, whose assumption is already $\bot$. The box 1–1 then ends on $\bot$ as soon as it opens, and ¬I gives $\neg\bot$:



So $\vdash \neg\bot$: "it is not the case that contradiction" is derivable from nothing, as it should be.

A noninstance by one change. Take a box opened by φ whose last line is $\neg\varphi$ instead of $\bot$. It looks as if the assumption has refuted itself, and ¬I still does not apply: the box must end on $\bot$. One more line, $\bot$ by ¬E citing the assumption and the line $\neg\varphi$, repairs it. The same holds if $\bot$ is reached only inside a smaller box within the box: it is not then a line of the box itself.

Why this rule. The statement $\neg\varphi$ says that φ fails, and the box shows what would happen if φ held: together with the lines above it, it would lead to $\bot$, which never holds. So in any situation where the lines above the box hold, φ cannot hold. That is exactly what $\neg\varphi$ claims. As for Section 2.1's rules, Chapter 3 turns this into a proof (Theorem 3.5.2).

The rules for ¬ have the same shapes as rules we already know, with $\bot$ in one slot. Put them beside the rules for →, applied to the statement $\varphi \to \bot$:

To ...	a negation $\neg\varphi$	an implication $\varphi \to \bot$
prove it	box from φ to $\bot$, then $\neg$I	box from φ to $\bot$, then $\to$I
use it, given φ	$\bot$ by $\neg$E	$\bot$ by $\to$E

Remark. Row by row, $\neg\varphi$ behaves exactly like $\varphi \to \bot$: "not φ" means "φ leads to contradiction". Some presentations of logic take this as the definition of $\neg$. This book keeps $\neg$ as a connective with its own rules, and Chapter 5 returns to the reading $\neg\varphi \approx \varphi \to \bot$ when it asks what a proof of a negation provides.

Warning. $\neg$I proves **negative** statements only. Its box opens with φ and ends by writing $\neg\varphi$. A proof that says "suppose x is **not** positive; ...; contradiction; hence x is positive" has a box opened by $\neg(x > 0)$, and $\neg$I applied to that box gives $\neg\neg(x > 0)$, not $x > 0$. Getting from there to $x > 0$ needs a rule we have not met. Section 2.5 is about that rule, and contrasts the two kinds of box in full.

Warning. Do not open the box with what you want to prove. To prove $\neg\varphi$, the box opens with φ. A box opened by $\neg\varphi$ and ending in $\bot$ gives $\neg\neg\varphi$ by $\neg$I, the negation of the goal, not the goal. This is the negation's version of the converse error that Section 1.4 found in Argument B.

2.2.2 First derivations with not-introduction

The goal-directed move of this section: *a goal $\neg\varphi$ opens a box with φ, and the new goal is $\bot$.* From there, Section 2.1's move takes over: *goal $\bot$, find a clash*, among the available lines or by manufacturing its missing half.

Example. Two negations from one premise

Show that (a) $P \vdash \neg\neg P$, and (b) $(P \vee Q) \to \neg P \vdash \neg P$.

Idea.

(a) The goal $\neg\neg P$ is $\neg(\neg P)$, a negation of $\neg P$. So the box opens with what is negated, $\neg P$, and aims at $\bot$. The clash is immediate: the premise P and the assumption $\neg P$. (b) The goal $\neg P$ opens a box on P. Inside it the premise is an implication, and the premise-directed move feeds it its hypothesis $P \vee Q$, which $\vee$I manufactures from P. The premise then hands back $\neg P$, and that clashes with the box's own assumption. The assumption is used twice: once to build the hypothesis, once in the clash.

Solution.

(a) Four lines.

1	P	
2	$\quad \neg P$	
3	$\quad \bot$	$\neg$E 1, 2
4	$\neg\neg P$	$\neg$I 2–3

(b) Six lines.

1	$(P \vee Q) \to \neg P$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad \neg P$	$\to$E 1, 3
5	$\quad \bot$	$\neg$E 2, 4
6	$\neg P$	$\neg$I 2–5

In (b), line 4 already says $\neg P$, but it lies inside the box of P and depends on that assumption, so it cannot be the conclusion. Line 6 is the same statement outside the box, where it depends on the premise alone.

Derivation (a) adds two negations. The reverse passage, from $\neg\neg P$ alone to P, would have to **remove** two, and no rule so far has a negation as its one given line and the unnegated statement as its new line: $\neg$E needs a second line and writes only $\bot$, and $\neg$I only adds a $\neg$. Section 2.5 returns to this.

Quick check. Is $\neg(P \vee Q) \vdash \neg P$? If so, give a derivation and name its rules. Compare it with Example (Manufacturing the other half of a clash), which has the same premise.

Solution. Yes. The goal $\neg P$ is a negation, so assume P and aim at $\bot$; $\vee$I turns P into $P \vee Q$, which clashes with the premise.

1	$\neg(P \vee Q)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg P$	$\neg$I 2–4

The rules are $\vee$I, $\neg$E, and $\neg$I. The box is the same as in Example (Manufacturing the other half of a clash) up to its line $\bot$. There the goal was $P \to R$, so the box had to end on R, written by $\bot$E, and closed by $\to$I. Here the goal is $\neg P$, so the box ends on $\bot$ itself and closes by $\neg$I. In words: if neither P nor Q holds, then in particular P does not.

In the next derivation the clash itself creates a goal that is a negation, and the move of this section is used a second time, one box inside another.

Example. Three negations become one

Show that $\neg\neg\neg P \vdash \neg P$.

Idea. The goal $\neg P$ opens a box on P, with goal $\bot$. Which clash? The only negation available is the premise $\neg(\neg\neg P)$, so the premise-directed question asks for its unnegated half, $\neg\neg P$: that becomes the subgoal. It is itself a negation, $\neg(\neg P)$, so the goal-directed move opens a second $\neg$I box, on $\neg P$, with goal $\bot$. Inside it, $\neg P$ clashes with the outer assumption P. This is the first derivation in which a $\neg$I box is opened for a subgoal that a clash created.

Solution. Seven lines.

1	$\neg\neg\neg P$	
2	$\quad P$	
3	$\quad\quad \neg P$	
4	$\quad\quad \bot$	$\neg$E 2, 3
5	$\quad \neg\neg P$	$\neg$I 3–4
6	$\quad \bot$	$\neg$E 5, 1
7	$\neg P$	$\neg$I 2–6

Line 4 may cite line 2, because the only box containing line 2, the box 2–6, also contains line 4 (Definition 1.4.2). The inner box 3–4 ends on $\bot$ as a line of its own, and so does the outer box 2–6, at line 6; each $\neg$I cites its own box. Notice what did **not** happen: no line removes a pair of negations. Three became one by adding them, twice, and clashing.

Warning. A box may reach $\bot$ without ever using its assumption. That is legal, but it is a sign: if no line of the box depends on φ, the same lines could have been written just above the box, where $\bot$ would then stand, and $\bot$E would have written $\neg\varphi$, or anything else, directly. The lines available above the box were already contradictory. When your $\neg$I box never uses its assumption, look for the error above the box, not in it.

The first example about numbers is the goal we could not reach at the start of the section.

Example. A negative statement proved by not-introduction

Let x be a real number. Prove that $x^2 + 1 \neq 0$, naming the rules.

Idea. The goal $x^2 + 1 \neq 0$ is $\neg(x^2 + 1 = 0)$, a negation, so open a box on $x^2 + 1 = 0$ and look for a clash. The school fact $x^2 \geq 0$, that is $\neg(x^2 < 0)$, is the negation available; its unnegated half $x^2 < 0$ is the subgoal, and algebra on the assumption produces it.

Solution. The goal $x^2 + 1 \neq 0$ is $\neg(x^2 + 1 = 0)$, so we use ¬I. Suppose $x^2 + 1 = 0$ [assumption for ¬I]. Then $x^2 = -1$, so $x^2 < 0$. But $x^2 \geq 0$, that is, $\neg(x^2 < 0)$, so we have a contradiction [¬E]. Hence $x^2 + 1 \neq 0$ [¬I].

2.2.3 How the rule looks in a mathematical proof

The proof that no rational number has square 2 has the shape of ¬I: the claim is negative, so we assume its opposite and look for ⊥. The tool that finds the clash is Theorem 2.1.3, and the school fact it needs is about writing fractions:

(L) Every rational number can be written as p/q with p and q integers, $q \neq 0$, and p and q **not both even**.

This holds because a fraction whose numerator and denominator are both even can be reduced by 2, and the reduction cannot go on for ever, since the denominator shrinks in absolute value each time. Making "cannot go on for ever" precise needs the well-ordering of the natural numbers, which comes in *Sets, Relations, and Cardinality*, Chapter 2; here we take (L) as known.

Theorem 2.2.1. Irrationality of the Square Root of Two

There is no rational number whose square is 2. In other words, $\sqrt{2}$ is irrational.

Idea. Fix a rational number r and prove the negative statement $r^2 \neq 2$ by ¬I: assume $r^2 = 2$ and hunt for ⊥. Write $r = p/q$ as in (L), so that the equation becomes $p^2 = 2q^2$. The only available negation is "not both even", so the clash must come from showing that p and q **are** both even; Theorem 2.1.3 is the tool for that, used twice. *Before reading on: the second use of Theorem 2.1.3 needs the line "q^2 is even". From which two earlier lines of the proof will it come?*

Proof. Let r be a rational number, and suppose $r^2 = 2$ [assumption for ¬I]. By (L), $r = p/q$ for integers p and q with $q \neq 0$, not both even. Then $p^2 = r^2q^2 = 2q^2$, so p^2 is even. By Theorem 2.1.3, p is even [→E], say $p = 2m$ for an integer m. Then $4m^2 = 2q^2$, so $q^2 = 2m^2$, and q^2 is even. By Theorem 2.1.3 again, q is even [→E]. So p and q are both even [∧I], while they were chosen not both even; this is a contradiction [¬E]. Hence $r^2 \neq 2$ [¬I].

Since r was an arbitrary rational number, no rational number has square 2. This proves the theorem.

■

Quick check. In the proof, which step is ¬E, and which two statements does it cite?

Solution. The step "this is a contradiction". It cites "p and q are both even", derived inside

the box of $r^2 = 2$ by $\wedge$I, and "p and q are not both even", which (L) supplied when p and q were chosen. In the derivation below these are lines 11 and 2, and the step is line 12.

The propositional part of the proof fits in a derivation. Fix r and the integers p, q that (L) supplies. Lines 1 and 2 record what (L) says about them; lines 3 and 4 are Theorem 2.1.3 for $n = p$ and $n = q$; the lines marked "algebra" are the computations of the proof.

1	$r = p/q,\ q \neq 0$	
2	$\neg(p \text{ even} \wedge q \text{ even})$	
3	p^2 even $\to p$ even	
4	q^2 even $\to q$ even	
5	$\quad r^2 = 2$	
6	$\quad p^2 = 2q^2$	algebra 1, 5
7	$\quad p^2$ even	def. of even 6
8	$\quad p$ even	$\to$E 3, 7
9	$\quad q^2$ even	algebra 6, 8
10	$\quad q$ even	$\to$E 4, 9
11	$\quad p$ even $\wedge\ q$ even	$\wedge$I 8, 10
12	$\quad \bot$	$\neg$E 11, 2
13	$\neg(r^2 = 2)$	$\neg$I 5–12

Line 13 is $r^2 \neq 2$. The box 5–12 is a $\neg$I box as the schema asks: opened by the unnegated statement $r^2 = 2$, and ending on $\bot$ as a line of its own. Two steps of the proof lie outside the derivation: choosing p, q, and m, and passing from the arbitrary r to every rational number. Both are quantifier rules, which come in Chapter 4.

Notice also what the proof does **not** need: that $\sqrt{2}$ exists as a real number. The theorem is a statement about rational numbers only. *Analysis on the Real Line*, Chapter 1 uses it in exactly this form, as evidence that the rational numbers have a gap where $\sqrt{2}$ should be.

And notice which rule closes it. "Suppose $r^2 = 2$" opens a box whose **negation** is the goal, so the proof is $\neg$I, even though it is what everyone calls a proof by contradiction. Section 2.5 meets a second kind of box, which looks the same on the page and is a different rule; the distinction starts from the warning above about $\neg(x > 0)$.

2.2.4 What the rule sounds like

Here is the row of the rule, phrase, and Lean table for this section's rule; the rows for $\neg$E and $\bot$E are in Section 2.1.

Rule	You write	Lean
$\neg$I	"Suppose φ. ... This is a contradiction. Hence not φ."	`intro hp` (goal `False`)

In Lean, the negation of `P` is defined to be `P → False`, with `False` playing the part of $\bot$: the reading of the remark above. So `intro` performs both $\to$I and $\neg$I.

2.2.5 Exercises

2.2.5.1 A. Check your understanding

Exercise. A1

(a) Write the schema of $\neg$I from memory, and say which statement opens its box.

(b) Which rule does this sentence use, and which box does it close? "Suppose 0 had a reciprocal y. Then $1 = 0 \cdot y = 0$, which is absurd. So 0 has no reciprocal."

(c) In the proof of Theorem 2.2.1, the sentence "p is even" follows from "p^2 is even". Which result and which rule justify it?

(d) Where in the proof of Theorem 2.2.1 is (L) used, and what would go wrong if (L) said only "$r = p/q$ with $q \neq 0$"?

(e) Run the proof of Theorem 2.2.1 with 4 in place of 2: assume $r^2 = 4$ and write $r = p/q$ as in (L). Which step can no longer be justified? (The conclusion "no rational number has square 4" is false, so some step must fail.)

Solution.

(a) The schema is the left one of Definition 2.2.1: a box opened by the assumption φ, ending on $\bot$ as a line of the box itself, then $\neg\varphi$ outside it, citing the box. The box opens with the **unnegated** statement φ.

(b) $\neg$I. The conclusion is the negative statement "0 has no reciprocal"; the box opens with "0 has a reciprocal y", and "which is absurd" is its line $\bot$, reached by $\neg$E from $1 = 0$ and the fact $1 \neq 0$.

(c) Theorem 2.1.3, for $n = p$, is the implication "if p^2 is even, then p is even"; together with "p^2 is even" it gives "p is even" by $\to$E (line 8 of the derivation, citing lines 3 and 7).

(d) The fact (L) supplies the line "p and q are not both even" (line 2 of the derivation), which the $\neg$E of line 12 cites. Without that clause the proof still derives "both even", but it contradicts nothing ($2 = 4/2$ has both even), and no $\bot$ is reached.

(e) With $r^2 = 4$ we get $p^2 = 4q^2 = 2(2q^2)$, so p^2 is even and p is even, $p = 2k$. Then $4k^2 = 4q^2$, so $q^2 = k^2$. This is where the proof stalls: nothing says q^2 is even, so Theorem 2.1.3 cannot be applied to q. Indeed for $r = 2 = 2/1$ we have $q = 1$, which is odd. Every step before the stall is valid; the proof simply fails to reach $\bot$.

2.2.5.2 B. Practice

Exercise. B1: A box inside a box

Show that $P \to \neg Q \vdash Q \to \neg P$. Name the move that opens each box.

Solution. The goal is an implication, so the outer box opens on Q, with goal $\neg P$. That goal is a negation, so the inner box opens on P, with goal $\bot$. Inside it, the premise fires on P

and gives $\neg Q$, which clashes with the outer assumption Q.

1	$P \to \neg Q$	
2	$\quad Q$	
3	$\quad\quad P$	
4	$\quad\quad \neg Q$	$\to$E 1, 3
5	$\quad\quad \bot$	$\neg$E 2, 4
6	$\quad \neg P$	$\neg$I 3–5
7	$Q \to \neg P$	$\to$I 2–6

Line 5 may cite line 2 because the box 2–6 contains line 5. Both boxes were read off the goal: the outer one by Chapter 1's move for an implication, the inner one by this section's move for a negation. In words: if P rules out Q, then Q rules out P.

Exercise. B2: A quadratic with no zero

Let x be a real number. Prove that $x^2 - x + 1 \neq 0$, naming the rules.

Hint: complete the square.

Solution. The goal is $\neg(x^2 - x + 1 = 0)$, a negation, so we use the rule $\neg$I. Suppose that $x^2 - x + 1 = 0$, as the assumption for $\neg$I. Completing the square,

$$x^2 - x + 1 = \left(x - \tfrac{1}{2}\right)^2 + \tfrac{3}{4},$$

so $(x - \frac{1}{2})^2 = -\frac{3}{4}$, and therefore $(x - \frac{1}{2})^2 < 0$. But the square of a real number is never negative, that is, $\neg((x - \frac{1}{2})^2 < 0)$; this is a contradiction [$\neg$E]. Hence $x^2 - x + 1 \neq 0$ [$\neg$I].

This is the pattern of Example (A negative statement proved by not-introduction), with the square $(x-\frac{1}{2})^2$ in place of x^2. The identity can be checked by expanding: $(x-\frac{1}{2})^2 = x^2-x+\frac{1}{4}$, and $\frac{1}{4} + \frac{3}{4} = 1$.

Exercise. B3: The square root of three

You may use that every integer can be written as $3k$, $3k+1$ or $3k+2$ with k an integer, in one way only, and that every rational number can be written as p/q with integers p, q, $q \neq 0$, and p, q not both divisible by 3.

(a) Prove that if n is an integer and n^2 is divisible by 3, then n is divisible by 3.

(b) Hence prove that no rational number has square 3.

Solution.

(a) Suppose n^2 is divisible by 3 [assumption for $\to$I], so n^2 has the form $3j$ with j an integer. The integer n has the form $3k$, $3k+1$ or $3k+2$, and we argue by cases [$\vee$E]; a three-way split is two uses of $\vee$E, one inside a box of the other.
Case $n = 3k$. Then n is divisible by 3, as required.
Cases $n = 3k+1$ *and* $n = 3k+2$. Then

$$(3k+1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1,$$
$$(3k+2)^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1,$$

so n^2 has the form $3j'+1$ as well as the form $3j$. An integer has only one of the three forms, so this is a contradiction [$\neg$E], and the case cannot occur. By $\bot$E, n is divisible by 3 in this case too.
All cases give that n is divisible by 3 [$\vee$E], which proves (a) [$\to$I]. The two impossible cases were closed exactly as the odd case in the proof of Theorem 2.1.3.

(b) Let r be a rational number and suppose $r^2 = 3$ [assumption for $\neg$I]. Write $r = p/q$ with p, q not both divisible by 3. Then $p^2 = 3q^2$, so 3 divides p^2, and by (a), $p = 3m$ for an integer m. Then $9m^2 = 3q^2$, so $q^2 = 3m^2$, and by (a) again, 3 divides q. So 3 divides both p and q [$\wedge$I]; this contradicts their choice [$\neg$E]. Hence $r^2 \neq 3$ [$\neg$I]. Since r was an arbitrary rational number, no rational number has square 3.

The form of (L) matters: the argument ends at "3 divides p and q", which clashes only with "not both divisible by 3". The form "not both even" of the text would give no clash, since $p = q = 3$ are both divisible by 3 and not both even.

Exercise. B4: A premise that turns on itself

Show that $P \to Q,\ Q \to \neg P \vdash \neg P$. Then say in one sentence what the sequent means.

Solution. The goal $\neg P$ is a negation, so assume P. The two premises, used in turn, turn P into $\neg P$, and the clash gives $\bot$.

1	$P \to Q$	
2	$Q \to \neg P$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad \neg P$	$\to$E 2, 4
6	$\quad \bot$	$\neg$E 3, 5
7	$\neg P$	$\neg$I 3–6

In words: if P leads to Q, and Q rules P out, then P refutes itself, and so P fails. Line 5 is the moment of self-refutation: the assumption has produced its own negation. Line 6 is still needed, because the box must end on $\bot$, not on $\neg P$.

2.2.5.3 C. Going deeper

Exercise. C1: The square root of six

Prove that no rational number has square 6. Use only the facts (E1)–(E3) of Section 2.1, (L), and Theorem 2.1.3, and name the rule at each logical step.

Solution (Hint 1). Follow the proof of Theorem 2.2.1. After writing $p = 2m$, you will need to show that q is even without knowing that q^2 is even.

Solution (Hint 2). You will reach $3q^2 = 2m^2$, so $3q^2$ is even. The proof of Theorem 2.1.3 passed from "n^2 is even" to "n is even"; imitate its shape for $3q^2$ and q.

Solution. Let r be a rational number and suppose $r^2 = 6$ [assumption for ¬I]. By (L), $r = p/q$ with integers p, q, $q \neq 0$, not both even. Then $p^2 = 6q^2 = 2(3q^2)$ is even, so p is even by Theorem 2.1.3 [→E]; write $p = 2m$. Then $4m^2 = 6q^2$, that is, $3q^2 = 2m^2$, so $3q^2$ is even.

We show that q is even. By (E1), q is even or odd, and we argue by cases [∨E]. *Case q even:* this is the goal. *Case q odd:* then q^2 is odd by (E3) [→E], say $q^2 = 2j + 1$, and then $3q^2 = 6j + 3 = 2(3j + 1) + 1$ is odd. So $3q^2$ is both even and odd [∧I]; this contradicts (E2) [¬E], so this case cannot occur, and in it q is even [⊥E]. In both cases q is even [∨E].

So p and q are both even [∧I]; this contradicts (L) [¬E]. Hence $r^2 \neq 6$ [¬I], and since r was arbitrary, no rational number has square 6.

Marking scheme. (10 points) - 2: opens with ¬I: a fixed rational r, the assumption $r^2 = 6$, and $r = p/q$ with p, q not both even from (L). - 2: derives p even from $p^2 = 2(3q^2)$ by Theorem 2.1.3. - 3: derives q even from $3q^2 = 2m^2$, with the odd case ruled out by a correct parity computation and (E2). - 2: reaches ⊥ by clashing "both even" with (L), and concludes $r^2 \neq 6$ by ¬I. - 1: rules named at the logical steps, and the passage from the arbitrary r to every rational number.

Fatal errors (0 for the step): applying Theorem 2.1.3 to q with only $3q^2$ known to be even; using "if 6 divides n^2, then 6 divides n" without proof.

Exercise. C2: A proof that proves too much

Here is a "proof" that $\sqrt{4}$ is irrational, modeled on Theorem 2.2.1.

> Suppose r is rational and $r^2 = 4$. Write $r = p/q$ with p, q integers, $q \neq 0$, not both divisible by 4. Then $p^2 = 4q^2$, so 4 divides p^2. Since n^2 divisible by 4 implies n divisible by 4, we get $p = 4m$. Then $16m^2 = 4q^2$, so $q^2 = 4m^2$, and as before 4 divides q. This contradicts the choice of p and q. Hence no rational number has square 4.

(a) Explain why the conclusion must be false.

(b) Find the step that fails, and show that it fails with a specific integer.

(c) Every other step is a correct use of a rule. Which rule turns the "contradiction" into

the conclusion, and why is the argument still wrong?

Solution.

(a) $2 = 2/1$ is rational and $2^2 = 4$.

(b) The step "n^2 divisible by 4 implies n divisible by 4" is a false lemma: for $n = 2$, $n^2 = 4$ is divisible by 4, but 2 is not. It fails exactly on the witness of (a): with $r = 2/1$ we have $p = 2$, and the lemma would claim that 4 divides 2.

(c) The last step is ¬I, applied to the box opened by "$r^2 = 4$". The rule is applied correctly, but the box contains a line that is not justified: the false lemma was used as if it were a premise available everywhere, through →E. A derivation is only as good as its premises. In Theorem 2.2.1 the corresponding lemma is Theorem 2.1.3, which we proved; here nothing proves the lemma, and nothing could. Compare part (e) of A1, where the honest proof stalls without reaching ⊥; here ⊥ is "reached" through a false line.

2.3 Derived Rules: Proved Sequents as New Rules

Section 2.1 closed an impossible case by hand, and Section 2.2 proved negations by hand, one box at a time. By now the same few patterns keep coming back: the impossible case of Example (An impossible case), and a box that turns a known implication round. Mathematicians give such patterns names ("by contraposition", "since P fails, Q") and use them in one breath. This section proves three of them once, from the rules we have, and then allows each to be used as a single line. The one new thing is that permission: a **derived rule**.

2.3.1 A box that keeps coming back

Here are two short arguments from different corners of mathematics.

> Let n be an integer whose square is not even. Suppose n is even. Then n^2 is even, which contradicts the hypothesis. Hence n is not even.
>
> Let x be a real number with $x^2 < 1$. Suppose $x \geq 1$. Then $x^2 \geq x \geq 1$, which contradicts $x^2 < 1$. Hence $x < 1$.

The subjects differ; the steps are the same. Each has a known implication ("if n is even, then n^2 is even"; "if $x \geq 1$, then $x^2 \geq 1$"), and a fact saying that the implication's conclusion fails. Each then opens a box on the implication's hypothesis, fires the implication inside it by →E, meets the clash with ¬E, and closes the box by ¬I (Definition 2.2.1). (For reals, $x^2 < 1$ and $x < 1$ are the exact negations of $x^2 \geq 1$ and $x \geq 1$.) The same four lines appear whenever an implication is used backwards, from the failure of its conclusion to the failure of its hypothesis.

A mathematician does not write these four lines. She writes "since n^2 is not even, n is not even", or "by contraposition", and moves on: the lines are the same every time. But a derivation is checked line by line against the rules, and "by contraposition" is not one of the rules. What licenses the short form? First we derive the pattern once, as a sequent, with the rules we have. Then we say exactly when a sequent already derived may be used as a single line.

2.3.2 Modus tollens, from the rules we have

The pattern of the two arguments is: from an implication and the negation of its conclusion, infer the negation of its hypothesis.

Proposition 2.3.1. Modus Tollens

$P \to Q,\ \neg Q \vdash \neg P.$

Idea. Read the skeleton off the goal. The goal $\neg P$ is a negation, and Section 2.2's move for a negated goal is a box opened on what is negated, here P, with goal $\bot$. Inside the box, look for a clash (Section 2.1's move for the goal $\bot$). The premise $\neg Q$ will clash with Q, so Q is the subgoal, and the premise $P \to Q$ gives it from the box's own assumption.

Proof. The following is a derivation with premises $P \to Q$ and $\neg Q$, and last line $\neg P$.

1	$P \to Q$	
2	$\neg Q$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad \bot$	$\neg$E 4, 2
6	$\neg P$	$\neg$I 3–5

Line 5 cites the premise on line 2, which lies in no box and so is available everywhere below it. The box 3–5 ends in $\bot$ as a line of its own, so $\neg$I closes it on line 6. This proves the proposition.

■

The name is Latin for "the mode that takes away": denying the conclusion takes away the hypothesis. Its partner, from $P \to Q$ and P to Q, is $\to$E, traditionally *modus ponens*, "the mode that puts" (Definition 1.3.1). The derivation uses $\to$E, $\neg$E, and $\neg$I, and nothing else.

A second form of the same fact packs the two premises into one implication. It is the form in which mathematicians state it: an implication **implies** its contrapositive.

Proposition 2.3.2. Contrapositive

$P \to Q \vdash \neg Q \to \neg P.$

Idea. The goal is an implication, so Chapter 1's first goal-directed move applies: open a box with its hypothesis $\neg Q$, and the new goal is $\neg P$. That goal is a negation, so open a second box, on P, inside the first. Inside both boxes we are exactly where the proof of Proposition 2.3.1 was: P and $P \to Q$ give Q, which clashes with $\neg Q$.

Proof. The following is a derivation with premise $P \to Q$ and last line $\neg Q \to \neg P$.

1	$P \to Q$	
2	$\quad \neg Q$	
3	$\quad\quad P$	
4	$\quad\quad Q$	$\to$E 1, 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad \neg P$	$\neg$I 3–5
7	$\neg Q \to \neg P$	$\to$I 2–6

Line 5 may cite line 2, because the only box containing line 2, the box 2–6, also contains line 5 (the scope rule of Definition 1.4.2). The inner box closes on line 6, a line of the outer box itself, and line 7 closes the outer box. This proves the proposition.

■

The statement $\neg Q \to \neg P$ is called the **contrapositive** of $P \to Q$. Like modus tollens, its derivation uses only the rules so far: $\to$I, $\to$E, $\neg$I, and $\neg$E. The passage back, from $\neg Q \to \neg P$ to $P \to Q$, is a different matter; the warnings below return to it.

Now compare the two derivations. Lines 3–6 of the derivation of Proposition 2.3.1 are, line for line and citation for citation, lines 3–6 of the derivation of Proposition 2.3.2: P, then Q by $\to$E 1, 3, then $\bot$ by $\neg$E 4, 2, then $\neg P$ by $\neg$I 3–5. And they are the four lines of the two arguments that opened this section. Writing them out every time buries the argument in bookkeeping.

2.3.3 What a derived rule is

A sequent that has been derived may be used as a rule.

This is what a mathematician does when she writes "by Lemma 3": the lemma was proved once, and each use of it is a single step. For derivations we must say exactly what that step may do.

Definition 2.3.1. Derived Rule

Let $\varphi_1, \ldots, \varphi_n \vdash \psi$ be a sequent, written with statement letters, that has **already been derived** earlier in the book. In a later derivation, replace each letter in $\varphi_1, \ldots, \varphi_n, \psi$ by a statement, the **same** statement for every occurrence of the same letter. If lines $i_1, \ldots, i_n$, all **available** at line k, hold the resulting instances of $\varphi_1, \ldots, \varphi_n$, we may write the resulting instance of ψ on line k. The justification names the result and cites $i_1, \ldots, i_n$. A sequent used in this way is a **derived rule**. For example, Proposition 2.3.1 becomes the rule MT:

i	$\varphi \to \psi$	
j	$\neg\psi$	
k	$\neg\varphi$	MT i, j

In words: a proved sequent is a new rule. Its premises become the cited lines, its conclusion the

line written, and its letters the metavariables of a schema (Definition 1.2.1). Read the definition clause by clause.

1. **"Already derived."** The sequent must have been derived **before** the line that uses it: earlier in the book, and never inside its own derivation. A derivation that used a sequent to derive that same sequent would prove nothing.
2. **"Replace each letter ... the same statement for every occurrence."** The letters of a derived sequent stood for arbitrary statements all along (Definition 1.2.3), so they may be replaced by any statements, provided the replacement is **uniform**. From $(A \wedge B) \to C$ and $\neg C$, the rule MT gives $\neg(A \wedge B)$, with φ replaced by $A \wedge B$ and ψ by C. Chapter 3 makes this replacement exact, under the name substitution.
3. **"Available at line k."** The scope rule of Definition 1.4.2 still governs every citation. A derived rule may not reach into a closed box, just as a primitive rule may not.
4. **"The resulting instance of ψ."** The rule writes the conclusion, with the same replacement, and nothing else. The derived rule does not open or close any box of the derivation that uses it.

We give the derived rules of this section short tags: MT for Proposition 2.3.1, CP for Proposition 2.3.2, and, once it is proved below, DS for Proposition 2.3.3. Every other derived sequent is a derived rule too, cited by its name: the five lines written out in Example (An implication with an impossible hypothesis), for instance, are now one line citing Proposition 2.1.2.

Convenience, not power. A derived-rule line abbreviates a stretch of derivation that we could write out in full. To expand it, repeat each cited line by (D3) of Definition 1.4.2, then copy the derivation of the sequent below the repetitions, with its letters replaced, its premises replaced by the repetitions, and its line numbers shifted. Every citation inside the copy points inside the copy or to a repetition, all in the current box, so every citation is available. The copy's boxes close before its last line, which is the instance of ψ. If the copy itself uses derived rules, expand those lines in the same way; this stops, because each derivation cites only results proved before it. So whatever can be derived with derived rules can be derived with the primitive rules alone.

The degenerate case is a sequent with nothing done to its premise, $P \vdash P$, whose derivation is the single premise line (part (a) of Example (Derivations, simplest first)): as a derived rule it copies an available line, which repetition already does. The first use with content is modus tollens proved a second time, with CP as a rule. The premise $P \to Q$ invites CP, and its conclusion $\neg Q \to \neg P$ is an implication whose hypothesis is the other premise.

1	$P \to Q$	
2	$\neg Q$	
3	$\neg Q \to \neg P$	CP 1
4	$\neg P$	$\to$E 3, 2

Line 3 is the instance of Proposition 2.3.2 with P and Q replaced by themselves; line 4 is modus ponens on it. Written out in full, line 3 would become a repetition of line 1 followed by lines 2–7 of the derivation of Proposition 2.3.2. This is honest because Proposition 2.3.2 was derived from the primitive rules, not from Proposition 2.3.1: the chain of derived rules ends in primitive rules, without a loop.

Warning. A derived rule must match its sequent exactly, under one uniform replacement of letters. From $P \to Q$ and $\neg R$, MT gives nothing: no single statement in place of ψ turns the pair into $\varphi \to \psi$ and $\neg\psi$. Equally, from $P \to \neg Q$ and Q, MT gives nothing yet: it needs

the negation of the conclusion, $\neg\neg Q$, and Q is not that line. (Here $\neg$I would give $\neg\neg Q$ first, as in Example (Two negations from one premise), and then MT applies.) Likewise DS, as stated below, refutes the **left** disjunct; applied blindly to $P \vee Q$ and $\neg Q$, it does not match.

2.3.4 Disjunctive syllogism: the impossible case, filed

The second pattern is the impossible case of Section 2.1. Every integer is even or odd; this one is not odd; so it is even. A disjunction, one side refuted, the other side left.

Proposition 2.3.3. Disjunctive Syllogism

$P \vee Q,\ \neg P \vdash Q$.

Idea. This is Example (An impossible case) with a simpler clash. There, the case P produced R by $\to$E, which clashed with $\neg R$. Here the case P clashes with the premise $\neg P$ at once. So: split the premise $P \vee Q$ by $\vee$E, both boxes ending in the current goal Q. The Q box is its own assumption. The P box cannot reach Q honestly, so it reaches a clash and closes by $\bot$E: an impossible case closes itself.

Proof. The following is a derivation with premises $P \vee Q$ and $\neg P$, and last line Q.

1	$P \vee Q$	
2	$\neg P$	
3	$\quad P$	
4	$\quad \bot$	$\neg$E 3, 2
5	$\quad Q$	$\bot$E 4
6	$\quad Q$	
7	Q	$\vee$E 1, 3–5, 6–6

The two boxes 3–5 and 6–6 are separate. The second consists of its assumption alone, which is already the statement Q that $\vee$E needs. This proves the proposition. ■

Set the derivation beside that of Example (An impossible case). Both split $P \vee Q$ into the same two boxes, and in both the Q box is one line long. Inside the P box, two lines reached $\bot$ there (lines 5–6) and one line does here (line 4); then $\bot$E closes the box in the same way. From now on this whole stretch is one line, tagged DS.

The rule as stated refutes the **left** disjunct. To refute the right one, first put the disjunction in the other order; that, too, is now one line, since Example (First derivations with "or") derived $P \vee Q \vdash Q \vee P$. (Exercise B1 derives the mirror image directly.)

Example. Derived rules in use

(a) Show that $P \to Q,\ Q \to R,\ \neg R \vdash \neg P$, and then write the derivation out with primitive rules only.

(b) Show that $P \vee Q,\ \neg P,\ Q \to R \vdash R$.

Solution. Idea. A derived rule is a **premise-directed shortcut**: a line $\varphi \to \psi$ with a line $\neg\psi$ invites MT; a disjunction with one side refuted invites DS. This is the move this section adds to the search for derivations: before opening a box by hand, ask whether a derived rule takes the step at once.

(a) Work backwards from the goal $\neg P$. The premise $P \to Q$ would give it by MT if $\neg Q$ were available. And $\neg Q$ comes by MT from $Q \to R$ and $\neg R$. So MT is used twice, the second time on the conclusion of the first.

1	$P \to Q$	
2	$Q \to R$	
3	$\neg R$	
4	$\neg Q$	MT 2, 3
5	$\neg P$	MT 1, 4

Line 4 is the instance with φ replaced by Q and ψ by R; line 5 the instance with P and Q. In primitive rules, each MT line becomes its four lines, as in lines 3–6 of the derivation of Proposition 2.3.1: a box on the hypothesis, →E, ¬E, ¬I. This natural expansion has eleven lines.

1	$P \to Q$	
2	$Q \to R$	
3	$\neg R$	
4	$\quad Q$	
5	$\quad R$	→E 2, 4
6	$\quad \bot$	¬E 5, 3
7	$\neg Q$	¬I 4–6
8	$\quad P$	
9	$\quad Q$	→E 1, 8
10	$\quad \bot$	¬E 9, 7
11	$\neg P$	¬I 8–10

The literal expansion of Definition 2.3.1 adds a repetition of each cited line and is longer; the natural one cites the available lines directly.

(b) The disjunction $P \vee Q$ has its left side refuted by $\neg P$, so DS gives Q, and then $Q \to R$ fires.

1	$P \lor Q$	
2	$\neg P$	
3	$Q \to R$	
4	Q	DS 1, 2
5	R	$\to$E 3, 4

Expanded, line 4 becomes the seven lines of the derivation of Proposition 2.3.3, with a case split on line 1 and the case P closed by $\bot$E.

Quick check. Which derived rule of this section, if any, gives a new line in one step from each pair of available lines?

(i) $P \to Q$ and $\neg Q$. (ii) $P \lor Q$ and $\neg P$. (iii) $P \lor Q$ and $Q \to R$. (iv) $P \to \neg Q$ and Q.

Solution.

(i) MT gives $\neg P$. (ii) DS gives Q. (iii) None: nothing is known about the case P. (iv) None in one step: MT needs the negation of the conclusion $\neg Q$, which is $\neg\neg Q$, not Q. A $\neg$I box gives $\neg\neg Q$ from Q first, as in Example (Two negations from one premise), and then MT gives $\neg P$.

2.3.5 Modus tollens on numbers, and the green card

In a mathematical proof, MT is the step "since the conclusion fails, the hypothesis fails". Here it is with numbers.

Example. Modus tollens on numbers

Let x be a real number with $x^2 \leq 4$. Use the implication "if $x > 2$, then $x^2 > 4$" to show that $x \leq 2$, naming the derived rule.

Solution. Take P: "$x > 2$" and Q: "$x^2 > 4$". The implication is $P \to Q$. (It holds because $x > 2 > 0$ gives $x^2 = x \cdot x > 2x > 4$.) The hypothesis $x^2 \leq 4$ is $\neg Q$, since for real numbers $x^2 \leq 4$ and $x^2 > 4$ are exact negations of each other. By MT (Proposition 2.3.1), $\neg P$, that is, $\neg(x > 2)$, which is $x \leq 2$.

The same step settles the card most people leave on the table in Example (Four cards) (Section 1.3): the cards show 6, 7, red, and green, and the claim is "if a card shows an even number on one side, then its other side is red". For the green card, let E be "this card's number is even" and R "its color side is red". The claim, for this card, is $E \to R$. The card shows green, so $\neg R$. If the claim holds, MT gives $\neg E$: the hidden number must be odd. Turning the card tests

exactly that line: an even number there would give E, and with $\neg R$ and $E \to R$, $\to$E and $\neg$E would reach $\bot$. The red card offers only R, and from $E \to R$ and R no rule gives anything about E; that was Section 1.3's affirming the consequent. So the green card must be turned and the red one need not be.

Modus tollens is the step people skip. CP turns the claim into its contrapositive $\neg R \to \neg E$, "a card whose color is not red has an odd number", which carries the claim's content to the green card; Section 2.5 derives the passage back, so the two say the same. The converse $R \to E$, "a red card has an even number", is a different claim, and the card rule never made it.

2.3.6 How this rule looks in a mathematical proof

Theorem 2.1.3 (Section 2.1) says that for an integer n, if n^2 is even then n is even. Section 2.1 proved it by cases, with the odd case shown impossible and closed by $\bot$E. Here is the same theorem argued as mathematicians usually write it, with the derived rules doing the work. We use the facts (E1)–(E3) of Section 2.1: every integer is even or odd; no integer is both; the square of an odd integer is odd.

Example. Even square, even number, by contraposition

Let n be an integer. Prove that if n^2 is even, then n is even, using CP and DS, and label each step with its rule.

Solution. Idea. The hypothesis "n^2 is even" hands us nothing to compute with, while "n is odd" hands us $n = 2k + 1$. So we turn (E3) round by CP: a number whose square is not odd is not odd itself. Then (E1), with one side refuted, leaves "even" by DS.

Suppose n^2 is even [assumption for $\to$I]. If n^2 were odd [assumption for $\neg$I], it would be both even and odd [$\wedge$I], contradicting (E2) [$\neg$E]; hence n^2 is not odd [$\neg$I]. By (E3), if n is odd then n^2 is odd, so by contraposition, if n^2 is not odd then n is not odd [CP]. Hence n is not odd [$\to$E]. By (E1), n is even or odd, so n is odd or even [or commutes], by the exchange of disjuncts of Example (First derivations with "or") used as a derived rule. Since n is not odd, n is even [DS]. This proves that if n^2 is even, then n is even [$\to$I].

With E: "n is even", O: "n is odd", S: "n^2 is even", T: "n^2 is odd", and the three facts as premises, the skeleton is this derivation.

1	$E \vee O$	
2	$\neg(S \wedge T)$	
3	$O \to T$	
4	$\quad S$	
5	$\quad\quad T$	
6	$\quad\quad S \wedge T$	$\wedge$I 4, 5
7	$\quad\quad \bot$	$\neg$E 6, 2
8	$\quad \neg T$	$\neg$I 5–7
9	$\quad \neg T \to \neg O$	CP 3
10	$\quad \neg O$	$\to$E 9, 8
11	$\quad O \vee E$	or commutes 1
12	$\quad E$	DS 11, 10
13	$S \to E$	$\to$I 4–12

Line 11 is Example (First derivations with "or") used as a derived rule, so that line 12 refutes the left disjunct, as DS requires.

Compare this with the twelve-line derivation after Theorem 2.1.3 in Section 2.1. There, line 11 split on (E1) at once, and the odd case, lines 6–10, got T from (E3), built $S \wedge T$, clashed with (E2), and closed by $\bot$E. Here the same work is spread over named steps. The $\neg$I box on lines 5–7 does the clash with (E2), which there was lines 8–9. Lines 9 and 10, CP and $\to$E, abbreviate a box that assumes O, gets T by (E3), as line 7 did there, and clashes with line 8. Line 12, DS, abbreviates a case split on the disjunction with the case O closed by $\bot$E, as lines 5–11 did there. The short version reads the way people talk.

Notice also what the argument never does. It feels like "suppose not", and written out in full it does contain one: CP assumes that n is odd, reaches $\bot$, and concludes that n is **not** odd. What it never does is assume that n is **not even** and conclude from a contradiction that n **is** even. Every negation in it is introduced; none is removed. The difference will matter in Section 2.5.

2.3.7 Common errors

Warning. A derived rule may be used only **after** its sequent has been derived, and never inside its own derivation. A "derivation" of Proposition 2.3.2 that justifies one of its lines by CP proves nothing: it assumes what it sets out to show. Exercise C2 is an example, and Section 2.6 returns to the error in general.

Warning. The converse of the contrapositive is a different sequent. CP passes from $P \to Q$ to $\neg Q \to \neg P$; it does not pass from $\neg Q \to \neg P$ back to $P \to Q$. Try it: in a box on P the goal is Q, a letter with no negation, and the natural box, on $\neg Q$, reaches $\bot$ and so closes to $\neg\neg Q$, not Q. Section 2.5 derives the passage back with a new rule. That no derivation with

the rules so far exists is true, but we state it without proof, and nothing depends on it.

Warning. "Suppose $x \geq 0$" inside a proof framed "Assume $x^3 < 0$; we show $x < 0$" is **not** the contrapositive. It is a box on the negation of the goal $x < 0$, and closing such a box on the unnegated goal needs the rule of Section 2.5. The contrapositive route replaces the frame: prove "if $x \geq 0$, then $x^3 \geq 0$", and CP turns it into "if $x^3 < 0$, then $x < 0$", since $x^3 < 0$ and $x < 0$ are the exact negations of $x^3 \geq 0$ and $x \geq 0$.

2.3.8 What the rules sound like

Here are the rows of the rule, phrase, and Lean table for this section's derived rules. As before, the last column is for readers who use a proof assistant, and nothing depends on it.

Rule	You write	Lean
MT	"Since Q fails, so does P."	`mt h hnq`
CP	"We prove the contrapositive: suppose not Q; ... so not P."	`mt`
DS	"P is impossible, so Q."	`h.resolve_left hnp`

2.3.9 Exercises

2.3.9.1 A. Check your understanding

Exercise. A1

(a) State Definition 2.3.1 in your own words, including what "already derived" and "the same statement" require.

(b) Here is a derivation that uses MT. Rewrite it with primitive rules only.

1	$R \to P$	
2	$\neg P$	
3	$\neg R$	MT 1, 2
4	$\neg R \vee S$	$\vee$I 3

(c) True or false, with a reason: "$\neg Q \to \neg P$ follows from $P \to Q$ by CP, so $P \to Q$ follows from $\neg Q \to \neg P$ by CP."

Solution.

(a) If a sequent has been derived earlier (not inside the derivation that uses it), then in a later derivation we may replace its letters by statements, each letter by one statement everywhere it occurs, and, when available lines hold the resulting instances of its premises, write the resulting instance of its conclusion on one line, citing the sequent and those lines.

(b) Line 3 becomes a box on R, then $\to$E, $\neg$E, and $\neg$I:

1	$R \to P$	
2	$\neg P$	
3	$\quad R$	
4	$\quad P$	$\to$E 1, 3
5	$\quad \bot$	$\neg$E 4, 2
6	$\neg R$	$\neg$I 3–5
7	$\neg R \vee S$	$\vee$I 6

(c) False. CP is the sequent $P \to Q \vdash \neg Q \to \neg P$. Applied to $\neg Q \to \neg P$, with P replaced by $\neg Q$ and Q by $\neg P$, it gives $\neg\neg P \to \neg\neg Q$, not $P \to Q$. The passage from $\neg Q \to \neg P$ to $P \to Q$ is the converse, a different sequent; Section 2.5 derives it with a new rule.

2.3.9.2 B. Practice

Exercise. B1: The other disjunctive syllogism

(a) Show that $P \vee Q,\ \neg Q \vdash P$ by a derivation from the primitive rules.

(b) Let x be a real number with $x^2 = 1$ and $x > 0$. You may use that $x^2 = 1$ implies $x = 1$ or $x = -1$. Deduce that $x = 1$, naming the result you use.

Solution.

(a) As in Proposition 2.3.3, with the roles of the boxes exchanged: now the case Q is the impossible one.

1	$P \vee Q$	
2	$\neg Q$	
3	$\quad P$	
4	$\quad Q$	
5	$\quad \bot$	$\neg$E 4, 2
6	$\quad P$	$\bot$E 5
7	P	$\vee$E 1, 3–3, 4–6

(b) Since $x^2 = 1$, we have $x = 1$ or $x = -1$. Since $x > 0$ and $-1 < 0$, we have $x \neq -1$. By part (a), with P: "$x = 1$" and Q: "$x = -1$", used as a derived rule (Definition 2.3.1), we conclude $x = 1$.

Exercise. B2: Putting a derivation in order

Here are nine lines, each a statement with its justification. Seven of them, in the right order and with the right boxes, form a derivation of $P \to Q \vdash \neg Q \to \neg P$ from the primitive rules. Give the order, and say where each box opens and closes. Then say why each of the other two lines cannot be used.

(i) $\neg Q \to \neg P$, by $\to$I on the box from $\neg Q$ to $\neg P$.
(ii) Q, by $\to$E from $P \to Q$ and P.
(iii) $P \to Q$, premise.
(iv) $\neg P$, by $\neg$I on the box from P to $\bot$.
(v) $\neg Q$, assumption.
(vi) $\bot$, by $\neg$E from Q and $\neg Q$.
(vii) P, assumption.
(viii) $\neg P$, assumption, opening the box for the $\neg$I that proves $\neg P$.
(ix) Q, by $\to$E from $P \to Q$ and the assumption P of (vii), written directly below (iv).

Solution. The order is (iii), (v), (vii), (ii), (vi), (iv), (i), with one box opened by (v) and closed after (iv), and a second box inside it, opened by (vii) and closed after (vi). This is the derivation of Proposition 2.3.2. It is the only admissible order. The premise comes first. (i) needs a box from $\neg Q$ to $\neg P$, so (v) opens a box whose last line is (iv). (iv) needs a box from P to $\bot$ inside it, so (vii) opens a box whose last line is (vi). Inside that box, (ii) needs (vii), and (vi) needs (ii) and (v).

(viii) cannot be used. $\neg$I proves $\neg P$ from a box opened by P, the statement being negated (Definition 2.2.1). A box opened by $\neg P$ and ending in $\bot$ would give $\neg\neg P$. (ix) cannot be used where it is placed. Directly below (iv), the box opened by (vii) has closed, so the assumption P is not available (Definition 1.4.2). Placed inside that box, the same line is (ii).

Exercise. B3

Show that $(P \vee Q) \to R,\ \neg R \vdash \neg P \wedge \neg Q$. You may use the derived rules of this section.

Solution. The premises are an implication and the negation of its conclusion, which invites MT: it gives $\neg(P \vee Q)$. The goal is a conjunction, so we need $\neg P$ and $\neg Q$, each a negation, each by a $\neg$I box. In the box of P, the clash is manufactured: $\vee$I gives $P \vee Q$, which clashes with line 3. The box of Q is the same.

1	$(P \lor Q) \to R$	
2	$\neg R$	
3	$\neg(P \lor Q)$	MT 1, 2
4	$\quad P$	
5	$\quad P \lor Q$	$\lor$I 4
6	$\quad \bot$	$\neg$E 5, 3
7	$\neg P$	$\neg$I 4–6
8	$\quad Q$	
9	$\quad P \lor Q$	$\lor$I 8
10	$\quad \bot$	$\neg$E 9, 3
11	$\neg Q$	$\neg$I 8–10
12	$\neg P \land \neg Q$	$\land$I 7, 11

Line 3 is MT with φ replaced by $P \lor Q$ and ψ by R.

2.3.9.3 C. Going deeper

Exercise. C1: Two premises that cannot both fail

(a) Show that $P \to Q$, $\neg P \to R$, $\neg Q$, $\neg R \vdash \bot$, using MT twice.

(b) Hence show that $P \to Q$, $\neg P \to R \vdash \neg(\neg Q \land \neg R)$ with the rules so far.

(c) In words, the conclusion of (b) says that Q and R do not both fail. Explain why a derivation with the rules so far naturally ends at $\neg(\neg Q \land \neg R)$ rather than at $Q \lor R$.

Solution.

(a) The first MT has $\varphi = P$, $\psi = Q$. The second has $\varphi = \neg P$, $\psi = R$, so it gives the negation of $\neg P$, which is $\neg\neg P$. Lines 5 and 6 then clash.

1	$P \to Q$	
2	$\neg P \to R$	
3	$\neg Q$	
4	$\neg R$	
5	$\neg P$	MT 1, 3
6	$\neg\neg P$	MT 2, 4
7	$\bot$	$\neg$E 5, 6

Line 7 cites $\neg P$ and then $\neg\neg P$: the unnegated line first, with φ of $\neg$E replaced by $\neg P$.

(b) The goal is a negation, so open a box on $\neg Q \land \neg R$ and aim at $\bot$. Inside, $\land$E supplies the two negated premises of (a), and the rest is (a).

1	$P \to Q$	
2	$\neg P \to R$	
3	$\quad \neg Q \wedge \neg R$	
4	$\quad \neg Q$	$\wedge$E 3
5	$\quad \neg R$	$\wedge$E 3
6	$\quad \neg P$	MT 1, 4
7	$\quad \neg\neg P$	MT 2, 5
8	$\quad \bot$	$\neg$E 6, 7
9	$\neg(\neg Q \wedge \neg R)$	$\neg$I 3–8

(c) To reach $Q \vee R$ by $\vee$I we would need Q or R on its own. The premises give Q once P is known and R once $\neg P$ is known, and nothing available says which of P, $\neg P$ holds. What the rules so far can do is refute the opposite, $\neg Q \wedge \neg R$, by a $\neg$I box, and that gives the negation in (b). The step from (b) to $Q \vee R$ uses the rule of Section 2.5.

Exercise. C2: A circular derivation

A student offers this derivation of $P \to Q \vdash \neg Q \to \neg P$.

1	$P \to Q$	
2	$\quad \neg Q$	
3	$\quad \neg Q \to \neg P$	CP 1
4	$\quad \neg P$	$\to$E 3, 2
5	$\neg Q \to \neg P$	$\to$I 2–4

(a) Every line is an instance of the rule it names. What is wrong?

(b) Give a correct derivation. May line 3 be replaced by a line "$\neg P$, MT 1, 2"?

Solution.

(a) Line 3 uses CP, the derived rule whose sequent, $P \to Q \vdash \neg Q \to \neg P$, is the very one being derived. Definition 2.3.1 allows a sequent to be used only when it has **already been derived**; here it has not, so line 3 has no justification. (The whole derivation is also pointless: line 3 alone is already the goal.)

(b) The derivation of Proposition 2.3.2, from the primitive rules:

1	$P \to Q$	
2	$\quad \neg Q$	
3	$\quad\quad P$	
4	$\quad\quad Q$	$\to$E 1, 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad \neg P$	$\neg$I 3–5
7	$\neg Q \to \neg P$	$\to$I 2–6

Yes: a line "$\neg P$, MT 1, 2" inside the box of $\neg Q$ is legitimate, because Proposition 2.3.1 was derived from the primitive rules, without CP. What matters is that the chain of derived rules ends in primitive rules, without a loop.

Exercise. C3: Exit check: labeling a proof

Here is a proof that for an integer n, if n^2 is odd, then n is odd.

> Suppose n^2 is odd. If n were even, $n = 2k$, then $n^2 = 2(2k^2)$ would be even; so, by contraposition, since n^2 is not even (it is odd, and no integer is both), n is not even. As every integer is even or odd, n is odd.

Label each sentence, or part of a sentence, with the rule or rules it uses, primitive or derived. Take as premises the facts (E1) and (E2) of Section 2.1 and the school computation "if n is even, then n^2 is even".

Solution. Write E: "n is even", O: "n is odd", S: "n^2 is even", T: "n^2 is odd".

- "Suppose n^2 is odd": the assumption T of the box for $\to$I, whose goal is O.
- "If n were even, $n = 2k$, then $n^2 = 2(2k^2)$ would be even": the implication $E \to S$, a premise (the school computation). Reading the sentence as a $\to$I box that proves $E \to S$, assuming E and computing, is equally correct.
- "n^2 is not even (it is odd, and no integer is both)": $\neg S$, by $\neg$I: a box on S builds $S \wedge T$ by $\wedge$I and clashes with (E2), $\neg(S \wedge T)$, by $\neg$E.
- "by contraposition ... n is not even": CP on $E \to S$ gives $\neg S \to \neg E$, and $\to$E with $\neg S$ gives $\neg E$. (One MT line does both.)
- "As every integer is even or odd, n is odd": DS on (E1), $E \vee O$, with the left side E refuted, gives O.
- The frame as a whole: $\to$I, closing the box of T on O, gives $T \to O$.

As a derivation:

1	$E \vee O$	
2	$E \to S$	
3	$\neg(S \wedge T)$	
4	$\quad T$	
5	$\quad\quad S$	
6	$\quad\quad S \wedge T$	$\wedge$I 5, 4
7	$\quad\quad \bot$	$\neg$E 6, 3
8	$\quad \neg S$	$\neg$I 5–7
9	$\quad \neg S \to \neg E$	CP 2
10	$\quad \neg E$	$\to$E 9, 8
11	$\quad O$	DS 1, 10
12	$T \to O$	$\to$I 4–11

The computation is right: $(2k)^2 = 4k^2 = 2(2k^2)$, and $2k^2$ is an integer.

Marking scheme. (10 points)

- 2: identifies the frame as $\to$I, with the box opened on the hypothesis "n^2 is odd".
- 2: identifies "n^2 is not even" as $\neg$I, naming the clash (with "no integer is both", by $\neg$E).
- 3: identifies CP (then $\to$E), or MT, applied to "if n is even, then n^2 is even", with that antecedent and consequent. Reading the "if n were even" sentence as a premise or as a $\to$I box that proves the implication earns these points equally.
- 2: identifies DS on "every integer is even or odd", naming "even" as the refuted side.
- 1: names the closing $\to$I.

Fatal errors (0 for the step): naming anything but $\neg$I for the "not even" step (it concludes a negation from a box on the unnegated statement); applying CP to the converse, "if n^2 is even, then n is even".

2.4 Iff, Equivalence, and What a Definition Says

One connective is still without rules: "if and only if". Theorems say "n is odd if and only if n^2 is odd", and every definition, though written with a plain "if", is used as an "if and only if". Section 1.1 warned that "if" means different things in a theorem and in a definition, and promised an explanation. The one new thing here is the pair of rules for $\leftrightarrow$; the rest of the section (provable equivalence, the layout of an "iff" proof, the convention about definitions, chains of equivalences) is what the rules are used for.

2.4.1 The rules for "if and only if"

"P if and only if Q" combines "P if Q", which is $Q \to P$, with "P only if Q", which is $P \to Q$, as Exercise (C1: "Only if" and "unless") explains. A proof of it travels both ways; a use of it, either way. An implication is proved by a box, opened on the hypothesis and closed on the conclusion (Definition 1.4.1); an "if and only if" needs two such boxes.

An "if and only if" is two implications, and proving it takes two proofs.

Definition 2.4.1. Rules for the Biconditional

i	φ		
	$\vdots$		
j	ψ		
k	ψ		
	$\vdots$		
l	φ		
m	$\varphi \leftrightarrow \psi$	$\leftrightarrow$I i–j, k–l	

i	$\varphi \leftrightarrow \psi$			i	$\varphi \leftrightarrow \psi$		
j	φ			j	ψ		
k	ψ	$\leftrightarrow$E i, j		k	φ	$\leftrightarrow$E i, j	

The first schema is the rule $\leftrightarrow$I (**iff-introduction**); its two assumptions are discharged when their boxes close, and each box's last line (ψ in the first, φ in the second) is a line **of that box itself**, not of a box inside it. The two schemas below it are the two forms of $\leftrightarrow$E (**iff-elimination**).

As in every schema (Definition 1.2.1), φ and ψ stand for any statements, replaced consistently, and i, j, k, l, m for line numbers. Read the rules clause by clause.

1. **Two boxes.** $\leftrightarrow$I needs two available boxes: one opened by φ and ending in ψ, the other opened by ψ and ending in φ, each last line a line of its own box. Each is exactly the box $\to$I would need for one direction. Their order does not matter, but both must be there: a single box from φ to ψ proves $\varphi \to \psi$ and nothing more.
2. **Separate boxes.** The boxes are siblings, not nested, so nothing in the second may cite a line of the first (the scope rule of Definition 1.4.2). Each direction is a proof on its own.
3. **Citation.** The justification cites both boxes as ranges, as in $\leftrightarrow$I 2–4, 5–7.
4. **Elimination, either way.** From $\varphi \leftrightarrow \psi$ and φ on available lines, $\leftrightarrow$E writes ψ; from $\varphi \leftrightarrow \psi$ and ψ, it writes φ. It is modus ponens in whichever direction we need, citing the biconditional first.

2.4.2 First derivations with iff

The section's goal-directed move: **a goal $\varphi \leftrightarrow \psi$ opens two boxes**, one on each side, and each box's goal is the other side.

Example. An iff read both ways

(a) Show that $P \leftrightarrow Q \vdash Q \leftrightarrow P$.

(b) Show that $P \leftrightarrow Q,\ \neg Q \vdash \neg P$.

Solution.

(a) *Idea.* The goal is an iff, so the last line is $\leftrightarrow$I, and we need a box from Q to P and a box from P to Q. Each box's goal is one $\leftrightarrow$E away: the premise crosses from its assumption to the other side.

1	$P \leftrightarrow Q$	
2	$\quad Q$	
3	$\quad P$	$\leftrightarrow$E 1, 2
4	$\quad P$	
5	$\quad Q$	$\leftrightarrow$E 1, 4
6	$Q \leftrightarrow P$	$\leftrightarrow$I 2–3, 4–5

Line 3 uses the second form of $\leftrightarrow$E (right to left), line 5 the first.

(b) *Idea.* The goal $\neg P$ is a negation, so open a box on P and aim at $\bot$ (Section 2.2). Inside, the premise $P \leftrightarrow Q$ turns P into Q, and Q clashes with $\neg Q$.

1	$P \leftrightarrow Q$	
2	$\neg Q$	
3	$\quad P$	
4	$\quad Q$	$\leftrightarrow$E 1, 3
5	$\quad \bot$	$\neg$E 4, 2
6	$\neg P$	$\neg$I 3–5

This is the derivation of Proposition 2.3.1 with $\leftrightarrow$E on line 4: if two statements are equivalent and one fails, so does the other.

The rules say nothing that the two implications do not say. The next result makes this exact.

Proposition 2.4.1. An Iff Is Two Implications

$P \leftrightarrow Q \vdash (P \to Q) \wedge (Q \to P)$, and $(P \to Q) \wedge (Q \to P) \vdash P \leftrightarrow Q$.

Idea. Each goal chooses its rule. For the first, the goal is a conjunction of two implications: $\wedge$I needs both, and each is built in its own $\to$I box, crossing the premise with $\leftrightarrow$E inside. For the second, the goal is an iff: open its two boxes. The premise is a conjunction, so take it apart first (premise-directed), and cross each box with one of its halves by $\to$E.

Proof. For the first claim, the following is a derivation with premise $P \leftrightarrow Q$ and last line $(P \to Q) \wedge (Q \to P)$.

1	$P \leftrightarrow Q$	
2	$\quad P$	
3	$\quad Q$	$\leftrightarrow$E 1, 2
4	$P \to Q$	$\to$I 2–3
5	$\quad Q$	
6	$\quad P$	$\leftrightarrow$E 1, 5
7	$Q \to P$	$\to$I 5–6
8	$(P \to Q) \wedge (Q \to P)$	$\wedge$I 4, 7

For the second claim, the following is a derivation with premise $(P \to Q) \wedge (Q \to P)$ and last line $P \leftrightarrow Q$.

1	$(P \to Q) \wedge (Q \to P)$	
2	$P \to Q$	$\wedge$E 1
3	$Q \to P$	$\wedge$E 1
4	$\quad P$	
5	$\quad Q$	$\to$E 2, 4
6	$\quad Q$	
7	$\quad P$	$\to$E 3, 6
8	$P \leftrightarrow Q$	$\leftrightarrow$I 4–5, 6–7

Lines 2 and 3 lie in no box, so both boxes may cite them. This proves both claims.

■

So $\leftrightarrow$ could have been an abbreviation for $(P \to Q) \wedge (Q \to P)$. We keep it as a connective with its own rules, so that every connective is proved by an introduction rule and used by an elimination rule.

Example. Negating both sides of an iff

Show that $P \leftrightarrow Q \vdash \neg P \leftrightarrow \neg Q$.

Solution. Idea. The goal is an iff, so there are two boxes, one opened by $\neg P$ and reaching $\neg Q$, the other the reverse. Each box's goal is a negation, so each contains a $\neg$I box: the goal-directed move applied twice, one inside the other. Inside the inner box, $\leftrightarrow$E crosses from one side of the premise to the other, and the result clashes with the outer assumption. Each half is part (b) of Example (An iff read both ways), run inside a box.

1	$P \leftrightarrow Q$	
2	$\quad \neg P$	
3	$\quad\quad Q$	
4	$\quad\quad P$	$\leftrightarrow$E 1, 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad \neg Q$	$\neg$I 3–5
7	$\quad \neg Q$	
8	$\quad\quad P$	
9	$\quad\quad Q$	$\leftrightarrow$E 1, 8
10	$\quad\quad \bot$	$\neg$E 9, 7
11	$\quad \neg P$	$\neg$I 8–10
12	$\neg P \leftrightarrow \neg Q$	$\leftrightarrow$I 2–6, 7–11

Line 5 cites line 2, the assumption of the outer box, still open there. The two halves are mirror images.

The derivation uses only the rules of this section and of Sections 2.1–2.2. Its converse, $\neg P \leftrightarrow \neg Q \vdash P \leftrightarrow Q$, is also derivable, with the rule of Section 2.5: in the box of P it must produce the **positive** statement Q, and the natural box, on $\neg Q$, closes to $\neg\neg Q$, not Q. That no derivation avoids the new rule is true, but we state it without proof, and nothing depends on it. Section 2.5 sets the converse as an exercise.

2.4.3 Provable equivalence

Proposition 2.4.1 is a pair of sequents, one in each direction. It is not the first such pair: Proposition 1.4.2 and Proposition 1.6.1 are pairs too. When two statements can each be derived from the other, either may stand in for the other in any derivation, and the situation deserves a name.

Two statements are provably equivalent when each can be derived from the other.

Definition 2.4.2. Provable Equivalence

Statements P and Q are **provably equivalent**, written $P \dashv\vdash Q$, if $P \vdash Q$ **and** $Q \vdash P$.

In words: there is a derivation with premise P and last line Q, and **another** with premise Q and last line P. The symbol is the turnstile of Definition 1.2.3 written twice, back to back; two derivations prove the claim.

The degenerate example is $P \dashv\vdash P$: the one-line derivation of part (a) of Example (Derivations, simplest first) serves in both directions. A first real example is $P \vee Q \dashv\vdash Q \vee P$: Example (First derivations with "or") shows $P \vee Q \vdash Q \vee P$, and the same derivation with the letters exchanged shows the converse. By the same exchange, part (a) of Example (An iff read both ways) gives $P \leftrightarrow Q \dashv\vdash Q \leftrightarrow P$.

A minimal change gives a one-way street. Example (Two negations from one premise) derived $P \vdash \neg\neg P$, but the converse, $\neg\neg P \vdash P$, would have to remove two negations, and no rule met so far removes one. So we cannot claim $P \dashv\vdash \neg\neg P$: the clause "$Q \vdash P$" is the one missing. (It holds once Section 2.5 adds its rule; that it cannot hold without that rule is true, and Chapter 3 proves it: Theorem 3.5.4.) Likewise Proposition 2.3.2 derives $\neg Q \to \neg P$ from $P \to Q$, but not yet the reverse.

The connective $\leftrightarrow$ and the relation $\dashv\vdash$ are two sides of one idea; the next result makes the link exact, and shows that $\dashv\vdash$ behaves as an equivalence should.

Proposition 2.4.2. Provable Equivalence and the Biconditional

For all statements P, Q, R:

(a) $P \dashv\vdash Q$ if and only if $\vdash P \leftrightarrow Q$;
(b) $P \dashv\vdash P$; if $P \dashv\vdash Q$, then $Q \dashv\vdash P$; and if $P \dashv\vdash Q$ and $Q \dashv\vdash R$, then $P \dashv\vdash R$.

Idea. Everything here moves whole derivations around. To prove $\vdash P \leftrightarrow Q$, the two boxes of $\leftrightarrow$I are the two given derivations, with their premises turned into the boxes' assumptions. To use $\vdash P \leftrightarrow Q$, write out its derivation and apply $\leftrightarrow$E. For transitivity, run one derivation into the next: the last line of the first is the premise of the second.

Proof. We use one observation twice. Let D be a derivation, and copy its lines, in order and with their numbers shifted, to some place in another derivation, with every line of the copy inside the same boxes as that place. Then every citation inside the copy is still available: if line i of D is available at line k of D, every box of D containing line i contains line k, and the boxes added around the copy contain both lines. Closed boxes cited in D are handled the same way. So each copied line is justified by the same rule as before.

($\Rightarrow$) in (a). Suppose $P \vdash Q$ and $Q \vdash P$, by derivations D_1 and D_2. Build a derivation with no premises as follows. Open a box with the assumption P and, below it, copy D_1, turning each of its premise lines P into a repetition of the assumption by (D3) of Definition 1.4.2. By the observation, the copy is justified, and its last line Q is a line of the box itself, since the last line of D_1 lies in none of D_1's boxes. Close the box. Open a second box with Q, copy D_2 in the same way, and close it on P. Then $\leftrightarrow$I on the two boxes gives $P \leftrightarrow Q$. So $\vdash P \leftrightarrow Q$.

($\Leftarrow$) in (a). Suppose $\vdash P \leftrightarrow Q$, by a derivation D with no premises. For $P \vdash Q$, start with the premise P, copy D below it, and write Q by $\leftrightarrow$E from the last line of the copy, $P \leftrightarrow Q$, and line 1. Both cited lines lie in no box, so both are available. For $Q \vdash P$, do the same with premise Q, using the other form of $\leftrightarrow$E. So $P \dashv\vdash Q$.

(b) The one-line derivation consisting of the premise P shows $P \vdash P$, in both directions, so $P \dashv\vdash P$. Symmetry holds because Definition 2.4.2 says the same thing about P, Q as about Q, P. For transitivity it is enough to show that $P \vdash Q$ and $Q \vdash R$ give $P \vdash R$, since the reverse direction is the same statement with P and R exchanged. Take a derivation D_1 of Q from P, and below its last line copy a derivation D_2 of R from Q, turning each premise line Q of D_2 into a repetition of the last line of D_1. That line lies in no box, so it is available everywhere below it, and by the observation the rest of the copy is justified. The result has premise P and last line R. This proves the

proposition.

■

Part (a) says that one derivation of $P \leftrightarrow Q$ from no premises carries the same information as two derivations, one each way. Part (b) says that $\dashv\vdash$ can be used like "=" in a calculation: chains of provable equivalences run either way and join end to end.

Here are equivalences derived so far. Each row gives two derived rules (Definition 2.3.1), one in each direction.

Provable equivalence	Derived in
$P \wedge Q \dashv\vdash Q \wedge P$	Proposition 1.2.1, with the letters exchanged for the converse
$P \vee Q \dashv\vdash Q \vee P$	Example (First derivations with "or"), likewise
$(P \wedge Q) \to R \dashv\vdash P \to (Q \to R)$	Proposition 1.4.2
$P \wedge (Q \vee R) \dashv\vdash (P \wedge Q) \vee (P \wedge R)$	Proposition 1.6.1
$(P \to R) \wedge (Q \to R) \dashv\vdash (P \vee Q) \to R$	Example (Cases on a hypothesis are two implications) and Example (A disjunctive hypothesis splits into two implications)
$P \leftrightarrow Q \dashv\vdash (P \to Q) \wedge (Q \to P)$	Proposition 2.4.1
$P \leftrightarrow Q \dashv\vdash Q \leftrightarrow P$	Example (An iff read both ways) (a), with the letters exchanged

The fourth row is distributivity of $\wedge$ over $\vee$ as a provable equivalence. Some familiar laws are not in the table, because the text has derived them in one direction only; associativity of $\vee$ is one (Example (Regrouping a disjunction)).

Warning. A sequent in one direction is not an equivalence. $P \wedge Q \vdash P$ holds by $\wedge$E, but $P \vdash P \wedge Q$ does not: take P: "$2 > 1$" and Q: "$1 > 2$", and the premise is true while the conclusion is false. (A derivation never leads from true premises to a false conclusion; Chapter 3 proves this, Theorem 3.5.2.) Before writing $\dashv\vdash$, check that **both** derivations exist.

2.4.4 Proving an iff: two directions

In a written proof, the two boxes of $\leftrightarrow$I become two paragraphs: ($\Rightarrow$) assumes the left side and reaches the right, ($\Leftarrow$) the reverse. Neither may borrow the other's assumption: they are two separate boxes.

The first example completes the story of parity. We use the facts (E1)–(E3) of Section 2.1: every integer is even or odd; no integer is both even and odd; the square of an odd integer is odd.

Theorem 2.4.3. Odd Square, Odd Number

Let n be an integer. Then n is odd if and only if n^2 is odd.

Idea. Two boxes. Forwards is (E3). Backwards, "n^2 is odd" gives nothing to compute with, so split on (E1) and show the even case impossible, as in Example (An impossible case).

Proof. ($\Rightarrow$) This is (E3).

($\Leftarrow$) Suppose n^2 is odd [assumption for the second box]. By (E1), n is even or odd, and we argue by cases [$\vee$E]. If n is odd, we are done. If n is even, say $n = 2k$ with k an integer, then $n^2 = 2(2k^2)$ is even; so n^2 is both even and odd [$\wedge$I], which contradicts (E2) [$\neg$E]. So this case cannot occur, and n is odd in it too [$\bot$E]. In both cases n is odd.

This proves the theorem [$\leftrightarrow$I].

■

With the derived rules of Section 2.3, ($\Leftarrow$) fits in one breath, "n cannot be even, since then n^2 would be even; so n is odd": $\neg$I with (E2), then MT, then DS on (E1).

The next example is about real numbers, and its two directions do different work. Each hypothesis is needed: dropping either one breaks one of the two directions, as the remark after the proof shows.

Theorem 2.4.4. Squaring Preserves Order

Let x and y be real numbers with $x \geq 0$ and $y \geq 0$. Then

$$x \leq y \iff x^2 \leq y^2.$$

Idea. Forwards, multiply $x \leq y$ by $x \geq 0$ and by $y \geq 0$, and chain. Backwards, nothing undoes squaring without square roots, so we split instead: either $x \leq y$, which is the goal, or $y < x$. In the second case the forward argument, run with a strict inequality, gives $y^2 < x^2$, which clashes with the hypothesis: an impossible case, closed by $\bot$E.

Proof. ($\Rightarrow$) Suppose $x \leq y$. Multiplying by $x \geq 0$ gives $x^2 \leq xy$, and multiplying by $y \geq 0$ gives $xy \leq y^2$. Hence $x^2 \leq y^2$.

($\Leftarrow$) Suppose $x^2 \leq y^2$. For real numbers, $x \leq y$ or $y < x$, and we argue by cases [$\vee$E]. If $x \leq y$, we are done. If $y < x$, then $x > y \geq 0$, so $x > 0$. Multiplying $y < x$ by $x > 0$ gives $xy < x^2$, and multiplying $y < x$ by $y \geq 0$ gives $y^2 \leq xy$. Hence $y^2 < x^2$, which contradicts $x^2 \leq y^2$ [$\neg$E]; so this case cannot occur, and $x \leq y$ holds in it [$\bot$E]. In both cases $x \leq y$.

This proves the theorem [$\leftrightarrow$I].

■

Remark. Each hypothesis has a job. The direction ($\Rightarrow$) needs $x \geq 0$, and fails without it: $x = -2$, $y = 1$ gives $x \leq y$ but $x^2 = 4 > 1 = y^2$. (Its proof also multiplied by $y \geq 0$, but

there $y \geq 0$ already follows from $y \geq x \geq 0$.) The direction ($\Leftarrow$) used only $y \geq 0$, and fails without it: $x = 1$, $y = -2$ gives $x^2 = 1 \leq 4 = y^2$ but $x > y$.

A payoff comes at once. For $a, b \geq 0$, apply Theorem 2.4.4 to $x = \sqrt{a}$ and $y = \sqrt{b}$: $\sqrt{a} \leq \sqrt{b}$ if and only if $a \leq b$. Taking square roots preserves order, a fact used constantly in estimates. (That square roots of nonnegative numbers exist is a school fact here; *Analysis on the Real Line*, Chapter 1 proves it from the least upper bound axiom.)

Quick check. Here is a proof of the statement "for a real number x, $x^2 = 1$ if and only if $x = 1$ or $x = -1$".

> If $x = 1$, then $x^2 = 1 \cdot 1 = 1$. If $x = -1$, then $x^2 = (-1)(-1) = 1$. This proves the claim.

Which direction has been proved, and which is missing? Supply the missing one.

Solution. The proof assumes the right side, "$x = 1$ or $x = -1$", splits it into its two cases, and reaches $x^2 = 1$ in both. So it proves ($\Leftarrow$), by $\vee$E. The direction ($\Rightarrow$) is missing.

($\Rightarrow$) Suppose $x^2 = 1$. Then $(x - 1)(x + 1) = x^2 - 1 = 0$. A product of real numbers is zero only if one of the factors is zero, so $x - 1 = 0$ or $x + 1 = 0$, that is, $x = 1$ or $x = -1$.

Warning. A proof of an iff can prove one direction twice and look complete. For "n is odd if and only if n^2 is odd", a proof that shows "if n is odd, then n^2 is odd" and then "if n^2 is not odd, then n is not odd" has proved ($\Rightarrow$) and its contrapositive (Proposition 2.3.2), which is ($\Rightarrow$) again. The direction ($\Leftarrow$), "if n^2 is odd, then n is odd", is still missing; its contrapositive would be "if n is not odd, then n^2 is not odd". Before writing "this proves the theorem", check that one paragraph assumes each side.

2.4.5 What a definition says

Here is a definition written the way this book and most others write definitions.

> A function $f : \mathbb{R} \to \mathbb{R}$ is **strictly increasing** if, for all real numbers x and y, $x < y$ implies $f(x) < f(y)$.

Grammatically the sentence is an implication, "f is strictly increasing **if** [condition]", and read as a theorem it would allow only one move, from the condition to the term. But a definition is not a theorem. It introduces a new term, and the new term means **exactly** the condition: no more, since nothing else was said about the term, and no less, since the condition is all the term was made to express. So the definition is used in both directions, as an $\leftrightarrow$ between the term and the condition, and $\leftrightarrow$E is how it is used.

In a definition, "if" means "if and only if".

This is a convention of mathematical language, not a rule of logic. Some books write "iff" or $:\Leftrightarrow$ in definitions; this book, like most, writes "if" and puts the term defined in **bold**, which signals that the "if" is an "iff". Here it is used three ways. The words "for all real numbers x and y"

are a quantifier, whose rules come in Chapter 4; here we use them informally.

Example. A definition used in both directions

(a) Show that $f(x) = 3x - 5$ defines a strictly increasing function $f\colon \mathbb{R} \to \mathbb{R}$.

(b) Suppose $g\colon \mathbb{R} \to \mathbb{R}$ is strictly increasing. Show that $g(1) < g(2)$.

(c) Show that $h(x) = x^2$ does not define a strictly increasing function $h\colon \mathbb{R} \to \mathbb{R}$.

Solution.

(a) Let x and y be real numbers with $x < y$. Multiplying by $3 > 0$ gives $3x < 3y$, and subtracting 5 gives $3x - 5 < 3y - 5$, that is, $f(x) < f(y)$. So f satisfies the condition, and by the definition, read from right to left, f is strictly increasing [$\leftrightarrow$E].

(b) Since g is strictly increasing, the definition, read from left to right, says that $x < y$ implies $g(x) < g(y)$ for all real x and y [$\leftrightarrow$E]. Taking $x = 1$ and $y = 2$, and using $1 < 2$, we get $g(1) < g(2)$ [$\to$E].

(c) Suppose h were strictly increasing [assumption for $\neg$I]. By the definition read from left to right, as in (b), $-1 < 0$ would give $h(-1) < h(0)$, that is, $1 < 0$, which is false [$\neg$E]. So h is not strictly increasing [$\neg$I].

Part (a) uses the definition from the condition to the term; parts (b) and (c) use it from the term to the condition. A definition read only as an "if" would permit (a) and nothing else.

Warning. The convention is for **definitions only**. In a theorem, "if" means "if", one direction. A familiar theorem of calculus, proved in *Analysis on the Real Line*, Chapter 7, says: if f is differentiable at a, then f is continuous at a. Its converse is false. The function $f(x) = |x|$ is continuous at 0, but it is not differentiable there, since its graph has a corner: the quotient $|h|/h$ is 1 for $h > 0$ and -1 for $h < 0$. Reading a theorem's "if" as "iff" means proving its converse without a proof, which is the error of Argument B in Section 1.1.

2.4.6 Chains of equivalences

"The following are equivalent: (a), (b), (c)" means that any two of the three are equivalent: three iff statements, or six implications. Three implications suffice, if they go round in a cycle:

$$\text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \Rightarrow \text{(a)}.$$

Once the cycle is proved, any condition leads to any other by following the arrows round. Here is the derivation behind that sentence.

Proposition 2.4.5. Proving Equivalence by a Cycle

$P \to Q$, $Q \to R$, $R \to P \vdash P \leftrightarrow Q$. Likewise, from the same three premises, $Q \leftrightarrow R$ and $R \leftrightarrow P$ are derivable.

Idea. The goal is an iff, so there are two boxes. From P, one step along the cycle reaches Q. From Q, the cycle reaches P the long way round, through R: a backward chain of $\to$E, as in Section 1.3. The other two equivalences need no new work, because the three premises do not change when the letters are rotated.

Proof. The following is a derivation with the three premises and last line $P \leftrightarrow Q$.

1	$P \to Q$	
2	$Q \to R$	
3	$R \to P$	
4	$\quad P$	
5	$\quad Q$	$\to$E 1, 4
6	$\quad Q$	
7	$\quad R$	$\to$E 2, 6
8	$\quad P$	$\to$E 3, 7
9	$P \leftrightarrow Q$	$\leftrightarrow$I 4–5, 6–8

Now replace P, Q, R throughout by Q, R, P respectively. The premises $P \to Q$, $Q \to R$, $R \to P$ become $Q \to R$, $R \to P$, $P \to Q$, the same three statements in a different order, and every line remains justified by the same rule. So the renamed derivation derives $Q \leftrightarrow R$ from the same premises. Rotating once more gives $R \leftrightarrow P$. This proves the proposition. ■

In a cycle every condition reaches every other along the arrows; longer cycles work the same way.

Warning. Two arrows back and forth are not a cycle through three conditions. Proving (a) $\Rightarrow$ (b) and (b) $\Rightarrow$ (a) shows that (a) and (b) are equivalent and says nothing about (c): no arrow enters or leaves it. For a real number x, take (a) "$x > 0$", (b) "$2x > 0$", and (c) "$x > 1$". Both arrows between (a) and (b) are true, and (c) is not equivalent to them: $x = \frac{1}{2}$ satisfies (a) and (b) but not (c). Every condition in the list must lie on the cycle.

Example. Three descriptions of the unit interval

Let x be a real number. Prove that the following are equivalent.

(a) $0 < x < 1$.
(b) $x^2 < x$.
(c) $x > 0$ and $1/x > 1$.

Solution. By Proposition 2.4.5 it suffices to prove (a) ⇒ (b) ⇒ (c) ⇒ (a).

(a) ⇒ (b). Suppose $0 < x < 1$. Multiplying $x < 1$ by $x > 0$ gives $x^2 < x$.

(b) ⇒ (c). Suppose $x^2 < x$. First, $x > 0$: otherwise $x \le 0$, and then $x \le 0 \le x^2$, so $x \le x^2$, which contradicts $x^2 < x$. (In rules: we have $x \le 0$ or $x > 0$; the first is refuted by a ¬I box, and DS (Proposition 2.3.3) leaves the second.) Since $x > 0$, we may divide $x^2 < x$ by x, which gives $x < 1$, and then divide $x < 1$ by x, which gives $1 < 1/x$. So $x > 0$ and $1/x > 1$.

(c) ⇒ (a). Suppose $x > 0$ and $1/x > 1$. Multiplying $1 < 1/x$ by $x > 0$ gives $x < 1$. So $0 < x < 1$.

The order of the cycle was a choice, made so that each arrow was short: (b) ⇒ (c) carried the only case split. The reverse cycle would also do, with different arrows to prove.

2.4.7 What the rules sound like

The rows for this section's rules complete the table for the connectives; Section 2.6 collects the rules in one place. The Lean names in the last column come from the two fields of an iff, `mp` ("modus ponens", left to right) and `mpr` (right to left).

Rule	You write	Lean
↔I	"(⇒) Suppose P. … so Q. (⇐) Suppose Q. … so P."	`constructor`
↔E	"Since P holds if and only if Q does, and P holds, so does Q."	`h.mp hp` / `h.mpr hq`

2.4.8 Exercises

2.4.8.1 A. Check your understanding

Exercise. A1

(a) State the rules ↔I and ↔E, and the definition of $P \dashv\vdash Q$.

(b) For a real number $x \ge 0$, consider "$x^2 = x$ if and only if $x = 0$ or $x = 1$". Write the directions (⇒) and (⇐) as implications. Which one would a proof begin with "Suppose $x^2 = x$"?

(c) True or false, with a reason: from the definition "an integer n is **even** if there is an integer k with $n = 2k$" and the fact that 10 is even, we may conclude that $10 = 2k$ for some integer k.

(d) True or false, with a reason: from the theorem "if $x > 1$, then $x^2 > 1$" and the fact that $x^2 > 1$, we may conclude that $x > 1$.

Solution.

(a) ↔I: from a box $\varphi \dots \psi$ and a box $\psi \dots \varphi$, both available, write $\varphi \leftrightarrow \psi$. ↔E: from $\varphi \leftrightarrow \psi$ and either side, write the other (Definition 2.4.1). $P \dashv\vdash Q$ means $P \vdash Q$ and $Q \vdash P$ (Definition 2.4.2).

(b) ($\Rightarrow$): if $x^2 = x$, then $x = 0$ or $x = 1$. ($\Leftarrow$): if $x = 0$ or $x = 1$, then $x^2 = x$. The direction ($\Rightarrow$) begins "Suppose $x^2 = x$", since it assumes the left side.

(c) True. A definition's "if" means "if and only if", so the definition may be used from the term "even" to the condition, by $\leftrightarrow$E. (Indeed $k = 5$.)

(d) False. A theorem's "if" is one direction only, and the conclusion would be its converse. The converse fails at $x = -2$: $x^2 = 4 > 1$ but $x < 1$.

2.4.8.2 B. Practice

Exercise. B1: Iff is transitive

Show that $P \leftrightarrow Q,\ Q \leftrightarrow R \vdash P \leftrightarrow R$.

Solution. The goal is an iff, so we open two boxes: P to R, and R to P. In each, the premises are used as bridges, one after the other, by $\leftrightarrow$E.

1	$P \leftrightarrow Q$	
2	$Q \leftrightarrow R$	
3	$\quad P$	
4	$\quad Q$	$\leftrightarrow$E 1, 3
5	$\quad R$	$\leftrightarrow$E 2, 4
6	$\quad R$	
7	$\quad Q$	$\leftrightarrow$E 2, 6
8	$\quad P$	$\leftrightarrow$E 1, 7
9	$P \leftrightarrow R$	$\leftrightarrow$I 3–5, 6–8

Lines 4 and 8 use $P \leftrightarrow Q$ in opposite directions, as do lines 5 and 7 with $Q \leftrightarrow R$.

Exercise. B2: A parity equivalence

Let n be an integer. Prove that n is even if and only if $n + 3$ is odd.

Solution. ($\Rightarrow$) Suppose n is even, say $n = 2k$ with k an integer. Then $n + 3 = 2k + 2 + 1 = 2(k + 1) + 1$, and $k + 1$ is an integer, so $n + 3$ is odd.

($\Leftarrow$) Suppose $n + 3$ is odd, say $n + 3 = 2j + 1$ with j an integer. Then $n = 2j - 2 = 2(j - 1)$, and $j - 1$ is an integer, so n is even.

This proves the equivalence.

Exercise. B3: Strict inequalities

Let x and y be real numbers with $x \geq 0$ and $y \geq 0$. Deduce from Theorem 2.4.4 and Example (Negating both sides of an iff) that $x < y$ if and only if $x^2 < y^2$. You may use that, for real numbers a and b, $a < b$ holds exactly when $b \leq a$ fails.

Solution. Apply Theorem 2.4.4 with the roles of x and y exchanged, which is allowed since both are nonnegative: $y \leq x$ if and only if $y^2 \leq x^2$. By Example (Negating both sides of an iff), with P: "$y \leq x$" and Q: "$y^2 \leq x^2$", used as a derived rule (Definition 2.3.1), we get: $y \leq x$ fails if and only if $y^2 \leq x^2$ fails. By the fact quoted, the left side says $x < y$ and the right side says $x^2 < y^2$. This proves that $x < y$ if and only if $x^2 < y^2$.

2.4.8.3 C. Going deeper

Exercise. C1: Divisibility by six

For integers d and n, write $d \mid n$ if $n = dk$ for some integer k. Let n be an integer. Prove that the following three conditions are equivalent, by a single cycle of three implications.

(i) $6 \mid n$;
(ii) $2 \mid n$ and $3 \mid n$;
(iii) $6 \mid n^2$.

You may use Theorem 2.1.3 and part (a) of Exercise (B3: The square root of three).

Solution (Hint 1). Choose the order of the cycle so that each arrow is short. The two results you may use both pass from a square to the number itself.

Solution (Hint 2). Try (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i). The last arrow is the one that needs an idea: the two halves of (ii) must be used together, not one at a time.

Solution. We prove (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii) $\Rightarrow$ (i), which suffices by Proposition 2.4.5.

(i) $\Rightarrow$ (iii). Suppose $n = 6k$. Then $n^2 = 36k^2 = 6(6k^2)$, so $6 \mid n^2$.

(iii) $\Rightarrow$ (ii). Suppose $n^2 = 6j$. Then $n^2 = 2(3j)$ is even, so n is even by Theorem 2.1.3, that is, $2 \mid n$. Also $n^2 = 3(2j)$, so $3 \mid n^2$, and by part (a) of Exercise (B3: The square root of three), $3 \mid n$. Hence $2 \mid n$ and $3 \mid n$ [$\wedge$I].

(ii) $\Rightarrow$ (i). Suppose $2 \mid n$ and $3 \mid n$, and write $n = 3b$ with b an integer. The integer b is odd or even. If b were odd, then $n = 3b$ would be odd: writing $b = 2c + 1$ gives $3b = 6c + 3 = 2(3c + 1) + 1$. But n is even, and no integer is both, so this case cannot occur [$\neg$E, $\bot$E]. Hence b is even, say $b = 2c$, and $n = 6c$, so $6 \mid n$.

Marking scheme. (10 points) - 1: names a cycle through all three conditions and cites Proposition 2.4.5 (or proves the needed implications explicitly). - 2: (i) $\Rightarrow$ (iii) by computation. - 3: (iii) $\Rightarrow$ (ii), citing Theorem 2.1.3 for the factor 2 and part (a) of Exercise (B3: The square root of three) for the factor 3. - 4: (ii) $\Rightarrow$ (i), combining the two factors correctly (ruling out b odd, or $n = 3n - 2n$, or otherwise).

Fatal errors (0 for the step): a "cycle" that omits a condition; using "$2 \mid n$ and $3 \mid n$ gives $6 \mid n$" without proof; using "$6 \mid n^2$ implies $6 \mid n$" as known. Any other cycle through the three conditions earns full marks if each of its arrows is proved: 1 point for the cycle and its citation, and 3 points for each of its three arrows.

Exercise. C2: Dropping a hypothesis, and repairing the statement

(a) Suppose only $y \geq 0$ is assumed in Theorem 2.4.4, and x is an arbitrary real number. The remark after it says that ($\Leftarrow$) survives. Which steps of its proof use $y \geq 0$, and why is $x \geq 0$ never needed? Refute ($\Rightarrow$) with a new pair.

(b) Prove that for **all** real numbers x and y, $|x| \leq |y|$ if and only if $x^2 \leq y^2$.

Hint: for (b), apply Theorem 2.4.4 to the right pair of numbers.

Solution.

(a) The proof of ($\Leftarrow$) uses $y \geq 0$ twice: for $x > 0$ from $x > y$, and for $y^2 \leq xy$. Its other multiplication is by the derived $x > 0$, so $x \geq 0$ is never used. For ($\Rightarrow$), take $x = -3$, $y = 1$: $x \leq y$ and $y \geq 0$, but $x^2 = 9 > 1 = y^2$.

(b) Let x and y be real numbers. Then $|x| \geq 0$ and $|y| \geq 0$, so Theorem 2.4.4 applies to the pair $|x|$, $|y|$ and gives $|x| \leq |y|$ if and only if $|x|^2 \leq |y|^2$. Finally $|x|^2 = x^2$, since $|x|$ is x or $-x$ and $(-x)^2 = x^2$, and likewise $|y|^2 = y^2$. This proves that $|x| \leq |y|$ if and only if $x^2 \leq y^2$.

2.5 Proof by Contradiction and the Classical Rules

Every rule so far either builds a statement or takes one apart. The rules for negation are no exception: ¬I (Definition 2.2.1) builds $\neg\varphi$ from a box that opens with φ and ends in $\bot$, and ¬E (Definition 2.1.1) uses $\neg\varphi$ against φ. Yet in three places this chapter has stopped one step short of a goal. This section adds the one rule that takes a negation away, reductio ad absurdum. It then derives from it the classical laws that mathematics uses without noticing, and shows that three of them are one principle in three forms.

2.5.1 Two boxes that look alike

Here are two arguments that a mathematician would both call "proof by contradiction".

The first is the proof of Theorem 2.2.1. It fixes a rational number r, supposes $r^2 = 2$, reaches a contradiction, and concludes that $r^2 \neq 2$. In the language of Definition 2.2.1, it opens a box with the assumption P ("$r^2 = 2$"), reaches $\bot$ inside it, and closes the box to get $\neg P$. That is ¬I.

The second is an argument of a kind that analysis uses constantly.

Example. A number below every positive number

Let x be a real number with $x \geq 0$, and suppose that $x \leq \varepsilon$ for **every** real number $\varepsilon > 0$. Prove that $x = 0$.

Solution. Suppose, for a contradiction, that $x \neq 0$. Since $x \geq 0$ and $x \neq 0$, we have $x > 0$. Then $\varepsilon := x/2$ is a real number with $\varepsilon > 0$, so the hypothesis applies to it and gives $x \leq x/2$. Subtracting $x/2$ from both sides gives $x/2 \leq 0$, and multiplying by 2 gives $x \leq 0$. This contradicts $x > 0$. Hence $x = 0$.

(The step "the hypothesis applies to it" uses a rule for "every", which Chapter 4 makes precise. The facts about $\leq$ are school algebra, which *Analysis on the Real Line*, Chapter 1 derives from axioms.)

Look at its shape. The box opens with the assumption $\neg P$, where P is "$x = 0$", reaches $\bot$, and closes to give P. If we tried to justify the last line by $\neg$I, we would get the negation of the box's assumption, $\neg\neg P$: "it is not the case that $x \neq 0$". That is not P. The two arguments put their negations in different places. $\neg$I puts a $\neg$ **into** the conclusion; the second argument takes one **out**.

2.5.2 Where the rules so far stop short

The second argument is not a curiosity. Three times already in this chapter a derivation has stopped short in the same way: it reached a negation where the goal was not one.

1. Section 2.2 warned against the argument "suppose x is not positive; ...; hence x is positive". The box opens with $\neg(x > 0)$ and reaches $\bot$, so $\neg$I gives $\neg\neg(x > 0)$, and no rule so far gets from there to $x > 0$.
2. In Exercise (C1: Two premises that cannot both fail), the premises $P \to Q$ and $\neg P \to R$ led to $\neg(\neg Q \wedge \neg R)$, "not both Q and R fail", and not to $Q \vee R$.
3. Section 2.4 noted, after Example (Negating both sides of an iff), that its converse, $\neg P \leftrightarrow \neg Q \vdash P \leftrightarrow Q$, resists the rules so far. The natural box on $\neg Q$, inside the box of P, closes to $\neg\neg Q$, not Q.

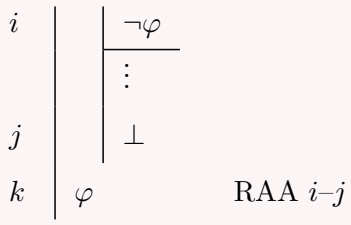
Try the smallest case, P from $\neg\neg P$. The goal P is a letter, so no introduction rule applies: the goal-directed moves have nothing to say. The premise $\neg\neg P$ is a negation, and a negation can be used only by $\neg$E, against the statement it negates, $\neg P$. So we assume $\neg P$ in a box. The box reaches $\bot$ at once, and $\neg$I closes it with $\neg\neg P$: the premise again. Every attempt goes round this circle.

In fact no derivation of P from $\neg\neg P$ uses only the rules of Chapter 1 and Sections 2.1–2.4. That is a claim about **all** derivations, and no search can prove it. Chapter 3 proves it (Theorem 3.5.4), by reading statements with more than two truth values. For the other two stalls the same is true, but we state it without proof, and nothing in this book depends on it. Here we name the missing step and add it as a rule.

2.5.3 Reductio ad absurdum

To prove φ, it is enough to show that "not φ" leads to a contradiction.

Definition 2.5.1. Reductio ad Absurdum



This is the rule RAA (**reductio ad absurdum**, "reduction to the absurd"). The line $\neg\varphi$ is an assumption, which opens the box and is **discharged** when the box closes. The line $\bot$ is a line **of that box itself**, not of a box inside it.

As in every schema (Definition 1.2.1), φ stands for any statement and i, j, k for line numbers. Read the rule clause by clause.

1. **The assumption is the exact negation of the goal.** To prove φ, the box opens with $\neg\varphi$: the negation of the **whole** of φ. If φ is $Q \vee R$, the box opens with $\neg(Q \vee R)$, not with $\neg Q \vee R$.
2. **The box must reach** $\bot$, as a line of the box itself: a $\bot$ inside a smaller box depends on a further assumption, as in ¬I.
3. **The result is** φ **itself**, not $\neg\neg\varphi$, which ¬I would give from the same box. This clause is the whole content of the rule.
4. **The box closes**, its assumption is discharged, and the justification cites the box as a range i–j.

In words: the box is a thought experiment in which φ fails. If the experiment ends in $\bot$, then φ cannot fail, and RAA lets us write φ. Set the new rule beside ¬I:

Rule	The box opens with	The box ends with	After the box
¬I	φ	$\bot$	$\neg\varphi$
RAA	$\neg\varphi$	$\bot$	φ

The rows differ only in where the ¬ sits, and that small difference makes RAA different in kind. Every other primitive rule is the introduction or the elimination rule of one connective ($\bot$E counts as the elimination rule of $\bot$), and repetition is a clause of Definition 1.4.2, not a rule for a connective. RAA belongs to no connective. It applies to every statement φ, whatever its main connective, and even to a letter, which has none.

For one kind of goal the new rule is never needed at the last step. Suppose the goal is itself a negation $\neg\psi$. An RAA box for it opens with $\neg\neg\psi$. Replace it by a ¬I box that opens with ψ; inside it, derive $\neg\neg\psi$ from ψ by the three lines of Example (Two negations from one premise) (a box on $\neg\psi$ that clashes with ψ); then continue as the RAA box did, citing that line wherever the RAA box cited its assumption. The box still ends in $\bot$, and ¬I closes it with $\neg\psi$. So the **last** step of a derivation of a negation never needs RAA, though steps inside its box still may. RAA earns its place on the other goals.

Warning. The assumption that opens an RAA box must be the **exact** negation of the goal. To prove $x \leq 1$ by contradiction, assume $x > 1$, since for real numbers "not $x \leq 1$" says $x > 1$. Assuming $x \geq 1$ instead is a common slip. It is not the negation: $x = 1$ satisfies both

$x \leq 1$ and $x \geq 1$. A box opened with $x \geq 1$ cannot honestly reach $\bot$ while $x = 1$ is possible, so a "contradiction" that seems to be reached hides an error.

Quick check. A box opens with the assumption $\neg(x < 0)$ and reaches $\bot$, as a line of the box itself. Which of the following may be written after it closes, and by which rule? (i) $x < 0$; (ii) $\neg\neg(x < 0)$; (iii) $x \geq 0$.

Solution.

(i) By RAA: the box opened with the negation of $x < 0$. (ii) By $\neg$I: $\neg$I writes the negation of the box's assumption, and the assumption is $\neg(x < 0)$. (iii) By neither. For real numbers $x \geq 0$ says the same as $\neg(x < 0)$, which is exactly the statement the box has refuted.

2.5.4 The move: when stuck, assume the negation of the goal

Chapter 1 and Sections 2.1–2.4 built two lists of moves. From the goal: its main connective chooses an introduction rule and so a box or two subgoals. From the premises: each available line's main connective chooses the elimination rule that uses it, and a goal $\bot$ asks for a clash. The new rule adds one move, for the moment when both lists are used up.

We are **stuck** at a gap when the goal-directed moves give nothing (the goal is a letter, or a disjunction with neither side in hand) and no premise-directed move brings the goal closer. Then open an RAA box on the negation of the goal. The new goal is $\bot$, so we hunt for a clash, and the box has given us one more line to clash with: $\neg\varphi$ itself.

When stuck, assume the negation of the goal, and hunt for a clash.

The move is a last resort, not a first one. Section 2.6 explains why it never does harm and still comes last. Four of the five derivations below reach the moment of being stuck, and their Ideas say where; the first shows the rule at its smallest.

2.5.5 Five derivations with the new rule

The degenerate case. The smallest use of RAA proves P from P.

Idea. There is nothing to find: the premise is the goal. The derivation is here only to show the rule at its smallest. The box opens with $\neg P$, and the one line it needs, $\bot$, comes from the clash of the assumption with the premise.

1	P	
2	$\quad\neg P$	
3	$\quad\bot$	$\neg$E 1, 2
4	P	RAA 2–3

It is legal and pointless, since line 1 already is the goal. It is the purest case of a pattern that turns up in real proofs: a contradiction box whose assumption does nothing except clash with a line we already had. The warnings at the end of this section return to it.

The stall, resolved. The next result is the one on which we got stuck.

Proposition 2.5.1. Double Negation

For every statement P:

(a) $P \vdash \neg\neg P$, by a derivation that does not use RAA;
(b) $\neg\neg P \vdash P$, by a derivation that uses RAA.

Idea. In (a) the goal $\neg\neg P$ is the negation of $\neg P$, so the goal-directed move of Section 2.2 opens a $\neg$I box with $\neg P$, and the premise clashes with it at once.

In (b) the goal P is a letter, so no introduction rule applies. The only premise is a negation, which can only clash; the attempt above showed that clashing it inside a $\neg$I box leads back to the premise. We are stuck, so we assume the negation of the goal: an RAA box on $\neg P$. The premise clashes with it at once.

Proof.

(a) This is the derivation of Example (Two negations from one premise) (a), which uses the rules of Definition 2.2.1 alone; we repeat it to set it beside (b).

1	P	
2	$\quad\neg P$	
3	$\quad\bot$	$\neg$E 1, 2
4	$\neg\neg P$	$\neg$I 2–3

(b) An RAA box on $\neg P$:

1	$\neg\neg P$	
2	$\quad\neg P$	
3	$\quad\bot$	$\neg$E 2, 1
4	P	RAA 2–3

Line 3 applies $\neg$E to the statement $\neg P$ on line 2 and to its negation $\neg\neg P$ on line 1. This proves both parts.

■

In both derivations, lines 2 and 3 form a box that opens with $\neg P$ and reaches $\bot$ by a clash with line 1. Only the rule that closes the box differs, and with it the last line. So the two directions are not mirror images: one needs only the rules for $\neg$, and the derivation of the other uses the new rule.

Nonexample by minimal change. Change only the last line of (a), from $\neg\neg P$ to P, keeping the justification ¬I 2–3. Lines 1–3 still check. What fails is the conclusion of ¬I, which must be the negation of the box's assumption, here $\neg\neg P$. To write P, the justification must be RAA.

From now on both parts are derived rules in the sense of Definition 2.3.1. A line $\neg\neg\varphi$ may be justified by "¬¬I i", citing a line φ, and a line φ by "¬¬E i", citing a line $\neg\neg\varphi$. The first stall of the list above is now one line: from $\neg\neg(x > 0)$, ¬¬E gives $x > 0$.

Contraposition, backwards. Proposition 2.3.2 derives $\neg Q \to \neg P$ from $P \to Q$ with the rules for $\to$ and $\neg$ alone. In practice we use it the other way round. To prove "if n^2 is even then n is even", one may prove "if n is not even then n^2 is not even", and the step from that back to the original goes from $\neg Q \to \neg P$ to $P \to Q$.

Proposition 2.5.2. Converse of the Contrapositive

For all statements P and Q, $\neg Q \to \neg P \vdash P \to Q$.

Idea. The goal is an implication, so the goal-directed move opens a box with P, and the new goal is Q, a letter. The premise needs $\neg Q$ before it gives anything, and $\neg Q$ is nowhere; the box's assumption P is a letter and cannot be taken apart. We are stuck at Q, so we assume its negation, $\neg Q$. That is exactly what the premise wanted: →E gives $\neg P$, which clashes with the P of the outer box, and RAA closes the inner box with Q.

Proof. The derivation is:

1	$\neg Q \to \neg P$	
2	$\quad P$	
3	$\quad\quad \neg Q$	
4	$\quad\quad \neg P$	→E 1, 3
5	$\quad\quad \bot$	¬E 2, 4
6	$\quad Q$	RAA 3–5
7	$P \to Q$	→I 2–6

Line 5 may cite line 2 because line 2 lies in a box that is still open. RAA enters at line 6. This proves the proposition.

■

So contraposition behaves like double negation: going from $P \to Q$ to $\neg Q \to \neg P$ needs no RAA, and our derivation of the way back uses one RAA. That no derivation of the way back can avoid it is true, but we state it without proof. When Q is itself a negation, the inner goal is a negation, and by the paragraph after Definition 2.5.1 that RAA can be replaced by ¬I.

Excluded middle. Every integer is even or odd. Every real number is ≥ 0 or < 0. The purely logical form of such facts is the law $P \vee \neg P$, which the next result derives from no premises at all.

Proposition 2.5.3. Excluded Middle

For every statement P, $\vdash P \vee \neg P$.

Idea. The goal is a disjunction, so Section 1.5's move postpones it: choose a side only when that side is in hand. But there are no premises, so no premise-directed move applies, and neither side can be derived on its own (informally: P may be true or false; Chapter 3 makes this exact). We are stuck, so we assume the negation of the whole goal, $\neg(P \vee \neg P)$, which says that neither side holds, and hunt for a clash. The new line can clash only with $P \vee \neg P$, which $\vee$I builds from either side. So we manufacture a side, as in Example (Manufacturing the other half of a clash). If P held, then $P \vee \neg P$ would hold, against the assumption; so $\neg P$, by $\neg$I. But then $P \vee \neg P$ holds after all, and that is the final clash.

Proof. The derivation is:

1	$\neg(P \vee \neg P)$	
2	$\quad P$	
3	$\quad P \vee \neg P$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg P$	$\neg$I 2–4
6	$P \vee \neg P$	$\vee$I 5
7	$\bot$	$\neg$E 6, 1
8	$P \vee \neg P$	RAA 1–7

The derivation has no premises, so it shows $\vdash P \vee \neg P$. RAA is used once, at line 8. This proves the proposition.

■

The assumption of line 1 is used twice, at lines 4 and 7. The first use refutes P; the second uses that refutation. Excluded middle now becomes a derived rule with no premises: a line $\varphi \vee \neg\varphi$ may be written anywhere, for any statement φ, justified by LEM with no line cited. Pasting in the eight lines above would replace it.

Its use is **proof by cases on a statement**. To derive χ, derive it once in a box that assumes φ and once in a box that assumes $\neg\varphi$, and combine the two boxes with the line $\varphi \vee \neg\varphi$ by $\vee$E. The paragraph after De Morgan's laws below uses it that way, and so does part (b) of the result after them.

A law with no negation. The last of the five is a law in which $\neg$ does not occur at all.

Proposition 2.5.4. Peirce's Law

For all statements P and Q, $\vdash ((P \to Q) \to P) \to P$.

Try it first (ten minutes): use the goal-directed and premise-directed moves of the earlier sections on this goal until they stall. At which gap do they stall, and which line, if it were available there, would let you finish?

Idea. The goal is an implication, so we open a box on $(P \to Q) \to P$ with the goal P, a letter. The only line that can produce P is the assumption itself, by $\to$E, once we have its hypothesis $P \to Q$. So $P \to Q$ becomes a subgoal, and the goal-directed move opens a box on P with the goal Q. There the moves stall: the resources are $(P \to Q) \to P$ and P, and neither says anything about Q. What the stalled gap lacks is a clash, which $\bot$E would turn into Q.

We are stuck, so the new move applies, but where? Assuming $\neg Q$ at the stalled gap gives nothing new: a clash with $\neg Q$ needs Q, the stalled goal itself. So apply the move one level out, to the goal P, and assume $\neg P$. That is exactly what the stalled gap lacked: inside the box of P, the pair P, $\neg P$ gives $\bot$, and $\bot$E gives Q. Then $P \to Q$, then P by $\to$E, which clashes with $\neg P$.

Proof. The derivation is:

1	$(P \to Q) \to P$	
2	$\quad \neg P$	
3	$\quad\quad P$	
4	$\quad\quad \bot$	$\neg$E 3, 2
5	$\quad\quad Q$	$\bot$E 4
6	$\quad P \to Q$	$\to$I 3–5
7	$\quad P$	$\to$E 1, 6
8	$\quad \bot$	$\neg$E 7, 2
9	P	RAA 2–8
10	$((P \to Q) \to P) \to P$	$\to$I 1–9

The derivation has no premises, and RAA enters at line 9. This proves the proposition.

■

The assumption $\neg P$ of line 2 is used twice: at line 4 it makes the innermost box contradictory, and at line 8 it clashes with the P that line 7 obtained. Notice where the new box went: around the stalled box, not inside it. An assumption made one level out is available in every gap inside its box, and so it could give the stalled gap the line it lacked. Peirce's law contains no $\neg$, yet our derivation uses RAA: the classical rule is about more than negation. (That every derivation of it must use RAA, or one of the two rules this section shows equivalent to it, is true, and Chapter 3 proves it in the remark after Theorem 3.5.4, by the same reading with more than two truth values that settles $\neg\neg P \vdash P$.)

2.5.6 De Morgan's laws

The negation of an "and" is an "or" of negations, and the negation of an "or" is an "and" of negations. For example, a real number x fails to lie in the interval $[0, 1]$, that is, $\neg(0 \leq x \wedge x \leq 1)$, exactly when $x < 0$ or $x > 1$. Each law has two directions, so there are four derivations to give.

Proposition 2.5.5. De Morgan's Laws

For all statements P and Q:

(a) $\neg(P \vee Q) \vdash \neg P \wedge \neg Q$;
(b) $\neg P \wedge \neg Q \vdash \neg(P \vee Q)$;
(c) $\neg P \vee \neg Q \vdash \neg(P \wedge Q)$;
(d) $\neg(P \wedge Q) \vdash \neg P \vee \neg Q$.

Try it first (ten minutes): look for a derivation of (d) with the rules of the earlier sections. Which rule would give its last line, and what would that rule need?

Idea. Parts (a)–(c) yield to their goals. In (a) the goal is a conjunction of negations: two ¬I boxes, one per conjunct. In (b) and (c) the goal is a negation: one ¬I box, with the disjunction used inside it by ∨E.

Part (d) resists. Its goal is a disjunction, so the natural last line is ∨I, from $\neg P$ or from $\neg Q$. But neither follows from the premise alone: with P true and Q false the premise is true and $\neg P$ false; with P false and Q true the premise is true and $\neg Q$ false; and the rules never lead from true premises to a false conclusion (Chapter 3 proves this). The premise itself can be used only by ¬E, against $P \wedge Q$, which we do not have. We are stuck at the whole goal, so we assume its negation, $\neg(\neg P \vee \neg Q)$, which says that neither $\neg P$ nor $\neg Q$ holds. It should give P and Q. Each is a letter, so each gets a small RAA box of its own, whose clash is manufactured by ∨I. Together P and Q clash with the premise.

Proof.

(a) Two ¬I boxes, one for each conjunct, then ∧I:

1	$\neg(P \vee Q)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg P$	$\neg$I 2–4
6	$\quad Q$	
7	$\quad P \vee Q$	$\vee$I 6
8	$\quad \bot$	$\neg$E 7, 1
9	$\neg Q$	$\neg$I 6–8
10	$\neg P \wedge \neg Q$	$\wedge$I 5, 9

(b) One $\neg$I box that opens with $P \vee Q$. Inside it, $\vee$E splits into two cases, and each case clashes with one half of the premise:

1	$\neg P \wedge \neg Q$	
2	$\neg P$	$\wedge$E 1
3	$\neg Q$	$\wedge$E 1
4	$\quad P \vee Q$	
5	$\quad\quad P$	
6	$\quad\quad \bot$	$\neg$E 5, 2
7	$\quad\quad Q$	
8	$\quad\quad \bot$	$\neg$E 7, 3
9	$\quad \bot$	$\vee$E 4, 5–6, 7–8
10	$\neg(P \vee Q)$	$\neg$I 4–9

(c) The same shape as (b), with the roles of $\wedge$ and $\vee$ exchanged:

1	$\neg P \vee \neg Q$	
2	$\quad P \wedge Q$	
3	$\quad P$	$\wedge$E 2
4	$\quad Q$	$\wedge$E 2
5	$\quad\quad \neg P$	
6	$\quad\quad \bot$	$\neg$E 3, 5
7	$\quad\quad \neg Q$	
8	$\quad\quad \bot$	$\neg$E 4, 7
9	$\quad \bot$	$\vee$E 1, 5–6, 7–8
10	$\neg(P \wedge Q)$	$\neg$I 2–9

(d) RAA on the goal, and inside it two smaller RAA boxes:

1	$\neg(P \wedge Q)$	
2	$\quad \neg(\neg P \vee \neg Q)$	
3	$\quad\quad \neg P$	
4	$\quad\quad \neg P \vee \neg Q$	$\vee$I 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad P$	RAA 3–5
7	$\quad\quad \neg Q$	
8	$\quad\quad \neg P \vee \neg Q$	$\vee$I 7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad Q$	RAA 7–9
11	$\quad P \wedge Q$	$\wedge$I 6, 10
12	$\quad \bot$	$\neg$E 11, 1
13	$\neg P \vee \neg Q$	RAA 2–12

The derivations of (a), (b), and (c) close every box by $\neg$I or $\vee$E and use no RAA. The derivation of (d) uses RAA at lines 6, 10, and 13. This proves all four laws.

■

Laws (a)–(c) conclude negations or a conjunction of negations, and $\neg$I builds negations. Law (d) concludes a **disjunction**, and $\vee$I needs to know **which** side holds, while the premise says only that P and Q do not both hold. RAA gets round the missing decision without making it.

Excluded middle makes the decision instead. Split on $P \vee \neg P$. In the case $\neg P$, $\vee$I gives $\neg P \vee \neg Q$ at once. In the case P, a $\neg$I box on Q builds $P \wedge Q$, which clashes with the premise, giving $\neg Q$; then $\vee$I gives $\neg P \vee \neg Q$. Written out, this route is 11 lines, with LEM as its one classical line. Both routes use a classical rule. That no derivation of (d) avoids one is true, but we state it without proof, and nothing in this book depends on it.

Quick check. Suppose RAA were deleted from the list of rules. Which of the four derivations in the proof of Proposition 2.5.5 would still be correct, and what is the first line of the others that would lose its justification?

Solution. The derivations of (a), (b), and (c) would still be correct: they use $\neg$I, $\neg$E, $\vee$I, $\vee$E, $\wedge$I, and $\wedge$E only. In the derivation of (d), line 6 is the first to lose its justification: it closes the box that opened with $\neg P$ and writes P, which is RAA. (Lines 10 and 13 lose theirs too.)

2.5.7 Implication as a disjunction

In everyday mathematics "if P then Q" and "either P fails or Q holds" are used interchangeably. One direction of that exchange needs only the earlier rules; our derivation of the other uses a

classical rule.

Proposition 2.5.6. Implication as a Disjunction

For all statements P and Q:

(a) $\neg P \vee Q \vdash P \to Q$, by a derivation that does not use RAA;

(b) $P \to Q \vdash \neg P \vee Q$, by a derivation that uses LEM.

Idea.

(a) The premise is a disjunction, so we use it by cases, each ending in the current goal $P \to Q$. In each case that goal is an implication, so we open a box with P. In the case $\neg P$, the box clashes at once and $\bot$E gives Q: this is the derivation of Proposition 2.1.2. In the case Q, the box repeats Q.

(b) The goal is a disjunction, so we postpone it. The premise is an implication, which gives Q only once we have P, and we do not. So we are stuck, and this time excluded middle is the better tool than an RAA box: we decide P first. In the case P the premise gives Q; in the case $\neg P$ the left side is in hand.

Proof.

(a) $\vee$E on the premise, with an implication built in each case:

1	$\neg P \vee Q$	
2	$\quad \neg P$	
3	$\quad\quad P$	
4	$\quad\quad \bot$	$\neg$E 3, 2
5	$\quad\quad Q$	$\bot$E 4
6	$\quad P \to Q$	$\to$I 3–5
7	$\quad Q$	
8	$\quad\quad P$	
9	$\quad\quad Q$	R 7
10	$\quad P \to Q$	$\to$I 8–9
11	$P \to Q$	$\vee$E 1, 2–6, 7–10

Line 9 repeats line 7 inside the box of line 8, because $\to$I needs Q as a line of that box itself.

(b) Cases on P:

1	$P \to Q$	
2	$P \vee \neg P$	LEM
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad \neg P \vee Q$	$\vee$I 4
6	$\quad \neg P$	
7	$\quad \neg P \vee Q$	$\vee$I 6
8	$\neg P \vee Q$	$\vee$E 2, 3–5, 6–7

Line 2 is Proposition 2.5.3 used as a derived rule, and the only RAA is inside it. This proves both parts.

■

The two routes of the section now stand side by side. RAA assumes the goal fails and hunts for a clash; LEM splits on a statement of our choosing and gives each case a resource it lacked. When a premise is waiting for a statement S (here $P \to Q$ waits for P), splitting on S is usually the shorter route.

2.5.8 One principle in three forms

RAA gave us ¬¬E and excluded middle. Had we added one of those two as the new primitive rule instead of RAA, would we have proved less?

To say this exactly, we need two rule forms and one notion. ¬¬E **taken as a rule** says: if $\neg\neg\varphi$ stands on an available line, we may write φ. LEM **taken as a rule** says: we may write $\varphi \vee \neg\varphi$ on any line, for any statement φ, citing nothing. Given a list of rules, a rule X **derives** a rule Y over that list if every use of Y in a derivation can be replaced by a fragment that uses only X and the rules of the list, without changing the premises or the last line. Then every sequent derivable with Y added to the list is derivable with X added instead.

Theorem 2.5.7. The Classical Rules Are Interderivable

Over the primitive rules other than RAA, each of the three rules RAA, ¬¬E, and LEM derives the other two. In particular, adding any one of the three to the other primitive rules gives the same derivable sequents.

Idea. Replacements can be chained, so it is enough to go once round a cycle. ① RAA derives the other two: that is Proposition 2.5.3 and Proposition 2.5.1 (b). ② ¬¬E derives RAA: ¬I closes an RAA box, which runs from $\neg\varphi$ to $\bot$, with $\neg\neg\varphi$, and ¬¬E finishes. ③ LEM derives ¬¬E: split on $\varphi \vee \neg\varphi$; the case φ is the goal, and the case $\neg\varphi$ clashes with $\neg\neg\varphi$.

Before reading on: in the fragment for ③ below, which rule does line $k+4$ use, and why may line $k+3$ cite line m?

Proof. ① Suppose a derivation uses ¬¬E at line k, writing φ and citing a line m that holds $\neg\neg\varphi$. Replace line k by the last three lines of the derivation in Proposition 2.5.1 (b), with its premise replaced by line m: a box opening with $\neg\varphi$, the line $\bot$ by ¬E citing the assumption and line m, and then φ by RAA. A use of LEM, writing $\varphi \vee \neg\varphi$, is replaced by the eight lines of the derivation in Proposition 2.5.3, with φ for P; that derivation has no premises. So RAA derives ¬¬E and LEM.

② Suppose a derivation uses RAA at line k, closing a box, lines i to j, that opens with $\neg\varphi$ and ends with $\bot$. Keep the box and replace line k by two lines:

i	$\neg\varphi$	
	⋮	
j	$\bot$	
k	$\neg\neg\varphi$	¬I i–j
$k+1$	φ	¬¬E k

Line k is correct by ¬I, since the box opens with $\neg\varphi$ and $\neg\neg\varphi$ is the negation of that assumption. Both rules ask for $\bot$ as a line of the box itself, so the box needs no change. So ¬¬E derives RAA.

③ Suppose a derivation uses ¬¬E at line k, writing φ and citing a line m that holds $\neg\neg\varphi$. Replace line k by six lines:

m	$\neg\neg\varphi$	
	⋮	
k	$\varphi \vee \neg\varphi$	LEM
$k+1$	φ	
$k+2$	$\neg\varphi$	
$k+3$	$\bot$	¬E $k+2$, m
$k+4$	φ	$\bot$E $k+3$
$k+5$	φ	$\vee$E k, $k+1$–$k+1$, $k+2$–$k+4$

The first case is a one-line box: it opens with φ and so already ends with the goal. In the second case, line $k+3$ may cite line m, because line m was available at the old line k and every new line sits in the same boxes as the old line k, or in boxes opened after it. No rule other than LEM and the rules of the list is used. So LEM derives ¬¬E.

In each replacement the later lines are renumbered, and a later citation of the old line k now cites the fragment's last line, which holds the same statement in the same boxes. The premises and the last line of the derivation are unchanged.

Finally, replacements can be chained. LEM derives RAA: replace each use of RAA by the fragment of ②, and then each use of ¬¬E so created by the fragment of ③. ¬¬E derives LEM: replace each use of LEM by the derivation of ①, and then each use of RAA in it by the fragment of ②. With ①, ②, and ③, this shows that each of the three rules derives the other two, as claimed.

■

So the classical part of our rules is **one** extra principle, whichever of its three forms we write down. Chapter 5 tells the story of the mathematicians who declined it, and shows what their mathematics looks like.

The base list matters. Part ③ uses ⊥E at line $k+4$, to close the impossible case as in Example (An impossible case), and without ⊥E among the base rules LEM no longer derives ¬¬E. We state this without proof; nothing depends on it.

Remark. Some books call exactly our RAA "proof by contradiction" or PBC, and some take ¬¬E as their primitive rule and derive RAA from it. By Theorem 2.5.7 nothing changes: all these choices prove the same sequents.

Remark. ⊥E could have been dropped from the primitive rules, since RAA and repetition derive it (Exercise B3 below asks for the derivation). We keep it so that the primitive rules other than RAA form a system of their own, the one Chapter 5 studies in its last section.

2.5.9 How reductio looks in a mathematical proof

Return to Example (A number below every positive number), and read its solution one sentence at a time, with the rule of each step in brackets.

1. "Suppose, for a contradiction, that $x \neq 0$." [An RAA box opens, with $\neg(x = 0)$, the exact negation of the goal $x = 0$.]
2. "Since $x \geq 0$ and $x \neq 0$, we have $x > 0$." Proposition 2.3.3
3. "Then $\varepsilon := x/2$ is a real number with $\varepsilon > 0$." [School algebra.]
4. "So the hypothesis applies to it and gives $x \leq x/2$." [The hypothesis is about **every** $\varepsilon > 0$; applying it to one is a rule for "every", from Chapter 4.]
5. "… gives $x/2 \leq 0$, and … $x \leq 0$." [School algebra.]
6. "This contradicts $x > 0$." [¬E: for real numbers $x > 0$ is $\neg(x \leq 0)$, the negation of the line $x \leq 0$.]
7. "Hence $x = 0$." [RAA closes the box.]

The words "for a contradiction" do not by themselves tell ¬I from RAA. Look at what is assumed and what is concluded. Here the box assumes a negation, $x \neq 0$, and the conclusion is the positive statement $x = 0$, so the step is RAA. The proof of Theorem 2.2.1 assumes the positive statement $r^2 = 2$ and concludes its negation, so it is ¬I. *Analysis on the Real Line*, Chapter 1 opens its proof that the natural numbers are not bounded above by supposing, for contradiction, that they are bounded above, and concludes that they are not: that is ¬I too, despite the phrase.

Warning. An RAA box on $\neg\varphi$ whose assumption is used **only at its last step**, against a φ derived without it, is legal, but it is a direct proof in disguise: the lines before the clash already derive φ, so the box can be deleted. The degenerate derivation of $P \vdash P$ above is the smallest case. Chapter 5 shows how to spot this costume in written proofs and take it off.

(A box that reaches $\bot$ without using its assumption at all is a different matter: the same lines, written before the box, derive $\bot$ without it, and $\bot$E then gives the goal at once.)

Warning. Not every "proof by contradiction" is RAA. Calling the proof of Theorem 2.2.1 an RAA is a mislabeling: its box opens with $r^2 = 2$, not with a negation, and it concludes $r^2 \neq 2$, which ¬I writes. The difference matters when you count classical steps: the box of that proof closes by ¬I and contributes none.

2.5.10 What the rules sound like

Written mathematics carries these rules in a handful of phrases. The Lean column is for readers who use that proof assistant; nothing later depends on it, and the full dictionary is in Chapter 5.

Rule	You write	Lean
RAA	"Suppose, for a contradiction, that φ fails. … This is impossible. Hence φ."	`by_contra h`
¬¬E	"It cannot be that φ fails; so φ."	`not_not.mp h`
LEM	"Either φ or not. If φ, … If not, … In either case, …"	`by_cases h : P`

2.5.11 Exercises

2.5.11.1 A. Check your understanding

Exercise. A1

(a) "Suppose 7 is even. Then $7 = 2k$ for some integer k, so $k = 7/2$, which is not an integer. This is impossible. Hence 7 is not even." Is the last step ¬I, RAA, or neither? What did the box open with?

(b) "Let x be real. Suppose $x^2 \geq 2x - 1$ fails, that is, $x^2 < 2x - 1$. Then $(x-1)^2 < 0$, which is impossible. Hence $x^2 \geq 2x - 1$." Same questions.

(c) "Assume P. Then, as shown above, Q. Hence $P \to Q$." Same questions.

(d) State RAA and ¬I side by side, and say which of the two the derivation of Proposition 2.5.1 (a) uses.

(e) In the derivation of Proposition 2.5.5 (d), why can the last line not be $\vee$I applied to a line derived from the premise alone?

(f) In the same derivation, the assumption $\neg(\neg P \vee \neg Q)$ on line 2 is cited twice. Name the two lines that cite it, and the rule each uses.

(g) In the fragment for ② in the proof of Theorem 2.5.7, which rule justifies line k, and why does the box of lines i to j need no change?

Solution.

(a) ¬I: the box opened with "7 is even", and the conclusion is its negation. ($7/2$ is not an integer because $3 < 7/2 < 4$.)

(b) RAA: the box opened with the negation of $x^2 \geq 2x-1$, and the conclusion is $x^2 \geq 2x-1$ itself. (The clash: $x^2 < 2x-1$ rearranges to $(x-1)^2 < 0$, and squares are nonnegative.)

(c) Neither: this is →I. The box opened with P and ended with Q; no $\bot$ appears.

(d) ¬I: a box that opens with φ and reaches $\bot$ gives $\neg\varphi$. RAA: a box that opens with $\neg\varphi$ and reaches $\bot$ gives φ. The derivation of Proposition 2.5.1 (a) uses ¬I, with $\neg P$ as the assumption, so its conclusion is the negation of $\neg P$, namely $\neg\neg P$.

(e) ∨I would need $\neg P$ or $\neg Q$ on a line derived from the premise alone. Neither can be: with P true and Q false the premise is true and $\neg P$ false, with P false and Q true the premise is true and $\neg Q$ false, and the rules never lead from true premises to a false conclusion (proved in Chapter 3, Theorem 3.5.2).

(f) Line 5 cites it, by ¬E against line 4, $\neg P \vee \neg Q$; and line 9 cites it, by ¬E against line 8, the same statement. Both lines lie in boxes inside the box of line 2, which is still open, so the citations are legal.

(g) ¬I: the box opens with $\neg\varphi$ and ends with $\bot$, so ¬I gives the negation of its assumption, $\neg\neg\varphi$. The box needs no change because ¬I, like RAA, asks only for a box that ends with $\bot$ as a line of the box itself; the two rules differ in what they write after the box, not in the box.

2.5.11.2 B. Practice

Exercise. B1: With and without the new rule

(a) Derive $\neg\neg P,\ \neg\neg(P \to Q) \vdash \neg\neg Q$ **without** RAA. Then say in one sentence why your derivation does not give Q.

(b) Derive $\neg P \to Q,\ \neg P \to \neg Q \vdash P$, using RAA.

Hint: in (a) the goal is a negation; in (b) it is a letter.

Solution.

(a) The goal $\neg\neg Q$ is a negation, so we open a ¬I box on $\neg Q$, with the goal $\bot$. The premises are negations, so each can only clash: $\neg\neg P$ against $\neg P$, and $\neg\neg(P \to Q)$ against $\neg(P \to Q)$. Both of those are negations, so each is itself a subgoal for a ¬I box. Make $\neg P$ the subgoal: a box on P. Inside it, make $\neg(P \to Q)$ the subgoal: a box on $P \to Q$, where →E gives Q, which clashes with $\neg Q$. The boxes then close from the inside out.

1	$\neg\neg P$	
2	$\neg\neg(P \to Q)$	
3	$\quad \neg Q$	
4	$\quad\quad P$	
5	$\quad\quad\quad P \to Q$	
6	$\quad\quad\quad Q$	$\to$E 5, 4
7	$\quad\quad\quad \bot$	$\neg$E 6, 3
8	$\quad\quad \neg(P \to Q)$	$\neg$I 5–7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad \neg P$	$\neg$I 4–9
11	$\quad \bot$	$\neg$E 10, 1
12	$\neg\neg Q$	$\neg$I 3–11

All three boxes close by $\neg$I, so no RAA is used. The derivation does not give Q because its last box opens with $\neg Q$, and $\neg$I can only write the negation of that assumption, $\neg\neg Q$; removing the two negations is $\neg\neg$E, a classical step.

(b) The goal P is a letter, and both premises want $\neg P$, which is nowhere. We are stuck, so we open an RAA box with $\neg P$, and it feeds both premises:

1	$\neg P \to Q$	
2	$\neg P \to \neg Q$	
3	$\quad \neg P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad \neg Q$	$\to$E 2, 3
6	$\quad \bot$	$\neg$E 4, 5
7	P	RAA 3–6

In words: "Suppose, for a contradiction, that P fails. Then the premises give both Q and $\neg Q$, which is impossible. Hence P."

Exercise. B2: Back from negated sides

Derive $\neg P \leftrightarrow \neg Q \vdash P \leftrightarrow Q$. Say where RAA enters and why.

Solution. The goal is a biconditional, so by Definition 2.4.1 we need two boxes, one from P to Q and one from Q to P. In the first, the goal Q is a letter and the premise speaks only about $\neg Q$, so we are stuck: we assume $\neg Q$ and argue by RAA. The second box is symmetric.

1	$\neg P \leftrightarrow \neg Q$	
2	$\quad P$	
3	$\quad\quad \neg Q$	
4	$\quad\quad \neg P$	$\leftrightarrow$E 1, 3
5	$\quad\quad \bot$	$\neg$E 2, 4
6	$\quad Q$	RAA 3–5
7	$\quad Q$	
8	$\quad\quad \neg P$	
9	$\quad\quad \neg Q$	$\leftrightarrow$E 1, 8
10	$\quad\quad \bot$	$\neg$E 7, 9
11	$\quad P$	RAA 8–10
12	$P \leftrightarrow Q$	$\leftrightarrow$I 2–6, 7–11

RAA enters at lines 6 and 11: each inner goal is a letter, and the premise produces only negations. In the converse, $P \leftrightarrow Q \vdash \neg P \leftrightarrow \neg Q$ (Example (Negating both sides of an iff)), the inner goals are negations, $\neg$I builds them, and no RAA is used.

Exercise. B3: The redundant rule

Show that $\bot$E is derivable from RAA and repetition: given a line $\bot$, derive an arbitrary statement P in one box, without $\bot$E.

Solution. Open an RAA box with $\neg P$ and repeat the $\bot$ inside it:

1	$\bot$	
2	$\quad \neg P$	
3	$\quad \bot$	R 1
4	P	RAA 2–3

The repetition at line 3 is needed because RAA asks for $\bot$ as a line of the box itself. So every use of $\bot$E can be replaced by these three lines; the book keeps the rule for the reason given in the remark after Theorem 2.5.7.

Exercise. B4: Part ② on an example

Here is a proof by RAA about a real number x.

> Suppose $x^2 \leq x$. Suppose, for a contradiction, that $x \leq 1$ fails, that is, $x > 1$. Then $x > 0$, and multiplying $x > 1$ by x gives $x^2 > x$, which contradicts $x^2 \leq x$. Hence $x \leq 1$.

Let P be the statement "$x \leq 1$".

(a) Which sentences of the argument form the RAA box, and what are its assumption and its last line?

(b) Apply part ② of the proof of Theorem 2.5.7 to this use of RAA. Write the statement that the new line k asserts, in symbols and in words.

(c) Which rule justifies the new line k, and which rule the line after it?

(d) Which lines of the original argument are unchanged by the replacement?

Solution.

(a) The box opens with "$x \leq 1$ fails", that is $\neg P$, and runs through the sentence ending "which contradicts $x^2 \leq x$"; its last line is $\bot$, the clash between $x^2 > x$ and the hypothesis $x^2 \leq x$. The last sentence, "Hence $x \leq 1$", is the RAA step.

(b) The new line k is $\neg\neg P$, that is $\neg\neg(x \leq 1)$: "it is not the case that $x \leq 1$ fails". The line after it is P, "$x \leq 1$".

(c) Line k is $\neg$I, closing the box that opened with $\neg P$; the next line is $\neg\neg$E, citing line k.

(d) Every line of the box is unchanged: the assumption $\neg P$ ("$x \leq 1$ fails", that is $x > 1$), the steps $x > 0$ and $x^2 > x$, and the clash. Only the closing step becomes two lines. For real numbers "not not $x \leq 1$" and "$x \leq 1$" mean the same, but for the rules they are different statements, joined by $\neg\neg$E.

2.5.11.3 C. Going deeper

Exercise. C1: Excluded middle does the work of reductio directly

Suppose that, instead of RAA, we had LEM as a rule: $\varphi \vee \neg\varphi$ may be written on any line, for any statement φ.

(a) Given a box that opens with $\neg\varphi$ and reaches $\bot$, derive φ using LEM, $\vee$E, $\bot$E, and the other primitive rules, but neither RAA nor $\neg\neg$E. Write the case $\neg\varphi$ as the given box, extended.

(b) Which two parts of the proof of Theorem 2.5.7 does your fragment shortcut, and which lines of their composite does it save?

Solution (Hint 1). Write $\varphi \vee \neg\varphi$ just before the given box. What must each of the two case boxes end with?

Solution (Hint 2). The first case box is a single line.

Solution.

(a) Write excluded middle for φ just before the given box, and use the box as the second case of $\vee$E, extended by one line:

h	$\varphi \vee \neg\varphi$	LEM
$h+1$	$\quad\vert\ \varphi$	
i	$\quad\vert\ \neg\varphi$	
	$\quad\vert\ \vdots$	
j	$\quad\vert\ \bot$	
$j+1$	$\quad\vert\ \varphi$	$\bot$E j
k	φ	$\vee$E h, $h+1$–$h+1$, i–$j+1$

Here $i = h + 2$. The first case is a one-line box that already ends with the goal. The second case is the given box with one more line, $\bot$E on its $\bot$; its lines are unchanged and cite what they cited before, since the box has not moved into or out of any other box. No RAA and no $\neg\neg$E is used.

(b) It shortcuts ② followed by ③. Part ② closes the box by $\neg$I, giving $\neg\neg\varphi$, then applies $\neg\neg$E; part ③ replaces that $\neg\neg$E by a split on $\varphi \vee \neg\varphi$ whose $\neg\varphi$ case is a new box that clashes with the line $\neg\neg\varphi$. The fragment in (a) uses the given box itself as the $\neg\varphi$ case, so it never writes $\neg\neg\varphi$. Counted exactly, it saves three lines: the line $\neg\neg\varphi$ of ②, and in ③ the assumption $\neg\varphi$ at line $k+2$ and the $\neg$E line $k+3$. The $\bot$E line $k+4$ of ③ survives, moved to the end of the given box.

Exercise. C2: Find the error

A student derives $\neg(\neg P \wedge \neg Q) \vdash P \vee Q$ as follows.

1	$\neg(\neg P \wedge \neg Q)$	
2	$P \vee \neg P$	LEM
3	$\quad\vert\ P$	
4	$\quad\vert\ P \vee Q$	$\vee$I 3
5	$\quad\vert\ \neg P$	
6	$\quad\vert\ \bot$	$\neg$E 3, 5
7	$\quad\vert\ Q$	$\bot$E 6
8	$\quad\vert\ P \vee Q$	$\vee$I 7
9	$P \vee Q$	$\vee$E 2, 3–4, 5–8

(a) Is the sequent $\neg(\neg P \wedge \neg Q) \vdash P \vee Q$ derivable?

(b) Does the student's derivation derive it? Name the first line that fails and the rule or clause it breaks.

(c) Show that the illegal step, if it were allowed, would prove too much: use the same move to "derive" $P \vee Q \vdash P$, and say why that sequent must not be derivable.

(d) Give a correct derivation.

Solution.

(a) Yes: part (d) below gives a derivation. The conclusion is true, and the argument is still wrong.

(b) No. Line 6 fails: it cites line 3, the assumption of the first case box, lines 3–4, which closed before line 5. By the scope rule of Definition 1.4.2, line 3 is not available at line 6. The form of ¬E is right; what is broken is availability. In words, the student used the assumption of one case inside the other case, where it has not been assumed.

(c) The same move derives $P \vee Q \vdash P$: split on the premise by ∨E, with a first box that opens with P and ends there, and a second box that opens with Q and "repeats" P from the first. That sequent must not be derivable: with P false and Q true the premise is true and the conclusion false, and Chapter 3 proves that no derivation leads from true premises to a false conclusion (Theorem 3.5.2). So citing a line of a sibling box is illegal in general, not just unlucky here.

(d) Keep lines 1–5, and in the case $\neg P$ we are stuck at the goal Q, so we argue for it by RAA, using the premise:

1	$\neg(\neg P \wedge \neg Q)$	
2	$P \vee \neg P$	LEM
3	$\quad P$	
4	$\quad P \vee Q$	∨I 3
5	$\quad \neg P$	
6	$\quad\quad \neg Q$	
7	$\quad\quad \neg P \wedge \neg Q$	∧I 5, 6
8	$\quad\quad \bot$	¬E 7, 1
9	$\quad Q$	RAA 6–8
10	$\quad P \vee Q$	∨I 9
11	$P \vee Q$	∨E 2, 3–4, 5–10

Line 7 cites line 5, the assumption of the enclosing case box, which is still open: the legal version of what the student wanted.

Exercise. C3: Exit check: a classical disjunction

Derive $\vdash P \vee (P \to Q)$ using the **primitive** rules only (those of Chapter 1, ¬E, ⊥E, ¬I, ↔I, ↔E, and RAA), with no derived rule, so neither LEM nor ¬¬E as a single line. Name the rule on every line, and say which line is classical.

Solution. The search. The goal is a disjunction, so we postpone it; there are no premises, so no premise-directed move applies; and neither side is derivable on its own (P may be false, and $P \to Q$ is false when P is true and Q false). We are stuck, so we assume the exact negation of the goal, $\neg(P \vee (P \to Q))$, and hunt for a clash. The assumption can clash only with $P \vee (P \to Q)$, which ∨I builds from either side. The side $P \to Q$ is an implication, so

we open a box on P with the goal Q. Inside it, $\vee$I gives $P \vee (P \to Q)$ from P, which clashes with the assumption, and $\bot$E writes Q. So $P \to Q$ holds, $\vee$I gives the goal, and that is the second clash.

1	$\neg(P \vee (P \to Q))$	
2	$\quad P$	
3	$\quad P \vee (P \to Q)$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\quad Q$	$\bot$E 4
6	$P \to Q$	$\to$I 2–5
7	$P \vee (P \to Q)$	$\vee$I 6
8	$\bot$	$\neg$E 7, 1
9	$P \vee (P \to Q)$	RAA 1–8

Lines 1 and 2 are assumptions, of the RAA box and of the $\to$I box; every other line names its rule on the right. The classical line is line 9; every other rule used is a rule for a connective. The shape is that of Proposition 2.5.3: there the inner box closed by $\neg$I to give $\neg P$; here it ends with $\bot$E and closes by $\to$I to give $P \to Q$.

A second correct route first derives $P \vee \neg P$ by the eight lines of Proposition 2.5.3, written out, and then splits on it: in the case P, $\vee$I gives the goal; in the case $\neg P$, a box on P clashes with $\neg P$, $\bot$E writes Q, $\to$I gives $P \to Q$, and $\vee$I the goal. Its classical line is the RAA inside the written-out excluded middle.

Marking scheme. (10 points)

- 2: an RAA box opened with $\neg(P \vee (P \to Q))$, the exact negation of the goal.
- 3: an inner box on P that reaches $\bot$ by $\vee$I and $\neg$E against the outer assumption, and then Q by $\bot$E.
- 2: $P \to Q$ by $\to$I, then $P \vee (P \to Q)$ by $\vee$I, and the second clash.
- 2: RAA closes the outer box, and it is named as the classical step.
- 1: every citation is available.

The second route earns the same points, step for step: 2 for the written-out excluded middle opened correctly (its RAA box on $\neg(P \vee \neg P)$); 3 for the box on P in the case $\neg P$, closed by $\bot$E; 2 for $\to$I, the two $\vee$I lines, and the $\vee$E; 2 for naming the RAA inside excluded middle as the classical step; 1 for availability. Fatal errors (0 for the step): opening the outer box with anything but $\neg(P \vee (P \to Q))$; closing the outer box by $\neg$I (that gives $\neg\neg(P \vee (P \to Q))$); a line justified by $\neg\neg$E or LEM taken as a rule, which the item excludes.

2.6 How to Find a Derivation

We now have every rule for the connectives. Checking a derivation is mechanical, as Definition 1.4.2 makes it: for each line, which rule, which lines, and are they available. Finding one is not, since an assumption may be any statement at all. Yet every Idea in Chapters 1 and 2 found its derivation with the same handful of moves. This section introduces no new rule. It collects those moves into one method, puts them in order, says why the order is right, and then

watches the method work: on four searches that succeed, one that fails, and one mathematical goal. It ends with every rule in one place.

2.6.1 The moves so far

Each section since Section 1.2 has added a move, either to the list read off the **goal** (goal-directed: which introduction rule builds the goal's main connective?) or to the list read off the **premises** (premise-directed: which elimination rule uses an available line's main connective?). Here they are together, with the section where each first appeared.

The situation	The move	First met
goal $\varphi \wedge \psi$	two subgoals, φ and ψ, joined by $\wedge$I	Section 1.2
resource $\varphi \wedge \psi$	take it apart by $\wedge$E	Section 1.2
resource $\varphi \to \psi$	fire it on φ by $\to$E; if φ is missing, make it a subgoal	Section 1.3
goal $\varphi \to \psi$	a box opened with φ, with the new goal ψ	Section 1.4
goal $\varphi \vee \psi$	postpone; choose a side only when it is in hand	Section 1.5
resource $\varphi \vee \psi$	two cases, each ending in the current goal	Section 1.6
goal $\bot$	find a clash, or manufacture one; an impossible case closes itself by $\bot$E	Section 2.1
goal $\neg\varphi$	a box opened with φ, with the new goal $\bot$	Section 2.2
a pattern with a name	use the derived rule as one line (MT, DS, ...)	Section 2.3
goal $\varphi \leftrightarrow \psi$	two boxes, one for each direction	Section 2.4
stuck	assume the negation of the goal (RAA), or split on $\chi \vee \neg\chi$ (LEM)	Section 2.5

Eleven moves is a lot to hold in mind at a blank page. The method below turns them into an order of attack. To state it we need a name for a derivation that is only half written.

2.6.2 Partial derivations

A search builds a derivation from the outside in: the last line first, then the boxes the goal asks for, then the lines in between. What exists halfway is not yet a derivation, but it is close to one.

Definition 2.6.1. Partial Derivation

A **partial derivation** is a list of lines laid out as a derivation (Definition 1.4.2), except that some stretches of lines are still missing. Each missing stretch is a **gap**, drawn as a row ⋮. The line just below a gap is its **current goal**; it is already written, but it has no justification

yet, since the lines of the gap are to supply one. The lines available at the current goal, in the sense of Definition 1.4.2, are the gap's **resources**.

In words: every line that has been written is a premise, an assumption, a current goal, or a line justified by a rule from lines and boxes available to it; the boxes are nested and closed as in any derivation; only the gaps are missing. A search starts with a single gap, below the premises and above the conclusion, and each step replaces a gap by lines, boxes, and smaller gaps. It ends when no gap is left, and then the partial derivation is a derivation.

The **degenerate** partial derivation is the starting point of every search: the premises, one gap, and the goal. Its resources are the premises. The snapshots in the worked searches below are all partial derivations; a line carries the number it will have in the finished derivation, so that snapshots can be compared, and a line whose place is not yet fixed carries no number. On paper you would number the lines at the end.

Nonexample. Take a snapshot and leave one of its boxes open at the bottom, so that the conclusion sits inside the box. That is not a partial derivation, even though every line is correct so far: Definition 1.4.2 asks that every box close and that the last line lie outside every box, and a gap cannot repair that, since filling a gap adds lines only above its current goal.

Quick check. In the second snapshot of Example (Splitting the goal, then using the premise) below, the second gap has the current goal $\bot$ on line 11. List its resources.

Solution. Lines 1, 7, 8, and 9: the premise $P \to \neg Q$, the line $\neg(P \wedge Q)$, and the assumptions Q and P of the two boxes that contain the gap. Line 7 lies in no box and above the gap, so it is available, though this gap will not need it. Lines 2–6 are not resources: they lie in a box that has closed.

2.6.3 The method

Work from both ends: let the goal choose the boxes, let the premises choose the eliminations, and meet in the middle.

Algorithm. **Searching for a derivation of $\Gamma \vdash \varphi$.**

1. **Set up.** Write the premises Γ at the top and φ at the bottom, with one gap between.
2. **Goal-directed step.** Take the main connective of the current goal (Definition 1.1.2) and set up its introduction rule:
   - $\psi \wedge \chi$: two gaps, with goals ψ and χ (Section 1.2);
   - $\psi \to \chi$: a box opened with ψ; inside it, the new goal is χ (Section 1.4);
   - $\neg\psi$: a box opened with ψ; inside it, the new goal is $\bot$ (Section 2.2);
   - $\psi \leftrightarrow \chi$: two boxes, one from ψ to χ and one from χ to ψ (Section 2.4);
   - $\psi \vee \chi$: **postpone**; do not choose a side yet (Section 1.5), unless the disjunction is being manufactured for a clash with its own negation (step 4), when either side may be chosen (see below).

   Repeat while the current goal is a $\wedge$, $\to$, $\neg$, or $\leftrightarrow$.
3. **Premise-directed step.** Take the main connective of a resource and apply its elimination rule: $\wedge$E freely (Section 1.2); $\to$E and $\leftrightarrow$E when the other statement they need is a resource, or can be made a new goal (Sections 1.3 and 2.4); $\vee$E by opening two

boxes, one per disjunct, **each ending in the current goal** (Section 1.6). A derived rule may do the work of several lines (Section 2.3): a resource $\psi \to \chi$ with $\neg\chi$ available gives $\neg\psi$ by MT; a resource $\psi \vee \chi$ with $\neg\psi$ available gives χ by DS.

4. **Clash.** If the current goal is $\bot$, look for a pair χ, $\neg\chi$ among the resources, or make χ the new goal for some resource $\neg\chi$ and manufacture it, starting again at step 2 for that goal. Whatever the current goal, if the resources clash, write $\bot$ and then the goal by $\bot$E: an impossible case closes itself (Section 2.1).
5. **Stuck.** Use RAA: open a box with the negation of the current goal, whose new goal is $\bot$; if that gives nothing, try the same with a goal further out whose box contains the stalled gap. Or use LEM on a well-chosen statement χ, and split on $\chi \vee \neg\chi$ by $\vee$E. Then return to step 3 (Section 2.5).

When no gap is left, number the lines and check the derivation as Definition 1.4.2 says.

The steps are listed in the order in which to try them, and the order has reasons.

The goal-directed steps lose nothing. Suppose the goal $\psi \to \chi$ is derivable from the resources Γ at all, by some derivation D. Then χ is derivable from Γ together with ψ: add ψ to the premises of D (Definition 1.2.3 asks only that the premises belong to the list), and append the line χ by $\to$E, citing the last line of D and the premise ψ. So opening a box on ψ never turns a derivable gap into an underivable one. With $\wedge$E, $\neg$E, or $\leftrightarrow$E as the appended line, the same argument covers $\wedge$, $\neg$, and $\leftrightarrow$. That is why step 2 may be applied without thinking.

Choosing a side of $\vee$ can lose everything. A disjunction can be derivable when neither side is. From $P \vee Q$ the goal $Q \vee P$ is derivable (Example (First derivations with "or")), but $P \vee Q$ does not say which side holds, so a last line $\vee$I from Q, or from P, leaves a gap that cannot be filled (Chapter 3 makes "cannot" exact, Corollary 3.5.3). Choosing a side is safe only once the resources decide it, typically inside a case of $\vee$E, or when the disjunction is being manufactured for a clash (step 4) with a resource that is its negation: then the resources derive $\bot$ as soon as they derive the disjunction, and from $\bot$ every statement, either side included, by $\bot$E.

Step 5 also loses nothing, and still comes last. If φ is derivable from the resources, then $\bot$ is derivable from them together with $\neg\varphi$, by one appended $\neg$E; so an RAA box is always safe. A split on $\chi \vee \neg\chi$ is safe too, since each case only adds a resource. Step 5 comes last because the goal $\bot$ gives step 2 nothing to work with, because a good χ must be chosen, and because a derivation that did not need the classical step reads worse with it (the costume of Section 2.5).

Quick check. For each goal, with no premises, name the first rule the method aims for and the box or gaps it sets up.

(a) $\neg(P \wedge \neg P)$
(b) $(P \vee Q) \to (Q \vee P)$
(c) $(P \to Q) \wedge (Q \to (P \to Q))$
(d) $P \vee \neg(P \wedge Q)$

Solution.

(a) $\neg$I: a box opened with $P \wedge \neg P$, goal $\bot$. (b) $\to$I: a box opened with $P \vee Q$, goal $Q \vee P$; that goal is a disjunction, so it is postponed, and step 3 uses the new resource $P \vee Q$ by $\vee$E. (c) $\wedge$I: two gaps, with goals $P \to Q$ and $Q \to (P \to Q)$, each then a box. (d)

Step 2 postpones the disjunction, and with no premises steps 3 and 4 have nothing to use, so the search is at step 5 at once. LEM on P works: in the case P, $\vee$I gives the left side; in the case $\neg P$, a $\neg$I box on $P \wedge Q$ clashes ($\wedge$E gives P) and gives $\neg(P \wedge Q)$, and $\vee$I gives the right side.

2.6.4 Splitting the goal

We now run the method. The first search uses only steps 2, 3, and 4. It is the full version of the chapter's main move, *read the skeleton off the goal, then work from both ends*, which comes back twice in the exercises with parts left blank.

Example. Splitting the goal, then using the premise

Find a derivation of $P \to \neg Q \vdash \neg(P \wedge Q) \wedge (Q \to \neg P)$.

Solution. Step 1. The premise on top, the goal at the bottom, one gap. The resources of the gap are the premise alone.

Step 2, first pass. The goal's main connective is $\wedge$, so it splits into two gaps, with goals $\neg(P \wedge Q)$ and $Q \to \neg P$, joined at the end by $\wedge$I:

1	$P \to \neg Q$	
	$\vdots$	
7	$\neg(P \wedge Q)$	
	$\vdots$	
13	$Q \to \neg P$	
14	$\neg(P \wedge Q) \wedge (Q \to \neg P)$	$\wedge$I 7, 13

Step 2, again. Both new goals have a connective, so step 2 applies to each. The goal $\neg(P \wedge Q)$ asks for a $\neg$I box opened with $P \wedge Q$ and ending in $\bot$. The goal $Q \to \neg P$ asks for a $\to$I box opened with Q; inside it the goal $\neg P$ asks, in turn, for a $\neg$I box opened with P. Every gap now has the goal $\bot$, and step 2 has nothing more to do:

1	$P \to \neg Q$	
2	$\quad P \wedge Q$	
	$\quad \vdots$	
6	$\quad \bot$	
7	$\neg(P \wedge Q)$	¬I 2–6
8	$\quad Q$	
9	$\quad\quad P$	
	$\quad\quad \vdots$	
11	$\quad\quad \bot$	
12	$\quad \neg P$	¬I 9–11
13	$Q \to \neg P$	→I 8–12
14	$\neg(P \wedge Q) \wedge (Q \to \neg P)$	∧I 7, 13

Steps 3 and 4, first gap. The resources are lines 1 and 2. Step 3 splits line 2 into P and Q by ∧E. Now P is the hypothesis of line 1, so →E gives $\neg Q$. Step 4 finds the pair Q, $\neg Q$.

Steps 3 and 4, second gap. The resources are lines 1, 7, 8, and 9, and line 7 is not needed. →E on lines 1 and 9 gives $\neg Q$, which clashes with line 8. Both gaps close:

1	$P \to \neg Q$	
2	$\quad P \wedge Q$	
3	$\quad P$	∧E 2
4	$\quad Q$	∧E 2
5	$\quad \neg Q$	→E 1, 3
6	$\quad \bot$	¬E 4, 5
7	$\neg(P \wedge Q)$	¬I 2–6
8	$\quad Q$	
9	$\quad\quad P$	
10	$\quad\quad \neg Q$	→E 1, 9
11	$\quad\quad \bot$	¬E 8, 10
12	$\quad \neg P$	¬I 9–11
13	$Q \to \neg P$	→I 8–12
14	$\neg(P \wedge Q) \wedge (Q \to \neg P)$	∧I 7, 13

What the method did. The goal-directed steps wrote nine of the fourteen lines, every box among them, before the premise was used; the premise-directed steps wrote four, lines 3, 4, 5, and 10. The premise itself was used once in each gap, by the same →E. Line 10 cites line 1 from inside two boxes, which is legal because line 1 is a premise and so is available

everywhere below it.

Notice how little thinking the search needed. Step 2 is mechanical, and it did most of the work: by the time the premise was consulted, each gap was small, with the goal $\bot$ and two or three resources. That is the typical shape of a successful search: **the goal fixes the skeleton, and the premises fill in the flesh.**

2.6.5 Cases must end in the goal

When the goal gives nothing, the premises must. The next search is driven by a disjunction among the premises.

Example. Proof by cases, found by the method

Find a derivation of $P \vee Q,\ P \to R,\ Q \to R \vdash R$.

Solution. The goal R is a letter, so step 2 has nothing to do. Step 3 looks at the resources. $\to$E on $P \to R$ needs P, and making P a new goal is hopeless: from $P \vee Q$ we know that one of P, Q holds, not which. The disjunction itself is the resource to use, by $\vee$E, with two boxes that each end in the current goal R:

1	$P \vee Q$	
2	$P \to R$	
3	$Q \to R$	
4	$\quad P$	
	$\quad \vdots$	
5	$\quad R$	
6	$\quad Q$	
	$\quad \vdots$	
7	$\quad R$	
8	R	$\vee$E 1, 4–5, 6–7

Inside the first box, P is a resource, so $\to$E on line 2 closes the gap at once; the second box is the same with line 3:

1	$P \vee Q$	
2	$P \to R$	
3	$Q \to R$	
4	$\quad P$	
5	$\quad R$	$\to$E 2, 4
6	$\quad Q$	
7	$\quad R$	$\to$E 3, 6
8	R	$\vee$E 1, 4–5, 6–7

This is the pattern that Section 1.6 called the skeleton of almost every proof by cases (compare Example (Cases on a hypothesis are two implications), which packages the same two boxes into one implication). The method found it without being told.

Why the boxes end in the current goal. $\vee$E (Definition 1.6.1) gives, after the two boxes, their common last line and nothing else. A box that ends in anything but the current goal leaves that goal still to be reached, and two boxes that end in different statements cannot be combined at all. The word **current** matters: had the goal been $S \to R$, step 2 would first open a box on S, the current goal inside it would be R, and the two case boxes, opened inside the box of S, would end in R, not in $S \to R$.

2.6.6 Postponing a disjunction

The third search has a disjunction as its goal. It shows why step 2 postpones one, and how a dead end can name the statement to split on.

Example. Choosing a side too early

Find a derivation of $P \to (Q \vee R) \vdash (P \to Q) \vee (P \to R)$.

Solution. The tempting start. The goal is a disjunction, and it is tempting to choose a side at once, say $P \to Q$, and finish with $\vee$I. The new goal is an implication, so we open a box on P; then $\to$E gives $Q \vee R$, and $\vee$E splits into the cases Q and R, each to end in the current goal Q. The first case ends at once. The second is a dead end:

1	$P \to (Q \vee R)$	
2	$\quad P$	
3	$\quad Q \vee R$	$\to$E 1, 2
4	$\quad\quad Q$	
5	$\quad\quad R$	
	$\quad\quad \vdots$	
	$\quad\quad Q$	
	$\quad Q$	$\vee$E
	$P \to Q$	$\to$I
	$(P \to Q) \vee (P \to R)$	$\vee$I

In the case R the resources are $P \to (Q \vee R)$, P, $Q \vee R$, and R, and none of them yields Q: no resource has Q as a conclusion except $Q \vee R$, which is already split, and nothing clashes. Choosing the right side instead fails in the mirror image, in the case Q. In fact neither side follows from the premise alone: with P and Q true and R false the premise is true and $P \to R$ is false, and with P and R true and Q false the premise is true and $P \to Q$ is false. (Chapter 3 turns such rows into proofs that no derivation exists: Corollary 3.5.3.)

The diagnosis. Each case of the dead end knew which side it wanted: the case Q wanted $P \to Q$, and the case R wanted $P \to R$. But those cases were opened inside the box of P, too late to choose the side of a disjunction that stands outside it. The side must be decided **before** any box is opened, and the premise does not decide it. Steps 2–4 are used up: we are at step 5.

The dead end also names the statement to split on. If Q holds, the left side is easy. If Q fails, then from P the premise gives $Q \vee R$ with its left side refuted, and DS gives R. So we use LEM on Q, and each case gets the whole goal:

1	$P \to (Q \vee R)$	
2	$Q \vee \neg Q$	LEM
3	$\quad Q$	
	$\quad \vdots$	
7	$\quad (P \to Q) \vee (P \to R)$	
8	$\quad \neg Q$	
	$\quad \vdots$	
13	$\quad (P \to Q) \vee (P \to R)$	
14	$(P \to Q) \vee (P \to R)$	$\vee$E 2, 3–7, 8–13

Now each case decides the side. In the case Q, the goal is $P \to Q$: a box on P whose last line repeats Q. In the case $\neg Q$, the goal is $P \to R$: a box on P, then $Q \vee R$ by $\to$E, then R by Proposition 2.3.3 used as a derived rule, citing $Q \vee R$ and $\neg Q$. Then $\vee$I, on the correct side each time:

1	$P \to (Q \lor R)$	
2	$Q \lor \neg Q$	LEM
3	$\quad Q$	
4	$\quad\quad P$	
5	$\quad\quad Q$	R 3
6	$\quad P \to Q$	$\to$I 4–5
7	$\quad (P \to Q) \lor (P \to R)$	$\lor$I 6
8	$\quad \neg Q$	
9	$\quad\quad P$	
10	$\quad\quad Q \lor R$	$\to$E 1, 9
11	$\quad\quad R$	DS 10, 8
12	$\quad P \to R$	$\to$I 9–11
13	$\quad (P \to Q) \lor (P \to R)$	$\lor$I 12
14	$(P \to Q) \lor (P \to R)$	$\lor$E 2, 3–7, 8–13

What the method did. Step 2 postponed the goal, the tempting start broke that rule, and its dead end showed both why the rule exists and what to do instead. Line 5 repeats line 3 because $\to$I needs Q as a line of the box of line 4 itself. LEM on P, the first split one thinks of, also succeeds, but less well: in the case P the premise gives $Q \lor R$, and a second $\lor$E, with two more implication boxes, is needed inside it; the finished derivation has 22 lines against these 14.

Warning. **Do not choose a disjunct early.** A goal $\psi \lor \chi$ tempts us to pick a side and prove it, and the choice usually fails, as the first attempt above did: neither $P \to Q$ nor $P \to R$ follows from the premise. Postpone the $\lor$ goal, use the resources first, and choose a side only inside a case that decides it.

2.6.7 When nothing else works

The fourth search reaches step 5 in the other form, RAA, and then needs steps 2 and 4 inside the box.

Example. A search that reaches step 5 at once

Find a derivation of $\neg(P \leftrightarrow Q) \vdash P \lor Q$.

Solution. Steps 1–4. The goal is a disjunction, so step 2 postpones it. The one resource is a negation, and a negation can be used only in a clash (step 4), which needs the goal $\bot$; the goal is $P \lor Q$. Neither side is in hand: the premise says only that P and Q differ, not which one holds. Steps 2–4 have nothing to offer.

Step 5. We assume the negation of the goal. The new goal is $\bot$, and the gap has one more resource:

1	$\neg(P \leftrightarrow Q)$	
2	$\quad\neg(P \vee Q)$	
	$\quad\vdots$	
12	$\quad\bot$	
13	$P \vee Q$	RAA 2–12

Step 4 inside the box. The resources are two negations, $\neg(P \leftrightarrow Q)$ and $\neg(P \vee Q)$, and no pair clashes yet. So we choose a statement to manufacture: $P \leftrightarrow Q$, to clash with line 1, or $P \vee Q$, to clash with line 2. The second is the goal we started from, a disjunction of two letters with no side in hand, so aiming at it leads back to the start. So we choose $P \leftrightarrow Q$, and it is a biconditional, so step 2 sets up two boxes:

1	$\neg(P \leftrightarrow Q)$	
2	$\quad\neg(P \vee Q)$	
3	$\quad\quad P$	
	$\quad\quad\vdots$	
6	$\quad\quad Q$	
7	$\quad\quad Q$	
	$\quad\quad\vdots$	
10	$\quad\quad P$	
11	$\quad P \leftrightarrow Q$	$\leftrightarrow$I 3–6, 7–10
12	$\quad\bot$	$\neg$E 11, 1
13	$P \vee Q$	RAA 2–12

The two small gaps. In the first, the resources are lines 1, 2, and 3, and the goal Q is a letter that no resource produces. But the resources clash once a line is manufactured: $\vee$I builds $P \vee Q$ from line 3, which clashes with line 2, and $\bot$E writes Q (step 4: an impossible case closes itself). The second gap is the mirror image:

1	$\neg(P \leftrightarrow Q)$	
2	$\quad\neg(P \vee Q)$	
3	$\quad\quad P$	
4	$\quad\quad P \vee Q$	$\vee$I 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad\quad Q$	$\bot$E 5
7	$\quad\quad Q$	
8	$\quad\quad P \vee Q$	$\vee$I 7
9	$\quad\quad \bot$	$\neg$E 8, 2
10	$\quad\quad P$	$\bot$E 9
11	$\quad P \leftrightarrow Q$	$\leftrightarrow$I 3–6, 7–10
12	$\quad \bot$	$\neg$E 11, 1
13	$P \vee Q$	RAA 2–12

What the method did. Step 5 came first, because nothing else applied, and after it steps 2 and 4 did the rest: step 4 chose which clash to aim for, step 2 read the two boxes off $P \leftrightarrow Q$, and step 4 closed each box with a manufactured clash and $\bot$E. In words: *suppose neither P nor Q holds; then each of them implies the other, since neither can hold at all; so $P \leftrightarrow Q$, against the premise.* The assumption of line 2 is used twice, at lines 5 and 9. A split by LEM on P also works, but less well: in the case $\neg P$ the goal side Q is a letter that the resources do not give, and the search is stuck there a second time.

The derivation of Peirce's law in Section 2.5 (Proposition 2.5.4) was found by the same method, and it adds one detail to step 5. There, step 2 opened a box with the goal P, step 3 made $P \to Q$ a subgoal, step 2 opened a box on P with the goal Q, and the search stalled at Q. Assuming $\neg Q$ at the stalled gap gives nothing new: a clash with $\neg Q$ needs Q, the stalled goal itself. Assuming $\neg P$ one level out does, because the new resource $\neg P$ is available in every gap inside its box, including the stalled one. That is why step 5 says to try a goal further out when the current one gives nothing.

2.6.8 A search that fails

The method is not a machine that always succeeds. Here it fails, and the failure is instructive.

Try $P \to Q \vdash Q \to P$. Step 2 opens a box on Q with the goal P:

1	$P \to Q$	
2	$\quad Q$	
	$\quad \vdots$	
	$\quad P$	
	$Q \to P$	$\to$I

The goal P is a letter. Step 3: $\to$E on line 1 needs P, the goal itself. Step 4: nothing clashes, and there is no negation among the resources to aim at. Step 5, RAA: assume $\neg P$; the only negation among the resources is now $\neg P$, and a clash with it needs P, the goal again. LEM on P: the case $\neg P$ needs P again. LEM on Q: the case $\neg Q$ clashes with line 2, and the case Q is where we started. Step 5 at the goal further out, an RAA box on $\neg(Q \to P)$ around the whole search: a clash with it needs $Q \to P$, the goal again, and step 2 opens the same box on Q with the same goal P. Every route returns to one gap, "P from $P \to Q$ and Q". That is affirming the consequent (Section 1.3), the step that would repair Argument B.

But notice what we have **not** shown. A search that fails proves nothing about derivations in general. An assumption may be any statement at all, by the assumption clause of Definition 1.4.2, so there are infinitely many derivations to consider, and a cleverer one might exist. To show that **no** derivation of $Q \to P$ from $P \to Q$ exists, we need a different tool.

That is the question this chapter leaves open. *We show that something is derivable by exhibiting a derivation; how could we ever show that something is not?* Chapter 3 answers it (Corollary 3.5.3). It gives each statement a truth value and proves that the rules never lead from true premises to a false conclusion; the assignment with P false and Q true, which makes $P \to Q$ true and $Q \to P$ false, then settles this search for good.

2.6.9 A mathematical goal

The method reads the frame of a proof off the goal whether the statements are letters or facts about numbers. In ordinary mathematics its payoff is that the first and last sentences of a proof, and its division into parts, are fixed before any mathematics is done.

Example. Building a skeleton

Let m and n be integers. Prove that if mn is odd, then m is odd and n is odd.

Solution. The skeleton. The main connective is "if ... then", so step 2 opens a $\to$I box: "Assume mn is odd. ... Hence m is odd and n is odd." The new goal is a conjunction, so step 2 again: two separate proofs, one that m is odd and one that n is odd. Each is a letter as far as logic can see, and the one resource, "mn is odd", cannot be taken apart. So step 3 brings in facts we can compute with: "if m is even then mn is even" is one, and CP (Proposition 2.3.2) turns it into "if mn is not even then m is not even", which the resource can feed. Then "every integer is even or odd" and DS finish.

The proof. Assume mn is odd. We show that m is odd and that n is odd.

First, if m is even, say $m = 2k$ with k an integer, then $mn = 2(kn)$ is even. By the contrapositive (Proposition 2.3.2), if mn is not even, then m is not even. Now mn is odd, so it is not even, since no integer is both; hence m is not even. Since every integer is even or odd, m is odd, by Proposition 2.3.3.

Exchanging the roles of m and n in the previous paragraph shows that n is odd. Hence m and n are both odd.

What the method did. It fixed the first sentence, the two paragraphs, and the last sentence; the only computation is $mn = 2(kn)$. No step uses RAA: the classical-looking move from "m is not even" to "m is odd" is disjunctive syllogism on the fact that every integer is even or odd. (The phrase "say $m = 2k$" names a witness, which is a rule for "there exists" from

Chapter 4.)

Warning. **Do not assume what you want to prove.** To derive a goal φ, it is tempting to open a box with φ. The box reaches φ at once, but after it closes it gives $\varphi \to \varphi$, not φ, and its assumption is gone. An assumption is useful when the goal asks for it: the hypothesis of an implication (step 2), the negated goal of RAA (step 5), a disjunct of a resource (step 3). Opening the box on the conclusion of an implication is the converse error of Argument B, diagnosed in Section 1.4.

2.6.10 The rules in one place

This is the reference sheet for Chapters 1 and 2: every primitive rule as its schema, with the label that defines it. As in every schema (Definition 1.2.1), φ, ψ, χ stand for any statements, replaced consistently, and $i, j, k, \ldots$ for line numbers. Each box rule needs the box's last line as a line of that box itself, not inside a further box, and every cited line or box must be available (Definition 1.4.2).

Conjunction (Definition 1.2.5): $\wedge$I, and the two forms of $\wedge$E.

i	φ	
j	ψ	
k	$\varphi \wedge \psi$	$\wedge$I i, j

i	$\varphi \wedge \psi$	
k	φ	$\wedge$E i

i	$\varphi \wedge \psi$	
k	ψ	$\wedge$E i

Implication (Definition 1.4.1): $\to$I and $\to$E.

i	φ	
	$\vdots$	
j	ψ	
k	$\varphi \to \psi$	$\to$I i–j

i	$\varphi \to \psi$	
j	φ	
k	ψ	$\to$E i, j

Disjunction (Definition 1.6.1): the two forms of $\vee$I, and $\vee$E, whose two boxes end in the same statement χ.

i	φ	
k	$\varphi \vee \psi$	$\vee$I i

i	φ	
k	$\psi \vee \varphi$	$\vee$I i

i	$\varphi \vee \psi$	
j	φ	
	$\vdots$	
k	χ	
l	ψ	
	$\vdots$	
m	χ	
n	χ	$\vee$E i, j–k, l–m

Contradiction and negation (Definition 2.1.1, Definition 2.2.1): ¬E, ⊥E (which may write **any** statement φ), and ¬I.

i	φ	
j	$\neg\varphi$	
k	$\bot$	¬E i, j

i	$\bot$	
k	φ	⊥E i

i	φ ⋮	
j	$\bot$	
k	$\neg\varphi$	¬I i–j

Biconditional (Definition 2.4.1): ↔I, and ↔E, which works in both directions.

i	φ ⋮	
j	ψ	
k	ψ ⋮	
l	φ	
m	$\varphi \leftrightarrow \psi$	↔I i–j, k–l

i	$\varphi \leftrightarrow \psi$	
j	φ	
k	ψ	↔E i, j

i	$\varphi \leftrightarrow \psi$	
j	ψ	
k	φ	↔E i, j

Reductio ad absurdum (Definition 2.5.1), the one classical rule, **and repetition**, clause (D3) of Definition 1.4.2, which copies an available line into the current box.

i	$\neg\varphi$ ⋮	
j	$\bot$	
k	φ	RAA i–j

i	φ ⋮	
k	φ	R i

Derived rules. Each is a proved sequent, used as one line that cites the lines holding its premises (Definition 2.3.1). The table lists every sequent that Chapters 1 and 2 name as a rule or use as one; every other sequent proved in the text may be used in the same way. Where a rule has no short name, the justification names the result, as "(ex falso)" or "VI".

In a justification	Sequent	Proved in	Our derivation uses RAA
(and commutes)	$P \wedge Q \vdash Q \wedge P$	Proposition 1.2.1	no
(implication is transitive)	$P \to Q,\ Q \to R \vdash P \to R$	Proposition 1.4.1	no
(curry)	$(P \wedge Q) \to R \dashv\vdash P \to (Q \to R)$	Proposition 1.4.2	no
(or commutes)	$P \vee Q \vdash Q \vee P$	Example (First derivations with "or")	no

In a justification	Sequent	Proved in	Our derivation uses RAA
(distributivity)	$P \wedge (Q \vee R) \dashv\vdash (P \wedge Q) \vee (P \wedge R)$	Proposition 1.6.1	no
(cases on a hypothesis)	$(P \to R) \wedge (Q \to R) \dashv\vdash (P \vee Q) \to R$	Example (Cases on a hypothesis are two implications); the converse Example (A disjunctive hypothesis splits into two implications)	no
(ex falso)	$P,\ \neg P \vdash Q$	Proposition 2.1.1	no
VI (vacuous implication)	$\neg P \vdash P \to Q$	Proposition 2.1.2	no
MT	$P \to Q,\ \neg Q \vdash \neg P$	Proposition 2.3.1	no
CP	$P \to Q \vdash \neg Q \to \neg P$	Proposition 2.3.2	no
DS	$P \vee Q,\ \neg P \vdash Q$	Proposition 2.3.3	no
(DS, other disjunct)	$P \vee Q,\ \neg Q \vdash P$	Exercise (B1: The other disjunctive syllogism) (a)	no
(iff symmetric)	$P \leftrightarrow Q \vdash Q \leftrightarrow P$	Example (An iff read both ways)	no
(iff as two implications)	$P \leftrightarrow Q \dashv\vdash (P \to Q) \wedge (Q \to P)$	Proposition 2.4.1	no
(iff of negations)	$P \leftrightarrow Q \vdash \neg P \leftrightarrow \neg Q$, and its converse	Example (Negating both sides of an iff); the converse Exercise (B2: Back from negated sides)	the converse only
(equivalence cycle)	$P \to Q,\ Q \to R,\ R \to P \vdash P \leftrightarrow Q$	Proposition 2.4.5	no
¬¬I	$P \vdash \neg\neg P$	Proposition 2.5.1 (a), first in Example (Two negations from one premise)	no
¬¬E	$\neg\neg P \vdash P$	Proposition 2.5.1 (b)	yes
(CP, converse)	$\neg Q \to \neg P \vdash P \to Q$	Proposition 2.5.2	yes
LEM	$\vdash P \vee \neg P$	Proposition 2.5.3	yes
(Peirce)	$\vdash ((P \to Q) \to P) \to P$	Proposition 2.5.4	yes
DM	the four laws of Proposition 2.5.5	Proposition 2.5.5	(d) only
(implication as disjunction)	$\neg P \vee Q \vdash P \to Q$, and its converse	Proposition 2.5.6	the converse only, through LEM

Two rows are proved in exercises, not in the text: the other form of DS and the converse of the iff of negations. Exercises may use them as rules; the text's own proofs do not.

Warning. **Cite only what is already proved.** A derived rule is an abbreviation for its derivation, so it may be used only after that derivation exists, and never inside it. For example, the second route to Proposition 2.5.5 (d) splits on excluded middle; a proof of excluded middle that used De Morgan's law (d) along that route would be circular. In this book the check is simple: cite only labels that appear earlier.

2.6.11 Exercises

2.6.11.1 A. Check your understanding

Exercise. A1

For (a)–(d), name the goal's main connective, the first rule the method aims for, and the first sentence of a proof. For (e)–(f), name the elimination rule the premise invites, and what it needs.

(a) For real x: if $x^3 < 0$ then $x < 0$.
(b) There is no integer n with $2n = 7$.
(c) For real x and y: $|xy| = |x||y|$ and $|x + y| \leq |x| + |y|$.
(d) For an integer n: n is odd if and only if n^2 is odd.
(e) The premise $(P \wedge Q) \to R$, with the goal R and the resources P and Q.
(f) The premise $\neg(P \vee Q)$, with the goal $\bot$ and the resource Q.

Solution.

(a) $\to$; $\to$I: "Assume $x^3 < 0$." (Starting "Suppose $x \geq 0$" inside that frame is an RAA box on the goal $x < 0$, not the contrapositive; a proof through the contrapositive replaces the frame, proving "if $x \geq 0$ then $x^3 \geq 0$" and then applying CP.)
(b) $\neg$, since the claim is "not: there is an integer n with $2n = 7$"; $\neg$I: "Suppose n is an integer with $2n = 7$." (Naming n uses a rule for "there exists", from Chapter 4.)
(c) $\wedge$; $\wedge$I: "We prove the two statements in turn."
(d) $\leftrightarrow$; $\leftrightarrow$I: "($\Rightarrow$) Assume n is odd."
(e) $\to$E, which needs the hypothesis $P \wedge Q$: make it a new goal, reached by $\wedge$I from P and Q; then $\to$E gives R.
(f) $\neg$E, which needs $P \vee Q$ beside $\neg(P \vee Q)$: make $P \vee Q$ the new goal (step 4), reached by $\vee$I from Q.

2.6.11.2 B. Practice

Exercise. B1: Three searches

For each sequent, name the first goal-directed step, then find a derivation.

(a) $P \wedge (Q \to R) \vdash (P \to Q) \to R$.
(b) $\neg(P \to Q) \vdash P \wedge \neg Q$. Which conjunct does your derivation obtain by RAA, and which by $\neg$I alone?

(c) $\neg P \to Q \vdash P \vee Q$.

Hint: in (c), look at what the premise is waiting for.

Solution.

(a) The goal is an implication: a box on $P \to Q$ with the goal R, a letter. Step 3 splits the premise; its half $Q \to R$ gives R once we have Q, and the box's assumption gives Q once we have P, which the premise's other half supplies. So the chain runs backwards from R to Q to P, and forwards it reads:

1	$P \wedge (Q \to R)$	
2	$\quad P \to Q$	
3	$\quad P$	$\wedge$E 1
4	$\quad Q$	$\to$E 2, 3
5	$\quad Q \to R$	$\wedge$E 1
6	$\quad R$	$\to$E 5, 4
7	$(P \to Q) \to R$	$\to$I 2–6

(b) The goal is a conjunction: two gaps, with goals P and $\neg Q$. The premise can be used only by $\neg$E, against $P \to Q$, so in each gap we aim to manufacture $P \to Q$. The goal $\neg Q$ is a negation: a box on Q, where $P \to Q$ follows by a box that repeats Q, and it clashes with the premise. The goal P is a letter, and the resources give nothing, so step 5: RAA on $\neg P$. Then $P \to Q$ follows by vacuous implication (Proposition 2.1.2, written VI), and it clashes with the premise. **The conjunct P is the one our derivation obtains by** RAA; $\neg Q$ needs only $\neg$I. (That no derivation of P from the premise avoids RAA is true, but we do not prove it.)

1	$\neg(P \to Q)$	
2	$\quad \neg P$	
3	$\quad P \to Q$	VI 2
4	$\quad \bot$	$\neg$E 3, 1
5	P	RAA 2–4
6	$\quad Q$	
7	$\quad\quad P$	
8	$\quad\quad Q$	R 6
9	$\quad P \to Q$	$\to$I 7–8
10	$\quad \bot$	$\neg$E 9, 1
11	$\neg Q$	$\neg$I 6–10
12	$P \wedge \neg Q$	$\wedge$I 5, 11

Written with primitive rules only, line 3 becomes the four lines of a box on P that clashes with $\neg P$, writes Q by $\bot$E, and closes by $\to$I; the derivation then has 15 lines instead of 12.

(c) The goal is a disjunction, so step 2 postpones it; the premise is waiting for $\neg P$, which is not a resource, and nothing clashes. Step 5, LEM on P, the statement the premise is waiting for: in the case P, $\vee$I gives the left side; in the case $\neg P$, the premise gives Q and $\vee$I the right side.

1	$\neg P \to Q$	
2	$P \vee \neg P$	LEM
3	$\quad P$	
4	$\quad P \vee Q$	$\vee$I 3
5	$\quad \neg P$	
6	$\quad Q$	$\to$E 1, 5
7	$\quad P \vee Q$	$\vee$I 6
8	$P \vee Q$	$\vee$E 2, 3–4, 5–7

The derivation has 8 lines. An RAA box on $\neg(P \vee Q)$ also works, with two manufactured clashes inside it.

Exercise. B2: The last step

A search for $P \to Q,\ R \to \neg Q \vdash P \to \neg R$ has reached the stage below; only the last line is missing. *Name the move, then complete it.*

1	$P \to Q$	
2	$R \to \neg Q$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad\quad R$	
6	$\quad\quad \neg Q$	$\to$E 2, 5
7	$\quad\quad \bot$	$\neg$E 4, 6
8	$\quad \neg R$	$\neg$I 5–7
9	**?**	?

Solution. The move is the chapter's main one: the skeleton was read off the goal, and the gap was filled from both ends. The goal $P \to \neg R$ is an implication, so the outer box opened with P (line 3); its new goal $\neg R$ is a negation, so the inner box opened with R (line 5). The premises then met the goal in the middle: $\to$E at lines 4 and 6 gave Q and $\neg Q$, and the clash at line 7. The inner box was closed at line 8. What remains is to close the outer box on its goal, by $\to$I 3–8:

1	$P \to Q$	
2	$R \to \neg Q$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad\quad R$	
6	$\quad\quad \neg Q$	$\to$E 2, 5
7	$\quad\quad \bot$	$\neg$E 4, 6
8	$\quad \neg R$	$\neg$I 5–7
9	$P \to \neg R$	$\to$I 3–8

The box runs from its assumption P, line 3, to its last line $\neg R$, line 8, which lies in the outer box itself and not in the inner one, as $\to$I requires.

Exercise. B3: Two boxes and a chain

Find a derivation of $P \to (Q \to R),\ P \to (R \to S) \vdash P \to (Q \to S)$, and describe the search in three sentences, naming its goal-directed and premise-directed steps.

Solution.

1	$P \to (Q \to R)$	
2	$P \to (R \to S)$	
3	$\quad P$	
4	$\quad Q \to R$	$\to$E 1, 3
5	$\quad R \to S$	$\to$E 2, 3
6	$\quad\quad Q$	
7	$\quad\quad R$	$\to$E 4, 6
8	$\quad\quad S$	$\to$E 5, 7
9	$\quad Q \to S$	$\to$I 6–8
10	$P \to (Q \to S)$	$\to$I 3–9

The search: the goal is an implication, so a goal-directed step opens a box on P with the goal $Q \to S$, and a second goal-directed step opens a box on Q with the goal S. With P a resource, premise-directed steps fire both premises by $\to$E, giving $Q \to R$ and $R \to S$. Then a backward chain from the goal S (it needs R, which needs Q, a resource) is written forwards as two more $\to$E lines, and the two boxes close in the reverse order of opening. (Lines 4 and 5 may equally be written inside the box of Q.)

2.6.11.3 C. Going deeper

Exercise. C1: The second half

A search for $P \vee (Q \wedge R) \vdash (P \vee Q) \wedge (P \vee R)$ has used the premise by $\vee$E and finished the first case; the second case is blank. *Name the move, then complete it.*

1	$P \vee (Q \wedge R)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad P \vee R$	$\vee$I 2
5	$\quad (P \vee Q) \wedge (P \vee R)$	$\wedge$I 3, 4
6	$\quad Q \wedge R$	
	$\quad \vdots$	
11	$\quad (P \vee Q) \wedge (P \vee R)$	
12	$(P \vee Q) \wedge (P \vee R)$	$\vee$E 1, 2–5, 6–11

Solution (Hint 1). The current goal of the blank box is a conjunction. What does step 2 do with it, and what does step 3 do with the box's assumption?

Solution (Hint 2). Line 7 is Q, by $\wedge$E from line 6.

Solution. The move is the chapter's main one again: read the skeleton off the goal, then work from both ends. The current goal $(P \vee Q) \wedge (P \vee R)$ is a conjunction, so step 2 splits it into the goals $P \vee Q$ and $P \vee R$, to be joined by $\wedge$I. Both are disjunctions, so they are postponed, and step 3 turns to the resource $Q \wedge R$: $\wedge$E gives Q and R. Now each disjunction has a side that is a resource, so $\vee$I is safe.

1	$P \vee (Q \wedge R)$	
2	$\quad P$	
3	$\quad P \vee Q$	$\vee$I 2
4	$\quad P \vee R$	$\vee$I 2
5	$\quad (P \vee Q) \wedge (P \vee R)$	$\wedge$I 3, 4
6	$\quad Q \wedge R$	
7	$\quad Q$	$\wedge$E 6
8	$\quad R$	$\wedge$E 6
9	$\quad P \vee Q$	$\vee$I 7
10	$\quad P \vee R$	$\vee$I 8
11	$\quad (P \vee Q) \wedge (P \vee R)$	$\wedge$I 9, 10
12	$(P \vee Q) \wedge (P \vee R)$	$\vee$E 1, 2–5, 6–11

Both boxes end in the current goal of the $\vee$E, as that rule requires. Here $\vee$I adds P on the left of Q and of R; in the first case it added Q and R on the right of P. Citing line 2 inside the second box would be the sibling-box error of Exercise (C2: Find the error).

Exercise. C2: A disjunction from nothing

(a) Derive $\vdash (P \to Q) \vee (Q \to P)$.

(b) In a paragraph, say which step of your derivation uses RAA or LEM, and why the method cannot avoid step 5 on this goal.

Solution (Hint 1). The goal is a disjunction and there are no premises. Which steps of the method does that leave?

Solution (Hint 2). Step 5 has two forms. Which statement would make each side of the goal easy, if you knew whether it held?

Solution.

(a) LEM on Q. In the case Q, the implication $P \to Q$ holds by a box that repeats Q. In the case $\neg Q$, the implication $Q \to P$ holds by a box on Q that clashes with $\neg Q$, followed by $\bot$E.

1	$Q \vee \neg Q$	LEM
2	$\quad Q$	
3	$\quad\quad P$	
4	$\quad\quad Q$	R 2
5	$\quad P \to Q$	$\to$I 3–4
6	$\quad (P \to Q) \vee (Q \to P)$	$\vee$I 5
7	$\quad \neg Q$	
8	$\quad\quad Q$	
9	$\quad\quad \bot$	$\neg$E 8, 7
10	$\quad\quad P$	$\bot$E 9
11	$\quad Q \to P$	$\to$I 8–10
12	$\quad (P \to Q) \vee (Q \to P)$	$\vee$I 11
13	$(P \to Q) \vee (Q \to P)$	$\vee$E 1, 2–6, 7–12

LEM on P works equally well, with the roles of the two sides exchanged.

(b) LEM is used at line 1, and RAA only inside it. The method cannot avoid step 5 here. Step 2 postpones the $\vee$ goal; with no premises, step 3 has nothing to eliminate and step 4 nothing to clash; and $\vee$I cannot come first, because neither side is derivable from no premises (with P true and Q false, $P \to Q$ is false; with P false and Q true, $Q \to P$ is false; Chapter 3 turns such rows into proofs, Corollary 3.5.3). So a resource must be created from nothing, and only step 5 does that. This shows that the **method** needs step 5; that every derivation of the sequent uses RAA or an equivalent is a stronger claim, which we do not prove.

Marking scheme. (10 points)

- 3: a correct split by LEM (on Q or on P), with two case boxes combined by $\vee$E, or a correct RAA box on the negated goal.
- 2: in one case, the implication by a box that repeats the case assumption.
- 2: in the other case, the implication by a box that clashes with the case assumption, then $\bot$E.
- 1: $\vee$I on the correct side in each case, every citation available.
- 2: the paragraph names the LEM line (or the RAA line) and explains why no other step of the method applies.

Fatal errors (0 for the step): choosing a side of the goal before the split; citing one case's assumption in the other case. A correct derivation by RAA on the negated goal earns full marks: its $2 + 2 + 1$ points go to the two implications derived inside the RAA box (one by a box that repeats an assumption, one by a clash and $\bot$E) and to the $\vee$I lines that clash with the RAA assumption.

Exercise. C3: Exit check: search

Find a derivation of $\neg(\neg P \wedge Q) \vdash P \vee \neg Q$ by the method of this section. Write one sentence

for each step of the method you take, and point to the classical step of your derivation.

Solution. The search.

1. *Step 1.* The premise $\neg(\neg P \wedge Q)$ on top, the goal $P \vee \neg Q$ at the bottom, one gap.
2. *Step 2.* The goal is a disjunction, so it is postponed: the premise does not say which side holds.
3. *Step 3.* Nothing can be eliminated. The premise is a negation, so it can be used only in a clash (step 4), against $\neg P \wedge Q$, and we do not have it.
4. *Step 4.* The goal is not $\bot$, and nothing clashes. We are stuck.
5. *Step 5.* We assume the negation of the goal, $\neg(P \vee \neg Q)$, with the new goal $\bot$.
6. *Step 4, inside the box.* The new assumption clashes with $P \vee \neg Q$, which $\vee$I builds from either side. The side P is a letter that nothing gives; the side $\neg Q$ is a negation, which step 2 can build. So $\neg Q$ becomes the goal: a box on Q, with the goal $\bot$. Choosing a side is safe here, since the disjunction is manufactured for a clash with a resource that is its negation (see the paragraph on choosing a side of $\vee$, after the method).
7. *Step 4, in the box on Q.* The resources are the premise, the assumption of step 5, and Q. We aim at the premise: $\neg P \wedge Q$ becomes the goal, and step 2 splits it. Its half Q is a resource; its half $\neg P$ is a negation: a box on P, with the goal $\bot$.
8. *Step 4, in the box on P.* $\vee$I builds $P \vee \neg Q$ from P, which clashes with the assumption of step 5. Now the boxes close from the inside out: $\neg P$, then $\neg P \wedge Q$, which clashes with the premise; then $\neg Q$, then $P \vee \neg Q$, which clashes with the assumption of step 5; and RAA closes the outer box.

1	$\neg(\neg P \wedge Q)$	
2	$\quad\neg(P \vee \neg Q)$	
3	$\quad\quad Q$	
4	$\quad\quad\quad P$	
5	$\quad\quad\quad P \vee \neg Q$	$\vee$I 4
6	$\quad\quad\quad \bot$	$\neg$E 5, 2
7	$\quad\quad \neg P$	$\neg$I 4–6
8	$\quad\quad \neg P \wedge Q$	$\wedge$I 7, 3
9	$\quad\quad \bot$	$\neg$E 8, 1
10	$\quad \neg Q$	$\neg$I 3–9
11	$\quad P \vee \neg Q$	$\vee$I 10
12	$\quad \bot$	$\neg$E 11, 2
13	$P \vee \neg Q$	RAA 2–12

The classical step is line 13, the RAA that closes the box of line 2; every other box closes by $\neg$I. The assumption of line 2 is used twice, at lines 6 and 12, just as in Example (A search that reaches step 5 at once).

A second route. Step 5 may take its other form: LEM on P. In the case P, $\vee$I gives the goal. In the case $\neg P$, the goal's side $\neg Q$ is a negation: a box on Q builds $\neg P \wedge Q$, which

clashes with the premise, so $\neg Q$, and $\vee$I. That derivation has 11 lines, and its classical step is the LEM line.

Marking scheme. (10 points)

- 2: step 2 postpones the disjunctive goal, with the reason (neither side is in hand).
- 2: steps 3 and 4 find that the premise can be used only by producing $\neg P \wedge Q$ for a clash.
- 2: step 5 is chosen: RAA on the whole goal, or LEM on P. (LEM on Q also works; our route with it then uses a second classical step, an RAA for P in the case Q; award it.)
- 3: the inner boxes and their clashes are correct, and every citation is available.
- 1: the classical line is named.

Fatal errors (0 for the step): $\vee$I as the last line, from a disjunct derived from the premise alone; an RAA box opened with anything but $\neg(P \vee \neg Q)$.

Exercise. C4: Two sequents that look alike

Derive each sequent, and say which of your two derivations needed the classical rule.

(a) $\neg(P \wedge Q),\ P \vdash \neg Q$.

(b) $\neg(P \wedge \neg Q),\ P \vdash Q$.

Then explain why, in (b), the natural $\neg$I box on $\neg Q$ ends at $\neg\neg Q$ and not at Q.

Solution.

(a) The goal is a negation: a $\neg$I box on Q, in which $P \wedge Q$ clashes with the first premise.

1	$\neg(P \wedge Q)$	
2	P	
3	$\quad Q$	
4	$\quad P \wedge Q$	$\wedge$I 2, 3
5	$\quad \bot$	$\neg$E 4, 1
6	$\neg Q$	$\neg$I 3–5

(b) The goal Q is a letter, and the resources give nothing directly, so step 5: RAA, a box on $\neg Q$, in which $P \wedge \neg Q$ clashes with the first premise.

1	$\neg(P \wedge \neg Q)$	
2	P	
3	$\quad \neg Q$	
4	$\quad P \wedge \neg Q$	$\wedge$I 2, 3
5	$\quad \bot$	$\neg$E 4, 1
6	Q	RAA 3–5

The derivation of (b) needed the classical rule, at line 6; the derivation of (a) did not. The two have the same shape, and the difference is the goal. In (a) it is a negation, which ¬I builds. In (b) it is a letter, and the box that reaches $\bot$ opens with $\neg Q$; ¬I writes the negation of a box's assumption, so closing that box by ¬I gives $\neg\neg Q$, and our derivation reaches Q with the classical rule instead: RAA in place of ¬I (or ¬¬E after it).

2.6.11.4 R. Review

Exercise. R1

For real numbers a and b, let $\max(a,b) \coloneqq a$ and $\min(a,b) \coloneqq b$ if $a \geq b$, and $\max(a,b) \coloneqq b$ and $\min(a,b) \coloneqq a$ if $a < b$. Prove that $\max(a,b) + \min(a,b) = a + b$.

Solution. Since $a \geq b$ or $a < b$, and these two cases cover every pair of real numbers, we argue by cases (∨E, Definition 1.6.1), as in Example (The absolute value of a product).

Case $a \geq b$. Then $\max(a,b) = a$ and $\min(a,b) = b$, so $\max(a,b) + \min(a,b) = a + b$.

Case $a < b$. Then $\max(a,b) = b$ and $\min(a,b) = a$, so $\max(a,b) + \min(a,b) = b + a = a + b$.

In either case the equation holds, which proves it. Both cases end in the same statement, the goal, as ∨E requires.

Exercise. R2

Let x be a real number. Write N for "$x < 0$", G for "$x > 1$", and S for "$x = 7$". Translate the argument below into a sequent in these letters, and derive it.

> x is neither negative nor greater than 1. Therefore x is not negative, and if x is greater than 1, then $x = 7$.

Solution. "Neither N nor G" says that $N \vee G$ fails, so the premise is $\neg(N \vee G)$, not $\neg N \vee G$. The conclusion is a conjunction whose second part is an implication. The sequent (Definition 1.2.4) is

$$\neg(N \vee G) \vdash \neg N \wedge (G \to S).$$

The goal is a conjunction; the first conjunct is a negation (a ¬I box on N), and the second an implication (a box on G). In both boxes ∨I builds $N \vee G$, which clashes with the premise; in the second box $\bot$E then gives S.

1	$\neg(N \vee G)$	
2	$\quad N$	
3	$\quad N \vee G$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg N$	$\neg$I 2–4
6	$\quad G$	
7	$\quad N \vee G$	$\vee$I 6
8	$\quad \bot$	$\neg$E 7, 1
9	$\quad S$	$\bot$E 8
10	$G \to S$	$\to$I 6–9
11	$\neg N \wedge (G \to S)$	$\wedge$I 5, 10

Lines 2–5 are half of Proposition 2.5.5 (a), which turns "neither . . . nor" into $\neg N \wedge \neg G$. The implication $G \to S$ is vacuously true, as in Proposition 2.1.2: the premise rules out G, and a promise never tested is never broken, whatever S says.

Exercise. R3

Is $(P \to Q) \to R \vdash P \to (Q \to R)$ derivable? And is its converse, $P \to (Q \to R) \vdash (P \to Q) \to R$?

Solution. The first is derivable, by the search method of this section. The goal gives a box on P, then a box on Q, with the goal R. The premise gives R by $\to$E once we produce $P \to Q$, a new goal: a box on P whose last line repeats Q.

1	$(P \to Q) \to R$	
2	$\quad P$	
3	$\quad\quad Q$	
4	$\quad\quad\quad P$	
5	$\quad\quad\quad Q$	R 3
6	$\quad\quad P \to Q$	$\to$I 4–5
7	$\quad\quad R$	$\to$E 1, 6
8	$\quad Q \to R$	$\to$I 3–7
9	$P \to (Q \to R)$	$\to$I 2–8

The repetition at line 5 puts Q inside the box of line 4, as $\to$I requires (clause (D3) of Definition 1.4.2). Two look-alike tools do not apply. Proposition 1.4.2 moves a **conjunction** in and out of a hypothesis, $(P \wedge Q) \to R \dashv\vdash P \to (Q \to R)$, and the premise here has $P \to Q$, not $P \wedge Q$, in that place. And Proposition 1.4.1 chains two implications, which we

do not have.

The converse is not derivable, but a search can only stall, not show it. With P and R false (and Q either way), the premise $P \to (Q \to R)$ is true because P is false, while $P \to Q$ is true and R is false, so the conclusion is false. Chapter 3 proves that such a row rules out every derivation (Corollary 3.5.3): the chapter's open question, in a new place.

Chapter 3

Truth, Soundness and Completeness

Chapters 1 and 2 built a calculus of derivations, and with it we can show that a sequent holds: write a derivation. What we cannot yet do is show that a sequent does **not** hold. Section 2.6 searched for a derivation of $P \to Q \vdash Q \to P$, and every route stalled; but a search that fails proves nothing, since a cleverer derivation might exist somewhere. The motivating problem of this chapter is to *prove that no derivation of $q \to p$ from $p \to q$ exists*, however long and however clever.

The answer is to give formulas a **meaning**. A valuation makes each atom true or false, a truth table computes the value of every formula from its atoms, and $\Gamma \models \psi$ says that every valuation making the premises true makes the conclusion true. Two theorems then tie meaning to derivation. **Soundness**: whatever can be derived is true in every row where the premises are true; so a single row with true premises and a false conclusion proves that no derivation exists. **Completeness**: whatever is true in every such row can be derived; so the rules of Chapters 1 and 2 are enough. To state and prove these theorems we first make formulas into mathematical objects, and we meet **induction**, on the natural numbers and on the shape of a formula, because both theorems are proved by it. These are the book's first proofs about proofs, and its first long proofs. The chapter ends with normal forms, the constructive content of completeness.

What you need. Chapters 1 and 2: the rules for every connective, the definition of a derivation with its boxes and its scope rule (Definition 1.4.2), derived rules (Definition 2.3.1), and the classical rules of Section 2.5, above all the law of excluded middle (Proposition 2.5.3). The arithmetic and order of the natural numbers are used as school facts.

Time. About 40 hours on the core track (all of A, one B and one C per section, the exit checks); about 55 hours for everything.

3.0.1 Roadmap

- **Formulas as objects.** The exact definition of a formula, parse trees, subformulas and height, the shorthand of Chapter 1 as a translation, and why a derivation stays a derivation when formulas are put in for its atoms.
- **Induction on numbers and on formulas.** The principle of induction, strong induction, and structural induction, and with them the proofs that every formula is read in only one way and that a definition by one clause per connective defines exactly one thing.
- **Valuations and truth tables.** Truth values for atoms, the value of every formula

computed along its parse tree, and why a table with 2^n rows tells everything about a formula with n atoms.
- **Tautologies, entailment, and equivalence.** The semantic counterpart $\Gamma \models \psi$ of a sequent, tautologies and satisfiability, the standard equivalences, and what may be replaced or substituted without changing truth.
- **Soundness.** The rules never lead from truth to falsehood, proved by induction on the lines of a derivation; so one refuting row shows that a sequent has no derivation.
- **Completeness.** Every truth-table consequence of finitely many premises has a derivation: the rules prove their own truth tables, one row at a time, and excluded middle glues the rows together.
- **Normal forms and adequate connectives.** Every formula is equivalent to one in disjunctive and in conjunctive normal form; validity read off a conjunctive normal form; which sets of connectives can express everything.

3.0.2 Named moves

- **Induction on the shape of a formula.** Prove it for atoms and $\bot$; prove it for each connective, assuming it for the parts.
- **Define by recursion on formulas.** One clause per way of building a formula; unique readability makes the definition well defined.
- **Refute with one row.** A valuation making every premise true and the conclusion false proves $\Gamma \nvdash \psi$, by soundness.
- **Induction on the lines of a derivation.** To prove that every derivable sequent has a property, show that every rule preserves it.
- **Split on an atom.** With $p \vee \neg p$ and $\vee$E, prove the goal once assuming p and once assuming $\neg p$.

3.1 Formulas as Objects

Chapter 2 ended on a question that no derivation can answer. We show that a sequent holds by writing a derivation of it; but how could we show that one does **not** hold, that no derivation of $Q \to P$ from $P \to Q$ exists, however long and however clever? This chapter answers the question, and the answer is a theorem about **every** derivation of **every** formula. A theorem about every formula needs, first of all, an exact description of what a formula is. This section gives it. It defines formulas, draws them as trees, names their parts, and says how the shorthand of Chapters 1 and 2 fits in. Nothing is yet proved about all formulas at once; the tool for that is the next section's.

3.1.1 Why formulas must become objects

Here is the theorem that Section 3.5 proves, stated loosely:

> If ψ can be derived from premises Γ, then ψ is true in every situation in which all the premises are true.

Read it as a mathematician reads a statement about all integers. "For every integer n, $n^2 - n$ is even" can be proved only by someone who knows what an integer is: every integer is even or odd, and each kind can be handled. A proof needs a way to take hold of an arbitrary object of the kind the statement is about. The loose theorem above is about arbitrary derivations, and a derivation is a list of formulas, so the proof will need a way to take hold of an arbitrary formula.

Chapters 1 and 2 never needed one. There we read statements by eye: we found the main connective of $\neg(P \wedge Q) \to R$ (Definition 1.1.2) by looking, and we drew its tree by hand. That is fine for one statement at a time. It is useless for a claim about all statements, because "all the strings we would recognize as statements" is not a description anyone can reason from. So we now do for formulas what *Analysis on the Real Line*, Chapter 1 will do for the real numbers: say exactly which objects we mean, in a form that later proofs can use.

The description has three parts. First, the **alphabet**, the symbols formulas are written with. Second, the **formation rules**, which say how formulas are built. Third, a closing clause saying that **nothing else** is a formula. The third is the one that is easy to forget and that makes everything else work.

3.1.2 The alphabet

Formulas are strings of symbols. A **string** is a finite list of symbols written one after another, and its **length** is the number of symbols in it. The symbols are these.

- **Atoms:** $p, q, r, p_1, p_2, p_3, \ldots$, infinitely many of them. Each atom is **one** symbol: p_1 is a single symbol whose name happens to carry a subscript, not the two symbols p and 1.
- **The constant** $\bot$, read "contradiction", as in Chapter 1.
- **The connectives:** $\neg$ (the only one that takes one part) and $\wedge, \vee, \to, \leftrightarrow$ (the **binary** connectives, which take two).
- **The parentheses** (and).

An atom plays the part that a letter played in Chapters 1 and 2: a statement whose inner structure we ignore. In those chapters the letters were capitals, P, Q, R, and they stood for **arbitrary** statements, atomic or compound. From now on we keep two kinds of letter apart, because the theorems of this chapter need the difference.

- An **atom** $p, q, r, \ldots$ is a symbol of the language. It occurs in formulas.
- A **metavariable** φ, ψ, χ (phi, psi, chi) is a name we use when talking **about** formulas: "let φ be a formula". It never occurs in a formula. Chapter 1 already used these three letters in exactly this way, in the schemas of its rules (Definition 1.2.1).

We write Γ and Δ (capital gamma and delta) for finite lists of formulas, as in the sequents of Chapter 1 (Definition 1.2.3). The capitals P, Q, R of Chapters 1 and 2 were stand-ins for arbitrary formulas; the last part of this section says exactly what that means.

We use $\circ$ as a metavariable for a binary connective: "$(\varphi \circ \psi)$" means any one of $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \to \psi)$, $(\varphi \leftrightarrow \psi)$. This saves writing four clauses where one will do.

3.1.3 Formulas

Chapter 1 described a statement as something "built from smaller ones by a connective". The definition below says exactly how the building goes. It is an **inductive definition**: it names the starting pieces, names the ways of building new formulas from old, and then closes the door on everything else.

A formula is a string built from atoms and $\bot$ by the connectives, one step at a time, with a pair of parentheses around every binary step.

Definition 3.1.1. Formula

The **formulas** are the strings given by the following clauses.

(F1) Every atom is a formula, and $\bot$ is a formula.

(F2) If φ is a formula, then $\neg\varphi$ is a formula.

(F3) If φ and ψ are formulas, then $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \to \psi)$ and $(\varphi \leftrightarrow \psi)$ are formulas.

(F4) **Nothing else** is a formula: a string is a formula **only if** it is the last entry of a **formation sequence**. A formation sequence is a finite list of strings $\chi_1, \ldots, \chi_m$, with $m \geq 1$, in which **every** entry χ_i is an atom or $\bot$, or is $\neg\chi_j$ for some **earlier** entry χ_j (so $j < i$), or is $(\chi_j \circ \chi_l)$ for some earlier entries χ_j and χ_l (so $j, l < i$; $j = l$ is allowed) and some binary connective $\circ$.

In words, clause by clause.

- *(F1), the starting pieces.* A single atom is a formula, and so is $\bot$. These are the only formulas of length one.
- *(F2), negation.* Writing $\neg$ in front of a formula gives a formula. No parentheses are added: $\neg p$ and $\neg\neg p$ are formulas, and $\neg\neg p$ is $\neg$ in front of the formula $\neg p$.
- *(F3), the binary connectives.* Writing a binary connective between two formulas **and** wrapping the result in one pair of parentheses gives a formula. The two formulas may be the same: $(p \wedge p)$ is a formula.
- *(F4), nothing else.* Clauses (F1)–(F3) say which strings **are** formulas. They do not say that other strings are **not**. Clause (F4) does: a formula is exactly something we can reach from atoms and $\bot$ by finitely many uses of (F2) and (F3), and the formation sequence is the written record of such a reaching.

Two facts about formation sequences are used constantly, and both are immediate from the definition.

First, **every entry of a formation sequence is a formula.** The first i entries $\chi_1, \ldots, \chi_i$ of a formation sequence are themselves a formation sequence (each entry was built from entries earlier still), and it ends in χ_i.

Second, **every formula was made by one of the clauses from formulas.** Take a formula χ and a formation sequence ending in it. Its last entry χ is an atom or $\bot$, or it is $\neg\chi_j$, or it is $(\chi_j \circ \chi_l)$, where the earlier entries χ_j, χ_l are formulas by the first fact. So every formula is an atom, or $\bot$, or $\neg\varphi$ for a formula φ, or $(\varphi \circ \psi)$ for formulas φ, ψ. We will call this **looking at the last step** of a formula.

Looking at the last step already tells us something about the first symbol. A formula of the form $\neg\varphi$ begins with $\neg$, and one of the form $(\varphi \circ \psi)$ begins with (and ends with). So:

> **The first-symbol test.** A formula of length one is an atom or $\bot$. A longer formula begins with $\neg$ or with (, and if it begins with (, it is $(\varphi \circ \psi)$ for formulas φ, ψ and ends with).

(A formula of the form $\neg\varphi$ has length at least 2, and one of the form $(\varphi \circ \psi)$ has length at least 5, since φ and ψ each have at least one symbol. So a formula of length one is made by (F1).)

Example. Formation sequences

Write down a formation sequence for each of the strings p, $\bot$, $\neg\neg q$ and $((p \wedge \neg q) \to p)$, which shows that each is a formula.

Solution. For p, the one-entry list p is a formation sequence: its only entry is an atom. Likewise the list $\bot$. These are the degenerate cases, and they matter: a formula need not have been built at all. Every structural argument in the next section has to deal with them.

For $\neg\neg q$:

$$q, \quad \neg q, \quad \neg\neg q.$$

The first entry is an atom; the second is $\neg$ in front of the first; the third is $\neg$ in front of the second.

For $((p \wedge \neg q) \to p)$:

$$p, \quad q, \quad \neg q, \quad (p \wedge \neg q), \quad ((p \wedge \neg q) \to p).$$

Entries 1 and 2 are atoms. Entry 3 is $\neg$ in front of entry 2. Entry 4 is $(\chi_1 \wedge \chi_3)$. Entry 5 is $(\chi_4 \to \chi_1)$: the atom p is used twice, and it need be listed only once.

A formula has many formation sequences. We could have listed q before p, or added entries that are never used, such as r or $\neg r$. The definition asks only that **some** formation sequence ends in the string, and nothing in this chapter depends on which one we take.

To show that a string **is** a formula, we exhibit a formation sequence, just as we show that a sequent holds by exhibiting a derivation. To show that a string is **not** a formula, exhibiting is not enough; we need a property that every formula has and the string lacks. The first-symbol test supplies such properties.

Example. Strings that are not formulas

Show that none of the strings $p\wedge \to q$, $(p \wedge q$, $(\neg p)$ and $p \wedge q$ is a formula.

Solution. $p\wedge \to q$: it has length 4, so by the first-symbol test a formula would begin with $\neg$ or (. It begins with p. Not a formula.

$(p \wedge q$: it begins with (, so by the first-symbol test a formula would end with). It ends with q. Not a formula. (The missing parenthesis is the whole problem: $(p \wedge q)$ is a formula.)

$(\neg p)$: it begins with (, so a formula would have to be $(\varphi \circ \psi)$, with the two symbols $\neg p$ between the parentheses equal to $\varphi \circ \psi$, and so containing a binary connective. The string $\neg p$ contains none. Not a formula. The official language puts no parentheses around a negation; (F2) adds none.

$p \wedge q$: it has length 3 and begins with p. Not a formula. This one is a surprise at first, since we have written $p \wedge q$ all along. It is **shorthand** for the formula $(p \wedge q)$, in the way explained later in this section.

The last two strings are ones Chapter 1 wrote freely, and that is no mistake in Chapter 1. Chapter 1 used a convenient shorthand; the official formulas carry every parenthesis that (F3) puts in and none that it does not. We come back to the shorthand below.

Quick check. Which of the following strings are formulas? For each, give a formation sequence or name the test it fails.

(a) $\neg(p \to \bot)$
(b) $(p \to \neg)$
(c) $\bot\bot$

Solution. (a) A formula: p, $\bot$, $(p \to \bot)$, $\neg(p \to \bot)$ is a formation sequence.

(b) Not a formula. It begins with (, so it would be $(\varphi \to \psi)$ with φ the string p and ψ the string $\neg$ (the only binary connective is the $\to$). But $\neg$ is a string of length one that is neither an atom nor $\bot$, so by the first-symbol test it is not a formula.

(c) Not a formula. It has length 2, so it would begin with $\neg$ or (; it begins with $\bot$.

3.1.4 Why this definition

Each feature of the definition is there because something breaks without it.

Without (F4), clauses (F1)–(F3) would be satisfied by far too many collections of strings. The collection of **all** strings contains every atom and $\bot$, and is closed under writing $\neg$ in front and under wrapping two strings with a connective in parentheses; so (F1)–(F3) alone cannot tell us that $p\wedge \to q$ is excluded. Clause (F4) says the formulas are only those strings that the building steps reach. In the language of sets, which *Sets, Relations, and Cardinality*, Chapter 1 introduces properly, the formulas are the **smallest** collection of strings closed under (F1)–(F3).

Without the parentheses in (F3), a string could be read in two ways. Take $p \wedge q \vee r$. Built one way, it is "$p \wedge q$" joined to r by $\vee$; built the other way, it is p joined to "$q \vee r$" by $\wedge$. The two constructions have different parse trees (defined below), and they say different things: if p is false and r is true, then "p and q, or r" is true, because r is, while "p, and q or r" is false, because p is. (Section 3.3 makes "true under an assignment of values" exact; the everyday reading of Chapter 1 is enough here.) With the parentheses of (F3) the two constructions give two **different** strings,

$$((p \wedge q) \vee r) \qquad \text{and} \qquad (p \wedge (q \vee r)),$$

and the ambiguity disappears. The next section proves that it disappears for every formula: a formula can be built in only one way (the unique readability theorem of Section 3.2).

Why not parentheses around a negation too? Nothing would break, but nothing would be gained. The symbol $\neg$ always stands in front of what it negates, so $\neg\varphi$ can be read in only one way without help. Keeping the official language as lean as possible makes the proofs about it shorter.

Why atoms, and not sentences like "$x > 0$"? Because the theorems of this chapter are about **forms**, and the form of an argument is what remains when the content of its atomic statements is forgotten (Chapter 1's slogan: logic studies the forms of arguments). An atom is a statement reduced to a name.

Warning. A formula is a string of symbols, and questions about formulas are questions about strings. "Is $(p \wedge p)$ the same formula as p?" has the answer **no**: the first has length 5 and the second length 1, even though they always have the same truth value (Section 3.3 gives the

means to check this). Sameness of formulas is letter-for-letter identity; sameness of meaning is a different relation, which Section 3.4 names.

3.1.5 Parse trees

A formation sequence records how a formula was built, but it is a list, and a list hides the shape of the building. Chapter 1 drew the shape as a tree, with the main connective at the top (the tree after Definition 1.1.2). We now make that drawing official.

A parse tree is a formula's construction drawn upside down: the last step at the root, the atoms at the leaves.

Definition 3.1.2. Parse Tree

A **parse tree for** a formula is defined along the clauses of Definition 3.1.1.

(T1) For an atom or $\bot$: a single node, labeled by that atom or by $\bot$.

(T2) If T is a parse tree for φ, then the tree whose **root** is labeled $\neg$ and has T hanging below it as its only subtree is a parse tree for $\neg\varphi$.

(T3) If T and T' are parse trees for φ and ψ, then the tree whose root is labeled $\circ$, with T hanging below it on the **left** and T' on the **right**, is a parse tree for $(\varphi \circ \psi)$.

The **leaves** are the nodes with nothing below them; a **branch** is a path from the root down to a leaf.

In words: (T1) is the tree of a formula that was not built at all, a single node. (T2) puts a new root $\neg$ on top of an existing tree. (T3) joins two existing trees under a new root carrying the connective, keeping left and right apart, because $(p \to q)$ and $(q \to p)$ are different formulas. Parentheses do not appear in a parse tree: their only job in the string was to show the grouping, and the tree shows the grouping by its shape.

Every formula has a parse tree. Take a formation sequence of the formula and build a tree for each entry in turn: an atom or $\bot$ gets (T1), and an entry $\neg\chi_j$ or $(\chi_j \circ \chi_l)$ gets (T2) or (T3) from the trees already built for the earlier entries. The tree built for the last entry is a parse tree for the formula.

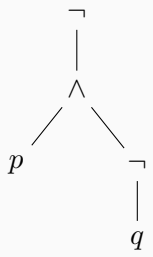
Could a formula have **two** different parse trees? That would happen if one string could be built in two different ways, as $p \wedge q \vee r$ could if parentheses were not written. Section 3.2 proves that it never happens (the unique readability theorem); until then we say "**a** parse tree". After that section, every formula has exactly one, and we speak of **the** parse tree.

Example. Three parse trees

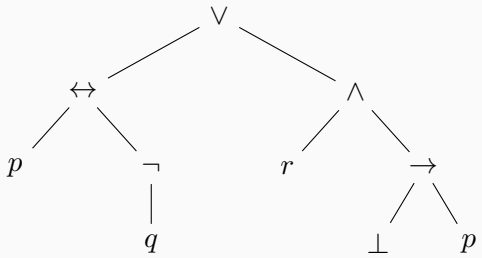
Draw a parse tree for each formula: (a) p; (b) $\neg(p \wedge \neg q)$; (c) $((p \leftrightarrow \neg q) \vee (r \wedge (\bot \to p)))$, which uses every connective and $\bot$.

Solution. (a) A single node labeled p, by (T1). This is the degenerate tree: its root is also its only leaf, and its only branch has one node.

(b) By (T1) the atoms p and q get one-node trees; (T2) puts $\neg$ above q; (T3) joins p and $\neg q$ under $\wedge$; (T2) puts $\neg$ on top.



(c) The main connective is the $\vee$: it is the last step, joining $(p \leftrightarrow \neg q)$ on the left to $(r \wedge (\bot \to p))$ on the right. Each side is taken apart in the same way.



The leaves are p, q, r, $\bot$ and p again: the atom p occurs twice in the formula, and each occurrence is its own leaf.

To draw a parse tree from a string, find the last step, draw it as the root, and repeat on the parts. For a formula $(\varphi \circ \psi)$, the last step is the binary connective that sits inside **exactly one** pair of parentheses; in (c), it is the $\vee$, and every other binary connective sits inside at least two pairs. Section 3.2 proves that this recipe always finds the main connective, and only it.

3.1.6 Subformulas and height

Each node of a parse tree is the root of the part of the tree below it, and that part is itself a parse tree, built by the same clauses, for a formula. The formulas that arise in this way are the pieces a formula is built from.

Definition 3.1.3. Subformula

The **subformulas** of a formula φ are the formulas whose parse trees hang from the nodes of a parse tree for φ: for each node, the formula of which the part of the tree from that node downwards is a parse tree. The formulas at the children of the root are the **immediate subformulas** of φ.

So the subformulas of φ include φ itself (at the root) and each atom that occurs in φ (at the leaves). A formula of the form $\neg\varphi$ has one immediate subformula, φ; a formula $(\varphi \circ \psi)$ has the two immediate subformulas φ and ψ (one, if φ and ψ are the same formula); an atom or $\bot$ has none. When we list the subformulas, a formula that sits at several nodes is listed once.

For the formula (c) of Example (Three parse trees), the subformulas are

$$((p \leftrightarrow \neg q) \vee (r \wedge (\bot \to p))), \quad (p \leftrightarrow \neg q), \quad (r \wedge (\bot \to p)),$$
$$\neg q, \quad (\bot \to p), \quad p, \quad q, \quad r, \quad \bot,$$

nine in all. The tree has ten nodes, $\vee$, $\leftrightarrow$, p, $\neg$, q, $\wedge$, r, $\to$, $\bot$ and p again, and only p sits at

two of them; so ten nodes give nine distinct subformulas. Counting by eye is easy to get wrong, and the safe method is to walk the tree level by level, as we just did.

A string that merely **appears inside** a formula need not be a subformula. The formula $((p \wedge q) \vee r)$ contains the string $q) \vee r$, which is not a formula at all, and it contains the string $p \wedge q$, which is only the shorthand of a formula. Its subformulas are the formulas at the nodes: $((p \wedge q) \vee r)$, $(p \wedge q)$, p, q and r.

The second measurement of a tree is how deep it goes.

Definition 3.1.4. Height

The **height** of a formula φ, written $\mathrm{ht}(\varphi)$, is the number of nodes on a **longest** branch of its parse tree.

Atoms and $\bot$ have height 1: their tree is one node, and its only branch has one node. (Some books count edges instead of nodes and give atoms height 0; with nodes, every formula has height at least 1.) In Example (Three parse trees), (b) has height 4, along the branch $\neg, \wedge, \neg, q$, and (c) has height 4 too, along $\vee, \leftrightarrow, \neg, q$ or along $\vee, \wedge, \rightarrow, \bot$.

The definition speaks of "its parse tree", and so it depends on the fact, proved only in Section 3.2, that a formula has one parse tree. Until then the height is a promise, not a number: nothing in Section 3.2 uses it before that fact is proved, and Section 3.2 then obtains the height a second way, by a recursive definition that needs no trees at all.

Quick check. Draw the parse tree of $(\neg p \vee q) \leftrightarrow \neg(r \wedge p)$ (in official form, $((\neg p \vee q) \leftrightarrow \neg(r \wedge p))$), list its subformulas, and give its height.

Solution. The root is $\leftrightarrow$. Its left subtree is $\vee$ over $\neg p$ and q, where $\neg p$ is $\neg$ over the leaf p. Its right subtree is $\neg$ over $\wedge$, and $\wedge$ has the leaves r and p.

The subformulas are the formula itself, $(\neg p \vee q)$, $\neg(r \wedge p)$, $\neg p$, q, $(r \wedge p)$, p and r: **eight**, with p listed once although it sits at two leaves.

The height is 4: the branches $\leftrightarrow, \vee, \neg, p$ and $\leftrightarrow, \neg, \wedge, r$ (and $\leftrightarrow, \neg, \wedge, p$) have four nodes, and no branch has more.

Warning. The height counts the nodes on the longest **branch**, not the symbols of the formula. The formula $\neg(p \wedge \neg q)$ has seven symbols, $\neg$, (, p, $\wedge$, $\neg$, q,), and height 4; the formula $((p \wedge q) \wedge (r \wedge p))$ has thirteen symbols and height only 3. Parentheses never count towards the height, and a wide tree can be shallow.

Warning. A parse tree is not a truth table. The tree records how a formula is **built**, and it is the same whatever the atoms stand for; nothing in it is true or false. Values come in Section 3.3, where they are computed **along** the tree, from the leaves up. Mixing the two, for instance writing "T" beside a node while drawing the tree, is a habit worth not forming: the same tree carries different values under different assignments.

3.1.7 Shorthand and official formulas

Nobody wants to write $(((p \wedge q) \to r) \leftrightarrow (p \to (q \to r)))$. Chapter 1 used conventions that let us drop parentheses (Section 1.1, the conventions before Definition 1.1.2), and we keep using them. What changes is their status: they are **not** part of the definition of a formula. A string written with the conventions is **shorthand**, and it stands for exactly one official formula, the one obtained by putting the dropped parentheses back.

The translation, from shorthand to official form, goes as follows.

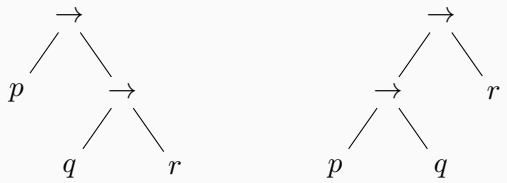
1. Put back the outer pair of parentheses of a formula whose main connective is binary: $p \wedge q$ stands for $(p \wedge q)$.
2. $\neg$ binds most tightly: it applies to the smallest formula that follows it. So $\neg p \wedge q$ stands for $(\neg p \wedge q)$, not for $\neg(p \wedge q)$.
3. $\wedge$ and $\vee$ bind more tightly than $\to$ and $\leftrightarrow$: $p \wedge q \to r$ stands for $((p \wedge q) \to r)$.
4. $\to$ groups to the right: $p \to q \to r$ stands for $(p \to (q \to r))$.
5. Chapter 1 sometimes wrote a negation in parentheses for emphasis, as in $(\neg P) \to Q$; such parentheses are simply removed.

In a mixture of $\wedge$ with $\vee$, or of $\to$ with $\leftrightarrow$, the shorthand keeps the parentheses (Chapter 1's rule for such mixtures), so no rule is needed for those.

Example. Two groupings of three implications

Write the official formula for the shorthand $p \to q \to r$, draw its parse tree beside that of $((p \to q) \to r)$, and find a situation in which the two formulas say different things.

Solution. By rule 4, $p \to q \to r$ stands for $(p \to (q \to r))$: the first $\to$ is the main connective, and its conclusion is $(q \to r)$. The other grouping, $((p \to q) \to r)$, has the second $\to$ as main connective, and its hypothesis is $(p \to q)$. Left, $(p \to (q \to r))$; right, $((p \to q) \to r)$.



The two trees are mirror images in shape, and they say different things. Suppose p, q and r are all false. The left formula is an implication whose hypothesis p is false, so it is true (vacuous truth, Proposition 2.1.2). The right formula has the true hypothesis $p \to q$ (again vacuous, since p is false) and the false conclusion r, so it is false. The everyday reading of Chapters 1 and 2 is enough to see this; Section 3.3 makes it a computation.

This is why the grouping convention matters. Chapter 1 chose to group to the right because statements of the form "if P, then if Q, then R" are common and $(P \to Q) \to R$ is rare.

Going the other way, from official to shorthand, we drop every pair of parentheses the rules would put back. $(\neg(p \vee q) \leftrightarrow (\neg p \wedge \neg q))$ becomes $\neg(p \vee q) \leftrightarrow \neg p \wedge \neg q$: the outer pair goes by rule 1, the pair around $\neg p \wedge \neg q$ by rule 3, and the pair after the first $\neg$ must stay, since without it rule 2 would read $\neg p \vee q$.

From now on we write formulas in shorthand whenever that is clearer, and every statement about "the formula $p \wedge q$" is about the official formula $(p \wedge q)$. Where the official form matters, as in counting parentheses or symbols, we write it out.

3.1.8 Letters, metavariables and substitution

We can now say exactly what the letters of Chapters 1 and 2 were.

Every rule was stated as a schema (Definition 1.2.1), with metavariables: $\wedge$I writes $\varphi \wedge \psi$ from φ and ψ, **for any** formulas φ and ψ. The derivations of those chapters were written with capital letters, as in the derivation of Proposition 1.4.1:

1	$P \to Q$	
2	$Q \to R$	
3	$\quad P$	
4	$\quad Q$	$\to$E 1, 3
5	$\quad R$	$\to$E 2, 4
6	$P \to R$	$\to$I 3–5

In the language of this chapter, read the letters P, Q, R as three atoms. Then this is a derivation of $p \to q,\ q \to r \vdash p \to r$ in the official sense, written in shorthand. But Chapter 1 meant more: the derivation was a **pattern**, good for any statements put in for the letters. Here is that promise made exact.

Substitution. Replace each atom of a derivation by a formula, **the same formula at every occurrence of the same atom**, and all such replacements at once. The result is again a derivation.

Here is why. First, each line becomes a formula. Replacing atoms by formulas in a formula χ gives a formula: take a formation sequence of χ, write in front of it a formation sequence of each formula put in for an atom, and replace the atoms inside the old entries. Each old entry that was an atom has become the last entry of one of the sequences written in front; each old entry built by (F2) or (F3) is now built by the same clause from the replaced versions of the same earlier entries. So the new list is a formation sequence, and its last entry is the replaced χ.

Second, each line is still justified. Each line of the old derivation was a premise, an assumption, a repetition, or an instance of a rule's schema (Definition 1.4.2, clauses (D1)–(D4)). Being an instance means: the line and the lines it cites are the schema with its metavariables replaced by certain formulas. The schema's own symbols are connectives, $\bot$ and parentheses, never atoms, so replacing atoms in the lines is the same as replacing atoms inside those certain formulas. The new lines are therefore the same schema with its metavariables replaced by the new formulas: an instance of the same rule.

Third, nothing about citations changes. The line numbers, the boxes, and which line cites which are untouched, and whether a line or a box is available depends on nothing else (Definition 1.4.2, the scope rule). So the result is a derivation, with the replaced premises and the replaced last line.

For instance, put $(p_1 \wedge p_2)$ for P, $\neg p_1$ for Q, and $(p_2 \vee p_1)$ for R in the derivation above:

1	$(p_1 \wedge p_2) \to \neg p_1$	
2	$\neg p_1 \to (p_2 \vee p_1)$	
3	$\quad p_1 \wedge p_2$	
4	$\quad \neg p_1$	$\to$E 1, 3
5	$\quad p_2 \vee p_1$	$\to$E 2, 4
6	$(p_1 \wedge p_2) \to (p_2 \vee p_1)$	$\to$I 3–5

Line 4 is still $\to$E: lines 1 and 3 are $\varphi \to \psi$ and φ with φ the formula $p_1 \wedge p_2$ and ψ the formula $\neg p_1$. Line 6 is still $\to$I with the same box.

The fact is immediate rule by rule, because every rule was stated for arbitrary formulas from the start. It is what the phrase "the letters stand for arbitrary statements" (Chapter 1, Section 1.2) and the clause "replace each letter ... the same statement for every occurrence" of Definition 2.3.1 were relying on. We use it in two places later in the chapter: the soundness proof of Section 3.5 reads a derived rule as its pasted derivation, and the completeness proof of Section 3.6 uses the sequents of Chapters 1 and 2 with their letters replaced by formulas. Its counterpart for truth values is a lemma of Section 3.4.

Remark. The replacement must be **uniform**. Replacing only the first P of the derivation above by $p_1 \wedge p_2$, and leaving the P on line 3 alone, would break line 4, which needs the hypothesis of line 1 to be literally the formula on line 3.

3.1.9 Exercises

3.1.9.1 A. Check your understanding

Exercise. A1

(a) Decide which of the following strings are formulas. Give a formation sequence for each formula, and for each nonformula name the test or clause it fails: (i) $(p \vee \bot)$; (ii) $\neg\neg p$; (iii) $(p\neg q)$; (iv) $((p \to q))$; (v) $(p \wedge q \wedge r)$.

(b) List the subformulas of $(\neg(p \to q) \vee q)$ and give its height.

(c) True or false, with a reason: every subformula of φ other than φ itself has smaller height than φ.

(d) Why does clause (F3) of Definition 3.1.1 put parentheses around its result, while (F2) does not?

(e) Explain why clauses (F1)–(F3) alone do not show that $\to p$ is not a formula, and which clause does.

Solution. (a) (i) A formula: p, $\bot$, $(p \vee \bot)$.

(ii) A formula: p, $\neg p$, $\neg\neg p$.

(iii) Not a formula. It begins with (, so by the first-symbol test it would be $(\varphi \circ \psi)$, and the string $p\neg q$ between the parentheses would contain a binary connective. It contains none.

(iv) Not a formula. It begins with (, so it would be $(\varphi \circ \psi)$ with $\varphi \circ \psi$ the string $(p \to q)$. The only binary connective in that string is the $\to$, so φ would be the string $(p$, of length 2. A formula of length 2 begins with $\neg$, by the first-symbol test (one beginning with (has length at least 5). So $(p$ is not a formula, and neither is the string. Doubled parentheses are never part of an official formula.

(v) Not a formula. It would be $(\varphi \wedge \psi)$ with the $\wedge$ one of the two. If it is the first, ψ is the string $q \wedge r$; if the second, φ is $p \wedge q$. Each has length 3 and begins with an atom, so by the first-symbol test neither is a formula. (Chapter 1's conventions give no reading of the shorthand $p \wedge q \wedge r$; Section 3.4 introduces it for the right-grouped $(p \wedge (q \wedge r))$ and explains why either grouping would do.)

(b) The formula is $(\neg(p \to q) \vee q)$. Its subformulas are $(\neg(p \to q) \vee q)$, $\neg(p \to q)$, $(p \to q)$, p and q: five, with q listed once although it sits at two leaves. The longest branch is $\vee, \neg, \to, p$ (or $\vee, \neg, \to, q$), so the height is 4.

(c) True. A subformula other than φ sits at a node below the root, and the part of the tree below that node is its parse tree. A longest branch of that part, with the path from the root of φ's tree down to the node added in front, is a branch of φ's tree with at least one more node. So φ's tree has a branch longer than every branch of the subformula's tree.

(d) A binary connective joins two formulas, and without parentheses a string such as $p \wedge q \vee r$ could be built in two ways, with different meanings (see "Why this definition"). A negation stands in front of the one formula it negates, so there is nothing to group.

(e) Clauses (F1)–(F3) say which strings are formulas and never which are not: the collection of all strings satisfies them. Clause (F4) says a formula is the last entry of a formation sequence, and from it follows the first-symbol test: $\to p$ has length 2 and does not begin with $\neg$ or (.

3.1.9.2 B. Practice

Exercise. B1: Trees from Chapter 1

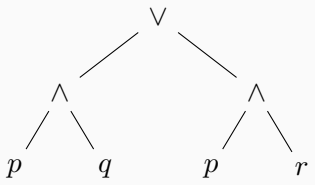
Each formula below appears, with capital letters, in a result of Chapter 1 (Proposition 1.4.2 and Proposition 1.6.1); here the letters are atoms. For each, write the official formula, draw its parse tree, list its subformulas, and give its height.

(a) $(p \wedge q) \to r$
(b) $p \to (q \to r)$
(c) $(p \wedge q) \vee (p \wedge r)$

Solution. *(a)* Official form $((p \wedge q) \to r)$. Root $\to$; left subtree $\wedge$ with leaves p, q; right subtree the leaf r. Subformulas: $((p \wedge q) \to r)$, $(p \wedge q)$, p, q, r. Height 3, along $\to, \wedge, p$.

(b) Official form $(p \to (q \to r))$. Root $\to$; left the leaf p; right subtree $\to$ with leaves q, r. Subformulas: $(p \to (q \to r))$, $(q \to r)$, p, q, r. Height 3, along $\to, \to, q$. (This is the left tree of Example (Two groupings of three implications). The formulas of (a) and (b) have different trees; Proposition 1.4.2 says that each can be derived from the other.)

(c) Official form $((p \wedge q) \vee (p \wedge r))$. Root $\vee$; left subtree $\wedge$ with leaves p, q; right subtree $\wedge$ with leaves p, r.



Subformulas: $((p \wedge q) \vee (p \wedge r))$, $(p \wedge q)$, $(p \wedge r)$, p, q, r: six, with p listed once although it sits at two leaves. Height 3.

Exercise. B2: Shorthand and back

(a) Write in shorthand, dropping every pair of parentheses the conventions allow: $((p \wedge q) \to (r \to p))$ and $(\neg(p \vee q) \to (\neg p \wedge \neg q))$.

(b) Write the official formula for: $p \to q \vee r \to \neg p$ and $\neg p \wedge q \leftrightarrow r$.

Solution. (a) $p \wedge q \to r \to p$. The outer pair goes by rule 1; $(p \wedge q)$ loses its pair because $\wedge$ binds more tightly than $\to$ (rule 3); $(r \to p)$ loses its pair because $\to$ groups to the right (rule 4).

$\neg(p \vee q) \to \neg p \wedge \neg q$. The outer pair goes; the pair around $\neg p \wedge \neg q$ goes by rule 3. The pair after the first $\neg$ stays: without it, rule 2 would make the $\neg$ apply to p alone.

(b) $(p \to ((q \vee r) \to \neg p))$. Rule 3 groups $q \vee r$ first; rule 4 then makes the first $\to$ the main connective, with conclusion $((q \vee r) \to \neg p)$.

$((\neg p \wedge q) \leftrightarrow r)$. Rule 2 makes $\neg$ apply to p alone; rule 3 groups $\neg p \wedge q$ before $\leftrightarrow$.

3.1.9.3 C. Going deeper

Exercise. C1: Forgetting the grouping convention

Suppose we forgot rule 4 of the shorthand, so that $p \to q \to r \to s$ could stand for any official formula obtained by inserting parentheses into it.

(a) List every official formula it could stand for, and draw the shape of each parse tree.

(b) How many are there? Say how the count is found, by splitting at the main connective.

(c) Which of them is the one rule 4 chooses?

Solution (Hint 1). The main connective is one of the three arrows. Choose it first.

Solution (Hint 2). Write c_n for the number of groupings of a chain of n atoms, and find c_1, c_2 and c_3 first.

Solution. (a) The main connective is one of the three arrows; what lies on each side is then grouped in every possible way.

- Main connective the **first** arrow: p on the left, and $q \to r \to s$ on the right, grouped in one of two ways. This gives $(p \to (q \to (r \to s)))$ and $(p \to ((q \to r) \to s))$.
- Main connective the **second** arrow: $(p \to q)$ on the left and $(r \to s)$ on the right, each grouped in one way. This gives $((p \to q) \to (r \to s))$.
- Main connective the **third** arrow: $p \to q \to r$ on the left, grouped in one of two ways, and s on the right. This gives $((p \to (q \to r)) \to s)$ and $(((p \to q) \to r) \to s)$.

Their parse trees have the root $\to$ in each case: in the first two, the leaf p on the left and a tree of height 3 on the right (a right-leaning chain, or a left-leaning one); in the third, two trees of height 2 side by side; in the last two, a tree of height 3 on the left and the leaf s on the right.

(b) There are 5. Write c_n for the number of groupings of a chain of n atoms joined by one binary connective. A chain of one or two atoms has one grouping: $c_1 = c_2 = 1$. For three atoms the main connective is the first or the second arrow, and the side with two atoms has one grouping: $c_3 = c_1c_2 + c_2c_1 = 2$. For four atoms, choosing the main connective splits the chain into a left part of k atoms and a right part of $4 - k$ atoms, for $k = 1, 2, 3$, and the two parts are grouped independently:

$$c_4 = c_1c_3 + c_2c_2 + c_3c_1 = 2 + 1 + 2 = 5.$$

The three terms are the three bullets of (a).

(c) Rule 4 groups to the right, so the main connective is the first arrow and its right side is grouped to the right again: $(p \to (q \to (r \to s)))$, the first formula of the list.

The numbers $1, 1, 2, 5, 14, 42, \ldots$ produced by this splitting are the **Catalan numbers**; they grow quickly: a chain of eleven atoms joined by ten connectives already has 16,796 readings. Parentheses, or a fixed convention, are not pedantry.

3.2 Induction on Numbers and on Formulas

Section 3.1 made formulas into objects: strings built from atoms and $\bot$ by five clauses, and by nothing else. There are infinitely many of them, and the theorems this chapter is heading for are about all of them. This section supplies the tool for proving anything about infinitely many objects at once. It is **induction**, first on the natural numbers, where it is familiar, then on formulas, where it takes the form this chapter uses again and again. With it we prove that every formula can be read in only one way, and that a definition given one clause per connective defines exactly one thing.

3.2.1 One argument for infinitely many cases

Here is a claim about every formula: *every formula has as many left parentheses as right ones.* It is believable. The clause (F3) of Definition 3.1.1 adds one of each, and the other clauses add none. But the claim is about infinitely many strings, and checking them one by one never ends.

The belief already contains the shape of a proof. Every formula is reached by finitely many building steps from atoms and $\bot$ (clause (F4)). The starting pieces have no parentheses, so they

have equally many of each kind. Each building step turns formulas that balance into a formula that balances. So, step by step along any formation sequence, balance is never lost, and every formula balances.

Exactly the same shape appears with numbers. To show that some property holds for every natural number, show that it holds for 0, and that it passes from each number to the next. Every natural number is reached from 0 by finitely many steps of "add one", so the property is never lost. This is **induction**. We state it for numbers first, because it is simpler there, then prove the version for formulas from it.

Two words about language before we start. Statements such as "for every $n \in \mathbb{N}$, ..." and "there is a k with ..." are used in this section in their everyday sense, as Chapters 1 and 2 used them. Their rules of proof are the subject of Chapter 4, and nothing here needs more than the everyday reading. Likewise, a "collection" of formulas or numbers is meant in its everyday sense; *Sets, Relations, and Cardinality*, Chapter 1 treats sets properly. As in the rest of the book, $\mathbb{N} = \{0, 1, 2, \dots\}$ starts at 0, and the arithmetic and order of the natural numbers are used as school facts.

3.2.2 The principle of induction

A **property of natural numbers** is an open sentence $P(n)$ about a natural number n, such as "$n^2 \geq n$" or "$2^n > n$": for each particular n it becomes a statement, true or false (compare the open sentence $x > 2$ of Example (Statements and nonstatements), which became a statement once x was fixed).

If a property holds for 0*, and passes from each number to the next, then it holds for every number.*

Axiom 3.2.1. Principle of Induction

Let $P(n)$ be a property of natural numbers. Suppose that

(i) $P(0)$ holds, and

(ii) **for every** $n \in \mathbb{N}$, if $P(n)$ holds, then $P(n+1)$ holds.

Then $P(n)$ holds **for every** $n \in \mathbb{N}$.

In symbols (the quantifier symbols are Chapter 4's; here they abbreviate the words):

$$\Big(P(0) \ \wedge \ \forall n \in \mathbb{N} \ \big(P(n) \to P(n+1)\big)\Big) \ \to \ \forall n \in \mathbb{N} \ P(n).$$

In words: hypothesis (i) is the **base case**. Hypothesis (ii) is the **inductive step**, and inside it, the assumption "$P(n)$ holds" is the **induction hypothesis**. The step is **one** implication, proved **once**, for a number n about which we assume nothing except that it is a natural number with $P(n)$. The conclusion is about every natural number at once.

Its status. We take Axiom 3.2.1 as a basic property of $\mathbb{N}$, given and not proved, in the same way that *Analysis on the Real Line*, Chapter 1 takes the least upper bound axiom as a basic property of $\mathbb{R}$. It is, in a precise sense, what the three dots in $\{0, 1, 2, \dots\}$ mean: the natural numbers are what we reach from 0 by adding one, again and again, and nothing else. *Sets, Relations, and Cardinality*, Chapter 2 proves that it is equivalent to the well-ordering principle (every nonempty collection of natural numbers has a least element), and *Number Systems: From*

$\mathbb{N}$ *to* $\mathbb{R}$, Chapter 1 derives it from the axioms of set theory. Nothing in this book depends on those two routes; every later use cites this principle.

Dominoes. The usual picture is a row of dominoes, one for each natural number. The base case knocks over the first. The inductive step says that each domino, if it falls, knocks over the next. Then every domino falls. The picture carries the one point that matters: the step must work for **every** n, the first transition from 0 to 1 included. A single domino standing too far from the next stops the whole row, however well the others are placed.

Not an infinite derivation. From $P(0)$ and the step we could derive $P(1)$ by $\to$E, then $P(2)$, and so on; each single $P(n)$ has a finite derivation, n lines long or so. But no finite derivation contains all of them, and "and so on" is not a rule. The principle of induction is what licenses the passage, in one step, from the two hypotheses to "for every n". It is a rule about $\mathbb{N}$, not a rule of logic.

The step as a box. In the language of Chapter 1, the inductive step is proved by a $\to$I box. We fix a natural number n, knowing nothing about it except what the step allows, and open a box assuming $P(n)$; inside the box we reach $P(n+1)$; closing the box gives the implication.

$$\begin{array}{lll} i & \vert & \quad \vert\, P(n) \\ & \vert & \quad \vert\, \vdots \\ j & \vert & \quad \vert\, P(n+1) \\ k & \vert\ P(n) \to P(n+1) & \qquad \to\text{I } i\text{–}j \end{array}$$

The passage from "this n was arbitrary" to "for every n" is a rule for "for every", and Chapter 4 states it. For now, "let $n \in \mathbb{N}$" means what Chapter 1 made it mean (Section 1.1): one fixed number, not further specified.

Starting from another number. Often a property is claimed only from some $b \in \mathbb{N}$ on, as in "for every $n \geq 1$". Then it suffices to prove $P(b)$, and, for every $n \in \mathbb{N}$ with $n \geq b$, that $P(n)$ implies $P(n+1)$. Indeed, apply Axiom 3.2.1 to the property $Q(m)$: "$P(b+m)$ holds". Its base case $Q(0)$ is $P(b)$, and its step from $Q(m)$ to $Q(m+1)$ is the step from $P(b+m)$ to $P(b+m+1)$, where $b+m \geq b$. So $Q(m)$ holds for every $m \in \mathbb{N}$, and every natural number $n \geq b$ is $b+m$ for the natural number $m = n-b$. We call this induction **started at** b; it is Axiom 3.2.1 applied to Q.

Here is the first proof by induction, with every part labeled.

Example. A sum of products of neighbors

Prove that for every natural number $n \geq 1$,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}.$$

Solution. Let $P(n)$ be the displayed equation. Its left side is the sum of the n terms $k(k+1)$ for $k = 1, \ldots, n$, so the left side for $n+1$ is the left side for n plus the one new term $(n+1)(n+2)$. Before proving anything, test small cases. For $n = 1, 2, 3$ the left side is 2, $2+6=8$ and $8+12=20$, and the right side is $\frac{1\cdot 2\cdot 3}{3} = 2$, $\frac{2\cdot 3\cdot 4}{3} = 8$ and $\frac{3\cdot 4\cdot 5}{3} = 20$. The claim is plausible. It starts at $n = 1$, so we use induction started at 1 (see above).

Base case. For $n = 1$ the left side is the single term $1 \cdot 2 = 2$, and the right side is $\frac{1 \cdot 2 \cdot 3}{3} = 2$. So $P(1)$ holds.

Inductive step. Let $n \in \mathbb{N}$ with $n \geq 1$, and suppose $P(n)$ holds (the induction hypothesis). Then

$$\begin{aligned} 1 \cdot 2 + \cdots + n(n+1) + (n+1)(n+2) &= \frac{n(n+1)(n+2)}{3} + (n+1)(n+2) \\ &= \frac{(n+1)(n+2)(n+3)}{3}, \end{aligned}$$

where the first equality is the induction hypothesis and the second takes out the common factor $(n+1)(n+2)$, leaving $\frac{n}{3} + 1 = \frac{n+3}{3}$. The last expression is the right side of $P(n+1)$. So $P(n)$ implies $P(n+1)$.

Conclusion. By Axiom 3.2.1, started at 1, $P(n)$ holds for every natural number $n \geq 1$.

Notice where the induction hypothesis entered: exactly once, to replace the sum of the first n terms by a closed expression. A proof by induction that never uses its induction hypothesis is either not an induction or is hiding a gap.

The inductive step of an inequality usually has a second ingredient besides the hypothesis: a crude estimate that finishes the job.

Example. An inequality by induction

Prove that $3^n \geq 2n + 1$ for every $n \in \mathbb{N}$.

Solution. Let $P(n)$ be "$3^n \geq 2n + 1$".

Base case. $3^0 = 1 = 2 \cdot 0 + 1$, so $P(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $3^n \geq 2n + 1$. Then

$$3^{n+1} = 3 \cdot 3^n \geq 3(2n+1) = 6n + 3 \geq 2n + 3 = 2(n+1) + 1,$$

where the first inequality multiplies the induction hypothesis by the positive number 3, and the second uses $6n \geq 2n$, true because $n \geq 0$. So $P(n+1)$ holds.

Conclusion. By Axiom 3.2.1, $3^n \geq 2n + 1$ for every $n \in \mathbb{N}$.

The estimate $6n + 3 \geq 2n + 3$ throws away a great deal. That is typical: the step needs only to reach $P(n+1)$, not to reach it tightly, and a generous estimate that is obviously true is better than a sharp one that needs its own proof.

Warning. The inductive step must hold for **every** n, including the first transition. Try to prove "$2^n \geq n^2$ for every $n \in \mathbb{N}$": the base case $1 \geq 0$ holds, and the step $2^{n+1} = 2 \cdot 2^n \geq 2n^2 \geq (n+1)^2$ looks fine, but its last inequality holds only for $n \geq 3$, so the steps from 0 to 1, from 1 to 2 and from 2 to 3 are not covered. The claim survives the first two gaps (at $n = 1$, $2 \geq 1$, and at $n = 2$ both sides are 4), but it is false at $n = 3$, where $2^3 = 8 < 9 = 3^2$;

it is true again from $n = 4$ on. *Sets, Relations, and Cardinality*, Chapter 2 shows a famous "proof" that fails in the same way at its very first step, from 1 to 2.

3.2.3 Strong induction

Sometimes the step from n to $n+1$ is the wrong shape. The immediate parts of a formula $(\varphi \circ \psi)$ are shorter than it, but not by exactly one symbol; a natural number may be best understood through a smaller one that is not its predecessor. In such cases we want to assume the property not just for n, but for **every** smaller number.

If a property holds for n whenever it holds for all smaller numbers, then it holds for every number.

Theorem 3.2.2. Strong Induction

Let $P(n)$ be a property of natural numbers. Suppose that **for every** $n \in \mathbb{N}$: if $P(k)$ holds for **every** $k \in \mathbb{N}$ with $k < n$, then $P(n)$ holds. Then $P(n)$ holds for every $n \in \mathbb{N}$.

This form is also called **course-of-values induction**: the hypothesis may use the whole course of values $P(0), \ldots, P(n-1)$, not just the last one. Its induction hypothesis is stronger (we assume more), so its step is easier to prove; the theorem says we lose nothing by it.

There seems to be no base case. There is one, hidden. For $n = 0$, the hypothesis says: if $P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$, then $P(0)$ holds. No natural number is less than 0, so "$P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$" is **vacuously** true: for each $k \in \mathbb{N}$, the implication "if $k < 0$, then $P(k)$" has a false hypothesis, and an implication with a false hypothesis holds (Proposition 2.1.2, $\neg P \vdash P \to Q$). So at $n = 0$ the hypothesis of the theorem hands us $P(0)$ outright. In practice, the proof of the step usually treats small n separately, and that is where the base cases live.

Idea. Strong induction is ordinary induction applied to a stronger property. Let $Q(n)$ say "$P(k)$ holds for **every** $k \leq n$". Then $Q(n)$ is exactly the hypothesis the theorem needs at $n+1$, since the numbers below $n+1$ are those up to n; so the theorem's hypothesis moves us from $Q(n)$ to $Q(n+1)$. Ordinary induction on Q does the rest, and $Q(n)$ contains $P(n)$.

Proof. Suppose the hypothesis of the theorem holds. For $n \in \mathbb{N}$, let $Q(n)$ be the property "$P(k)$ holds for every $k \in \mathbb{N}$ with $k \leq n$". We prove $Q(n)$ for every n by Axiom 3.2.1.

Base case. No natural number is less than 0, so $P(k)$ holds for every $k \in \mathbb{N}$ with $k < 0$, vacuously. By the hypothesis of the theorem at $n = 0$, $P(0)$ holds. The only $k \in \mathbb{N}$ with $k \leq 0$ is 0, so $Q(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $Q(n)$. The natural numbers $k < n+1$ are exactly those with $k \leq n$, since no natural number lies strictly between n and $n+1$. So $Q(n)$ says that $P(k)$ holds for every $k \in \mathbb{N}$ with $k < n+1$, and the hypothesis of the theorem at $n+1$ gives $P(n+1)$. A natural number $k \leq n+1$ satisfies $k \leq n$ or $k = n+1$, and $P(k)$ holds in both cases. Hence $Q(n+1)$.

By Axiom 3.2.1, $Q(n)$ holds for every $n \in \mathbb{N}$. Given $n \in \mathbb{N}$, $Q(n)$ applied to $k = n$ gives $P(n)$. This proves the theorem.

■

Quick check. In the proof of Theorem 3.2.2, to which property is Axiom 3.2.1 applied, and why is it not applied to $P(n)$ itself?

Solution. To $Q(n)$: "$P(k)$ holds for every $k \in \mathbb{N}$ with $k \leq n$". Applied to $P(n)$ itself, ordinary induction would give us only $P(n)$ as induction hypothesis, while the hypothesis of the theorem at $n+1$ asks for $P(k)$ at **every** $k \leq n$. The property Q carries exactly that information from one step to the next.

The first use shows the typical shape: the step splits into a few small cases, handled directly, and a general case that reaches back further than one.

Example. Threes and fives

Prove that every natural number $n \geq 8$ is a sum of threes and fives: $n = 3a + 5b$ for some integers $a, b \geq 0$.

Solution. Ordinary induction is awkward here. From $n = 3a + 5b$ we cannot easily reach $n+1$: we would have to trade some threes for fives or the other way round, and there may be none of the kind we want to trade. Going back **three** steps is easy instead: if $n-3$ is a sum of threes and fives, add one more three. So we use strong induction.

For $n \in \mathbb{N}$, let $P(n)$ be "if $n \geq 8$, then $n = 3a + 5b$ for some integers $a, b \geq 0$". Let $n \in \mathbb{N}$, and suppose $P(k)$ holds for every $k \in \mathbb{N}$ with $k < n$. We show $P(n)$.

Case $n < 8$. Then the hypothesis "$n \geq 8$" of $P(n)$ is false, and $P(n)$ holds vacuously.

Cases $n = 8, 9, 10$. Directly: $8 = 3 + 5$, $9 = 3 \cdot 3$, $10 = 5 \cdot 2$.

Case $n \geq 11$. Let $k = n - 3$. Then $k \geq 8$, so $k \in \mathbb{N}$, and $k < n$. By the induction hypothesis $P(k)$, there are integers $a, b \geq 0$ with $n - 3 = 3a + 5b$. Then $n = 3(a+1) + 5b$, with $a + 1 \geq 0$ and $b \geq 0$.

In every case $P(n)$ holds. By Theorem 3.2.2, $P(n)$ holds for every $n \in \mathbb{N}$: every $n \geq 8$ is a sum of threes and fives.

The three direct cases are the base cases. They are needed because the general case reaches back three steps: for $n = 8, 9, 10$, the number $n - 3$ is $5, 6, 7$, which is below 8, and there the induction hypothesis $P(n-3)$ is vacuous and gives us nothing. The bound 8 cannot be lowered: 7 is not a sum of threes and fives, since neither 7 nor $7 - 5 = 2$ is a multiple of 3.

3.2.4 Structural induction

We return to formulas. The formation sequence of a formula plays the part that "$0, 1, 2, \ldots$" plays for numbers: every formula is reached from atoms and $\bot$ by finitely many uses of the clauses. So a property of formulas can be proved in the same two parts.

A **property of formulas** is a statement $P(\varphi)$ about a formula φ, true or false for each particular formula: "φ has as many left as right parentheses", "φ contains the atom p".

If every atom and $\bot$ has a property, and each way of building a formula passes the property from the parts to the whole, then every formula has it.

Theorem 3.2.3. Structural Induction

Let $P(\varphi)$ be a property of formulas. Suppose that

(i) $P(p)$ holds for every atom p, and $P(\bot)$ holds;

(ii) **for every** formula φ, if $P(\varphi)$ holds, then $P(\neg\varphi)$ holds;

(iii) **for all** formulas φ, ψ and **each** binary connective $\circ$, if $P(\varphi)$ and $P(\psi)$ hold, then $P((\varphi \circ \psi))$ holds.

Then $P(\varphi)$ holds for every formula φ.

In words: (i) is the base case, one for each kind of formula of length one. (ii) and (iii) are the inductive steps, one for each building clause of Definition 3.1.1, and inside them "$P(\varphi)$ and $P(\psi)$" is the induction hypothesis, assumed for the **immediate parts** of the formula being built. There are six cases in all when the four binary connectives are counted separately; often the four are handled at once, with $\circ$ left unspecified.

Idea. Take any formula and a formation sequence ending in it. Walk along the sequence: each entry is an atom or $\bot$, which has the property by (i), or is built from **earlier** entries, which have it already, by (ii) or (iii). "Earlier entries, any of them" is exactly the shape of strong induction, applied to the position in the list.

Proof. Suppose (i)–(iii) hold, and let χ be a formula. By Definition 3.1.1 (F4) there is a formation sequence $\chi_1, \ldots, \chi_m$ with $\chi_m = \chi$. For $i \in \mathbb{N}$, let $Q(i)$ be the property "if $1 \leq i \leq m$, then $P(\chi_i)$ holds". We prove $Q(i)$ for every $i \in \mathbb{N}$ by Theorem 3.2.2.

Let $i \in \mathbb{N}$, and suppose $Q(k)$ holds for every $k \in \mathbb{N}$ with $k < i$. If $i = 0$ or $i > m$, then $Q(i)$ holds vacuously. So let $1 \leq i \leq m$. By the definition of a formation sequence, one of three things is true.

- χ_i is an atom or $\bot$. Then $P(\chi_i)$ holds by (i).
- $\chi_i = \neg\chi_j$ for some $j < i$. Then $1 \leq j \leq m$, and $Q(j)$ gives $P(\chi_j)$. The entry χ_j is a formula (every entry of a formation sequence is one, Section 3.1), so (ii) applies and gives $P(\chi_i)$.
- $\chi_i = (\chi_j \circ \chi_l)$ for some $j, l < i$ and a binary connective $\circ$. Then $1 \leq j, l \leq m$, and $Q(j)$ and $Q(l)$ give $P(\chi_j)$ and $P(\chi_l)$, both entries are formulas, and (iii) gives $P(\chi_i)$.

So $Q(i)$ holds. By Theorem 3.2.2, $Q(i)$ holds for every $i \in \mathbb{N}$; in particular $Q(m)$ gives $P(\chi_m)$, that is, $P(\chi)$. Since χ was an arbitrary formula, this proves the theorem.

■

The proof uses **strong** induction, because an entry of a formation sequence may be built from any earlier entries, not only from the one just before it.

Remark. A tempting alternative is induction on the **height** of a formula: prove the property for all formulas of height 1, then for height $h+1$ assuming it for smaller heights. That route is circular here. The height was defined through "the" parse tree (Definition 3.1.4), and a formula has only one parse tree only because of the unique readability theorem below, whose proof uses structural induction. Inducting on the position in a formation sequence needs nothing but Definition 3.1.1.

Warning. A structural induction must cover **every** clause, the base case $\bot$ and the step for $\neg$ included. Try to prove "every formula contains an atom": it holds for atoms, and the steps for $\neg$ and for $(\varphi \circ \psi)$ keep it, since the atoms of the parts are atoms of the whole. But the case $\bot$ fails, and so does the claim: $\bot$, $\neg\bot$ and $(\bot \to \bot)$ contain no atom. A proof that "forgets" $\bot$ proves something false.

Quick check. Which clause of Definition 3.1.1 makes step (ii) of Theorem 3.2.3 necessary? Give a property of formulas that holds for every atom and for $\bot$, passes from φ and ψ to $(\varphi \circ \psi)$ for every binary connective, and yet fails for some formula.

Solution. Clause (F2), which builds $\neg\varphi$ from φ: step (ii) is the step for that clause. Without it, the formulas made by (F2) would be missed.

One such property: "φ is an atom, or is $\bot$, or begins with (". Atoms and $\bot$ have it, and every formula $(\varphi \circ \psi)$ begins with (, whatever its parts are. But $\neg p$ does not have it. So (i) and (iii) alone do not prove a property of every formula; the property fails exactly at a formula made by the forgotten clause.

3.2.5 Counting parentheses

Our first structural induction proves the claim this section began with, together with two refinements: one that the next theorem needs, and one that Exercise C1 uses. We count parentheses in strings, and for this some words about strings.

An **initial segment** of a string s is the string of the first k symbols of s, for some k with $0 \le k \le$ the length of s. For $k = 0$ it is the **empty** string, with no symbols; for k equal to the length it is s itself. An initial segment is **proper** if it is shorter than s. The **excess** of a string s is

$$\mathrm{ex}(s) := (\text{number of (in } s) - (\text{number of) in } s).$$

So a left parenthesis has excess 1, a right parenthesis -1, and any other symbol 0; and the excess of a string written as two strings one after the other is the sum of their excesses, since the counts add. For example, $\mathrm{ex}(\,(p \wedge (q\,) = 2$: after reading $(p \wedge (q$ we are inside two pairs of parentheses that have not yet closed. That is the picture to keep: the excess of an initial segment is the **depth** we are at when we have read that far.

Lemma 3.2.4. Parenthesis Count

Let χ be a formula.

(a) $\mathrm{ex}(\chi) = 0$: χ has as many left as right parentheses.

(b) Every initial segment u of χ has $\mathrm{ex}(u) \geq 0$. If moreover χ has the form $(\varphi \circ \psi)$ and u is proper and nonempty, then $\mathrm{ex}(u) \geq 1$.

(c) For every occurrence of a binary connective in χ, the initial segment of χ that ends just before that occurrence has excess at least 1.

In words: (a) is balance. (b) says that reading a formula from the left, we never close more parentheses than we have opened, and that inside a formula $(\varphi \circ \psi)$ we stay at depth at least 1 until the very last symbol. (c) says that no binary connective lies outside all parentheses. Part (a) is the fact we set out to prove; (c), with (a), is what the unique readability theorem needs, and (b) shows that no proper initial segment of a formula is a formula (Exercise C1). The steps for (b) and (c) use (a) for the parts, so the three are proved together.

Idea. Induction on the shape of a formula, all three parts at once. For $\neg\varphi$, the extra $\neg$ changes no count. For $(\varphi \circ \psi)$, list its initial segments by where they stop: before anything, inside φ, at the connective, inside ψ, at the end. Each is "(" followed by pieces whose excess the induction hypothesis controls.

Proof. We prove, by Theorem 3.2.3, that every formula χ has the property $P(\chi)$: "(a), (b), and (c) hold for χ".

Atoms and $\bot$. The formula is one symbol, not a parenthesis. (a) Its excess is 0. (b) Its initial segments are the empty string and the formula itself, both of excess 0; and it does not have the form $(\varphi \circ \psi)$, which has length at least 5. (c) It contains no binary connective.

Negation. Let φ be a formula with $P(\varphi)$; we show $P(\neg\varphi)$. (a) $\mathrm{ex}(\neg\varphi) = 0 + \mathrm{ex}(\varphi) = 0$, by (a) for φ. (b) An initial segment of $\neg\varphi$ is either empty, of excess 0, or $\neg$ followed by an initial segment u' of φ, of excess $\mathrm{ex}(u') \geq 0$ by (b) for φ. The formula $\neg\varphi$ begins with $\neg$, so it does not have the form $(\varphi' \circ \psi')$, and the second sentence of (b) says nothing about it. (c) A binary connective in $\neg\varphi$ lies in φ, and the initial segment of $\neg\varphi$ before it is $\neg$ followed by the initial segment of φ before it, whose excess is at least 1 by (c) for φ.

Binary connectives. Let φ, ψ be formulas with $P(\varphi)$ and $P(\psi)$, let $\circ$ be a binary connective, and let $\chi = (\varphi \circ \psi)$.

(a) $\mathrm{ex}(\chi) = 1 + \mathrm{ex}(\varphi) + 0 + \mathrm{ex}(\psi) - 1 = 0$, by (a) for φ and for ψ.

(b) Every initial segment of χ is of exactly one of the following five kinds, and its excess is as shown:

- the empty string: excess 0;
- (followed by an initial segment u' of φ: excess $1 + \mathrm{ex}(u') \geq 1$, by (b) for φ;
- $(\varphi \circ$: excess $1 + \mathrm{ex}(\varphi) + 0 = 1$, by (a) for φ;
- $(\varphi \circ$ followed by a nonempty initial segment v' of ψ: excess $1 + 0 + \mathrm{ex}(v') \geq 1$, by (a) for φ and (b) for ψ;
- χ itself: excess 0, by (a) for χ.

All five have excess at least 0. The proper nonempty ones are those of the middle three kinds, and they have excess at least 1.

(c) An occurrence of a binary connective in χ is the connective $\circ$ between φ and ψ, or lies in

φ, or lies in ψ. Before $\circ$ stands (φ, of excess 1. Before an occurrence in φ stands (followed by the initial segment of φ before it, of excess at least $1+1=2$ by (c) for φ. Before an occurrence in ψ stands ($\varphi\circ$ followed by the initial segment of ψ before it, of excess at least $1+0+1=2$ by (a) for φ and (c) for ψ. In all three cases the excess is at least 1.

So $P(\chi)$ holds. By Theorem 3.2.3, every formula has the property P, which proves the lemma.

■

The proof of (c) gave more than (c) states, and the next proof uses the extra. In a formula $(\varphi\circ\psi)$, the initial segment before the connective $\circ$ has excess exactly 1, and the initial segment before **any other** binary connective has excess at least 2. In the language of depth: the main connective is the only binary connective at depth one. That is the recipe of Section 3.1 for finding the root of a parse tree.

The three parts were proved together for a reason. The second sentence of (b), alone, cannot be proved by structural induction. For $(\neg(p\wedge q)\wedge r)$, the initial segment $(\neg(p$ is (followed by the initial segment $\neg(p$ of the formula $\neg(p\wedge q)$, and its excess is $1+\mathrm{ex}(\neg(p)$; to know that this is at least 1, we need $\mathrm{ex}(\neg(p)\geq 0$, a fact about an initial segment of a **negation**, which the second sentence of (b) does not cover. The first sentence of (b) does. A property that is too weak to pass through the induction must be strengthened until it does, and here the strengthening is (b)'s first sentence.

3.2.6 Unique readability

We can now prove what Section 3.1 promised: a formula can be built in only one way.

Theorem 3.2.5. Unique Readability

Let χ be a formula that is neither an atom nor $\bot$. Then **exactly one** of the following holds:

(i) $\chi=\neg\varphi$ for a formula φ;
(ii) $\chi=(\varphi\circ\psi)$ for formulas φ,ψ and a binary connective $\circ$.

In case (i), φ is **uniquely** determined by χ; in case (ii), $\circ$, φ and ψ are uniquely determined by χ.

In words: every compound formula has one main connective, and one way of being split into its immediate parts. This is the precise form of Chapter 1's claim that a statement built from letters by connectives can be read in only one way. Existence is the easy half; the content is **uniqueness**. The danger it excludes is a string such as $p\wedge q\vee r$, which, if it were a formula, could be split at $\wedge$ or at $\vee$.

Idea. Some form holds, by looking at the last step (Section 3.1); the first symbol tells the two forms apart. For case (ii), suppose $\chi=(\varphi\circ\psi)=(\varphi'\circ'\psi')$. By the remark after Lemma 3.2.4, each main connective is the only binary connective at depth one; so both sit at the same position of the string, and everything else is forced.

Proof. By Section 3.1 (looking at the last step), χ is an atom, $\bot$, $\neg\varphi$ for a formula φ, or $(\varphi \circ \psi)$ for formulas φ, ψ; by hypothesis it is one of the last two. It cannot be both, since a string of the form $\neg\varphi$ begins with $\neg$ and one of the form $(\varphi \circ \psi)$ begins with (.

Case (i). If $\chi = \neg\varphi = \neg\varphi'$, removing the first symbol gives $\varphi = \varphi'$.

Case (ii). Let $\chi = (\varphi \circ \psi)$. Before the connective $\circ$ stands the initial segment (φ, of excess $1 + \mathrm{ex}(\varphi) = 1$ by Lemma 3.2.4 (a). Before a binary connective inside φ stands (followed by an initial segment of φ of excess at least 1, by Lemma 3.2.4 (c); so the excess is at least 2. Before a binary connective inside ψ stands $(\varphi \circ$ followed by an initial segment of ψ of excess at least 1, by (c) for ψ and (a) for φ; so again at least 2. Hence the position of $\circ$ in χ is **the only** position of a binary connective in χ whose initial segment before it has excess 1, and this description mentions only the string χ.

Now suppose also $\chi = (\varphi' \circ' \psi')$ for formulas φ', ψ' and a binary connective $\circ'$. By the same argument, the position of $\circ'$ is the only position with that description. So $\circ$ and $\circ'$ stand at the same position of χ, and $\circ = \circ'$. That position is the one just after (and φ, and also the one just after (and φ', so φ and φ' have the same length; both are the string of symbols of χ between the first symbol and that position, so $\varphi = \varphi'$. Likewise ψ and ψ' are both the string of symbols between that position and the last symbol, so $\psi = \psi'$. This proves the theorem. ■

The theorem settles the question Section 3.1 left open about parse trees.

Every formula has exactly one parse tree. We know it has one. That it has only one is proved by structural induction. An atom or $\bot$ has only the one-node tree, since a tree made by (T2) or (T3) of Definition 3.1.2 is a tree for a string beginning with $\neg$ or (. A parse tree for $\neg\varphi$ must have been made by (T2), from a parse tree for some φ' with $\neg\varphi' = \neg\varphi$; so $\varphi' = \varphi$, and by the induction hypothesis the tree below the root is the one parse tree of φ. A parse tree for $(\varphi \circ \psi)$ must have been made by (T3), from parse trees for some φ', ψ' and a connective $\circ'$ with $(\varphi' \circ' \psi') = (\varphi \circ \psi)$; by Theorem 3.2.5, $\circ' = \circ$, $\varphi' = \varphi$ and $\psi' = \psi$, and by the induction hypothesis the two subtrees are the parse trees of φ and ψ. From now on we speak of **the** parse tree of a formula, and the height of Definition 3.1.4 is a well-defined number.

3.2.7 Definition by recursion

Height, the set of atoms, the number of connectives: each is naturally defined **one clause per way of building a formula**. The height of $\neg\varphi$ is one more than that of φ; the atoms of $(\varphi \circ \psi)$ are those of φ together with those of ψ. Such a definition is called **recursive**: it defines the value at a formula through the values at its immediate parts. The question is whether it really defines something. For a string that could be split in two ways, the clauses might assign two different values, and then they would define nothing. Unique readability says this never happens.

We say **function on formulas** for a rule that assigns to each formula exactly one value; the values may be numbers, truth values, formulas, or collections of atoms. (*Sets, Relations, and Cardinality*, Chapter 1 makes "function" precise; nothing here needs more than the everyday sense.)

Give a value for each atom and for $\bot$, and say how to get the value of a compound formula from its parts; then exactly one function on formulas obeys these clauses.

Theorem 3.2.6. Structural Recursion

Suppose we are given

- a value a_p for each atom p, and a value $a_\bot$;
- a rule $g_\neg$ that produces a value $g_\neg(\varphi, x)$ from any formula φ and any value x;
- for each binary connective $\circ$, a rule $g_\circ$ that produces a value $g_\circ(\varphi, \psi, x, y)$ from any formulas φ, ψ and any values x, y.

Then there is **exactly one** function f on formulas such that, for every atom p, all formulas φ, ψ and every binary connective $\circ$,

$$\begin{aligned} f(p) &= a_p, \qquad f(\bot) = a_\bot, \\ f(\neg\varphi) &= g_\neg(\varphi, f(\varphi)), \\ f((\varphi \circ \psi)) &= g_\circ(\varphi, \psi, f(\varphi), f(\psi)). \end{aligned}$$

In words: the three lines are the **clauses** of a recursive definition, one for each clause of Definition 3.1.1. The rules $g_\neg$ and $g_\circ$ may look at the immediate parts themselves as well as at their values; most definitions use only the values, and the list of subformulas below uses the parts. The theorem says the clauses define a function (**existence**) and only one (**uniqueness**).

Idea. Build the function on short formulas first. A formula with one symbol is an atom or $\bot$, and the first clauses fix its value. A longer formula has immediate parts that are shorter, whose values are already fixed, and unique readability says which clause applies and to which parts, with no choice left. So the value at each formula is forced, length by length. Formally: an ordinary induction on n, proving that there is exactly one function on the formulas of length at most n that obeys the clauses.

Proof. For $n \in \mathbb{N}$ with $n \geq 1$, call a function h, defined on the formulas of length at most n, **n-good** if it obeys the three clauses at every formula of length at most n. This makes sense: the immediate parts of a formula are shorter than it, so the clauses at a formula of length at most n mention values of h only at formulas of length at most n. Let $S(n)$ be "there is exactly one n-good function"; we prove $S(n)$ for every $n \geq 1$ by Axiom 3.2.1, started at 1.

Base case. By the first-symbol test (Section 3.1), the formulas of length 1 are the atoms and $\bot$. A 1-good function must give each atom p the value a_p and $\bot$ the value $a_\bot$, and the function that does so is 1-good, since the other two clauses concern no formula of length 1. So $S(1)$ holds.

Inductive step. Let $n \in \mathbb{N}$ with $n \geq 1$, suppose $S(n)$, and let h_n be the n-good function. Define h on the formulas of length at most $n+1$ as follows. At a formula of length at most n, let h agree with h_n. Let χ have length $n+1 \geq 2$; it is neither an atom nor $\bot$, so by Theorem 3.2.5 it is **either** $\neg\varphi$ for exactly one formula φ, **or** $(\varphi \circ \psi)$ for exactly one $\circ$, φ, ψ; the parts have length at most n. Let $h(\chi) := g_\neg(\varphi, h_n(\varphi))$ in the first case and $h(\chi) := g_\circ(\varphi, \psi, h_n(\varphi), h_n(\psi))$ in the second. Unique readability leaves no choice of case or of parts, so this is one value. The function h is $(n+1)$-good: at formulas of length at most n it agrees with the n-good h_n, including at their parts, and at formulas of length $n+1$ it obeys the clauses by construction.

For uniqueness, let h' be any $(n+1)$-good function. Its restriction to the formulas of length

at most n is n-good, so it is h_n, by $S(n)$. At a formula $\chi = \neg\varphi$ of length $n+1$, the clause for $\neg$ gives $h'(\chi) = g_\neg(\varphi, h'(\varphi)) = g_\neg(\varphi, h_n(\varphi)) = h(\chi)$; at $\chi = (\varphi \circ \psi)$ the same argument with $g_\circ$ gives $h'(\chi) = h(\chi)$. So $h' = h$, and $S(n+1)$ holds. By Axiom 3.2.1, started at 1, $S(n)$ holds for every $n \geq 1$; let h_n denote the n-good function.

The function f. If $m \geq n$, the restriction of h_m to the formulas of length at most n is n-good, so it is h_n. For a formula χ of length n, let $f(\chi) := h_n(\chi)$; then $f(\chi) = h_m(\chi)$ for every $m \geq n$. The function f obeys the clauses: for $\chi = \neg\varphi$ of length m, $f(\chi) = h_m(\chi) = g_\neg(\varphi, h_m(\varphi)) = g_\neg(\varphi, f(\varphi))$, and likewise for the other two clauses. And it is the only one: a function f' on formulas obeying the clauses restricts, for each n, to an n-good function, which is h_n; so $f'(\chi) = h_n(\chi) = f(\chi)$ for a formula χ of length n. This proves the theorem.

■

Unique readability was used at one place: to define $h(\chi)$ by exactly one clause, applied to exactly one choice of parts. Without it, the clauses could demand two different values at one string, and then no function would obey them.

Example. Four recursive definitions

Each of the following is a function on formulas by Theorem 3.2.6.

- **Height:** $\mathrm{ht}(p) = \mathrm{ht}(\bot) = 1$, $\mathrm{ht}(\neg\varphi) = \mathrm{ht}(\varphi) + 1$, and
$$\mathrm{ht}((\varphi \circ \psi)) = 1 + \max\{\mathrm{ht}(\varphi), \mathrm{ht}(\psi)\}.$$
- **Atoms:** $\mathrm{atoms}(p) = \{p\}$; $\mathrm{atoms}(\bot)$ has no members; $\mathrm{atoms}(\neg\varphi) = \mathrm{atoms}(\varphi)$; and $\mathrm{atoms}((\varphi \circ \psi))$ consists of the members of $\mathrm{atoms}(\varphi)$ together with those of $\mathrm{atoms}(\psi)$.
- **Number of connectives:** 0 for atoms and $\bot$; one more than for φ at $\neg\varphi$; one more than the sum for φ and ψ at $(\varphi \circ \psi)$.
- **Subformulas:** the list of subformulas of p is p, and of $\bot$ is $\bot$; of $\neg\varphi$, the formula $\neg\varphi$ followed by the subformulas of φ; of $(\varphi \circ \psi)$, the formula $(\varphi \circ \psi)$ followed by the subformulas of φ and then those of ψ, each formula kept only at its first appearance.

Compute all four for $\chi = (\neg(p \wedge q) \to (q \vee \bot))$, and also its length.

Solution. Work up from the leaves. For the parts: $\mathrm{ht}(p \wedge q) = 1 + \max\{1,1\} = 2$, so $\mathrm{ht}(\neg(p \wedge q)) = 3$; $\mathrm{ht}(q \vee \bot) = 2$. Hence

$$\mathrm{ht}(\chi) = 1 + \max\{3,2\} = 4.$$

The atoms of $\neg(p \wedge q)$ are p and q, those of $q \vee \bot$ are just q ($\bot$ contributes none), so $\mathrm{atoms}(\chi) = \{p, q\}$. The connectives: $\wedge$ and $\neg$ on the left, $\vee$ on the right, $\to$ at the root, four in all. The subformulas, in the order the clause lists them: χ, $\neg(p \wedge q)$, $(p \wedge q)$, p, q, $(q \vee \bot)$, $\bot$; the second q is dropped, being already listed. The length is 14: the official formula consists of the fourteen symbols

$$(\ \neg\ (\ p\ \wedge\ q\ )\ \to\ (\ q\ \vee\ \bot\ )\ ).$$

Two checks connect these definitions with Section 3.1. The **recursive height agrees with the tree height**: the parse tree of $\neg\varphi$ is that of φ with one node on top, so each branch gains one node; the tree of $(\varphi \circ \psi)$ is a new root over the trees of φ and ψ, so a longest branch is the root

followed by a longest branch of the taller of the two. A structural induction whose cases are these two sentences (and the one-node tree for atoms and $\bot$) shows the two heights are equal. In the same way, the recursive list of subformulas is the list of formulas at the nodes of the parse tree (Definition 3.1.3): the nodes of the tree of $\neg\varphi$ are its root, carrying $\neg\varphi$, and the nodes of the tree of φ.

The length of a formula also satisfies clauses of this shape, by counting symbols: a formula of length one for atoms and $\bot$, one more at $\neg\varphi$, and at $(\varphi \circ \psi)$ the lengths of φ and ψ plus 3 (two parentheses and the connective). For χ above: $(p \wedge q)$ has $1+1+3=5$ symbols, so $\neg(p \wedge q)$ has 6; $(q \vee \bot)$ has 5; so χ has $6+5+3=14$, as counted.

Warning. A recursive definition must have a clause for **every** way of building a formula, and each clause may use only the **immediate parts**. "Let $f(\varphi)$ be f of the longest subformula of φ plus one" is not a recursive definition: at φ itself the longest subformula is φ, and the "definition" refers to the value it is trying to define. The theorem applies only to clauses of the form it states.

3.2.8 Exercises

3.2.8.1 A. Check your understanding

Exercise. A1

(a) For each statement, say which kind of induction fits best (ordinary, strong, structural), and write down its base case(s) and the statement of its inductive step, without proving them: (i) $4^n - 1$ is divisible by 3 for every $n \in \mathbb{N}$; (ii) every natural number $n \geq 12$ is $4a + 5b$ for some integers $a, b \geq 0$; (iii) every formula contains an atom or $\bot$.

(b) True or false, with a reason: to apply Theorem 3.2.2, one must first check $P(0)$ separately.

(c) In the proof of Theorem 3.2.3, on what is the strong induction performed?

(d) Which parts of Lemma 3.2.4 does the proof of Theorem 3.2.5 use, and for which formulas?

(e) Where in the proof of Theorem 3.2.6 is Theorem 3.2.5 used, and why is it not needed for uniqueness?

Solution. (a) (i) Ordinary induction. Base case: $4^0 - 1 = 0$ is divisible by 3. Step: for every $n \in \mathbb{N}$, if $3 \mid 4^n - 1$, then $3 \mid 4^{n+1} - 1$. (It holds because $4^{n+1} - 1 = 4(4^n - 1) + 3$.)

(ii) Strong induction, reaching back four: from $n - 4 = 4a + 5b$ we get $n = 4(a+1) + 5b$. With $P(n)$ "if $n \geq 12$, then $n = 4a + 5b$ for some integers $a, b \geq 0$", the step is: for every $n \in \mathbb{N}$, if $P(k)$ holds for every $k < n$, then $P(n)$. Inside it, $n < 12$ is vacuous, and the base cases are $n = 12, 13, 14, 15$: $12 = 4 \cdot 3$, $13 = 4 \cdot 2 + 5$, $14 = 4 + 5 \cdot 2$, $15 = 5 \cdot 3$. For $n \geq 16$, $k = n - 4 \geq 12$ and $k < n$, and the hypothesis $P(n-4)$ is used.

(iii) Structural induction. Base cases: an atom contains itself, and $\bot$ contains $\bot$. Steps: if φ contains an atom or $\bot$, so does $\neg\varphi$; if φ and ψ do, so does $(\varphi \circ \psi)$, for each binary $\circ$. (Compare the warning in the section: with "an atom" alone, the case $\bot$ fails.)

(b) False. At $n = 0$ the hypothesis of Theorem 3.2.2 reads "if $P(k)$ holds for every $k \in \mathbb{N}$

with $k < 0$, then $P(0)$", and its "if" part holds vacuously, since no natural number is less than 0. So proving the hypothesis for every n includes proving $P(0)$; in practice small n are treated as separate cases inside the step, as in Example (Threes and fives).

(c) On the position i of an entry in a formation sequence $\chi_1, \dots, \chi_m$ of the given formula, with the property "if $1 \le i \le m$, then χ_i has P". Not on the height, which at that point is not yet known to be well defined.

(d) Part (a) for φ, to get excess 1 before the main connective and to get excess $1+0$ before the part ψ; part (c) for φ and for ψ, to get excess at least 2 before every other binary connective. Here $\chi = (\varphi \circ \psi)$ is the formula being read. Part (b) is not used in that proof.

(e) In the inductive step, to define $h(\chi)$ for a formula χ of length $n+1$: unique readability says which clause applies and to which parts, so the clauses give exactly one value. That two functions obeying the clauses agree needs no unique readability: by Theorem 3.2.3, if f and f' both obey the clauses, they agree at atoms and $\bot$; and if they agree at φ (and ψ), then at $\neg\varphi$ both equal $g_\neg(\varphi, f(\varphi))$, and at $(\varphi \circ \psi)$ both equal $g_\circ(\varphi, \psi, f(\varphi), f(\psi))$, whichever way the string may be split.

3.2.8.2 B. Practice

Exercise. B1

Prove that $2^n \ge n+1$ for every $n \in \mathbb{N}$.

Solution. Let $P(n)$ be "$2^n \ge n+1$".

Base case. $2^0 = 1 = 0+1$, so $P(0)$ holds.

Inductive step. Let $n \in \mathbb{N}$ and suppose $2^n \ge n+1$. Then

$$2^{n+1} = 2 \cdot 2^n \ge 2(n+1) = 2n+2 \ge n+2 = (n+1)+1,$$

where the first inequality multiplies the induction hypothesis by $2 > 0$, and the second uses $n \ge 0$. So $P(n+1)$ holds.

By Axiom 3.2.1, $2^n \ge n+1$ for every $n \in \mathbb{N}$.

Exercise. B2

Prove by structural induction that every formula has at most as many distinct subformulas as it has symbols. Use the recursive description of the subformulas in Example (Four recursive definitions) and the clauses for the length given after it.

Solution. For a formula φ, write $s(\varphi)$ for the number of distinct subformulas of φ and $\mathrm{len}(\varphi)$ for its length. We prove $s(\varphi) \le \mathrm{len}(\varphi)$ by Theorem 3.2.3.

Atoms and $\bot$. The only subformula is the formula itself: $s = 1 = \mathrm{len}$.

Negation. Suppose $s(\varphi) \le \mathrm{len}(\varphi)$. The subformulas of $\neg\varphi$ are $\neg\varphi$ and those of φ, so there are at most $1 + s(\varphi)$ of them (fewer if one repeats). Hence $s(\neg\varphi) \le 1 + s(\varphi) \le 1 + \mathrm{len}(\varphi) =$

$\mathrm{len}(\neg\varphi)$.

Binary connectives. Suppose $s(\varphi) \le \mathrm{len}(\varphi)$ and $s(\psi) \le \mathrm{len}(\psi)$. The subformulas of $(\varphi \circ \psi)$ are the formula itself and those of φ and of ψ, so

$$s((\varphi \circ \psi)) \le 1 + s(\varphi) + s(\psi) \le 1 + \mathrm{len}(\varphi) + \mathrm{len}(\psi) < \mathrm{len}(\varphi) + \mathrm{len}(\psi) + 3,$$

and the right side is $\mathrm{len}((\varphi \circ \psi))$.

By Theorem 3.2.3, $s(\varphi) \le \mathrm{len}(\varphi)$ for every formula φ. (The bound is far from tight: the parentheses never start a subformula of their own.)

Exercise. B3

The number of nodes of the parse tree of a formula is given by the recursion $\mathrm{nodes}(p) = \mathrm{nodes}(\bot) = 1$, $\mathrm{nodes}(\neg\varphi) = \mathrm{nodes}(\varphi) + 1$, $\mathrm{nodes}((\varphi \circ \psi)) = \mathrm{nodes}(\varphi) + \mathrm{nodes}(\psi) + 1$. Prove that every formula φ of height h satisfies $\mathrm{nodes}(\varphi) \le 2^h - 1$, and give a formula of height 3 for which equality holds.

Solution. We prove $\mathrm{nodes}(\varphi) \le 2^{\mathrm{ht}(\varphi)} - 1$ by Theorem 3.2.3, using the recursive height of Example (Four recursive definitions).

Atoms and $\bot$. $\mathrm{nodes} = 1 = 2^1 - 1$.

Negation. Let $h = \mathrm{ht}(\varphi)$ and suppose $\mathrm{nodes}(\varphi) \le 2^h - 1$. Then $\mathrm{ht}(\neg\varphi) = h + 1$ and

$$\mathrm{nodes}(\neg\varphi) \le 2^h - 1 + 1 = 2^h \le 2^{h+1} - 1,$$

where the last step is $1 \le 2^h$.

Binary connectives. Let $a = \mathrm{ht}(\varphi)$, $b = \mathrm{ht}(\psi)$, and suppose the bound for both. Then $h \coloneqq \mathrm{ht}((\varphi \circ \psi)) = 1 + \max\{a, b\}$, so $a \le h - 1$ and $b \le h - 1$, and $2^a, 2^b \le 2^{h-1}$. Hence

$$\mathrm{nodes}((\varphi \circ \psi)) \le (2^a - 1) + (2^b - 1) + 1 \le 2 \cdot 2^{h-1} - 1 = 2^h - 1.$$

By Theorem 3.2.3, the bound holds for every formula. Equality at height 3: $((p \wedge q) \wedge (r \wedge p))$ has $7 = 2^3 - 1$ nodes (a root, two $\wedge$ nodes, and four leaves). Equality needs every node above the bottom level to have two children.

3.2.8.3 C. Going deeper

Exercise. C1: Initial segments are not formulas

(a) Prove that no proper initial segment of a formula is a formula.

(b) Deduce, in a way different from the proof in the text, the uniqueness part of case (ii) of Theorem 3.2.5: if $(\varphi \circ \psi) = (\varphi' \circ' \psi')$ for formulas $\varphi, \psi, \varphi', \psi'$ and binary connectives $\circ, \circ'$, then $\varphi = \varphi'$, $\circ = \circ'$ and $\psi = \psi'$.

Solution (Hint 1). Structural induction on the formula whose initial segments you consider. Which part of Lemma 3.2.4 settles the case $(\varphi \circ \psi)$ at once?

Solution (Hint 2). For (b): remove the first symbol from both sides. Now both φ and φ' are initial segments of the same string.

Solution. (a) The empty string is not a formula, since every formula has at least one symbol. So it is enough to show that no proper **nonempty** initial segment of a formula χ is a formula. We prove this by Theorem 3.2.3.

Atoms and $\bot$. A formula of length 1 has no proper nonempty initial segment.

Negation. Suppose no proper nonempty initial segment of φ is a formula, and let u be a proper nonempty initial segment of $\neg\varphi$. Then u is $\neg$ followed by a proper initial segment u' of φ. Suppose, for a contradiction, that u is a formula. It begins with $\neg$, so it is not an atom or $\bot$ and not of the form $(\varphi'' \circ \psi'')$; by looking at the last step (Section 3.1), $u = \neg\varphi''$ for a formula φ'', and removing the first symbol gives $u' = \varphi''$. So u' is a formula, hence nonempty, and it is a proper nonempty initial segment of φ, contradicting the induction hypothesis.

Binary connectives. Let u be a proper nonempty initial segment of $(\varphi \circ \psi)$. By Lemma 3.2.4 (b), $\mathrm{ex}(u) \geq 1$. If u were a formula, Lemma 3.2.4 (a) would give $\mathrm{ex}(u) = 0$. So u is not a formula. (This case needs no induction hypothesis.)

By Theorem 3.2.3, no formula has a proper nonempty initial segment that is a formula, which proves (a).

(b) Suppose $(\varphi \circ \psi) = (\varphi' \circ' \psi')$. Removing the first symbol, the strings $\varphi \circ \psi)$ and $\varphi' \circ' \psi')$ are equal. Both φ and φ' are initial segments of this string, so the shorter of the two is an initial segment of the longer. If their lengths differed, the shorter would be a proper initial segment of the longer, and a formula, contradicting (a). So they have the same length, and being initial segments of the same string, $\varphi = \varphi'$. The next symbol is then $\circ$ on one side and $\circ'$ on the other, so $\circ = \circ'$; what remains is $\psi) = \psi')$, and removing the last symbol gives $\psi = \psi'$.

Marking scheme. (10 points)

- 1: reduces (a) to nonempty segments (the empty string is not a formula).
- 2: sets up a structural induction with the property "no proper nonempty initial segment is a formula", with the atom and $\bot$ case.
- 3: negation case: reads the segment as $\neg$ followed by a proper initial segment of φ, and uses the last step of a formula beginning with $\neg$ to reach a contradiction with the hypothesis.
- 2: binary case: excess at least 1 by Lemma 3.2.4 (b) against excess 0 by (a).
- 2: (b): compares φ and φ' as initial segments of one string, concludes equal length by (a), then reads off $\circ$ and ψ.

Fatal errors (0 for the step): using Theorem 3.2.5 to prove (b); assuming the lengths of φ and φ' are equal.

Exercise. C2: Find the error

The following argument claims to prove that every formula contains an **even** number of parentheses.

(1) We prove, by strong induction on n, that every formula with n symbols has an even number of parentheses.
(2) Let $n \in \mathbb{N}$, let χ be a formula with n symbols, and suppose that every formula with fewer than n symbols has an even number of parentheses.
(3) If χ is an atom or $\bot$, it has no parentheses, and 0 is even.
(4) If $\chi = \neg\varphi$, then φ is a formula with $n-1$ symbols, so by hypothesis it has an even number of parentheses; and χ has the same parentheses as φ.
(5) If $\chi = (\varphi \circ \psi)$, remove the outer pair of parentheses. The string $\varphi \circ \psi$ has $n-2$ symbols, so by hypothesis it has an even number of parentheses.
(6) Putting the outer pair back adds 2, so χ has an even number of parentheses.
(7) By Theorem 3.2.2, every formula has an even number of parentheses.

(a) Is the conclusion true?
(b) Which line fails, and why?
(c) Show that the move made in that line is illegal by applying the same move to a different property that every formula has, and reaching a false statement.
(d) Repair the argument.

Solution. *(a)* Yes. By Lemma 3.2.4 (a), a formula has as many left as right parentheses, say k of each, so $2k$ in all.

(b) Line 5. The induction hypothesis of line 2 is about **formulas** with fewer than n symbols, and line 5 applies it to the string $\varphi \circ \psi$, which is not a formula: for $\chi = (p \wedge q)$, it is $p \wedge q$, which fails the first-symbol test of Section 3.1 (length 3, beginning with an atom). The hypothesis says nothing about such strings. Every other line is correct: line 4 applies the hypothesis to the formula φ, and lines 1, 2, 3, 6 and 7 are a correct frame for a strong induction on the length.

(c) Take the property of Lemma 3.2.4 (c), that no binary connective lies outside all parentheses; every formula has it. Run the same argument for it: in the case $\chi = (\varphi \circ \psi)$, line 5's move applies the hypothesis to the string $\varphi \circ \psi$ and asserts that no binary connective of $\varphi \circ \psi$ lies outside all parentheses. For $\chi = (p \wedge q)$ this asserts it of $p \wedge q$, whose $\wedge$ lies outside all parentheses. So the move leads from true hypotheses to a false statement; it is not a valid step, even though in the original argument its conclusion happens to be true.

(d) In line 5, apply the hypothesis to the formulas φ and ψ, each with fewer than n symbols: they have $2a$ and $2b$ parentheses for some integers $a, b \geq 0$, and χ has $2a + 2b + 2$, which is even. (Alternatively, prove the claim by Theorem 3.2.3, with the same three cases.)

Exercise. C3: Exit check: structural induction

Say that a formula is a **$\neg\wedge$-formula** if its only connectives are $\neg$ and $\wedge$ and it does not contain $\bot$. Define, by Theorem 3.2.6, the number of nodes of the parse tree and the number

of negations:

$$\begin{aligned} \text{nodes}(p) &= 1, \\ \text{nodes}(\neg\varphi) &= \text{nodes}(\varphi) + 1, \\ \text{nodes}((\varphi \wedge \psi)) &= \text{nodes}(\varphi) + \text{nodes}(\psi) + 1, \\ \text{neg}(p) &= 0, \\ \text{neg}(\neg\varphi) &= \text{neg}(\varphi) + 1, \\ \text{neg}((\varphi \wedge \psi)) &= \text{neg}(\varphi) + \text{neg}(\psi), \end{aligned}$$

with the clauses for $\bot$ and the other connectives chosen in any way (say, as for $\wedge$). Prove by structural induction that every $\neg\wedge$-formula φ satisfies $\text{nodes}(\varphi) > \text{neg}(\varphi)$.

Solution (Hint 1). Theorem 3.2.3 is about all formulas. Put the restriction to $\neg\wedge$-formulas inside the property.

Solution (Hint 2). Let $P(\varphi)$ be "if φ is a $\neg\wedge$-formula, then $\text{nodes}(\varphi) > \text{neg}(\varphi)$". The immediate parts of a $\neg\wedge$-formula are $\neg\wedge$-formulas.

Solution. For a formula φ, let $P(\varphi)$ be: "if φ is a $\neg\wedge$-formula, then $\text{nodes}(\varphi) > \text{neg}(\varphi)$". We prove $P(\varphi)$ for every formula φ by Theorem 3.2.3; for a $\neg\wedge$-formula it is the claim.

The immediate parts of a $\neg\wedge$-formula are $\neg\wedge$-formulas, since every symbol of a part is a symbol of the whole.

Atoms. $\text{nodes}(p) = 1 > 0 = \text{neg}(p)$.

$\bot$, and the connectives $\vee, \to, \leftrightarrow$. The formulas $\bot$ and $(\varphi \circ \psi)$ with $\circ$ one of $\vee, \to, \leftrightarrow$ are not $\neg\wedge$-formulas, so for them P holds vacuously, whatever the clauses chosen for them.

Negation. Suppose $P(\varphi)$, and let $\neg\varphi$ be a $\neg\wedge$-formula. Then φ is one, so $\text{nodes}(\varphi) > \text{neg}(\varphi)$ by $P(\varphi)$, and adding 1 to both sides gives $\text{nodes}(\neg\varphi) > \text{neg}(\neg\varphi)$.

Conjunction. Suppose $P(\varphi)$ and $P(\psi)$, and let $(\varphi \wedge \psi)$ be a $\neg\wedge$-formula. Then φ and ψ are, so $\text{nodes}(\varphi) > \text{neg}(\varphi)$ and $\text{nodes}(\psi) > \text{neg}(\psi)$. Adding the two inequalities and then 1,

$$\text{nodes}((\varphi \wedge \psi)) = \text{nodes}(\varphi) + \text{nodes}(\psi) + 1 > \text{neg}(\varphi) + \text{neg}(\psi) = \text{neg}((\varphi \wedge \psi)).$$

(Since the counts are integers, each hypothesis gives at least 1 to spare, so the left side exceeds the right by at least 3.)

By Theorem 3.2.3, $P(\varphi)$ holds for every formula, which proves the claim.

Marking scheme. (10 points)

- 2: states the property for every formula, in a form that covers exactly the $\neg\wedge$-formulas, with the recursive counts.
- 1: atom case.
- 2: negation case, from the hypothesis for φ, after checking that φ is a $\neg\wedge$-formula.
- 3: conjunction case, using the hypotheses for **both** parts.
- 2: names the induction principle and says why $\bot$ and the other connectives are harmless

(vacuous).

Fatal errors (0 for the step): applying the hypothesis to the formula being proved; a conjunction case that uses one part only and says "similarly" for the other.

3.3 Valuations and Truth Tables

Sections 3.1 and 3.2 made formulas into objects: every formula is built from atoms and $\bot$ by the connectives in exactly one way, and so a function on formulas can be defined one clause per connective. So far a formula is only a string of symbols. It says nothing, and it is neither true nor false. This section gives formulas a meaning. We say what it takes to fix the truth or falsity of every atom, and then compute the truth value of every formula from the values of its atoms, one connective at a time. The result is the truth table, the second way of looking at a formula that Chapters 1 and 2 kept promising.

Throughout, formulas are those of Definition 3.1.1, with atoms $p, q, r, p_1, p_2, \ldots$, and we use Chapter 1's shorthand for them (outer parentheses dropped, the usual precedence, $\to$ grouped to the right), as Section 3.1 allows.

3.3.1 Reading a table off the rules

Before any definition, a question. Suppose we want to call each statement true or false, and we want the rules of Chapters 1 and 2 to be **safe**: whenever the lines a rule cites are true, the line it writes is true. How must "true" behave for each connective? The rules answer almost completely. This subsection is motivation only; nothing later rests on it, and Section 3.6 makes it exact.

Conjunction. Take statements φ and ψ. If both are true, $\wedge$I writes $\varphi \wedge \psi$ from them, so $\varphi \wedge \psi$ must be true. If $\varphi \wedge \psi$ is true, $\wedge$E writes φ and writes ψ, so both must be true; turned around, if either of them is false, then $\varphi \wedge \psi$ must be false. All four combinations of values are settled:

φ	ψ	$\varphi \wedge \psi$
T	T	T
T	F	F
F	T	F
F	F	F

Here T stands for "true" and F for "false".

Contradiction. The rule $\bot$E writes **any** statement from $\bot$. If $\bot$ were ever true, $\bot$E would lead from a true line to every statement, false ones included. So $\bot$ must be false, always.

Negation. The rule $\neg$E writes $\bot$ from φ and $\neg\varphi$. Since $\bot$ is false, φ and $\neg\varphi$ are never both true: if φ is true, $\neg\varphi$ is false. And Proposition 2.5.3 derives $\varphi \vee \neg\varphi$ from no premises at all, so at least one of the two is true: if φ is false, $\neg\varphi$ is true. Negation flips the value.

Implication. The rule $\to$E writes ψ from $\varphi \to \psi$ and φ. So in the combination "φ true, ψ false", the implication cannot be true: that is the one row where $\varphi \to \psi$ is false. The other three rows are forced by derivations. Proposition 2.1.2 derives $P \to Q$ from $\neg P$, so whenever P is false (and $\neg P$ therefore true), $P \to Q$ must be true, whatever Q is. And $P \to Q$ is derivable from Q alone (Exercise (B2: An assumption that is not used) (a) of Section 1.4):

1	Q	
2	$\quad P$	
3	$\quad Q$	R 1
4	$P \to Q$	$\to$I 2–3

so whenever Q is true, $P \to Q$ must be true. That settles the rows with P false (by the first derivation) and the row with both true (by the second).

The vacuous rows of $\to$, "if P is false, then $P \to Q$ is true", are often presented as a convention to be swallowed. Here they are forced by $\bot$E, the rule that closes an impossible case, exactly as Section 2.1 said they would be. The rows of $\vee$ and $\leftrightarrow$ can be read off their rules in the same way, and Exercise C1 below asks for a sharper version of the argument for $\to$. We now turn the answers into a definition.

3.3.2 Valuations

A formula such as $p \wedge \neg q$ cannot be true or false on its own, because p and q are atoms: letters with no content. What decides its value is a choice of which atoms are true. We need a word for one such choice.

A valuation is one possible situation, described by saying which atoms are true and which are false.

Definition 3.3.1. Valuation

A **valuation** v assigns to **every** atom p **exactly one** of the two truth values T and F. We write $v(p)$ for the value it assigns to p, and say that p is **true under** v if $v(p) = \mathrm{T}$, **false under** v if $v(p) = \mathrm{F}$.

In words: a valuation answers, for each atom separately, the question "true or false?", and it answers every such question. Two valuations are equal when they give every atom the same value. Nothing links the values of different atoms: $v(p)$ and $v(q)$ may be chosen freely, and that freedom is the whole point.

Some valuations, simplest first.

- The **all-T valuation** gives every atom the value T; the **all-F valuation** gives every atom F. They are the two extreme cases, and both will return as tests.
- The valuation with $v(p) = \mathrm{T}$, $v(q) = \mathrm{F}$ and $v(r) = \mathrm{F}$ for every other atom r. This is the typical way to describe a valuation: give the values of the few atoms we care about, and fix the rest once and for all.
- A valuation may treat infinitely many atoms differently: $v(p_n) = \mathrm{T}$ for odd n and $v(p_n) = \mathrm{F}$ for even n, with every other atom false. Formulas are finite, so no formula will ever see more than finitely many of these values. Making this precise is the job of the lemma in the next subsection but one.

Nonexample. "p is true" and "p is false" together do not describe a valuation: the definition asks for **exactly one** value per atom. And a list that gives p the value T and says nothing about the other atoms is not a valuation either, since it fails "every atom". We never need more than such a partial list, but we complete it (by F, say) so that the definition applies.

3.3.3 The value of a formula

A valuation fixes the atoms. The value of every other formula should then follow, computed from the values of its parts by the tables of the first subsection. Since formulas are built by the clauses of Definition 3.1.1, this is a definition by recursion.

The value of a formula is computed from the bottom of its parse tree upwards, one connective at a time.

Definition 3.3.2. Truth Value of a Formula

Let v be a valuation. The **truth value** $v(\varphi)$ of a formula φ **under** v is defined by recursion on φ:

(V1) for an atom p, $v(p)$ is the value the valuation assigns to p; and $v(\bot) \coloneqq \mathrm{F}$;
(V2) $v(\neg\varphi) \coloneqq \mathrm{T}$ if $v(\varphi) = \mathrm{F}$, and $v(\neg\varphi) \coloneqq \mathrm{F}$ if $v(\varphi) = \mathrm{T}$;
(V3) $v(\varphi \wedge \psi) \coloneqq \mathrm{T}$ if $v(\varphi) = \mathrm{T}$ **and** $v(\psi) = \mathrm{T}$; otherwise F;
(V4) $v(\varphi \vee \psi) \coloneqq \mathrm{T}$ if $v(\varphi) = \mathrm{T}$ **or** $v(\psi) = \mathrm{T}$ (or both); otherwise F;
(V5) $v(\varphi \to \psi) \coloneqq \mathrm{F}$ if $v(\varphi) = \mathrm{T}$ and $v(\psi) = \mathrm{F}$; otherwise T;
(V6) $v(\varphi \leftrightarrow \psi) \coloneqq \mathrm{T}$ if $v(\varphi) = v(\psi)$; otherwise F.

We say φ is **true under** v, or v **makes** φ true, if $v(\varphi) = \mathrm{T}$; otherwise φ is **false under** v.

Clauses (V2)–(V6) say that the value of a compound formula depends **only** on the values of its immediate parts, and they are summed up in the tables of the five connectives:

φ	$\neg\varphi$
T	F
F	T

φ	ψ	$\varphi \wedge \psi$	$\varphi \vee \psi$	$\varphi \to \psi$	$\varphi \leftrightarrow \psi$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Read them clause by clause. (V1): an atom has the value the valuation gives it, and $\bot$ is false under every valuation. (V2): negation flips. (V3): a conjunction needs both parts. (V4): a disjunction needs at least one part; "or" is **inclusive**, as the rule $\vee$I of Section 1.5 already said, so the row with both parts true gives T. (V5): an implication is false in exactly one case, a true hypothesis with a false conclusion. (V6): a biconditional says the two sides agree.

The columns for $\neg$, $\wedge$ and $\to$, and the value of $\bot$, are the ones the first subsection read off the rules. Section 3.6 proves that every row of every table is, in a precise sense, a derivable sequent.

Well defined. Clause (V1) gives a value to every atom and to $\bot$, and each of (V2)–(V6) gives a rule computing the value of a compound formula from the values of its immediate parts. By Theorem 3.2.6, such clauses define exactly one function on all formulas. This is what Section 3.2 was for. Without unique readability a formula could be split into immediate parts in two ways, and the two splittings could give two values. The string $p \wedge q \vee r$, with no parentheses and no convention, shows the danger: its two readings are $(p \wedge q) \vee r$ and $p \wedge (q \vee r)$, and under a valuation with $v(p) = \mathrm{F}$, $v(q) = \mathrm{F}$ and $v(r) = \mathrm{T}$,

$$v((p \wedge q) \vee r) = \mathrm{T}, \qquad v(p \wedge (q \vee r)) = \mathrm{F}.$$

In the first, $p \wedge q$ is F and r is T, so the disjunction is T. In the second, $q \vee r$ is T but p is F, so the conjunction is F. A formula of Definition 3.1.1 carries its parentheses, and has one value.

Example. Computing a value

Let v be a valuation with $v(p) = \mathrm{F}$, $v(q) = \mathrm{T}$ and $v(r) = \mathrm{F}$. Compute $v(\varphi)$ for

$$\varphi = (p \vee \neg q) \to (q \wedge r).$$

Solution. Work from the leaves of the parse tree up to its root, one node at a time, each time using one clause of Definition 3.3.2. List the subformulas so that each comes after its immediate parts:

Subformula	Clause	Value
p, q, r	(V1)	F, T, F
$\neg q$	(V2): $v(q) = \mathrm{T}$	F
$p \vee \neg q$	(V4): both parts F	F
$q \wedge r$	(V3): $v(r) = \mathrm{F}$	F
$(p \vee \neg q) \to (q \wedge r)$	(V5): the hypothesis is F	T

So $v(\varphi) = \mathrm{T}$. The last step did not need the value of $q \wedge r$: a false hypothesis makes an implication true whatever its conclusion is. Computing by hand, one may stop a branch as soon as its value is decided.

Nonexample. The rule "every formula gets the value T" agrees with the all-T valuation on every atom, but it is not the truth value of Definition 3.3.2 under any valuation. It breaks (V1) at $\bot$, and it breaks (V2) at $\neg p$, where p and $\neg p$ would both be T. The truth value under the all-T valuation is the function that (V1)–(V6) produce from it, and it makes $\neg p$ false.

Warning. Values do not distribute over connectives the way symbols seem to. The value of $\neg(p \wedge q)$ is not computed by negating p and q and keeping the $\wedge$: under a valuation with $v(p) = \mathrm{T}$ and $v(q) = \mathrm{F}$, the formula $\neg(p \wedge q)$ is T (since $p \wedge q$ is F), while $\neg p \wedge \neg q$ is F (since $\neg p$ is F). Always follow the parse tree: the main connective is applied last.

Quick check. Let $v(p) = \mathrm{F}$, $v(q) = \mathrm{T}$, $v(r) = \mathrm{T}$. What is $v((p \vee q) \to \neg r)$?

Solution. F. Bottom up: $p \vee q$ is T, by (V4), since q is T; $\neg r$ is F, by (V2); and an implication with a true hypothesis and a false conclusion is F, by (V5).

3.3.4 Only the atoms of a formula matter

A valuation fixes infinitely many values, but a formula contains only finitely many atoms. We expect the value of a formula to depend only on the values of its own atoms. That is the fact that makes truth tables finite, and it is our first proof by structural induction about meaning.

Recall from Section 3.2 that atoms(φ), the set of atoms occurring in φ, is defined by recursion:

$\text{atoms}(p) = \{p\}$ for an atom p, $\text{atoms}(\bot)$ is empty, $\text{atoms}(\neg\varphi) = \text{atoms}(\varphi)$, and the atoms of $\varphi \circ \psi$, for each binary connective $\circ$, are those of φ together with those of ψ. Two valuations v and w **agree on** a set of atoms if $v(p) = w(p)$ for every atom p in it.

Lemma 3.3.1. Coincidence Lemma

Let φ be a formula, and let v and w be valuations that agree on $\text{atoms}(\varphi)$. Then $v(\varphi) = w(\varphi)$.

Idea. Structural induction on φ, with the property "for **all** valuations v, w that agree on the atoms of φ, $v(\varphi) = w(\varphi)$". At an atom, the two valuations agree by hypothesis. At a compound formula, both values are computed by the same clause from the values of the immediate parts, and the parts have fewer atoms to disagree on, so the hypothesis for the parts applies.

Proof. We apply Theorem 3.2.3 to the property $C(\varphi)$: *for all valuations v and w that agree on* $\text{atoms}(\varphi)$, $v(\varphi) = w(\varphi)$.

Atoms. Let p be an atom and let v, w agree on $\text{atoms}(p) = \{p\}$. Then $v(p) = w(p)$, which is $C(p)$.

$\bot$. By (V1), $v(\bot) = \text{F} = w(\bot)$ for all valuations v, w. So $C(\bot)$ holds.

Negation. Assume $C(\varphi)$, and let v, w agree on $\text{atoms}(\neg\varphi)$. Since $\text{atoms}(\neg\varphi) = \text{atoms}(\varphi)$, they agree on $\text{atoms}(\varphi)$, so $v(\varphi) = w(\varphi)$ by $C(\varphi)$. By (V2), $v(\neg\varphi)$ is determined by $v(\varphi)$ and $w(\neg\varphi)$ by $w(\varphi)$ through the same rule, so $v(\neg\varphi) = w(\neg\varphi)$. This proves $C(\neg\varphi)$.

Conjunction. Assume $C(\varphi)$ and $C(\psi)$, and let v, w agree on $\text{atoms}(\varphi \wedge \psi)$. Every atom of φ is an atom of $\varphi \wedge \psi$, so v and w agree on $\text{atoms}(\varphi)$, and $v(\varphi) = w(\varphi)$ by $C(\varphi)$. In the same way $v(\psi) = w(\psi)$ by $C(\psi)$. By (V3), $v(\varphi \wedge \psi) = \text{T}$ exactly when $v(\varphi) = v(\psi) = \text{T}$, that is, exactly when $w(\varphi) = w(\psi) = \text{T}$, that is, exactly when $w(\varphi \wedge \psi) = \text{T}$. Hence $v(\varphi \wedge \psi) = w(\varphi \wedge \psi)$, which is $C(\varphi \wedge \psi)$.

Disjunction. Assume $C(\varphi)$ and $C(\psi)$, and let v, w agree on $\text{atoms}(\varphi \vee \psi)$. As in the previous case, $v(\varphi) = w(\varphi)$ and $v(\psi) = w(\psi)$. By (V4), $v(\varphi \vee \psi) = \text{T}$ exactly when one of $v(\varphi)$, $v(\psi)$ is T, that is, when one of $w(\varphi)$, $w(\psi)$ is T, that is, when $w(\varphi \vee \psi) = \text{T}$. So $C(\varphi \vee \psi)$.

Implication. Assume $C(\varphi)$ and $C(\psi)$, and let v, w agree on $\text{atoms}(\varphi \to \psi)$. Again $v(\varphi) = w(\varphi)$ and $v(\psi) = w(\psi)$. By (V5), $v(\varphi \to \psi) = \text{F}$ exactly when $v(\varphi) = \text{T}$ and $v(\psi) = \text{F}$, that is, when $w(\varphi) = \text{T}$ and $w(\psi) = \text{F}$, that is, when $w(\varphi \to \psi) = \text{F}$. So $C(\varphi \to \psi)$.

Biconditional. Assume $C(\varphi)$ and $C(\psi)$, and let v, w agree on $\text{atoms}(\varphi \leftrightarrow \psi)$. Again $v(\varphi) = w(\varphi)$ and $v(\psi) = w(\psi)$. By (V6), $v(\varphi \leftrightarrow \psi) = \text{T}$ exactly when $v(\varphi) = v(\psi)$, that is, when $w(\varphi) = w(\psi)$, that is, when $w(\varphi \leftrightarrow \psi) = \text{T}$. So $C(\varphi \leftrightarrow \psi)$.

Every clause of Definition 3.1.1 has been covered, so by Theorem 3.2.3, $C(\varphi)$ holds for every formula φ. This proves the lemma.

■

The four binary cases are the same argument four times: the values of the parts agree, and one clause turns equal inputs into equal outputs. Here they are written out so that the pattern can

be seen in full; later proofs write such a repeated case once, and say which cases it stands for.

Two consequences. First, to know $v(\varphi)$ we need only the values of v on the finitely many atoms of φ: if φ has n atoms, it has at most 2^n different "situations", one for each way of giving its atoms the values T and F. (Each atom has two choices, and the choices are independent, so the number of ways doubles with each new atom: $2, 4, 8, \ldots, 2^n$. This count is itself a small induction on n, by Axiom 3.2.1.) Second, the extra atoms a valuation happens to fix are irrelevant: two valuations that differ only on atoms not in φ give φ the same value.

Quick check. Let $\varphi = p \to (q \vee p)$. Two valuations v and w agree on p and q but disagree on r and on every p_n. What does Lemma 3.3.1 say about $v(\varphi)$ and $w(\varphi)$? Does it say anything about $v(\varphi \wedge r)$ and $w(\varphi \wedge r)$?

Solution. $\text{atoms}(\varphi) = \{p, q\}$, and v, w agree there, so $v(\varphi) = w(\varphi)$. For $\varphi \wedge r$ the atoms are p, q, r, and the valuations disagree on r, so the lemma does not apply. Indeed the values can differ: φ is true under every valuation (if p is true, so is $q \vee p$), so $v(\varphi \wedge r) = v(r)$ and $w(\varphi \wedge r) = w(r)$, which differ.

3.3.5 Truth tables

Lemma 3.3.1 says that a formula's value is decided by finitely many choices, and so it can be tabulated once and for all.

A truth table lists the value of a formula in every possible situation, one row per way of giving its atoms values.

Definition 3.3.3. Truth Table

Let φ be a formula and let $p_1, \ldots, p_n$ be a list of distinct atoms that **includes every atom of** φ. A **row** is an assignment of a truth value to each of $p_1, \ldots, p_n$. The **truth table** of φ (on $p_1, \ldots, p_n$) lists, for each of the 2^n rows, the value $v(\varphi)$ for a valuation v that **agrees with the row** on $p_1, \ldots, p_n$. We say φ is true (or false) **in** the row.

Well defined. Two things must be checked. **Existence:** every row comes from a valuation, for instance the one that gives $p_1, \ldots, p_n$ the row's values and every other atom F. **Independence of the choice:** two valuations that agree with the same row agree on $p_1, \ldots, p_n$, hence on $\text{atoms}(\varphi)$, so by Lemma 3.3.1 they give φ the same value. Without the lemma, "the value in a row" would not make sense, since a row says nothing about the infinitely many other atoms.

The list $p_1, \ldots, p_n$ is allowed to contain atoms that do not occur in φ; then pairs of rows that differ only there show the same value. This freedom lets us compare several formulas in one table, over the atoms of all of them, and Section 3.4 does that constantly. When φ has no atoms at all, as $\neg\bot$ or $\bot \to \bot$ do, the list may be empty, and the table has $2^0 = 1$ row: such a formula has the same value under every valuation.

Conventions. The atom columns come first. The rows are listed with T before F, the last atom alternating fastest: for two atoms, TT, TF, FT, FF. In practice we add a column for each subformula, building up to φ in the order of the parse tree, as in Example (Computing a value), but now for every row at once.

Example. The truth table of modus ponens

Make the truth table of $((p \to q) \wedge p) \to q$ on p, q, and read off from it what $\to$E does.

Solution. Columns for the two atoms, then for each subformula in turn:

p	q	$p \to q$	$(p \to q) \wedge p$	$((p \to q) \wedge p) \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

The third column is (V5): F only in the row TF. The fourth is (V3) applied to the third column and the p column: T only in the first row. The last is (V5) again, with the fourth column as hypothesis and q as conclusion: the hypothesis is false in rows 2–4, so those rows are T, and in row 1 the conclusion q is T.

So the formula is true in every row. Read as a statement about $\to$E: the only row in which both given lines $p \to q$ and p are true is row 1, and in that row the line the rule writes, q, is true as well. The rule never leads from true lines to a false one, at least when its lines are atoms. Section 3.5 proves this for every rule and every formula.

Now a table built column by column for a biconditional, whose two sides Chapter 2 related by derivations.

Example. An implication and its contrapositive

Make the truth table of $(p \to q) \leftrightarrow (\neg q \to \neg p)$.

Solution.

p	q	$p \to q$	$\neg q$	$\neg p$	$\neg q \to \neg p$	$(p \to q) \leftrightarrow (\neg q \to \neg p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

The columns $p \to q$ and $\neg q \to \neg p$ are identical: each is F only in row 2. For $\neg q \to \neg p$ that is the row where the hypothesis $\neg q$ is T and the conclusion $\neg p$ is F, that is, q false and p true, which is the same row TF. So the biconditional is T in every row.

Compare Chapter 2. There Proposition 2.3.2 derived $\neg Q \to \neg P$ from $P \to Q$ with the rules for implication and negation, and Proposition 2.5.2 derived the way back with a derivation that uses RAA. The table sees no difference between the two directions; it only sees that the two columns agree. What a table can and cannot tell about derivations is the subject of Sections 3.5 and 3.6.

A table with n atoms has 2^n rows, so three atoms give 8 and five give 32. Often we do not need all of them.

Example. Working backwards from F

Find every row in which

$$\varphi = (p \to q) \to (p \to (q \wedge r))$$

is false: first with the full table on p, q, r, then without it.

Solution. The full table.

p	q	r	$p \to q$	$q \wedge r$	$p \to (q \wedge r)$	$(p \to q) \to (p \to (q \wedge r))$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	T
T	F	F	F	F	F	T
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	T	F	T	T
F	F	F	T	F	T	T

The formula is false in exactly one row, row 2: p and q true, r false.

Backwards from F. An implication is false in exactly one way, by (V5): hypothesis true, conclusion false. Apply that to the main connective of φ, and then to whatever it forces.

1. φ is F exactly when $p \to q$ is T and $p \to (q \wedge r)$ is F.
2. $p \to (q \wedge r)$ is F exactly when p is T and $q \wedge r$ is F.
3. With p true, $p \to q$ is T exactly when q is T.
4. With q true, $q \wedge r$ is F exactly when r is F.

Each step was forced, so there is exactly one row making φ false: $p = \text{T}$, $q = \text{T}$, $r = \text{F}$. Four short steps replace eight rows.

The second method has a name in this book: **working backwards from F**. Assume the formula is false, and ask what each part is forced to be, starting from the main connective. Implications and disjunctions are false in only one way, and conjunctions are true in only one way, so they force values; a false conjunction, a true disjunction, or a true implication leaves a choice, and then we split into cases. If the forcing ends in a contradiction (some atom would need both values), the formula is false in **no** row. If it ends in a consistent assignment, that assignment is a row making the formula false, and the cases that were split give all such rows.

Warning. A truth table sees only the connectives. Everything inside an atom is invisible to it. The statement "for every real x, if $x > 2$, then $x^2 > 4$" is true, but its propositional form, with the atoms a: "$x > 2$" and b: "$x^2 > 4$", is $a \to b$, which is false in the row with a true and b false. That row cannot happen for real numbers, and the reason lies in algebra, not in the connectives. Likewise "for every $\varepsilon > 0$ there is an N such that ..." is a single atom for a truth table, with two rows and nothing to compute. Chapter 4 looks inside atoms.

Quick check. Why does Definition 3.3.3 need Lemma 3.3.1? What would go wrong in a table for $p \vee q$ if the value of $p \vee q$ could depend on the atom r?

Solution. A row gives values to the listed atoms only, and many valuations agree with it, differing on the other atoms. The table records "the" value of the formula in the row, so all those valuations must give the same value; Lemma 3.3.1 guarantees it. If $v(p \vee q)$ could depend on r, the row $p = \mathrm{F}$, $q = \mathrm{F}$ would contain valuations giving different values, and there would be no single entry to write.

3.3.6 Exercises

3.3.6.1 A. Check your understanding

Exercise. A1

(a) Make the truth tables of $\neg(p \wedge \neg q)$ and of $(p \vee q) \to p$.

(b) Make the truth table of $p \leftrightarrow \neg p$. Is it true in any row?

(c) Compute the value of $(p \to \bot) \vee (\bot \to p)$ under the all-F valuation, and under the all-T valuation.

(d) How many rows has the truth table of a formula on the atoms p, q, r, s?

(e) True or false, with a reason: if $v(p) \neq w(p)$, then $v(p \to (q \to p)) \neq w(p \to (q \to p))$.

Solution.

(a)

p	q	$\neg q$	$p \wedge \neg q$	$\neg(p \wedge \neg q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

p	q	$p \vee q$	$(p \vee q) \to p$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	T

The first formula has the same column as $p \to q$: false only in row TF.

(b) In the row $p = \mathrm{T}$, $\neg p$ is F, and in the row $p = \mathrm{F}$, $\neg p$ is T; the two sides always differ, so by (V6) the formula is F in both rows. It is true in no row.

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

(c) The disjunct $\bot \to p$ has a false hypothesis, since $v(\bot) = \mathrm{F}$ under every valuation (V1), so it is T by (V5), and the disjunction is T by (V4). This holds under both valuations (and under every other).

(d) $2^4 = 16$.

(e) False. The formula $p \to (q \to p)$ is true under every valuation: if $v(p) = \mathrm{F}$, the hypothesis

is false; if $v(p) = \mathrm{T}$, the conclusion $q \to p$ has a true conclusion. So v and w give it the value T however they differ. Lemma 3.3.1 says that agreement on the atoms forces agreement on the formula; it says nothing in the other direction.

3.3.6.2 B. Practice

Exercise. B1: Backwards from F

For each formula, work backwards from F to find every row on its atoms in which it is false, or show that there is none.

(a) $((p \vee q) \wedge ((p \to r) \wedge (q \to r))) \to r$.
(b) $(p \to (q \to r)) \to ((p \to q) \to (p \to r))$.
(c) $((p \to r) \vee (q \to r)) \to ((p \vee q) \to r)$.
(d) $((p \to q) \wedge (q \to r)) \to (r \to p)$.

Solution. *(a)* Suppose the formula is F. Then r is F, and the hypothesis is T, so each of $p \vee q$, $p \to r$, $q \to r$ is T (V3). With r false, $p \to r$ true forces $p = \mathrm{F}$, and $q \to r$ true forces $q = \mathrm{F}$. Then $p \vee q$ is F, a contradiction. So the formula is false in no row. It is the table's version of proof by cases: if r follows from each of p and q, and one of them holds, then r holds.

(b) Suppose it is F. Then $p \to (q \to r)$ is T and $(p \to q) \to (p \to r)$ is F; the latter forces $p \to q$ to be T and $p \to r$ to be F, and the last forces $p = \mathrm{T}$, $r = \mathrm{F}$. Then $p \to q$ true with p true forces $q = \mathrm{T}$. Now $p \to (q \to r)$ has a true hypothesis and the conclusion $q \to r$, which is F (true q, false r), so it is F, a contradiction. False in no row.

(c) Suppose it is F. Then $(p \vee q) \to r$ is F: $r = \mathrm{F}$ and $p \vee q$ is T. And $(p \to r) \vee (q \to r)$ is T: with r false, $p \to r$ is T exactly when $p = \mathrm{F}$, and $q \to r$ exactly when $q = \mathrm{F}$; so at least one of p, q is F. Together with $p \vee q$ true: exactly one of p, q is T. This leaves a choice, so split. The formula is false in exactly two rows: $p = \mathrm{T}, q = \mathrm{F}, r = \mathrm{F}$ and $p = \mathrm{F}, q = \mathrm{T}, r = \mathrm{F}$. Check the first: $q \to r$ is T, so the hypothesis is T; $p \vee q$ is T and r is F, so the conclusion is F.

(d) Suppose it is F. Then $r \to p$ is F: $r = \mathrm{T}$, $p = \mathrm{F}$. The hypothesis must be T: $p \to q$ is T because p is F, and $q \to r$ is T because r is T, whatever q is. Nothing forces q. So the formula is false in exactly two rows: $p = \mathrm{F}$, $r = \mathrm{T}$, and q either value.

Exercise. B2: Counting atoms, the last step blank

Call a formula **binary-built** if it is built from atoms by the four binary connectives only, so that neither $\neg$ nor $\bot$ occurs in it; the immediate parts of a binary-built formula are again binary-built. Define by recursion (Theorem 3.2.6) the number $\mathrm{occ}(\varphi)$ of atom occurrences and the number $\mathrm{bin}(\varphi)$ of binary connectives:

$$\begin{aligned} \mathrm{occ}(p) &= 1, \quad \mathrm{occ}(\varphi \circ \psi) = \mathrm{occ}(\varphi) + \mathrm{occ}(\psi), \\ \mathrm{bin}(p) &= 0, \quad \mathrm{bin}(\varphi \circ \psi) = \mathrm{bin}(\varphi) + \mathrm{bin}(\psi) + 1, \end{aligned}$$

for every atom p and every binary connective $\circ$. (For $\bot$ and $\neg$ give both functions any values;

they play no part below.) Here is a proof, by Theorem 3.2.3, that every binary-built formula φ has

$$\operatorname{occ}(\varphi) = \operatorname{bin}(\varphi) + 1.$$

The property is $C(\varphi)$: *if φ is binary-built, then* $\operatorname{occ}(\varphi) = \operatorname{bin}(\varphi) + 1$.

$\bot$ and negation. Neither $\bot$ nor a formula $\neg\varphi$ is binary-built, so C holds for them: its "if" is never met.

Atoms. $\operatorname{occ}(p) = 1 = 0 + 1 = \operatorname{bin}(p) + 1$.

Conjunction. Assume $C(\varphi)$ and $C(\psi)$, and let $\varphi \wedge \psi$ be binary-built. Then φ and ψ are binary-built, so

$$\begin{aligned}\operatorname{occ}(\varphi \wedge \psi) &= \operatorname{occ}(\varphi) + \operatorname{occ}(\psi) = \big(\operatorname{bin}(\varphi) + 1\big) + \big(\operatorname{bin}(\psi) + 1\big) \\ &= \big(\operatorname{bin}(\varphi) + \operatorname{bin}(\psi) + 1\big) + 1 = \operatorname{bin}(\varphi \wedge \psi) + 1.\end{aligned}$$

Disjunction and biconditional. The computation is the same with $\vee$, and with $\leftrightarrow$, in place of $\wedge$.

Implication. _____

Name the move, then complete it.

Solution. The move is **induction on the shape of a formula** (Theorem 3.2.3): the case for a compound formula uses the property for its immediate parts, and nothing else.

Implication. Assume $C(\varphi)$ and $C(\psi)$, and let $\varphi \to \psi$ be binary-built. Its immediate parts φ and ψ are binary-built, so $\operatorname{occ}(\varphi) = \operatorname{bin}(\varphi) + 1$ and $\operatorname{occ}(\psi) = \operatorname{bin}(\psi) + 1$. Hence

$$\begin{aligned}\operatorname{occ}(\varphi \to \psi) &= \operatorname{occ}(\varphi) + \operatorname{occ}(\psi) = \big(\operatorname{bin}(\varphi) + 1\big) + \big(\operatorname{bin}(\psi) + 1\big) \\ &= \big(\operatorname{bin}(\varphi) + \operatorname{bin}(\psi) + 1\big) + 1 = \operatorname{bin}(\varphi \to \psi) + 1,\end{aligned}$$

which is $C(\varphi \to \psi)$. All clauses of Definition 3.1.1 are covered, so Theorem 3.2.3 gives $C(\varphi)$ for every formula, and in particular the equation for every binary-built formula.

For instance $(p \to q) \wedge (p \vee r)$ has four atom occurrences (p counted twice) and three binary connectives.

Exercise. B3: A table for a named pattern

Make the truth table of $((p \vee q) \wedge \neg p) \to q$, and say which derived rule of Chapter 2 it corresponds to, as Example (The truth table of modus ponens) did for $\to$E.

Solution.

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$((p \vee q) \wedge \neg p) \to q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

True in every row. The only row in which both $p \vee q$ and $\neg p$ are true is row 3, and there q is true. This is disjunctive syllogism, DS (Proposition 2.3.3): from $P \vee Q$ and $\neg P$, infer Q.

3.3.6.3 C. Going deeper

Exercise. C1: The rows of implication are forced

Pretend we do not know the table of $\to$. Say only that there is **some** table: a rule t giving a truth value $t(a, b)$ for each pair of truth values a, b, with

$$v(\varphi \to \psi) = t(v(\varphi), v(\psi))$$

for every valuation v and all formulas φ, ψ. (The other connectives keep their clauses.) Suppose the two rules for implication preserve truth in the following sense.

- (E) For every valuation v and all formulas φ, ψ: if $v(\varphi \to \psi) = \mathrm{T}$ and $v(\varphi) = \mathrm{T}$, then $v(\psi) = \mathrm{T}$.
- (I) For every list of formulas Γ and all formulas φ, ψ: if every valuation that makes all of Γ and φ true makes ψ true, then every valuation that makes all of Γ true makes $\varphi \to \psi$ true.

Prove that t is the table of (V5): $t(\mathrm{T}, \mathrm{F}) = \mathrm{F}$, and $t(a, b) = \mathrm{T}$ for the other three pairs.

Solution (Hint 1). Each of the four values of t is one instance of (E) or of (I), with atoms for φ and ψ and a well-chosen valuation.

Solution (Hint 2). For $t(\mathrm{T}, \mathrm{T})$ and $t(\mathrm{F}, \mathrm{F})$, try (I) with Γ empty and $\varphi = \psi = p$.

Solution. Let p and q be distinct atoms.

$t(\mathrm{T}, \mathrm{F}) = \mathrm{F}$. Take a valuation v with $v(p) = \mathrm{T}$, $v(q) = \mathrm{F}$. If $t(\mathrm{T}, \mathrm{F})$ were T, then $v(p \to q) = t(\mathrm{T}, \mathrm{F}) = \mathrm{T}$ and $v(p) = \mathrm{T}$, so (E) would give $v(q) = \mathrm{T}$, which is false. So $t(\mathrm{T}, \mathrm{F}) = \mathrm{F}$.

$t(\mathrm{T}, \mathrm{T}) = t(\mathrm{F}, \mathrm{F}) = \mathrm{T}$. Apply (I) with Γ empty and $\varphi = \psi = p$. Its hypothesis, "every valuation making p true makes p true", holds. So every valuation makes $p \to p$ true. A valuation with $v(p) = \mathrm{T}$ gives $t(\mathrm{T}, \mathrm{T}) = v(p \to p) = \mathrm{T}$, and one with $v(p) = \mathrm{F}$ gives $t(\mathrm{F}, \mathrm{F}) = \mathrm{T}$.

$t(\mathrm{F}, \mathrm{T}) = \mathrm{T}$. Apply (I) with Γ the list q, $\varphi = p$ and $\psi = q$. Every valuation making q and p true makes q true, so every valuation making q true makes $p \to q$ true. A valuation with $v(p) = \mathrm{F}$, $v(q) = \mathrm{T}$ makes q true, so $t(\mathrm{F}, \mathrm{T}) = v(p \to q) = \mathrm{T}$.

No step used a value of t that had not yet been found, since the formulas in Γ, φ and ψ contain no $\to$. So the only table of $\to$ under which both rules preserve truth is the table of (V5). Conditions (E) and (I) are what Section 3.5 proves about the rules $\to$E and $\to$I; the exercise shows that nothing but the standard table could have passed.

Marking scheme. (10 points)

- 3: the row (T, F) from (E), with a valuation making p true and q false, and the argument by contradiction.
- 3: the rows (T, T) and (F, F) from (I), with the hypothesis of (I) checked, and a valuation for each value of p.
- 3: the row (F, T) from (I) with a nonempty Γ (or any other correct instance), hypothesis checked.
- 1: says why the argument is not circular (no $\to$ in the formulas used).

Fatal errors (0 for the step): using the standard table of $\to$ to justify a value of t; an instance of (I) whose hypothesis is not checked.

Exercise. C2: Find the error

Here is a claim and a "proof".

> **Claim.** Let v_{T} be the all-T valuation. Every formula built from atoms by $\wedge$, $\vee$ and $\to$ only is true under v_{T}.
>
> **Proof.** By Theorem 3.2.3, with the property "if φ is built from atoms by $\wedge$, $\vee$, $\to$ only, then $v_{\mathrm{T}}(\varphi) = \mathrm{T}$". The formulas $\bot$, $\neg\varphi$ and $\varphi \leftrightarrow \psi$ are not built in this way, so the property holds for them. If a formula built in this way is $\varphi \wedge \psi$, $\varphi \vee \psi$ or $\varphi \to \psi$, then φ and ψ are built in this way too, so the hypothesis applies to them. (1) For an atom p, $v_{\mathrm{T}}(p) = \mathrm{T}$ by the definition of v_{T}. (2) If $v_{\mathrm{T}}(\varphi) = \mathrm{T}$ and $v_{\mathrm{T}}(\psi) = \mathrm{T}$, then $v_{\mathrm{T}}(\varphi \wedge \psi) = \mathrm{T}$ by (V3). (3) If $v_{\mathrm{T}}(\varphi) = \mathrm{T}$ and $v_{\mathrm{T}}(\psi) = \mathrm{T}$, then $v_{\mathrm{T}}(\varphi \vee \psi) = \mathrm{T}$ by (V4), since at least one part is true. (4) If $v_{\mathrm{T}}(\varphi) = \mathrm{T}$ and $v_{\mathrm{T}}(\psi) = \mathrm{T}$, then $v_{\mathrm{T}}(\varphi \to \psi) = \mathrm{T}$ by (V5), because $v_{\mathrm{T}}(\varphi) = \mathrm{F}$ and an implication with a false hypothesis is true. This covers every clause, and proves the claim.

(a) Is the claim true?

(b) Name the line that fails and say exactly what is wrong with it.

(c) Give a formula on which the reason given in that line is false.

(d) Repair the line.

Solution. (a) Yes. The claim is true, and the repaired proof in (d) proves it. So no formula is a counterexample to the claim; the error is in the argument only.

(b) Line (4). Its hypothesis states $v_{\mathrm{T}}(\varphi) = \mathrm{T}$, and its reason states $v_{\mathrm{T}}(\varphi) = \mathrm{F}$. The reason contradicts the induction hypothesis it is supposed to use: it quotes the vacuous row of (V5), which does not apply here. The conclusion $v_{\mathrm{T}}(\varphi \to \psi) = \mathrm{T}$ is correct, but not for the reason given.

(c) $p \to q$. Under v_{T}, $v_{\mathrm{T}}(p) = \mathrm{T}$, so the stated reason "$v_{\mathrm{T}}(p) = \mathrm{F}$" is false, although $v_{\mathrm{T}}(p \to q) = \mathrm{T}$ holds.

(d) (4) If $v_{\mathrm{T}}(\varphi) = \mathrm{T}$ and $v_{\mathrm{T}}(\psi) = \mathrm{T}$, then $v_{\mathrm{T}}(\varphi \to \psi) = \mathrm{T}$ by (V5), because the conclusion ψ is true, so we are not in the one row (true hypothesis, false conclusion) where an implication is false. With this line the induction is correct, and the claim follows. Section 3.7 uses this fact to show that $\wedge$, $\vee$ and $\to$ cannot express negation.

Marking scheme. (10 points)

- 2: the claim is true (with the reason: the repaired proof).
- 3: names line (4), and says its reason contradicts the hypothesis $v_{\mathrm{T}}(\varphi) = \mathrm{T}$ of that very case.
- 2: a formula, such as $p \to q$, on which the stated reason fails.
- 3: the repair, citing the row of (V5) that actually applies (conclusion true).

Fatal errors (0 for the step): calling the claim false; locating the error in line (3) (at least one part true is correct) or in the handling of $\bot$, $\neg$, $\leftrightarrow$.

3.4 Tautologies, Entailment and Equivalence

Section 3.3 gave every formula a value under every valuation, and collected the values in a truth table. One formula at a time, though, is not what Chapters 1 and 2 were about. They were about **sequents**: premises on the left, a conclusion on the right, and the claim that the conclusion can be derived. This section defines the counterpart of a sequent in terms of truth, $\Gamma \models \psi$: every valuation that makes the premises true makes the conclusion true. On the way it names the formulas that are true in every row, and the pairs of formulas that are true in the same rows. It ends with two lemmas about changing a formula, by replacing one part or by putting formulas in for its atoms, that later sections use.

3.4.1 Tautologies and contradictions

Some tables of Section 3.3 had a last column made entirely of T: the formula $((p \to q) \wedge p) \to q$ of Example (The truth table of modus ponens), and the biconditional of Example (An implication and its contrapositive). Their truth does not depend on the situation. One table in Exercise A1 there, that of $p \leftrightarrow \neg p$, was all F. These extremes deserve names.

A tautology is true for reasons of form alone: whatever the atoms mean, it comes out true.

Definition 3.4.1. Tautology and Contradiction

A formula φ is a **tautology** if $v(\varphi) = \mathrm{T}$ for **every** valuation v. It is a **contradiction** if $v(\varphi) = \mathrm{F}$ for **every** valuation v.

By Lemma 3.3.1, the value of φ under a valuation depends only on the row the valuation gives to the atoms of φ. So φ is a tautology exactly when its truth table has T in every row, and a contradiction exactly when it has F in every row. Infinitely many valuations, finitely many rows to check.

Examples, simplest first.

- $p \to p$ and $p \vee \neg p$ are tautologies: in the row $p = \mathrm{T}$ both are true, and in the row $p = \mathrm{F}$, $p \to p$ has a false hypothesis and $\neg p$ is true.
- $p \wedge \neg p$ and $p \leftrightarrow \neg p$ are contradictions: in each row one of p, $\neg p$ is false, and the two sides of the biconditional differ.
- A formula with no atoms, such as $\bot$ or $\neg\bot$, has the same value under every valuation (its table has one row). So it is either a contradiction, like $\bot$, or a tautology, like $\neg\bot$. This degenerate case is the only one in which "not a tautology" forces "a contradiction".
- *Nonexample.* $p \to q$ is neither: it is T in the row TT and F in the row TF. Most formulas

are like this. Changing one symbol of the tautology $p \to p$, to get $p \to q$, loses the property: the rows in which the hypothesis and conclusion differ are no longer excluded by their being the same atom.

The formulas in between also need a word, and so do **lists** of formulas, because a list of premises can fail to be true together even when each premise alone can be true.

Definition 3.4.2. Satisfiable

A formula φ is **satisfiable** if **there is** a valuation v with $v(\varphi) = \mathrm{T}$. A list of formulas $\Gamma = \varphi_1, \ldots, \varphi_n$, where $n \geq 0$, is **satisfiable** if **there is** a valuation that makes **every** φ_i true. A formula or list that is not satisfiable is **unsatisfiable**.

In words: a formula is satisfiable if some row makes it true, and a list is satisfiable if some **single** row makes all its members true at once. So a formula is unsatisfiable exactly when it is a contradiction. The **empty list** ($n = 0$) is satisfiable: every valuation makes all of its members true, since it has none to check.

The list $p, \neg p$ is unsatisfiable, although each of its members is satisfiable: a row making p true makes $\neg p$ false. Satisfiability of a list is about a common row, not about each formula separately. And the list $p \to q,\ q \to r,\ r \to \neg p$ is satisfiable, by the row with p false and q, r true (the first two have true conclusions, and the third has conclusion $\neg p$, which is true); in fact p is false in every row that makes all three true.

Tautologies and unsatisfiable formulas are two sides of one coin. For every formula ψ,

$$\psi \text{ is a tautology} \iff \neg\psi \text{ is unsatisfiable.}$$

Indeed, by (V2) of Definition 3.3.2, $v(\neg\psi) = \mathrm{T}$ exactly when $v(\psi) = \mathrm{F}$. So "no valuation makes $\neg\psi$ true" says "no valuation makes ψ false", which is "every valuation makes ψ true". Section 3.7 uses this to turn a question about tautologies into one about satisfiability.

Warning. "Not a tautology" does not mean "a contradiction". The formula $p \to q$ is neither: some rows make it true and some false. To show a formula is **not** a tautology, one row making it false is enough; to show it **is** a contradiction, every row must make it false.

3.4.2 Semantic entailment

Now the counterpart of a sequent. A derivation from premises Γ to a conclusion ψ is a promise that ψ follows from Γ. In terms of truth, "follows" should mean: no situation makes the premises true and the conclusion false. We want to prove, in the next two sections, that the rules of Chapters 1 and 2 derive exactly the sequents that pass this test; for that we need a precise name for the test.

An argument is valid when its conclusion is true in every row in which all its premises are true.

Definition 3.4.3. Semantic Entailment

Let $\Gamma = \varphi_1, \ldots, \varphi_n$ be a list of formulas, where $n \geq 0$, and let ψ be a formula. We say Γ **semantically entails** ψ, and write

$$\Gamma \models \psi,$$

if **every** valuation that makes **all** of $\varphi_1, \ldots, \varphi_n$ true also makes ψ true. We then also call the argument from Γ to ψ **valid**. We write $\varphi_1, \ldots, \varphi_n \models \psi$ for $\Gamma \models \psi$, and $\models \psi$ when Γ is empty. If $\Gamma \models \psi$ fails, we write $\Gamma \not\models \psi$, and a valuation that makes every formula of Γ true and ψ false is a **refuting valuation**, or **refuting row**.

Clause by clause. "Every valuation": the condition is checked in all situations, not in the actual one; a valid argument cannot be spoiled by what the atoms happen to mean. "That makes all of $\varphi_1, \ldots, \varphi_n$ true": rows in which some premise is false are ignored, whatever the conclusion does there. "Also makes ψ true": in the remaining rows, the conclusion must hold. In symbols, for every valuation v,

$$(v(\varphi_1) = \mathrm{T} \text{ and } \ldots \text{ and } v(\varphi_n) = \mathrm{T}) \;\Rightarrow\; v(\psi) = \mathrm{T}.$$

And $\Gamma \not\models \psi$ says: **there is** a valuation making every premise true and the conclusion false. One refuting row is enough.

Checking it. By Lemma 3.3.1, whether a valuation makes the premises and the conclusion true depends only on its row on the atoms occurring in Γ and ψ. So $\Gamma \models \psi$ can be decided by one truth table on those atoms (Definition 3.3.3 allows extra atoms in the list): find the rows in which every premise is T, and look at the conclusion there.

First examples. Row 1 of Example (The truth table of modus ponens) is the only row making $p \to q$ and p true, and q is true there, so $p \to q,\ p \models q$. By Example (An implication and its contrapositive), $p \to q \models \neg q \to \neg p$ and $\neg q \to \neg p \models p \to q$, since the two columns are equal.

The empty list. When Γ is empty, every valuation makes "all of Γ" true, so $\models \psi$ says that every valuation makes ψ true: $\models \psi$ holds exactly when ψ is a tautology.

Unsatisfiable premises. If Γ is unsatisfiable, no valuation makes all of Γ true, so the condition has no row to check and holds: $\Gamma \models \psi$ for **every** formula ψ. For example $p, \neg p \models q$ and $\bot \models q$. This is the table's version of Proposition 2.1.1, $P, \neg P \vdash Q$, and of $\bot$E.

More premises. If $\Gamma \models \psi$, and Δ is a list containing every formula of Γ (and perhaps others), then $\Delta \models \psi$: a valuation making all of Δ true makes all of Γ true, and so makes ψ true. Adding premises never destroys an entailment. In the same way, the order of the premises and repetitions among them do not matter: only which formulas occur in the list does.

A test by satisfiability. Finally, for every list Γ and formula ψ,

$$\Gamma \models \psi \iff \text{the list } \Gamma, \neg\psi \text{ is unsatisfiable.}$$

Both sides say that no valuation makes every formula of Γ true and ψ false, because $v(\neg\psi) = \mathrm{T}$ exactly when $v(\psi) = \mathrm{F}$. With Γ empty this is the equivalence between tautologies and unsatisfiable negations above.

Example. Affirming the consequent is not valid

Show that $p \to q,\ q \not\models p$.

Solution. We need one row making both premises true and the conclusion false. The conclusion p must be F, and the premise q must be T. In the row $p = \mathrm{F}$, $q = \mathrm{T}$, the premise $p \to q$ is T, since its hypothesis is false. So this row makes $p \to q$ and q true and p false: it is a refuting row.

The pattern is the one of Section 1.3, "from $\varphi \to \psi$ and ψ, infer φ", which is not an instance

of →E. Section 1.1's Quick check gave the row a face: with P: "n is divisible by 4" and Q: "n is even", at $n = 2$ the premises $P \to Q$ and Q are true and P is false. It is also the step that would repair Argument B of Section 1.1: from "if n is even, then n^2 is even" and "n^2 is even", conclude "n is even". The conclusion of that instance happens to be true for every integer n, as Argument A proves; the refuting row shows that the **form** does not guarantee it, so the step proves nothing. Section 3.5 upgrades this row from "not valid" to "not derivable by any derivation".

A valid argument with several premises is checked in the same way, and working backwards from F often settles it without a table.

Example. Two cases, two conclusions

Show that $p \lor q,\ p \to r,\ q \to s \models r \lor s$.

Solution. Suppose a valuation v makes all three premises true and the conclusion false; we derive a contradiction. Since $r \lor s$ is F, both r and s are F (V4). Since $p \to r$ is T and r is F, the hypothesis p is F (V5). In the same way $q \to s$ forces q to be F. But then $p \lor q$ is F, contradicting the first premise. So no refuting valuation exists, and the entailment holds.

The search replaced a table of $2^4 = 16$ rows by four forced steps. The entailment is the table's picture of a proof by cases with ∨E: in the case p, the second premise gives r, and ∨I gives $r \lor s$; in the case q, the third gives s, and ∨I again.

Changing the grouping of a formula can change what it entails.

Example. One grouping entails the other, not conversely

Decide whether $(p \to q) \to r \models p \to (q \to r)$, and whether $p \to (q \to r) \models (p \to q) \to r$.

Solution. The first holds. Suppose a valuation makes $(p \to q) \to r$ true and $p \to (q \to r)$ false. The latter forces $p =$ T and $q \to r$ false, so $q =$ T and $r =$ F. Then $p \to q$ is T, and $(p \to q) \to r$ has a true hypothesis and a false conclusion, so it is F, a contradiction.

The second fails. Take $p =$ F and $r =$ F, with q either value. Then $p \to (q \to r)$ is T, since its hypothesis is false. But $p \to q$ is T for the same reason, so $(p \to q) \to r$ is T → F, which is F. These two rows are refuting rows.

So the two groupings are not interchangeable, which is why Chapter 1 fixed a convention (→ groups to the right) and why the convention matters.

Warning. $\models$ is not →. The formula $\varphi \to \psi$ is a formula: it has a value in each row and can be a part of larger formulas. The statement $\varphi \models \psi$ is not a formula at all. It is a claim **about** two formulas, true or false once and for all, like $\varphi \vdash \psi$. They are related (Exercise B3 shows $\varphi \models \psi$ holds exactly when $\varphi \to \psi$ is a tautology), but writing $p \models q \models r$, or $\neg(p \models q)$ inside a formula, mixes the two levels and means nothing.

Quick check. Is $p, \neg p \models q$? Is $p \to q,\ \neg p \models \neg q$?

Solution. The first holds: the list $p, \neg p$ is unsatisfiable, so there is no row to check. The second fails: the row $p = \mathrm{F}$, $q = \mathrm{T}$ makes $p \to q$ true (false hypothesis) and $\neg p$ true, and $\neg q$ false. It is the pattern "denying the hypothesis", the mirror image of affirming the consequent.

3.4.3 Semantic equivalence

Two formulas can be different strings and still say the same thing in every situation, as $p \to q$ and $\neg q \to \neg p$ did in Example (An implication and its contrapositive). Chapter 2 had a word for formulas that derive each other, provable equivalence (Definition 2.4.2). Here is the counterpart in terms of truth.

Two formulas are equivalent when they have the same truth table.

Definition 3.4.4. Semantic Equivalence

Formulas φ and ψ are **semantically equivalent**, written $\varphi \equiv \psi$, if $\varphi \models \psi$ **and** $\psi \models \varphi$.

In words, each entails the other. Equivalently, $v(\varphi) = v(\psi)$ for **every** valuation v. Indeed, if $\varphi \equiv \psi$ and $v(\varphi) = \mathrm{T}$, then $v(\psi) = \mathrm{T}$ by $\varphi \models \psi$; if $v(\varphi) = \mathrm{F}$, then $v(\psi)$ cannot be T, since $\psi \models \varphi$ would make $v(\varphi) = \mathrm{T}$. The converse is plain. By (V6) of Definition 3.3.2, "$v(\varphi) = v(\psi)$ for every v" says that $\varphi \leftrightarrow \psi$ is true under every v, so

$$\varphi \equiv \psi \iff \models \varphi \leftrightarrow \psi.$$

From "same value under every valuation" it follows at once that $\equiv$ is reflexive, symmetric, and transitive: $\varphi \equiv \varphi$; if $\varphi \equiv \psi$, then $\psi \equiv \varphi$; if $\varphi \equiv \psi$ and $\psi \equiv \chi$, then $\varphi \equiv \chi$.

Examples: $p \wedge q \equiv q \wedge p$ (the clause (V3) does not care about the order); $p \equiv \neg\neg p$ (two flips); $p \vee \neg p \equiv \neg\bot$ (both are true under every valuation). The degenerate case: any two tautologies are equivalent, and so are any two contradictions. *Nonexample:* $p \wedge q$ and p are not equivalent. One direction holds, $p \wedge q \models p$, but the other fails in the row $p = \mathrm{T}$, $q = \mathrm{F}$. Entailment in one direction is not equivalence. Nor is equivalence identity: $p \wedge q$ and $q \wedge p$ are different formulas, with different parse trees, and $\to$E treats them as different (Section 1.3).

The equivalences used most often are collected below. They are stated for arbitrary formulas φ, ψ, χ, not just atoms, since Section 3.7 applies them to compound formulas.

Proposition 3.4.1. Standard Equivalences

For all formulas φ, ψ and χ:

(a) (contrapositive) $\varphi \to \psi \equiv \neg\psi \to \neg\varphi$;
(b) (De Morgan) $\neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi$ and $\neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi$;
(c) (implication as a disjunction) $\varphi \to \psi \equiv \neg\varphi \vee \psi$;
(d) (double negation) $\neg\neg\varphi \equiv \varphi$;

(e) (biconditional) $\varphi \leftrightarrow \psi \equiv (\varphi \to \psi) \wedge (\psi \to \varphi)$;
(f) (commutativity) $\varphi \wedge \psi \equiv \psi \wedge \varphi$ and $\varphi \vee \psi \equiv \psi \vee \varphi$;
(g) (distributivity) $\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi)$ and $\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi)$;
(h) (associativity) $(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi)$ and $(\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi)$.

Idea. For atoms these are truth-table checks. For arbitrary formulas the same tables work, with φ, ψ, χ in the place of atoms: under a fixed valuation, each of φ, ψ, χ has some value, and both sides are computed from those values by the clauses (V2)–(V6), exactly as in a table whose first columns are φ, ψ, χ.

Proof. Fix a valuation v, and let $a = v(\varphi)$, $b = v(\psi)$, $c = v(\chi)$. Each side of each equivalence is built from φ, ψ, χ by connectives, so by (V2)–(V6) of Definition 3.3.2 its value under v is computed from a, b, c by the clauses of those connectives. The combination (a, b), or (a, b, c), is one row of the tables below, whose first columns list all 4 (or 8) combinations. In each table the columns of the two sides of each equivalence agree in every row, as the reader can check column by column. Hence both sides have the same value under v, and since v was arbitrary, they are equivalent by the paraphrase of Definition 3.4.4.

Parts (a), (b), (c), and (e):

φ	ψ	$\varphi \to \psi$	$\neg\psi \to \neg\varphi$	$\neg\varphi \vee \psi$	$(\varphi \to \psi) \wedge (\psi \to \varphi)$
T	T	T	T	T	T
T	F	F	F	F	F
F	T	T	T	T	F
F	F	T	T	T	T

The first three formula columns agree, which gives (a) and (c). The last agrees with the column of $\varphi \leftrightarrow \psi$ in the table of Definition 3.3.2 (T exactly when $a = b$), which gives (e).

φ	ψ	$\neg(\varphi \wedge \psi)$	$\neg\varphi \vee \neg\psi$	$\neg(\varphi \vee \psi)$	$\neg\varphi \wedge \neg\psi$
T	T	F	F	F	F
T	F	T	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

This gives (b). For (d): if $a = \mathrm{T}$, then $v(\neg\varphi) = \mathrm{F}$ and $v(\neg\neg\varphi) = \mathrm{T}$; if $a = \mathrm{F}$, then $v(\neg\varphi) = \mathrm{T}$ and $v(\neg\neg\varphi) = \mathrm{F}$; in both cases $v(\neg\neg\varphi) = a$. For (f): clause (V3) makes $v(\varphi \wedge \psi) = \mathrm{T}$ exactly when $a = b = \mathrm{T}$, a condition that does not change when a and b are swapped, so $v(\varphi \wedge \psi) = v(\psi \wedge \varphi)$; likewise (V4) for $\vee$.

Parts (g) and (h):

φ	ψ	χ	$\varphi \wedge (\psi \vee \chi)$	$(\varphi \wedge \psi) \vee (\varphi \wedge \chi)$	$\varphi \vee (\psi \wedge \chi)$	$(\varphi \vee \psi) \wedge (\varphi \vee \chi)$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	F	F	T	T
F	T	T	F	F	T	T
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

φ	ψ	χ	$(\varphi \wedge \psi) \wedge \chi$	$\varphi \wedge (\psi \wedge \chi)$	$(\varphi \vee \psi) \vee \chi$	$\varphi \vee (\psi \vee \chi)$
T	T	T	T	T	T	T
T	T	F	F	F	T	T
T	F	T	F	F	T	T
T	F	F	F	F	T	T
F	T	T	F	F	T	T
F	T	F	F	F	T	T
F	F	T	F	F	T	T
F	F	F	F	F	F	F

In each of the two tables the first two formula columns agree, and so do the last two. This proves all parts.

■

Chapters 1 and 2 derived many of these laws as pairs of sequents, some directions with derivations that use RAA or LEM (Proposition 2.5.5 (d), Proposition 2.5.1, Proposition 2.5.6) and others without. That some of them cannot do without a classical rule, Chapter 2 stated without proof (for De Morgan (d), for instance); for $\neg\neg P \vdash P$, Section 3.5 proves it in an optional result. The tables do not see that difference: they see only that the two sides agree in every row. Section 3.6 shows that every such agreement can be turned into derivations, with RAA among the rules.

Long conjunctions and disjunctions. Associativity lets us stop worrying about grouping. We write $\varphi_1 \wedge \varphi_2 \wedge \cdots \wedge \varphi_n$ for the conjunction grouped to the right, $\varphi_1 \wedge (\varphi_2 \wedge (\cdots \wedge \varphi_n))$, and likewise for $\vee$; for $n = 1$ it is φ_1 itself. Then, for every valuation v and every $n \geq 1$,

$$v(\varphi_1 \wedge \cdots \wedge \varphi_n) = \mathrm{T} \iff v(\varphi_i) = \mathrm{T} \text{ for every } i \text{ with } 1 \leq i \leq n,$$
$$v(\varphi_1 \vee \cdots \vee \varphi_n) = \mathrm{T} \iff v(\varphi_i) = \mathrm{T} \text{ for some } i \text{ with } 1 \leq i \leq n.$$

For the conjunction, we prove by Axiom 3.2.1 on n, started at 1, that the first line holds for **every** list of n formulas. For $n = 1$ both sides say $v(\varphi_1) = \mathrm{T}$. If it holds for every list of n formulas, take a list of $n + 1$. The conjunction is $\varphi_1 \wedge \theta$, where $\theta = \varphi_2 \wedge \cdots \wedge \varphi_{n+1}$ has n terms. By (V3), $v(\varphi_1 \wedge \theta) = \mathrm{T}$ exactly when $v(\varphi_1) = \mathrm{T}$ and $v(\theta) = \mathrm{T}$, and by the hypothesis for n, $v(\theta) = \mathrm{T}$ exactly when $v(\varphi_i) = \mathrm{T}$ for $2 \leq i \leq n + 1$. Together: exactly when every $v(\varphi_i) = \mathrm{T}$. The disjunction is the same with (V4) and "some". Two consequences follow at once, since the right-hand sides do not care about order or grouping. Reordering the terms of a long conjunction, or grouping them differently, gives an equivalent formula; likewise for disjunctions. And De Morgan's laws extend to many terms:

$$\neg(\varphi_1 \wedge \cdots \wedge \varphi_n) \equiv \neg\varphi_1 \vee \cdots \vee \neg\varphi_n, \qquad \neg(\varphi_1 \vee \cdots \vee \varphi_n) \equiv \neg\varphi_1 \wedge \cdots \wedge \neg\varphi_n,$$

since, for instance, both sides of the first are true exactly when some φ_i is false.

Quick check. Is $p \to (q \to r) \equiv (p \wedge q) \to r$? And is $p \to (q \to r) \equiv (p \to q) \to r$?

Solution. The first holds. Each side is false in exactly one row, $p = q = \mathrm{T}$, $r = \mathrm{F}$: the left needs p true and $q \to r$ false, the right needs $p \wedge q$ true and r false. (Chapter 1 derived the two from each other, Proposition 1.4.2.) The second fails: by Example (One grouping entails the other, not conversely), the row $p = r = \mathrm{F}$ makes $p \to (q \to r)$ true and $(p \to q) \to r$ false.

3.4.4 Replacing a part by an equivalent part

In algebra, if $a = b$, then a may be replaced by b inside any expression. Semantic equivalence behaves in the same way, because the value of a formula depends only on the values of its parts. This is what makes chains of equivalences, one standard equivalence applied deep inside a formula at a time, a legitimate way of computing.

Recall from Section 3.1 that the subformulas of χ are the formulas at the nodes of its parse tree (Definition 3.1.3). An **occurrence** of φ in χ is a node whose subtree is a parse tree of φ; **replacing** that occurrence by φ' means putting a parse tree of φ' in place of that subtree, which gives the parse tree of a new formula χ'.

Lemma 3.4.2. Replacement

Let $\varphi \equiv \varphi'$. If χ' is obtained from a formula χ by replacing one occurrence of φ in χ by φ', then $\chi' \equiv \chi$.

Idea. Induct on the shape of χ. Either the occurrence is all of χ, and then χ is φ and χ' is φ', equivalent by hypothesis; or it sits inside one immediate part of χ, the other part is untouched, and the value of χ is computed from the values of its parts, one of which has not changed and one of which has not changed its value.

Proof. We apply Theorem 3.2.3 to the property $C(\chi)$: *for every formula χ' obtained from χ by replacing one occurrence of φ by φ', and every valuation v, $v(\chi') = v(\chi)$.* By the paraphrase of Definition 3.4.4, $C(\chi)$ says that each such χ' is equivalent to χ.

In every case, the replaced occurrence sits either at the top node of the tree of χ or at a node below it; in the second case the node lies in the subtree of exactly one immediate part of χ, since those subtrees are the parse trees of the immediate parts (Definition 3.1.2). If the occurrence is the root, then $\chi = \varphi$ and $\chi' = \varphi'$, and $v(\chi') = v(\chi)$ because $\varphi \equiv \varphi'$. It remains to treat an occurrence below the root.

Atoms and $\bot$. The parse tree is one node, so there is no node below the root.

Negation. Let $\chi = \neg\theta$ and assume $C(\theta)$. An occurrence below the root lies in θ, so $\chi' = \neg\theta'$, where θ' is obtained from θ by replacing that occurrence. By $C(\theta)$, $v(\theta') = v(\theta)$, and (V2) gives $v(\neg\theta') = v(\neg\theta)$.

Binary connectives. Let $\chi = \theta_1 \circ \theta_2$, with $\circ$ one of $\wedge, \vee, \to, \leftrightarrow$, and assume $C(\theta_1)$ and $C(\theta_2)$. If the occurrence lies in θ_1, then $\chi' = \theta_1' \circ \theta_2$, with θ_1' obtained from θ_1 by the replacement, and $v(\theta_1') = v(\theta_1)$ by $C(\theta_1)$. The clause of Definition 3.3.2 for $\circ$ computes $v(\chi')$ from $v(\theta_1')$ and $v(\theta_2)$, and $v(\chi)$ from $v(\theta_1)$ and $v(\theta_2)$, by the same rule, so $v(\chi') = v(\chi)$. An occurrence in θ_2 is the same with the roles of the parts exchanged.

This covers every clause of Definition 3.1.1, so $C(\chi)$ holds for every χ, which proves the lemma.

■

Replacing several occurrences, one after another, keeps equivalence at each step, and $\equiv$ is transitive, so the final formula is equivalent to the first. For instance, with (b), (d), and then (c) of Proposition 3.4.1, each applied inside the formula:

$$\begin{aligned} p \to \neg(q \wedge \neg r) &\equiv p \to (\neg q \vee \neg\neg r) \\ &\equiv p \to (\neg q \vee r) \\ &\equiv \neg p \vee (\neg q \vee r). \end{aligned}$$

The first step replaces the subformula $\neg(q \wedge \neg r)$ by an equivalent one (De Morgan with $\varphi = q$, $\psi = \neg r$); the second replaces $\neg\neg r$ by r; the third replaces the whole formula, by (c) with $\varphi = p$ and $\psi = \neg q \vee r$. Section 3.7 computes normal forms this way.

Warning. Replacement needs **equivalence**; entailment in one direction is not enough. We have $p \wedge q \models p$, but replacing $p \wedge q$ by p inside $\neg(p \wedge q)$ gives $\neg p$, and the row $p = \mathrm{T}$, $q = \mathrm{F}$ makes $\neg(p \wedge q)$ true and $\neg p$ false. A negation reverses the direction of an entailment, so one-way information cannot pass through it.

3.4.5 Substituting formulas for atoms

The letters P, Q of Chapters 1 and 2 stood for arbitrary statements, and Section 3.1 turned that into a remark about derivations: a derivation written with letters is a pattern, and putting formulas in for the letters, uniformly, gives a derivation again. The truth table of Example (The truth table of modus ponens), though, was computed for the atoms p and q. Does it tell us anything about $(r \vee s) \to \neg t$ in place of p? It should: the table only used the values of p and q, and a compound formula also has a value in each row. This subsection makes that precise.

Let $p_1, \ldots, p_n$ be distinct atoms and $\varphi_1, \ldots, \varphi_n$ formulas. The **substitution** σ of $\varphi_1, \ldots, \varphi_n$ for $p_1, \ldots, p_n$ is the function on formulas defined by recursion (Theorem 3.2.6):

$$\begin{aligned} &\sigma(p_i) = \varphi_i \ (1 \le i \le n), \qquad \sigma(q) = q \ \text{ for every other atom } q, \qquad \sigma(\bot) = \bot, \\ &\sigma(\neg\chi) = \neg\sigma(\chi), \qquad \sigma(\chi \circ \theta) = \sigma(\chi) \circ \sigma(\theta) \ \text{ for each binary } \circ\,. \end{aligned}$$

In words: $\sigma(\chi)$ replaces **every** occurrence of each p_i in χ by φ_i, all at once, and changes nothing else. For a list $\Gamma = \gamma_1, \ldots, \gamma_m$, we write $\sigma(\Gamma)$ for the list $\sigma(\gamma_1), \ldots, \sigma(\gamma_m)$. For example, with σ putting $q \wedge r$ for p and p for q,

$$\sigma\big((p \to q) \vee \neg p\big) = \big((q \wedge r) \to p\big) \vee \neg(q \wedge r).$$

The p that σ puts in for q is not replaced again: the replacements happen simultaneously, in one pass over the parse tree.

Lemma 3.4.3. Substitution

Let σ be the substitution of $\varphi_1, \ldots, \varphi_n$ for the distinct atoms $p_1, \ldots, p_n$.

(a) If $\models \chi$, then $\models \sigma(\chi)$.
(b) If $\Gamma \models \psi$, then $\sigma(\Gamma) \models \sigma(\psi)$.

Idea. Under a valuation v, the formula $\sigma(\chi)$ behaves exactly like χ itself under a different valuation w: the one that gives each atom p_i the value that v gives to φ_i. So every row for $\sigma(\chi)$ is, in disguise, a row for χ, and whatever holds in all rows of χ holds in all rows of $\sigma(\chi)$. The disguise is proved by induction on the shape of χ.

Proof. Let v be a valuation, and let w be the valuation with $w(p) := v(\sigma(p))$ for every atom p; so $w(p_i) = v(\varphi_i)$ and $w(q) = v(q)$ for every other atom q.

Claim. $v(\sigma(\chi)) = w(\chi)$ for every formula χ.

Proof of Claim. By Theorem 3.2.3. For an atom p, $v(\sigma(p)) = w(p)$ is the definition of w. For $\bot$, $v(\sigma(\bot)) = v(\bot) = \mathrm{F} = w(\bot)$ by (V1). For a negation, assume $v(\sigma(\chi)) = w(\chi)$; then $v(\sigma(\neg\chi)) = v(\neg\sigma(\chi))$, which by (V2) is the flip of $v(\sigma(\chi)) = w(\chi)$, and so equals $w(\neg\chi)$. For a binary connective $\circ$, assume the claim for χ and θ; then $v(\sigma(\chi \circ \theta)) = v(\sigma(\chi) \circ \sigma(\theta))$ is computed by the clause for $\circ$ from $v(\sigma(\chi)) = w(\chi)$ and $v(\sigma(\theta)) = w(\theta)$, and so equals $w(\chi \circ \theta)$, which the same clause computes from the same two values.

■

(a) Let $\models \chi$, and let v be any valuation. By the claim, $v(\sigma(\chi)) = w(\chi) = \mathrm{T}$, since χ is a tautology. So $\sigma(\chi)$ is true under every valuation.

(b) Let $\Gamma \models \psi$, and let v make every formula of $\sigma(\Gamma)$ true. For each γ in Γ, the claim gives $w(\gamma) = v(\sigma(\gamma)) = \mathrm{T}$, so w makes every formula of Γ true. By $\Gamma \models \psi$, $w(\psi) = \mathrm{T}$, and by the claim again $v(\sigma(\psi)) = w(\psi) = \mathrm{T}$. This proves $\sigma(\Gamma) \models \sigma(\psi)$.

■

So a table computed once for atoms settles infinitely many instances. From $\models p \vee \neg p$ we get $\models (q \to r) \vee \neg(q \to r)$, and every other instance of excluded middle. From $p \to q$, $p \models q$ (Example (The truth table of modus ponens)) we get $\varphi \to \psi$, $\varphi \models \psi$ for all formulas φ, ψ, which is the table's version of $\to$E as a schema. And from Example (Affirming the consequent is not valid)'s refuting row we learn something about all instances too, but only in one direction; the next warning says which.

Warning. Substitution must be **uniform**: every occurrence of p_i is replaced by the same φ_i. Replacing only the first p of the tautology $p \to p$ by q gives $q \to p$, which is false in the row $q = \mathrm{T}$, $p = \mathrm{F}$. And the lemma runs one way only. An instance can be a tautology when the pattern is not: $p \vee q$ is not a tautology, but its instance with $\neg p$ put in for q is $p \vee \neg p$, which is. So a refuting row for a pattern with atoms shows that the **pattern** is not valid, not that every instance fails.

3.4.6 Two questions, side by side

We now have two ways of saying that ψ follows from Γ. They were built from different materials, and it is worth setting them next to each other before the next two sections connect them.

	$\Gamma \vdash \psi$	$\Gamma \models \psi$
What it says	**there is** a derivation from premises in Γ to ψ	**every** valuation making all of Γ true makes ψ true
What it is about	syntax: lines, rules, boxes	meaning: truth values
How to show it holds	exhibit one derivation	check every row (or work backwards from F)
How to show it fails	no method yet: a failed search proves nothing	exhibit one refuting row
Defined in	Definition 1.2.3 (Section 1.2)	Definition 3.4.3 (this section)

Look at the fourth row. To show that something is **not** derivable, we would have to survey infinitely many derivations; to show that something is **not** valid, one row suffices. And to show that something **is** valid takes a finite, mechanical check, while showing it derivable may need an idea. Each column is easy exactly where the other is hard.

The next two sections prove that the two columns always give the same answer. Section 3.5 proves **soundness**: if $\Gamma \vdash \psi$, then $\Gamma \models \psi$; so a refuting row proves that no derivation exists, which answers the question Chapter 2 ended on. Section 3.6 proves **completeness**: if $\Gamma \models \psi$, then $\Gamma \vdash \psi$; so a successful table check guarantees that a derivation exists, even before anyone has found it.

3.4.7 Exercises

3.4.7.1 A. Check your understanding

Exercise. A1

(a) Classify each formula as a tautology, a contradiction, or satisfiable but not a tautology: $p \to (q \to p)$; $p \wedge \neg(p \vee q)$; $(p \to q) \wedge (q \to p)$; $\neg(p \to p)$; $(p \wedge q) \to (p \vee r)$.

(b) For each, decide whether it holds, and give a refuting row if it does not: $p \wedge q \models p \leftrightarrow q$; $p \leftrightarrow q \models p \wedge q$; $\neg p \to p \models p$.

(c) True or false, with a reason: if $\Gamma \models \psi$ and $\Gamma \models \neg\psi$, then Γ is unsatisfiable.

(d) True or false, with a reason: $p \vee q \models p$ or $p \vee q \models q$.

Solution. *(a)* $p \to (q \to p)$: a tautology (if p is T, the conclusion $q \to p$ is T; if p is F, the hypothesis is false). $p \wedge \neg(p \vee q)$: a contradiction (if p is T, so is $p \vee q$, and $\neg(p \vee q)$ is F). $(p \to q) \wedge (q \to p)$: satisfiable, not a tautology (T in the row TT, F in TF). $\neg(p \to p)$: a contradiction, since $p \to p$ is a tautology. $(p \wedge q) \to (p \vee r)$: a tautology (if the hypothesis is T, then p is T, so is $p \vee r$).

(b) $p \wedge q \models p \leftrightarrow q$ holds: the only row making $p \wedge q$ true has $p = q =$ T, where $p \leftrightarrow q$ is true. $p \leftrightarrow q \models p \wedge q$ fails: the row $p = q =$ F makes $p \leftrightarrow q$ true and $p \wedge q$ false. $\neg p \to p \models p$ holds: in the row $p =$ F, the premise $\neg p \to p$ has a true hypothesis and a false conclusion, so it is false; the only row making the premise true has $p =$ T.

(c) True. If a valuation made all of Γ true, it would make both ψ and $\neg\psi$ true, which (V2) forbids. So no valuation makes all of Γ true.

(d) False. The row $p = \mathrm{F}$, $q = \mathrm{T}$ refutes $p \vee q \models p$, and the row $p = \mathrm{T}$, $q = \mathrm{F}$ refutes $p \vee q \models q$. Each disjunct fails in some row where the disjunction holds, although in every such row one of them holds: $\models$ must work row by row with a single conclusion.

3.4.7.2 B. Practice

Exercise. B1: Valid or not

Decide whether each entailment holds, by a table or by working backwards from F. If it holds, also give a derivation of the corresponding sequent with $\vdash$ in place of $\models$, using the rules of Chapters 1 and 2; if it fails, give a refuting row.

(a) $p \to q,\ \neg q \models \neg p$.
(b) $p \to (q \vee r),\ \neg q \models p \to r$.
(c) $p \to q \models \neg p \to \neg q$.
(d) $p \vee q,\ p \to r,\ q \to r \models r$.

Solution. *(a)* Holds. Backwards from F: $\neg p$ false means $p = \mathrm{T}$; $\neg q$ true means $q = \mathrm{F}$; then $p \to q$ is F, so no row makes the premises true and the conclusion false. A derivation, with modus tollens (Proposition 2.3.1) as a derived rule:

1	$p \to q$	
2	$\neg q$	
3	$\neg p$	MT 1, 2

(b) Holds. Backwards from F: $p \to r$ false means $p = \mathrm{T}$, $r = \mathrm{F}$; $\neg q$ true means $q = \mathrm{F}$; then $q \vee r$ is F, and $p \to (q \vee r)$ is F, contradicting the first premise. A derivation, with disjunctive syllogism (Proposition 2.3.3), which refutes the left disjunct:

1	$p \to (q \vee r)$	
2	$\neg q$	
3	$\quad p$	
4	$\quad q \vee r$	$\to$E 1, 3
5	$\quad r$	DS 4, 2
6	$p \to r$	$\to$I 3–5

(c) Fails. The row $p = \mathrm{F}$, $q = \mathrm{T}$ makes $p \to q$ true (false hypothesis) and $\neg p \to \neg q$ false ($\neg p$ true, $\neg q$ false). It is the pattern of the Quick check, "denying the hypothesis", in the form of an implication.

(d) Holds. Backwards from F: $r = \mathrm{F}$; then $p \to r$ true forces $p = \mathrm{F}$, and $q \to r$ true forces $q = \mathrm{F}$; then $p \vee q$ is F, a contradiction. A derivation by cases:

1	$p \vee q$	
2	$p \to r$	
3	$q \to r$	
4	$\quad p$	
5	$\quad r$	$\to$E 2, 4
6	$\quad q$	
7	$\quad r$	$\to$E 3, 6
8	r	$\vee$E 1, 4–5, 6–7

Exercise. B2: A biconditional as a disjunction

Show by a truth table that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$. Which theorem of this chapter will guarantee, without any search, that each side can be derived from the other?

Solution.

p	q	$p \leftrightarrow q$	$p \wedge q$	$\neg p \wedge \neg q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
T	T	T	T	F	T
T	F	F	F	F	F
F	T	F	F	F	F
F	F	T	F	T	T

The columns of $p \leftrightarrow q$ and of the disjunction agree in every row, so the two formulas are equivalent: each side entails the other. The theorem that turns each entailment into a derivation is completeness, in Section 3.6: if $\Gamma \models \psi$, then $\Gamma \vdash \psi$. So $p \leftrightarrow q \vdash (p \wedge q) \vee (\neg p \wedge \neg q)$ and its converse are both derivable, before any derivation has been written.

Exercise. B3: Entailment and implication

Let Γ be a list of formulas and φ, ψ formulas. Prove:

(a) $\varphi \models \psi$ if and only if $\models \varphi \to \psi$;
(b) $\Gamma, \varphi \models \psi$ if and only if $\Gamma \models \varphi \to \psi$.

Solution. We prove (b); then (a) is (b) with Γ empty, since Γ, φ is then the one-formula list φ.

($\Rightarrow$) Suppose $\Gamma, \varphi \models \psi$, and let v make every formula of Γ true. If $v(\varphi) = \mathrm{F}$, then $v(\varphi \to \psi) = \mathrm{T}$ by (V5). If $v(\varphi) = \mathrm{T}$, then v makes every formula of the list Γ, φ true, so $v(\psi) = \mathrm{T}$ by hypothesis, and again $v(\varphi \to \psi) = \mathrm{T}$. So $\Gamma \models \varphi \to \psi$.

($\Leftarrow$) Suppose $\Gamma \models \varphi \to \psi$, and let v make every formula of Γ, φ true. Then v makes every formula of Γ true, so $v(\varphi \to \psi) = \mathrm{T}$, and $v(\varphi) = \mathrm{T}$. By (V5), an implication with a true hypothesis is true only if its conclusion is, so $v(\psi) = \mathrm{T}$. So $\Gamma, \varphi \models \psi$.

Part (b) is the table's version of →I (the direction ⇒) and of →E (the direction ⇐).

3.4.7.3 C. Going deeper

Exercise. C1: Substitution with a new connective, the last half blank

Extend the formulas by a new binary connective | ("not both", often called NAND): add to Definition 3.1.1 the clause "if φ and ψ are formulas, so is $(\varphi \mid \psi)$", and to Definition 3.3.2 the clause

(V7) $v(\varphi \mid \psi) := \mathrm{F}$ if $v(\varphi) = v(\psi) = \mathrm{T}$; otherwise T.

Assume that structural induction and recursion work for the extended formulas as before, with one more clause. A substitution σ of $\varphi_1, \ldots, \varphi_n$ for $p_1, \ldots, p_n$ is defined by the clauses in the text, with $\sigma(\chi \mid \theta) := \sigma(\chi) \mid \sigma(\theta)$ added. Prove that if χ is a tautology of the extended language, so is $\sigma(\chi)$.

Here is the first half of a proof. *Let v be a valuation, and let w be the valuation with $w(p) := v(\sigma(p))$ for every atom p. We show $v(\sigma(\chi)) = w(\chi)$ for every formula χ of the extended language, by structural induction. Atoms: $v(\sigma(p)) = w(p)$ is the definition of w.*

Name the move, then complete it.

Solution. The move is **induction on the shape of a formula**, now with eight cases: atoms, $\bot$, $\neg$, and five binary connectives. The property is $C(\chi)$: $v(\sigma(\chi)) = w(\chi)$.

$\bot$. $\sigma(\bot) = \bot$, so $v(\sigma(\bot)) = \mathrm{F} = w(\bot)$ by (V1).

Negation. Assume $C(\chi)$. Then $v(\sigma(\neg\chi)) = v(\neg\sigma(\chi))$, which by (V2) is T exactly when $v(\sigma(\chi)) = \mathrm{F}$, that is, when $w(\chi) = \mathrm{F}$, that is, when $w(\neg\chi) = \mathrm{T}$.

The four old binary connectives. Assume $C(\chi)$ and $C(\theta)$, and let $\circ$ be one of $\wedge, \vee, \to, \leftrightarrow$. Then $v(\sigma(\chi \circ \theta)) = v(\sigma(\chi) \circ \sigma(\theta))$ is computed by the clause for $\circ$ from the two values $v(\sigma(\chi)) = w(\chi)$ and $v(\sigma(\theta)) = w(\theta)$; the same clause computes $w(\chi \circ \theta)$ from the same two values. So they are equal.

The new connective. Assume $C(\chi)$ and $C(\theta)$. By (V7), $v(\sigma(\chi|\theta)) = v(\sigma(\chi)|\sigma(\theta))$ is F exactly when $v(\sigma(\chi)) = v(\sigma(\theta)) = \mathrm{T}$, that is, when $w(\chi) = w(\theta) = \mathrm{T}$, that is, when $w(\chi \mid \theta) = \mathrm{F}$.

This covers every clause, so $C(\chi)$ holds for every formula. Now let χ be a tautology and v any valuation: $v(\sigma(\chi)) = w(\chi) = \mathrm{T}$. So $\sigma(\chi)$ is a tautology.

Only one case is new. The proof of Lemma 3.4.3 never looked at what the clauses say, only that each computes the value of a compound from the values of its parts; any connective defined by a table passes.

Marking scheme. (10 points)

- 1: names the move and states the property $v(\sigma(\chi)) = w(\chi)$ for all χ.
- 1: the $\bot$ case.
- 2: the $\neg$ case, from the hypothesis for χ.
- 2: the old binary cases, using the hypotheses for both parts.
- 3: the | case, with (V7) and both hypotheses.
- 1: the conclusion for tautologies.

Fatal errors (0 for the step): omitting the | case; using w where v is meant in the conclusion.

Exercise. C2: Is equivalence preserved by substitution?

Let σ be a substitution and suppose $\varphi \equiv \psi$. Is $\sigma(\varphi) \equiv \sigma(\psi)$? Prove it, or give a counterexample. Then suppose that De Morgan's law (b) of Proposition 3.4.1 had been checked only for atoms, $\neg(p \wedge q) \equiv \neg p \vee \neg q$, by its four-row table. Use your answer to deduce from this, in one line, that $\neg((r \to s) \wedge \neg t) \equiv \neg(r \to s) \vee \neg\neg t$.

Solution. Yes. By the paraphrase of Definition 3.4.4, $\varphi \equiv \psi$ means $\models \varphi \leftrightarrow \psi$. By Lemma 3.4.3 (a), $\models \sigma(\varphi \leftrightarrow \psi)$, and by the definition of σ, $\sigma(\varphi \leftrightarrow \psi) = \sigma(\varphi) \leftrightarrow \sigma(\psi)$. So $\models \sigma(\varphi) \leftrightarrow \sigma(\psi)$, that is, $\sigma(\varphi) \equiv \sigma(\psi)$. (Alternatively, apply Lemma 3.4.3 (b) to $\varphi \models \psi$ and to $\psi \models \varphi$.)

For the instance: the substitution of $r \to s$ for p and $\neg t$ for q turns the two sides of $\neg(p \wedge q) \equiv \neg p \vee \neg q$ into $\neg((r \to s) \wedge \neg t)$ and $\neg(r \to s) \vee \neg\neg t$, so these are equivalent by the first part. Checking a law for atoms, by one small table, is therefore enough.

A tempting counterexample fails for a reason worth seeing. With σ putting $\neg p$ for q, the nonequivalent formulas p and $\neg q$ become the equivalent p and $\neg\neg p$: substitution can create equivalences. It cannot destroy one, which is what the question asks.

Marking scheme. (10 points)

- 3: reduces $\equiv$ to a tautology $\models \varphi \leftrightarrow \psi$, or to two entailments, citing the definition.
- 3: applies Lemma 3.4.3 correctly.
- 2: computes $\sigma(\varphi \leftrightarrow \psi) = \sigma(\varphi) \leftrightarrow \sigma(\psi)$ from the definition of σ.
- 2: the instance, with the substitution named.

Fatal errors (0 for the step): answering "no" by a pair that is not equivalent before the substitution; using a nonuniform replacement.

3.5 Soundness

Chapter 2 ended on a question it could not answer. Section 2.6 searched for a derivation of $P \to Q \vdash Q \to P$, and every route came back to the same gap; but a search that fails proves nothing, since a cleverer derivation might exist. The row that refutes affirming the consequent in Section 3.4, $p = \mathrm{F}$ and $q = \mathrm{T}$, finds what the search could not: it makes $p \to q$ true and $q \to p$ false. This section proves the theorem that turns such a row into a proof that **no** derivation exists. It is called soundness: whatever the rules derive is true in every row where the premises are true. The proof is the book's first long proof about proofs, and it is an induction on the lines of a derivation.

3.5.1 What must be proved, and why it is not obvious

Chapters 1 and 2 checked each rule as it arrived. $\wedge$I is safe because two true statements make a true conjunction (Section 1.2); $\to$E is safe because that is what "if ... then" asserts (Section

1.3). Section 1.2 said once that Chapter 3 would prove it for every rule of Chapters 1 and 2 at once. Why is that more than collecting the separate checks?

The first attempt at a proof is: go down a derivation, line by line, and show that each line is true in every row in which the premises are true. For a derivation without boxes this works. Each line is a premise, or comes from earlier lines by a rule, and the rule turns true lines into a true line. But it fails at the first box. Here is the derivation of Proposition 2.3.2, written with atoms:

1	$p \to q$	
2	$\quad \neg q$	
3	$\quad\quad p$	
4	$\quad\quad q$	$\to$E 1, 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad \neg p$	$\neg$I 3–5
7	$\neg q \to \neg p$	$\to$I 2–6

Line 3 is p, and the row $p = \mathrm{F}$, $q = \mathrm{F}$ makes the premise $p \to q$ true and line 3 false. Line 5 is $\bot$, which is false in **every** row. So the lines inside a box are not true in the rows where the premises are true, and the first attempt collapses. Yet the derivation is correct, and its last line $\neg q \to \neg p$ is true in every row making $p \to q$ true (Example (An implication and its contrapositive)).

What goes wrong is that a line inside a box is not claimed on the strength of the premises alone. Line 4 says: *if* $p \to q$, $\neg q$ and p all hold, then q holds. That is a true claim about rows. Every line depends on the premises **and** on the assumptions of the boxes around it, and a correct statement about each line must mention both. Chapter 1 gave this list a name.

3.5.2 Derivations of formulas

From now on the lines of a derivation carry **formulas** in the sense of Definition 3.1.1, with atoms $p, q, r, \ldots$. Nothing in the rules changes. Every rule was stated as a schema, whose letters φ, ψ, χ stand for arbitrary statements (Definition 1.2.1), so it applies to formulas as it stands, and Definition 1.4.2 and Definition 1.2.3 read the same with "formula" for "statement". As Section 3.1 said, a derivation written with Chapter 2's letters P, Q is a pattern, and reading its letters as atoms, or as any formulas, gives a derivation of formulas.

Two conventions for the rest of the section. The **primitive rules** are those of Chapters 1 and 2 collected in Section 2.6: R, $\wedge$I, $\wedge$E, $\to$I, $\to$E, $\vee$I, $\vee$E, $\neg$I, $\neg$E, $\bot$E, RAA, $\leftrightarrow$I and $\leftrightarrow$E. The **box rules** among them are the five that cite a box: $\to$I, $\neg$I, RAA, $\vee$E and $\leftrightarrow$I. The other eight, R included, cite only lines; we call them the **one-step rules**.

A derivation may also use derived rules such as MT and LEM. A line citing one abbreviates a stretch of derivation in rules proved earlier, with the letters replaced by formulas (Definition 2.3.1), and Section 2.3 showed how to write it out: repeat the cited lines, then paste the derivation of the derived rule below them. Every chain of derived rules ends in primitive rules. So whenever $\Gamma \vdash \psi$, there is also a derivation of ψ from premises in Γ that uses **primitive rules only**. That is the only kind of derivation the proof below has to consider.

3.5.3 The context of a line

Let a derivation have lines $1, \dots, n$. We write φ_k for the formula on line k. The **length** of a derivation is its number of lines. The **context** of line k is the list of the premises together with the assumptions of every box containing line k: the statements the line depends on. We write it Γ_k. The context of the last line is just the premises, since by Definition 1.4.2 the last line lies in no box. For lists of formulas Δ and Δ', we write $\Delta \subseteq \Delta'$ when every formula of Δ occurs in Δ'.

The context of a line is everything that line depends on: the premises, and every assumption still open where the line stands.

In the derivation of Proposition 2.3.2 above, the boxes are 2–6 and 3–5, and the contexts are:

Line	Formula	Boxes containing it	Context
1	$p \to q$	none	$p \to q$
2	$\neg q$	2–6	$p \to q,\ \neg q$
3, 4, 5	$p,\ q,\ \bot$	2–6, 3–5	$p \to q,\ \neg q,\ p$
6	$\neg p$	2–6	$p \to q,\ \neg q$
7	$\neg q \to \neg p$	none	$p \to q$

Now look at each line through its context. Line 3 is one of the formulas in its own context, so every row making the context true makes line 3 true. Line 4: every row making $p \to q$ and p true makes q true. Line 5: no row makes $p \to q$, $\neg q$, and p all true (the first and third force q), so the context is unsatisfiable and it entails even $\bot$ (Section 3.4). Line 6: every row making $p \to q$ and $\neg q$ true makes p false. Line 7: its context is the premise, and we know $p \to q \models \neg q \to \neg p$. So on this derivation, every line k satisfies

$$\Gamma_k \models \varphi_k.$$

This is the statement that survives the boxes, and it is the one we shall prove for every line of every derivation.

Two facts about contexts, read off the scope rule, do the bookkeeping. The first says that citing an available line never involves an assumption the citing line does not have. The second says exactly what the context of a box's last line is, when the box is cited by a box rule.

Lemma 3.5.1. Contexts in a Derivation

Let lines i, k and $a \le b$ belong to a derivation, and let Γ_j denote the context of line j.

(a) If line i is available at line k, then $\Gamma_i \subseteq \Gamma_k$.

(b) If line k is justified by a box rule citing the box whose first line is a and whose last line is b, then the formulas of Γ_b are exactly those of Γ_k together with φ_a, the assumption of that box. We write this $\Gamma_b = \Gamma_k, \varphi_a$.

Idea. Both parts compare two sets of boxes, since contexts differ only in the assumptions of the boxes around a line. For (a), the scope rule says that the boxes around line i are among the boxes around line k. For (b), the boxes around line b should be the cited box itself and

the boxes around line k; the danger is a further box inside the cited box that also ends on line b, and the box rules forbid exactly that.

Proof. *(a)* The premises belong to every context. Any other formula of Γ_i is the assumption of a box containing line i. Since line i is available at line k, every box containing line i also contains line k, by the scope rule of Definition 1.4.2. So that assumption belongs to Γ_k.

(b) Let B be the box from line a to line b. Since line k cites B, the box B is available at line k. By the scope rule, it ends before line k, and the boxes containing B are exactly the boxes containing line k.

We find the boxes containing line b. Every box containing line k contains B, and so contains line b; and B contains line b. Conversely, let C be a box containing line b. By Definition 1.4.2, two boxes are either disjoint or one inside the other; C and B share line b, so $C = B$, or C contains B, or C lies inside B. The last is impossible: then line b, the last line of B, would be a line of a box inside B, and every box rule requires the last line of its box to be a line of that box itself, not of a box inside it (Definition 1.4.1, Definition 2.2.1, Definition 2.5.1, Definition 1.6.1, Definition 2.4.1). So $C = B$, or C is one of the boxes containing B, which are the boxes containing line k.

Hence the boxes containing line b are B and the boxes containing line k. The assumption of B is φ_a, so Γ_b consists of the premises, the assumptions of the boxes containing line k, and φ_a, which is Γ_k together with φ_a.

■

In the derivation of Proposition 2.3.2, line 6 cites the box 3–5 by ¬I, and $\Gamma_5 = p \to q,\ \neg q,\ p$ is $\Gamma_6 = p \to q,\ \neg q$ with the assumption p added, as (b) says. Line 5 cites lines 4 and 2, both available there, and Γ_4 and Γ_2 are contained in Γ_5, as (a) says.

Warning. Part (b) needs the box to be **cited by a box rule**. Definition 1.4.2 allows a box to end on the same line as a box inside it, and then its last line has a larger context. In the derivation below, the box 2–4 is available at line 5, but $\Gamma_4 = p,\ q,\ r$ contains r, the assumption of the inner box, as well as q. No box rule may cite 2–4: its last line lies inside the box 3–4, not in 2–4 itself. (This is why the derivation of Example (Regrouping a disjunction) needs its line 11.)

1	p	
2	$\quad q$	
3	$\quad\quad r$	
4	$\quad\quad p$	R 1
5	p	R 1

Quick check. In the derivation of Proposition 2.3.2 above, which case of the soundness claim handles line 5? Which lines' contexts are involved, and why does $\Gamma_5 \models \bot$ hold?

Solution. Line 5 is $\bot$, by $\neg$E citing lines 4 and 2: a one-step rule. Lines 4 and 2 are available at line 5, so by Lemma 3.5.1 (a), $\Gamma_4 = p \to q,\ \neg q,\ p$ and $\Gamma_2 = p \to q,\ \neg q$ are contained in $\Gamma_5 = p \to q,\ \neg q,\ p$. The claim for the earlier lines gives $\Gamma_4 \models q$ and $\Gamma_2 \models \neg q$. So a valuation making Γ_5 true would make both q and $\neg q$ true, which clause (V2) of Definition 3.3.2 forbids. No valuation makes Γ_5 true, and $\Gamma_5 \models \bot$ holds because there is no row to check.

3.5.4 The soundness theorem

We can now state the theorem that Chapters 1 and 2 kept pointing to.

Theorem 3.5.2. Soundness

Let Γ be a list of formulas and ψ a formula. If $\Gamma \vdash \psi$, then $\Gamma \models \psi$.

Try it first (ten minutes): suppose line k is $\varphi \to \psi$, by $\to$I from a box that opens with φ and ends with ψ. Suppose you already know that every row making the context of line k and φ true makes ψ true. Why is $\varphi \to \psi$ true in every row making the context of line k true? What happens in the rows where φ is false?

Idea. An attempt at the $\to$I case runs into two things. First, knowledge about the box's last line is knowledge about rows making its **own** context true, which contains the assumption φ; that is why the claim must carry contexts, and why Lemma 3.5.1 (b) is needed. Second, the rows in which φ is false are not covered by the box at all, and they need nothing: there $\varphi \to \psi$ is true by (V5). The same two moves handle every box rule. The Step roadmap:

① Prove the stronger claim $\Gamma_k \models \varphi_k$ for **every** line k of a derivation in primitive rules.

② Use strong induction on k (Theorem 3.2.2): a rule may cite **any** earlier line, not only the line just above, so the hypothesis must be available for all earlier lines at once.

③ Split on how line k is justified. A premise or an assumption belongs to its own context. A one-step rule: apply the hypothesis to the cited lines, move their contexts into Γ_k by Lemma 3.5.1 (a), and use one clause of Definition 3.3.2. A box rule: apply the hypothesis to each box's last line, whose context is Γ_k plus the box's assumption by Lemma 3.5.1 (b), and argue about the rows where that assumption is true or false.

④ The last line lies in no box, so its context is the premises, all in Γ; adding the other formulas of Γ keeps the entailment.

Before reading on: in the case of RAA *(Claim 3 below), where does the proof use that $v(\neg\varphi)$ is* F *exactly when $v(\varphi)$ is* T*, and which half of that fact does it use?*

Proof. **Step 1 (reduction to primitive rules).** Suppose $\Gamma \vdash \psi$. As shown in the subsection on derivations of formulas, there is a derivation of ψ whose premises all belong to Γ and which uses primitive rules only. Fix one, with lines $1, \ldots, n$; let φ_k be the formula on line k and Γ_k its context.

Step 2 (the induction). For $k \in \mathbb{N}$, let $S(k)$ be the statement: *if* $1 \le k \le n$, *then* $\Gamma_k \models \varphi_k$. We prove $S(k)$ for every k by Theorem 3.2.2. So let $k \in \mathbb{N}$, and assume $S(j)$ for every $j < k$.

If $k = 0$ or $k > n$, there is nothing to prove, so let $1 \leq k \leq n$. Every line and every box that line k cites is available at line k, and so lies before line k (Definition 1.4.2). Hence the induction hypothesis $\Gamma_j \models \varphi_j$ holds for every line j that line k cites, and for the last line of every box it cites.

Let v be a valuation that makes every formula of Γ_k true. We show $v(\varphi_k) = \mathrm{T}$; since v is arbitrary, this proves $\Gamma_k \models \varphi_k$. By Definition 1.4.2, line k is a premise, an assumption, or the conclusion of one primitive rule, and the three claims below treat these cases.

Step 3 (the cases).

Claim. If line k is a premise or an assumption, then $v(\varphi_k) = \mathrm{T}$.

Proof of Claim. A premise belongs to every context. An assumption opens a box, that box contains line k, and so its assumption φ_k belongs to Γ_k. In both cases φ_k is one of the formulas that v makes true.

■

For the one-step rules we first record what the hypothesis gives for a cited line. If line k cites line i, then line i is available at line k, so $\Gamma_i \subseteq \Gamma_k$ by Lemma 3.5.1 (a). Hence v makes every formula of Γ_i true, and the induction hypothesis $\Gamma_i \models \varphi_i$ gives

$$v(\varphi_i) = \mathrm{T} \quad \text{for every line } i \text{ cited by line } k. \tag{$*$}$$

Claim. If line k is justified by a one-step rule, then $v(\varphi_k) = \mathrm{T}$.

Proof of Claim. We take the rules one at a time; in each, the cited lines are true under v by $(*)$.

R. Line k repeats a line i, so $\varphi_k = \varphi_i$, and $v(\varphi_k) = \mathrm{T}$.

$\wedge$I. Line k is $\varphi \wedge \psi$, citing lines holding φ and ψ. Both are true under v, so $\varphi \wedge \psi$ is true by (V3).

$\wedge$E. Line k is φ, citing a line holding $\varphi \wedge \psi$. That line is true under v, so by (V3) both conjuncts are true, and in particular φ. The second form, which writes ψ, is the same with the other conjunct, and we skip it.

$\to$E. Line k is ψ, citing lines holding $\varphi \to \psi$ and φ. Both are true under v. By (V5), an implication with a true hypothesis and a false conclusion is false; since $\varphi \to \psi$ is true and φ is true, ψ is true.

$\vee$I. Line k is $\varphi \vee \psi$ or $\psi \vee \varphi$, citing a line holding φ. That line is true under v, so both disjunctions are true by (V4).

$\neg$E. Line k is $\bot$, citing lines holding φ and $\neg\varphi$. By $(*)$ both would be true under v, and by (V2) $\neg\varphi$ is false whenever φ is true. So no valuation makes every formula of Γ_k true, and there is no v to consider: $\Gamma_k \models \bot$ holds because there is no row to check.

$\bot$E. Line k is any formula φ, citing a line holding $\bot$. By $(*)$, $v(\bot) = \mathrm{T}$, which (V1) forbids. So again no valuation makes every formula of Γ_k true, and $\Gamma_k \models \varphi$ holds.

$\leftrightarrow$E. Line k is ψ, citing lines holding $\varphi \leftrightarrow \psi$ and φ. Both are true under v; by (V6), $v(\psi) = v(\varphi) = \mathrm{T}$. The second form, which writes φ from $\varphi \leftrightarrow \psi$ and ψ, is the same with the sides exchanged, and we skip it.

■

For the box rules we record what the hypothesis gives for a cited box. If line k cites, by a box rule, the box from line a to line b, then $\Gamma_b = \Gamma_k, \varphi_a$ by Lemma 3.5.1 (b), and the induction hypothesis for line b says

$$\Gamma_k, \varphi_a \models \varphi_b : \quad \text{every valuation making } \Gamma_k \text{ and } \varphi_a \text{ true makes } \varphi_b \text{ true.} \qquad (**)$$

Claim. If line k is justified by a box rule, then $v(\varphi_k) = \mathrm{T}$.

Proof of Claim. Recall that v makes every formula of Γ_k true.

$\to$I. Line k is $\varphi \to \psi$, citing a box that opens with φ and ends with ψ. By $(**)$, every valuation making Γ_k and φ true makes ψ true. If $v(\varphi) = \mathrm{F}$, then $v(\varphi \to \psi) = \mathrm{T}$ by (V5). If $v(\varphi) = \mathrm{T}$, then v makes Γ_k and φ true, so $v(\psi) = \mathrm{T}$, and again $v(\varphi \to \psi) = \mathrm{T}$ by (V5).

$\neg$I. Line k is $\neg\varphi$, citing a box that opens with φ and ends with $\bot$. By $(**)$, every valuation making Γ_k and φ true makes $\bot$ true; since (V1) makes $\bot$ false under every valuation, no valuation makes Γ_k and φ true. As v makes Γ_k true, it must make φ false, and then $v(\neg\varphi) = \mathrm{T}$ by (V2).

RAA. Line k is φ, citing a box that opens with $\neg\varphi$ and ends with $\bot$. By $(**)$ and (V1), as in the previous case, no valuation makes Γ_k and $\neg\varphi$ true. Suppose $v(\varphi) = \mathrm{F}$. Then $v(\neg\varphi) = \mathrm{T}$ by (V2), so v makes Γ_k and $\neg\varphi$ true, which is impossible. Hence $v(\varphi) = \mathrm{T}$.

$\vee$E. Line k is χ, citing a line i holding $\varphi \vee \psi$, a box that opens with φ and ends with χ, and a box that opens with ψ and ends with χ. By $(*)$, $v(\varphi \vee \psi) = \mathrm{T}$, so by (V4), $v(\varphi) = \mathrm{T}$ or $v(\psi) = \mathrm{T}$. By $(**)$ for the first box, every valuation making Γ_k and φ true makes χ true; for the second, every valuation making Γ_k and ψ true makes χ true. If $v(\varphi) = \mathrm{T}$, then v makes Γ_k and φ true, so $v(\chi) = \mathrm{T}$. If $v(\psi) = \mathrm{T}$, the second box gives $v(\chi) = \mathrm{T}$ in the same way.

$\leftrightarrow$I. Line k is $\varphi \leftrightarrow \psi$, citing a box that opens with φ and ends with ψ, and one that opens with ψ and ends with φ. By $(**)$, every valuation making Γ_k and φ true makes ψ true, and every valuation making Γ_k and ψ true makes φ true. If $v(\varphi) = \mathrm{T}$, the first box gives $v(\psi) = \mathrm{T}$. If $v(\varphi) = \mathrm{F}$, then $v(\psi) = \mathrm{F}$ too, since $v(\psi) = \mathrm{T}$ would give $v(\varphi) = \mathrm{T}$ by the second box. In both cases $v(\varphi) = v(\psi)$, so $v(\varphi \leftrightarrow \psi) = \mathrm{T}$ by (V6).

■

The three claims cover every way line k can be justified, so $S(k)$ holds. By Theorem 3.2.2, $S(k)$ holds for every $k \in \mathbb{N}$: every line k of the derivation satisfies $\Gamma_k \models \varphi_k$.

Step 4 (the last line). The last line n of a derivation lies in no box (Definition 1.4.2), so its context Γ_n consists of the premises alone, and $\varphi_n = \psi$. By Step 3, $\Gamma_n \models \psi$. Every premise belongs to Γ (Definition 1.2.3), so a valuation making every formula of Γ true makes every formula of Γ_n true, and hence makes ψ true. This proves $\Gamma \models \psi$.

■

The proof is long, but each case is one step of the truth-table reasoning of Section 3.3, placed in the right context. Every primitive rule has passed. So has every derived rule, since each abbreviates primitive ones. *The rules never lead from truth to falsehood*, not merely line by line but for every derivation that can ever be written.

Why strong induction and the position k of a line? A rule may cite any available line, however

far back: line 7 of the derivation of Proposition 2.3.2 cites the box that begins at line 2. So the hypothesis is needed for all earlier lines at once. A different route would induct on the length of the whole derivation, removing its last line. But the lines above the last line need not form a derivation, since a box may still be open there (in the derivation of Proposition 2.3.2, lines 1–6 end inside the box 2–6), and that route has to cut boxes open. Carrying the context of every line avoids the surgery.

Example. Running the claim on a case split

Here is a derivation of $p \vee q \vdash q \vee p$. List the context of each line, and check $\Gamma_k \models \varphi_k$ for each k, naming the case of the proof that handles it.

1	$p \vee q$	
2	$\quad p$	
3	$\quad q \vee p$	$\vee$I 2
4	$\quad q$	
5	$\quad q \vee p$	$\vee$I 4
6	$q \vee p$	$\vee$E 1, 2–3, 4–5

Solution. The boxes are 2–3 and 4–5.

Line	Context Γ_k	Case	Why $\Gamma_k \models \varphi_k$
1	$p \vee q$	premise	φ_1 is in its context
2	$p \vee q,\ p$	assumption	φ_2 is in its context
3	$p \vee q,\ p$	$\vee$I citing 2	$\Gamma_2 \subseteq \Gamma_3$; a row making p true makes $q \vee p$ true
4	$p \vee q,\ q$	assumption	φ_4 is in its context
5	$p \vee q,\ q$	$\vee$I citing 4	as for line 3, with q
6	$p \vee q$	$\vee$E citing 1 and both boxes	see below

For line 6: by Lemma 3.5.1 (b), $\Gamma_3 = \Gamma_6, p$ and $\Gamma_5 = \Gamma_6, q$, and the claim for lines 3 and 5 says that rows making $p \vee q$ and p true, and rows making $p \vee q$ and q true, make $q \vee p$ true. A row making Γ_6 true makes $p \vee q$ true, so it makes p or q true, and one of the two boxes applies. Line 6 is the last line, and its context is the premise: so $p \vee q \models q \vee p$, as soundness promises.

3.5.5 Refuting with one row

Soundness is most often used backwards. If the conclusion can be false while the premises are true, no derivation can exist.

Corollary 3.5.3. Refuting Row

Let Γ be a list of formulas and ψ a formula. If some valuation makes every formula of Γ true and ψ false, then $\Gamma \nvdash \psi$.

Proof. Let v make every formula of Γ true and ψ false. Then $\Gamma \models \psi$ fails (Definition 3.4.3). If $\Gamma \vdash \psi$ held, Theorem 3.5.2 would give $\Gamma \models \psi$. So $\Gamma \vdash \psi$ does not hold. ■

Here $\Gamma \nvdash \psi$ means that there is **no** derivation of ψ from premises in Γ: not with the primitive rules, not with any derived rule, not in a thousand lines. One row has settled a question about infinitely many derivations.

Chapter 2's question. Section 2.6 searched for a derivation of $P \to Q \vdash Q \to P$ and stalled. The row $p = \mathrm{F}$, $q = \mathrm{T}$ makes $p \to q$ true (false hypothesis) and $q \to p$ false (true hypothesis, false conclusion). By Corollary 3.5.3,

$$p \to q \nvdash q \to p.$$

The letters of Chapter 2 stood for arbitrary statements, and the search asked for a derivation of the **pattern**: one derivation, written with P and Q, that works whatever they are. Such a derivation, with P and Q read as the atoms p and q, would derive $p \to q \vdash q \to p$, which we have just ruled out. So the pattern has no derivation, and the search was right to fail. (Particular instances can be derivable: with P and Q the same statement, $P \to P \vdash P \to P$ takes one line. A refuting row for atoms speaks about the pattern, as the warning after Lemma 3.4.3 said.)

Affirming the consequent. Chapter 1 promised, in Section 1.3, that no derivation leads from $P \to Q$ and Q to P, whichever rules of Chapters 1 and 2 it uses. The refuting row of Example (Affirming the consequent is not valid), $p = \mathrm{F}$ and $q = \mathrm{T}$, makes $p \to q$ and q true and p false, so by Corollary 3.5.3,

$$p \to q,\ q \nvdash p.$$

Chapter 1 could already see that the step "from $\varphi \to \psi$ and ψ, infer φ" is not an instance of $\to$E. That did not exclude a longer derivation reaching p some other way. The row excludes every derivation at once. So the repair of Argument B that Section 1.4 considered is not merely unjustified: it is impossible.

A stall in Chapter 2. Section 2.5 found that the De Morgan direction $\neg(P \wedge Q) \vdash \neg P \vee \neg Q$ cannot end with $\vee$I, because neither $\neg P$ nor $\neg Q$ follows from the premise alone, and it said "the rules never lead from true premises to a false conclusion" as its reason. Now that reason is a theorem: the row $p = \mathrm{T}$, $q = \mathrm{F}$ makes $\neg(p \wedge q)$ true and $\neg p$ false, so $\neg(p \wedge q) \nvdash \neg p$, and the row $p = \mathrm{F}$, $q = \mathrm{T}$ gives $\neg(p \wedge q) \nvdash \neg q$.

Warning. A refuting row must make **every** premise true. For $p \to q,\ p \vdash q$, the row $p = \mathrm{F}$, $q = \mathrm{F}$ makes the conclusion false, but it also makes the premise p false, so it refutes nothing; and the sequent is derivable, by one line of $\to$E. A row in which some premise fails is simply not one of the rows Definition 3.4.3 looks at.

Warning. Soundness goes one way. It says that derivable sequents are valid; it does not say that a formula true in the row you happened to check is derivable. The atom p is true under the all-T valuation, but $\nvdash p$: the all-F valuation makes it false, so Corollary 3.5.3 (with Γ empty) applies. Whether **every** valid sequent is derivable is a separate question, answered in Section 3.6.

Quick check. Give a row showing $p \to r \nvdash (p \vee q) \to r$.

Solution. Take $p = \mathrm{F}$, $q = \mathrm{T}$, $r = \mathrm{F}$. The premise $p \to r$ is true, since p is false. The conclusion has hypothesis $p \vee q$, which is true, and conclusion r, which is false, so it is false. By Corollary 3.5.3, no derivation exists. (The rows with p true fail: they make the premise force r true, or make the premise false.)

3.5.6 Reductio says something new

Chapter 2 introduced RAA in Section 2.5 because the rules before it stalled on $\neg\neg P \vdash P$, and said that Chapter 3 would prove that no derivation of P from $\neg\neg P$ avoids RAA. Truth tables cannot do it. The row method only proves nonderivability from **all** the rules, and with RAA the sequent is derivable, so no two-valued row refutes it. The trick is to run the proof of Theorem 3.5.2 with a different notion of truth, one with three values, under which every rule except RAA still passes. This result is optional: it answers a question Chapter 2 raised, and nothing later uses it.

A **three-valued valuation** u assigns to each atom one of the values 0, $\frac{1}{2}$, 1, ordered $0 < \frac{1}{2} < 1$. Think of 1 as "true", 0 as "false", and $\frac{1}{2}$ as "not yet settled". It is extended to all formulas by recursion (Theorem 3.2.6):

$$
\begin{aligned}
u(\bot) &= 0, \qquad u(\neg\varphi) = u(\varphi \to \bot), \\
u(\varphi \wedge \psi) &= \min\{u(\varphi), u(\psi)\}, \qquad u(\varphi \vee \psi) = \max\{u(\varphi), u(\psi)\}, \\
u(\varphi \to \psi) &= \begin{cases} 1 & \text{if } u(\varphi) \le u(\psi), \\ u(\psi) & \text{otherwise,} \end{cases} \\
u(\varphi \leftrightarrow \psi) &= \min\{u(\varphi \to \psi), u(\psi \to \varphi)\}.
\end{aligned}
$$

So $u(\neg\varphi) = 1$ if $u(\varphi) = 0$, and $u(\neg\varphi) = 0$ otherwise. On the values 0 and 1 alone these are exactly the tables of Definition 3.3.2, with 1 for T and 0 for F. For a list Γ, let $u(\Gamma)$ be the least of the values $u(\gamma)$ for γ in Γ, and $u(\Gamma) = 1$ if Γ is empty. Write $\Gamma \models_3 \psi$ if $u(\Gamma) \le u(\psi)$ for every three-valued valuation u: the conclusion is at least as true as the weakest premise.

Theorem 3.5.4. Reductio Is Not Derivable from the Other Rules

Optional: nothing later depends on this.

Neither $\neg\neg p \vdash p$ nor $\vdash p \vee \neg p$ has a derivation that uses only the primitive rules other than RAA (and derived rules derived from those).

Idea. Repeat the proof of Theorem 3.5.2 with $\models_3$ in place of $\models$: every line of a derivation without RAA satisfies $\Gamma_k \models_3 \varphi_k$. Then show that both sequents fail for $\models_3$, at $u(p) = \frac{1}{2}$. The one case that breaks is RAA, and that is the point.

Proof. As in Step 1 of the proof of Theorem 3.5.2, a derived rule derived without RAA abbreviates a pasted derivation in the primitive rules other than RAA, so it is enough to consider those. Fix a derivation that uses only the primitive rules other than RAA, with lines $1, \dots, n$, formulas φ_k and contexts Γ_k. We show $\Gamma_k \models_3 \varphi_k$ for every k with $1 \le k \le n$, by Theorem 3.2.2 exactly as in Step 2 of the proof of Theorem 3.5.2, the cases $k = 0$ and $k > n$ being vacuous. Fix k, assume the claim for all earlier lines, and fix a three-valued valuation u; write $m = u(\Gamma_k)$. We must show $m \le u(\varphi_k)$. Two facts do the bookkeeping. If line k cites line i, then $\Gamma_i \subseteq \Gamma_k$ (Lemma 3.5.1 (a)), and the least value over a longer list is at most the least over a shorter one, so $m \le u(\Gamma_i) \le u(\varphi_i)$. If line k cites a box from line a to line b by a box rule, then $\Gamma_b = \Gamma_k, \varphi_a$ (Lemma 3.5.1 (b)), so the hypothesis for line b says $\min\{m, u(\varphi_a)\} \le u(\varphi_b)$. Below, $x = u(\varphi)$, $y = u(\psi)$, $z = u(\chi)$.

Premises and assumptions: φ_k is in Γ_k, so $m \le u(\varphi_k)$. R*:* $m \le u(\varphi_i) = u(\varphi_k)$. $\wedge$I*:* $m \le x$ and $m \le y$, so $m \le \min\{x, y\}$. $\wedge$E*:* $m \le \min\{x, y\} \le x$, and likewise $\le y$. $\vee$I*:* $m \le x \le \max\{x, y\}$.

$\to$E*:* $m \le u(\varphi \to \psi)$ and $m \le x$. If $x \le y$, then $m \le x \le y$. Otherwise $u(\varphi \to \psi) = y$, so $m \le y$. $\leftrightarrow$E*:* $m \le u(\varphi \leftrightarrow \psi) \le u(\varphi \to \psi)$ and $m \le x$, so $m \le y$ as for $\to$E; the second form is symmetric.

$\neg$E*:* $m \le x$ and $m \le u(\neg\varphi)$. If $x = 0$, then $m \le 0$; if $x > 0$, then $u(\neg\varphi) = 0$ and $m \le 0$. So $m = 0 = u(\bot)$. $\bot$E*:* $m \le u(\bot) = 0$, so $m = 0 \le u(\varphi_k)$.

$\to$I*:* the hypothesis gives $\min\{m, x\} \le y$. If $x \le y$, then $u(\varphi \to \psi) = 1 \ge m$. Otherwise $y < x$ and $u(\varphi \to \psi) = y$; since $\min\{m, x\} \le y < x$, the minimum is not x, so it is m, and $m \le y$. $\leftrightarrow$I*:* the same argument for each box gives $m \le u(\varphi \to \psi)$ and $m \le u(\psi \to \varphi)$, so m is at most their minimum.

$\neg$I*:* the hypothesis gives $\min\{m, x\} \le u(\bot) = 0$. If $x = 0$, then $u(\neg\varphi) = 1 \ge m$. If $x > 0$, the minimum is 0 and is not x, so $m = 0 \le u(\neg\varphi)$.

$\vee$E*:* $m \le \max\{x, y\}$, and the boxes give $\min\{m, x\} \le z$ and $\min\{m, y\} \le z$. If $x \ge y$, then $m \le x$, so $\min\{m, x\} = m$ and $m \le z$. If $y > x$, then $m \le y$ and $m \le z$ from the second box.

This covers every rule used, so $\Gamma_k \models_3 \varphi_k$ for every line, and in particular for the last line, whose context is the premises. As in Step 4 of the proof of Theorem 3.5.2, a list containing all the premises has least value at most theirs, so every sequent $\Gamma \vdash \psi$ derived without RAA satisfies $\Gamma \models_3 \psi$.

Now let $u(p) = \frac{1}{2}$. Then $u(\neg p) = 0$, since $\frac{1}{2} > 0$, and $u(\neg\neg p) = 1$, since $u(\neg p) = 0$. So $u(\neg\neg p) = 1 > \frac{1}{2} = u(p)$, and $\neg\neg p \models_3 p$ fails. Also $u(p \vee \neg p) = \max\{\frac{1}{2}, 0\} = \frac{1}{2} < 1$, the value of the empty list, so $\models_3 p \vee \neg p$ fails. Hence neither sequent is derivable without RAA.

■

The case that was not written is RAA itself. Its box would give $\min\{m, u(\neg\varphi)\} \le 0$, and we

would need $m \leq x$. At $u(\varphi) = \frac{1}{2}$ the box gives nothing, since $u(\neg\varphi) = 0$ already, while m may be 1. This is the three-valued face of the question asked in the Idea of Theorem 3.5.2: the two-valued RAA case used that $v(\varphi) = \mathrm{F}$ forces $v(\neg\varphi) = \mathrm{T}$, and with a third value, φ and $\neg\varphi$ can both fail to be true.

Remark. The same values settle Peirce's law, $((p \to q) \to p) \to p$, which Section 2.5 derived with RAA (Proposition 2.5.4): at $u(p) = \frac{1}{2}$, $u(q) = 0$, we get $u(p \to q) = 0$, $u((p \to q) \to p) = 1$, and the whole formula has value $\frac{1}{2}$, so it too needs RAA. The values do not refute everything Chapter 2 derived with RAA: the De Morgan direction $\neg(p \wedge q) \vdash \neg p \vee \neg q$ passes this test, and so does $\vdash (p \to q) \vee (q \to p)$. Neither of these is derivable without RAA either, but showing that needs a different kind of model, and this book states it without proof, as Chapter 2 did for De Morgan's law. RAA says something the other rules cannot, and Chapter 5 says what it is and who objected to it.

3.5.7 Exercises

3.5.7.1 A. Check your understanding

Exercise. A1

These parts are about the proof of Theorem 3.5.2. Part (c) uses the derivation of Proposition 2.1.2, written with atoms:

1	$\neg p$	
2	$\quad p$	
3	$\quad \bot$	$\neg$E 2, 1
4	$\quad q$	$\bot$E 3
5	$p \to q$	$\to$I 2–4

(a) Why does the proof use strong induction on k, rather than ordinary induction from line k to line $k+1$?

(b) In the $\vee$E case, the induction hypothesis is applied to the last lines b and d of the two boxes. What does it say about each, and which part of Lemma 3.5.1 identifies the contexts?

(c) List the context of each of the five lines of the derivation above, and check $\Gamma_k \models \varphi_k$ for each.

(d) Suppose a rule "from $\varphi \to \psi$ and ψ, infer φ" were added to the primitive rules. Which part of the proof would fail? Give a valuation showing it.

(e) Where does Step 4 use that the last line of a derivation lies in no box?

Solution. (a) A rule may cite any available line, however far back, and the last line of a box that closed long ago. The hypothesis must therefore be available for every earlier line, not just for line $k-1$. For example, line 7 of the derivation of Proposition 2.3.2 uses the claim for line 6, and line 6 uses it for line 5; but line 5 uses it for line 2, three lines back.

(b) It says $\Gamma_b \models \chi$ and $\Gamma_d \models \chi$. By Lemma 3.5.1 (b), $\Gamma_b = \Gamma_k, \varphi$ and $\Gamma_d = \Gamma_k, \psi$: every row

making Γ_k and φ true makes χ true, and every row making Γ_k and ψ true makes χ true. It is **not** $\Gamma_k \models \chi$, which is what we are trying to prove.

(c) The box is 2–4.

Line	Context	Check
1	$\neg p$	premise, in its context
2	$\neg p,\ p$	assumption, in its context
3	$\neg p,\ p$	no row makes $\neg p$ and p true, so it entails $\bot$
4	$\neg p,\ p$	the same unsatisfiable context entails q
5	$\neg p$	a row making $\neg p$ true makes p false, so $p \to q$ true

Line 3 is the $\neg$E case, line 4 the $\bot$E case, and line 5 the $\to$I case, with $\Gamma_4 = \Gamma_5, p$ by Lemma 3.5.1 (b).

(d) Claim 2, the one-step rules: it would need a new case, and the case is false. From $v(\varphi \to \psi) = \mathrm{T}$ and $v(\psi) = \mathrm{T}$ nothing follows about $v(\varphi)$. The valuation with $v(p) = \mathrm{F}$, $v(q) = \mathrm{T}$ makes $p \to q$ and q true and p false; with the new rule, the three-line derivation of p from $p \to q$ and q would have a last line whose context does not entail it.

(e) It concludes that the context of the last line is just the premises. Without that, the last line could depend on an undischarged assumption, and $\Gamma_n \models \psi$ would say nothing about the rows making the premises true.

3.5.7.2 B. Practice

Exercise. B1: Derivable or not

Determine which of the following are derivable. Justify your answer.

(a) $(p \wedge q) \to r \vdash p \to r$.
(b) $\neg(p \wedge q) \vdash \neg p$.
(c) $p \leftrightarrow q,\ q \vee r \vdash p$.
(d) $p \to (q \vee r) \vdash (p \to q) \vee (p \to r)$.

Solution. *(a)* Not derivable. The row $p = \mathrm{T}$, $q = \mathrm{F}$, $r = \mathrm{F}$ makes the premise true (its hypothesis $p \wedge q$ is false) and the conclusion $p \to r$ false. By Corollary 3.5.3 there is no derivation.

(b) Not derivable. The row $p = \mathrm{T}$, $q = \mathrm{F}$ makes $p \wedge q$ false, so $\neg(p \wedge q)$ true, and $\neg p$ false. By Corollary 3.5.3 there is no derivation.

(c) Not derivable. The row $p = \mathrm{F}$, $q = \mathrm{F}$, $r = \mathrm{T}$ makes $p \leftrightarrow q$ true (both sides false) and $q \vee r$ true, and p false.

(d) Derivable: Example (Choosing a side too early) in Section 2.6 derives it, with LEM on

q. So no row can refute it: by Theorem 3.5.2 it is valid. A search for a refuting row confirms this. The conclusion is false only if both $p \to q$ and $p \to r$ are false, that is, $p = \mathrm{T}$ and $q = r = \mathrm{F}$; but then $q \vee r$ is false and the premise is false.

The trap is (d). Its conclusion **looks** stronger than the premise, since it seems to decide in advance which disjunct follows from p; but a derivation exists, and trying to refute it by a row is bound to fail.

Exercise. B2: Ordering the $\vee$E case

Here are nine lines. Seven of them, in the right order, form the $\vee$E case of the proof of Theorem 3.5.2. Line k is χ, by $\vee$E citing line i (holding $\varphi \vee \psi$), the box from line a to line b (opening with φ, ending with χ), and the box from line c to line d (opening with ψ, ending with χ). Give an admissible order, and say why each of the other two lines cannot be used.

(i) By the induction hypothesis for lines b and d, which come before line k, $\Gamma_b \models \chi$ and $\Gamma_d \models \chi$.

(ii) Hence $v(\chi) = \mathrm{T}$ in either case; since v was arbitrary, $\Gamma_k \models \chi$.

(iii) Let v be a valuation making every formula of Γ_k true.

(iv) If $v(\psi) = \mathrm{T}$, then v makes every formula of Γ_d true, so $v(\chi) = \mathrm{T}$.

(v) By Lemma 3.5.1 (a), $\Gamma_i \subseteq \Gamma_k$, so v makes every formula of Γ_i true; by the induction hypothesis for line i, $v(\varphi \vee \psi) = \mathrm{T}$, so $v(\varphi) = \mathrm{T}$ or $v(\psi) = \mathrm{T}$.

(vi) By the induction hypothesis for line k, $\Gamma_k \models \chi$.

(vii) By Lemma 3.5.1 (b), Γ_b is Γ_k with φ added, and Γ_d is Γ_k with ψ added.

(viii) If $v(\varphi) = \mathrm{T}$, then v makes every formula of Γ_b true, so $v(\chi) = \mathrm{T}$.

(ix) Line b is inside the box, and the box was cited at line k, so the context of line b is Γ_k; the induction hypothesis for line b then gives $\Gamma_k \models \chi$ directly.

Solution. An admissible order: (iii), (v), (vii), (i), (viii), (iv), (ii).

What each line needs: (v) and (viii), (iv) use the valuation fixed in (iii); (viii) and (iv) use (vii) and (i); (ii) uses the case split of (v) and both of (viii) and (iv). So any order is admissible in which (iii) comes before (v), (viii), and (iv); (vii) and (i) come before (viii) and (iv); (v) comes before (ii); and (viii) and (iv) come before (ii). For instance (vii), (i), (iii), (v), (iv), (viii), (ii) is also admissible: (vii) and (i) do not mention v, and the two cases may be taken in either order.

Line (vi) applies the induction hypothesis to line k itself. The hypothesis of Theorem 3.2.2 covers only the lines $j < k$; (vi) assumes what is to be proved.

Line (ix) uses the context of the citing line for the box's last line. It is false: line b lies inside the box, whose assumption φ is part of its context, so $\Gamma_b = \Gamma_k, \varphi$, not Γ_k. In the derivation of Example (Running the claim on a case split), line 3 has context $p \vee q,\ p$, while line 6 has context $p \vee q$, and $p \vee q \models q \vee p$ could not be read off from line 3 directly.

3.5.7.3 C. Going deeper

Exercise. C1: A safe rule and an unsafe one

(a) Add to the primitive rules a one-step rule X: from a line $\varphi \wedge \psi$, infer $\varphi \vee \psi$. Show that Theorem 3.5.2 still holds for derivations that may use X.

(b) Add instead a one-step rule Y: from a line $\varphi \vee \psi$, infer $\varphi \wedge \psi$. Show that soundness fails: name the case of the proof that breaks, give a derivation using Y of a sequent with a refuting row, and give the row.

Solution (Hint 1). The proof of Theorem 3.5.2 treats one rule per case. Adding a rule adds one case to Claim 2 and changes nothing else.

Solution (Hint 2). For (b), start a derivation from the premise p, and look for a way to write $p \vee q$ first.

Solution. (a) The proof of Theorem 3.5.2 goes through with one more case in Claim 2. Every other part is unchanged: Step 1 and Step 4 do not look at the rules, Lemma 3.5.1 (a) holds for any cited line, and (b) concerns only box rules, and X is not one. *The case* X. Line k is $\varphi \vee \psi$, citing a line i holding $\varphi \wedge \psi$. By $(*)$, $v(\varphi \wedge \psi) = \mathrm{T}$, so $v(\varphi) = \mathrm{T}$ by (V3), and $v(\varphi \vee \psi) = \mathrm{T}$ by (V4). So every line of a derivation with X still satisfies $\Gamma_k \models \varphi_k$, and soundness holds.

(b) The new case of Claim 2 fails: from $v(\varphi \vee \psi) = \mathrm{T}$, (V4) gives only that one disjunct is true, and (V3) needs both. With Y, this is a derivation of $p \vdash q$:

1	p	
2	$p \vee q$	$\vee$I 1
3	$p \wedge q$	Y 2
4	q	$\wedge$E 3

The row $p = \mathrm{T}$, $q = \mathrm{F}$ makes the premise true and the conclusion false, so $p \not\models q$, although $p \vdash q$ with Y. The failure is at line 3: its context is p, and the row makes p and line 2 true but line 3 false.

Marking scheme. (10 points)

- 2: (a) says that only one case is added, and why the rest of the proof (Steps 1 and 4, Lemma 3.5.1) is unaffected.
- 3: (a) the case for X, using the hypothesis for the cited line through Lemma 3.5.1 (a) and the clauses (V3), (V4).
- 2: (b) names the failing case, with the reason (one true disjunct does not give a true conjunction).
- 3: (b) a correct derivation with Y and a refuting row for its sequent.

Fatal errors (0 for the step): a "refuting row" that makes a premise false; in (a), checking X only on atoms without the context argument.

Exercise. C2: Exit check: soundness

Close the section first, and write from memory.

Write out, in full, the →I, ¬I and ∨E cases of the proof of Theorem 3.5.2: state the claim being proved for every line, set up the induction, and prove the three cases, naming the reason for each step.

Solution. The claim. Fix a derivation in primitive rules with lines $1, \dots, n$, and write φ_k for the formula on line k and Γ_k for its context (Definition 1.4.2). We prove $\Gamma_k \models \varphi_k$ for every k with $1 \le k \le n$, by Theorem 3.2.2 applied to "if $1 \le k \le n$, then $\Gamma_k \models \varphi_k$" (for $k = 0$ it holds vacuously): fix k with $1 \le k \le n$, assume $\Gamma_j \models \varphi_j$ for every line $j < k$, and let v be a valuation making every formula of Γ_k true. Every line and box that line k cites lies before line k, so the hypothesis applies to the cited lines and to the last lines of the cited boxes. We show $v(\varphi_k) = \mathrm{T}$.

→I. Line k is $\varphi \to \psi$, citing the box from line a (the assumption φ) to line b (the formula ψ). By Lemma 3.5.1 (b), $\Gamma_b = \Gamma_k, \varphi$. Since $b < k$, the hypothesis gives $\Gamma_k, \varphi \models \psi$. If $v(\varphi) = \mathrm{F}$, then $v(\varphi \to \psi) = \mathrm{T}$ by (V5). If $v(\varphi) = \mathrm{T}$, then v makes every formula of Γ_k, φ true, so $v(\psi) = \mathrm{T}$, and $v(\varphi \to \psi) = \mathrm{T}$ by (V5).

¬I. Line k is $\neg\varphi$, citing the box from line a (the assumption φ) to line b (the formula $\bot$). By Lemma 3.5.1 (b) and the hypothesis for line b, $\Gamma_k, \varphi \models \bot$. Since $v(\bot) = \mathrm{F}$ under every valuation (V1), no valuation makes all of Γ_k, φ true. As v makes Γ_k true, $v(\varphi) = \mathrm{F}$, so $v(\neg\varphi) = \mathrm{T}$ by (V2).

∨E. Line k is χ, citing line i (holding $\varphi \vee \psi$) and the boxes from a to b (φ to χ) and from c to d (ψ to χ). Line i is available at line k, so $\Gamma_i \subseteq \Gamma_k$ by Lemma 3.5.1 (a); v makes Γ_i true, and the hypothesis for line i gives $v(\varphi \vee \psi) = \mathrm{T}$, so $v(\varphi) = \mathrm{T}$ or $v(\psi) = \mathrm{T}$ by (V4). By Lemma 3.5.1 (b), $\Gamma_b = \Gamma_k, \varphi$ and $\Gamma_d = \Gamma_k, \psi$, and the hypothesis for lines b and d gives $\Gamma_k, \varphi \models \chi$ and $\Gamma_k, \psi \models \chi$. If $v(\varphi) = \mathrm{T}$, then v makes Γ_k, φ true, so $v(\chi) = \mathrm{T}$; if $v(\psi) = \mathrm{T}$, likewise by the second box.

Conclusion. With the other cases, the induction gives $\Gamma_k \models \varphi_k$ for every line. The last line lies in no box, so its context is the premises, which all belong to Γ; a valuation making Γ true makes the premises true, and so makes the last line ψ true: $\Gamma \models \psi$.

Marking scheme. (10 points)

- 2: states the claim $\Gamma_k \models \varphi_k$ for **every** line, with contexts.
- 2: uses the course-of-values hypothesis on earlier lines only, and says why the cited lines and boxes are earlier.
- 2: identifies the context of each box's last line as Γ_k plus the box's assumption, citing Lemma 3.5.1 (b) (and (a) for the cited line in ∨E).
- 1 + 1 + 1: the semantic step in each of the three cases (the rows where the assumption is false for →I; no row satisfies the box's context for ¬I; the split on which disjunct is true for ∨E).
- 1: the conclusion for the last line.

A proof that inducts on the length of the derivation instead, and handles open boxes correctly, earns the same points step for step. Fatal errors (0 for the step): applying the hypothesis to line k itself; using Γ_k as the context of a box's last line.

3.6 Completeness

Soundness (Theorem 3.5.2) says that the rules of Chapters 1 and 2 never lead from true premises to a false conclusion: whatever can be derived is true in every row where the premises are. That answers half of the chapter's question. The other half is the converse. Could the rules be too **weak**, so that some sequent is true in every row and yet no derivation of it exists? This section proves that they are not: every truth-table consequence has a derivation. The proof reads a derivation off the truth table, one row at a time, and then glues the rows together with excluded middle.

3.6.1 The question soundness leaves open

Recall the two relations from Section 3.4. $\Gamma \vdash \psi$ says that **there is** a derivation of ψ from premises in Γ (Definition 1.2.3). $\Gamma \models \psi$ says that **every** valuation making all of Γ true makes ψ true (Definition 3.4.3). Soundness is the implication

$$\Gamma \vdash \psi \;\Rightarrow\; \Gamma \models \psi.$$

Its converse is not automatic. The rules are a list that people wrote down, and a list can leave something out. Suppose, as a thought experiment, that we struck RAA (Definition 2.5.1) from the list. Every derivation that remains is still a derivation, so soundness still holds. But the tautology $p \vee \neg p$ would then have no derivation (this is the optional Theorem 3.5.4 of Section 3.5). So a sound list of rules can be too short. The question is whether **our** list is long enough.

Here is a sequent to keep in mind. The formula $(p \to q) \vee (q \to p)$ is a tautology: if q is true then $p \to q$ is true, and if q is false then $q \to p$ is true. Chapter 2 found a derivation of it (Exercise (C2: A disjunction from nothing)), but only after reaching for LEM, and nothing in Chapter 2 promised that a derivation would be there to find. We want a theorem that promises it in advance, for every tautology and every valid sequent.

We are going to prove:

> If Γ is a list of formulas and $\Gamma \models \psi$, then $\Gamma \vdash \psi$.

The plan has two ideas, one per row and one for all the rows together.

1. **One row gives one derivation.** Fix a row of the truth table, that is, a valuation v. Write each atom as it stands in that row: p if v makes p true, $\neg p$ if v makes it false. From these signed atoms we will derive the formula ψ if ψ is true in the row, and $\neg\psi$ if it is false. The derivation climbs the parse tree of ψ, one connective at a time.
2. **The rows are glued by excluded middle.** If $\Gamma \models \psi$, the first idea gives ψ from Γ and the signed atoms of **each** row. Two rows that differ only at the atom p have derivations from "$\ldots, p$" and from "$\ldots, \neg p$". The rule $\vee$E applied to $p \vee \neg p$ merges them into one derivation that no longer mentions p. Removing the atoms one at a time leaves a derivation from Γ alone.

The first idea is known as Kalmár's lemma, and gluing rows by excluded middle is the standard way to finish; the route is the classical one. We keep the premises Γ throughout, and we isolate the two ingredients as lemmas so that each can be checked on its own. Before either, we need a little bookkeeping about derivations.

3.6.2 Bookkeeping: adding premises and chaining derivations

In Chapter 2 we learned to use a proved sequent as a single line (Definition 2.3.1): its derivation could be pasted in, with its letters replaced by formulas. The completeness proof uses this move dozens of times, so we now state exactly what it gives and check that pasting really produces a derivation.

Lemma 3.6.1. Weakening and Cut

Let Γ and Δ be lists of formulas, and let φ, ψ, $\alpha_1, \dots, \alpha_m$ ($m \geq 1$), and β be formulas.

(a) **Weakening.** If $\Gamma \vdash \psi$ and every member of Γ is a member of Δ, then $\Delta \vdash \psi$. In particular, $\Gamma \vdash \psi$ implies $\Gamma, \varphi \vdash \psi$.

(b) **Cut.** If $\Gamma \vdash \varphi$ and $\Gamma, \varphi \vdash \psi$, then $\Gamma \vdash \psi$.

(c) **Chaining.** If $\Gamma \vdash \alpha_i$ for **every** $i = 1, \dots, m$, and $\alpha_1, \dots, \alpha_m \vdash \beta$, then $\Gamma \vdash \beta$.

In words: (a) extra premises never hurt; (b) a formula that has been derived may be used as a premise; (c) a sequent already proved can be applied to formulas already derived. Part (c) is Definition 2.3.1 for sequents: applying a proved sequent to formulas already derived, outside every box. Inside a box, Section 2.3's expansion does the same job.

Idea. Part (a) is read off Definition 1.2.3, which asks only that the premises of the derivation **belong to** the list. For (b) and (c), write the derivations one after another and let the last one cite, instead of its premise α_i, the last line of the derivation of α_i. The only thing to check is the scope rule of Definition 1.4.2: the last line of a derivation lies in no box, so it stays available to everything written after it.

Proof. (a) A derivation witnessing $\Gamma \vdash \psi$ has premises that belong to Γ, hence to Δ, and its last line is ψ. By Definition 1.2.3 it witnesses $\Delta \vdash \psi$. The second sentence is the case where Δ is Γ, φ.

For (b) and (c) we prove one statement that contains both:

($*$) If $\Gamma \vdash \alpha_i$ for every $i = 1, \dots, m$, and $\Gamma, \alpha_1, \dots, \alpha_m \vdash \beta$, then $\Gamma \vdash \beta$.

Part (b) is ($*$) with $m = 1$, $\alpha_1 = \varphi$ and $\beta = \psi$. Part (c) follows from ($*$), since $\alpha_1, \dots, \alpha_m \vdash \beta$ gives $\Gamma, \alpha_1, \dots, \alpha_m \vdash \beta$ by (a).

To prove ($*$), let D_i be a derivation of α_i with premises in Γ, for each i, and let E be a derivation of β with premises in $\Gamma, \alpha_1, \dots, \alpha_m$. If the last line of E is one of its premises, then β is a member of Γ, and the one-line derivation whose only line is the premise β shows $\Gamma \vdash \beta$; or β is some α_i, and D_i shows $\Gamma \vdash \beta$. Otherwise, build a list of lines F as follows.

1. First, as premises, every premise of $D_1, \dots, D_m$, and every premise of E that is a member of Γ.
2. Then the lines of D_1 that are not premises, then those of D_2, and so on up to D_m, each with its boxes.
3. Then the lines of E that are not premises, with its boxes.

Renumber the lines, and change each citation as follows. A citation of a premise that is in Γ now cites the premise line of step 1 carrying that formula. A citation in E of a premise α_i that is not in Γ now cites the last line of D_i; that line is not a premise of D_i, because the premises of D_i are in Γ and α_i is not, so it was copied in step 2. Every other citation follows its line to its new number.

We check that F is a derivation (Definition 1.4.2). The premises come first and lie in no box. The boxes are those of the D_i and of E, moved as blocks, so two boxes are still disjoint or nested, each still begins with its assumption, and each line lies in exactly the boxes it lay in before. Hence a line and a box that were available at a citing line are still available there, since the boxes containing each are unchanged. A premise line of step 1 lies in no box and comes first, so it is available at every later line. The last line of D_i lies in no box (it was the last line of a derivation) and comes before every line of E, so it is available at every line of E. Each line carries the same formula and the same justification as before, citing lines with the same formulas, so it is an instance of the same rule. Finally, the last line of F is the last line of E, namely β, and it lies in no box. The premises of F are all in Γ. Therefore $\Gamma \vdash \beta$, which proves $(*)$.

■

The lemma is pure bookkeeping, but it is what makes "cite a proved sequent as one line" honest. From now on, when a derivation cites a sequent proved earlier as a single line, the lemma (for a line in no box) or Definition 2.3.1 (for a line inside a box, where the cited lines are repeated inside the box first) is the reason the result is a derivation in the primitive rules. As Section 3.5 said, soundness was proved for the primitive rules, and a line citing a derived rule abbreviates a pasted derivation; for whole derivations, part (c) is the precise form of "abbreviates", and inside a box the pasting of Definition 2.3.1 is the same bookkeeping.

Quick check. Suppose $\Gamma \vdash \alpha$ and $\Gamma \vdash \neg\alpha$. Which part of Lemma 3.6.1, applied to which proved sequent, gives $\Gamma \vdash \chi$ for every formula χ?

Solution. Part (c) with $m = 2$, $\alpha_1 = \alpha$, $\alpha_2 = \neg\alpha$ and $\beta = \chi$, applied to Proposition 2.1.1, $P, \neg P \vdash Q$, with α for P and χ for Q.

3.6.3 The rules prove their own truth tables

Each line of the truth table of a connective says something like "with α true and β false, $\alpha \to \beta$ comes out false". Read with signs, that line is a sequent: from α and $\neg\beta$, conclude $\neg(\alpha \to \beta)$. If every such sequent is derivable, then the rules know the truth tables. This is the key to the first idea of the plan. Section 3.3 began by reading the tables off the rules; the lemma below makes that reading exact: every line of every table is a derivable sequent.

Lemma 3.6.2. The Rules Prove the Truth Tables

For all formulas α and β, the following sequents are derivable, by derivations that do not use RAA.

(i) $(\neg 1)$ $\alpha \vdash \neg\neg\alpha$, and $(\bot 1) \vdash \neg\bot$.

(ii) $(\wedge 1)\ \alpha, \beta \vdash \alpha \wedge \beta$; $(\wedge 2)\ \neg\alpha \vdash \neg(\alpha \wedge \beta)$; $(\wedge 3)\ \neg\beta \vdash \neg(\alpha \wedge \beta)$.

(iii) $(\vee 1)\ \alpha \vdash \alpha \vee \beta$; $(\vee 2)\ \beta \vdash \alpha \vee \beta$; $(\vee 3)\ \neg\alpha, \neg\beta \vdash \neg(\alpha \vee \beta)$.

(iv) $(\to 1)\ \neg\alpha \vdash \alpha \to \beta$; $(\to 2)\ \beta \vdash \alpha \to \beta$; $(\to 3)\ \alpha, \neg\beta \vdash \neg(\alpha \to \beta)$.

(v) $(\leftrightarrow 1)\ \alpha, \beta \vdash \alpha \leftrightarrow \beta$; $(\leftrightarrow 2)\ \neg\alpha, \neg\beta \vdash \alpha \leftrightarrow \beta$; $(\leftrightarrow 3)\ \alpha, \neg\beta \vdash \neg(\alpha \leftrightarrow \beta)$; $(\leftrightarrow 4)\ \neg\alpha, \beta \vdash \neg(\alpha \leftrightarrow \beta)$.

Each sequent has as premises **only the parts it needs**, each as itself or negated. For instance, $\alpha \wedge \beta$ is false as soon as α is false, whatever β is, and $(\wedge 2)$ has the single premise $\neg\alpha$. This economy will matter: when the proof of the next lemma uses a row, it needs to know something only about the parts the row mentions.

Idea. Most of the sequents are already in the book: rules of Chapter 1, or results of Chapter 2 with α and β put in for the letters. The others are short derivations of the kinds Chapter 2 practiced: a box that repeats a premise, or a $\neg$I box that reaches a clash. The only thing to watch is that no step, and no cited result, uses RAA.

Proof. The table lists, for each sequent, the rows of the truth table it covers (by Definition 3.3.2) and where its derivation comes from. The letters of the Chapter 2 results are replaced by α and β, as Definition 2.3.1 allows.

Tag	Rows covered	Sequent	Derivation
$(\neg 1)$	α T, so $\neg\alpha$ F	$\alpha \vdash \neg\neg\alpha$	Example (Two negations from one premise) (a)
$(\bot 1)$	$\bot$ is always F	$\vdash \neg\bot$	printed below
$(\wedge 1)$	T, T give T	$\alpha, \beta \vdash \alpha \wedge \beta$	$\wedge$I (Definition 1.2.5)
$(\wedge 2)$	F, anything give F	$\neg\alpha \vdash \neg(\alpha \wedge \beta)$	$\vee$I gives $\neg\alpha \vee \neg\beta$, then Proposition 2.5.5 (c)
$(\wedge 3)$	anything, F give F	$\neg\beta \vdash \neg(\alpha \wedge \beta)$	$\vee$I gives $\neg\alpha \vee \neg\beta$, then Proposition 2.5.5 (c)
$(\vee 1)$	T, anything give T	$\alpha \vdash \alpha \vee \beta$	$\vee$I (Definition 1.5.1)
$(\vee 2)$	anything, T give T	$\beta \vdash \alpha \vee \beta$	$\vee$I
$(\vee 3)$	F, F give F	$\neg\alpha, \neg\beta \vdash \neg(\alpha \vee \beta)$	$\wedge$I gives $\neg\alpha \wedge \neg\beta$, then Proposition 2.5.5 (b)
$(\to 1)$	F, anything give T	$\neg\alpha \vdash \alpha \to \beta$	Proposition 2.1.2
$(\to 2)$	anything, T give T	$\beta \vdash \alpha \to \beta$	printed below
$(\to 3)$	T, F give F	$\alpha, \neg\beta \vdash \neg(\alpha \to \beta)$	printed below
$(\leftrightarrow 1)$	T, T give T	$\alpha, \beta \vdash \alpha \leftrightarrow \beta$	printed below
$(\leftrightarrow 2)$	F, F give T	$\neg\alpha, \neg\beta \vdash \alpha \leftrightarrow \beta$	printed below
$(\leftrightarrow 3)$	T, F give F	$\alpha, \neg\beta \vdash \neg(\alpha \leftrightarrow \beta)$	printed below
$(\leftrightarrow 4)$	F, T give F	$\neg\alpha, \beta \vdash \neg(\alpha \leftrightarrow \beta)$	printed below

In the second column, "F, anything give F" means: if α is F, then $\alpha \wedge \beta$ is F whatever the value of β.

The seven printed derivations follow. First ($\to$2) and ($\to$3) (the first is the derivation of Exercise (B2: An assumption that is not used) (a), printed again because a proof does not rest on an exercise): an implication whose conclusion we already have, and the refutation of an implication by a true hypothesis with a false conclusion.

1	β	
2	$\quad\alpha$	
3	$\quad\beta$	R 1
4	$\alpha \to \beta$	$\to$I 2–3

1	α	
2	$\neg\beta$	
3	$\quad\alpha \to \beta$	
4	$\quad\beta$	$\to$E 3, 1
5	$\quad\bot$	$\neg$E 4, 2
6	$\neg(\alpha \to \beta)$	$\neg$I 3–5

Next ($\leftrightarrow$1) and ($\leftrightarrow$2). In the first, each box repeats the premise it needs; in the second, each box reaches a clash and closes by $\bot$E.

1	α	
2	β	
3	$\quad\alpha$	
4	$\quad\beta$	R 2
5	$\quad\beta$	
6	$\quad\alpha$	R 1
7	$\alpha \leftrightarrow \beta$	$\leftrightarrow$I 3–4, 5–6

1	$\neg\alpha$	
2	$\neg\beta$	
3	$\quad\alpha$	
4	$\quad\bot$	$\neg$E 3, 1
5	$\quad\beta$	$\bot$E 4
6	$\quad\beta$	
7	$\quad\bot$	$\neg$E 6, 2
8	$\quad\alpha$	$\bot$E 7
9	$\alpha \leftrightarrow \beta$	$\leftrightarrow$I 3–5, 6–8

Then ($\leftrightarrow$3), ($\leftrightarrow$4) and ($\bot$1). The first two assume the biconditional and use it in the direction that clashes with a premise; the last is a box whose assumption $\bot$ is already the $\bot$ that $\neg$I asks for, the degenerate box of Section 2.2.

1	α	
2	$\neg\beta$	
3	$\quad\alpha \leftrightarrow \beta$	
4	$\quad\beta$	$\leftrightarrow$E 3, 1
5	$\quad\bot$	$\neg$E 4, 2
6	$\neg(\alpha \leftrightarrow \beta)$	$\neg$I 3–5

1	$\neg\alpha$	
2	β	
3	$\quad\alpha \leftrightarrow \beta$	
4	$\quad\alpha$	$\leftrightarrow$E 3, 2
5	$\quad\bot$	$\neg$E 4, 1
6	$\neg(\alpha \leftrightarrow \beta)$	$\neg$I 3–5

1	$\quad\bot$	
2	$\neg\bot$	$\neg$I 1–1

None of these derivations uses RAA: their boxes close by $\to$I, $\neg$I and $\leftrightarrow$I. Nor do the cited results: Proposition 2.1.2 and Example (Two negations from one premise) come from Sections 2.1 and 2.2, before RAA was introduced, and Chapter 2 derived De Morgan's laws (b) and (c) without it (it says so right after the derivations of Proposition 2.5.5). This proves the lemma.

■

The $\neg$ connective needs only one sequent. When α is false, $\neg\alpha$ is true, and the "sequent" $\neg\alpha \vdash \neg\alpha$ is a one-line derivation; only the other row, ($\neg$1), needs a proof.

Warning. The sequents of Lemma 3.6.2 go **from the parts to the whole**. Read backwards they are false. For instance $\alpha \vee \beta \vdash \alpha$ is not derivable in general: the row $p = \mathrm{F}$, $q = \mathrm{T}$ makes $p \vee q$ true and p false, so $p \vee q \nvdash p$ by Corollary 3.5.3. A row of a table tells us the value of the whole from the values of the parts, never the other way round.

3.6.4 One row, one derivation

We now name the formulas that "write down a row".

A signed formula is a formula as it stands in one row: itself if the row makes it true, its negation if the row makes it false.

For a valuation v and a formula φ, the **signed formula** φ^v is

$$\varphi^v := \begin{cases} \varphi & \text{if } v(\varphi) = \mathrm{T}, \\ \neg\varphi & \text{if } v(\varphi) = \mathrm{F}. \end{cases}$$

In particular, for an atom p, the **signed atom** p^v is p or $\neg p$. For example, if $v(p) = \mathrm{T}$ and $v(q) = \mathrm{F}$, then $p^v = p$, $q^v = \neg q$, $(p \to q)^v = \neg(p \to q)$ and $(q \to p)^v = q \to p$. Whatever v is, $v(\varphi^v) = \mathrm{T}$: the signed formula is always the true one of φ and $\neg\varphi$, by the clause for $\neg$ in Definition 3.3.2.

The signed atoms $p_1^v, \ldots, p_n^v$ describe the row of v completely, as far as $p_1, \ldots, p_n$ go. The next lemma says they are enough premises to derive the signed version of any formula built from these atoms.

Lemma 3.6.3. One Row of the Truth Table Is Derivable

Let φ be a formula, let $p_1, \ldots, p_n$ ($n \geq 0$) be atoms among which are **all** the atoms of φ, and let v be a valuation. Then

$$p_1^v, \ldots, p_n^v \vdash \varphi^v.$$

In words: in each row of the truth table of φ, the value of φ can be derived from the values of its atoms, where "value T" is written as the formula and "value F" as its negation.

Idea. Climb the parse tree. At an atom, φ^v is one of the premises. At a connective, the induction hypothesis gives the signed parts α^v, β^v, and Lemma 3.6.2 has a sequent from exactly those signed parts to the signed whole. So structural induction (Theorem 3.2.3), with Lemma 3.6.1 (c) to chain, does the whole job. The only care is to pick, for each pair of

values, the row whose premises are the signs we actually have.

Proof. Fix the atoms $p_1, \dots, p_n$ and the valuation v, and write Σ for the list $p_1^v, \dots, p_n^v$. We prove, by structural induction (Theorem 3.2.3), that every formula φ has the property

> if every atom of φ is among $p_1, \dots, p_n$, then $\Sigma \vdash \varphi^v$.

Every atom of an immediate part α or β of a formula is an atom of the formula itself, so when φ satisfies the "if", so do its immediate parts, and the induction hypothesis applies to them.

Atoms. Let φ be an atom among $p_1, \dots, p_n$, say $\varphi = p_i$. Then $\varphi^v = p_i^v$ is a member of Σ, and the one-line derivation whose only line is the premise p_i^v shows $\Sigma \vdash \varphi^v$.

The formula $\bot$. By Definition 3.3.2, $v(\bot) = \mathrm{F}$, so $\bot^v = \neg\bot$. By ($\bot$1) of Lemma 3.6.2, $\vdash \neg\bot$, and weakening (Lemma 3.6.1 (a)) gives $\Sigma \vdash \neg\bot$.

Negation. Let $\varphi = \neg\alpha$, and assume $\Sigma \vdash \alpha^v$.

- *Case* $v(\alpha) = \mathrm{T}$. Then $\alpha^v = \alpha$, and $v(\neg\alpha) = \mathrm{F}$, so $\varphi^v = \neg\neg\alpha$. From $\Sigma \vdash \alpha$ and ($\neg$1), $\alpha \vdash \neg\neg\alpha$, chaining (Lemma 3.6.1 (c)) gives $\Sigma \vdash \neg\neg\alpha$.
- *Case* $v(\alpha) = \mathrm{F}$. Then $\alpha^v = \neg\alpha$, and $v(\neg\alpha) = \mathrm{T}$, so $\varphi^v = \neg\alpha = \alpha^v$. The hypothesis is already the claim.

Conjunction. Let $\varphi = \alpha \wedge \beta$, and assume $\Sigma \vdash \alpha^v$ and $\Sigma \vdash \beta^v$. By Definition 3.3.2, $v(\alpha \wedge \beta) = \mathrm{T}$ exactly when $v(\alpha) = v(\beta) = \mathrm{T}$.

- *Case* $v(\alpha) = v(\beta) = \mathrm{T}$. Then $\varphi^v = \alpha \wedge \beta$, $\alpha^v = \alpha$, $\beta^v = \beta$. Chaining $\Sigma \vdash \alpha$ and $\Sigma \vdash \beta$ with ($\wedge$1), $\alpha, \beta \vdash \alpha \wedge \beta$, gives $\Sigma \vdash \alpha \wedge \beta$.
- *Case* $v(\alpha) = \mathrm{F}$. Then $\varphi^v = \neg(\alpha \wedge \beta)$ and $\alpha^v = \neg\alpha$. Chaining $\Sigma \vdash \neg\alpha$ with ($\wedge$2), $\neg\alpha \vdash \neg(\alpha \wedge \beta)$, gives $\Sigma \vdash \neg(\alpha \wedge \beta)$. The hypothesis for β is not used.
- *Case* $v(\alpha) = \mathrm{T}$, $v(\beta) = \mathrm{F}$. Then $\varphi^v = \neg(\alpha \wedge \beta)$ and $\beta^v = \neg\beta$, and chaining $\Sigma \vdash \neg\beta$ with ($\wedge$3) gives $\Sigma \vdash \neg(\alpha \wedge \beta)$.

Implication. Let $\varphi = \alpha \to \beta$, and assume $\Sigma \vdash \alpha^v$ and $\Sigma \vdash \beta^v$. By Definition 3.3.2, $v(\alpha \to \beta) = \mathrm{F}$ exactly when $v(\alpha) = \mathrm{T}$ and $v(\beta) = \mathrm{F}$.

- *Case* $v(\alpha) = \mathrm{F}$. Then $\varphi^v = \alpha \to \beta$ and $\alpha^v = \neg\alpha$. Chaining $\Sigma \vdash \neg\alpha$ with ($\to$1), $\neg\alpha \vdash \alpha \to \beta$, gives $\Sigma \vdash \alpha \to \beta$.
- *Case* $v(\alpha) = \mathrm{T}$, $v(\beta) = \mathrm{T}$. Then $\varphi^v = \alpha \to \beta$ and $\beta^v = \beta$. Chaining $\Sigma \vdash \beta$ with ($\to$2), $\beta \vdash \alpha \to \beta$, gives $\Sigma \vdash \alpha \to \beta$.
- *Case* $v(\alpha) = \mathrm{T}$, $v(\beta) = \mathrm{F}$. Then $\varphi^v = \neg(\alpha \to \beta)$, $\alpha^v = \alpha$ and $\beta^v = \neg\beta$. Chaining $\Sigma \vdash \alpha$ and $\Sigma \vdash \neg\beta$ with ($\to$3), $\alpha, \neg\beta \vdash \neg(\alpha \to \beta)$, gives $\Sigma \vdash \neg(\alpha \to \beta)$.

Disjunction and biconditional. These go as for implication. For each pair of values of α and β, the table in the proof of Lemma 3.6.2 names a sequent whose premises are among α^v, β^v and whose conclusion is φ^v: for $\vee$, ($\vee$1) if $v(\alpha) = \mathrm{T}$, ($\vee$2) if $v(\alpha) = \mathrm{F}$ and $v(\beta) = \mathrm{T}$, ($\vee$3) if both are F; for $\leftrightarrow$, ($\leftrightarrow$1) to ($\leftrightarrow$4) for the value pairs T T, F F, T F, F T. Chaining the induction hypotheses for those premises with that sequent gives $\Sigma \vdash \varphi^v$.

The atoms and $\bot$ have the property, and each formation clause preserves it, so by Theorem 3.2.3 every formula has it. This proves the lemma.

■

Each case used the induction hypothesis only for the parts named in the premises of its sequent: the conjunction case with α false never looked at β. That is the economy of Lemma 3.6.2 at work. The disjunction and biconditional cases are written out in Exercise (C1: The remaining cases).

Here is the lemma run on one formula, with every chained sequent written as a line.

Example. Deriving one row

Let $\varphi = p \to (q \wedge \neg p)$ and let $v(p) = v(q) = \mathrm{T}$. Find φ^v, and write the derivation of $p^v, q^v \vdash \varphi^v$ that the proof of Lemma 3.6.3 produces.

Solution. Compute up the parse tree: $v(\neg p) = \mathrm{F}$, so $v(q \wedge \neg p) = \mathrm{F}$, and since $v(p) = \mathrm{T}$, $v(\varphi) = \mathrm{F}$. So $\varphi^v = \neg(p \to (q \wedge \neg p))$, and the premises are $p^v = p$ and $q^v = q$.

The proof climbs the same tree. At $\neg p$ with $v(p) = \mathrm{T}$, it is the negation case with α true: $(\neg 1)$ gives $\neg\neg p$. At $q \wedge \neg p$, the values are T and F: $(\wedge 3)$ gives $\neg(q \wedge \neg p)$ from $(\neg p)^v = \neg\neg p$. At the root, the values are T and F: $(\to 3)$ gives the goal from p and $\neg(q \wedge \neg p)$. Written as a derivation, with each sequent of Lemma 3.6.2 cited as a derived rule:

1	p	
2	q	
3	$\neg\neg p$	$(\neg 1)$ 1
4	$\neg(q \wedge \neg p)$	$(\wedge 3)$ 3
5	$\neg(p \to (q \wedge \neg p))$	$(\to 3)$ 1, 4

The premise q is never cited. It was used only in computing which case applies: the $\wedge$ case with $\beta = \neg p$ false needs only β's sign, not q's. A derivation need not use every premise (Definition 1.2.3).

Quick check. Let $\varphi = p \to \neg q$ and $v(p) = v(q) = \mathrm{T}$. What sequent does Lemma 3.6.3 provide, and which sequent of Lemma 3.6.2 does the proof use at the root?

Solution. Here $v(\neg q) = \mathrm{F}$, so $v(\varphi) = \mathrm{F}$, and the lemma gives $p, q \vdash \neg(p \to \neg q)$. At the root the values are T and F, so the proof uses $(\to 3)$, $\alpha, \neg\beta \vdash \neg(\alpha \to \beta)$, with $\alpha = p$ and $\beta = \neg q$. Its premise $\neg\beta = \neg\neg q$ comes from the negation case below it: $q \vdash \neg\neg q$, by $(\neg 1)$.

3.6.5 Gluing the rows: splitting on an atom

Lemma 3.6.3 derives a formula from the signed atoms of **one** row. A sequent $\Gamma \vdash \psi$ mentions no row at all. The signed atoms must be removed, and the tool for that is excluded middle.

Lemma 3.6.4. Splitting on an Atom

Let Δ be a list of formulas, p an atom and ψ a formula. If $\Delta, p \vdash \psi$ and $\Delta, \neg p \vdash \psi$, then $\Delta \vdash \psi$.

Idea. This is proof by cases on $p \vee \neg p$. Proposition 2.5.3 gives $p \vee \neg p$ with no premises, and $\vee$E (Definition 1.6.1) needs a box that reaches ψ from p and a box that reaches ψ from $\neg p$. The two given derivations are exactly the contents of those boxes, once their premise p (or $\neg p$) is replaced by the box's assumption.

Before reading on: this is the only place in the completeness proof where a classical rule enters. Where exactly, and why could the $\neg$I *of Chapter 2 not do the job?*

Proof. Let D be a derivation of ψ with premises in Δ, p, and E one with premises in $\Delta, \neg p$. Build a list of lines as follows.

1. As premises, every premise of D or of E that is a member of Δ.
2. The line $p \vee \neg p$, justified by Proposition 2.5.3 (LEM) with p for P; it cites nothing.
3. A box that opens with the assumption p, followed by the lines of D that are not premises, with their boxes. A citation of a premise of D that is in Δ now cites the premise line of step 1; a citation of the premise p now cites the assumption. If the last line of D is itself a premise, the box is shorter: if that premise is p, the box is the assumption alone, which is then its last line; otherwise the box is the assumption followed by ψ, by R, citing the premise line carrying ψ.
4. A second box, built from E in the same way, opening with the assumption $\neg p$.
5. The line ψ, by $\vee$E, citing line 2 of this list and the two boxes.

This is a derivation (Definition 1.4.2). The premises lie in no box and come first, so they are available everywhere. Inside the first box, the lines of D keep their boxes, which now sit inside the new box, and they cite the same lines and boxes as before, or a premise line, or the assumption, which is available at every line of its own box. As in the proof of Lemma 3.6.1, each line is still an instance of its rule. The last line of the first box is ψ, and it is a line of that box itself and not of a box inside it: it was the last line of D, which lies in no box of D (or it is the assumption itself, or the repetition line). The same holds for the second box. Both boxes and the line $p \vee \neg p$ lie in no box, and neither does the last line, so they are available at it. Hence the last line is an instance of $\vee$E, and the list is a derivation of ψ whose premises are all in Δ. So $\Delta \vdash \psi$.

■

Here is its shape, with Δ standing for the premise lines:

h	Δ	
i	$p \vee \neg p$	LEM
a	$\quad p$	
	$\quad \vdots$	
b	$\quad \psi$	
c	$\quad \neg p$	
	$\quad \vdots$	
d	$\quad \psi$	
k	ψ	$\vee$E i, a–b, c–d

The answer to the prompt: the classical rule enters at line i. Proposition 2.5.3 was derived in Chapter 2 with RAA, and nothing else in the proof of this lemma is classical. ¬I could not replace it: ¬I produces a negation, and here we need a disjunction with no premises to split on. Nothing in the proof uses that p is an atom; the lemma holds for any formula in place of p, but atoms are all we shall split on.

3.6.6 The completeness theorem

Now the plan of the opening can be carried out.

Theorem 3.6.5. Completeness

Let Γ be a list of formulas and ψ a formula. If $\Gamma \models \psi$, then $\Gamma \vdash \psi$.

A list is finite, as in Definition 1.2.3. That matters: the proof works with the atoms that occur in Γ and ψ, and it needs them to be finitely many.

Idea. **Step roadmap.** ① List the atoms $p_1, \ldots, p_n$ of Γ and ψ. ② **Claim:** for **every** valuation v, $\Gamma, p_1^v, \ldots, p_n^v \vdash \psi$. If v makes all of Γ true, then ψ is true in that row, and Lemma 3.6.3 derives it. If v makes some premise γ false, the lemma derives $\neg\gamma$, which clashes with the premise γ. ③ Remove the signed atoms, last first, by Lemma 3.6.4. To remove p_j we need derivations from both p_j and $\neg p_j$ with the same other signed atoms: two rows that differ only at p_j. That is why the Claim must hold for every valuation, not just one.

Proof. **Step 1.** The list Γ is finite and each formula has finitely many symbols, so only finitely many atoms occur in Γ or ψ. List them without repetition as $p_1, \ldots, p_n$, where $n \geq 0$.

Step 2.

Claim. For every valuation v, $\Gamma, p_1^v, \ldots, p_n^v \vdash \psi$.

Proof of Claim. Let v be a valuation, and write Σ for $p_1^v, \ldots, p_n^v$.

Case 1: $v(\gamma) = \mathrm{T}$ for every member γ of Γ. Since $\Gamma \models \psi$, by Definition 3.4.3 $v(\psi) = \mathrm{T}$, so

$\psi^v = \psi$. Every atom of ψ is among $p_1, \ldots, p_n$, so Lemma 3.6.3 gives $\Sigma \vdash \psi$, and weakening (Lemma 3.6.1 (a)) gives $\Gamma, \Sigma \vdash \psi$.

Case 2: $v(\gamma) = \mathrm{F}$ for some member γ of Γ. Then $\gamma^v = \neg\gamma$. Every atom of γ is among $p_1, \ldots, p_n$, so Lemma 3.6.3 gives $\Sigma \vdash \neg\gamma$, and by weakening $\Gamma, \Sigma \vdash \neg\gamma$. Also $\Gamma, \Sigma \vdash \gamma$, by the one-line derivation whose only line is the premise γ. By Proposition 2.1.1, $\gamma, \neg\gamma \vdash \psi$. Chaining (Lemma 3.6.1 (c)) gives $\Gamma, \Sigma \vdash \psi$.

■

Step 3. For each integer j with $0 \le j \le n$, let $S(j)$ be the statement

$$\text{for every valuation } v, \quad \Gamma, p_1^v, \ldots, p_j^v \vdash \psi,$$

where for $j = 0$ the list of signed atoms is empty. The Claim is $S(n)$.

Claim. If $1 \le j \le n$ and $S(j)$ holds, then $S(j-1)$ holds.

Proof of Claim. Let v be a valuation. Let v^+ be the valuation that agrees with v at every atom except that $v^+(p_j) = \mathrm{T}$, and v^- the one that agrees with v except that $v^-(p_j) = \mathrm{F}$; by Definition 3.3.1, a valuation may assign any value to each atom, so both exist. The atoms $p_1, \ldots, p_{j-1}$ are different from p_j, so v^+ and v^- agree with v on them, and $p_i^{v^+} = p_i^{v^-} = p_i^v$ for $i < j$. Also $p_j^{v^+} = p_j$ and $p_j^{v^-} = \neg p_j$. Applying $S(j)$ to v^+ and to v^- gives, with Δ standing for the list $\Gamma, p_1^v, \ldots, p_{j-1}^v$,

$$\Delta, p_j \vdash \psi \quad \text{and} \quad \Delta, \neg p_j \vdash \psi.$$

By Lemma 3.6.4, $\Delta \vdash \psi$. Since v was arbitrary, this is $S(j-1)$.

■

It remains to go down from $S(n)$ to $S(0)$, which takes n applications of the Claim. We count the applications made: for $m \in \mathbb{N}$, let $T(m)$ say "if $m \le n$, then $S(n-m)$ holds". $T(0)$ says that $S(n)$ holds, which is Step 2. Suppose $T(m)$ holds, and let $m+1 \le n$. Then $m \le n$, so $S(n-m)$ holds by $T(m)$; and $j := n-m$ satisfies $1 \le j \le n$, because $0 \le m \le n-1$. The Claim gives $S(j-1) = S(n-(m+1))$, which is $T(m+1)$. By Axiom 3.2.1, $T(m)$ holds for every $m \in \mathbb{N}$, and $T(n)$ is $S(0)$.

Step 4. $S(0)$ says that $\Gamma \vdash \psi$, for every valuation v; there is at least one valuation (for instance the one making every atom true), so $\Gamma \vdash \psi$. This proves the theorem.

■

If $n = 0$, the formulas of Γ and ψ are built from $\bot$ alone, and Step 3 does nothing: $S(0)$ is the Claim itself.

A common alternative route first turns $\Gamma \models \psi$ into a single tautology, $\models \gamma_1 \to (\gamma_2 \to \cdots \to (\gamma_k \to \psi))$, proves completeness for tautologies, and then unpacks the implications with $\to$E. Keeping Γ as premises, as Step 2 does, costs one extra case (a false premise) and saves that detour.

Let us run the proof on the tautology of the opening.

Example. The method on a tautology

Run the proof of Theorem 3.6.5 on $\models (p \to q) \lor (q \to p)$, and write the derivation it produces, citing the sequents of Lemma 3.6.2 as derived rules.

Solution. Write χ for $(p \to q) \lor (q \to p)$. The list Γ is empty and the atoms are $p_1 = p$, $p_2 = q$. Every valuation makes χ true (if $v(q) = \mathrm{T}$ the left side is true; otherwise the right side is), so Step 2 is always Case 1, and Lemma 3.6.3 produces four short derivations, one per row:

- $p, q \vdash \chi$: $p \to q$ by $(\to 2)$ from q, then $\lor$I;
- $p, \neg q \vdash \chi$: $q \to p$ by $(\to 1)$ from $\neg q$, then $\lor$I;
- $\neg p, q \vdash \chi$ and $\neg p, \neg q \vdash \chi$: $p \to q$ by $(\to 1)$ from $\neg p$, then $\lor$I.

(In the row $p = \mathrm{F}$, $q = \mathrm{T}$, both $(\to 1)$ and $(\to 2)$ would do; the proof of Lemma 3.6.3 uses $(\to 1)$ whenever α is false.) Step 3 first splits on $q = p_2$, twice: once with p and once with $\neg p$ among the premises. Then it splits on $p = p_1$, once. Three uses of Lemma 3.6.4 in two rounds. Pasting as in its proof, the last split is the outermost:

1	$p \lor \neg p$	LEM
2	$\quad p$	
3	$\quad q \lor \neg q$	LEM
4	$\quad\quad q$	
5	$\quad\quad p \to q$	$(\to 2)$ 4
6	$\quad\quad (p \to q) \lor (q \to p)$	$\lor$I 5
7	$\quad\quad \neg q$	
8	$\quad\quad q \to p$	$(\to 1)$ 7
9	$\quad\quad (p \to q) \lor (q \to p)$	$\lor$I 8
10	$\quad (p \to q) \lor (q \to p)$	$\lor$E 3, 4–6, 7–9
11	$\quad \neg p$	
12	$\quad q \lor \neg q$	LEM
13	$\quad\quad q$	
14	$\quad\quad p \to q$	$(\to 1)$ 11
15	$\quad\quad (p \to q) \lor (q \to p)$	$\lor$I 14
16	$\quad\quad \neg q$	
17	$\quad\quad p \to q$	$(\to 1)$ 11
18	$\quad\quad (p \to q) \lor (q \to p)$	$\lor$I 17
19	$\quad (p \to q) \lor (q \to p)$	$\lor$E 12, 13–15, 16–18
20	$(p \to q) \lor (q \to p)$	$\lor$E 1, 2–10, 11–19

The derivation has 20 lines. The one found by search in Exercise (C2: A disjunction from nothing) has 13, and splits only once. Here the split on q inside the box of $\neg p$ (lines 12–19)

is wasted: both of its boxes do the same thing, because in the rows with p false the value of q does not matter. The proof is a **method** that always works, not a recipe for short derivations; to find a short one, search as in Section 2.6.

Warning. Completeness promises a derivation, not a short one. For a sequent with n atoms, the proof builds 2^n row derivations and glues them with $2^n - 1$ splittings: with $n = 10$ atoms, that is 1024 row derivations and 1023 uses of LEM. The theorem answers "is there a derivation?", not "how short is the shortest?".

With soundness, completeness closes the circle.

Corollary 3.6.6. Derivable Means Valid

For every list of formulas Γ and every formula ψ,

$$\Gamma \vdash \psi \iff \Gamma \models \psi.$$

In particular, whether $\Gamma \vdash \psi$ holds can be decided by checking the 2^n rows of a truth table, where n is the number of atoms occurring in Γ and ψ.

Proof. ($\Rightarrow$) is Theorem 3.5.2 and ($\Leftarrow$) is Theorem 3.6.5. For the second sentence, let $p_1, \ldots, p_n$ be the atoms occurring in Γ and ψ. By Lemma 3.3.1, the values of the members of Γ and of ψ under a valuation depend only on its values at $p_1, \ldots, p_n$, so $\Gamma \models \psi$ holds exactly when it holds in each of the 2^n rows of the joint truth table. That is a finite check, and by the first sentence it decides $\Gamma \vdash \psi$. ■

So a question that looks as if it needs a search through infinitely many possible derivations can be settled by a finite table. The search for a derivation may still be the quicker way to a **yes**, and one row is the quickest way to a **no** (Corollary 3.5.3), but neither is needed.

Remark. We defined $\Gamma \vdash \psi$ only for a list, that is, for finitely many premises. One may allow an infinite collection Γ of premises and say $\Gamma \vdash \psi$ when ψ is derivable from finitely many members of Γ (a derivation has finitely many lines, so it can only use finitely many). The equivalence with $\Gamma \models \psi$ still holds for infinite Γ; this is the compactness theorem of propositional logic. We do not prove it, and nothing in the book depends on it. Our proof does not reach it: Step 1 needs finitely many atoms.

3.6.7 Two roads to every answer

We now have two independent ways to settle any question "is $\Gamma \vdash \psi$?". One is to **search** for a derivation, as in Section 2.6. The other is to **check a table**, and Corollary 3.6.6 says the two always agree. Here are two of Chapter 2's classical laws settled by the table first.

De Morgan's law (d). Is $\neg(p \wedge q) \models \neg p \vee \neg q$? The four rows:

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

In every row where the premise is T, so is the conclusion. So $\neg(p \wedge q) \models \neg p \vee \neg q$, and by Theorem 3.6.5, $\neg(p \wedge q) \vdash \neg p \vee \neg q$.

Peirce's law. Is $(p \to q) \to p \models p$? Work backwards from F (Section 3.3). A refuting row would need p false. Then $p \to q$ is true, so $(p \to q) \to p$ is "true implies false", which is false. So no row makes the premise true and the conclusion false: $(p \to q) \to p \models p$, and by Theorem 3.6.5, $(p \to q) \to p \vdash p$.

Both answers came before anyone found a derivation. Chapter 2 found both by hand (Proposition 2.5.5 (d), and Proposition 2.5.4, whose $\vdash ((P \to Q) \to P) \to P$ gives the sequent by $\to$E), and the derivations there use RAA. Completeness explains why a search had to succeed; it does not say how.

Section 3.4 remarked that the tables do not see which of Chapter 2's laws needed RAA. Now we can say why they cannot. A table decides $\Gamma \models \psi$, and by Corollary 3.6.6 that is the same as derivability with **all** the rules, RAA included. A table has no column recording which rules a derivation uses. Whether a sequent can be derived without RAA is a question about a shorter list of rules, and for that list the completeness theorem is false, as the next paragraph explains. So no truth-table check can tell the laws that need RAA from those that do not.

Which rules did the proof of completeness itself use? Lemma 3.6.1 only pastes derivations. Lemma 3.6.2 uses no RAA, as its proof checks. Lemma 3.6.3 and the Claim chain those sequents, together with Proposition 2.1.1, which Chapter 2 derived before RAA appeared. The one classical step is the line $p \vee \neg p$ of Lemma 3.6.4. So the theorem is really about the full list of rules, RAA included. Without RAA the list is sound but not complete, as the optional Theorem 3.5.4 shows for $\vdash p \vee \neg p$ and $\neg\neg p \vdash p$; the proof above breaks exactly at the splitting.

Warning. Completeness does not say that a tautology has a derivation **without** RAA. It says there is a derivation from the rules of Chapters 1 and 2, all of them. The formula $(p \to q) \vee (q \to p)$ is a tautology, and every derivation found for it so far uses LEM or RAA. Whether a particular tautology can do without them is a different question, and a table cannot answer it.

3.6.8 Exercises

3.6.8.1 A. Check your understanding

Exercise. A1

These questions are about Lemma 3.6.3 and Theorem 3.6.5.

(a) In the conjunction case of the proof of Lemma 3.6.3, suppose $v(\alpha) = \text{F}$ and $v(\beta) = \text{T}$. Which sequent of Lemma 3.6.2 is used, and which induction hypotheses?

(b) Where is RAA used in the proof of Theorem 3.6.5, including inside the results it cites?

(c) State the idea of the proof of Theorem 3.6.5 in two sentences.

(d) Run Step 2 of the proof for $\Gamma = p$, $\psi = p \vee q$ and the valuation with $v(p) = \mathrm{F}$, $v(q) = \mathrm{T}$: which case applies, and which sequent does it prove?

(e) Step 1 lists the atoms **without repetition**. Where in the proof is that used?

(f) Which step of the proof fails if Γ is allowed to be an infinite collection of formulas, each with its own atoms?

Solution. *(a)* ($\wedge 2$), $\neg\alpha \vdash \neg(\alpha \wedge \beta)$, and only the hypothesis for α, $\Sigma \vdash \neg\alpha$. The hypothesis for β is not needed, because $\alpha \wedge \beta$ is false once α is.

(b) Only inside Lemma 3.6.4, at the line $p \vee \neg p$: Proposition 2.5.3 was derived with RAA. The sequents of Lemma 3.6.2 are derived without it, Lemma 3.6.1 only pastes derivations, and Proposition 2.1.1 comes before RAA in Chapter 2.

(c) Each row of the truth table gives a derivation of the conclusion from the premises and that row's signed atoms: by Lemma 3.6.3 if the row makes every premise true, and from a clash with a false premise otherwise. Two rows that differ at one atom are merged by excluded middle and proof by cases, and after all the atoms are merged the signed atoms are gone.

(d) The valuation makes the premise p false, so Case 2 applies with $\gamma = p$. Lemma 3.6.3 gives $\neg p, q \vdash \neg p$ (the atom case), the premise gives p, and Proposition 2.1.1 gives $p \vee q$. The sequent proved is $p, \neg p, q \vdash p \vee q$.

(e) In the Claim of Step 3: v^+ and v^- change only the value at p_j, and they agree with v on $p_1, \ldots, p_{j-1}$ only because those atoms differ from p_j. If p_j also appeared as some p_i with $i < j$, then $p_i^{v^+} = p_j$ and $p_i^{v^-} = \neg p_j$ would differ, and the two sequents would not share the list Δ that Lemma 3.6.4 needs.

(f) Step 1. Infinitely many atoms may occur, so there is no finite list $p_1, \ldots, p_n$: the Claim would need infinitely many signed atoms as premises, which is not a list, and Step 3's descent from $S(n)$ has no n to start from.

3.6.8.2 B. Practice

Exercise. B1: Three rows from the primitive rules

The table in the proof of Lemma 3.6.2 obtains ($\wedge 2$), ($\wedge 3$) and ($\vee 3$) from De Morgan's laws. Derive

$$\neg\alpha \vdash \neg(\alpha \wedge \beta), \qquad \neg\beta \vdash \neg(\alpha \wedge \beta), \qquad \neg\alpha, \neg\beta \vdash \neg(\alpha \vee \beta)$$

directly from the primitive rules, and say which of your derivations use RAA.

Solution. Each goal is a negation, so each derivation opens a $\neg$I box on the unnegated formula and hunts for a clash.

1	$\neg\alpha$	
2	$\quad\alpha \wedge \beta$	
3	$\quad\alpha$	$\wedge$E 2
4	$\quad\bot$	$\neg$E 3, 1
5	$\neg(\alpha \wedge \beta)$	$\neg$I 2–4

1	$\neg\beta$	
2	$\quad\alpha \wedge \beta$	
3	$\quad\beta$	$\wedge$E 2
4	$\quad\bot$	$\neg$E 3, 1
5	$\neg(\alpha \wedge \beta)$	$\neg$I 2–4

For the disjunction, the assumption $\alpha \vee \beta$ is used by $\vee$E with the goal $\bot$, and each case clashes with one premise.

1	$\neg\alpha$	
2	$\neg\beta$	
3	$\quad\alpha \vee \beta$	
4	$\quad\quad\alpha$	
5	$\quad\quad\bot$	$\neg$E 4, 1
6	$\quad\quad\beta$	
7	$\quad\quad\bot$	$\neg$E 6, 2
8	$\quad\bot$	$\vee$E 3, 4–5, 6–7
9	$\neg(\alpha \vee \beta)$	$\neg$I 3–8

The three derivations have 5, 5 and 9 lines, and none uses RAA: every box closes by $\neg$I or $\vee$E.

Exercise. B2: The method on modus ponens

Apply the method of Theorem 3.6.5 to $p,\ p \to q \models q$: list the four rows and the case of Step 2 that each falls under, and write the derivation that the proof produces, citing LEM and the sequents of Lemma 3.6.2 as single lines. (It is long; that is expected, and the usual length of an exercise answer does not apply.) Compare it with the derivation by $\to$E.

Solution. The atoms are $p_1 = p$ and $p_2 = q$. The rows:

- $p = \mathrm{T}$, $q = \mathrm{T}$: both premises true, Case 1; $q^v = q$ is itself a premise.
- $p = \mathrm{T}$, $q = \mathrm{F}$: the premise $p \to q$ is false, Case 2 with $\gamma = p \to q$; Lemma 3.6.3 gives $p, \neg q \vdash \neg(p \to q)$ by $(\to 3)$.
- $p = \mathrm{F}$, q either value: the premise p is false, Case 2 with $\gamma = p$; the lemma gives $\neg p$ (an atom case).

In Case 2, the clash is $\neg$E followed by $\bot$E, which is the derivation of Proposition 2.1.1 written out. Splitting on q inside each case of a split on p:

1	p	
2	$p \to q$	
3	$p \vee \neg p$	LEM
4	$\quad p$	
5	$\quad q \vee \neg q$	LEM
6	$\quad\quad q$	
7	$\quad\quad \neg q$	
8	$\quad\quad \neg(p \to q)$	$(\to 3)$ 4, 7
9	$\quad\quad \bot$	$\neg$E 2, 8
10	$\quad\quad q$	$\bot$E 9
11	$\quad q$	$\vee$E 5, 6–6, 7–10
12	$\quad \neg p$	
13	$\quad q \vee \neg q$	LEM
14	$\quad\quad q$	
15	$\quad\quad \bot$	$\neg$E 1, 12
16	$\quad\quad q$	$\bot$E 15
17	$\quad\quad \neg q$	
18	$\quad\quad \bot$	$\neg$E 1, 12
19	$\quad\quad q$	$\bot$E 18
20	$\quad q$	$\vee$E 13, 14–16, 17–19
21	q	$\vee$E 3, 4–11, 12–20

The box at line 6 has one line: in the row $p = q = \mathrm{T}$ the goal q is the signed atom itself, so, as in the proof of Lemma 3.6.4, the box is its assumption alone. The result has 21 lines. The derivation by $\to$E has 3: the two premises and q by $\to$E citing lines 2 and 1. The method is uniform, not short: it does the same thing for every valid sequent, and pays for that with a split on every atom.

Exercise. B3: Table first, then a derivation

Decide by a truth table whether $\neg(p \leftrightarrow q) \models \neg p \leftrightarrow q$. If it holds, find a derivation of $\neg(p \leftrightarrow q) \vdash \neg p \leftrightarrow q$ by search, where you may cite the sequents of Lemma 3.6.2 as derived rules. Which rule of your derivation is classical?

Solution. The table:

p	q	$\neg(p \leftrightarrow q)$	$\neg p \leftrightarrow q$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	F	F

The premise is true in rows 2 and 3, and so is the conclusion. So the entailment holds, and by Theorem 3.6.5 a derivation exists.

The search: the goal is a biconditional, so two boxes, one from $\neg p$ to q and one from q to $\neg p$. In the second, the goal $\neg p$ is a negation: a box on p, and p, q give $p \leftrightarrow q$ by ($\leftrightarrow$1), which clashes with the premise. In the first, the goal q is an atom, the only premise is a negation, and nothing else applies: we are stuck, and assume $\neg q$. Then $\neg p$, $\neg q$ give $p \leftrightarrow q$ by ($\leftrightarrow$2), which clashes with the premise.

1	$\neg(p \leftrightarrow q)$	
2	$\quad \neg p$	
3	$\quad\quad \neg q$	
4	$\quad\quad p \leftrightarrow q$	($\leftrightarrow$2) 2, 3
5	$\quad\quad \bot$	$\neg$E 4, 1
6	$\quad q$	RAA 3–5
7	$\quad q$	
8	$\quad\quad p$	
9	$\quad\quad p \leftrightarrow q$	($\leftrightarrow$1) 8, 7
10	$\quad\quad \bot$	$\neg$E 9, 1
11	$\quad \neg p$	$\neg$I 8–10
12	$\neg p \leftrightarrow q$	$\leftrightarrow$I 2–6, 7–11

The classical rule is the RAA at line 6. (Pasting in the derivations of the two cited sequents gives a derivation in the primitive rules of 22 lines.) Whether some other derivation could avoid RAA is a question the table cannot answer. In fact none can; we state that without proof, as Chapter 2 did for similar sequents, and nothing depends on it.

3.6.8.3 C. Going deeper

Exercise. C1: The remaining cases

Write out in full the disjunction case and the biconditional case of the proof of Lemma 3.6.3, in the style of the implication case.

Solution. Throughout, Σ is $p_1^v, \ldots, p_n^v$, and the induction hypotheses are $\Sigma \vdash \alpha^v$ and $\Sigma \vdash \beta^v$. "Chaining" is Lemma 3.6.1 (c).

Disjunction, $\varphi = \alpha \vee \beta$. By Definition 3.3.2, $v(\alpha \vee \beta) = \mathrm{F}$ exactly when $v(\alpha) = v(\beta) = \mathrm{F}$.

- *Case* $v(\alpha) = \mathrm{T}$. Then $\varphi^v = \alpha \vee \beta$ and $\alpha^v = \alpha$. Chaining $\Sigma \vdash \alpha$ with $(\vee 1)$, $\alpha \vdash \alpha \vee \beta$, gives $\Sigma \vdash \alpha \vee \beta$.
- *Case* $v(\alpha) = \mathrm{F}$, $v(\beta) = \mathrm{T}$. Then $\varphi^v = \alpha \vee \beta$ and $\beta^v = \beta$. Chaining $\Sigma \vdash \beta$ with $(\vee 2)$ gives $\Sigma \vdash \alpha \vee \beta$.
- *Case* $v(\alpha) = v(\beta) = \mathrm{F}$. Then $\varphi^v = \neg(\alpha \vee \beta)$, $\alpha^v = \neg\alpha$ and $\beta^v = \neg\beta$. Chaining both hypotheses with $(\vee 3)$, $\neg\alpha, \neg\beta \vdash \neg(\alpha \vee \beta)$, gives $\Sigma \vdash \neg(\alpha \vee \beta)$.

Biconditional, $\varphi = \alpha \leftrightarrow \beta$. By Definition 3.3.2, $v(\alpha \leftrightarrow \beta) = \mathrm{T}$ exactly when $v(\alpha) = v(\beta)$. Each case uses both hypotheses.

- *Case T, T.* $\varphi^v = \alpha \leftrightarrow \beta$; chain $\Sigma \vdash \alpha$, $\Sigma \vdash \beta$ with $(\leftrightarrow 1)$.
- *Case F, F.* $\varphi^v = \alpha \leftrightarrow \beta$; chain $\Sigma \vdash \neg\alpha$, $\Sigma \vdash \neg\beta$ with $(\leftrightarrow 2)$.
- *Case T, F.* $\varphi^v = \neg(\alpha \leftrightarrow \beta)$; chain $\Sigma \vdash \alpha$, $\Sigma \vdash \neg\beta$ with $(\leftrightarrow 3)$.
- *Case F, T.* $\varphi^v = \neg(\alpha \leftrightarrow \beta)$; chain $\Sigma \vdash \neg\alpha$, $\Sigma \vdash \beta$ with $(\leftrightarrow 4)$.

The cases cover all four pairs of values, so in each the property holds for φ.

Marking scheme. (10 points)

- 4: the $\vee$ case: the three value cases, each with the right sequent of Lemma 3.6.2 and only the hypotheses its premises need.
- 4: the $\leftrightarrow$ case: the four value cases, each with the right sequent and both hypotheses.
- 1: names chaining (Lemma 3.6.1 (c)) as the step that combines the hypotheses with the sequent.
- 1: says that the cases cover every pair of values.

Fatal errors (0 for the step): using a sequent whose premises are not the signed parts α^v, β^v of that case.

Exercise. C2: Exit check: completeness

Close the section first, and write from memory.

(a) Prove Lemma 3.6.4: if $\Delta, p \vdash \psi$ and $\Delta, \neg p \vdash \psi$, then $\Delta \vdash \psi$.

(b) Prove Step 3 of the proof of Theorem 3.6.5 in full: from the Claim of Step 2, for every valuation v, $\Gamma, p_1^v, \ldots, p_n^v \vdash \psi$, deduce $\Gamma \vdash \psi$. State the statement you prove by induction, and say which principle of induction you use.

Solution (Hint 1). For (a): excluded middle and proof by cases. For (b): remove the signed atoms from the end of the list, one at a time.

Solution (Hint 2). For (b): let $S(j)$ say "for every valuation v, $\Gamma, p_1^v, \ldots, p_j^v \vdash \psi$", and show that $S(j)$ implies $S(j-1)$ for $1 \leq j \leq n$.

Solution. (a) Let D derive ψ from premises in Δ, p, and E from premises in $\Delta, \neg p$. Write, as premises, those of D and E that lie in Δ; then $p \vee \neg p$ by LEM (Proposition 2.5.3, a derived

rule with no premises); then a box opening with p containing the nonpremise lines of D, in which citations of the premise p now cite the assumption and citations of premises in Δ cite the premise lines; then a box opening with $\neg p$, built from E in the same way; then ψ by $\vee$E, citing the line $p \vee \neg p$ and the two boxes. (If D or E ends on a premise, its box is the assumption alone when that premise is the assumption's formula, and otherwise the assumption followed by ψ by R.) Every copied line keeps its boxes, now inside the new box, and cites lines still available to it. The last line of each box is ψ, as a line of that box itself, because the last line of D or E lay in no box. The two boxes and the LEM line lie in no box, so they are available at the final line. So the result is a derivation of ψ from premises in Δ, and $\Delta \vdash \psi$.

(b) For $0 \leq j \leq n$, let $S(j)$ be: for every valuation v, $\Gamma, p_1^v, \ldots, p_j^v \vdash \psi$. The Claim of Step 2 is $S(n)$.

The step. Let $1 \leq j \leq n$ and assume $S(j)$. Let v be any valuation, and let v^+, v^- agree with v except that $v^+(p_j) = \mathrm{T}$ and $v^-(p_j) = \mathrm{F}$. Since the p_i are distinct, both agree with v on $p_1, \ldots, p_{j-1}$. By $S(j)$ for v^+ and for v^-, with $\Delta \coloneqq \Gamma, p_1^v, \ldots, p_{j-1}^v$: $\Delta, p_j \vdash \psi$ and $\Delta, \neg p_j \vdash \psi$. By (a), $\Delta \vdash \psi$. As v was arbitrary, $S(j-1)$ holds.

The induction. We use Axiom 3.2.1 on $m \in \mathbb{N}$ with the statement $T(m)$: "if $m \leq n$, then $S(n-m)$". $T(0)$ is $S(n)$. If $T(m)$ holds and $m+1 \leq n$, then $S(n-m)$ holds and $j = n-m$ lies between 1 and n, so the step gives $S(n-m-1)$, which is $T(m+1)$. Hence $T(n)$, which is $S(0)$: for every valuation, $\Gamma \vdash \psi$. Valuations exist, so $\Gamma \vdash \psi$.

(Inducting downwards on j, or on the number of atoms removed, is the same argument; either earns full marks.)

Marking scheme. (10 points)

- 3: (a) pastes both derivations into the two boxes of a $\vee$E on $p \vee \neg p$, with the premise p (or $\neg p$) redirected to the box's assumption, and says why every citation is still available.
- 1: (a) cites LEM (Proposition 2.5.3) as a derived rule.
- 3: (b) sets up an induction on the number of atoms removed (or downward on j) with a statement quantified over **every** valuation, and names the induction principle.
- 2: (b) in the step, applies the hypothesis to **both** valuations that differ at p_j, and notes that they agree with v on the other atoms.
- 1: (b) concludes $\Gamma \vdash \psi$ from $S(0)$.

Fatal errors (0 for the step): splitting on an atom using the Claim for a single valuation only; an induction statement that fixes v before the induction, so that the two valuations of the step are not covered by the hypothesis.

3.7 Normal Forms and Adequate Connectives

The proof of completeness wrote each row of a truth table as a list of signed atoms $p_1^v, \ldots, p_n^v$ (Section 3.6). Join the signed atoms of one row by $\wedge$, and you get a formula that is true in that row and in no other. Join those formulas, one for each row where a given formula φ is true, by $\vee$, and you get a formula with the same truth table as φ. This section turns that observation into two standard shapes for formulas, the disjunctive and the conjunctive normal form. It then reads validity off the conjunctive shape at a glance, and asks how few connectives are really

needed.

3.7.1 Conjunctions and disjunctions of many formulas

Normal forms are long conjunctions and disjunctions. Section 3.4 fixed how to write them, in the paragraph on long conjunctions after Proposition 3.4.1. For formulas $\chi_1, \dots, \chi_k$ with $k \geq 1$, $\chi_1 \wedge \chi_2 \wedge \cdots \wedge \chi_k$ is the conjunction **grouped to the right**, $\chi_1 \wedge (\chi_2 \wedge (\cdots \wedge \chi_k))$, and likewise for $\vee$; for $k = 1$ it is χ_1 itself. That paragraph proved, by induction on the number of terms, how such formulas are evaluated: for every valuation v,

$$\begin{aligned} v(\chi_1 \wedge \cdots \wedge \chi_k) = \mathrm{T} &\iff v(\chi_i) = \mathrm{T} \text{ for every } i = 1, \dots, k, \\ v(\chi_1 \vee \cdots \vee \chi_k) = \mathrm{T} &\iff v(\chi_i) = \mathrm{T} \text{ for some } i = 1, \dots, k. \end{aligned} \tag{3.7.1}$$

We use these two lines constantly below, and we cite them by this display.

So a long conjunction is true when **all** its parts are, and a long disjunction when **at least one** is, just as the words say.

3.7.2 Literals, clauses and the two normal forms

The building blocks are the smallest formulas that still carry a sign.

Definition 3.7.1. Literal

A **literal** is an atom or the negation of an atom.

So p, $\neg p$, q_7 and $\neg q_7$ are literals. The signed atom p^v of Section 3.6 is always a literal. The formulas $\neg\neg p$, $\neg(p \wedge q)$ and $\bot$ are not: in the first two, $\neg$ is applied to something other than an atom, and $\bot$ is neither an atom nor a negation. The **opposite** of a literal is $\neg p$ if the literal is p, and p if it is $\neg p$. By the clause for $\neg$ in Definition 3.3.2, a literal and its opposite take opposite values under every valuation.

From literals we build two kinds of blocks and two kinds of formulas.

- A **clause** joins one or more literals by $\vee$, such as $p \vee \neg q \vee r$, or $\neg q$ alone.
- A **conjunct** (in this section, short for "conjunctive clause") joins one or more literals by $\wedge$, such as $p \wedge \neg q \wedge r$, or $\neg q$ alone.

A normal form is a formula whose connectives come in fixed layers: negations only on atoms, then one layer of one connective, then one layer of the other.

Definition 3.7.2. Disjunctive Normal Form

A formula **has disjunctive normal form** (DNF), or is **in DNF**, when it has the shape $C_1 \vee \cdots \vee C_k$, with $k \geq 1$, where every C_i joins **one or more literals** by $\wedge$.

Definition 3.7.3. Conjunctive Normal Form

A formula **has conjunctive normal form** (CNF), or is **in CNF**, when it has the shape

$K_1 \wedge \cdots \wedge K_k$, with $k \geq 1$, where every K_i joins **one or more literals** by $\vee$.

In words: a DNF is an "or of ands" and a CNF an "and of ors", and in both the only negations sit directly on atoms. Read a DNF as a list of situations, any one of which makes it true; read a CNF as a list of requirements, all of which must be met.

Check some formulas against the definitions, clause by clause.

1. $(p \wedge \neg q) \vee (\neg p \wedge q)$ is in DNF: two conjuncts, each a conjunction of literals. It is not in CNF: as a conjunction of clauses it would have to be a single clause (it is not a conjunction), and a clause cannot contain $\wedge$.
2. $(p \vee q) \wedge (\neg p \vee \neg q)$ is in CNF, with two clauses, and not in DNF.
3. $p \wedge \neg q$ is in both: in DNF with one conjunct, and in CNF with two clauses of one literal each. Likewise $p \vee \neg q$ is in both.
4. A single literal, $\neg r$, is in both forms: one conjunct of one literal, and one clause of one literal. This degenerate case matters, because a formula such as $\neg r$ must count as already normal.

A nonexample by a minimal change: $(p \wedge \neg q) \vee r$ is in DNF, but $\neg(p \wedge q) \vee r$ is not in either form. Only one symbol moved, and the literal condition fails: $\neg$ now stands on the conjunction $p \wedge q$, not on an atom. The formula $p \to q$ is in neither form either, since $\to$ is not allowed at all.

Why these layers? Because they are the shapes that the truth table hands us, as the next theorem shows, and because in these shapes some questions become easy: the subsection "Reading validity off a CNF" below decides validity of a CNF without a table.

Warning. A literal is an atom with **at most one** $\neg$, and nothing else. $\neg\neg p$ is equivalent to the literal p but is not a literal, so $\neg\neg p \vee q$ is not in CNF as written. Likewise $\bot$ is not a literal, so a "normal form" containing $\bot$ is not one; that is why the constructions below never use it.

Quick check. Which of the following are in CNF, and which are in DNF? (i) $(p \vee \neg q) \wedge r$; (ii) $\neg(p \vee q)$; (iii) $p \vee (\neg q \wedge r)$.

Solution. (i) CNF, with clauses $p \vee \neg q$ and r; not DNF, since the conjunct would have to be the whole formula and $p \vee \neg q$ is not a literal. (ii) Neither: $\neg$ stands on a disjunction. (iii) DNF, with conjuncts p and $\neg q \wedge r$; not CNF, since the one clause would contain $\wedge$.

3.7.3 Reading a DNF off the truth table

Take a valuation v and atoms $p_1, \ldots, p_n$ with $n \geq 1$, and form the conjunct of the signed atoms of the row of v,

$$C_v := p_1^v \wedge \cdots \wedge p_n^v.$$

For every valuation w, each p_i^v is true under w exactly when $w(p_i) = v(p_i)$: if $v(p_i) = \mathrm{T}$, then $p_i^v = p_i$; if $v(p_i) = \mathrm{F}$, then $p_i^v = \neg p_i$, which is true under w exactly when $w(p_i) = \mathrm{F}$. By Eq. 3.7.1,

$$w(C_v) = \mathrm{T} \iff w(p_i) = v(p_i) \text{ for every } i = 1, \dots, n. \tag{3.7.2}$$

So C_v is true in the row of v and in no other row. It is a formula that "names" one row.

Example. Exactly one of two

Find a formula in DNF that is true exactly when exactly one of p, q is true, and check it.

Solution. The rows with exactly one of p, q true are $(p, q) = (\mathrm{T}, \mathrm{F})$ and (F, T). Their conjuncts are $p \wedge \neg q$ and $\neg p \wedge q$, so the answer is

$$(p \wedge \neg q) \vee (\neg p \wedge q).$$

Check against the table: in the row T, F the first conjunct is true; in F, T the second is; in T, T each conjunct has a false literal ($\neg q$, $\neg p$); in F, F likewise (p, q). So the formula is true in exactly the two rows wanted.

The same recipe works for any formula.

Theorem 3.7.1. Disjunctive Normal Form

Every formula φ is equivalent to a formula in DNF. If φ has at least one atom, the DNF can be chosen so that every atom in it is an atom of φ.

Idea. List the rows where φ is true and disjoin their conjuncts C_v. A valuation makes the disjunction true exactly when it agrees with one of those rows, which by Lemma 3.3.1 is exactly when it makes φ true. Two degenerate cases need a convention: a formula with no T-row, whose disjunction would be empty, and a formula with no atoms, which has no rows to speak of.

Proof. By the paraphrase of Definition 3.4.4, two formulas φ and ψ are equivalent exactly when $v(\varphi) = v(\psi)$ for every valuation v.

Case 1: φ has at least one atom. Let $p_1, \dots, p_n$ ($n \geq 1$) be the atoms of φ. A **row** is a choice of a value for each p_i; there are 2^n rows, and for each we fix a valuation that takes those values at $p_1, \dots, p_n$ (and, say, T at every other atom). By Lemma 3.3.1, $v(\varphi)$ depends only on the row of v. Let $v_1, \dots, v_k$ be the fixed valuations of the rows in which φ is T.

If $k \geq 1$, let $D := C_{v_1} \vee \cdots \vee C_{v_k}$, which is in DNF by Definition 3.7.2, with atoms among $p_1, \dots, p_n$. Let w be any valuation. By Eq. 3.7.1 and Eq. 3.7.2, $w(D) = \mathrm{T}$ exactly when w agrees with some v_j on $p_1, \dots, p_n$. If it does, then $w(\varphi) = v_j(\varphi) = \mathrm{T}$ by Lemma 3.3.1. Conversely, if $w(\varphi) = \mathrm{T}$, then the row of w is one where φ is T, so w agrees on $p_1, \dots, p_n$ with the valuation v_j fixed for that row. Hence $w(D) = w(\varphi)$ for every w, and $\varphi \equiv D$.

If $k = 0$, then φ is F in every row, hence under every valuation. The formula $p_1 \wedge \neg p_1$ is in DNF (one conjunct of two literals) and is F under every valuation by Definition 3.3.2. So $\varphi \equiv p_1 \wedge \neg p_1$.

Case 2: φ has no atoms. Any two valuations agree on all the atoms of φ (there are none), so by Lemma 3.3.1 $v(\varphi)$ is the same for every v. Let p be an atom. If that value is T, then $\varphi \equiv p \vee \neg p$, which is true under every valuation and is in DNF (two conjuncts of one literal); if it is F, then $\varphi \equiv p \wedge \neg p$. This proves the theorem.

■

The conjuncts C_v are the signed atoms of Section 3.6 joined by $\wedge$. There the signed atoms of a row were premises from which φ^v was derived; here they are joined into a formula that is true in that row only. The proof is the proof of completeness with the derivations taken out.

For $p \wedge \neg p$ we could have written $\bot$, which is also F under every valuation. We did not, because $\bot$ is not a literal, and because the adequacy theorem at the end of this section reuses the construction in a language without $\bot$.

3.7.4 Conjunctive normal form

A DNF lists the rows where a formula is true. A CNF does the opposite: it lists the rows where the formula is **false**, one clause per row, each clause being false in its row and true in every other.

Theorem 3.7.2. Conjunctive Normal Form

Every formula φ is equivalent to a formula in CNF. If φ has at least one atom, the CNF can be chosen so that every atom in it is an atom of φ.

Idea. Apply Theorem 3.7.1 to $\neg\varphi$ and negate the result: $\varphi \equiv \neg\neg\varphi \equiv \neg(C_1 \vee \cdots \vee C_k)$. Now push the $\neg$ inwards with De Morgan's laws and double negation (Proposition 3.4.1): $\neg(C_1 \vee \cdots \vee C_k)$ becomes $\neg C_1 \wedge \cdots \wedge \neg C_k$, and each $\neg C_i$, the negation of a conjunction of literals, becomes the disjunction of the opposite literals. The proof checks the outcome directly with Eq. 3.7.1, which avoids De Morgan's laws for many terms.

Proof. By Theorem 3.7.1, $\neg\varphi \equiv C_1 \vee \cdots \vee C_k$ for some conjuncts $C_1, \ldots, C_k$ (with atoms among those of φ if it has any, since $\neg\varphi$ has the same atoms). For each i, let K_i be the disjunction of the opposites of the literals of C_i, in the same order, and let $K := K_1 \wedge \cdots \wedge K_k$. Then K is in CNF, with the same atoms as $C_1 \vee \cdots \vee C_k$.

Let w be a valuation. Each literal of K_i is true under w exactly when the corresponding literal of C_i is false. By Eq. 3.7.1, used three times:

$$\begin{aligned} w(K) = \mathrm{T} &\iff \text{for every } i,\ w(K_i) = \mathrm{T} \\ &\iff \text{for every } i,\ \text{some literal of } C_i \text{ is false under } w \\ &\iff \text{for every } i,\ w(C_i) = \mathrm{F} \\ &\iff w(C_1 \vee \cdots \vee C_k) = \mathrm{F}. \end{aligned}$$

Since $C_1 \vee \cdots \vee C_k \equiv \neg\varphi$, the last line holds exactly when $w(\neg\varphi) = \mathrm{F}$, that is, when $w(\varphi) = \mathrm{T}$. So $w(K) = w(\varphi)$ for every w, and $\varphi \equiv K$.

■

Unwinding the construction: the conjuncts C_i of the DNF of $\neg\varphi$ are the rows where φ is **false**, and K_i is the clause that is false exactly in the row of C_i. So the CNF can be read straight off the table: **one clause per F-row**, built from the opposites of that row's signed atoms. In the row $p = \mathrm{T}$, $q = \mathrm{F}$, for instance, the clause is $\neg p \vee q$. A formula with no F-row gets, by the proof, the clause of a contradiction's DNF negated, $\neg p \vee p$ (from $C_1 = p \wedge \neg p$), which is true in every row.

Alternatively, a CNF can be reached by **rewriting**. First replace each $\bot$, which is not a literal, by $p \wedge \neg p$ for an atom p; both are false in every row. Replace $\alpha \leftrightarrow \beta$ by $(\alpha \to \beta) \wedge (\beta \to \alpha)$ and $\alpha \to \beta$ by $\neg\alpha \vee \beta$; push each $\neg$ inwards by De Morgan's laws, and remove $\neg\neg$; finally distribute $\vee$ over $\wedge$, replacing $\alpha \vee (\beta \wedge \gamma)$ by $(\alpha \vee \beta) \wedge (\alpha \vee \gamma)$. Each step replaces a part of the formula by an equivalent one: the equivalences are $\bot \equiv p \wedge \neg p$ and those of Proposition 3.4.1, which hold for arbitrary formulas, and Lemma 3.4.2 says the whole formula stays equivalent; since $\equiv$ is transitive (Section 3.4), the steps chain. Stopping before the distribution step, and distributing $\wedge$ over $\vee$ instead, gives a DNF. The rewriting route is often much shorter than the table, as the next example shows, but it can also be much longer, as the warning after it shows.

Example. Two routes to a normal form

Put $\varphi = (p \to q) \to (\neg q \to r)$ into DNF and into CNF, first from its truth table and then by rewriting.

Solution. The table route. Work backwards from F (Section 3.3): φ is false exactly when $p \to q$ is T and $\neg q \to r$ is F. The latter needs $\neg q$ true and r false, so $q = r = \mathrm{F}$; then $p \to q$ true needs $p = \mathrm{F}$. So φ has exactly one F-row, $p = q = r = \mathrm{F}$, as the table confirms:

p	q	r	$p \to q$	$\neg q \to r$	$(p \to q) \to (\neg q \to r)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	F

The DNF of Theorem 3.7.1 has one conjunct for each of the seven T-rows:

$$(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \\ \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r).$$

The CNF of Theorem 3.7.2 has one clause for the single F-row, built from the opposites of $\neg p, \neg q, \neg r$: it is the single clause $p \vee q \vee r$.

The rewriting route. Eliminate the three implications, then push the negation in:

$$\begin{aligned} \varphi &\equiv \neg(\neg p \vee q) \vee (\neg\neg q \vee r) \\ &\equiv (\neg\neg p \wedge \neg q) \vee (q \vee r) \\ &\equiv (p \wedge \neg q) \vee q \vee r, \end{aligned}$$

using (c) of Proposition 3.4.1 three times, then De Morgan's law (b) for $\neg(\alpha \vee \beta)$, then double negation (d) twice. This is already a DNF, with three conjuncts. For a CNF, distribute, by (g) together with commutativity (f) and associativity (h): $(p \wedge \neg q) \vee (q \vee r) \equiv (p \vee q \vee r) \wedge (\neg q \vee q \vee r)$. The second clause contains q and $\neg q$, so it is true in every row, and a conjunction with a part that is true in every row is equivalent to its other part. So $\varphi \equiv p \vee q \vee r$: one clause, as the table route found.

The formula $p \vee q \vee r$ is also a DNF, with three conjuncts of one literal each. So the table's seven conjuncts and the rewriting's three can both be replaced by three one-literal conjuncts. Both routes are correct; they differ in length, and neither is guaranteed to give the shortest normal form.

Warning. A formula has **many** equivalent CNFs and DNFs; "normal form" does not mean "unique form". $p \vee q \vee r$ and $(p \vee q \vee r) \wedge (\neg q \vee q \vee r)$ are both CNFs of the formula above. And the two routes have opposite weaknesses. The table route needs all 2^n rows. The rewriting route can explode at the distribution step: distributing $(p_1 \wedge q_1) \vee (p_2 \wedge q_2) \vee (p_3 \wedge q_3)$ gives 8 clauses, one for each way of picking p_i or q_i from every conjunct, and n such conjuncts give 2^n clauses.

3.7.5 Reading validity off a CNF

A CNF is a list of requirements, all of which must hold. That makes one question easy.

Proposition 3.7.3. Validity of a CNF

(a) A formula in CNF is a tautology exactly when **every** clause contains some atom together with its negation.

(b) A formula in DNF is satisfiable exactly when **some** conjunct contains no atom together with its negation.

Try it first (ten minutes): take a clause in which no atom occurs both with and without $\neg$, such as $p \vee \neg q \vee r$. Can you find a row that makes it false? What does that row do to a CNF containing this clause?

Idea. The direction "every clause has a pair $\Rightarrow$ tautology" is the easy one: $p \vee \neg p$ is true in every row, so a clause containing both is too, and a conjunction of clauses true in every row is true in every row. The attempt above is the other direction. To make a CNF false, it is enough to make **one** clause false, and a clause is false only when **all** its literals are. So set every atom of the clause against its literal: F for a plain atom, T for a negated one. The one thing that could stop us is an atom that appears both plain and negated, since it would have to be both F and T. That is exactly the pair the proposition talks about.

Proof. (a) Let $K = K_1 \wedge \cdots \wedge K_k$ be in CNF.

($\Leftarrow$) Suppose every clause K_i contains some atom p and its negation $\neg p$. Let v be a valuation. One of p, $\neg p$ is true under v, by Definition 3.3.2, so $v(K_i) = \mathrm{T}$ by Eq. 3.7.1. This holds for

every i, so $v(K) = \mathrm{T}$ by Eq. 3.7.1. As v was arbitrary, K is a tautology (Definition 3.4.1).

($\Rightarrow$) We prove the contrapositive. Suppose some clause K_i contains no atom together with its negation. Define a valuation v by: $v(p) := \mathrm{F}$ if the literal p occurs in K_i; $v(p) := \mathrm{T}$ if the literal $\neg p$ occurs in K_i; $v(p) := \mathrm{T}$ for every other atom. This is well defined, because no atom occurs in K_i both as p and as $\neg p$. Every literal of K_i is false under v, so $v(K_i) = \mathrm{F}$ by Eq. 3.7.1, and hence $v(K) = \mathrm{F}$, again by Eq. 3.7.1. So K is not a tautology.

(b) Let $D = C_1 \vee \cdots \vee C_k$ be in DNF.

($\Leftarrow$) Suppose C_i contains no atom together with its negation. Define v by $v(p) := \mathrm{T}$ if p occurs in C_i, $v(p) := \mathrm{F}$ if $\neg p$ occurs in C_i, and $v(p) := \mathrm{T}$ otherwise; this is well defined as in (a). Every literal of C_i is true under v, so $v(C_i) = \mathrm{T}$ and $v(D) = \mathrm{T}$ by Eq. 3.7.1. So D is satisfiable (Definition 3.4.2).

($\Rightarrow$) If every conjunct contains some p and $\neg p$, then under every valuation one of these two literals is false, so every conjunct is false and so is D, by Eq. 3.7.1. So D is not satisfiable. This proves the proposition.

■

The two parts are two faces of one fact. Section 3.4 showed that a formula is a tautology exactly when its negation is unsatisfiable (after Definition 3.4.2). Let D be in DNF, and let K be the CNF whose clauses are the conjuncts of D with every literal replaced by its opposite. The proof of Theorem 3.7.2, applied to $\varphi = \neg D$ with D itself as the DNF of $\neg\varphi \equiv D$, shows $K \equiv \neg D$. A clause of K has an atom together with its negation exactly when the corresponding conjunct of D has. So D is unsatisfiable exactly when $\neg D$, and so K, is a tautology, which by (a) happens exactly when every conjunct of D has such a pair. That is (b) again.

The proposition trades one kind of work for another. Checking a CNF for validity is now a glance, but getting the CNF in the first place may take the 2^n rows of the table route or the explosion of the rewriting route. Nothing is free: the work has moved, not vanished.

Example. Validity at a glance

Decide whether each formula is a tautology, without a table.

(a) $(\neg p \vee q \vee p) \wedge (r \vee \neg q \vee s \vee q) \wedge (\neg s \vee \neg r \vee s) \wedge (p \vee \neg r \vee r \vee q)$.

(b) The same formula with the third clause changed to $\neg s \vee \neg r \vee q$.

Solution. (a) Each clause has a pair: $\neg p, p$ in the first; $\neg q, q$ in the second; $\neg s, s$ in the third; $\neg r, r$ in the fourth. By Proposition 3.7.3 (a), it is a tautology. A table would have needed 16 rows.

(b) The new third clause $\neg s \vee \neg r \vee q$ has the atoms s, r, q, each once, so it has no pair, and the formula is not a tautology. The proof of Proposition 3.7.3 names a refuting row: $s = \mathrm{T}$, $r = \mathrm{T}$, $q = \mathrm{F}$, and $p = \mathrm{T}$, like every atom not in the clause; in fact either value of p works. In that row the third clause is F, so the conjunction is F.

3.7.6 Adequate sets of connectives

Our formulas use five connectives and $\bot$. Chapter 2 already showed that some of them can be traded for others: an implication for a disjunction (Proposition 2.5.6), a biconditional for two implications (Proposition 2.4.1). How far can this go? Before we can answer, we need to say what a set of connectives must be able to "say".

Every formula whose atoms are among $p_1, \ldots, p_n$ has a truth table: a value for each of the 2^n rows. Turn this around, and ask whether **every** table is the table of some formula.

Definition 3.7.4. Truth Function

Let $n \geq 1$. A **truth function** with n inputs assigns to **each** list $(a_1, \ldots, a_n)$ of truth values (each a_i is T or F) a truth value $f(a_1, \ldots, a_n)$. A formula φ **expresses** f if

$$v(\varphi) = f(v(p_1), \ldots, v(p_n)) \quad \text{for every valuation } v.$$

In words: a truth function is a truth table with n input columns and one output column, with no formula attached. There are 2^n lists of inputs and two choices of output for each, so there are 2^{2^n} truth functions with n inputs: 4 with one input and 16 with two. Examples: the **negation** function with one input, $\mathrm{T} \mapsto \mathrm{F}$, $\mathrm{F} \mapsto \mathrm{T}$, expressed by $\neg p_1$; the two-input function that is T only at (T, T), expressed by $p_1 \wedge p_2$; the three-input **majority** function, T when at least two inputs are T, expressed by $(p_1 \wedge p_2) \vee (p_1 \wedge p_3) \vee (p_2 \wedge p_3)$. A degenerate case: the one-input function that is always T is expressed by $p_1 \vee \neg p_1$, and also by $p_2 \to p_2$, which does not mention p_1 at all. The definition allows that.

We say a formula is **in** a set S of connectives if every connective occurring in it belongs to S, and $\bot$ occurs in it only if $\bot$ belongs to S (for this purpose $\bot$ counts as a connective with no inputs).

Definition 3.7.5. Adequate Set of Connectives

A set S of connectives is **adequate** if **every** truth function, with **any** number $n \geq 1$ of inputs, is expressed by some formula in S.

A set containing an adequate set is adequate, since a formula in the smaller set is also in the larger one. The theorem below shows that $\{\neg, \wedge, \vee\}$ is adequate, so the full set $\{\neg, \wedge, \vee, \to, \leftrightarrow, \bot\}$ is too. The question is which smaller sets are.

Theorem 3.7.4. Adequate Sets

(a) The sets $\{\neg, \wedge, \vee\}$, $\{\neg, \wedge\}$, $\{\neg, \vee\}$ and $\{\to, \bot\}$ are adequate.

(b) The set $\{\wedge, \vee, \to\}$ is not adequate: no formula in it expresses the negation function.

Idea. For (a), the construction of Theorem 3.7.1 works for any table, not only for the table of a formula, and its output uses only $\neg$, $\wedge$, $\vee$. To shrink the set further, rewrite each missing connective in terms of the others, for instance $\alpha \vee \beta$ as $\neg(\neg\alpha \wedge \neg\beta)$, recursively through the formula. For (b), look at the row where every atom is true: $\wedge$, $\vee$ and $\to$ all give T on T inputs, so every formula in them is T there, while negation must give F.

Proof. *(a) The set* $\{\neg, \wedge, \vee\}$. Let f be a truth function with $n \geq 1$ inputs. For each of the 2^n lists of inputs, fix a valuation taking those values at $p_1, \ldots, p_n$, and let $v_1, \ldots, v_k$ be those fixed for the lists where f is T. If $k \geq 1$, let $D := C_{v_1} \vee \cdots \vee C_{v_k}$, with the conjuncts C_v formed from $p_1, \ldots, p_n$; if $k = 0$, let $D := p_1 \wedge \neg p_1$. By Eq. 3.7.1 and Eq. 3.7.2, a valuation w makes D true exactly when $(w(p_1), \ldots, w(p_n))$ is a list where f is T, that is, when $f(w(p_1), \ldots, w(p_n)) = \mathrm{T}$. So D expresses f, and D is in $\{\neg, \wedge, \vee\}$.

The other three sets. By Theorem 3.2.6, the following clauses define three functions t_1, t_2, t_3 on all formulas. Each has $t_i(p) := p$ for every atom p, and $t_i(\bot) := \bot$, $t_i(\alpha \to \beta) := t_i(\alpha) \to t_i(\beta)$ and $t_i(\alpha \leftrightarrow \beta) := t_i(\alpha) \leftrightarrow t_i(\beta)$ (these last clauses are never used below). The other clauses rewrite the connectives $\neg$, $\wedge$, $\vee$:

- t_1, into $\{\neg, \wedge\}$: $t_1(\neg\alpha) := \neg t_1(\alpha)$, $t_1(\alpha \wedge \beta) := t_1(\alpha) \wedge t_1(\beta)$, and $t_1(\alpha \vee \beta) := \neg(\neg t_1(\alpha) \wedge \neg t_1(\beta))$;
- t_2, into $\{\neg, \vee\}$: $t_2(\neg\alpha) := \neg t_2(\alpha)$, $t_2(\alpha \wedge \beta) := \neg(\neg t_2(\alpha) \vee \neg t_2(\beta))$, and $t_2(\alpha \vee \beta) := t_2(\alpha) \vee t_2(\beta)$;
- t_3, into $\{\to, \bot\}$: $t_3(\neg\alpha) := t_3(\alpha) \to \bot$, $t_3(\alpha \wedge \beta) := (t_3(\alpha) \to (t_3(\beta) \to \bot)) \to \bot$, and $t_3(\alpha \vee \beta) := (t_3(\alpha) \to \bot) \to t_3(\beta)$.

Claim. Let $S_1 = \{\neg, \wedge\}$, $S_2 = \{\neg, \vee\}$, $S_3 = \{\to, \bot\}$. For every formula φ in $\{\neg, \wedge, \vee\}$ and each i, the formula $t_i(\varphi)$ is in S_i, and $v(t_i(\varphi)) = v(\varphi)$ for every valuation v.

Proof of Claim. By structural induction (Theorem 3.2.3) on φ, with the property "if φ is in $\{\neg, \wedge, \vee\}$, then for each i, $t_i(\varphi)$ is in S_i and has the same value as φ under every valuation". The immediate parts of a formula in $\{\neg, \wedge, \vee\}$ are in $\{\neg, \wedge, \vee\}$, so the induction hypothesis applies to them. For $\bot$, $\alpha \to \beta$ and $\alpha \leftrightarrow \beta$ the property holds vacuously, since these formulas are not in $\{\neg, \wedge, \vee\}$. For an atom, $t_i(p) = p$. In the other cases, write $a = v(t_i(\alpha)) = v(\alpha)$ and $b = v(t_i(\beta)) = v(\beta)$, by the induction hypothesis. Each clause for t_i above uses only the connectives of S_i, applied to formulas in S_i, so $t_i(\varphi)$ is in S_i. For the values, use Definition 3.3.2, and that $\chi \to \bot$ is true exactly when χ is false ($\bot$ is always F):

- $\neg t_i(\alpha)$ and $t_3(\alpha) \to \bot$ are true exactly when $a = \mathrm{F}$, as is $\neg\alpha$.
- $\neg(\neg t_2(\alpha) \vee \neg t_2(\beta))$ is true exactly when neither $\neg t_2(\alpha)$ nor $\neg t_2(\beta)$ is, that is, when $a = b = \mathrm{T}$. And $(t_3(\alpha) \to (t_3(\beta) \to \bot)) \to \bot$ is true exactly when $t_3(\alpha) \to (t_3(\beta) \to \bot)$ is false, that is, when $a = \mathrm{T}$ and $t_3(\beta) \to \bot$ is false, that is, when $a = b = \mathrm{T}$. Both agree with $\alpha \wedge \beta$.
- $\neg(\neg t_1(\alpha) \wedge \neg t_1(\beta))$ is true exactly when a and b are not both F. And $(t_3(\alpha) \to \bot) \to t_3(\beta)$ is true exactly when $t_3(\alpha) \to \bot$ is false or $b = \mathrm{T}$, that is, when $a = \mathrm{T}$ or $b = \mathrm{T}$. Both agree with $\alpha \vee \beta$.

The remaining clauses, for $t_1(\alpha \wedge \beta)$ and $t_2(\alpha \vee \beta)$, keep the connective and have the right value by the induction hypothesis.

■

Now let f be any truth function, and D the formula in $\{\neg, \wedge, \vee\}$ that expresses it. By the Claim, $t_i(D)$ is in S_i and $v(t_i(D)) = v(D) = f(v(p_1), \ldots, v(p_n))$ for every valuation v. So each S_i is adequate.

(b) Let v_{T} be the valuation with $v_{\mathrm{T}}(p) = \mathrm{T}$ for every atom p. We show by structural induction that every formula in $\{\wedge, \vee, \to\}$ is true under v_{T}; the property is "if φ is in $\{\wedge, \vee, \to\}$, then $v_{\mathrm{T}}(\varphi) = \mathrm{T}$". For an atom, $v_{\mathrm{T}}(p) = \mathrm{T}$. The formulas $\bot$, $\neg\alpha$ and $\alpha \leftrightarrow \beta$ are not in the set, so for them the property holds vacuously. If φ is $\alpha \wedge \beta$, $\alpha \vee \beta$ or $\alpha \to \beta$ and is in the set, then so are α and β, and by the induction hypothesis $v_{\mathrm{T}}(\alpha) = v_{\mathrm{T}}(\beta) = \mathrm{T}$. By Definition 3.3.2,

$v_{\mathrm{T}}(\alpha \wedge \beta) = \mathrm{T}$ and $v_{\mathrm{T}}(\alpha \vee \beta) = \mathrm{T}$; and $v_{\mathrm{T}}(\alpha \to \beta) = \mathrm{T}$ because its conclusion β is true. By Theorem 3.2.3, every formula in the set is true under v_{T}.

Now suppose some φ in $\{\wedge, \vee, \to\}$ expressed the negation function f. Then $v_{\mathrm{T}}(\varphi) = f(v_{\mathrm{T}}(p_1)) = f(\mathrm{T}) = \mathrm{F}$, contradicting $v_{\mathrm{T}}(\varphi) = \mathrm{T}$. So no such φ exists, and the set is not adequate. This proves the theorem.

■

In the implication case of (b), the reason is that the **conclusion** β is true. That is where a careless argument goes wrong, and Exercise (C2: Find the error) asked you to find a "proof" that goes wrong there.

The five connectives were chosen because each matches a word mathematicians use, not because they are needed: two suffice, and Exercise (C1: One connective is enough) shows that one well-chosen connective suffices.

Warning. The $\bot$ in $\{\to, \bot\}$ is essential. Without it, $\{\to\}$ is not adequate: every formula in $\{\to\}$ is also in $\{\wedge, \vee, \to\}$, so by (b) of Theorem 3.7.4 none expresses negation. Implication alone cannot say "false"; implication together with a formula that is always false can.

Quick check. Use the translation t_3 to express $p \wedge q$ with $\to$ and $\bot$ only, and check your formula in the row $p = \mathrm{T}$, $q = \mathrm{F}$.

Solution. $t_3(p \wedge q) = (p \to (q \to \bot)) \to \bot$. In the row $p = \mathrm{T}$, $q = \mathrm{F}$: $q \to \bot$ is $\mathrm{F} \to \mathrm{F}$, so T; then $p \to (q \to \bot)$ is T; then the whole is $\mathrm{T} \to \bot$, so F, which agrees with $p \wedge q$.

3.7.7 Exercises

3.7.7.1 A. Check your understanding

Exercise. A1

(a) Which of $\neg p$, $\neg\neg q$, $\bot$, r and $\neg(p \vee q)$ are literals?

(b) Classify each as a clause, a conjunct, both, or neither: $p \vee \neg q$; $\neg r$; $p \wedge (q \vee r)$.

(c) Is $(p \vee q) \wedge (\neg p \vee p \vee r)$ a tautology? Is $(q \vee \neg q) \wedge (\neg r \vee p \vee r)$? Give a refuting row where the answer is no.

(d) Give a formula in DNF that is not satisfiable.

(e) True or false: every formula in CNF with only one clause is also in DNF.

Solution. *(a)* $\neg p$ and r. $\neg\neg q$ has two negations, $\bot$ is not an atom, and $\neg(p \vee q)$ negates a disjunction.

(b) $p \vee \neg q$ is a clause. $\neg r$ is both (one literal). $p \wedge (q \vee r)$ is neither: its part $q \vee r$ is not a literal.

(c) The first is not a tautology: its first clause $p \vee q$ has no pair, and the row $p = q = \mathrm{F}$ (any r) makes it false, hence the whole conjunction. The second is: each clause has a pair ($q, \neg q$ and $r, \neg r$), by Proposition 3.7.3 (a).

(d) $p \wedge \neg p$: its one conjunct contains a pair, so by Proposition 3.7.3 (b) it is not satisfiable.

(e) True. A single clause $\ell_1 \vee \cdots \vee \ell_m$ is a disjunction of m conjuncts of one literal each.

3.7.7.2 B. Practice

Exercise. B1: Normal forms of three formulas

Find a DNF and a CNF of each formula, by either route, and check one of them in two rows.

(a) $p \leftrightarrow \neg q$.
(b) $(p \wedge q) \rightarrow r$.
(c) $\neg(p \vee (q \rightarrow r))$.

Solution. *(a)* By the table: $p \leftrightarrow \neg q$ is true when p and q have opposite values, so its T-rows are T, F and F, T, and its F-rows are T, T and F, F. DNF: $(p \wedge \neg q) \vee (\neg p \wedge q)$, as in Example (Exactly one of two). CNF, one clause per F-row from the opposite literals: the row T, T gives $\neg p \vee \neg q$ and the row F, F gives $p \vee q$, so $(\neg p \vee \neg q) \wedge (p \vee q)$. Check the CNF: in the row T, F both clauses are true, and so is $p \leftrightarrow \neg q$; in the row T, T the first clause is false, and so is $p \leftrightarrow \neg q$.

(b) By rewriting: $(p \wedge q) \rightarrow r \equiv \neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$. This is a single clause, so it is a CNF, and it is also a DNF with three one-literal conjuncts. Check: in the row T, T, F it is false (every literal false), as is $(p \wedge q) \rightarrow r$; in the row F, T, F it is true, as is the original (its hypothesis is false).

(c) By rewriting: $\neg(p \vee (q \rightarrow r)) \equiv \neg p \wedge \neg(q \rightarrow r) \equiv \neg p \wedge \neg(\neg q \vee r) \equiv \neg p \wedge q \wedge \neg r$. This single conjunct is a DNF, and it is also a CNF with three one-literal clauses. Check: in the row F, T, F it is true, and so is the original ($q \rightarrow r$ is false and p is false); in the row F, T, T it is false, and so is the original ($q \rightarrow r$ is true).

Exercise. B2: Two connectives from two

Find formulas in $\{\neg, \wedge\}$, with atoms p and q, equivalent to $p \vee q$ and to $p \leftrightarrow q$. Check the second by a truth table.

Solution. $p \vee q \equiv \neg(\neg p \wedge \neg q)$: this is the translation t_1 in the proof of Theorem 3.7.4, and the right side is F exactly when $\neg p$ and $\neg q$ are both true, that is, when $p \vee q$ is F.

For the biconditional, $p \leftrightarrow q$ says $p \rightarrow q$ and $q \rightarrow p$, and $p \rightarrow q$ says "not p without q", that

is, $\neg(p \wedge \neg q)$. So $p \leftrightarrow q \equiv \neg(p \wedge \neg q) \wedge \neg(\neg p \wedge q)$. The table:

p	q	$\neg(p \wedge \neg q)$	$\neg(\neg p \wedge q)$	$\neg(p \wedge \neg q) \wedge \neg(\neg p \wedge q)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

The last two columns agree in every row.

Exercise. B3: Odd parity

Give an example of a formula in DNF, with atoms p, q, r, that is true in exactly the rows where an **odd** number of p, q, r are true. List the properties your answer must have, and check it against them.

Solution. The rows with an odd number of T's are those with three T's, T T T, and those with one, T F F, F T F, and F F T. One conjunct per row, as in Theorem 3.7.1:

$$(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r).$$

The properties an answer must have: (i) it is in DNF (a disjunction of conjunctions of literals); (ii) its atoms are among p, q, r; (iii) it is T in the four rows T T T, T F F, F T F, F F T; (iv) it is F in the other four, T T F, T F T, F T T, F F F. Check: (i) and (ii) hold by inspection. By Eq. 3.7.2, each conjunct is T in its own row only, so the disjunction is T exactly in the four rows listed, which gives (iii) and (iv).

A different correct answer, for instance the same conjuncts in another order, or with the literals of a conjunct reordered, passes the same checks. No DNF with fewer conjuncts works: a conjunct omitting an atom, say $p \wedge q$, would be T in two rows that differ at that atom, and changing one atom always changes the parity.

3.7.7.3 C. Going deeper

Exercise. C1: One connective is enough

Add to the language two binary connectives: $\alpha \mid \beta$ ("not both", NAND), which is F exactly when α and β are both T, and $\alpha \downarrow \beta$ ("neither", NOR), which is T exactly when α and β are both F.

(a) Prove that $\{|\}$ is adequate.
(b) Prove that $\{\downarrow\}$ is adequate.

Solution. (a) First, two equivalences, checked from the table of $|$: $\alpha \,|\, \alpha$ is F exactly when α is T, so $\alpha|\alpha \equiv \neg\alpha$; and $(\alpha|\beta)|(\alpha|\beta) \equiv \neg(\alpha|\beta)$, which is T exactly when α and β are both T, so it is equivalent to $\alpha \wedge \beta$. Now define, by recursion on formulas as in Theorem 3.2.6, $s(p) := p$, $s(\neg\alpha) := s(\alpha) \,|\, s(\alpha)$, $s(\alpha \wedge \beta) := (s(\alpha) \,|\, s(\beta)) \,|\, (s(\alpha) \,|\, s(\beta))$ (the other clauses do not matter). By structural induction, exactly as in the Claim in the proof of Theorem 3.7.4, every formula

φ in $\{\neg, \wedge\}$ has $s(\varphi)$ in $\{|\}$ and $v(s(\varphi)) = v(\varphi)$ for every v: the atom case is immediate, and the two equivalences above give the $\neg$ and $\wedge$ cases from the induction hypothesis. Given a truth function f, Theorem 3.7.4 (a) gives a formula φ in $\{\neg, \wedge\}$ expressing it, and then $s(\varphi)$ expresses f and is in $\{|\}$.

(b) The same, with $\{\neg, \vee\}$: $\alpha \downarrow \alpha$ is T exactly when α is F, so $\alpha \downarrow \alpha \equiv \neg\alpha$; and $(\alpha \downarrow \beta) \downarrow (\alpha \downarrow \beta) \equiv \neg(\alpha \downarrow \beta) \equiv \alpha \vee \beta$. The recursive translation $s(\neg\alpha) := s(\alpha) \downarrow s(\alpha)$, $s(\alpha \vee \beta) := (s(\alpha) \downarrow s(\beta)) \downarrow (s(\alpha) \downarrow s(\beta))$ turns a formula in $\{\neg, \vee\}$ expressing f into one in $\{\downarrow\}$.

(Recursion and structural induction for the language with the new connectives hold by the same proofs as Theorem 3.2.6 and Theorem 3.2.3, with one more formation clause.)

Marking scheme. (10 points)

- 3: (a) the two equivalences $\alpha \mid \alpha \equiv \neg\alpha$ and $(\alpha \mid \beta) \mid (\alpha \mid \beta) \equiv \alpha \wedge \beta$, read off the table of $|$.
- 3: (a) the recursive translation and the structural induction showing that it keeps the value under every valuation.
- 1: (a) the conclusion, via Theorem 3.7.4 (a).
- 3: (b) the same for $\downarrow$, with $\{\neg, \vee\}$.

Fatal errors (0 for the step): checking the translation on examples only, with no induction.

Exercise. C2: Two sets that are not adequate

(a) Prove that $\{\wedge, \vee, \to, \leftrightarrow\}$ is not adequate.

(b) Add to the language a three-place connective $\mathsf{M}(\alpha, \beta, \gamma)$, **majority**, which is T exactly when at least two of α, β, γ are T. Prove that $\{\neg, \mathsf{M}\}$ is not adequate.

Solution (Hint 1). For (b): compare each row with the row in which every atom's value is flipped.

Solution (Hint 2). For (b): show that every formula in $\neg$ and M takes opposite values in a row and in its flipped row.

Solution. *(a)* Let v_{T} make every atom true. Every formula in $\{\wedge, \vee, \to, \leftrightarrow\}$ is true under v_{T}, by the structural induction in the proof of Theorem 3.7.4 (b) with one more case: if $v_{\mathrm{T}}(\alpha) = v_{\mathrm{T}}(\beta) = \mathrm{T}$, then $v_{\mathrm{T}}(\alpha \leftrightarrow \beta) = \mathrm{T}$, since the two values are equal. As there, no such formula expresses negation, which must give F at the input T.

(b) For a valuation v, let $\bar{v}$ be the **flipped** valuation: $\bar{v}(p)$ is the opposite of $v(p)$, for every atom p. We show by structural induction that every formula φ in $\{\neg, \mathsf{M}\}$ satisfies: for every valuation v, $\bar{v}(\varphi)$ is the opposite of $v(\varphi)$. For an atom this is the definition of $\bar{v}$. If $\bar{v}(\alpha)$ is the opposite of $v(\alpha)$, then $\bar{v}(\neg\alpha)$, the opposite of $\bar{v}(\alpha)$, is $v(\alpha)$, which is the opposite of $v(\neg\alpha)$. If the property holds for α, β, γ, then flipping v flips all three values; at least two of them are T after the flip exactly when at least two were F before, that is, exactly when

at most one was T, that is, exactly when $\mathsf{M}(\alpha,\beta,\gamma)$ was F. So $\bar{v}(\mathsf{M}(\alpha,\beta,\gamma))$ is the opposite of $v(\mathsf{M}(\alpha,\beta,\gamma))$. The other connectives and $\bot$ do not occur in formulas of $\{\neg,\mathsf{M}\}$, so those cases are vacuous. (Structural induction for the extended language holds by the proof of Theorem 3.2.3, with one more formation clause.)

Now the two-input function of $p_1 \wedge p_2$ is F at (T, F) and at the flipped input (F, T). If a formula φ in $\{\neg,\mathsf{M}\}$ expressed it, then for a valuation v with $v(p_1) = \mathrm{T}$, $v(p_2) = \mathrm{F}$ we would have $v(\varphi) = \mathrm{F}$ and $\bar{v}(\varphi) = \mathrm{F}$, since $\bar{v}(p_1) = \mathrm{F}$ and $\bar{v}(p_2) = \mathrm{T}$. But these values must be opposite. So no formula in $\{\neg,\mathsf{M}\}$ expresses it, and the set is not adequate.

Marking scheme. (10 points)

- 3: (a) states the all-T valuation and runs the induction, with the $\leftrightarrow$ case and the vacuous cases ($\neg$, $\bot$) named.
- 1: (a) names a truth function that is not expressed (negation), with the reason.
- 4: (b) states the flipped-row property for **every** formula in $\{\neg,\mathsf{M}\}$ and proves it by structural induction, including the M case.
- 2: (b) exhibits a truth function that is F on a row and on its flipped row, and concludes.

Fatal errors (0 for the step): in (b), checking the flipped-row property on a few formulas instead of proving it for all; in (a), an implication case justified by the hypothesis being true rather than by the conclusion.

Exercise. C3: Exit check: deciding sequents

For each sequent, decide whether it holds. If it does, justify this by a derivation, or by a truth table and the theorem that turns the table into a derivation; if it does not, give a refuting row and say which theorem turns the row into a proof that no derivation exists.

(a) $(p \to q) \vee (p \to r) \vdash p \to (q \vee r)$.

(b) $(p \to q) \to r \vdash p \to r$.

(c) $p \vee (q \wedge r) \vdash (p \vee q) \wedge r$.

(d) $(p \vee q) \wedge \neg p \vdash q \vee r$.

Solution. *(a)* Derivable. The goal is an implication: a box on p, with the goal $q \vee r$; the premise is a disjunction, used by $\vee$E inside the box; in each case $\to$E with p gives one side, and $\vee$I gives the goal.

1	$(p \to q) \vee (p \to r)$	
2	$\quad p$	
3	$\quad\quad p \to q$	
4	$\quad\quad q$	$\to$E 3, 2
5	$\quad\quad q \vee r$	$\vee$I 4
6	$\quad\quad p \to r$	
7	$\quad\quad r$	$\to$E 6, 2
8	$\quad\quad q \vee r$	$\vee$I 7
9	$\quad q \vee r$	$\vee$E 1, 3–5, 6–8
10	$p \to (q \vee r)$	$\to$I 2–9

(b) Not derivable. In the row $p = \mathrm{T}$, $q = \mathrm{F}$, $r = \mathrm{F}$, the premise is true ($p \to q$ is F, so $(p \to q) \to r$ is T) and the conclusion $p \to r$ is F. By Corollary 3.5.3, $(p \to q) \to r \nvdash p \to r$.

(c) Not derivable. In the row $p = \mathrm{T}$, $q = \mathrm{T}$, $r = \mathrm{F}$ (the value of q does not matter), the premise is true because p is, and the conclusion is false because r is. By Corollary 3.5.3, the sequent fails.

(d) Derivable. Split the premise by $\wedge$E, then use $p \vee q$ by $\vee$E: the case p clashes with $\neg p$ and closes by $\bot$E; the case q gives the goal by $\vee$I.

1	$(p \vee q) \wedge \neg p$	
2	$p \vee q$	$\wedge$E 1
3	$\neg p$	$\wedge$E 1
4	$\quad p$	
5	$\quad \bot$	$\neg$E 4, 3
6	$\quad q \vee r$	$\bot$E 5
7	$\quad q$	
8	$\quad q \vee r$	$\vee$I 7
9	$q \vee r$	$\vee$E 2, 4–6, 7–8

Neither derivation uses RAA. As the problem allows, (a) and (d) may instead be justified by a table (or backwards from F) showing the entailment, followed by Theorem 3.6.5.

Marking scheme. (10 points; 2.5 per sequent)

- 1 per sequent: the correct verdict.
- 1.5 per sequent: its justification: a derivation in which every line is justified and every citation available, or a table check with Theorem 3.6.5; or a row that makes **every** premise true and the conclusion false, checked, with Corollary 3.5.3 (or Theorem 3.5.2) cited.

Fatal errors (0 for the sequent's justification): a "refuting row" that makes some premise

false; a derivation citing a line that is not available.

Exercise. C4: Exit check: normal forms

Put $(p \to q) \to (q \wedge r)$ into DNF and into CNF, by either route, and check one of your answers against the truth table.

Solution. By rewriting. Eliminate the outer implication and then the inner one, and push the negation in:

$$\begin{aligned}(p \to q) \to (q \wedge r) &\equiv \neg(p \to q) \vee (q \wedge r)\\ &\equiv \neg(\neg p \vee q) \vee (q \wedge r)\\ &\equiv (p \wedge \neg q) \vee (q \wedge r).\end{aligned}$$

The last line is a DNF. Here $\neg(p \to q) \equiv p \wedge \neg q$, since an implication is false exactly when its hypothesis is true and its conclusion false. For a CNF, distribute:

$$(p \wedge \neg q) \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \wedge (\neg q \vee q) \wedge (\neg q \vee r),$$

and drop the clause $\neg q \vee q$, which is true in every row: the CNF is $(p \vee q) \wedge (p \vee r) \wedge (\neg q \vee r)$. (The clause $p \vee r$ is redundant too: $(p \vee q) \wedge (\neg q \vee r)$ is an equivalent CNF, as splitting on q shows. Either is correct.)

The check. The formula is false exactly when $p \to q$ holds and $q \wedge r$ fails. Its table:

p	q	r	$p \to q$	$q \wedge r$	$(p \to q) \to (q \wedge r)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	T	F	F

The T-rows are T T T, T F T, T F F, and F T T. The DNF $(p \wedge \neg q) \vee (q \wedge r)$ is true in exactly these: $p \wedge \neg q$ in T F T and T F F, and $q \wedge r$ in T T T and F T T. In each of the four F-rows the CNF has a false clause: $\neg q \vee r$ in T T F and F T F, $p \vee q$ in F F T and F F F.

By the table route, the DNF has four conjuncts, one per T-row, and the CNF four clauses, one per F-row, for instance $\neg p \vee \neg q \vee r$ for the row T T F. Any correct DNF and CNF earns full marks; a CNF is not unique.

Marking scheme. (10 points)

- 2: eliminates $\to$ correctly, with $\neg(p \to q) \equiv p \wedge \neg q$ and its reason.
- 3: a correct DNF.
- 3: a correct CNF, by distribution carried out in full or by one clause per F-row.
- 2: the check against the table, covering at least the four F-rows (or the four T-rows).

Fatal errors (0 for the step): $\neg(p \to q)$ rewritten as $\neg p \to \neg q$; a "CNF" with a conjunction inside a clause.

3.7.7.4 R. Review

Exercise. R1

Find a derivation of $p \lor q,\ \neg(p \land r),\ r \vdash q$.

Solution. The method of Section 2.6 (Example (Proof by cases, found by the method) shows it on a proof by cases). The goal q is an atom, so no introduction rule applies; the premise $p \lor q$ is a disjunction, so we use it by $\lor$E, with the goal q in both boxes. The case q is the goal. In the case p, nothing gives q directly, so we look for a clash: p and the premise r build $p \land r$ by $\land$I, which clashes with $\neg(p \land r)$, and $\bot$E closes the impossible case (Example (Manufacturing the other half of a clash)).

1	$p \lor q$	
2	$\neg(p \land r)$	
3	r	
4	$\quad p$	
5	$\quad p \land r$	$\land$I 4, 3
6	$\quad \bot$	$\neg$E 5, 2
7	$\quad q$	$\bot$E 6
8	$\quad q$	
9	q	$\lor$E 1, 4–7, 8–8

The second box is its assumption alone, which is already the goal. No RAA is used.

Exercise. R2

Let m and n be integers, and consider the argument: "$m + n$ is even only if m and n are both even or both odd. m is even and n is odd. Therefore $m + n$ is not even." Translate it into a sequent, using atoms e: "$m + n$ is even", a: "m is even", b: "n is even", and decide whether the premises entail the conclusion.

Solution. "P only if Q" is $P \to Q$ (Section 1.1, and Exercise (C1: "Only if" and "unless")). "Both odd" is $\neg a \land \neg b$, since an integer is odd exactly when it is not even. So the sequent is

$$e \to ((a \land b) \lor (\neg a \land \neg b)),\quad a \land \neg b \ \models\ \neg e.$$

Work backwards from F. A refuting row needs $\neg e$ false, so $e = \mathrm{T}$, and the second premise true, so $a = \mathrm{T}$, $b = \mathrm{F}$. In that row $a \land b$ and $\neg a \land \neg b$ are both F, so the first premise is $\mathrm{T} \to \mathrm{F}$, which is F. So no row makes both premises true and the conclusion false: the entailment holds. By Theorem 3.6.5, the sequent is also derivable. (The eight-row table gives the same

answer; working backwards found the only candidate row at once.)

Exercise. R3

Is $(p \wedge (q \vee r)) \to ((p \wedge q) \vee r)$ a tautology? Is it derivable without RAA?

Solution. A tautology. Work backwards from F (Section 3.3). The formula is false only if $p \wedge (q \vee r)$ is true and $(p \wedge q) \vee r$ is false. The second needs r false and $p \wedge q$ false; the first needs p true and $q \vee r$ true, so with r false, q true. But then $p \wedge q$ is true. No row makes the formula false, so it is a tautology (Definition 3.4.1).

Derivable without RAA. Completeness (Theorem 3.6.5) says it is derivable, but not how, and its proof uses LEM in Lemma 3.6.4, so it does not answer the second question. A search does: the goal is an implication, so a box on $p \wedge (q \vee r)$; split it by $\wedge$E and use $q \vee r$ by $\vee$E.

1	$p \wedge (q \vee r)$	
2	p	$\wedge$E 1
3	$q \vee r$	$\wedge$E 1
4	q	
5	$p \wedge q$	$\wedge$I 2, 4
6	$(p \wedge q) \vee r$	$\vee$I 5
7	r	
8	$(p \wedge q) \vee r$	$\vee$I 7
9	$(p \wedge q) \vee r$	$\vee$E 3, 4–6, 7–8
10	$(p \wedge (q \vee r)) \to ((p \wedge q) \vee r)$	$\to$I 1–9

No RAA is used: the boxes close by $\vee$E and $\to$I. The tools: truth values (Definition 3.3.2) for the first question, the search of Chapter 2 for the second. The look-alike tool, completeness, answers only "derivable with all the rules".

Chapter 4

Predicate Logic and the Quantifier Rules

"$x^2 \geq 0$ for every real x." "There is no largest prime." "For every $\varepsilon > 0$ there is an N such that $|a_n - L| < \varepsilon$ for every $n \geq N$." The statements of analysis are made of the words "for every" and "there exists", and Chapters 1–3 cannot see them: to a truth table, each of these sentences is a single atom, and the argument "every positive real has a square root; 2 is positive; so 2 has a square root" has the invalid form $p, q \therefore r$. This chapter adds the missing grammar (predicates, terms, and the two quantifiers) and then, exactly as Chapters 1 and 2 did for the connectives, gives the rules **one per section**: how to use a "for every", how to prove one, how to prove a "there exists", how to use one, and the rules for equality. Each quantifier rule comes with a side condition about names, and the side conditions turn out to be exactly the rules that analysts break when a δ secretly depends on an x, or when a witness is treated as if it were arbitrary.

The motivating problem is this. A function f is continuous when for every x and every $\varepsilon > 0$ there is a $\delta > 0$ that works; it is uniformly continuous when for every $\varepsilon > 0$ there is a $\delta > 0$ that works for every x. The two definitions differ only in the order of two quantifiers. *Why does "uniformly continuous implies continuous" have a proof a few lines long, while its converse is false?* The answer is a fact about the order in which names are introduced in a derivation, and Section 4.10 reads it off rule by rule. Along the way the chapter translates mathematical definitions into formulas, negates them quantifier by quantifier, gives formulas a meaning in **models**, and uses models to show that some searches for a derivation must fail. The last section, which is optional, proves that the extended rules are sound.

What you need. Chapters 1–3. From Chapters 1 and 2: the rules for every connective, the definition of a derivation with its boxes and its scope rule (Definition 1.4.2), derived rules (Definition 2.3.1), the classical rules of Section 2.5, and the search method of Section 2.6, which each rule section here extends by one step. From Chapter 3: formulas as objects (Definition 3.1.1), structural induction and recursion (Theorem 3.2.3, Theorem 3.2.6), truth tables, and soundness with its refuting rows (Theorem 3.5.2, Corollary 3.5.3). School arithmetic and the order of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are used in examples as facts.

Time. About 70 hours on the core track (all of A, one B, and one C per section, the exit checks); about 90 hours for everything.

4.0.1 Roadmap

- **Predicates and quantifiers.** Predicates, variables, and the domain; how to read "for every" and "there exists", bounded quantifiers, and why the order of two quantifiers can change everything.
- **Terms, formulas, free variables, and substitution.** The exact language, which letters are still holes, the operation "put t in for x" that every rule uses, and the one way it goes wrong: capture.
- **Using "for every": instantiation.** The rule ∀E: a universal statement may be applied to any term, and the term is read off the goal.
- **Proving "for every": let x be arbitrary.** The rule ∀I and the flagged box: prove the statement for a fresh name about which nothing is assumed.
- **Proving "there exists": exhibit a witness.** The rule ∃I, and why the witness is chosen only when an available line shows it.
- **Using "there exists": name a witness.** The rule ∃E, its box, and its fresh-name condition, and the first derivation that moves an ∃ past a ∀ (and where the reverse move breaks down).
- **Equality and unique existence.** The rules $=$ I and $=$ E, calculation chains, "there is exactly one", and the completed table of rules, phrases, and proof-assistant commands.
- **Translating mathematics.** Definitions from analysis written as formulas with nested and bounded quantifiers, the bounded forms of the four rules, and the search method in one place.
- **Negating and moving quantifiers.** "Not every" is "some not", negation pushed inward quantifier by quantifier, and the laws that move a quantifier across a connective.
- **Uniformly continuous implies continuous, rule by rule.** The chapter's centerpiece: the full derivation, and the exact lines at which the converse breaks down.
- **Models and counter-models.** A meaning for every formula, in a structure with an assignment; a counter-model as a proof that a sentence does not follow.
- **Soundness for predicate logic** (optional). The soundness theorem of Chapter 3 extended to the quantifier and equality rules, with each side condition used in its own case.

4.0.2 Named moves

- **Instantiate at the term you need** (∀E): read the term off the goal, not off the order of the premises.
- **Let x be arbitrary** (∀I): a fresh name, about which nothing is assumed.
- **Postpone the witness** (∃I): choose the term only when an available line shows it.
- **Name the witness early** (∃E): "choose x_0 with …", open the box at once, and never generalize over it.
- **Fix, obtain, choose:** ∀I, then ∀E and ∃E on the hypothesis, then ∃I with a term built from what is available. The shape of most proofs in analysis.
- **Negate quantifier by quantifier:** push ¬ inward, turning ∀ into ∃ and back, and keep the bounds.
- **Refute with a model:** a structure in which the premises are true and the conclusion false.
- **Read the dependency off the order:** a variable may depend on the variables quantified to its left, and only on them.

4.1 Predicates and Quantifiers

Chapters 1–3 built a complete account of the connectives: rules for using and proving "and", "or", "implies", and "not", a meaning for every formula through truth tables, and the theorems that the rules prove exactly what the tables say. In all of that, an atomic statement was a closed box. The statements of analysis are not boxes. "For every $\varepsilon > 0$ there is an N such that ..." has its whole content inside, in the words "for every" and "there is" and in the letters they govern. This section opens the box. It introduces predicates, variables, the domain a statement talks about, and the two quantifiers $\forall$ and $\exists$, and it teaches how to **read** a quantified sentence. It proves no theorem and states no rule: the four quantifier rules come one per section, in Sections 4.3 to 4.6.

4.1.1 An argument the truth table rejects

Here is an argument that nobody would doubt.

> Every positive real number has a square root. The number 2 is positive. Therefore 2 has a square root.

It is correct in the plainest sense: if the two premises are true, the conclusion must be. Now try to check it with the tools of Chapters 1–3. The three sentences contain no connective at all, so each is atomic. Give them atoms:

- p: "every positive real number has a square root";
- q: "2 is positive";
- r: "2 has a square root".

The argument has the form $p, q \therefore r$. Three different atoms, and nothing connects them. The valuation that makes p and q true and r false makes both premises true and the conclusion false. So by Corollary 3.5.3,

$$p,\ q \nvdash r.$$

No derivation, with any rule of Chapters 1 and 2, leads from p and q to r.

There is no contradiction here, and it is worth seeing exactly why. Corollary 3.5.3 is about the **form** $p, q \therefore r$: it says that no derivation leads from two arbitrary statements to a third, unrelated one. That is true. Our argument is correct for a reason that the form cannot see. The first premise speaks about **every** positive real number. The second says that 2 is **one** of the things the first premise speaks about. The conclusion says of 2 what the first premise said of all of them. The three statements share parts: a property ("is positive"), another property ("has a square root"), an object (2), and the word "every". Chopped into atoms, the shared parts disappear.

This is the warning of Section 3.3 come true: a truth table sees only the connectives, and everything inside an atom is invisible to it. Section 3.3 ended that warning with the remark that this chapter looks inside atoms. We now do. The first job is a vocabulary for the parts inside: the properties, the objects, and the words "every" and "some". Section 4.3 then gives the rule that makes the argument above a derivation, four lines long.

4.1.2 Predicates: statements with holes

Chapter 1 met the sentence "$x > 2$" and called it an **open sentence** (item (e) of Example (Statements and nonstatements)). It is not a statement, because its truth depends on what x is: it is true when x is 3 and false when x is 0. Put a number in for x and it becomes a statement.

The argument of the last subsection is built from two open sentences of this kind: "x is positive" and "x has a square root". Each has one hole, marked by x. The premise "2 is positive" is the first open sentence with 2 put into the hole; the conclusion is the second, with 2 put into its hole. The first premise does something else with the holes, which we come to in the next subsection. Open sentences are the missing parts, and they deserve a name.

A predicate is a statement with holes in it; filling every hole with a name of an object gives a statement.

A **predicate** is an open sentence with one or more marked holes. The number of holes is its **arity**: a predicate with one hole is **one-place** (or unary), with two holes **two-place** (binary), and so on. We write a predicate as a capital letter followed by its holes in parentheses: $P(x)$ for "x is positive", $S(x)$ for "x has a square root", $D(x, y)$ for "x divides y". The letters that mark the holes are **variables**: x, y, z, u, and in analysis also $\varepsilon, \delta, N, n$, and for primes and other natural numbers p, q. A **constant** is a name of one particular object: 0, 1, 2, or a letter a, b, c that has been given a fixed meaning.

In words: a predicate says something **about** objects without saying which objects. The variables are placeholders, not objects. "x is positive" is about whatever will be put in for x, and until something is, it is neither true nor false.

Many predicates of mathematics are written with a symbol between their two holes rather than a letter in front: $x < y$, $x \leq y$, $x = y$, $x \mid y$. These are two-place predicates like any other; only the way of writing them differs. Membership $x \in A$, for a fixed set A, is a one-place predicate of x: "x is an element of A". *Sets, Relations, and Cardinality*, Chapter 1 develops sets and membership properly. Until then a set is understood informally, as a collection of objects (usually numbers), and nothing in this section uses more about a set A than whether a given object is an element of it. For sets A and B, $A \subseteq B$ means that every element of A is an element of B; so $A \subseteq \mathbb{R}$ says that A is a set of real numbers.

Objects can also be built from other objects: $x + 1$, x^2, $|x|$, $2 \cdot y$. The symbols that build them ($+$, squaring, $|\cdot|$, $\cdot$) are **function symbols**, and an expression such as $x^2 + 1$ names an object as soon as x does. A function symbol may also be a letter, $f(x)$. *Sets, Relations, and Cardinality*, Chapter 1 defines functions as sets of pairs; here a function symbol is simply a way of naming one object in terms of others. Section 4.2 makes all of this exact, under the name **term**. For now, a term is any expression that names an object: a variable, a constant, or a function symbol applied to terms.

Filling holes is the first way to turn a predicate into a statement.

- $P(2)$: "2 is positive". A statement, and true.
- $P(-3)$: "-3 is positive". A statement, and false.
- $D(3, 12)$: "3 divides 12". True.
- $D(3, y)$: "3 divides y". Still open: one hole is filled and one is not. It is a one-place predicate of y.
- $P(x^2 + 1)$: "$x^2 + 1$ is positive". Still open, in x. (It happens to be true whatever real number x is, but that is a fact about the numbers, and the sentence still has a hole.)

The last two items show that filling holes is not all or nothing. A predicate with some holes filled is a predicate with fewer holes. Only when **every** hole is filled, or closed in the way described next, do we have a statement.

Remark. A clash of letters to keep apart. In Chapters 1 and 2, capital letters P, Q, R stood for whole statements, and Chapter 3 read them as atoms or as arbitrary formulas. In this

chapter P, Q, R are **predicate symbols**, followed by their holes: $P(x)$, $R(x, y)$. A bare P or Q in this chapter, in a reference back to Chapters 1–3 or in a review exercise, is a statement letter as before. The parentheses tell the two uses apart. The small letters p, q, r were atoms in the argument that opened this section, which was written in the language of Chapter 3, and they are atoms again wherever a later section reads a formula with a truth table (Section 4.11). A formula of this chapter has no atoms (Section 4.2 replaces them by predicates with their holes), so inside one, p and q are variables, used for primes and other natural numbers. A property $P(n)$ of natural numbers, as in the principle of induction (Axiom 3.2.1), was already a one-place predicate in this sense.

4.1.3 Quantifiers and the domain

There is a second way to turn "x is positive" into a statement, and the first premise of our argument uses it. The premise does not put a particular number into the hole of "x has a square root". It says that the open sentence "if x is positive, then x has a square root" is true **whatever** real number is put in. The word "every" closes the hole without filling it.

Two questions have to be answered before "whatever is put in" makes sense. What may be put in? And how many of those things must make the sentence true?

The first question is answered by a **domain of discourse**, or **domain**: the collection of objects that the variables range over. For the argument above, the domain is the real numbers, $\mathbb{R}$. In number theory it is often $\mathbb{Z}$ or $\mathbb{N}$ (in this book $\mathbb{N} = \{0, 1, 2, \dots\}$, with 0). A domain always has at least one element; Section 4.11 explains why the book insists on this, and how it handles the empty set instead.

The second question is answered by the **quantifier**.

"For every" promises something of each object in the domain; "there exists" promises it of at least one.

Let φ be an open sentence in which the variable x marks a hole (it may have other holes too). There are two quantifiers.

- The **universal quantifier**: $\forall x\, \varphi$, read "**for every** x, φ", "for all x, φ", or "for each x, φ". It is **true** when **every** element of the domain, put in for x, makes φ true.
- The **existential quantifier**: $\exists x\, \varphi$, read "**there exists** an x such that φ", "there is an x with φ", or "for **some** x, φ". It is **true** when **at least one** element of the domain, put in for x, makes φ true.

In words: $\forall x$ and $\exists x$ each close the hole marked x, so they remove x from the list of holes. $\forall x\, \varphi$ makes one claim about the whole domain at once: no element is an exception. $\exists x\, \varphi$ claims that there is at least one element for which φ holds, and it does not say which, or how many. "At least one" includes the case of exactly one and the case of all of them. The word "some" in mathematics never means "some but not all".

With these, the first premise of our argument, over the domain $\mathbb{R}$, is

$$\forall x\, (P(x) \to S(x)),$$

"for every x, if x is positive, then x has a square root". The second premise is $P(2)$ and the conclusion is $S(2)$. Now the shared parts are visible: P occurs in both premises, S in the first premise and the conclusion, and 2 in the second premise and the conclusion. The rule of Section 4.3 uses the first premise with x replaced by 2, giving $P(2) \to S(2)$, and $\to$E finishes.

Notice the $\to$ inside the quantifier. The premise does not say that every real number is positive and has a square root; it says that every real number which is positive has one. The subsection on bounded quantifiers below explains why a "for every" restricted to part of the domain always comes with an implication.

Remark. "True" in this section, and up to Section 4.11, means what it means in ordinary mathematics: we look at the domain and ask whether every element, or some element, makes the sentence true. Section 4.11 turns this reading into a definition, with models. Until then the reading is used only to understand sentences and to decide whether they are true as statements about numbers. Every claim that a sentence can be **derived** from others will be proved by writing a derivation, never by this reading.

4.1.3.1 The same sentence over different domains

A quantified sentence is a claim about its domain, so changing the domain can change its truth value. Two examples show how.

Take $\forall x\,(x \geq 0)$, "every x is at least 0".

- Over $\mathbb{N}$ it is **true**: every natural number is at least 0.
- Over $\mathbb{Z}$ it is **false**: the integer -1 is in the domain, and $-1 \geq 0$ is false.

One element of the domain that makes φ false is enough to make $\forall x\,\varphi$ false. Such an element is a **counterexample**. To show that the sentence is true over $\mathbb{N}$, on the other hand, we had to say something about every natural number at once.

Take $\exists x\,(x^2 = 2)$, "some x has square 2".

- Over $\mathbb{R}$ it is **true**: the real number $\sqrt{2}$ has square 2. (Its existence is a school fact here; *Analysis on the Real Line*, Chapter 1 proves it from the axioms of the real numbers.)
- Over $\mathbb{Q}$ it is **false**: that no rational number has square 2 is exactly Theorem 2.2.1.

One element that makes φ true is enough to make $\exists x\,\varphi$ true. Such an element is a **witness**. To show that the sentence is false over $\mathbb{Q}$, on the other hand, a theorem about every rational number was needed, and Chapter 2 proved it by contradiction.

The two quantifiers are mirror images. For $\forall$, a single counterexample refutes and a general argument confirms. For $\exists$, a single witness confirms and a general argument refutes. Section 4.9 makes the mirror exact: "not every x satisfies φ" says the same as "some x fails φ".

Quick check. Over which of the domains $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$, $\mathbb{Z}$, $\mathbb{R}$ is the sentence $\exists x\,(x + x = x)$ true? Give a witness where it is true, and a reason where it is false.

Solution. Only $x = 0$ satisfies $x + x = x$, since subtracting x from both sides gives $x = 0$. So the sentence is true over $\mathbb{Z}$ and over $\mathbb{R}$, with the witness 0. It is false over $\mathbb{Z}_{>0}$, which does not contain 0: every $n \in \mathbb{Z}_{>0}$ has $n + n > n$, since $n > 0$. (Over $\mathbb{N} = \{0, 1, 2, \dots\}$ it would be true, again with the witness 0.)

4.1.3.2 Reading a quantified sentence

Quantified sentences are read from the outside in: the first quantifier governs everything after it, and inside it may sit another quantifier, a connective, or a predicate. Reading aloud slowly, one symbol at a time, is the most reliable way to understand one. Here are three, simplest first.

Example. Reading three sentences

Read each sentence aloud, say what it claims, and decide whether it is true over the domain given.

(a) $\forall x\,(x^2 \geq 0)$, over $\mathbb{R}$.

(b) $\exists x\,\forall y\,(x \leq y)$, over $\mathbb{N}$.

(c) $\neg\exists p\,(\text{Prime}(p) \wedge \forall q\,(\text{Prime}(q) \to q \leq p))$, over $\mathbb{N}$, where $\text{Prime}(p)$ is the one-place predicate "p is prime".

Solution. (a) "For every x, x^2 is at least 0." The claim is about every real number, and it is true: the square of a real number is never negative (a school fact about signs: $x \cdot x$ is a product of two numbers of the same sign, or 0).

(b) "There is an x such that for every y, $x \leq y$." Unpack from the outside. The outer quantifier $\exists x$ asks for one natural number x. What must be true of it is the inside, $\forall y\,(x \leq y)$: every natural number y is at least x. So the sentence says that $\mathbb{N}$ has a **least element**. It is true, with the witness $x = 0$: every natural number is at least 0.

(c) The outer symbol is $\neg$, so the sentence denies what follows. What follows is $\exists p\,(\dots)$: there is a p that is prime and such that every prime q is at most p. Such a p would be a **largest prime**. So the sentence says "there is no largest prime". It is true: this is the ancient theorem that there are infinitely many primes. Nothing in this book depends on that theorem; here it is only something to read.

Notice the shape inside (c): $\forall q\,(\text{Prime}(q) \to q \leq p)$, "every prime is at most p", has an implication under the $\forall$. It speaks about primes only, and the implication is how a "for every" is restricted to them. The next subsection makes this a rule of writing.

The sentence in (b) has two quantifiers, and they come in a definite order: first $\exists x$, then $\forall y$. The order matters, and the subsection "The order of quantifiers" below is about it. First, restricted quantifiers.

4.1.4 Bounded quantifiers

Mathematics rarely says "for every real number". It says "for every $\varepsilon > 0$", "for every $x \in A$", "there is an $n \geq N$". The quantifier ranges over a **part** of the domain, described by a condition. There is no new quantifier for this; the condition is written into the sentence with a connective. Which connective turns out to matter.

A "for every" restricted to a condition is an implication; a "there exists" restricted to a condition is a conjunction.

Definition 4.1.1. Bounded Quantifier

Let A be a set, x a variable, and φ an open sentence. The **bounded quantifiers** are the abbreviations

$$\forall x \in A\, \varphi \quad \text{stands for} \quad \forall x\, (x \in A \to \varphi),$$
$$\exists x \in A\, \varphi \quad \text{stands for} \quad \exists x\, (x \in A \wedge \varphi).$$

In the same way, for a term a in which x does not occur, $\forall x > a\, \varphi$ stands for $\forall x\, (x > a \to \varphi)$ and $\exists x > a\, \varphi$ for $\exists x\, (x > a \wedge \varphi)$, and likewise with $\geq$, $<$, $\leq$ in place of $>$.

In words: "for every x in A, φ" means "for every x whatsoever, **if** x is in A, **then** φ". "There is an x in A with φ" means "there is an x that is in A **and** satisfies φ". The condition $x \in A$, or $x > a$, is the **bound**. A bounded quantifier is not a new symbol of the language: it is shorthand, and every statement about it is a statement about the formula it stands for.

The display after Axiom 3.2.1, in Section 3.2, used $\forall n \in \mathbb{N}$ as an abbreviation of the words "for every natural number n". It was already this bounded form: $\forall n \in \mathbb{N}\, \big(P(n) \to P(n+1)\big)$ stands for $\forall n\, \big(n \in \mathbb{N} \to (P(n) \to P(n+1))\big)$.

Why $\to$ under $\forall$, and $\wedge$ under $\exists$. Try the other choices and see what they say. Take the domain $\mathbb{R}$, the set $A = \{1, 2\}$, and the open sentence $x > 5$.

- "Every element of A is greater than 5" is false: 1 is not. The correct form $\forall x\, (x \in A \to x > 5)$ is false too, with the counterexample $x = 1$, which is in A and not greater than 5. The wrong form $\forall x\, (x \in A \wedge x > 5)$ says that **every real number** is in A and greater than 5. It is false as well, but for a reason that has nothing to do with A's elements: $x = 0$ is not in A. Change A to $\{6, 7\}$ and the correct form becomes true while the wrong one stays false.
- "Some element of A is greater than 5" is false for $A = \{1, 2\}$. The correct form $\exists x\, (x \in A \wedge x > 5)$ is false too: no real number is both in A and greater than 5. The wrong form $\exists x\, (x \in A \to x > 5)$ is **true**, with the witness $x = 0$: since $0 \notin A$, the implication "if $0 \in A$ then $0 > 5$" has a false hypothesis and is true (vacuous truth, the phenomenon behind Proposition 2.1.2).

So a $\forall$ with $\wedge$ inside claims too much: it says the whole domain lies in A. An $\exists$ with $\to$ inside claims too little: anything outside A is a witness. Neither says what was meant.

Warning. An existential quantifier with an implication directly inside, $\exists x\, (\alpha \to \beta)$, is almost always a mistranslation. Over $\mathbb{R}$, "some element of $\{1, 2\}$ exceeds 5" is false, but $\exists x\, (x \in \{1, 2\} \to x > 5)$ is true, because $x = 0$ makes the hypothesis false. When you write $\exists$ and then $\to$, check whether you meant $\wedge$.

One domain per formula. The logic of this book has a single domain for all the variables of a formula. Mathematics often mixes kinds of objects in one sentence: in "for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that for every $n \geq N$, ...", the letter ε is a real number, while N and n are natural numbers. The convention is: **the domain is the larger kind** (here $\mathbb{R}$), and **the smaller kind enters as a bound** ($\exists N \in \mathbb{N}$, $\forall n \in \mathbb{N}$). So the sentence begins

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \in \mathbb{N}\, \big(n \geq N \to \dots\big),$$

and n ranges over real numbers that happen to be required, by the bound, to be natural numbers. A sequence (a_n) becomes a function symbol a, with $a(n)$, written a_n, the n-th term. Its values at real numbers that are not natural numbers are never used, because a is only ever applied to variables bounded by $n \in \mathbb{N}$. We state this convention once, here; every translation in this chapter follows it.

The empty set. Bounded quantifiers make sense even when the bound is never satisfied. Let $\emptyset$ be the **empty set**, the set with no elements (*Sets, Relations, and Cardinality*, Chapter 1 introduces it properly), so that $x \in \emptyset$ is false for every x.

- $\forall x \in \emptyset\ \varphi$ stands for $\forall x\,(x \in \emptyset \to \varphi)$. For every x, the hypothesis $x \in \emptyset$ is false, so the implication is true. The sentence is **true**, whatever φ says. For example, "every element of $\emptyset$ is greater than 5" and "every element of $\emptyset$ is less than 5" are both true.
- $\exists x \in \emptyset\ \varphi$ stands for $\exists x\,(x \in \emptyset \wedge \varphi)$. No x makes the first conjunct true, so the sentence is **false**, whatever φ says.

Section 2.1 met this phenomenon for implications: a statement whose hypothesis never holds is true, and says nothing. A universal statement about the elements of an empty set is true for the same reason, because there is nothing to test. This is the reading; Section 4.4 derives the universal half with the rules.

Warning. "Every element of A has property φ" does **not** imply "some element of A has property φ". It fails when A is empty: "every element of $\emptyset$ is greater than 5" is true, and "some element of $\emptyset$ is greater than 5" is false. The step from "every" to "some" needs an element, and the bound can remove all of them.

4.1.5 The order of quantifiers

A sentence with two quantifiers of different kinds can change its meaning completely when they are swapped. This is the single most important fact about reading quantifiers, and the definitions of analysis depend on it.

Example. Swapping two quantifiers

Over the domain $\mathbb{R}$, compare

$$\text{(i)}\ \ \forall x\,\exists y\,(x + y = 0) \qquad \text{and} \qquad \text{(ii)}\ \ \exists y\,\forall x\,(x + y = 0).$$

Read each aloud, and decide which is true.

Solution. (i) "For every x there is a y with $x + y = 0$": every real number has an additive inverse. It is **true**. Given any real x, the number $y = -x$ satisfies $x + y = x + (-x) = 0$.

(ii) "There is a y such that for every x, $x + y = 0$": one single number y is an inverse of every real number at once. It is **false**. Whatever real number y is proposed, take $x = 1 - y$; then $x + y = 1 \neq 0$. So no y works.

The two sentences have the same letters, the same quantifiers and the same equation. Only the order differs. In (i), the y may be chosen **after** x is known, so it may depend on x: the witness $-x$ is a different number for each x. In (ii), y must be chosen **first**, once and for all, and then it must work for every x that comes after.

What makes the pair work as an example? Two properties, and any other pair with the same shape would do as well. First, for each x separately there is a suitable y. Second, no one y is suitable for all x together, because each y serves only one x (here, only $x = -y$).

Quantifiers as a game. The difference in Example (Swapping two quantifiers) can be felt by reading a sentence as a game between two players. The reading applies to a sentence whose quantifiers all stand at the front, unnegated, followed by an inside with no quantifiers, as in (i) and (ii); a bounded quantifier such as $\forall \varepsilon > 0$ counts, its bound being checked with the inside. The **challenger** chooses a value for each $\forall$ variable and tries to make the sentence false. The **defender** chooses a value for each $\exists$ variable and tries to make it true. They move in the order in which the quantifiers are written, left to right, and each player sees every move made so far. When all variables have values, the inside is checked. The sentence is true exactly when the defender has a way of playing that wins against every choice of the challenger. A quantifier under a $\neg$, or in the hypothesis of an implication, changes sides: $\neg\forall x\, \varphi$ is played as $\exists x\, \neg\varphi$, with the defender moving. Section 4.9 proves the laws that push a $\neg$ inward in this way.

- In (i), the challenger moves first and picks some x. The defender sees x and answers $y = -x$. The defender always wins.
- In (ii), the defender moves first and must commit to some y without knowing x. The challenger sees y and answers $x = 1 - y$. The challenger always wins.

Read the dependency off the order: a variable may depend on the variables quantified to its left, and only on them. In (i), y is to the right of x, so the choice of y may use x. In (ii), y is to the left, so it may not. Sections 4.4 to 4.6 will show this same dependence as a fact about the boxes of a derivation, and Section 4.10 shows that it is the whole difference between continuity and uniform continuity.

When swapping is harmless. Two quantifiers of the **same** kind may be swapped. $\forall x\, \forall y\, \varphi$ and $\forall y\, \forall x\, \varphi$ both say that φ holds for every choice of the pair; in the game, the challenger makes both moves, and the order of the challenger's own moves gives nothing away. Likewise $\exists x\, \exists y\, \varphi$ and $\exists y\, \exists x\, \varphi$ both say that some pair works. Section 4.4 derives the first swap with the rules. For quantifiers of **different** kinds, one direction is always safe: if one y works for every x, then for every x some y works (the same one). Section 4.6 derives this, $\exists y\, \forall x\, \varphi$ to $\forall x\, \exists y\, \varphi$, and Example (Swapping two quantifiers) shows that the other direction fails.

Quick check. Over the domain $\mathbb{N}$, swap the two quantifiers in "for every n there is a prime larger than n", that is, in $\forall n\, \exists p\, (\text{Prime}(p) \wedge p > n)$. Read the result aloud. Is it true?

Solution. The swapped sentence is $\exists p\, \forall n\, (\text{Prime}(p) \wedge p > n)$: "there is a prime that is larger than every natural number". It is **false**. Whatever p is proposed, take $n = p$: then $p > n$ is false, so the inside fails. (The original sentence is true; it is another form of the theorem that there are infinitely many primes, which we do not use.) As in Example (Swapping two quantifiers), the original lets the prime depend on n, and the swapped sentence demands one prime for all n at once.

4.1.6 Words that hide quantifiers

English marks quantifiers loosely, and mathematical English has habits that hide them altogether. Three traps are worth naming now, before they cause trouble in a translation.

Warning. The word **"any"** can mean "every" or "some", and only the sentence around it decides. "Any x in A is positive" means every element of A is positive: $\forall x \in A\ (x > 0)$. "If

any x in A is positive, then A is nonempty" means if **some** element is positive: $\exists x \in A\,(x > 0) \to \ldots$. When you meet "any", rewrite the sentence with "every" and with "some", and keep the one that says what was meant.

Hidden universal quantifiers. A theorem stated as "if n is even, then n^2 is even" has no quantifier word in it, and yet it is not about one number. It means: for every integer n, if n is even, then n^2 is even. A letter that is not introduced anywhere, in the statement of a theorem, is read with a "for every" in front. Chapter 5 names this convention and lists the others that written mathematics relies on. When translating, put the hidden quantifier back first.

An implication inside $\exists$. This is the warning after Definition 4.1.1, and it deserves repeating because it comes up in translations of analysis: "there is an N such that, if $n \geq N$, then ..." is not the shape of the sentence displayed under **One domain per formula** above, where the implication $n \geq N \to \ldots$ sits under $\forall n \in \mathbb{N}$. The implication belongs under a $\forall$: "there is an N such that **for every** n, if $n \geq N$, then ...".

4.1.7 What comes next

We can now read quantified sentences and decide, as mathematicians, whether they are true. We cannot yet **reason** with them in the sense of Chapters 1 and 2: there is no rule for using a "for every", for proving one, for proving a "there exists", or for using one. Section 4.2 first makes the language exact, because the rules will need to say precisely what "put 2 in for x" means. Sections 4.3 to 4.6 then give the four rules, one per section, in the same way as Chapters 1 and 2 gave the rules for the connectives.

4.1.8 Exercises

4.1.8.1 A. Check your understanding

Exercise. A1

(a) Read each sentence aloud, and say over which of the domains $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$ it is true: (i) $\forall x\,(x{+}1 > x)$; (ii) $\exists x\,(x{+}1 = 0)$; (iii) $\forall x\,\exists y\,(y < x)$; (iv) $\exists x\,(2{\cdot}x = 1)$; (v) $\forall x\,(x{\cdot}x \geq x)$.

(b) With domain $\mathbb{R}$, which of the following translates "some element of A is negative"? (i) $\exists x\,(x \in A \to x < 0)$; (ii) $\exists x\,(x \in A \wedge x < 0)$; (iii) $\forall x\,(x \in A \wedge x < 0)$. For each of the two others, say what it claims instead.

(c) What is the arity of the predicate "$x + y = z$", and how many holes does "$3 + y = z$" have left?

(d) True or false, with a reason: $\forall x \in \emptyset\,(x \neq x)$ is true.

Solution. (a) (i) "For every x, $x + 1$ is greater than x." True over all three domains: adding 1 always gives a larger number.

(ii) "There is an x with $x + 1 = 0$." The only solution is $x = -1$. False over $\mathbb{N}$, which does not contain -1; true over $\mathbb{Z}$ and $\mathbb{R}$, with the witness -1.

(iii) "For every x there is a y smaller than x." False over $\mathbb{N}$: for $x = 0$ there is no natural number $y < 0$. True over $\mathbb{Z}$ and $\mathbb{R}$: given x, take $y = x - 1$.

(iv) "There is an x with $2x = 1$." The only solution is $x = \frac{1}{2}$. False over $\mathbb{N}$ and $\mathbb{Z}$, true over $\mathbb{R}$ with the witness $\frac{1}{2}$.

(v) "For every x, $x \cdot x$ is at least x." True over $\mathbb{N}$: if $x = 0$, then $x \cdot x = 0 = x$; if $x \geq 1$, then $x \cdot x \geq 1 \cdot x = x$. True over $\mathbb{Z}$: an integer is either at least 1, as before, or at most 0, and then $x \cdot x \geq 0 \geq x$. False over $\mathbb{R}$: the counterexample $x = \frac{1}{2}$ has $x \cdot x = \frac{1}{4} < \frac{1}{2}$.

(b) (ii) is the translation (Definition 4.1.1: $\wedge$ under $\exists$). (i) is true as soon as some real number lies outside A, since that number makes the implication vacuously true; so it can be true when no element of A is negative (for $A = \{1\}$, say). (iii) claims that every real number is in A and negative, which is false whatever A is (take $x = 0$).

(c) It has three holes, x, y, z, so it is three-place. Filling x with 3 leaves two holes, y and z.

(d) True. It stands for $\forall x\,(x \in \emptyset \to x \neq x)$. For every x the hypothesis $x \in \emptyset$ is false, so the implication is true, even though its conclusion $x \neq x$ is false for every x.

4.1.8.2 B. Practice

Exercise. B1: Ordered quantifiers

Determine which of the following sentences are true over the domain given. Justify your answer: give the witness a true $\exists$ asks for (as a function of the earlier variables, if there are any), or the counterexample that refutes a $\forall$.

(a) $\forall x\,\exists y\,(x < y)$, over $\mathbb{N}$.
(b) $\forall x\,\exists y\,(x + y = 1)$, over $\mathbb{N}$.
(c) $\exists x\,\forall y\,(x \cdot y = y)$, over $\mathbb{R}$.
(d) $\forall x\,\exists y\,(y \cdot y = x)$, over $\mathbb{R}$.

Solution. *(a)* True. Given any natural number x, the natural number $y = x + 1$ satisfies $x < y$. The witness depends on x, which the order $\forall x\,\exists y$ allows.

(b) False. The challenger chooses $x = 2$. Then $x + y = 1$ forces $y = -1$, which is not a natural number; every natural y gives $2 + y \geq 2$. So for $x = 2$ no y works. (For $x = 0$ and $x = 1$ the witnesses $y = 1$ and $y = 0$ do exist, so the challenger must choose with care.) (Over $\mathbb{Z}$ the sentence would be true, with $y = 1 - x$.)

(c) True, with the witness $x = 1$: for every real y, $1 \cdot y = y$. The witness is chosen first and does not depend on y, as the order $\exists x\,\forall y$ demands.

(d) False, with the counterexample $x = -1$: for every real y, $y \cdot y \geq 0 > -1$, so no y has $y \cdot y = -1$.

Exercise. B2: Order matters

Give an example of a two-place predicate $R(x, y)$ on the domain $\mathbb{N}$ such that $\forall x\,\exists y\,R(x, y)$ is true and $\exists y\,\forall x\,R(x, y)$ is false. Check both claims.

Solution. Take $R(x, y)$: "$y > x$".

$\forall x\,\exists y\,(y > x)$ *is true.* Given any natural number x, the natural number $y = x + 1$ satisfies

$y > x$.

$\exists y \forall x\,(y > x)$ *is false.* Whatever natural number y is proposed, take $x = y$: then $y > x$ is false. So no y is greater than every x.

Any answer must have two properties, and a different answer can be checked against them.

1. **Every x has a partner:** for each $x \in \mathbb{N}$ there is at least one y with $R(x, y)$.
2. **No y is a partner of every x:** for each $y \in \mathbb{N}$ there is at least one x with $R(x, y)$ false.

For example, "$y \neq x$" also works: each x has the partner $x + 1$, and each y fails to be a partner of $x = y$. So does "$y \geq x$": each x is its own partner, and each y fails to be a partner of $x = y + 1$. A predicate that does **not** work is "$y \geq 1$": property 1 holds, but property 2 fails, because $y = 1$ is a partner of every x, so $\exists y \forall x\,(y \geq 1)$ is true.

4.1.8.3 C. Going deeper

Exercise. C1: A least element

Decide whether $\exists x \forall y\,(x \leq y)$ is true over each of the domains $\mathbb{N}$, $\mathbb{Z}$, $[0, 1]$ and $(0, 1]$. Give the witness where it is true, and where it is false show that every proposed x is defeated by some y.

Solution. The sentence says that the domain has a **least element**: one x that is at most every element y.

Over $\mathbb{N}$: true, with the witness $x = 0$. Every natural number y satisfies $0 \leq y$.

Over $\mathbb{Z}$: false. Let x be any integer. The integer $y = x - 1$ satisfies $y < x$, so $x \leq y$ fails. No integer is least.

Over $[0, 1]$: true, with the witness $x = 0$. Every $y \in [0, 1]$ satisfies $0 \leq y$, by the definition of the interval.

Over $(0, 1]$: false. Let $x \in (0, 1]$, so $0 < x \leq 1$. The number $y = \frac{x}{2}$ satisfies $0 < \frac{x}{2} < x \leq 1$, so $y \in (0, 1]$ and $y < x$; hence $x \leq y$ fails. The defeating y depends on the proposed x, which is allowed, because in the game the challenger chooses y after seeing x.

The last two domains differ by a single point, 0, and that point is exactly the least element of $[0, 1]$. Removing it leaves a set with no least element, although every element of $(0, 1]$ is still at least 0. *Analysis on the Real Line*, Chapter 1 returns to this distinction between a least element and a greatest lower bound.

Exercise. C2: The wrong connective

Take the domain $\mathbb{R}$.

(a) The sentence "every element of A is positive" is mistranslated as $\forall x\,(x \in A \wedge x > 0)$, and "some element of A is positive" as $\exists x\,(x \in A \to x > 0)$. For each mistranslation, give a set $A \subseteq \mathbb{R}$ for which it has a different truth value from the English sentence.

(b) Show that the first mistranslation is false for **every** set $A \subseteq \mathbb{R}$, and the second is true

for **every** set $A \subseteq \mathbb{R}$. What does this say about how much each one tells us about A?

Solution. *(a)* Take $A = \{1\}$ for the first. "Every element of $\{1\}$ is positive" is true. The mistranslation claims that every real x is in $\{1\}$ and positive; $x = 0$ is neither, so it is false.

Take $A = \{-1\}$ for the second. "Some element of $\{-1\}$ is positive" is false, since its only element is negative. The mistranslation is true, with the witness $x = 0$: since $0 \notin \{-1\}$, the implication "if $0 \in A$, then $0 > 0$" has a false hypothesis and is true.

(b) For the first, let $A \subseteq \mathbb{R}$ be any set. The real number $x = 0$ does not satisfy $x > 0$, so it does not satisfy $x \in A \wedge x > 0$, whatever A is. It is a counterexample to the $\forall$, so the sentence is false.

For the second, let $A \subseteq \mathbb{R}$ be any set, and consider the real number $x = 1$. The conclusion $1 > 0$ is true, so the implication $1 \in A \to 1 > 0$ is true whether or not $1 \in A$. Hence $x = 1$ is a witness, whatever A is, and the sentence is true.

So neither mistranslation tells us anything about A at all: the first is false and the second is true, whichever set A is. A sentence whose truth value does not depend on A cannot express a property of A. This is the precise sense in which the wrong connective "claims too much" under $\forall$ and "too little" under $\exists$.

Marking scheme. (10 points) - 2: a set for the first mistranslation, with both truth values checked. - 2: a set for the second mistranslation, with both truth values checked, and the implication at the witness justified (its hypothesis is false, or its conclusion is true). - 3: for the first, for an arbitrary A, a real number (the same for every A, or chosen by cases covering every A) that fails $x \in A \wedge x > 0$, with the check. - 3: for the second, for an arbitrary A, a witness (the same for every A, or chosen by cases covering every A) at which $x \in A \to x > 0$ holds, with a reason that covers every A (its conclusion is true, or a check in every case). Fatal errors (0 for the step): a witness or counterexample that depends on A in (b) without a case split covering every A; reading $\exists x\,(x \in A \to x > 0)$ as "some element of A is positive".

4.2 Terms, Formulas, Free Variables and Substitution

Section 4.1 read quantified sentences the way a mathematician does, by eye. That was enough to understand them, and it is not enough to reason with them. Each quantifier rule of Sections 4.3 to 4.6, and the rule for using an equation in Section 4.7, has the form "put the term t in for the variable x in the formula φ", written $\varphi[t/x]$, and a rule has to say exactly which symbols change and which do not. This section makes the language of Section 4.1 exact, in the way Section 3.1 made the propositional language exact, and then defines the operation $\varphi[t/x]$. The one new thing is substitution, together with the single way in which it can go wrong, called **capture**. Terms, formulas, and free variables are the smaller definitions it needs, and each comes just before it is used.

4.2.1 Which letters are holes

Three sentences, over the domain $\mathbb{N}$:

- "$x > 2$" is not a statement. It is the open sentence of item (e) of Example (Statements and nonstatements), true for some values of x and false for others.
- "$\forall x\,(x^2 \geq 0)$" is a statement, and a true one. The letter x still appears, but the quantifier has closed its hole: the sentence does not ask which x we mean.
- "$\forall x\,(x \geq y)$" is neither true nor false as it stands. It is true when y is 0 (every natural number is at least 0) and false when y is 1 (the number 0 is not at least 1). It is an open sentence **in** y, although it begins with a quantifier.

So the question "is this a statement?" is not answered by looking for quantifiers. It is answered by asking which letters are still holes. In the third sentence x is closed by $\forall x$ and y is open. The open letters will be called **free**, and the closed ones **bound**.

Here is why the difference matters for reasoning. Over $\mathbb{N}$, the sentence

$$\exists y\,(y \neq x)$$

is an open sentence in x, and it is true for every x: given x, the number $y = x+1$ differs from it. Now suppose that we want to say the same thing about a particular number named by another letter, and we put y in for x without thinking. The result is

$$\exists y\,(y \neq y),$$

"some number differs from itself", which is false. Nothing about numbers went wrong. The y that we put in landed inside the scope of $\exists y$, and the quantifier took hold of it: the new y was meant to name a number from outside, and it has become the quantifier's own placeholder. This is **capture**.

Capture is the one trap of substitution, and the rules of Sections 4.3 to 4.7 substitute at almost every step. So we need exact answers to three questions: what the expressions are that can be put in; which occurrences of x get replaced; and when putting t in for x is safe.

(A remark on the choice of example. The formula $\forall y\,(y \neq x)$ looks just as natural, and putting y in for x there gives $\forall y\,(y \neq y)$. But $\forall y\,(y \neq x)$ is false for every x already, since $y = x$ is a counterexample, and a false sentence turned into a false sentence shows nothing.)

4.2.2 Terms

Section 3.1 described the propositional language by its alphabet and its formation rules. The language of predicate logic has more symbols, of more kinds.

- **Variables:** x, y, z, u, with subscripts $x_0, x_1, y_0, \ldots$, and in analysis also $\varepsilon, \delta, N, n, k, L$, and for primes and other natural numbers p, q. There are infinitely many, so that a fresh one is always available. Each is one symbol.
- **Constant symbols**, such as 0, 1, 2, a, b, c. (A letter may be a constant in one language and a function symbol in another: in a language with a sequence, a is the sequence's one-place function symbol, as below, and not a constant.)
- **Function symbols**, each with a number of places $n \geq 1$, its **arity**: f and g, the arithmetic symbols $+$, $\cdot$ and $-$ (two places), and $-$ in front of a number, halving $\cdot/2$, squaring and $|\cdot|$ (one place). The one-place and the two-place minus are different symbols written alike; the number of terms they are applied to shows which is meant.
- **Predicate symbols**, each with an arity $n \geq 1$: P, Q, R, S, $<$, $\leq$, $>$, $\geq$, Prime, and $\in A$ for a fixed set A (one place, as in Section 4.1).
- **Equality** $=$, which every language has.
- The constant $\bot$, the connectives $\neg, \wedge, \vee, \to, \leftrightarrow$ of Chapter 3, and the **quantifier symbols** $\forall$ and $\exists$.

- Parentheses (,), and the comma.

Which constant, function, and predicate symbols are available depends on the subject. For the argument of Section 4.1 we need the constant 2 and the one-place predicate symbols P and S. For arithmetic we need 0, 1, $+$, $\cdot$, and $<$. Every formula uses only finitely many symbols, and we never need to fix the whole list in advance.

The first things to build are the names of objects.

A term is a name of an object, built from variables and constants by function symbols.

Definition 4.2.1. Term

The **terms** are the strings given by the following clauses.

(Tm1) Every variable is a term, and every constant symbol is a term.

(Tm2) If f is a function symbol of arity n and $t_1, \ldots, t_n$ are terms, then $f(t_1, \ldots, t_n)$ is a term.

(Tm3) **Nothing else** is a term: a string is a term **only if** it is the last entry of a finite list of strings in which every entry is a variable, a constant symbol, or $f(s_1, \ldots, s_n)$ for some function symbol f of arity n and **earlier** entries $s_1, \ldots, s_n$.

In words: (Tm1) gives the starting pieces, the names that are not built from anything. (Tm2) applies a function symbol to as many terms as it has places, with the terms in parentheses and separated by commas. (Tm3) closes the door, exactly as clause (F4) of Definition 3.1.1 did for propositional formulas: a term is something reached from variables and constants by finitely many uses of (Tm2).

A **variable of** a term t is a variable that occurs in t. A term with no variables, such as 2 or $f(c)$, names one fixed object once the symbols are interpreted; a term with variables, such as $x + 1$, names an object only once its variables have values.

Shorthand. As with formulas in Section 3.1, we write terms in the familiar way and let the official form be understood. So $x+y$ is shorthand for $+(x, y)$, $2 \cdot y$ for $\cdot(2, y)$, x^2 for the squaring symbol applied to x, $-x$ for the one-place minus applied to x, $x - y$ for the two-place minus applied to x and y, $x/2$ for halving applied to x, $|x|$ for $|\cdot|$ applied to x, and a_n for $a(n)$, with a a one-place function symbol (the convention of Section 4.1 for sequences). The school conventions about grouping ($\cdot$ before $+$, and so on) say which official term is meant.

Here are terms, simplest first.

- x and 2: by (Tm1). The degenerate terms, built in no steps; every argument about all terms has to handle them.
- $x + 1$, officially $+(x, 1)$: by (Tm2) from the terms x and 1.
- $f(f(c))$: by (Tm2) twice, from c, for a one-place f. Its only building blocks are constants, so it has no variables.
- $|a_n - L|$, officially $|\cdot|(-(a(n), L))$, with the two-place minus: by (Tm2) three times, from the variables n and L. Its variables are n and L. The letter a is a function symbol, not a variable.

And strings that are **not** terms, each by one clause:

- $x < 1$ is not a term. It does not name an object; it says something about one. It is built with a predicate symbol, and (Tm2) uses only function symbols. It will be a formula.
- $f(x, y)$, for a one-place f, is not a term: (Tm2) applies f to exactly one term.

Terms can be drawn as trees, exactly as formulas were in Section 3.1: a variable or a constant is a single node, and $f(t_1, \ldots, t_n)$ is a node labeled f with the trees of $t_1, \ldots, t_n$ below it, in order. And a term can be read in only one way. The proof is the proof of unique readability for formulas in Section 3.2 (Theorem 3.2.5), with the parentheses and commas of (Tm2) doing the work that the parentheses of (F3) did there. We do not repeat it.

4.2.3 Formulas

Next, the statements and open sentences. Atomic formulas say something about terms; everything else is built from them by connectives and quantifiers.

A first-order formula is built from predicates applied to terms, by the connectives and the two quantifiers.

Definition 4.2.2. First-Order Formula

The **formulas** (of predicate logic, or **first-order formulas**) are the strings given by the following clauses.

(Fm1) **Atomic formulas.** If R is a predicate symbol of arity n and $t_1, \ldots, t_n$ are terms, then $R(t_1, \ldots, t_n)$ is a formula. If t_1 and t_2 are terms, then $t_1 = t_2$ is a formula. And $\bot$ is a formula.

(Fm2) If φ is a formula, then $\neg\varphi$ is a formula.

(Fm3) If φ and ψ are formulas, then $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$, $(\varphi \to \psi)$, and $(\varphi \leftrightarrow \psi)$ are formulas.

(Fm4) If φ is a formula and x is a **variable**, then $\forall x\, \varphi$ and $\exists x\, \varphi$ are formulas.

(Fm5) **Nothing else** is a formula: a string is a formula only if it is the last entry of a finite list in which every entry is an atomic formula, or is built by (Fm2), (Fm3) or (Fm4) from **earlier** entries.

In words, clause by clause.

- *(Fm1).* The atomic formulas are the new starting pieces. They replace the atoms p, q, r of Chapter 3, and they have an inside: a predicate symbol and its terms. Equality is always there. The constant $\bot$ is kept.
- *(Fm2), (Fm3).* The connectives work exactly as in Definition 3.1.1, parentheses included.
- *(Fm4).* The new clause. A quantifier symbol followed by a **variable** is put in front of a formula, with no parentheses, as $\neg$ is. Only a variable may follow $\forall$ or $\exists$: $\forall 2\, P(2)$ and $\forall f(x)\, P(x)$ are not formulas. The variable need not occur in φ: $\forall x\, P(c)$ is a formula, and says no more than $P(c)$ (a quantifier with nothing to bind is harmless and useless).
- *(Fm5).* The closing clause, as before.

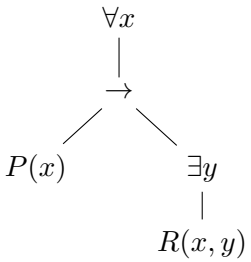
Structural induction and structural recursion (Theorem 3.2.3, Theorem 3.2.6) carry over to first-order formulas with one change in the list of cases. The base case covers every **atomic** formula (and $\bot$), and there are two new building steps, from φ to $\forall x\, \varphi$ and to $\exists x\, \varphi$, for every variable x, each with one part, like the step to $\neg\varphi$. So, to prove that every formula has a property, prove it for the atomic formulas and $\bot$, and prove that it passes from φ to $\neg\varphi$, $\forall x\, \varphi$, and $\exists x\, \varphi$, and from φ, ψ to $(\varphi \circ \psi)$. To define a function on formulas, give one clause for each of these cases. The proofs in Section 3.2 go through word for word with the new cases added, and unique readability holds for the same reason; we use all three without repeating them. The same two principles hold for terms, with (Tm1) as the base case and (Tm2) as the one building step.

Shorthand. The conventions of Section 3.1 for dropping parentheses stay, with three additions.

1. Two-place predicate symbols such as $<, \leq$ are written between their terms: $x < y$ is shorthand for $<(x, y)$. The one-place symbol $\in A$ is written after its term: $t \in A$ is shorthand for $(\in A)(t)$. (When sets are themselves objects of the domain, as in *Sets, Relations, and Cardinality*, Chapter 1, $\in$ is a two-place predicate symbol and $x \in A$ is $\in (x, A)$; in this chapter A is always a fixed set, so we treat $\in A$ as one-place.) We write $t_1 \neq t_2$ for $\neg(t_1 = t_2)$.
2. **A quantifier binds as tightly as** $\neg$: it applies to the smallest formula that follows it. So $\forall x\, P(x) \to Q(c)$ stands for $(\forall x\, P(x) \to Q(c))$, an implication whose hypothesis is $\forall x\, P(x)$. For the quantifier to govern the whole implication, write $\forall x\, (P(x) \to Q(x))$, where the parentheses are the official ones around the implication.
3. The bounded quantifiers of Definition 4.1.1, $\forall x \in A\ \varphi$ and the rest, stand for the formulas given there.

Rule 2 matters in every rule section. The formula $\forall x\, P(x) \to Q(c)$ has main connective $\to$, while $\forall x\, (P(x) \to Q(x))$ has main symbol $\forall x$. The rule for using a "for every" (Section 4.3) applies only to the second.

Parse trees. A parse tree for a first-order formula is drawn as in Section 3.1, with one new kind of node: a **quantifier node**, labeled $\forall x$ or $\exists x$, with the tree of the formula it governs as its only subtree. The leaves are atomic formulas (or $\bot$); their terms have trees of their own, which we do not draw. Here is the tree of $\forall x\, (P(x) \to \exists y\, R(x, y))$.



The root is the quantifier node $\forall x$, so the formula's main symbol is $\forall x$. Below it is the implication, with the atomic formula $P(x)$ on the left and the quantifier node $\exists y$ on the right, which governs the atomic formula $R(x, y)$.

Quick check. Draw the parse tree of $\forall x\, P(x) \to \exists y\, \forall x\, R(x, y)$, written in shorthand. What is its main symbol?

Solution. By shorthand rule 2, each quantifier applies to the smallest formula after it, so the official formula is $(\forall x\, P(x) \to \exists y\, \forall x\, R(x, y))$ and its main symbol is $\to$. The root is $\to$. Its left subtree is the node $\forall x$ above the leaf $P(x)$. Its right subtree is the node $\exists y$, below it the node $\forall x$, and below that the leaf $R(x, y)$. The two $\forall x$ nodes are on different branches.

4.2.4 Free and bound occurrences

The parse tree shows which quantifier governs which part of a formula. That is exactly what decides whether a letter is a hole.

An occurrence of a variable is bound if a quantifier on that variable stands above it in the parse tree; otherwise it is free.

Definition 4.2.3. Free and Bound Variables

Let φ be a formula and x a variable. An **occurrence** of x in φ is an appearance of x inside one of the terms of an atomic formula of φ. (The x written just after a quantifier symbol, as in $\forall x$, is part of the quantifier node and is not an occurrence.)

- An occurrence of x is **bound** if the path from its leaf up to the root of the parse tree passes through a quantifier node $\forall x$ or $\exists x$, with the **same** variable x. The lowest such node **binds** the occurrence.
- An occurrence that is not bound is **free**.
- The **scope** of a quantifier node is the part of the tree below it: the formula it governs.
- The variable x is a **free variable of** φ, or is **free in** φ, if at least one occurrence of x in φ is free.

In words: an occurrence of x is bound when it lies in the scope of a quantifier on x; quantifiers on other variables do not bind it. If several quantifiers on x stand above it, the nearest one, the lowest in the tree, is the one that binds it. A free occurrence is a hole still open. A variable is free in a formula if it has **at least one** free occurrence, even if it has bound occurrences too.

One more word, for the times when a variable must be new to a formula. A variable **appears in** φ if it is written anywhere in φ: inside a term, or just after a quantifier symbol. So y appears in $\forall y\, P(x)$, although it has no occurrence there. A variable that does not appear in φ has no occurrence in φ, and no quantifier of φ is on it; every freshness condition of this chapter is stated with "appear" for that reason.

Definition 4.2.4. Sentence

A **sentence** is a formula with no free variables.

A sentence is what Chapter 1 called a statement, once the domain and the symbols are fixed: every hole is closed, so it is true or false. A formula with free variables is an open sentence, true or false only once values are given to its free variables.

Examples, simplest first. The domain is irrelevant to all of them: being free or bound is a matter of the string, not of what it means.

- $\bot$ and $P(c)$ are sentences: they contain no variables at all. So is $\forall x\, P(c)$. These are the degenerate cases.
- $\forall x\, (x^2 \geq 1)$: the single occurrence of x lies below the node $\forall x$, so it is bound. A sentence.
- $\forall x\, (x \geq y)$: the occurrence of x is bound by $\forall x$; the occurrence of y has no quantifier on y above it, so it is free. The free variables are y alone. Not a sentence.
- $\forall x\, \big(P(x) \to \exists y\, R(x, y)\big)$, drawn above: both occurrences of x are bound by the root; the occurrence of y in $R(x, y)$ is bound by $\exists y$. A sentence.
- $P(x) \wedge \forall x\, Q(x)$: the occurrence of x in $P(x)$ is **free** (no quantifier stands above it: the $\forall x$ governs only $Q(x)$), and the occurrence in $Q(x)$ is bound. So x is a free variable of this formula, and it is not a sentence.
- $\forall x\, \exists x\, P(x)$: the occurrence of x has two quantifiers on x above it, and the lower one, $\exists x$, binds it. The outer $\forall x$ binds nothing. A sentence.

The recursive description. In proofs about all formulas we need "x is free in φ" in the form that structural induction can use, one case per clause of Definition 4.2.2.

- x is free in an atomic formula exactly when x occurs in one of its terms; x is never free in

$\bot$.

- x is free in $\neg\varphi$ exactly when x is free in φ.
- x is free in $(\varphi \circ \psi)$ exactly when x is free in φ or in ψ.
- x is free in $\forall y\,\varphi$, and in $\exists y\,\varphi$, exactly when x is a different variable from y **and** x is free in φ.

Each line is read off the tree. In an atomic formula there are no quantifier nodes, so every occurrence is free. The tree of $\neg\varphi$ or $(\varphi \circ \psi)$ adds a root that is not a quantifier node, so it binds nothing. The tree of $\forall y\,\varphi$ adds the root $\forall y$ on top of the tree of φ: an occurrence of x inside φ keeps its path, with one more node at the top, so it is free in $\forall y\,\varphi$ exactly when it was free in φ and the new node is not a quantifier on x, that is, when x and y are different variables.

Example. The free variables of a limit statement

With the conventions of Section 4.1 (domain $\mathbb{R}$, the sequence as a function symbol a), "the sequence (a_n) converges to L" is the formula

$$\forall\varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon).$$

Write it without bounded quantifiers, and find its free variables. Is it a sentence?

Solution. Unfolding the three bounded quantifiers by Definition 4.1.1 ($\to$ under each $\forall$, $\wedge$ under the $\exists$) gives

$$\forall\varepsilon\Big(\varepsilon > 0 \to \exists N\,(N \in \mathbb{N} \wedge \forall n\,(n \in \mathbb{N} \to (n \geq N \to |a_n - L| < \varepsilon)))\Big).$$

The variables that occur are ε, N, n, and L. Check each.

- Every occurrence of ε lies below the root $\forall\varepsilon$: bound.
- Every occurrence of N lies below $\exists N$: bound.
- Every occurrence of n (in $n \in \mathbb{N}$, in $n \geq N$, and in the term a_n) lies below $\forall n$: bound.
- The occurrence of L has no quantifier on L above it: free.

So L is the only free variable, and the formula is **not** a sentence. It is an open sentence in L, as it should be: until L is given a value, "(a_n) converges to L" is neither true nor false, and which values make it true depends on the sequence (for $a_n = 1/(n+1)$ the value $L = 0$ does and $L = 1$ does not; for $a_n = (-1)^n$ no value does). Putting $\exists L$ in front gives the sentence "(a_n) converges".

The other symbols in it are not variables, and "free" and "bound" do not apply to them. The letter a is a function symbol; the formula is certainly about the sequence a, but a is fixed by the language, like $+$, and cannot be quantified in this logic. The symbol $\in \mathbb{N}$ is a one-place predicate symbol, 0 is a constant, and $>$, $\geq$, $<$, $-$ and $|\cdot|$ are predicate and function symbols of the language.

Warning. The same letter can occur free and bound in one formula. In $P(x) \wedge \forall x\,Q(x)$, the first x is a hole and the second is the quantifier's placeholder, and the two have nothing to do with each other. This is legal, and it is a trap for the reader: a proof that "uses x" from such a formula has to say which x it means. The book avoids it by renaming the bound variable (the last part of this section explains how): $P(x) \wedge \forall z\,Q(z)$ says the same thing, and

the question does not arise.

Quick check. Is x free in $\forall y\,(x < y) \to \exists x\,(x = 0)$? At which occurrences?

Solution. Yes. By shorthand rule 2 the formula is $(\forall y\,(x < y) \to \exists x\,(x = 0))$. The occurrence of x in $x < y$ lies below $\forall y$ only, which is a quantifier on y, so it is **free**. The occurrence of x in $x = 0$ lies below $\exists x$, so it is bound. One free occurrence is enough: x is a free variable of the formula. The variable y, by contrast, is not free: its only occurrence lies below $\forall y$.

4.2.5 Substitution

We now come to the operation that every quantifier rule uses. The rule of Section 4.3 will take the premise $\forall x\,(P(x) \to S(x))$ of Section 4.1's argument and produce $P(2) \to S(2)$: it strips the quantifier and puts 2 in for x. The rule of Section 4.5 goes the other way. Both need to say exactly what "put 2 in for x" does. The answer has one principle: only the **free** occurrences are replaced, because only they are holes. A bound occurrence belongs to its quantifier, and the quantifier is not asking for a value.

$\varphi[t/x]$ is φ with t put in for every free occurrence of x, and nothing else changed.

Definition 4.2.5. Substitution of a Term for a Variable

Let t be a term and x a variable. For a term s, the term $s[t/x]$ is s with every occurrence of x replaced by t; by recursion on terms,

$$x[t/x] = t, \qquad y[t/x] = y \ \text{ for a variable } y \neq x, \qquad c[t/x] = c,$$
$$f(s_1, \ldots, s_n)[t/x] = f(s_1[t/x], \ldots, s_n[t/x]).$$

For a formula φ, the formula $\varphi[t/x]$, read "φ with t for x", is defined by recursion on φ:

$$\begin{aligned} R(s_1, \ldots, s_n)[t/x] &= R(s_1[t/x], \ldots, s_n[t/x]), \\ (s_1 = s_2)[t/x] &= (s_1[t/x] = s_2[t/x]), \qquad \bot[t/x] = \bot, \\ (\neg\varphi)[t/x] &= \neg(\varphi[t/x]), \\ (\varphi \circ \psi)[t/x] &= (\varphi[t/x] \circ \psi[t/x]), \\ (\forall x\,\varphi)[t/x] &= \forall x\,\varphi, \\ (\forall y\,\varphi)[t/x] &= \forall y\,(\varphi[t/x]) \ \text{ for a variable } y \neq x, \end{aligned}$$

and the last two lines likewise with $\exists$ in place of $\forall$.

In words, clause by clause: in a term, every x is replaced, since a term has no quantifiers. In an atomic formula, the replacement happens inside its terms. The connectives pass the substitution to their parts. A quantifier on x **stops** it: below $\forall x$ there is no free occurrence of x to replace. A quantifier on another variable y passes it on.

In $[t/x]$ the slash is a separator, not division: t is the whole term written before the slash, and x the variable after it. So $[y + 1/x]$ puts $y + 1$ in for x.

Well defined. The clauses give one value for each case of Definition 4.2.2, so structural recursion (carried over as above) says that they define exactly one formula $\varphi[t/x]$ for each φ. And the result is a formula: each clause builds its value by the clauses of Definition 4.2.1 and Definition 4.2.2 from values already known to be terms and formulas.

This is a different operation from the substitution σ of Section 3.4 (the one in Lemma 3.4.3). There, formulas were put in for **atoms**, at every occurrence. Here a **term** is put in for a **variable**, and bound occurrences are left alone.

Example. Four substitutions

Compute each substitution from Definition 4.2.5.

(a) $(P(x) \to S(x))[2/x]$.
(b) $(x < y \wedge \forall x\,(x \geq 0))[y+1/x]$.
(c) $(\forall y\, R(x,y))[f(z)/x]$.
(d) $(\forall x\, P(x))[c/x]$.

Solution. *(a)* The implication passes the substitution to both sides, and in each atomic formula the term x becomes 2: the result is $P(2) \to S(2)$. This is the step that Section 4.3's rule makes in the argument of Section 4.1.

(b) The conjunction passes it to both parts. In $x < y$ the x becomes $y+1$, giving $y+1 < y$. In $\forall x\,(x \geq 0)$ the quantifier is on x, so the substitution stops. The result is

$$y+1 < y \ \wedge\ \forall x\,(x \geq 0).$$

Only the free occurrence of x changed. The bound one, in $x \geq 0$, was the quantifier's placeholder, and replacing it would have turned the sentence "every x is at least 0" into a claim about $y+1$.

(c) The quantifier is on y, a variable different from x, so it passes the substitution on: the result is $\forall y\, R(f(z), y)$.

(d) The quantifier is on x, so nothing changes: the result is $\forall x\, P(x)$. There was no free x to replace.

Two facts follow from the definition at once, and both are used again and again.

First, $\varphi[x/x] = \varphi$: putting x in for itself changes nothing. In each clause the replaced symbol is x again.

Second, and more useful: **if x is not free in φ, then** $\varphi[t/x] = \varphi$, for every term t. Example (d) was one instance. Section 4.9 uses this fact in its proofs, so we prove it here.

Claim. Let x be a variable and t a term. For every formula φ: if x is not free in φ, then $\varphi[t/x] = \varphi$.

Idea. Substitution changes only free occurrences of x, and there are none. Formally, structural induction on φ, using the recursive description of "free" case by case. The only case with any content is a quantifier on a variable other than x, where the substitution is passed on, and the induction hypothesis applies because x is still not free inside.

Proof. For terms first: if x does not occur in a term s, then $s[t/x] = s$, by induction on s (a variable other than x and a constant are unchanged by the clauses, and $f(s_1, \ldots, s_n)[t/x] = f(s_1, \ldots, s_n)$ when each $s_i[t/x] = s_i$). Now we prove the claim by structural induction on φ.

Atomic formulas and $\bot$. If x is not free in $R(s_1, \ldots, s_n)$, then x occurs in none of the s_i (the recursive description), so each $s_i[t/x] = s_i$, and the clause for atomic formulas returns $R(s_1, \ldots, s_n)$. The same holds for $s_1 = s_2$, and $\bot[t/x] = \bot$.

Connectives. If x is not free in $\neg\varphi$, it is not free in φ, so $(\neg\varphi)[t/x] = \neg(\varphi[t/x]) = \neg\varphi$ by the induction hypothesis. If x is not free in $(\varphi \circ \psi)$, it is free in neither part, and the same argument applies to both.

Quantifiers. Consider $\forall y\, \varphi$. If y is x, then $(\forall x\, \varphi)[t/x] = \forall x\, \varphi$ by the definition, whatever φ is. If y is a different variable and x is not free in $\forall y\, \varphi$, then x is not free in φ (the recursive description), so $(\forall y\, \varphi)[t/x] = \forall y\, (\varphi[t/x]) = \forall y\, \varphi$ by the induction hypothesis. The case of $\exists y\, \varphi$ is the same.

This proves the claim for every formula φ.

■

4.2.6 Capture, and when a substitution is safe

The definition of substitution never refuses. Given any φ, t and x, it returns a formula. Sometimes that formula does not say what we meant.

Take $\varphi = \exists y\, (y \neq x)$, the example at the start of the section, and $t = y$. The quantifier $\exists y$ is on a variable different from x, so the definition passes the substitution through, and

$$(\exists y\, (y \neq x))[y/x] = \exists y\, (y \neq y).$$

The definition was followed correctly. The meaning is wrong. The formula $\exists y\, (y \neq x)$ says something about the number x: some number differs from it, which over $\mathbb{N}$ is true whatever x is. We wanted to say the same thing about the number y, and we got "some number differs from itself", which is false. The y we put in was supposed to name a number from outside, and it landed in the scope of $\exists y$ and was **captured**.

Here is a second example, with a compound term. Over $\mathbb{R}$, $\exists y\, (y > x)$ is true for every x: there is a number larger than x. Put $y + 1$ in for x:

$$(\exists y\, (y > x))[y + 1/x] = \exists y\, (y > y + 1).$$

The result says that some number is larger than its own successor, which is false. It should have said that some number is larger than $y + 1$, which is true.

Compare a substitution that is safe. Over $\mathbb{R}$, $\forall z\, (x < z)$ says that x is smaller than every number (false, as it happens, for every x; but what matters is what it says). Put $y + 1$ in for x: the

quantifier $\forall z$ passes the substitution on, and

$$(\forall z\,(x < z))[y + 1/x] = \forall z\,(y + 1 < z),$$

which says exactly that $y + 1$ is smaller than every number. Nothing was captured, because the term $y + 1$ does not contain z, the only variable quantified above the occurrence of x.

The difference between the safe case and the two unsafe ones is exactly this: whether a variable of t is quantified somewhere above a free occurrence of x.

A term is free for x when putting it in for x lets none of its variables be captured.

Definition 4.2.6. Free For

Let t be a term, x a variable, and φ a formula. The term t is **free for** x in φ if **no** free occurrence of x in φ lies in the scope of a quantifier $\forall y$ or $\exists y$ such that y is a variable of t.

In words: look at each place where t will go, that is, at each free occurrence of x. Look at the quantifiers above it in the tree. If none of them is on a variable of t, the substitution is safe there. The term is free for x when it is safe at every such place. Informally: **no variable of t becomes bound** when t is put in.

Four special cases, each immediate from the definition, cover most uses.

- A term with **no variables** (2, c, $f(c)$) is free for every variable in every formula: there is no y to capture.
- The variable x is free for x in every formula. A free occurrence of x never lies in the scope of a quantifier on x; if it did, it would be bound.
- A variable z that does **not appear** in φ is free for every variable in φ: no quantifier of φ is on z.
- If x is **not free** in φ, every term is free for x in φ: there are no free occurrences to check. (And then $\varphi[t/x] = \varphi$ by the claim above.)

In our examples: y is **not** free for x in $\exists y\,(y \neq x)$, because the free occurrence of x lies in the scope of $\exists y$ and y is a variable of the term y. The term $y + 1$ is not free for x in $\exists y\,(y > x)$, for the same reason. And $y + 1$ **is** free for x in $\forall z\,(x < z)$: the only quantifier above the free occurrence of x is on z, which is not a variable of $y + 1$.

The recursive description. For induction proofs, "free for" can be read off case by case, like "free":

- t is free for x in every atomic formula and in $\bot$;
- t is free for x in $\neg\varphi$ exactly when it is free for x in φ, and in $(\varphi \circ \psi)$ exactly when it is free for x in both φ and ψ;
- t is free for x in $\forall y\,\varphi$ (and in $\exists y\,\varphi$) exactly when **either** x is not free in $\forall y\,\varphi$, **or** y is not a variable of t and t is free for x in φ.

The first two lines hold because an atomic formula has no quantifiers and a connective node is not a quantifier. For the third: if x is not free in $\forall y\,\varphi$, there is nothing to check. If it is free, then y is a different variable and the free occurrences of x in $\forall y\,\varphi$ are those in φ; each lies in the scope of the new root $\forall y$ and of the quantifiers inside φ above it. The root is harmless exactly when y is not a variable of t, and the quantifiers inside are harmless exactly when t is free for x in φ.

Why the definition. "Free for" is the precise condition under which $\varphi[t/x]$ says about t what φ says about x. Each rule of Sections 4.3, 4.5, and 4.7 will carry it as a side condition, and Section

4.3 shows the capture example of this section turning a true premise into a false conclusion when the side condition is ignored. The definition asks for no more than it needs: a variable of t may be quantified elsewhere in φ, as long as no free occurrence of x lies in its scope. For instance, y is free for x in $P(x) \wedge \exists y\, Q(y)$, and the result $P(y) \wedge \exists y\, Q(y)$ says of y what the original said of x.

Warning. Capture changes meaning, and it gives no warning: the definition of $\varphi[t/x]$ returns a formula either way. Over $\mathbb{R}$, $\exists y\,(y > x)$ is true for every x, and $(\exists y\,(y > x))[y + 1/x]$ is $\exists y\,(y > y+1)$, which is false. Before putting t in for x, check that no variable of t is quantified above a free occurrence of x.

Quick check. Is $y + 1$ free for x in $\forall z\,(x < z)$? Is it free for x in $\forall y\,(x < y)$? Compute both substitutions and say what each result claims.

Solution. In $\forall z\,(x < z)$: yes. The only quantifier above the occurrence of x is on z, and z is not a variable of $y + 1$. The result is $\forall z\,(y + 1 < z)$, "$y + 1$ is smaller than every number", which is what the original said about x.

In $\forall y\,(x < y)$: no. The occurrence of x lies in the scope of $\forall y$, and y is a variable of $y + 1$. The definition gives $\forall y\,(y + 1 < y)$: "every number is larger than its own successor". That is a sentence about no particular number at all, and a false one; the original said something about x, and the result should have said the same about $y+1$. The y of $y+1$ was captured.

4.2.7 Renaming bound variables

What should we do when t is not free for x in φ? A bound variable is a placeholder, and the name of a placeholder does not matter. "Some number differs from x" can be written $\exists y\,(y \neq x)$ or $\exists z\,(z \neq x)$, just as $\int_0^1 s^2\,ds$ and $\int_0^1 u^2\,du$ are the same integral. So the way out is: **rename the bound variable first, then substitute.**

Renaming. Let φ be a formula and y, z variables, with z **not appearing** in φ. The formula $\forall z\,\varphi[z/y]$ is obtained from $\forall y\,\varphi$ by **renaming** the bound variable y to z; likewise $\exists z\,\varphi[z/y]$ from $\exists y\,\varphi$. The same may be done to a quantifier anywhere inside a larger formula.

In words: change the variable after the quantifier, and change the occurrences it binds, to a letter that appears nowhere in the formula. The condition that z does not appear in φ does two jobs. It makes z free for y in φ (the third special case above), so no new z is captured by a quantifier inside φ. And it ensures that no free z already in φ is captured by the new quantifier $\forall z$. Without it, renaming can change the meaning: $\forall y\,(y \geq z)$ says something about z (over $\mathbb{N}$, that $z = 0$), and "renaming" y to z would give $\forall z\,(z \geq z)$, a sentence that is simply true.

Its status. Renaming is not a rule of the calculus, and we have not proved that it keeps meaning. Here is where it stands. For a renaming at the front of a formula, $\forall y\,\varphi$ and $\forall z\,\varphi[z/y]$ can each be derived from the other in three lines, once the rules of Section 4.4 are available; that section does it, and Section 4.6 does the same for $\exists$. A renaming **inside** a larger formula we take as a convention of this book: formulas that differ only by renaming bound variables are treated as saying the same thing. Its justification in terms of meaning follows from a lemma of Section 4.12 (Lemma 4.12.1, with the variable z as the term), together with Lemma 4.11.1 and

an induction on the surrounding formula, which we do not carry out. Section 4.12 is optional, and nothing outside it depends on that lemma.

The recipe. To put t in for x in φ when t is not free for x: rename each quantifier of φ that is on a variable of t and has a free occurrence of x in its scope, each to a different variable that appears neither in φ nor in t (each such step is a renaming in the sense above); then substitute. After the renaming, t is free for x.

- $\exists y\,(y \neq x)$ with y for x: rename y to z, giving $\exists z\,(z \neq x)$; then substitute, giving $\exists z\,(z \neq y)$, "some number differs from y". True over $\mathbb{N}$ for every y, as intended.
- $\exists y\,(y > x)$ with $y + 1$ for x: rename to $\exists z\,(z > x)$; substitute, giving $\exists z\,(z > y + 1)$. True over $\mathbb{R}$, as intended.

Which fresh letter we choose does not matter: two choices give formulas that differ only by a renaming. From now on, whenever a rule writes $\varphi[t/x]$, it will require t to be free for x in φ, and if we want a t that is not, we rename first.

4.2.8 How substitution looks in a mathematical proof

Mathematicians substitute constantly and never write $[t/x]$. Two examples show the operation, and its trap, in their natural habitat.

"Apply the hypothesis with $\varepsilon/2$ in place of ε." Suppose we know that (a_n) converges to L, and we have fixed an $\varepsilon > 0$ for some other purpose. A proof says: "applying the hypothesis with $\varepsilon/2$, there is an $N \in \mathbb{N}$ with $|a_n - L| < \varepsilon/2$ for all $n \geq N$". The hypothesis is $\forall\varepsilon\,\varphi$, with

$$\varphi \;=\; \Big(\varepsilon > 0 \to \exists N \in \mathbb{N}\; \forall n \in \mathbb{N}\; \big(n \geq N \to |a_n - L| < \varepsilon\big)\Big),$$

and the proof uses the instance $\varphi[t/\varepsilon]$ with the term $t = \varepsilon/2$:

$$\varepsilon/2 > 0 \to \exists N \in \mathbb{N}\; \forall n \in \mathbb{N}\; \big(n \geq N \to |a_n - L| < \varepsilon/2\big).$$

The term $\varepsilon/2$ contains the variable ε: the fixed ε of the proof, not the bound one of the hypothesis. It is free for ε in φ, because the only quantifiers above the free occurrences of ε in φ are on N and n. Section 4.3's rule makes this step, and "$\varepsilon/2 > 0$" then comes from $\varepsilon > 0$ by arithmetic.

"Take $N = n$." Now suppose the proof has chosen N, so it knows

$$\psi \;=\; \forall n \in \mathbb{N}\; \big(n \geq N \to |a_n - L| < \varepsilon\big),$$

with N free. A tempting step is to "apply this with n in place of N". No rule allows this: N has been chosen, not quantified, so no term may be put in for it. Writing the step out shows how badly it goes wrong, because n is not even free for N in ψ: the occurrence of N lies in the scope of $\forall n$. The substitution produces

$$\forall n \in \mathbb{N}\; \big(n \geq n \to |a_n - L| < \varepsilon\big),$$

which says that **every** term of the sequence is within ε of L, a far stronger claim that does not follow. In words, the error is to let the threshold N be "the n we are looking at", when n is the placeholder of "for all $n \geq N$". A proof that needs a statement about a particular index names it with a new letter, say $m \geq N$, and substitutes m, which is free for n.

Induction, as a formula. The display after Axiom 3.2.1 in Section 3.2 used a property $P(n)$. With substitution it can be written for any formula φ in which the variable n is free, as the formula

$$\Big(\varphi[0/n] \;\wedge\; \forall n \in \mathbb{N}\,(\varphi \to \varphi[n+1/n])\Big) \;\to\; \forall n \in \mathbb{N}\,\varphi,$$

where $\varphi[0/n]$ plays the part of $P(0)$ and $\varphi[n+1/n]$ the part of $P(n+1)$. The term $n+1$ is free for n in φ: its only variable is n, and a free occurrence of n never lies in the scope of a quantifier on n. This is how the principle of induction reads in the language of this chapter; its status (a basic property of $\mathbb{N}$) is unchanged.

4.2.9 Exercises

4.2.9.1 A. Check your understanding

Exercise. A1

(a) For each formula, mark each occurrence of each variable as free or bound, list the free variables, and say whether it is a sentence: (i) $\forall x\,(P(x) \to Q(y))$; (ii) $\exists x\,\forall y\,R(x,y)$; (iii) $\forall x\,(x<y) \wedge \exists y\,(y<x)$; (iv) $\forall \varepsilon\,(\varepsilon > 0 \to \exists \delta\,(\delta > 0 \wedge \delta < \varepsilon))$.

(b) Which of the following are terms, and which are formulas, for one-place f, two-place g and two-place predicate R? (i) $g(f(x),c)$; (ii) $R(f(x),c)$; (iii) $f(R(x,c))$; (iv) $\forall c\,R(c,c)$.

(c) True or false, with a reason: if t is a variable, then t is free for x in every formula.

(d) True or false, with a reason: $(\forall x\,P(x) \wedge Q(x))[c/x]$ is $\forall x\,P(x) \wedge Q(c)$.

Solution. (a) (i) The single occurrence of x, in $P(x)$, is bound by $\forall x$ (the x after $\forall$ belongs to the quantifier node and is not an occurrence); the occurrence of y in $Q(y)$ is free (no quantifier on y). Free variables: y. Not a sentence.

(ii) The occurrence of x is bound by $\exists x$, that of y by $\forall y$. No free variables: a sentence.

(iii) By shorthand rule 2 this is $(\forall x\,(x<y) \wedge \exists y\,(y<x))$. In the left part, x is bound by $\forall x$ and y is free. In the right part, y is bound by $\exists y$ and x is free (the $\forall x$ governs only the left part). Free variables: x and y, each with one free and one bound occurrence. Not a sentence.

(iv) Every occurrence of ε lies below $\forall\varepsilon$, and both occurrences of δ lie below $\exists\delta$; 0 is a constant. A sentence.

(b) (i) A term, by (Tm2) twice. (ii) A formula, atomic by (Fm1): the predicate R applied to the terms $f(x)$ and c. (iii) Neither: $R(x,c)$ is a formula, not a term, and (Tm2) applies f only to a term. (iv) Neither: (Fm4) puts a **variable** after $\forall$, and c is a constant symbol.

(c) False. The variable y is not free for x in $\exists y\,(y \neq x)$, since the free occurrence of x lies in the scope of $\exists y$. (A variable t **is** free for x when t is x itself, or when t does not appear in the formula.)

(d) True. The formula is $(\forall x\,P(x) \wedge Q(x))$ by shorthand rule 2. The $\wedge$ passes the substitution to both parts; $\forall x\,P(x)$ is unchanged because its quantifier is on x, and $Q(x)$ becomes $Q(c)$.

4.2.9.2 B. Practice

Exercise. B1: Computing substitutions

Compute each substitution. Where the term is not free for the variable, say which quantifier would capture it, rename first, and then substitute.

(a) $(P(x) \wedge R(x,y))[f(y)/x]$.
(b) $(\forall x\, R(x,y) \to P(x))[c/x]$.
(c) $(\forall y\, (x < y))[y+1/x]$.
(d) $(\exists x\, (x = 2 \cdot y))[x/y]$.
(e) $(\exists y\, (y > x) \wedge Q(y))[x+1/y]$.

Solution. *(a)* No quantifiers, so $f(y)$ is free for x. Every occurrence of x is replaced: $P(f(y)) \wedge R(f(y), y)$. The y inside $f(y)$ and the y already in $R(x,y)$ now stand side by side; with no quantifier around, both are the same free y, as intended.

(b) By shorthand rule 2 the formula is $(\forall x\, R(x,y) \to P(x))$. The constant c has no variables, so it is free for x. The left part is unchanged (quantifier on x); the free x in $P(x)$ becomes c: $\forall x\, R(x,y) \to P(c)$.

(c) Not free for: the free occurrence of x lies in the scope of $\forall y$, and y is a variable of $y+1$. Rename y to z, which appears nowhere: $\forall z\, (x < z)$. Then substitute: $\forall z\, (y+1 < z)$. (Without renaming, the definition would give $\forall y\, (y+1 < y)$, which says nothing about $y+1$.)

(d) Not free for: the free occurrence of y lies in the scope of $\exists x$, and the term is x. Rename x to z: $\exists z\, (z = 2 \cdot y)$, "$y$ doubled is something", then substitute: $\exists z\, (z = 2 \cdot x)$. (Without renaming, the result would be $\exists x\, (x = 2 \cdot x)$, "some number is its own double", a claim about no particular number.)

(e) By shorthand rule 2 the formula is $(\exists y\, (y > x) \wedge Q(y))$. The occurrence of y in $y > x$ is bound, and the one in $Q(y)$ is free and not in the scope of any quantifier; so $x+1$ is free for y (the $\exists y$ above the bound y is irrelevant, since that occurrence is not replaced). The result is $\exists y\, (y > x) \wedge Q(x+1)$.

Exercise. B2: The set of free variables

For a term t, let $\mathrm{Var}(t)$ be the set of variables of t, and for a formula φ, let $\mathrm{FV}(\varphi)$ be the set of its free variables.

(a) Write down clauses that define $\mathrm{Var}(t)$ by recursion on terms and $\mathrm{FV}(\varphi)$ by recursion on formulas, one clause for each case of Definition 4.2.1 and of Definition 4.2.2.
(b) Use your clauses to compute FV of $\forall x\, (x < y) \wedge \exists y\, (y < x)$ and of $\forall x\, \exists y\, (x + y = z)$, showing each step.

Solution. *(a)* For terms:

$$\mathrm{Var}(x) = \{x\} \text{ for a variable } x, \qquad \mathrm{Var}(c) = \emptyset,$$
$$\mathrm{Var}(f(t_1, \ldots, t_n)) = \mathrm{Var}(t_1) \cup \cdots \cup \mathrm{Var}(t_n).$$

For formulas:

$$\begin{aligned} \mathrm{FV}(R(t_1,\dots,t_n)) &= \mathrm{Var}(t_1)\cup\cdots\cup\mathrm{Var}(t_n),\\ \mathrm{FV}(t_1=t_2) &= \mathrm{Var}(t_1)\cup\mathrm{Var}(t_2), \qquad \mathrm{FV}(\bot)=\emptyset,\\ \mathrm{FV}(\neg\varphi) &= \mathrm{FV}(\varphi), \qquad \mathrm{FV}((\varphi\circ\psi)) = \mathrm{FV}(\varphi)\cup\mathrm{FV}(\psi),\\ \mathrm{FV}(\forall x\,\varphi) &= \mathrm{FV}(\exists x\,\varphi) = \mathrm{FV}(\varphi)\setminus\{x\}. \end{aligned}$$

These are the four lines of the recursive description of "free" in the text, rewritten as sets: $x \in \mathrm{FV}(\varphi)$ exactly when x is free in φ. By structural recursion (carried over to terms and to first-order formulas), each set of clauses defines exactly one function.

(b) First formula. $\mathrm{FV}(x<y)=\{x,y\}$, so $\mathrm{FV}(\forall x\,(x<y))=\{x,y\}\setminus\{x\}=\{y\}$. $\mathrm{FV}(y<x)=\{x,y\}$, so $\mathrm{FV}(\exists y\,(y<x))=\{x\}$. The conjunction gives $\{y\}\cup\{x\}=\{x,y\}$, which agrees with A1(a)(iii).

Second formula. $\mathrm{FV}(x+y=z)=\mathrm{Var}(x+y)\cup\mathrm{Var}(z)=\{x,y,z\}$. Then $\mathrm{FV}(\exists y\,(x+y=z))=\{x,z\}$, and $\mathrm{FV}(\forall x\,\exists y\,(x+y=z))=\{z\}$. So the formula is an open sentence in z alone (over $\mathbb{R}$ it is true for every z: given x, take $y=z-x$).

4.2.9.3 C. Going deeper

Exercise. C1: What substitution does to the free variables

Let φ be a formula, x a variable and t a term, with t free for x in φ, and with x free in φ.

(a) Prove that the free variables of $\varphi[t/x]$ are the free variables of φ other than x, together with the variables of t:

$$\mathrm{FV}(\varphi[t/x]) = (\mathrm{FV}(\varphi)\setminus\{x\})\cup\mathrm{Var}(t),$$

with the notation of B2.

(b) Show, with the capture example $\exists y\,(y>x)$, $t=y+1$, that the conclusion can fail when t is not free for x. Show also, with an example of your own, that it can fail when x is not free in φ.

Solution (Hint 1). Structural induction on φ. The hypothesis "x free in φ" does not pass to both parts of $(\varphi\circ\psi)$, so prove a statement that also covers formulas in which x is not free.

Solution (Hint 2). Prove, for every φ in which t is free for x: if x is free in φ, the displayed equation holds; if not, $\varphi[t/x]=\varphi$. Start with the same statement for terms: $\mathrm{Var}(s[t/x]) = (\mathrm{Var}(s)\setminus\{x\})\cup\mathrm{Var}(t)$ when x occurs in s.

Solution. (a) **Terms.** Let s be a term in which x occurs. Then $\mathrm{Var}(s[t/x]) = (\mathrm{Var}(s)\setminus\{x\})\cup\mathrm{Var}(t)$. By induction on s: if s is the variable x, then $s[t/x]=t$ and both sides are $\mathrm{Var}(t)$. (A constant, or a variable other than x, does not contain x.) If $s=f(s_1,\dots,s_n)$, then $s[t/x]=f(s_1[t/x],\dots,s_n[t/x])$; for each s_i containing x, the induction hypothesis gives $\mathrm{Var}(s_i[t/x]) = (\mathrm{Var}(s_i)\setminus\{x\})\cup\mathrm{Var}(t)$, and for each s_i not containing x, $s_i[t/x]=s_i$ (as

in the proof of the claim in the text). At least one s_i contains x, so the union over i is $(\mathrm{Var}(s) \setminus \{x\}) \cup \mathrm{Var}(t)$.

Formulas. We prove, by structural induction on φ, the statement $S(\varphi)$: *if t is free for x in φ, then* $\mathrm{FV}(\varphi[t/x]) = (\mathrm{FV}(\varphi) \setminus \{x\}) \cup \mathrm{Var}(t)$ *when* $x \in \mathrm{FV}(\varphi)$*, and* $\varphi[t/x] = \varphi$ *when* $x \notin \mathrm{FV}(\varphi)$. The second half is the claim proved in the text, for every φ; so in each case below we assume $x \in \mathrm{FV}(\varphi)$ and prove the equation. Write $E(\chi)$ for the set $(\mathrm{FV}(\chi) \setminus \{x\}) \cup \mathrm{Var}(t)$.

Atomic. Let $\varphi = R(s_1, \ldots, s_n)$ with x occurring in some s_i. By the clause for atomic formulas, $\mathrm{FV}(\varphi[t/x])$ is the union over i of the sets $\mathrm{Var}(s_i[t/x])$. Exactly as in the term case for $f(s_1, \ldots, s_n)$ (the result for terms for each s_i containing x, $s_i[t/x] = s_i$ for the others, and at least one s_i containing x), this union is $E(\varphi)$. The same for $s_1 = s_2$. ($\bot$ has no free variables.)

Negation. $\mathrm{FV}((\neg\varphi)[t/x]) = \mathrm{FV}(\varphi[t/x]) = E(\varphi) = E(\neg\varphi)$, by the induction hypothesis (t is free for x in φ, and x is free in φ).

Binary connective. Let x be free in $(\varphi \circ \psi)$; then t is free for x in both parts, and x is free in at least one. If it is free in both, the induction hypothesis gives $E(\varphi) \cup E(\psi) = E((\varphi \circ \psi))$. If it is free in φ only, then $\psi[t/x] = \psi$, and $E(\varphi) \cup \mathrm{FV}(\psi) = E((\varphi \circ \psi))$, because $x \notin \mathrm{FV}(\psi)$. Symmetrically if it is free in ψ only.

Quantifier. Let x be free in $\forall y\, \varphi$. Then y is not x and x is free in φ; and since t is free for x in $\forall y\, \varphi$ while x is free there, $y \notin \mathrm{Var}(t)$ and t is free for x in φ (the recursive description of "free for"). So

$$\begin{aligned}\mathrm{FV}\big((\forall y\, \varphi)[t/x]\big) &= \mathrm{FV}\big(\varphi[t/x]\big) \setminus \{y\} = \big((\mathrm{FV}(\varphi) \setminus \{x\}) \cup \mathrm{Var}(t)\big) \setminus \{y\} \\ &= (\mathrm{FV}(\varphi) \setminus \{x, y\}) \cup \mathrm{Var}(t) = E(\forall y\, \varphi),\end{aligned}$$

where the third equality uses $y \notin \mathrm{Var}(t)$, and the last uses $\mathrm{FV}(\forall y\, \varphi) = \mathrm{FV}(\varphi) \setminus \{y\}$. The case of $\exists y$ is the same.

This proves $S(\varphi)$ for every formula, and (a) is its first half.

(b) With $\varphi = \exists y\, (y > x)$ and $t = y + 1$: $\mathrm{FV}(\varphi) = \{x\}$, so the equation would give $\mathrm{FV}(\varphi[t/x]) = \{y\}$. But $\varphi[t/x] = \exists y\, (y > y + 1)$, whose set of free variables is $\emptyset$. The y of the term was captured, so it is not free in the result; the quantifier case of the proof fails because $y \in \mathrm{Var}(t)$.

For the second hypothesis, take $\varphi = P(z)$ and $t = y$. Then x is not free in φ, and t is free for x (there are no quantifiers). Now $\varphi[t/x] = P(z)$, with free variables $\{z\}$, while the equation would give $\{z\} \cup \{y\} = \{y, z\}$. Without a free x, nothing is put in, so the variables of t do not appear.

Marking scheme. (10 points) - 2: proves the statement for terms, or states it and handles the variable case and the case of a compound term. - 2: sets up an induction that survives the connective case (strengthened statement, or a separate treatment of parts in which x is not free, using $\varphi[t/x] = \varphi$). - 3: the quantifier case, using that "free for" gives $y \notin \mathrm{Var}(t)$ and naming where that fact is used. - 1: the atomic and negation cases. - 2: both counterexamples in (b), each checked by computing both sides. Fatal errors (0 for the step): using the equation $\mathrm{FV}(\chi[t/x]) = (\mathrm{FV}(\chi) \setminus \{x\}) \cup \mathrm{Var}(t)$ for a part χ in which x is not free (it fails there; see the second example in (b)); removing y from $\mathrm{Var}(t)$ without justification.

Exercise. C2: Renaming is not swapping

(a) Show that $\forall x\,\exists y\,(x < y)$ and $\forall y\,\exists x\,(y < x)$ are the same sentence up to renaming bound variables: give a sequence of renamings, each allowed by the condition of the text, leading from the first to the second.

(b) Show that $\forall x\,\exists y\,(x < y)$ and $\exists y\,\forall x\,(x < y)$ are **not** the same up to renaming. *Hint: what does a renaming leave unchanged?*

Solution. (a) A renaming needs a variable that does not appear in the formula, so the two letters cannot be exchanged in one step; go through a third letter z.

1. In $\forall x\,\exists y\,(x < y)$, rename the bound x to z, which does not appear: $\forall z\,\exists y\,(z < y)$.
2. Rename the bound y to x, which no longer appears: $\forall z\,\exists x\,(z < x)$.
3. Rename the bound z to y, which no longer appears: $\forall y\,\exists x\,(y < x)$.

Each step renames one quantifier and the occurrences it binds, to a letter absent from the formula at that moment. The result is the second sentence. (Over $\mathbb{N}$ both say "there is no largest natural number", in two spellings.)

(b) A renaming changes only which **letters** are used. It does not change the parse tree's shape, which quantifier symbol ($\forall$ or $\exists$) sits at which node, or which quantifier node binds which occurrence. In $\forall x\,\exists y\,(x < y)$ the root is a $\forall$ node; in $\exists y\,\forall x\,(x < y)$ the root is an $\exists$ node. No sequence of renamings turns a $\forall$ node into an $\exists$ node, so no sequence of renamings leads from one sentence to the other.

The two sentences also say different things: over $\mathbb{N}$, the first is true (take $y = x + 1$) and the second false (for any y, take $x = y$). This is Example (Swapping two quantifiers) again. Renaming changes the **names** of the placeholders; swapping changes the **order** in which they are chosen, and the order is what carries the meaning.

4.3 Using "For Every": Instantiation

Section 4.2 made the language exact and defined the operation $\varphi[t/x]$, "put the term t in for the variable x", with its one hazard, capture. We can now give the first quantifier rule. It is the rule for **using** a universal statement: whatever holds for every object holds for the object named by any term we care to write. The rule is called $\forall$E. It is the rule behind every "by the theorem, applied to $x = 2$". With it alone, universal statements can be used but not proved, and the section ends at exactly that gap, which Section 4.4 fills.

4.3.1 Applying a statement about everything

Every application of a general fact to a particular case is one step of logic, and so far that step has been taken on trust. Three earlier places took it.

- Section 1.3 used the theorem "if an integer is divisible by 2 and by 3, then it is divisible by 6" at $n = 42$, taking the step from "every integer" to the one integer 42 on trust.
- In Example (A number below every positive number) (Section 2.5) the hypothesis was "$x \leq \varepsilon$ for every $\varepsilon > 0$", and the proof applied it to the one number $\varepsilon = x/2$.
- Section 4.1 opened with the argument "every positive real has a square root; 2 is positive; so 2 has a square root", which a truth table calls invalid because it sees only three unrelated

atoms.

The same step happens in each: a statement about **every** object is applied to **one** object, named by a term. In the language of Section 4.2, the first premise of the square-root argument is $\forall x\,(P(x) \to S(x))$, and applying it to 2 gives

$$(P(x) \to S(x))[2/x] \;=\; P(2) \to S(2),$$

which is part (a) of Example (Four substitutions). From there →E finishes. All that is missing is a rule that licenses the first step.

4.3.2 The rule

A universal statement may be applied to any term: remove the quantifier and put the term in for the variable.

Definition 4.3.1. Universal Elimination

i	$\forall x\,\varphi$	
k	$\varphi[t/x]$	$\forall$E i

Here φ is a formula, x a variable, and t **any** term, subject to one side condition: t **is free for** x **in** φ (Definition 4.2.6). Line i must be available at line k (the scope rule of Definition 1.4.2). This rule is called ∀E (**universal elimination**, or **instantiation**), and $\varphi[t/x]$ is an **instance** of $\forall x\,\varphi$.

In words: if a line says that φ holds for every x, then any later line, where that line is still available, may write what φ says about t. The rule cites one line and no box.

Read the definition clause by clause.

1. **The cited line begins with $\forall x$.** Its main symbol must be the quantifier, with the whole of φ as its scope. By shorthand rule 2 of Section 4.2, $\forall x\,P(x) \to Q(c)$ is an implication whose hypothesis is $\forall x\,P(x)$; its main symbol is →, so ∀E does not apply to it. A ∀ buried inside another connective is not available to the rule until the rules for that connective have dug it out.
2. **Any term.** The term may be a constant (c, 2), a variable (y, or x itself), or a compound term ($f(c)$, $a-1$, $\varepsilon/2$). It may be a variable that already occurs elsewhere in the derivation. Nothing about the term has to be established first: ∀E needs no line saying that t "exists" or "is in the domain", because every term names an element of the domain.
3. **Every free occurrence, by the same term.** The new line is $\varphi[t/x]$, so by Definition 4.2.5 every free occurrence of x in φ is replaced, and all of them by the same t. We do not choose which occurrences change.
4. **Free for x.** This is the condition of Definition 4.2.6, and it is the rule's only restriction. It forbids exactly the substitutions in which a variable of t would be captured by a quantifier of φ.
5. **Line i is available.** Line i must be available at line k (the scope rule of Definition 1.4.2). Every rule asks this; it is not a restriction special to ∀E.

A degenerate case. If x is not free in φ, then $\varphi[t/x] = \varphi$ for every term t (the claim proved in Section 4.2), and ∀E simply removes a quantifier that bound nothing: from $\forall x\,P(c)$ it writes $P(c)$. Legal, and dull.

One line, many uses. The cited line stays where it is, available as before. A derivation may instantiate the same universal line as often as it likes, at different terms or the same one. Worked derivation (d) below uses one premise twice.

Why this rule. It says no more than the word "every". If every object in the domain satisfies φ, then so does the object that t names, whichever object that is. A term always names something, since the domain is never empty (Section 4.1), so the instance always has something to be about.

Why "free for". The instance must say about t what φ says about x. Section 4.2 showed that when a variable of t is captured, $\varphi[t/x]$ can say something quite different. The condition rules capture out, and the subsection on common errors below shows the rule going wrong without it. Section 4.11 states that the rule, with its condition, never leads from true premises to a false line, and the optional Section 4.12 proves it.

Quick check. Line 1 of a derivation is $\forall x\,\exists z\,(z > x + y)$. For which terms t may ∀E write $(\exists z\,(z > x + y))[t/x]$ on line 2? Is $t = y$ allowed? Is $t = z + 1$?

Solution. The only free occurrence of x in $\exists z\,(z > x + y)$ lies in the scope of $\exists z$, and that is the only quantifier above it. So t is free for x exactly when z is not a variable of t: **any term not containing** z is allowed. The term y is allowed, and gives $\exists z\,(z > y + y)$; the variable y already occurs free in the line, and that does no harm. The term $z + 1$ is not allowed: it would give $\exists z\,(z > z + 1 + y)$, where the z of the term has been captured. Nor is z itself. To instantiate at z, rename the bound z first (Section 4.2), say to w: from $\forall x\,\exists w\,(w > x + y)$, ∀E at z gives $\exists w\,(w > z + y)$.

4.3.3 Four derivations

The worked derivations below use ∀E and the rules of Chapter 1. Each has an Idea that says why each rule was chosen, continuing the search method of Section 2.6: a goal-directed step looks at the goal's main symbol, a premise-directed step at a resource's. The new question is the one that ∀E raises every time: **at which term?**

Example. Instantiation at work

Give a derivation of each sequent.

(a) $\forall x\,(P(x) \to Q(x)),\ P(c) \vdash Q(c)$.
(b) $\forall x\,\forall y\,R(x, y) \vdash R(c, c)$.
(c) $\forall x\,\forall y\,(R(x, y) \to R(y, x)),\ R(a, b) \vdash R(b, a)$.
(d) $\forall x\,(P(x) \to P(f(x))),\ P(c) \vdash P(f(f(c)))$.

Solution.

(a) *Idea.* The goal $Q(c)$ has no connective or quantifier, so no introduction rule produces it, and the search turns to the premises. The premise $P(c)$ cannot be taken apart. The premise $\forall x\,(\dots)$ begins with ∀, so it can be used only by ∀E. At which term? The goal mentions the term c, so instantiate at c. That gives an implication whose hypothesis

is a premise and whose conclusion is the goal.

1	$\forall x\,(P(x) \to Q(x))$	
2	$P(c)$	
3	$P(c) \to Q(c)$	$\forall$E 1
4	$Q(c)$	$\to$E 3, 2

Line 3 is $(P(x) \to Q(x))[c/x]$; the constant c has no variables, so it is free for x. Read c as 2, $P(x)$ as "x is positive", and $Q(x)$ as "x has a square root" (Section 4.1 wrote S for this predicate), and the derivation is the square-root argument of Section 4.1, in the four lines promised there. The argument that truth tables rejected is derivable once the rules can see inside its sentences.

(b) *Idea.* The premise has two quantifiers, and $\forall$E removes one at a time, the outermost first: the main symbol of the premise is $\forall x$, and only after it is gone is $\forall y$ the main symbol. The goal $R(c, c)$ shows the term for both.

1	$\forall x\,\forall y\,R(x, y)$	
2	$\forall y\,R(c, y)$	$\forall$E 1
3	$R(c, c)$	$\forall$E 2

Line 2 is $(\forall y\,R(x, y))[c/x]$: the quantifier $\forall y$ is on a variable other than x, so the substitution passes through it (Definition 4.2.5). Line 3 instantiates y at c as well. Nothing forbids putting the same term in for two different variables; "for all x and y" includes the case $x = y$.

(c) *Idea.* Work backwards from the goal $R(b, a)$. The only line that can produce it is the first premise, through an instance of its inside, $R(x, y) \to R(y, x)$, followed by $\to$E. Which instance has conclusion $R(b, a)$? Matching $R(y, x)$ with $R(b, a)$ gives $y := b$ and $x := a$. Then the hypothesis of that instance is $R(a, b)$, which is the second premise. So the terms are read off the goal: a for x first (the outer quantifier), then b for y.

1	$\forall x\,\forall y\,(R(x, y) \to R(y, x))$	
2	$R(a, b)$	
3	$\forall y\,(R(a, y) \to R(y, a))$	$\forall$E 1
4	$R(a, b) \to R(b, a)$	$\forall$E 3
5	$R(b, a)$	$\to$E 4, 2

A reader who matches the goal $R(b, a)$ against the hypothesis $R(x, y)$ instead of the conclusion $R(y, x)$ puts b for x and a for y, and gets $R(b, a) \to R(a, b)$. That is a legal line, but its hypothesis is the goal itself, so $\to$E cannot fire. Match the goal against the conclusion of the implication, since that is the part $\to$E delivers.

(d) *Idea, forwards.* The premise $P(c)$ is the hypothesis of the instance of the first premise at c, so instantiating at c and firing gives $P(f(c))$. Now $P(f(c))$ is the hypothesis of the instance at $f(c)$, so the same premise, instantiated again, fires again and gives $P(f(f(c)))$, the goal.

Idea, backwards. The goal $P(f(f(c)))$ has the shape $P(f(x))$, the conclusion of the inside of the first premise, with $x := f(c)$. So the last $\forall$E must be at the **compound** term $f(c)$, and its hypothesis $P(f(c))$ becomes the new goal. That has the shape $P(f(x))$ with $x := c$, and its hypothesis is the premise $P(c)$.

1	$\forall x\,(P(x) \to P(f(x)))$	
2	$P(c)$	
3	$P(c) \to P(f(c))$	$\forall$E 1
4	$P(f(c))$	$\to$E 3, 2
5	$P(f(c)) \to P(f(f(c)))$	$\forall$E 1
6	$P(f(f(c)))$	$\to$E 5, 4

Line 5 cites line 1 again: a universal line can be instantiated as often as needed. Its term $f(c)$ contains no variable, so it is free for x, and $(P(x) \to P(f(x)))[f(c)/x]$ is $P(f(c)) \to P(f(f(c)))$: **every** x is replaced by the whole term $f(c)$, including the one inside $f(x)$.

An arithmetic reading: over $\mathbb{N}$, let $f(n)$ be $n+2$, let $P(n)$ say "n is even", and let c be 0. The first premise says that adding 2 to an even number gives an even number, the second that 0 is even, and the derivation concludes that $f(f(0)) = 4$ is even. Each further step up the even numbers would cost two more lines. Reaching "every n of the form $f(\cdots f(0)\cdots)$" at once is a job for induction, not for this rule.

In each derivation, the Idea chose the term of every $\forall$E by matching: the goal, or a hypothesis still needed, has a definite shape, and the instance is chosen to have that shape. Instantiating at a term that nothing asks for is legal, and it fills a derivation with lines that lead nowhere.

4.3.4 Checking a line of instantiation

Choosing the term can take thought; checking a line never does. For a line justified "$\forall$E i", ask four questions.

1. Is line i available here (Definition 1.4.2)?
2. Is the main symbol of line i a universal quantifier? If so, read off its variable x and its scope φ.
3. Is this line $\varphi[t/x]$ for some term t? Find t by comparing the two lines at a free occurrence of x in φ; then check that **every** free occurrence of x became that same t, and that nothing else changed.
4. Is t free for x in φ?

Run on line 5 of (d): line 1 is available, since no box contains it; its main symbol is $\forall x$, with scope $P(x) \to P(f(x))$; comparing gives $t = f(c)$, and both occurrences of x became $f(c)$; and $f(c)$ has no variables, so it is free for x. The line passes.

4.3.5 How this rule looks in a mathematical proof

Let a be a real number. A proof might say:

> Since $y^2 \geq 0$ for every real y, applying this to $a-1$ gives $(a-1)^2 \geq 0$, that is, $a^2 - 2a + 1 \geq 0$.

The first half of the sentence is one $\forall$E, at the compound term $a-1$. The second half rewrites $(a-1)^2$ by school algebra. In a derivation such a step is written as one line justified "arith", for **arithmetic**:

1	$\forall y\,(y^2 \geq 0)$	
2	$(a-1)^2 \geq 0$	$\forall$E 1
3	$a^2 - 2\cdot a + 1 \geq 0$	arith 2

Line	Sentence
1	"Since $y^2 \geq 0$ for every real y, ..." (the general fact)
2	"... applying this to $a-1$ gives $(a-1)^2 \geq 0$, ..." ($\forall$E at $a-1$)
3	"... that is, $a^2 - 2a + 1 \geq 0$." (arith)

A line justified "arith" records a fact of school arithmetic or algebra, used on the lines it cites; here the fact is that $(u-1)^2$ and $u^2 - 2u + 1$ are the same number for every real u. It is **not a new rule**. Written out in full, it is a universally quantified arithmetic statement, here

$$\forall u\,((u-1)^2 \geq 0 \to u^2 - 2\cdot u + 1 \geq 0),$$

taken as one more premise, instantiated by $\forall$E at the term on the line (here a) and applied by $\to$E to the cited line. That is exactly the pattern of (a) above. The book writes it as one line because spelling it out every time would bury the logic under algebra; its bound variables are letters that appear nowhere else in the derivation, and that keeps every term free for them. As in the earlier chapters, the facts of school algebra are used here as facts.

Section 4.4 will turn this three-line derivation into a proof of $x^2 - 2x + 1 \geq 0$ for **every** real x. That needs a rule this section does not have.

4.3.6 Common errors

Three mistakes are worth seeing written out, each with a reading in which the premise is true and the offending line false.

Capture. The premise $\forall x\,\exists y\,(y \neq x)$ says, over $\mathbb{N}$, that for every number there is a different one; it is true, since $x + 1 \neq x$. Instantiating "at y" without looking writes this:

1	$\forall x\,\exists y\,(y)$	$\neg$E **??**
2	$\exists y\,(y)$	$\neg$E **??**

Line 2 says that some number differs from itself, which is false. The step breaks the side condition: the free occurrence of x in $\exists y\,(y \neq x)$ lies in the scope of $\exists y$, and y is a variable of the term y, so y is not free for x. This is the capture example of Section 4.2, now inside a derivation. The repair is to rename first: $\forall x\,\exists z\,(z \neq x)$ is the same premise up to renaming, and $\forall$E at y gives $\exists z\,(z \neq y)$, "some number differs from y", which is what was meant.

Warning. $\forall$E at a term that is not free for x can turn a true premise into a false line, and the definition of $\varphi[t/x]$ gives no signal: it returns a formula either way. Over $\mathbb{N}$, from the true $\forall x\,\exists y\,(y \neq x)$, the "instance at y" is $\exists y\,(y \neq y)$, which is false. Before instantiating at a term with variables, look for quantifiers on those variables inside φ.

A buried quantifier. From the line $\forall x\, P(x) \to Q(c)$ one may **not** write $P(c) \to Q(c)$. The main symbol of the line is $\to$, and the $\forall x$ governs only the hypothesis $\forall x\, P(x)$. The line says "if **every** x has P, then $Q(c)$"; the false step reads it as "if c has P, then $Q(c)$", which is far stronger. Over $\mathbb{N}$, read $P(x)$ as $x \geq 1$, $Q(x)$ as $x = 0$, and c as 1. The line is true, because its hypothesis "every natural number is at least 1" is false (0 is not). The false step's result, "if $1 \geq 1$, then $1 = 0$", is false.

Replacing only some occurrences. Over $\mathbb{R}$, $\forall x\, (x \leq x^2 \vee x < 1)$ is true: a real $x \geq 1$ has $x^2 = x \cdot x \geq x$, and every other real has $x < 1$. Writing $2 \leq x^2 \vee x < 1$ from it replaces the first x and leaves the others. That is not $\varphi[2/x]$, which is $2 \leq 2^2 \vee 2 < 1$: the substitution replaces every free occurrence. The half-replaced line still has x free, and when x names 1 it reads $2 \leq 1 \vee 1 < 1$, which is false.

4.3.7 A stall

Try the sequent

$$\forall x\, (P(x) \wedge Q(x)) \vdash \forall x\, P(x).$$

It is plainly correct: if every x has both properties, then every x has the first. Search for a derivation. The goal begins with $\forall$, and no rule so far is made for proving a $\forall$. Apart from $\bot$E (Definition 2.1.1) and RAA (Definition 2.5.1), which reach a line only by way of a contradiction, a line beginning with $\forall$ can only be taken out of a line or box that already contains it. So the goal-directed step has nothing to offer, and we turn to the premise. $\forall$E at c gives $P(c) \wedge Q(c)$, and $\wedge$E gives $P(c)$. The same works at d, at $f(c)$, at any term we name. But each line is an instance. No finite list of instances is the statement $\forall x\, P(x)$, which speaks about every object at once, including all the objects that none of our terms happen to name.

So with $\forall$E alone, a universal statement can be **used** but not **proved**. Leaving aside a detour through a contradiction, a derivation can write $\forall x\, \psi$ only by taking it out of a line or box that already contains it, as line 2 of (b) took $\forall y\, R(c, y)$ out of the premise. Section 1.3 was in the same position with implications, and Section 1.4 filled the gap with a box. Section 4.4 does the same for $\forall$: it proves P of one name about which nothing is known, and that box finishes this derivation in four lines.

4.3.8 What the rule sounds like

The table of Sections 1.2–1.6 and 2.1–2.5 gains a row for $\forall$E.

Rule	You write	Lean
$\forall$E	"In particular, for $x = t$, ..."; "applying the hypothesis to t, ..."	`h t` (or `specialize h t`)

Written mathematics has many forms of the phrase: "in particular, for $x = 3$, ..."; "applying the hypothesis to $x_0 - 1$, ..."; "by the theorem with $\varepsilon/2$ in place of ε, ...". In the Lean column, `h` is the universal statement and `t` the term; applying `h` to `t` gives the instance, just as applying an implication to a proof of its hypothesis gave its conclusion in Section 1.3. In Lean a universal statement behaves like a function that takes a term and returns a proof of the instance.

4.3.9 The search method: one more step

The search method of Section 2.6 had a goal-directed step and a premise-directed step for each connective. Instantiation adds a premise-directed step.

- **A resource $\forall x\,\varphi$.** Do not instantiate blindly. Instantiate at the term that the goal, or another resource, asks for: match the shape of the goal (or of a hypothesis still needed) against φ, and read off the term. If the term is not free for x in φ, rename the bound variables of φ first.

Instantiating is always **safe**, in the sense of Section 2.6: the new line is added and nothing is lost, since the universal line stays available. What can go wrong is only that the derivation fills with instances nobody needs, as a forward search filled with useless lines in Section 1.3.

4.3.10 Exercises

4.3.10.1 A. Check your understanding

Exercise. A1

(a) Write the schema of ∀E and its side condition from memory.

(b) For each pair, decide whether line 2 follows from line 1 by one application of ∀E. If not, name the clause of Definition 4.3.1 that fails.

- (i) Line 1: $\forall x\,\exists y\,(y > x)$; line 2: $\exists y\,(y > y)$.
- (i) Line 1: $\forall x\,(x \cdot x \geq 0)$; line 2: $3 \cdot x \geq 0$.
- (i) Line 1: $\forall x\,P(x) \to P(c)$; line 2: $P(a) \to P(c)$.
- (i) Line 1: $\forall x\,(P(x) \to P(f(x)))$; line 2: $P(f(c)) \to P(f(f(c)))$.

(d) True or false, with a reason: from $\forall x\,\forall y\,R(x,y)$, one application of ∀E gives $\forall x\,R(x,c)$.

(e) Why does ∀E not need a line saying that the term t names an element of the domain?

Solution.

(a) From line i: $\forall x\,\varphi$, line k may be $\varphi[t/x]$, justified "∀E i", for any term t that is free for x in φ, with line i available at line k.

(b) (i) Not legal: the term y is not free for x in $\exists y\,(y > x)$, since the free x lies in the scope of $\exists y$ (clause 4, capture). Over $\mathbb{Z}$, line 1 is true and line 2 is false.

(ii) Not legal: only the first x was replaced, so line 2 is not $\varphi[t/x]$ for any t (clause 3). The instance at 3 is $3 \cdot 3 \geq 0$. Line 2 still has x free, and over $\mathbb{R}$ it is false when x names -1.

(iii) Not legal: the main symbol of line 1 is $\to$, not $\forall$ (clause 1); the $\forall x$ governs only $P(x)$. Over $\mathbb{N}$ with $P(x)$ read as $x \geq 1$, c as 0, and a as 1: line 1 says "if every natural number is at least 1, then $0 \geq 1$", true because its hypothesis is false; line 2 says "if $1 \geq 1$, then $0 \geq 1$", which is false.

(iv) Legal: it is $\varphi[f(c)/x]$ with $\varphi = P(x) \to P(f(x))$, both occurrences of x replaced by $f(c)$, and $f(c)$ has no variables, so it is free for x (clause 2: any term, compound ones included).

(v) False. The main symbol is $\forall x$, so one application removes $\forall x$ and gives $\forall y\,R(t,y)$ for some term t. The line $\forall x\,R(x,c)$ would need the inner quantifier removed while the outer one stays, and the rule cannot reach inside the outer quantifier. (The sequent $\forall x\,\forall y\,R(x,y) \vdash \forall x\,R(x,c)$ is derivable, but it needs the rule of Section 4.4.)

(vi) Every term names an element of the domain: once the symbols are interpreted, a constant names one, and a compound term names the value of a function at elements. And the domain is never empty (Section 4.1). So "every element satisfies φ" applies to

whatever t names, with nothing further to check. (Bounded quantifiers are different: to use $\forall x \in A\ \varphi$ at t one also needs $t \in A$, and Section 4.8 builds that into a rule.)

4.3.10.2 B. Practice

Exercise. B1: Two links

Give a derivation of

$$\forall x\,(P(x) \to Q(x)),\ \forall x\,(Q(x) \to R(x)),\ P(a) \vdash R(a),$$

and say, for each ∀E line, which line of the search asked for its term.

Solution. The goal $R(a)$ is the conclusion of the instance of the second premise at a, so it suffices to reach $Q(a)$; that is the conclusion of the instance of the first premise at a, so it suffices to reach $P(a)$, a premise. Both terms come from the goal: the second ∀E (line 6) from $R(a)$, the first (line 4) from the subgoal $Q(a)$.

1	$\forall x\,(P(x) \to Q(x))$	
2	$\forall x\,(Q(x) \to R(x))$	
3	$P(a)$	
4	$P(a) \to Q(a)$	∀E 1
5	$Q(a)$	→E 4, 3
6	$Q(a) \to R(a)$	∀E 2
7	$R(a)$	→E 6, 5

Exercise. B2: An increasing function

Read f as a function from $\mathbb{R}$ to $\mathbb{R}$ and a, b, c as real numbers. Give a derivation of

$$\forall x\,\forall y\,(x \le y \to f(x) \le f(y)),\ a \le b,\ b \le c\ \vdash\ f(a) \le f(b) \wedge f(b) \le f(c),$$

and then write the argument as one sentence of prose.

Solution. The goal is a conjunction, so its last line is ∧I, with subgoals $f(a) \le f(b)$ and $f(b) \le f(c)$. Each is the conclusion of an instance of the first premise: matching $f(x) \le f(y)$ with $f(a) \le f(b)$ gives $x := a$, $y := b$, and with $f(b) \le f(c)$ gives $x := b$, $y := c$. The hypotheses of the two instances are the other two premises.

1	$\forall x\, \forall y\, (x \le y \to f(x) \le f(y))$	
2	$a \le b$	
3	$b \le c$	
4	$\forall y\, (a \le y \to f(a) \le f(y))$	$\forall$E 1
5	$a \le b \to f(a) \le f(b)$	$\forall$E 4
6	$f(a) \le f(b)$	$\to$E 5, 2
7	$\forall y\, (b \le y \to f(b) \le f(y))$	$\forall$E 1
8	$b \le c \to f(b) \le f(c)$	$\forall$E 7
9	$f(b) \le f(c)$	$\to$E 8, 3
10	$f(a) \le f(b) \wedge f(b) \le f(c)$	$\wedge$I 6, 9

The premise is instantiated twice, along two different paths: once at a and b, once at b and c. In prose: *Since f is increasing and $a \le b$, we have $f(a) \le f(b)$, and likewise $b \le c$ gives $f(b) \le f(c)$.* The word "increasing" carries the first premise, and the two applications of it are the two pairs of $\forall$E lines.

Exercise. B3: Squares are not negative

Let a and b be real numbers. Give a derivation of

$$\forall x\, (x^2 \ge 0) \vdash (a+b)^2 \ge 0 \wedge (a-b)^2 \ge 0.$$

Then extend it by one line justified "arith" to a derivation of $a^2 + b^2 \ge 2ab$ from the same premise, and say which earlier line the arith line uses.

Solution. The goal is a conjunction of two instances of the premise, at the compound terms $a+b$ and $a-b$. Both terms are free for x in $x^2 \ge 0$, which has no quantifier.

1	$\forall x\, (x^2 \ge 0)$	
2	$(a+b)^2 \ge 0$	$\forall$E 1
3	$(a-b)^2 \ge 0$	$\forall$E 1
4	$(a+b)^2 \ge 0 \wedge (a-b)^2 \ge 0$	$\wedge$I 2, 3
5	$a^2 + b^2 \ge 2 \cdot a \cdot b$	arith 3

Lines 1–4 are the first derivation. For the second goal only line 3 is needed: $(a-b)^2 = a^2 - 2ab + b^2$, so $(a-b)^2 \ge 0$ says $a^2 - 2ab + b^2 \ge 0$, and adding $2ab$ to both sides gives $a^2 + b^2 \ge 2ab$. That algebra is the arith line 5. (Lines 2 and 4 are then unused, which is allowed, and deleting them leaves a three-line derivation of $a^2 + b^2 \ge 2ab$. Line 2 would give $a^2 + b^2 \ge -2ab$ in the same way.)

4.3.10.3 C. Going deeper

Exercise. C1: A partner makes a loop

A relation R is **symmetric** if $R(x, y)$ gives $R(y, x)$, and **transitive** if $R(x, y)$ and $R(y, z)$ give $R(x, z)$. Show that in a symmetric and transitive relation, every element that is related to something is related to itself, by giving a derivation of

$$\forall x\, \forall y\, (R(x, y) \to R(y, x)),$$
$$\forall x\, \forall y\, \forall z\, ((R(x, y) \land R(y, z)) \to R(x, z)),\ R(a, b) \ \vdash\ R(a, a).$$

Explain how each term of each ∀E was chosen.

Solution (Hint 1). The goal $R(a, a)$ can only come from the transitivity premise. Match its conclusion $R(x, z)$ with the goal.

Solution (Hint 2). The match gives $x \coloneqq a$ and $z \coloneqq a$, and leaves y open. Which y makes the hypothesis $R(a, y) \land R(y, a)$ reachable from the premises?

Solution. The search. The goal $R(a, a)$ is atomic, so it must be the conclusion of an instance of a premise. Symmetry ends in $R(y, x)$, and its instance at a, a is $R(a, a) \to R(a, a)$, whose hypothesis is the goal itself: a dead end. Transitivity ends in $R(x, z)$. Matching gives $x \coloneqq a$ and $z \coloneqq a$. The variable y does not occur in the conclusion, so the goal says nothing about it; the hypothesis $R(a, y) \land R(y, a)$ must decide. The premise $R(a, b)$ offers $R(a, y)$ with $y \coloneqq b$. Then the second conjunct is $R(b, a)$, which symmetry gives from $R(a, b)$: the instance of the symmetry premise at $x \coloneqq a$, $y \coloneqq b$ (read off its needed conclusion $R(b, a)$, as in Example (Instantiation at work) (c)).

1	$\forall x\, \forall y\, (R(x, y) \to R(y, x))$	
2	$\forall x\, \forall y\, \forall z\, ((R(x, y) \land R(y, z)) \to R(x, z))$	
3	$R(a, b)$	
4	$\forall y\, (R(a, y) \to R(y, a))$	∀E 1
5	$R(a, b) \to R(b, a)$	∀E 4
6	$R(b, a)$	→E 5, 3
7	$\forall y\, \forall z\, ((R(a, y) \land R(y, z)) \to R(a, z))$	∀E 2
8	$\forall z\, ((R(a, b) \land R(b, z)) \to R(a, z))$	∀E 7
9	$(R(a, b) \land R(b, a)) \to R(a, a)$	∀E 8
10	$R(a, b) \land R(b, a)$	∧I 3, 6
11	$R(a, a)$	→E 9, 10

The terms. Lines 7 and 9 put a in for x and for z: both read off the goal. Line 8 puts b in for y: read off the premise $R(a, b)$, the only available fact of the form $R(a, \cdot)$. Lines 4 and 5 put in a and b: read off the needed line $R(b, a)$. All terms are constants, so every one is free for its variable.

The hypothesis "related to something" is needed. In a symmetric and transitive relation an element related to nothing need not be related to itself: on $\{0,1\}$, the relation that holds only between 0 and 0 is symmetric and transitive, and $R(1,1)$ fails. Without a premise like $R(a,b)$ the search has no term for y that makes the hypothesis of line 9 reachable.

Marking scheme. (10 points) - 2: identifies transitivity as the source of the goal and instantiates it with a for x and z. - 2: chooses b for y, with the reason (the premise $R(a,b)$). - 2: obtains $R(b,a)$ from symmetry at a, b, with $\to$E. - 2: builds the conjunction in the right order and fires the transitivity instance. - 2: every $\forall$E removes the outermost quantifier of a cited line and replaces every occurrence of its variable. Fatal errors (0 for the step): removing an inner quantifier before an outer one; instantiating transitivity at a term for y for which neither conjunct of the hypothesis is available.

Exercise. C2: Instantiating at the variable itself

(a) Show that $\forall x\, P(x) \vdash P(x)$, by a derivation whose one $\forall$E line you check against every clause of Definition 4.3.1.

(b) The reverse step, from a line $P(x)$ to $\forall x\, P(x)$, is an instance of no rule so far. Show that it must not be allowed without restriction: give a reading over $\mathbb{Z}$, with x naming one particular integer, in which $P(x)$ is true and $\forall x\, P(x)$ is false.

(c) Section 4.4 will allow the step from a line $P(x_0)$ to $\forall x\, P(x)$ under conditions on the name x_0: the line $P(x_0)$ must close a special box opened for x_0, and x_0 must appear in no premise, in no assumption still in force, and not in the conclusion $\forall x\, P(x)$. Using this description, explain why the route "$\forall x\, P(x)$, so $P(x)$, so $\forall x\, P(x)$" cannot use the letter x as the name, what it must do instead, and why no name at all will do when $P(x)$ is itself the premise.

Solution.

(a)

1	$\forall x\, P(x)$	
2	$P(x)$	$\forall$E 1

Clause 1: the main symbol of line 1 is $\forall x$, with scope $P(x)$. Clause 2: the term is the variable x itself, and any term is allowed. Clause 3: $P(x)[x/x] = P(x)$, so every free occurrence was replaced, by x. Clause 4: x is free for x in every formula (Section 4.2: a free occurrence of x never lies in the scope of a quantifier on x). Clause 5: line 1 is a premise and lies in no box, so it is available at line 2. So line 2 is legal. Its x is now free: line 2 says "P holds of whatever x names", which is true whatever x names, since line 1 says that P holds of everything.

(b) Read $P(x)$ as "x is even" and let x name 2. Then $P(x)$ is true, and $\forall x\, P(x)$, "every integer is even", is false, since 1 is odd. A rule allowing the step from $P(x)$ to $\forall x\, P(x)$ in general would lead from a true line to a false one.

(c) With the letter x as the name, the conclusion $\forall x\, P(x)$ contains x (the variable after the quantifier counts, which is what "appear" means), so the condition "not in the conclusion" fails, whatever else holds. The route must instantiate at a **different** letter instead: open the box for a new name x_0, write $P(x_0)$ by $\forall$E at x_0 from line 1, and close the box with $\forall x\, P(x)$. Now x_0 appears nowhere in the premise $\forall x\, P(x)$, there is no other assumption, and x_0 is not in the conclusion; every condition holds.

If $P(x)$ is itself the premise, take any name. If the name is x, the line $P(x)$ is available but x appears in the premise and in the conclusion. If the name is some x_0 other than x, the box needs the line $P(x_0)$, and the search finds nothing that gives it: the premise is about x, and no line mentions x_0. A stalled search proves nothing by itself, since a cleverer derivation might exist. By (b), none should exist: $P(x) \vdash \forall x\, P(x)$ would lead from a true premise to a false conclusion. Section 4.11 turns this reading into a counter-model for the sequent $P(x_0) \vdash \forall x\, P(x)$, with x_0 naming 2; the same reading with x naming 2 is a counter-model for $P(x) \vdash \forall x\, P(x)$, and the soundness statement given there (proved in the optional Section 4.12) then shows that $P(x) \nvdash \forall x\, P(x)$.

Marking scheme. (10 points) - 3: (a) the derivation, with each of the five clauses checked, including that x is free for x. - 2: (b) a reading with $P(x)$ true and $\forall x\, P(x)$ false, the value of x stated. - 3: (c) the flag x fails "not in the conclusion"; the fix instantiates at a fresh x_0 inside the box. - 2: (c) for the premise $P(x)$: the name x fails the side condition, and for a different name the search stalls, with the pointer to Section 4.11 for non-derivability. Fatal errors (0 for the step): claiming the flag x is fine because x is bound in the conclusion; treating $P(x)$ and $P(x_0)$ as the same line.

4.4 Proving "For Every": Let x Be Arbitrary

Section 4.3 gave the rule for using a universal statement, and ended at a stall: from $\forall x\,(P(x) \wedge Q(x))$ it could derive $P(c)$ for every term c we cared to name, but never $\forall x\, P(x)$ itself. No list of instances is a statement about everything. This section gives the rule that proves a universal statement, $\forall$I. It needs one new device, a box that carries a **name** instead of an assumption, and one new side condition, which turns the mathematician's "let x be arbitrary" into something a reader can check line by line.

4.4.1 One number we know nothing about

Here is a fact of school algebra.

> For every real number x, $x^2 - 2x + 1 \geq 0$.

We cannot check it number by number; there are too many numbers. The proof everyone writes takes **one** real number and argues about it: *Let $x \in \mathbb{R}$. Then $x^2 - 2x + 1 = (x-1)^2$, and a square is never negative. So $x^2 - 2x + 1 \geq 0$.* Then, often without saying so: *since x was arbitrary, this holds for every real x.*

Chapter 1 adopted a convention for sentences like "let x be a real number": the letter names **one fixed** number, not further specified (Section 1.1). It promised that Chapter 4 would explain why an argument of that kind proves a statement about every x at once. Section 2.2 met the same step at the end of the proof of Theorem 2.2.1, passing from an arbitrary rational r to every rational; Section 3.2 met it in induction, passing from "let $n \in \mathbb{N}$" to "for every n". Each time the step was taken on trust. Here is why it is correct, and when it is not.

The argument about x used **nothing** about x except that it is a real number. It never said "suppose $x > 0$" or "take $x = 3$". So it could be repeated word for word with 0, with -7, with π, with any real number in place of x, and it would prove the claim for that number. One argument about a number we know nothing about is a recipe for an argument about each number, and

that is what "for every" asks for.

Everything depends on "nothing". If the argument had used a fact about x that not every number has, the recipe would break for some numbers. "Let x be a real number with $x > 0$. Then $x > 0$. Since x was arbitrary, every real number is positive" is the same shape, and its conclusion is false. The step "since x was arbitrary" is legitimate only when x **was** arbitrary: when no premise and no assumption still in force says anything about it. The rule of this section makes that condition exact.

4.4.2 The flagged box

To prove "for every x", take a fresh name that nothing is known about, prove the statement for that name, and generalize.

In a derivation the argument about the arbitrary x is drawn as a box, as Chapter 1 drew the argument under an assumption. The box opens with no assumption at all. Instead it carries the arbitrary name, written at its top left, and is called a **flagged box**; the name is its **flag**.

Definition 4.4.1. Rules for the Universal Quantifier

$$\begin{array}{lll|l} i & x_0 & \vdots & \\ j & & \varphi[x_0/x] & \\ k & \forall x\,\varphi & & \forall\text{I } i\text{–}j \end{array} \qquad\qquad \begin{array}{ll|ll} i & \forall x\,\varphi & & \\ k & \varphi[t/x] & & \forall\text{E } i \end{array}$$

The rule on the left is ∀I (**universal introduction**, or **generalization**). It cites a flagged box, with flag x_0, from its first line i to its last line j, and writes $\forall x\,\varphi$ on line k. The flag x_0 is a **variable**. The box's last line j is $\varphi[x_0/x]$, and it is a line of the flagged box **itself**, not of a box inside it. The side condition is on the name: at line k,

- x_0 appears in **no premise**,
- x_0 appears in **no assumption of a box still open at line** k, and
- x_0 does **not** appear in $\forall x\,\varphi$.

The rule on the right is ∀E, as in Definition 4.3.1: t is any term free for x in φ.

In words: to prove that φ holds for every x, open a box flagged with a new name x_0, prove $\varphi[x_0/x]$, "φ holds of x_0", as the box's last line, close the box, and write $\forall x\,\varphi$. The side condition says that x_0 really was arbitrary.

What a flagged box is. Definition 1.4.2 (Section 1.4) said that a box is a block of consecutive lines whose first line is an assumption. We now extend that definition, and two things change. First, a box may be **flagged**: it carries a name, a variable written at its top, and a flagged box need **not** open with an assumption. The box of ∀I has none; the box of the rule for using "there exists" in Section 4.6 is flagged and has one. Second, the rules that close a flagged box carry side conditions about its name, such as the one above. Everything else in Definition 1.4.2 stays word for word: two boxes are disjoint or one is inside the other, a box ends at one of its lines, and the scope rule decides which lines and boxes are available where. A flagged box with no assumption adds nothing to what is assumed; it only marks where the name is in use.

Here are the two boxes side by side: the box of →I from Definition 1.4.1, and the flagged box.

i	φ		
	$\vdots$		
j	ψ		
k	$\varphi \to \psi$	$\to$I i–j	

i	x_0	$\vdots$	
j		$\varphi[x_0/x]$	
k	$\forall x\, \varphi$		$\forall$I i–j

Both draw a boundary and both are cited as a range i–j. The $\to$I box opens with what we suppose; the flagged box opens with the name we have fixed. In the schema the flagged box begins with a gap $\vdots$ on line i, only so that the range has a first line; in a real derivation line i is whatever the box's first line is. When a flagged box immediately contains another box, as in the swap of two quantifiers in the examples below, the range starts at the first line of that inner box.

Read the rule part by part: the flag, the three clauses of the side condition, and the last line.

- **The flag.** Opening the box is the written "let x_0 be arbitrary". The flag is always a variable, never a constant: a constant has a fixed meaning, while the whole point of x_0 is that it may stand for anything. The book writes flags with a subscript 0, x_0, y_0, ε_0, so that they stand out.
- **No premise, no open assumption** (the first and second clauses). This is "we know nothing about x_0". What is known at line k is the premises and the assumptions of the boxes still open there. If none of them mentions x_0, then nothing that was used about x_0 inside the box is special to x_0.
- **Not in the conclusion** (the third clause). The name must not leak out. Inside the box it stood for an arbitrary object, and $\forall x\, \varphi$ is meant to speak about all objects through the bound x. A conclusion that still mentions x_0 ties what it says to the one object x_0 names, while claiming it of everything.
- **The last line.** $\varphi[x_0/x]$ is what was proved about x_0, and it must be a line of the flagged box itself, just as the last line of a $\to$I box must be (Section 1.4). A line inside a further box would still depend on that box's assumption.

"Appear" is the word of Section 4.2: written anywhere in the formula, including just after a quantifier symbol. So a box flagged x can never end in $\forall x\, \varphi$, since x appears in $\forall x\, \varphi$. That is one more reason the book never uses the quantified letter itself as the flag.

Fresh is not the same as new. The condition asks about three places only: the premises, the open assumptions, and the conclusion. Lines **outside** the box may mention x_0, before it or after it. An earlier line $\forall y\, R(x_0, y)$, obtained by $\forall$E from a premise, is harmless; Exercise C1 builds a derivation with such a line. And once a flagged box has closed, its name may flag a later box. A name that satisfies the side condition at the line that closes its box is called **fresh** there. It need not be new to the whole derivation.

Why each clause. Each clause stops a derivation of a false conclusion from true premises. Here is one failure for each.

A name in a premise. If $\forall$I could generalize over a name in a premise, then $P(x_0) \vdash \forall x\, P(x)$:

1	$P(x_0)$		
2	x_0	$P(x_0)$	R 1
3	$\forall x\, P(x)$		$\forall$I 2–2

Over $\mathbb{Z}$, read $P(x)$ as "x is even" and let x_0 name 2. The premise is true and line 3, "every

integer is even", is false. Line 3 breaks the first clause: x_0 appears in the premise on line 1. The premise **does** say something about x_0, so x_0 is not arbitrary.

A name in an open assumption. Suppose $x_0 > 0$, and inside that supposition generalize over x_0:

1	x_0	$x_0 > 0$	
2		$x_0 \mid x_0 > 0$	R 1
3		$\forall x\,(x > 0)$	$\forall$I 2–2
4		$x_0 > 0 \to \forall x\,(x > 0)$	$\to$I 1–3
5	$\forall y\,(y > 0 \to \forall x\,(x > 0))$		$\forall$I 1–4

Line 3 breaks the second clause: x_0 appears in the assumption on line 1, whose box is still open at line 3. Every other line is legal, including line 5, where the box of line 1 has closed. The inner box on line 2 is flagged x_0 while the outer box flagged x_0 is still open; the definition allows this, since its side condition asks only about premises, open assumptions and the conclusion. And line 5 says "every positive number makes every number positive", which over $\mathbb{Z}$ is false: at $y = 1$ the hypothesis $1 > 0$ holds and $\forall x\,(x > 0)$ fails at $x = 0$. This is the "let x be a real number with $x > 0$ ... since x was arbitrary" of the opening, made into lines.

A name in the conclusion. From the premise $\forall x\,(x \le x)$, true over $\mathbb{N}$, open a box flagged y_0:

1	$\forall x\,(x \le x)$		
2	y_0	$y_0 \le y_0$	$\forall$E 1
3	$\forall x\,(x \le y_0)$		$\forall$I 2–2

Line 2 is $(x \le y_0)[y_0/x]$, so line 3 has the shape of $\forall$I with $\varphi = (x \le y_0)$. But y_0 appears in the conclusion, which breaks the third clause. Line 3 says "every natural number is at most y_0", and that is false whatever y_0 names: $y_0 + 1$ is a counterexample. (Exercise C1 of Section 4.5 turns this into a sentence, "there is a largest natural number".)

The requirement on the last line is the one Chapter 1 already imposed on every box rule, for the same reason.

4.4.3 Derivations with flagged boxes

The search method of Section 2.6 gets a new goal-directed step: when the goal is $\forall x\,\varphi$, open a flagged box with a fresh name x_0, and make $\varphi[x_0/x]$ the box's last line, the new goal. The examples below use it, each with an Idea.

Example. Flagged boxes at work

Give a derivation of each sequent; in (b), of each direction.

(a) $\forall x\,\forall y\,R(x,y) \vdash \forall x\,R(x,x)$.

(b) $\forall y\,\varphi \dashv\vdash \forall z\,\varphi[z/y]$, for a formula φ and variables y, z with z not appearing in φ.

(c) $\forall x\,\forall y\,R(x,y) \vdash \forall y\,\forall x\,R(x,y)$.

Solution.

(a) *Idea.* The goal begins with $\forall x$, so the goal-directed step opens a box flagged x_0, and the new goal is $R(x,x)[x_0/x] = R(x_0,x_0)$. That goal names the term for ∀E (Section 4.3's step): instantiate the premise at x_0, twice.

1	$\forall x\,\forall y\,R(x,y)$	
2	x_0 | $\forall y\,R(x_0,y)$	∀E 1
3	| $R(x_0,x_0)$	∀E 2
4	$\forall x\,R(x,x)$	∀I 2–3

The side condition at line 4: x_0 does not appear in the premise, no box is open at line 4, and x_0 does not appear in $\forall x\,R(x,x)$. Lines 2 and 3 instantiate at the variable x_0, which is free for x and for y, since the formulas contain no quantifier on x_0. Compare Section 4.3, (b), which reached only $R(c,c)$: there the term was a constant, a name fixed by the language, and nothing can be generalized over a constant; here it is a flag, a variable that no premise mentions.

(b) *Idea.* This is the renaming of a bound variable at the front of a formula, which Section 4.2 took as a convention and promised to derive. Each direction has a ∀ goal, so a flagged box with a name that appears in neither formula, say z_0; inside, one ∀E at z_0 gives the instance the box needs. The only thought is checking that the two instances are the same formula.

1	$\forall y\,\varphi$		1	$\forall z\,\varphi[z/y]$	
2	z_0 | $\varphi[z_0/y]$	∀E 1	2	z_0 | $\varphi[z_0/y]$	∀E 1
3	$\forall z\,\varphi[z/y]$	∀I 2–2	3	$\forall y\,\varphi$	∀I 2–2

Take z_0 to be a variable that does not appear in φ and is different from y and z. Since z does not appear in φ, the formula $\varphi[z/y]$ is φ with each free y changed to z, and changing those z's on to z_0 gives the same result as changing each free y to z_0 at once: $(\varphi[z/y])[z_0/z] = \varphi[z_0/y]$. So in the left derivation, line 2 is a legal ∀E (z_0 does not appear in φ, so it is free for y), and it is the instance $(\varphi[z/y])[z_0/z]$ that the ∀I on line 3 needs. In the right derivation, line 2 is $(\varphi[z/y])[z_0/z]$, a legal ∀E for the same reason, and it is $\varphi[z_0/y]$, the instance that $\forall y\,\varphi$ needs. At line 3 of each, z_0 appears neither in the premise nor in the conclusion, and no box is open. Each direction has three lines.

(c) *Idea.* Read the goal's quantifiers from the outside: $\forall y$ first, so open a box flagged y_0; the new goal $\forall x\,R(x,y_0)$ is again universal, so open a box flagged x_0 inside it; the innermost goal is $R(x_0,y_0)$. Two boxes, opened in the goal's order. Now the premise's quantifiers are removed in **its** order: ∀E at x_0, then at y_0.

1	$\forall x\,\forall y\,R(x,y)$		
2	y_0 | x_0 | $\forall y\,R(x_0,y)$		∀E 1
3	| | $R(x_0,y_0)$		∀E 2
4	| $\forall x\,R(x,y_0)$		∀I 2–3
5	$\forall y\,\forall x\,R(x,y)$		∀I 2–4

The box flagged y_0 immediately contains the box flagged x_0, so it has no line of its own before the inner box; following the convention after Definition 4.4.1, line 5 cites it as 2–4, from the first line of the inner box. Its last line is line 4, a line of the outer box itself. At

line 4, x_0 is in no premise, the only open box (flagged y_0) has no assumption, and x_0 does not appear in $\forall x\, R(x, y_0)$. At line 5 the same holds for y_0.

Nothing in (c) used that the formula under the quantifiers is $R(x, y)$. For any formula ψ and different variables x and y, with flags x_0 and y_0 that are different from each other and from x and y and that do not appear in ψ, the same five lines derive $\forall x\, \forall y\, \psi \vdash \forall y\, \forall x\, \psi$: line 3 becomes $(\psi[x_0/x])[y_0/y]$, which equals $(\psi[y_0/y])[x_0/x]$, since the two substitutions put two different fresh names in for two different variables and so do not interfere. Exchanging the letters x and y gives the other direction. So **two adjacent universal quantifiers may always be swapped**: $\forall x\, \forall y\, \psi \dashv\vdash \forall y\, \forall x\, \psi$. Section 4.1 read this off the game, where the challenger makes both moves; here it is derived.

Now the sequent at which Section 4.3 stalled, together with its converse. "For every x, $P(x)$ and $Q(x)$" says the same as "$P(x)$ for every x, and $Q(x)$ for every x":

Proposition 4.4.1. Universal Quantifier and Conjunction

$$\forall x\, (P(x) \wedge Q(x)) \dashv\vdash \forall x\, P(x) \wedge \forall x\, Q(x).$$

Idea. ($\vdash$) The goal is a conjunction, so (Section 1.2) two subgoals, $\forall x\, P(x)$ and $\forall x\, Q(x)$. Each is universal, so each gets its own flagged box, and inside each, $\forall$E at the flag and $\wedge$E give the box's last line. ($\dashv$) The goal is universal, so one flagged box, with the goal $P(x_0) \wedge Q(x_0)$; take the premise apart by $\wedge$E and instantiate each half at x_0.

Proof. ($\vdash$)

1		$\forall x\, (P(x) \wedge Q(x))$	
2	x_0	$P(x_0) \wedge Q(x_0)$	$\forall$E 1
3	x_0	$P(x_0)$	$\wedge$E 2
4		$\forall x\, P(x)$	$\forall$I 2–3
5	x_0	$P(x_0) \wedge Q(x_0)$	$\forall$E 1
6	x_0	$Q(x_0)$	$\wedge$E 5
7		$\forall x\, Q(x)$	$\forall$I 5–6
8		$\forall x\, P(x) \wedge \forall x\, Q(x)$	$\wedge$I 4, 7

At lines 4 and 7, x_0 appears neither in the premise nor in the conclusion of that line, and no box is open. The second flagged box reuses the name x_0. That is allowed: the side condition at line 7 asks only about the premise, the open assumptions, and $\forall x\, Q(x)$, and the earlier box, with its lines 2 and 3, closed at line 3 and is not available at line 7. (A fresh y_0 would do equally well.)

($\dashv$)

1	$\forall x\, P(x) \wedge \forall x\, Q(x)$		
2	$\forall x\, P(x)$		$\wedge$E 1
3	$\forall x\, Q(x)$		$\wedge$E 1
4	x_0 | $P(x_0)$		$\forall$E 2
5	| $Q(x_0)$		$\forall$E 3
6	| $P(x_0) \wedge Q(x_0)$		$\wedge$I 4, 5
7	$\forall x\, (P(x) \wedge Q(x))$		$\forall$I 4–6

At line 7, x_0 is in neither the premise nor the conclusion, and no box is open. The two derivations prove the two sequents, as claimed.

■

The first derivation is the answer to Section 4.3's stall: its lines 1–4 derive $\forall x\, P(x)$ from $\forall x\, (P(x) \wedge Q(x))$. The proposition says that $\forall$ distributes over $\wedge$. Section 4.9 collects it with the other laws that move a quantifier across a connective, and there the corresponding statement for $\forall$ and $\vee$ holds in one direction only.

4.4.4 Every element of the empty set

Section 4.1 read $\forall x \in \emptyset\ \varphi$ as true whatever φ says, because there is no element to test, and promised a derivation. Section 2.1 met the same phenomenon for implications: $\neg P \vdash P \to Q$ (Proposition 2.1.2, the derived rule written VI). With $\forall$I the two meet.

Let A be a set with no elements. What that means is the premise $\forall x\, \neg(x \in A)$: no object is in A. From it, "every element of A is greater than 5", which by Definition 4.1.1 is $\forall x\, (x \in A \to x > 5)$, has a four-line derivation:

1	$\forall x\, \neg(x \in A)$		
2	x_0 | $\neg(x_0 \in A)$		$\forall$E 1
3	| $x_0 \in A \to x_0 > 5$		VI 2
4	$\forall x\, (x \in A \to x > 5)$		$\forall$I 2–3

The goal is universal, so a flagged box; the box's goal $x_0 \in A \to x_0 > 5$ is an implication whose hypothesis is refuted by the instance of the premise at x_0, and VI turns the refutation into the implication. Line 3 uses VI as a derived rule (Definition 2.3.1), with the formulas $x_0 \in A$ and $x_0 > 5$ in place of the letters P and Q. This is allowed because the derivation of Proposition 2.1.2 uses only $\neg$E, $\bot$E and $\to$I, whose schemas apply to any formulas (Definition 1.2.1), and it has no flagged box, so pasting it into the box flagged x_0 gives a derivation in the rules themselves. From now on, any derived rule of Chapter 2 is used in this way on first-order formulas. Nothing in the derivation used $x > 5$: with $x < 5$, or with any formula in place of $x > 5$ in which x_0 does not appear, the same four lines go through. So, from the premise that A is empty, "every element of A has property φ" is derivable for every φ, as Section 4.1 said; the set $\emptyset$ is the case $A = \emptyset$. One more flagged box gives a statement about upper bounds:

1	$\forall x\, \neg(x \in A)$		
2	u_0 | x_0 | $\neg(x_0 \in A)$		$\forall$E 1
3	| | $x_0 \in A \to x_0 \leq u_0$		VI 2
4	| $\forall x\, (x \in A \to x \leq u_0)$		$\forall$I 2–3
5	$\forall u\, \forall x\, (x \in A \to x \leq u)$		$\forall$I 2–4

Over $\mathbb{R}$, line 5 says that every real number u is an upper bound of A: every element of A is at most u. For an empty set this holds for every u, because there are no elements to check. At line 4, x_0 appears in neither the premise nor $\forall x\, (x \in A \to x \leq u_0)$, and the only open box (flagged u_0) has no assumption. The name u_0 does appear in that conclusion, which is allowed, since the condition at line 4 is about x_0 only. At line 5, u_0 appears in neither the premise nor $\forall u\, \forall x\, (x \in A \to x \leq u)$, and no box is open.

4.4.5 How this rule looks in a mathematical proof

Here is the opening proof as a derivation, with the general fact about squares as a premise and the algebra as one line justified "arith" (Section 4.3).

1	$\forall y\, (y^2 \geq 0)$		
2	x_0 | $(x_0 - 1)^2 \geq 0$		$\forall$E 1
3	| $x_0^2 - 2 \cdot x_0 + 1 \geq 0$		arith 2
4	$\forall x\, (x^2 - 2 \cdot x + 1 \geq 0)$		$\forall$I 2–3

Line	Sentence
box x_0	"Let $x \in \mathbb{R}$."
2	"Then $(x-1)^2 \geq 0$, since a square is never negative." ($\forall$E at $x_0 - 1$)
3	"That is, $x^2 - 2x + 1 \geq 0$." (arith)
4	"Since x was arbitrary, $x^2 - 2x + 1 \geq 0$ for every real x." ($\forall$I)

Lines 2 and 3 are the derivation of Section 4.3 with the flag x_0 in place of the constant a. What changed is only the name: a was a constant that the language fixed, so nothing could be generalized over it; x_0 is a variable that no premise mentions, so line 4 may generalize. The written proof uses the same letter x for the flag and for the bound variable, and says "let $x \in \mathbb{R}$" where the derivation opens the box. Readers of mathematics are expected to supply the box.

Since the domain is $\mathbb{R}$, "let $x \in \mathbb{R}$" asks nothing of x. A proof that says "let $\varepsilon > 0$" does ask something, and its box opens with the assumption $\varepsilon_0 > 0$; Section 4.8 gives the bounded form of the rule that handles it.

4.4.6 Common errors

Three errors with $\forall$I are common enough to name now. The first of them opens Chapter 5's catalogue of steps that are not allowed.

- **Checking instances.** Verifying φ at $x = 0, 1, 2$, or at a thousand values, proves $\varphi[0/x]$, $\varphi[1/x]$, $\varphi[2/x]$, and so on, and not $\forall x\, \varphi$. The rule needs one proof about a fresh name, not many proofs about particular terms. Exercise C2 shows how plausible this error can look.
- **Generalizing over a name in an open assumption.** Inside "suppose $x_0 > 0$", the name x_0 is not arbitrary. The second derivation of "Why each clause" shows where this leads.
- **Generalizing over a name in a premise.** A premise about x_0, such as "x_0 is even", makes x_0 special. The first derivation of "Why each clause" shows it.

Warning. "Let x be arbitrary" is a promise about every line that follows, and the most common way to break it is an assumption made on the way. "Let $n \in \mathbb{N}$ and suppose n is even. Then n^2 is even. Since n was arbitrary, every n^2 is even" is false at $n = 1$. What the argument proves is "for every n, **if** n is even, then n^2 is even": the $\to$I box on "n is even" must close **before** the $\forall$I step, so that the assumption is no longer open.

Quick check. In the derivation below, line 6 is justified by $\forall$I. Is it legal? If not, which clause of the side condition fails, and can you find a reading in which the premises are true and line 6 is false?

1		$R(c, y_0)$	
2		$\forall x\, \forall y\, (R(x, y) \to R(y, x))$	
3	y_0	$\forall y\, (R(c, y) \to R(y, c))$	$\forall$E 2
4		$R(c, y_0) \to R(y_0, c)$	$\forall$E 3
5		$R(y_0, c)$	$\to$E 4, 1
6		$\forall y\, R(y, c)$	$\forall$I 3–5

Solution. No. Lines 1–5 are legal; in particular line 5 may cite line 1, which lies in no box. But at line 6 the flag y_0 appears in the premise on line 1, so the first clause fails: line 5 was proved about y_0 using a fact about y_0, and y_0 was not arbitrary. A reading: over $\mathbb{Z}$, let $R(x, y)$ be $x \neq y$, which is symmetric, so premise 2 is true; let c name 0 and y_0 name 1, so premise 1, $0 \neq 1$, is true. Line 6 says that every integer differs from 0, which is false at $y = 0$.

4.4.7 A foreshadowing

The rules for $\forall$ are now complete, and a natural sequent is still out of reach:

$$\forall x\, P(x) \vdash \exists x\, P(x).$$

If everything has P, then something has P, since the domain is not empty. But the goal is an existential statement, and no rule yet produces an $\exists$. Section 4.5 gives the rule for proving "there exists", and this sequent is its first example.

4.4.8 What the rule sounds like

The table gains a row for ∀I, below the row for ∀E from Section 4.3.

Rule	You write	Lean
∀E	"In particular, for $x = t$, …"; "applying the hypothesis to t, …"	`h t` (or `specialize h t`)
∀I	"Let x be arbitrary. … Since x was arbitrary, …"	`intro x`

Other forms of the opening phrase are "let $x \in \mathbb{R}$", "fix $\varepsilon > 0$", "take any $n \in \mathbb{N}$"; the closing phrase is often left out, since the reader knows that a proof which began by fixing x proves a statement about every x. In Lean, `intro x` turns the goal `forall x, P x` into the goal `P x` with a new name `x` in context, about which nothing is assumed: Lean's version of opening the flagged box. The same tactic opened the →I box in Section 1.4, and the two boxes are, to Lean, one kind of thing.

4.4.9 The search method: one more step

- **A goal $\forall x\, \varphi$.** Open a flagged box with a fresh name x_0, one that appears in no premise, in no open assumption, and not in the goal. The new goal, the box's last line, is $\varphi[x_0/x]$.

This step is **safe**, in the sense of Section 2.6. If $\forall x\, \varphi$ can be derived at all, then so can $\varphi[x_0/x]$, by one more ∀E at x_0; so opening the box never closes off a derivation that would otherwise exist. Take the step as soon as the goal is universal, and choose a flag that appears nowhere yet, so that the side condition cannot fail.

4.4.10 Exercises

4.4.10.1 A. Check your understanding

Exercise. A1

Three derivations, D1, D2 and D3, are used in part (b).

D1:

1		$\forall x\,(P(x) \to Q(x))$	
2		$P(x_0)$	
3	x_0	$P(x_0) \to Q(x_0)$	∀E 1
4		$Q(x_0)$	→E 3, 2
5		$\forall x\, Q(x)$	∀I 3–4

D2:

1	$\forall x\,\forall y\,R(x,y)$	
2	$\forall y\,R(x_0,y)$	$\forall$E 1
3	x_0 | $R(x_0,x_0)$	$\forall$E 2
4	$\forall x\,R(x,x)$	$\forall$I 3–3

D3:

1	$P(x_0)$	
2	x_0 | $P(x_0)$	R 1
3	$\forall x\,P(x)$	$\forall$I 2–2
4	$P(x_0) \to \forall x\,P(x)$	$\to$I 1–3

(a) Write the schema of $\forall$I with its side condition.

(b) For each of D1, D2, D3, decide whether its $\forall$I line is legal. If not, name the clause of the side condition that fails.

(c) True or false, with a reason: "fresh" means that x_0 appears nowhere in the derivation outside its flagged box.

(d) In the derivation of the ($\vdash$) half of Proposition 4.4.1, the name x_0 flags two boxes. Why does the side condition allow this?

Solution.

(a) A box flagged x_0, with no assumption, from line i to line j, whose last line j is $\varphi[x_0/x]$, a line of the flagged box itself; then line k: $\forall x\,\varphi$, justified "$\forall$I i–j". At line k the name x_0 appears in no premise, in no assumption of a box still open at line k, and not in $\forall x\,\varphi$ (Definition 4.4.1).

(b) *D1:* not legal. At line 5 the flag x_0 appears in the premise on line 2 (the first clause). Over $\mathbb{Z}$, with $P(x)$ and $Q(x)$ both read as "x is even" and x_0 naming 2, both premises are true and line 5, "every integer is even", is false.
D2: legal. Line 2 mentions x_0 outside the box, and that is allowed: it is neither a premise nor an assumption, and the conclusion $\forall x\,R(x,x)$ does not contain x_0. Line 3 cites line 2, which is available, since no box contains it. (It is also correct in meaning: line 2 came from a premise that says nothing about x_0, so it holds whatever x_0 names.)
D3: not legal. At line 3 the assumption $P(x_0)$ on line 1 is still open, and x_0 appears in it (the second clause). Line 4, which closes the box legally, then says "if x_0 has P, everything has P", false over $\mathbb{Z}$ with P "even" and x_0 naming 2.

(c) False. The side condition asks about the premises, the assumptions of boxes still open at the $\forall$I line, and the conclusion of that line. Other lines may mention x_0, as line 2 of D2 does, and a closed flagged box may share its name with a later one.

(d) At line 7, which closes the second box, the side condition asks whether x_0 appears in the premise (no), in an assumption of an open box (no box is open), and in $\forall x\,Q(x)$ (no). The first box's lines mention x_0, but they are neither premises nor open assumptions, and the box closed at line 3, so its lines are not even available at line 7.

4.4.10.2 B. Practice

Exercise. B1: A chain, for every x

Give a derivation of

$$\forall x\,(P(x) \to Q(x)),\ \forall x\,(Q(x) \to R(x)) \vdash \forall x\,(P(x) \to R(x)),$$

and check the side condition at its ∀I line.

Solution. The goal is universal, so open a box flagged x_0; the new goal $P(x_0) \to R(x_0)$ is an implication, so open a →I box on $P(x_0)$; the new goal is $R(x_0)$. Now Section 4.3's exercise B1 runs inside the boxes, with x_0 for a.

1		$\forall x\,(P(x) \to Q(x))$	
2		$\forall x\,(Q(x) \to R(x))$	
3	x_0	$P(x_0)$	
4		$P(x_0) \to Q(x_0)$	∀E 1
5		$Q(x_0)$	→E 4, 3
6		$Q(x_0) \to R(x_0)$	∀E 2
7		$R(x_0)$	→E 6, 5
8		$P(x_0) \to R(x_0)$	→I 3–7
9		$\forall x\,(P(x) \to R(x))$	∀I 3–8

The flagged box immediately contains the →I box, so line 9 cites it from line 3. At line 9: x_0 appears in neither premise; the assumption $P(x_0)$ mentions x_0, but its box closed at line 7, so it is not open at line 9; and x_0 does not appear in $\forall x\,(P(x) \to R(x))$. The order of the boxes matters: the →I box is inside the flagged box and closes first.

Exercise. B2: Contraposition under a quantifier

(a) Give a derivation of $\forall x\,(P(x) \to Q(x)) \vdash \forall x\,\neg Q(x) \to \forall x\,\neg P(x)$.

(b) Try to derive the converse, $\forall x\,\neg Q(x) \to \forall x\,\neg P(x) \vdash \forall x\,(P(x) \to Q(x))$. Say where the search stalls, and which side condition stops it.

Solution.

(a) The goal is an implication: a →I box on $\forall x\,\neg Q(x)$, with goal $\forall x\,\neg P(x)$. That is universal: a box flagged x_0, with goal $\neg P(x_0)$. That is a negation: a ¬I box on $P(x_0)$, with goal $\bot$. Inside, instantiate both universal lines at x_0 and look for a clash: $P(x_0)$ and the instance of the premise give $Q(x_0)$, and the assumption $\forall x\,\neg Q(x)$ at x_0 gives $\neg Q(x_0)$.

1	$\forall x\,(P(x) \to Q(x))$	
2	$\quad \forall x\,\neg Q(x)$	
3	$\quad x_0 \quad\quad P(x_0)$	
4	$\quad\quad\quad P(x_0) \to Q(x_0)$	$\forall$E 1
5	$\quad\quad\quad Q(x_0)$	$\to$E 4, 3
6	$\quad\quad\quad \neg Q(x_0)$	$\forall$E 2
7	$\quad\quad\quad \bot$	$\neg$E 5, 6
8	$\quad\quad \neg P(x_0)$	$\neg$I 3–7
9	$\quad \forall x\,\neg P(x)$	$\forall$I 3–8
10	$\forall x\,\neg Q(x) \to \forall x\,\neg P(x)$	$\to$I 2–9

At line 9, the open box is the $\to$I box of line 2, whose assumption $\forall x\,\neg Q(x)$ does not contain x_0; the premise and the conclusion do not contain it either. So the side condition holds, even though the flagged box sits inside an open box.

(b) The goal is universal, so open a box flagged x_0, then a $\to$I box on $P(x_0)$, with goal $Q(x_0)$. The only premise is an implication, $\forall x\,\neg Q(x) \to \forall x\,\neg P(x)$. Using it needs its hypothesis $\forall x\,\neg Q(x)$, which nothing supplies; and its conclusion, $\forall x\,\neg P(x)$, would clash with $P(x_0)$ and close the case by $\bot$E. So everything turns on deriving $\forall x\,\neg Q(x)$. The goal $Q(x_0)$ is atomic and nothing produces it, so the method's last resort is an RAA box on $\neg Q(x_0)$. Inside, the available facts about Q are $\neg Q(x_0)$, and to reach $\forall x\,\neg Q(x)$ from it we would close a flagged box over x_0. The side condition forbids it: x_0 appears in the open assumptions $P(x_0)$ and $\neg Q(x_0)$. A fresh name y_0 does not help either, since nothing tells us anything about $\neg Q(y_0)$. The search stalls at the second clause of the side condition.

A stalled search proves nothing by itself; a cleverer derivation might exist. Section 4.11 settles the question with a structure in which the premise is true and the conclusion false, and shows that the converse does not follow.

Exercise. B3: A number is at most its absolute value, for every number

Example (A number is at most its absolute value) (Section 1.6) derived $|x| > x \vee |x| = x$ for one fixed real x from three premises about that x: $x \geq 0 \vee x < 0$, $x \geq 0 \to |x| = x$, and $x < 0 \to |x| > x$. Turn it into a derivation of

$$\forall x\,(|x| > x \vee |x| = x)$$

from premises that are universal statements. Explain why the premises of Example (A number is at most its absolute value), copied with x_0 in place of x, will not do.

Solution. The goal is universal, so the derivation of Section 1.6 must run inside a box flagged x_0. Its three premises mention the number it is about. So here the premises are the three facts **for every** x, and inside the box $\forall$E at x_0 gives each fact about x_0.

1		$\forall x\,(x \geq 0 \lor x < 0)$	
2		$\forall x\,(x \geq 0 \to \lvert x\rvert = x)$	
3		$\forall x\,(x < 0 \to \lvert x\rvert > x)$	
4	x_0	$x_0 \geq 0 \lor x_0 < 0$	$\forall$E 1
5		$x_0 \geq 0 \to \lvert x_0\rvert = x_0$	$\forall$E 2
6		$x_0 < 0 \to \lvert x_0\rvert > x_0$	$\forall$E 3
7		$x_0 \geq 0$	
8		$\lvert x_0\rvert = x_0$	$\to$E 5, 7
9		$\lvert x_0\rvert > x_0 \lor \lvert x_0\rvert = x_0$	$\lor$I 8
10		$x_0 < 0$	
11		$\lvert x_0\rvert > x_0$	$\to$E 6, 10
12		$\lvert x_0\rvert > x_0 \lor \lvert x_0\rvert = x_0$	$\lor$I 11
13		$\lvert x_0\rvert > x_0 \lor \lvert x_0\rvert = x_0$	$\lor$E 4, 7–9, 10–12
14		$\forall x\,(\lvert x\rvert > x \lor \lvert x\rvert = x)$	$\forall$I 4–13

Lines 7–13 are the derivation of Example (A number is at most its absolute value) with x_0 for x, its premises replaced by lines 4–6. At line 14, x_0 appears in none of the premises, no box is open (the two case boxes closed at lines 9 and 12), and x_0 is not in the conclusion. The derivation has 14 lines.

If instead the premises were $x_0 \geq 0 \lor x_0 < 0$, $x_0 \geq 0 \to \lvert x_0\rvert = x_0$ and $x_0 < 0 \to \lvert x_0\rvert > x_0$, the box could reach line 13 just as well, but line 14 would break the first clause: x_0 would appear in a premise. And rightly so. Premises about x_0 are facts about one number, and a derivation from them proves something about that number only. The universal premises are true facts about every real number, and that is what entitles the conclusion to speak about every real number.

4.4.10.3 C. Going deeper

Exercise. C1: How fresh is fresh?

(a) Give a derivation of $\forall x\,\forall y\,R(x,y) \vdash R(x_0,x_0) \land \forall x\,R(x,x)$ in which the line $R(x_0,x_0)$ is written **before** a flagged box that uses the same name x_0. Check every clause of the side condition at the $\forall$I line.

(b) A stronger condition is sometimes proposed: "x_0 occurs nowhere in the derivation outside its box". Does your derivation meet it? Explain why the conclusion of your derivation is nonetheless correct, so that the stronger condition forbids more than it needs to.

(c) Could the last line of the flagged box in (a) be obtained by R, repeating the earlier line $R(x_0,x_0)$? Is the result a derivation?

Solution (Hint 1). The goal is a conjunction. Derive the left conjunct first, by two ∀E lines outside any box.

Solution (Hint 2). For the right conjunct, open a box flagged x_0 and instantiate the premise at x_0 again inside it. At the ∀I line, list the premises, the open assumptions and the conclusion, and look for x_0 in each.

Solution.

	1	$\forall x\,\forall y\,R(x,y)$	
	2	$\forall y\,R(x_0,y)$	∀E 1
	3	$R(x_0,x_0)$	∀E 2
(a)	4	x_0 | $\forall y\,R(x_0,y)$	∀E 1
	5	| $R(x_0,x_0)$	∀E 4
	6	$\forall x\,R(x,x)$	∀I 4–5
	7	$R(x_0,x_0)\wedge\forall x\,R(x,x)$	∧I 3, 6

The side condition at line 6. *Premises:* the only premise is line 1, $\forall x\,\forall y\,R(x,y)$, and x_0 does not appear in it. *Open assumptions:* no box is open at line 6, so there are none. *Conclusion:* $\forall x\,R(x,x)$ does not contain x_0. The last line of the box, line 5, is $R(x,x)[x_0/x]$ and is a line of the box itself. So line 6 is legal, and the derivation has 7 lines. Lines 2, 3 and 7 mention x_0 outside the box, and the side condition does not ask about them.

(b) No: lines 2, 3 and 7 mention x_0 outside its box. Yet the conclusion is correct. The reason the side condition looks only at premises and open assumptions is that these are the only lines taken without proof, so they are the only lines that can single out what x_0 names; every other available line is derived from them. Lines 2 and 3 were derived from a premise that does not mention x_0; they hold whatever x_0 names. So the argument in the box really is about an arbitrary object, and $\forall x\,R(x,x)$ follows. The stronger condition would reject this derivation, although renaming the box's flag to a new letter z_0 (lines 4–5 becoming $\forall y\,R(z_0,y)$ and $R(z_0,z_0)$) gives a derivation that meets it and has the same last line. A condition that rejects a correct derivation for the spelling of a name forbids more than safety needs.

(c) Yes. R may copy into the box any line available there, and line 3 lies in no box, so it is available at every later line. With line 5 justified "R 3" (and line 4 deleted), the box is a single line $R(x_0,x_0)$, and the side condition at the ∀I line is checked exactly as in (a): nothing in it looks at where the box's last line came from. The derivation is legal, and it is correct for the reason in (b): line 3 was derived from a premise that says nothing about x_0.

Marking scheme. (10 points) - 2: the left conjunct by two ∀E lines at x_0, outside any box. - 2: a flagged box with ∀E at the flag and ∀I citing it. - 2: the side condition checked clause by clause: premise, open assumptions, conclusion. - 2: (b) explains that lines outside the box that do not rest on premises or assumptions about x_0 cannot make x_0 special. Alternatively, 2 for showing that renaming the box's flag to a new letter gives a derivation that meets the

stronger condition and has the same last line. - 2: (c) correct answer with the availability check for the repeated line. Fatal errors (0 for the step): claiming that line 3 makes line 6 illegal; checking the side condition against lines that are not premises, open assumptions or the conclusion.

Exercise. C2: Find the error

A student offers this proof.

> **Claim.** For every real number x, $x^2 - 2x + 1 \geq 0$.
>
> **Proof.** (1) Let x be a real number. (2) At $x = 0$: $0 - 0 + 1 = 1 \geq 0$. (3) At $x = 1$: $1 - 2 + 1 = 0 \geq 0$. (4) At $x = 2$: $4 - 4 + 1 = 1 \geq 0$. (5) So $x^2 - 2x + 1 \geq 0$, and since x was arbitrary, the claim holds for every real x.

(a) Is the claim true?

(b) Does the argument prove it? Name the step that fails and the rule it misuses.

(c) Show that the step can lead to a false conclusion: give a polynomial p for which the same checks at $x = 0, 1, 2$ succeed and $p(x) \geq 0$ fails for some real x.

(d) Point to a correct proof.

Solution (Hint 1). Write lines (2)–(4) as formulas. Which terms do they mention?

Solution (Hint 2). Write the student's proof as a derivation: open a box flagged x_0 for line (1), and write lines (2)–(4) as lines inside it. What would line (5) have to cite?

Solution.

(a) Yes. For every real x, $x^2 - 2x + 1 = (x - 1)^2 \geq 0$, since a square is never negative.

(b) No. Lines (2)–(4) are true, and line (1) opens a flagged box, "let x be arbitrary". The error is in line (5). With φ the formula $x^2 - 2 \cdot x + 1 \geq 0$, lines (2)–(4) establish $\varphi[0/x]$, $\varphi[1/x]$ and $\varphi[2/x]$: statements about the constants 0, 1, 2. None of them is $\varphi[x_0/x]$ for the flag. Inside the box, nothing about the arbitrary x was ever proved. The first sentence of (5), "so $x^2 - 2x + 1 \geq 0$", has the right form: it is $\varphi[x_0/x]$, with the student's x as the flag. But no rule writes it: it is not a premise, an assumption or a repetition, and no rule takes $\varphi[0/x]$, $\varphi[1/x]$ and $\varphi[2/x]$ to $\varphi[x_0/x]$. So the closing "since x was arbitrary" applies ∀I to a box whose last line was never derived: the lines the box legally derives are all instances at constants, not the instance at the flag. Line (1) was idle: the argument never used the x it introduced.

(c) Take $p(x) = x(x - 1)(x - 2)$. The same checks succeed: $p(0) = 0 \geq 0$, $p(1) = 0 \geq 0$, $p(2) = 0 \geq 0$. But $p(-1) = (-1)(-2)(-3) = -6 < 0$, so "$p(x) \geq 0$ for every real x" is false. So the pattern of (5), concluding "for every x" from checks at a few constants, can lead from true checks to a false conclusion. It is not an instance of ∀I, as (b) showed. (Section 4.11 states, and the optional Section 4.12 proves, that legal steps never lead from true premises to a false conclusion.) The claim in (a) is true for a different reason, one that the argument never gave.

(d) The derivation under "How this rule looks in a mathematical proof": inside the box flagged x_0, $\forall$E applies "squares are not negative" at $x_0 - 1$, an arith line rewrites it as $x_0^2 - 2x_0 + 1 \geq 0$, and that line **is** $\varphi[x_0/x]$, so $\forall$I applies.

Marking scheme. (10 points) - 2: (a) the claim is true, with a reason (it is a square). - 3: (b) names step (5) and $\forall$I; says that lines (2)–(4) are instances at constants, not the instance at the flag. - 3: (c) a polynomial that passes the three checks and fails somewhere, with both computations shown. - 2: (d) a correct proof, generalizing over a statement proved about the arbitrary x. Fatal errors (0 for the step): answering (b) with "the claim is false"; a "counterexample" in (c) that fails one of the three checks or is non-negative everywhere.

4.5 Proving "There Exists": Exhibit a Witness

Sections 4.3 and 4.4 gave the two rules for "for every": $\forall$E uses a universal statement, and $\forall$I proves one. This section starts on the other quantifier. It gives the rule that **proves** an existential statement, $\exists$I, and the habit that goes with it: when the goal is "there exists", do not guess the object, but wait until a line of the derivation shows it. Section 4.4 left open a sequent that no rule so far could finish, $\forall x\, P(x) \vdash \exists x\, P(x)$; it is finished here in three lines.

4.5.1 Showing that something exists

How does a mathematician prove that there is a natural number N with $1/(N+1) < 1/100$? By naming one. "Take $N = 100$: then $1/(N+1) = 1/101$, and $1/101 < 1/100$." Nobody asks for more, and nobody objects that $N = 1000$ would also have done. The proof exhibits a **witness**, checks that it has the property, and stops.

Look at what the proof does with the witness at the end. The statement proved, "there is an N with $1/(N+1) < 1/100$", does not say which N; the 100 still in it belongs to the bound $1/100$, not to N. The number was needed to establish the claim and is then forgotten: a reader of the conclusion learns that some N works, not which one.

Chapter 1 used the same pattern from its first page. It recalled that an integer n is **even** when $n = 2k$ for **some** integer k, and said that the word "some" has rules of its own, to come in Chapter 4. Here is the first of them. To show that 6 is even, take $k = 3$: then $6 = 2 \cdot 3$. The statement "6 is even" is $\exists k\, (6 = 2 \cdot k)$, and the proof passes from the instance $6 = 2 \cdot 3$ to it. The second rule for "some", how to **use** "$n = 2k$ for some k", is the subject of Section 4.6.

There is a helpful way to see the step, which Section 4.1's reading of $\exists$ suggests. Over a domain with three elements named a, b, c, the sentence $\exists x\, P(x)$ says what the disjunction

$$P(a) \vee P(b) \vee P(c)$$

says (grouped to the right, as in Section 3.4): at least one of the three has P. An existential is a long "or", with one disjunct for each object of the domain. Proving a disjunction from one of its sides is $\vee$I (Definition 1.5.1), and proving an existential from one instance is the rule of this section. The analogy is only a guide (a domain such as $\mathbb{N}$ has infinitely many elements, and a formula cannot be infinitely long), but it predicts the rule's behavior exactly: it throws information away, and it must never be run backwards.

4.5.2 The rule of existential introduction

To prove that something has a property, show that one particular thing has it.

Definition 4.5.1. Existential Introduction

i	$\varphi[t/x]$	
k	$\exists x\,\varphi$	$\exists\mathrm{I}\ i$

The rule given by this schema is called ∃I (**existential introduction**). Here φ is **any** formula, x any variable, and t **any** term, subject to one side condition: t **is free for** x **in** φ (Definition 4.2.6). Line i must be available at line k (Definition 1.4.2).

In words, clause by clause.

1. **One given line, no box.** The rule cites one available line, as ∨I does. Like ∀E of Section 4.3, it opens no box and introduces no fresh name.
2. **The cited line is an instance.** Line i must have the form $\varphi[t/x]$: the formula φ with the term t put in for every free occurrence of x (Definition 4.2.5). The term t is the **witness**.
3. **The rule reads the cited line backwards.** The line $\varphi[t/x]$ is given, and φ is not: we choose it. In practice we look at the cited line, choose a variable x that does not appear in it (Section 4.2), and replace some of the occurrences of t by x; the result is φ, and $\varphi[t/x]$ is then the cited line. If x does appear in the line, this recipe can fail (from $R(x,c)$, marking c gives $\varphi = R(x,x)$, and $\varphi[c/x] = R(c,c)$ is not the line), so check directly that $\varphi[t/x]$ is exactly the cited line. In every case check clause 5. One line is an instance of many formulas, so one line has many existential conclusions. The table below shows this.
4. **Any term.** The witness may be a constant (3, c), a compound term ($f(c)$, $2k+1$), or a **variable**: a variable that appears elsewhere in the derivation, a fresh name of a flagged box such as x_0, or even the variable x itself. No box is needed for any of these. The rule never asks where t came from.
5. **Free for x.** The only restriction is the condition of Definition 4.2.6, the same one that ∀E carries. Its job is shown by the nonexample below.

One line, several readings. Let line i be $R(c,c)$, with R a two-place predicate symbol and c a constant. The witness is c, and c occurs twice. Each occurrence may be marked with x or left alone, and each choice is a different φ:

φ	$\varphi[c/x]$ (the cited line)	the new line $\exists x\,\varphi$
$R(x,c)$	$R(c,c)$	$\exists x\,R(x,c)$
$R(c,x)$	$R(c,c)$	$\exists x\,R(c,x)$
$R(x,x)$	$R(c,c)$	$\exists x\,R(x,x)$
$R(c,c)$	$R(c,c)$	$\exists x\,R(c,c)$

All four are legal steps. Read over $\mathbb{N}$ with $R(x,y)$ as "$x \le y$" and c as 5: the cited line says $5 \le 5$, and the three useful conclusions say that something is at most 5, that 5 is at most something, and that something is at most itself. Each is a true and weaker statement. The last row is the **degenerate** choice: φ has no free x, so $\varphi[c/x]$ is φ (Section 4.2) and the new line $\exists x\,R(c,c)$ quantifies over nothing. It is legal, and it says nothing that line i did not.

Why this rule. Under the reading of Section 4.1, $\exists x\,\varphi$ is true when at least one element of

the domain, put in for x, makes φ true. If $\varphi[t/x]$ is true, then the object that t names is such an element, so $\exists x\,\varphi$ is true. The side condition is what makes "$\varphi[t/x]$ says about t what φ says about x" correct: that was the content of "free for" in Section 4.2. Section 4.12 turns this sentence into a proof.

$\exists$I *forgets the witness, and keeps the truth.*

A nonexample: capture. Let line 1 be $\forall y\,R(y,y)$. With $\varphi = \forall y\,R(x,y)$ and $t = y$, the definition of substitution gives $\varphi[y/x] = \forall y\,R(y,y)$, which is line 1. So it is tempting to write $\exists x\,\forall y\,R(x,y)$ by $\exists$I.

1	$\forall y\,R(y,y)$	
2	$\exists x\,\forall y\,R(x,y)$	$\exists$I 1

Line 2 breaks the side condition: the free occurrence of x in φ lies in the scope of $\forall y$, and y is the variable of the term t, so y is **not** free for x in φ. The step would be wrong. Read over $\mathbb{Z}$ with $R(x,y)$ as "$x \le y$": line 1 says that every integer is at most itself, which is true, and line 2 says that some integer is at most every integer, that is, that there is a least integer, which is false. The y put in for x was meant to name one object, and it was captured by $\forall y$ and made to range over all of them.

Warning. The witness must be **written on a line**. "There is an N with $1/(N+1) < 1/100$: take N large enough" is not an application of $\exists$I, because "large enough" is not a term, and the rule asks for an instance $\varphi[t/x]$ on an available line. In a written proof, "large enough" is a promise to name the witness, and the promise is kept by a term such as $N = 100$, followed by the check that it works ($1/101 < 1/100$). Until then nothing has been proved.

Quick check. Let line 1 be $2 + 2 = 4$. Which of

$$\exists x\,(x + x = 4), \qquad \exists x\,(x + 2 = 4), \qquad \exists x\,(2 + 2 = x)$$

may line 2 be, justified by one application of $\exists$I to line 1? For each that may, name φ and the witness t.

Solution. All three. For $\exists x\,(x + x = 4)$, take $\varphi = (x + x = 4)$ and $t = 2$: both 2's of line 1 are marked. For $\exists x\,(x + 2 = 4)$, take $\varphi = (x + 2 = 4)$ and $t = 2$: only the first 2 is marked, and the second is written out in φ. For $\exists x\,(2 + 2 = x)$, take $\varphi = (2 + 2 = x)$ and $t = 4$: the witness is a different term. In each case $\varphi[t/x]$ is exactly line 1, and φ has no quantifier, so t is free for x.

4.5.3 From "for every" to "there exists"

Section 4.4 left $\forall x\,P(x) \vdash \exists x\,P(x)$ open. It looks like the easiest sequent in the chapter, and it has a small puzzle in it. To use $\exists$I, the goal needs an instance $P(t)$ on a line. The premise gives an instance for every term, by $\forall$E. But the sequent mentions no constant, so which term?

Proposition 4.5.1. Every Implies Some

$\forall x\, P(x) \vdash \exists x\, P(x)$.

Proof. The following is a derivation.

1	$\forall x\, P(x)$	
2	$P(x)$	$\forall$E 1
3	$\exists x\, P(x)$	$\exists$I 2

Line 2 is $\forall$E at the term x: with $\varphi = P(x)$, the instance $\varphi[x/x]$ is $P(x)$, and the variable x is free for x in every formula (Section 4.2). Line 3 is $\exists$I with the same φ and the witness x. This proves the proposition.

■

The puzzle is answered by clause 4 of the reading of Definition 4.5.1: a variable is a term, so no constant is needed. The line $P(x)$ has x free; it says that P holds of whatever x names. Nothing in the derivation says what that is, and nothing needs to. The same three lines prove $\forall x\, \varphi \vdash \exists x\, \varphi$ for every formula φ, since $\varphi[x/x] = \varphi$ (Section 4.2).

Remark. The rules accept this derivation in any case; it leads from a true premise to a true conclusion only because every domain has **at least one** element, the convention of Section 4.1. Over an empty domain, $\forall x\, P(x)$ would be true with nothing to check and $\exists x\, P(x)$ false with no witness; the convention enters at line 2, where $P(x)$ treats x as naming an object, and there must be one to name. Section 4.11 explains why the book keeps the convention.

This does not clash with the empty set of Section 4.1, where $\forall x \in \emptyset\ \varphi$ is true and $\exists x \in \emptyset\ \varphi$ is false. A bounded quantifier is an abbreviation (Definition 4.1.1): $\forall x \in \emptyset\ \varphi$ is $\forall x\,(x \in \emptyset \to \varphi)$, and $\exists x \in \emptyset\ \varphi$ is $\exists x\,(x \in \emptyset \wedge \varphi)$. The domain is still non-empty; it is the set $\emptyset$ that has no elements. Running the three lines on the bounded premise gives $\forall$E at x, the line $x \in \emptyset \to \varphi$, and then $\exists$I gives only $\exists x\,(x \in \emptyset \to \varphi)$, an existential with $\to$ inside. No line $x \in \emptyset$ is available to fire the implication, so the bounded "every implies some" is not derived, and it should not be.

4.5.4 An existential goal: postpone the witness

The search method of Section 2.6 asks two questions at each gap: what is the goal, and what are the resources? Sections 4.3 and 4.4 added a step for each side of $\forall$. Now $\exists$ can be a goal.

If the goal is $\exists x\, \varphi$, its last line can be $\exists$I, and then the new goal is $\varphi[t/x]$ for a term t **of our choosing**. That is the same situation as a goal $\varphi \vee \psi$, where $\vee$I left us to choose a side, and the danger is the same. A wrong t can leave a goal that does not follow from the resources, although the existential does. Section 2.6 met the danger for $\vee$ and answered it by postponing; the answer here is the same.

Postpone the witness: choose the term t of a goal $\exists x\, \varphi$ only when an available line shows an instance $\varphi[t/x]$.

Until then, work on the premises and on the other goals, and watch what terms appear. Often the witness is a term that only appears late, built by ∀E or →E from the premises. Here are four derivations, in order of difficulty, each with an Idea that says why each rule was chosen.

Example. Derivations with existential introduction

Show that:

(a) $P(c) \wedge Q(c) \vdash \exists x\,(P(x) \wedge Q(x))$;

(b) $R(c,c) \vdash \exists x\,R(x,c) \wedge \exists x\,R(x,x)$; and decide whether one ∃I step leads from $\exists x\,R(x,x)$ to $\exists x\,R(x,c)$;

(c) $\forall x\,R(x,x) \vdash \forall x\,\exists y\,R(x,y)$;

(d) $\forall x\,(P(x) \to Q(f(x))),\ P(c) \vdash \exists y\,Q(y)$.

Solution.

(a) *Idea.* The goal is an existential, and the premise already is an instance of a suitable φ: mark both occurrences of c. One goal-directed step finishes.

1	$P(c) \wedge Q(c)$	
2	$\exists x\,(P(x) \wedge Q(x))$	∃I 1

Here $\varphi = P(x) \wedge Q(x)$ and $t = c$. Marking only the first c, with $\varphi = P(x) \wedge Q(c)$, would give the legal but weaker line $\exists x\,(P(x) \wedge Q(c))$, which is not the goal.

(b) *Idea.* The goal is a conjunction, so it splits into two goals (Section 1.2), and each is an existential with the line $R(c,c)$ as an instance, read in two different ways (the first and third rows of the table above).

1	$R(c,c)$	
2	$\exists x\,R(x,c)$	∃I 1
3	$\exists x\,R(x,x)$	∃I 1
4	$\exists x\,R(x,c) \wedge \exists x\,R(x,x)$	∧I 2, 3

Line 2 uses $\varphi = R(x,c)$ and line 3 uses $\varphi = R(x,x)$, both with the witness c.
Now the second question. From the line $\exists x\,R(x,x)$, the step to $\exists x\,R(x,c)$ is **not** an ∃I step. The new line is $\exists x\,\varphi$ with $\varphi = R(x,c)$, so the cited line would have to be $R(t,c)$ for some term t. The line $\exists x\,R(x,x)$ is not of that form: its main symbol is $\exists x$, not R. The rule passes from an **instance** to an existential, never from one existential to another. Nor is the step safe by any other route: read over $\mathbb{Z}$ with $R(x,y)$ as "$x \cdot y = 1$" and c as 0, the line $\exists x\,R(x,x)$ is true (take $x = 1$), and $\exists x\,R(x,c)$, "some integer times 0 is 1", is false.

(c) *Idea.* The goal is a ∀, so the step of Section 4.4 opens a flagged box with a fresh name x_0, and the new goal is $\exists y\,R(x_0,y)$. That is an existential: postpone it and look at the premise. The premise is a ∀; instantiate it at the term the goal mentions, x_0. The result, $R(x_0,x_0)$, shows the witness: it is x_0 itself, a variable used as a term, as x was in Proposition 4.5.1.

1	$\forall x\, R(x,x)$		
2	x_0 | $R(x_0,x_0)$		$\forall$E 1
3	$\exists y\, R(x_0,y)$		$\exists$I 2
4	$\forall x\, \exists y\, R(x,y)$		$\forall$I 2–3

Line 3 uses $\varphi = R(x_0, y)$, the variable y, and the witness x_0; it marks the second occurrence of x_0 and keeps the first. Line 4 is $\forall$I with the flag x_0 (Definition 4.4.1): x_0 appears in no premise, there is no open assumption, and x_0 does not appear in line 4. The $\exists$I inside the box needed no box of its own, although its witness is a fresh name. Read over $\mathbb{N}$ with $R(x, y)$ as "$x \leq y$": every number is at most itself, so every number is at most something.

(d) *Idea.* The goal $\exists y\, Q(y)$ is an existential, and no line yet has the form $Q(t)$. Postpone, and work forwards. The premise $\forall x\, (\dots)$ is a $\forall$; the only term in sight is c, so instantiate at c (Section 4.3), and $\to$E with the premise $P(c)$ fires. The result $Q(f(c))$ shows the witness: the compound term $f(c)$.

1	$\forall x\, (P(x) \to Q(f(x)))$	
2	$P(c)$	
3	$P(c) \to Q(f(c))$	$\forall$E 1
4	$Q(f(c))$	$\to$E 3, 2
5	$\exists y\, Q(y)$	$\exists$I 4

Line 5 uses $\varphi = Q(y)$ and $t = f(c)$. Now see what choosing the witness first would have done. The term in sight at the start is c, and "take $y = c$" leaves the goal $Q(c)$. Nothing gives it: the premise speaks of Q only at terms of the form $f(\dots)$, and $Q(c)$ need not hold at all. Read over $\mathbb{Z}$ with P as "odd", Q as "even", $f(x)$ as $x + 1$, and c as 3: the premises say that the successor of every odd integer is even and that 3 is odd, the conclusion says that some integer is even, and the witness the derivation finds is $f(3) = 4$, while $Q(c)$ says that 3 is even. The search stalls exactly because it chose the witness before the derivation had produced it.

The four Ideas share a pattern. In (a) and (b) the witness stood on a premise line from the start; in (c) and (d) it appeared only after other rules had run, and in (d) it was a term that no line mentioned at the start. Postponing costs nothing in (a) to (c), where the right witness was already a term in sight when the existential goal appeared (in (c) the goal names x_0), and it is necessary in (d), where the only term in sight at the start, c, is the wrong witness.

4.5.5 How this rule looks in a mathematical proof

In a written proof, $\exists$I is the word "take", followed by a check. "6 is even: take $k = 3$; then $6 = 2 \cdot 3$." Over $\mathbb{Z}$, with one line of school arithmetic justified "arith" as in Section 4.3:

1	$6 = 2 \cdot 3$	arith
2	$\exists k\, (6 = 2 \cdot k)$	$\exists$I 1

Line 2 marks the 3 with k, and keeps the 6 and the 2: $\varphi = (6 = 2 \cdot k)$. In the same way "7 is odd: take $k = 3$, since $7 = 2 \cdot 3 + 1$" is an $\exists$I with $\varphi = (7 = 2 \cdot k + 1)$.

A bounded witness. Analysis states its existentials with bounds. "There is an $N \in \mathbb{N}$ with $1/(N+1) < 1/100$" abbreviates $\exists N\,(N \in \mathbb{N} \wedge 1/(N+1) < 1/100)$ (Definition 4.1.1), so the witness must satisfy both conjuncts. Taking $N = 100$ is two steps: $\wedge$I puts the bound and the inequality together, and $\exists$I forgets the witness.

1	$100 \in \mathbb{N}$	arith
2	$1/(100+1) < 1/100$	arith
3	$100 \in \mathbb{N} \wedge 1/(100+1) < 1/100$	$\wedge$I 1, 2
4	$\exists N\,(N \in \mathbb{N} \wedge 1/(N+1) < 1/100)$	$\exists$I 3

Line 4 marks two of the three occurrences of 100 in line 3 and keeps the one on the right. Section 4.8 folds lines 3 and 4 into a single bounded step. The witness 100 is the least that works, since $N = 99$ gives $1/100$, which is not less than $1/100$; but the proof does not need the least witness, only one.

A witness built from what is available. Often the witness is a term made from other names. To show that for every integer n there is an odd integer greater than $2n$, a proof says "let n be an integer; $2n+1$ will do, since it is odd and greater than $2n$". The words "let n be an integer" open the flagged box of Section 4.4, and "$2n+1$ will do" is $\exists$I with the compound witness $2n+1$, as in (d) above. In general, the witness may be any term the reader can check.

Warning. $\exists$I runs one way only. From $P(c)$ it gives $\exists x\,P(x)$; from $\exists x\,P(x)$ it gives nothing, and in particular not $P(c)$. "Some integer is even" is true and "3 is even" is false: the existential does not say **which** object has the property, so it cannot be read back as a statement about a named one. Using an existential takes a rule of its own, with a box, and that is Section 4.6.

4.5.6 What the rule sounds like

Here is the row of the rule, phrase and Lean table for $\exists$I; nothing later depends on the last column, and Section 4.7 collects the whole table.

Rule	You write	Lean
$\exists$I	"Take $x = t$. Then …"; "$2k+1$ will do."	`use t` (or `Exists.intro t h`)

The phrase names the witness and then checks it. A proof that names the witness without the check ("take $N = 100$.") has written the line $\exists x\,\varphi$ without the instance $\varphi[t/x]$ it cites; the reader must supply the instance, and should be able to.

The search method: one more step. Each rule section adds a step to the method of Section 2.6, and Section 4.8 collects them.

- **Goal $\exists x\,\varphi$.** Postpone. Work on the premises and on other goals until an available line is an instance $\varphi[t/x]$, with t free for x in φ; then $\exists$I. The witness may be a compound term that appears only after other rules have run, as $f(c)$ did in (d).

4.5.7 Exercises

4.5.7.1 A. Check your understanding

Exercise. A1

(a) State ∃I as a schema, with its side condition.

(b) Line 1 of a derivation is $R(c, f(c))$. Which of the following may line 2 be, by one application of ∃I to line 1? For each that may, give φ and the witness t; for each that may not, say why.
(i) $\exists x\, R(x, f(c))$; (ii) $\exists x\, R(c, x)$; (iii) $\exists x\, R(x, f(x))$; (iv) $\exists x\, R(x, x)$; (v) $\exists x\, R(f(c), x)$.

(c) Line 1 of a derivation is $\forall y\,(y \leq y \cdot y)$. Is $\exists x\, \forall y\,(y \leq x \cdot x)$ a legal ∃I step from line 1? Read both lines over $\mathbb{Z}$: which is true?

(d) True or false, with a reason: from the line $\exists x\, P(x)$, one application of ∃I gives $P(c)$.

(e) The current goal of a search is $\exists x\, \varphi$ and no available line is an instance of φ. What does the search method say to do, and why not choose a witness now?

Solution. *(a)* From $\varphi[t/x]$ on an available line i, write $\exists x\, \varphi$ on line k, justified "∃I i". Side condition: t is free for x in φ.

(b) (i) Yes: $\varphi = R(x, f(c))$, $t = c$; only the first c is marked. (ii) Yes: $\varphi = R(c, x)$, $t = f(c)$; the witness is the compound term. (iii) Yes: $\varphi = R(x, f(x))$, $t = c$; both occurrences of c are marked, and $\varphi[c/x]$ is $R(c, f(c))$. (iv) No: the conclusion fixes $\varphi = R(x, x)$, whose instances $R(t, t)$ have equal arguments, while the arguments c and $f(c)$ of line 1 differ. (v) No: the conclusion fixes $\varphi = R(f(c), x)$, whose instances $R(f(c), t)$ have $f(c)$ as first argument, while line 1 has c there. (Read over $\mathbb{N}$ with $R(x, y)$ as "$x < y$", $f(x)$ as $x + 1$, and c as 0: line 1 says $0 < 1$, and (v) says that something exceeds 1, true but not by this step; (iv) says that some number is less than itself, which is false.)

(c) No. With $\varphi = \forall y\,(y \leq x \cdot x)$, the formula $\varphi[y/x]$ is line 1, so the witness would be the variable y; but the free occurrences of x in φ lie in the scope of $\forall y$, so y is not free for x in φ. Over $\mathbb{Z}$, line 1 is true: for $y \leq 0$, $y \leq 0 \leq y \cdot y$, and for $y \geq 1$, $y \leq y \cdot y$ after multiplying $1 \leq y$ by $y > 0$. The other line is false: whatever integer x is proposed, $y = x \cdot x + 1$ gives $y > x \cdot x$. So the step would lead from a true line to a false one, which is what the side condition prevents.

(d) False. ∃I concludes an existential from an instance; it never concludes an instance. The line $P(c)$ does not follow from $\exists x\, P(x)$ at all: over $\mathbb{Z}$ with P as "even" and c as 3, the first is true and the second false.

(e) Postpone the goal: work on the premises (∀E at the terms the goal mentions, →E, and so on) until a line $\varphi[t/x]$ appears, then use ∃I. A witness chosen now leaves the goal $\varphi[t/x]$, which may not follow from the resources even when $\exists x\, \varphi$ does, as "take $y = c$" did in Example (Derivations with existential introduction) (d).

4.5.7.2 B. Practice

Exercise. B1

Derive $\forall x\,(P(x) \to Q(x)),\ P(a) \vdash \exists x\,(P(x) \wedge Q(x))$. In the Idea, say which rule each line

was chosen by: the goal or a premise.

Solution. Idea. The goal is an existential: postpone it. The premise $\forall x\,(\dots)$ is instantiated at the only term in sight, a, and $\to$E with $P(a)$ gives $Q(a)$; both steps are premise-directed. Now the witness a is visible, and the instance the goal needs is $P(a) \wedge Q(a)$, which $\wedge$I builds. The last step is the postponed $\exists$I.

1	$\forall x\,(P(x) \to Q(x))$	
2	$P(a)$	
3	$P(a) \to Q(a)$	$\forall$E 1
4	$Q(a)$	$\to$E 3, 2
5	$P(a) \wedge Q(a)$	$\wedge$I 2, 4
6	$\exists x\,(P(x) \wedge Q(x))$	$\exists$I 5

Line 6 uses $\varphi = P(x) \wedge Q(x)$ and the witness a.

Exercise. B2

Over $\mathbb{R}$, with the premise $\forall x\,(x + (-x) = 0)$, derive $\forall x\,\exists y\,(x + y = 0)$, the true sentence (i) of Example (Swapping two quantifiers). Say what the witness is, and what it depends on.

Solution. Idea. The goal is a $\forall$: open a flagged box with a fresh name x_0 (Section 4.4); the new goal is $\exists y\,(x_0 + y = 0)$. Postpone it, and instantiate the premise at the term the goal mentions, x_0. The line $x_0 + (-x_0) = 0$ shows the witness: $-x_0$.

1	$\forall x\,(x + (-x) = 0)$		
2	x_0	$x_0 + (-x_0) = 0$	$\forall$E 1
3		$\exists y\,(x_0 + y = 0)$	$\exists$I 2
4	$\forall x\,\exists y\,(x + y = 0)$		$\forall$I 2–3

Line 3 uses $\varphi = (x_0 + y = 0)$ and the witness $-x_0$, which contains no variable bound in φ. At line 4, x_0 appears in no premise, there is no open assumption, and x_0 does not appear in the conclusion, so $\forall$I applies (Definition 4.4.1).

The witness $-x_0$ is built from x_0, so it depends on x_0: a different number for each x. That is legal, because $\exists$I is applied **inside** the box, where x_0 is available as a term. It is the box picture of the game of Section 4.1: the $\exists y$ is played after the $\forall x$, and may use it.

Exercise. B3

Over $\mathbb{Z}$, prove "there is an integer n with $n^2 = n + 2$" as a derivation whose only line not

justified by a rule is one line of arithmetic justified "arith"; write n^2 as $n \cdot n$. Then write the proof as one sentence.

Solution. The witness 2 works, since $2 \cdot 2 = 4 = 2 + 2$.

1	$2 \cdot 2 = 2 + 2$	arith
2	$\exists n\,(n \cdot n = n + 2)$	$\exists$I 1

Line 2 uses $\varphi = (n \cdot n = n + 2)$ with the witness 2. Line 1 contains four 2's, and φ marks three of them: both factors on the left and the first term of the sum on the right. The last 2 is the constant of the statement and stays. Marking all four would give $\exists n\,(n \cdot n = n + n)$, a different statement.

In one sentence: *Take $n = 2$; then $n^2 = 4 = n + 2$.*

The witness -1 works too, since $(-1) \cdot (-1) = 1 = -1 + 2$; a proof needs only one.

4.5.7.3 C. Going deeper

Exercise. C1: The conclusion clause of for-all-introduction

The side condition of $\forall$I (Definition 4.4.1) has three clauses about the flag x_0: it appears in no premise, in no assumption of a box still open at the $\forall$I line, and not in the conclusion $\forall x\,\varphi$. Show that the third clause is needed: give a "derivation" that breaks **only** that clause, from a premise that is true over $\mathbb{N}$ to a sentence (Definition 4.2.4) that is false over $\mathbb{N}$. Check that every other step is legal.

Solution (Hint 1). The conclusion of the illegal $\forall$I still mentions the flag, so it is not yet a sentence. Which rule of this section can replace a name on a line by a bound variable?

Solution (Hint 2). Start from the premise $\forall x\,(x \le x)$, and open a flagged box with the name y_0.

Solution.

1	$\forall x\,(x \le x)$	
2	y_0 | $y_0 \le y_0$	$\forall$E 1
3	$\forall x\,(x \le y_0)$	$\forall$I 2–2
4	$\exists y\,\forall x\,(x \le y)$	$\exists$I 3

The illegal line. Line 3 is $\forall$I with the flag y_0 and $\varphi = (x \le y_0)$: the instance $\varphi[y_0/x]$ is $y_0 \le y_0$, the last line of the box. The flag y_0 appears in no premise (line 1 has no y_0), and there is no open assumption at all. But y_0 appears in the conclusion $\forall x\,(x \le y_0)$, so the third clause fails, and only it.

The other lines are legal. Line 2 is ∀E at the term y_0, which is free for x in $x \leq x$ since that formula has no quantifier. Line 4 is ∃I with $\varphi = \forall x\,(x \leq y)$ and the witness y_0: $\varphi[y_0/y]$ is line 3, and y_0 is free for y in φ, since the only quantifier of φ is on x.

Over $\mathbb{N}$. Line 1 says that every natural number is at most itself: true. Line 4 says that some natural number is at least every natural number, that is, that there is a largest natural number: false, since for any y, $x = y + 1$ is larger.

What the clause prevents. The box proved $y_0 \leq y_0$, a fact about y_0 in which the two sides are the same name. Line 3 generalized only one side and kept the other, so it claimed that **one** fixed y_0 is at least every x: a claim about y_0 that the box never proved. The flag is "let y_0 be arbitrary"; it must not survive into the statement that "since y_0 was arbitrary" produces. Line 4 then shows the damage, by turning the leaked name into a witness.

Marking scheme. (10 points) - 3: a derivation whose only illegal line is a ∀I whose flag appears in its conclusion. - 2: checks the other two clauses at that line (no premise, no open assumption contains the flag). - 2: checks the remaining lines (here ∀E and ∃I, with free-for). - 3: shows the premise true and the conclusion false over $\mathbb{N}$, with the witness of the falsity $(y + 1)$. Fatal errors (0 for the step): a second illegal line (for instance a capture in ∃I); a premise that is false over $\mathbb{N}$.

Exercise. C2: A witness is not a name

Explain why the rules so far (Chapters 1–2 and ∀E, ∀I, ∃I) cannot derive $\exists x\,P(x) \vdash P(c)$. Go through the rules that could produce the last line $P(c)$, and say what each would need. Then say, in one sentence, why no rule for using ∃ **should** give $P(c)$.

Solution. The argument below is informal: it explains why every search stalls, and it does not by itself prove that no derivation exists. Section 4.11 gives a model in which the premise is true and $P(c)$ is false. With the soundness theorem stated there (and proved in the optional Section 4.12), that model shows no derivation exists.

Which rules could write $P(c)$. The line $P(c)$ is atomic, so no introduction rule writes it: ∧I, →I, ∨I, ¬I, ↔I, ∀I and ∃I all write a line whose main symbol is their own connective or quantifier. The remaining rules take $P(c)$ out of something larger: ∧E needs a conjunction with $P(c)$ as a side; →E needs an implication $\psi \to P(c)$ and ψ; ↔E a biconditional with $P(c)$ as a side; ∨E a disjunction and two boxes each ending in $P(c)$; ∀E a universal $\forall x\,\psi$ with $\psi[c/x] = P(c)$; ⊥E the line ⊥; RAA a box from $\neg P(c)$ to ⊥; and R the line $P(c)$ itself.

The derived rules of Chapter 2 abbreviate derivations in these rules (Definition 2.3.1), so they add no new way to write $P(c)$.

What the premise can supply. The only premise is $\exists x\,P(x)$, whose main symbol is ∃. No rule so far looks inside an existential: each elimination rule needs a cited line that is ⊥ (for ⊥E) or whose main symbol is ∧, →, ∨, ↔, ¬ or ∀, and ∃I produces existentials rather than using them. So the premise can enter a derivation only as a whole: copied, built into a larger formula, set against a line $\neg\exists x\,P(x)$ by ¬E, or cited as the side ψ beside a line $\psi \to \chi$ or $\psi \leftrightarrow \chi$ by →E or ↔E. None of these reaches the information inside it, and the lines the rules above need (a conjunction containing $P(c)$, an implication ending in it, a biconditional with it as a side, a disjunction for ∨E, a universal with it as an instance, a contradiction) have nowhere to come from. A search therefore stalls at once.

Why it should stall. The premise says that **something** has P and does not say what; the conclusion says that the object named c has P. Over $\mathbb{Z}$ with P as "even" and c as 3, the premise is true and the conclusion false. So no rule for using an existential may give $P(c)$, and the rule of Section 4.6 does not: it names the witness with a **fresh** variable, about which nothing else is known, instead of with c.

Marking scheme. (10 points) - 3: identifies that $P(c)$ is atomic, so only elimination rules, $\bot$E, RAA or R can write it, and names what each needs. - 3: observes that no rule so far uses a line whose main symbol is $\exists$, so the premise cannot supply what those rules need. - 2: says that this explains the stall but is not a proof of non-derivability (that needs a counter-model together with soundness, Sections 4.11–4.12). - 2: gives a reading with the premise true and $P(c)$ false, and the one-sentence reason (the premise does not say which object has P). Fatal errors (0 for the step): claiming that $\exists$I read backwards gives $P(c)$; claiming that a failed search proves the sequent underivable.

4.6 Using "There Exists": Name a Witness

Section 4.5 gave the rule that proves an existential statement: exhibit a witness, and forget it. This section gives the rule that **uses** one, $\exists$E. It is the last of the four quantifier rules, and the subtlest, because an existential says that a witness exists without saying which one it is. The rule handles this with a box and a fresh name, as $\forall$I did, but the name now stands for an object that is **not** arbitrary, and the side condition keeps it from being treated as if it were. With the rule in hand, the section moves an $\exists$ past a $\forall$, and shows exactly where the reverse move breaks down.

4.6.1 Using a witness whose identity is unknown

Chapter 2 proved that if n^2 is even then n is even (Theorem 2.1.3). One of the facts it rested on, (E3), says that the square of an odd integer is odd, and it was shown by taking "an integer k with $n = 2k + 1$" and computing with it. Chapter 2 noted that naming the k of (E3) lay outside its derivation: it is a quantifier step, whose rule had to wait. Here is the simplest proof with the same step.

> *If n is even, then $n + 2$ is even.* Since n is even, choose an integer k with $n = 2k$. Then $n + 2 = 2k + 2 = 2(k + 1)$, so $n + 2$ is even.

The hypothesis is $\exists k\,(n = 2 \cdot k)$. The proof does not know which integer k is; it does not need to. It gives the unknown witness a name, uses the one fact it has about it, $n = 2k$, and reaches a conclusion, "$n + 2$ is even", in which the name no longer appears. The witness $k + 1$ of the last step is forgotten by $\exists$I, as in Section 4.5.

Why is this correct? Two tempting shortcuts show what the proof is careful **not** to do.

- It does not name the witness by a term that already means something. From "some integer is even" we may not conclude "3 is even": the existential does not say that the witness is 3, and, as the warning "$\exists$I runs one way only" of Section 4.5 noted, it cannot be read back as a statement about a named object.
- It does not let the conclusion mention the name. If it did, the conclusion would be a statement about k, and outside the argument k refers to nothing; the existential promised only that **some** k works.

So the name must be new, the argument may use only what the existential says about it, and

the conclusion must not mention it. Then the argument works for the witness, whichever integer it is, because it never used anything else about it.

The long "or" of Section 4.5 gives the same picture from another side. Over a domain with three elements named a, b, c, the line $\exists x\, P(x)$ says $P(a) \vee P(b) \vee P(c)$, and $\vee$E (Definition 1.6.1) would use it by proving a goal χ three times, once in each case. The three cases look alike, so they can be done once, with a name that could be any of a, b, c. All three cases must end in the **same** χ, so χ must not say which case it came from. That is the box of the rule, and its side condition.

4.6.2 The rule of existential elimination

To use "there exists", give the witness a new name, assume only what the existential says about it, and conclude something that does not mention it.

Definition 4.6.1. Rules for the Existential Quantifier

$$\begin{array}{lll|ll} h & \exists x\,\varphi & & & \\ i & x_0 \mid \varphi[x_0/x] & & & \\ & \quad\vdots & & & \\ j & \quad \chi & & & \\ k & \chi & \exists\text{E } h,\ i\text{–}j & & \end{array} \qquad \begin{array}{ll|ll} i & \varphi[t/x] & & \\ k & \exists x\,\varphi & \exists\text{I } i & \end{array}$$

The rule on the left is $\exists$E (**existential elimination**). It cites a line h holding $\exists x\,\varphi$ and a **flagged box** with the name x_0, a variable, whose first line is the assumption $\varphi[x_0/x]$ and whose last line j, a line of that box itself, is χ; line h and the box must be available at line k (Definition 1.4.2, with the flagged boxes of Section 4.4). The new line k repeats χ. Side condition, at line k: the name x_0 does **not appear**

- in χ,
- in $\exists x\,\varphi$,
- in any premise, or
- in any assumption of a box **still open at line** k.

The rule on the right is $\exists$I, recalled from Definition 4.5.1: t is any term free for x in φ.

The box of $\exists$E is the flagged box of Section 4.4 with one difference: it **opens with an assumption**, $\varphi[x_0/x]$, "x_0 is a witness". The flagged box of $\forall$I had no assumption, because "let x_0 be arbitrary" assumes nothing about x_0. The citation "$\exists$E h, i–j" names the line of the existential and the box, as $\vee$E names its disjunction and its boxes. "Appear" is the word of Section 4.2: written anywhere in the formula, after a quantifier symbol included.

Read the rule clause by clause.

1. **The assumption.** The box assumes $\varphi[x_0/x]$: the formula of the existential, with the new name in place of x. This is "choose x_0 with φ". It is the **only** thing the box knows about x_0.
2. **The last line.** The box ends with χ, a line of the box itself, not of a box inside it, as for $\vee$E and $\forall$I. It is "whatever the witness is, χ follows". The new line k then asserts χ outside the box, where x_0 is no longer assumed to be anything.

3. x_0 **does not appear in** χ**.** The box shows that χ follows from "x_0 is a witness"; if χ mentioned x_0, line k would make a claim about a name that, outside the box, is tied to nothing. The derivation after this list shows the damage.
4. x_0 **does not appear in a premise or in an assumption of an open box.** The name must be new to everything we know at line k. If a premise or an open assumption already said something about x_0, the assumption $\varphi[x_0/x]$ would add to it a claim that x_0 **is** the witness, which the existential never made.
5. x_0 **does not appear in** $\exists x\,\varphi$**.** The existential must not itself mention the name chosen for its witness. This clause also makes x_0 free for x in φ (Section 4.2: a variable that does not appear in φ is free for every variable there), so the assumption $\varphi[x_0/x]$ says of x_0 exactly what φ says of x.
6. **"Still open at line** k**".** The box's own assumption $\varphi[x_0/x]$ is closed at line k, so it is not among the assumptions checked. This matters: read inside the box, "an open assumption" would include $\varphi[x_0/x]$, which contains x_0 whenever x is free in φ, and the rule could then never be applied in the cases that matter.

Clauses 3, 4 and 5 can each be seen failing. Each derivation below breaks exactly one of them and leads from premises that are true, under the reading given, to a false line.

The conclusion clause (3). Over $\mathbb{N}$, "some number exceeds 5" is true. Let the $\exists$E box end in the line $x_0 > 5$, which mentions its own witness:

1	$\exists x\,(x > 5)$	
2	x_0 | $x_0 > 5$	
3	$x_0 > 5$	$\exists$E 1, 2–2
4	x_0 | $x_0 > 5$	R 3
5	$\forall x\,(x > 5)$	$\forall$I 4–4

Line 3 breaks clause 3. Every other line is legal: in particular line 5 is a correct $\forall$I, since x_0 appears in no premise and in no open assumption (there is none), and not in line 5. Over $\mathbb{N}$ the conclusion says that every number exceeds 5, which is false at 0. Once the name leaves its box, it can be generalized over as if it were arbitrary.

The premise clause (4). Over $\mathbb{N}$, with P read as "even" and the variable x_0 read as 1, both premises below are true: some number is even, and 1 is not even.

1	$\exists x\,P(x)$	
2	$\neg P(x_0)$	
3	x_0 | $P(x_0)$	
4	| $\bot$	$\neg$E 3, 2
5	$\bot$	$\exists$E 1, 3–4

Line 5 breaks clause 4: x_0 appears in the premise on line 2. The box assumed that x_0, which the premise had already described as odd, is the even witness, and derived a contradiction from that assumption, not from the premises.

The existential clause (5). Over $\mathbb{Z}$, with $R(x, y)$ read as "$x < y$", the premise below says that every integer has a smaller one: true.

1	$\forall y\, \exists x\, R(x, y)$	
2	$\exists x\, R(x, x_0)$	∀E 1
3	x_0 | $R(x_0, x_0)$	
4	| $\exists z\, R(z, z)$	∃I 3
5	$\exists z\, R(z, z)$	∃E 2, 3–4

Line 2 is a legal ∀E at the term x_0. Line 5 breaks clause 5: the existential on line 2 is $\exists x\, \varphi$ with $\varphi = R(x, x_0)$, and x_0 appears in it. The assumption $\varphi[x_0/x]$ on line 3 is $R(x_0, x_0)$: the box has assumed that the witness smaller than x_0 is x_0 itself. The conclusion says that some integer is less than itself, which is false. The other clauses hold at line 5: x_0 is not in $\exists z\, R(z, z)$, not in the premise, and no box is open.

A fourth illegal pattern, a second witness given a name that an open assumption already uses, is the error of the warning "Two witnesses, one name" below.

Warning. **The witness is not arbitrary.** The flagged boxes of ∀I and ∃E look alike: both introduce a fresh name. Their names mean opposite things. In a ∀I box, x_0 is **any** object, and what is proved about it holds for all. In an ∃E box, x_0 is **one particular** object, the witness, about which we know $\varphi[x_0/x]$ and nothing else; what is proved about it holds for it alone. From "some integer is even" nothing follows about every integer, and the rules agree:

1	$\exists x\, P(x)$	
2	x_0 | $P(x_0)$	
3	| x_0 | $P(x_0)$	R 2
4	| $\forall x\, P(x)$	∀I 3–3
5	$\forall x\, P(x)$	∃E 1, 2–4

Line 4 is blocked: x_0 appears in the assumption $P(x_0)$ of the ∃E box, which is still open at line 4. That clause of ∀I is what keeps a witness from being generalized over inside its box; clause 3 of ∃E, that x_0 not appear in χ, keeps it from leaving the box (the derivation for clause 3 above).

4.6.3 Using an existential: open the box early

When a resource is $\exists x\, \varphi$, the premise-directed step of the search method is to open its ∃E box at once, with a fresh name x_0, and to make the box's last line the **current goal**. Inside the box, the assumption $\varphi[x_0/x]$ is a new resource, and the name x_0 is a new term for later steps to use.

Name the witness early: open the ∃E *box as soon as the existential is available, and aim it at the current goal.*

Why early? The box takes nothing away: every line available before it opens is available inside it, so every resource is still there, and the assumption $\varphi[x_0/x]$ is one more. And a box aimed at the current goal χ can always be closed once χ is reached: x_0 is chosen, when the box opens, to appear in no premise, in no open assumption, not in $\exists x\, \varphi$ and not in the current goal χ.

The assumptions still open at the line that closes the box are the ones that were open when it opened, so every clause of the side condition holds there. (One caution: a ∀I inside the box needs a flag that does not appear in the new assumption; a flag that is fresh at that point has this property.) What the early box gives is the witness. The name x_0 is often exactly the term that a postponed ∃ goal is waiting for (Section 4.5), and the sooner it exists, the sooner the search can see it.

The first result is the derivation that this step writes almost by itself.

Proposition 4.6.1. Some Distributes Over And, One Way

$\exists x\,(P(x) \wedge Q(x)) \vdash \exists x\,P(x) \wedge \exists x\,Q(x)$.

Idea. The premise is an existential, so open its box first, with a fresh x_0, and aim it at the goal. Inside, the assumption $P(x_0) \wedge Q(x_0)$ splits by ∧E, and each half is an instance of one conjunct of the goal, with the witness x_0. The goal does not mention x_0, so the box closes.

Proof. The following is a derivation.

1		$\exists x\,(P(x) \wedge Q(x))$	
2	x_0	$P(x_0) \wedge Q(x_0)$	
3		$P(x_0)$	∧E 2
4		$\exists x\,P(x)$	∃I 3
5		$Q(x_0)$	∧E 2
6		$\exists x\,Q(x)$	∃I 5
7		$\exists x\,P(x) \wedge \exists x\,Q(x)$	∧I 4, 6
8		$\exists x\,P(x) \wedge \exists x\,Q(x)$	∃E 1, 2–7

Line 2 is the assumption $\varphi[x_0/x]$ for $\varphi = P(x) \wedge Q(x)$. Lines 4 and 6 are ∃I with the witness x_0. At line 8, x_0 does not appear in line 7, in line 1 (which is also the only premise), or in any assumption of an open box (none is open). So line 8 is a correct ∃E, and the derivation has 8 lines.

■

Notice where the witness went. It was used twice inside the box, as the witness of two different existentials, and then forgotten twice. The conclusion records that something has P and something has Q, and no longer that it is the same thing.

Quick check. Is $\exists x\,P(x) \wedge \exists x\,Q(x) \vdash \exists x\,(P(x) \wedge Q(x))$? Run the search, and say where it stalls.

Solution. The premise splits by $\wedge$E into $\exists x\, P(x)$ and $\exists x\, Q(x)$. Opening both boxes early gives the assumptions $P(x_0)$ and, for the second box, a **different** name, $Q(y_0)$: the second box is opened while the first is open, so x_0 appears in an open assumption and may not be reused. The postponed goal $\exists x\,(P(x) \wedge Q(x))$ needs one term t with $P(t) \wedge Q(t)$, and the resources speak of two names, with nothing to say that they are equal. The search stalls there. It must: over $\mathbb{Z}$, with P as "even" and Q as "odd", the premise is true and the conclusion, "some integer is both even and odd", is false. Section 4.11 gives a counter-model, and with the soundness statement stated there (proved in Section 4.12) this shows that no derivation exists.

Here are three more derivations, each with an Idea saying why each rule was chosen: two for part (a), one for each direction, and one for part (b).

Example. Derivations with existential elimination

Show that:

(a) $\exists x\,(P(x) \vee Q(x)) \dashv\vdash \exists x\, P(x) \vee \exists x\, Q(x)$;
(b) $\exists x\, P(x),\ \forall x\,(P(x) \to \neg Q(x)) \vdash \neg\forall x\, Q(x)$.

Solution.

(a) ($\vdash$) *Idea.* The premise is an existential: open its box first. Inside, the assumption $P(x_0) \vee Q(x_0)$ is a disjunction, and the premise-directed step for $\vee$ (Section 1.6) opens two cases, each ending in the current goal. In each case one side of the goal has an instance with the witness x_0, so $\exists$I and $\vee$I finish it.

1		$\exists x\,(P(x) \vee Q(x))$	
2	x_0	$P(x_0) \vee Q(x_0)$	
3		$P(x_0)$	
4		$\exists x\, P(x)$	$\exists$I 3
5		$\exists x\, P(x) \vee \exists x\, Q(x)$	$\vee$I 4
6		$Q(x_0)$	
7		$\exists x\, Q(x)$	$\exists$I 6
8		$\exists x\, P(x) \vee \exists x\, Q(x)$	$\vee$I 7
9		$\exists x\, P(x) \vee \exists x\, Q(x)$	$\vee$E 2, 3–5, 6–8
10		$\exists x\, P(x) \vee \exists x\, Q(x)$	$\exists$E 1, 2–9

Line 9 is $\vee$E (Definition 1.6.1), and it is the last line of the $\exists$E box itself. At line 10, x_0 is in neither line 9 nor line 1, and no box is open.

($\dashv$) *Idea.* Now the premise is a disjunction: $\vee$E first, with two cases, each ending in the goal. In each case the resource is an existential, so its box opens at once, and inside it $\vee$I and $\exists$I build the goal.

1	$\exists x\, P(x) \lor \exists x\, Q(x)$	
2	$\quad \exists x\, P(x)$	
3	$\quad x_0 \mid P(x_0)$	
4	$\quad\quad P(x_0) \lor Q(x_0)$	$\lor$I 3
5	$\quad\quad \exists x\,(P(x) \lor Q(x))$	$\exists$I 4
6	$\quad \exists x\,(P(x) \lor Q(x))$	$\exists$E 2, 3–5
7	$\quad \exists x\, Q(x)$	
8	$\quad x_0 \mid Q(x_0)$	
9	$\quad\quad P(x_0) \lor Q(x_0)$	$\lor$I 8
10	$\quad\quad \exists x\,(P(x) \lor Q(x))$	$\exists$I 9
11	$\quad \exists x\,(P(x) \lor Q(x))$	$\exists$E 7, 8–10
12	$\exists x\,(P(x) \lor Q(x))$	$\lor$E 1, 2–6, 7–11

At line 6, the open assumption is line 2, which does not contain x_0; at line 11 it is line 7, which does not either. The name x_0 is used in both cases. That is allowed: when the second box opens, the first has closed, and the side condition looks only at the premises, the boxes still open, the existential and the conclusion. Lines 6 and 11 are lines of the two case boxes themselves, as $\lor$E requires.

The two derivations mirror each other: $\exists$E outside and $\lor$E inside in one, the other way round in the other. Over a domain, both sides say that some element has P or Q. Contrast Proposition 4.6.1, where only one direction holds: $\exists$ passes through $\lor$ in both directions and through $\land$ in one.

(b) *Idea.* The goal is a negation, so the goal-directed step of Section 2.2 opens a $\neg$I box on $\forall x\, Q(x)$ (Definition 2.2.1), with the goal $\bot$. Then the premise $\exists x\, P(x)$ opens its box, aimed at $\bot$. Inside, the witness x_0 is the term at which to instantiate both universal statements: the premise gives $\neg Q(x_0)$, and the assumption gives $Q(x_0)$, a clash.

1	$\exists x\, P(x)$	
2	$\forall x\,(P(x) \to \neg Q(x))$	
3	$\quad \forall x\, Q(x)$	
4	$\quad x_0 \mid P(x_0)$	
5	$\quad\quad P(x_0) \to \neg Q(x_0)$	$\forall$E 2
6	$\quad\quad \neg Q(x_0)$	$\to$E 5, 4
7	$\quad\quad Q(x_0)$	$\forall$E 3
8	$\quad\quad \bot$	$\neg$E 7, 6
9	$\quad \bot$	$\exists$E 1, 4–8
10	$\neg\forall x\, Q(x)$	$\neg$I 3–9

At line 9, the box of line 3 is still open, so its assumption $\forall x\, Q(x)$ is checked: x_0 does not appear in it. Nor is x_0 in $\bot$, in line 1, or in line 2. Read over $\mathbb{Z}$ with P as "odd" and Q as

"divisible by 4": some integer is odd, no odd integer is divisible by 4, so not every integer is divisible by 4. The witness, whichever odd integer it is, is a counterexample to "every integer is divisible by 4", and the derivation never needs to know which.

4.6.4 Moving one quantifier past another

Section 4.1 read $\exists y\,\forall x\,\varphi$ and $\forall x\,\exists y\,\varphi$ as games, and observed that the first gives the second: if one y works for every x, then for every x some y works, namely the same one. Here is that observation as a derivation, with the letters of a two-place predicate. It shows, written out in full, a pattern that later derivations of the chapter use again and again and that later exercises leave more and more of to the reader.

Proposition 4.6.2. Some-All Gives All-Some

$\exists x\,\forall y\,R(x,y) \vdash \forall y\,\exists x\,R(x,y)$.

Idea. Three moves, read off the goal and the premise.

① **Fix.** The goal is $\forall y$: open a flagged box with a fresh y_0 (Section 4.4); the new goal is $\exists x\,R(x,y_0)$.

② **Obtain.** That goal is an existential: postpone it. The premise is an existential: open its box early, with a fresh x_0, assuming $\forall y\,R(x_0,y)$, and aim it at the current goal $\exists x\,R(x,y_0)$. Instantiate the assumption at the term the goal mentions, y_0: this gives $R(x_0,y_0)$.

③ **Choose.** That line is an instance of the postponed goal, with the witness x_0: $\exists$I.

Before reading on: at the line that closes the $\exists$E *box, check each clause of its side condition. Where could* x_0 *still appear?*

Proof. The following is a derivation.

1	$\exists x\,\forall y\,R(x,y)$			
2	y_0	x_0	$\forall y\,R(x_0,y)$	
3			$R(x_0,y_0)$	$\forall$E 2
4			$\exists x\,R(x,y_0)$	$\exists$I 3
5		$\exists x\,R(x,y_0)$		$\exists$E 1, 2–4
6	$\forall y\,\exists x\,R(x,y)$			$\forall$I 2–5

Line 2 is the assumption $\varphi[x_0/x]$ for $\varphi = \forall y\,R(x,y)$. Line 3 is $\forall$E at y_0. Line 4 is $\exists$I with $\varphi = R(x,y_0)$ and the witness x_0, which is free for x since $R(x,y_0)$ has no quantifier.

Line 5 closes the $\exists$E box, whose last line, line 4, is a line of the box itself. Its side condition holds: x_0 does not appear in $\chi = \exists x\,R(x,y_0)$, nor in the existential $\exists x\,\forall y\,R(x,y)$ of line 1, which is also the only premise; and the one box still open at line 5, the flagged box of y_0, has no assumption.

Line 6 closes the flagged box of y_0 by $\forall$I (Definition 4.4.1). That box immediately contains the $\exists$E box, so its range starts at line 2 (the convention of Section 4.4), and its last line, line 5, is a line of the box itself, the instance $(\exists x\, R(x,y))[y_0/y]$. The name y_0 appears neither in the premise nor in line 6, and at line 6 no box is open. This proves the proposition.

■

The two names played opposite parts. The name y_0 is arbitrary: nothing was assumed about it, and line 6 generalizes over it. The name x_0 is the witness: line 2 assumed everything the premise says about it, and line 5 forgets it. The witness x_0 works for y_0 because it works for every y, which line 2 says. This is the shape of a great many proofs in analysis, and it has a name.

Fix, obtain, choose: fix the arbitrary objects of the goal ($\forall$I); obtain witnesses from the hypotheses ($\forall$E and $\exists$E); choose the goal's witness from what is now available ($\exists$I).

Nothing in the proof used that the formula under the quantifiers is $R(x,y)$. For any formula φ and different variables x and y, with different names x_0 and y_0, other than x and y, that do not appear in φ, the same six lines derive $\exists x\,\forall y\,\varphi \vdash \forall y\,\exists x\,\varphi$. Line 2 is $\forall y\,(\varphi[x_0/x])$. Line 3 is $(\varphi[x_0/x])[y_0/y]$, which equals $(\varphi[y_0/y])[x_0/x]$, since the two substitutions put fresh names in for two different variables and so do not interfere; so line 4 is $\exists$I with the formula $\varphi[y_0/y]$ and the witness x_0. The fresh names are free for their variables, since neither appears in φ, and the side conditions are checked as before. With the letters x and y exchanged, this is the derivation of $\exists y\,\forall x\,\varphi$ to $\forall x\,\exists y\,\varphi$ promised in Section 4.1.

Either nesting works. The derivation opened the flagged box of y_0 first and the $\exists$E box inside it. The other order is just as good.

1	$\exists x\,\forall y\,R(x,y)$		
2	x_0 | $\forall y\,R(x_0,y)$		
3	y_0 | $R(x_0,y_0)$		$\forall$E 2
4	| $\exists x\,R(x,y_0)$		$\exists$I 3
5	$\forall y\,\exists x\,R(x,y)$		$\forall$I 3–4
6	$\forall y\,\exists x\,R(x,y)$		$\exists$E 1, 2–5

At line 5, y_0 appears neither in the premise, nor in the open assumption $\forall y\,R(x_0,y)$ (its y is a bound variable, not y_0), nor in line 5. At line 6, x_0 is not in line 5, not in line 1, and no box is open. In this order the $\exists$E box is aimed at the whole goal; in the first, at the goal left inside the flagged box. Both are the search method's "open the box early": early relative to the goal that the box must reach.

The converse search stalls at a side condition. Try $\forall y\,\exists x\,R(x,y) \vdash \exists x\,\forall y\,R(x,y)$. The goal is an existential: postpone it; it will need a witness x_0 and a line $\forall y\,R(x_0,y)$. The premise is a universal, and to get an existential out of it we must instantiate it at some term. The natural routes then meet a side condition. Here are two of them.

First route: obtain the witness, then try to generalize. Instantiate at a name y_0, open the $\exists$E box on the result, and try to prove $\forall y\,R(x_0,y)$ inside it by $\forall$I:

1	$\forall y\,\exists x\,R(x,y)$	
2	$\exists x\,R(x,y_0)$	$\forall$E 1
3	$x_0 \mid R(x_0,y_0)$	
4	$\quad y_0 \mid R(x_0,y_0)$	R 3
5	$\quad \forall y\,R(x_0,y)$	$\forall$I 4–4
6	$\quad \exists x\,\forall y\,R(x,y)$	$\exists$I 5
7	$\exists x\,\forall y\,R(x,y)$	$\exists$E 2, 3–6

Line 5 is blocked: y_0 appears in the assumption $R(x_0,y_0)$ of the $\exists$E box, which is still open. The witness x_0 was obtained **for** y_0, and the assumption says so. Choosing a different flag, say z_0, does not help: the goal inside its box would be $R(x_0,z_0)$, and nothing gives it.

Second route: fix first, obtain inside, and carry the witness out. Open the flagged box of y_0 first, obtain x_0 inside it, and try to close the $\exists$E box with a line about x_0, so that $\forall$I can then generalize over y_0:

1	$\forall y\,\exists x\,R(x,y)$	
2	$y_0 \mid \exists x\,R(x,y_0)$	$\forall$E 1
3	$\quad x_0 \mid R(x_0,y_0)$	
4	$\quad R(x_0,y_0)$	$\exists$E 2, 3–3
5	$\forall y\,R(x_0,y)$	$\forall$I 2–4
6	$\exists x\,\forall y\,R(x,y)$	$\exists$I 5

Line 4 is blocked: x_0 appears in $\chi = R(x_0,y_0)$. Lines 5 and 6 would be legal on their own; the derivation fails only because the witness escaped its box at line 4.

The two blocked lines are the two halves of one fact. The witness x_0 is obtained after y_0 is fixed, so it may depend on y_0; the conclusion asks for an x_0 chosen before any y, which may not. This is the dependence that Section 4.1 read off the order of the quantifiers, now visible as the order of the boxes. A stalled search is not a proof that no derivation exists. Section 4.11 gives a structure on $\mathbb{Z}$ in which the premise is true and the conclusion false, and the soundness statement stated there (proved in Section 4.12) turns it into a proof that no derivation exists.

4.6.5 Renaming the bound variable of an existential

Section 4.2 promised that a bound variable at the front of a formula may be renamed, and Section 4.4 derived the renaming for $\forall$ (Example (Flagged boxes at work) (b)). Here is the derivation for $\exists$. Let φ be a formula and y, z variables, with z not appearing in φ. Then

$$\exists y\,\varphi \;\dashv\vdash\; \exists z\,\varphi[z/y].$$

Take a fresh name z_0, a variable different from y and z that does not appear in φ, as in Section 4.4. Both directions use one fact about substitution: since z does not appear in φ, each free z of $\varphi[z/y]$ stands where φ had a free y, and nowhere else, so putting a term t in for those z's gives $\varphi[t/y]$; that is, $(\varphi[z/y])[t/z] = \varphi[t/y]$. (It is a structural induction of the kind in Section 4.2, one clause at a time.)

1	$\exists y\,\varphi$			1	$\exists z\,\varphi[z/y]$		
2	z_0 | $\varphi[z_0/y]$			2	z_0 | $\varphi[z_0/y]$		
3	| $\exists z\,\varphi[z/y]$	$\exists$I 2		3	| $\exists y\,\varphi$	$\exists$I 2	
4	$\exists z\,\varphi[z/y]$	$\exists$E 1, 2–3		4	$\exists y\,\varphi$	$\exists$E 1, 2–3	

In the left derivation, line 3 is $\exists$I with the formula $\varphi[z/y]$, the variable z and the witness z_0: by the fact, $(\varphi[z/y])[z_0/z]$ is line 2. The witness is free for z, because the quantifiers of $\varphi[z/y]$ are those of φ, and none of them is on z_0. In the right derivation, line 2 is the assumption of the box for the existential $\exists z\,\varphi[z/y]$, which is $(\varphi[z/y])[z_0/z]$, that is $\varphi[z_0/y]$ by the fact, and line 3 is $\exists$I with φ, y and the witness z_0, which is free for y since it does not appear in φ. In both, line 4 is a correct $\exists$E: z_0 does not appear in φ, so it is in neither line 1 nor line 3, and no box is open.

So $\exists y\,(y \neq x)$ and $\exists z\,(z \neq x)$ can each be derived from the other, as the recipe of Section 4.2 assumed.

4.6.6 Two witnesses

A proof often obtains several witnesses, one from each existential hypothesis. Each needs its own box and its own name.

Example. The sum of two even integers is even

Over $\mathbb{Z}$, with "n is even" written $\exists k\,(n = 2 \cdot k)$, derive

$$\exists k\,(n = 2 \cdot k),\ \exists k\,(m = 2 \cdot k) \vdash \exists k\,(n + m = 2 \cdot k).$$

Solution. Idea. Both premises are existentials: open both boxes early, the second inside the first, with **different** names k_0 and l_0. The goal is an existential: postpone it. Inside both boxes the resources are $n = 2 \cdot k_0$ and $m = 2 \cdot l_0$, and arithmetic gives $n + m = 2 \cdot (k_0 + l_0)$, which shows the witness: the compound term $k_0 + l_0$.

1	$\exists k\,(n = 2 \cdot k)$		
2	$\exists k\,(m = 2 \cdot k)$		
3	k_0 | $n = 2 \cdot k_0$		
4	| l_0 | $m = 2 \cdot l_0$		
5	| | $n + m = 2 \cdot (k_0 + l_0)$	arith 3, 4	
6	| | $\exists k\,(n + m = 2 \cdot k)$	$\exists$I 5	
7	| $\exists k\,(n + m = 2 \cdot k)$	$\exists$E 2, 4–6	
8	$\exists k\,(n + m = 2 \cdot k)$	$\exists$E 1, 3–7	

Line 5 is one line of arithmetic: from $n = 2k_0$ and $m = 2l_0$, substituting and factoring gives $n + m = 2k_0 + 2l_0 = 2(k_0 + l_0)$. The substitutions are steps of the rule =E, which arrives only in Section 4.7, so here the whole calculation is one line justified "arith", citing the two lines it uses. Line 6 is $\exists$I with $\varphi = (n + m = 2 \cdot k)$ and the witness $k_0 + l_0$.

At line 7 the only box still open is the outer one, whose assumption is line 3; l_0 appears neither in it, nor in the premises on lines 1 and 2 (line 2 is also the existential cited), nor in line 6. At line 8, k_0 appears neither in line 7, nor in the premises (line 1 is the existential cited), and no box is open. The derivation has 8 lines. The written proof is three sentences: *Choose k with $n = 2k$, and l with $m = 2l$. Then $n + m = 2(k + l)$. So $n + m$ is even.*

Quick check. Why must the second witness in Example (The sum of two even integers is even) be called l_0, and not k_0 again? Name the line and the clause of the side condition that a second k_0 would break.

Solution. The inner box would then be flagged k_0 with the assumption $m = 2 \cdot k_0$, and line 7 would close it by ∃E. At line 7 the outer box is still open, and its assumption $n = 2 \cdot k_0$ (line 3) contains k_0. That breaks the clause “x_0 does not appear in any assumption of a box still open at line k”. In words: the second witness would be assumed to be the same integer as the first, so n and m would be forced equal, which the premises never said.

4.6.7 How this rule looks in a mathematical proof

In a written proof, the ∃E box is announced by “choose”, “let x_0 be such a point”, “say $n = 2k$”, “fix such a k”. The assumption of the box is the property in the sentence, and the rest of the paragraph is the inside of the box. The claim the paragraph ends on, usually the clause after the final “so”, must not mention the chosen name, though the computation leading up to it may. If that claim still mentions the name, the paragraph has not finished: a later step must reach a claim without the name before the box can close, often an ∃I that forgets it (as in the example below), or a contradiction $\bot$ (as in Example (Derivations with existential elimination) (b)).

Here is an instance from analysis. A sequence (a_n) is **bounded** if there is an M with $|a_n| \leq M$ for every $n \in \mathbb{N}$. A proof says:

> *Since (a_n) is bounded, choose M with $|a_n| \leq M$ for all $n \in \mathbb{N}$. Then $|2a_n| = 2|a_n| \leq 2M$ for all $n \in \mathbb{N}$, so $(2a_n)$ is bounded.*

With the domain $\mathbb{R}$, the sequence as a function symbol a, and the bound on n written out by Definition 4.1.1, the proof is this derivation.

1		$\exists M\,\forall n\,(n \in \mathbb{N} \to \lvert a_n\rvert \le M)$	
2	M_0	$\forall n\,(n \in \mathbb{N} \to \lvert a_n\rvert \le M_0)$	
3		n_0 | $n_0 \in \mathbb{N}$	
4		$n_0 \in \mathbb{N} \to \lvert a_{n_0}\rvert \le M_0$	$\forall$E 2
5		$\lvert a_{n_0}\rvert \le M_0$	$\to$E 4, 3
6		$\lvert 2 \cdot a_{n_0}\rvert \le 2 \cdot M_0$	arith 5
7		$n_0 \in \mathbb{N} \to \lvert 2 \cdot a_{n_0}\rvert \le 2 \cdot M_0$	$\to$I 3–6
8		$\forall n\,(n \in \mathbb{N} \to \lvert 2 \cdot a_n\rvert \le 2 \cdot M_0)$	$\forall$I 3–7
9		$\exists M\,\forall n\,(n \in \mathbb{N} \to \lvert 2 \cdot a_n\rvert \le M)$	$\exists$I 8
10	$\exists M\,\forall n\,(n \in \mathbb{N} \to \lvert 2 \cdot a_n\rvert \le M)$		$\exists$E 1, 2–9

The sentences match the boxes. "Choose M" opens the $\exists$E box (line 2). "For all $n \in \mathbb{N}$" in the second sentence is the flagged box of n_0 with its $\to$I box (lines 3–7, closed by line 8); the arithmetic $|2a_n| = 2|a_n| \le 2M$ is line 6. "So $(2a_n)$ is bounded" is two steps: $\exists$I with the witness $2 \cdot M_0$ (line 9), which forgets the bound $2M$, and the close of the box (line 10). The conclusion mentions neither M_0 nor $2 \cdot M_0$, as the side condition requires. Section 4.8 shortens the bounded steps.

Warning. **Two witnesses, one name.** "Let n and m be even, say $n = 2k$ and $m = 2k$" is a common error in parity proofs. It gives two unknown witnesses the same name, and the rest of the proof then uses an equation the premises never gave. For instance it "proves" that the sum of two even integers is divisible by 4:

1		$\exists k\,(n = 2 \cdot k)$	
2		$\exists k\,(m = 2 \cdot k)$	
3	k_0	$n = 2 \cdot k_0$	
4		k_0 | $m = 2 \cdot k_0$	
5		$n + m = 4 \cdot k_0$	arith 3, 4
6		$\exists k\,(n + m = 4 \cdot k)$	$\exists$I 5
7		$\exists k\,(n + m = 4 \cdot k)$	$\exists$E 2, 4–6
8		$\exists k\,(n + m = 4 \cdot k)$	$\exists$E 1, 3–7

Line 7 is blocked: k_0 appears in the assumption $n = 2 \cdot k_0$ of the outer box, still open at line 7. With $n = 2$ and $m = 4$, both premises are true and $n + m = 6$ is not divisible by 4.

4.6.8 What the rule sounds like

Here is the row of the rule, phrase and Lean table for $\exists$E; Section 4.7 collects the whole table, and nothing later depends on the last column.

Rule	You write	Lean
∃E	"Choose k with $n = 2k$. Then …"; "let x_0 be such a point."	`obtain` (or `Exists.elim h`)

The phrase "choose" is apt and slightly misleading. We do not choose the witness, since we cannot see it; we choose a **name** for it. A written proof that says "choose k" promises the reader that nothing more about k will be used than the property stated, and that k will be gone by the end of the argument.

The search method: one more step. With this rule, each of the four quantifier rules has added a step, and Section 4.8 collects them with the steps of Section 2.6.

- **Premise $\exists x\,\varphi$.** Open the ∃E box early, as soon as the existential is available, with a fresh name x_0 and the assumption $\varphi[x_0/x]$; make its last line the current goal, which must not mention x_0. Give every new witness a new name.

4.6.9 Exercises

4.6.9.1 A. Check your understanding

Exercise. A1

These parts are about the proof of Proposition 4.6.2.

(a) State ∃E as a schema, with every clause of its side condition.

(b) The proof opened the flagged box of y_0 first and the ∃E box of x_0 inside it. Could they be opened in the other order? If so, does every side condition still hold?

(c) Answer the *Before reading on* prompt of the proof: at line 5, where could x_0 still appear, and does it?

(d) In the two attempts at the converse, $\forall y\,\exists x\,R(x,y) \vdash \exists x\,\forall y\,R(x,y)$, which line is blocked in each, and by which clause of which rule?

(e) Give the idea of the proof in two sentences.

Solution. *(a)* From $\exists x\,\varphi$ on line h and a flagged box with name x_0, from the assumption $\varphi[x_0/x]$ on line i to χ on line j, a line of the box itself, write χ on line k, justified "∃E h, i–j". Side condition at line k: x_0 does not appear in χ, in $\exists x\,\varphi$, in any premise, or in any assumption of a box still open at line k.

(b) Yes, either order works. In the other order (the derivation under **Either nesting works**, right after the proof), the ∃E box encloses the flagged box. At the ∀I line, y_0 appears in no premise, not in the open assumption $\forall y\,R(x_0,y)$, and not in the conclusion; at the ∃E line, x_0 is not in $\forall y\,\exists x\,R(x,y)$, not in the premise, and no box is open.

(c) At line 5, the clauses list four places: χ, which is line 4, $\exists x\,R(x,y_0)$; the existential of line 1, $\exists x\,\forall y\,R(x,y)$; the premise, which is line 1 again; and the assumptions of boxes open at line 5. The only box open at line 5 is the flagged box of y_0, which has no assumption. The name x_0 appears in none of these, so line 5 is correct. It does appear in lines 2–4, inside the box, which the side condition allows.

(d) In the first attempt, line 5 (∀I over y_0) is blocked by the clause "y_0 appears in no

assumption of a box still open", since the ∃E box's assumption $R(x_0, y_0)$ contains y_0. In the second, line 4 (∃E) is blocked by the clause "x_0 does not appear in χ", since $\chi = R(x_0, y_0)$.

(e) Fix an arbitrary y_0 and obtain the witness x_0 of the premise, which satisfies $R(x_0, y)$ for every y, in particular for y_0. So x_0 is a witness for $\exists x\, R(x, y_0)$, and since y_0 was arbitrary, $\forall y\, \exists x\, R(x, y)$ follows.

4.6.9.2 B. Practice

Exercise. B1

Derive $\forall x\, \neg(P(x) \wedge Q(x)),\ \exists x\, P(x) \vdash \exists x\, \neg Q(x)$. Read the sequent over $\mathbb{Z}$ with P as "positive" and Q as "negative".

Solution. Idea. The premise $\exists x\, P(x)$ opens its box first, with a witness x_0, aimed at the goal. The goal $\exists x\, \neg Q(x)$ is postponed; the witness x_0 is the natural candidate, so the new goal inside is $\neg Q(x_0)$, a negation: a ¬I box on $Q(x_0)$. Inside, $P(x_0) \wedge Q(x_0)$ clashes with the first premise at x_0.

1	$\forall x\, \neg(P(x) \wedge Q(x))$	
2	$\exists x\, P(x)$	
3	x_0 | $P(x_0)$	
4	| | $Q(x_0)$	
5	| | $P(x_0) \wedge Q(x_0)$	∧I 3, 4
6	| | $\neg(P(x_0) \wedge Q(x_0))$	∀E 1
7	| | $\bot$	¬E 5, 6
8	| $\neg Q(x_0)$	¬I 4–7
9	| $\exists x\, \neg Q(x)$	∃I 8
10	$\exists x\, \neg Q(x)$	∃E 2, 3–9

At line 10, x_0 is not in line 9, not in line 2, not in either premise, and no box is open. Over $\mathbb{Z}$: no integer is both positive and negative, and some integer is positive, so some integer is not negative. The witness of the conclusion is the positive witness of the premise.

Exercise. B2: Fix, obtain, choose, faded

Here is a derivation of

$$\forall x\, \exists y\, R(x, y),\ \forall x\, \forall y\, (R(x, y) \to Q(y)) \ \vdash\ \forall x\, \exists y\, (R(x, y) \wedge Q(y))$$

with its last line missing.

1		$\forall x\, \exists y\, R(x, y)$	
2		$\forall x\, \forall y\, (R(x, y) \to Q(y))$	
3	x_0	$\exists y\, R(x_0, y)$	$\forall$E 1
4		y_0 | $R(x_0, y_0)$	
5			$\forall y\, (R(x_0, y) \to Q(y))$ $\forall$E 2
6			$R(x_0, y_0) \to Q(y_0)$ $\forall$E 5
7			$Q(y_0)$ $\to$E 6, 4
8			$R(x_0, y_0) \wedge Q(y_0)$ $\wedge$I 4, 7
9			$\exists y\, (R(x_0, y) \wedge Q(y))$ $\exists$I 8
10		$\exists y\, (R(x_0, y) \wedge Q(y))$	$\exists$E 3, 4–9

Name the move, then complete it. Then mark which lines are the "fix", which the "obtain", and which the "choose".

Solution. The missing line closes the flagged box of x_0 by $\forall$I; the move is the "fix" of fix, obtain, choose, completed: "since x_0 was arbitrary".

1		$\forall x\, \exists y\, R(x, y)$	
2		$\forall x\, \forall y\, (R(x, y) \to Q(y))$	
3	x_0	$\exists y\, R(x_0, y)$	$\forall$E 1
4		y_0 | $R(x_0, y_0)$	
5			$\forall y\, (R(x_0, y) \to Q(y))$ $\forall$E 2
6			$R(x_0, y_0) \to Q(y_0)$ $\forall$E 5
7			$Q(y_0)$ $\to$E 6, 4
8			$R(x_0, y_0) \wedge Q(y_0)$ $\wedge$I 4, 7
9			$\exists y\, (R(x_0, y) \wedge Q(y))$ $\exists$I 8
10		$\exists y\, (R(x_0, y) \wedge Q(y))$	$\exists$E 3, 4–9
11		$\forall x\, \exists y\, (R(x, y) \wedge Q(y))$	$\forall$I 3–10

Line 11 cites the flagged box 3–10; its last line, line 10, is a line of that box itself and is the instance $\psi[x_0/x]$ for $\psi = \exists y\, (R(x, y) \wedge Q(y))$. The name x_0 appears in neither premise and not in line 11, and at line 11 no box is open. (At line 10, y_0 is not in line 9, not in line 3, not in the premises, and the only open box, that of x_0, has no assumption.)

The moves: **fix** is the flagged box of x_0 (opened before line 3, closed at line 11); **obtain** is line 3 ($\forall$E at x_0) and the $\exists$E box of y_0 (lines 4–10), with lines 5–7 using the second premise on the witness; **choose** is line 9, $\exists$I with the witness y_0.

Exercise. B3

An integer n is **odd** when $n = 2k + 1$ for some integer k. Over $\mathbb{Z}$, derive

$$\exists k\,(m = 2 \cdot k + 1),\ \exists k\,(n = 2 \cdot k + 1) \vdash \exists k\,(m \cdot n = 2 \cdot k + 1),$$

with one line justified "arith". Then write the proof as three sentences of prose.

Solution. Idea. As in Example (The sum of two even integers is even): open both boxes early, with different names, and postpone the goal until the arithmetic shows the witness.

1	$\exists k\,(m = 2 \cdot k + 1)$	
2	$\exists k\,(n = 2 \cdot k + 1)$	
3	k_0 | $m = 2 \cdot k_0 + 1$	
4	l_0 | $n = 2 \cdot l_0 + 1$	
5	$m \cdot n = 2 \cdot (2 \cdot k_0 \cdot l_0 + k_0 + l_0) + 1$	arith 3, 4
6	$\exists k\,(m \cdot n = 2 \cdot k + 1)$	$\exists$I 5
7	$\exists k\,(m \cdot n = 2 \cdot k + 1)$	$\exists$E 2, 4–6
8	$\exists k\,(m \cdot n = 2 \cdot k + 1)$	$\exists$E 1, 3–7

Line 5: $(2k_0 + 1)(2l_0 + 1) = 4k_0l_0 + 2k_0 + 2l_0 + 1 = 2(2k_0l_0 + k_0 + l_0) + 1$. Line 6 is $\exists$I with $\varphi = (m \cdot n = 2 \cdot k + 1)$ and the witness $2 \cdot k_0 \cdot l_0 + k_0 + l_0$. At line 7, l_0 is in none of line 6, line 2, the premises, and the open assumption on line 3; at line 8, k_0 is in none of line 7, line 1, the premises, and no box is open.

In prose: *Choose integers k and l with $m = 2k + 1$ and $n = 2l + 1$. Then $mn = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1$. Since $2kl + k + l$ is an integer, mn is odd.*

4.6.9.3 C. Going deeper

Exercise. C1

Derive

$$\exists x\,\forall y\,R(x, y),\ \forall x\,\forall y\,(R(x, y) \to R(y, x)) \vdash \forall y\,\exists x\,R(y, x),$$

and check the side condition of every $\exists$E and $\forall$I line. Then give a reading over $\mathbb{Z}$ in which both premises are true, and say which integer the derivation's witness is.

Solution (Hint 1). The shape is the one of Proposition 4.6.2: fix, obtain, choose. Which line does symmetry have to turn round?

Solution (Hint 2). Open a flagged box for y_0, and inside it the $\exists$E box of the first premise, with the assumption $\forall y\,R(x_0, y)$.

Solution. Idea. The goal is $\forall y$: fix y_0. The new goal $\exists x\, R(y_0, x)$ is postponed. The first premise is an existential: obtain x_0 with $\forall y\, R(x_0, y)$, and instantiate it at the term the goal mentions, y_0: $R(x_0, y_0)$. The goal needs $R(y_0, t)$, with y_0 first, so the second premise turns the line round, instantiated at x_0 and y_0 in that order (read off the line to be turned). Then choose the witness x_0.

1			$\exists x\, \forall y\, R(x, y)$	
2			$\forall x\, \forall y\, (R(x, y) \to R(y, x))$	
3	y_0	x_0	$\forall y\, R(x_0, y)$	
4			$R(x_0, y_0)$	$\forall$E 3
5			$\forall y\, (R(x_0, y) \to R(y, x_0))$	$\forall$E 2
6			$R(x_0, y_0) \to R(y_0, x_0)$	$\forall$E 5
7			$R(y_0, x_0)$	$\to$E 6, 4
8			$\exists x\, R(y_0, x)$	$\exists$I 7
9			$\exists x\, R(y_0, x)$	$\exists$E 1, 3–8
10			$\forall y\, \exists x\, R(y, x)$	$\forall$I 3–9

Line 9 ($\exists$E): x_0 does not appear in $\chi = \exists x\, R(y_0, x)$, nor in the existential of line 1, nor in either premise; the only box open at line 9 is the flagged box of y_0, which has no assumption. Line 8 is a line of the $\exists$E box itself.

Line 10 ($\forall$I): the flagged box immediately contains the $\exists$E box, so its range starts at line 3. Its last line, line 9, is a line of the box itself and is $(\exists x\, R(y, x))[y_0/y]$. The name y_0 appears in neither premise and not in line 10, and no box is open.

A reading. Over $\mathbb{Z}$, let $R(x, y)$ mean "$x \cdot y = 0$". The second premise holds since $x \cdot y = y \cdot x$. The first holds with the witness 0, and only with it: for $x \neq 0$, the instance $y = 1$ gives $x \cdot 1 = x \neq 0$. The conclusion says that every integer has a partner with product 0, which is true. The derivation's x_0 names the witness of the first premise, so under this reading it can only be 0, and the derivation gives every y the partner 0; the derivation itself never uses which integer it is.

Marking scheme. (10 points) - 2: opens a flagged box for the goal's $\forall y$ with a fresh name, and postpones the existential. - 2: opens the $\exists$E box of the first premise with a **fresh** witness, aimed at the current goal. - 2: instantiates the assumption at y_0 and the symmetry premise at x_0, y_0 in that order, and fires it. - 2: $\exists$I with the witness x_0, inside the $\exists$E box. - 1: checks the side conditions of the $\exists$E and $\forall$I lines clause by clause. - 1: a reading over $\mathbb{Z}$ in which both premises hold, with the witness identified (or, if the reading does not determine it, correctly described as whichever witness the first premise has). Fatal errors (0 for the step): an $\exists$E box closed on a line containing its witness; $\forall$I over a name in an open assumption; the symmetry premise instantiated in the wrong order and then used as if it fired.

Exercise. C2: Find the error

Here is a "derivation" of $\exists x\, (P(x) \wedge Q(x)) \vdash \exists x\, P(x)$.

1	$\exists x\,(P(x) \wedge Q(x))$	
2	x_0 | $P(x_0) \wedge Q(x_0)$	
3	$P(x_0)$	$\wedge$E 2
4	$P(x_0)$	$\exists$E 1, 2–3
5	$\exists x\,P(x)$	$\exists$I 4

(a) Is the conclusion derivable from the premise? Justify your answer.

(b) Does this derivation prove it? If not, name the exact line that fails, the rule, and the clause of its side condition that is broken.

(c) Show that the illegal step itself is unsound: use the same step to "derive", from a premise true over $\mathbb{Z}$, a sentence false over $\mathbb{Z}$.

Solution (Hint 1). For (a), the conclusion is half of the conclusion of a result proved in this section.

Solution (Hint 2). For (c), drop Q: start from the premise $\exists x\,P(x)$ and take the same illegal step as line 4.

Solution. *(a)* Yes, the conclusion is derivable: by Proposition 4.6.1 the premise gives $\exists x\,P(x) \wedge \exists x\,Q(x)$, and $\wedge$E gives $\exists x\,P(x)$. Directly, moving the $\exists$I inside the box repairs the derivation:

1	$\exists x\,(P(x) \wedge Q(x))$	
2	x_0 | $P(x_0) \wedge Q(x_0)$	
3	$P(x_0)$	$\wedge$E 2
4	$\exists x\,P(x)$	$\exists$I 3
5	$\exists x\,P(x)$	$\exists$E 1, 2–4

At line 5, x_0 is not in line 4, not in line 1, and no box is open.

(b) No. Line 4 fails: it is $\exists$E with $\chi = P(x_0)$, and x_0 appears in χ, which breaks the clause "x_0 does not appear in χ" of Definition 4.6.1. The witness has escaped its box. Lines 1–3 are correct, and line 5 would be a correct $\exists$I if line 4 were a legitimate line. A true conclusion does not make the argument correct: the conclusion is derivable, but not by this derivation.

(c) The same step, from $\exists x\,P(x)$ alone:

1	$\exists x\, P(x)$	
2	$x_0 \mid P(x_0)$	
3	$P(x_0)$	$\exists$E 1, 2–2
4	$x_0 \mid P(x_0)$	R 3
5	$\forall x\, P(x)$	$\forall$I 4–4

Line 3 is the illegal step of line 4 above. Line 4 repeats line 3 into a flagged box, and line 5 is a correct $\forall$I: x_0 appears in no premise, in no open assumption, and not in line 5. Over $\mathbb{Z}$ with P as "even", the premise "some integer is even" is true and the conclusion "every integer is even" is false. So the step from $\exists x\, P(x)$ to $P(x_0)$ can lead from truth to falsehood; it is a counterexample to the **step**, while the conclusion of the original sequent has no counterexample at all. (This is the same failure as the derivation after Definition 4.6.1 that "proves" every natural number exceeds 5.)

Marking scheme. (10 points) - 2: (a) says the conclusion is derivable and gives a correct derivation or cites Proposition 4.6.1 with $\wedge$E. - 3: (b) names line 4, the rule $\exists$E, and the clause "x_0 not in χ". - 1: (b) says that the other lines are correct, so the error is in that one line. - 4: (c) gives a derivation that uses the same illegal step and is otherwise legal, with a reading in which the premise is true and the conclusion false. Fatal errors (0 for the step): answering (a) "no" because the derivation is wrong; giving a counterexample to the conclusion of the original sequent (there is none); blaming line 5.

4.7 Equality and Unique Existence

Sections 4.3 to 4.6 gave the four quantifier rules, and with them every logical symbol of the language of Section 4.2 (the connectives, $\bot$, the quantifiers, and equality) has rules, except one. So far the equals sign has been treated like a predicate symbol: $a = b$ was an atomic formula like $P(a)$, and no rule looked inside it. Yet almost every calculation in mathematics is a chain of equations, and every use of "the" (the limit, the supremum, the inverse) quietly asserts that something is unique. This section adds the two rules for equality, =I and =E, which say together that *equals may be replaced by equals*. Its first application is the phrase "there is exactly one", which is an existential statement together with an equation. The section ends by completing the table of rules, phrases, and Lean tactics that Chapter 1 began.

4.7.1 What the rules so far cannot do with an equation

Here is the first line of a calculation any reader has written:

> Since $a = b$ and $a + a = 2a$, we have $b + b = 2b$.

Nothing could be simpler, and nothing in Sections 4.3–4.6 derives it. The line $a = b$ is an atomic formula, and so far the rules have treated $=$ exactly as they treat a two-place predicate symbol written between its terms, like the R in front of $R(a, b)$. From the premises $R(a, b)$ and $P(a)$ no rule gives $P(b)$: a relation between a and b says nothing, by itself, about the properties of either. The quantifier rules do not help either, since neither premise begins with a quantifier. As far as the rules of Chapters 1–2 and Sections 4.3–4.6 are concerned, $a = b$ could mean "a is less than b" or "a and b are coprime", and under those readings the general pattern behind the

step, from $R(a,b)$ and $P(a)$ to $P(b)$, fails: reading $R(a,b)$ as $a < b$ and $P(x)$ as $x < 1$ over $\mathbb{Z}$, with $a = 0$ and $b = 5$, both premises are true and $P(b)$ is false.

What makes the step correct is what $=$ means: a and b are names of **the same object**. Any statement true of the object is true of it whichever name is used. Two things follow, and they are the two rules of this section.

1. Every term names the same object as itself. Nothing needs to be assumed to write $t = t$.
2. If t_1 and t_2 name the same object, then a statement about t_1 may be rewritten as the same statement about t_2.

The first rule is unusual: it cites no earlier line at all. Chapter 1 anticipated this when it defined rules (Section 1.2): the definition of a rule schema (Definition 1.2.1) was worded so that a rule of Chapter 4 could have no given lines at all. This is that rule.

4.7.2 The rules for equality

Everything is equal to itself, and equals may be replaced by equals.

Definition 4.7.1. Rules for Equality

$$\begin{array}{lll|l|ll} & & & i & t_1 = t_2 & \\ k & t = t & =\text{I} & j & \varphi[t_1/x] & \\ & & & k & \varphi[t_2/x] & =\text{E } i, j \end{array}$$

The rule $=$I (**equality introduction**, or reflexivity) writes $t = t$ for **any** term t, at any line, citing nothing. The rule $=$E (**equality elimination**, or substitution of equals) cites an equation $t_1 = t_2$ on line i and a line j of the form $\varphi[t_1/x]$, both available at line k, and writes $\varphi[t_2/x]$ on line k. Here φ is **any** formula and x any variable, subject to one side condition: **both t_1 and t_2 are free for x in φ** (Definition 4.2.6).

In words: $=$I says that every term equals itself. $=$E says that when $t_1 = t_2$ is known, a line that says something about t_1 may be rewritten as the same thing said about t_2. The formula φ is the "something": a statement with a hole x, which line j fills with t_1 and line k fills with t_2.

Read the definition clause by clause.

1. **$=$I cites nothing.** Its justification is "$=$I" alone. It is the first primitive rule of the book with no given lines (derived rules from results with no premises, such as LEM of Section 2.5, already cite nothing), and so it may stand anywhere: as the first line of a derivation with no premises, or deep inside a box.
2. **The equation is read from left to right.** The cited equation $t_1 = t_2$ licenses replacing t_1 by t_2, and not the other way round. From $a = b$ and $P(b)$, the rule does **not** give $P(a)$, because $P(b)$ is $\varphi[b/x]$ for $\varphi = P(x)$, and the equation would have to be $b = a$. The reverse direction needs one more step, symmetry, which we derive below.
3. **Some or all occurrences.** The choice of φ decides which occurrences of t_1 are replaced: exactly those that φ marks with x. Occurrences of t_1 that φ writes out are left alone. Choosing φ is the whole art of using the rule.
4. **Free for x.** Neither term may have a variable captured by a quantifier of φ, the condition of Definition 4.2.6 that the rules of Sections 4.3 and 4.5 also carry. The nonexample below shows what goes wrong without it.

One line, many readings. Take the equation $a = b$ and the line $a + a = 2 \cdot a$. The term a occurs three times in the line, and φ may mark any of them. Some of the choices:

φ	$\varphi[a/x]$ (the cited line)	$\varphi[b/x]$ (the new line)
$x + a = 2 \cdot a$	$a + a = 2 \cdot a$	$b + a = 2 \cdot a$
$a + x = 2 \cdot a$	$a + a = 2 \cdot a$	$a + b = 2 \cdot a$
$a + a = 2 \cdot x$	$a + a = 2 \cdot a$	$a + a = 2 \cdot b$
$x + x = 2 \cdot x$	$a + a = 2 \cdot a$	$b + b = 2 \cdot b$
$a + a = 2 \cdot a$	$a + a = 2 \cdot a$	$a + a = 2 \cdot a$

Each of the three occurrences is marked or not, so there are $2^3 = 8$ choices in all, and each is a legal step. The fourth row is the step of the opening calculation. The last row is the **degenerate** choice: φ contains no x, so $\varphi[a/x]$ and $\varphi[b/x]$ are both φ, and the rule copies the line unchanged. It is legal and useless, and it shows that "some occurrences" includes "none". Another degenerate case: the equation may be $t = t$ itself, and then =E again returns the cited line.

Quick check. From the equation $x_0 = 3$ and the line $x_0 \cdot x_0 > x_0$, which formula φ makes =E give $3 \cdot 3 > x_0$? Which gives $x_0 \cdot 3 > 3$?

Solution. For $3 \cdot 3 > x_0$, take $\varphi = (z \cdot z > x_0)$, with a variable z that appears nowhere else: $\varphi[x_0/z]$ is the cited line and $\varphi[3/z]$ is $3 \cdot 3 > x_0$. For $x_0 \cdot 3 > 3$, take $\varphi = (x_0 \cdot z > z)$. The hole has to be a variable other than x_0, since the occurrences of x_0 that stay must be written out in φ.

A nonexample: capture. Take the equation $a = y$, with y a variable, and the line $\exists y\,(y < a)$. It is tempting to "substitute y for a" and write $\exists y\,(y < y)$. With $\varphi = \exists y\,(y < x)$, the cited line is $\varphi[a/x]$, and the new line would be $\varphi[y/x]$. But y is **not** free for x in φ: the x sits inside $\exists y$, and the y put there would be captured. The side condition fails, and rightly. Read over $\mathbb{Z}$, with a and y both naming 0, the premises say "$a = y$" (true) and "some integer is less than 0" (true), while $\exists y\,(y < y)$ says that some integer is less than itself (false).

1	$a = y$	
2	$\exists y\,(y < a)$	
3	$\exists y\,(y < y)$	=E 1, 2

Line 3 breaks the side condition of Definition 4.7.1. What the argument meant to say needs a different name for the bound variable: $\exists z\,(z < a)$, which is the same sentence up to renaming (Section 4.2), becomes $\exists z\,(z < y)$, which is true when y names 0.

Why these two rules. They are exactly the two facts about "same object" listed above, and nothing else about equality needs to be assumed. Symmetry ($s = t$ gives $t = s$), transitivity ($s = t$ and $t = u$ give $s = u$), and "applying the same function to both sides" are all derived below. So the two rules play for $=$ the part that the introduction and elimination rules played for each connective: one builds an equation, one uses it. The name of =E is less natural than the others, since the equation is used to rewrite some **other** line; but "elimination" fits, since

the rule takes an equation apart into its use.

Warning. **A bound variable is not a place to substitute.** From $x = 0$ and $\forall x\,(x = x)$, the line $\forall x\,(0 = x)$ does **not** follow by =E, although it looks like "replace the first x by 0". The x's of $\forall x\,(x = x)$ are bound, and substitution replaces only free occurrences (Definition 4.2.5). To write $\forall x\,(x = x)$ as $\varphi[x/z]$, either z does not occur free in φ, and then $\varphi[0/z]$ is the unchanged line, or z occurs free in φ (for instance, φ is $\forall x\,(z = x)$), and then every free z lies inside $\forall x$, so x is not free for z in φ and the side condition fails. Over $\mathbb{Z}$, with x naming 0, the premises are true and $\forall x\,(0 = x)$ is false.

4.7.3 Symmetry, transitivity and congruence

Everything we expect of equality now follows from the two rules. Each derivation below is short, and the only thought in each is the choice of φ.

It turns out that the equation in =E may be read in either direction after all, at the cost of two extra lines (an =I and an =E; after the abbreviation (sym) below, just one), and that equations chain:

Proposition 4.7.1. Equality Is Symmetric and Transitive

For all terms s, t, u:

(a) $s = t \vdash t = s$;

(b) $s = t,\ t = u \vdash s = u$.

Idea. For (a), the goal $t = s$ must be the last line of an =E that uses the premise $s = t$ (it is the only equation available), so it has the form $\varphi[t/x]$, and the cited line $\varphi[s/x]$ must already be on a line. Try $\varphi = (x = s)$: then $\varphi[t/x]$ is $t = s$, the goal, and $\varphi[s/x]$ is $s = s$, which =I supplies for free. For (b), rewrite the t of the first premise using the second: $\varphi = (s = x)$.

Proof.

(a) With $\varphi := (x = s)$, for a variable x that occurs in neither s nor t, the line $s = s$ is $\varphi[s/x]$ and the goal $t = s$ is $\varphi[t/x]$. Both terms are free for x in φ, since φ has no quantifier. So the following is a derivation:

1	$s = t$	
2	$s = s$	=I
3	$t = s$	=E 1, 2

(b) With $\varphi := (s = x)$, for a variable x that does not occur in s, the first premise $s = t$ is $\varphi[t/x]$ and the goal $s = u$ is $\varphi[u/x]$; again φ has no quantifier. So the following is a derivation:

1	$s = t$	
2	$t = u$	
3	$s = u$	=E 2, 1

Line 3 of (b) cites the equation $t = u$ on line 2 as the rule's line i, and $s = t$ on line 1 as its line j. This proves (a) and (b).

■

Notice where the "reflexivity" line went in (a): the equation $s = t$ was not rewritten at all. It was the **tool**, and the line it rewrote, $s = s$, was manufactured for the purpose. This is the typical use of =I: it supplies a line for =E to work on.

From now on, symmetry and transitivity may be used as one line, with the justifications "(sym)" and "(trans)": each abbreviates the three-line derivation above with the given terms in place of s, t, u. This is the license of Definition 2.3.1, extended from statement letters to terms. The extension is harmless for the same reason: the derivations above used nothing about s, t, u except that they are terms, so they can be pasted in with any terms.

The next derivation is the rule behind "apply f to both sides". Let f be a one-place function symbol, and a, b constants.

1	$a = b$	
2	$f(a) = f(a)$	=I
3	$f(a) = f(b)$	=E 1, 2

Here $\varphi = (f(a) = f(x))$: it keeps the first a and marks the second. With $\varphi = (f(x) = f(x))$ instead, the same two lines would give $f(b) = f(b)$, which is true and useless. Nothing is special about f: the same three lines give $a^2 = b^2$ from $a = b$, or $a + c = b + c$, or $|a| = |b|$. Every "do the same thing to both sides of an equation" is this derivation.

The last derivation of this group has no premises at all. It says that equals share every property P.

Claim. $\vdash \forall x\, \forall y\, (x = y \to (P(x) \to P(y)))$.

Finding it. The goal is $\forall x$, so the search method of Section 2.6, with the step of Section 4.4, opens a flagged box for x_0; the new goal $\forall y\,(\dots)$ opens a second flagged box, for y_0. The goal $x_0 = y_0 \to (P(x_0) \to P(y_0))$ is an implication, so a box opens with $x_0 = y_0$, and the new goal $P(x_0) \to P(y_0)$ opens another, with $P(x_0)$. Now the goal is $P(y_0)$, the resources are $x_0 = y_0$ and $P(x_0)$, and one =E with $\varphi = P(x)$ finishes. Every line but one was written by a goal-directed step.

1	x_0	y_0		$x_0 = y_0$	
2				$P(x_0)$	
3				$P(y_0)$	=E 1, 2
4				$P(x_0) \to P(y_0)$	→I 2–3
5			$x_0 = y_0 \to (P(x_0) \to P(y_0))$		→I 1–4
6		$\forall y\,(x_0 = y \to (P(x_0) \to P(y)))$			∀I 1–5
7	$\forall x\,\forall y\,(x = y \to (P(x) \to P(y)))$				∀I 1–6

The two ∀I lines cite the flagged boxes as 1–5 and 1–6: each flagged box immediately contains another box, so its range starts at that box's first line (the convention of Section 4.4). At line 6 the name y_0 appears in no premise (there are none), in no open assumption (the boxes on lines 1 and 2 have closed), and not in the conclusion; at line 7 the same holds for x_0. The derivation has 7 lines.

The claim is one of the classical "axioms of equality", often taken as a starting point in place of =E. Here it is a consequence. Conversely, with the claim as a premise, ∀E twice and →E twice would do the work of =E for the formula $P(x)$; the rule does it for every formula at once.

4.7.4 Calculations are chains of substitutions

A calculation in a proof looks like this:

$$a + b = 2 \cdot b + b = 2 \cdot 3 + 3 = 9,$$

written under the premises $a = 2 \cdot b$ and $b = 3$. The chain is a single statement to the eye, but logically it is several equations, each obtained from the last. Here is the derivation it abbreviates, with one line of arithmetic, justified "arith" as in Section 4.3.

A chain as a derivation. From $b = 3$ and $a = 2 \cdot b$, derive $a + b = 9$.

1	$b = 3$	
2	$a = 2 \cdot b$	
3	$a + b = a + b$	=I
4	$a + b = 2 \cdot b + b$	=E 2, 3
5	$a + b = 2 \cdot 3 + 3$	=E 1, 4
6	$2 \cdot 3 + 3 = 9$	arith
7	$a + b = 9$	=E 6, 5

The choices of φ, with z a variable not occurring in the terms:

- line 4: $\varphi = (a + b = z + b)$, rewriting the a on the **right** of line 3 and keeping the one on the left;
- line 5: $\varphi = (a + b = 2 \cdot z + z)$, rewriting **both** remaining b's on the right at once;
- line 7: $\varphi = (a + b = z)$, rewriting the whole right-hand side by the equation of line 6.

The pattern is general. A chain $t_1 = t_2 = \cdots = t_n$ starts with $t_1 = t_1$ by =I, keeps t_1 fixed on

the left, and rewrites the right-hand side once per link, each time by =E with the equation that justifies the link. Each link is either a premise, as on lines 4 and 5, or an arithmetic fact, as on line 7. The last line is $t_1 = t_n$. Transitivity (Proposition 4.7.1 (b)) is the same move made once; a chain makes it $n - 1$ times.

A **mixed chain**, such as $a^2 + 1 = 10 \le 10 + b^2$ with $a = 3$, is not made of =E alone. Its = links are =E steps as above, and a $\le$ link needs a fact about $\le$: here that $b^2 \ge 0$, so that adding b^2 does not decrease a number. The = link before it then carries over by one more =E, with $\varphi = (z \le 10 + b^2)$ and the equation $a^2 + 1 = 10$ turned round by symmetry (Proposition 4.7.1 (a)); transitivity of $\le$ is needed only where two $\le$ links meet, as in $x \le y \le z$. Those are facts about order, not about equality. *Analysis on the Real Line*, Chapter 1 derives them from the axioms of an ordered field; until then they enter derivations as "arith" lines, one per link.

Quick check. Line 5 of the chain rewrites both b's on the right of line 4 at once. Which φ would rewrite only the b inside $2 \cdot b$, and what line would it give? How would the chain then reach line 5?

Solution. Take $\varphi = (a + b = 2 \cdot z + b)$. Then line 4 is $\varphi[b/z]$, and =E 1, 4 gives $\varphi[3/z]$, the line $a + b = 2 \cdot 3 + b$. One more =E with the same equation $b = 3$ and $\varphi = (a + b = 2 \cdot 3 + z)$ then gives $a + b = 2 \cdot 3 + 3$, which is line 5. Marking both occurrences at once saves that line.

4.7.5 What equality sounds like in a proof

In a written proof, =E is the word "substituting", or no word at all: the chain $a+b = 2b+b = 9$ is read as a sequence of substitutions, and the reader supplies the φ of each link. "By symmetry" or "rewriting the equation as $t = s$" is Proposition 4.7.1 (a). "Applying f to both sides", "squaring both sides", "adding c to both sides" are the congruence derivation above. And =I is almost never written, because $t = t$ goes without saying; it is visible only in the first line of a formal chain, and in a proof assistant, where `rfl` proves $t = t$.

Here is a proof from school algebra, read link by link.

> **Claim.** If $a + b = 0$ and $a + c = 0$, then $b = c$.
>
> *Proof.* $b = b + 0 = b + (a + c) = (b + a) + c = (a + b) + c = 0 + c = c$.

The first link is the fact $y + 0 = y$ for every y, applied at $y = b$ (an instance by $\forall$E) and read from right to left: the chain needs $b = b + 0$, which is the instance $b + 0 = b$ turned round by symmetry. The second link rewrites the 0 by the premise $a + c = 0$, again turned round, since the chain needs 0 to become $a + c$. The third and fourth are instances of associativity and commutativity. The fifth rewrites $a + b$ by the premise $a + b = 0$ in the direction written. The last is the instance $0 + c = c$. So the claim says that a number has **at most one** additive inverse; the facts it uses are the field axioms, which *Analysis on the Real Line*, Chapter 1 states as axioms and which here are school algebra.

4.7.6 Unique existence

Mathematics says "**the** limit of the sequence", "**the** supremum of A", "**the** additive inverse of a". Each "the" makes two claims. There is such an object, and there is only one, so that the name picks it out. The second claim is the one that is easy to forget, and the one the claim

above was about.

"There is exactly one x with φ": there is one, and any other is it.

Definition 4.7.2. Unique Existence

Let φ be a formula, x a variable, and y a variable **other than x that does not appear in φ** (Section 4.2). Then

$$\exists!x\,\varphi \quad \text{abbreviates} \quad \exists x\,(\varphi \wedge \forall y\,(\varphi[y/x] \to y = x)).$$

We read $\exists!x\,\varphi$ as "there is **exactly one** x such that φ", or "there is a **unique** x with φ".

In words: there is an x that satisfies φ, and every y that satisfies φ (that is, satisfies φ with y in place of x) is equal to that x. The first conjunct is existence. The second is uniqueness, and it is a statement about the x already found: any other candidate coincides with it. Like the bounded quantifiers of Definition 4.1.1, $\exists!$ is not a new symbol of the language but shorthand, and every rule applies to the formula it abbreviates.

Well-definedness. The abbreviation seems to depend on the choice of y. It does not, up to renaming a bound variable (Section 4.2). The argument rests on one fact about substitution, used again below: *if y does not appear in φ, then replacing x by y and then y by a term t is the same as replacing x by t*, that is, $(\varphi[y/x])[t/y]$ is $\varphi[t/x]$. This is a structural induction on φ of the kind in Section 4.2. In a term, the y's of $s[y/x]$ are exactly the former x's, so $(s[y/x])[t/y] = s[t/x]$; atomic formulas, $\bot$ and the connectives follow clause by clause. For $\forall x\,\psi$, both sides are $\forall x\,\psi$, since substitution for x stops at $\forall x$ and y does not appear in ψ (the second fact of Section 4.2). For $\forall z\,\psi$ with z not x, also z is not y, because y does not appear in φ; so both substitutions pass under the quantifier and the induction hypothesis applies, and $\exists$ is the same. Now let y and y' be two admissible choices. If they are the same variable there is nothing to show. Otherwise y' does not appear in $\varphi[y/x] \to y = x$, since it is neither x nor y and does not appear in φ; so the bound y of $\forall y\,(\varphi[y/x] \to y = x)$ may be renamed y', which gives $\forall y'\,((\varphi[y/x])[y'/y] \to y' = x)$. By the fact with $t = y'$, this is $\forall y'\,(\varphi[y'/x] \to y' = x)$, the formula written with y'.

Why y must not appear in φ. The condition keeps the new quantifier $\forall y$ from capturing anything of φ's own. Take $\varphi = (x \le y)$, in which y is free, and read over $\mathbb{N}$ with y naming 0. "There is exactly one natural number x with $x \le 0$" is true: $x = 0$. The formula written with the forbidden bound variable y is

$$\exists x\,(x \le y \wedge \forall y\,(y \le y \to y = x)),$$

because $(x \le y)[y/x]$ is $y \le y$. Its second conjunct says that **every** natural number equals x, which is false. The free y of φ, which was to name 0, has been captured by $\forall y$. With an admissible variable, say z, the formula is $\exists x\,(x \le y \wedge \forall z\,(z \le y \to z = x))$, which is true.

Examples. Simplest first; each is read over the domain named, with the informal truth of Section 4.1.

1. Over $\mathbb{R}$, $\exists!x\,(x + 2 = 5)$ is true. *Existence:* $3 + 2 = 5$. *Uniqueness:* if $y + 2 = 5$, then subtracting 2 gives $y = 3$, so y is the x already found.
2. Over $\mathbb{N}$, $\exists!x\,(x \cdot x = 4)$ is true. *Existence:* $2 \cdot 2 = 4$. *Uniqueness:* for a natural number y, if $y \le 1$ then $y \cdot y \le 1$, and if $y \ge 3$ then $y \cdot y \ge 9$; so $y \cdot y = 4$ forces $y = 2$.
3. Over $\mathbb{R}$, $\exists!x\,\forall y\,(x + y = y)$ is true: 0 is the **unique** additive identity. *Existence:* $0 + y = y$ for every y. *Uniqueness:* suppose $z + y = y$ for every y. Apply this to the one instance

$y = 0$ (by $\forall$E): $z + 0 = 0$. Since also $z + 0 = z$, symmetry and transitivity give $z = 0$. The whole uniqueness proof is the choice of the instance $y = 0$.

4. **Degenerate.** For any constant c, $\exists!x\,(x = c)$ holds in every domain, and it is derivable from no premises (the derivation follows). By contrast, $\exists!x\,(x = x)$ says that the domain has exactly one element, which is false in $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}$: existence holds and uniqueness fails.

Nonexample by minimal change. Keep example 2 and change the domain to $\mathbb{Z}$. Existence still holds, with the same witness 2. Uniqueness fails: $y = -2$ also satisfies $y \cdot y = 4$, and $-2 \neq 2$. So over $\mathbb{Z}$, $\exists!x\,(x \cdot x = 4)$ is false, and the failing clause is exactly the second conjunct.

The derivation of $\exists!x\,(x = c)$, with the abbreviation written out, shows that $\exists!$ needs no rules of its own:

1	$c = c$	=I
2	y_0 | $y_0 = c$	
3	$y_0 = c \to y_0 = c$	$\to$I 2–2
4	$\forall y\,(y = c \to y = c)$	$\forall$I 2–3
5	$c = c \wedge \forall y\,(y = c \to y = c)$	$\wedge$I 1, 4
6	$\exists x\,(x = c \wedge \forall y\,(y = c \to y = x))$	$\exists$I 5

Line 3 closes a box whose only line is its assumption: the box shows $y_0 = c \to y_0 = c$, as $P \to P$ is shown in Chapter 1. Line 6 is $\exists$I with the witness c, and it reads line 5 as $\psi[c/x]$ for $\psi = (x = c \wedge \forall y\,(y = c \to y = x))$, which marks with x the first c and the last one, and keeps the other two. The choice of which occurrences to mark is the one discussed with Definition 4.5.1.

Warning. **Uniqueness is not existence.** A uniqueness argument ("suppose y also satisfies φ; then $y = x$") proves nothing about whether anything satisfies φ. Over $\mathbb{R}$, every two solutions of $x^2 = -1$ are equal, vacuously, because there are none; yet no real number is "the" square root of -1. Conversely, an existence proof does not give uniqueness: $x^2 = 1$ has the solution 1, and "the solution" would be wrong, since -1 is another. Before writing "the", check both halves.

4.7.7 How uniqueness is used and proved

A proof of "exactly one" rarely follows the shape of Definition 4.7.2, which ties the uniqueness clause to the witness just found. It proves two separate statements: there is at least one, and there is at most one. The second is "any two objects with the property are equal", and it makes no mention of a witness. The next result says that the two shapes say the same thing.

Proposition 4.7.2. Two Halves of Unique Existence

Let φ be a formula, x a variable, and y a variable other than x that does not appear in φ. Then

$$\exists!x\,\varphi \;\dashv\vdash\; \exists x\,\varphi \wedge \forall x\,\forall y\,(\varphi \wedge \varphi[y/x] \to x = y).$$

The right-hand side is "existence and at most one". Its second conjunct is what a mathematician proves under the heading "uniqueness": take two objects x and y with the property, and show $x = y$.

Idea. Both directions are searches in the style of Section 2.6 with the quantifier steps. For $\vdash$, the premise is an $\exists$, so open its $\exists$E box first, with a witness x_0: inside it, $\varphi[x_0/x]$ gives existence at once by $\exists$I, and "at most one" is a $\forall\forall$ goal, which opens two flagged boxes u_0, v_0. Two objects u_0, v_0 with the property are each equal to the witness x_0, by the uniqueness clause, and so equal to each other: the witness is the "third point" that links them. For $\dashv$, open the $\exists$E box on $\exists x\,\varphi$ with a witness x_0; the goal inside is $\exists! x\,\varphi$, whose witness will be x_0 again, and the uniqueness clause for x_0 comes from "at most one" applied to the pair x_0, y_0.

Proof. Choose four distinct variables x_0, u_0, v_0, y_0 that do not appear in φ, and none of which is x or y. Three facts about substitution are used throughout. First, since y does not appear in φ, $(\varphi[y/x])[t/y]$ is $\varphi[t/x]$ for every term t, the fact proved for the well-definedness of Definition 4.7.2. Second, $\varphi[y/x]$ has no free x (every free x became y), so it is unchanged by a substitution for x. Third, for a term t that does not contain y, $\varphi[t/x]$ contains no y, so it is unchanged by a substitution for y.

($\vdash$) The following is a derivation with premise $\exists! x\,\varphi$, written out.

1	$\exists x\,(\varphi \wedge \forall y\,(\varphi[y/x] \to y = x))$					
2	x_0	$\varphi[x_0/x] \wedge \forall y\,(\varphi[y/x] \to y = x_0)$				
3		$\varphi[x_0/x]$				$\wedge$E 2
4		$\forall y\,(\varphi[y/x] \to y = x_0)$				$\wedge$E 2
5		$\exists x\,\varphi$				$\exists$I 3
6		u_0	v_0	$\varphi[u_0/x] \wedge \varphi[v_0/x]$		
7				$\varphi[u_0/x]$		$\wedge$E 6
8				$\varphi[v_0/x]$		$\wedge$E 6
9				$\varphi[u_0/x] \to u_0 = x_0$		$\forall$E 4
10				$u_0 = x_0$		$\to$E 9, 7
11				$\varphi[v_0/x] \to v_0 = x_0$		$\forall$E 4
12				$v_0 = x_0$		$\to$E 11, 8
13				$v_0 = v_0$		=I
14				$x_0 = v_0$		=E 12, 13
15				$u_0 = v_0$		=E 14, 10
16			$\varphi[u_0/x] \wedge \varphi[v_0/x] \to u_0 = v_0$			$\to$I 6–15
17		$\forall y\,(\varphi[u_0/x] \wedge \varphi[y/x] \to u_0 = y)$				$\forall$I 6–16
18		$\forall x\,\forall y\,(\varphi \wedge \varphi[y/x] \to x = y)$				$\forall$I 6–17
19		$\exists x\,\varphi \wedge \forall x\,\forall y\,(\varphi \wedge \varphi[y/x] \to x = y)$				$\wedge$I 5, 18
20	$\exists x\,\varphi \wedge \forall x\,\forall y\,(\varphi \wedge \varphi[y/x] \to x = y)$					$\exists$E 1, 2–19

Lines 9 and 11 are ∀E on line 4 at u_0 and at v_0, by the first substitution fact; each term is a variable not bound in $\varphi[y/x] \to y = x_0$, so it is free for y there. Lines 13 and 14 are the symmetry of Proposition 4.7.1 (a), written out; line 15 is =E with the equation $x_0 = v_0$ and $\psi = (u_0 = z)$, which turns line 10 into $u_0 = v_0$. Line 17 is ∀I with the flag v_0: line 16 is its instance at v_0 (by the third and first substitution facts), and v_0 appears in no premise, in no assumption still open at line 17 (only line 2 is, and v_0 is not in it), and not in line 17. Line 18 is ∀I with the flag u_0 in the same way; its instance at u_0 is line 17, by the second substitution fact. Finally, line 20 closes the ∃E box: its last line 19 does not contain x_0, and x_0 is in neither line 1 nor any open assumption.

(⊣) The following is a derivation with the right-hand side as premise.

1	$\exists x\, \varphi \land \forall x\, \forall y\, (\varphi \land \varphi[y/x] \to x = y)$	
2	$\exists x\, \varphi$	∧E 1
3	$\forall x\, \forall y\, (\varphi \land \varphi[y/x] \to x = y)$	∧E 1
4	x_0 | $\varphi[x_0/x]$	
5	y_0 | | $\varphi[y_0/x]$	
6	$\forall y\, (\varphi[x_0/x] \land \varphi[y/x] \to x_0 = y)$	∀E 3
7	$\varphi[x_0/x] \land \varphi[y_0/x] \to x_0 = y_0$	∀E 6
8	$\varphi[x_0/x] \land \varphi[y_0/x]$	∧I 4, 5
9	$x_0 = y_0$	→E 7, 8
10	$x_0 = x_0$	=I
11	$y_0 = x_0$	=E 9, 10
12	$\varphi[y_0/x] \to y_0 = x_0$	→I 5–11
13	$\forall y\, (\varphi[y/x] \to y = x_0)$	∀I 5–12
14	$\varphi[x_0/x] \land \forall y\, (\varphi[y/x] \to y = x_0)$	∧I 4, 13
15	$\exists x\, (\varphi \land \forall y\, (\varphi[y/x] \to y = x))$	∃I 14
16	$\exists x\, (\varphi \land \forall y\, (\varphi[y/x] \to y = x))$	∃E 2, 4–15

Line 6 is ∀E at x_0 on line 3, by the second substitution fact, and line 7 is ∀E at y_0 on line 6, by the third and first. Lines 10 and 11 are symmetry, written out. Line 13 is ∀I with the flag y_0: line 12 is its instance at y_0, by the first substitution fact; at line 13 the open assumptions are line 4, which does not contain y_0, the premise does not contain it either, and neither does line 13. Line 15 is ∃I with the witness x_0, which is free for x since it does not appear in φ; line 14 is its instance at x_0, by the second substitution fact. Line 16 closes the ∃E box: line 15 does not contain x_0, and x_0 is in neither line 2 nor the premise, which is the only assumption open outside the box. This proves both directions.

■

In practice this is how "there is exactly one" is proved: an existence proof, then a separate "suppose x and y both satisfy φ; then $x = y$". And it is how it is used: from $\exists!x\, \varphi$ we may take a witness, and we may identify any two objects with the property. The claim about additive

inverses above was the "at most one" half; existence of an inverse is a field axiom.

One more instance: "the maximum". A **maximum** of a set A of reals is an element $m \in A$ with $a \leq m$ for every $a \in A$. If m and m' are both maxima, then $m' \leq m$ (apply the property of m to the element $m' \in A$) and $m \leq m'$ (apply the property of m' to $m \in A$), so $m = m'$ by the order fact that $a \leq b$ and $b \leq a$ give $a = b$. That is "at most one"; a set may have no maximum at all, such as $(0, 1)$, so existence is a separate question. Only when both halves are known may we write "the maximum of A".

4.7.8 The search method: one more step

Each of Sections 4.3 to 4.6 added a step to the search method of Section 2.6. Equality adds one.

- **An equation.** When an equation $t_1 = t_2$ is available and the goal differs from an available line only in some occurrences of t_1 and t_2, look for the φ that makes one =E turn the line into the goal: mark with x exactly the places where they differ. If the equation points the wrong way, turn it round first (sym). If no available line will do, make one with =I, as the proof of Proposition 4.7.1 did.

Section 4.8 collects all the steps into one method.

4.7.9 The rule, phrase and Lean table, completed

Chapter 1 began a table that pairs each rule with the sentence that performs it in a written proof and, for readers who use a proof assistant, with the Lean code that does the same (Sections 1.2–1.6), and Chapter 2 added the rows for negation, the biconditional, and the classical rules (Sections 2.1–2.5). Sections 4.3–4.6 added a row for each quantifier rule, and this section adds the two rows for equality. Here is the whole table for the calculus of this volume. Nothing later depends on the last column, and Chapter 5 turns the middle column into a full dictionary of proof phrases.

Rule	You write	Lean
$\wedge$I	"We show P, and we show Q."	`constructor`
$\wedge$E	"In particular, P."	`h.1` (and `h.2`)
$\to$E	"Since P implies Q, and P holds, Q."	`exact h hp`
$\to$I	"Assume P. … Hence Q."	`intro hp`
R	"As shown above, P."	`exact hp`
$\vee$I	"So P or Q holds."	`left` (or `Or.inl hp`); `right`
$\vee$E	"Case 1: P. … so R. Case 2: Q. … so R. In either case, R."	`Or.elim h` (or `cases h`)
$\neg$E	"…, which contradicts …"	`exact hn hp` (goal `False`)
$\bot$E	"This case cannot occur." "So there is nothing to prove."	`exfalso`
$\neg$I	"Suppose φ. … This is a contradiction. Hence not φ."	`intro hp` (goal `False`)
$\leftrightarrow$I	"($\Rightarrow$) Suppose P. … so Q. ($\Leftarrow$) Suppose Q. … so P."	`constructor`
$\leftrightarrow$E	"Since P holds if and only if Q does, and P holds, so does Q."	`h.mp hp` / `h.mpr hq`
RAA	"Suppose, for a contradiction, that φ fails. … This is impossible. Hence φ."	`by_contra h`

Rule	You write	Lean
∀E	"In particular, for $x = t$, …"; "applying the hypothesis to t, …"	`h t` (or `specialize h t`)
∀I	"Let x be arbitrary. … Since x was arbitrary, …"	`intro x`
∃I	"Take $x = t$. Then …"	`use t` (or `Exists.intro t h`)
∃E	"Choose x with … . Then …"	`obtain` (or `Exists.elim h`)
=I	(usually unwritten) "$t = t$."	`rfl`
=E	"Substituting t_2 for t_1, …"; a chain $t_1 = t_2 = \cdots = t_n$	`rw [h]`; chains `calc`

Of the derived rules of Chapter 2, MT, CP and DS have their rows in Section 2.3, and ¬¬E and LEM in Section 2.5; all the derived sequents are collected in Section 2.6. With the two equality rows, every rule of the calculus has its phrase.

4.7.10 Exercises

4.7.10.1 A. Check your understanding

Exercise. A1

(a) State =I and =E as schemas, with the side condition of =E.

(b) In the derivation below, name for each of lines 4, 5, 6, and 8 the rule, the cited lines, and the formula φ (with a variable z as its hole).

1	$a = b + 1$	
2	$b = 2$	
3	$a \cdot a = a \cdot a$	=I
4	$a \cdot a = (b + 1) \cdot a$	=E 1, 3
5	$a \cdot a = (b + 1) \cdot (b + 1)$	=E 1, 4
6	$a \cdot a = (2 + 1) \cdot (2 + 1)$	=E 2, 5
7	$(2 + 1) \cdot (2 + 1) = 9$	arith
8	$a \cdot a = 9$	=E 7, 6

(c) In the same derivation, could lines 4 and 5 be replaced by one line? With which φ?

(d) True or false, with a reason: from the lines $a = b$ and $P(b)$, one application of =E gives $P(a)$.

(e) Write $\exists!x\,(x \cdot x = 9)$ without the symbol $\exists!$, and decide whether it is true over $\mathbb{N}$ and over $\mathbb{Z}$.

Solution. (a) =I: at any line k, write $t = t$, for any term t, citing nothing. =E: from $t_1 = t_2$ on line i and $\varphi[t_1/x]$ on line j, both available at line k, write $\varphi[t_2/x]$ on line k; side condition: t_1 and t_2 are both free for x in φ.

(b) Line 4: =E 1, 3, with $\varphi = (a \cdot a = z \cdot a)$, which rewrites the first a on the right. Line 5:

=E 1, 4, with $\varphi = (a \cdot a = (b+1) \cdot z)$. Line 6: =E 2, 5, with $\varphi = (a \cdot a = (z+1) \cdot (z+1))$, which rewrites both b's at once. Line 8: =E 7, 6, with $\varphi = (a \cdot a = z)$. In each case the left-hand side $a \cdot a$ is written out in φ, so it is kept.

(c) Yes: =E 1, 3 with $\varphi = (a \cdot a = z \cdot z)$ gives $a \cdot a = (b+1) \cdot (b+1)$ in one step, rewriting both a's on the right and keeping the two on the left.

(d) False. With the equation $a = b$, =E passes from $\varphi[a/x]$ to $\varphi[b/x]$. The line $P(b)$ is $\varphi[b/x]$ for $\varphi = P(x)$, so the rule would need the equation $b = a$. One application of symmetry first (Proposition 4.7.1 (a)) gives $b = a$, and then =E gives $P(a)$.

(e) $\exists x\,(x \cdot x = 9 \wedge \forall y\,(y \cdot y = 9 \to y = x))$. Over $\mathbb{N}$ it is true: $x = 3$ works, and a natural number y with $y \cdot y = 9$ is 3, since $y \le 2$ gives $y \cdot y \le 4$ and $y \ge 4$ gives $y \cdot y \ge 16$. Over $\mathbb{Z}$ it is false: whatever x is proposed, 3 and -3 both satisfy $y \cdot y = 9$, and they cannot both equal x.

4.7.10.2 B. Practice

Exercise. B1: The identity is unique

A set with an operation $\cdot$ has an **identity** e if $e \cdot x = x$ and $x \cdot e = x$ for every x. Derive

$$\forall x\,(e \cdot x = x \wedge x \cdot e = x),\ \ \forall x\,(e' \cdot x = x \wedge x \cdot e' = x) \ \vdash\ e = e',$$

where e and e' are constants. Which half of "there is exactly one identity" does this prove?

Hint: look at the one element that both premises can say something about.

Solution. The goal is an equation between e and e', and the only term that both premises can connect to both is the product $e \cdot e'$. Instantiate the first premise at e' and the second at e (∀E, reading the terms off the goal):

1	$\forall x\,(e \cdot x = x \wedge x \cdot e = x)$	
2	$\forall x\,(e' \cdot x = x \wedge x \cdot e' = x)$	
3	$e \cdot e' = e' \wedge e' \cdot e = e'$	∀E 1
4	$e \cdot e' = e'$	∧E 3
5	$e' \cdot e = e \wedge e \cdot e' = e$	∀E 2
6	$e \cdot e' = e$	∧E 5
7	$e = e'$	=E 6, 4

Line 7 uses the equation $e \cdot e' = e$ of line 6 to rewrite the left side of line 4, with $\varphi = (z = e')$: $\varphi[e \cdot e'/z]$ is line 4, and $\varphi[e/z]$ is $e = e'$. In prose: $e = e \cdot e' = e'$, the first equation because e' is an identity and the second because e is.

This is the "at most one" half: any two identities are equal. It says nothing about whether an identity exists. To use it in Proposition 4.7.2, take $\varphi = \forall z\,(x \cdot z = z \wedge z \cdot x = z)$, and run lines 3–7 again with variables u_0, v_0 in place of e, e': inside flagged boxes for u_0 and v_0 and a →I box on $\varphi[u_0/x] \wedge \varphi[v_0/x]$, the two premises (with z as their bound variable) come from that assumption by ∧E, and the same steps give $u_0 = v_0$. Then →I and ∀I twice give

$\forall x\,\forall y\,(\varphi \wedge \varphi[y/x] \to x = y)$, the second conjunct of the proposition. With the existence of one identity as well, Proposition 4.7.2 gives "there is exactly one".

Exercise. B2: A unique square root of zero

Over $\mathbb{R}$, write $\exists!x\,(x \cdot x = 0)$ without $\exists!$, and prove it: existence, then uniqueness in the form "suppose y also satisfies ...".

Hint: a product of two nonzero reals is nonzero.

Solution. Written out, the statement is $\exists x\,(x \cdot x = 0 \wedge \forall y\,(y \cdot y = 0 \to y = x))$. We take the witness $x = 0$ and prove the two conjuncts.

Existence. $0 \cdot 0 = 0$.

Uniqueness. Let y be a real number, and suppose $y \cdot y = 0$. If $y \neq 0$, then $y \cdot y$ is a product of two nonzero reals and so is nonzero, which contradicts $y \cdot y = 0$. Hence $y = 0$, which is the witness. Since y was arbitrary, every y with $y \cdot y = 0$ equals 0.

So 0 witnesses the existential, and $\exists!x\,(x \cdot x = 0)$ holds. The rules behind the words: "we take the witness $x = 0$" is $\exists$I, postponed to the end as usual; "let y be a real number" is a flagged box for $\forall$I; "suppose $y \cdot y = 0$" is a $\to$I box; "if $y \neq 0$ ... contradicts" is an RAA box on $y \neq 0$. (That box gives $y = 0$ directly; $\neg$I would give only $\neg(y \neq 0)$. Since for reals $y = 0$ or $y \neq 0$ is also a school fact, a proof by cases would do as well.)

Exercise. B3: Exactly one fixed point

A **fixed point** of a function $f\colon \mathbb{R} \to \mathbb{R}$ is a real x with $f(x) = x$.

(a) Formalize "f has exactly one fixed point" over $\mathbb{R}$, first with $\exists!$ and then without it.

(b) Write a formula without $\exists!$ that says "f does **not** have exactly one fixed point", in the form "there is none, or there are two different ones".

(c) Decide the statement of (a) for $f(x) = x \cdot x$, for $f(x) = x + 1$ and for $f(x) = 2 \cdot x$, and check your answer against the formula of (b).

Solution. (a) With $\exists!$: $\exists!x\,(f(x) = x)$. Written out, with y as the new variable: $\exists x\,(f(x) = x \wedge \forall y\,(f(y) = y \to y = x))$.

(b) $\neg\exists x\,(f(x) = x) \vee \exists x\,\exists y\,(f(x) = x \wedge f(y) = y \wedge x \neq y)$. Read informally over $\mathbb{R}$: either no real number is fixed, or two different ones are. That is exactly the failure of "exactly one". If no x is fixed, the first disjunct holds. If some x is fixed but the uniqueness clause fails for it, then some y is fixed and differs from x. Conversely, either disjunct rules out a unique fixed point: with none there is no witness, and two different fixed points $x \neq y$ would both have to equal any proposed witness w, giving $x = y$. (Section 4.9 derives such negations mechanically, one quantifier at a time.)

(c) For $f(x) = x \cdot x$: $x \cdot x = x$ holds for $x = 0$ and for $x = 1$, so there are two different fixed points; (a) is false, and the second disjunct of (b) is true with $x = 0$, $y = 1$. For $f(x) = x + 1$:

$x+1=x$ has no solution, so (a) is false and the first disjunct of (b) is true. For $f(x)=2\cdot x$: $2x=x$ gives $x=0$, and 0 is fixed; so (a) is true with the witness 0, and both disjuncts of (b) are false.

4.7.10.3 C. Going deeper

Exercise. C1: Two unique quantifiers do not commute

Show that $\exists!x\,\exists!y\,R(x,y)$ and $\exists!y\,\exists!x\,R(x,y)$ can have different truth values, by giving a finite domain D and a two-place relation R on it for which the first is true and the second false. Check both, reading $\exists!$ informally.

Hint: first say in words what each sentence claims about the rows and columns of a table of R.

Solution. **What any answer must satisfy.** Call x *single* if exactly one y has $R(x,y)$, and call y *co-single* if exactly one x has $R(x,y)$. The first sentence says that **exactly one** element is single. The second says that exactly one element is co-single. So an answer needs exactly one single element and a number of co-single elements other than one: none, or two or more.

An example. Take $D=\{0,1,2\}$ and $R=\{(1,2),(2,0),(2,1)\}$.

- Single elements: 0 has no partner, 1 has exactly one (2), and 2 has two (0 and 1). So 1 is the only single element, and $\exists!x\,\exists!y\,R(x,y)$ is true, with the witness 1.
- Co-single elements: $y=0$ is reached only from $x=2$; $y=1$ only from $x=2$; $y=2$ only from $x=1$. All three are co-single, so there is not exactly one, and $\exists!y\,\exists!x\,R(x,y)$ is false.

No example exists on one or two elements (there are only $2+16=18$ relations on such domains). On one element, 0 is single and co-single exactly when $R(0,0)$ holds, so the two sentences agree. On two elements, suppose exactly one element is single; after relabeling, call it 0, and let c be its only partner. Then 1 is not single, so it has no partners or two. If it has none, column c has one entry and the other column has none. If it has two, column c has two entries and the other column has one. Either way exactly one element is co-single. Exchanging the roles of rows and columns gives the converse, so the two sentences have the same truth value. Three elements is therefore the smallest possible domain.

The moral: $\exists!$ is not a quantifier that commutes with its own kind, as two $\exists$'s do. "Exactly one" hides a $\forall$ (the uniqueness clause), and the order of a $\forall$ and an $\exists$ matters.

Exercise. C2: "At most one", two ways

Let P be a one-place predicate symbol. Derive both directions of

$$\forall x\,\forall y\,(P(x)\wedge P(y)\to x=y)\ \dashv\vdash\ \neg\exists x\,\exists y\,((P(x)\wedge P(y))\wedge x\neq y),$$

and say which of your two derivations uses RAA, and at which line.

Solution (Hint 1). The right-hand side is a negation: for $\vdash$, open a $\neg$I box, and inside it two $\exists$E boxes. For $\dashv$, the goal is $\forall\forall$, then an implication, then an equation.

Solution (Hint 2). In the $\dashv$ direction, open flagged boxes for x_0 and y_0 and a $\to$I box on $P(x_0) \wedge P(y_0)$. The goal is now the atomic formula $x_0 = y_0$. Which resources can give it?

Solution. ($\vdash$) The goal is a negation, so a $\neg$I box opens on $\exists x\, \exists y\, (\dots)$; inside, the premise-directed step opens an $\exists$E box for each $\exists$, with the goal $\bot$ as each box's last line. The premise, instantiated at the two witnesses, gives $x_0 = y_0$, which clashes with $x_0 \neq y_0$.

1	$\forall x\, \forall y\, (P(x) \wedge P(y) \to x = y)$	
2	$\quad \exists x\, \exists y\, ((P(x) \wedge P(y)) \wedge x \neq y)$	
3	$\quad x_0 \quad \exists y\, ((P(x_0) \wedge P(y)) \wedge x_0 \neq y)$	
4	$\quad\quad y_0 \quad (P(x_0) \wedge P(y_0)) \wedge x_0 \neq y_0$	
5	$\quad\quad\quad P(x_0) \wedge P(y_0)$	$\wedge$E 4
6	$\quad\quad\quad x_0 \neq y_0$	$\wedge$E 4
7	$\quad\quad\quad \forall y\, (P(x_0) \wedge P(y) \to x_0 = y)$	$\forall$E 1
8	$\quad\quad\quad P(x_0) \wedge P(y_0) \to x_0 = y_0$	$\forall$E 7
9	$\quad\quad\quad x_0 = y_0$	$\to$E 8, 5
10	$\quad\quad\quad \bot$	$\neg$E 9, 6
11	$\quad\quad \bot$	$\exists$E 3, 4–10
12	$\quad \bot$	$\exists$E 2, 3–11
13	$\neg\exists x\, \exists y\, ((P(x) \wedge P(y)) \wedge x \neq y)$	$\neg$I 2–12

The brackets in the statement are written out because the default groups a longer conjunction to the right (Section 3.4); with the grouping $(P(x_0) \wedge P(y_0)) \wedge x_0 \neq y_0$ on line 4, $\wedge$E takes out its two halves on lines 5 and 6. At line 11, y_0 is not in $\bot$, not in line 3, not in the premise, and not in the assumptions still open (lines 2 and 3). At line 12 the same holds for x_0, with line 2 the only open assumption.

($\dashv$) The goal-directed steps open flagged boxes for x_0 and y_0 and a $\to$I box on $P(x_0) \wedge P(y_0)$, leaving the goal $x_0 = y_0$. No resource gives an equation, so the search is stuck, and step 5 of the method applies: an RAA box on $x_0 \neq y_0$. Inside, $\wedge$I and two $\exists$I steps rebuild the formula that the premise denies.

1	$\neg\exists x\,\exists y\,((P(x) \wedge P(y)) \wedge x \neq y)$	
2	x_0 | y_0 | $P(x_0) \wedge P(y_0)$	
3	$x_0 \neq y_0$	
4	$(P(x_0) \wedge P(y_0)) \wedge x_0 \neq y_0$	$\wedge$I 2, 3
5	$\exists y\,((P(x_0) \wedge P(y)) \wedge x_0 \neq y)$	$\exists$I 4
6	$\exists x\,\exists y\,((P(x) \wedge P(y)) \wedge x \neq y)$	$\exists$I 5
7	$\bot$	$\neg$E 6, 1
8	$x_0 = y_0$	RAA 3–7
9	$P(x_0) \wedge P(y_0) \to x_0 = y_0$	$\to$I 2–8
10	$\forall y\,(P(x_0) \wedge P(y) \to x_0 = y)$	$\forall$I 2–9
11	$\forall x\,\forall y\,(P(x) \wedge P(y) \to x = y)$	$\forall$I 2–10

The $\dashv$ derivation uses RAA, at line 8: the box opens with $x_0 \neq y_0$, that is $\neg(x_0 = y_0)$, and closes with $x_0 = y_0$. Closing it by $\neg$I instead would give $\neg\neg(x_0 = y_0)$, and no rule of Chapter 1 or Sections 2.1–2.4 removes the double negation. (With the derived rule $\neg\neg$E of Section 2.5, that $\neg$I box followed by $\neg\neg$E is an equally good derivation; its classical step is the $\neg\neg$E line, which hides an RAA.) The $\vdash$ derivation uses no classical rule.

Marking scheme. (10 points) - 2: $\vdash$: opens a $\neg$I box on the existential, with the goal $\bot$. - 2: $\vdash$: two $\exists$E boxes with **fresh** witnesses, each ending in $\bot$, and the premise instantiated at those witnesses. - 2: $\dashv$: two flagged boxes and a $\to$I box, in the order read off the goal. - 2: $\dashv$: an RAA box on $x_0 \neq y_0$, with the clash obtained by $\wedge$I and two $\exists$I steps. - 2: names the $\dashv$ direction as the one using RAA, and the line. - A $\neg$I box on $x_0 \neq y_0$ followed by $\neg\neg$E, or an LEM split on $x_0 = y_0$ whose second case reaches $\bot$ the same way (then $\bot$E), earns the 2 points of the RAA-box line; for the naming line, the reader names the $\neg\neg$E or LEM line as the classical step, a derived rule whose derivation uses RAA (Section 2.5). Fatal errors (0 for the step): an $\exists$E box closing on a line that contains its witness; a $\forall$I over a name in an open assumption; $\neg$I claimed to give $x_0 = y_0$.

4.8 Translating Mathematics: Nested and Bounded Quantifiers

The calculus is now complete: Sections 4.3–4.7 gave a rule for using and a rule for proving each quantifier, and the two rules for equality. Before any of those rules can act on a definition from analysis, the definition has to be written as a formula, and written correctly. The definitions of analysis have three or four quantifiers stacked in a fixed order, most of them carrying a bound such as $\varepsilon > 0$ or $n \geq N$, and every bound costs extra lines in a derivation. This section does three things. It translates the standard definitions into formulas, by a procedure that works from the outside in. It turns each quantifier rule into a **bounded** rule that handles the bound in the same step. And it collects the steps of Sections 4.3–4.7 into the search method of Section 2.6, extended to quantifiers and equality.

4.8.1 Four pieces in one sentence

"The sequence (a_n) converges to L" is written $a_n \to L$, and a first course defines it in one breath: for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N$. The sentence has three quantifiers, each with a bound ($\varepsilon > 0$, $N \in \mathbb{N}$, $n \in \mathbb{N}$), and a fourth condition, $n \geq N$, hidden in "for all $n \geq N$", which will turn out to sit under an implication. Each piece has a job, and a slip in any piece changes the meaning:

- drop the bound $\varepsilon > 0$, and the challenger may choose $\varepsilon = -1$; then $|a_n - L| < -1$ is impossible, and **no** sequence converges;
- swap "for every ε" and "there is N", and one N must serve every ε; then only sequences that are eventually equal to L converge;
- read "for all $n \geq N$" as "for all n", and the first terms must already be close to L; then $1, \frac{1}{2}, \frac{1}{3}, \ldots$ converges to nothing.

So translation deserves a procedure, and the rules deserve a form that handles bounds without fuss. We take them in that order.

4.8.2 Translation from the outside in

Find the outermost claim, name its quantifier and its bound, then translate what it claims about each object, in the same way.

The procedure, in three steps:

1. **Fix the domain.** State what the variables range over. By the one-domain convention of Section 4.1 (after Definition 4.1.1), a sentence that mixes kinds of object uses the largest kind as the domain, and each smaller kind enters as a bound: $N \in \mathbb{N}$ inside the domain $\mathbb{R}$. A sequence (a_n) is a function symbol a, with a_n its value at n.
2. **Find the outermost claim.** Ask what the sentence says first, before any detail: "every $\varepsilon > 0$ has a property", "there is an M with a property", "no prime is largest". That gives the outermost quantifier (or connective) and its bound.
3. **Recurse.** Translate the property, which is a sentence about the object just quantified, by the same two steps. Stop at atomic formulas: equations, inequalities, membership, and predicate symbols such as $\mathrm{Prime}(p)$.

A bound is attached to its own quantifier with the connective of Definition 4.1.1: $\to$ under $\forall$, $\wedge$ under $\exists$. The bounded notation $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ does this automatically, which is why we use it.

Example. Five definitions, translated

Translate each sentence into a formula, stating the domain.

(a) $a_n \to L$: for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for every $n \geq N$.

(b) The set A of real numbers is bounded above.

(c) The function $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at a: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ for every real x with $|x - a| < \delta$.

(d) There is no largest prime.

(e) Every integer greater than 1 has a prime divisor.

Solution. *(a)* Domain $\mathbb{R}$. The outermost claim is about every $\varepsilon > 0$: $\forall \varepsilon > 0$. What it says about ε is that some natural number N works: $\exists N \in \mathbb{N}$. What it says about N is that every natural number n from N on is good: $\forall n \in \mathbb{N}$, with the further condition $n \geq N$ under an implication. So

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \in \mathbb{N} \; \big(n \geq N \to |a_n - L| < \varepsilon\big).$$

The phrase "for every $n \geq N$" carries two bounds on n, membership in $\mathbb{N}$ and $n \geq N$; the first is a bounded quantifier, the second the hypothesis of the implication. The free variable is L (and the statement is also about the function symbol a, which is not a variable).

(b) Domain $\mathbb{R}$; A names a set of reals, and $x \in A$ is a one-place predicate of x (Section 4.1). "Bounded above" means that some real number is at least every element of A. Outermost claim: there is an M. Its property: every element of A is at most M:

$$\exists M \; \forall x \in A \; (x \leq M).$$

The bound $x \in A$ belongs to the $\forall$; there is no bound on M, which may be any real number.

(c) Domain $\mathbb{R}$, with f a one-place function symbol and a a variable, free in the formula:

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \; \big(|x - a| < \delta \to |f(x) - f(a)| < \varepsilon\big).$$

The last quantifier has no bound of the form "$x \in$ something", because f is defined on all of $\mathbb{R}$; its restriction $|x - a| < \delta$ goes under the implication, like $n \geq N$ in (a). For f defined on a set A, the last quantifier would be $\forall x \in A$.

(d) Domain $\mathbb{N}$, with $\text{Prime}(p)$ the predicate "p is prime". Section 4.1 read the sentence as $\neg\exists p \, (\text{Prime}(p) \wedge \forall q \, (\text{Prime}(q) \to q \leq p))$. A mathematician more often states it positively, "for every prime there is a larger prime":

$$\forall p \; \big(\text{Prime}(p) \to \exists q \; (\text{Prime}(q) \wedge p < q)\big).$$

The two forms say the same thing over $\mathbb{N}$: the laws of Section 4.9 push the $\neg$ inside and give $\forall p \, (\text{Prime}(p) \to \exists q \, (\text{Prime}(q) \wedge \neg(q \leq p)))$, and the order fact that $\neg(q \leq p)$ is the same as $p < q$ finishes the job. (They are not provably equivalent in logic alone: that last step is a fact about the order of $\mathbb{N}$, not a law of logic.)

(e) Domain $\mathbb{Z}$. Outermost claim: every integer greater than 1 has a property, $\forall n > 1$. The property: some p is prime and divides n. "Divides" is itself a quantifier: $p \mid n$ means $n = p \cdot k$ for some integer k. So

$$\forall n > 1 \; \exists p \; \big(\text{Prime}(p) \wedge \exists k \; (n = p \cdot k)\big).$$

Unfolded, the first bound is $\forall n \, (n > 1 \to \ldots)$. The innermost quantifier came from a **defined** word, "divides", which is typical: a definition in words often hides quantifiers, and a translation unfolds them.

Two remarks on the example. First, the order of the quantifiers in (a) and (c) is the order in which the objects are **chosen**: ε first, by a challenger; then N or δ, by the defender, knowing ε; then n or x, by the challenger again, knowing both. That is the game reading of Example (Swapping two quantifiers). Second, in (c) the letter a is free, so the formula is a statement about a; "f is continuous at every point" would add $\forall a$ in front.

Quick check. Translate "A has a maximum" (an element of A that is at least every element of A) over $\mathbb{R}$. Which quantifier carries the bound $x \in A$, and does the other quantifier carry a bound too?

Solution. $\exists m \in A\, \forall x \in A\, (x \leq m)$. The bound $x \in A$ belongs to the $\forall$: "every element of A". The $\exists$ carries a bound too, $m \in A$, because a maximum must itself lie in A; dropping it gives $\exists m\, \forall x \in A\, (x \leq m)$, which is "$A$ is bounded above", a weaker statement. For $A = (0, 1)$ the weaker statement is true ($m = 1$) and the stronger one is false.

Warning. **Three mistranslations to avoid.** (1) An $\exists$ with $\to$ under it, such as $\exists x\, (x \in A \to x > 5)$: it is true as soon as something lies outside A (Section 4.1). (2) A bound on the wrong quantifier: "f is bounded on A" is $\exists M\, \forall x \in A\, (|f(x)| \leq M)$; the form $\exists M \in A\, \forall x\, (|f(x)| \leq M)$ asks for a bound that lies in A and works on all of $\mathbb{R}$, and it fails for $f(x) = x$ on $A = [0, 1]$, which is bounded on A by 1. (3) "For all large n" read as "for all n": "$a_n > 0$ for all large n" is $\exists N \in \mathbb{N}\, \forall n \in \mathbb{N}\, (n \geq N \to a_n > 0)$, not $\forall n \in \mathbb{N}\, (a_n > 0)$. The sequence $a_n = n - 3$ satisfies the first (take $N = 4$) and not the second ($a_0 = -3$).

4.8.3 Rules for bounded quantifiers

Every analysis formula has bounds, and each bound doubles the work of a rule. To use $\forall \varepsilon > 0\, \varphi$ at the term 1, $\forall$E gives $1 > 0 \to \varphi[1/\varepsilon]$, and a second step, $\to$E with a line $1 > 0$, gives $\varphi[1/\varepsilon]$. To prove $\exists N \in \mathbb{N}\, \psi$ with the witness 7, $\wedge$I builds $7 \in \mathbb{N} \wedge \psi[7/N]$ and $\exists$I finishes. Every step is legal and every step is predictable, and a derivation about limits drowns in them. So we fold each pair into one step.

A bounded quantifier gets a rule of its own: the bound travels with the quantifier.

Proposition 4.8.1. Rules for Bounded Quantifiers

Let A be a fixed set, so that $\in A$ is a one-place predicate symbol (Section 4.2), x a variable and φ a formula. The following four rules, the **bounded rules**, may be used in derivations, with the rule names of the unbounded rules and the same side conditions: in $\forall x \in A$ E and $\exists x \in A$ I, the term t is free for x in φ; in $\forall x \in A$ I, the name x_0 appears in no premise, in no assumption of a box still open at line k, and not in $\forall x \in A\, \varphi$; in $\exists x \in A$ E, it appears in no premise, in no assumption of a box still open at line k, and not in χ or in $\exists x \in A\, \varphi$. The last line of each box is a line of that box itself.

i	$\forall x \in A\, \varphi$		i	$t \in A$	
j	$t \in A$		j	$\varphi[t/x]$	
k	$\varphi[t/x]$	$\forall$E i, j	k	$\exists x \in A\, \varphi$	$\exists$I i, j

$$\begin{array}{lll|l} i & x_0 & x_0 \in A & \\ & & \vdots & \\ j & & \varphi[x_0/x] & \\ \hline k & \forall x \in A\ \varphi & & \forall\text{I } i\text{–}j \end{array} \qquad \begin{array}{lll|l} h & & \exists x \in A\ \varphi & \\ i & x_0 & x_0 \in A & \\ j & & \varphi[x_0/x] & \\ & & \vdots & \\ l & & \chi & \\ \hline k & \chi & & \exists\text{E } h,\ i\text{–}l \end{array}$$

Every use of a bounded rule can be replaced by a block of steps of the unbounded rules (Sections 4.3–4.6) and of Chapter 1, with the same premises, the same assumptions open, and the same last line. Hence any derivation that uses the bounded rules converts into one that does not. The same holds with $x \in A$ and $t \in A$ replaced by $x > a$ and $t > a$, or by the bounds $\geq a$, $< a$, $\leq a$, for a term a in which x does not occur.

In words: to **use** "every x in A has φ", produce an element t of A and conclude φ for t. To **prove** "some x in A has φ", exhibit an element t of A with φ. To **prove** "every x in A has φ", let x_0 be an arbitrary element of A (the box opens with the assumption $x_0 \in A$) and prove φ for it. To **use** "some x in A has φ", take such an element x_0, with both facts about it as assumptions, and derive something that does not mention it. These are exactly the sentences a mathematician writes: "let $\varepsilon > 0$", "since $\delta > 0$, the hypothesis applies to δ", "take $N = 7 \in \mathbb{N}$", "choose $\delta > 0$ such that ...".

The bounded $\forall$I box is the only flagged box with an assumption that closes by $\forall$I. Its assumption $x_0 \in A$ mentions the flag, and that is allowed: the side condition asks about assumptions of boxes **still open at line** k, and the box's own assumption has closed there.

Why is this a proposition and not simply a definition? Because the calculus is fixed: its rules are those of Chapters 1–2 and Sections 4.3–4.7, and a new rule is admissible only if it adds no derivable sequents. Definition 2.3.1 gave one way to show that: a proved sequent, written with statement letters, may be used as one line. It does not cover these rules. The bounded $\forall$E and $\exists$I are one line each, but they are schemas in a formula, a variable and a term, not in statement letters; the bounded $\forall$I and $\exists$E close boxes, which no proved sequent can. So the proposition proves admissibility the direct way: each bounded step is shown to be an abbreviation of a block of unbounded steps. In this sense the bounded rules are derived rules, in a looser sense that extends Chapter 2's: a line citing one abbreviates a block of primitive steps, not a proved sequent.

Idea. A bounded quantifier is an abbreviation (Definition 4.1.1): $\forall x \in A\ \varphi$ **is** the formula $\forall x\,(x \in A \to \varphi)$, and $\exists x \in A\ \varphi$ **is** $\exists x\,(x \in A \wedge \varphi)$. So each bounded rule should be an unbounded rule on the unfolded formula, plus the connective rule that handles $\to$ or $\wedge$. Write the block, then check two things: the block is a correct piece of derivation wherever the bounded step was, and its side conditions hold exactly when the bounded ones do.

Proof. We give the replacement block for each rule and check it. In each block, the lines of the bounded version keep their labels, and new lines get new letters.

Bounded $\forall$E. Line i is $\forall x\,(x \in A \to \varphi)$. Replace line k by two lines:

i	$\forall x\,(x \in A \to \varphi)$	
j	$t \in A$	
m	$t \in A \to \varphi[t/x]$	$\forall$E i
k	$\varphi[t/x]$	$\to$E m, j

Line m is $\forall$E at t: the formula $(x \in A \to \varphi)[t/x]$ is $t \in A \to \varphi[t/x]$ by Definition 4.2.5, since A is not a variable. Its side condition asks that t be free for x in $x \in A \to \varphi$ (Definition 4.2.6), which holds exactly when t is free for x in φ, since $x \in A$ has no quantifier. Lines i and j are available at m and k because they were available at k.

Bounded $\exists$I. Replace line k by:

i	$t \in A$	
j	$\varphi[t/x]$	
m	$t \in A \wedge \varphi[t/x]$	$\wedge$I i, j
k	$\exists x\,(x \in A \wedge \varphi)$	$\exists$I m

Line m is the instance at t of $x \in A \wedge \varphi$, and t is free for x there exactly when it is free for x in φ.

Bounded $\forall$I. Replace the box and line k by a flagged box with no assumption, which contains a $\to$I box holding the old lines:

i	x_0		$x_0 \in A$	
			$\vdots$	
j			$\varphi[x_0/x]$	
m		$x_0 \in A \to \varphi[x_0/x]$		$\to$I i–j
k	$\forall x\,(x \in A \to \varphi)$			$\forall$I i–m

The lines between i and j are unchanged. Each of them lies inside both new boxes, so every line it cited is still available (Definition 1.4.2, with the flagged boxes of Section 4.4): a line outside the old box is outside both new ones, and a line inside the old box is inside the new $\to$I box. The side conditions inside the old box are unaffected too: a line inside it has the same open assumptions as before, $x_0 \in A$ (now the assumption of the $\to$I box) together with the outer ones, because the flagged box adds none; so every flagged box that closes inside the old box still meets its side condition. Line m is $\to$I, and it is the last line of the flagged box, as $\forall$I requires; it is the instance at x_0 of $x \in A \to \varphi$. Now the side condition of $\forall$I at line k: the premises are the same as before; the boxes still open at line k are the same as those open at line k in the bounded version, because the $\to$I box on $x_0 \in A$ has **closed** before line k; and the conclusion is the same formula. So the unbounded side condition holds exactly when the bounded one does.

Bounded $\exists$E. Line h is $\exists x\,(x \in A \wedge \varphi)$. Replace the box's two assumptions by one assumption and two lines of $\wedge$E:

h	$\exists x\,(x \in A \wedge \varphi)$		
g	x_0 | $x_0 \in A \wedge \varphi[x_0/x]$		
i		$x_0 \in A$	$\wedge$E g
j		$\varphi[x_0/x]$	$\wedge$E g
		$\vdots$	
l		χ	
k	χ		$\exists$E h, g–l

Lines i and j now hold the same formulas as the old assumptions, inside the same box, so every line that cited them may still cite them. At a line inside the box, the open assumptions are now g together with the outer ones, where before they were $x_0 \in A$ and $\varphi[x_0/x]$ together with the outer ones. A name appears in g exactly when it appears in $x_0 \in A$ or in $\varphi[x_0/x]$, so every flagged box that closes inside the box still meets its side condition. The box assumes the instance at x_0 of $x \in A \wedge \varphi$, as $\exists$E requires. Its side condition at line k names χ, the formula $\exists x\,(x \in A \wedge \varphi)$, the premises and the assumptions open at line k: the same four things as the bounded side condition, with the same result.

For the bounds $x > a$ and the others, the same four blocks work with $x > a$ and $t > a$ in place of $x \in A$ and $t \in A$. One point needs the hypothesis that x does not occur in a: then $(x > a)[t/x]$ is $t > a$, and the bound contains no bound variable, so free-for conditions are unaffected by it.

Replacing the bounded steps one at a time, each by its block, removes them all; the premises and the last line do not change. This proves the proposition.

■

Each block is one line longer than the bounded step it replaces: two lines for one in $\forall$E and $\exists$I; the extra $\to$I line in $\forall$I; three lines (an assumption and two $\wedge$E) for two assumptions in $\exists$E. So **each bounded step saves exactly one line**, and a derivation with many bounds saves many. More than the count, the bounded rules keep the derivation at the level at which a mathematician thinks: "let $\varepsilon > 0$" is one step, not two.

Quick check. Unfolded, $\exists N \in \mathbb{N}\ \psi$ is $\exists N\,(N \in \mathbb{N} \wedge \psi)$. A student uses the bounded $\exists$E on it and writes the box with the single assumption $N_0 \in \mathbb{N} \wedge \psi[N_0/N]$. Is that wrong?

Solution. No, it is the unbounded $\exists$E, which is always available, since the bounded formula **is** the unfolded one. It costs two $\wedge$E lines inside the box to separate $N_0 \in \mathbb{N}$ from $\psi[N_0/N]$. The bounded rule opens the box with the two assumptions already separated. Both are correct; the bounded one is shorter by one line.

4.8.4 A limit derivation with bounds

Here is the first derivation in the book about a limit. The claim is the school fact "if $a_n \to L$, then $2a_n \to 2L$", with the definition of Example (Five definitions, translated) (a).

Claim. Over $\mathbb{R}$,

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon)$$

derives the same formula with $2 \cdot a_n$ and $2 \cdot L$ in place of a_n and L, using two lines of arithmetic, justified "arith": $\varepsilon_0 > 0$ gives $\varepsilon_0/2 > 0$, and $|a_{n_0} - L| < \varepsilon_0/2$ gives $|2 \cdot a_{n_0} - 2 \cdot L| < \varepsilon_0$ (because $|2u - 2v| = 2|u - v|$).

Idea. Run the search. The goal starts with $\forall \varepsilon > 0$: open a bounded $\forall$I box, "let $\varepsilon_0 > 0$". The new goal starts with $\exists N \in \mathbb{N}$: **postpone** it (the step of Section 4.5), since no witness is in sight. Turn to the premise: it starts with $\forall \varepsilon > 0$, and the step of Section 4.3 says to instantiate it at the term the goal needs. Which term? The goal will need $|2a_n - 2L| < \varepsilon_0$, which follows from $|a_n - L| < \varepsilon_0/2$: so instantiate at $\varepsilon_0/2$, after an "arith" line $\varepsilon_0/2 > 0$ for the bound. That gives an $\exists N \in \mathbb{N}$ line: open its $\exists$E box **at once**, with a witness N_0 (the step of Section 4.6), and let the box's last line be the current goal. Now the postponed goal has its witness: N_0 itself. The remaining goal $\forall n \in \mathbb{N}\,(\dots)$ opens a bounded $\forall$I box for n_0, and a $\to$I box on $n_0 \geq N_0$ finishes with one $\forall$E, one $\to$E, and the second "arith" line. This is the pattern of Proposition 4.6.2: **fix** ε_0, **obtain** N_0, **choose** N_0 as the witness.

1		$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert a_n - L\rvert < \varepsilon)$	
2	ε_0	$\varepsilon_0 > 0$	
3		$\varepsilon_0/2 > 0$	arith 2
4		$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert a_n - L\rvert < \varepsilon_0/2)$	$\forall$E 1, 3
5	N_0	$N_0 \in \mathbb{N}$	
6		$\forall n \in \mathbb{N}\ (n \geq N_0 \to \lvert a_n - L\rvert < \varepsilon_0/2)$	
7	n_0	$n_0 \in \mathbb{N}$	
8		$n_0 \geq N_0 \to \lvert a_{n_0} - L\rvert < \varepsilon_0/2$	$\forall$E 6, 7
9		$n_0 \geq N_0$	
10		$\lvert a_{n_0} - L\rvert < \varepsilon_0/2$	$\to$E 8, 9
11		$\lvert 2 \cdot a_{n_0} - 2 \cdot L\rvert < \varepsilon_0$	arith 10
12		$n_0 \geq N_0 \to \lvert 2 \cdot a_{n_0} - 2 \cdot L\rvert < \varepsilon_0$	$\to$I 9–11
13		$\forall n \in \mathbb{N}\ (n \geq N_0 \to \lvert 2 \cdot a_n - 2 \cdot L\rvert < \varepsilon_0)$	$\forall$I 7–12
14		$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert 2 \cdot a_n - 2 \cdot L\rvert < \varepsilon_0)$	$\exists$I 5, 13
15		$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert 2 \cdot a_n - 2 \cdot L\rvert < \varepsilon_0)$	$\exists$E 4, 5–14
16		$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert 2 \cdot a_n - 2 \cdot L\rvert < \varepsilon)$	$\forall$I 2–15

Check the side conditions at the three lines that close flagged boxes, and the witness line.

- Line 13 ($\forall n \in \mathbb{N}$ I, flag n_0): n_0 is not in the premise, not in line 13, and not in the assumptions still open at line 13, which are lines 2, 5, and 6.
- Line 14 ($\exists N \in \mathbb{N}$ I): the witness N_0, with its bound on line 5 and the instance on line 13.
- Line 15 ($\exists N \in \mathbb{N}$ E, flag N_0): N_0 is not in line 15, the conclusion (the witness has been forgotten by line 14), not in line 4, not in the premise, and not in the open assumption on line 2.

- Line 16 ($\forall\varepsilon > 0$ I, flag ε_0): ε_0 is not in the premise and not in line 16, and no assumption is open at line 16.

Line 4 is where N comes to depend on ε: the witness N_0 is obtained from the premise **at** $\varepsilon_0/2$, so it depends on ε_0, exactly as the order $\forall\varepsilon\,\exists N$ allows. Section 4.10 makes this reading of dependence precise.

In prose, line by line:

> Let $\varepsilon > 0$ [lines 2, 16]. Then $\varepsilon/2 > 0$ [3], so, since $a_n \to L$, there is $N \in \mathbb{N}$ with $|a_n - L| < \varepsilon/2$ for every $n \geq N$ [4]; choose such an N [5, 6, 15]. Let $n \in \mathbb{N}$ with $n \geq N$ [7, 9]. Then $|a_n - L| < \varepsilon/2$ [8, 10], so $|2a_n - 2L| = 2|a_n - L| < \varepsilon$ [11]. So this N works [12–14].

Written with the unbounded rules, the same derivation has 22 lines, since each of its six bounded steps (lines 4, 8, 13, 14, 15, and 16) costs one line more. The claim "$\frac{1}{n} \to 0$" (for $n \geq 1$) is not derived here: its proof needs a fact about $\mathbb{N}$ inside $\mathbb{R}$, the Archimedean property, which is a theorem of *Analysis on the Real Line*, Chapter 1, not of logic.

4.8.5 The search method, extended

Section 2.6 put the moves for the connectives in order. Each of Sections 4.3–4.7 added a step. Here is the method with all of them.

Algorithm. **Searching for a derivation of $\Gamma \vdash \varphi$ in predicate logic.**

1. **Set up.** Write the premises Γ at the top and φ at the bottom, with one gap between. Translate any definition in words into a formula first, as in Example (Five definitions, translated).
2. **Goal-directed step.** Take the current goal's main symbol and set up its introduction rule:
   - $\wedge$, $\to$, $\neg$, $\leftrightarrow$: as in Section 2.6;
   - $\forall x\,\psi$: open a **flagged box** with a fresh name x_0; the new goal, the box's last line, is $\psi[x_0/x]$ (Section 4.4). For $\forall x \in A\ \psi$, the box opens with the assumption $x_0 \in A$;
   - $\vee$: **postpone**, as in Section 2.6;
   - $\exists x\,\psi$: **postpone**; work on the premises until an available line shows a term t with $\psi[t/x]$, then use $\exists$I (Section 4.5). For $\exists x \in A\ \psi$, the line $t \in A$ is needed too.

   Repeat while the goal is $\wedge$, $\to$, $\neg$, $\leftrightarrow$, or $\forall$.
3. **Premise-directed step.** Take the main symbol of a resource and apply its elimination rule:
   - $\wedge$, $\to$, $\vee$, $\leftrightarrow$, and the derived rules: as in Section 2.6;
   - $\exists x\,\psi$: open the $\exists$E box **early**, as soon as the line is available, with a fresh witness x_0, and make its last line the current goal, which must not mention x_0 (Section 4.6). An early box lets the witness serve the goals that come later;
   - $\forall x\,\psi$: do not instantiate blindly; instantiate at the term that the goal, or another resource, mentions (Section 4.3). For $\forall x \in A\ \psi$, the line $t \in A$ is needed too;
   - an equation $t_1 = t_2$: use $=$E with a φ chosen so that one substitution turns an available line into the goal, turning the equation round first if needed (Section 4.7).
4. **Clash.** As in Section 2.6: for the goal $\bot$, find or manufacture a pair χ, $\neg\chi$; an impossible case closes itself by $\bot$E.
5. **Stuck.** As in Section 2.6: RAA on the current goal, or LEM on a well-chosen statement.

When no gap is left, number the lines and check the derivation as Definition 1.4.2, extended in Section 4.4, says: every citation is available, and every side condition holds (t free for x at $\forall$E and $\exists$I, bounded or not; t_1 and t_2 free for x at $=$E; a fresh name at each line that closes a flagged box).

Two of the new steps deserve a word on their place in the order. The goal step for $\forall$ is safe in the same way as the goal steps for $\to$ and $\neg$ (Section 2.6): if $\forall x\,\psi$ is derivable at all, then so is $\psi[x_0/x]$ for a fresh x_0, by one more $\forall$E; so opening the box loses nothing. The goal step for $\exists$ is unsafe in the same way as the one for $\vee$: a witness chosen too early may be the wrong one, as in Example (Derivations with existential introduction), where "take $y = c$" stalled. That is why $\exists$ is postponed and the $\exists$E boxes of the premises are opened early: the witnesses they supply are often the terms the postponed goal is waiting for, as N_0 was in the limit derivation above.

4.8.6 Exercises

4.8.6.1 A. Check your understanding

Exercise. A1

(a) Translate into formulas, stating the domain: (i) every real number has a cube root; (ii) the function f is bounded on A; (iii) between any two different real numbers there is a rational number; (iv) the sequence (a_n) is eventually constant; (v) some natural number divides every natural number.

(b) Three students translate "some element of A is larger than every element of B", over $\mathbb{R}$:
   (i) $\exists x \in A\ \forall y \in B\ (x > y)$; (ii) $\exists x\,(x \in A \to \forall y \in B\ (x > y))$; (iii) $\forall y \in B\ \exists x \in A\ (x > y)$. Which are correct? For each wrong one, give sets A, B for which it has the wrong truth value.

(c) Which variables are free in your answer to (a)(iv)?

(d) Write the bounded rule $\exists x \in A$ I as a block of two unbounded steps, and name the two rules.

Solution. *(a)* (i) Over $\mathbb{R}$: $\forall x\,\exists y\,(y \cdot y \cdot y = x)$. (ii) Over $\mathbb{R}$: $\exists M\,\forall x \in A\,(|f(x)| \le M)$. (iii) Over $\mathbb{R}$, with $\in \mathbb{Q}$ a one-place predicate symbol: $\forall x\,\forall y\,(x < y \to \exists q \in \mathbb{Q}\ (x < q \wedge q < y))$; "different" is carried by $x < y$, which names the smaller number x (the case $y < x$ is the same statement with the letters exchanged). (iv) Over $\mathbb{R}$, with a a function symbol: $\exists N \in \mathbb{N}\,\forall n \in \mathbb{N}\,(n \ge N \to a_n = a_N)$. (v) Over $\mathbb{N}$: $\exists d\,\forall n\,\exists k\,(n = d \cdot k)$; it is true, with $d = 1$.

(b) Only (i) is correct. (ii) has $\to$ under $\exists$: with $A = \{0\}$ and $B = \{5\}$, the sentence in words is false, but (ii) is true, because $x = 7 \notin A$ makes the implication vacuously true. (iii) has the quantifiers in the wrong order: it lets the element of A depend on y. With $A = B = \mathbb{N}$, (iii) is true (for each y, take $x = y + 1$), and the sentence in words is false, since no natural number exceeds every natural number.

(c) None of the variables: N and n are bound. The formula is a statement about the function symbol a, which is not a variable.

(d) From $t \in A$ and $\varphi[t/x]$: first $t \in A \wedge \varphi[t/x]$ by $\wedge$I, then $\exists x\,(x \in A \wedge \varphi)$ by $\exists$I, which is $\exists x \in A\ \varphi$ unfolded.

4.8.6.2 B. Practice

Exercise. B1: Three definitions

Formalize each statement, stating the domain and the kind of each symbol (variable, constant, function symbol, predicate symbol; membership $\in A$ in a fixed set A is a one-place predicate symbol, as in Section 4.2).

(a) The set A of real numbers is bounded (above and below).

(b) The function $f\colon \mathbb{R} \to \mathbb{R}$ is increasing on A: for all $x, y \in A$, $x < y$ implies $f(x) \leq f(y)$.

(c) The sequence (a_n) is bounded: there is a real M with $|a_n| \leq M$ for all n.

Then, for (c), write the formula with the two quantifiers in the other order, and say why it holds for every sequence.

Solution. No constants occur in any of the three parts.

(a) Domain $\mathbb{R}$; m, M, x variables (all bound); $\in A$ a one-place and $\leq$ a two-place predicate symbol. $\exists m\, \exists M\, \forall x \in A\, (m \leq x \wedge x \leq M)$. The two witnesses are chosen before x, so they must work for every element of A.

(b) Domain $\mathbb{R}$; x, y variables (bound); f a one-place function symbol; $\in A$ a one-place and $<$, $\leq$ two-place predicate symbols. $\forall x \in A\, \forall y \in A\, (x < y \to f(x) \leq f(y))$.

(c) Domain $\mathbb{R}$; M, n variables (bound); a a one-place function symbol (the one-domain convention of Section 4.1), and $|\cdot|$ a one-place function symbol; $\in \mathbb{N}$ a one-place and $\leq$ a two-place predicate symbol. $\exists M\, \forall n \in \mathbb{N}\, (|a_n| \leq M)$.

Swapped: $\forall n \in \mathbb{N}\, \exists M\, (|a_n| \leq M)$. It holds for every sequence, because the witness may depend on n: take $M = |a_n|$. The sequence $a_n = n$, which is not bounded, satisfies it. As in Example (Swapping two quantifiers), the swap turns a statement about one M for all n into a statement about each term separately.

Exercise. B2: A bounded rehearsal

Using the bounded rules of Proposition 4.8.1, derive

$$\forall x \in A\, (P(x) \to Q(x)),\ \ \exists x \in A\, P(x) \ \vdash\ \exists x \in A\, Q(x).$$

Then say how many lines the derivation would have with the unbounded rules only.

Solution. The goal is an $\exists$: postpone it. The second premise is an $\exists$: open its bounded $\exists$E box at once, with a witness x_0 and the two assumptions $x_0 \in A$ and $P(x_0)$, and aim for the goal as the box's last line. Inside, the first premise is a $\forall x \in A$, and the term the box offers is x_0, with its bound on a line. The witness for the goal is x_0 again.

1		$\forall x \in A\,(P(x) \to Q(x))$	
2		$\exists x \in A\,P(x)$	
3	x_0	$x_0 \in A$	
4		$P(x_0)$	
5		$P(x_0) \to Q(x_0)$	$\forall$E 1, 3
6		$Q(x_0)$	$\to$E 5, 4
7		$\exists x \in A\,Q(x)$	$\exists$I 3, 6
8		$\exists x \in A\,Q(x)$	$\exists$E 2, 3–7

At line 8, x_0 is not in line 8, not in line 2, and not in either premise; no other box is open. The derivation uses three bounded steps (lines 5, 7 and 8), each of which costs one more line unbounded, so the unbounded derivation has $8 + 3 = 11$ lines: the $\exists$E box assumes $x_0 \in A \wedge P(x_0)$ and splits it with two $\wedge$E lines; line 5 becomes $\forall$E then $\to$E; line 7 becomes $\wedge$I then $\exists$I.

The bound is the whole point of the exercise: the witness x_0 must carry its membership in A from the premise to the conclusion, and the bounded rules make that one assumption and one citation.

4.8.6.3 C. Going deeper

Exercise. C1: Where the $\forall x$ stands

Let $f\colon \mathbb{R} \to \mathbb{R}$ and $A \subseteq \mathbb{R}$. Translate, over $\mathbb{R}$:

(a) f is **uniformly continuous** on A: for every $\varepsilon > 0$ there is $\delta > 0$ such that for all $x, y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$;

(b) f is continuous at every point of A: for every $x \in A$ and every $\varepsilon > 0$ there is $\delta > 0$ such that for every $y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$.

Then describe, in one sentence, the only difference between the two formulas, and say what it means for the choice of δ.

Solution. *(a)* $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon)$.

(b) $\forall x \in A\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon)$.

The only difference is the position of $\forall x \in A$: in (a) it stands after $\exists \delta > 0$, in (b) before it. In (b) the defender chooses δ knowing x, so δ may depend on x and on ε; in (a) δ is chosen before any x is given, so one δ must serve every $x \in A$ at once, and it depends on ε only. Section 4.10 derives (b) from (a) and shows that x^2 on $\mathbb{R}$ satisfies (b) but not (a); Section 4.11 turns this example into a counter-model, so (b) does not derive (a).

Exercise. C2: Six orders, four statements

Fix a formula μ (the "matrix") in the variables x, y, z, and consider the six sentences

obtained by putting the prefixes $\forall x$, $\exists y$, $\forall z$ in front of μ in the six possible orders: $\forall x\,\exists y\,\forall z\,\mu$, $\forall z\,\exists y\,\forall x\,\mu$, and so on. Call two of these orders **essentially the same** if the two sentences are provably equivalent ($\dashv\vdash$) for every matrix μ, and **essentially different** if some matrix over $\mathbb{Z}$ gives them different truth values (read informally, as in Section 4.1).

(a) Show that the six orders fall into four groups, with any two orders in one group essentially the same.

(b) Using the matrices $y > x$ and $y > z$ over $\mathbb{Z}$, show that any two orders from different groups are essentially different. So the six orders say four different things.

Solution (Hint 1). Two $\forall$'s side by side can be exchanged: that was one of the derivations of Example (Flagged boxes at work). What is left to decide is where $\exists y$ stands.

Solution (Hint 2). List the six orders by the position of $\exists y$: first, middle or last. When it is first or last, the two $\forall$'s are side by side.

Solution. (a) Sort the orders by the position of $\exists y$.

- $\exists y$ first: $\exists y\,\forall x\,\forall z\,\mu$ and $\exists y\,\forall z\,\forall x\,\mu$.
- $\exists y$ last: $\forall x\,\forall z\,\exists y\,\mu$ and $\forall z\,\forall x\,\exists y\,\mu$.
- $\exists y$ in the middle: $\forall x\,\exists y\,\forall z\,\mu$, and $\forall z\,\exists y\,\forall x\,\mu$.

Two adjacent $\forall$'s may be exchanged: $\forall x\,\forall z\,\psi \dashv\vdash \forall z\,\forall x\,\psi$ for every formula ψ, by the derivation of Example (Flagged boxes at work) that exchanges two universal quantifiers (it uses nothing about its matrix but that it is a formula). This gives the "last" pair at once. For the "first" pair the $\forall$'s sit under $\exists y$: open the $\exists$E box on $\exists y\,\forall x\,\forall z\,\mu$ with a witness y_0, exchange the two $\forall$'s inside it, and close with $\exists$I at y_0; the other direction is the same. So there are at most four groups: $\{\exists y\,\forall x\,\forall z,\ \exists y\,\forall z\,\forall x\}$, $\{\forall x\,\forall z\,\exists y,\ \forall z\,\forall x\,\exists y\}$, $\{\forall x\,\exists y\,\forall z\}$ and $\{\forall z\,\exists y\,\forall x\}$.

(b) Call the groups F ($\exists y$ first), L (last), M_x ($\forall x\,\exists y\,\forall z$) and M_z ($\forall z\,\exists y\,\forall x$). Over $\mathbb{Z}$, evaluate each group with the two matrices, reading informally (Section 4.1):

Group	$\mu = (y > x)$	$\mu = (y > z)$
F	false: no y exceeds every x	false: no y exceeds every z
L	true: given x, z, take $y = x + 1$	true: take $y = z + 1$
M_x	true: given x, take $y = x + 1$	false: no y exceeds every z
M_z	false: no y exceeds every x	true: given z, take $y = z + 1$

Every two rows differ in at least one column, so every two groups are separated by one of the two matrices: F from L, from M_x and from M_z by the column in which the other group has "true"; L from M_x by $y > z$ and from M_z by $y > x$; M_x from M_z by either. So the six orders say four different things: within a group the sentences are provably equivalent by (a), and across groups some matrix over $\mathbb{Z}$ tells them apart.

Marking scheme. (10 points) - 2: lists the six orders and sorts them by the position of $\exists y$. - 3: justifies the two collapses, citing the exchange of adjacent $\forall$'s, and handles the pair under $\exists y$ (by an $\exists$E box, or equivalently). - 3: gives, for each of the six pairs of groups, a matrix and the two truth values, with witnesses for the true sentences. - 2: concludes that the six orders say four different things, combining (a) and (b). Fatal errors (0 for the step): exchanging $\forall$ with $\exists$; a "separation" in which the two sentences have the same truth value.

4.9 Negating and Moving Quantifiers

To prove that a sequence does **not** converge, or that a function is **not** uniformly continuous, we must use the negation of a definition, and a negation is only usable once the $\neg$ has been pushed all the way in. "It is not the case that for every $\varepsilon > 0$ there is an N ..." gives a proof nothing to hold. "There is an $\varepsilon > 0$ such that for every N there is an $n \geq N$ with ..." tells it what to find. Section 4.1 read the step from the first form to the second informally, as the challenger and the defender changing sides. This section derives the laws that make it exact, with the rules of Sections 4.3–4.6. Once they are proved, negating a definition is mechanical: one law per symbol, from the outside in. The section also derives the laws that move a quantifier across a connective, and it ends with a short analysis proof in which a negation and a well-chosen ε do the work.

4.9.1 Not every, and some not

Section 4.1 called the two quantifiers mirror images: a single counterexample refutes a "for every", and a single witness confirms a "there exists". The mirror suggests four sequents. "Not every x satisfies φ" should say the same as "some x fails φ", and "no x satisfies φ" the same as "every x fails φ". Three of the four directions come straight out of the search method. The fourth does not, and the reason it does not is worth a close look.

To negate a quantifier, flip it and move the $\neg$ inside.

In the derivations of this section, $\neg\varphi[x_0/x]$ means $(\neg\varphi)[x_0/x]$, which is $\neg(\varphi[x_0/x])$ by Definition 4.2.5; the two readings give the same formula. The fresh name x_0 is always a variable different from x that does not appear in φ (or in ψ, when there is one); so it does not appear in $\forall x\,\varphi$ or $\exists x\,\varphi$ either. The side conditions of $\forall$I and $\exists$E need this, and it also makes x_0 free for x in φ (Section 4.2).

Proposition 4.9.1. Quantifier Negation

Let φ be a formula and x a variable.

(a) $\neg\forall x\,\varphi \vdash \exists x\,\neg\varphi$, by a derivation that uses RAA;

(b) $\exists x\,\neg\varphi \vdash \neg\forall x\,\varphi$, by a derivation that does not;

(c) $\neg\exists x\,\varphi \dashv\vdash \forall x\,\neg\varphi$, both directions by derivations that do not use RAA.

Try it first (ten minutes): run the search method of Section 4.8 on (a). Which rule should give the last line, and what would that rule need from the premise?

Idea. Start with (a), and with the obstacle. The goal $\exists x\,\neg\varphi$ is an $\exists$, so the search postpones it until an available line shows an instance $\neg\varphi[t/x]$. None ever does. The premise $\neg\forall x\,\varphi$ is a negation, and a negation is used only by clashing it with what it negates, $\forall x\,\varphi$. It says that a counterexample exists and names none, so $\exists$I has nothing to act on. We are stuck, and the last step of the method applies: assume $\neg\exists x\,\neg\varphi$, "no x fails φ", and hunt for a clash. The clash will be with the premise, so the new goal is $\forall x\,\varphi$. Open a flagged box for it, with the goal $\varphi[x_0/x]$. That goal is again stuck (we know nothing about φ), so a second RAA assumes $\neg\varphi[x_0/x]$; and now an instance of $\neg\varphi$ is on a line, so $\exists$I produces $\exists x\,\neg\varphi$, which clashes with the outer assumption.

For (b) and (c) the method runs without getting stuck. A negated goal opens a $\neg$I box (Section 2.2); an $\exists$ premise opens an $\exists$E box early (Section 4.6); a $\forall$ goal opens a flagged box (Section 4.4). The clash comes in one of two ways. In (b), and in (c) right to left, the instance assumed in the $\exists$E box clashes with the matching instance that $\forall$E takes from the $\forall$ line. In (c) left to right, $\exists$I turns the instance into $\exists x\,\varphi$, which meets the premise $\neg\exists x\,\varphi$.

Proof. For (a), the derivation:

1	$\neg\forall x\,\varphi$	
2	$\quad\neg\exists x\,\neg\varphi$	
3	$\quad x_0 \quad\quad \neg\varphi[x_0/x]$	
4	$\quad\quad\quad \exists x\,\neg\varphi$	$\exists$I 3
5	$\quad\quad\quad \bot$	$\neg$E 4, 2
6	$\quad\quad \varphi[x_0/x]$	RAA 3–5
7	$\quad\forall x\,\varphi$	$\forall$I 3–6
8	$\quad\bot$	$\neg$E 7, 1
9	$\exists x\,\neg\varphi$	RAA 2–8

Line 4 is $\exists$I at the term x_0, which is free for x in $\neg\varphi$. At line 7 the side condition of $\forall$I holds: x_0 does not appear in the premise, in the assumption $\neg\exists x\,\neg\varphi$ still open at line 7, or in $\forall x\,\varphi$, since it is not x and does not appear in φ; and line 6 is the last line of the flagged box.

For (b):

1	$\exists x\,\neg\varphi$	
2	$\quad\forall x\,\varphi$	
3	$\quad x_0 \mid \neg\varphi[x_0/x]$	
4	$\quad\quad\varphi[x_0/x]$	$\forall$E 2
5	$\quad\quad\bot$	$\neg$E 4, 3
6	$\quad\bot$	$\exists$E 1, 3–5
7	$\neg\forall x\,\varphi$	$\neg$I 2–6

At line 6, x_0 does not appear in $\bot$, in $\exists x\,\neg\varphi$, in the premise, or in the assumption $\forall x\,\varphi$ still open there.

For (c), left to right and right to left:

1	$\neg\exists x\,\varphi$	
2	$x_0 \mid \varphi[x_0/x]$	
3	$\quad\exists x\,\varphi$	$\exists$I 2
4	$\quad\bot$	$\neg$E 3, 1
5	$\neg\varphi[x_0/x]$	$\neg$I 2–4
6	$\forall x\,\neg\varphi$	$\forall$I 2–5

1	$\forall x\,\neg\varphi$	
2	$\quad\exists x\,\varphi$	
3	$\quad x_0 \mid \varphi[x_0/x]$	
4	$\quad\quad\neg\varphi[x_0/x]$	$\forall$E 1
5	$\quad\quad\bot$	$\neg$E 3, 4
6	$\quad\bot$	$\exists$E 2, 3–5
7	$\neg\exists x\,\varphi$	$\neg$I 2–6

In the first, line 5 is the instance of $\neg\varphi$ at x_0, as $\forall$I requires, and x_0 appears neither in the premise nor in $\forall x\,\neg\varphi$; no assumption is open at line 6. In the second, the side condition of $\exists$E at line 6 is checked as in (b). No line of the derivations of (b) and (c) is justified by RAA, and the derivation of (a) uses it twice, at lines 6 and 9. This proves the proposition.

■

So "not every" and "some not" are interderivable, and so are "none" and "every not". The direction that cost two RAA boxes is the one that **produces** an existential statement from a negative one. That is not an accident of our derivation: no derivation of (a) avoids RAA. This is a claim about all derivations, which no search can establish, and we state it without proof (Chapter 3's optional Theorem 3.5.4 proves the propositional analogue for $\neg\neg P \vdash P$). Chapter 5 returns to what is at stake: the conclusion of (a) asserts that something fails φ, and the derivation never says what. This is the law that *Analysis on the Real Line*, Chapter 1 uses at the end of the proof of the Archimedean property, where "not every n satisfies $n \leq x$" becomes "some n satisfies $n > x$".

Quick check. From the line $\neg\forall x\,P(x)$, may a derivation write $\neg P(c)$ for a constant c? If not, what may it write, and what does a derivation do next to use it?

Solution. No. $\neg\forall x\,P(x)$ says that **some** element fails P, not that the element named c does: over $\mathbb{Z}$, with P read as "is even" and c as 0, $\neg\forall x\,P(x)$ is true and $\neg P(0)$ is false. By (a) the

derivation may write $\exists x\, \neg P(x)$. To use that line it opens an $\exists$E box with a **fresh** witness x_0 and the assumption $\neg P(x_0)$ (Section 4.6), and nothing about c follows.

The laws as one line. From now on a derivation may use any law of this section as one line, citing it, in the way Proposition 4.8.1 allows a bounded rule: the line abbreviates the derivation printed here, with φ (and ψ) replaced by the formulas at hand and the fresh name chosen so that it appears nowhere in the larger derivation. Then every side condition inside the pasted block still holds, since the block's premises become lines of the larger derivation and the name is new to all of them. In a derivation the justification is a short tag: QN (a) for Proposition 4.9.1 (a), NIm for Proposition 4.9.2 below, and VI for Proposition 2.1.2.

4.9.2 The negation of an implication

Every definition with a bounded $\forall$ contains an implication, $\forall x\,(x \in A \to \varphi)$, so negating it means negating an implication. The answer is the one Section 4.1 found for the wrong translation of a bounded $\exists$: an implication fails exactly when its hypothesis holds and its conclusion does not. Chapter 2 met this law only as an exercise (Exercise (B1: Three searches) (b)), and a proof in the text may not rest on an exercise, so we prove it here.

Proposition 4.9.2. Negated Implication

For all formulas φ and ψ,

$$\neg(\varphi \to \psi) \dashv\vdash \varphi \wedge \neg\psi.$$

The direction $\vdash$ is derived with RAA, and the direction $\dashv$ without it.

Idea. For $\vdash$ the goal is a conjunction, so $\wedge$I asks for φ and $\neg\psi$ separately. The premise is a negation, usable only to clash with $\varphi \to \psi$, so in each part we manufacture $\varphi \to \psi$. For $\neg\psi$: open a $\neg$I box on ψ; then $\varphi \to \psi$ follows by $\to$I, whose box repeats ψ. For φ we are stuck, since φ has no known main connective; so RAA assumes $\neg\varphi$, and $\neg\varphi$ gives $\varphi \to \psi$ by Proposition 2.1.2. For $\dashv$, the goal is a negation: assume $\varphi \to \psi$ and clash ψ with $\neg\psi$.

Proof. The two derivations:

1	$\neg(\varphi \to \psi)$	
2	$\quad\neg\varphi$	
3	$\quad\varphi \to \psi$	VI 2
4	$\quad\bot$	$\neg$E 3, 1
5	φ	RAA 2–4
6	$\quad\psi$	
7	$\quad\quad\varphi$	
8	$\quad\quad\psi$	R 6
9	$\quad\varphi \to \psi$	$\to$I 7–8
10	$\quad\bot$	$\neg$E 9, 1
11	$\neg\psi$	$\neg$I 6–10
12	$\varphi \wedge \neg\psi$	$\wedge$I 5, 11

1	$\varphi \wedge \neg\psi$	
2	$\quad\varphi \to \psi$	
3	$\quad\varphi$	$\wedge$E 1
4	$\quad\psi$	$\to$E 2, 3
5	$\quad\neg\psi$	$\wedge$E 1
6	$\quad\bot$	$\neg$E 4, 5
7	$\neg(\varphi \to \psi)$	$\neg$I 2–6

Line 3 of the first derivation uses Proposition 2.1.2 as a derived rule (Definition 2.3.1), and line 8 repeats line 6 into the box, since $\to$I needs ψ as a line of its own box. The only RAA is at line 5.

■

The derivations use only the rules of Chapters 1 and 2, so the law holds for statements as well as formulas, and it may be used as a derived rule in the sense of Definition 2.3.1. Over the real numbers: "if $x > 2$ then $x^2 > 4$" fails for a given x exactly when $x > 2$ and $x^2 \leq 4$.

4.9.3 Bounded quantifiers keep their bounds

Analysis negates bounded quantifiers: $\neg\forall\varepsilon > 0$, $\neg\exists N \in \mathbb{N}$. The two laws so far combine into laws for them, and the combination has one surprise in it, which is no surprise once the bound is unfolded: **the bound stays as it is**.

Negating a bounded quantifier flips the quantifier and keeps the bound.

Proposition 4.9.3. Negating Bounded Quantifiers

Let A be a fixed set, so that $\in A$ is a one-place predicate symbol, let x be a variable and φ a formula. Then

(a) $\neg\forall x \in A\ \varphi \dashv\vdash \exists x \in A\ \neg\varphi$, where $\vdash$ is derived with RAA and $\dashv$ without;

(b) $\neg\exists x \in A\ \varphi \dashv\vdash \forall x \in A\ \neg\varphi$, both directions without RAA.

The same holds with $x \in A$ replaced by $x > a$, or by $x \geq a$, $x < a$, $x \leq a$, for a term a in which x does not occur.

So $\neg\forall\varepsilon > 0\ \varphi$ becomes $\exists\varepsilon > 0\ \neg\varphi$, and never $\exists\varepsilon \leq 0\ \neg\varphi$.

Idea. Unfold the bounds (Definition 4.1.1): (a) is $\neg\forall x\,(x \in A \to \varphi) \dashv\vdash \exists x\,(x \in A \wedge \neg\varphi)$. Push the $\neg$ through the quantifier by Proposition 4.9.1, and then through the implication by Proposition 4.9.2; the second step must happen under the $\exists$, so it happens inside an $\exists$E box. The negation lands on φ, and the bound $x \in A$ comes out of the implication unnegated. For (b), the derivations of Proposition 4.9.1 (c) go through with the bounded rules of Proposition 4.8.1 in place of the unbounded ones.

Proof. For (a), $\vdash$ and $\dashv$, with the bounds unfolded:

1	$\neg\forall x\,(x \in A \to \varphi)$	
2	$\exists x\,\neg(x \in A \to \varphi)$	QN (a) 1
3	x_0 | $\neg(x_0 \in A \to \varphi[x_0/x])$	
4	$x_0 \in A \wedge \neg\varphi[x_0/x]$	NIm 3
5	$\exists x\,(x \in A \wedge \neg\varphi)$	$\exists$I 4
6	$\exists x\,(x \in A \wedge \neg\varphi)$	$\exists$E 2, 3–5

1	$\exists x\,(x \in A \wedge \neg\varphi)$	
2	x_0 | $x_0 \in A \wedge \neg\varphi[x_0/x]$	
3	$\neg(x_0 \in A \to \varphi[x_0/x])$	NIm 2
4	$\exists x\,\neg(x \in A \to \varphi)$	$\exists$I 3
5	$\exists x\,\neg(x \in A \to \varphi)$	$\exists$E 1, 2–4
6	$\neg\forall x\,(x \in A \to \varphi)$	QN (b) 5

Line 3 on the left is the instance at x_0 of $\neg(x \in A \to \varphi)$, since A is not a variable; line 4 is Proposition 4.9.2 with $x_0 \in A$ and $\varphi[x_0/x]$ for φ and ψ. At line 6, x_0 does not appear in the conclusion, in line 2 or in the premise, and no assumption is open. The right-hand derivation is checked the same way. The only use of RAA is inside the two cited laws on the left: in QN (a) and in the direction $\vdash$ of NIm.

For (b), with the bounded rules:

1	$\neg\exists x \in A\,\varphi$	
2	x_0 | $x_0 \in A$	
3	| $\varphi[x_0/x]$	
4	| $\exists x \in A\,\varphi$	$\exists$I 2, 3
5	| $\bot$	$\neg$E 4, 1
6	$\neg\varphi[x_0/x]$	$\neg$I 3–5
7	$\forall x \in A\,\neg\varphi$	$\forall$I 2–6

1	$\forall x \in A\,\neg\varphi$	
2	| $\exists x \in A\,\varphi$	
3	| x_0 | $x_0 \in A$	
4	| | $\varphi[x_0/x]$	
5	| | $\neg\varphi[x_0/x]$	$\forall$E 1, 3
6	| | $\bot$	$\neg$E 4, 5
7	| $\bot$	$\exists$E 2, 3–6
8	$\neg\exists x \in A\,\varphi$	$\neg$I 2–7

At line 7 on the left, the box's own assumption $x_0 \in A$ has closed, and x_0 is in neither the

premise nor the conclusion; at line 7 on the right, x_0 is not in $\bot$, in line 2, in the premise, or in the open assumption on line 2. Neither derivation uses RAA.

For the bounds $x > a$ and the others, the same four derivations work with $x > a$ and $x_0 > a$ in place of $x \in A$ and $x_0 \in A$, choosing x_0 not to appear in a; since x does not occur in a, $(x > a)[x_0/x]$ is $x_0 > a$. This proves the proposition.

■

Warning. **Negate the quantifier, not the bound** (Chapter 5 lists this among the wrong negations it catalogues). The negation of "$a_n \to L$" begins $\exists \varepsilon > 0$, not $\exists \varepsilon \leq 0$. The wrong form, $\exists \varepsilon \leq 0\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ (n \geq N \wedge |a_n - L| \geq \varepsilon)$, is true for **every** sequence: $\varepsilon = 0$ works, since $|a_n - L| \geq 0$ always. So it would "prove" that the constant sequence $a_n = 0$ does not converge to 0.

4.9.4 Moving a quantifier across a connective

Negation is not the only thing that needs moving. A formula such as $\exists x\, P(x) \to \psi$ has its quantifier buried in a hypothesis, and proofs often want it outside, or the reverse. Two groups of laws do this. The **distribution** laws move a quantifier across $\wedge$ or $\vee$ when both sides mention x; the **passage** laws move it across a connective when one side does not.

Proposition 4.9.4. Distribution of Quantifiers

For all formulas φ, ψ and every variable x:

(a) $\forall x\,(\varphi \wedge \psi) \dashv\vdash \forall x\,\varphi \wedge \forall x\,\psi$;
(b) $\exists x\,(\varphi \vee \psi) \dashv\vdash \exists x\,\varphi \vee \exists x\,\psi$;
(c) $\forall x\,\varphi \vee \forall x\,\psi \vdash \forall x\,(\varphi \vee \psi)$;
(d) $\exists x\,(\varphi \wedge \psi) \vdash \exists x\,\varphi \wedge \exists x\,\psi$.

All four are derived without RAA.

The proposition collects laws that are mostly proved already. Its point is the comparison: $\forall$ goes across $\wedge$ and $\exists$ across $\vee$ in both directions, while $\forall$ across $\vee$ and $\exists$ across $\wedge$ go one way only.

Idea. Three of the four laws were derived in Sections 4.4 and 4.6 for $P(x)$ and $Q(x)$; only the fresh name's instances were used there, so formulas can take their place. For (c), the goal is a $\forall$ and the premise a disjunction: open the flagged box first, then split on the premise inside it.

Proof.

(a) is Proposition 4.4.1, (d) is Proposition 4.6.1, and (b) is the derivation of $\exists$ over $\vee$ in Example (Derivations with existential elimination). Those derivations are written with $P(x)$ and $Q(x)$; they work word for word with φ and ψ in their place, because they use

nothing about $P(x)$ and $Q(x)$ beyond the instances at the fresh name, which is chosen different from x and not appearing in φ or ψ.

For (c), the goal is a $\forall$: open a flagged box. Inside it the premise is a disjunction, so split on it; in each case one $\forall$E at x_0 and one $\vee$I reach the instance of $\varphi \vee \psi$:

1	$\forall x\, \varphi \vee \forall x\, \psi$	
2	x_0 | $\forall x\, \varphi$	
3	$\varphi[x_0/x]$	$\forall$E 2
4	$\varphi[x_0/x] \vee \psi[x_0/x]$	$\vee$I 3
5	$\forall x\, \psi$	
6	$\psi[x_0/x]$	$\forall$E 5
7	$\varphi[x_0/x] \vee \psi[x_0/x]$	$\vee$I 6
8	$\varphi[x_0/x] \vee \psi[x_0/x]$	$\vee$E 1, 2–4, 5–7
9	$\forall x\, (\varphi \vee \psi)$	$\forall$I 2–8

Line 8 is the instance of $\varphi \vee \psi$ at x_0 and the last line of the flagged box; at line 9, x_0 is in neither the premise nor the conclusion, and no assumption is open. This proves the proposition.

■

The converses of (c) and (d) are false. Over $\mathbb{Z}$, every integer is even or odd, but neither "every integer is even" nor "every integer is odd" holds; some integer is even and some integer is odd, but none is both. Section 4.11 turns these two observations into proofs that the converses are not derivable.

The passage laws need a condition: the quantified variable must not be free in the formula that the quantifier moves away from. Section 4.2 proved the fact that makes this work: if x is not free in ψ, then $\psi[t/x] = \psi$ for every term t. So an instance of $\psi \to \varphi$ at x_0 is $\psi \to \varphi[x_0/x]$, with ψ untouched.

Proposition 4.9.5. Passage Laws

Let φ and ψ be formulas and x a variable that is **not free in** ψ. Then

(a) $\forall x\, (\psi \to \varphi) \dashv\vdash \psi \to \forall x\, \varphi$;
(b) $\forall x\, (\varphi \to \psi) \dashv\vdash \exists x\, \varphi \to \psi$;
(c) $\exists x\, (\varphi \wedge \psi) \dashv\vdash \exists x\, \varphi \wedge \psi$;
(d) $\exists x\, (\varphi \to \psi) \dashv\vdash \forall x\, \varphi \to \psi$;
(e) $\exists x\, (\psi \to \varphi) \dashv\vdash \psi \to \exists x\, \varphi$.

Of the ten directions, two are derived here with RAA: $\dashv$ in (d), through LEM and the cited QN (a), and $\dashv$ in (e), through LEM. The other eight are derived without it.

Read (b) aloud: "everything with φ has ψ" says the same as "if something has φ, then ψ". In (b) and (d) the quantifier changes from $\forall$ to $\exists$ or back as it leaves the hypothesis of an implication.

That is the game reading of Section 4.1 once more: a quantifier in a hypothesis changes sides.

Idea. Each intuitionistic direction is a short run of the search method. A goal $\to$ opens a $\to$I box; a goal $\forall$ a flagged box; a premise $\exists$ an $\exists$E box, early; a premise $\forall$ is instantiated at x_0, where the condition on ψ keeps ψ unchanged; and the $\exists$ goal waits for its witness x_0. The two classical directions get stuck. In (d), the premise $\forall x\,\varphi \to \psi$ is usable only once $\forall x\,\varphi$ is in hand, and it may not be; so decide it, by LEM. If $\forall x\,\varphi$ holds, then ψ holds, and $\varphi \to \psi$ holds at every x. If not, (a) of Proposition 4.9.1 supplies an x_0 with $\neg\varphi[x_0/x]$, and $\varphi[x_0/x] \to \psi$ is vacuously true. In (e), decide ψ: if ψ holds, the premise supplies a witness; if not, $\psi \to \varphi$ is vacuously true at every x.

Proof. In every derivation below, x_0 is a variable other than x that does not appear in φ or ψ, and $\psi[x_0/x] = \psi$ because x is not free in ψ (Section 4.2). So the instance at x_0 of $\psi \to \varphi$ is $\psi \to \varphi[x_0/x]$, and likewise for the other combinations.

(a), $\vdash$ and $\dashv$:

1	$\forall x\,(\psi \to \varphi)$	
2	$\quad\psi$	
3	$\quad x_0 \quad \psi \to \varphi[x_0/x]$	$\forall$E 1
4	$\quad\quad \varphi[x_0/x]$	$\to$E 3, 2
5	$\quad\forall x\,\varphi$	$\forall$I 3–4
6	$\psi \to \forall x\,\varphi$	$\to$I 2–5

1	$\psi \to \forall x\,\varphi$	
2	$x_0 \quad \psi$	
3	$\quad\quad \forall x\,\varphi$	$\to$E 1, 2
4	$\quad\quad \varphi[x_0/x]$	$\forall$E 3
5	$\quad \psi \to \varphi[x_0/x]$	$\to$I 2–4
6	$\forall x\,(\psi \to \varphi)$	$\forall$I 2–5

At line 5 on the left the assumption ψ is open, and x_0 does not appear in it: this is where the fresh name must avoid ψ.

(b), $\vdash$ and $\dashv$:

1	$\forall x\,(\varphi \to \psi)$	
2	$\quad\exists x\,\varphi$	
3	$\quad x_0 \quad \varphi[x_0/x]$	
4	$\quad\quad \varphi[x_0/x] \to \psi$	$\forall$E 1
5	$\quad\quad \psi$	$\to$E 4, 3
6	$\quad\psi$	$\exists$E 2, 3–5
7	$\exists x\,\varphi \to \psi$	$\to$I 2–6

1	$\exists x\,\varphi \to \psi$	
2	$x_0 \quad \varphi[x_0/x]$	
3	$\quad\quad \exists x\,\varphi$	$\exists$I 2
4	$\quad\quad \psi$	$\to$E 1, 3
5	$\quad \varphi[x_0/x] \to \psi$	$\to$I 2–4
6	$\forall x\,(\varphi \to \psi)$	$\forall$I 2–5

At line 6 on the left, the conclusion ψ does not mention the witness x_0: this is where $\exists$E's side condition meets the condition on ψ.

(c), $\vdash$ and $\dashv$:

1	$\exists x\,(\varphi \wedge \psi)$			1	$\exists x\,\varphi \wedge \psi$		
2	x_0 | $\varphi[x_0/x] \wedge \psi$			2	$\exists x\,\varphi$		$\wedge$E 1
3	$\varphi[x_0/x]$		$\wedge$E 2	3	ψ		$\wedge$E 1
4	$\exists x\,\varphi$		$\exists$I 3	4	x_0 | $\varphi[x_0/x]$		
5	ψ		$\wedge$E 2	5	$\varphi[x_0/x] \wedge \psi$		$\wedge$I 4, 3
6	$\exists x\,\varphi \wedge \psi$		$\wedge$I 4, 5	6	$\exists x\,(\varphi \wedge \psi)$		$\exists$I 5
7	$\exists x\,\varphi \wedge \psi$		$\exists$E 1, 2–6	7	$\exists x\,(\varphi \wedge \psi)$		$\exists$E 2, 4–6

(d), $\vdash$:

1	$\exists x\,(\varphi \to \psi)$		
2	$\forall x\,\varphi$		
3	x_0 | $\varphi[x_0/x] \to \psi$		
4	$\varphi[x_0/x]$		$\forall$E 2
5	ψ		$\to$E 3, 4
6	ψ		$\exists$E 1, 3–5
7	$\forall x\,\varphi \to \psi$		$\to$I 2–6

(d), $\dashv$, with LEM on $\forall x\,\varphi$:

1	$\forall x\,\varphi \to \psi$		
2	$\forall x\,\varphi \vee \neg\forall x\,\varphi$		LEM
3	$\forall x\,\varphi$		
4	ψ		$\to$E 1, 3
5	φ		
6	ψ		R 4
7	$\varphi \to \psi$		$\to$I 5–6
8	$\exists x\,(\varphi \to \psi)$		$\exists$I 7
9	$\neg\forall x\,\varphi$		
10	$\exists x\,\neg\varphi$		QN (a) 9
11	x_0 | $\neg\varphi[x_0/x]$		
12	$\varphi[x_0/x] \to \psi$		VI 11
13	$\exists x\,(\varphi \to \psi)$		$\exists$I 12
14	$\exists x\,(\varphi \to \psi)$		$\exists$E 10, 11–13
15	$\exists x\,(\varphi \to \psi)$		$\vee$E 2, 3–8, 9–14

Line 8 is $\exists$I at the term x itself: $(\varphi \to \psi)[x/x]$ is $\varphi \to \psi$, as in Proposition 4.5.1, and as

there the step is sound because every domain is non-empty (Section 4.11). At line 14, x_0 is in neither the conclusion nor line 10, nor in the premise or the open assumption $\neg\forall x\,\varphi$.

(e), $\vdash$ and $\dashv$, the second with LEM on ψ:

1	$\exists x\,(\psi \to \varphi)$	
2	$\quad\psi$	
3	$\quad x_0 \mid \psi \to \varphi[x_0/x]$	
4	$\quad\quad\varphi[x_0/x]$	$\to$E 3, 2
5	$\quad\quad\exists x\,\varphi$	$\exists$I 4
6	$\quad\exists x\,\varphi$	$\exists$E 1, 3–5
7	$\psi \to \exists x\,\varphi$	$\to$I 2–6

1	$\psi \to \exists x\,\varphi$	
2	$\psi \vee \neg\psi$	LEM
3	$\quad\psi$	
4	$\quad\exists x\,\varphi$	$\to$E 1, 3
5	$\quad x_0 \mid \varphi[x_0/x]$	
6	$\quad\quad\psi$	
7	$\quad\quad\varphi[x_0/x]$	R 5
8	$\quad\psi \to \varphi[x_0/x]$	$\to$I 6–7
9	$\quad\exists x\,(\psi \to \varphi)$	$\exists$I 8
10	$\quad\exists x\,(\psi \to \varphi)$	$\exists$E 4, 5–9
11	$\quad\neg\psi$	
12	$\quad\psi \to \varphi$	VI 11
13	$\quad\exists x\,(\psi \to \varphi)$	$\exists$I 12
14	$\exists x\,(\psi \to \varphi)$	$\vee$E 2, 3–10, 11–13

Line 13 is again $\exists$I at the variable x. At line 10, x_0 is in neither the conclusion $\exists x\,(\psi \to \varphi)$ nor line 4, $\exists x\,\varphi$, nor in the premise or the open assumption ψ of line 3. In each of the eight other derivations, the side conditions are checked as in Proposition 4.9.1, and LEM and RAA do not occur. This proves the proposition.

■

Why the condition on ψ. Drop it, and (b) fails. Take φ and ψ both to be $P(x)$. Then $\forall x\,(P(x) \to P(x))$ is true in every domain, while $\exists x\,P(x) \to P(x)$, over $\mathbb{Z}$ with P read as "is even" and x read as 1, says "if some integer is even, then 1 is even", which is false. The derivation of (b) shows where the condition was used. Without it, line 4 would be $P(x_0) \to P(x_0)$, line 5 would be $P(x_0)$, and the $\exists$E box could not close on a line that mentions its witness.

Quick check. A student writes $\forall x\,(x > 0 \to \exists y\,(y \cdot y = x))$ as $\forall x\,\exists y\,(x > 0 \to y \cdot y = x)$, citing a passage law. Which law, and is the condition met?

Solution. Law (e) of Proposition 4.9.5, applied under $\forall x$ (a law may be used on a part under a quantifier; the next subsection says why), with ψ the formula $x > 0$, φ the formula $y \cdot y = x$, and y as the moving variable. The condition asks that y not be free in ψ, and y does not occur in $x > 0$; so the step is correct. It is the classical direction of (e), the one derived with LEM.

4.9.5 Negating a definition, mechanically

With the laws in hand, negating a definition is a procedure.

Algorithm. **Negating a formula.**

1. **Rewrite** $\leftrightarrow$ as two implications, by Proposition 2.4.1.
2. **Push** $\neg$ **inward**, one symbol at a time, from the outside, citing a law at each step:
   - $\neg\forall x$ becomes $\exists x\,\neg$, and $\neg\exists x$ becomes $\forall x\,\neg$ (Proposition 4.9.1); a bounded quantifier keeps its bound (Proposition 4.9.3);
   - $\neg(\alpha \to \beta)$ becomes $\alpha \wedge \neg\beta$ (Proposition 4.9.2);
   - $\neg(\alpha \wedge \beta)$ becomes $\neg\alpha \vee \neg\beta$, and $\neg(\alpha \vee \beta)$ becomes $\neg\alpha \wedge \neg\beta$ (De Morgan, Proposition 2.5.5);
   - $\neg\neg\alpha$ becomes α (Proposition 2.5.1).
3. **Stop at atomic formulas.** In an ordered domain, replace $\neg(s < t)$ by $s \geq t$ and $\neg(s \leq t)$ by $s > t$; these are facts about the order, not laws of logic.

Each step of 1 and 2 replaces a part of the formula by a part provably equivalent to it. Write that chain with $\dashv\vdash$ between the stages, not $\equiv$. Step 3 is not a law of logic: make its replacements after the chain, citing the order fact.

Why may a step act on a part buried inside the formula? Because provable equivalence passes through every symbol. If $\alpha \dashv\vdash \beta$, then $\neg\alpha \dashv\vdash \neg\beta$: from $\neg\alpha$, a $\neg$I box on β derives α from β and clashes it with $\neg\alpha$, and the other direction is the same with the letters exchanged. Also $\alpha \wedge \gamma \dashv\vdash \beta \wedge \gamma$, and the same holds for the other connectives, with the part on either side, by pasting the given derivations into the boxes of the connective's rules. Under a quantifier, $\forall x\,\alpha \vdash \forall x\,\beta$ by a flagged box: $\forall$E at x_0, the derivation of $\alpha[x_0/x] \vdash \beta[x_0/x]$, and $\forall$I; and $\exists x\,\alpha \vdash \exists x\,\beta$ by an $\exists$E box ending in $\exists$I. Here x_0 is fresh: a variable other than x that appears in neither α nor β. The derivation of $\alpha[x_0/x] \vdash \beta[x_0/x]$ is available because each law used holds for **every** formula that meets its condition, in particular for the instances at x_0. The laws this book applies to parts have only two kinds of condition, and both survive, for the same reason. A passage law of Proposition 4.9.5 asks that its moving variable y not be free in ψ; since y appears in β, the fresh x_0 is not y. If y is x itself, substitution stops at the quantifier on x, and $\psi[x_0/x]$ is ψ because x is not free in ψ; the pair is then the law's own instance, unchanged. Otherwise substituting x_0 for x passes the quantifier on y unchanged, and $\psi[x_0/x]$ has no free variable that ψ lacks except x_0, and y is not free in it either. A law of Proposition 4.9.3 with a bound such as $z > a$ asks that its bound variable z not occur in the term a; since z appears in β, again x_0 is not z. If z is x, substitution stops at the quantifier on x and the pair is unchanged; otherwise it

passes the quantifier on z, and $a[x_0/x]$ contains no variable that a lacks except x_0, so z does not occur in it. So the pair $\alpha[x_0/x]$, $\beta[x_0/x]$ is again an instance of the same law, with its condition met. A part sitting under several symbols is reached one symbol at a time.

Example. Three negations from analysis

Negate each statement into positive form: no $\neg$ in front of a quantifier, a connective or an inequality. Work over $\mathbb{R}$.

(a) $a_n \to L$: $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon)$.

(b) s is the supremum of A, in the two-clause form "s is an upper bound of A, and for every $\varepsilon > 0$ some element of A exceeds $s - \varepsilon$":

$$\forall x \in A\ (x \leq s)\ \wedge\ \forall \varepsilon > 0\ \exists x \in A\ (s - \varepsilon < x).$$

(c) f is uniformly continuous on A: $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon)$.

Solution. (a) Three quantifiers, then an implication, then an inequality:

$$\begin{aligned}
&\neg \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon) \\
\dashv\vdash\ &\exists \varepsilon > 0\ \neg \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon) \\
\dashv\vdash\ &\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \neg \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon) \\
\dashv\vdash\ &\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ \neg (n \geq N \to |a_n - L| < \varepsilon) \\
\dashv\vdash\ &\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ (n \geq N \wedge \neg (|a_n - L| < \varepsilon)).
\end{aligned}$$

The first three steps are Proposition 4.9.3, the fourth is Proposition 4.9.2. Last, the order fact turns $\neg(|a_n - L| < \varepsilon)$ into $|a_n - L| \geq \varepsilon$:

$$\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ (n \geq N \wedge |a_n - L| \geq \varepsilon).$$

In words: some $\varepsilon > 0$ is such that, however far out we start, some later term is at least ε away from L. Every bound stayed, and the order of the quantifiers stayed: ε, then N, then n.

(b) The outer symbol is $\wedge$. By De Morgan the negation becomes a disjunction, and each side is negated on its own:

$$\begin{aligned}
&\neg (\forall x \in A\ (x \leq s)\ \wedge\ \forall \varepsilon > 0\ \exists x \in A\ (s - \varepsilon < x)) \\
\dashv\vdash\ &\neg \forall x \in A\ (x \leq s)\ \vee\ \neg \forall \varepsilon > 0\ \exists x \in A\ (s - \varepsilon < x) \\
\dashv\vdash\ &\exists x \in A\ \neg (x \leq s)\ \vee\ \exists \varepsilon > 0\ \forall x \in A\ \neg (s - \varepsilon < x).
\end{aligned}$$

The first step is Proposition 2.5.5, (d) for $\vdash$ and (c) for $\dashv$; the second is Proposition 4.9.3, (a) on the left and (a) then (b) on the right. Last, the order facts turn $\neg(x \leq s)$ into $s < x$ and $\neg(s - \varepsilon < x)$ into $x \leq s - \varepsilon$:

$$\exists x \in A\ (s < x)\ \vee\ \exists \varepsilon > 0\ \forall x \in A\ (x \leq s - \varepsilon).$$

In words: s fails to be the supremum when some element of A lies above it (it is not an upper bound), or when some $s - \varepsilon$ with $\varepsilon > 0$ is still an upper bound (it is not the least).

(c) Four quantifiers and an implication:

$$\begin{aligned}
&\neg\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ \big(|x-y| < \delta \to |f(x)-f(y)| < \varepsilon\big)\\
\dashv\vdash\ &\exists\varepsilon > 0\ \forall\delta > 0\ \exists x \in A\ \exists y \in A\ \neg\big(|x-y| < \delta \to |f(x)-f(y)| < \varepsilon\big)\\
\dashv\vdash\ &\exists\varepsilon > 0\ \forall\delta > 0\ \exists x \in A\ \exists y \in A\ \big(|x-y| < \delta \wedge \neg(|f(x)-f(y)| < \varepsilon)\big),
\end{aligned}$$

where the first step is four uses of Proposition 4.9.3, and the second is Proposition 4.9.2. Last, the order fact turns $\neg(|f(x)-f(y)| < \varepsilon)$ into $|f(x)-f(y)| \geq \varepsilon$:

$$\exists\varepsilon > 0\ \forall\delta > 0\ \exists x \in A\ \exists y \in A\ \big(|x-y| < \delta \wedge |f(x)-f(y)| \geq \varepsilon\big).$$

In words: one $\varepsilon > 0$ defeats every δ: however small $\delta > 0$ is, two points of A closer than δ have values at least ε apart. To prove that a function is **not** uniformly continuous, this is the statement to prove, and it says exactly what to exhibit: first ε; then, for an arbitrary δ, the two points x and y, which may depend on δ. Section 4.10 does this for x^2 on $\mathbb{R}$.

The same procedure finishes the rewriting promised in Section 4.8 for "there is no largest prime". By Proposition 4.9.1 (c),

$$\begin{aligned}
&\neg\exists p\ \big(\mathrm{Prime}(p) \wedge \forall q\ (\mathrm{Prime}(q) \to q \leq p)\big)\\
\dashv\vdash\ &\forall p\ \neg\big(\mathrm{Prime}(p) \wedge \forall q\ (\mathrm{Prime}(q) \to q \leq p)\big).
\end{aligned}$$

A negated conjunction $\neg(\alpha \wedge \beta)$ is interderivable with $\alpha \to \neg\beta$: De Morgan's law (Proposition 2.5.5 (d)) and Proposition 2.5.6 (a) give one direction, and (b) of the latter with Proposition 2.5.5 (c) the other. Then Proposition 4.9.1 (a) and (b) turn $\neg\forall q\,(\mathrm{Prime}(q) \to q \leq p)$ into $\exists q\,\neg(\mathrm{Prime}(q) \to q \leq p)$, and Proposition 4.9.2, used under $\exists q$, pushes the last $\neg$ in; both steps act on a part, as the paragraph before Example (Three negations from analysis) allows:

$$\forall p\ \Big(\mathrm{Prime}(p) \to \exists q\ \big(\mathrm{Prime}(q) \wedge \neg(q \leq p)\big)\Big).$$

The quantifier $\forall q\,(\mathrm{Prime}(q) \to \dots)$ works like a bounded one, with the bound $\mathrm{Prime}(q)$. That bound is not one that Proposition 4.9.3 covers, but the same two steps work for it, and the bound stays unnegated.

Quick check. Negate "every non-empty set of reals that is bounded above has a supremum", writing $E(A)$ for "A is non-empty", $B(A)$ for "A is bounded above" and $S(A)$ for "A has a supremum", with the variable A ranging over sets of reals.

Solution. The statement is $\forall A\,\big((E(A) \wedge B(A)) \to S(A)\big)$. By Proposition 4.9.1 (a) and (b) and then Proposition 4.9.2, its negation is

$$\exists A\ \big((E(A) \wedge B(A)) \wedge \neg S(A)\big)\ :$$

there is a non-empty set of reals, bounded above, with no supremum. The two conditions are not negated: they stay as conditions on the counterexample, just as a bound stays with its quantifier.

Warning. **The negation of an implication is not an implication.** "If $x > 2$, then $x^2 > 4$" is not negated by "if $x > 2$, then $x^2 \leq 4$", nor by "if $x \leq 2$, then $x^2 \leq 4$". For $x = 0$ the original is true (its hypothesis fails), and so is the second candidate, so the candidate cannot be its negation. The negation is $x > 2 \wedge x^2 \leq 4$ (Proposition 4.9.2), and it is false for every real x. Chapter 5 catalogues this slip among the wrong negations.

4.9.6 How the laws look in a mathematical proof

Chapter 2 proved, in Example (A number below every positive number), that a number $x \geq 0$ below every positive number is 0. Its key step chose $\varepsilon := x/2$, and the reading after it said that "the hypothesis applies to it" is a rule for "every" that Chapter 4 would make precise. The rule is ∀E, in its bounded form (Proposition 4.8.1), at a term built from the data. Here is the same move once more, the way it appears in analysis.

Two real numbers that are closer than every positive number are equal.

Proposition 4.9.6. Arbitrarily Close Numbers Are Equal

Let a and b be real numbers. If $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.

Idea. The goal $a = b$ is atomic, so no introduction rule applies, and the hypothesis is a ∀ whose instances are all inequalities, never an equation. We are stuck, so suppose $a \neq b$. The hypothesis may be instantiated at any positive term, so choose the term that makes the numbers collide: $\varepsilon := |a - b|$, which is positive exactly because $a \neq b$.

Proof. Suppose that $|a - b| < \varepsilon$ for every $\varepsilon > 0$, and suppose, for a contradiction, that $a \neq b$ [an RAA box]. Then $a - b \neq 0$, so $|a - b| > 0$, by the school fact that $|u| > 0$ for every real $u \neq 0$. So the hypothesis applies to the term $|a - b|$ [bounded ∀E, citing the line $|a - b| > 0$] and gives $|a - b| < |a - b|$. No real number is less than itself, so this is a contradiction [¬E]. Hence $a = b$ [RAA].

■

The last step is a genuine RAA: the box assumed $\neg(a = b)$, written $a \neq b$, and concluded $a = b$; ¬I would only have given $\neg\neg(a = b)$. The hypothesis cannot be weakened to "for **some** $\varepsilon > 0$": $a = 0$, $b = \frac{1}{2}$, $\varepsilon = 1$ satisfy $|a - b| < \varepsilon$ and $a \neq b$. The facts about $|\cdot|$ used here are school algebra; *Analysis on the Real Line*, Chapter 1 proves them from the axioms of $\mathbb{R}$.

Read the result through the negation laws. Its contrapositive says: if $a \neq b$, then $\neg\forall\varepsilon > 0\,(|a - b| < \varepsilon)$, which by Proposition 4.9.3 is interderivable with $\exists\varepsilon > 0\,\neg(|a - b| < \varepsilon)$, that is, by the order fact, $\exists\varepsilon > 0\,(|a - b| \geq \varepsilon)$. The proof did not search for that ε: it named it, $|a - b|$, from the data. A proof by contradiction in analysis is usually this shape. Negate the goal; the negation hands over an object (here the positive number $|a - b|$); feed it to a hypothesis that holds for **every** object of its kind.

4.9.7 Exercises

4.9.7.1 A. Check your understanding

Exercise. A1

These parts are about the derivation of Proposition 4.9.1 (a).

(a) Which lines are justified by RAA, and what does each of the two boxes assume and conclude?

(b) Why can the derivation not reach $\exists x\, \neg\varphi$ by one $\exists$I directly from the premise?

(c) Line 4 is an $\exists$I at x_0. Why is its conclusion not the conclusion of the derivation, although the two formulas are the same?

(d) Run the derivation with φ the formula $x \cdot x \neq 2$, over $\mathbb{R}$. Write lines 1 and 9 in words. Does any line name the number $\sqrt{2}$? What more would be needed to reach $\exists x\, (x \cdot x = 2)$?

(e) True or false, with a reason: $\exists \varepsilon \leq 0\ \neg\varphi$ is a correct way to write the negation of $\forall \varepsilon > 0\ \varphi$.

Solution. *(a)* Lines 6 and 9. The inner box (lines 3–5) assumes $\neg\varphi[x_0/x]$, reaches $\bot$, and line 6 concludes $\varphi[x_0/x]$. The outer box (lines 2–8) assumes $\neg\exists x\, \neg\varphi$, reaches $\bot$, and line 9 concludes $\exists x\, \neg\varphi$.

(b) $\exists$I needs a line holding an instance $\neg\varphi[t/x]$. The premise $\neg\forall x\, \varphi$ is not such an instance (its main symbol is $\neg$ followed by $\forall$), and no rule takes it apart except $\neg$E, which needs $\forall x\, \varphi$. The premise says a counterexample exists without naming one.

(c) Line 4 lies inside three boxes: the box opened at line 2, and the flagged box of x_0 and the inner box, both opened at line 3. It depends on the assumptions on lines 2 and 3, so it is not available at the end. It is used only to clash with line 2. The conclusion on line 9 is reached by closing the outer box with RAA.

(d) Line 1: "not every real number has a square different from 2". Line 9: "some real number does not have a square different from 2", that is, $\exists x\, \neg(x \cdot x \neq 2)$. No line names $\sqrt{2}$: the only witness on any line is the fresh name x_0, inside boxes that closed, and line 9 comes from RAA, not from $\exists$I. To reach $\exists x\, (x \cdot x = 2)$, open an $\exists$E box on line 9 with the assumption $\neg(x_1 \cdot x_1 \neq 2)$, apply $\neg\neg$E (the formula $x_1 \cdot x_1 \neq 2$ is $\neg(x_1 \cdot x_1 = 2)$) to get $x_1 \cdot x_1 = 2$, and close with $\exists$I.

(e) False. By Proposition 4.9.3 the negation is $\exists \varepsilon > 0\ \neg\varphi$: the bound stays. The wrong form can be true together with the statement it claims to negate. Over $\mathbb{R}$, take φ to be $\varepsilon > 0$: then $\forall \varepsilon > 0\, (\varepsilon > 0)$ is true, and $\exists \varepsilon \leq 0\ \neg(\varepsilon > 0)$ is true as well, with the witness $\varepsilon = 0$. A statement and its negation are never both true.

4.9.7.2 B. Practice

Exercise. B1: Four negations

Negate each definition into positive form, citing the law used at each step. Work over $\mathbb{R}$, with (a_n) a sequence and f a function.

(a) (a_n) is bounded: $\exists M\ \forall n \in \mathbb{N}\ (|a_n| \le M)$.
(b) f is increasing on A: $\forall x \in A\ \forall y \in A\ (x < y \to f(x) \le f(y))$.
(c) $\lim_{x\to a} f(x) = L$: $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x\ ((0 < |x-a| \wedge |x-a| < \delta) \to |f(x) - L| < \varepsilon)$.
(d) (a_n) is a Cauchy sequence: $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m \in \mathbb{N}\ \forall n \in \mathbb{N}\ ((m \ge N \wedge n \ge N) \to |a_m - a_n| < \varepsilon)$.

Solution. *(a)* By Proposition 4.9.1 (c) (the $\exists M$ has no bound), then Proposition 4.9.3 (a) under $\forall M$:

$$\neg\exists M\ \forall n \in \mathbb{N}\ (|a_n| \le M) \dashv\vdash \forall M\ \neg\forall n \in \mathbb{N}\ (|a_n| \le M) \dashv\vdash \forall M\ \exists n \in \mathbb{N}\ \neg(|a_n| \le M),$$

and the order fact turns $\neg(|a_n| \le M)$ into $|a_n| > M$:

$$\forall M\ \exists n \in \mathbb{N}\ (|a_n| > M).$$

Every real number is exceeded by the size of some term.

(b) Two uses of Proposition 4.9.3, then Proposition 4.9.2 and the order fact:

$$\exists x \in A\ \exists y \in A\ (x < y \wedge f(x) > f(y)).$$

Two points of A, in increasing order, whose values are in decreasing order.

(c) Proposition 4.9.3 twice, Proposition 4.9.1 (a) and (b) for the unbounded $\forall x$, Proposition 4.9.2, and the order fact:

$$\exists \varepsilon > 0\ \forall \delta > 0\ \exists x\ ((0 < |x-a| \wedge |x-a| < \delta) \wedge |f(x) - L| \ge \varepsilon).$$

The hypothesis $0 < |x-a| < \delta$ is not negated: it moves, unchanged, into a conjunction.

(d) Proposition 4.9.3 four times, Proposition 4.9.2, the order fact:

$$\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists m \in \mathbb{N}\ \exists n \in \mathbb{N}\ ((m \ge N \wedge n \ge N) \wedge |a_m - a_n| \ge \varepsilon).$$

For some $\varepsilon > 0$, however far out we start, two later terms are at least ε apart.

Exercise. B2: Two passage laws in one derivation

Derive $\exists x\, P(x) \to \forall y\, Q(y) \vdash \forall x\, \forall y\, (P(x) \to Q(y))$ without RAA. Then name the two laws of Proposition 4.9.5 that the sequent combines, and say how.

Solution. The goal is $\forall x$, then $\forall y$, then $\to$: open two flagged boxes and a $\to$I box, in the goal's order. Inside, the premise is an implication whose hypothesis $\exists x\, P(x)$ comes from the assumption $P(x_0)$ by $\exists$I.

1	$\exists x\, P(x) \to \forall y\, Q(y)$	
2	x_0 | y_0 | $P(x_0)$	
3	$\exists x\, P(x)$	$\exists$I 2
4	$\forall y\, Q(y)$	$\to$E 1, 3
5	$Q(y_0)$	$\forall$E 4
6	$P(x_0) \to Q(y_0)$	$\to$I 2–5
7	$\forall y\,(P(x_0) \to Q(y))$	$\forall$I 2–6
8	$\forall x\, \forall y\,(P(x) \to Q(y))$	$\forall$I 2–7

At line 7, y_0 is not in the premise or in line 7, and no assumption is open (the box of line 2 has closed). At line 8 the same holds for x_0. No RAA is used.

The sequent combines (b) and (a) of Proposition 4.9.5. Law (b), with φ the formula $P(x)$ and ψ the formula $\forall y\, Q(y)$ (in which x is not free), turns the premise into $\forall x\,(P(x) \to \forall y\, Q(y))$. Then law (a), applied under $\forall x$ (a provable equivalence may be used under a quantifier, the paragraph after the negation procedure) with ψ the formula $P(x)$, φ the formula $Q(y)$ and y as the moving variable (y is not free in $P(x)$), turns $P(x) \to \forall y\, Q(y)$ into $\forall y\,(P(x) \to Q(y))$.

Exercise. B3: Choosing ε from the data

Let a and b be real numbers with $a < b + \varepsilon^2$ for every $\varepsilon > 0$. Prove that $a \le b$, by the method of Proposition 4.9.6: suppose not, and choose an ε built from the data that makes the numbers collide. Then explain why the choice $\varepsilon = a - b$ does not work.

Solution. Suppose $a < b + \varepsilon^2$ for every $\varepsilon > 0$, and suppose, for a contradiction, that $a > b$ (the negation of $a \le b$). Let $\varepsilon := \min(1, a - b)$. Then $\varepsilon > 0$, since $1 > 0$ and $a - b > 0$. Since $0 < \varepsilon \le 1$, $\varepsilon^2 = \varepsilon \cdot \varepsilon \le \varepsilon \cdot 1 = \varepsilon$, and $\varepsilon \le a - b$. The hypothesis applies to this ε and gives

$$a < b + \varepsilon^2 \le b + \varepsilon \le b + (a - b) = a,$$

so $a < a$, a contradiction. Hence $a \le b$.

The choice $\varepsilon = a - b$, copied from the text, gives $a < b + (a - b)^2$, that is, $a - b < (a - b)^2$. When $a - b > 1$ this is true (for $a - b = 2$: $2 < 4$), so there is no contradiction. The square shrinks a number only when it is at most 1, and the min with 1 is what puts ε there.

4.9.7.3 C. Going deeper

Exercise. C1: Two prenex forms

A formula is in **prenex form** when all its quantifiers stand at the front, before a part with no quantifier. Using the laws of Proposition 4.9.5, bring $\forall x\, P(x) \to \exists y\, Q(y)$ into prenex form in two ways, ending with $\exists x\, \exists y\,(P(x) \to Q(y))$ and with $\exists y\, \exists x\,(P(x) \to Q(y))$. Justify each step, and say which directions of the laws are classical.

Solution (Hint 1). The formula is an implication with a quantifier on each side. Law (d) moves the quantifier out of the hypothesis, and law (e) moves it out of the conclusion.

Solution (Hint 2). For the first route, start with law (d), taking ψ to be the whole conclusion $\exists y\, Q(y)$.

Solution. First route: x out first. Law (d) of Proposition 4.9.5, with φ the formula $P(x)$ and ψ the formula $\exists y\, Q(y)$, in which x is not free:

$$\forall x\, P(x) \to \exists y\, Q(y) \;\dashv\vdash\; \exists x\, (P(x) \to \exists y\, Q(y)).$$

Then law (e) under $\exists x$, with ψ the formula $P(x)$, φ the formula $Q(y)$, and y as the moving variable (y is not free in $P(x)$): $P(x) \to \exists y\, Q(y) \;\dashv\vdash\; \exists y\, (P(x) \to Q(y))$. A provable equivalence may be used under a quantifier (the paragraph after the negation procedure), so

$$\exists x\, (P(x) \to \exists y\, Q(y)) \;\dashv\vdash\; \exists x\, \exists y\, (P(x) \to Q(y)).$$

Second route: y out first. Law (e), with ψ the formula $\forall x\, P(x)$, φ the formula $Q(y)$, and y as the moving variable (y is not free in $\forall x\, P(x)$):

$$\forall x\, P(x) \to \exists y\, Q(y) \;\dashv\vdash\; \exists y\, (\forall x\, P(x) \to Q(y)).$$

Then law (d) under $\exists y$, with φ the formula $P(x)$ and ψ the formula $Q(y)$ (x is not free in $Q(y)$):

$$\exists y\, (\forall x\, P(x) \to Q(y)) \;\dashv\vdash\; \exists y\, \exists x\, (P(x) \to Q(y)).$$

Classical steps. Going from the given formula towards the prenex form, every step of both routes uses the direction $\dashv$ of (d) or of (e): the direction from the implication to the $\exists$. These are exactly the two directions that Proposition 4.9.5 derives with LEM. The way back uses only the intuitionistic directions.

Marking scheme. (10 points) - 3: first route, with law (d) then law (e), each with φ, ψ and the moving variable named, and the "not free" condition checked. - 3: second route, likewise with (e) then (d). - 2: says why a law may be applied under $\exists x$ or $\exists y$ (an equivalence used under a quantifier, by an $\exists$E box closed with $\exists$I). - 2: identifies the $\dashv$ directions of (d) and (e) as the classical ones, used in every step towards prenex form. Fatal errors (0 for the step): moving $\forall x$ out of the hypothesis as $\forall x$ (it becomes $\exists x$); applying a passage law when the moving variable is free in ψ.

Exercise. C2: Exit check: negations

(a) A sequence of functions f_n converges to f **uniformly** on A when

$$\forall \varepsilon > 0\; \exists N \in \mathbb{N}\; \forall n \in \mathbb{N}\; \forall x \in A\; (n \geq N \to |f_n(x) - f(x)| < \varepsilon),$$

where $f_n(x)$ is a term in the two variables n and x. Negate this into positive form, citing the laws.

(b) Using the negation of uniform continuity in Example (Three negations from analysis) (c), prove that $f(x) = 1/x$ is not uniformly continuous on $(0, 1)$. Give the witnesses that the negation asks for, each as a function of what is chosen before it, and check every condition.

Solution (Hint 1). For (b), the negation asks for one ε first. Near 0, the graph of $1/x$ is steep: two points close together have values far apart.

Solution (Hint 2). Take $\varepsilon = 1$ and let $\delta > 0$ be arbitrary. You must now find $x, y \in (0, 1)$, depending on δ, with $|x - y| < \delta$ and $|1/x - 1/y| \geq 1$.

Solution. *(a)* Proposition 4.9.3 four times, then Proposition 4.9.2 and the order fact:

$$\exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \in \mathbb{N}\ \exists x \in A\ \big(n \geq N \wedge |f_n(x) - f(x)| \geq \varepsilon\big).$$

In words: for some $\varepsilon > 0$, however far out we start, some later f_n is at least ε away from f at some point of A. The point may change with n.

(b) By Example (Three negations from analysis) (c), we must prove

$$\exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in (0,1)\ \exists y \in (0,1)\ \Big(|x - y| < \delta \wedge \Big|\frac{1}{x} - \frac{1}{y}\Big| \geq \varepsilon\Big).$$

Take $\varepsilon = 1$, which is positive. Let $\delta > 0$ be arbitrary, and let $t := \min(\delta, \frac{1}{2})$, so $0 < t \leq \frac{1}{2}$ and $t \leq \delta$; then take $x := t$ and $y := t/2$, which depend on δ. Check:

- $x, y \in (0, 1)$: $0 < t/2 < t \leq \frac{1}{2} < 1$.
- $|x - y| = t/2 < t \leq \delta$, since $t > 0$.
- $|1/x - 1/y| = |1/t - 2/t| = 1/t \geq 2 \geq 1 = \varepsilon$, since $t \leq \frac{1}{2}$.

Since $\delta > 0$ was arbitrary, the negation of uniform continuity holds, so $1/x$ is not uniformly continuous on $(0, 1)$.

Marking scheme. (10 points) - 3: (a) the negation, with every bound kept, $\wedge$ in place of $\to$, and the quantifiers in the order $\exists\varepsilon\ \forall N\ \exists n\ \exists x \in A$. - 2: (b) the witness ε, stated before δ is given. - 2: for an arbitrary $\delta > 0$, points x and y given as functions of δ. - 3: the three checks: $x, y \in (0, 1)$; $|x - y| < \delta$; $|f(x) - f(y)| \geq \varepsilon$. A different correct choice earns full marks (for instance $x = t$, $y = t/3$); the scheme rewards the order of the choices and the checks. Fatal errors (0 for the step): ε chosen depending on δ; points chosen before δ is given, or not depending on it where they must; a bound negated instead of the quantifier ($\exists\varepsilon \leq 0$, $\exists n < N$).

4.10 Uniformly Continuous Implies Continuous, Rule by Rule

This chapter opened with a question. Continuity and uniform continuity differ only in where one quantifier stands; why does one imply the other in a few lines, while the converse is false? Section 4.1 gave the informal answer: a variable may depend on the variables quantified to its left, and only on them. Exercise (C1: Where the $\forall x$ stands) of Section 4.8 asked for both translations into formulas, and Section 4.9 negated uniform continuity (Example (Three negations from analysis) (c)). This section answers the question with the rules. It derives "uniformly continuous implies continuous" in full, line by line, with the bounded rules of Proposition 4.8.1. It then reads off the derivation the one thing this section adds: **which boxes are open where a constant is chosen**, and what that says about what the constant may depend on. The same reading locates the exact lines at which the converse breaks down, and the function x^2 shows that the converse is not merely unproved but false.

4.10.1 Two definitions that differ in one place

Let A be a set of real numbers and f a function defined on A. We work over the domain $\mathbb{R}$, with f a one-place function symbol, $\in A$ a one-place predicate symbol (Section 4.2), and $|\cdot|$ the absolute value. *Analysis on the Real Line*, Chapters 5 and 6 define continuity and uniform continuity properly and develop them. Here the two definitions are only **quantifier patterns**, and nothing is used about f except that $f(x)$ is a term. (A function symbol has a value at every real number; the values outside A are never used, because f is only applied to variables bounded by $\in A$, as with sequences in Section 4.1.)

f is **uniformly continuous on** A (UC) when

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon).$$

f is **continuous at every point of** A (C) when

$$\forall x \in A\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon).$$

The two formulas have the same matrix and the same four bounded quantifiers. Only $\forall x \in A$ has moved: in C it stands **before** $\exists \delta > 0$, in UC **after** it. In the game of Section 4.1, the defender chooses δ in C after seeing x and ε, and in UC after seeing only ε. So in UC a single δ must serve every point of A at once: that is what **uniformly** means.

Quick check. Swap the first two quantifiers of UC, to get $\exists \delta > 0\ \forall \varepsilon > 0\ \forall x \in A\ \forall y \in A\ (|x - y| < \delta \to |f(x) - f(y)| < \varepsilon)$. What does the new formula say about two points of A closer than δ? Does $f(x) = x$ on $\mathbb{R}$ satisfy it?

Solution. Now δ is chosen before ε, so it must work for every $\varepsilon > 0$ at once. For two points $x, y \in A$ with $|x - y| < \delta$, this says $|f(x) - f(y)| < \varepsilon$ for every $\varepsilon > 0$, and by Proposition 4.9.6, $f(x) = f(y)$. So the formula says: for some $\delta > 0$, any two points of A closer than δ have the **same** value. The function $f(x) = x$ on $\mathbb{R}$ fails it: whatever $\delta > 0$ is proposed, the points 0 and $\delta/2$ are closer than δ and have different values. Yet $f(x) = x$ is uniformly continuous, with $\delta = \varepsilon$. Swapping $\forall \varepsilon$ and $\exists \delta$ asks far too much.

4.10.2 The derivation

Example. Uniformly continuous implies continuous

Derive C from UC, with the bounded rules of Proposition 4.8.1.

Idea. Run the search method of Section 4.8. The goal C begins $\forall x \in A$, then $\forall \varepsilon > 0$: the **goal-directed** steps open two bounded $\forall$I boxes, "let $x_0 \in A$" and "let $\varepsilon_0 > 0$", and write the last two lines. The new goal begins $\exists \delta > 0$, and no witness is in sight, so it is **postponed** (Section 4.5). Turn to the premise UC. It begins $\forall \varepsilon > 0$, and the term the goal mentions is ε_0, whose bound $\varepsilon_0 > 0$ is on a line: $\forall$E at ε_0 (Section 4.3). That gives an $\exists \delta > 0$ line, and its $\exists$E box is opened **at once**, with a witness δ_0 (Section 4.6), aiming for the postponed goal as its last line. Inside, the premise has become "δ_0 works for all $x, y \in A$"; instantiate it at the point we have, x_0. Now the postponed goal has its witness: δ_0 itself.

Before reading on: which lines does the line introducing δ_0 rest on, and does any of them mention x_0?

Solution.

1		$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y \rvert < \delta \to \lvert f(x) - f(y) \rvert < \varepsilon)$	
2	x_0	$x_0 \in A$	
3	ε_0	$\varepsilon_0 > 0$	
4		$\exists \delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y \rvert < \delta \to \lvert f(x) - f(y) \rvert < \varepsilon_0)$	$\forall$E 1, 3
5	δ_0	$\delta_0 > 0$	
6		$\forall x \in A\ \forall y \in A\ (\lvert x - y \rvert < \delta_0 \to \lvert f(x) - f(y) \rvert < \varepsilon_0)$	
7		$\forall y \in A\ (\lvert x_0 - y \rvert < \delta_0 \to \lvert f(x_0) - f(y) \rvert < \varepsilon_0)$	$\forall$E 6, 2
8		$\exists \delta > 0\ \forall y \in A\ (\lvert x_0 - y \rvert < \delta \to \lvert f(x_0) - f(y) \rvert < \varepsilon_0)$	$\exists$I 5, 7
9		$\exists \delta > 0\ \forall y \in A\ (\lvert x_0 - y \rvert < \delta \to \lvert f(x_0) - f(y) \rvert < \varepsilon_0)$	$\exists$E 4, 5–8
10		$\forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in A\ (\lvert x_0 - y \rvert < \delta \to \lvert f(x_0) - f(y) \rvert < \varepsilon)$	$\forall$I 3–9
11		$\forall x \in A\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall y \in A\ (\lvert x - y \rvert < \delta \to \lvert f(x) - f(y) \rvert < \varepsilon)$	$\forall$I 2–10

Every line is tagged with its rule; lines 4, 7 and 8 are the one-line bounded rules, and lines 9, 10 and 11 close bounded boxes. The terms and the side conditions:

- Line 4: $\forall$E at ε_0, with its bound on line 3. The term ε_0 is a variable that does not appear in line 1, so it is free for ε.
- Line 7: $\forall$E at x_0, with its bound on line 2, which is available because the box of line 2 is still open.
- Line 8: $\exists$I with the witness δ_0, its bound on line 5 and the instance on line 7.
- Line 9 ($\exists$E, flag δ_0): δ_0 does not appear in line 9 (the witness was forgotten at line 8), in line 4, in the premise, or in the assumptions still open at line 9, lines 2 and 3.
- Line 10 ($\forall$I, flag ε_0): ε_0 does not appear in the premise, in line 10, or in the assumption still open at line 10, line 2.

- Line 11 (∀I, flag x_0): x_0 does not appear in the premise or in line 11, and no assumption is open.

The answer to the prompt: line 5 is the assumption of the ∃E box, so it rests on line 4, which rests on lines 1 and 3. None of the three mentions x_0. The witness δ_0 was obtained from the premise at ε_0 alone, before x_0 played any part. That is uniformity, read off the lines.

The derivation has 11 lines. Written with the unbounded rules it has 17, since each of its six bounded steps (lines 4, 7, 8, 9, 10 and 11) costs one line more (Section 4.8). The bounded rules keep it at the level at which a mathematician thinks, and the same argument in prose is three sentences, each aligned with the lines it abbreviates:

> Let $x \in A$ and $\varepsilon > 0$ [lines 2, 3, 10, 11]. By uniform continuity there is $\delta > 0$ such that $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon$ for all $u, v \in A$; choose such a δ [4, 5, 6, 9]. Applied with $u = x$, this gives $|f(x) - f(y)| < \varepsilon$ for every $y \in A$ with $|x - y| < \delta$, so this δ works for x and ε [7, 8].

The prose renames the bound variables of UC to u, v, so that "x" can name the fixed point; a derivation needs no such care, because ∀E handles bound variables by substitution. Chapter 5 studies this passage from a derivation to prose, at several levels of detail.

4.10.3 Reading dependence off the boxes

The derivation shows something the informal proof leaves unsaid: the exact place where δ is chosen, and what surrounds it there.

A constant chosen in a box may depend on the names of the boxes open around that box, and it does depend only on the names in the line it comes from.

Quick check. In the derivation above, which flagged boxes surround the box of δ_0, opened at line 5? What may δ_0 therefore depend on, and what does it in fact depend on?

Solution. The boxes of x_0 (opened at line 2) and of ε_0 (opened at line 3). So δ_0 may depend on x_0 and on ε_0, as C allows. In fact it depends only on ε_0: it comes from line 4, which mentions ε_0 and not x_0.

Why "may"? Inside the boxes of x_0 and ε_0, every term built from x_0 and ε_0 is available, so a premise could have been instantiated at such a term, and a witness obtained from the resulting ∃ line would depend on both names. The side conditions permit that here, because the goal C puts $\exists\delta$ inside both ∀'s: the witness is forgotten by ∃I (line 8) before either box closes. Why "does"? Because everything known about δ_0 is what its ∃ line says, and line 4 mentions ε_0 alone. So δ_0 could have been obtained before x_0 entered at all: fix ε_0, obtain δ_0, and only then let $x_0 \in A$. That is the order of UC itself. A derivation in that order nests the boxes as ε_0, δ_0, x_0 (the box of δ_0 rests on line 4, so it stays inside the box of ε_0), and it ends in $\forall\varepsilon > 0\ \forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\dots)$, which is C with its first two quantifiers exchanged. C follows by exchanging them back, as in the swap of Example (Flagged boxes at work) (c), here with the bounded rules: boxes for $x_1 \in A$ and $\varepsilon_1 > 0$, ∀E at ε_1 and then at x_1, and two ∀I lines. C's order of choices allows more dependence than this proof uses.

The limit derivation of Section 4.8 showed the same thing on a smaller scale: its witness N_0 came from a line obtained at $\varepsilon_0/2$, inside the box of ε_0 only, and so it depended on ε_0.

4.10.4 The converse breaks at a side condition

The converse, from C to UC, is the interesting direction. The search for it fails, and the failure has an exact address.

Example. The search from C to UC

Run the search method on the sequent with premise C and goal UC, and find where it stops.

Try it first (ten minutes): start the search. Where does a δ come from, and at what moment can you get one?

Idea. The obvious plan copies the derivation above: fix ε_0, obtain δ_0 from the premise, choose it as the witness. It runs into an obstacle at the second step. In C, $\exists\delta$ stands behind $\forall x \in A$, so to reach it the premise must first be instantiated at a point, and the bounded $\forall$E needs a line $t \in A$. No such line exists until we assume one, in a box for some x_0. The δ_0 obtained inside that box comes from a line that mentions x_0, and both side conditions then hold it in: it cannot leave its own box while a line about it still mentions it, and x_0 cannot be generalized while a line about δ_0 still mentions x_0.

Solution. The goal UC begins $\forall\varepsilon > 0$, so open a box for ε_0. The new goal $\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\dots)$ is postponed. The premise C begins $\forall x \in A$, and no line $t \in A$ is available; the only way to get one is to assume it, so open a bounded box "let $x_0 \in A$". Then the premise yields a δ. Here is how far the search gets, every line legal:

1	$\forall x \in A\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$		
2	ε_0 | $\varepsilon_0 > 0$		
3	x_0 | $x_0 \in A$		
4	$\forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon)$		$\forall$E 1, 3
5	$\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$		$\forall$E 4, 2
6	δ_0 | $\delta_0 > 0$		
7	$\forall y \in A\ (\lvert x_0-y\rvert < \delta_0 \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$		
8	$\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$		$\exists$I 6, 7
9	$\exists\delta > 0\ \forall y \in A\ (\lvert x_0-y\rvert < \delta \to \lvert f(x_0)-f(y)\rvert < \varepsilon_0)$		$\exists$E 5, 6–8
10	$\forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon_0)$		$\forall$I 3–9
11	$\forall\varepsilon > 0\ \forall x \in A\ \exists\delta > 0\ \forall y \in A\ (\lvert x-y\rvert < \delta \to \lvert f(x)-f(y)\rvert < \varepsilon)$		$\forall$I 2–10

The goal inside the box of ε_0 needs a δ that works for **all** points: the line $\forall x \in A\ \forall y \in A\ (\lvert x-y\rvert < \delta_0 \to \lvert f(x)-f(y)\rvert < \varepsilon_0)$, so that $\exists$I at δ_0 could finish. Line 7 says this only for the point x_0. There are two ways to try to generalize it, and each is stopped by a side condition.

- **Generalize inside the box of δ_0.** After line 7, open a new flagged box with the name x_0 and the assumption $x_0 \in A$, repeat line 7 in it, and close it by $\forall$I with the conclusion $\forall x \in A\ \forall y \in A\ (|x - y| < \delta_0 \to \dots)$. That line breaks the side condition of $\forall$I: x_0 appears in assumptions still open there, line 3 and line 7. (With a different name x_1 the box is legal, but line 7 says nothing about x_1, so the box cannot reach its last line.)
- **Leave the box of δ_0 first, carrying line 7.** Close the $\exists$E box with line 7 itself as its conclusion, in order to generalize over x_0 outside it. That line breaks the side condition of $\exists$E: its conclusion mentions the witness δ_0.

The only legal way out of the box of δ_0 is to forget δ_0, as line 8 does with $\exists$I before line 9 closes the box. Then $\forall$I on x_0 is legal, and line 10 is what it gives: $\forall x \in A\ \exists \delta > 0$, with δ still behind x. That is C at ε_0 again, not UC, and closing the box of ε_0 gives line 11, which is C with its first two quantifiers exchanged. The search has gone round in a circle.

These are the same two blocks that stopped the converse of Proposition 4.6.2: a $\forall$I blocked by an open assumption that mentions its name, and an $\exists$E box that cannot close on a line mentioning its witness. That is no coincidence: with the bounds and the matrix stripped away, C to UC is the swap $\forall x\, \exists \delta$ to $\exists \delta\, \forall x$.

A failed search proves nothing about all derivations: a cleverer route might exist. For this sequent none does. The evidence is a function for which C is true and UC is false, and Section 4.11 explains why such a function settles the question.

4.10.5 A function that is continuous and not uniformly continuous

Example. The square function on the real line

Let $f(x) = x^2$ on $A = \mathbb{R}$. Show that f satisfies C and does not satisfy UC.

Solution. C holds. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. Let

$$\delta := \min\Big(1,\ \frac{\varepsilon}{2|x| + 1}\Big),$$

which is positive, since $\varepsilon > 0$ and $2|x| + 1 \geq 1 > 0$. It depends on ε **and on** x. Let $y \in \mathbb{R}$ with $|x - y| < \delta$. By the triangle inequality (Example (The triangle inequality by cases)), used twice, and since $\delta \leq 1$, $|y| \leq |x| + |y - x| < |x| + 1$, so $|x + y| \leq |x| + |y| < 2|x| + 1$. Hence

$$|x^2 - y^2| = |x - y| \cdot |x + y| \leq |x - y|\,(2|x| + 1) < \delta\,(2|x| + 1) \leq \varepsilon,$$

where the strict step uses $2|x| + 1 > 0$, and the last step uses $\delta \leq \varepsilon/(2|x| + 1)$. So this δ works for x and ε.

UC fails. By the negation of uniform continuity in Example (Three negations from analysis) (c), with the bounds $x \in \mathbb{R}$ and $y \in \mathbb{R}$ dropped because they always hold, we must show

$$\exists \varepsilon > 0\ \forall \delta > 0\ \exists x\ \exists y\ (|x - y| < \delta \wedge |x^2 - y^2| \geq \varepsilon).$$

Take $\varepsilon = 1$. Let $\delta > 0$ be arbitrary, and take $x := 1/\delta$ and $y := 1/\delta + \delta/2$, which depend on

δ. Then $|x-y| = \delta/2 < \delta$, and, since $y > x > 0$ (so $y-x$ and $y+x$ are both positive),

$$|x^2 - y^2| = (y-x)(y+x) = \frac{\delta}{2}\left(\frac{2}{\delta} + \frac{\delta}{2}\right) = 1 + \frac{\delta^2}{4} \geq 1.$$

Since δ was arbitrary, UC fails.

The δ for C grows smaller as $|x|$ grows, and no single δ serves the whole line. The proof of failure says why: however small δ is, far enough out (at $x = 1/\delta$) the graph is so steep that two points $\delta/2$ apart have values more than 1 apart. The function is the semantic side of Example (The search from C to UC). Read in the real numbers, with f as squaring, the premise C is true and the conclusion UC is false; Section 4.11 calls such a reading a **counter-model**, and explains why a counter-model rules out every derivation, so that no derivation of UC from C exists.

Warning. **"Depending on ε" is a claim, and it can be false.** When a proof says "choose δ (depending on ε)", the parenthesis records which boxes are open where δ is chosen, and which names its formula uses. This is the logical origin of the book's rule that every chosen constant says what it depends on: the dependence is a record of the open boxes, and a reader can check it. A proof of uniform continuity must be able to write "depending on ε" truthfully. A "proof" that x^2 is uniformly continuous on $\mathbb{R}$ that takes $\delta = \min(1, \varepsilon/(2|x|+1))$ "(depending on ε)" has a formula for δ that mentions x: the δ was chosen inside the box of x, so the proof shows only C, and Example (The square function on the real line) shows that UC is false.

4.10.6 Exercises

4.10.6.1 A. Check your understanding

Exercise. A1

These parts are about the derivation of Example (Uniformly continuous implies continuous).

(a) Why is line 7 legal? Name the rule, the term, and the lines it cites, and say why those lines are available.

(b) At which line is the premise's "for all $x \in A$" used, and at which term?

(c) State the idea of the derivation in two sentences.

(d) Run the derivation for $f(x) = 3x + 1$ on $A = \mathbb{R}$, where UC holds with $\delta = \varepsilon/3$. For a given x_0 and $\varepsilon_0 > 0$, which δ does the derivation hand to C at line 8, and does it depend on x_0? Check that it works.

Solution. *(a)* Line 7 is the bounded $\forall$E at the term x_0. It cites line 6, $\forall x \in A\, \forall y \in A\, (\dots)$, and line 2, $x_0 \in A$, the bound's instance. Line 6 is the assumption of the $\exists$E box that contains line 7, and line 2 is the assumption of the box of x_0, which also contains line 7; both boxes are still open at line 7. The term x_0 is a variable not appearing in line 6, so it is free for x.

(b) At line 7, at the term x_0. Line 6 says that δ_0 works for **all** pairs of points of A, and line 7 uses that for pairs with first point x_0.

(c) Uniform continuity at ε supplies one δ that works for all points at once, and it is supplied before any point is named. Continuity at a point x then takes that same δ, specialized to the point x.

(d) Line 4 says that some $\delta > 0$ works at ε_0, and line 8 hands C that witness δ_0, whatever it is; the derivation knows only lines 5 and 6 about it. Here $\varepsilon_0/3$ is such a witness (so is any δ with $0 < \delta \leq \varepsilon_0/3$). Reading the derivation with the witness the exercise names, $\delta_0 = \varepsilon_0/3$, the δ handed to C is $\varepsilon_0/3$. It does not depend on x_0. It works: for every real y with $|x_0 - y| < \varepsilon_0/3$,

$$|f(x_0) - f(y)| = |(3x_0 + 1) - (3y + 1)| = 3|x_0 - y| < 3 \cdot \frac{\varepsilon_0}{3} = \varepsilon_0.$$

4.10.6.2 B. Practice

Exercise. B1: Lipschitz implies uniformly continuous

A function f is **Lipschitz** on A when

$$\exists L \geq 0\ \forall x \in A\ \forall y \in A\ (|f(x) - f(y)| \leq L \cdot |x - y|).$$

Prove that a function that is Lipschitz on A is uniformly continuous on A, first as a derivation with the bounded rules (lines of arithmetic justified "arith"), then as a short proof in prose. Say what your δ depends on. (The bound $L \geq 0$ is part of the definition: if A has at most one point, then $x = y$ in the matrix, and any L, even $L = -1$, would satisfy it. The bound rules this out, so a proof may use $L \geq 0$, for instance to divide by $L + 1$.)

Solution. The goal UC begins $\forall \varepsilon > 0$: open a box for ε_0. The new goal $\exists \delta > 0 \ldots$ is postponed until a witness is in hand. The premise begins $\exists L \geq 0$: open its $\exists$E box at once, with a witness L_0, aiming for the postponed goal, which does not mention L_0. Work backwards from what the boxes for x_0 and y_0 will need: from $|x_0 - y_0| < \delta$ we want $|f(x_0) - f(y_0)| < \varepsilon_0$, and the premise gives $|f(x_0) - f(y_0)| \leq L_0|x_0 - y_0| < L_0\delta$ when $L_0 > 0$. So $\delta = \varepsilon_0/L_0$ is the natural guess, but $L_0 = 0$ is allowed and then ε_0/L_0 is undefined. Take $\delta = \varepsilon_0/(L_0 + 1)$ instead: $L_0 + 1 \geq 1 > 0$, and $L_0\delta = \varepsilon_0 \cdot L_0/(L_0 + 1) < \varepsilon_0$.

1		$\exists L \geq 0\, \forall x \in A\, \forall y \in A\, (\lvert f(x) - f(y)\rvert \leq L \cdot \lvert x - y\rvert)$	
2	ε_0	$\varepsilon_0 > 0$	
3		L_0 $L_0 \geq 0$	
4		$\forall x \in A\, \forall y \in A\, (\lvert f(x) - f(y)\rvert \leq L_0 \cdot \lvert x - y\rvert)$	
5		$\varepsilon_0/(L_0{+}1) > 0$	arith 2, 3
6		x_0 $x_0 \in A$	
7		y_0 $y_0 \in A$	
8		$\lvert x_0 - y_0\rvert < \varepsilon_0/(L_0{+}1)$	
9		$\forall y \in A\, (\lvert f(x_0) - f(y)\rvert \leq L_0 \cdot \lvert x_0 - y\rvert)$	∀E 4, 6
10		$\lvert f(x_0) - f(y_0)\rvert \leq L_0 \cdot \lvert x_0 - y_0\rvert$	∀E 9, 7
11		$\lvert f(x_0) - f(y_0)\rvert < \varepsilon_0$	arith 2, 3, 8, 10
12		$\lvert x_0 - y_0\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x_0) - f(y_0)\rvert < \varepsilon_0$	→I 8–11
13		$\forall y \in A\, (\lvert x_0 - y\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∀I 7–12
14		$\forall x \in A\, \forall y \in A\, (\lvert x - y\rvert < \varepsilon_0/(L_0{+}1) \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∀I 6–13
15		$\exists \delta > 0\, \forall x \in A\, \forall y \in A\, (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∃I 5, 14
16		$\exists \delta > 0\, \forall x \in A\, \forall y \in A\, (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∃E 1, 3–15
17		$\forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in A\, \forall y \in A\, (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	∀I 2–16

Line 11 is arithmetic from lines 2, 3, 8 and 10: $|f(x_0)-f(y_0)| \leq L_0|x_0-y_0| \leq L_0 \cdot \varepsilon_0/(L_0+1)$, where the second step uses $L_0 \geq 0$, and $L_0/(L_0+1) < 1$ with $\varepsilon_0 > 0$ gives $L_0\varepsilon_0/(L_0+1) < \varepsilon_0$. Line 15 is the bounded ∃I at the term $\varepsilon_0/(L_0+1)$, with its bound on line 5. At line 16, L_0 does not appear in line 16 (the witness term was forgotten at line 15), in line 1, which is the premise, or in the assumption $\varepsilon_0 > 0$ still open there. At lines 13, 14 and 17 the flags y_0, x_0, ε_0 do not appear in the premise, in the line itself, or in the assumptions then open (among lines 2, 3, 4 and 6).

In prose: *Let $L \geq 0$ be a constant with $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in A$. Let $\varepsilon > 0$, and let $\delta := \varepsilon/(L+1)$, which is positive and depends on ε (and on the constant L, which is fixed before ε), but not on any point. For $x, y \in A$ with $|x-y| < \delta$, $|f(x)-f(y)| \leq L|x-y| \leq L\delta = \varepsilon L/(L+1) < \varepsilon$. So f is uniformly continuous on A.* The box of L_0 is open where δ is chosen, so δ may depend on L_0. That is harmless: L_0 is a constant of the function, not a point, and its box could as well have been opened first, outside the box of ε_0.

Exercise. B2: Put the proof in order

Below are ten sentences. Eight of them, in a suitable order, form a proof that a function f uniformly continuous on A is continuous at every point of A; two are distractors. Write down an order for the eight, and say why each distractor is wrong.

(i) Since x and y lie in A and $|x - y| < \delta$, the choice of δ, applied to $u = x$ and $v = y$, gives $|f(x) - f(y)| < \varepsilon$.

(ii) Let $\delta > 0$ be arbitrary.

(iii) Let $\varepsilon > 0$.

(iv) Since $\varepsilon > 0$ and $x \in A$ were arbitrary, f is continuous at every point of A.

(v) Suppose f is uniformly continuous on A.

(vi) By uniform continuity, applied to ε, choose $\delta > 0$ such that for all $u, v \in A$, $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon$.

(vii) Choose $\delta > 0$ for this x and this ε, and then let $x \in A$ be arbitrary.

(viii) Let $x \in A$.

(ix) So for every $y \in A$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$: this δ works for x and ε.

(x) Let $y \in A$ with $|x - y| < \delta$.

Solution. An admissible order: (v), (viii), (iii), (vi), (x), (i), (ix), (iv).

(v) is the hypothesis of the implication being proved, so it opens the proof (its $\to$I box is the outermost one). After it, other orders are admissible as long as each sentence comes after what it uses. (vi) uses the ε of (iii); (x) uses the δ of (vi) and the x of (viii); then (i), (ix), (iv) follow in that order. So exactly three orders are admissible: (viii), (iii), (vi); or (iii), (viii), (vi); or (iii), (vi), (viii), each preceded by (v) and followed by (x), (i), (ix), (iv). Only the first matches the derivation of Example (Uniformly continuous implies continuous) directly. The second and the third fix ε before x; the third even chooses δ before any point is given, which is possible because uniform continuity supplies δ from ε alone. They correspond to variants of that derivation in which the box of ε_0 (in the third, also that of δ_0) is opened outside the box of x_0. Both variants end in $\forall \varepsilon > 0\ \forall x \in A\ \exists \delta > 0\ \forall y \in A\ (\dots)$, and C follows by exchanging the two bounded universal quantifiers (the swap of Example (Flagged boxes at work) (c), with the bounded rules); sentence (iv), "since ε and x were arbitrary", covers both steps in prose.

Distractor (ii) introduces δ as an arbitrary object, by $\forall$I. But in "continuous at x" the δ is under $\exists$: the proof must **produce** it, by a choice from the hypothesis, not receive it from a challenger. A proof that took δ arbitrary would have to show that every $\delta > 0$ works, which is false in general.

Distractor (vii) chooses δ and only afterwards lets x be arbitrary. It looks legitimate, because when (vii) is read after (viii), x is already fixed, and continuity at x does allow δ to depend on x. But the sentence reverses the order of the two steps: a δ fitted to one particular x is followed by "let x be arbitrary", which claims that the same δ works for every x. In the rules: if δ really was chosen for this x, its $\exists$E box assumes a line that mentions x, and generalizing over x while that assumption is open breaks the side condition of $\forall$I, the first blocked line of Example (The search from C to UC). If δ does not depend on x, as here, then "for this x" is idle and the sentence should be (vi).

Exercise. B3: The square root

Prove that $f(x) = \sqrt{x}$ is uniformly continuous on $A = [0, \infty)$. You may use the school inequality $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for all $x, y \ge 0$. State what your δ depends on.

Solution. Let $\varepsilon > 0$, and let $\delta := \varepsilon^2$, which is positive and depends on ε only. Let $x, y \in [0, \infty)$ with $|x - y| < \delta$. Then, by the inequality given and because the square root is strictly increasing on $[0, \infty)$,

$$|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|} < \sqrt{\varepsilon^2} = \varepsilon.$$

Since x and y were arbitrary, this δ works for all pairs of points; since ε was arbitrary, f is uniformly continuous on $[0, \infty)$. The δ was chosen before the points: in a derivation of this proof, its $\exists$I line would lie inside the box of ε_0 only.

4.10.6.3 C. Going deeper

Exercise. C1: Composition, half done

Let f be uniformly continuous on A, with $f(x) \in B$ for every $x \in A$, and let g be uniformly continuous on B. Here is the start of a proof that the composite $x \mapsto g(f(x))$ is uniformly continuous on A:

> Fix $\varepsilon > 0$. By the uniform continuity of g, choose $\eta > 0$ such that $|u - v| < \eta$ implies $|g(u) - g(v)| < \varepsilon$ for all $u, v \in B$.

The rest is blank. *Name the move, then complete it.*

Solution (Hint 1). The goal asks for a δ. Which hypothesis can produce one, and at which positive number should it be applied?

Solution (Hint 2). "By the uniform continuity of f, applied to η, choose $\delta > 0$ such that ..."

Solution. *The move* is fix, obtain, choose: ε was fixed ($\forall$I), η obtained from the hypothesis on g ($\forall$E at ε, then $\exists$E); now obtain δ from the hypothesis on f at the term the goal needs, η ($\forall$E at η, then $\exists$E), and choose that δ as the witness ($\exists$I).

Completion. Since $\eta > 0$, the uniform continuity of f, applied to η, gives $\delta > 0$ such that $|x-y| < \delta$ implies $|f(x)-f(y)| < \eta$ for all $x, y \in A$; choose such a δ. It depends on η, hence on ε, and on no point. Let $x, y \in A$ with $|x-y| < \delta$. Then $|f(x)-f(y)| < \eta$, and $f(x), f(y) \in B$, so the choice of η, applied to $u = f(x)$ and $v = f(y)$, gives $|g(f(x)) - g(f(y))| < \varepsilon$. Since x, y were arbitrary, this δ works for all pairs of points of A; since $\varepsilon > 0$ was arbitrary, $x \mapsto g(f(x))$ is uniformly continuous on A.

Marking scheme. (10 points) - 2: names the move (fix, obtain, choose), or the rules $\forall$E at η, $\exists$E, $\exists$I. - 3: applies the uniform continuity of f at η (not at ε), using $\eta > 0$, and chooses δ before any point. - 3: for arbitrary $x, y \in A$ with $|x - y| < \delta$, gets $|f(x) - f(y)| < \eta$ and then $|g(f(x)) - g(f(y))| < \varepsilon$, using $f(x), f(y) \in B$. - 2: concludes with the quantifiers in the order ε, δ, points, and says δ depends on ε only. Fatal errors (0 for the step): δ chosen after x or depending on x; the uniform continuity of g applied to points not known to lie in B.

Exercise. C2: Gluing two sets

Let A and B be sets of real numbers and S the set of numbers that lie in A or in B. Suppose f is uniformly continuous on A and uniformly continuous on B.

(a) Show that f need not be uniformly continuous on S: take $A = (0, 1)$, $B = (1, 2)$, and $f(x) = 0$ for $x < 1$, $f(x) = 1$ for $x \geq 1$.

(b) Prove that f is uniformly continuous on S when $A = [a, b]$ and $B = [b, c]$ with $a < b < c$.

Solution (Hint 1). For (b), the two sets share the point b. Two points on opposite sides of b are each close to b.

Solution (Hint 2). For (b), apply the two hypotheses at $\varepsilon/2$ and take $\delta = \min(\delta_1, \delta_2)$.

Solution. *(a)* On $A = (0, 1)$, f is constantly 0, so every $\delta > 0$ works for every $\varepsilon > 0$: $|f(x) - f(y)| = 0 < \varepsilon$. The same holds on $B = (1, 2)$, where f is constantly 1. On S we prove the negation of uniform continuity (Example (Three negations from analysis) (c)). Take $\varepsilon = 1$. Let $\delta > 0$, and let $\eta := \min(\delta/4, \frac{1}{2})$, $x := 1 - \eta$, $y := 1 + \eta$. Then $\frac{1}{2} \leq x < 1$, so $x \in A \subseteq S$; $1 < y \leq \frac{3}{2}$, so $y \in B \subseteq S$; $|x - y| = 2\eta \leq \delta/2 < \delta$; and $|f(x) - f(y)| = |0 - 1| = 1 \geq \varepsilon$. So f is not uniformly continuous on S. The gluing fails because the two sets come arbitrarily close at 1 but have no point in common: points close to 1 on opposite sides are each controlled only by their own set, and no common point links them.

(b) Let $\varepsilon > 0$. By the uniform continuity on A at $\varepsilon/2$, choose $\delta_1 > 0$ such that $|f(x) - f(y)| < \varepsilon/2$ for all $x, y \in A$ with $|x - y| < \delta_1$; likewise choose $\delta_2 > 0$ for B. Let $\delta := \min(\delta_1, \delta_2) > 0$, which depends on ε only. Let $x, y \in S$ with $|x - y| < \delta$. There are three cases.

- *Case 1:* $x, y \in A$. Then $|f(x) - f(y)| < \varepsilon/2 < \varepsilon$, since $|x - y| < \delta \leq \delta_1$.
- *Case 2:* $x, y \in B$. The same, with δ_2.
- *Case 3: one point in each set*, say $x \in A$ and $y \in B$ (the other way round is the same with the letters exchanged). Then $x \leq b \leq y$, so $|x - b| = b - x \leq y - x = |x - y| < \delta \leq \delta_1$, and $b \in A$; hence $|f(x) - f(b)| < \varepsilon/2$. Likewise $|b - y| \leq |x - y| < \delta_2$ and $b \in B$, so $|f(b) - f(y)| < \varepsilon/2$. By the triangle inequality (Example (The triangle inequality by cases)),
$$|f(x) - f(y)| \leq |f(x) - f(b)| + |f(b) - f(y)| < \varepsilon.$$

Every point of S lies in A or in B, so the cases cover all pairs. So f is uniformly continuous on S. The common point b is what (a) lacked: there, A and B came arbitrarily close at 1 but had no point in common.

Marking scheme. (10 points) - 3: (a) uniform continuity on A and on B checked, and the negation on S proved: ε first, then for arbitrary δ two points depending on δ, with all three conditions checked. - 2: (b) δ_1 for A at some $\varepsilon_1 > 0$ and δ_2 for B at some $\varepsilon_2 > 0$ with $\varepsilon_1 + \varepsilon_2 \leq \varepsilon$ (the solution takes $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$), and $\delta = \min(\delta_1, \delta_2)$, depending on ε only. - 2: the cases with both points in one set. - 3: the mixed case, through the point b, with $|x - b|, |b - y| \leq |x - y|$ justified and the triangle inequality cited. Fatal errors (0 for the step): a δ depending on x or y; the mixed case omitted; in (a), points chosen before δ.

Exercise. C3: Exit check: a quantifier derivation

(a) Give a full derivation of

$$\exists x\,(P(x) \wedge \forall y\,R(x,y)) \vdash \forall y\,\exists x\,(P(x) \wedge R(x,y)),$$

and check the side condition of every line that closes a flagged box.

(b) For the converse, $\forall y\,\exists x\,(P(x) \wedge R(x,y)) \vdash \exists x\,(P(x) \wedge \forall y\,R(x,y))$, give a domain with two elements and a reading of P and R under which the premise is true and the conclusion false, checking both with the informal reading of Section 4.1.

Solution. (a) The goal is a $\forall$: open a flagged box for y_0. The premise is an $\exists$: open its $\exists$E box at once, with a witness x_0, aiming for the goal's instance. Inside, take the conjunction apart, instantiate $\forall y$ at y_0, and choose x_0 as the witness:

1		$\exists x\,(P(x) \wedge \forall y\,R(x,y))$		
2	y_0	x_0	$P(x_0) \wedge \forall y\,R(x_0,y)$	
3			$P(x_0)$	$\wedge$E 2
4			$\forall y\,R(x_0,y)$	$\wedge$E 2
5			$R(x_0,y_0)$	$\forall$E 4
6			$P(x_0) \wedge R(x_0,y_0)$	$\wedge$I 3, 5
7			$\exists x\,(P(x) \wedge R(x,y_0))$	$\exists$I 6
8		$\exists x\,(P(x) \wedge R(x,y_0))$		$\exists$E 1, 2–7
9	$\forall y\,\exists x\,(P(x) \wedge R(x,y))$			$\forall$I 2–8

- Line 8 ($\exists$E, flag x_0): x_0 does not appear in line 8, in line 1 (the premise), or in any assumption still open at line 8 (there is none: the box of y_0 has no assumption). Line 7, the last line of the box of x_0, is a line of that box itself and is the formula $\exists x\,(P(x) \wedge R(x,y_0))$ that line 8 repeats (Definition 4.6.1).
- Line 9 ($\forall$I, flag y_0): y_0 does not appear in the premise or in line 9, and no assumption is open. Line 8, the last line of the box of y_0, is the instance at y_0 of $\exists x\,(P(x) \wedge R(x,y))$.

The other nesting, with the box of y_0 inside the box of x_0, works too: $\forall$I at the end of the inner box (its name y_0 is in no open assumption, since the assumption on line 2 does not mention it), then $\exists$E on a conclusion without x_0.

(b) Take a domain with two elements a, b; let P be true of both, and let $R(x,y)$ mean "x is y". The premise is true: for $y = a$ the witness $x = a$ has $P(a)$ and $R(a,a)$, and for $y = b$ the witness $x = b$ works. The conclusion is false: $x = a$ fails $\forall y\,R(a,y)$ at $y = b$, and $x = b$ fails it at $y = a$. The witness in the premise depends on y; the conclusion asks for one x that serves every y. (Section 4.11 makes this informal reading a structure, and shows that it proves the converse is not derivable.)

Marking scheme. (10 points) - 2: a flagged box with a fresh name y_0 for the $\forall$I, closed correctly. - 3: an $\exists$E box with a fresh witness x_0, whose conclusion does not contain x_0, with the side condition checked. - 2: $\wedge$E, $\forall$E at y_0 and $\exists$I with the witness x_0, with the correct terms. - 3: the two-element reading: domain, meaning of P and R, the premise checked

for each y and the conclusion refuted for each x. Either nesting of the two boxes earns full marks. Fatal errors (0 for the step): $\forall$I on a name that appears in an open assumption; an $\exists$E conclusion containing the witness.

4.11 Models and Counter-models

Sections 4.3 to 4.10 derived a good deal, and some searches stalled. The converse of Proposition 4.6.2 could not be finished: every attempt ran into a side condition. The converse of Proposition 4.6.1 stalled in the same way, and so did the search for $\exists x\, P(x) \vdash P(c)$ in Exercise (C2: A witness is not a name). But a search that fails proves nothing, since a cleverer derivation might exist. Chapter 3 met the same problem for the connectives and solved it with meaning: a row of a truth table that makes the premises true and the conclusion false rules out every derivation at once (Corollary 3.5.3). This section builds the meaning of a quantified formula. A row of T and F is replaced by a **model**: a domain of objects, with a meaning for every symbol. A model that makes the premises true and the conclusion false is a **counter-model**, and it is the tool that settles the stalled searches.

4.11.1 Why a row is not enough

Chapter 3's semantics gave each atom a truth value and computed the rest. Try it on the stalled sequent of Proposition 4.6.2's converse,

$$\forall y\, \exists x\, R(x, y) \vdash \exists x\, \forall y\, R(x, y).$$

Neither side has a main connective among $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$: each is a quantified formula, and a truth table would have to treat each as one atom. The sequent would become $p \vdash q$, refuted by the row $p = \mathrm{T}$, $q = \mathrm{F}$. That "refutation" is worthless. The same reading refutes $\forall x\, P(x) \vdash \exists x\, P(x)$, which is derivable (Proposition 4.5.1). Treating a quantified formula as an atom throws away exactly the structure that the quantifier rules use.

So what does make $\forall y\, \exists x\, R(x, y)$ true or false? Section 4.1 read such sentences informally, **over a domain**: "for every y in the domain there is an x in the domain with $R(x, y)$". To decide it we need three things.

- **A domain**: the objects the variables range over. Over $\mathbb{Z}$ the sentence may be true and over a three-element set false.
- **A meaning for each symbol.** R is a predicate symbol, a name with two slots and no meaning of its own (Section 4.2). Read as $>$ on $\mathbb{Z}$, the sentence says "every integer has an integer above it", which is true; read as $<$ on $\mathbb{N}$, it says "every natural number has a natural number below it", which is false at $y = 0$.
- **Values for the free variables**, if there are any. The formula $\exists y\, (y < x)$ is neither true nor false on $\mathbb{N}$ until we know which number x is.

The first two together are a **structure**; the third is an **assignment**. A structure with an assignment decides every formula, by clauses that read the quantifiers as Section 4.1 read them. That is all this section does, and then it uses the result.

4.11.2 Structures

Situate the notion first. A formula is built from **symbols**: constants such as c and 0, function symbols such as f and $+$, predicate symbols such as P, R and $<$, each function and predicate

symbol with its number of argument places, its **arity** (Section 4.2, Definition 4.2.1 and Definition 4.2.2). The symbol $=$ is special: it is part of the logic, with rules of its own (Definition 4.7.1), and it will always mean the same thing. A structure gives every other symbol a meaning.

A structure is a non-empty set of objects together with a meaning for each symbol: an object for each constant, an operation for each function symbol, and a relation for each predicate symbol.

Definition 4.11.1. Structure

Fix a collection of constants, function symbols and predicate symbols, each function and predicate symbol with its arity $n \geq 1$. A **structure** $\mathfrak{M}$ (also called a **model**) for these symbols consists of

(M1) a **non-empty** set D, the **domain** of $\mathfrak{M}$;

(M2) for each constant c, an element $c^{\mathfrak{M}} \in D$;

(M3) for each function symbol f of arity n, a function $f^{\mathfrak{M}} \colon D^n \to D$, defined at **every** n-tuple of elements of D;

(M4) for each predicate symbol R of arity n, a set $R^{\mathfrak{M}}$ of n-tuples of elements of D.

The symbol $=$ is not interpreted by $\mathfrak{M}$: in every structure it means **identity**, "is the same element of D as".

In words. (M1) says what the variables range over, and insists that there is at least one object. (M2) says which object each name names; two constants may name the same object. (M3) makes each function symbol an operation that can be applied to **any** objects of the domain and always returns an object of the domain. (M4) makes each predicate symbol a relation: $R^{\mathfrak{M}}$ is the set of tuples for which R "holds". A one-place predicate symbol P becomes a subset $P^{\mathfrak{M}} \subseteq D$, a two-place one a set of ordered pairs, and so on. Here D^n is the set of n-tuples $(d_1, \ldots, d_n)$ of elements of D; *Sets, Relations, and Cardinality*, Chapter 1 develops sets, pairs and functions properly, and this section uses only their everyday meaning.

We usually describe a structure in one line, naming the domain and the meaning of each symbol, and write "$\mathfrak{M} = (\mathbb{Z}, <)$" for the structure with domain $\mathbb{Z}$ in which the symbol $<$ means the usual order. When a symbol has its usual meaning on a familiar set, we say so and do not write $<^{\mathfrak{M}}$.

Examples, simplest first.

- *One element.* $D = \{a\}$, with $P^{\mathfrak{M}} = \{a\}$. This is the degenerate structure. Every variable must take the value a, so "for every element" and "there is an element" both mean "a". It will reappear as a test case: a sentence true here and false in some other structure cannot be derivable (by the soundness statement at the end of this section), and some of the chapter's non-derivable sequents need at least two elements to refute.
- *Two elements, one relation.* $D = \{a, b\}$, with $R^{\mathfrak{M}} = \{(a, a), (b, b)\}$: the relation "is the same as", written out as a set of pairs. A finite structure can be given completely by listing.
- *The integers with their order.* $D = \mathbb{Z}$, with $R^{\mathfrak{M}} = \{(x, y) \in \mathbb{Z}^2 : x > y\}$. Here the relation is described by a condition rather than a list.
- *Arithmetic.* $D = \mathbb{R}$, with the constants 0 and 1 naming zero and one, the function symbols $+$, $\cdot$ and two-place $-$ meaning addition, multiplication and subtraction, the one-place symbols $-$, $(\cdot)^2$ and $|\cdot|$ meaning negation, squaring and absolute value, and $<$, $\leq$, $>$, $\geq$ the usual order. Extended by the other familiar symbols with their usual meanings (numerals, halving, the predicates $\in A$, division with its value at a zero denominator fixed arbitrarily as in the non-example below, and a function symbol for each sequence

or function an example uses), this is the structure in which the analysis examples of this chapter have been read all along.

Non-example by minimal change. Take the arithmetic structure and add a one-place function symbol g, meant to be read as $x \mapsto 1/x$. This is **not** a structure: (M3) asks for a function defined at every element of D, and $1/0$ is not defined. Everything else in the description is fine; exactly clause (M3) fails, at the single point 0. There are two repairs. Shrink the domain to $(0, \infty)$, where $1/x$ is always defined; or keep $\mathbb{R}$ and give g any value at 0, say $g^{\mathfrak{M}}(0) := 0$. The second repair is the one this chapter uses for terms such as $1/n$: the value at 0 is fixed arbitrarily and is never looked at, because such terms are used only under a bound $n \in \mathbb{Z}_{>0}$ (as in Exercise (C3: Exit check: formalize, negate, decide)), exactly as a sequence a_n was read in Section 4.1 as a function symbol whose values off $\mathbb{N}$ are never used.

Why this definition. Each clause earns its place.

- *Non-empty.* This is a genuine choice, and it is made because of Proposition 4.5.1. A later subsection explains it.
- *Total functions.* Clause (M3) is what makes every term name an object: the term $g(0)$ must have a value, or the formula $g(0) = g(0)$, which =I derives from nothing, would have no truth value.
- *Identity for =.* The rule =E replaces equals by equals **everywhere**. If a structure could read $=$ as some other relation, =E would fail. On $\mathbb{Z}$, read $x = y$ as "x and y have the same parity" and P as "is positive": then $-2 = 2$ and $P(2)$ would both hold, while $P(-2)$, which =E derives from them, fails. With $=$ fixed as identity, no such reading exists.

Warning. A structure must interpret **every** symbol that occurs in the formulas at hand. "Take $D = \mathbb{Z}$" is not yet a structure for $\forall y \exists x\, R(x, y)$: until R is given a meaning, the sentence has no truth value, and the same domain makes it true with R read as $>$ and false with R read as "$x \cdot x = y$" (at $y = 2$). Always write the domain **and** the meaning of each symbol.

4.11.3 Assignments and the value of a term

A structure names the objects that **constants** stand for. Variables are different: their whole purpose is to range over the domain, so a structure gives them no fixed value. A separate object records what they stand for at a given moment.

Definition 4.11.2. Assignment

Let $\mathfrak{M}$ be a structure with domain D. An **assignment** in $\mathfrak{M}$ is a function ℓ that gives **every** variable x a value $\ell(x) \in D$. For an assignment ℓ, a variable x and an element $d \in D$, the assignment $\ell[x \mapsto d]$ is the one that gives x the value d and every other variable the value ℓ gives it:

$$\ell[x \mapsto d](y) := \begin{cases} d & \text{if } y \text{ is the variable } x, \\ \ell(y) & \text{if } y \text{ is any other variable.} \end{cases}$$

In words: an assignment is a choice of value for every variable at once, and $\ell[x \mapsto d]$ changes that choice at the one variable x, leaving the others alone. The assignment $\ell[x \mapsto d]$ is what the quantifiers will use: "for every x" will mean "for every $d \in D$, with x set to d". Changes can be stacked: $\ell[x \mapsto d][y \mapsto e]$ first sets x to d and then y to e. When x and y are different variables

the order does not matter, since each change touches only its own variable; when they are the same variable, the later change wins.

There are always assignments, because D is non-empty: pick any $d_0 \in D$ and let every variable have the value d_0. Like a valuation in Chapter 3 (Definition 3.3.1), an assignment fixes infinitely many values, and a formula will only ever look at finitely many of them.

With a structure and an assignment, every term names an element of the domain. Terms are built by the clauses of Definition 4.2.1, so the value is defined by recursion on terms, in the pattern of Theorem 3.2.6.

The value of a term is computed from its variables and constants upwards, one function symbol at a time.

For a structure $\mathfrak{M}$ and an assignment ℓ, the **value** $t^{\mathfrak{M},\ell} \in D$ of a term t is:

- $x^{\mathfrak{M},\ell} := \ell(x)$ for a variable x;
- $c^{\mathfrak{M},\ell} := c^{\mathfrak{M}}$ for a constant c;
- $f(t_1,\dots,t_n)^{\mathfrak{M},\ell} := f^{\mathfrak{M}}(t_1^{\mathfrak{M},\ell},\dots,t_n^{\mathfrak{M},\ell})$ for a function symbol f of arity n.

The third clause is where (M3) is needed: $f^{\mathfrak{M}}$ must accept every tuple the terms $t_1,\dots,t_n$ might produce.

For example, in the arithmetic structure on $\mathbb{R}$, let $\ell(x) = 2$ and $\ell(y) = -3$. The term $x \cdot x + y$, which is $+(\cdot(x,x),y)$ written in the usual way, has value $2 \cdot 2 + (-3) = 1$. Under $\ell[y \mapsto 1]$ it has value $2 \cdot 2 + 1 = 5$, and under $\ell[x \mapsto 0]$ it has value -3. The term $1 + 1$ has value 2 under every assignment, since it contains no variable.

4.11.4 Satisfaction

Now the central definition. We want to say when a structure and an assignment make a formula true. Formulas are built by the clauses of Definition 4.2.2, so we define it by recursion, one clause per way of building a formula, and each clause reads its connective or quantifier as Chapter 3 and Section 4.1 read it.

A formula is satisfied when the objects its terms name stand in the relations its predicates name, combined by the connectives as in a truth table, with "for every x" and "there is an x" ranging over the domain.

Definition 4.11.3. Satisfaction

Let $\mathfrak{M}$ be a structure with domain D and ℓ an assignment in $\mathfrak{M}$. The relation $\mathfrak{M},\ell \models \varphi$, read "$\mathfrak{M}$ **satisfies** φ under ℓ", is defined by recursion on the formula φ:

(S1) $\mathfrak{M},\ell \models R(t_1,\dots,t_n)$ if $(t_1^{\mathfrak{M},\ell},\dots,t_n^{\mathfrak{M},\ell}) \in R^{\mathfrak{M}}$;

(S2) $\mathfrak{M},\ell \models t_1 = t_2$ if $t_1^{\mathfrak{M},\ell}$ and $t_2^{\mathfrak{M},\ell}$ are the **same** element of D;

(S3) $\mathfrak{M},\ell \models \bot$ **never** holds;

(S4) $\mathfrak{M},\ell \models \neg\varphi$ if $\mathfrak{M},\ell \models \varphi$ does not hold;

(S5) $\mathfrak{M},\ell \models \varphi \wedge \psi$ if both $\mathfrak{M},\ell \models \varphi$ **and** $\mathfrak{M},\ell \models \psi$; $\mathfrak{M},\ell \models \varphi \vee \psi$ if at least one of them holds; $\mathfrak{M},\ell \models \varphi \to \psi$ unless $\mathfrak{M},\ell \models \varphi$ holds and $\mathfrak{M},\ell \models \psi$ does not; $\mathfrak{M},\ell \models \varphi \leftrightarrow \psi$ if both hold or neither does;

(S6) $\mathfrak{M},\ell \models \forall x\,\varphi$ if $\mathfrak{M},\ell[x \mapsto d] \models \varphi$ **for every** $d \in D$;

(S7) $\mathfrak{M},\ell \models \exists x\,\varphi$ if $\mathfrak{M},\ell[x \mapsto d] \models \varphi$ for **at least one** $d \in D$.

When $\mathfrak{M}, \ell \models \varphi$ does not hold we write $\mathfrak{M}, \ell \not\models \varphi$. For a list Γ of formulas, $\mathfrak{M}, \ell \models \Gamma$ means that $\mathfrak{M}, \ell \models \gamma$ for **every** formula γ of Γ.

Read it clause by clause. (S1): an atomic formula is satisfied exactly when the values of its terms form a tuple in the relation that interprets its predicate symbol. (S2): an equation is satisfied when both sides name the same object; this is where "$=$ means identity" enters. (S3)–(S5) are the $\bot$ half of (V1) and clauses (V2)–(V6) of Chapter 3's Definition 3.3.2, with "is true under v" replaced by "is satisfied by $\mathfrak{M}$ under ℓ": $\bot$ is never satisfied, $\neg$ flips, $\wedge$ needs both, $\vee$ needs one, $\to$ fails only with a satisfied hypothesis and an unsatisfied conclusion, $\leftrightarrow$ needs agreement. (S6) and (S7) are the new clauses. To decide $\forall x\, \varphi$, we run through the elements d of the domain, set x to d and leave every other variable alone, and ask whether φ is satisfied each time. For $\exists x\, \varphi$ one success is enough. In symbols, for (S6) and (S7):

$$\begin{aligned} \mathfrak{M}, \ell \models \forall x\, \varphi \quad &\text{iff} \quad \text{for every } d \in D, \quad \mathfrak{M}, \ell[x \mapsto d] \models \varphi, \\ \mathfrak{M}, \ell \models \exists x\, \varphi \quad &\text{iff} \quad \text{there is } d \in D \text{ with } \mathfrak{M}, \ell[x \mapsto d] \models \varphi. \end{aligned}$$

Well defined. The clauses define, for each formula, which pairs $(\mathfrak{M}, \ell)$ satisfy it, by recursion on the formula: (S1)–(S3) settle the atomic formulas and $\bot$, and each of (S4)–(S7) settles a compound formula from its immediate parts. One point needs a word. Clauses (S6) and (S7) consult the immediate part φ under **other** assignments $\ell[x \mapsto d]$, not under ℓ. So the recursion must define satisfaction of each formula under **all** assignments at once, and then (S6) reads off the answer for φ at every $\ell[x \mapsto d]$. The recursion is still on the formula alone, and Theorem 3.2.6, applied to formulas built by the clauses of Definition 4.2.2, gives exactly one such relation.

A trace, step by step. Decide $\forall x\, \exists y\, (x < y)$ in two structures, both with $<$ the usual order: $\mathfrak{M}_1$ with domain $\mathbb{N}$, and $\mathfrak{M}_2$ with domain $\{1, 2, 3\}$. Let ℓ be any assignment.

In $\mathfrak{M}_1$: by (S6) we must check, for every $d \in \mathbb{N}$, that $\mathfrak{M}_1, \ell[x \mapsto d] \models \exists y\, (x < y)$. Fix d. By (S7) it is enough to find one $e \in \mathbb{N}$ with $\mathfrak{M}_1, \ell[x \mapsto d][y \mapsto e] \models x < y$. Take $e = d + 1$. Under $\ell[x \mapsto d][y \mapsto d+1]$ the terms x and y have the values d and $d+1$, and $d < d+1$, so (S1) holds. Hence the sentence is satisfied. Notice that the witness $e = d + 1$ depends on d: (S7) is checked **inside** (S6), once for each d.

In $\mathfrak{M}_2$: (S6) asks the same for $d = 1, 2, 3$. At $d = 1$ and $d = 2$ the witnesses $e = 2$ and $e = 3$ work. At $d = 3$ no $e \in \{1, 2, 3\}$ has $3 < e$, so (S7) fails for $\ell[x \mapsto 3]$, and (S6) fails. So $\mathfrak{M}_2 \not\models \forall x\, \exists y\, (x < y)$, and the failure has a name: the element 3 has nothing above it.

The assignment matters for free variables. In $\mathfrak{M}_1 = (\mathbb{N}, <)$, with $\leq$ also read as usual, take the formula $\forall y\, (x \leq y)$, in which x is free. Under an assignment with $\ell(x) = 0$ it is satisfied, since every $e \in \mathbb{N}$ has $0 \leq e$ (recall that $\mathbb{N} = \{0, 1, 2, \dots\}$). Under an assignment with $\ell(x) = 1$ it is not: $e = 0$ gives $1 \leq 0$, which fails. The same formula in the same structure is satisfied under one assignment and not under another; it is a property of x, "x is least", not a statement.

Degenerate cases. In the one-element structure $D = \{a\}$, every assignment gives every variable the value a, so $\ell[x \mapsto a] = \ell$ and (S6) and (S7) both reduce to "$\mathfrak{M}, \ell \models \varphi$". So $\forall x\, \varphi$ and $\exists x\, \varphi$ are satisfied together or not at all, and every equation $x = y$ is satisfied. In any structure, $\forall x\, (x = x)$ is satisfied, since every d is the same element as itself; and $\exists x\, \neg(x = x)$ never is.

Quick check. Is $\forall x\, \exists y\, (y \cdot y = x)$ satisfied in $\mathbb{R}$, in $[0, \infty)$, and in $\mathbb{C}$, each with $\cdot$ the usual product?

Solution. In $\mathbb{R}$, no: at $d = -1$, (S7) asks for a real e with $e \cdot e = -1$, and every real square is ≥ 0. In $[0, \infty)$, yes: for each $d \geq 0$, $e = \sqrt{d}$ is in the domain and $e \cdot e = d$; and $[0, \infty)$ with $\cdot$ is a structure, since a product of non-negative reals is non-negative (M3). In $\mathbb{C}$, yes: every complex number has a complex square root (school algebra of complex numbers: $r(\cos\theta + i\sin\theta)$ with $r \geq 0$ has the square root $\sqrt{r}\,(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})$). The sentence is the same all three times; only the structure changes.

4.11.5 Only the free variables matter

The example of $\forall y\,(x \leq y)$ above showed ℓ mattering. In the first example it did not: the verdict on $\forall x\,\exists y\,(x < y)$ was reached for "any assignment ℓ". That is no accident. A sentence has no free variables (Definition 4.2.4), and every variable it contains is bound, so every value ℓ gives is overwritten by (S6) or (S7) before it is looked at. We prove the general fact, which is the first-order version of Chapter 3's Lemma 3.3.1: satisfaction depends only on the values of the **free** variables.

Say that two assignments ℓ and ℓ' **agree on** a set of variables if $\ell(y) = \ell'(y)$ for every variable y in it. We need the free variables of a formula, recursively defined in Definition 4.2.3: every variable occurring in an atomic formula is free there; $\neg\varphi$ has the free variables of φ; $\varphi \circ \psi$, for a binary connective $\circ$, has those of φ and those of ψ; and $\forall x\,\varphi$ and $\exists x\,\varphi$ have those of φ other than x.

Lemma 4.11.1. Free Variables Coincidence Lemma

Let $\mathfrak{M}$ be a structure and φ a formula, and let ℓ and ℓ' be assignments in $\mathfrak{M}$ that agree on every free variable of φ. Then

$$\mathfrak{M}, \ell \models \varphi \quad \text{if and only if} \quad \mathfrak{M}, \ell' \models \varphi.$$

Idea. Structural induction on φ, as for Lemma 3.3.1, after a first induction on terms: a term's value depends only on the values of its variables. The connective cases are Chapter 3's. The quantifier case is the only new one, and it explains why the property must be stated **for all** pairs of assignments: (S6) compares $\ell[x \mapsto d]$ with $\ell'[x \mapsto d]$, which are new assignments, so the induction hypothesis must apply to them. They agree at x, since both give it d, and at every other free variable of φ, since those are free in $\forall x\,\varphi$.

Proof. Terms first. We show: if ℓ and ℓ' agree on every variable occurring in a term t, then $t^{\mathfrak{M},\ell} = t^{\mathfrak{M},\ell'}$. We use structural induction on terms (Definition 4.2.1; the pattern of Theorem 3.2.3). For a variable x, the variable x occurs in x, so $x^{\mathfrak{M},\ell} = \ell(x) = \ell'(x) = x^{\mathfrak{M},\ell'}$. For a constant c, both values are $c^{\mathfrak{M}}$. For $f(t_1, \ldots, t_n)$, every variable of t_i occurs in $f(t_1, \ldots, t_n)$, so the hypothesis for t_i gives $t_i^{\mathfrak{M},\ell} = t_i^{\mathfrak{M},\ell'}$ for each i; applying $f^{\mathfrak{M}}$ to equal tuples gives equal values.

Formulas. We apply structural induction to the property $C(\varphi)$: *for all assignments ℓ, ℓ' in $\mathfrak{M}$ that agree on the free variables of φ, $\mathfrak{M}, \ell \models \varphi$ if and only if $\mathfrak{M}, \ell' \models \varphi$.*

Atomic formulas. Let φ be $R(t_1, \ldots, t_n)$, and let ℓ, ℓ' agree on its free variables. Every variable occurring in φ is free in it (Definition 4.2.3), so ℓ and ℓ' agree on the variables of

each t_i, and by the claim for terms $t_i^{\mathfrak{M},\ell} = t_i^{\mathfrak{M},\ell'}$. The two tuples in (S1) are equal, so one lies in $R^{\mathfrak{M}}$ exactly when the other does. The case $t_1 = t_2$ is the same with (S2). For $\bot$, neither side holds, by (S3).

Connectives. Assume $C(\varphi)$ and $C(\psi)$, and let ℓ, ℓ' agree on the free variables of $\varphi \wedge \psi$. These include the free variables of φ and those of ψ, so $\mathfrak{M}, \ell \models \varphi$ if and only if $\mathfrak{M}, \ell' \models \varphi$, and likewise for ψ. By (S5), satisfaction of $\varphi \wedge \psi$ is decided by these two answers through the same rule for ℓ and for ℓ', so $C(\varphi \wedge \psi)$ holds. The cases $\neg\varphi$, $\varphi \vee \psi$, $\varphi \to \psi$, and $\varphi \leftrightarrow \psi$ are the same argument with (S4) and the other parts of (S5), as in the proof of Lemma 3.3.1.

Universal quantifier. Assume $C(\varphi)$, let x be a variable, and let ℓ, ℓ' agree on the free variables of $\forall x\, \varphi$. Let $d \in D$. We claim that $\ell[x \mapsto d]$ and $\ell'[x \mapsto d]$ agree on the free variables of φ. At x, both give d. A free variable y of φ other than x is a free variable of $\forall x\, \varphi$ (Definition 4.2.3), so $\ell[x \mapsto d](y) = \ell(y) = \ell'(y) = \ell'[x \mapsto d](y)$. By $C(\varphi)$, applied to these two assignments,

$$\mathfrak{M}, \ell[x \mapsto d] \models \varphi \quad \text{iff} \quad \mathfrak{M}, \ell'[x \mapsto d] \models \varphi.$$

This holds for every $d \in D$, so by (S6) $\mathfrak{M}, \ell \models \forall x\, \varphi$ if and only if $\mathfrak{M}, \ell' \models \forall x\, \varphi$. This proves $C(\forall x\, \varphi)$.

Existential quantifier. The same argument, with (S7) in place of (S6): the displayed equivalence holds for each d, so there is a d making the left side hold exactly when there is one making the right side hold.

Every clause of Definition 4.2.2 has been covered, so $C(\varphi)$ holds for every formula φ. This proves the lemma.

■

The lemma has a consequence that the rest of the section uses constantly.

Sentences. If φ is a sentence, it has no free variables, so **any** two assignments agree on its free variables, and the lemma says that $\mathfrak{M}, \ell \models \varphi$ either holds for every assignment ℓ or for none. We then write $\mathfrak{M} \models \varphi$, and say that φ is **true in** $\mathfrak{M}$ (or that $\mathfrak{M}$ is a model of φ), in the first case, and $\mathfrak{M} \not\models \varphi$, φ is **false in** $\mathfrak{M}$, in the second. At least one assignment exists because D is non-empty, so a sentence is always exactly one of true and false in $\mathfrak{M}$. This is the precise form of Section 4.1's informal "true over a domain".

Only finitely many values matter. A formula has finitely many free variables. To decide $\mathfrak{M}, \ell \models \varphi$, it is enough to know ℓ on those. So we describe an assignment by the values that matter ("$\ell(x) = 2$"), exactly as Chapter 3 described a valuation by the values of a few atoms.

Warning. The same sentence can be true in one structure and false in another, and "true" without a structure means nothing. $\forall x\, \exists y\, (x < y)$ is true in $(\mathbb{N}, <)$ and false in $(\{1, 2, 3\}, <)$, as traced above. A sentence of predicate logic is like a formula of Chapter 3 with atoms: it has a truth value only once the meaning of its symbols has been fixed.

4.11.6 Semantic entailment

Chapter 3 said that Γ entails ψ when every row that makes Γ true makes ψ true (Definition 3.4.3). The row is now a structure with an assignment.

An argument is valid when every structure, under every assignment, that satisfies all the premises also satisfies the conclusion.

Definition 4.11.4. Semantic Entailment in Predicate Logic

Let Γ be a list of formulas and φ a formula. We say Γ **semantically entails** φ, and write $\Gamma \models \varphi$, if for **every** structure $\mathfrak{M}$ for the symbols occurring in Γ and φ, and **every** assignment ℓ in $\mathfrak{M}$,

$$\mathfrak{M}, \ell \models \Gamma \quad \Rightarrow \quad \mathfrak{M}, \ell \models \varphi.$$

We write $\models \varphi$ when Γ is empty. If $\Gamma \models \varphi$ fails, we write $\Gamma \not\models \varphi$; a structure $\mathfrak{M}$ with an assignment ℓ such that $\mathfrak{M}, \ell \models \Gamma$ and $\mathfrak{M}, \ell \not\models \varphi$ is a **counter-model** to $\Gamma \models \varphi$.

In words: "every structure" means every domain and every meaning of the symbols, so an entailment holds for reasons of form and not because of what R happens to mean. "Every assignment" covers every value of the free variables; when Γ and φ are sentences, the assignment makes no difference, by Lemma 4.11.1. And a counter-model is the analogue of Chapter 3's refuting row: one structure, with one assignment, where the premises hold and the conclusion does not. The symbol $\models$ is the one of Chapter 3, and the context always shows which is meant: a sequent of formulas with quantifiers or predicate symbols is read with this definition.

A structure may interpret symbols that do not occur in Γ or φ. The clauses of Definition 4.11.3 consult only the symbols of the formula at hand, so such extra meanings change nothing.

First examples of entailment. The argument Section 4.1 opened with, "every positive real has a square root; 2 is positive; so 2 has a square root", is $\forall x\,(P(x) \to Q(x)),\ P(c) \vdash Q(c)$, derived in Section 4.3. Truth tables called it invalid, since they saw three unrelated atoms. It is valid in the new sense:

$$\forall x\,(P(x) \to Q(x)),\ P(c) \models Q(c).$$

Let $\mathfrak{M}, \ell$ satisfy both premises, and let $a = c^{\mathfrak{M}}$. By (S6) for the first premise, with $d = a$, we get $\mathfrak{M}, \ell[x \mapsto a] \models P(x) \to Q(x)$. Under $\ell[x \mapsto a]$ the term x has value a, and under ℓ the term c has value a; so by (S1), $\mathfrak{M}, \ell[x \mapsto a] \models P(x)$ holds if and only if $a \in P^{\mathfrak{M}}$ if and only if $\mathfrak{M}, \ell \models P(c)$, which is the second premise. So by (S5) $\mathfrak{M}, \ell[x \mapsto a] \models Q(x)$, that is, $a \in Q^{\mathfrak{M}}$, and by (S1) again $\mathfrak{M}, \ell \models Q(c)$. Chapter 3's objection to this argument is gone.

4.11.7 Why domains are non-empty

Clause (M1) of Definition 4.11.1 demands a non-empty domain. Here is the reason. Proposition 4.5.1 derives $\forall x\,P(x) \vdash \exists x\,P(x)$ in three lines: $\forall$E at the variable x, then $\exists$I. The matching entailment holds for our structures:

$$\forall x\,P(x) \models \exists x\,P(x).$$

Let $\mathfrak{M}, \ell \models \forall x\,P(x)$. Since D is non-empty, choose some $d \in D$. By (S6), $\mathfrak{M}, \ell[x \mapsto d] \models P(x)$, and this d is the one element (S7) asks for, so $\mathfrak{M}, \ell \models \exists x\,P(x)$.

Now suppose empty domains were allowed. In a structure with $D = \emptyset$, (S6) says $\forall x\,P(x)$ is satisfied, since "for every $d \in \emptyset$" has nothing to check; and (S7) says $\exists x\,P(x)$ is not, since there is no d at all. So the sequent would be derivable and yet fail in a structure, and the rules would lead from a true premise to a false conclusion. (Worse, no assignment into an empty domain exists, so even the clauses could not be applied as written.) One of two things had to give: the rule $\forall$E at a variable, or empty domains. The book, like most of mathematics, keeps the rules and makes domains non-empty.

Nothing is lost. Statements about an empty **set** are handled by the bounded quantifiers of Definition 4.1.1, inside a non-empty domain. The bounded forms $\forall x \in \emptyset\ \varphi$ and $\exists x \in \emptyset\ \varphi$ abbreviate $\forall x\,(x \in \emptyset \to \varphi)$ and $\exists x\,(x \in \emptyset \wedge \varphi)$, and in a structure with domain $\mathbb{R}$ in which the one-place predicate symbol $\in \emptyset$ (Section 4.2) is interpreted, by (M4), as the empty subset of $\mathbb{R}$, the first is true (every hypothesis $d \in \emptyset$ is false) and the second is false (no d is in $\emptyset$). Vacuous truth lives in the bound, not in the domain.

4.11.8 Counter-models

A counter-model is found the way a refuting row was found in Chapter 3, by working backwards from what must fail. The conclusion must be false, so read off what that demands; the premises must be true, so read off what they demand; and then build the smallest structure meeting both. The worked examples that follow (the swap, the converses of Proposition 4.6.1 and of the $\forall/\vee$ law, $\exists x\,P(x) \vdash P(c)$, and $P(x_0) \vdash \forall x\,P(x)$) are sequents whose derivations stalled or were blocked in earlier sections.

Example. The swap of $\exists$ and $\forall$

Show that

$$\forall y\,\exists x\,R(x,y) \not\models \exists x\,\forall y\,R(x,y),$$

first by a structure on $\mathbb{Z}$, then by a structure with two elements.

Solution. Both formulas are sentences, so by Lemma 4.11.1 we need only a structure; any assignment will do.

A structure on $\mathbb{Z}$. Let $\mathfrak{M}$ have domain $\mathbb{Z}$ and $R^{\mathfrak{M}} = \{(x,y) : x > y\}$, so that $R(x,y)$ says "$x > y$". The premise says "every integer has an integer above it". To check it by (S6) and (S7): for each $e \in \mathbb{Z}$ we need some $d \in \mathbb{Z}$ with $(d,e) \in R^{\mathfrak{M}}$, that is, $d > e$; $d = e + 1$ works. The conclusion says "some integer is above every integer". By (S7) and (S6) it would need a $d \in \mathbb{Z}$ with $d > e$ for **every** $e \in \mathbb{Z}$; but $e = d$ gives $d > d$, which is false. So $\mathfrak{M} \models \forall y\,\exists x\,R(x,y)$ and $\mathfrak{M} \not\models \exists x\,\forall y\,R(x,y)$.

A structure with two elements. Let $D = \{a,b\}$ and $R^{\mathfrak{M}} = \{(a,a),(b,b)\}$, so that $R(x,y)$ says "x is y". Premise: for $e = a$ take $d = a$, and for $e = b$ take $d = b$; both pairs are in $R^{\mathfrak{M}}$. Conclusion: $d = a$ fails at $e = b$, since $(a,b) \notin R^{\mathfrak{M}}$, and $d = b$ fails at $e = a$. So the premise is true and the conclusion false.

The pattern is the one Section 4.1 read as a game. In the premise the witness x is chosen **after** y and may depend on it ($x = y + 1$, or $x = y$); the conclusion asks for one x that works for every y in advance. A one-element structure cannot separate the two sentences: with $D = \{a\}$, both are true if $(a,a) \in R^{\mathfrak{M}}$ and both false otherwise. Two elements are the least that will do.

The two-element structure is the $\mathbb{Z}$ structure's idea at its smallest: each element is related to something, and nothing is related to everything. Finite counter-models are convenient because every clause can be checked by listing. They are not always available: Exercise C1 below gives a sentence that is true in every finite structure and false in an infinite one, and Exercise C2 a sentence with only infinite models.

The converse of Proposition 4.6.1. Section 4.6 derived $\exists x\,(P(x) \wedge Q(x)) \vdash \exists x\,P(x) \wedge \exists x\,Q(x)$

and asked about the converse. We show

$$\exists x\, P(x) \land \exists x\, Q(x) \not\models \exists x\, (P(x) \land Q(x)).$$

Backwards from the conclusion: it is false when no element is in both $P^{\mathfrak{M}}$ and $Q^{\mathfrak{M}}$. The premise needs $P^{\mathfrak{M}}$ and $Q^{\mathfrak{M}}$ non-empty. So take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \{b\}$. The premise holds by (S5) and (S7), with witnesses a for P and b for Q. The conclusion fails: at $d = a$, $a \notin Q^{\mathfrak{M}}$; at $d = b$, $b \notin P^{\mathfrak{M}}$. In arithmetic: in $\mathbb{Z}$, some integer is even and some integer is odd, but no integer is both. The two witnesses of the premise are different objects, and the conclusion needs one.

The converse of the $\forall/\lor$ law. The one-way law $\forall x\, P(x) \lor \forall x\, Q(x) \vdash \forall x\, (P(x) \lor Q(x))$ of Proposition 4.9.4 has a false converse, and the same structure shows it:

$$\forall x\, (P(x) \lor Q(x)) \not\models \forall x\, P(x) \lor \forall x\, Q(x).$$

With $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \{b\}$: each element is in $P^{\mathfrak{M}}$ or in $Q^{\mathfrak{M}}$, so the premise holds; $\forall x\, P(x)$ fails at b and $\forall x\, Q(x)$ fails at a, so by (S5) the disjunction fails. In $\mathbb{Z}$: every integer is even or odd, but it is false that every integer is even, and false that every integer is odd.

A witness is not a named constant. Exercise (C2: A witness is not a name) asked why $\exists x\, P(x) \vdash P(c)$ cannot be derived. Take $D = \{a, b\}$, $c^{\mathfrak{M}} = a$ and $P^{\mathfrak{M}} = \{b\}$. The premise holds with the witness b; the conclusion $P(c)$ fails, since the value of c is $a \notin P^{\mathfrak{M}}$. So $\exists x\, P(x) \not\models P(c)$. The premise says that something has P, and nothing says that the object named c is that something.

The assignment is part of a counter-model. Section 4.4 showed why $\forall$I may not generalize over a name in a premise: from $P(x_0)$ one would "derive" $\forall x\, P(x)$. Here x_0 is a **variable**, free in the premise, so a counter-model needs an assignment as well as a structure. Let $\mathfrak{M}$ have domain $\mathbb{Z}$ with $P^{\mathfrak{M}}$ the set of even integers, and let $\ell(x_0) = 2$. Then $\mathfrak{M}, \ell \models P(x_0)$, since 2 is even, and $\mathfrak{M}, \ell \not\models \forall x\, P(x)$, since (S6) fails at $d = 1$. So $P(x_0) \not\models \forall x\, P(x)$. Under the assignment with $\ell(x_0) = 1$ the premise is false, and that assignment refutes nothing; the counter-model is the structure **with** $\ell(x_0) = 2$.

Warning. A "counter-model" that falsifies a premise refutes nothing. For $\forall y\, \exists x\, R(x, y) \vdash \exists x\, \forall y\, R(x, y)$, the structure $(\mathbb{N}, <)$, with $R(x, y)$ read as "$x < y$", makes the conclusion false, since no natural number is below every natural number; but it also makes the premise false, since nothing in $\mathbb{N}$ is below $y = 0$. It is not a counter-model, and it proves nothing about the sequent. As with rows in Chapter 3, check the premises **first**.

A model of a list of sentences. Counter-models are one use of structures; describing a whole kind of object is another. Consider the three sentences

$$\forall x\, \neg(x < x),$$
$$\forall x\, \forall y\, \forall z\, \big((x < y \land y < z) \to x < z\big),$$
$$\forall x\, \exists y\, (x < y),$$

which say that $<$ is a **strict order** (never $x < x$; $x < y$ and $y < z$ give $x < z$) **with no largest element**. The structure $(\mathbb{N}, <)$ is a model of all three: no number is less than itself, the order is transitive, and $d + 1 > d$ is a witness for the third sentence. It is the one the sentences were written with in mind, but not the only one: in $(\mathbb{Z}, <)$ and $(\mathbb{Q}, <)$ all three are true, by the same checks. These models differ in ways the three sentences cannot see: $\mathbb{Z}$ has no least element, and $\mathbb{Q}$ has a third element between any two. And the one-element structure $D = \{a\}$ is not a model of the three sentences: if $(a, a) \in <^{\mathfrak{M}}$ the first sentence fails, and if not, the third fails at $d = a$. Exercise C2 shows that no **finite** structure is a model of them.

Quick check. Give a counter-model to $\forall x\, P(x) \to \forall x\, Q(x) \models \forall x\, (P(x) \to Q(x))$. (Think about what makes an implication with a universal hypothesis true cheaply.)

Solution. Take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \emptyset$. The premise is an implication whose hypothesis $\forall x\, P(x)$ is false (it fails at b), so by (S5) the premise is true. The conclusion fails at $d = a$: $a \in P^{\mathfrak{M}}$ and $a \notin Q^{\mathfrak{M}}$, so $P(x) \to Q(x)$ is not satisfied under $\ell[x \mapsto a]$. (Any structure with $P^{\mathfrak{M}} \neq D$ and some element in $P^{\mathfrak{M}}$ but not in $Q^{\mathfrak{M}}$ works.)

4.11.9 From counter-models to non-derivability

A counter-model shows $\Gamma \not\models \varphi$. The stalled searches asked something else: whether $\Gamma \vdash \varphi$. The bridge is the same as in Chapter 3, where Theorem 3.5.2 turned a refuting row into a proof that no derivation exists (Corollary 3.5.3). For predicate logic it reads:

Soundness for predicate logic. If $\Gamma \vdash \varphi$ with the rules of Chapters 1 and 2 and the quantifier and equality rules of Sections 4.3 to 4.7, then $\Gamma \models \varphi$ in the sense of Definition 4.11.4. Consequently, a counter-model to $\Gamma \models \varphi$ shows $\Gamma \nvdash \varphi$.

This statement is **not proved in this section**. Its proof extends Chapter 3's to the new rules, and it is the subject of Section 4.12, which is optional. The proof is where each side condition of the quantifier rules is used: the fresh-name conditions of $\forall$I and $\exists$E, and "free for" in $\forall$E, $\exists$I, and $=$E. A reader who skips Section 4.12 may take the statement on trust, as the rest of the book does; a reader who reads it will see why the side conditions are exactly as strict as they are.

With the statement, the examples above become non-derivability results. $\forall y\, \exists x\, R(x, y) \nvdash \exists x\, \forall y\, R(x, y)$: the converse search of Proposition 4.6.2 had to stall, and the two lines where it stalled, the blocked $\forall$I and the blocked close of the $\exists$E box, are not a lack of cleverness. Likewise the converse of Proposition 4.6.1, the converse of the $\forall/\vee$ law, $\exists x\, P(x) \vdash P(c)$, and $P(x_0) \vdash \forall x\, P(x)$ have no derivations. And the same reasoning is the semantic side of Section 4.10: there, Example (The square function on the real line) showed that x^2 on $\mathbb{R}$ is continuous and not uniformly continuous, so the arithmetic structure on $\mathbb{R}$, with f read as squaring, is a counter-model to "continuous entails uniformly continuous".

Warning. A counter-model proves **non**-derivability; it never proves derivability, and a structure in which a sentence is true says nothing about whether the sentence is derivable. $\forall x\, \exists y\, (x < y)$ is true in $(\mathbb{N}, <)$, yet it is not derivable: it is false in $(\{1, 2, 3\}, <)$. To show $\Gamma \vdash \varphi$, write a derivation.

4.11.10 Exercises

4.11.10.1 A. Check your understanding

Exercise. A1

Let $\mathfrak{M}_{\mathbb{N}}$, $\mathfrak{M}_{\mathbb{Z}}$, $\mathfrak{M}_{\mathbb{Q}}$ be the structures with domains $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, each with $<$ the usual order. For each sentence in (a)–(d), say in which of the three structures it is true, with a witness or a

failing element for each verdict.

(a) $\forall x\, \exists y\, (x < y)$.

(b) $\exists x\, \forall y\, \neg(y < x)$.

(c) $\forall x\, \forall y\, (x < y \to \exists z\, (x < z \wedge z < y))$.

(d) $\forall x\, \exists y\, (y < x)$.

(e) True or false: if a sentence is true in some structure, then it is derivable.

(f) Why must the domain of a structure be non-empty? Name the derivable sequent that would otherwise fail.

Solution. *(a)* True in all three: for each d, take $y = d+1$, which lies in the same domain and is larger.

(b) This says "there is a least element". True in $\mathfrak{M}_{\mathbb{N}}$, with $x = 0$: no natural number is below 0. False in $\mathfrak{M}_{\mathbb{Z}}$ and $\mathfrak{M}_{\mathbb{Q}}$: for any d, the element $e = d-1$ has $e < d$, so (S6) fails for $\neg(y < x)$ under $\ell[x \mapsto d]$.

(c) This says "between any two elements there is a third" (density). False in $\mathfrak{M}_{\mathbb{N}}$ and $\mathfrak{M}_{\mathbb{Z}}$: at $x = 1$, $y = 2$ there is no integer strictly between. True in $\mathfrak{M}_{\mathbb{Q}}$: for $d < e$ in $\mathbb{Q}$, $z = (d+e)/2$ is rational and $d < z < e$.

(d) This says "no element is least". False in $\mathfrak{M}_{\mathbb{N}}$, at $x = 0$. True in $\mathfrak{M}_{\mathbb{Z}}$ and $\mathfrak{M}_{\mathbb{Q}}$, with $y = d - 1$.

	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$
(a)	true	true	true
(b)	true	false	false
(c)	false	false	true
(d)	false	true	true

(e) False. By the soundness statement, a derivable sentence is true in **every** structure. Sentence (a) is true in $\mathfrak{M}_{\mathbb{N}}$ but false in the structure $(\{1, 2, 3\}, <)$, so it is not derivable.

(f) The sequent is $\forall x\, P(x) \vdash \exists x\, P(x)$, derived in Proposition 4.5.1 by $\forall$E at the variable x and $\exists$I. In an empty domain $\forall x\, P(x)$ would be true, with nothing to check, and $\exists x\, P(x)$ false, with no witness; so the rules would lead from a true premise to a false conclusion. (There would also be no assignments at all.)

4.11.10.2 B. Practice

Exercise. B1: Three counter-models

Give a counter-model for each of the following, and check each premise and the conclusion against Definition 4.11.3.

(i) $\exists x\, P(x) \not\models \forall x\, P(x)$.

(ii) $\forall x\, (P(x) \to Q(x)) \not\models \exists x\, P(x) \to \forall x\, Q(x)$.

(iii) $\not\models \exists x\, P(x) \to P(x)$. (Here x is free in the last $P(x)$; the assignment is part of the

answer.)

Solution. *(i)* Take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$. The premise holds by (S7) with $d = a$. The conclusion fails by (S6) at $d = b$, since $b \notin P^{\mathfrak{M}}$.

(ii) Take $D = \{a, b\}$, $P^{\mathfrak{M}} = Q^{\mathfrak{M}} = \{a\}$. Premise: at $d = a$, $P(x)$ and $Q(x)$ are both satisfied, so $P(x) \to Q(x)$ is; at $d = b$, $P(x)$ is not satisfied, so $P(x) \to Q(x)$ is, by (S5). Hence the premise holds by (S6). Conclusion: $\exists x\, P(x)$ holds ($d = a$) and $\forall x\, Q(x)$ fails (at $d = b$), so by (S5) the implication fails. In words: whatever has P has Q, and something has P; that does not make everything have Q.

(iii) Take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, and an assignment with $\ell(x) = b$. The hypothesis $\exists x\, P(x)$ holds by (S7) with $d = a$; its x is bound, and the value $\ell(x)$ is overwritten. The conclusion $P(x)$ has x free, with value $\ell(x) = b \notin P^{\mathfrak{M}}$, so it fails. By (S5) the implication fails under ℓ. Under an assignment with $\ell(x) = a$ the implication holds; so the formula is satisfied under some assignments and not others, and a counter-model must name ℓ. Now take φ and ψ both to be $P(x)$. The formula $\forall x\, (P(x) \to P(x))$ is satisfied in every structure under every assignment, since $P(x) \to P(x)$ is satisfied at every d by (S5) and (S6). So this structure, with ℓ, is a counter-model to $\forall x\, (P(x) \to P(x)) \models \exists x\, P(x) \to P(x)$, and by the soundness statement of this section that sequent is not derivable. This is why the passage law $\forall x\, (\varphi \to \psi) \dashv\vdash \exists x\, \varphi \to \psi$ of Proposition 4.9.5 needs x not free in ψ.

Exercise. B2: A model on three elements

Give an example of a structure with exactly three elements in which both

$$\forall x\, \exists y\, R(x, y) \qquad \text{and} \qquad \forall x\, \neg R(x, x)$$

are true. Check both sentences. Which properties must every such structure have?

Solution. Take $D = \{0, 1, 2\}$ and $R^{\mathfrak{M}} = \{(0, 1), (1, 2), (2, 0)\}$: each element is related to the next one around a circle. First sentence: for $d = 0, 1, 2$, the witnesses $e = 1, 2, 0$ give pairs in $R^{\mathfrak{M}}$. Second sentence: none of $(0, 0)$, $(1, 1)$, $(2, 2)$ is in $R^{\mathfrak{M}}$, so $\neg R(x, x)$ is satisfied under $\ell[x \mapsto d]$ for each d.

The properties every answer must have: the domain has three elements; every element d has **some** e with $(d, e) \in R^{\mathfrak{M}}$; and **no** pair (d, d) is in $R^{\mathfrak{M}}$, so that e is always different from d. Any $R^{\mathfrak{M}}$ meeting these works, for instance $\{(0, 1), (1, 0), (2, 0)\}$ or the set of all pairs (d, e) with $d \neq e$.

Exercise. B3: The converse from Section 4.4

Exercise (B2: Contraposition under a quantifier) derived $\forall x\, (P(x) \to Q(x)) \vdash \forall x\, \neg Q(x) \to \forall x\, \neg P(x)$, and its search for the converse stalled. Show

$$\forall x\, \neg Q(x) \to \forall x\, \neg P(x) \not\models \forall x\, (P(x) \to Q(x)).$$

Solution. Work backwards. The conclusion fails when some d has $d \in P^{\mathfrak{M}}$ and $d \notin Q^{\mathfrak{M}}$. The premise is cheapest to make true through a false hypothesis: make $\forall x\, \neg Q(x)$ false by putting some element in $Q^{\mathfrak{M}}$. Take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \{b\}$.

Premise: $\forall x\, \neg Q(x)$ fails at $d = b$, since $b \in Q^{\mathfrak{M}}$; so the implication holds by (S5). Conclusion: at $d = a$, $a \in P^{\mathfrak{M}}$ and $a \notin Q^{\mathfrak{M}}$, so $P(x) \to Q(x)$ fails under $\ell[x \mapsto a]$, and (S6) fails. By the soundness statement of this section, the converse is not derivable, and the stalled search could not have succeeded.

4.11.10.3 C. Going deeper

Exercise. C1: True in every finite model

Let f be a one-place function symbol, and let σ be the sentence

$$\forall x\, \forall y\, \big(f(x) = f(y) \to x = y\big) \to \forall y\, \exists x\, \big(f(x) = y\big),$$

"if f is injective, then f is surjective".

(a) Prove that σ is true in every structure whose domain is finite.

(b) Give a structure with an infinite domain in which σ is false.

(c) Deduce that σ is not derivable, although no finite structure is a counter-model.

Solution (Hint 1). For (a), suppose $f^{\mathfrak{M}}$ is injective on a finite D with n elements, and count the elements of its image.

Solution (Hint 2). For (b), look for an operation on $\mathbb{N}$ that never repeats a value but misses a value.

Solution. (a) Let $\mathfrak{M}$ have a finite domain D with $n \geq 1$ elements. If the hypothesis of σ is false in $\mathfrak{M}$, then σ is true by (S5). So suppose it is true: by (S6), (S5) and (S2), for all $d, e \in D$, $f^{\mathfrak{M}}(d) = f^{\mathfrak{M}}(e)$ implies $d = e$; that is, $f^{\mathfrak{M}}$ takes different values at different elements. Then the n elements of D have n different values, so the set of values $\{f^{\mathfrak{M}}(d) : d \in D\}$ is a subset of D with n elements. A subset of an n-element set with n elements is the whole set (a counting fact from school, proved in *Sets, Relations, and Cardinality*, Chapter 3). So every $e \in D$ is $f^{\mathfrak{M}}(d)$ for some d, which is the conclusion of σ by (S6), (S7) and (S2). Hence σ is true in $\mathfrak{M}$.

(b) Take $D = \mathbb{N}$ and $f^{\mathfrak{M}}(d) = d + 1$. The hypothesis holds: $d + 1 = e + 1$ gives $d = e$. The conclusion fails at $y = 0$: there is no $d \in \mathbb{N}$ with $d + 1 = 0$, since $d \geq 0$ gives $d + 1 \geq 1$. So σ is false in $\mathfrak{M}$.

(c) The structure of (b) is a counter-model to $\models \sigma$ (with Γ empty). By the soundness statement of this section, σ is not derivable. By (a), no finite structure could have shown this: a search for counter-models that lists finite structures of size 1, 2, 3, ... would never stop.

Marking scheme. (10 points)

- 1: (a) disposes of the case where the hypothesis of σ is false, by the clause for $\to$.
- 3: (a) reads the hypothesis as injectivity of $f^{\mathfrak{M}}$ and uses finiteness to get surjectivity, naming the counting fact.
- 3: (b) a structure with infinite domain, with the hypothesis checked and the conclusion refuted at a named element.
- 3: (c) concludes non-derivability through soundness and says why finite structures could not have done it.

Fatal errors (0 for the step): in (b), a "counter-model" in which $f^{\mathfrak{M}}$ is not injective; in (a), an argument that never uses finiteness.

Exercise. C2: A sentence with no finite model

Let τ be the conjunction of the three sentences of the text saying that $<$ is a strict order with no largest element:

$$\forall x\, \neg(x < x) \;\wedge\; \forall x\, \forall y\, \forall z\, ((x < y \wedge y < z) \to x < z) \;\wedge\; \forall x\, \exists y\, (x < y).$$

Prove that every model of τ has an infinite domain. Where does your proof use that domains are non-empty?

Solution (Hint 1). Start from one element and keep climbing with the third conjunct.

Solution (Hint 2). Prove by induction on k that there are elements $d_1 < d_2 < \cdots < d_k$ of D with $d_i < d_j$ whenever $i < j$.

Solution. Let $\mathfrak{M}$ be a model of τ, with domain D, and write $d < e$ for $(d, e) \in <^{\mathfrak{M}}$. We prove, for every $k \geq 1$, the statement $A(k)$: *there are elements $d_1, \ldots, d_k$ of D with $d_i < d_j$ whenever $1 \leq i < j \leq k$.* We use Axiom 3.2.1, applied to the property $A(m+1)$ of $m \in \mathbb{N}$, so that the base case is $A(1)$.

Base. D is non-empty (Definition 4.11.1, (M1)), so there is some $d_1 \in D$, and $A(1)$ holds, with no pairs $i < j$ to check. This is the one place the proof uses non-emptiness. If an empty domain were allowed, with "for every $d \in \emptyset$" read as having nothing to check, all three conjuncts would hold in it (each begins with $\forall$), and it has no elements at all: the statement "every model is infinite" would fail.

Step. Suppose $d_1, \ldots, d_k$ are as in $A(k)$. By the third conjunct and (S6), (S7), there is $d_{k+1} \in D$ with $d_k < d_{k+1}$. For $i < k$, we have $d_i < d_k$ by $A(k)$, and then $d_i < d_{k+1}$ by the second conjunct (transitivity). So $A(k+1)$ holds.

Now the elements $d_1, \ldots, d_k$ are all different: if $d_i = d_j$ with $i < j$, then $d_i < d_j$ says $d_i < d_i$, which the first conjunct forbids. So for every $k \geq 1$, the domain contains k different elements. A finite set with n elements does not contain $n+1$ different elements, so D is not finite. Since $(\mathbb{N}, <)$ is a model of τ (as the text checked), τ has models, and all of them are infinite.

Marking scheme. (10 points)

- 2: names the non-empty domain as the start of the construction, and says what fails without it.
- 3: an induction producing, for every k, a chain of k elements, using the third conjunct for the step.
- 2: uses transitivity to keep the whole chain increasing, not only neighbors.
- 3: uses irreflexivity to show the elements are different, and concludes that D is infinite.

A second route: a proof that every non-empty finite subset of D has a $<$-maximal element (induction on its size, with irreflexivity in the base and transitivity in the step), applied to D, which is non-empty, and contradicted by the third conjunct, earns 2 for using D non-empty, 3 for the induction, 2 for transitivity in the step, and 3 for irreflexivity plus the contradiction with the third conjunct.

Fatal errors (0 for the step): building the chain without showing its elements are distinct; an "infinite chain" asserted without an induction or a recursion.

Exercise. C3: Exit check: formalize, negate, decide

Work over the domain $\mathbb{R}$, with $n \in \mathbb{Z}_{>0}$ as a bound as in Section 4.1, and $1/n$ as a term.

(a) Formalize, with bounded quantifiers: "for every $\varepsilon > 0$ there is $N \in \mathbb{Z}_{>0}$ such that $1/n < \varepsilon$ for every $n \in \mathbb{Z}_{>0}$ with $n \geq N$."

(b) Formalize the sentence obtained from (a) by moving "there is $N \in \mathbb{Z}_{>0}$" to the front.

(c) Negate each of (a) and (b) into positive form: no $\neg$ left in front of a quantifier or an implication, and $\neg(s < t)$ written as $s \geq t$.

(d) Decide which of (a) and (b) is true in $\mathbb{R}$. For the true sentence give a proof, saying what each chosen number depends on; for the false one, prove its negation, giving the witnesses it asks for.

You may use that for every real number there is a natural number larger than it.

Solution. A remark on the term $1/n$. In the structure $\mathbb{R}$, the symbol / must be a function on all of $\mathbb{R}^2$ (Definition 4.11.1, (M3)); give $x/0$ a value for every $x \in \mathbb{R}$, say $x/0 \coloneqq 0$. The bound $n \in \mathbb{Z}_{>0}$ keeps $n \geq 1$, so those values are never used.

(a) $\forall \varepsilon > 0\ \exists N \in \mathbb{Z}_{>0}\ \forall n \in \mathbb{Z}_{>0}\ (n \geq N \to 1/n < \varepsilon)$.

(b) $\exists N \in \mathbb{Z}_{>0}\ \forall \varepsilon > 0\ \forall n \in \mathbb{Z}_{>0}\ (n \geq N \to 1/n < \varepsilon)$.

(c) Push $\neg$ inward one quantifier at a time, keeping each bound (Proposition 4.9.3), then negate the implication (Proposition 4.9.2):

$$\begin{aligned} \neg(a): &\quad \exists \varepsilon > 0\ \forall N \in \mathbb{Z}_{>0}\ \exists n \in \mathbb{Z}_{>0}\ (n \geq N \wedge 1/n \geq \varepsilon), \\ \neg(b): &\quad \forall N \in \mathbb{Z}_{>0}\ \exists \varepsilon > 0\ \exists n \in \mathbb{Z}_{>0}\ (n \geq N \wedge 1/n \geq \varepsilon). \end{aligned}$$

(d) **(a) is true.** Let $\varepsilon > 0$. Choose a natural number N with $N > 1/\varepsilon$, by the fact allowed in the problem; since $1/\varepsilon > 0$, $N \in \mathbb{Z}_{>0}$, and N depends on ε. Let $n \in \mathbb{Z}_{>0}$ with $n \geq N$. Then $n \geq N > 1/\varepsilon > 0$, and multiplying $n > 1/\varepsilon$ by the positive number ε/n gives $\varepsilon > 1/n$. Since n was arbitrary, and then ε, (a) holds.

(b) is false; its negation is true. Let $N \in \mathbb{Z}_{>0}$. Take $\varepsilon = 1/N$, which is positive, and $n = N$, which is in $\mathbb{Z}_{>0}$ with $n \geq N$. Then $1/n = 1/N \geq 1/N = \varepsilon$. Both witnesses depend on N, as the order $\forall N\ \exists \varepsilon\ \exists n$ allows. Since N was arbitrary, $\neg(b)$ holds, so (b) is false. In words: no single N works for every ε, because $\varepsilon = 1/N$ defeats that N.

The two sentences differ only in the order of $\forall \varepsilon$ and $\exists N$. In (a) the N is chosen after ε and may depend on it; in (b) it must be chosen first, which is the swap of Example (The swap of ∃ and ∀) in an analysis setting.

Marking scheme. (10 points)

- 2: both formalizations, with the bounds $\varepsilon > 0$, $N \in \mathbb{Z}_{>0}$, $n \in \mathbb{Z}_{>0}$ attached to the right quantifiers and $\to$ under the $\forall n$.
- 3: both negations in positive form, with every bound kept, the quantifier order of each preserved, and $\wedge$ under the $\exists n$.
- 2: verdict "true" for (a), with a proof in which N is chosen after ε and said to depend on it.
- 3: verdict "false" for (b), with ε and n given as functions of N and the three facts $\varepsilon > 0$, $n \geq N$, $1/n \geq \varepsilon$ checked.

Fatal errors (0 for the step): in the negation of (b), ε chosen before N (a single ε for all N); a bound negated instead of the quantifier ($\exists \varepsilon \leq 0$, or $\exists n < N$).

4.11.10.4 R. Review

Exercise. R1

Decide whether each sequent is derivable, and justify your answer.

(a) $(p \to q) \to r \vdash p \to r$.
(b) $(p \to q) \to r \vdash q \to r$.

Solution. (a) Not derivable. Look for a refuting row, backwards from the conclusion: $p \to r$ is false only when $p = \mathrm{T}$ and $r = \mathrm{F}$. The premise must then be true, so its hypothesis $p \to q$ must be false, which needs $q = \mathrm{F}$. In the row $p = \mathrm{T}$, $q = \mathrm{F}$, $r = \mathrm{F}$, the premise is $\mathrm{F} \to \mathrm{F}$, true, and the conclusion is $\mathrm{T} \to \mathrm{F}$, false. By Corollary 3.5.3 there is no derivation.

(b) Derivable. The goal is an implication, so open a box on q; the premise is waiting for $p \to q$, which a box on p produces by repeating q (the goal-directed and premise-directed steps of the search method of Section 2.6).

1	$(p \to q) \to r$	
2	$\quad q$	
3	$\quad\quad p$	
4	$\quad\quad q$	R 2
5	$\quad p \to q$	$\to$I 3–4
6	$\quad r$	$\to$E 1, 5
7	$q \to r$	$\to$I 2–6

The row method of (a) could not settle (b): no row refutes it, and only a derivation (or Theorem 3.6.5) shows it is derivable.

Exercise. R2

Derive $\neg(P \to \neg Q) \vdash \neg\neg P \wedge \neg\neg Q$ without RAA and without derived rules that use it. Which rule would you need to derive $P \wedge Q$ itself?

Solution. The goal is a conjunction, so derive each conjunct. Each is a negation, so each gets a $\neg$I box, whose goal is $\bot$. The only way to use the premise $\neg(P \to \neg Q)$ is to clash with it, so inside each box we build $P \to \neg Q$. In the first box, from $\neg P$: a $\to$I box on P clashes P with $\neg P$ and uses $\bot$E (the pattern of Proposition 2.1.2). In the second, from $\neg Q$: a $\to$I box on P simply repeats $\neg Q$. These are the rules of Chapters 1 and 2 (Definition 2.2.1, Definition 2.1.1, Definition 1.4.1).

1	$\neg(P \to \neg Q)$	
2	$\quad \neg P$	
3	$\quad\quad P$	
4	$\quad\quad \bot$	$\neg$E 3, 2
5	$\quad\quad \neg Q$	$\bot$E 4
6	$\quad P \to \neg Q$	$\to$I 3–5
7	$\quad \bot$	$\neg$E 6, 1
8	$\neg\neg P$	$\neg$I 2–7
9	$\quad \neg Q$	
10	$\quad\quad P$	
11	$\quad\quad \neg Q$	R 9
12	$\quad P \to \neg Q$	$\to$I 10–11
13	$\quad \bot$	$\neg$E 12, 1
14	$\neg\neg Q$	$\neg$I 9–13
15	$\neg\neg P \wedge \neg\neg Q$	$\wedge$I 8, 14

No RAA is used. To reach $P \wedge Q$ we would need P from $\neg\neg P$ and Q from $\neg\neg Q$, that is, $\neg\neg$E, which is derived with RAA (Proposition 2.5.1), or an RAA box directly (Definition 2.5.1). The look-alike is Exercise (B1: Three searches) (b), $\neg(P \to Q) \vdash P \wedge \neg Q$: there the conjunct P needed RAA, while $\neg Q$ came by $\neg$I alone. Here both conjuncts stop at a double negation. That no derivation of P (or of $P \wedge Q$) from the premise avoids RAA is true, but we state it without proof, and nothing depends on it.

Exercise. R3

Derive $\neg\exists x\,(P(x) \to \neg Q(x)) \vdash \forall x\,(\neg\neg P(x) \land \neg\neg Q(x))$ without RAA.

Solution. The goal is $\forall x$, so open a flagged box with a fresh name x_0 (Definition 4.4.1); the new goal is $\neg\neg P(x_0) \land \neg\neg Q(x_0)$. Inside, run the derivation of R2 (Exercise (R2)) with $P(x_0)$ and $Q(x_0)$ in place of P and Q. One change: the premise is now $\neg\exists x\,(P(x) \to \neg Q(x))$, not $\neg(P \to \neg Q)$, so each clash first turns $P(x_0) \to \neg Q(x_0)$ into $\exists x\,(P(x) \to \neg Q(x))$ by $\exists$I at x_0 (Definition 4.5.1).

1	$\neg\exists x\,(P(x) \to \neg Q(x))$	
2	x_0 | $\neg P(x_0)$	
3	| | $P(x_0)$	
4	| | $\bot$	$\neg$E 3, 2
5	| | $\neg Q(x_0)$	$\bot$E 4
6	| $P(x_0) \to \neg Q(x_0)$	$\to$I 3–5
7	| $\exists x\,(P(x) \to \neg Q(x))$	$\exists$I 6
8	| $\bot$	$\neg$E 7, 1
9	$\neg\neg P(x_0)$	$\neg$I 2–8
10	| $\neg Q(x_0)$	
11	| | $P(x_0)$	
12	| | $\neg Q(x_0)$	R 10
13	| $P(x_0) \to \neg Q(x_0)$	$\to$I 11–12
14	| $\exists x\,(P(x) \to \neg Q(x))$	$\exists$I 13
15	| $\bot$	$\neg$E 14, 1
16	$\neg\neg Q(x_0)$	$\neg$I 10–15
17	$\neg\neg P(x_0) \land \neg\neg Q(x_0)$	$\land$I 9, 16
18	$\forall x\,(\neg\neg P(x) \land \neg\neg Q(x))$	$\forall$I 2–17

The side condition of $\forall$I at line 18: x_0 does not appear in the premise on line 1, no box is open at line 18, and x_0 does not appear in the conclusion. The flagged box immediately contains the $\neg$I box of line 2, so it is cited from line 2, the first line of that inner box. Lines 7 and 14 are legal $\exists$I steps: $P(x_0) \to \neg Q(x_0)$ is $(P(x) \to \neg Q(x))[x_0/x]$. The whole derivation is R2's, run inside a flagged box, with an $\exists$I before each clash.

Exercise. R4

Is the sentence $\forall x\,\exists y\,(x < y) \to \exists y\,\forall x\,(x < y)$ true in $(\mathbb{N}, <)$? Is it derivable?

Solution. True in $(\mathbb{N}, <)$*?* No. The hypothesis $\forall x\, \exists y\, (x < y)$ is true in $(\mathbb{N}, <)$: each d has $d + 1$ above it. The conclusion $\exists y\, \forall x\, (x < y)$ is false: a candidate e fails at $x = e$, since $e < e$ is false. By (S5) of Definition 4.11.3, an implication with a true hypothesis and a false conclusion is false.

Derivable? No. By the soundness statement of this section, a derivable sentence is true in every structure; this one is false in $(\mathbb{N}, <)$, so no derivation exists.

Why these tools. A structure settles the question in two lines, while a derivation search could only fail, and a failed search proves nothing. A truth table is the wrong look-alike: it reads the sentence as $p \to q$ with two atoms, and a refuting row for $p \to q$ says nothing here, since the sentence $\forall x\, P(x) \to \exists x\, P(x)$ has the same shape and is derivable (Proposition 4.5.1, then $\to$I). The soundness statement used here is proved in Section 4.12.

4.12 Soundness for Predicate Logic

This section is optional, and nothing outside it depends on it: Section 4.11 stated soundness for predicate logic without proof, and the rest of the book uses that statement. The results below build on one another, so they carry no optional marker of their own; the section as a whole is the unit a reader may skip. Here is the proof. It is Chapter 3's proof of Theorem 3.5.2 with six new cases, one for each quantifier and equality rule, and it is worth reading for one reason above all: each side condition of Sections 4.3 to 4.7 is used in the case of its own rule, and there one can see what would go wrong without it. The section proves a lemma about substitution and its corollary on renaming bound variables, then the theorem, then the corollary that turns every counter-model of Section 4.11 into a proof that no derivation exists.

4.12.1 What carries over, and what is new

Chapter 3 proved that whatever the rules of Chapters 1 and 2 derive is true in every row where the premises are true. The proof had four steps (Steps 1–4 of the proof of Theorem 3.5.2). It reduced to derivations in the primitive rules; it proved a claim for **every** line, not only the last, namely $\Gamma_k \models \varphi_k$, where Γ_k is the context of line k; it did so by strong induction on k, one case per rule; and it read off the claim for the last line. We follow the same four steps. Three things change.

The formulas. Lines now carry first-order formulas (Definition 4.2.2), and $\models$ is the entailment of Definition 4.11.4: $\Gamma_k \models \varphi_k$ says that every structure $\mathfrak{M}$ and assignment ℓ with $\mathfrak{M}, \ell \models \Gamma_k$ satisfy φ_k. Assignments are not optional here. Lines inside a flagged box can mention the box's name, a **variable** such as x_0, free; such a line is then not a sentence, and its truth depends on the value of x_0. This is why Section 4.4 insisted that fresh names are always variables, never constants: the proof below changes their values.

The rules. The primitive rules are the thirteen of Chapter 3's list (Section 3.5) and six new ones: $\forall$E (Definition 4.3.1), $\forall$I (Definition 4.4.1), $\exists$I (Definition 4.5.1), $\exists$E (Definition 4.6.1), $=$I and $=$E (Definition 4.7.1). Two of the new rules cite a **box**, the flagged box of $\forall$I and the $\exists$E box, so the box rules are now seven: Chapter 3's five, and $\forall$I and $\exists$E. The other four new rules cite only lines and join the one-step rules.

Derived rules and "arith". As in Step 1 of Theorem 3.5.2, a line citing a derived rule abbreviates a pasted derivation in primitive rules. This covers Chapter 2's derived rules (Defi-

nition 2.3.1), whose derivations have no flagged boxes and apply to first-order formulas as they stand, and the bounded rules of Proposition 4.8.1, each of which Section 4.8 expanded into a displayed block of unbounded steps whose side conditions hold whenever the bounded ones do, and the one-line "(sym)" and "(trans)" of Section 4.7 (Proposition 4.7.1), each of which abbreviates its three-line derivation from =I and =E with the given terms pasted in. Those derivations have no boxes, and each φ they use is an equation, with no quantifier, so every term is free for its variable and the side conditions of Definition 4.7.1 hold. A line justified "arith" is not a rule at all: it records an arithmetic fact. Written out, it is a universally quantified arithmetic sentence taken as an extra **premise**, whose bound variables are distinct and appear nowhere else in the derivation, instantiated by ∀E at the terms on the line and applied by →E, as Section 4.4 did explicitly when it turned the premises of a Chapter 1 derivation into universal statements. That choice of bound variables means that no fresh name of the derivation appears in the new premise, so the side conditions of ∀I and ∃E still hold, and that every term on the line is free for the variable it replaces at each ∀E step. So a derivation with "arith" lines is a derivation in primitive rules from its premises together with those arithmetic sentences, and soundness speaks about that derivation.

Renaming. Section 4.2 took, as a convention, that formulas differing only by renaming bound variables say the same thing, so a derivation may write a line, or apply a rule to a cited line, in a renamed form. This is not a rule either. Corollary 4.12.2 below shows that a formula and any renamed form of it are satisfied by exactly the same structures and assignments, so such a step changes nothing in the proof, which may take every line in the form its rule displays.

4.12.2 Contexts, once more

The context Γ_k of line k is still the list of the premises together with the assumptions of every box containing line k (Section 3.5, before Lemma 3.5.1). A flagged box of ∀I has **no** assumption, so it adds nothing to the context of the lines inside it. The ∃E box opens with its assumption $\varphi[x_0/x]$, which does join the context.

Section 4.4 extended Definition 1.4.2 to flagged boxes and kept its scope rule and its notion of availability word for word; two boxes are still disjoint or one inside the other. So Lemma 3.5.1 (a), whose proof uses only the scope rule, holds as it stands: if line i is available at line k, then $\Gamma_i \subseteq \Gamma_k$.

Part (b) needs one extension, since its statement names "the assumption of that box". It becomes: *if line k is justified by a box rule citing the box from line a to line b, then Γ_b is Γ_k together with the assumption of that box if it has one, and Γ_b is Γ_k if it is a flagged box without an assumption.* The proof of (b) in Section 3.5 is unchanged. It showed that the boxes containing line b are exactly the cited box and the boxes containing line k, and for that it used one fact about box rules: the last line of the cited box is a line of that box itself, not of a box inside it. The two new box rules demand the same: ∀I requires $\varphi[x_0/x]$ to be the last line of the flagged box, a line of that box itself (Definition 4.4.1), and ∃E requires χ to be the last line of the box itself (Definition 4.6.1). So the boxes containing line b are the cited box and those containing line k, and the context of line b adds to Γ_k only the cited box's assumption, if any. When a flagged box immediately contains another box and is cited from the first line of that inner box, as the citation convention of Section 4.4 says, the argument is the same: the cited box runs from that line to its last line, and its last line lies in no box inside it.

Here is the derivation of Proposition 4.6.2, with its contexts.

1	$\exists x\,\forall y\,R(x,y)$	
2	$y_0 \mid x_0 \mid \forall y\,R(x_0,y)$	
3	$R(x_0,y_0)$	$\forall$E 2
4	$\exists x\,R(x,y_0)$	$\exists$I 3
5	$\exists x\,R(x,y_0)$	$\exists$E 1, 2–4
6	$\forall y\,\exists x\,R(x,y)$	$\forall$I 2–5

Line	Boxes containing it	Context
1	none	$\exists x\,\forall y\,R(x,y)$
2, 3, 4	flagged y_0 (2–5); $\exists$E box x_0 (2–4)	$\exists x\,\forall y\,R(x,y)$, $\forall y\,R(x_0,y)$
5	flagged y_0 (2–5)	$\exists x\,\forall y\,R(x,y)$
6	none	$\exists x\,\forall y\,R(x,y)$

Line 6 cites the flagged box 2–5, which has no assumption, and indeed $\Gamma_5 = \Gamma_6$. Line 5 cites the $\exists$E box 2–4, and Γ_4 is Γ_5 with its assumption $\forall y\,R(x_0,y)$ added. Line 3 mentions x_0 and y_0 free, and line 4 mentions y_0 free, while its context mentions x_0 free, through the assumption $\forall y\,R(x_0,y)$: the claim $\Gamma_4 \models \exists x\,R(x,y_0)$ is about every structure **and every value** of x_0 and y_0 that makes the context true.

4.12.3 Substitution and satisfaction

Five of the six new rules write a substitution: $\forall$E and $\exists$I write $\varphi[t/x]$ for a term t, in their conclusion or premise; $=$E writes one in both; and $\forall$I and $\exists$E write $\varphi[x_0/x]$ for the fresh name. To handle them we need to know what $\varphi[t/x]$ **means** in a structure. The natural guess is that substituting the term t for x has the same effect as giving x the value of t. Section 4.2 showed that this can fail: in $\exists y\,(y > x)$, putting $y+1$ for x blindly gives $\exists y\,(y > y+1)$, which is false in $\mathbb{Z}$, while giving x the value of $y+1$, whatever it is, leaves $\exists y\,(y > x)$ true. The y of the term was captured. The lemma says that capture is the only thing that can go wrong.

Lemma 4.12.1. Substitution Lemma

Let $\mathfrak{M}$ be a structure, ℓ an assignment in $\mathfrak{M}$, φ a formula, x a variable, and t a term that is free for x in φ (Definition 4.2.6). Write $a := t^{\mathfrak{M},\ell}$. Then

$$\mathfrak{M},\ell \models \varphi[t/x] \quad \text{if and only if} \quad \mathfrak{M},\ell[x \mapsto a] \models \varphi.$$

Idea. First terms: substituting t for x in a term s and then evaluating is the same as evaluating s with x set to the value of t. Then structural induction on φ, for all assignments at once, since the quantifier clauses change the assignment. Atomic formulas follow from the claim for terms, and connectives pass straight through. The only real case is $\forall y\,\varphi$ with x free in it. The substitution goes inside, and (S6) sets y to each d in turn. For the induction hypothesis to help, the value of t must not change when y does, and that is exactly what "free for" guarantees: y does not occur in t.

Proof. Fix $\mathfrak{M}$, x and t. For an assignment ℓ, write $a_\ell := t^{\mathfrak{M},\ell}$.

Terms. We show, for every term s and every assignment ℓ, that $(s[t/x])^{\mathfrak{M},\ell} = s^{\mathfrak{M},\ell[x\mapsto a_\ell]}$, by structural induction on s (Definition 4.2.1). If s is the variable x, then $s[t/x]$ is t, with value a_ℓ, and $s^{\mathfrak{M},\ell[x\mapsto a_\ell]} = \ell[x \mapsto a_\ell](x) = a_\ell$ (Definition 4.11.2). If s is another variable y, then $s[t/x]$ is y, and both sides are $\ell(y)$, since $\ell[x \mapsto a_\ell]$ changes only x. If s is a constant c, both sides are $c^{\mathfrak{M}}$. If s is $f(s_1, \ldots, s_n)$, then $s[t/x]$ is $f(s_1[t/x], \ldots, s_n[t/x])$ (Definition 4.2.5), and applying $f^{\mathfrak{M}}$ to the values, which agree by the hypothesis for $s_1, \ldots, s_n$, gives equal results.

Formulas. We prove, by structural induction on φ, the property $B(\varphi)$: *if t is free for x in φ, then for every assignment ℓ, $\mathfrak{M}, \ell \models \varphi[t/x]$ if and only if $\mathfrak{M}, \ell[x \mapsto a_\ell] \models \varphi$.* Two facts are used throughout. If t is free for x in $\neg\varphi$ or in $\varphi \circ \psi$, it is free for x in the parts, by the recursive description of "free for" after Definition 4.2.6 in Section 4.2. And by Definition 4.2.5, substitution passes through connectives: $(\neg\varphi)[t/x]$ is $\neg(\varphi[t/x])$, and $(\varphi \circ \psi)[t/x]$ is $\varphi[t/x] \circ \psi[t/x]$.

Atomic formulas. $R(s_1, \ldots, s_n)[t/x]$ is $R(s_1[t/x], \ldots, s_n[t/x])$. By the claim for terms, the tuple of values of the $s_i[t/x]$ under ℓ is the tuple of values of the s_i under $\ell[x \mapsto a_\ell]$, so (S1) of Definition 4.11.3 gives the same answer on both sides. Equations $s_1 = s_2$ are the same with (S2). For $\bot$, $\bot[t/x]$ is $\bot$, and neither side holds (S3).

Connectives. For $\varphi \wedge \psi$: by $B(\varphi)$ and $B(\psi)$, $\mathfrak{M}, \ell \models \varphi[t/x]$ if and only if $\mathfrak{M}, \ell[x \mapsto a_\ell] \models \varphi$, and likewise for ψ; (S5) combines the two answers in the same way on both sides. The cases $\neg$, $\vee$, $\to$, $\leftrightarrow$ are the same with (S4) and the other parts of (S5).

Universal quantifier, x not free in $\forall y\, \varphi$. This happens when y is the variable x, or when x is not free in φ. Then $(\forall y\, \varphi)[t/x]$ is $\forall y\, \varphi$ itself, since there is no free occurrence of x to replace (Section 4.2 proved that $\psi[t/x] = \psi$ whenever x is not free in ψ). The assignments ℓ and $\ell[x \mapsto a_\ell]$ differ at most at x, which is not free in $\forall y\, \varphi$; so by Lemma 4.11.1, $\mathfrak{M}, \ell \models \forall y\, \varphi$ if and only if $\mathfrak{M}, \ell[x \mapsto a_\ell] \models \forall y\, \varphi$, which is $B(\forall y\, \varphi)$.

Universal quantifier, x free in $\forall y\, \varphi$. Then y is not x, and x is free in φ. Every free occurrence of x in $\forall y\, \varphi$ lies in the scope of $\forall y$, so, as t is free for x in $\forall y\, \varphi$, **the variable y does not occur in t**, and t is free for x in φ (Definition 4.2.6, and its recursive description in Section 4.2). By Definition 4.2.5, $(\forall y\, \varphi)[t/x]$ is $\forall y\, (\varphi[t/x])$. Let $d \in D$. Since y does not occur in t, the assignments ℓ and $\ell[y \mapsto d]$ agree on every variable of t, so $a_{\ell[y\mapsto d]} = a_\ell$, by the claim for terms in the proof of Lemma 4.11.1. Since x and y are different variables, $\ell[y \mapsto d][x \mapsto a_\ell] = \ell[x \mapsto a_\ell][y \mapsto d]$. Hence, using $B(\varphi)$ for the assignment $\ell[y \mapsto d]$ in the second step,

$$\begin{aligned}
\mathfrak{M}, \ell \models \forall y\, (\varphi[t/x]) &\iff \text{for every } d \in D,\ \mathfrak{M}, \ell[y \mapsto d] \models \varphi[t/x] \\
&\iff \text{for every } d \in D,\ \mathfrak{M}, \ell[y \mapsto d][x \mapsto a_\ell] \models \varphi \\
&\iff \text{for every } d \in D,\ \mathfrak{M}, \ell[x \mapsto a_\ell][y \mapsto d] \models \varphi \\
&\iff \mathfrak{M}, \ell[x \mapsto a_\ell] \models \forall y\, \varphi,
\end{aligned}$$

where the first and last steps are (S6). This is $B(\forall y\, \varphi)$.

Existential quantifier. The same two subcases, with (S7) in place of (S6) and "for some d" in place of "for every d".

Every clause of Definition 4.2.2 is covered, so $B(\varphi)$ holds for every formula. This proves the lemma.

■

The hypothesis "free for" was used once, in the second quantifier case, to get $a_{\ell[y\mapsto d]} = a_\ell$. The capture example shows what happens without it. In the structure on $\mathbb{Z}$ with $>$, $+$, and 1 as usual, take $\varphi = \exists y\,(y > x)$ and $t = y + 1$, which is **not** free for x in φ. The blind substitution is $\exists y\,(y > y + 1)$, false under every assignment. But $\mathfrak{M}, \ell[x \mapsto a] \models \exists y\,(y > x)$ holds for every $a \in \mathbb{Z}$, with witness $a + 1$. The two sides of the lemma disagree, and the reason is visible in the displayed chain: when (S7) sets y to a new value, the value of $y + 1$ moves with it.

Quick check. Let $\mathfrak{M}$ be the structure on $\mathbb{Z}$ with $<$, $+$, and 1 as usual, $\varphi = \forall z\,(x < z \to y < z)$, $t = y{+}1$, and let $\ell(x) = 3$, $\ell(y) = 5$. Is t free for x in φ? Compute both sides of Lemma 4.12.1.

Solution. Yes: the only quantifier in φ is $\forall z$, and z does not occur in $y{+}1$. Here $a = t^{\mathfrak{M},\ell} = 6$. The right side asks whether $\mathfrak{M}, \ell[x \mapsto 6] \models \varphi$: every $z > 6$ is > 5, so yes. The left side is $\varphi[t/x] = \forall z\,(y + 1 < z \to y < z)$ under ℓ: every $z > 6$ is > 5, so yes. Both sides hold, as the lemma says. (Under ℓ itself, φ is false: $z = 4$ has $3 < 4$ and not $5 < 4$. The lemma compares $\varphi[t/x]$ with φ at the **new** value of x, not at the old one.)

The lemma also pays a debt of Section 4.2, which took renaming of bound variables inside a formula as a convention and left its justification in terms of meaning to this section.

Corollary 4.12.2. Renaming Keeps Meaning

Let φ be a formula and y, z variables, with z not appearing in φ (Section 4.2). Let $\mathfrak{M}$ be a structure for the symbols of φ, with domain D.

(a) For every assignment ℓ, $\mathfrak{M}, \ell \models \forall y\,\varphi$ if and only if $\mathfrak{M}, \ell \models \forall z\,\varphi[z/y]$, and $\mathfrak{M}, \ell \models \exists y\,\varphi$ if and only if $\mathfrak{M}, \ell \models \exists z\,\varphi[z/y]$.

(b) If ψ' arises from a formula ψ by such a renaming of one quantifier anywhere inside ψ (Section 4.2), then for every structure $\mathfrak{M}$ for the symbols of ψ and every assignment ℓ in it, $\mathfrak{M}, \ell \models \psi$ if and only if $\mathfrak{M}, \ell \models \psi'$.

Proof. *(a)* If z is y, then $\varphi[y/y]$ is φ and there is nothing to prove. Let z be a variable other than y. Since z does not appear in φ, no quantifier of φ is on z, so z is free for y in φ (Definition 4.2.6). Let $d \in D$. Under $\ell[z \mapsto d]$ the term z has value d, so Lemma 4.12.1 gives: $\mathfrak{M}, \ell[z \mapsto d] \models \varphi[z/y]$ if and only if $\mathfrak{M}, \ell[z \mapsto d][y \mapsto d] \models \varphi$. The assignments $\ell[z \mapsto d][y \mapsto d]$ and $\ell[y \mapsto d]$ differ at most at z, which does not appear in φ; by Lemma 4.11.1, the last statement holds if and only if $\mathfrak{M}, \ell[y \mapsto d] \models \varphi$. So $\mathfrak{M}, \ell[z \mapsto d] \models \varphi[z/y]$ if and only if $\mathfrak{M}, \ell[y \mapsto d] \models \varphi$, for every $d \in D$. Now (S6) gives the statement for $\forall$, and (S7) the statement for $\exists$.

(b) By structural induction on ψ, for all assignments at once, since the quantifier clauses change the assignment. If the renamed quantifier is the outermost symbol of ψ, this is (a). Otherwise ψ is not atomic, and the renaming takes place inside one immediate part of ψ, which it turns into a formula satisfied by the same assignments, by the induction hypothesis; the other parts are unchanged. If ψ is $\neg\alpha$ or $\alpha \circ \beta$, (S4) or (S5) then gives the same answer for ψ and ψ' under every ℓ. If ψ is $\forall w\,\alpha$ or $\exists w\,\alpha$, with the renaming inside α, then ψ' is

$\forall w\,\alpha'$ or $\exists w\,\alpha'$, and α, α' are satisfied by the same assignments $\ell[w \mapsto d]$, so (S6) or (S7) gives the same answer for ψ and ψ'.

■

Several renamings one after another are covered by applying (b) once for each.

4.12.4 The soundness theorem

We can now state the theorem that Section 4.11 took on trust.

Theorem 4.12.3. Soundness for Predicate Logic

Let Γ be a list of formulas and ψ a formula. If $\Gamma \vdash \psi$ with the rules of Chapters 1 and 2 and the quantifier and equality rules of Sections 4.3 to 4.7, then $\Gamma \models \psi$ (Definition 4.11.4).

Idea. The steps are those of Chapter 3's proof (Steps 1–4 of Theorem 3.5.2).

① Reduce to derivations in the primitive rules.

② Prove $\Gamma_k \models \varphi_k$ for every line k, by strong induction on k (Theorem 3.2.2): fix a structure $\mathfrak{M}$ and an assignment ℓ satisfying Γ_k, and show that they satisfy φ_k.

③ The cases. Premises, assumptions, and the thirteen propositional rules go through word for word, because (S3)–(S5) are Chapter 3's clauses. $\forall$E, $\exists$I, and $=$E are Lemma 4.12.1, used once or twice. $=$I is (S2). The two box rules are the heart of the proof, and they both use the same trick: **change the value of the fresh name**. In $\forall$I, to show $\forall x\,\varphi$ under ℓ, take any d, and set x_0 to d; the context does not notice, because x_0 is not in it, so the box's last line $\varphi[x_0/x]$ holds there, and moving d from x_0 to x gives φ under $\ell[x \mapsto d]$. In $\exists$E, set x_0 to the witness d that $\exists x\,\varphi$ provides; the box's assumption holds there, so its last line χ holds there, and since χ does not mention x_0, it holds under ℓ as well.

④ The last line lies in no box, so its context is the premises.

Before reading on: in the $\forall$I case below, the assignment $\ell[x_0 \mapsto d]$ must still satisfy the context Γ_k, and at the end the conclusion must come back from $\ell[x_0 \mapsto d][x \mapsto d]$ to $\ell[x \mapsto d]$. Which clause of the side condition of $\forall$I does each step use?

Proof. **Step 1 (reduction).** Suppose $\Gamma \vdash \psi$. By the subsection "What carries over, and what is new", there is a derivation of ψ in the primitive rules whose premises all belong to Γ. (A derivation with "arith" lines is of this kind only for the list of its premises together with the arithmetic sentences A those lines stand for, as explained there. The theorem applies to it with the list Γ, A in place of Γ, and gives $\Gamma, A \models \psi$: truth in every structure that satisfies Γ and A. The sentences A hold, for instance, when the arithmetic symbols have their usual meaning on $\mathbb{R}$, as in Section 4.11.) Fix one, with lines $1, \ldots, n$, formulas φ_k, and contexts Γ_k. Let Σ be the finite set of constants, function symbols and predicate symbols that occur in some line of it. A line may contain symbols that occur neither in Γ nor in ψ, as c does in the line $P(c)$ between $\forall x\,P(x)$ and $\exists x\,P(x)$; so the induction is run over structures that interpret every symbol of Σ, and Step 4 returns to the structures of Definition 4.11.4.

Step 2 (the induction). For $k \in \mathbb{N}$, let $S(k)$ be: *if* $1 \le k \le n$, *then every structure* $\mathfrak{M}$ *that interprets every symbol of* Σ *and every assignment* ℓ *in* $\mathfrak{M}$ *with* $\mathfrak{M}, \ell \models \Gamma_k$ *satisfy* $\mathfrak{M}, \ell \models \varphi_k$. In Steps 2 and 3, "structure" means a structure that interprets every symbol of Σ, and possibly more. We prove $S(k)$ for all k by Theorem 3.2.2. For $k = 0$ and $k > n$, $S(k)$ holds vacuously. Let $1 \le k \le n$ and assume $S(j)$ for every $j < k$. Every line and box that line k cites is available at line k, so lies before it (Definition 1.4.2); hence the hypothesis holds for every cited line and for the last line of every cited box. Let $\mathfrak{M}$ be a structure, with domain D, and ℓ an assignment with $\mathfrak{M}, \ell \models \Gamma_k$. We show $\mathfrak{M}, \ell \models \varphi_k$. Two consequences of the hypothesis do the bookkeeping, as in the proof of Theorem 3.5.2. If line k cites line i, then $\Gamma_i \subseteq \Gamma_k$ by Lemma 3.5.1 (a), so $\mathfrak{M}, \ell \models \Gamma_i$, and $S(i)$ gives

$$\mathfrak{M}, \ell \models \varphi_i \quad \text{for every line } i \text{ cited by line } k. \tag{$*$}$$

If line k cites, by a box rule, the box from line a to line b, then by Lemma 3.5.1 (b), extended in the subsection "Contexts, once more", and $S(b)$:

every structure and assignment satisfying Γ_k and the box's assumption, if any, satisfy φ_b. (**)

Step 3 (the cases).

Claim. If line k is a premise, an assumption, or justified by one of the thirteen primitive rules of Chapters 1 and 2, then $\mathfrak{M}, \ell \models \varphi_k$.

Proof of Claim. Claims 1–3 in the proof of Theorem 3.5.2 treat these cases. Each step there uses only $(*)$, $(**)$, the fact that a premise or an assumption belongs to its own context, and the clauses (V1)–(V6) of Definition 3.3.2. Clauses (S3)–(S5) of Definition 4.11.3 are (V1)–(V6) with "v makes φ true" replaced by "$\mathfrak{M}, \ell \models \varphi$". Making that replacement throughout, the proofs of those claims prove this one, case by case.

■

Claim. If line k is justified by ∀E, ∃I, =I, or =E, then $\mathfrak{M}, \ell \models \varphi_k$.

Proof of Claim. ∀E. Line k is $\varphi[t/x]$, citing a line holding $\forall x\, \varphi$, with t free for x in φ. By $(*)$, $\mathfrak{M}, \ell \models \forall x\, \varphi$, so by (S6), $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$ for every $d \in D$, in particular for $d = t^{\mathfrak{M},\ell}$. By Lemma 4.12.1, which applies because t is free for x in φ, $\mathfrak{M}, \ell \models \varphi[t/x]$.

∃I. Line k is $\exists x\, \varphi$, citing a line holding $\varphi[t/x]$, with t free for x in φ. By $(*)$, $\mathfrak{M}, \ell \models \varphi[t/x]$. By Lemma 4.12.1, $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$ with $d = t^{\mathfrak{M},\ell} \in D$. So d is the element (S7) asks for, and $\mathfrak{M}, \ell \models \exists x\, \varphi$.

=I. Line k is $t = t$. The two sides have the same value $t^{\mathfrak{M},\ell}$, so (S2) holds.

=E. Line k is $\varphi[t_2/x]$, citing lines holding $t_1 = t_2$ and $\varphi[t_1/x]$, with t_1 and t_2 free for x in φ. By $(*)$ and (S2), $t_1^{\mathfrak{M},\ell}$ and $t_2^{\mathfrak{M},\ell}$ are the same element d; and $\mathfrak{M}, \ell \models \varphi[t_1/x]$. By Lemma 4.12.1 for t_1, $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. By Lemma 4.12.1 for t_2, read from right to left, $\mathfrak{M}, \ell \models \varphi[t_2/x]$.

■

Claim. If line k is justified by ∀I or ∃E, then $\mathfrak{M}, \ell \models \varphi_k$.

Proof of Claim. ∀I. Line k is $\forall x\, \varphi$, citing a flagged box with name x_0 and no assumption, whose last line b holds $\varphi[x_0/x]$. The side condition of Definition 4.4.1 says that at line k the name x_0 appears in no premise and in no assumption of a box still open at line k, and

not in $\forall x\,\varphi$. The premises and the assumptions of the boxes open at line k are exactly the formulas of Γ_k, so

(i) x_0 appears in no formula of Γ_k; and

(ii) x_0 does not appear in $\forall x\,\varphi$, so x_0 is not the variable x, and x_0 does not appear in φ.

The flagged box has no assumption, so $(**)$ says: every structure and assignment satisfying Γ_k satisfy $\varphi[x_0/x]$.

Let $d \in D$; we show $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. Put $\ell' := \ell[x_0 \mapsto d]$. It differs from ℓ at most at x_0, which by (i) is free in no formula of Γ_k; so by Lemma 4.11.1, $\mathfrak{M}, \ell' \models \Gamma_k$. **This is where the fresh-name condition is used:** changing the value of x_0 cannot disturb anything we assumed. By $(**)$, $\mathfrak{M}, \ell' \models \varphi[x_0/x]$. By (ii), x_0 does not appear in φ, so no quantifier of φ is on x_0, and the term x_0 is free for x in φ (Definition 4.2.6). Its value under ℓ' is d, so Lemma 4.12.1 gives

$$\mathfrak{M}, \ell[x_0 \mapsto d][x \mapsto d] \models \varphi.$$

The assignments $\ell[x_0 \mapsto d][x \mapsto d]$ and $\ell[x \mapsto d]$ agree at x, where both give d, and at every variable other than x and x_0, where both give the value ℓ gives. They can differ only at x_0, which by (ii) does not appear in φ. By Lemma 4.11.1, $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. Since $d \in D$ was arbitrary, (S6) gives $\mathfrak{M}, \ell \models \forall x\,\varphi$.

$\exists$E. Line k is χ, citing a line h holding $\exists x\,\varphi$ and a box with name x_0 that opens with the assumption $\varphi[x_0/x]$ and ends with χ. The side condition of Definition 4.6.1 says that x_0 appears neither in χ, nor in $\exists x\,\varphi$, nor in any premise, nor in any assumption of a box still open at line k. So

(i) x_0 appears in no formula of Γ_k;

(ii) x_0 is not x, and does not appear in φ;

(iii) x_0 does not appear in χ.

By $(**)$, every structure and assignment satisfying Γ_k and $\varphi[x_0/x]$ satisfy χ.

By $(*)$ for line h, $\mathfrak{M}, \ell \models \exists x\,\varphi$; by (S7), choose $d \in D$ with $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. Put $\ell' := \ell[x_0 \mapsto d]$. First, $\mathfrak{M}, \ell' \models \Gamma_k$, by (i) and Lemma 4.11.1, as in the $\forall$I case. Second, $\mathfrak{M}, \ell' \models \varphi[x_0/x]$. Indeed, $\ell'[x \mapsto d] = \ell[x_0 \mapsto d][x \mapsto d]$ and $\ell[x \mapsto d]$ differ at most at x_0, which by (ii) does not appear in φ, so $\mathfrak{M}, \ell'[x \mapsto d] \models \varphi$ by Lemma 4.11.1; the term x_0 is free for x in φ by (ii), and has value d under ℓ', so Lemma 4.12.1 gives $\mathfrak{M}, \ell' \models \varphi[x_0/x]$. **This is where "$x_0$ not in $\exists x\,\varphi$" is used.** By $(**)$, $\mathfrak{M}, \ell' \models \chi$. Finally, ℓ' and ℓ differ at most at x_0, which by (iii) does not appear in χ, so $\mathfrak{M}, \ell \models \chi$ by Lemma 4.11.1. **This is where "x_0 not in χ" is used.**

■

The three claims cover every way a line can be justified, so $S(k)$ holds, and by Theorem 3.2.2 $S(k)$ holds for every line k of the derivation.

Step 4 (the last line). The last line n lies in no box (Definition 1.4.2), so Γ_n consists of premises only, and $\varphi_n = \psi$. Let $\mathfrak{M}$ be a structure for the symbols occurring in Γ and ψ, with domain D, and ℓ an assignment with $\mathfrak{M}, \ell \models \Gamma$. Expand $\mathfrak{M}$ to a structure $\mathfrak{M}'$ with the same domain and the same meanings, which also interprets the symbols of Σ that $\mathfrak{M}$ does not, so that $\mathfrak{M}'$ interprets every symbol of Σ: fix some $e \in D$, which exists because D is non-empty (M1), and give each new constant the value e, each new function symbol, of arity m say, the function $D^m \to D$ with constant value e, and each new predicate symbol the empty relation.

Then ℓ is an assignment in $\mathfrak{M}'$, and by the remark after Definition 4.11.4 (the clauses of Definition 4.11.3 consult only the symbols of the formula at hand), each formula of Γ, and ψ, is satisfied by $\mathfrak{M}', \ell$ exactly when it is satisfied by $\mathfrak{M}, \ell$. Every premise belongs to Γ, so $\mathfrak{M}', \ell \models \Gamma_n$; by $S(n)$, $\mathfrak{M}', \ell \models \psi$; and so $\mathfrak{M}, \ell \models \psi$. This proves $\Gamma \models \psi$.

■

The answer to the prompt: the first step uses the clauses "no premise" and "no open assumption", which together say that x_0 is not in Γ_k; the second uses "not in $\forall x\, \varphi$". The last line $\varphi[x_0/x]$ of the box, which may mention x_0 as often as it likes, plays no part in either. That is the precise sense in which "let x_0 be arbitrary" works: whatever value x_0 is given, what we know still holds, so what we proved about x_0 holds at that value.

The $\exists$E case uses the same move with the opposite purpose. There the value of x_0 is **not** arbitrary; it is set to the witness d. What makes that legitimate is that nothing else knows about x_0 (condition (i)), that the assumption $\varphi[x_0/x]$ then holds at the witness (condition (ii)), and that the result χ forgets it again (condition (iii)).

4.12.5 Counter-models prove nonderivability

The corollary is the one Section 4.11 used, now with a proof.

Corollary 4.12.4. Counter-models

Let Γ be a list of formulas and ψ a formula. If some structure $\mathfrak{M}$ and assignment ℓ satisfy every formula of Γ and do not satisfy ψ, then $\Gamma \nvdash \psi$.

Proof. The pair $\mathfrak{M}, \ell$ shows that $\Gamma \models \psi$ fails (Definition 4.11.4). If $\Gamma \vdash \psi$ held, Theorem 4.12.3 would give $\Gamma \models \psi$. So $\Gamma \vdash \psi$ does not hold.

■

If the derivations in question may use "arith" lines, a counter-model must also satisfy the arithmetic sentences those lines stand for (Step 1 of the proof of Theorem 4.12.3), for instance by giving the arithmetic symbols their usual meaning.

Every counter-model of Section 4.11 is now a proof that no derivation exists.

- By Example (The swap of $\exists$ and $\forall$), $\forall y\, \exists x\, R(x, y) \nvdash \exists x\, \forall y\, R(x, y)$. The converse search of Proposition 4.6.2 stalled at two lines, a $\forall$I blocked by an open assumption and an $\exists$E blocked by its conclusion; now we know that no other route could have succeeded.
- The two-element structure with $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \{b\}$ gives $\exists x\, P(x) \wedge \exists x\, Q(x) \nvdash \exists x\, (P(x) \wedge Q(x))$ and $\forall x\, (P(x) \vee Q(x)) \nvdash \forall x\, P(x) \vee \forall x\, Q(x)$: the one-way laws of Proposition 4.9.4 are one-way.
- $\exists x\, P(x) \nvdash P(c)$, which Exercise (C2: A witness is not a name) could only explain informally.
- $P(x_0) \nvdash \forall x\, P(x)$, by the structure on $\mathbb{Z}$ with $P^{\mathfrak{M}}$ the even integers and $\ell(x_0) = 2$: generalizing over a name in a premise is not merely forbidden, it would be unsound.
- The passage laws of Proposition 4.9.5 need x not free in ψ: with x free, $\nvdash \exists x\, P(x) \to P(x)$, by the counter-model of Exercise (B1: Three counter-models) (iii), which names an assignment.

Remark. The converse of Theorem 4.12.3 is also true: if $\Gamma \models \psi$, then $\Gamma \vdash \psi$. This is the **completeness theorem** for predicate logic, proved by Gödel in 1929. Its proof needs to build a structure out of the formulas themselves, and it is beyond this book; nothing here depends on it.

Remark. One difference from Chapter 3 remains even with completeness: there is no truth table to check. Whether a sentence of predicate logic is derivable cannot, in general, be decided by any mechanical procedure (Church and Turing, 1936). A failed search and a missing counter-model can both go on for ever.

4.12.6 Every side condition is a case of the proof

The side conditions of Sections 4.3 to 4.7 were introduced one at a time, each with a counterexample. The proof above shows that they are all the same kind of thing: each is used in the case of its own rule, and dropping it breaks a step of that case. Each counterexample of Sections 4.3 to 4.7 can be read as a failed case:

- generalizing over a name in a premise, as in "$P(x_0) \vdash \forall x\, P(x)$" (Section 4.4): the ∀I case fails at "$\mathfrak{M}, \ell[x_0 \mapsto d] \models \Gamma_k$", since changing x_0 changes the premise $P(x_0)$;
- generalizing over a name in an open assumption, as in "from $x_0 > 0$, $\forall x\,(x > 0)$" (Section 4.4): the same step, with the assumption $x_0 > 0$ in Γ_k;
- a name that leaks into the conclusion of ∀I, as in Exercise (C1: The conclusion clause of for-all-introduction): the last step of the ∀I case, since x_0 now occurs in φ;
- treating the witness as arbitrary, by ∀I over its name inside its ∃E box (Section 4.6): the ∀I case again, since the box's assumption is in Γ_k;
- letting the witness escape its box, or giving a second witness a name already in an open assumption (Section 4.6): the ∃E case, at condition (iii) or (i);
- a capture in ∀E (Section 4.3) or ∃I (Section 4.5): the ∀E or ∃I case, where Lemma 4.12.1 does not apply; Exercise C2 and the solution to Exercise B1 give false conclusions;
- a capture in =E, as in $a = y$, $\exists y\,(y < a) \vdash \exists y\,(y < y)$ (Section 4.7): the =E case, where Lemma 4.12.1 does not apply to t_2.

Warning. A side condition is not mere bookkeeping: a derivation that breaks one is not covered by Theorem 4.12.3, and it can reach a false line, as the counter-model to $P(x_0) \vdash \forall x\, P(x)$ in the previous subsection shows for the first item of the list.

Quick check. Exercise (C2: Find the error) gives a derivation of $\exists x\,(P(x) \wedge Q(x)) \vdash \exists x\, P(x)$ whose ∃E box ends in $P(x_0)$; the line after the box is $P(x_0)$ by ∃E, and the last line is $\exists x\, P(x)$ by ∃I. Which case of the proof of Theorem 4.12.3 fails, at which step, and for which line? Give a structure and an assignment showing that the claim $\Gamma_k \models \varphi_k$ is false for that line.

Solution. The ∃E case fails, at its last step: $\chi = P(x_0)$ contains x_0, so condition (iii) is false, and "$\mathfrak{M}, \ell' \models \chi$" cannot be carried back from $\ell' = \ell[x_0 \mapsto d]$ to ℓ. The line concerned is the

illegal ∃E line $P(x_0)$, whose context is the premise alone. Take $D = \{a, b\}$, $P^{\mathfrak{M}} = Q^{\mathfrak{M}} = \{a\}$ and $\ell(x_0) = b$: the premise $\exists x\,(P(x) \wedge Q(x))$ holds (witness a), and $P(x_0)$ fails, since $b \notin P^{\mathfrak{M}}$. The final conclusion $\exists x\,P(x)$ is still true in this structure, as it is in every structure satisfying the premise; the fault is in the line before it.

4.12.7 Exercises

4.12.7.1 A. Check your understanding

Exercise. A1

These parts are about the proof of Theorem 4.12.3.

(a) In the ∀I case, name the two places where the side condition of ∀I is used, and say which clause each uses.

(b) The derivation below breaks the side condition of ∃E: its conclusion $P(x_0)$ contains the fresh name. Which step of the ∃E case fails? Give a structure and an assignment in which the premise holds and line 3 does not.

(c) Give the idea of the proof in two sentences.

(d) Why does the claim $\Gamma_k \models \varphi_k$ quantify over assignments, and not only over structures? Point to a line of the derivation of Proposition 4.6.2 printed in this section.

1	$\exists x\,P(x)$	
2	x_0 | $P(x_0)$	
3	$P(x_0)$	∃E 1, 2–2

Solution. *(a)* First, to show that $\ell[x_0 \mapsto d]$ still satisfies Γ_k: this uses "x_0 appears in no premise and in no assumption of a box open at line k", which together say that x_0 is in no formula of Γ_k, so Lemma 4.11.1 applies. Second, to pass from $\ell[x_0 \mapsto d][x \mapsto d]$ to $\ell[x \mapsto d]$ (and, before that, to know that x_0 is free for x in φ): this uses "x_0 does not appear in $\forall x\,\varphi$".

(b) The last step: from $\mathfrak{M}, \ell[x_0 \mapsto d] \models \chi$ to $\mathfrak{M}, \ell \models \chi$, which needs x_0 not in χ. Take $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $\ell(x_0) = b$. The premise $\exists x\,P(x)$ holds, with witness a; line 3, $P(x_0)$, fails, since $\ell(x_0) = b \notin P^{\mathfrak{M}}$. Inside the box, under $\ell[x_0 \mapsto a]$, line 2 holds; the proof would carry it back to ℓ, and that is exactly what fails.

(c) Every line of a derivation is satisfied by every structure and assignment that satisfy its context, by strong induction on the line, with one case per rule. The quantifier box rules are handled by changing the value of the fresh name, which the side conditions guarantee the context and the conclusion cannot notice.

(d) Lines inside boxes contain free variables, the fresh names, and their truth depends on the values of those variables. Line 3, $R(x_0, y_0)$, is not a sentence: in the structure on $\mathbb{Z}$ with R read as $<$, it is satisfied when $\ell(x_0) = 1$, $\ell(y_0) = 2$, and not when $\ell(x_0) = 2$, $\ell(y_0) = 1$. The ∀I and ∃E cases work precisely by changing such values.

4.12.7.2 B. Practice

Exercise. B1: The ∃I case

Write out the ∃I case of the proof of Theorem 4.12.3 in full: state what is known about the cited line, and show $\mathfrak{M}, \ell \models \exists x\, \varphi$, naming the reason for each step. Where is "t free for x in φ" used?

Solution. Line k is $\exists x\, \varphi$, by ∃I citing a line i holding $\varphi[t/x]$, where t is free for x in φ (Definition 4.5.1). Let $\mathfrak{M}$ be a structure interpreting every symbol of the derivation (as in Step 2), with domain D, and ℓ an assignment with $\mathfrak{M}, \ell \models \Gamma_k$. Line i is available at line k, so $\Gamma_i \subseteq \Gamma_k$ by Lemma 3.5.1 (a), and $\mathfrak{M}, \ell \models \Gamma_i$. Since $i < k$, the induction hypothesis $S(i)$ gives $\mathfrak{M}, \ell \models \varphi[t/x]$. Let $d := t^{\mathfrak{M},\ell}$, an element of D. Because t is free for x in φ, Lemma 4.12.1 applies and gives $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. By (S7) of Definition 4.11.3, with the element d, $\mathfrak{M}, \ell \models \exists x\, \varphi$.

"Free for" is used exactly once, to apply Lemma 4.12.1. Without it the step fails: from $\forall y\, R(y, y)$, read as $(\forall y\, R(x, y))[y/x]$, an "∃I" would give $\exists x\, \forall y\, R(x, y)$, and in the structure on $\mathbb{Z}$ with R read as $\leq$ the first is true and the second ("there is a least integer") false.

Exercise. B2: Three nonderivability results

Use the following counter-models from Section 4.11 to state three nonderivability results, citing Corollary 4.12.4, and check in each case that the structure (and assignment) satisfy every premise and not the conclusion.

(a) The two-element structure of Example (The swap of ∃ and ∀).

(b) The counter-model of Exercise (B3: The converse from Section 4.4).

(c) The counter-model of Exercise (B1: Three counter-models) (iii).

Solution. *(a)* $D = \{a, b\}$, $R^{\mathfrak{M}} = \{(a, a), (b, b)\}$. The premise $\forall y\, \exists x\, R(x, y)$ holds (for $y = a$ take $x = a$, for $y = b$ take $x = b$); the conclusion $\exists x\, \forall y\, R(x, y)$ fails ($x = a$ fails at $y = b$, $x = b$ at $y = a$). Both are sentences, so any assignment will do. By Corollary 4.12.4, $\forall y\, \exists x\, R(x, y) \nvdash \exists x\, \forall y\, R(x, y)$.

(b) $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, $Q^{\mathfrak{M}} = \{b\}$. The premise $\forall x\, \neg Q(x) \to \forall x\, \neg P(x)$ holds, since its hypothesis fails at b; the conclusion $\forall x\, (P(x) \to Q(x))$ fails at a. By Corollary 4.12.4, $\forall x\, \neg Q(x) \to \forall x\, \neg P(x) \nvdash \forall x\, (P(x) \to Q(x))$.

(c) $D = \{a, b\}$, $P^{\mathfrak{M}} = \{a\}$, with an assignment $\ell(x) = b$. There are no premises, so the condition on premises holds trivially. The formula $\exists x\, P(x) \to P(x)$ fails under ℓ: its hypothesis holds (witness a) and its conclusion, with x free, has value $b \notin P^{\mathfrak{M}}$. By Corollary 4.12.4, with Γ empty, $\nvdash \exists x\, P(x) \to P(x)$. Here the assignment is essential: under $\ell(x) = a$ the formula holds, and the structure alone would not have been a counter-model.

4.12.7.3 C. Going deeper

Exercise. C1: The ∃E case

Write out the ∃E case of the proof of Theorem 4.12.3 in full, from memory: the data of line

k, the side condition, what the induction hypothesis gives for the cited line and the cited box, and the argument, naming at each step which clause of the side condition it uses.

Solution (Hint 1). The tool is the assignment $\ell[x_0 \mapsto d]$, where d is a witness for $\exists x\,\varphi$ under ℓ. You will need Lemma 4.11.1 three times and Lemma 4.12.1 once.

Solution (Hint 2). Start: "Let $\mathfrak{M}$ be a structure interpreting every symbol of the derivation, with domain D, and ℓ an assignment with $\mathfrak{M}, \ell \models \Gamma_k$. By the induction hypothesis for line h and Lemma 3.5.1 (a), $\mathfrak{M}, \ell \models \exists x\,\varphi$, so choose $d \in D$ with $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$."

Solution. Data. Line k is χ, by $\exists$E citing a line h holding $\exists x\,\varphi$ and a box with name x_0, from line a (the assumption $\varphi[x_0/x]$) to line b (the formula χ). Side condition (Definition 4.6.1): x_0 appears neither in χ, nor in $\exists x\,\varphi$, nor in a premise, nor in an assumption of a box still open at line k.

What the hypothesis gives. Line h is available at line k, so $\Gamma_h \subseteq \Gamma_k$ (Lemma 3.5.1 (a)), and $S(h)$ holds: every structure interpreting every symbol of the derivation, and every assignment satisfying Γ_h, satisfy $\exists x\,\varphi$. The box is cited by a box rule, so its last line has context $\Gamma_k, \varphi[x_0/x]$ (Lemma 3.5.1 (b), extended in the subsection "Contexts, once more"), and $S(b)$ says that every such structure and assignment satisfying Γ_k and $\varphi[x_0/x]$ satisfy χ.

The argument. Let $\mathfrak{M}$ be a structure interpreting every symbol of the derivation (as in Step 2), with domain D, and ℓ an assignment with $\mathfrak{M}, \ell \models \Gamma_k$. Then $\mathfrak{M}, \ell \models \Gamma_h$, so $\mathfrak{M}, \ell \models \exists x\,\varphi$; by (S7), choose $d \in D$ with $\mathfrak{M}, \ell[x \mapsto d] \models \varphi$. Let $\ell' := \ell[x_0 \mapsto d]$.

1. $\mathfrak{M}, \ell' \models \Gamma_k$. The formulas of Γ_k are the premises and the assumptions of the boxes open at line k; by the side condition, x_0 appears in none of them. ℓ' differs from ℓ only at x_0, so Lemma 4.11.1 applies to each formula of Γ_k.
2. $\mathfrak{M}, \ell' \models \varphi[x_0/x]$. Since x_0 does not appear in $\exists x\,\varphi$, it is not x and does not appear in φ. So $\ell'[x \mapsto d]$ and $\ell[x \mapsto d]$ differ at most at x_0, which is not in φ, and $\mathfrak{M}, \ell'[x \mapsto d] \models \varphi$ by Lemma 4.11.1. The term x_0 is free for x in φ, as no quantifier of φ is on x_0, and $x_0^{\mathfrak{M},\ell'} = d$; so Lemma 4.12.1 gives $\mathfrak{M}, \ell' \models \varphi[x_0/x]$.
3. By 1, 2 and the hypothesis for the box, $\mathfrak{M}, \ell' \models \chi$.
4. ℓ' and ℓ differ only at x_0, which does not appear in χ; by Lemma 4.11.1, $\mathfrak{M}, \ell \models \chi$.

Since $\mathfrak{M}$ and ℓ were arbitrary, this proves $S(k)$.

Marking scheme. (10 points)

- 2: states the data and the full side condition, and the context $\Gamma_k, \varphi[x_0/x]$ of the box's last line.
- 2: obtains a witness d from the cited line through the induction hypothesis and (S7), and sets x_0 to d.
- 2: shows the new assignment still satisfies Γ_k, using "x_0 in no premise and no open assumption".
- 2: shows the new assignment satisfies $\varphi[x_0/x]$, using "x_0 not in $\exists x\,\varphi$" and the substitution lemma.
- 2: applies the hypothesis for the box, and returns to ℓ using "x_0 not in χ".

A proof that treats the box's assumption and the context in the other order earns the same points. Fatal errors (0 for the step): setting x_0 to an arbitrary element rather than the witness; concluding $\mathfrak{M}, \ell \models \chi$ directly from $\mathfrak{M}, \ell' \models \chi$ without the condition on χ.

Exercise. C2: "Free for" is needed

Show that the side condition "t is free for x in φ" of ∀E cannot be dropped. Give a two-line "derivation" that uses ∀E with a term that is not free for x, a structure in which its premise is true and its conclusion false, and the step of the proof of Lemma 4.12.1 that fails.

Solution (Hint 1). Look for a premise $\forall x\, \varphi$ in which φ contains a quantifier $\forall y$ and x occurs free inside its scope; instantiate at $t = y$.

Solution (Hint 2). Try $\varphi = \neg\forall y\, (y \leq x)$, "$x$ is not an upper bound of everything", in a structure with no largest element.

Solution. Take $\varphi = \neg\forall y\, (y \leq x)$ and $t = y$. The free occurrence of x in φ lies in the scope of $\forall y$, and y occurs in t, so t is **not** free for x in φ. Ignoring this, ∀E would allow:

1	$\forall x\, \neg\forall y\, (y \leq x)$	
2	$\neg\forall y\, (y \leq y)$	∀E 1

In $(\mathbb{Z}, \leq)$, the premise says "for every integer x, not every integer y is $\leq x$", that is, no integer is largest: for each x, the integer $x + 1$ is not $\leq x$. It is true. The conclusion says "not every integer y has $y \leq y$", which is false, since $\leq$ is reflexive. So the premise is true and the conclusion false, and a rule allowing this step would be unsound.

The step of Lemma 4.12.1 that fails is the second quantifier case, applied to $\forall y\, (y \leq x)$: it needs y not in t, so that the value of t does not change when (S6) sets y to each d. Here $t = y$, and its value is the new d each time: $\forall y\, (y \leq y)$ compares each d with itself, while $\forall y\, (y \leq x)$ at the value a of t compares each d with the fixed a. The legal route renames first (Section 4.2): $\forall x\, \neg\forall z\, (z \leq x)$, and then ∀E at y gives $\neg\forall z\, (z \leq y)$, which is true in $(\mathbb{Z}, \leq)$ under every assignment.

Marking scheme. (10 points)

- 3: a premise and a ∀E step at a term that is not free for x, with the capture identified.
- 4: a structure, with the premise checked true and the conclusion checked false.
- 3: names the quantifier case of Lemma 4.12.1 and the fact it needs (y not in t).

Fatal errors (0 for the step): a "counter-model" in which the premise is false; a step in which t is in fact free for x.

Chapter 5

From Derivations to Mathematical Prose

Here is a complete proof, as a textbook of analysis would print it, that a function uniformly continuous on a set A is continuous at every point of A:

> Let $x \in A$ and $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ (depending on ε only) such that $|f(u) - f(v)| < \varepsilon$ for all $u, v \in A$ with $|u - v| < \delta$. In particular $|f(x) - f(y)| < \varepsilon$ for every $y \in A$ with $|x - y| < \delta$, so this δ works for x and ε.

Section 4.10 proved the same fact as an eleven-line derivation (Example (Uniformly continuous implies continuous)), with three nested boxes: one for the point x_0, one for ε_0, one for the chosen δ_0. The paragraph has no boxes and no line numbers. *Where did the boxes go?* Nowhere. "Let $x \in A$ and $\varepsilon > 0$" opens the first two, "choose" opens the third, and the parenthesis "(depending on ε only)" records that δ came from a line that mentions ε and not x, which is exactly where uniform continuity and continuity differ. Words such as "let", "choose", "suppose" and "in either case" are the boxes, spoken. This chapter makes that correspondence exact in both directions. Learning to hear the boxes is what lets a reader working alone check a proof, by restoring the derivation underneath wherever a step is in doubt; learning to speak them is what makes a proof checkable by someone else.

The chapter builds a dictionary from rules to phrases, writes two proofs at four levels from full derivation to plain prose, fixes once and for all which steps plain prose may leave unsaid, and turns all of this into habits for reading, writing and checking proofs, with a catalogue of the errors that recur. From *Analysis on the Real Line*, Chapter 1 on, proofs are written in plain prose, and a rule tag appears only where the rule is itself the point, as in a review exercise that looks back at this volume; *Sets, Relations, and Cardinality*, Chapters 1–3 may still tag some proofs. The chapter ends with the one rule that mathematicians have seriously argued about: reductio ad absurdum, the rule behind excluded middle, what analysis would lose without it, and where this book stands.

What you need. Chapters 1–4. From Chapters 1 and 2: the rules for every connective, derivations with their boxes and scope rule (Definition 1.4.2), derived rules (Definition 2.3.1), the classical rules of Section 2.5 and their interderivability (Theorem 2.5.7), and the two running examples Theorem 2.1.3 and Theorem 2.2.1. From Chapter 3: induction (Axiom 3.2.1), the cut rule (Lemma 3.6.1), truth tables and refuting rows (Example (Working backwards from F), Definition 3.4.3, Corollary 3.5.3), soundness (Theorem 3.5.2), completeness and the proof of Lemma 3.6.4 (Theorem 3.6.5, Corollary 3.6.6), and, for Section 5.7 only, the optional Theorem 3.5.4. From Chapter 4: the quantifier rules and their side conditions (Definition 4.4.1,

Definition 4.6.1), the bounded rules (Proposition 4.8.1), definitions of analysis translated into formulas (Section 4.8), the laws for negating quantifiers (Proposition 4.9.1), and the derivation that uniform continuity implies continuity together with the place where its converse breaks down (Example (Uniformly continuous implies continuous), Example (The search from C to UC), and the counter-model Example (The square function on the real line)). School arithmetic and the order of $\mathbb{R}$ are used in examples as facts.

Time. About 35 hours on the core track (all of A, one B and one C per section, the exit checks); about 45 hours for everything.

5.0.1 Roadmap

- **The connective rules in prose.** The dictionary rows for the connectives and the classical rules, each with an analysis proof annotated phrase by phrase; the three faces of "proof by contradiction"; and the contradiction proofs that are direct or contrapositive proofs in costume.
- **The quantifier rules in prose.** The rows for "for every", "there exists", equality and unique existence, and for induction, "it suffices to show" and "without loss of generality"; the dependency parenthesis; and the quantifiers that English leaves unwritten or puts in the wrong place.
- **Four levels of one proof.** Full derivation, skeleton, annotated prose and plain prose; the convention that says which steps plain prose may leave silent; the irrationality of $\sqrt{2}$ and "uniformly continuous implies continuous", each written four times; and how to check a doubtful sentence by expanding it.
- **Reading a proof: self-explanation.** The book's training passage for the habit of naming, at each line, what makes it true; four questions to ask of every line; and five strategies that good readers apply to a proof as a whole.
- **Writing a proof.** A seven-step procedure from the statement to plain prose, the conventions of written mathematics as rules made audible, and three worked examples: labeling a proof, turning a derivation into a paragraph, and writing a new proof from its goal.
- **Checking proofs, and what is not allowed.** Two questions for every line, the catalogue (N1)–(N11) of errors with the rule each one breaks, and four student proofs checked line by line.
- **The dispute over excluded middle.** Intuitionistic logic as the rules without reductio; why excluded middle cannot be refuted; what a proof provides under the BHK reading; an irrational power of an irrational number that is rational, with and without a witness; a short history; what analysis would lose; proofs as programs; and the book's position.

5.0.2 Named moves

- **Hear the rule in the phrase.** "Let $x \in A$" opens a $\forall$I box and "suppose" an assumption; "choose" is $\exists$E; "take" and "let $\delta := \ldots$" are $\exists$I; "suppose, for contradiction" is RAA or $\neg$I; both boxes end in $\bot$, and what the box assumes and the line written after it closes tell which.
- **Expand to check.** When a sentence is doubtful, rewrite that stretch, and only that stretch, as a skeleton, and look for the line that breaks a rule.
- **Skeleton first.** Write the first and the last sentence of every box before filling any.
- **Self-explain each line** by earlier lines, definitions and named results, and the rule that combines them; never by paraphrase.
- **Two columns when checking:** *is the line true?* and *does it follow, and by which rule?* The first column finds where to look; the second decides.
- **Say what depends on what.** Every chosen object carries its dependencies: the names

in the line it comes from, among the boxes open when it was chosen.

5.1 The Connective Rules in Prose

Chapters 1–4 built a calculus in which every step of a proof is one rule applied to lines that are available. Mathematicians do not write Fitch boxes, and from *Analysis on the Real Line* on neither does this series; yet every proof in it is a derivation underneath, written in sentences. This section and the next compile the dictionary between the two: for each rule, the phrases that carry it in written mathematics, a sentence of analysis in which it does its work, and whether a writer may leave it unsaid. This section covers the connectives and the classical rules; Section 5.2 covers the quantifiers, equality and induction.

5.1.1 The same proof, twice

Section 2.1 proved Theorem 2.1.3 ("if n^2 is even, then n is even") by cases, and printed a table that matched each line of its derivation with one sentence of the proof. Section 2.3 proved the same theorem a second way, in Example (Even square, even number, by contraposition), with the derived rules CP and DS doing the work. Here is that second derivation again, with E: "n is even", O: "n is odd", S: "n^2 is even", T: "n^2 is odd", and the facts (E1)–(E3) of Section 2.1 (every integer is even or odd; no integer is both; the square of an odd integer is odd) as its premises.

1	$E \vee O$	
2	$\neg(S \wedge T)$	
3	$O \to T$	
4	$\quad S$	
5	$\quad\quad T$	
6	$\quad\quad S \wedge T$	$\wedge$I 4, 5
7	$\quad\quad \bot$	$\neg$E 6, 2
8	$\quad \neg T$	$\neg$I 5–7
9	$\quad \neg T \to \neg O$	CP 3
10	$\quad \neg O$	$\to$E 9, 8
11	$\quad O \vee E$	or commutes 1
12	$\quad E$	DS 11, 10
13	$S \to E$	$\to$I 4–12

A mathematician would write the same proof in four sentences:

> Suppose n^2 is even. If n^2 were odd, it would be both even and odd, which (E2) forbids; so n^2 is not odd. By (E3), read contrapositively, n is not odd. Since n is even or odd, by (E1), n is even.

The paragraph accounts for every line of the derivation, though some lines have no words of their own.

Lines	Words that carry them	Rule
4	"Suppose n^2 is even."	$\to$I, the box opens
5	"If n^2 were odd"	$\neg$I, the box opens
6	"it would be both even and odd"	$\wedge$I
7	"which (E2) forbids"	$\neg$E, citing premise 2
8	"so n^2 is not odd"	$\neg$I, the box closes
9	"By (E3), read contrapositively"	CP, citing premise 3
10	"n is not odd"	$\to$E
11–12	"Since n is even or odd, by (E1), n is even."	DS, citing premise 1
13	nothing	$\to$I, the box closes

Three features of the table recur through the whole dictionary. The **openings** of boxes always have words, which name what is assumed: "suppose", "if ... were". Some lines have **no words**: line 13 closes the box of line 4, and the paragraph simply stops; line 11 only reorders the disjuncts of (E1) for the schema of Proposition 2.3.3, and prose does not see the order of a disjunction (the convention on silent steps of Section 5.3 lets the exchange be silent). And a **premise** is never a sentence of its own: it enters through its name, "(E2)" or "by (E1)", at the step that uses it.

Quick check. Which word of the paragraph tells the reader that line 5 is an assumption and not a claim, and what would the sentence assert if that word were replaced by "is"?

Solution. The word "were" (with "if"): "if n^2 were odd" supposes something in order to refute it, which is the opening of a $\neg$I box. With "is", the sentence "n^2 is odd, so it is both even and odd" would assert T as a claim rather than an assumption: it would sit, unjustified, directly in the box of line 4, where it is in fact false (n^2 is even there), and the proof would be wrong.

5.1.2 How to read the dictionary

The dictionary is a set of rows, one per rule, grouped by connective. Each row has four entries.

- **The rule**, as Chapters 1, 2 and 4 named it, primitive or derived.
- **Phrases**, in the order a writer should prefer them; the later ones are variants a reader must recognize. They are not a fixed vocabulary: any wording that does the same job is correct.
- **An analysis sentence**, from a proof the reader has met or will meet in *Analysis on the Real Line*, with the premises of the rule visible.
- **Whether the rule may be silent in plain prose.** Section 5.3 names four levels of writing, from the full derivation down to the plain prose of *Analysis on the Real Line* onward, and turns this column into a convention. "Never" means that a proof which leaves the step out is incomplete, however obvious the step looks.

Readers who use the Lean proof assistant will find the Lean command for each primitive rule in the rule, phrase and Lean table that ends Section 4.7; nothing in this chapter depends on it.

Each group of rows is followed by a worked example, a short proof annotated phrase by phrase. A rule tag such as [∧E] after a phrase says which rule the phrase carries; proofs in this volume and in *Sets, Relations, and Cardinality* may carry such tags, and from *Analysis on the Real Line* on they are dropped.

5.1.3 Conjunction

Rule	Phrases	Silent in plain prose?
∧I	"…, and …"; "we show (a) … and (b) …"; "both … hold"	often, when both conjuncts are displayed
∧E	"in particular"; "so $a > 0$"; using one clause of a hypothesis without comment	yes

In analysis, ∧I is the shape of every definition with two clauses. "s is a least upper bound of A" will mean that s is an upper bound of A **and** no smaller number is an upper bound of A; a proof of it has two parts, labeled (a) and (b), and ends by putting them together. (*Analysis on the Real Line*, Chapter 1 states the second clause as "$s \le M$ for every upper bound M of A", which says the same: by trichotomy, $s \le M$ holds exactly when $M < s$ fails, and an upper bound M with $M < s$ would be a smaller one.) ∧E is the step from a chain to one of its links: "since $x \in (0,1)$, in particular $x > 0$" takes the left conjunct of $0 < x \wedge x < 1$.

Example. Two clauses of a least upper bound

Let $A \coloneqq \{x \in \mathbb{R} : 0 < x < 2\}$. Prove that 2 is an upper bound of A, and that no real number $t < 2$ is an upper bound of A. Tag each step with its rule.

Solution. (a) Let $a \in A$ [∀I]. Then $0 < a < 2$, by the definition of A; in particular $a < 2$ [∧E], so $a \le 2$ [∨I]. So 2 is an upper bound of A.

(b) Let t be a real number with $t < 2$ [∀I, →I], and suppose t is an upper bound of A [assumption for ¬I]. Let $m \coloneqq \max\{t, 0\}$ and $a \coloneqq (m+2)/2$. We have $t \ge 0$ or $t < 0$, and by the definition of the maximum we consider the two cases [∨E]: if $t \ge 0$, then $m = t$, so $m \ge t$, $m \ge 0$ and $m < 2$; if $t < 0$, then $m = 0$, so $m > t$, $m \ge 0$ and $m < 2$. So in either case $m \ge t$ and $0 \le m < 2$. Hence $1 \le a < 2$; thus $0 < a$ and $a < 2$, that is, $a \in A$ [∧I]. As t is an upper bound of A, $a \le t$ [∀E at a]. But $a - m = (2-m)/2 > 0$, so $a > m \ge t$, and $a > t$. This contradicts $a \le t$ [¬E]. Hence t is not an upper bound of A [¬I].

So 2 is an upper bound of A, and no smaller number is [∧I].

The ∧E of (a) could have been silent ("then $a < 2$, so $a \le 2$"); the ∨I after it weakened a strict inequality, since $a \le 2$ means $a < 2 \vee a = 2$. In (b), the two conjuncts $0 < a$ and $a < 2$ were displayed and then merged by "that is, $a \in A$". The bound $m < 2$ needed the case split hidden in max, and it is written out, as the warning below requires. The final sentence is

the ∧I of the two clauses. And the clashing lines $a \le t$ and $a > t$ are both named before "contradicts", as the ¬E row below requires.

5.1.4 Implication

Rule	Phrases	Silent in plain prose?
→I	"assume …; then …; hence if … then …"; "suppose …"; "if …, then …" as the frame of a whole proof	never the assumption; the closing "hence" may be silent
→E	"since …, …"; "by (hypothesis, lemma), …"; "hence"	the citation, never
R	"recall that"; "as noted above"	yes
MT, CP	"…; so P cannot hold"; "we prove the contrapositive"; "equivalently, if not Q then not P"	the announcement "we prove the contrapositive", never

In analysis, →I is a proof that begins "Suppose $|x-1|<1$" and ends "so $x>0$"; the example below writes it out. →E is every citation of a theorem: "since f is continuous at a and $a_n \to a$, we have $f(a_n) \to f(a)$" applies an implication proved in *Analysis on the Real Line*, Chapter 5 to two facts in hand, and the word "since" introduces those facts. For CP, Section 2.3's Example (Even square, even number, by contraposition) is the model, as the opening of this section showed. (Theorem 2.1.3 itself is proved by cases, with no contraposition.)

Example. A hypothesis assumed, a lemma cited, a contrapositive announced

Let x be a real number. (a) Prove that if $|x-1|<1$, then $x>0$. (b) Deduce that if $x \le 0$, then $|x-1| \ge 1$. Tag each step with its rule.

Solution. We use the school fact that, for real u and r, $|u|<r$ holds exactly when $-r<u<r$.

(a) Suppose $|x-1|<1$ [assumption for →I]. By the school fact, $-1<x-1<1$ [↔E], in particular $-1<x-1$ [∧E], so $x>0$ [arith].

(b) Suppose $x \le 0$ [assumption for →I]. Recall from (a) that $|x-1|<1$ implies $x>0$ [R]. Since $x \le 0$, that is, $x>0$ fails, $|x-1|<1$ cannot hold [MT], so $|x-1| \ge 1$.

Part (a) ends at "$x>0$": the closing line of →I, "hence, if $|x-1|<1$, then $x>0$", is silent. The assumption could not be: without "suppose", the sentence "$|x-1|<1$" would be a claim about every real x, and a false one.

In (b), "recall" is R, and it could be dropped. "Cannot hold" is MT, with P the statement $|x-1|<1$ and Q the statement $x>0$; the last step reads $\neg(|x-1|<1)$ as $|x-1| \ge 1$, by the order fact of Section 2.1. An equivalent (b) announces the derived rule instead: "This is the contrapositive of (a)", which is CP applied to (a), followed by →E. That announcement

may never be silent: a reader who sees "suppose $x \leq 0$" and then a citation of (a) must be told that (a) is being used backwards.

5.1.5 Disjunction and cases

Rule	Phrases	Silent in plain prose?
$\vee$I	"…, so certainly P or Q"; "in particular $x \neq 0$ or …"	yes
$\vee$E	"we consider two cases"; "Case 1 … Case 2 …"; "in either case"; "otherwise"	never the split; the "in either case" may be silent
DS	"since $x \neq 0$, the first case is impossible, so …"	the rule's name, yes; the fact $x \neq 0$ it uses, never
"or" and "and" commute (Example (First derivations with "or"), Proposition 1.2.1)	"P or Q" read as "Q or P", with no words of its own	yes

In analysis, $\vee$I hides inside every weakening of a strict inequality: "$x > 0$, so $x \geq 0$". $\vee$E is the proof of the triangle inequality in Section 1.6 (Example (The triangle inequality by cases)), which splits on the sign of $x + y$ and reaches $|x + y| \leq |x| + |y|$ in both cases. DS is the step "$x^2 = x$, so $x = 0$ or $x = 1$; since $x \neq 0$, $x = 1$".

Example. Two cases, a weakening, a refuted disjunct

Let x be a real number. (a) Prove that $|x| + x \geq 0$. (b) Prove that if $x^2 = x$ and $x \neq 0$, then $x = 1$. Tag each step with its rule.

Solution. (a) Every real number satisfies $x \geq 0$ or $x < 0$, and we consider the two cases [$\vee$E].

Case $x \geq 0$. Then $|x| = x$, so $|x| + x = 2x \geq 0$.

Case $x < 0$. Then $|x| = -x$, so $|x| + x = 0$, and certainly $|x| + x \geq 0$ [$\vee$I].

So $|x| + x \geq 0$ in either case [$\vee$E, closing].

(b) Suppose $x^2 = x$ and $x \neq 0$ [assumption for $\to$I]. Then $x(x - 1) = x^2 - x = 0$, and a product of two real numbers is 0 only if one factor is, so $x = 0$ or $x - 1 = 0$ [$\to$E]. Since $x \neq 0$, the first case is impossible, so $x - 1 = 0$ [DS], that is, $x = 1$.

In (a), the split was announced with the disjunction that makes it exhaust; $\vee$E needs that disjunction on a line, and a split without it is the missing-case error that Section 5.6 catalogues. The second case weakened $|x| + x = 0$ to $|x| + x \geq 0$, which is $\vee$I. The closing "in either case" could be silent. In (b), the words "the first case is impossible" carry the fact $x \neq 0$ that DS uses: the rule's name is not needed, but that fact is.

Warning. "Clearly" often hides a case split. "Clearly $\max\{a, b\} \geq (a + b)/2$" is true for all real a, b, but its proof is $\vee$E on "$a \leq b$ or $b \leq a$": if $a \leq b$, then $\max\{a, b\} = b = (b + b)/2 \geq$

$(a+b)/2$, and the other case is the same with the letters exchanged. The case split is never silent; write it, or write "by cases on whether $a \le b$" and the one computation.

5.1.6 Negation and contradiction

Rule	Phrases	Silent in plain prose?
¬I	"suppose P; ... a contradiction; hence $\neg P$"; "P is impossible, since ..."	never
¬E	"this contradicts ..."; "which is impossible"; "but ..."	never: the clashing pair is named
⊥E	"this case cannot occur"; "vacuously"; "there is nothing to prove"	yes, once the contradiction or the empty case is stated
DM, quantifier negation	"not all ... so some ... fails"; "it is not the case that ..., so ..."	the negated form must be written

In analysis, ¬I is the proof of Theorem 2.2.1: "suppose some rational r has $r^2 = 2$; ...; this is a contradiction; hence no rational number has square 2". ¬E is a sentence such as "then $n \le 1/\varepsilon < n$, a contradiction": the chain gives $n < n$, which clashes with the fact that no number is less than itself. One strict link is essential, since $n \le 1/\varepsilon \le n$ gives only $n \le n$, which clashes with nothing. ⊥E is "if $A = \emptyset$, there is nothing to prove", said of a claim about every element of A: with no element to check, the claim is vacuously true (Section 4.4).

The last row needs more room. To negate "s is an upper bound of A", that is, $\forall a \in A\ (a \le s)$, a proof writes the negated form: **some** $a \in A$ has $a > s$. The passage uses Proposition 4.9.3 (a), derived with RAA, and the order fact that $\neg(a \le s)$ says $a > s$. *Analysis on the Real Line*, Chapter 1's proof that $\mathbb{N}$ is not bounded above is the instructive case. It opens "suppose, for contradiction, that $\mathbb{N}$ is bounded above" and closes with "therefore $\mathbb{N}$ is not bounded above"; that is ¬I, despite the phrase. Its one classical step comes after the box: from "not every $n \in \mathbb{N}$ has $n \le x$" it passes to "some $n \in \mathbb{N}$ has $n > x$", by Proposition 4.9.3 (a), which Section 4.9 derives with RAA through Proposition 4.9.1 (a) and Proposition 4.9.2, and then by trichotomy for $\neg(n \le x)$, giving $n > x$. That step produces an n without naming it, and the proof needs no name.

Example. A negative claim, its positive form, and an empty case

(a) Prove that no real number is an upper bound of $\mathbb{R}$. (b) Prove the positive form: for every real s there is a real x with $x > s$. (c) Let A be an empty set of real numbers. Prove that every real number is an upper bound of A. Tag each step with its rule.

Solution. (a) Let s be a real number [∀I], and suppose s is an upper bound of $\mathbb{R}$ [assumption for ¬I]. Since $s+1$ is a real number, $s+1 \le s$ [∀E at $s+1$], that is, $1 \le 0$. This contradicts $0 < 1$ [¬E]. Hence s is not an upper bound of $\mathbb{R}$ [¬I].

(b) Let s be a real number [∀I]. Take $x := s+1$ [∃I]; then $x - s = 1 > 0$, so $x > s$.

(c) Let s be a real number and $a \in A$ [$\forall$I, $\to$I]. Since A has no elements, $a \in A$ is impossible [$\neg$E, against "$a \notin A$"], so there is nothing to prove: $a \leq s$ [$\bot$E]. Hence s is an upper bound of A.

Claim (a) is $\forall s\, \neg\forall x\ (x \leq s)$, and its proof writes the $\neg$I box in full: the assumption, the named clash, the conclusion. The positive form (b), $\forall s\, \exists x\ (x > s)$, says more: it hands over the number that beats s. Given (b) as a premise, (a) follows with no classical step: if s were an upper bound of $\mathbb{R}$, the x that (b) provides for s would satisfy $x > s$ and $x \leq s$. The general passage from the shape of (a) to the shape of (b), with no witness in hand, goes through Proposition 4.9.1 (a), derived with RAA, and trichotomy. So when both forms are within reach, prove the positive one. In (c), once the clash and the empty set are named, the $\bot$E line itself may be silent. Part (c) is the prose form of the derivation in Section 4.4 that concludes $\forall u\, \forall x\ (x \in A \to x \leq u)$ from $\forall x\, \neg(x \in A)$, with its vacuous step unpacked into the $\neg$E and $\bot$E lines.

5.1.7 Proof by contradiction

Rule	Phrases	Silent in plain prose?
RAA	"suppose, for contradiction, that $\neg P$"; "suppose not"; "assume the contrary"	never
$\neg\neg$E, LEM	"either P or not"; "P must hold, since ..."; "otherwise"	never: written as an explicit case split or an explicit removal of a double negation ("P must hold, since ...")

In analysis, RAA is the proof of Proposition 4.9.6: "suppose, for a contradiction, that $a \neq b$; then $|a-b| > 0$, so the hypothesis gives $|a-b| < |a-b|$, a contradiction; hence $a = b$". The box assumes $\neg(a = b)$ and concludes $a = b$, so it is genuine RAA; $\neg$I would give only $\neg\neg(a = b)$. LEM is a split such as "either the sequence has infinitely many terms in the left half of the interval, or it does not", the step of the bisection proof of the Bolzano–Weierstrass theorem (*Analysis on the Real Line*, Chapter 2 proves the theorem). That split is genuinely non-constructive: nobody can decide in general which case holds. Section 5.7 explains why that matters, and a proof that makes such a step says so in one clause. A split such as "$x \geq 0$ or $x < 0$" is also excluded middle in logical form, but it is an instance of the trichotomy of the order of $\mathbb{R}$, which *Analysis on the Real Line*, Chapter 1 takes as an axiom, and it is never flagged.

Example. One proof by contradiction, read three ways

Let a and b be real numbers with $a \leq b + \varepsilon$ for every $\varepsilon > 0$. Prove that $a \leq b$, first by RAA; then rewrite the proof with $\neg$I and $\neg\neg$E, and with a case split.

Solution. By RAA. Suppose, for a contradiction, that $a > b$ [assumption for RAA]. Then $\varepsilon := (a-b)/2$ is positive, so the hypothesis gives $a \leq b + (a-b)/2$ [$\forall$E at $(a-b)/2$], that is, $2a \leq a + b$, so $a \leq b$. This contradicts $a > b$ [$\neg$E]. Hence $a \leq b$ [RAA].

The box assumes $a > b$, which is $\neg(a \leq b)$ by the order fact of Section 2.1, and the proof concludes $a \leq b$: a negation goes **out**.

By ¬I and ¬¬E. Close the same box with "hence $a > b$ is impossible" [¬I], the line $\neg(a > b)$, which is $\neg\neg(a \le b)$ under the same reading of $a > b$ as $\neg(a \le b)$; then "so $a \le b$ must hold" [¬¬E]. The two steps do what RAA did in one, and show that the box itself uses no classical rule.

By cases. By trichotomy, $a \le b$ or $a > b$ [∨E]. In the first case we are done. In the second, the lines of the box above reach $a \le b$ against $a > b$, so this case cannot occur, and in it $a \le b$ [¬E, ⊥E]. In either case $a \le b$.

The three proofs share their mathematics, the instance $\varepsilon = (a - b)/2$, and differ only in how the classical step is written: as RAA, as ¬¬E, or as a case split. All three also read $a > b$ as $\neg(a \le b)$ by the order fact of Section 2.1, itself an instance of trichotomy; read the other way, that fact gives $a \le b$ from $\neg(a > b)$ directly, so the first two put an excluded-middle step on top of it. Over $\mathbb{R}$ trichotomy is an axiom, so no step here is flagged.

5.1.8 The biconditional

Rule	Phrases	Silent in plain prose?
↔I	"(⇒) … (⇐) …"; "conversely"; "we prove both directions"	never either direction
↔E	"by definition"; "that is"; "i.e."; "equivalently"	yes, for a definitional "if and only if"

In analysis, Theorem 2.4.4 of Section 2.4 (for $x, y \ge 0$, $x \le y \iff x^2 \le y^2$) is the model of ↔I: two labeled directions, each a →I box. ↔E is every unwinding of a definition: "f is strictly increasing, that is, $x < y$ implies $f(x) < f(y)$" (Example (A definition used in both directions)). A definition is an "if and only if" (Section 2.4), and the words "that is" use it in one direction or the other.

Example. Both directions, the second by cases

Let x be a real number with $x \ge 0$. Prove that $x \le 1$ if and only if $x^2 \le x$. Tag each step with its rule.

Solution. (⇒) Suppose $x \le 1$ [assumption for →I]. Multiplying by $x \ge 0$ gives $x^2 \le x$.

(⇐) Conversely, suppose $x^2 \le x$ [assumption for →I]. Since $x \ge 0$, either $x = 0$ or $x > 0$ [∨E]. If $x = 0$, then $x \le 1$. If $x > 0$, dividing $x^2 \le x$ by x gives $x \le 1$.

So $x \le 1$ if and only if $x^2 \le x$ [↔I].

The labels (⇒) and (⇐) are the two boxes of ↔I, and "conversely" opens the second. The hypothesis $x \ge 0$ does a job in each: it keeps the inequality's direction when we multiply, and it reduces the cases to $x = 0$ and $x > 0$, by ↔E on the definition of $\ge$. Without it, (⇐) still holds, since a negative x has $x^2 \ge 0 > x$, so $x^2 \le x$ forces $x \ge 0$ and the proof above applies; but (⇒) fails: $x = -1$ has $x \le 1$ and $x^2 = 1 > x$.

5.1.9 The three faces of "by contradiction"

"Proof by contradiction" is used in practice for three different arguments. Each opens a box and finds $\bot$ inside it; they differ in what the box assumes and what the proof concludes, and only face (i) is free of a classical step: face (iii) ends with RAA, and face (ii) ends with Proposition 2.5.2, which uses RAA unless Q is itself a negation.

Face	The box assumes	The proof concludes	Last step
(i) refuting	P	$\neg P$	$\neg$I
(ii) contrapositive	$\neg Q$, inside "suppose P"	$P \to Q$, from $\neg Q \to \neg P$	Proposition 2.5.2
(iii) genuine	$\neg P$	P	RAA

(i) Refuting a statement. The claim is negative: irrational, "no largest", $x \neq 5$. The box assumes the positive statement, as the proof of Theorem 2.2.1 assumes $r^2 = 2$, and no classical rule is used.

(ii) The contrapositive. The claim is $P \to Q$, and the proof derives $\neg P$ from $\neg Q$, with P playing no part until the end. The passage back to $P \to Q$ is Proposition 2.5.2, which Section 2.5 derives with RAA (avoidable when Q is itself a negation, as Section 2.5 notes after the proof). Written as "suppose P; suppose, for contradiction, $\neg Q$; ... $\neg P$, a contradiction", it is that derivation done inline.

(iii) Genuine reductio. The claim P is positive, and $\neg P$ is used throughout the box, typically because it hands over an object: $a \neq b$ makes $|a - b|$ a positive number to feed to a hypothesis about every $\varepsilon > 0$.

To tell them apart, read the first sentence of the box and the sentence after it: what does it assume, and what does the proof write after it closes? The words "for contradiction" do not decide it; *Analysis on the Real Line*, Chapter 1 uses them for a face (i) proof.

The style rule: use the weakest form that works, and say which one you use. A direct proof, when one exists, beats all three. A negative claim is proved by face (i). Face (ii) is announced as "we prove the contrapositive", which tells the reader the goal $\neg Q \to \neg P$ at once. Face (iii) is for a positive claim that resists the other routes, with its classical step made visible as in Example (One proof by contradiction, read three ways).

5.1.10 Costumes

A proof written as "suppose, for contradiction, ..." sometimes uses its assumption for nothing except the final clash. It is then a direct proof or a contrapositive proof **in costume**, and it reads better with the costume off. Section 2.5 met the smallest case; here are the two costumes as they appear in prose, each with a test to run on a finished proof.

Example. A direct proof in costume

Here is a proof that $x^2 - 2x > 3$ for every real $x > 3$. Decide what it really uses, and rewrite it.

> Let $x > 3$. Suppose, for a contradiction, that $x^2 - 2x \leq 3$. Since $x > 3$, both $x - 3$ and $x + 1$ are positive, so $(x - 3)(x + 1) > 0$, that is, $x^2 - 2x - 3 > 0$. So

> $x^2 - 2x > 3$. This contradicts our assumption. Hence $x^2 - 2x > 3$.

Solution. The assumption "$x^2 - 2x \leq 3$" is cited exactly once, in "This contradicts our assumption". The sentences before it prove $x^2 - 2x > 3$ from $x > 3$ alone, so the RAA box contains a complete derivation of the goal. Deleting the box leaves a direct proof:

> Let $x > 3$. Then $x - 3 > 0$ and $x + 1 > 0$, so $x^2 - 2x - 3 = (x - 3)(x + 1) > 0$, that is, $x^2 - 2x > 3$.

Nothing is lost, the RAA is gone, and the reader no longer holds a false assumption in mind for three sentences.

The test: if the assumption of the box is cited only in the line that produces $\bot$, the box proved the goal without it. Delete the box.

Example. A contrapositive in costume

Here is a proof that, for an integer n, if n^2 is even then n is even. Decide what it really uses, and rewrite it.

> Suppose n^2 is even. Suppose, for a contradiction, that n is not even. Then n is odd, say $n = 2k + 1$ with k an integer. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd, so n^2 is not even. This contradicts the fact that n^2 is even. Hence n is even.

Solution. Here the assumption of the box, "n is not even", is used heavily; what is used only once is the **hypothesis** "n^2 is even", in "This contradicts ...". The sentences between derive "n^2 is not even" from "n is not even" alone. So the real content is $\neg Q \to \neg P$, with P the statement "n^2 is even" and Q the statement "n is even", and the frame around it is Proposition 2.5.2 done inline: face (ii). Written openly:

> We prove the contrapositive: if n is not even, then n^2 is not even. Suppose n is not even. Then n is odd, say $n = 2k+1$ with k an integer, and $n^2 = 2(2k^2+2k)+1$ is odd, so it is not even.

The rewrite tells the reader at the start what will be shown, and the final clash disappears. The classical step has not: it is now inside the announcement, which uses Proposition 2.5.2. (The proof by cases of Theorem 2.1.3 in Section 2.1 needs no RAA at all.)

The test: if everything inside the box follows from its assumption $\neg Q$ alone, and the hypothesis P of the theorem enters only in the line that produces $\bot$, then the box proved $\neg Q \to \neg P$. Present it as a contrapositive.

The two tests share one smell: *the clash at the end is with a statement the proof already had, or would have had without the box.* A genuine reductio uses its assumed negation throughout, and the clash comes somewhere the reader did not see coming; in Example (One proof by contradiction, read three ways), the assumption $a > b$ was what made $(a - b)/2$ positive.

Warning. "Assume P" means different things at the top of different proofs. At the top of a proof of $P \to Q$, it is the opening of the $\to$I box, and it is exactly right. At the top of a proof of P itself, it assumes the goal, and everything after it is circular: the last line "hence P" cites a line that is not available outside the box. For instance, "assume $\sqrt{2}$ is irrational; then ... ; hence $\sqrt{2}$ is irrational" proves nothing, however correct the middle: it is entry (N9) of the catalogue in Section 5.6.

5.1.11 Exercises

5.1.11.1 A. Check your understanding

Exercise. A1

(a) Name the rule that each phrase most often carries: (i) "in particular"; (ii) "conversely"; (iii) "in either case"; (iv) "this case cannot occur"; (v) "suppose, for contradiction, that x is not positive"; (vi) "we prove the contrapositive"; (vii) "that is"; (viii) "since $x \neq 0$, the first case is impossible"; (ix) "so $x \geq 0$", after "$x > 0$"; (x) "which contradicts $x^2 \geq 0$".

(b) Which of the phrases in (a) carry steps that may be silent in plain prose?

(c) A proof ends "Suppose $7 = 2k$ for an integer k. Then $k = 7/2$, which is not an integer. Hence 7 is not even." Which face of "by contradiction" is this?

(d) True or false, with a reason: a proof that begins "suppose, for contradiction" always uses RAA.

Solution. (a) (i) $\wedge$E. (ii) $\leftrightarrow$I, opening the second direction. (iii) $\vee$E, closing. (iv) $\bot$E. (v) RAA, opening its box (the goal is "x is positive", and the box assumes its negation). (vi) CP, or, as the frame of a whole proof, Proposition 2.5.2. (vii) $\leftrightarrow$E, on a definition. (viii) DS. (ix) $\vee$I. (x) $\neg$E.

(b) (i), (iii) (the closing "in either case"), (iv) (once the contradiction is stated), (vii) (for a definition), and (ix). In (viii) the name of DS may be dropped, but not the fact $x \neq 0$. The openings (ii), (v), (vi) and the clash (x) are never silent.

(c) Face (i), refuting: the box assumes the positive statement "7 is even", that is, $7 = 2k$ for some integer k, and the proof concludes its negation, by $\neg$I. (The value $7/2$ is not an integer because $3 < 7/2 < 4$.)

(d) False. The phrase also opens $\neg$I boxes. *Analysis on the Real Line*, Chapter 1's proof that $\mathbb{N}$ is not bounded above begins "suppose, for contradiction, that $\mathbb{N}$ is bounded above" and concludes the negation, by $\neg$I. What decides is what the box assumes and what the proof concludes.

5.1.11.2 B. Practice

Exercise. B1: Take off the costume

Here is a proof that $\frac{a+b}{2} \geq \sqrt{ab}$ for all real $a > 0$ and $b > 0$.

> Let $a, b > 0$. Suppose, for a contradiction, that $\frac{a+b}{2} < \sqrt{ab}$. Since $(\sqrt{a}-\sqrt{b})^2 \geq 0$, expanding gives $a-2\sqrt{ab}+b \geq 0$, so $\frac{a+b}{2} \geq \sqrt{ab}$. This contradicts our assumption. Hence $\frac{a+b}{2} \geq \sqrt{ab}$.

(a) Which costume is this? Justify your answer by saying where the assumption of the box is cited.

(b) Rewrite the proof without the contradiction.

Solution. *(a)* A direct proof in costume (Example (A direct proof in costume)). The assumption $\frac{a+b}{2} < \sqrt{ab}$ is cited only in "This contradicts our assumption", the line that produces $\bot$. The sentence beginning "Since" proves the goal from $a, b > 0$ alone, without using the assumption.

(b) Let $a, b > 0$, so that $\sqrt{a}$ and $\sqrt{b}$ are defined and $\sqrt{a}\sqrt{b} = \sqrt{ab}$. Since $(\sqrt{a} - \sqrt{b})^2 \geq 0$, expanding gives $a - 2\sqrt{ab} + b \geq 0$. Adding $2\sqrt{ab}$ to both sides and dividing by 2 gives $\frac{a+b}{2} \geq \sqrt{ab}$, as claimed.

Exercise. B2: Where the classical step sits

Here is a proof that if x is a real number with $|x| \leq |y|$ for every real $y \neq 0$, then $x = 0$.

> Suppose $|x| \leq |y|$ for every real $y \neq 0$. Suppose, for a contradiction, that $x \neq 0$. Then $y := x/2$ is a non-zero real number, so the hypothesis gives $|x| \leq |x/2| = |x|/2$, and hence $|x| \leq 0$. But $|x| > 0$, since $x \neq 0$. This is a contradiction. Hence $x = 0$.

(a) Is the last step RAA or ¬I? Is the proof in costume?

(b) Rewrite it so that the box closes by ¬I and one further step removes a double negation. Tag the two steps.

(c) Which of the two versions tells the reader more, and what?

Solution. *(a)* RAA: the box assumes $\neg(x = 0)$, written $x \neq 0$, and the proof concludes the positive statement $x = 0$. It is not in costume. The assumption does work inside the box (it makes $x/2$ non-zero, so that the hypothesis applies to it), and the hypothesis is used in the middle of the box, not only at the clash; neither test of this section is met.

(b) Suppose $|x| \leq |y|$ for every real $y \neq 0$. If $x \neq 0$, then $x/2 \neq 0$, so the hypothesis gives $|x| \leq |x|/2$, that is, $|x| \leq 0$, against $|x| > 0$. Hence $x \neq 0$ is impossible [¬I], so $x = 0$ must hold [¬¬E].

(c) The second. It shows that the box uses no classical rule: without one, the argument proves $\neg\neg(x = 0)$, "x cannot be non-zero", and the classical rule enters only in the final step, "so $x = 0$ must hold" [¬¬E]. The first version does both steps at once in the single RAA, and so hides where the classical rule enters.

5.1.11.3 C. Going deeper

Exercise. C1: Euclid without the contradiction

Here is a common proof that there are infinitely many primes. It uses two facts about integers that we take as known here: every integer $m \geq 2$ has a prime divisor (*Sets, Relations, and Cardinality*, Chapter 2 proves it); and if an integer d divides two integers, it divides their difference (if $u = dk$ and $v = dl$, then $u - v = d(k - l)$).

> Suppose, for a contradiction, that there are only finitely many primes, say $p_1, \ldots, p_k$. Let $N \coloneqq p_1 p_2 \cdots p_k + 1$. Then $N \geq 2$, so N has a prime divisor p. Since $p_1, \ldots, p_k$ are all the primes, $p = p_i$ for some i. Then p_i divides both N and $p_1 p_2 \cdots p_k$, so it divides their difference 1, which is impossible for a prime. This is a contradiction. Hence there are infinitely many primes.

(a) "There are infinitely many primes" means "the set of primes is **not** finite". Is the last step of the proof RAA or ¬I?

(b) Where is the assumption "$p_1, \ldots, p_k$ are all the primes" used? Which smell from this section does that suggest?

(c) Rewrite the argument as a direct proof of: *for every finite list $p_1, \ldots, p_k$ of primes, with $k \geq 1$, there is a prime not in the list.*

(d) Say what the rewrite gains over the original.

Solution. (a) ¬I. The box opens with "the set of primes is finite", the positive statement, and the conclusion is its negation: face (i). No RAA is used.

(b) Once, in "$p = p_i$ for some i", where the clash is manufactured. Everything else holds for **any** finite list of primes, complete or not: the smell of a costume.

(c) Let $p_1, \ldots, p_k$ be a finite list of primes, $k \geq 1$, and let $N \coloneqq p_1 p_2 \cdots p_k + 1$. Each $p_i \geq 2$, so $N \geq 3$, and N has a prime divisor p. Let i be an index with $1 \leq i \leq k$ [∀I, opening], and suppose $p = p_i$ [assumption for ¬I]. Then p divides N and $p_1 p_2 \cdots p_k$, so it divides their difference 1, say $1 = pm$ with m an integer. Since $p > 0$, $m > 0$, so $m \geq 1$ and $1 = pm \geq p \geq 2$, a contradiction [¬E]. Hence $p \neq p_i$ [¬I]. Since i was arbitrary, $p \neq p_i$ for every i [∀I, closing], and p is a prime not in the list.

(d) The rewrite proves a stronger statement: given any finite list of primes, it **produces** a prime outside it, any prime divisor of N; the original shows only that a list of all the primes is impossible. Its one ¬I box, for the negative claim $p \neq p_i$, is honest, since its assumption is what the clash uses. The original theorem follows at once: a finite list of all the primes is non-empty, since 2 is prime, so by (c) it would miss a prime.

Marking scheme. (10 points) - 2: (a) ¬I, with the reason (the box assumes the positive statement and the proof concludes its negation). - 2: (b) locates the single use of the assumption and names the costume smell. - 4: (c) a proof for an arbitrary finite list: N defined, $N \geq 2$ justified, a prime divisor p taken, $p \neq p_i$ for every i shown. - 2: (d) names the gain: a prime outside any given list is produced. Fatal errors (0 for the step): in (c), a proof that still assumes the list contains all the primes; in (a), calling the step RAA because of the words "for a contradiction".

Exercise. C2: Which face, and the weakest form

Here is a proof that if x is a real number with $x^2 < \varepsilon$ for every $\varepsilon > 0$, then $x = 0$.

> Suppose $x^2 < \varepsilon$ for every $\varepsilon > 0$. Suppose, for a contradiction, that $x \neq 0$. Then $x^2 > 0$, so we may take $\varepsilon = x^2$, and the hypothesis gives $x^2 < x^2$, which is impossible. Hence $x = 0$.

(a) Which face of "by contradiction" is this, as written?

(b) Rewrite the proof in the weakest form you can, and say which rules it uses at its logical steps, and which school fact.

Solution (Hint 1). The goal $x = 0$ is positive. Look for a school fact about x^2 that is a disjunction: one case leads to the goal through another school fact, and the hypothesis rules out the other.

Solution (Hint 2). Start: "Since $x^2 \geq 0$, either $x^2 > 0$ or $x^2 = 0$."

Solution. (a) Face (iii), genuine reductio. The box assumes $\neg(x = 0)$, written $x \neq 0$, uses it to make x^2 a positive number to which the hypothesis applies, and concludes the positive statement $x = 0$. The last step is RAA. It is not in costume: the assumption does work inside the box.

(b) Suppose $x^2 < \varepsilon$ for every $\varepsilon > 0$ [assumption for $\to$I]. Since $x^2 \geq 0$, either $x^2 > 0$ or $x^2 = 0$, and we consider the two cases [$\vee$E].

Case $x^2 > 0$. Then the hypothesis applies to $\varepsilon = x^2$ [$\forall$E] and gives $x^2 < x^2$, which contradicts the fact that no number is less than itself [$\neg$E]. So this case cannot occur, and in it $x = 0$ [$\bot$E].

Case $x^2 = 0$. Then $x \cdot x = 0$, so $x = 0$ [$\to$E], since a product of two real numbers is 0 only if a factor is.

In either case $x = 0$ [$\vee$E].

The logical steps are $\to$I, $\vee$E, $\forall$E, $\neg$E, $\to$E (citing the second school fact, that a product is 0 only if a factor is) and $\bot$E; no classical rule is used. The classical content has moved into the school fact "$x^2 > 0$ or $x^2 = 0$", which rests on the trichotomy of the order, an axiom of *Analysis on the Real Line*, Chapter 1; Section 5.7 explains why that axiom is where the book makes its classical commitment.

Marking scheme. (10 points) - 3: (a) identifies RAA (face (iii)), with the reason: the assumption is a negation and the conclusion is positive. - 2: (b) splits on a disjunction that is justified (from $x^2 \geq 0$, or from trichotomy). - 3: (b) closes the case $x^2 > 0$ by the instance $\varepsilon = x^2$, a named clash, and $\bot$E. - 2: (b) settles the case $x^2 = 0$ with its reason and concludes "in either case". A rewrite with $\neg$I and $\neg\neg$E in place of the case split earns the points of (b) except the 2 for the split, provided it names $\neg\neg$E as its one classical step. Fatal errors (0 for the step): calling (a) face (i) because the word "contradiction" appears; splitting on "$x^2 > 0$ or $x^2 < 0$", which does not exhaust.

5.2 The Quantifier Rules in Prose

Section 5.1 read the connective rules in the sentences of written proofs. The statements of analysis, though, are made of "for every" and "there exists", and most of the words in an analysis proof carry a quantifier rule: "let", "fix", "choose", "take", "will do". This section completes the dictionary with the rows for the quantifiers, equality and unique existence, and for the three patterns that are not single rules but are written as if they were: induction, the lemma ("it suffices to show"), and symmetry ("without loss of generality"). It also makes exact one habit of analysis prose, the parenthesis "(depending on ε)".

5.2.1 Three sentences, three rules

Section 4.8 derived "if $a_n \to L$, then $2a_n \to 2L$" in sixteen lines. In a proof in *Analysis on the Real Line* the same argument takes three sentences:

> Let $\varepsilon > 0$. Since $a_n \to L$ and $\varepsilon/2 > 0$, we may choose $N \in \mathbb{N}$ (depending on ε) such that $|a_n - L| < \varepsilon/2$ for every $n \geq N$. Then for every $n \geq N$, $|2a_n - 2L| = 2|a_n - L| < \varepsilon$.

Each sentence opens a box, and the boxes are nested in the order of the sentences.

- "Let $\varepsilon > 0$" opens the bounded $\forall$I box of lines 2–15 in the derivation of Section 4.8, with the fresh name ε_0. Its closing line, line 16, is not written: the paragraph simply ends, and a reader who heard "let" knows that the box closes there.
- "Choose N ..." is two rules. The words "since $a_n \to L$ and $\varepsilon/2 > 0$" are the $\forall$E of line 4, at the term $\varepsilon_0/2$, with its bound from line 3. The word "choose" opens the $\exists$E box of lines 5–14, with the fresh name N_0.
- "Then for every $n \geq N$" opens the $\forall$I box of n_0 and the $\to$I box of $n_0 \geq N_0$ (lines 7 and 9), and the computation is lines 8, 10 and 11. The words "for every $n \geq N$" also announce the lines that close these two boxes, 12 and 13.
- The $\exists$I of line 14, which writes "some N works" from "this N works", and the closing line 15 of the $\exists$E box, are silent. They are the end of the paragraph.

The one word in the paragraph that no rule asks for is the parenthesis "(depending on ε)". It is the subject of a subsection below. First, the rows.

5.2.2 The quantifier rows

Rule	Phrases	Silent in plain prose?
$\forall$I	"let $x \in A$"; "fix $\varepsilon > 0$"; "let $\varepsilon > 0$ be given"; "since x was arbitrary"	never the introduction; "since x was arbitrary" may be silent
$\forall$E	"in particular, for $x = \ldots$"; "applying ... with $\varepsilon/2$ in place of ε"	the instance must be named when it is not the obvious one
$\exists$I	"take $N = \ldots$"; "let $\delta := \ldots$; then ..."; "... will do"	never; its dependencies are stated
$\exists$E	"choose N such that"; "by hypothesis there is k with $n = 2k$"; "let x_0 be such a point"	never; the name must be fresh

The phrases in the table are mostly the bounded forms of the rules (Proposition 4.8.1): "let $x \in A$" opens a $\forall$I box whose assumption is $x \in A$, and "choose $N \in \mathbb{N}$ such that ..." opens an $\exists$E box with the two assumptions $N \in \mathbb{N}$ and "...". A bound is part of the phrase: "let $\varepsilon > 0$" is one step, not "let ε" followed by "suppose $\varepsilon > 0$".

In analysis, $\forall$I is the first sentence of every limit proof: "Let $\varepsilon > 0$." $\forall$E is "apply the definition of $a_n \to L$ with $\varepsilon/2$": the instance $\varepsilon/2$ is not the obvious one, so it is named. $\exists$I is the line that places a constant, "let $\delta := \min\{1, \varepsilon/10\}$", written before the computation that shows it works. $\exists$E is "choose N (depending on ε) such that ...", in which the name N has not been used before in the proof.

Example. A constant sequence

Let c be a real number and let $a_n := c$ for every $n \in \mathbb{N}$. Prove that $a_n \to c$, and annotate each phrase with its rule.

Solution. Let $\varepsilon > 0$ [$\forall$I, opening]. Take $N := 0$ [$\exists$I]. Let $n \in \mathbb{N}$ with $n \geq N$ [$\forall$I, $\to$I, opening]. Then

$$|a_n - c| = |c - c| = 0 < \varepsilon,$$

where the first equality substitutes $a_n = c$ [=E] and the last step uses $\varepsilon > 0$, the assumption of the first box. This proves $a_n \to c$.

The last sentence stands for three closing lines, all silent: the $\to$I and $\forall$I for n ("for every $n \geq N$"), and the $\forall$I for ε ("since ε was arbitrary"). The witness "take $N := 0$" is placed before the work that justifies it.

Two things here are degenerate, and both are worth seeing once. The witness N depends on nothing: any $N \in \mathbb{N}$ works. And the assumption $n \geq N$ is never used. The box is still opened, because the goal $\forall n \in \mathbb{N}\ (n \geq N \to \dots)$ asks for it; without "let $n \geq N$", the line "$|a_n - c| = 0$" would not say which n it is about.

5.2.3 The dependency parenthesis

In the paragraph that opened this section, N was chosen "(depending on ε)". Section 4.10 read off a derivation the rule behind the parenthesis:

A constant chosen in a box may depend on the names of the boxes open around that box, and it does depend only on the names in the line it comes from.

In the derivation of Section 4.8, the witness N_0 is obtained at line 5 from line 4, which mentions ε_0, inside the box of ε_0 only. So N_0 may depend on ε_0, and it does. In the derivation of Example (Uniformly continuous implies continuous), the witness δ_0 is obtained inside the boxes of x_0 and ε_0, so it **may** depend on both; but it comes from a line that mentions ε_0 alone, so it **does** depend only on ε_0, and that is uniformity.

The parenthesis is the prose record of this. "Choose N (depending on ε)" says: the boxes open here are those of ε (besides the data of the theorem, such as the sequence), and the line from which N comes mentions ε. "Choose δ (depending on ε and on x)" says that δ came from a line mentioning both, inside both boxes. A witness that came from a line mentioning neither is a constant, and the proof may say "(depending on neither)" or simply "a fixed δ". This is why proofs in analysis say what each chosen constant depends on. A missing parenthesis is not a

wrong step, but it hides the one fact a reader most needs in order to check the quantifier order, and many false proofs in analysis go wrong exactly there.

Example. The reciprocals tend to zero

Prove that $1/n \to 0$, for the sequence $(1/n)_{n\geq 1}$. Annotate each phrase with its rule, and state what the chosen constant depends on.

Solution. The sequence starts at $n = 1$, so the definition of Section 4.8 is read with N and n in $\mathbb{Z}_{>0}$ instead of $\mathbb{N}$: for every $\varepsilon > 0$ there is $N \in \mathbb{Z}_{>0}$ such that $|1/n - 0| < \varepsilon$ for every $n \in \mathbb{Z}_{>0}$ with $n \geq N$. We use one school fact about the real numbers: for every real number u there is an integer greater than u. (*Analysis on the Real Line*, Chapter 1 proves it from the axioms of $\mathbb{R}$.)

Let $\varepsilon > 0$ [$\forall$I, opening]. It suffices to find $N \in \mathbb{Z}_{>0}$ with

$$\frac{1}{N} < \varepsilon, \tag{$*$}$$

since then every $n \in \mathbb{Z}_{>0}$ with $n \geq N$ has $0 < 1/n \leq 1/N < \varepsilon$, so $|1/n - 0| < \varepsilon$ [cut: the claim $(*)$ displayed, then used].

By the school fact, applied to $u = 1/\varepsilon$ [$\forall$E], there is an integer greater than $1/\varepsilon$; choose such an integer N [$\exists$E], which depends on ε. Since $\varepsilon > 0$, we have $1/\varepsilon > 0$, so $N > 0$ and $N \in \mathbb{Z}_{>0}$. From $N > 1/\varepsilon > 0$, taking reciprocals reverses the inequality: $1/N < \varepsilon$, which is $(*)$. So this N will do [$\exists$I], and $1/n \to 0$.

Read the proof as a skeleton. The first sentence opens the box of ε. The second is a cut, in the sense of the row below: it states a simpler claim $(*)$, shows on the spot that $(*)$ is enough, and leaves $(*)$ to be proved. The next paragraph proves $(*)$. It contains two different quantifier steps that both concern N: "choose" is the $\exists$E on the school fact, which hands over some integer beyond $1/\varepsilon$ and calls it N; "will do" is the $\exists$I that offers this N as the witness the definition asks for. Between them, the clause "so $N > 0$ and $N \in \mathbb{Z}_{>0}$" checks the bound $N \in \mathbb{Z}_{>0}$ of the witness, which the bounded $\exists$I needs as a line.

The constant N depends on ε only: it came from the school fact at $1/\varepsilon$, inside the box of ε. For instance, $\varepsilon = 0.03$ has $1/\varepsilon = 33.3\ldots$, and $N = 34$ is a choice that works; so is any larger integer.

Quick check. "For every $\varepsilon > 0$, $|a_n - L| < \varepsilon$ for all large n." Where is the hidden $\exists N$, and between which two quantifiers does it stand?

Solution. "For all large n" is "there is an N such that for every $n \geq N$" (Section 4.8): the hidden $\exists N$ stands after $\forall\varepsilon > 0$ and before $\forall n \geq N$. Written out, the sentence is $\forall\varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to |a_n - L| < \varepsilon)$, the definition of $a_n \to L$. Since $\exists N$ comes after $\forall\varepsilon$, the N may depend on ε.

5.2.4 Equality and unique existence

Rule	Phrases	Silent in plain prose?
=I, =E	"substituting"; chains "$\cdots = \cdots = \ldots$"; "by (1)"	yes, inside chains
$\exists!$	"there is exactly one"; "existence ... uniqueness ..."; "suppose y also ..."	never either half

In analysis, =E is at work in every chain of equalities and inequalities, such as

$$|x^2 - 4| = |x - 2|\,|x + 2| \le 5|x - 2| \quad \text{for } |x - 2| < 1,$$

where each $=$ link substitutes equals for equals and each inequality link applies one fact about order (Section 4.7). (Here $|x - 2| < 1$ gives $1 < x < 3$, so $3 < x + 2 < 5$.) The =I lines $t = t$ that such chains rest on are never written. Unique existence (Definition 4.7.2) has two halves, which Proposition 4.7.2 separates: "$2x + 3 = 7$ has exactly one real solution: $x = 2$ is one, and if $2y + 3 = 7$, then $y = 2$". The first clause is existence, by $\exists$I; the second is uniqueness, a $\forall$I and $\to$I box on a second candidate. Neither half implies the other, so neither may be silent.

The uniqueness half often stands alone, when existence is not claimed or is proved elsewhere. Uniqueness of limits is a case of this kind.

Example. A sequence has at most one limit

Let (a_n) be a sequence of real numbers with $a_n \to L$ and $a_n \to M$. Prove that $L = M$, and annotate each phrase with its rule.

Solution. Let $\varepsilon > 0$ [$\forall$I, opening]. Then $\varepsilon/2 > 0$, so, applying the definition of $a_n \to L$ with $\varepsilon/2$ in place of ε [$\forall$E], we may choose $N_1 \in \mathbb{N}$ (depending on ε) such that $|a_n - L| < \varepsilon/2$ for every $n \ge N_1$ [$\exists$E, fresh name N_1]. Likewise, from $a_n \to M$, choose $N_2 \in \mathbb{N}$ (depending on ε) such that $|a_n - M| < \varepsilon/2$ for every $n \ge N_2$ [$\forall$E, $\exists$E, fresh name N_2]. Let $n := \max\{N_1, N_2\}$. Then $n \ge N_1$ and $n \ge N_2$, so both bounds apply at n [$\forall$E at n, twice, with $\to$E], and

$$\begin{aligned} |L - M| &= |(L - a_n) + (a_n - M)| \\ &\le |a_n - L| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where the inequality is the triangle inequality of Example (The triangle inequality by cases), with $|L - a_n| = |a_n - L|$. Since $\varepsilon > 0$ was arbitrary, $|L - M| < \varepsilon$ for every $\varepsilon > 0$ [$\forall$I, closing], and Proposition 4.9.6 gives $L = M$ [$\to$E, citing it].

Four things in this proof are never silent, and each is visible. The instance $\varepsilon/2$ is named twice, because it is not the obvious instance. The two witnesses get two names, N_1 and N_2: the hypotheses promise a threshold for L and a threshold for M, and nothing says they are the same number, so calling both N would assert an equality nobody proved (Section 5.6 catalogues the error). The point $n = \max\{N_1, N_2\}$ is named, since the bounds are applied at it. And the closing result is cited by name.

What is silent is the bookkeeping of the two $\exists$E boxes. The line $|L - M| < \varepsilon$ mentions neither N_1, N_2 nor n, so the side condition of $\exists$E lets it leave both boxes, and then the box

of ε closes on it. The calculation chain hides its $=$E steps, as the row allows. The proof shows that a sequence has **at most one** limit: the uniqueness half of $\exists!$. Whether a given sequence has a limit at all is a separate question, which this proof does not touch.

5.2.5 Induction, lemmas and symmetry

The last three rows are not rules of the calculus. Each is a pattern of rules that proofs announce with a phrase of its own.

Pattern	Phrases	Silent in plain prose?
induction (Axiom 3.2.1, Theorem 3.2.2)	"we argue by induction on n"; "base case"; "suppose the claim holds for n"	never either the base case or the step; the induction hypothesis is stated
cut, lemma (Lemma 3.6.1)	"we first show (Claim)"; "it suffices to show"; "we claim"	never the claim itself; "it suffices" may be dropped if the claim is displayed and then used
symmetry	"without loss of generality"; "similarly"; "by symmetry"	allowed only when the swap turns the statement and the hypotheses into themselves, up to identities such as $\lvert b-a\rvert = \lvert a-b\rvert$

An induction is $\to$E applied to Axiom 3.2.1, whose hypothesis is a conjunction: the base case $P(0)$, and the step "**for every** $n \in \mathbb{N}$, if $P(n)$ then $P(n+1)$". So the step is a $\forall$I box with a $\to$I box inside it, and the phrase "suppose the claim holds for n" opens the inner box; its assumption is the **induction hypothesis**, and it must be stated, because the step uses it. A lemma is a cut: Lemma 3.6.1 (b) says that once φ is derived, it may be used as a premise. "It suffices to show φ" is the same cut seen from the goal, as Section 1.3 observed: it promises a derivation of the goal from φ and leaves φ to be proved. In analysis, "it suffices to show $\lvert a_n - L\rvert < \varepsilon$ for $n \geq N$" is the cut that reduces a limit to a single estimate; Example (The reciprocals tend to zero) made one with the claim $(*)$. Symmetry is a case split $\vee$E whose second box is the first with two letters exchanged, up to identities such as $\lvert b-a\rvert = \lvert a-b\rvert$; "without loss of generality $x \leq y$" writes the first box and leaves that check to the reader.

Example. An induction, and a legitimate symmetry

(a) Let a be a real number with $0 < a \leq 1$. Prove that $0 < a^n \leq 1$ for every $n \in \mathbb{N}$, and annotate each phrase with its rule.

(b) Let a and b be real numbers. Prove that $\lvert a-b\rvert = \max\{a,b\} - \min\{a,b\}$, with a "without loss of generality", and write out the case split it abbreviates.

Solution. *(a)* For $n \in \mathbb{N}$, let $P(n)$ be the statement "$0 < a^n$ and $a^n \leq 1$". We argue by induction on n, using Axiom 3.2.1.

Base case. $a^0 = 1$, and $0 < 1$ and $1 \leq 1$, so $P(0)$ holds [$\wedge$I].

Step. Let $n \in \mathbb{N}$ [$\forall$I, opening], and suppose $P(n)$ holds, that is, $0 < a^n$ and $a^n \leq 1$ [$\to$I,

opening: the induction hypothesis]. Then $a^{n+1} = a^n a$. Since $a^n > 0$ [$\wedge$E] and $a > 0$, the product $a^n a$ is positive, so $0 < a^{n+1}$. Since $a \leq 1$ and $a^n > 0$, multiplying gives $a^n a \leq a^n$, and $a^n \leq 1$ [$\wedge$E], so $a^{n+1} \leq 1$. Hence $P(n+1)$ holds [$\wedge$I].

By induction, $P(n)$ holds for every $n \in \mathbb{N}$ Axiom 3.2.1.

The base case is the first hypothesis of Axiom 3.2.1, written as its own paragraph. The step is the second: "let $n \in \mathbb{N}$" and "suppose $P(n)$ holds" open its two boxes, and "hence $P(n+1)$" is the last line of the inner one; the closing lines "so $P(n) \to P(n+1)$, for every n" are silent. The induction hypothesis is stated in full, and each of its halves is taken out by $\wedge$E and used: $a^n > 0$ twice (for the sign of a^{n+1}, and to multiply $a \leq 1$ by a^n without reversing it), and $a^n \leq 1$ once, at the end. The statement was chosen with care: the tempting "$a^n \leq a$" fails at $n = 0$ whenever $a < 1$, since $a^0 = 1$, and an induction for it would have to start at $n = 1$.

(b) Without loss of generality, $a \leq b$. Then $\max\{a, b\} = b$ and $\min\{a, b\} = a$, and since $b - a \geq 0$,

$$|a - b| = |b - a| = b - a = \max\{a, b\} - \min\{a, b\}.$$

The phrase abbreviates a $\vee$E on "$a \leq b$ or $b \leq a$", which holds for all real a, b by trichotomy. The first box is the paragraph above. The second box, on $b \leq a$, reads: then $\max\{a, b\} = a$ and $\min\{a, b\} = b$, and since $a - b \geq 0$, $|a - b| = a - b = \max\{a, b\} - \min\{a, b\}$. That is the first box with a and b exchanged throughout, read through three identities: $|b - a| = |a - b|$, $\max\{b, a\} = \max\{a, b\}$, and $\min\{b, a\} = \min\{a, b\}$. These say that the **statement** is unchanged when a and b are exchanged, and there are no hypotheses to check. This is what makes the swap exact, and so what licenses "without loss of generality".

Warning. "Without loss of generality" needs a statement and hypotheses that are unchanged by the swap. Here is a false "proof" that $\max\{a, b\} = b$ for all real a, b: "Without loss of generality $a \leq b$; then $\max\{a, b\} = b$." Exchanging a and b turns the claim into $\max\{b, a\} = a$, which is a different claim, so the second case is not the first one in disguise; and indeed $a = 2$, $b = 1$ gives $\max\{a, b\} = 2 \neq 1$. Before writing "without loss of generality", swap the letters in the statement and the hypotheses, and check that each comes back unchanged, up to identities such as $|b - a| = |a - b|$.

5.2.6 What the sentence does not say

Some quantifiers in mathematical English are not written at all, and some are written where they do not apply. Four habits of mathematical English cover most of them.

The hidden universal closure. A theorem stated as "if n is even, then n^2 is even" means "for every integer n, if n is even, then n^2 is even". Section 4.1 introduced this convention: *a letter that is not introduced anywhere, in the statement of a theorem, is read with a "for every" in front of the whole statement.* It applies to the statement of a theorem, lemma or exercise, not to a line inside a proof: inside a proof every letter has been introduced by a "let" or a "choose", and its box tells the reader which. The proof of such a theorem begins by opening the hidden box: "let n be an integer".

"For all large n" hides an $\exists N$ and a bound, as the Quick check above found. "Eventually" means the same. The negation of "$P(n)$ for all large n" is "for every N there is an $n \geq N$ for which $P(n)$ fails", usually said "$P(n)$ fails for infinitely many n".

"Any" can mean "every" or "some", and only the rest of the sentence decides (Section 4.1). "Any $x \in A$ is positive" means every element of A is positive. "If any $x \in A$ is positive, then A is non-empty" means: if **some** element of A is positive. When "any" appears, rewrite the sentence once with "every" and once with "some", and keep the one that was meant.

Quantifiers at the end of a sentence. English lets a writer put "for some x" or "for all y" after the formula they govern, and then the order of two such phrases is unclear. "$x^2 > y$ for some x for all y", over $\mathbb{R}$, has two readings:

$$\forall y\, \exists x\, (x^2 > y) \qquad \text{and} \qquad \exists x\, \forall y\, (x^2 > y).$$

The first is true (for $y \geq 0$ take $x = y + 1$, since $(y+1)^2 = y^2 + 2y + 1 > y$; for $y < 0$ take $x = 0$). The second is false (whatever x is proposed, $y = x^2$ refutes it). The remedy is to write the quantifiers of such a statement **in front**, in the order in which they apply. A trailing phrase such as "for every $n \geq N$" is safe only when it is the last quantifier of the sentence, as in the definition of $a_n \to L$.

Warning. Moving "for all x" to the end of a sentence can change continuity into uniform continuity. "$|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$, for all x", written after "for every $\varepsilon > 0$ there is $\delta > 0$ such that", does not say whether $\forall x$ comes before $\exists \delta$ (continuity at every point) or after it (uniform continuity). The two formulas of Section 4.10 differ exactly there, and Example (The square function on the real line) shows that the difference is real: $f(x) = x^2$ on $\mathbb{R}$ satisfies the first and not the second. Write the quantifiers in front.

5.2.7 Exercises

5.2.7.1 A. Check your understanding

Exercise. A1

Let (a_n) be a sequence of real numbers and L a real number. Here is a proof that if $a_n \to L$ and $L > 0$, then $a_n > 0$ for all large n. Its sentences are numbered.

> (1) Suppose $a_n \to L$ and $L > 0$. (2) Then $L/2 > 0$. (3) Choose $N \in \mathbb{N}$ such that $|a_n - L| < L/2$ for every $n \geq N$. (4) Let $n \in \mathbb{N}$ with $n \geq N$. (5) Then $L - a_n \leq |a_n - L| < L/2$, so $a_n > L/2 > 0$. (6) Hence $a_n > 0$ for every $n \geq N$, that is, $a_n > 0$ for all large n.

(a) Name the rule or rules carried by each sentence.

(b) Sentence (3) has no dependency parenthesis. What should it say, and why does N not depend on n?

(c) Which instance of ∀E in the proof is not the obvious one?

(d) Which closing lines are silent?

Solution. (a) (1) →I, opening: the hypothesis of the theorem is assumed (its two conjuncts are then available by ∧E). (2) An arithmetic step from $L > 0$. (3) ∀E on the definition of $a_n \to L$ at $L/2$, with its bound from (2), then ∃E, opening a box with the fresh name N. (4) ∀I and →I, opening the boxes of n and of $n \geq N$. (5) ∀E at n and →E on the line from (3), giving $|a_n - L| < L/2$; then the school fact $-u \leq |u|$ with $u = a_n - L$, and arithmetic.

(6) $\to$I and $\forall$I, closing ("for every $n \geq N$"), then $\exists$I with the witness N ("for all large n" is $\exists N\, \forall n \geq N$), and the $\exists$E box of (3) closes, since the conclusion does not mention N.

(b) "Choose $N \in \mathbb{N}$ (depending on L) such that ...". N is obtained at line (3), from the definition instantiated at $L/2$, a term that mentions L (and the sequence, which is data). The box of n is opened only at (4), after N is chosen, so N cannot depend on n.

(c) The instance $\varepsilon = L/2$ in (3). It is the instance the proof is built around: any ε with $0 < \varepsilon \leq L$ would do, and $L/2$ leaves the margin $a_n > L/2$.

(d) The closing lines of the $\exists$E box (3) and of the $\to$I box (1), "hence, if $a_n \to L$ and $L > 0$, then ...". Sentence (6) writes the closings of the boxes of (4) in words, and the $\exists$I through "that is".

5.2.7.2 B. Practice

Exercise. B1: Ambiguous English

Each sentence has a quantifier phrase at its end. Write both readings in symbols, with the quantifiers in front, and decide whether each reading is true for the function given.

(a) "$f(x) \leq M$ for every real x, for some real M", for $f(x) = x$.
(b) "$x + y = 0$ for some real y, for every real x."
(c) "$|x - y| < \delta$ implies $|f(x) - f(y)| < 1$ for all real x, y, for some $\delta > 0$", for $f(x) = x^2$.

Solution. *(a)* $\exists M\, \forall x\ (f(x) \leq M)$, "$f$ is bounded above", and $\forall x\, \exists M\ (f(x) \leq M)$. For $f(x) = x$ the first is false: whatever M is proposed, $x = M + 1$ has $f(x) = M + 1 > M$. The second is true for every function: given x, take $M = f(x)$.

(b) $\forall x\, \exists y\ (x + y = 0)$ and $\exists y\, \forall x\ (x + y = 0)$. The first is true: given x, take $y = -x$. The second is false: for any y, the two values $x = -y$ and $x = 1 - y$ give $x + y = 0$ and $x + y = 1$, so not every x works.

(c) $\exists \delta > 0\ \forall x \forall y\ (|x - y| < \delta \to |f(x) - f(y)| < 1)$ and $\forall x \forall y\ \exists \delta > 0\ (|x - y| < \delta \to |f(x) - f(y)| < 1)$. For $f(x) = x^2$ the first is false: it is the uniform continuity condition at $\varepsilon = 1$, and Example (The square function on the real line) refutes it with $x = 1/\delta$, $y = 1/\delta + \delta/2$, for which $|x - y| < \delta$ and $|x^2 - y^2| = 1 + \delta^2/4 \geq 1$. The second is true for every function: given x and y, if $x = y$ then $|f(x) - f(y)| = 0 < 1$ for any δ, say $\delta = 1$; if $x \neq y$, take $\delta = |x - y|$, so that $|x - y| < \delta$ is false and the implication holds vacuously.

In (a) and (c), the reading with $\exists$ in front is the one a writer means; the other is true of every function, so it says nothing about f. In (b), the intended reading is $\forall x\, \exists y$. Written with the quantifiers in front, the sentences say what they mean: "there is M such that $f(x) \leq M$ for every x", "for every x there is y with $x + y = 0$", "there is $\delta > 0$ such that for all $x, y, \ldots$".

Exercise. B2: An induction, annotated

Prove that $5^n - 1$ is divisible by 4 for every $n \in \mathbb{N}$. Annotate each phrase of your proof with its rule, and say which hypothesis of Axiom 3.2.1 each part of the proof establishes.

Solution. For $n \in \mathbb{N}$, let $P(n)$ be "$4 \mid 5^n - 1$", that is, "there is an integer k with $5^n - 1 = 4k$". We argue by induction on n.

Base case. $5^0 - 1 = 0 = 4 \cdot 0$, so $P(0)$ holds, with the witness $k = 0$ [$\exists$I]. This is hypothesis (i) of Axiom 3.2.1.

Step. Let $n \in \mathbb{N}$ [$\forall$I, opening], and suppose $P(n)$ holds [$\to$I, opening: the induction hypothesis]. By the induction hypothesis there is an integer k with $5^n - 1 = 4k$; choose such a k (depending on n) [$\exists$E, fresh name k]. Then

$$5^{n+1} - 1 = 5 \cdot 5^n - 1 = 5(5^n - 1) + 4 = 5 \cdot 4k + 4 = 4(5k + 1),$$

where the third equality substitutes $5^n - 1 = 4k$ [=E]. Since $5k + 1$ is an integer, $P(n + 1)$ holds, with the witness $5k + 1$ [$\exists$I]. The step is hypothesis (ii) of Axiom 3.2.1: the $\forall$I on n and the $\to$I on $P(n)$.

By induction, $4 \mid 5^n - 1$ for every $n \in \mathbb{N}$ Axiom 3.2.1.

The $\exists$E box of k closes on "$P(n+1)$", which does not mention k, as its side condition requires; its closing line, and those of the two boxes of the step, are silent.

5.2.7.3 C. Going deeper

Exercise. C1: When "similarly" is honest

(a) Prove that $\max\{x, y\} + \min\{x, y\} = x + y$ for all real x and y, with a "without loss of generality". Then write out the $\vee$E it abbreviates, and check that its second box is the first with the letters exchanged.

(b) Let $x \geq 0$ and y be real numbers. A student proves $x \max\{x, y\} \geq xy$ as follows: "Without loss of generality $x \leq y$. Then $\max\{x, y\} = y$, and $xy \geq xy$." Explain why "without loss of generality" is not legitimate here, prove the case the student skipped, and show by an example that the hypothesis $x \geq 0$ cannot be dropped from the statement.

Solution (Hint 1). In (b), exchange x and y in the statement and in the hypothesis, and compare with the original.

Solution (Hint 2). In the case $y \leq x$, $\max\{x, y\} = x$, and the claim becomes $x^2 \geq xy$.

Solution. (a) Without loss of generality $x \leq y$. Then $\max\{x, y\} = y$ and $\min\{x, y\} = x$, so $\max\{x, y\} + \min\{x, y\} = y + x = x + y$.

The phrase abbreviates $\vee$E on "$x \leq y$ or $y \leq x$", true for all real x, y by trichotomy. Box 1, on $x \leq y$, is the paragraph above. Box 2, on $y \leq x$: then $\max\{x, y\} = x$ and $\min\{x, y\} = y$, so $\max\{x, y\} + \min\{x, y\} = x + y$. Exchanging x and y in box 1 gives "$y \leq x$; $\max\{y, x\} = x$, $\min\{y, x\} = y$; $\max\{y, x\} + \min\{y, x\} = x + y = y + x$", which is box 2 read through $\max\{y, x\} = \max\{x, y\}$, $\min\{y, x\} = \min\{x, y\}$ and $y + x = x + y$. These identities say that

the statement is unchanged by the exchange, and there are no hypotheses, so the swap is exact.

(b) Exchanging x and y changes both the hypothesis and the claim: it gives "$y \geq 0$, and $y \max\{y, x\} \geq yx$", which is not the original statement. So the case $y \leq x$ is not the case $x \leq y$ in disguise, and it needs its own argument.

Case $y \leq x$. Then $\max\{x, y\} = x$, and multiplying $x \geq y$ by $x \geq 0$ gives $x \max\{x, y\} = x^2 \geq xy$.

With the student's case ($x \leq y$: $x \max\{x, y\} = xy \geq xy$), this proves the claim in either case. The hypothesis $x \geq 0$ is used only in the skipped case. It cannot be dropped: $x = -1$, $y = -2$ has $\max\{x, y\} = -1$, so $x \max\{x, y\} = 1$, while $xy = 2$, and $1 \geq 2$ is false.

Marking scheme. (10 points) - 2: (a) a correct computation in the case $x \leq y$. - 3: (a) the $\vee$E written out, with its disjunction justified, and the exchange checked against the statement. - 2: (b) the reason: the exchange changes the hypothesis and the claim. - 2: (b) the case $y \leq x$ proved, using $x \geq 0$. - 1: (b) a counterexample without $x \geq 0$, checked. Fatal errors (0 for the step): in (b), "the other case is similar" without computing it; a counterexample that satisfies $x \geq 0$.

Exercise. C2: Moving a quantifier changes the meaning

Give an example of a sentence of analysis, in English, that begins "for every x ...", whose meaning changes when the phrase "for every x" is moved to the end of the sentence. Write both readings in symbols and show that one is true and the other false.

Solution. An example. "For every real $x > 0$, there is $n \in \mathbb{Z}_{>0}$ with $1/n < x$." Moved: "There is $n \in \mathbb{Z}_{>0}$ with $1/n < x$, for every real $x > 0$." The readings are

$$\forall x > 0\ \exists n \in \mathbb{Z}_{>0}\ (1/n < x) \qquad \text{and} \qquad \exists n \in \mathbb{Z}_{>0}\ \forall x > 0\ (1/n < x).$$

The first is true: given $x > 0$, there is an integer $n > 1/x$ (the school fact of Example (The reciprocals tend to zero)), and $n > 1/x > 0$ gives $n \in \mathbb{Z}_{>0}$ and $1/n < x$. The second is false: whatever $n \in \mathbb{Z}_{>0}$ is proposed, $x = 1/n$ is positive and $1/n < x$ fails. (It is natural to read the moved sentence as the second formula, since its "for every x" now comes after "there is n"; whichever reading a reader picks, the sentence no longer says one thing.)

What a different answer must have, so that it can be checked against this one: - the sentence contains "for every x" and at least one existential quantifier ("there is", "for some") whose scope contains a formula mentioning x; - moved to the end, "for every x" can be read as coming after the existential quantifier, so that the witness may no longer depend on x; - both readings are written in symbols with all quantifiers in front, and there is a specific interpretation (here $\mathbb{R}$ with its usual order) in which one reading is true and the other false, each claim checked; - the false reading is refuted by naming, for each candidate witness, a value of x that defeats it. Another correct answer, for $f(x) = x^2$ on $\mathbb{R}$, is "for every x, for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$" (continuity, true); with "for every x" moved to the end, it becomes the sentence of the warning, whose uniform-continuity reading Example (The square function on the real line) refutes.

5.3 Four Levels of One Proof

Sections 5.1 and 5.2 matched each rule with the phrases that carry it. A dictionary works one word at a time; a proof is read and written whole. This section puts the dictionary to work on two complete proofs, the irrationality of $\sqrt{2}$ and "uniformly continuous implies continuous", and writes each of them four times: as a full derivation, as a skeleton, as prose with every rule tagged, and as the plain prose of a textbook. The four texts are one proof written for four different readers. The section names the four levels, fixes exactly which steps the last level may leave unsaid, and ends with the use of the levels that matters most to a reader working alone: when a sentence looks doubtful, rewrite it, and only it, at level 1, as a skeleton, and look.

5.3.1 One proof, four texts

The book has already printed some proofs twice. In Section 2.2 the proof of Theorem 2.2.1 appears as a paragraph with a rule in brackets after several of its steps, and then, below it, as a 13-line derivation of its propositional part. In Section 4.10, Example (Uniformly continuous implies continuous) is an 11-line derivation, followed by three sentences of prose with line numbers in brackets. In both places the derivation and the paragraph say the same thing, and the reader is asked to match them. What was not said is how many such texts there are, what each leaves out, and who each is for.

There are four, and the differences between them are not matters of taste.

A derivation, a skeleton and a paragraph are the same proof; they differ in what the reader is trusted to restore.

Definition 5.3.1. Levels of Proof

A proof can be written at four **levels**.

- **Level 0, the full derivation.** A derivation in the sense of Definition 1.4.2, with the rules of Chapters 1, 2 and 4: **every line** is a premise, an assumption, or **one** rule applied to available lines. A derived rule used as one line (Definition 2.3.1), such as the bounded rules of Proposition 4.8.1, counts as one rule. Facts about numbers enter as premises, or as lines justified "arith", each of which records **one** such fact applied to the lines it cites.
- **Level 1, the skeleton.** The boxes and the logical lines only: every assumption, every flagged name, every line that opens or closes a box, with the rule that closes it, and every line that such a closing line cites. Each stretch of mathematics between them is compressed into one line, justified "arith", "by def", or by a cited result; a stretch of routine logical steps inside one box is compressed into one line justified by the names of its rules in order, such as "∀E, ∃I".
- **Level 2, annotated prose.** Sentences, each tagged in brackets with the rule or rules it carries, such as [∀I] or [∃E].
- **Level 3, plain prose.** Sentences without tags, in which some steps are left unsaid. Which steps may be left unsaid is fixed by the convention that follows, not by the writer.

In words: level 0 hides no logic. Every line other than an arith line can be checked mechanically against the schema of its rule, and each arith line records one fact about numbers that can be checked by hand. Level 1 keeps the logical architecture and lets the arithmetic run in blocks. Level 2 is level 1 turned into sentences, with the architecture still labeled. Level 3 removes the

labels and some of the lines, following a fixed convention.

The levels are already on the page in earlier chapters. The derivation in Section 4.10 is level 0. The paragraph after Theorem 2.2.1, with its tags [→E], [∧I], [¬E], [¬I], is level 2 for the rules Chapter 2 had; its quantifier steps carry no tags. The derivation printed below that paragraph is not quite level 0, and the reason is instructive: it fixes r, p and q in advance and leaves out the two quantifier steps, choosing p, q (and later m) and passing from one r to every r, because Chapter 2 had no rules for them yet. Section 2.2 says so in the paragraph after the derivation. With Chapter 4's rules those steps can be written. The example below writes the choices of p and q and the step to every r; the choice of m stays inside the arithmetic fact of its line 15, since evenness is treated there as a single predicate E.

A **non-example**, by one change: take a level-1 skeleton and delete one of its boxes, say the box in which a name N_0 is chosen, keeping the lines inside. The result is not a shorter skeleton. It is a list in which N_0 appears from nowhere, and no reader can tell whether it was chosen, given, or assumed. A skeleton may compress the work inside a box; it may not compress the boxes.

Why four levels and not two? Because each serves a task. Level 0 is for settling a doubt completely: nothing in it is taken on trust. Level 1 is for planning a proof and for reading one: it is what you write on scratch paper before the first sentence (Section 5.5 turns this into a procedure), and what you reconstruct from a textbook when a proof will not come clear (Section 5.4). Level 2 is for learning, which is why this volume and *Sets, Relations, and Cardinality*, Chapters 1–3 use it and later volumes do not: from *Analysis on the Real Line*, Chapter 1 on, proofs are written at level 3, in plain prose, and a rule tag appears only where the rule is itself the point, as in a review exercise that looks back at those chapters. Level 3 is what mathematicians read and write.

Warning. **A skeleton is not a finished proof.** A level-1 text with lines such as "$|x_0 - y| < \delta_0 \to |f(x_0) - f(y)| < \varepsilon_0$, arith" is a plan: it says that a computation goes there, and it can hide a computation that does not work. For $f(x) = x^2$ on $\mathbb{R}$, a skeleton of "f is uniformly continuous" with one line "arith" at that point looks exactly like a correct skeleton for $f(x) = 3x + 1$, where $\delta = \varepsilon/3$ works because $|(3x+1) - (3y+1)| = 3|x - y|$; the claim is false for the first function (Example (The square function on the real line)), and only writing out the arithmetic shows where it fails. Expand every "arith" line before calling the proof done.

5.3.2 Which steps may be silent

Level 3 leaves things out. If it left out whatever a writer happened to find obvious, a reader could never tell an omission from a gap. So the book fixes the omissions once, by rule. The principle behind the list: a step may go unsaid when a reader who knows the rules can restore it uniquely from what is written. A step that **creates** something (a name, a witness, a case split), **assumes** something, or **cites** something cannot be restored that way, so it is always written.

Silent steps are the ones whose result is forced by the lines around them; every choice, assumption and citation is spoken.

Definition 5.3.2. Silent Steps

In level-3 prose the following steps **may be silent**, that is, left unwritten:

- ∧I when both conjuncts are displayed, and ∧E;

- R;
- $\vee$I;
- $\bot$E, once the contradiction or the empty case has been stated;
- the **name** of DS (not the fact it uses);
- the exchange of the two members of a disjunction or of a conjunction (Example (First derivations with "or"), Proposition 1.2.1), made only to match the schema of a rule, with no citation;
- $\leftrightarrow$E with a definition ("that is", "by definition", or nothing);
- $\forall$E at the obvious term;
- $\rightarrow$E whose implication is a line already in the proof that is neither a hypothesis of the statement, nor a cited result, nor an instance of one (for instance, an instance of the property of a chosen name), when its antecedent is displayed or has just been assumed;
- $=$I and $=$E inside a chain of equalities or inequalities;
- the phrase "it suffices to show", when the claim of the cut is displayed and then used;
- the **closing** lines of $\rightarrow$I ("hence if P then Q"), of $\forall$I ("since x was arbitrary"), of $\vee$E ("in either case") and of $\exists$E (the conclusion repeated outside the box of the chosen name).

The following are **never silent**:

- the openings of $\rightarrow$I, $\neg$I, RAA and $\forall$I, and of $\exists$E **with a fresh name**;
- the conclusions of $\neg$I and of RAA ("hence $\neg P$", "hence P");
- a $\neg\neg$E or LEM step, written as an explicit case split or as an explicit removal of a double negation (Proposition 2.5.1 (b); "P must hold, since ..."), with the flag of Section 5.7 when the step is genuinely non-constructive;
- the case split of $\vee$E, with the reason the cases exhaust;
- the clash of $\neg$E: both members of the clashing pair are named;
- the witness of $\exists$I, with what it depends on;
- an instance of $\forall$E at a term that is not the obvious one;
- every citation, and so every $\rightarrow$E whose implication is a hypothesis of the statement or a cited result, or an instance of one;
- both directions of $\leftrightarrow$I, and both halves of an $\exists!$ proof;
- the base case and the step of an induction, with the induction hypothesis stated;
- the announcement of a contrapositive;
- the negated form, after De Morgan (Proposition 2.5.5) or a quantifier negation (Proposition 4.9.1, Proposition 4.9.3);
- the claim of a cut;
- a "without loss of generality", together with the symmetry that licenses it.

The two lists collect the last column of the dictionary in Sections 5.1 and 5.2 in one place, together with the requirements stated in the text under its tables: the reason a case split exhausts and the flag on a genuinely non-constructive step (Section 5.1), and chains of inequalities as well as equalities (Section 5.2). The dictionary gives each row its phrase and an example, and this definition gives the reader one list to check against. Two items are made explicit here that the dictionary leaves implicit. The first is the $\rightarrow$E whose implication is a line already in the proof rather than a citation, as in "for every $n \geq N$" after N is chosen. The second is the closing line of $\exists$E, which the dictionary's row for $\exists$E leaves implicit: when a proof says "choose N such that ..." and goes on to prove something that does not mention N, the $\exists$E line that carries that conclusion outside the box of N is never written in prose. What is written is the choice, with its fresh name.

Read the two lists against the principle. Everything in the first list is forced. $\wedge$E has one premise

and two possible conclusions, and the next sentence shows which was used. The closing line of a box is determined by the box. $\forall$E at the obvious term x, when x is the name in sight, is forced by the sentence that uses it. Everything in the second list carries information that the surrounding lines do not determine. Which name is chosen, and that it is fresh; which witness, and what it depends on; which cases, and why they are all the cases; which result is cited. A reader who meets "Then p is even" with no citation has to guess what licensed it, and a guess is not a check.

A small example. In "since $x \in (0,1)$, in particular $x > 0$", the step from the conjunction "$0 < x$ and $x < 1$" to its first conjunct is $\wedge$E, and the unwinding of $x \in (0,1)$ into that conjunction is $\leftrightarrow$E with the definition of an interval. Both may be silent, and here the writer kept the phrase "in particular" anyway, which is allowed: the convention says what **may** be omitted, never what must be. On the other side, "suppose $r^2 = 2$" can never be omitted from the proof of Theorem 2.2.1: without it, the reader does not know that a box has opened, and the final "$r^2 \neq 2$" has no visible source.

Warning. **Level 3 is not level 0 with steps missing at random.** A proof that skips the reason a case split is exhaustive, or chooses δ without saying what it depends on, or uses a lemma without naming it, is not "written at level 3": it has a gap, because those steps are in the never-silent list. Conversely, a proof that omits "since ε was arbitrary" has no gap. When you judge whether a written proof is complete, judge it against Definition 5.3.2, not against your sense of what is obvious.

5.3.3 The irrationality of the square root of two, four times

The first example is the proof of Theorem 2.2.1, now with every quantifier step written. It uses the fact that Section 2.2 calls (L): every rational number is p/q for integers p, q with $q \neq 0$, not both even. Write $E(x)$ for "x is even". The proof uses only two clauses of (L), and in a convenient form: $r = p/q$ with $q \neq 0$ gives $p = r \cdot q$, which avoids division; and "not both even" is $\neg(E(p) \wedge E(q))$. The clause $q \neq 0$ is then not needed at all, since $q = 0$ would force $p = 0$, and both would be even. It also uses Theorem 2.1.3, which we write as the formula $\forall n \in \mathbb{Z}\ (E(n \cdot n) \to E(n))$.

Example. The irrationality of the square root of two at four levels

Write the proof of Theorem 2.2.1, that no rational number has square 2, at levels 0, 1, 2 and 3 of Definition 5.3.1, and match the levels line by line.

Idea. The move this example shows in full is **skeleton first**: before any arithmetic, write the first and the last line of every box. The goal $\forall r \in \mathbb{Q}\ \neg(r \cdot r = 2)$ dictates the outer box ($\forall$I, "let $r \in \mathbb{Q}$") and the next one ($\neg$I, "suppose $r^2 = 2$", ending in $\bot$). Inside, (L) instantiated at r_0 is an existential, so its $\exists$E boxes open at once, with fresh names p_0, q_0, and they too end in $\bot$. Only then is there something to compute: the arithmetic fills the innermost box, and it has a target, the clash with "not both even". Sections 5.5 and 5.6 come back to this move twice, with parts of the proof left blank.

Before reading on: the level-3 proof below has four sentences. Which lines of the level-0

derivation does its second sentence, "By (L), write $r = p/q$ ... ", stand for?

Solution. **Level 0.** Premise 1 is (L) in the form above, premise 2 is Theorem 2.1.3. The derivation has 24 lines.

1	$\forall r \in \mathbb{Q}\ \exists p \in \mathbb{Z}\ \exists q \in \mathbb{Z}\ (p = r \cdot q \wedge \neg(E(p) \wedge E(q)))$	
2	$\forall n \in \mathbb{Z}\ (E(n \cdot n) \to E(n))$	
3	r_0 | $r_0 \in \mathbb{Q}$	
4	| | $r_0 \cdot r_0 = 2$	
5	| | $\exists p \in \mathbb{Z}\ \exists q \in \mathbb{Z}\ (p = r_0 \cdot q \wedge \neg(E(p) \wedge E(q)))$	$\forall$E 1, 3
6	| | p_0 | $p_0 \in \mathbb{Z}$	
7	| | | $\exists q \in \mathbb{Z}\ (p_0 = r_0 \cdot q \wedge \neg(E(p_0) \wedge E(q)))$	
8	| | | q_0 | $q_0 \in \mathbb{Z}$	
9	| | | | $p_0 = r_0 \cdot q_0 \wedge \neg(E(p_0) \wedge E(q_0))$	
10	| | | | $p_0 = r_0 \cdot q_0$	$\wedge$E 9
11	| | | | $p_0 \cdot p_0 = 2 \cdot (q_0 \cdot q_0)$	arith 4, 10
12	| | | | $E(p_0 \cdot p_0)$	arith 8, 11
13	| | | | $E(p_0 \cdot p_0) \to E(p_0)$	$\forall$E 2, 6
14	| | | | $E(p_0)$	$\to$E 13, 12
15	| | | | $E(q_0 \cdot q_0)$	arith 6, 8, 11, 14
16	| | | | $E(q_0 \cdot q_0) \to E(q_0)$	$\forall$E 2, 8
17	| | | | $E(q_0)$	$\to$E 16, 15
18	| | | | $E(p_0) \wedge E(q_0)$	$\wedge$I 14, 17
19	| | | | $\neg(E(p_0) \wedge E(q_0))$	$\wedge$E 9
20	| | | | $\bot$	$\neg$E 18, 19
21	| | | $\bot$	$\exists$E 7, 8–20
22	| | $\bot$	$\exists$E 5, 6–21
23	| $\neg(r_0 \cdot r_0 = 2)$	$\neg$I 4–22
24	$\forall r \in \mathbb{Q}\ \neg(r \cdot r = 2)$	$\forall$I 3–23

The three "arith" lines are facts about integers, each checkable by hand. Line 11: $p_0 \cdot p_0 = (r_0 q_0)(r_0 q_0) = (r_0 \cdot r_0)(q_0 \cdot q_0) = 2 \cdot (q_0 \cdot q_0)$, by lines 10 and 4. Line 12: $p_0 \cdot p_0$ is 2 times the integer $q_0 \cdot q_0$, so it is even. Line 15: by line 14, $p_0 = 2m$ for an integer m; then line 11 reads $4m^2 = 2q_0^2$, so $q_0^2 = 2m^2$ is even. The side conditions at the flagged lines hold: at line 21 the flag q_0 is not in $\bot$, in line 7, in a premise, or in the assumptions still open (lines 3, 4, 6, 7); at line 22 the flag p_0 is not in $\bot$, in line 5, in a premise, or in lines 3 and 4; at line 24 the flag r_0 is in no premise and not in line 24.

Level 1. Lines 10–17, the ∧E that extracts $p_0 = r_0 \cdot q_0$, the arithmetic and the two uses of Theorem 2.1.3, cited as "lemma", are compressed into two lines, and the logical steps of lines 18–20 into one line; every box, every assumption and every closing line is kept.

1	$\forall r \in \mathbb{Q}\ \exists p \in \mathbb{Z}\ \exists q \in \mathbb{Z}\ (p = r \cdot q \wedge \neg(E(p) \wedge E(q)))$	
2	r_0 | $r_0 \in \mathbb{Q}$	
3	| | $r_0 \cdot r_0 = 2$	
4	| | $\exists p \in \mathbb{Z}\ \exists q \in \mathbb{Z}\ (p = r_0 \cdot q \wedge \neg(E(p) \wedge E(q)))$	∀E 1, 2
5	| | p_0 | $p_0 \in \mathbb{Z}$	
6	| | | $\exists q \in \mathbb{Z}\ (p_0 = r_0 \cdot q \wedge \neg(E(p_0) \wedge E(q)))$	
7	| | | q_0 | $q_0 \in \mathbb{Z}$	
8	| | | | $p_0 = r_0 \cdot q_0 \wedge \neg(E(p_0) \wedge E(q_0))$	
9	| | | | $E(p_0)$	∧E; arith; lemma 3, 5, 7, 8
10	| | | | $E(q_0)$	arith; lemma 3, 5, 7, 8, 9
11	| | | | $\bot$	∧E, ∧I, ¬E 8, 9, 10
12	| | | $\bot$	∃E 6, 7–11
13	| | $\bot$	∃E 4, 5–12
14	| $\neg(r_0 \cdot r_0 = 2)$	¬I 3–13
15	$\forall r \in \mathbb{Q}\ \neg(r \cdot r = 2)$	∀I 2–14

Level-1 line 9 is level-0 lines 10–14; line 10 is lines 15–17; line 11 is lines 18–20. Premise 2 is not a line of level 1; it is cited as "lemma". Every other line of level 1 is a line of level 0, renumbered.

Level 2.

> Let $r \in \mathbb{Q}$ [∀I: the box of r_0], and suppose $r^2 = 2$ [assumption for ¬I]. By (L), there are integers p and q with $p = rq$, not both even; choose such p and q [∀E at r, then ∃E twice, with fresh names]. Then $p^2 = r^2q^2 = 2q^2$ [arith], so p^2 is even [arith], and by Theorem 2.1.3 p is even [∀E at p, →E]. Writing $p = 2m$, we get $4m^2 = 2q^2$, so $q^2 = 2m^2$ is even [arith], and by the same theorem q is even [∀E at q, →E]. So p and q are both even [∧I], while they were chosen not both even: a contradiction [¬E; then ∃E twice, since $\bot$ mentions neither p nor q]. Hence $r^2 \neq 2$ [¬I]. Since r was arbitrary, no rational number has square 2 [∀I].

Level 3.

> Let r be rational, and suppose $r^2 = 2$. By (L), write $r = p/q$ with integers p, q, not both even. Then $p^2 = 2q^2$, so p^2 is even, and p is even by Theorem 2.1.3; writing $p = 2m$ gives $q^2 = 2m^2$, so q is even by the same theorem. Thus p and q are both even, a contradiction; hence $r^2 \neq 2$.

The answer to the prompt: "By (L), write $r = p/q$ with integers p, q, not both even" stands for line 5 (∀E of premise 1 at r_0) and for the openings of the two ∃E boxes, lines 6–9. It also

displays the conjunct, in the form $r = p/q$, on its own, so the $\wedge$E of line 10 that extracts it has nothing left to say and can be silent.

Look at what the four texts share. Every level has the box of r, the box of the assumption $r^2 = 2$, the choice of p and q, two uses of the lemma, and a contradiction followed by "hence $r^2 \neq 2$". That common part is the proof; the rest is how much of the routine work each reader is trusted to do. The skeleton is the place where you can see it most clearly: fifteen lines, of which only lines 9 and 10 carry the mathematics; every other line is a decision about logic.

Quick check. Which lines of the level-0 derivation in Example (The irrationality of the square root of two at four levels) have no counterpart in the level-3 text, and which item of Definition 5.3.2 lets each go?

Solution. Eight lines. Line 5 is $\forall$E of premise 1 at the obvious term r_0, the name in sight; "By (L)" cites premise 1, and the instance is not written. Lines 10 and 19 are $\wedge$E. Lines 13 and 16 are $\forall$E of the lemma at the obvious terms p_0 and q_0; the citations "by Theorem 2.1.3" and "by the same theorem" remain, and they are the $\rightarrow$E lines 14 and 17. Lines 21 and 22 are the closing lines of the two $\exists$E boxes. Line 24 is the closing line of $\forall$I: level 3 never says "since r was arbitrary". Every other line has a phrase: lines 1 and 2, the premises, are the citations "By (L)" and "by Theorem 2.1.3"; lines 3, 4, 6–9 the openings; lines 11, 12, 15 the arithmetic; lines 14, 17 the citations; lines 18 and 20 the clash, "both even, a contradiction", which names the pair through "not both even" two sentences earlier; and line 23, "hence $r^2 \neq 2$", the conclusion of $\neg$I, which is never silent.

5.3.4 Uniform continuity, four times

The second example has no arithmetic at all. That makes it a good test of the levels: what is left when there is nothing to compress?

Example. Uniformly continuous implies continuous, at four levels

Write the proof of Example (Uniformly continuous implies continuous), that a function uniformly continuous on A is continuous at every point of A, at levels 0, 1, 2 and 3, and say where each level records what δ depends on.

Solution. **Level 0** is the derivation of Example (Uniformly continuous implies continuous), copied here with the same line numbers. Line 1 is uniform continuity (UC), line 11 is continuity at every point (C).

1		$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	
2	x_0	$x_0 \in A$	
3	ε_0	$\varepsilon_0 > 0$	
4		$\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∀E 1, 3
5	δ_0	$\delta_0 > 0$	
6		$\forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta_0 \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	
7		$\forall y \in A\ (\lvert x_0 - y\rvert < \delta_0 \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∀E 6, 2
8		$\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∃I 5, 7
9		$\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∃E 4, 5–8
10		$\forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon)$	∀I 3–9
11		$\forall x \in A\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	∀I 2–10

Level 1. Every line of level 0 other than the premise opens or closes a box, or is cited by a closing line, except line 7: lines 7 and 8, which specialize the chosen δ_0 to the point x_0 and repackage the result, are the one stretch of work inside a box. They become one line.

1		$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	
2	x_0	$x_0 \in A$	
3	ε_0	$\varepsilon_0 > 0$	
4		$\exists\delta > 0\ \forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∀E 1, 3
5	δ_0	$\delta_0 > 0$	
6		$\forall x \in A\ \forall y \in A\ (\lvert x - y\rvert < \delta_0 \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	
7		$\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∀E, ∃I 2, 5, 6
8		$\exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∃E 4, 5–7
9		$\forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon)$	∀I 3–8
10		$\forall x \in A\ \forall\varepsilon > 0\ \exists\delta > 0\ \forall y \in A\ (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	∀I 2–9

Level 2.

> Let $x \in A$ [∀I] and let $\varepsilon > 0$ [∀I]. By uniform continuity, applied at ε [∀E], there is $\delta > 0$ such that $|u - v| < \delta$ implies $|f(u) - f(v)| < \varepsilon$ for all $u, v \in A$; choose such a δ, which depends on ε only [∃E]. Taking $u = x$ [∀E], every $y \in A$ with $|x - y| < \delta$ has $|f(x) - f(y)| < \varepsilon$. So this δ works for x and ε [∃I; then ∃E closes, since "some δ works" does not mention the chosen one]. Since ε and x were arbitrary, f is continuous at every point of A [∀I twice].

As in Section 4.10, the prose renames the bound variables of UC to u, v, so that x can name the fixed point.

Level 3.

> Let $x \in A$ and $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ (depending on ε

only) such that $|f(u) - f(v)| < \varepsilon$ for all $u, v \in A$ with $|u - v| < \delta$. In particular $|f(x) - f(y)| < \varepsilon$ for every $y \in A$ with $|x - y| < \delta$, so this δ works for x and ε.

Where the dependence of δ is recorded. At level 0, in two places: the box of δ_0 lies inside the boxes of x_0 and ε_0, so δ_0 **may** depend on both; and the line it comes from, line 4, mentions ε_0 and not x_0, so it **does** depend only on ε_0 (Section 4.10). At level 1 the record is the same, lines 3 to 5 being unchanged. At level 2 it is the clause "which depends on ε only", placed at the ∃E tag, after the ∀E at ε that produced it. At level 3 it is the parenthesis "(depending on ε only)", and nothing else: the parenthesis is the whole of the box structure that survives.

Here the gap between level 0 and level 1 is one line, because there is no arithmetic to compress. The large step is from level 1 to level 3: ten lines become three sentences. What the level-3 text keeps is exactly the never-silent part of Definition 5.3.2: the openings of the two ∀I boxes ("let $x \in A$ and $\varepsilon > 0$"), the citation and the choice with its fresh name ("by uniform continuity, choose δ"), the dependence of the witness, and the witness itself ("this δ works"). The ∀E at $u = x$ is the obvious instance and is reduced to "in particular". The ∃E closing line 9 and the two ∀I closing lines 10 and 11 of level 0 are silent.

5.3.5 Expand to check

The levels are not only a way to write. They are a way to check. When a sentence of a level-3 proof leaves you uneasy, rewrite that stretch, and only that stretch, at level 1. Either every line of the expansion is legal, and the doubt is settled, or one line breaks a rule, and you have found the error and its exact address.

Here is a sentence that sounds like the level-3 proof above run backwards, from continuity to uniform continuity.

> Let $\varepsilon > 0$. For each $x \in A$, continuity at x gives some $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for every $y \in A$ with $|x - y| < \delta$. Hence, choosing δ as above for every x, we have $|f(x) - f(y)| < \varepsilon$ whenever $x, y \in A$ and $|x - y| < \delta$, so f is uniformly continuous on A.

Expand it. "Let $\varepsilon > 0$" opens a box for ε_0. "For each $x \in A$" opens a box for x_0. "Continuity at x gives some δ" is ∀E twice and the opening of an ∃E box for δ_0. At level 1, with the line the sentence needs as line 7, its rule marked with a question mark:

1			$\forall x \in A\, \forall \varepsilon > 0\, \exists \delta > 0\, \forall y \in A\, (\lvert x - y\rvert < \delta \to \lvert f(x) - f(y)\rvert < \varepsilon)$	
2	ε_0		$\varepsilon_0 > 0$	
3		x_0	$x_0 \in A$	
4			$\exists \delta > 0\, \forall y \in A\, (\lvert x_0 - y\rvert < \delta \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	∀E twice 1, 3, 2
5		δ_0	$\delta_0 > 0$	
6			$\forall y \in A\, (\lvert x_0 - y\rvert < \delta_0 \to \lvert f(x_0) - f(y)\rvert < \varepsilon_0)$	
7			$\forall x \in A\, \forall y \in A\, (\lvert x - y\rvert < \delta_0 \to \lvert f(x) - f(y)\rvert < \varepsilon_0)$	∀I? 3–6

Line 7 is what the phrase "choosing δ as above for every x" needs: one δ_0 that works for all x and y, so that ∃I can then produce UC's $\exists\delta$ inside the box of ε_0. But line 7 would be a ∀I over

x_0, and no such step is legal here. The box of x_0 opened at line 3 is still open, so ∀I cannot cite it; and a new box with the flag x_0, opened inside the box of δ_0, breaks the side condition of ∀I (Definition 4.4.1), because x_0 appears in the open assumptions on lines 3 and 6. The phrase "for every x" has generalized over a name on which the chosen δ_0 depends. This is the first blocked line of Example (The search from C to UC), met there in the search; here it was found by expanding one English sentence. Section 5.6 catalogues the error, and Example (The square function on the real line) shows that no repair is possible: for x^2 on $\mathbb{R}$ the conclusion is false.

The expansion took six lines, and it was enough to expand the one doubtful stretch. That is the general pattern: you never need to rewrite a whole proof at level 0 to check it. Find the sentence you do not trust, find the boxes open at that sentence, and write the few lines it stands for.

5.3.6 Exercises

5.3.6.1 A. Check your understanding

Exercise. A1

Parts (a)–(c) are about Example (Uniformly continuous implies continuous, at four levels), parts (d) and (e) about Example (The irrationality of the square root of two at four levels).

(a) Which lines of the level-0 derivation does the level-3 sentence "Let $x \in A$ and $\varepsilon > 0$" stand for? Which of them open a box and which close one, and which are written at level 3?

(b) Where is the dependence of δ recorded at each of the four levels?

(c) State the idea of the proof in two sentences.

(d) At which line of each of the four levels is the "not both even" clause of (L) used to reach the contradiction?

(e) True or false, with a reason: at level 3, the sentence "hence $r^2 \neq 2$" may be left out of the proof of Theorem 2.2.1, since the reader has just been told that a contradiction was reached.

Solution. *(a)* Lines 2, 3, 10 and 11. Lines 2 and 3 open the flagged boxes of x_0 and ε_0, with their bounds $x_0 \in A$ and $\varepsilon_0 > 0$; lines 10 and 11 close them by ∀I. The openings are written ("let"); the closings are silent at level 3 (Definition 5.3.2), and at level 2 they are the sentence "since ε and x were arbitrary".

(b) At level 0 and level 1: the box of δ_0 sits inside the boxes of x_0 and ε_0, and the line it comes from (line 4) mentions ε_0 only. At level 2: the clause "which depends on ε only" at the ∃E tag. At level 3: the parenthesis "(depending on ε only)".

(c) Uniform continuity at ε provides one δ that serves all pairs of points of A at once, and it provides it before any point is named. Continuity at a point x asks only for the pairs whose first point is x, so the same δ works.

(d) Level 0: the clause is line 9's second conjunct, taken out by ∧E at line 19 and used in the ¬E of line 20. Level 1: it is the second conjunct of the assumption on line 8, used at line 11. Level 2: "while they were chosen not both even: a contradiction". Level 3: "Thus p and q are both even, a contradiction", where the clash refers back to "not both even" in the sentence that chose p and q.

(e) False. "Hence $r^2 \neq 2$" is the conclusion of ¬I, which Definition 5.3.2 puts among the steps that are never silent. A contradiction alone does not say which assumption it refutes; in this proof the boxes of p, q and of $r^2 = 2$ are all open when it is reached, and the sentence says which box closes and what it gives.

5.3.6.2 B. Practice

Exercise. B1: Some-all gives all-some, on the natural numbers

Read Proposition 4.6.2 over the domain $\mathbb{N}$, with $R(x, y)$ as "$x \leq y$": from $\exists x \, \forall y \, (x \leq y)$, conclude $\forall y \, \exists x \, (x \leq y)$. The premise is true, with the witness $x = 0$. Write the proof at levels 1, 2 and 3. Your proof must use the premise, although this instance of the conclusion also has a one-line proof, with $x = y$.

Solution. **Level 1.** The proof has no arithmetic, so the skeleton is the level-0 derivation of Proposition 4.6.2 with $R(x, y)$ read as $x \leq y$:

1	$\exists x \, \forall y \, (x \leq y)$		
2	y_0 | x_0 | $\forall y \, (x_0 \leq y)$		
3		$x_0 \leq y_0$	∀E 2
4		$\exists x \, (x \leq y_0)$	∃I 3
5		$\exists x \, (x \leq y_0)$	∃E 1, 2–4
6	$\forall y \, \exists x \, (x \leq y)$		∀I 2–5

Level 2.

> Let $y \in \mathbb{N}$ [∀I]. By hypothesis, some natural number is at most every natural number; choose such a number x_0, which depends on nothing, since the hypothesis mentions no y [∃E, x_0 fresh]. In particular $x_0 \leq y$ [∀E at y], so some natural number is at most y [∃I with witness x_0; then ∃E closes, since this does not mention x_0]. Since y was arbitrary, every natural number has a natural number at most it [∀I].

Level 3.

> Let $y \in \mathbb{N}$. By hypothesis, choose $x_0 \in \mathbb{N}$ with $x_0 \leq z$ for every $z \in \mathbb{N}$; the hypothesis does not mention y, so x_0 does not depend on y. In particular $x_0 \leq y$, so x_0 is a natural number at most y.

The level-3 text renames the bound variable of the hypothesis to z, so that y can name the fixed number. The witness x_0 was chosen inside the box of y_0, but it comes from line 1, which mentions no y; so it does not depend on y. The one-line proof, "$y \leq y$, so take $x = y$", is ∃I with the witness y, which **does** depend on y. Both prove the conclusion; only the first is an instance of Proposition 4.6.2, and only the first shows that a single x serves every y.

Exercise. B2: Back from prose to a skeleton

Here is a level-3 proof.

> **Claim.** If n is an odd integer, then $8 \mid n^2 - 1$.
>
> Let n be odd, say $n = 2k + 1$ with k an integer. Then $n^2 - 1 = 4k(k + 1)$, so it suffices to show that $k(k + 1)$ is even. If $k = 2j$ for an integer j, then $k(k + 1) = 2 \cdot j(2j + 1)$; if $k = 2j + 1$, then $k(k + 1) = 2 \cdot (2j + 1)(j + 1)$. Either way $k(k + 1) = 2i$ for an integer i, and then $n^2 - 1 = 8i$.

Reconstruct its level-1 skeleton. Work over the domain $\mathbb{Z}$, with "n is odd" as $\exists k\,(n = 2k+1)$ and $8 \mid m$ as $\exists t\,(m = 8t)$. Name the rule that closes each box, and say where the proof uses the fact (E1) of Section 2.1, that every integer is even or odd.

Solution. The goal is $\forall n\,(\exists k\,(n = 2k + 1) \to \exists t\,(n^2 - 1 = 8t))$: its skeleton is a $\forall$I box for n_0, and inside it a $\to$I box. The hypothesis is an existential, so its $\exists$E box (name k_0) opens at once. "It suffices to show that $k(k + 1)$ is even" is a cut: the claim $\exists i\,(k_0(k_0 + 1) = 2i)$ is proved by $\vee$E on (E1) for k_0, with an $\exists$E and an $\exists$I in each case, and is then used through one more $\exists$E and the final $\exists$I.

1	n_0		$\exists k\,(n_0 = 2\cdot k+1)$	
2		k_0	$n_0 = 2\cdot k_0+1$	
3			$n_0\cdot n_0 - 1 = 4\cdot(k_0\cdot(k_0+1))$	arith 2
4			$\exists j\,(k_0 = 2\cdot j)\vee\exists j\,(k_0 = 2\cdot j+1)$	(E1)
5			$\exists j\,(k_0 = 2\cdot j)$	
6			j_0 $k_0 = 2\cdot j_0$	
7			$k_0\cdot(k_0+1) = 2\cdot(j_0\cdot(2\cdot j_0+1))$	arith 6
8			$\exists i\,(k_0\cdot(k_0+1) = 2\cdot i)$	$\exists$I 7
9			$\exists i\,(k_0\cdot(k_0+1) = 2\cdot i)$	$\exists$E 5, 6–8
10			$\exists j\,(k_0 = 2\cdot j+1)$	
11			j_1 $k_0 = 2\cdot j_1+1$	
12			$k_0\cdot(k_0+1) = 2\cdot((2\cdot j_1+1)\cdot(j_1+1))$	arith 11
13			$\exists i\,(k_0\cdot(k_0+1) = 2\cdot i)$	$\exists$I 12
14			$\exists i\,(k_0\cdot(k_0+1) = 2\cdot i)$	$\exists$E 10, 11–13
15			$\exists i\,(k_0\cdot(k_0+1) = 2\cdot i)$	$\vee$E 4, 5–9, 10–14
16			i_0 $k_0\cdot(k_0+1) = 2\cdot i_0$	
17			$n_0\cdot n_0 - 1 = 8\cdot i_0$	arith 3, 16
18			$\exists t\,(n_0\cdot n_0 - 1 = 8\cdot t)$	$\exists$I 17
19			$\exists t\,(n_0\cdot n_0 - 1 = 8\cdot t)$	$\exists$E 15, 16–18
20			$\exists t\,(n_0\cdot n_0 - 1 = 8\cdot t)$	$\exists$E 1, 2–19
21			$\exists k\,(n_0 = 2\cdot k+1)\to\exists t\,(n_0\cdot n_0 - 1 = 8\cdot t)$	$\to$I 1–20
22			$\forall n\,(\exists k\,(n = 2\cdot k+1)\to\exists t\,(n\cdot n - 1 = 8\cdot t))$	$\forall$I 1–21

The arithmetic lines: $(2k_0+1)^2 - 1 = 4k_0^2 + 4k_0 = 4k_0(k_0+1)$ (line 3); $2j_0(2j_0+1)$ for $k_0 = 2j_0$ (line 7); $(2j_1+1)(2j_1+2) = 2(2j_1+1)(j_1+1)$ (line 12); $4\cdot 2i_0 = 8i_0$ (line 17). The boxes close by $\exists$E (lines 9, 14, 19, 20), $\vee$E (line 15), $\to$I (line 21) and $\forall$I (line 22). The fact (E1) enters at line 4, instantiated at k_0; in the prose it is the words "if $k = 2j$ … if $k = 2j+1$", whose reason for exhausting the cases is left to the reader. By Definition 5.3.2 that reason should be written ("since every integer is even or odd"), so the given proof is slightly short of level 3 at exactly this point. The two case names may both be j in the prose, because their boxes are disjoint; here they are j_0 and j_1 to keep the lines apart.

5.3.6.3 C. Going deeper

Exercise. C1: Arbitrarily close numbers, at two levels

Consider Proposition 4.9.6: for real numbers a and b, if $|a-b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$.

(a) Write its proof at level 1, as a skeleton in which school facts about $|\cdot|$ and $<$ are lines justified "arith".

(b) Write it at level 3.

(c) Which lines of your skeleton are silent at level 3? Which instance of ∀E must the level-3 text name, and why?

Solution. (a) The statement is an implication, so the outer box is →I; the goal $a = b$ is atomic, and the proof in Section 4.9 opens an RAA box on $a \neq b$.

1	$\forall \varepsilon > 0\ (\lvert a-b\rvert < \varepsilon)$	
2	$\quad \neg(a = b)$	
3	$\quad \lvert a-b\rvert > 0$	arith 2
4	$\quad \lvert a-b\rvert < \lvert a-b\rvert$	∀E 1, 3
5	$\quad \neg(\lvert a-b\rvert < \lvert a-b\rvert)$	arith
6	$\quad \bot$	¬E 4, 5
7	$a = b$	RAA 2–6
8	$\forall \varepsilon > 0\ (\lvert a-b\rvert < \varepsilon) \to a = b$	→I 1–7

Line 4 is the bounded ∀E at the term $|a-b|$, whose bound $|a-b| > 0$ is line 3. Line 5 is the school fact that no real number is less than itself.

(b)

> Suppose $|a-b| < \varepsilon$ for every $\varepsilon > 0$, and suppose, for a contradiction, that $a \neq b$. Then $|a-b| > 0$, so the hypothesis applies with $\varepsilon = |a-b|$ and gives $|a-b| < |a-b|$; but no real number is less than itself. Hence $a = b$.

(c) Only line 8, the closing line of →I, is silent: "hence if ... then $a = b$" is not written. Line 5 is not silent: it is one member of the clashing pair of the ¬E on line 6, and level 3 names it ("no real number is less than itself"). The instance of line 4 must be named, $\varepsilon = |a-b|$, because it is not the obvious term: no name in sight is the obvious candidate for ε, and the whole proof turns on choosing this one. Saying only "by the hypothesis" would leave the reader to find the instance, and that instance is the idea of the proof. The RAA opening, "suppose, for a contradiction, that $a \neq b$", and its conclusion, "hence $a = b$", are also never silent.

Marking scheme. (10 points) - 2: (a) the →I box on the hypothesis, and inside it a box opened on $a \neq b$. - 2: (a) the box closed by RAA (¬I alone would give only $\neg\neg(a = b)$), with $\bot$ as its last line. - 2: (a) the ∀E at $|a-b|$, citing a line $|a-b| > 0$. - 2: (b) a level-3 text with the opening of the contradiction, the instance, the clash and "hence $a = b$", and no tags. - 2: (c) the silent line identified, and the instance $\varepsilon = |a-b|$ said to be named because it is not the obvious term. - Other routes: a ¬I box on $a \neq b$ followed by ¬¬E, or a split on $|a-b| > 0$ or $|a-b| = 0$ (arith) whose first case reaches $\bot$ by the same ∀E and clash and then $a = b$ by $\bot$E, while the second gives $a = b$ by arith, earns the 4 points of the first two lines, provided the ¬¬E line is written explicitly, or the disjunction of the split is justified (from $|a-b| \geq 0$, or by trichotomy). In the split, the case assumption $|a-b| > 0$ is the line the ∀E cites for the third line, and for (b) and (c) the opening of the ¬I box or of the split

stands in for the opening of the contradiction. Fatal errors (0 for the step): the box closed by $\neg I$ with the conclusion $a = b$; the hypothesis applied at an ε not known to be positive.

Exercise. C2: Exit check: levels

Let (a_n) be a sequence of real numbers with $a_n \to L$, and let $c \neq 0$ be a real number. Here is a level-0 derivation that $c\,a_n \to cL$, with the definition of a limit as in Example (Five definitions, translated) (a), the hypothesis $c \neq 0$ as premise 2, and two lines of school arithmetic justified "arith".

1	$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert a_n - L\rvert < \varepsilon)$	
2	$\neg(c = 0)$	
3	$\varepsilon_0 \mid \varepsilon_0 > 0$	
4	$\varepsilon_0/\lvert c\rvert > 0$	arith 2, 3
5	$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert a_n - L\rvert < \varepsilon_0/\lvert c\rvert)$	$\forall$E 1, 4
6	$N_0 \mid N_0 \in \mathbb{N}$	
7	$\forall n \in \mathbb{N}\ (n \geq N_0 \to \lvert a_n - L\rvert < \varepsilon_0/\lvert c\rvert)$	
8	$n_0 \mid n_0 \in \mathbb{N}$	
9	$n_0 \geq N_0 \to \lvert a_{n_0} - L\rvert < \varepsilon_0/\lvert c\rvert$	$\forall$E 7, 8
10	$n_0 \geq N_0$	
11	$\lvert a_{n_0} - L\rvert < \varepsilon_0/\lvert c\rvert$	$\to$E 9, 10
12	$\lvert c \cdot a_{n_0} - c \cdot L\rvert < \varepsilon_0$	arith 2, 11
13	$n_0 \geq N_0 \to \lvert c \cdot a_{n_0} - c \cdot L\rvert < \varepsilon_0$	$\to$I 10–12
14	$\forall n \in \mathbb{N}\ (n \geq N_0 \to \lvert c \cdot a_n - c \cdot L\rvert < \varepsilon_0)$	$\forall$I 8–13
15	$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert c \cdot a_n - c \cdot L\rvert < \varepsilon_0)$	$\exists$I 6, 14
16	$\exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert c \cdot a_n - c \cdot L\rvert < \varepsilon_0)$	$\exists$E 5, 6–15
17	$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \in \mathbb{N}\ (n \geq N \to \lvert c \cdot a_n - c \cdot L\rvert < \varepsilon)$	$\forall$I 3–16

Write the proof at level 2 and at level 3. Then say in one sentence how the case $c = 0$ goes.

Solution. **Level 2.**

Let $\varepsilon > 0$ [$\forall$I]. Since $c \neq 0$, $\lvert c\rvert > 0$, so $\varepsilon/\lvert c\rvert > 0$ [arith]. Applying the definition of $a_n \to L$ with $\varepsilon/\lvert c\rvert$ in place of ε [$\forall$E], choose $N \in \mathbb{N}$, depending on ε (and on the fixed number c), such that $\lvert a_n - L\rvert < \varepsilon/\lvert c\rvert$ for every $n \geq N$ [$\exists$E, N fresh]. Let $n \in \mathbb{N}$ [$\forall$I] with $n \geq N$ [assumption for $\to$I]. Then $\lvert a_n - L\rvert < \varepsilon/\lvert c\rvert$ [$\forall$E, $\to$E], so

$$\lvert c\,a_n - cL\rvert = \lvert c\rvert\,\lvert a_n - L\rvert < \lvert c\rvert \cdot \frac{\varepsilon}{\lvert c\rvert} = \varepsilon$$

[arith]. So this N works for ε [$\to$I, $\forall$I, $\exists$I with witness N; then $\exists$E closes]. Since

$\varepsilon > 0$ was arbitrary, $c\,a_n \to cL$ [$\forall$I].

Level 3.

Let $\varepsilon > 0$. Since $c \neq 0$, we have $\varepsilon/|c| > 0$, so by the definition of $a_n \to L$, applied with $\varepsilon/|c|$, there is $N \in \mathbb{N}$ (depending on ε and c) such that $|a_n - L| < \varepsilon/|c|$ for every $n \geq N$. For every $n \geq N$,

$$|c\,a_n - cL| = |c|\,|a_n - L| < |c| \cdot \frac{\varepsilon}{|c|} = \varepsilon.$$

So this N works for ε, and $c\,a_n \to cL$.

The case $c = 0$. Then $c\,a_n = 0 = cL$ for every n, so $|c\,a_n - cL| = 0 < \varepsilon$ for every $\varepsilon > 0$ and every n, and any N works, for instance $N = 0$; the hypothesis $a_n \to L$ is not needed.

At level 3, the silent steps are those Definition 5.3.2 allows: the $\forall$E and $\to$E of lines 9 and 11 at the obvious n (the implication of line 11 is line 9, an instance of the property of the chosen N_0, not of a hypothesis or a result), carried by "for every $n \geq N$", and the closing lines 13, 14, 16 and 17. The $\forall$E of line 5 is **not** silent, because its instance $\varepsilon/|c|$ is not the obvious one; it is named ("applied with $\varepsilon/|c|$"). The witness of line 15 is named ("this N"), and its dependence is stated.

Marking scheme. (10 points) - 4: every never-silent step is present at level 3, one point each: $\varepsilon > 0$ fixed first ($\forall$I); the hypothesis applied at $\varepsilon/|c|$, with its positivity justified, and N chosen from it ($\exists$E); the witness N produced with its dependence on ε stated ($\exists$I); $n \geq N$ taken arbitrary ($\forall$I). - 3: the level-2 tags are correct: $\forall$I for ε and for n, $\forall$E at $\varepsilon/|c|$, $\exists$E for N, $\to$I for $n \geq N$, $\exists$I for the witness. Deduct one point per wrong or missing tag, down to 0. - 2: no step silent at level 3 beyond those Definition 5.3.2 allows, steps already scored in the first line not counted again; in particular the estimate $|c\,a_n - cL| = |c|\,|a_n - L| < |c| \cdot \varepsilon/|c| = \varepsilon$ is written out for an arbitrary $n \geq N$. - 1: the case $c = 0$ handled, or excluded explicitly with a reason. A level 3 that writes "for every $n \geq N$" as the single phrase for $\forall$I together with $\to$I earns the $\forall$I point, as in the solution. Fatal errors (0 for the step): N chosen before ε is fixed; dividing by $|c|$ in the case $c = 0$.

5.4 Reading a Proof: Self-Explanation

Section 5.3 showed that a level-3 proof leaves steps unsaid by a fixed convention, and that any doubtful stretch can be expanded and checked. That presupposes a reader who notices the doubtful stretch. This section is about becoming that reader. It teaches one habit, **self-explanation**: for each line of a proof, say what makes it true. The habit is taught here once, on one short proof, and the rest of the book relies on it. When a later proof ends its Idea with "*Before reading on: ...*" or "*Self-explain lines 3–5*", this is the passage it means.

5.4.1 Two readers of one proof

Two people read the same proof, line by line, to the end. Both say they understood it. Asked afterwards whether it was correct, one can answer and point to the reason each line holds; the other can only repeat the proof in other words. What separates them is not intelligence or background. It is what each did at every line.

The first asked, at each line, *what makes this true?* and answered by pointing: to earlier lines, to a definition, to a named result, and to the rule that combines them. The second asked *what*

does this say? and answered by restating it. Only the first answer can come out wrong, and that is exactly why it is useful. A restatement always succeeds, so it tests nothing.

To self-explain a line is to name its warrant: the earlier lines, definitions and results it comes from, and the rule that gets it from them.

The answer has the shape of a justification column in a derivation. In the derivations of Chapters 1–4 that column was printed beside every line. In level-3 prose it is gone, and self-explanation is the reader writing it back in, at level 1 or 2 of Definition 5.3.1, one line at a time.

5.4.2 The training passage

Here is a short proof. Read it once, then read the two explanations that follow.

Example. Explaining a proof, well and badly

Claim. For every real number $x > 0$, $x + 1/x \geq 2$.

Proof. Let $x > 0$.

1. $(x-1)^2 \geq 0$.
2. So $x^2 - 2x + 1 \geq 0$.
3. Dividing by x, $x - 2 + 1/x \geq 0$.
4. Hence $x + 1/x \geq 2$.

Self-explain each line of the proof.

Solution. **Explained well.**

- *The opening.* "Let $x > 0$" opens a box for an arbitrary x with the assumption $x > 0$: the claim is $\forall x > 0$ $(x + 1/x \geq 2)$, and this is its bounded $\forall$I. Everything below may use $x > 0$; so far nothing has.
- *Line 1* is an instance of the school fact that the square of every real number is at least 0, at the term $x - 1$ (a $\forall$E). It uses no earlier line, and in particular not $x > 0$: it holds for every real x.
- *Line 2* follows from line 1 because $(x-1)^2$ and $x^2 - 2x + 1$ are the same number, by expanding the square; one is replaced by the other (=E).
- *Line 3* follows from line 2 by dividing both sides by x. Two things need $x > 0$ here. Dividing by x needs $x \neq 0$, and dividing both sides of an inequality by a number is guaranteed to keep its direction only when that number is positive. Both come from the assumption $x > 0$, and this is the **only** place in the proof where the assumption is used. The left side becomes $(x^2 - 2x + 1)/x = x - 2 + 1/x$, and the right side $0/x = 0$.
- *Line 4* follows from line 3 by adding 2 to both sides, which keeps any inequality. It is the claim for this x; since x was arbitrary, the claim holds for every $x > 0$ (the closing $\forall$I, silent in the proof as written).

Explained badly.

- *Line 1:* we start with a square, which is positive.
- *Line 2:* multiply it out.
- *Line 3:* now we divide by x and it still works.
- *Line 4:* rearranging gives the answer.

Both explanations follow the order of the proof, and the second is not even wrong everywhere; it is shorter, and every sentence after the first describes what happened (the first we return to below). But compare what each one did with line 3. The first says *why* the inequality survives the division, and in saying it, finds the one place where the hypothesis $x > 0$ earns its keep. The second says "it still works", which is a description of the outcome, not a reason.

The difference shows as soon as the proof is wrong. Change the claim to "for every real $x \neq 0$, $x+1/x \geq 2$", and keep all four lines. The bad explanation reads the new proof exactly as happily as the old one: we start with a square, multiply out, divide by x and it still works, rearrange. The good explanation stops at line 3. It needs "the inequality keeps its direction because $x > 0$", and the hypothesis now says only $x \neq 0$; for negative x, division by x reverses the inequality, and line 3 does not follow. The claim is in fact false: at $x = -1$, $x + 1/x = -2 < 2$. A reader who explains warrants finds the gap; a reader who paraphrases cannot, because a paraphrase of a wrong step reads exactly like a paraphrase of a right one.

The bad explanation also slipped in its first line. A square is not always positive, only at least 0: $(x-1)^2 = 0$ at $x = 1$, where the claim holds with equality. Restating invites such slips, because nothing is being checked. Explaining a warrant forces the exact fact, "every square is at least 0", because the explanation has to cite it.

So the first kind of reading does three things the second does not. It **checks** each step, since a warrant can fail to exist. It **locates the hypotheses**, since every use of a hypothesis is a warrant. And it **prepares the reader to reuse the proof**, since the warrants, not the restatements, are what transfer to the next problem.

5.4.3 How to explain a line

For each line, ask four questions, in this order.

1. **What kind of line is it?** An assumption that opens a box ("let", "suppose", "choose ... such that"), a line derived from earlier ones, or the closing of a box ("hence", "so", "since x was arbitrary").
2. **Where does it come from?** Name the earlier lines, the definitions and the results it uses. "Line 2", "the definition of even", "Theorem 2.1.3", "the hypothesis $x > 0$". A result is named, not paraphrased.
3. **By which step?** A rule of Chapters 1–4, or a piece of arithmetic that you can say in one clause ("divide both sides by x", "expand the square").
4. **Is it legitimate here?** Is every cited line still available, inside the boxes that are open (Definition 1.4.2)? Is a chosen name fresh? Does a division have a non-zero divisor, and does an inequality keep its direction?

You do not need to write out a derivation; a phrase per line is enough, at the level of the "explained well" column above. When you cannot answer one of the four questions, you have found what to look at: either a step you do not yet understand, or a gap in the proof. Section 5.3 says what to do next: expand that stretch to level 1.

Warning. **Self-explanation is not re-reading, and not summarizing.** Reading a proof twice more makes it familiar, not checked: familiarity is what makes the bad explanation of Example (Explaining a proof, well and badly) feel adequate. A summary ("the proof expands a square and divides by x") is useful afterwards, but it skips exactly the warrants that matter, such as why the division by x is allowed and keeps the inequality's direction. Self-explanation is local and backward-looking: each line, against what came before it.

Quick check. Here is a six-line proof that for real numbers x and y, if $x + y > 2$, then $x > 1$ or $y > 1$.

1. We prove the contrapositive: if it is not the case that $x > 1$ or $y > 1$, then $x + y > 2$ fails.
2. Suppose it is not the case that $x > 1$ or $y > 1$.
3. Then $x \leq 1$ and $y \leq 1$.
4. Adding these, $x + y \leq 2$.
5. So $x + y > 2$ fails.
6. This proves the contrapositive, and hence the claim.

Self-explain lines 2–4.

Solution. Line 2 is an assumption: it opens the $\to$I box for the contrapositive announced in line 1, whose hypothesis is $\neg(x > 1 \vee y > 1)$.

Line 3 comes from line 2 in two steps. First Proposition 2.5.5 (a), $\neg(P \vee Q) \vdash \neg P \wedge \neg Q$, gives "not $x > 1$, and not $y > 1$". Then the order fact that $\neg(u > 1)$ means $u \leq 1$, for every real u, turns each conjunct into the form written.

Line 4 uses both conjuncts of line 3 ($\wedge$E, twice) and the order fact that inequalities may be added: $x \leq 1$ and $y \leq 1$ give $x + y \leq 1 + y \leq 1 + 1 = 2$.

(For the record: line 5 is the order fact $x + y \leq 2$ means $\neg(x + y > 2)$; line 6 closes the box by $\to$I and then passes from the contrapositive to the claim by Proposition 2.5.2, a classical step.)

5.4.4 Strategies of good readers

Self-explanation is the habit for single lines. Readers who are good at proofs also do a few things with a proof as a whole. Here are five, shown on one short proof.

Claim. For real numbers $x > 0$ and $y > 0$, $\sqrt{xy} \leq (x + y)/2$.

Proof. Let $x > 0$ and $y > 0$. Since $x = (\sqrt{x})^2$, $y = (\sqrt{y})^2$ and $\sqrt{xy} = \sqrt{x}\,\sqrt{y}$,

$$x + y - 2\sqrt{xy} = (\sqrt{x} - \sqrt{y})^2 \geq 0.$$

Hence $2\sqrt{xy} \leq x + y$, and dividing by 2 gives the claim.

1. **Try it first.** Before reading the proof, spend a few minutes on the claim yourself. A natural attempt squares both sides: $xy \leq (x + y)^2/4$ is $4xy \leq x^2 + 2xy + y^2$, that is, $0 \leq (x - y)^2$, which is true. Already you know what the difficulty is (the square root) and what your attempt rests on (that a square is at least 0).
2. **Identify the framework.** Find the skeleton, level 1 of Definition 5.3.1. Here: a bounded $\forall$I on x and y (opened by "Let", closed silently) with the bounds $x > 0$, $y > 0$, then a direct chain. There is no case split, no contradiction, and no chosen witness. Knowing that tells you where to look: every line is a computation, so every warrant is an algebraic fact.
3. **Break it into sub-proofs.** The proof has one claim inside it, the displayed identity, and then a short finish. A claim inside a proof is a cut (the cut row of Section 5.2's dictionary): it is proved once and then used. Here the claim is proved inside the box on x and y, so it

is simply a line that later lines of the box may cite (Definition 1.4.2); Lemma 3.6.1 (b) is the same move for sequents proved outside every box. Read the claim on its own: it needs $\sqrt{xy} = \sqrt{x}\sqrt{y}$ (true for $x, y \geq 0$) and $(\sqrt{x})^2 = x$.

4. **Illustrate a hard line with an example.** If the identity is not clear, try numbers. With $x = 1$ and $y = 4$: the left side is $1 + 4 - 2 \cdot 2 = 1$, the right side $(1 - 2)^2 = 1$. An example checks nothing in general, but it shows what the line claims.
5. **Compare with your own attempt.** Your squaring route proves the claim too, and it hides a warrant the given proof avoids: passing back from $xy \leq (x + y)^2/4$ to $\sqrt{xy} \leq (x + y)/2$ uses the converse of "squaring preserves order": from $a^2 \leq b^2$ conclude $a \leq b$. This holds when $b \geq 0$, because then $|a| \leq b$; here $b = (x + y)/2 > 0$. Without that hypothesis it fails: $a = 1$, $b = -2$. Comparing the two routes shows why the given proof introduces $\sqrt{x}$ and $\sqrt{y}$: it never has to undo a squaring. The comparison also links this claim to Example (Explaining a proof, well and badly): put $y = 1/x$, and $\sqrt{xy} = 1$, so the claim becomes $x + 1/x \geq 2$.

The five strategies work at the level of the whole proof and self-explanation at the level of the line, and they support each other. Trying first tells you which line will be the hard one; the framework tells you which boxes are open at each line, which is what the fourth question of self-explanation needs.

5.4.5 Exercises

5.4.5.1 A. Check your understanding

Exercise. A1

Here is a proof that if $a > 0$, $b > 0$ and $a^2 = b^2$, then $a = b$.

(1) $a^2 - b^2 = 0$. (2) $(a - b)(a + b) = 0$. (3) $a + b > 0$. (4) Hence $a - b = 0$, that is, $a = b$.

Classify each explanation as one that **justifies** its line or one that **paraphrases** it.

(a) "Line 1 follows from the hypothesis $a^2 = b^2$ by subtracting b^2 from both sides."
(b) "Line 2: then the factored form is zero as well."
(c) "Line 2 is line 1 rewritten with the identity $u^2 - v^2 = (u - v)(u + v)$, at $u = a$ and $v = b$."
(d) "Line 3: now we know that $a + b$ is positive."
(e) "Line 3 holds because $a > 0$ and $b > 0$ are hypotheses, and a sum of two positive numbers is positive."
(f) "Line 4: so $a - b$ is zero, which is what we wanted."

Solution.

(a) Justifies: it names the source (the hypothesis) and the step (subtracting b^2).
(b) Paraphrases: "is zero as well" restates line 2 without saying why the product equals the left side of line 1.
(c) Justifies: it names the identity and the instance.
(d) Paraphrases: it restates line 3 and gives no reason.
(e) Justifies: it names both hypotheses and the fact that combines them.
(f) Paraphrases: it gives no warrant. A justification would say: by line 2 a product is zero,

so one of its factors is zero; line 3 says the factor $a+b$ is not zero, being positive; so $a-b=0$ (Proposition 2.3.3, applied to "$a+b=0$ or $a-b=0$" and $a+b\neq 0$).

5.4.5.2 B. Practice

Exercise. B1: Self-explain a counterexample

Here, with its lines numbered, is the half of Example (The square function on the real line) that shows $f(x)=x^2$ is **not** uniformly continuous on $\mathbb{R}$.

(1) By the negation of uniform continuity, with the bounds $x\in\mathbb{R}$ and $y\in\mathbb{R}$ dropped because they always hold, we must show

$$\exists\varepsilon>0\ \forall\delta>0\ \exists x\ \exists y\ (|x-y|<\delta\wedge|x^2-y^2|\geq\varepsilon).$$

(2) Take $\varepsilon=1$. (3) Let $\delta>0$ be arbitrary, and take $x:=1/\delta$ and $y:=1/\delta+\delta/2$, which depend on δ. (4) Then $|x-y|=\delta/2<\delta$, (5) and, since $y>x>0$,

$$|x^2-y^2|=(y-x)(y+x)=\frac{\delta}{2}\Big(\frac{2}{\delta}+\frac{\delta}{2}\Big)=1+\frac{\delta^2}{4}\geq 1.$$

(3) Since δ was arbitrary, uniform continuity fails.

Self-explain each line, citing labels where a line uses an earlier result. Say at which lines $\delta>0$ is used.

Solution. (1) A cut: the goal "f is not uniformly continuous" is replaced by the displayed formula, and the proof then proves the formula. The formula is the positive form of the negation of uniform continuity, Example (Three negations from analysis) (c), read with $A=\mathbb{R}$; there the two are shown interderivable, by Proposition 4.9.3, Proposition 4.9.2 and the order fact $\neg(u<\varepsilon)$ iff $u\geq\varepsilon$. The direction used here is from the displayed formula to the negation. The bounds can be dropped because a bounded quantifier $\exists x\in\mathbb{R}$ abbreviates $\exists x\,(x\in\mathbb{R}\wedge\dots)$ (Definition 4.1.1), and over the domain $\mathbb{R}$ the conjunct $x\in\mathbb{R}$ always holds.

(2) The witness for $\exists\varepsilon>0$: a bounded $\exists$I at the term 1, with its bound $1>0$. It is written first, but as a rule it is applied last: the instance it needs, the formula after $\exists\varepsilon>0$ with 1 for ε, is what lines (3)–(6) prove.

(3) "Let $\delta>0$ be arbitrary" opens a bounded $\forall$I box flagged δ_0 with the assumption $\delta_0>0$. As in Section 4.4, the prose uses the letter δ both for the flag and for the bound variable; below, δ in the arithmetic is the flag δ_0. The terms $1/\delta$ and $1/\delta+\delta/2$ are the witnesses for $\exists x$ and $\exists y$ (two $\exists$I, applied at the end of the box). They may mention δ because δ's box is open; the clause "which depend on δ" records this. The term $1/\delta$ exists because $\delta\neq 0$, from $\delta>0$.

(4) Arithmetic: $y-x=\delta/2$, which is positive since $\delta>0$, so $|x-y|=\delta/2$; and $\delta/2<\delta$ since $\delta>0$. This is the first conjunct.

(5) Arithmetic. Since $\delta>0$, $x=1/\delta>0$ and $y=x+\delta/2>x$. Then $x^2<y^2$, so $|x^2-y^2|=y^2-x^2=(y-x)(y+x)$, with $y-x=\delta/2$ and $y+x=2/\delta+\delta/2$; multiplying out gives $1+\delta^2/4$, which is at least 1 because $\delta^2/4\geq 0$. This is the second conjunct; with (4) it gives the conjunction ($\wedge$I), and the two $\exists$I of (3) then give $\exists x\,\exists y\,(\dots)$.

(6) The closing line of the bounded ∀I box flagged δ_0 (Proposition 4.8.1). The side condition holds: the proof has no premises; the only assumption mentioning δ_0 is $\delta_0 > 0$, the assumption of the box this line closes, and no outer box is open; and δ_0 does not appear in the conclusion $\forall\delta > 0\,\exists x\,\exists y\,(\dots)$, whose δ is a bound variable different from the flag. With (2) this proves the displayed formula, and by the cut (1) uniform continuity fails.

$\delta > 0$ is used in (3) (so that $1/\delta$ exists), in (4) (twice) and in (5) (for $y > x > 0$).

Exercise. B2: The unjustified line

Here is a proof that for real numbers $x < y$, $x^3 < y^3$.

(1) Let $x < y$. (2) Then $y^3 - x^3 = (y - x)(y^2 + xy + x^2)$. (3) Since $x < y$, $y - x > 0$. (4) Also $y^2 + xy + x^2 > 0$. (5) So $y^3 - x^3 > 0$, as a product of two positive numbers. (6) Hence $x^3 < y^3$.

Every line is true. Line 1 opens the box; of the rest, all but one are instances of a routine algebra or order fact, cited or not; one line is not an instance of any fact stated so far, and it needs an argument of its own. Self-explain the proof line by line to find it, and supply that argument as its warrant.

Solution. Line 1 opens the box, with the assumption $x < y$. Line 2 is the identity $(y - x)(y^2 + xy + x^2) = y^3 - x^3$, checked by multiplying out. Line 3 cites line 1 and the order fact $u < v$ iff $v - u > 0$. Line 5 cites lines 2, 3 and 4 and the fact that a product of two positive numbers is positive. Line 6 is line 5 rearranged by the same order fact.

Line 4 is the one without a warrant: it cites nothing, and it is not an instance of any fact stated so far. It is not obvious either, since xy can be negative. A warrant: complete the square,

$$y^2 + xy + x^2 = \left(y + \frac{x}{2}\right)^2 + \frac{3x^2}{4} \geq 0,$$

with equality only if $x = 0$ and $y + x/2 = 0$, that is, only if $x = y = 0$. Line 1 rules that out, since $x < y$ gives $x \neq y$. So $y^2 + xy + x^2 > 0$. The repaired line 4 uses the hypothesis $x < y$ a second time, which the proof as printed hides; a reader who explains only "the bracket is positive" has paraphrased line 4 and missed this.

5.4.5.3 C. Going deeper

Exercise. C1: A longer proof

Self-explain the proof of Lemma 3.6.4 in Section 3.6: for each of its five numbered steps, and for each claim in the paragraph that follows them, say what it uses and why that is legitimate. Then state the idea of the proof in two sentences.

Solution (Hint 1). Keep Definition 1.4.2 open beside you: the paragraph after the five steps checks, in turn, the premises, the scope rule for the transplanted citations, the rule schemas, the last line of each box, and the availability of what the last line cites.

Solution (Hint 2). Start with step 2. Which result supplies $p \vee \neg p$, what is substituted for its letter, and why may that line cite nothing?

Solution. Step 1. The premises of the new list are the members of Δ that D or E uses. Clause (D1) of Definition 1.4.2 places premises first, above every box, and puts no restriction on which statements they are. The conclusion will be $\Delta \vdash \psi$, so only members of Δ are taken. The premises p, $\neg p$ of D and E are therefore left out, unless they also belong to Δ, which does no harm.

Step 2. The line $p \vee \neg p$ is Proposition 2.5.3 with the atom p for the letter P, used as a derived rule (Definition 2.3.1) with no premises, so it cites nothing. This is the one classical step, through LEM.

Step 3. The box opened by p contains the non-premise lines of D with their own boxes; every citation of the premise p is redirected to the assumption, and every citation of a premise in Δ to the premise line of step 1. The two degenerate cases (the last line of D is a premise) are handled separately so that the box still ends in ψ, by repetition (R, clause (D3)) where needed.

Step 4. The same, for E with the assumption $\neg p$.

Step 5. The last line ψ is $\vee$E (Definition 1.6.1), citing the disjunction of step 2 and the two boxes, each opened by one disjunct and ending in ψ.

The verification paragraph. It checks that the list is a derivation. (i) Premises are first and in no box, so they are available everywhere. (ii) Inside each new box, a line of D cites the same lines and boxes as before, renumbered, or a premise line, or the assumption, and each of those is available by the scope rule of Definition 1.4.2: the old boxes have simply been placed inside one new box, which contains both the citing line and the cited one. (iii) Each line is still an instance of its rule's schema, since its statement and the statements it cites are unchanged, as in the proof of Lemma 3.6.1. (iv) The last line of each new box is ψ, as a line of that box itself, which $\vee$E requires. (v) The disjunction and the two boxes are available at the last line, which lies in no box. So the last line is a correct $\vee$E, and $\Delta \vdash \psi$.

The idea in two sentences. The lemma is proof by cases on $p \vee \neg p$: excluded middle supplies the disjunction, and the two given derivations, with their premise p or $\neg p$ turned into the assumption of a box, are the two cases of $\vee$E. All the work is in checking that the transplanted lines still cite only available lines.

Marking scheme. (10 points) - 2: step 2 explained: Proposition 2.5.3 as a derived rule at the atom p, citing nothing, and identified as the classical step. - 2: steps 3 and 4 explained: the redirection of citations of p (or $\neg p$) to the assumption and of members of Δ to the premise lines, and the degenerate cases. - 1: step 5 named as $\vee$E with its three citations. - 3: the verification paragraph explained against Definition 1.4.2: availability by the scope rule, schemas unchanged, each box ending in ψ as its own line. - 2: an idea in two sentences that names both the case split on $p \vee \neg p$ and the transplanting of D and E into the two boxes. Fatal errors (0 for the step): explanations that restate the steps without naming what they use; calling the case split $\neg$I or RAA.

5.5 Writing a Proof

Section 5.4 trained a reader to say what licenses each line of a proof, as Example (Explaining a proof, well and badly) did line by line. This section turns to the writer. A proof is written for a reader who will check it in exactly that way, so the writer's job is to make each line's warrant visible. The section gives a procedure that leads from a statement to plain prose, the conventions of written mathematics, each a rule of Chapters 1–4 made audible, and three worked examples.

5.5.1 A procedure

A reader checks a proof forwards, one line at a time. A writer finds a proof mostly backwards, from the goal. The two directions are not in conflict: Section 2.6 showed that the search for a derivation is goal-directed, and Definition 1.4.2 says that the derivation found is a list in which each line cites only lines above it. The procedure below keeps both, the search in scratch work and the derivation on the page. Its steps 3 and 6 use the levels of Definition 5.3.1 and the convention of Definition 5.3.2 from Section 5.3.

Algorithm. **Writing a proof of a statement.**

1. **Write the goal in symbols.** Translate the statement, its hypotheses and its conclusion into a formula, as in Section 4.8, with every quantifier and its bound. Read a letter that is not introduced as universally quantified (Section 4.1).
2. **Unwind the definitions.** Replace each defined word ("bounded", "converges", "irrational", "$\subseteq$") by its definition, in the hypotheses and in the goal. This is $\leftrightarrow$E on a definition.
3. **Read the skeleton off the goal** (Section 2.6, extended in Section 4.8), and write it at level 1: the boxes, their first lines and their last lines. Open the $\exists$E boxes of existential hypotheses early.
4. **Fill the boxes:** backwards in scratch, forwards on the page. The search runs from the goal; the written proof runs from the hypotheses, each line citing lines above it. What the search found goes into the Idea, not into the Proof.
5. **Choose constants last in the search, and write them first on the page,** each with what it depends on.
6. **Write level 3,** leaving silent exactly the steps that Definition 5.3.2 allows, and no others.
7. **Reread as a checker:** for each line, is it true, and by which rule does it follow from which available lines? (Section 5.6 makes this a two-column check.)

Steps 1–3 involve no mathematics, only logical form, and they fix the first and last sentence of every box before anything is invented ("skeleton first"). Steps 4 and 5 need a remark each.

Step 4 as a logical fact. The search of Section 2.6 opens a box for the goal before it knows how the box will be filled, and it postpones an existential goal until a witness appears; that is why it works backwards. But a derivation is a list in which line k cites only lines available at k (Definition 1.4.2), and a written proof is that list in sentences. So the order of the page is forced by the rules, not by taste: "$|f(x) - f(a)| < \varepsilon$ holds if $|x - a| < \varepsilon/2$" points from a later line to an earlier one. The backward reasoning lives in the Idea block.

Step 5 as a logical fact. A chosen constant, such as "let $\delta := \varepsilon/2$", is the witness of an $\exists$I line. The search finds it last, because only the end of the search shows what it must do. On the page it comes first, right after the names it is built from, because later lines use it. Those

names are its dependencies; they must be among the names of the boxes open where it is chosen (Section 4.10).

Quick check. Which step of the procedure produces the sentence "Let $\varepsilon > 0$" at the start of a proof that $a_n \to L$, and which step keeps it at level 3?

Solution. Step 3. Unwound in step 2, the goal begins $\forall \varepsilon > 0$, so its skeleton is a bounded $\forall$I box whose assumption is $\varepsilon_0 > 0$; "Let $\varepsilon > 0$" is the first line of that box. Step 6 keeps it, because the opening of a $\forall$I box is never silent (Definition 5.3.2). The closing "since ε was arbitrary" may be silent.

5.5.2 Conventions of written mathematics

A handful of conventions make a proof checkable. Each is a rule of Chapters 1–4 heard in a sentence.

- **Announce the skeleton.** "We prove the contrapositive", "by induction on n", "we consider two cases", "suppose, for a contradiction, that ...". An announcement tells the reader which box is opening, and so what its last line must be.
- **Name every object when it is introduced,** by "let" (an arbitrary object, $\forall$I), "choose" (a witness, $\exists$E, with a fresh name), or "let ... := ..." (a constant, the witness of $\exists$I), and say what a chosen object depends on.
- **One claim per sentence.** A sentence carries one line, or a short chain of lines joined by "so", each step of which the reader can name. Two unrelated claims joined by "and" hide which one is used later.
- **End with what was shown.** The last sentence of a proof is the last line of the derivation, outside every box. If it is not the statement to be proved, the reader should be told how the two are related.
- **Label displayed claims that will be cited,** $(*)$ or (1), and cite them by the label. A citation is a line number in prose.

None of these is a matter of style. An object used without being introduced is a line citing nothing, and an announcement that does not match its box (assuming the conclusion of an implication) proves the converse; these are entries (N9) and (N5) of the catalogue in Section 5.6.

5.5.3 Matching sentences and rules

The first two examples go in opposite directions.

Example. Label every sentence

Here is a short proof. Label each sentence with the rule it uses.

> **Claim.** For a real number x, if $|x - 1| < 1$, then $x > 0$.
>
> *Proof.* (1) Assume $|x - 1| < 1$. (2) Either $x \geq 1$ or $x < 1$. (3) Suppose first that $x \geq 1$. (4) Then $x \geq 1 > 0$. (5) Suppose instead that $x < 1$. (6) Then $x - 1 < 0$, so $|x - 1| = 1 - x$. (7) Hence $1 - x < 1$, that is, $x > 0$. (8) In either case $x > 0$, which proves the claim.

Solution.

Sentence	What it does	Rule
(1)	opens a box with the hypothesis	assumption for $\to$I
(2)	states a fact about real numbers	a school fact, entered as a line
(3)	opens the first case box	assumption for $\vee$E
(4)	$x \geq 1$ and $1 > 0$ give $x > 0$	$\to$E, with the fact "$a \geq b$ and $b > c$ imply $a > c$"
(5)	opens the second case box	assumption for $\vee$E
(6)	computes $\lvert x - 1 \rvert$	$\to$E twice, with the facts "$x < 1$ implies $x - 1 < 0$" and "$a < 0$ implies $\lvert a \rvert = -a$"
(7)	uses the assumption of (1)	=E, putting (6) into (1), then $\to$E with "$1 - x < 1$ implies $x > 0$"
(8)	combines the cases, then closes the outer box	$\vee$E on (2), (3)–(4), (5)–(7); then $\to$I on (1)–(8)

Sentence (7) uses the assumption made in (1), and it may: that box is still open. The facts of school algebra enter as implications, each used by $\to$E. Sentence (8) closes two boxes, which is normal.

Measured against Definition 5.3.2, the proof leaves out nothing that must be written: the split in (2) is on the page, and the disjunction it splits is a school fact (every real number is at least 1 or below it), so it covers every case; each case box opens with its assumption; and every step it leaves unwritten is of a kind that may be silent, for instance the $\forall$E instances of the school facts at the obvious terms, the conclusion $x > 0$ of case 1, which is left to the chain $x \geq 1 > 0$ of (4), the $\wedge$I behind (4), whose conjuncts are displayed, the =E of (7), and the closing of the $\to$I box, for which "which proves the claim" stands. Labeling also shows where the work is. The case split exists only because $\lvert \cdot \rvert$ is defined by cases (Chapter 1); the school fact $-\lvert u \rvert \leq u$, used with $u = x - 1$, gives $-1 < x - 1$ in one step, and a proof written that way would have no $\vee$E at all.

The second direction starts from a derivation whose premises are facts about integers.

Example. From a derivation to a paragraph

Let n be an integer. Write E for "n is even", O for "n is odd", F for "$n + 1$ is even", and G for "$n(n + 1)$ is even". The derivation below uses four facts about integers as premises. Write the paragraph a mathematician would write for it.

1	$E \vee O$	(every integer is even or odd)
2	$E \to G$	(an even factor makes a product even)
3	$O \to F$	(odd plus one is even)
4	$F \to G$	(an even factor again)
5	$\quad E$	
6	$\quad G$	$\to$E 2, 5
7	$\quad O$	
8	$\quad F$	$\to$E 3, 7
9	$\quad G$	$\to$E 4, 8
10	G	$\vee$E 1, 5–6, 7–9

Solution.

> Every integer is even or odd, so we consider two cases. If n is even, then $n(n+1)$ has an even factor, and so it is even. If n is odd, then $n+1$ is even, since an odd number plus one is even; so again $n(n+1)$ has an even factor and is even. In either case $n(n+1)$ is even.

Compare the two. The paragraph keeps one sentence for each box and each rule that matters: line 1 becomes "we consider two cases", with its reason; the two boxes become the two "if" sentences; line 10 becomes "in either case". It drops the line numbers and the rule names, and it merges the two $\to$E steps of lines 8 and 9 into one sentence joined by "so". It never skips a box: the reader still sees both cases and sees them combined.

By Definition 5.3.2, the closing "in either case ..." could be dropped at level 3; the split of line 1, with the reason it covers every integer, could not. The paragraph leaves the premises unproved. Premises 2–4 are one line of algebra each (for instance, $n = 2k$ gives $n(n+1) = 2(k(n+1))$), and a careful write-up includes them; premise 1, that every integer is even or odd, is not, and *Sets, Relations, and Cardinality*, Chapter 2 proves it.

5.5.4 From the goal to prose

The third example runs all seven steps on a statement not seen before. Sets of reals are used here as Chapter 4 used them, with "$a \in A$" a one-place predicate and $A \subseteq B$ read as "every element of A is an element of B"; *Sets, Relations, and Cardinality*, Chapter 1 develops sets properly.

Example. A subset of a bounded set

Let A and B be sets of real numbers with $A \subseteq B$. Prove that if B is bounded above, then A is bounded above.

Solution. Steps 1 and 2: the goal in symbols, unwound. Over the domain $\mathbb{R}$, "B is bounded above" is $\exists M\ \forall b \in B\ (b \leq M)$ (Section 4.8), and $A \subseteq B$ is $\forall x \in A\ (x \in B)$. The statement is an implication, so the derivation below is the inside of a $\to$I box: its assumption is the first formula (line 2), the hypothesis is the second, and its goal is $\exists M\ \forall a \in A\ (a \leq M)$.

Step 3: the skeleton. The goal is an existential, so it is postponed: no witness is in sight yet. The assumption "B is bounded above" (line 2) is an existential, so its $\exists$E box opens at once, with a fresh name M_0 and the assumption $\forall b \in B\ (b \leq M_0)$; the box's last line is the goal.

Step 4: filling the box. Backwards: the goal needs a witness M and a proof of $\forall a \in A\ (a \leq M)$. The only bound in sight is M_0, so try it. Then the new goal is a bounded universal: a box "let $a_0 \in A$" whose last line is $a_0 \leq M_0$. That line comes from the bound on B, instantiated at a_0, which needs $a_0 \in B$; and $a_0 \in B$ comes from $A \subseteq B$ instantiated at a_0. Forwards, the same lines in the order of the list are the derivation below (with the bounded rules of Proposition 4.8.1; each bounded $\forall$E abbreviates a $\forall$E and a $\to$E, and the bounded $\forall$I box a $\forall$I box with a $\to$I box inside).

1		$\forall x \in A\ x \in B$	
2		$\exists M\ \forall b \in B\ b \leq M$	
3	M_0	$\forall b \in B\ b \leq M_0$	
4		a_0 | $a_0 \in A$	
5		$a_0 \in B$	$\forall$E 1, 4
6		$a_0 \leq M_0$	$\forall$E 3, 5
7		$\forall a \in A\ a \leq M_0$	$\forall$I 4–6
8		$\exists M\ \forall a \in A\ a \leq M$	$\exists$I 7
9		$\exists M\ \forall a \in A\ a \leq M$	$\exists$E 2, 3–8

The side conditions hold: a_0 appears in neither line 1 nor line 2, not in line 7 and not in the open assumption on line 3; M_0 appears in none of lines 1, 2 and 9.

Step 5: the constant. The witness for the goal is the bound M_0. The search could choose it only once the $\exists$E box supplied M_0, and confirm it only at line 6; the page writes it first, in the sentence that chooses it. It depends on B only: its box is opened before the box of a_0, so it cannot depend on a_0, and it must not, because a single bound has to serve every element of A.

Step 6: level 3.

> Suppose B is bounded above, and choose an upper bound M of B (depending on B only). Let $a \in A$. Since $A \subseteq B$, we have $a \in B$, and so $a \leq M$. Hence M is an upper bound of A, and A is bounded above.

The two instances of $\forall$E (lines 5 and 6) are at the obvious term a, and each appears only as the citation that licenses it ("since $A \subseteq B$", "so", which cites the choice of M). The closing of the $\forall$I box, "since a was arbitrary", is silent. Two steps that are never silent are on the page: the $\exists$E with its fresh name ("choose an upper bound M"), and the $\exists$I with its witness and dependence ("M is an upper bound of A"; "depending on B only"). "Suppose B is bounded above" opens the $\to$I box; its closing is silent.

Step 7: reread. The order of the first two sentences matters. With "let $a \in A$" first and M chosen inside, the ∃I would fall inside the box of a_0, giving only $\forall a \in A\ \exists M\ (a \le M)$, which says nothing.

The example used $A \subseteq B$ once, at line 5, and "B is bounded above" once, at line 9, by ∃E, whose box assumption is used at line 6. The two warnings below concern steps 4 and 5.

Warning. **The search is not the proof.** A page that reads "we want $x^2 < 4$; this holds if $x < 2$; so let $x < 2$" has copied the search, and each "holds if" points backwards. A backward step can be false, and the page hides it: "$x < 2$ implies $x^2 < 4$" fails at $x = -3$. In scratch work every backward step must be checked as an implication pointing forwards; on the page only the forward chain appears, each line citing lines above it.

Warning. **A constant without its dependencies is a claim nobody can check.** In a proof that x^2 is uniformly continuous on $\mathbb{R}$, the sentence "let $\delta := \min\{1, \varepsilon/(2|x|+1)\}$" looks innocent. Asked "what does δ depend on?", the formula answers: on ε **and on** x, so the δ was chosen inside the box of x, and the proof shows only continuity at every point. Uniform continuity is false here (Example (The square function on the real line)). Writing "(depending on ε only)" after the constant is what lets a reader catch the error.

5.5.5 Exercises

5.5.5.1 A. Check your understanding

Exercise. A1

Here is a proof.

> **Claim.** For real numbers x and y, if $x + y > 2$, then $x > 1$ or $y > 1$.
>
> (1) Assume $x + y > 2$. (2) Suppose, for a contradiction, that it is not the case that $x > 1$ or $y > 1$. (3) Then $x > 1$ fails and $y > 1$ fails. (4) So $x \le 1$ and $y \le 1$. (5) Adding, $x + y \le 2$. (6) This contradicts $x + y > 2$. (7) Hence $x > 1$ or $y > 1$, which proves the claim.

(a) Label each sentence with the rule it uses, as in Example (Label every sentence).

(b) Which sentence closes an RAA box, and which uses one of De Morgan's laws (Proposition 2.5.5)? Which law?

(c) Everything inside the box of (2) follows from (2) alone, until the clash in (6). Which face of "by contradiction" (Section 5.1) does the proof wear, and which face is it really? Write the first sentence of the rewritten proof.

(d) Sentence (7) closes two boxes. Which of the two closings may be silent at level 3, by Definition 5.3.2?

Solution. (a)

Sentence	Rule
(1)	assumption, opening the $\to$I box
(2)	assumption $\neg(x > 1 \lor y > 1)$, opening an RAA box
(3)	Proposition 2.5.5 (a), $\neg(P \lor Q) \vdash \neg P \land \neg Q$, applied to (2)
(4)	$\land$E on (3), twice; $\to$E with the order fact "$\neg(a > 1)$ implies $a \leq 1$", twice; then $\land$I
(5)	$\land$E on (4), then $\to$E with "$a \leq 1$ and $b \leq 1$ imply $a + b \leq 2$"
(6)	$\to$E with the order fact "$a \leq 2$ implies $\neg(a > 2)$" on (5), then $\neg$E on (1) and the result
(7)	RAA on the box (2)–(6); then $\to$I on (1)–(7)

(b) Sentence (7) closes the RAA box: the box opened with the negation of the goal $x > 1 \lor y > 1$ and closes with the goal itself. Sentence (3) uses De Morgan's law (a), a direction derived without RAA.

(c) It wears face (iii), genuine RAA, but it is a contrapositive in costume, as in Example (A contrapositive in costume): the hypothesis (1) is used only at the clash, and everything else derives $x + y \leq 2$ from $x \leq 1$ and $y \leq 1$. The rewrite opens: "We prove the contrapositive: if $x \leq 1$ and $y \leq 1$, then $x + y \leq 2$." The rewrite is clearer, but it does not remove the classical step: passing from the contrapositive back to the claim is Proposition 2.5.2, which uses RAA.

(d) The closing of the $\to$I box ("so if $x + y > 2$, then ...") may be silent. The conclusion of the RAA box, "hence $x > 1$ or $y > 1$", may not.

5.5.5.2 B. Practice

Exercise. B1: From skeleton to prose

Each part gives a statement and a level-1 skeleton of its proof. Write the proof at level 3, keeping every step that Definition 5.3.2 says is never silent.

(a) For every integer n, if n is odd, then $3n + 5$ is even. (Domain $\mathbb{Z}$.)

1	n_0	$\exists k\,(n_0 = 2 \cdot k + 1)$	
2		k_0 | $n_0 = 2 \cdot k_0 + 1$	
3		$3 \cdot n_0 + 5 = 2 \cdot (3 \cdot k_0 + 4)$	arith 2
4		$\exists k\,(3 \cdot n_0 + 5 = 2 \cdot k)$	$\exists$I 3
5		$\exists k\,(3 \cdot n_0 + 5 = 2 \cdot k)$	$\exists$E 1, 2–4
6		$\exists k\,(n_0 = 2 \cdot k + 1) \to \exists k\,(3 \cdot n_0 + 5 = 2 \cdot k)$	$\to$I 1–5
7		$\forall n\,(\exists k\,(n = 2 \cdot k + 1) \to \exists k\,(3 \cdot n + 5 = 2 \cdot k))$	$\forall$I 1–6

(b) No real number x satisfies $x^2 + 1 = 2x - 1$. (Domain $\mathbb{R}$.)

1		$\exists x\,(x^2+1=2\cdot x-1)$	
2		x_0 | $x_0^2+1=2\cdot x_0-1$	
3		$(x_0-1)^2+1=0$	arith 2
4		$(x_0-1)^2\geq 0$	arith
5		$1\leq 0$	arith 3, 4
6		$\neg(1\leq 0)$	arith
7		$\bot$	$\neg$E 5, 6
8		$\bot$	$\exists$E 1, 2–7
9		$\neg\exists x\,(x^2+1=2\cdot x-1)$	$\neg$I 1–8

(c) For a real number x: $x^3=x$ if and only if $x=-1$, $x=0$ or $x=1$. (Here x is fixed; "or" chains group to the right.)

1		$x^3=x$	
2		$(x+1)\cdot(x\cdot(x-1))=0$	arith 1
3		$x=-1\vee(x=0\vee x=1)$	product zero, arith 2
4		$x=-1\vee(x=0\vee x=1)$	
5		$x=-1$	
6		$x^3=x$	arith 5
7		$x=0\vee x=1$	
8		$x=0$	
9		$x^3=x$	arith 8
10		$x=1$	
11		$x^3=x$	arith 10
12		$x^3=x$	$\vee$E 7, 8–9, 10–11
13		$x^3=x$	$\vee$E 4, 5–6, 7–12
14		$x^3=x\leftrightarrow(x=-1\vee(x=0\vee x=1))$	$\leftrightarrow$I 1–3, 4–13

Hint: line 3 of (c) uses the school fact that a product of real numbers is zero only if one of its factors is zero.

Solution. (a) Let n be an integer, and suppose n is odd. Choose an integer k with $n=2k+1$. Then

$$3n+5=6k+3+5=2(3k+4),$$

and $3k+4$ is an integer, so $3n+5$ is even.

"Choose an integer k" is the $\exists$E box, with a fresh name; the witness $3k+4$ of line 4 is on the page, and it depends on k, hence on n. Lines 5, 6 and 7 close boxes and are silent.

(b) Suppose, to the contrary, that some real x satisfies $x^2+1=2x-1$, and fix such an x.

Rearranging, $x^2 - 2x + 2 = 0$, that is, $(x-1)^2 + 1 = 0$. But $(x-1)^2 \geq 0$, since a square is non-negative, so $1 \leq 0$, which is false. Hence no real x satisfies the equation.

"Suppose, to the contrary" opens the ¬I box of line 1 (the goal is a negation, so no classical rule is used), and "fix such an x" the ∃E box. The clash of lines 5–7 is named, and the conclusion of the ¬I box is written; line 8 is silent.

(c) Let x be real.

($\Rightarrow$) Suppose $x^3 = x$. Then $(x+1)\,x\,(x-1) = x^3 - x = 0$. A product of real numbers is zero only if a factor is zero, so $x + 1 = 0$, $x = 0$ or $x - 1 = 0$; that is, $x = -1$, $x = 0$ or $x = 1$.

($\Leftarrow$) Suppose $x = -1$, $x = 0$ or $x = 1$. In the three cases, $x^3 = -1 = x$, $x^3 = 0 = x$ and $x^3 = 1 = x$.

This proves the equivalence.

Both directions are written, and the citation in line 3 is visible. The two splits of ($\Leftarrow$) are merged into "in the three cases"; the disjunction split is the hypothesis itself, and the closing lines 12 and 13 are silent.

Exercise. B2: Put the proof in order

Let (a_n) and (b_n) be sequences of real numbers with $a_n \to L$ and $b_n \to M$, where "$a_n \to L$" is the pattern of Section 4.8: for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for every $n \geq N$. Eight of the ten sentences below, in a suitable order, form a level-3 proof that $a_n + b_n \to L + M$; two are distractors. Write down an order for the eight, say which other orders are also admissible, and say why each distractor is wrong.

(i) Let $n \in \mathbb{N}$ with $n \geq N$.

(ii) Since $b_n \to M$ and $\varepsilon/2 > 0$, choose $N_2 \in \mathbb{N}$ (depending on ε) such that $|b_n - M| < \varepsilon/2$ for every $n \geq N_2$.

(iii) Let $N := \max\{N_1, N_2\}$, where N_1 and N_2 serve every ε at once.

(iv) Then $n \geq N_1$ and $n \geq N_2$.

(v) Let $\varepsilon > 0$.

(vi) Since $n \geq N$ and $\varepsilon > 0$ were arbitrary, $a_n + b_n \to L + M$.

(vii) Choose $N_1 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for every $n \geq N_1$.

(viii) Let $N := \max\{N_1, N_2\}$.

(ix) So $|(a_n + b_n) - (L + M)| \leq |a_n - L| + |b_n - M| < \varepsilon/2 + \varepsilon/2 = \varepsilon$.

(x) Since $a_n \to L$ and $\varepsilon/2 > 0$, choose $N_1 \in \mathbb{N}$ (depending on ε) such that $|a_n - L| < \varepsilon/2$ for every $n \geq N_1$.

Solution. An admissible order: (v), (x), (ii), (viii), (i), (iv), (ix), (vi).

The only other admissible order swaps (x) and (ii), which use only ε and one hypothesis each. Nothing else moves: (viii) uses N_1 and N_2; (i) uses N; (iv) uses $n \geq N$ and the definition of N; (ix) uses (iv), (x) and (ii) at this n, and the triangle inequality (Example (The triangle inequality by cases)); (vi) closes the boxes of n and ε, with the ∃I of the witness N inside "$a_n + b_n \to L + M$". The maximum exists because there are finitely many numbers.

Distractor (iii) claims that N_1 and N_2 serve every ε at once. They were chosen from $\forall\varepsilon\,\exists N$, inside the box of ε, so they depend on ε and serve only the ε they were chosen for; a uniform N would need $\exists N\,\forall\varepsilon$, which the hypothesis does not give. This is the error that Section 5.6 catalogues as (N2). For $a_n = 1/(n+1)$, $L = 0$, no single N_1 serves every $\varepsilon > 0$: at $\varepsilon = 1/(N_1+2)$ the term $a_{N_1} = 1/(N_1+1)$ is too far from 0.

Distractor (vii) applies the hypothesis at ε instead of $\varepsilon/2$. It is a legal step, but with it in place of (x), the bound in (ix) becomes $|(a_n+b_n)-(L+M)| < \varepsilon + \varepsilon/2$, not $< \varepsilon$, so (ix) would no longer follow. It is an ε of the wrong kind for the goal.

Exercise. B3: The last step

Here is a level-1 skeleton for the statement "for real numbers x and y, if x is rational and y is irrational, then $x+y$ is irrational", with x and y fixed:

1	$x \in \mathbb{Q} \wedge \neg(y \in \mathbb{Q})$	
2	$x \in \mathbb{Q}$	$\wedge$E 1
3	$\neg(y \in \mathbb{Q})$	$\wedge$E 1
4	$\quad x+y \in \mathbb{Q}$	
5	$\quad (x+y)-x \in \mathbb{Q}$	difference of rationals 4, 2
6	$\quad y \in \mathbb{Q}$	arith 5
7	$\quad \bot$	$\neg$E 6, 3
8	$\neg(x+y \in \mathbb{Q})$	$\neg$I 4–7
9	$x \in \mathbb{Q} \wedge \neg(y \in \mathbb{Q}) \to \neg(x+y \in \mathbb{Q})$	$\to$I 1–8

and its level-3 prose, with the closing sentence of the ¬I box left blank:

> Let x be rational and y irrational. Suppose $x+y$ is rational. Then $y = (x+y)-x$ is a difference of two rational numbers, so y is rational. This contradicts the irrationality of y. _____

Name the move, then complete it.

Solution. The move is skeleton first: the blank is line 8, the line written after the ¬I box of lines 4–7 closes, a line the skeleton fixes before any arithmetic. That box assumed "$x+y$ is rational" and reached $\bot$ at line 7, so line 8 concludes its negation. It is the first face of "by contradiction" (Section 5.1), proof of a negative statement, and uses no classical rule: "irrational" means "not rational", so the goal of the outer box, line 8, is a negation, and the inner box is opened on $x+y \in \mathbb{Q}$ with goal $\bot$.

Completion. "Hence $x+y$ is not rational, that is, $x+y$ is irrational."

The conclusion of a ¬I box is never silent (Definition 5.3.2), so this sentence must be written even though the reader can guess it. Line 9, the closing of the →I box, may stay silent. The sentence before the blank, "this contradicts the irrationality of y", is line 7, the clash of ¬E, which names both lines it cites.

5.5.5.3 C. Going deeper

Exercise. C1: A constant found backwards

Let $f(x) = 2x + 3$ and let a be a fixed real number. Prove that for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ for every real x with $|x - a| < \delta$. Write your answer as an Idea, which finds δ backwards, followed by a Proof, which runs forwards. Say what δ depends on.

Solution (Hint 1). Compute $|f(x) - f(a)|$ in terms of $|x - a|$ before choosing anything.

Solution (Hint 2). The Proof begins: "Let $\varepsilon > 0$, and let $\delta \coloneqq \dots$".

Solution. Idea. We want $|f(x) - f(a)| < \varepsilon$. Compute: $f(x) - f(a) = (2x + 3) - (2a + 3) = 2(x - a)$, so $|f(x) - f(a)| = 2|x - a|$. This is below ε exactly when $|x - a| < \varepsilon/2$. So $\delta = \varepsilon/2$ should work, and it does not involve x or a.

Proof. Let $\varepsilon > 0$, and let $\delta \coloneqq \varepsilon/2$, which is positive and depends on ε only. Let x be a real number with $|x - a| < \delta$. Then

$$|f(x) - f(a)| = |2(x - a)| = 2|x - a| < 2\delta = \varepsilon.$$

Since x was arbitrary, δ works for ε; since $\varepsilon > 0$ was arbitrary, this proves the claim.

The Idea reads the computation from the goal back to $|x - a|$; the Proof places δ right after ε, before any x appears, and justifies it by the final display. Because δ does not depend on a either, the same proof shows that one δ serves every point: f is uniformly continuous on $\mathbb{R}$, in the sense of Section 4.10.

Marking scheme. (10 points) - 2: an Idea that computes $|f(x) - f(a)| = 2|x - a|$ and reads off δ from the goal. - 2: the Proof opens with $\varepsilon > 0$ arbitrary and introduces δ before any x, with $\delta > 0$ checked. - 2: says what δ depends on (ε only, or ε and a if the choice involves a). - 3: for arbitrary x with $|x - a| < \delta$, the forward chain to $|f(x) - f(a)| < \varepsilon$, each step justified. - 1: a closing sentence with the quantifiers in the order ε, δ, x. Any δ with $0 < \delta \leq \varepsilon/2$ earns full marks. Fatal errors (0 for the step): δ depending on x; a Proof that runs backwards ("this holds if ...").

Exercise. C2: Rewrite it properly

A sequence (a_n) is **bounded** if there is $M \in \mathbb{R}$ with $|a_n| \leq M$ for every $n \in \mathbb{N}$ (Section 4.6). The following proof that a sequence (a_n) with $a_n \to L$ is bounded is correct in substance but badly written.

> We want some M with $|a_n| \leq M$. For big n, a_n is near L, so $|a_n|$ is less than $|L| + 1$ if we use 1 for ε. The small n are only finitely many, so take the biggest. So M is the max of all of them and it works.

(a) Name three faults of the text, each by the convention it breaks (this section) or the step that is not written.

(b) Rewrite the proof at level 3.

Solution (Hint 1). Which objects does the text use without introducing them, and what does each depend on?

Solution (Hint 2). The rewrite begins: "Apply the definition of $a_n \to L$ with $\varepsilon = 1$, and choose $N \in \mathbb{N}$ such that ...".

Solution. *(a)* (i) It writes the search: "we want some M" and "if we use 1 for ε" describe what is being looked for, not lines that follow. (ii) It uses objects without introducing them: "big n" hides an N, obtained from the hypothesis at $\varepsilon = 1$ by ∀E and ∃E, which is never named; "the biggest" does not say of which numbers. (iii) It never says what M is or what it depends on: M is built from the finitely many terms before N, so it depends on N. (Also acceptable: the split into small and big n has no stated reason; the last sentence does not say what was shown.)

(b) Apply the definition of $a_n \to L$ with $\varepsilon = 1$, and choose $N \in \mathbb{N}$ such that $|a_n - L| < 1$ for every $n \geq N$. Let

$$M := \max\{|a_0|, |a_1|, \ldots, |a_{N-1}|, |L| + 1\},$$

the largest of finitely many numbers, so it exists; it depends on N (when $N = 0$ the list is just $|L| + 1$). Let $n \in \mathbb{N}$. Either $n < N$ or $n \geq N$, since every natural number is one or the other. If $n < N$, then $|a_n|$ is in the list, so $|a_n| \leq M$. If $n \geq N$, then by the triangle inequality (Example (The triangle inequality by cases))

$$|a_n| = |(a_n - L) + L| \leq |a_n - L| + |L| < 1 + |L| \leq M.$$

In either case $|a_n| \leq M$. Hence (a_n) is bounded.

The ∀E instance $\varepsilon = 1$ is named, because it is not the obvious one; the witness M of the final ∃I is defined before it is used, with its dependence; the split on $n < N$ or $n \geq N$ carries its reason (every natural number is one or the other).

Marking scheme. (10 points) - 3: (a) three genuine faults, each tied to a convention or an unwritten step (1 each). - 2: (b) applies the hypothesis at a named positive ε and chooses N from it. - 2: defines M before using it, with its dependence on N, bounding the early terms and $|L| + \varepsilon$ (a maximum or a sum). - 2: the two cases $n < N$ and $n \geq N$, the second by the triangle inequality (cited) or an equivalent estimate from $|a_n - L| < \varepsilon$, such as $L - \varepsilon < a_n < L + \varepsilon$ and a split on the sign of a_n. - 1: ends with what was shown. Fatal errors (0 for the step): M chosen before N or as a maximum or sum of infinitely many numbers; the early terms omitted.

5.6 Checking Proofs, and What Is Not Allowed

A reader working alone must judge proofs, including their own, and is the person least able to do it. The remedy is procedural. This section gives two questions to ask of every line and a catalogue of the errors that recur, each tied to the rule of Chapters 1–4 that it breaks, and then checks four short proofs line by line.

5.6.1 Two questions for every line

A proof is correct when every line is licensed: by Definition 1.4.2, each line is a premise (a hypothesis, a proved result or a true school fact), an assumption, or one rule applied to available lines. So checking a proof means asking two questions of every line, with the answers in two columns.

- **Is it true?** Is the line correct whenever the hypotheses and the assumptions of the open boxes are true?
- **Does it follow, and by which rule?** From which available lines, and by which rule?

The first column finds where to look; the second decides.

The two questions are different. A line can be **true and not follow**: a correct fact the proof has not earned. This is the dangerous case, because nothing about the line looks wrong. A line can also **follow and be false**: false in a situation where the hypotheses and open assumptions are true. Since the rules never lead from true lines to a false one (Theorem 3.5.2; Section 4.11 for predicate logic), some line it rests on, neither a hypothesis nor an open assumption, is false there too; follow it back until the second column fails. (Proof C's "1 is even" is true given its box's assumption.)

Quick check. A proof that $n^2 \geq n$ for every integer n opens with the line "$n^2 - n = n(n-1) \geq 0$". Is the line true? Does it follow? Would the answers change if n ranged over the reals?

Solution. For an integer n the line is true: no integer lies strictly between 0 and 1, so $n \leq 0$ (both factors ≤ 0) or $n \geq 1$ (both ≥ 0). But as written it follows from nothing: it is the goal rearranged, without the case split that makes it true. For real n it is false at $n = 1/2$, where $n(n-1) = -1/4$. The first column depends on the situation; the second fails in both.

5.6.2 The catalogue

Most wrong proofs fail in one of a few ways, each breaking a particular rule.

Example. What is not allowed

Each entry names the error, the rule it breaks and where that rule lives, and a failing instance whose conclusion is marked **true** or **false**: a wrong argument can reach either.

(N1) **Verifying instances and concluding "for all".** Breaks ∀I, which needs one proof

about a fresh name (Definition 4.4.1, Section 4.4). Instance: $n^2 - n + 11$ is prime for $n = 0, 1, \ldots, 10$ (the values are $11, 11, 13, 17, 23, 31, 41, 53, 67, 83, 101$), "so it is prime for every $n \in \mathbb{N}$". **False:** at $n = 11$ the value is $121 = 11^2$.

(N2) **A witness that depends on a variable introduced later, while uniformity is claimed.** Breaks the side conditions of ∃E and ∀I: a witness δ_0 obtained inside the box of x_0 may not leave it (Section 4.10, Example (The search from C to UC)). Instance: "x^2 is continuous at every x, with $\delta = \min\{1, \varepsilon/(2|x| + 1)\}$; so one δ for each ε serves every x, and x^2 is uniformly continuous on $\mathbb{R}$". **False** (Example (The square function on the real line)).

(N3) **Treating the name from ∃E as arbitrary.** Breaks the side condition of ∀I: the witness appears in the open assumption of its box (Definition 4.6.1, Section 4.6). Instance: "Some element of A is positive; let x be such an element. Since x was arbitrary, every element of A is positive." **False** for $A = \{-1, 1\}$.

(N4) **Reusing one witness name for two ∃E's.** Breaks the fresh-name condition: the second name may not appear in an open assumption, and the first box's assumption mentions it (Section 4.6, Example (The sum of two even integers is even)). Sibling boxes may share a name. Instance: "m and n are even, so $m = 2k$ and $n = 2k$ for an integer k; hence $m = n$". **False:** $m = 2$, $n = 4$.

(N5) **Proving the converse.** Breaks →I, whose box opens with the hypothesis, not the conclusion (Definition 1.4.1, Section 1.4). Instance: "If $x^2 > 1$, then $x > 1$: suppose $x > 1$; then $x^2 > x > 1$." **False:** $x = -2$. Argument B of Section 1.1 makes the same move with a **true** conclusion.

(N6) **Affirming the consequent; denying the antecedent.** Breaks Definition 1.4.2 (Section 1.4), being no rule: $P \to Q$ with Q gives no P, and with $\neg P$ gives no $\neg Q$. Each pattern has a refuting row, so no derivation exists (Corollary 3.5.3, Section 3.5): Example (Affirming the consequent is not valid), and for denying the row P false, Q true. Instances: "a multiple of 4 is even and 6 is even, so 6 is a multiple of 4"; "... and 6 is not a multiple of 4, so 6 is not even". Both **false**.

(N7) **A missing case.** Breaks ∨E, which needs one box for each disjunct (Definition 1.6.1, Section 1.6). Instance: "$|x| > 0$ for every real x: every real x has $x > 0$, $x = 0$ or $x < 0$; if $x > 0$, $|x| = x > 0$; if $x < 0$, $|x| = -x > 0$". **False:** the case $x = 0$ was dropped, and $|0| = 0$.

(N8) **Cases that do not exhaust.** Breaks ∨E, whose first input must be a disjunction already derived (Section 1.6). Instance: "$x^2 \geq x$ for every real x: either $x \geq 1$ or $x \leq 0$; in the first case $x^2 = x \cdot x \geq x$, in the second $x^2 \geq 0 \geq x$". **False:** $x = 1/2$ satisfies neither case, and $1/4 < 1/2$.

(N9) **Citing a line that is not available:** the goal itself (circular reasoning), a result proved later, or a general "fact" neither proved nor a true school fact (a false "fact" never qualifies). Breaks Definition 1.4.2: every line is a premise, an assumption, or a rule applied to available lines (Section 1.4). Instances: "if n^2 is even, then n is even: suppose n^2 is even; since n is even, ..." (**true**, and circular); "$x < 1$, so $x^2 < 1$, since squaring preserves $<$" (**false:** $x = -2$).

(N10) **Using an assumption after its box has closed.** Breaks the scope rule of Definition 1.4.2 (Section 1.4). Instance: "Suppose n is even, say $n = 2k$; then $n^2 = 4k^2$. Hence if n is even, $4 \mid n^2$. Now $n^2 = 4k^2$, so $4 \mid n^2$ for every integer n." **False:** $n = 1$. Proof D below makes the same move with a **true** conclusion.

(N11) **A wrong negation.** An RAA box opened with a statement that is not the negation of the goal; a bound negated instead of a quantifier (Proposition 4.9.3); the negation of $\varphi \to \psi$ written as an implication, $\neg\varphi \to \neg\psi$ or $\varphi \to \neg\psi$, instead of $\varphi \wedge \neg\psi$ (Proposi-

tion 4.9.2). Breaks RAA, which closes only a box opened with $\neg\varphi$ (Section 2.5), or the laws of Section 4.9. Instances: "if $x > 2$, then $x^2 > 4$" is "refuted" because "if $x \leq 2$, then $x^2 \leq 4$" holds at $x = 0$ (**false**: both are true at $x = 0$, so one is not the negation of the other); "$a_n = 0$ does not converge to 0, since some $\varepsilon \leq 0$ has $|a_n - 0| \geq \varepsilon$ for all n" (**false**).

An entry diagnoses a step, not the conclusion: a **false** instance shows the step leading from true premises to a false conclusion, and a **true** one that a correct conclusion does not excuse it. From *Sets, Relations, and Cardinality*, Chapter 1 on, a warning about one of these errors cites the entry by number, as in "(N3) of Example (What is not allowed)". Some entries name errors met before: (N1) is the first error of Section 4.4's list, and two parts of (N11) are warnings of Section 4.9.

Quick check. Which entry does the sentence "choose N larger than every N_ε, one for each $\varepsilon > 0$" commit, and when is the same move **not** an error?

Solution. (N2). Each N_ε comes from $\forall\varepsilon\,\exists N$, inside the box of its own ε; one N above all of them would be uniform, as if the hypothesis were $\exists N\,\forall\varepsilon$, and usually none exists, since the N_ε can grow without bound as ε shrinks. The move is legitimate when ε ranges over **finitely many** values $\varepsilon_1, \ldots, \varepsilon_k$: then $\max\{N_{\varepsilon_1}, \ldots, N_{\varepsilon_k}\}$ exists, because a finite set of natural numbers has a maximum.

Warning. **A true conclusion does not make a proof correct.** "I know the result is true" answers the first column for the last line and nothing else. Argument B of Section 1.1 has a true conclusion and only true sentences, and it is wrong (Proof B below).

5.6.3 Four proofs, checked

Here are four short proofs as a student might write them; two are correct and two are not.

Example. Proof A

Claim. For a real number x, if $x^2 - 3x + 2 < 0$, then $1 < x < 2$.

(1) Assume $x^2 - 3x + 2 < 0$. (2) Then $(x - 1)(x - 2) < 0$. (3) A product of two real numbers is negative only when one factor is positive and the other negative, so either $x - 1 > 0$ and $x - 2 < 0$, or $x - 1 < 0$ and $x - 2 > 0$. (4) In the first case, $1 < x < 2$. (5) In the second case, $x < 1$ and $x > 2$, which is impossible. (6) Hence $1 < x < 2$.

Solution.

Line	True?	Follows? By which rule?
(1)	assumption	opens the $\to$I box

(2)	yes: $(x-1)(x-2) = x^2 - 3x + 2$	=E with the algebraic identity
(3)	yes	→E with the stated fact about signs; a disjunction of two conjunctions
(4)	yes, in this case	∨E, case 1: ∧E gives $x > 1$ and $x < 2$
(5)	the case is impossible	∨E, case 2: $x > 2$ and $x < 1$ give $2 < 1$, which clashes with $\neg(2 < 1)$ (¬E); the case reaches $1 < x < 2$ by ⊥E
(6)	yes	∨E on (3), (4), (5); then →I on (1)–(6)

Verdict: correct. The disjunction of line (3) is derived from a stated fact, so the cases exhaust and (N8) does not apply. Line (5) is the one to look at twice: "which is impossible" is ⊥E, which lets a case that cannot occur reach the goal, so that ∨E can combine the cases. By Definition 5.3.2 it could be silent, since the clash is stated.

Example. Proof B

Claim. For an integer n, if n^2 is even, then n is even.

(1) Suppose n is even. (2) Then $n = 2k$ for some integer k. (3) So $n^2 = 4k^2 = 2(2k^2)$. (4) Hence n^2 is even. (5) Therefore, if n^2 is even, then n is even.

Solution.

Line	True?	Follows? By which rule?
(1)	assumption	opens a box, with "n is even"
(2)	yes, given (1)	the definition of "even", then ∃E with a fresh k
(3)	yes	algebra
(4)	yes, given (1)	∃I with the witness $2k^2$: the definition of "even"
(5)	the statement is true, but ...	**does not follow**: →I on the box (1)–(4) gives "if n is even, then n^2 is even", not (5)

Verdict: wrong; line (5) fails; entry (N5). Every line is true, and lines (1)–(4) prove the converse. $\to$I turns a box from P to Q into $P \to Q$, with the box's assumption on the **left**; writing the claim instead is the step from $P \to Q$ to $Q \to P$. That step has a refuting row, P false and Q true, so no derivation makes it (Corollary 3.5.3); and it fails in practice: "if $x > 1$, then $x^2 > 1$" is true, its converse false at $x = -2$. This is Argument B of Section 1.1, diagnosed in Section 1.4. The claim is true: Theorem 2.1.3 proves it in a box that opens with "n^2 is even".

Example. Proof C

Claim. For an integer n, the number $n^2 - 2$ is not divisible by 4.

(1) Suppose $n^2 - 2$ is divisible by 4, say $n^2 - 2 = 4k$ with k an integer. (2) Then $n^2 = 4k + 2 = 2(2k + 1)$, so n^2 is even. (3) By Theorem 2.1.3, n is even, say $n = 2m$ with m an integer. (4) Then $n^2 = 4m^2$, so $4m^2 = 4k + 2$, and hence $1 = 2(m^2 - k)$. (5) So 1 is even. (6) But 1 is odd, and no integer is both even and odd; this is a contradiction. (7) Hence $n^2 - 2$ is not divisible by 4.

Solution.

Line	True?	Follows? By which rule?
(1)	assumption	opens the ¬I box, with "4 divides $n^2 - 2$"; then ∃E with a fresh k
(2)	yes, given (1)	algebra, then ∃I with witness $2k + 1$
(3)	yes, given (2)	$\to$E with Theorem 2.1.3 and (2); then ∃E with a fresh m
(4)	yes, given (1) and (3)	algebra: subtract $4k$, divide by 2
(5)	yes, given (4)	∃I with witness $m^2 - k$, an integer
(6)	yes	∧I of (5) and "1 is odd", then ¬E against "no integer is both even and odd"
(7)	yes	¬I on the box (1)–(6)

Verdict: correct. The goal is a negation, and the ¬I box opens with the positive statement: a proof "by contradiction" of face (i) of Section 5.1, with no RAA. The witness names k and m differ, as (N4) requires.

Example. Proof D

Claim. For a real number x: if $x > 1$, then $x^2 > x$; and $x^2 + 1 > x$.

(1) Suppose $x > 1$. (2) Multiplying both sides by x, which is positive, gives

$x^2 > x$. (3) Hence if $x > 1$, then $x^2 > x$. (4) For the second part, $x^2+1 > x^2$, and $x^2 > x$ by (2), so $x^2 + 1 > x$.

Solution.

Line	True?	Follows? By which rule?
(1)	assumption	opens the →I box for the first conjunct
(2)	yes, given (1)	→E with "multiplying by a positive number keeps >"; $x > 0$ since $x > 1 > 0$
(3)	yes	→I on the box (1)–(2)
(4)	"$x^2 > x$" is **false** for $x = 1/2$	**does not follow**: it cites (2), which lies inside the box closed at (3)

Verdict: wrong; line (4) fails; entry (N10). The claim is a conjunction, so the skeleton is ∧I with two separate proofs. The second cites line (2), derived inside the box that assumed $x > 1$ and closed at line (3), so line (2) is not available. The first column confirms it: at $x = 1/2$, $x^2 = 1/4 < 1/2 = x$, a counterexample to the illegal step, though not to the claim.

The claim is true. For every real x,

$$x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0,$$

since a square is non-negative; so $x^2 + 1 > x$, with nothing from the closed box.

In Proofs B and D every sentence sounds right, and in Proof B every sentence is true. The errors are in the connections, which the rules govern: "which rule, from which available lines?" catches both.

5.6.4 Exercises

5.6.4.1 A. Check your understanding

Exercise. A1

Each line below comes from a proof. Name the catalogue entry of Example (What is not allowed) that it commits, and the rule it breaks.

(a) "We checked $3^n > n^2$ for $n = 0, 1, 2, 3, 4$, so it holds for every $n \in \mathbb{N}$."

(b) "By hypothesis some $x \in A$ has $f(x) = 0$; fix such an x. Since x was arbitrary, $f(x) = 0$ for every $x \in A$."

(c) "To prove that $f(x_1) = f(x_2)$ implies $x_1 = x_2$, suppose $x_1 = x_2$; then $f(x_1) = f(x_2)$."

(d) "To prove $x \geq 0$, suppose, for a contradiction, that $x \leq 0$."

(e) "If n is divisible by 6, then n is even. Since n is even, n is divisible by 6."

(f) "Every real x satisfies $x > 0$ or $x < 0$; we consider the two cases."

Solution. (a) (N1): ∀I needs one proof about a fresh name, not five instances.

(b) (N3): x was chosen by ∃E and appears in the open assumption $f(x) = 0$, so ∀I on it breaks the side condition. For $A = \{0, 1\}$ and $f(x) = x$ the conclusion fails.

(c) (N5): the →I box opens with the conclusion, so it proves the converse, which every function satisfies.

(d) (N11): the negation of $x \geq 0$ is $x < 0$, not $x \leq 0$, so RAA cannot close this box with $x \geq 0$. Closed by ¬I it proves $x > 0$, more than the goal, so it closes only if the hypotheses exclude $x = 0$.

(e) (N6), affirming the consequent. No rule licenses it, so it breaks Definition 1.4.2: $P \to Q$ with Q gives no P. Here $n = 4$ is even and not divisible by 6.

(f) (N8): the disjunction is not derived, and it is false at $x = 0$, so ∨E has no valid first input.

5.6.4.2 B. Practice

Exercise. B1: Two proofs in two columns

Check each proof in two columns. For a wrong proof, name the first line that fails and its catalogue entry, and say whether the claim is true.

(i) **Claim.** For integers m and n, if m and n are odd, then $m + n$ is even. (1) Assume m and n are odd. (2) Choose integers k and l with $m = 2k + 1$ and $n = 2l + 1$. (3) Then $m + n = 2k + 2l + 2 = 2(k + l + 1)$. (4) Since $k + l + 1$ is an integer, $m + n$ is even.

(ii) **Claim.** For a real number x, if $x^2 < 1$, then $x < 1$. (1) Assume $x^2 < 1$. (2) Suppose, for a contradiction, that $x > 1$. (3) Then $x > 0$, so $x^2 = x \cdot x > x > 1$. (4) This contradicts $x^2 < 1$. (5) Hence $x < 1$.

Solution. (i)

Line	True?	Follows? By which rule?
(1)	assumption	opens the →I box
(2)	yes, given (1)	∧E, the definition of "odd", and two ∃E's with the **different** fresh names k and l
(3)	yes	algebra, =E in a chain
(4)	yes	∃I with the witness $k + l + 1$; then the boxes close, and →I

Verdict: correct. In (2) the second witness has a new name, as (N4) of Example (What is not allowed) requires.

(ii)

Line	True?	Follows? By which rule?
(1)	assumption	opens the →I box
(2)	assumption	opens a box with "$x > 1$", announced as RAA
(3)	yes, given (2)	→E with order facts
(4)	yes	(3) gives $\neg(x^2 < 1)$ by order facts; ¬E on (1) and that line
(5)	true, given (1), but ...	**does not follow**: the box assumed $x > 1$, which is not the negation of $x < 1$; ¬I on it gives only $\neg(x > 1)$, that is, $x \leq 1$

Verdict: wrong; line (5) fails, because the box of line (2) opened with the wrong statement; entry (N11). The negation of $x < 1$ is $x \geq 1$, and the case $x = 1$ is never examined: $x = 1$ has $\neg(x > 1)$ and not $x < 1$, a counterexample to the step. The claim is true. Repair: suppose, for a contradiction, that $x \geq 1$; then $x^2 = x \cdot x \geq x \geq 1$, which contradicts $x^2 < 1$; hence $x < 1$.

Exercise. B2: Find the error

Find the error in the following proof, name its catalogue entry, and decide whether the claim is true.

> **Claim.** For all real numbers x and y, $|x| + |y| = |x + y|$.
>
> (1) Let x and y be real numbers. (2) Either x and y are both ≥ 0, or they are both < 0. (3) If both are ≥ 0, then $x + y \geq 0$, so $|x| + |y| = x + y = |x + y|$. (4) If both are < 0, then $x + y < 0$, so $|x| + |y| = -x - y = -(x + y) = |x + y|$. (5) In either case $|x| + |y| = |x + y|$.

Solution. Line (2) fails. Its disjunction is not derived, and it is false when x and y have opposite signs, so line (5) combines cases that do not exhaust: entry (N8) of Example (What is not allowed). Lines (3) and (4) are correct in their cases.

The claim is **false**: for $x = 1$, $y = -1$, $|x| + |y| = 2$ and $|x + y| = 0$; this pair satisfies neither case of line (2). What is true is $|x + y| \leq |x| + |y|$ (Example (The triangle inequality by cases)).

5.6.4.3 C. Going deeper

Exercise. C1: The second half

Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous at a, as the quantifier pattern of Section 4.8: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ for every real x with $|x - a| < \delta$. Suppose $f(a) > 0$. Here is a level-1 skeleton for the statement "there is $\delta > 0$ such that $f(x) > 0$ for every x with $|x - a| < \delta$":

1		$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x\ (\lvert x - a\rvert < \delta \to \lvert f(x) - f(a)\rvert < \varepsilon)$	
2		$f(a) > 0$	
3		$f(a)/2 > 0$	arith 2
4		$\exists \delta > 0\ \forall x\ (\lvert x - a\rvert < \delta \to \lvert f(x) - f(a)\rvert < f(a)/2)$	$\forall$E 1, 3
5	δ_0	$\delta_0 > 0$	
6		$\forall x\ (\lvert x - a\rvert < \delta_0 \to \lvert f(x) - f(a)\rvert < f(a)/2)$	
7		x_0 | $\lvert x_0 - a\rvert < \delta_0$	
8		$\lvert x_0 - a\rvert < \delta_0 \to \lvert f(x_0) - f(a)\rvert < f(a)/2$	$\forall$E 6
9		$\lvert f(x_0) - f(a)\rvert < f(a)/2$	$\to$E 8, 7
10		$f(x_0) > 0$	arith 2, 9
11		$\lvert x_0 - a\rvert < \delta_0 \to f(x_0) > 0$	$\to$I 7–10
12		$\forall x\ (\lvert x - a\rvert < \delta_0 \to f(x) > 0)$	$\forall$I 7–11
13		$\exists \delta > 0\ \forall x\ (\lvert x - a\rvert < \delta \to f(x) > 0)$	$\exists$I 5, 12
14		$\exists \delta > 0\ \forall x\ (\lvert x - a\rvert < \delta \to f(x) > 0)$	$\exists$E 4, 5–13

Its level-3 prose begins:

> Since $f(a) > 0$, the number $f(a)/2$ is positive. By the continuity of f at a, applied with $\varepsilon = f(a)/2$, choose $\delta > 0$ such that $|f(x) - f(a)| < f(a)/2$ for every real x with $|x - a| < \delta$.

The rest is blank. *Name the move, then complete it.* Say which skeleton lines your sentences carry, and what δ depends on.

Solution (Hint 1). What is left is the box of x_0 and the witness.

Solution (Hint 2). "Let x be a real number with $|x - a| < \delta$. Then . . . "

Solution. The move: skeleton first: the goal fixes the $\exists$I of line 13 and the box of x_0. Filling it: $\forall$E at the non-obvious instance $\varepsilon = f(a)/2$ (line 4), $\exists$E on its δ (line 5), and that δ as the witness.

Completion.

> Let x be a real number with $|x-a|<\delta$. Then $|f(x)-f(a)|<f(a)/2$, so, as $-|u|\le u$, $f(x)-f(a)>-f(a)/2$, that is, $f(x)>f(a)/2>0$. So this δ, which depends only on a, is as required.

"Let $x\ldots$ with $|x-a|<\delta$" opens the two boxes at line 7; "then $|f(x)-f(a)|<f(a)/2$" is lines 8–9, the $\forall$E at the obvious term x silent; "so $\ldots$ $f(x)>f(a)/2>0$" is line 10, which uses line 2. Lines 11 and 12 are silent closings, and the last sentence is lines 13–14, with the δ already chosen as witness. It depends on a (through $f(a)$ and the continuity at a) and on no x: the box of δ_0 encloses the box of x_0.

Marking scheme. (10 points) - 2: names the move (skeleton first; a hypothesis applied at a chosen ε, then its δ reused as the witness), or the rules $\forall$E, $\exists$E, $\exists$I. - 3: for an arbitrary x with $|x-a|<\delta$, derives $f(x)>f(a)/2$, with the step from $|u|<r$ to $u>-r$ justified. - 2: concludes $f(x)>0$ using $f(a)>0$, and states the conclusion with δ before x. - 2: matches the sentences to the skeleton's lines. - 1: says δ depends on a and not on x. Second route: refuting $f(x)\le 0$ by $|f(x)-f(a)|\ge f(a)-f(x)\ge f(a)>f(a)/2$ earns the second line and the $f(x)>0$ part of the third. Fatal errors (0 for the step): choosing δ after x; applying continuity at an ε not shown to be positive.

Exercise. C2: A true conclusion

Here is a proof that $\sqrt{2}+\sqrt{3}$ is irrational, a true conclusion. Find the line that fails and its catalogue entry, give a counterexample to the illegal step, and write a correct proof.

> (1) Suppose $\sqrt{2}+\sqrt{3}=r$ for some rational number r. (2) Squaring, $r^2=2+2\sqrt{6}+3=5+2\sqrt{6}$. (3) Hence $\sqrt{6}=(r^2-5)/2$, which is rational, since r is. (4) But $\sqrt{6}=\sqrt{2}\cdot\sqrt{3}$ is irrational, being a product of two irrational numbers. (5) This contradiction shows that $\sqrt{2}+\sqrt{3}$ is irrational.

Solution (Hint 1). Run the second column on every line.

Solution (Hint 2). For a correct proof, isolate $\sqrt{3}=r-\sqrt{2}$ before squaring.

Solution. The failing line is (4). Its first half is true, $\sqrt{6}=\sqrt{2}\cdot\sqrt{3}$, but its reason is not available. "A product of two irrational numbers is irrational" is neither proved nor a school fact: it is false, since $\sqrt{2}$ is irrational (Theorem 2.2.1), and $\sqrt{2}\cdot\sqrt{2}=2$ is rational. So the line cites nothing: entry (N9) of Example (What is not allowed). The product $\sqrt{2}\cdot\sqrt{2}=2$ is a counterexample to the illegal step; there is none to the conclusion, which is true. Lines (1)–(3) and (5), and the ¬I skeleton, are correct.

A correct proof. Suppose $\sqrt{2}+\sqrt{3}=r$ with r rational. Then $r>0$, since both square roots are positive. Isolating, $\sqrt{3}=r-\sqrt{2}$, and squaring,

$$3=r^2-2r\sqrt{2}+2, \qquad \text{so} \qquad \sqrt{2}=\frac{r^2-1}{2r},$$

where dividing by $2r$ is legitimate because $r > 0$. The right-hand side is rational, which contradicts Theorem 2.2.1. Hence $\sqrt{2} + \sqrt{3}$ is irrational.

Marking scheme. (10 points) - 3: names line (4) as the failing line, and the illegal step as the unproved general "fact", entry (N9). - 2: a counterexample to that step ($\sqrt{2} \cdot \sqrt{2} = 2$, or another product of two irrationals that is rational). - 1: says the conclusion is true, so no counterexample to it exists. - 4: a correct proof: the ¬I box on "r rational", the algebra, the division justified by $r \neq 0$, and the clash with Theorem 2.2.1 (or with $\sqrt{3}$ or $\sqrt{6}$, its irrationality proved). Fatal errors (0 for the step): "correcting" line (4) by another unproved fact; dividing by r without $r \neq 0$.

Exercise. C3: Exit check: four student proofs

Judge each proof. If it is wrong, give the line that fails, its entry in Example (What is not allowed), and a counterexample to the illegal step. If it is correct, name the rule of its most delicate step and check that step.

(1) **Claim.** For every integer n, if n is a multiple of 4 or a multiple of 6, then n is even. (1) Let n be an integer that is a multiple of 4 or of 6. (2) Case 1: n is a multiple of 4. Choose an integer k with $n = 4k$; then $n = 2(2k)$, so n is even. (3) Case 2: n is a multiple of 6. Choose an integer k with $n = 6k$; then $n = 2(3k)$, so n is even. (4) In either case, n is even.

(2) **Claim.** For integers m and n, if m and n are even, then mn is divisible by 4. (1) Assume m and n are even. (2) Then $m = 2k$ and $n = 2k$ for an integer k. (3) So $mn = 4k^2$, which is divisible by 4.

(3) **Claim.** For every real x, $|x - 1| \leq |x| + 1$. (1) Either $x \geq 1$ or $x \leq 0$. (2) If $x \geq 1$, then $|x - 1| = x - 1 \leq x + 1 = |x| + 1$. (3) If $x \leq 0$, then $|x - 1| = 1 - x = 1 + |x|$. (4) In either case, $|x - 1| \leq |x| + 1$.

(4) **Claim.** For every real x, if $x^2 = 2x$, then $x = 2$. (1) Assume $x^2 = 2x$. (2) Canceling x from both sides, $x = 2$. (3) Hence if $x^2 = 2x$, then $x = 2$.

Solution (Hint 1). Run the two-column check on every line, and keep the two questions apart: a true claim can have a wrong proof.

Solution (Hint 2). In (1), ask which boxes are open when the second k is chosen. In (3), ask what ∨E needs from line (1).

Solution. (1) **Correct.** The skeleton is ∀I and →I, then ∨E on the hypothesis itself, so the cases exhaust. The delicate step is the second "choose an integer k", an ∃E reusing a name. It is legal: the first box of k lies inside Case 1, which closed before Case 2 opened, so k appears in no open assumption, not in the existential and not in the conclusion. Sibling boxes may share a name (Section 4.6); (N4) forbids only a name still in use.

(2) **Wrong; line (2) fails; entry (N4) of Example (What is not allowed).** The second witness is named k while the box of $m = 2k$ is open. Counterexample to the step: $m = 2$, $n = 4$; no integer k has $2 = 2k$ and $4 = 2k$. The conclusion is true: with $m = 2k$ and $n = 2l$,

$mn = 4kl$.

(3) **Wrong; line (1) fails; entry (N8).** "Either $x \geq 1$ or $x \leq 0$" is not derived, and is false at $x = 1/2$, a counterexample to the step. The conclusion is true; the missing case repairs the proof: if $0 < x < 1$, then $|x - 1| = 1 - x < 1 < |x| + 1$.

(4) **Wrong; line (2) fails; entry (N9); the claim is false.** Canceling uses the "fact" "if $ax = bx$, then $a = b$", which is neither proved nor a school fact, and is false: $a = 1$, $b = 2$, $x = 0$. It needs $x \neq 0$, which no line provides. The claim fails at $x = 0$; what is true is "$x = 0$ or $x = 2$", from $x(x - 2) = 0$.

Marking scheme. (10 points: 2.5 per proof) - For each of (2), (3), (4): the verdict "wrong" (1); the failing line and its entry, (N4), (N8), (N9) (1); a counterexample to the illegal step (0.5). - For (1): the verdict "correct" (1); the delicate step named as an $\exists$E reusing a name in a sibling box (or as the $\vee$E on the hypothesis) (1); that step checked against the side condition (0.5). For (4), "$\rightarrow$E without its antecedent $x \neq 0$" is also accepted. Fatal errors (0 for that proof): calling proof (1) wrong as an instance of (N4) because the name k recurs; judging (2) or (3) correct because its conclusion is true.

5.7 The Dispute over Excluded Middle

Chapter 2 found that one rule stands apart. RAA belongs to no connective, and over the other primitive rules it derives $\neg\neg$E and LEM and is derived by each of them (Theorem 2.5.7). Chapter 3's optional Theorem 3.5.4 showed that the other propositional rules cannot derive it. Some mathematicians have refused to use it, for a century now and for reasons that are mathematical rather than merely fussy. This section asks what those reasons are, what can still be proved without the rule, and what analysis would lose.

5.7.1 One extra rule

Recall what Section 2.5 established. Over the primitive rules other than RAA, the three rules RAA, $\neg\neg$E, and LEM each derive the other two, so adding any one of them gives the same derivable sequents. There is one classical principle, and it has three faces. Nothing needs re-proving here; what is new is the decision to leave the principle out and see what remains.

Call a derivation **intuitionistic** if no line is justified by RAA and no line uses a derived rule or a cited result whose own derivation uses RAA. **Intuitionistic logic** is the system of these derivations. Its rules are the introduction and elimination rules for $\wedge$, $\rightarrow$, and $\vee$ (Chapter 1); $\neg$I, $\neg$E, $\bot$E, $\leftrightarrow$I, and $\leftrightarrow$E (Chapter 2); the rules for $\forall$, $\exists$, and $=$ (Chapter 4); and repetition. **Classical logic**, the logic of Chapters 1–4, is the same list with RAA added.

This is why Chapter 2 kept $\bot$E as a primitive rule although RAA and repetition derive it (the remark after Theorem 2.5.7). Had $\bot$E been dropped, striking RAA would have taken it away too, and the remaining system would be weaker than the one intuitionists actually use: it could no longer derive $\bot \vdash P$. That is a claim about all derivations, and we state it without proof; nothing depends on it. With $\bot$E kept, the rules without RAA are exactly intuitionistic logic.

System	Rules	$\vdash P \vee \neg P$?
intuitionistic	the primitive rules other than RAA	not derivable (Theorem 3.5.4 for the propositional rules; the rules for $\forall$, $\exists$ and $=$ do not change this, stated without proof)
classical	the same rules and RAA	derivable (Proposition 2.5.3)

Every intuitionistic derivation is a derivation, so everything intuitionistic logic derives, classical logic derives too, and soundness applies to it: Theorem 3.5.2, and, for derivations whose lines contain quantifiers or equations, the soundness statement for predicate logic in Section 4.11. The difference is a list of sequents that only the classical system reaches. Chapters 2–4 flag the derivations that use RAA as they go, so the results on that list can be read off the text; the Quick check asks for them.

Quick check.

(a) Which labeled results of Chapter 2 (propositions and theorems) were derived with RAA, directly or through $\neg\neg$E, LEM, or a cited result that uses one of them? (b) The same question for Chapters 3 and 4.

Solution.

(a) Seven: Proposition 2.5.1 (b); Proposition 2.5.2; Proposition 2.5.3; Proposition 2.5.5 (d); Proposition 2.5.6 (b), through LEM; Theorem 2.5.7, whose proof uses all three: RAA in ①, $\neg\neg$E in ②, LEM in ③; and Proposition 2.5.4. Every other proposition and theorem of Chapter 2 uses no classical rule, including Theorem 2.2.1, whose box closes by $\neg$I.

(b) In Chapter 3: Lemma 3.6.4, through LEM, and with it Theorem 3.6.5 and the direction ($\Leftarrow$) of Corollary 3.6.6. In Chapter 4: Proposition 4.9.1 (a); the direction $\vdash$ of Proposition 4.9.2; the direction $\vdash$ of Proposition 4.9.3 (a); the two directions $\dashv$ of Proposition 4.9.5, in (d) and in (e), both through LEM; and Proposition 4.9.6, as proved, since its box assumes $a \neq b$ and concludes $a = b$.

5.7.2 Excluded middle cannot be refuted

If intuitionists do not assert $P \vee \neg P$, perhaps they deny it. They do not, and they could not: its double negation needs no classical rule at all.

Proposition 5.7.1. Excluded Middle Cannot Be Refuted

Optional: nothing later depends on this.

For every statement P, $\vdash \neg\neg(P \vee \neg P)$, by a derivation that uses neither RAA nor $\bot$E.

Try it first (ten minutes): the goal is a negation. Open the box that the goal asks for, hunt for

a clash, and note which rule closes each box you open.

Idea. An attempt that starts from $P \vee \neg P$ stalls at once: $\vee$I needs one side, and neither side is in hand. But the goal is not $P \vee \neg P$. It is $\neg(\neg(P \vee \neg P))$, a negation, so the goal-directed move of Section 2.2 opens a $\neg$I box on $\neg(P \vee \neg P)$ with the goal $\bot$. That assumption can clash only with $P \vee \neg P$, which $\vee$I builds from either side, so we manufacture a side as in Example (Manufacturing the other half of a clash). A box on P gives $P \vee \neg P$, a clash, and $\neg$I closes it with $\neg P$. Then $\vee$I builds $P \vee \neg P$ from $\neg P$, and that is the second clash. Every box ends in $\bot$ and closes by $\neg$I.

Before reading on: self-explain lines 5 and 6.

Proof. The derivation is:

1	$\neg(P \vee \neg P)$	
2	$\quad P$	
3	$\quad P \vee \neg P$	$\vee$I 2
4	$\quad \bot$	$\neg$E 3, 1
5	$\neg P$	$\neg$I 2–4
6	$P \vee \neg P$	$\vee$I 5
7	$\bot$	$\neg$E 6, 1
8	$\neg\neg(P \vee \neg P)$	$\neg$I 1–7

The two boxes close by $\neg$I, at lines 5 and 8. The other rules used are $\vee$I (lines 3 and 6) and $\neg$E (lines 4 and 7). The derivation has no premises. This proves the proposition.

■

Set it beside the derivation of Proposition 2.5.3. Lines 1–7 are the same, line for line. Only line 8 differs: there RAA closed the outer box and wrote $P \vee \neg P$; here $\neg$I closes it and writes the negation of its assumption. So the whole classical content of Proposition 2.5.3 sits in its last line.

Non-example by minimal change. Keep lines 1–7, and write $P \vee \neg P$ at line 8 with the justification $\neg$I 1–7. The box still checks. What fails is the conclusion of $\neg$I, which must be the negation of the box's assumption, $\neg\neg(P \vee \neg P)$. Writing $P \vee \neg P$ there needs RAA.

What the proposition says is this: **nobody can refute excluded middle**. Suppose some derivation, intuitionistic or not, gave $\neg(P \vee \neg P)$ from no premises. Then $\neg$E against the line $\neg\neg(P \vee \neg P)$ would give $\bot$ from no premises, and soundness forbids that: Theorem 3.5.2, or the soundness statement for predicate logic in Section 4.11 when P or the derivation involves quantifiers or equality, since no valuation, and no structure and assignment, makes $\bot$ true. The intuitionist does not claim that excluded middle fails. The intuitionist declines to **assert** it, because asserting it would mean having a proof of one side, and for many statements P, an open conjecture for instance, nobody has one.

Remark. The proposition is one instance of a general fact, due to Glivenko (1929) and stated here without proof: if a statement built from letters by the connectives is derivable with all the rules, then its double negation is derivable without RAA. Classical propositional logic survives inside intuitionistic logic, behind two negations. The fact fails once quantifiers are allowed; $\neg\neg\forall x\,(\varphi \vee \neg\varphi)$ is in general not derivable without RAA (also stated without proof).

Warning. Intuitionistic logic does not say that excluded middle is false. No instance $P \vee \neg P$ can be refuted, since $\neg\neg(P \vee \neg P)$ is derivable without RAA; dropping a rule is not adding its negation. (Brouwer's own analysis adds axioms about the real numbers under which the **universal** statement "every real number is 0 or not 0" is refuted, but no single instance is.)

5.7.3 What a proof provides

Chapter 3 gave the connectives a meaning through truth values: every letter is true or false, and a table computes the rest. That semantics already takes a side, since a row decides every letter. The intuitionist explains a statement instead by saying what counts as a **proof** of it. The resulting clauses are called the **BHK reading**, after Brouwer, Heyting, and Kolmogorov.

To know a statement is to have a proof of it, and a proof of a compound statement is built from proofs of its parts.

- There is **no** proof of $\bot$.
- A proof of $P \wedge Q$ is a pair: a proof of P and a proof of Q.
- A proof of $P \vee Q$ is a proof of P or a proof of Q, **with a label saying which**.
- A proof of $P \to Q$ is a method that takes **any** proof of P and returns a proof of Q.
- A proof of $\neg P$ is a method that takes any proof of P and returns a proof of $\bot$. So $\neg P$ is read as $P \to \bot$, exactly the reading that the table in Section 2.2, after Definition 2.2.1, found row by row.
- A proof of $\exists x\,\varphi$ is an object t **together with** a proof of $\varphi[t/x]$.
- A proof of $\forall x\,\varphi$ is a method that produces, for **every** object t, a proof of $\varphi[t/x]$.

In words: a disjunction must say which side, an existence claim must name the object, and an implication or a universal statement must come with a method that works on every input. A proof of a negation provides no object at all; it is a guarantee that no proof of the negated statement will ever arrive.

Each intuitionistic rule builds exactly what its clause asks for. $\vee$I writes $P \vee Q$ from a line P, so it knows which side it used. $\exists$I writes $\exists x\,\varphi$ from an instance $\varphi[t/x]$, so the witness t is on the page. A $\to$I box is a method: given the assumption, it produces the conclusion. RAA is the exception. Its box shows that $\neg\varphi$ leads to $\bot$, which is a proof of $\neg\neg\varphi$, and then it writes φ. Under the BHK reading a method for refuting every refutation of φ is not a proof of φ.

Excluded middle under BHK. A proof of $P \vee \neg P$ for **every** statement P would be a single method that decides each statement, says which side holds, and proves that side. Nobody has such a method. Take Goldbach's conjecture, that every even integer greater than 2 is a sum of two primes: nobody has proved it and nobody has refuted it, so nobody can produce a labeled proof of "Goldbach's conjecture or its negation".

The proposition under BHK. The derivation of Proposition 5.7.1 is such a method, and it decides nothing. Its conclusion asks for a method that takes any f, a method sending proofs

of $P \vee \neg P$ to proofs of $\bot$, and returns a proof of $\bot$. Here is one. Given a proof p of P, label it "left" and apply f: that turns proofs of P into proofs of $\bot$, which is a proof g of $\neg P$ (lines 2–5). Now label g "right" and apply f once more (lines 6–7). At no point did the method need to know whether P holds.

Not every, and some not. Chapter 4 derived $\neg\forall x\,\varphi \vdash \exists x\,\neg\varphi$ (Proposition 4.9.1 (a)) with two RAA boxes, and noted that the derivation never says which x fails φ. The BHK reading says why that matters. A proof of the premise is a method that refutes every proof of $\forall x\,\varphi$; a proof of the conclusion must name an object. Nothing in the first produces the second, and the derivation does not try: its conclusion comes from RAA, not from an $\exists$I applied to an available instance. Classical mathematics accepts the step, and *Analysis on the Real Line*, Chapter 1 uses it at the end of its proof of the Archimedean property. A constructive mathematician accepts that conclusion only with an object in hand.

Quick check. Describe a BHK proof of $\neg(P \wedge \neg P)$. Does it decide whether P holds?

Solution. We need a method that takes any proof of $P \wedge \neg P$ and returns a proof of $\bot$. A proof of $P \wedge \neg P$ is a pair: a proof p of P and a method f turning proofs of P into proofs of $\bot$. Apply f to p. That is the whole method, and it never decides P: it only uses a pair that someone else supplies. (It is the BHK reading of a $\neg$I box on $P \wedge \neg P$ whose clash is $\neg$E on its two halves.)

5.7.4 Irrational powers, with and without a witness

Here is the standard example of a proof that establishes existence without saying what exists.

Theorem 5.7.2. An Irrational Power of an Irrational Number Can Be Rational

Optional: nothing later depends on this.

There are irrational numbers a and b such that a^b is rational.

Try it first (ten minutes): find such a and b explicitly, and prove that your a and b are irrational and that a^b is rational.

Idea. The first candidates anyone tries are $a = b = \sqrt{2}$. Then the question is whether $\sqrt{2}^{\sqrt{2}}$ is rational, and the attempt stalls there: nothing in this book decides it. There are two ways out. The classical one splits on the undecided question with LEM. If $\sqrt{2}^{\sqrt{2}}$ is rational, the first candidates work. If not, use it as the new a and raise it to $\sqrt{2}$ once more, which turns the exponent into $\sqrt{2} \cdot \sqrt{2} = 2$. The proof below takes this way. The other way chooses an exponent that undoes the square root, and the proposition after the proof takes it.

Proof. Let $s \coloneqq \sqrt{2}^{\sqrt{2}}$, a positive real number. By excluded middle, s is rational or s is not rational [LEM], and we argue by cases [$\vee$E].

Case 1: s is rational. Take $a \coloneqq \sqrt{2}$ and $b \coloneqq \sqrt{2}$. Both are irrational by Theorem 2.2.1, and $a^b = s$ is rational [$\exists$I].

Case 2: s is not rational. Take $a \coloneqq s$ and $b \coloneqq \sqrt{2}$. Then a is irrational by the case assumption and b by Theorem 2.2.1, and

$$a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^{\,2} = 2,$$

which is rational [$\exists$I].

In either case there are irrational a and b with a^b rational. This proves the theorem. ∎

The classical step is the first line, LEM on the statement "s is rational". Every other step is intuitionistic. The proof does **not** tell us which pair works. It hands over two candidate pairs, $(\sqrt{2}, \sqrt{2})$ and $(\sqrt{2}^{\sqrt{2}}, \sqrt{2})$, and a guarantee that one of them succeeds. Under the BHK reading a proof of an existence claim must name the object, so this proof is not one. Section 2.5 described excluded middle as making a decision; here the decision about "s is rational" is made, and nobody learns its outcome.

Remark. In fact the second case holds: $\sqrt{2}^{\sqrt{2}}$ is transcendental, so in particular irrational, by the Gelfond–Schneider theorem, proved independently by Gelfond and by Schneider in 1934. That theorem lies far beyond this book, and nothing here uses it. It also does not repair the proof above, which never needed to know the answer.

The other way out finds the witness. Since $\sqrt{2}^{\,b} = 2^{b/2}$, the exponent $b = 2\log_2 m = \log_2(m^2)$ gives $\sqrt{2}^{\,b} = m$ for every integer $m \geq 2$. The choice $m = 2$ gives $b = 2$, which is rational, so m must avoid the powers of 2; the first choice left is $m = 3$.

Proposition 5.7.3. A Witness for an Irrational Power

Optional: nothing later depends on this.

The number $\log_2 9$ is irrational. Consequently $a \coloneqq \sqrt{2}$ and $b \coloneqq \log_2 9$ are irrational numbers with $a^b = 3$.

Idea. The irrationality goes as for $\sqrt{2}$: suppose $\log_2 9$ is a fraction p/q, clear the denominator, and look at parity. Here the clash is quicker than for $\sqrt{2}$: a power of 2 would equal a power of 9, and one is even while the other is odd. The positivity of $\log_2 9$ is what makes the power of 2 even.

Proof. Suppose that $\log_2 9 = p/q$ for integers p and q with $q \neq 0$; replacing p, q by $-p$, $-q$ if necessary, we may take $q \geq 1$. Since $2^0 = 1 < 9$ and $t \mapsto 2^t$ is increasing, $\log_2 9 > 0$, so $p > 0$, that is, $p \geq 1$. From $2^{p/q} = 9$, raising both sides to the power q gives

$$2^p = 9^q.$$

The left side is even, since $2^p = 2 \cdot 2^{p-1}$ with $p - 1 \geq 0$. The right side is odd, being a product of q odd factors. No integer is both even and odd, by (E2) of Section 2.1, so this is a contradiction. Hence $\log_2 9$ is not rational [¬I].

For the consequence, $a = \sqrt{2}$ is irrational by Theorem 2.2.1, and $b = \log_2 9$ is irrational by what was just shown. Since $\log_2 9 = \log_2(3^2) = 2\log_2 3$,

$$a^b = \left(2^{1/2}\right)^{\log_2 9} = 2^{(\log_2 9)/2} = 2^{\log_2 3} = 3.$$

This proves the proposition. ∎

This proof uses no classical rule. The irrationality of $\log_2 9$ is a negative statement, "no fraction equals $\log_2 9$", and its box closes by ¬I, like the proof of Theorem 2.2.1. The witness $(\sqrt{2}, \log_2 9)$ is written down. So Proposition 5.7.3 gives a second proof of Theorem 5.7.2, and this one provides what the BHK reading asks for. (The exponent laws for real exponents are school facts here; *Analysis on the Real Line*, Chapter 10 defines a^x for irrational x and proves them.)

5.7.5 A short history

The dispute has dates.

In 1908 L. E. J. Brouwer published "De onbetrouwbaarheid der logische principes" ("The unreliability of the logical principles") in the *Tijdschrift voor Wijsbegeerte*. His point was that excluded middle had been abstracted from reasoning about finite collections, where every case can in principle be checked, and then applied without justification to infinite ones, where it cannot. Over the next two decades he rebuilt parts of analysis on this view, and the movement took the name **intuitionism**.

In 1925 A. N. Kolmogorov showed that classical reasoning can be imitated inside a logic without excluded middle by putting a double negation in front of every subformula: the first **double-negation translation** (Exercise C2 below checks one step of it).

In July 1927 David Hilbert lectured in Hamburg on "Die Grundlagen der Mathematik", printed in 1928. He defended excluded middle in a much-repeated comparison: to take it from the mathematician would be like forbidding the astronomer the telescope or the boxer the use of his fists. The quarrel between his program and Brouwer's became known as the **Grundlagenstreit**, the dispute over foundations. It turned personal in 1928, when Hilbert had Brouwer removed from the editorial board of the *Mathematische Annalen*; the board was then dissolved and re-formed without him.

In 1929 V. Glivenko proved the theorem of the remark above: classical propositional logic sits inside intuitionistic logic behind a double negation. In 1930 Arend Heyting published "Die formalen Regeln der intuitionistischen Logik" in the *Sitzungsberichte der Preußischen Akademie der Wissenschaften*. It wrote down rules for intuitionistic logic, in a format different from ours, and from then on the logic could be studied as a formal system like any other.

In 1933 Kurt Gödel published a double-negation translation of arithmetic, which shows that classical arithmetic is consistent if intuitionistic arithmetic is. Gerhard Gentzen had found an equivalent translation independently and withdrew his paper of 1933 when Gödel's appeared.

In 1967 Errett Bishop published *Foundations of Constructive Analysis*. It developed a large part of analysis without excluded middle, and also without Brouwer's philosophy and his special axioms about the real numbers. Most of what is now called constructive analysis descends from it.

5.7.6 What analysis would lose

What does analysis look like without the rule? Here are four previews, stated informally and without proof. The book takes the first as an axiom and proves the classical versions of the other three; each is a place where classical reasoning, by the rule or by an axiom, does work that a constructive proof cannot.

- **The supremum.** *Analysis on the Real Line*, Chapter 1 takes as an axiom that every non-empty set of real numbers that is bounded above has a supremum. Constructively, a non-empty set A bounded above has a supremum **if and only if** it is **order-located**: for all real numbers $x < y$, either y is an upper bound of A or some element of A exceeds x (Bishop's form of the least upper bound principle). The classical axiom asserts the supremum without requiring anyone to make these decisions.
- **Monotone sequences.** E. Specker (1949) gave an increasing, bounded sequence of rational numbers that a computer program can list term by term, but whose limit is not a computable real number: no program can print its decimal digits. So the convergence of bounded monotone sequences, which *Analysis on the Real Line*, Chapter 2 proves from the axiom, cannot come with a method for computing the limit.
- **The intermediate value theorem.** Let f be continuous on $[a, b]$ with $f(a) < 0 < f(b)$. Classically, *Analysis on the Real Line*, Chapter 6 proves that $f(x) = 0$ for some $x \in [a, b]$. Constructively, what holds in general is the **approximate** version: for every $\varepsilon > 0$ some $x \in [a, b]$ has $|f(x)| < \varepsilon$. The exact version holds under extra hypotheses, for example when f is strictly increasing. A bisection needs the sign of f at a midpoint m, and finitely many digits of $f(m)$ cannot always tell whether $f(m)$ is negative, zero, or positive: the digits of a value equal to 0 begin like those of a tiny positive or negative one.
- **Bolzano–Weierstrass.** A bisection proof of the theorem (*Analysis on the Real Line*, Chapter 2 proves it by another route) says: one of the two halves of an interval contains infinitely many terms of the sequence; continue with that half. That step is LEM on "the left half contains infinitely many terms", a statement that nobody can decide in general, and the proof does not say which half.

5.7.7 Proofs as programs

The BHK clauses read like the description of a programming language. A proof of $P \wedge Q$ is a pair; a proof of $P \vee Q$ is a value tagged "left" or "right"; a proof of $P \to Q$ is a function; a proof of $\exists x\, \varphi$ is a pair of an object and a proof; a proof of $\forall x\, \varphi$ is a function that accepts any object. This match between propositions and types, and between proofs and programs, is the **Curry–Howard correspondence**. It is what a proof assistant such as Lean is built on: checking a proof is checking that a program has the right type.

Intuitionistic logic fits this picture exactly. LEM does not, since no program decides every proposition. Lean adds classical logic by an axiom, `Classical.choice`, a form of the axiom of choice. Excluded middle, `Classical.em`, is then a **theorem**, derived from choice together

with two extensionality principles by Diaconescu's theorem. Lean draws one line with care. A definition that uses classical logic to produce **data**, for instance a number chosen by deciding an arbitrary proposition, must be marked `noncomputable`, since no program computes it. A proof that uses `Classical.em` needs no mark, because proofs carry no data.

```
-- excluded middle is a theorem, proved from the axiom of choice
example (p : Prop) : Or p (Not p) := Classical.em p

-- a number defined by deciding an arbitrary proposition
open Classical in
noncomputable def indicator (p : Prop) : Nat := if p then 1 else 0
```

Warning. "Constructive" does not mean "computable in practice". A constructive proof provides a method, and the method may be hopeless. Deciding whether a whole number is prime by trying every smaller divisor is constructive, and for a number with a thousand digits no computer could finish it. Constructive mathematics asks whether a method exists, not whether it is fast.

5.7.8 The book's position

The main line of this book stays classical. The decision is made once, in the axioms. The ordered-field axioms of *Analysis on the Real Line*, Chapter 1 include trichotomy: for real numbers a and b, exactly one of $a < b$, $a = b$, $b < a$ holds. Constructive analysis does not accept trichotomy for real numbers, because it would decide the sign of every real number. So a case split such as "$x \geq 0$ or $x < 0$" rests on an axiom, and the book does not flag it.

What the book does flag is a genuinely non-constructive step of the kind discussed above, such as the bisection step of Bolzano–Weierstrass, and every use of the axiom of choice. One clause says so, and nothing more. Such a step is never silent anyway: by Definition 5.3.2, a ¬¬E or LEM step is written out as an explicit case split or an explicit removal of a double negation. A reader who wants to know where the classical rule enters a proof can always find the place.

5.7.9 Exercises

5.7.9.1 A. Check your understanding

Exercise. A1

Parts (a)–(d) are about Proposition 5.7.1 and its derivation; part (e) is about the proof of Theorem 5.7.2.

(a) Which rule closes each of the two boxes, and at which line?

(b) The outer box opens with a negation and ends in $\bot$, the shape of an RAA box. Why is no RAA used?

(c) Give the BHK reading of the conclusion $\neg\neg(P \vee \neg P)$.

(d) True or false, with a reason: for some statement P, intuitionistic logic derives $\neg(P \vee \neg P)$ from no premises.

(e) Which line of that proof uses LEM, and on which statement? Does the proof say which pair (a, b) works?

Solution.

(a) The inner box, lines 2–4, opens with P and closes by ¬I at line 5, giving $\neg P$. The outer box, lines 1–7, opens with $\neg(P \vee \neg P)$ and closes by ¬I at line 8.

(b) The rule that closes a box is decided by what is written after it, not by the box. Line 8 writes $\neg\neg(P \vee \neg P)$, the negation of the box's assumption $\neg(P \vee \neg P)$, and that is what ¬I writes. RAA would be needed only to write $P \vee \neg P$ after the same box.

(c) A method that takes any method f, sending proofs of $P \vee \neg P$ to proofs of $\bot$, and returns a proof of $\bot$. The derivation provides one: label a proof of P "left" and apply f, which gives a proof of $\neg P$; label that proof "right" and apply f again.

(d) False. Such a derivation would also be a derivation with all the rules. Together with the derivation of $P \vee \neg P$ from no premises (Proposition 2.5.3), ¬E would derive $\bot$ from no premises. No derivation does that: by Theorem 3.5.2, or by the soundness statement for predicate logic in Section 4.11 when P or the derivation involves quantifiers or equality, $\bot$ would be true under every valuation (in every structure under every assignment), and no valuation, and no structure and assignment, makes $\bot$ true.

(e) The first line, which splits on "$\sqrt{2}^{\sqrt{2}}$ is rational" by LEM. The proof does not say which pair works: it shows that $(\sqrt{2}, \sqrt{2})$ works in the first case and $(\sqrt{2}^{\sqrt{2}}, \sqrt{2})$ in the second, and does not decide which case holds.

5.7.9.2 B. Practice

Exercise. B1: Double negation of double negation elimination

Derive $\vdash \neg\neg(\neg\neg P \to P)$ without RAA, and name the rule on every line.

Hint: the goal is a negation; compare the derivation of the proposition in this section.

Solution. The goal is a negation, so we open a ¬I box on $\neg(\neg\neg P \to P)$, with the goal $\bot$. The assumption clashes only with $\neg\neg P \to P$, an implication, which →I builds from a box on $\neg\neg P$ that ends in P. The first way to have P is to assume it: inside a box on P, the implication holds by →I with P repeated, and it clashes with the assumption, so $\neg P$ follows by ¬I. Now $\neg P$ clashes with any $\neg\neg P$, so a second box on $\neg\neg P$ reaches $\bot$, $\bot$E writes P, and the implication is built a second time.

1	$\neg(\neg\neg P \to P)$	
2	$\quad P$	
3	$\quad\quad \neg\neg P$	
4	$\quad\quad P$	R 2
5	$\quad \neg\neg P \to P$	$\to$I 3–4
6	$\quad \bot$	$\neg$E 5, 1
7	$\neg P$	$\neg$I 2–6
8	$\quad \neg\neg P$	
9	$\quad \bot$	$\neg$E 7, 8
10	$\quad P$	$\bot$E 9
11	$\neg\neg P \to P$	$\to$I 8–10
12	$\bot$	$\neg$E 11, 1
13	$\neg\neg(\neg\neg P \to P)$	$\neg$I 1–12

Lines 1, 2, 3, and 8 are assumptions. Line 4 repeats line 2 into the box of line 3, since $\to$I needs P as a line of that box itself. Line 9 is $\neg$E on $\neg P$ (line 7) and its negation $\neg\neg P$ (line 8). Every box closes by $\to$I or $\neg$I, so no RAA is used. Unlike the derivation of the proposition in this section, this one uses $\bot$E, at line 10. The sequent $\vdash \neg\neg P \to P$ itself has no derivation without RAA (we state this without proof); its double negation has one.

Exercise. B2: Reading proofs as methods

(a) Give the BHK reading of a proof of $P \to \neg\neg P$, and describe such a proof.

(b) The same for $(P \to Q) \to (\neg Q \to \neg P)$.

(c) The same for $\forall n \in \mathbb{N}\ \exists m \in \mathbb{N}\ (m > n)$, over the natural numbers.

(d) A proof of an existence claim $\exists x\, \varphi$ opens with "suppose that no x satisfies φ" and reaches a contradiction. Without a classical rule, which of $\neg\neg\exists x\, \varphi$ and $\exists x\, \varphi$ does it prove? What does it provide under the BHK reading?

Solution.

(a) A method that takes any proof p of P and returns a proof of $\neg\neg P$, that is, a method that takes any proof f of $\neg P$ and returns a proof of $\bot$. The method: given p, and then given f, apply f to p. This is the BHK reading of Proposition 2.5.1 (a).

(b) A method that takes any method h (from proofs of P to proofs of Q) and returns a proof of $\neg Q \to \neg P$: a method that takes any g (from proofs of Q to proofs of $\bot$) and returns a proof of $\neg P$, that is, a method from proofs of P to proofs of $\bot$. The method: given h, g, and a proof p of P, compute $g(h(p))$. This is the BHK reading of Proposition 2.3.2, whose derivation uses no RAA.

(c) Unfolding the bounded quantifiers (Definition 4.1.1): a method that takes any object n and a proof that $n \in \mathbb{N}$, and returns an object m together with a proof of $m \in \mathbb{N} \wedge m > n$.

The method: given n and the proof of $n \in \mathbb{N}$, output $m = n + 1$, with the proofs that $n + 1 \in \mathbb{N}$ and $n + 1 > n$.

(d) The box opens with $\neg\exists x\,\varphi$ and ends in $\bot$, so $\neg$I closes it with $\neg\neg\exists x\,\varphi$. Writing $\exists x\,\varphi$ instead is RAA (or $\neg\neg$E after $\neg$I), a classical step. Under the BHK reading the proof provides a method that takes any proof that nothing satisfies φ and returns a proof of $\bot$. It provides no object: a proof of $\exists x\,\varphi$ would have to name one.

Exercise. B3: Where the witness goes missing

In the derivation of Proposition 4.9.1 (a), which step makes the existence claim without a witness? Why does no witness appear, although line 4 is an $\exists$I?

Solution. Two lines are classical: line 6, an RAA that writes the instance $\varphi[x_0/x]$ after a box on its negation, and line 9, the RAA that writes the conclusion $\exists x\,\neg\varphi$. Line 6 produces no existence claim. Line 9 is the step that asserts existence: it closes the box on $\neg\exists x\,\neg\varphi$, which ends in $\bot$, and writes $\exists x\,\neg\varphi$. Closed by $\neg$I, the same box would give only $\neg\neg\exists x\,\neg\varphi$.

No witness appears because no $\exists$I produces the conclusion. Line 4 does derive $\exists x\,\neg\varphi$ from the instance $\neg\varphi[x_0/x]$, but that instance is the assumption of the box 3–5, used only to clash with line 2 and discharged at line 6. The name x_0 is the flagged name of the $\forall$I box 3–6: an arbitrary object, not a particular one. Line 4 is unavailable at line 9, and the conclusion there is justified by RAA from the box 2–8, which names nothing. Under the BHK reading, line 9 asserts "something fails φ" and hands over no object that fails it.

5.7.9.3 C. Going deeper

Exercise. C1: More witnesses

This section proved that $\log_2 9$ is irrational; here are variants.

(a) Show that $a = \sqrt{2}$ and $b = \log_2 25$ are irrational numbers with a^b rational.

(b) For which integers $m \geq 2$ are $a = \sqrt{2}$ and $b = \log_2(m^2)$ irrational numbers with a^b rational? Prove your answer.

Solution (Hint 1). Compute a^b first; it does not depend on whether b is rational. Then write $m = 2^k r$ with r odd.

Solution (Hint 2). Suppose $\log_2(m^2) = p/q$ with integers p and $q \geq 1$; then $2^p = m^{2q}$.

Solution.

(a) First, $a^b = (2^{1/2})^{\log_2 25} = 2^{(\log_2 25)/2} = 2^{\log_2 5} = 5$, since $\log_2 25 = 2\log_2 5$; this is rational. Next, $a = \sqrt{2}$ is irrational by Theorem 2.2.1. Finally, suppose that $\log_2 25 = p/q$ with integers p and $q \geq 1$. Since $25 > 1 = 2^0$ and $t \mapsto 2^t$ is increasing, $\log_2 25 > 0$,

so $p \geq 1$. Raising $2^{p/q} = 25$ to the power q gives $2^p = 25^q$. The left side is even because $p \geq 1$, and the right side is odd, being a product of odd factors. No integer is both, so $\log_2 25$ is irrational.

(b) **Answer:** exactly the integers $m \geq 2$ that are not powers of 2.

For every $m \geq 2$, $a = \sqrt{2}$ is irrational and

$$a^b = \left(2^{1/2}\right)^{\log_2(m^2)} = 2^{\log_2 m} = m,$$

which is rational. So the question is only when $b = \log_2(m^2) = 2\log_2 m$ is irrational. Write $m = 2^k r$ with $k \in \mathbb{N}$ and r odd, $r \geq 1$.

If $r = 1$, then $m = 2^k$ with $k \geq 1$, and $b = 2k$ is rational: these m fail.

If $r \geq 3$, suppose that $b = p/q$ with integers p and $q \geq 1$. Since $m \geq 2 > 1$, $b > 0$, so $p \geq 1$. Raising $2^{p/q} = m^2$ to the power q gives

$$2^p = m^{2q} = 2^{2kq} r^{2q}.$$

Since $r^{2q} \geq 3 > 1$, the right side exceeds 2^{2kq}, so $p > 2kq$. Dividing by 2^{2kq} gives $r^{2q} = 2^{p-2kq}$ with $p - 2kq \geq 1$. The left side is odd, a product of odd factors, and the right side is even. No integer is both, so b is irrational: these m work. The smallest are $m = 3, 5, 6, 7$. The cases $m = 3$ and $m = 5$ are the example of this section and part (a).

Marking scheme. (10 points) - 2: (a) $a^b = 5$, with the exponent laws used. - 2: (a) $\log_2 25$ irrational, with $p \geq 1$ justified by $\log_2 25 > 0$ and the parity clash $2^p = 25^q$. - 2: (b) $a^b = m$ for every m, so only the rationality of b matters. - 1: (b) powers of 2 excluded, with $b = 2k$ rational. - 3: (b) for $m = 2^k r$, $r \geq 3$ odd, the clash $r^{2q} = 2^{p-2kq}$ with $p - 2kq \geq 1$ (or an equivalent comparison of the odd parts of 2^p and m^{2q}). Fatal errors (0 for the step): concluding irrationality of b from "m is irrational" or "$\log_2 m$ is not an integer"; a parity clash with p possibly 0 or negative left unaddressed.

Exercise. C2: Modus ponens survives Kolmogorov's translation

For statements built from letters by $\wedge$, $\vee$, $\to$, and $\neg$, define the translation K by putting $\neg\neg$ in front of every subformula:

$$KP = \neg\neg P \text{ for a letter } P, \qquad K(\varphi \circ \psi) = \neg\neg(K\varphi \circ K\psi), \qquad K(\neg\varphi) = \neg\neg\neg K\varphi,$$

where $\circ$ is any of $\wedge$, $\vee$, $\to$. Chapter 2's Exercise (B1: With and without the new rule) (a) derives $\neg\neg P,\ \neg\neg(P \to Q) \vdash \neg\neg Q$ without RAA (in its printed solution).

(a) Show that every $K\psi$ has the form $\neg\neg X$ for some statement X.

(b) Deduce that $K(\varphi \to \psi),\ K\varphi \vdash K\psi$ without RAA, for all statements φ and ψ.

Solution (Hint 1). Put $K\varphi$ and $K\psi$ in for P and Q in the Chapter 2 sequent. Which of its two premises is $K(\varphi \to \psi)$?

Solution (Hint 2). The conclusion you get has two negations too many. Look for a result of Section 2.2 that removes negations without ¬¬E.

Solution.

(a) By the definition, each of the three clauses produces a statement whose first two symbols are ¬¬: $KP = \neg\neg P$, $K(\varphi \circ \psi) = \neg\neg(K\varphi \circ K\psi)$, and $K(\neg\varphi) = \neg\neg(\neg K\varphi)$. So $K\psi = \neg\neg X$, with X the letter, $K\varphi \circ K\psi$, or $\neg K\varphi$, respectively. No induction is needed, since the outermost clause alone decides the first two symbols.

(b) Three steps, each without RAA.

① The derivation of Exercise (B1: With and without the new rule) (a) uses no RAA, so its sequent may be used as a derived rule (Definition 2.3.1) with the statements $K\varphi$ and $K\psi$ in place of P and Q:

$$\neg\neg K\varphi,\ \neg\neg(K\varphi \to K\psi) \vdash \neg\neg K\psi.$$

Its second premise is $K(\varphi \to \psi)$ itself, by the clause for →.

② The first premise follows from $K\varphi$ by ¬¬I (Proposition 2.5.1 (a), derived without RAA): a box on $\neg K\varphi$ clashes at once with $K\varphi$.

③ By (a), $K\psi = \neg\neg X$, so the conclusion of ① is $\neg\neg\neg\neg X = \neg\neg\neg(\neg X)$. Example (Three negations become one) derives $\neg\neg\neg P \vdash \neg P$ without RAA; with $\neg X$ in place of P it gives $\neg\neg\neg(\neg X) \vdash \neg(\neg X)$, that is, $\neg\neg K\psi \vdash K\psi$.

Chaining ②, ①, and ③ gives $K(\varphi \to \psi),\ K\varphi \vdash K\psi$, and no step is RAA: the derived rules used were all derived without it. In words, modus ponens survives the translation intuitionistically. (This is the translation that puts ¬¬ in front of every subformula. In the Gödel–Gentzen translation of the history above, $\varphi \to \psi$ translates without an outer ¬¬, and modus ponens survives trivially.)

Marking scheme. (10 points) - 2: (a) every clause of K begins with ¬¬, read off the outermost clause. - 3: (b) the Chapter 2 sequent instantiated with $K\varphi$, $K\psi$, and its second premise recognized as $K(\varphi \to \psi)$. - 2: (b) $\neg\neg K\varphi$ from $K\varphi$ by ¬¬I, named as RAA-free. - 3: (b) $\neg\neg K\psi \vdash K\psi$ by the triple-negation result applied to $\neg X$, using (a). Fatal errors (0 for the step): removing the extra negations by ¬¬E (a classical step); assuming $K\psi$ is a letter.

5.7.9.4 R. Review

Exercise. R1

A set A of real numbers **has a maximum** if some element m of A satisfies $a \le m$ for every $a \in A$. Write "A has no maximum" in positive form, with no ¬ in front of a quantifier, and prove that the interval $(0, 1)$ has no maximum.

Solution. This uses the method of Chapter 4, Section 4.9: negate quantifier by quantifier, keeping the bounds (Proposition 4.9.3; compare Example (Three negations from analysis)). In symbols, "A has a maximum" is $\exists m \in A\ \forall a \in A\ (a \le m)$. Its negation becomes, step by

step,

$$\neg\exists m \in A\, \forall a \in A\, (a \leq m) \dashv\vdash \forall m \in A\, \neg\forall a \in A\, (a \leq m)$$
$$\dashv\vdash \forall m \in A\, \exists a \in A\, \neg(a \leq m),$$

by part (b) of Proposition 4.9.3 and then part (a) under the $\forall$. For real numbers $\neg(a \leq m)$ says $a > m$. So the positive form is

$$\forall m \in A\, \exists a \in A\, (a > m) :$$

every element of A is exceeded by some element of A.

Now let $A = (0, 1)$ and let $m \in (0, 1)$. Take $a := (m + 1)/2$, which depends on m. Then

$$a - m = \frac{1 - m}{2} > 0 \quad \text{and} \quad 1 - a = \frac{1 - m}{2} > 0,$$

since $m < 1$, so $m < a < 1$; with $m > 0$ this gives $0 < a < 1$. So $a \in (0, 1)$ and $a > m$. Since $m \in (0, 1)$ was arbitrary, the positive form holds for $(0, 1)$, and by the interderivability above, $(0, 1)$ has no maximum. The directions used to go back from the positive form, $\dashv$ in (a) and in (b), are derived without RAA, and $a > m$ gives $\neg(a \leq m)$ by trichotomy; so this half of the argument uses no classical step beyond the order axioms.

Exercise. R2

Give a Fitch derivation of $P \to (Q \vee R)$, $Q \to S$, $R \to S \vdash P \to S$, and write it as one sentence of prose.

Solution. This uses Chapter 1's rules: $\to$I (Definition 1.4.1), $\to$E (Definition 1.3.1), and $\vee$E (Definition 1.6.1). The goal is an implication, so a box on P with the goal S. Inside it, the first premise gives $Q \vee R$, which we use by cases; each case reaches S through one of the other premises.

1	$P \to (Q \vee R)$	
2	$Q \to S$	
3	$R \to S$	
4	$\quad P$	
5	$\quad Q \vee R$	$\to$E 1, 4
6	$\quad\quad Q$	
7	$\quad\quad S$	$\to$E 2, 6
8	$\quad\quad R$	
9	$\quad\quad S$	$\to$E 3, 8
10	$\quad S$	$\vee$E 5, 6–7, 8–9
11	$P \to S$	$\to$I 4–10

Line 10 may cite line 5, since both lie in the box of line 4, which is still open; the case boxes

end with S as lines of their own. In prose: "Suppose P; then Q or R holds by the first premise, and S follows in the case Q by the second premise and in the case R by the third, so P implies S." Each premise is cited where it is used, as Definition 5.3.2 requires for every $\to$E on a hypothesis; the first premise is also the reason the two cases exhaust. The closing "so P implies S" is the closing line of $\to$I, which Definition 5.3.2 allows to go unsaid.

Exercise. R3

Decide with a truth table whether $(P \to Q) \to Q \models P \vee Q$, reading P and Q as atoms. If the entailment holds, give a derivation of $(P \to Q) \to Q \vdash P \vee Q$.

Solution. This item reaches back within this volume, since there is no earlier one: it uses the tables of Chapter 3 and the derivations of Chapter 2.

The table. Work backwards from F, as in Example (Working backwards from F) and the entailments of Definition 3.4.3. A refuting row needs $P \vee Q$ false, so P and Q are both false. In that row $P \to Q$ is true, so the premise $(P \to Q) \to Q$ is "T $\to$ F", which is false. So no row makes the premise true and the conclusion false, and the entailment holds. The full table agrees:

P	Q	$P \to Q$	$(P \to Q) \to Q$	$P \vee Q$
T	T	T	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

The derivation. The goal is a disjunction with neither side in hand, and the premise gives Q only once $P \to Q$ is available. We are stuck, so we open an RAA box on $\neg(P \vee Q)$ (Definition 2.5.1). Inside it, $P \to Q$ is built by a box on P: $\vee$I gives $P \vee Q$, which clashes with the assumption, and $\bot$E writes Q. Then the premise gives Q, and $\vee$I clashes again.

1	$(P \to Q) \to Q$	
2	$\quad \neg(P \vee Q)$	
3	$\quad\quad P$	
4	$\quad\quad P \vee Q$	$\vee$I 3
5	$\quad\quad \bot$	$\neg$E 4, 2
6	$\quad\quad Q$	$\bot$E 5
7	$\quad P \to Q$	$\to$I 3–6
8	$\quad Q$	$\to$E 1, 7
9	$\quad P \vee Q$	$\vee$I 8
10	$\quad \bot$	$\neg$E 9, 2
11	$P \vee Q$	RAA 2–10

The one classical line is line 11. Closing the same box by $\neg$I instead gives $\neg\neg(P \vee Q)$, and

then the derivation is intuitionistic: the premise yields the double negation of its classical conclusion, the pattern of Glivenko's theorem in the remark of this section.

No derivation of this sequent avoids RAA or one of its equivalents; we state this without proof, as Chapter 2 did for similar sequents. (With $\bot$ in place of Q, the sequent would turn $\neg\neg P$ into P without any further classical step, and Chapter 3's optional theorem in Section 3.5 shows that $\neg\neg P \vdash P$ needs RAA.)

Exercise. R4

Prove that for every integer n, $n^3 - n$ is divisible by 3.

Solution. The tool is proof by cases on the remainder of n on division by 3: every integer n is $3k$, $3k+1$, or $3k+2$ for some integer k. This is a school fact, of the same kind as "every integer is even or odd" in Chapter 2. It is $\vee$E with three cases (Definition 1.6.1, used twice), and in each case $\exists$I (Definition 4.5.1) with an explicit witness. First factor:

$$n^3 - n = (n-1)\,n\,(n+1).$$

Let n be an integer, and let k be an integer with $n = 3k$, $n = 3k+1$, or $n = 3k+2$.

Case $n = 3k$. Then $n^3 - n = 3 \cdot \big((3k-1)\,k\,(3k+1)\big)$.

Case $n = 3k+1$. Then $n^3 - n = (3k)(3k+1)(3k+2) = 3 \cdot \big(k\,(3k+1)(3k+2)\big)$.

Case $n = 3k+2$. Then $n^3 - n = (3k+1)(3k+2)(3k+3) = 3 \cdot \big((3k+1)(3k+2)(k+1)\big)$.

In each case $n^3 - n = 3j$ for the integer j in brackets, so $3 \mid n^3 - n$. Since n was arbitrary, this holds for every integer n.

Why the look-alikes are worse. A proof by contradiction ("suppose $3 \nmid n^3 - n$ ...") would have to reach exactly this case analysis and then clash with its assumption at the last line: a direct proof in costume (Example (A direct proof in costume)). Induction is closer, but it fits the statement badly: with base case $P(0)$ and the step from n to $n+1$ (Axiom 3.2.1) it proves the claim only for $n \in \mathbb{N}$, and the negative integers $n = -m$, $m \in \mathbb{Z}_{>0}$, need a second argument, for instance $(-m)^3 - (-m) = -(m^3 - m)$. The case split covers all integers at once.